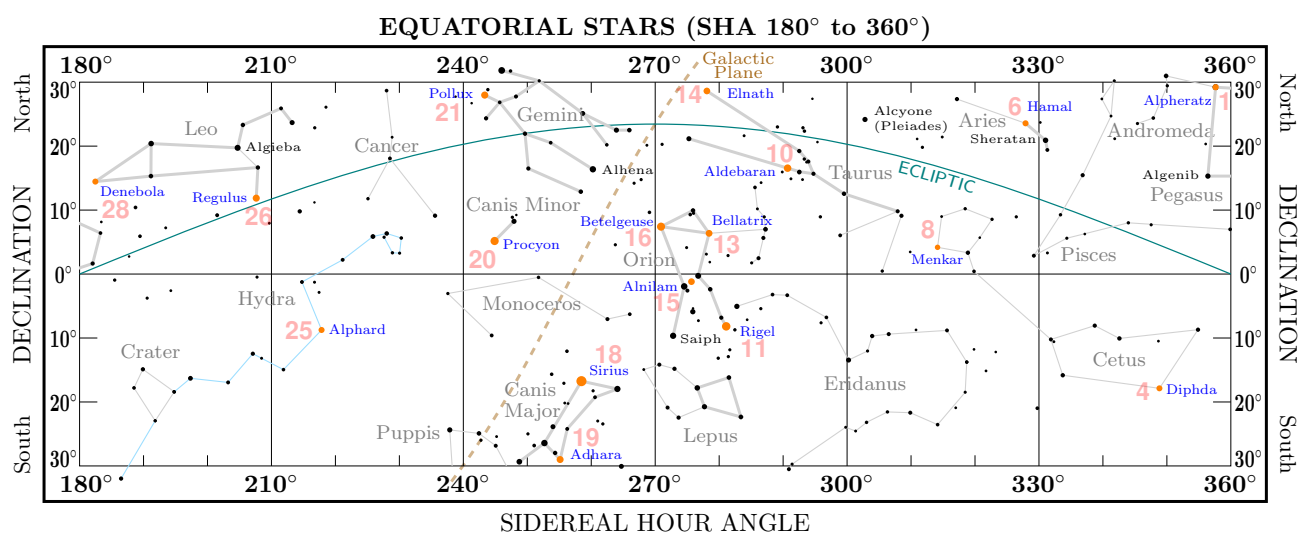
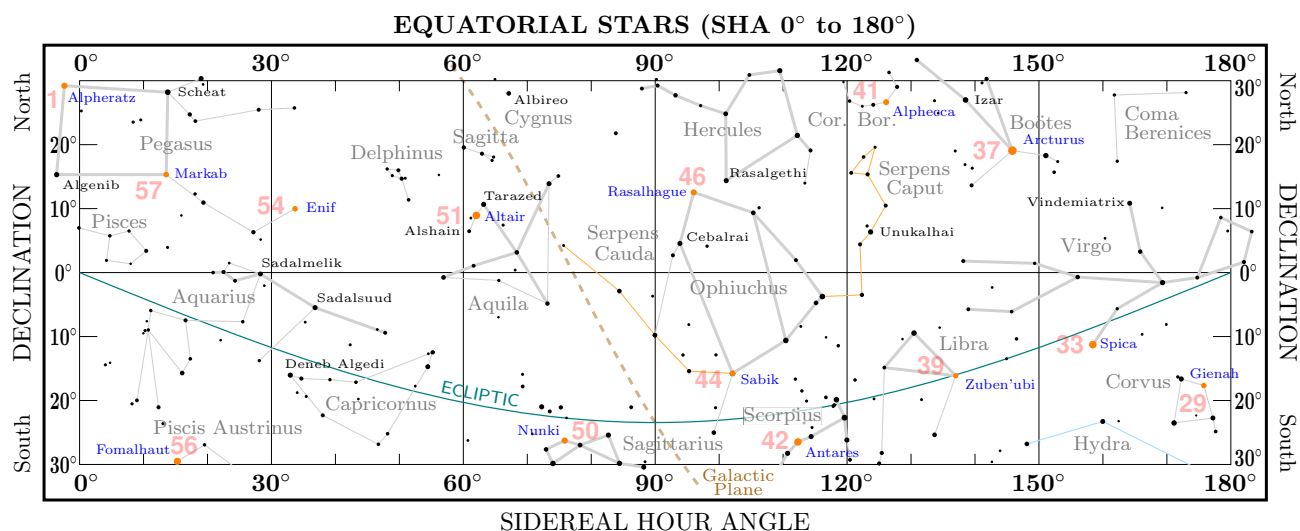




THE NAUTICAL ALMANAC

2025

Author: Andrew BAUER
Original concept from: Enno RODEGERDTS

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Disclaimer: These are computer generated tables - use them at your own risk. The accuracy has been randomly checked with JPL HORIZONS System, but cannot be guaranteed. The author claims no liability for any consequences arising from use of these tables. Besides, this publication only contains the 'daily pages' of the Nautical Almanac: an official version of the Nautical Almanac is indispensable.

Information in the data page footers

Information pertaining to the IERS EOP data has been added to the odd data page footers if using MiKTeX or TeX Live (2020 or later). The International Earth Rotation Service (IERS) provides accurate data (updated weekly) on the Earth Orientation Parameters (EOP).

Earth's speed of rotation is not constant, i.e. the day length fluctuates.¹ This is due to *internal torques* caused by relative movements and mass redistribution of Earth's core, mantle, oceans, atmosphere, and cryosphere. This has an immediate impact on the GHA values of all celestial objects.

The IERS monitors and measures several parameters taking the actual speed of Earth's rotation into account. Their measured data begins on 2nd January 1973. Predictive data begins following the last day of (obtained) data and extends about 360 days into the future. (The IERS results are published with a delay of about 18-hours between the date of publication and the last available date with measured EOP.²) These Nautical Almanac daily pages take the (measured or predicted) UT1-UTC values into account providing highly accurate navigational data especially if the predictions are fairly recent.

As long as either measured or predicted data is available the footer will show:

[IERS Earth Orientation data as of dd-mmm-yyyy](#)

This indicates that IERS EOP data is in use - older dates are measured; newer dates are predictions.

If the final date of IERS prediction data is on the current data page, the footer shows:

[IERS Earth Orientation predictions end dd-mmm-yyyy](#)

Pages with dates beyond the final date of IERS prediction data have the following footer:

[No IERS EOP prediction data available](#)

Skyfield then defaults to using the ΔT and leap second files that ship with Skyfield internally.

The footers mentioned are only displayed as long as `'useIERS = True'` is set in `config.py` to enable use of IERS EOP data.

Brief historical overview

The story begins with the XEphem astronomical library, which is declared 'end of life' by its author, Elwood Charles Downey, as no further updates are planned. He generously gave permission for use of XEphem code in Ephem (also known as Pyephem), an astronomical library authored by Brandon Rhodes. Enno Rodegerdts (<https://sv-inua.net/>) created the original Nautical Almanac 'daily pages' in Pyalmanac using Python 2 and LaTeX. After contacting him I obtained permission for its future enhancement and maintenance. Pyalmanac uses Ephem.

Meanwhile Brandon Rhodes was working on a far more sophisticated astronomical library, Skyfield. This was 'state of the art' and clearly surpassed the 'Jean Meeus'-based Pyephem/Ephem. Skyfield uses NASA's NAIF (Navigation and Ancillary Information Facility) SPICE algorithms. The results agree with those from the HORIZONS System (*operated by NASA JPL (Jet Propulsion Laboratory) SSD (Solar System Dynamics) group, not by NAIF*). This in turn implies that celestial positions calculated by Skyfield agree with those generated by the United States Naval Observatory and their *Astronomical Almanac* to within 0.0005 arcseconds (half a milliarcsecond).

Pyephem was then in 'maintenance mode'. Clearly Pyalmanac needed adaptation to use Skyfield, and thus SFalmanac was born. However its performance was poor regarding the calculation of 'events' such as: sunrise, sunset, moonrise, moonset, civil twilight start/end and nautical twilight start/end. An interim (faster) solution was required.

A hybrid application, originally named Skyalmanac, was developed using Ephem to calculate 'events' and Skyfield for the rest. This was indeed much faster at the cost of poorer 'event time' data. It took a while to find a better solution: multiprocessing, which was built into SFalmanac. This now could compare to the execution times in Pyalmanac but with improved data.

New functionality was added to SFalmanac, e.g. lunar phase as a graphic; Lunar Distance tables and charts. The original Skyalmanac is deprecated and has now been replaced with the latest SFalmanac code, so Skyalmanac and SFalmanac are now identical apart from the name. Since April 2019 <http://thenauticalalmanac.com> has been publishing Celestial Navigation related material with software provided here.

¹https://en.wikipedia.org/wiki/Day_length_fluctuations

²<https://hpiers.obspm.fr/eoppc/bul/bulb/explanatory.html>

January 01, 02, 03 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	100°54.0	130°30.3	S13°35.2	335°46.4	N23°32.7	29°01.0	N21°47.2	114°23.0	S07°55.0	Alpheratz	357°34.8	29°13.9
1	115°56.4	145°30.2	34.1	350°49.7	33.0	44°03.8	47.2	129°25.3	54.9	Ankaa	353°07.2	-42°10.4
2	130°58.9	160°30.2	33.0	5°53.1	33.3	59°06.5	47.2	144°27.6	54.8	Schedar	349°31.1	56°40.7
3	146°01.4	175°30.1	•• 31.9	20°56.4	•• 33.5	74°09.3	•• 47.2	159°29.9	•• 54.8	Diphda	348°47.3	-17°51.1
4	161°03.8	190°30.0	30.8	35°59.7	33.8	89°12.0	47.1	174°32.2	54.7	Achernar	335°20.0	-57°06.9
5	176°06.3	205°30.0	29.7	51°03.0	34.1	104°14.8	47.1	189°34.4	54.6	Hamal	327°51.1	23°34.9
6	191°08.8	220°29.9	S13°28.6	66°06.3	N23°34.4	119°17.5	N21°47.1	204°36.7	S07°54.5	Polaris	313°44.0	89°22.4
7	206°11.2	235°29.9	27.5	81°09.6	34.6	134°20.3	47.1	219°39.0	54.5	Acamar	315°11.5	-40°12.4
8	221°13.7	250°29.8	26.4	96°13.0	34.9	149°23.0	47.0	234°41.3	54.4	Menkar	314°06.0	4°11.3
9	236°16.2	265°29.7	•• 25.4	111°16.3	•• 35.2	164°25.8	•• 47.0	249°43.6	•• 54.3	Mirfak	308°27.9	49°57.2
10	251°18.6	280°29.7	24.3	126°19.6	35.5	179°28.5	47.0	264°45.9	54.2	Aldebaran	290°39.3	16°33.6
11	266°21.1	295°29.6	23.2	141°22.9	35.7	194°31.3	47.0	279°48.2	54.1	Rigel	281°03.6	-8°10.4
12	281°23.5	310°29.6	S13°22.1	156°26.2	N23°36.0	209°34.0	N21°46.9	294°50.5	S07°54.1	Capella	280°21.4	46°01.4
13	296°26.0	325°29.5	21.0	171°29.6	36.3	224°36.8	46.9	309°52.8	54.0	Bellatrix	278°22.6	6°22.3
14	311°28.5	340°29.5	19.9	186°32.9	36.6	239°39.5	46.9	324°55.0	53.9	Elnath	278°01.5	28°37.7
15	326°30.9	355°29.4	•• 18.8	201°36.2	•• 36.8	254°42.3	•• 46.9	339°57.3	•• 53.8	Alnilam	275°37.4	-1°11.2
16	341°33.4	10°29.4	17.7	216°39.5	37.1	269°45.0	46.9	354°59.6	53.7	Betelgeuse	270°51.8	7°24.7
17	356°35.9	25°29.3	16.6	231°42.9	37.4	284°47.7	46.8	10°01.9	53.7	Canopus	263°51.8	-52°42.5
18	11°38.3	40°29.3	S13°15.5	246°46.2	N23°37.7	299°50.5	N21°46.8	25°04.2	S07°53.6	Sirius	258°25.9	-16°45.0
19	26°40.8	55°29.2	14.4	261°49.5	37.9	314°53.2	46.8	40°06.5	53.5	Adhara	255°05.5	-29°00.3
20	41°43.3	70°29.2	13.3	276°52.9	38.2	329°56.0	46.8	55°08.8	53.4	Procyon	244°50.5	5°09.6
21	56°45.7	85°29.1	•• 12.2	291°56.2	•• 38.5	344°58.7	•• 46.7	70°11.1	•• 53.4	Pollux	243°16.9	27°57.9
22	71°48.2	100°29.1	11.1	306°59.5	38.8	0°01.5	46.7	85°13.4	53.3	Avior	234°14.1	-59°35.2
23	86°50.7	115°29.1	10.0	322°02.9	39.0	15°04.2	46.7	100°15.6	53.2	Suhail	222°46.0	-43°31.9
Mer.pass. 17:14		ν -0.1' d -1.1' m -4.45		ν 3.3' d 0.3' m -1.21		ν 2.7' d -0.0' m -2.74		ν 2.3' d -0.1' m 1.05		Miaplacidus	221°37.5	-69°48.9
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	101°53.1	130°29.0	S13°08.9	337°06.2	N23°39.3	30°07.0	N21°46.7	115°17.9	S07°53.1	Alphard	217°47.5	-8°46.0
1	116°55.6	145°29.0	07.8	352°09.6	39.6	45°09.7	46.6	130°20.2	53.0	Regulus	207°34.2	11°50.6
2	131°58.0	160°28.9	06.7	7°12.9	39.9	60°12.4	46.6	145°22.5	53.0	Dubhe	193°40.7	61°36.7
3	147°00.5	175°28.9	•• 05.6	22°16.2	•• 40.1	75°15.2	•• 46.6	160°24.8	•• 52.9	Denebola	182°24.9	14°25.8
4	162°03.0	190°28.9	04.5	37°19.6	40.4	90°17.9	46.6	175°27.1	52.8	Gienah	175°43.6	-17°40.8
5	177°05.4	205°28.8	03.4	52°22.9	40.7	105°20.7	46.5	190°29.4	52.7	Acrux	173°00.1	-63°14.0
6	192°07.9	220°28.8	S13°02.3	67°26.3	N23°41.0	120°23.4	N21°46.5	205°31.6	S07°52.6	Gacrux	171°51.7	-57°14.9
7	207°10.4	235°28.7	01.2	82°29.6	41.3	135°26.2	46.5	220°33.9	52.6	Alioth	166°13.0	55°49.1
8	222°12.8	250°28.7	13°00.1	97°33.0	41.5	150°28.9	46.5	235°36.2	52.5	Spica	158°22.4	-11°17.5
9	237°15.3	265°28.7	12°59.0	112°36.3	•• 41.8	165°31.6	•• 46.4	250°38.5	•• 52.4	Alkaid	152°52.1	49°11.0
10	252°17.8	280°28.6	57.9	127°39.7	42.1	180°34.4	46.4	265°40.8	52.3	Hadar	148°36.3	-60°29.3
11	267°20.2	295°28.6	56.8	142°43.0	42.4	195°37.1	46.4	280°43.1	52.2	Menkent	147°57.8	-36°29.4
12	282°22.7	310°28.6	S12°55.7	157°46.4	N23°42.6	210°39.9	N21°46.4	295°45.4	S07°52.2	Arcturus	145°48.1	19°03.0
13	297°25.2	325°28.6	54.5	172°49.7	42.9	225°42.6	46.3	310°47.7	52.1	Rigil Kent.	139°40.7	-60°56.1
14	312°27.6	340°28.5	53.4	187°53.1	43.2	240°45.3	46.3	325°49.9	52.0	Kochab	137°20.4	74°02.8
15	327°30.1	355°28.5	•• 52.3	202°56.4	•• 43.5	255°48.1	•• 46.3	340°52.2	•• 51.9	Zuben'ubi	136°56.2	-16°08.7
16	342°32.5	10°28.5	51.2	217°59.8	43.7	270°50.8	46.3	355°54.5	51.8	Alphecca	126°04.0	26°37.7
17	357°35.0	25°28.4	50.1	233°03.1	44.0	285°53.6	46.3	10°56.8	51.8	Antares	112°16.2	-26°29.2
18	12°37.5	40°28.4	S12°49.0	248°06.5	N23°44.3	300°56.3	N21°46.2	25°59.1	S07°51.7	Atria	107°10.9	-69°04.2
19	27°39.9	55°28.4	47.9	263°09.8	44.6	315°59.0	46.2	41°01.4	51.6	Sabik	102°03.1	-15°45.4
20	42°42.4	70°28.4	46.8	278°13.2	44.8	331°01.8	46.2	56°03.6	51.5	Shaula	96°10.8	-37°07.3
21	57°44.9	85°28.3	•• 45.7	293°16.6	•• 45.1	346°04.5	•• 46.2	71°05.9	•• 51.4	Rasalhague	95°58.9	12°32.4
22	72°47.3	100°28.3	44.6	308°19.9	45.4	1°07.2	46.1	86°08.2	51.4	Eltanin	90°42.7	51°29.0
23	87°49.8	115°28.3	43.5	323°23.3	45.7	16°10.0	46.1	101°10.5	51.3	Kaus Aust.	83°32.9	-34°22.4
Mer.pass. 17:10		ν -0.0' d -1.1' m -4.46		ν 3.3' d 0.3' m -1.24		ν 2.7' d -0.0' m -2.74		ν 2.3' d -0.1' m 1.06		Vega	80°33.7	38°48.3
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	102°52.3	130°28.3	S12°42.4	338°26.6	N23°45.9	31°12.7	N21°46.1	116°12.8	S07°51.2	Nunki	75°48.2	-26°16.0
1	117°54.7	145°28.3	41.2	353°30.0	46.2	46°15.5	46.1	131°15.1	51.1	Altair	62°00.3	8°56.0
2	132°57.2	160°28.3	40.1	8°33.4	46.5	61°18.2	46.0	146°17.4	51.0	Peacock	53°06.3	-56°39.4
3	147°59.6	175°28.2	•• 39.0	23°36.7	•• 46.8	76°20.9	•• 46.0	161°19.6	•• 50.9	Deneb	49°26.2	45°22.2
4	163°02.1	190°28.2	37.9	38°40.1	47.1	91°23.7	46.0	176°21.9	50.9	Enif	33°39.1	9°59.4
5	178°04.6	205°28.2	36.8	53°43.5	47.3	106°26.4	46.0	191°24.2	50.8	Al Na'ir	27°33.2	-46°50.6
6	193°07.0	220°28.2	S12°35.7	68°46.8	N23°47.6	121°29.1	N21°45.9	206°26.5	S07°50.7	Fomalhaut	15°14.7	-29°29.6
7	208°09.5	235°28.2	34.6	83°50.2	47.9	136°31.9	45.9	221°28.8	50.6	Scheat	13°45.4	28°13.2
8	223°12.0	250°28.2	33.5	98°53.6	48.2	151°34.6	45.9	236°31.1	50.5	Markab	13°30.1	25°20.4
9	238°14.4	265°28.2	•• 32.3	113°57.0	•• 48.4	166°37.3	•• 45.9	251°33.3	•• 50.5	Jan 01 Wed		
10	253°16.9	280°28.1	31.2	129°00.3	48.7	181°40.1	45.9	266°35.6	50.4	Venus	29°36.3	15:18
11	268°19.4	295°28.1	30.1	144°03.7	49.0	196°42.8	45.8	281°37.9	50.3	Mars	234°52.5	01:37
12	283°21.8	310°28.1	S12°29.0	159°07.1	N23°49.3	211°45.5	N21°45.8	296°40.2	S07°50.2	Jupiter	288°07.1	22:00
13	298°24.3	325°28.1	27.9	174°10.5	49.5	226°48.3	45.8	311°42.5	50.1	Saturn	13°29.0	16:20
14	313°26.8	340°28.1	26.8	189°13.8	49.8	241°51.0	45.8	326°44.8	50.1	Jan 02 Thu		
15	328°29.2	355°28.1	•• 25.7	204°17.2	•• 50.1	256°53.7	•• 45.7	341°47.0	•• 50.0	Venus	28°35.9	15:18
16	343°31.7	10°28.1	24.5	219°20.6	50.4	271°56.5	45.7	356°49.3	49.9	Mars	235°13.1	01:31
17	358°34.1	25°28.1	23.4	234°24.0	50.6	286°59.2	45.7	11°51.6	49.8	Jupiter	288°13.8	21:56
18	13°36.6	40°28.1	S12°22.3	249°27.4	N23°50.9	302°01.9	N21°45.7	26°53.9	S07°49.7	Saturn	13°24.8	16:16
19	28°39.1	55°28.1	21.2	264°30.8	51.2	317°04.7	45.6	41°56.2	49.6	Jan 03 Fri		
20	43°41.5	70°28.1	20.1	279°34.1	51.5	332°07.4	45.6	56°58.4	49.6	Venus	27°36.0	15:18
21	58°44.0	85°28.1	•• 19.0	29								

Sun				Moon				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	179°08.4	S22°59.9		164°13.2	7.1'	S25°51.6	-7.2'	57.4'
1	194°08.1	59.7		178°39.2	7.1'	25°44.4	-7.4'	57.4'
2	209°07.8	59.5		193°05.4	7.2'	25°37.1	-7.5'	57.4'
3	224°07.5	· · 59.3		207°31.6	7.2'	25°29.5	-7.7'	57.4'
4	239°07.2	59.1		221°57.8	7.3'	25°21.9	-7.8'	57.5'
5	254°06.9	58.8		236°24.1	7.3'	25°14.0	-8.0'	57.5'
6	269°06.6	S22°58.6		250°50.4	7.4'	S25°06.1	-8.1'	57.5'
7	284°06.3	58.4		265°16.8	7.5'	24°57.9	-8.3'	57.5'
8	299°06.0	58.2		279°43.3	7.5'	24°49.7	-8.4'	57.5'
9	314°05.7	· · 58.0		294°09.8	7.6'	24°41.2	-8.6'	57.6'
10	329°05.4	57.8		308°36.4	7.6'	24°32.6	-8.7'	57.6'
11	344°05.1	57.6		323°03.0	7.7'	24°23.9	-8.9'	57.6'
12	359°04.8	S22°57.4		337°29.7	7.8'	S24°15.0	-9.0'	57.6'
13	14°04.5	57.1		351°56.5	7.8'	24°06.0	-9.2'	57.6'
14	29°04.2	56.9		6°23.3	7.9'	23°56.9	-9.3'	57.7'
15	44°03.9	· · 56.7		20°50.1	7.9'	23°47.6	-9.4'	57.7'
16	59°03.7	56.5		35°17.1	8.0'	23°38.1	-9.6'	57.7'
17	74°03.4	56.3		49°44.1	8.1'	23°28.5	-9.7'	57.7'
18	89°03.1	S22°56.0		64°11.2	8.1'	S23°18.8	-9.9'	57.7'
19	104°02.8	55.8		78°38.3	8.2'	23°08.9	-10.0'	57.8'
20	119°02.5	55.6		93°05.5	8.3'	22°58.9	-10.1'	57.8'
21	134°02.2	· · 55.4		107°32.8	8.3'	22°48.8	-10.3'	57.8'
22	149°01.9	55.2		122°00.1	8.4'	22°38.5	-10.4'	57.8'
23	164°01.6	54.9		136°27.5	8.5'	22°28.1	-10.5'	57.8'
SD = 16.3' d = -0.2'				SD = 15.6'				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	179°01.3	S22°54.7		150°55.0	8.5'	S22°17.6	-10.7'	57.8'
1	194°01.0	54.5		165°22.5	8.6'	22°06.9	-10.8'	57.9'
2	209°00.7	54.3		179°50.1	8.7'	21°56.1	-10.9'	57.9'
3	224°00.4	· · 54.0		194°17.8	8.7'	21°45.1	-11.1'	57.9'
4	239°00.1	53.8		208°45.5	8.8'	21°34.1	-11.2'	57.9'
5	253°59.9	53.6		223°13.3	8.9'	21°22.9	-11.3'	57.9'
6	268°59.6	S22°53.3		237°41.2	9.0'	S21°11.6	-11.4'	58.0'
7	283°59.3	53.1		252°09.2	9.0'	21°00.2	-11.6'	58.0'
8	298°59.0	52.9		266°37.2	9.1'	20°48.6	-11.7'	58.0'
9	313°58.7	· · 52.6		281°05.3	9.2'	20°36.9	-11.8'	58.0'
10	328°58.4	52.4		295°33.4	9.2'	20°25.1	-11.9'	58.0'
11	343°58.1	52.2		310°01.7	9.3'	20°13.2	-12.0'	58.0'
12	358°57.8	S22°51.9		324°30.0	9.4'	S20°01.2	-12.1'	58.1'
13	13°57.5	51.7		338°58.3	9.4'	19°49.0	-12.3'	58.1'
14	28°57.2	51.5		353°26.8	9.5'	19°36.8	-12.4'	58.1'
15	43°56.9	· · 51.2		7°55.3	9.6'	19°24.4	-12.5'	58.1'
16	58°56.7	51.0		22°23.9	9.7'	19°11.9	-12.6'	58.1'
17	73°56.4	50.8		36°52.5	9.7'	18°59.3	-12.7'	58.1'
18	88°56.1	S22°50.5		51°21.3	9.8'	S18°46.6	-12.8'	58.2'
19	103°55.8	50.3		65°50.0	9.9'	18°33.8	-12.9'	58.2'
20	118°55.5	50.0		80°18.9	9.9'	18°20.9	-13.0'	58.2'
21	133°55.2	· · 49.8		94°47.8	10.0'	18°07.8	-13.1'	58.2'
22	148°54.9	49.5		109°16.8	10.1'	17°54.7	-13.2'	58.2'
23	163°54.6	49.3		123°45.9	10.1'	17°41.5	-13.3'	58.2'
SD = 16.3' d = -0.2'				SD = 15.8'				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	178°54.3	S22°49.1		138°15.0	10.2'	S17°28.2	-13.4'	58.3'
1	193°54.1	48.8		152°44.2	10.3'	17°14.7	-13.5'	58.3'
2	208°53.8	48.6		167°13.5	10.3'	17°01.2	-13.6'	58.3'
3	223°53.5	· · 48.3		181°42.8	10.4'	16°47.6	-13.7'	58.3'
4	238°53.2	48.1		196°12.2	10.5'	16°33.9	-13.8'	58.3'
5	253°52.9	47.8		210°41.7	10.5'	16°20.1	-13.9'	58.3'
6	268°52.6	S22°47.6		225°11.2	10.6'	S16°06.2	-14.0'	58.3'
7	283°52.3	47.3		239°40.8	10.7'	15°52.2	-14.1'	58.4'
8	298°52.0	47.1		254°10.4	10.7'	15°38.1	-14.2'	58.4'
9	313°51.8	· · 46.8		268°40.1	10.8'	15°24.0	-14.2'	58.4'
10	328°51.5	46.6		283°09.9	10.8'	15°09.7	-14.3'	58.4'
11	343°51.2	46.3		297°39.8	10.9'	14°55.4	-14.4'	58.4'
12	358°50.9	S22°46.1		312°09.6	11.0'	S14°41.0	-14.5'	58.4'
13	13°50.6	45.8		326°39.6	11.0'	14°26.5	-14.6'	58.4'
14	28°50.3	45.6		341°09.6	11.1'	14°11.9	-14.7'	58.5'
15	43°50.0	· · 45.3		355°39.7	11.1'	13°57.2	-14.7'	58.5'
16	58°49.7	45.0		10°09.8	11.2'	13°42.5	-14.8'	58.5'
17	73°49.5	44.8		24°40.0	11.2'	13°27.7	-14.9'	58.5'
18	88°49.2	S22°44.5		39°10.2	11.3'	S13°12.8	-15.0'	58.5'
19	103°48.9	44.3		53°40.5	11.3'	12°57.8	-15.0'	58.5'
20	118°48.6	44.0		68°10.9	11.4'	12°42.8	-15.1'	58.5'
21	133°48.3	· · 43.7		82°41.3	11.4'	12°27.7	-15.2'	58.5'
22	148°48.0	43.5		97°11.7	11.5'	12°12.5	-15.2'	58.6'
23	163°47.8	43.2		111°42.2	11.5'	11°57.3	-15.3'	58.6'
SD = 16.3' d = -0.2'				SD = 15.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:23	10:39	■	■	13:30	15:46
N 70°	08:04	09:47	■	■	14:21	16:05
68°	07:49	09:16	■	■	14:53	16:20
66°	07:37	08:52	10:25	13:43	15:17	16:32
64°	07:26	08:33	09:48	14:20	15:35	16:42
62°	07:17	08:18	09:22	14:47	15:50	16:52
60°	07:09	08:05	09:02	15:07	16:03	17:00
N 58°	07:02	07:54	08:45	15:24	16:15	17:07
56°	06:55	07:44	08:31	15:38	16:24	17:13
54°	06:50	07:35	08:19	15:50	16:33	17:19
52°	06:44	07:27	08:08	16:01	16:41	17:24
50°	06:39	07:20	07:58	16:10	16:48	17:29
45°	06:28	07:05	07:38	16:30	17:04	17:41
N 40°	06:18	06:52	07:22	16:46	17:17	17:51
35°	06:09	06:40	07:08	17:00	17:28	18:00
30°	06:00	06:30	06:56	17:12	17:38	18:08
20°	05:44	06:12	06:36	17:33	17:57	18:24
N 10°	05:28	05:55	06:17	17:51	18:14	18:40
0°	05:12	05:38	06:00	18:08	18:30	18:56
S 10°	04:53	05:20	05:43	18:25	18:48	19:15
20°	04:31	05:00	05:25	18:43	19:08	19:37
30°	04:03	04:36	05:03	19:05	19:32	20:05
35°	03:45	04:21	04:51	19:18	19:47	20:23
40°	03:22	04:03	04:36	19:32	20:05	20:46
45°	02:53	03:41	04:18	19:50	20:26	21:15
S 50°	02:09	03:13	03:57	20:11	20:55	21:58
52°	01:43	02:58	03:46	20:22	21:10	22:24
54°	01:04	02:41	03:34	20:34	21:27	23:02
56°	—:—	02:20	03:20	20:47	21:47	—:—
58°	—:—	01:52	03:04	21:03	22:15	—:—
S 60°	—:—	01:10	02:45	21:22	22:56	—:—

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	■	■	12:47	■	■	18:17
N 70°	■	13:51	12:17	■	15:25	18:45
68°	■	12:41	11:55	■	16:33	19:05
66°	■	12:04	11:38	■	17:09	19:21
64°	12:03	11:38	11:23	15:16	17:34	19:34
62°	11:24	11:17	11:11	15:55	17:54	19:44
60°	10:57	11:00	11:01	16:21	18:09	19:54
N 58°	10:35	10:46	10:52	16:42	18:23	20:02
56°	10:18	10:33	10:43	16:59	18:34	20:09
54°	10:03	10:22	10:36	17:14	18:45	20:15
52°	09:50	10:13	10:30	17:26	18:53	20:20
50°	09:38	10:04	10:24	17:37	19:01	20:25
45°	09:14	09:46	10:11	18:00	19:18	20:36
N 40°	08:55	09:31	10:01	18:19	19:32	20:45
35°	08:39	09:18	09:52	18:34	19:44	20:53
30°	08:25	09:07	09:44	18:47	19:54	20:59
20°	08:01	08:47	09:30	19:09	20:11	21:11
N 10°	07:40	08:30	09:18	19:29	20:26	21:20
0°	07:20	08:15	09:06	19:47	20:40	21:30
S 10°	07:01	07:59	08:55	20:04	20:53	21:39

January 04, 05, 06 UT (Sat., Sun., Mon.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	103°51.4	130°28.1	S12°15.6	339°47.7	N23°52.6	32°18.3	N21°45.5	117°07.6	S07°49.2					Alpheratz	357°34.9	29°13.8	
1	118°53.9	145°28.1	14.5	354°51.1	52.8	47°21.1	45.5	132°09.9	49.2					Ankaa	353°07.2	-42°10.4	
2	133°56.3	160°28.1	13.4	9°54.5	53.1	62°23.8	45.5	147°12.1	49.1					Schedar	349°31.1	56°40.7	
3	148°58.8	175°28.1	•• 12.2	24°57.9	•• 53.4	77°26.5	•• 45.5	162°14.4	•• 49.0					Diphda	348°47.3	-17°51.1	
4	164°01.3	190°28.1	11.1	40°01.2	53.7	92°29.2	45.4	177°16.7	48.9					Achernar	335°20.0	-57°06.9	
5	179°03.7	205°28.1	10.0	55°04.6	53.9	107°32.0	45.4	192°19.0	48.8					Hamal	327°51.1	23°34.9	
6	194°06.2	220°28.2	S12°08.9	70°08.0	N23°54.2	122°34.7	N21°45.4	207°21.3	S07°48.7					Polaris	313°45.3	89°22.5	
7	209°08.6	235°28.2	07.8	85°11.4	54.5	137°37.4	45.4	222°23.5	48.7					Acamar	315°11.5	-40°12.4	
8	224°11.1	250°28.2	06.6	100°14.8	54.8	152°40.2	45.3	237°25.8	48.6					Menkar	314°06.0	4°11.3	
9	239°13.6	265°28.2	•• 05.5	115°18.2	•• 55.1	167°42.9	•• 45.3	252°28.1	•• 48.5					Mirfak	308°27.9	49°57.2	
10	254°16.0	280°28.2	04.4	130°21.6	55.3	182°45.6	45.3	267°30.4	48.4					Aldebaran	290°39.3	16°33.6	
11	269°18.5	295°28.2	03.3	145°25.0	55.6	197°48.3	45.3	282°32.7	48.3					Rigel	281°03.6	-8°10.4	
12	284°21.0	310°28.2	S12°02.1	160°28.4	N23°55.9	212°51.1	N21°45.3	297°34.9	S07°48.3					Capella	280°21.4	46°01.5	
13	299°23.4	325°28.3	12°01.0	175°31.8	56.2	227°53.8	45.2	312°37.2	48.2					Bellatrix	278°22.6	6°22.3	
14	314°25.9	340°28.3	11°59.9	190°35.2	56.4	242°56.5	45.2	327°39.5	48.1					Elnath	278°01.5	28°37.7	
15	329°28.4	355°28.3	•• 58.8	205°38.6	•• 56.7	257°59.2	•• 45.2	342°41.8	•• 48.0					Alnilam	275°37.4	-1°11.2	
16	344°30.8	10°28.3	57.6	220°42.0	57.0	273°02.0	45.2	357°44.1	47.9					Betelgeuse	270°51.8	7°24.7	
17	359°33.3	25°28.3	56.5	235°45.4	57.3	288°04.7	45.1	12°46.3	47.8					Canopus	263°51.8	-52°42.5	
18	14°35.7	40°28.3	S11°55.4	250°48.8	N23°57.5	303°07.4	N21°45.1	27°48.6	S07°47.8					Sirius	258°25.9	-16°45.0	
19	29°38.2	55°28.4	54.3	265°52.2	57.8	318°10.1	45.1	42°50.9	47.7					Adhara	255°05.5	-29°00.3	
20	44°40.7	70°28.4	53.1	280°55.7	58.1	333°12.9	45.1	57°53.2	47.6					Procyon	244°50.5	5°09.6	
21	59°43.1	85°28.4	•• 52.0	295°59.1	•• 58.4	348°15.6	•• 45.1	72°55.5	•• 47.5					Pollux	243°16.9	27°57.9	
22	74°45.6	100°28.4	50.9	311°02.5	58.6	3°18.3	45.0	87°57.7	47.4					Avior	234°14.1	-59°35.2	
23	89°48.1	115°28.5	49.8	326°05.9	58.9	18°21.0	45.0	103°00.0	47.3					Suhail	222°45.9	-43°31.9	
Mer.pass. 17:02			ν 0.0' d -1.1' m -4.48			ν 3.4' d 0.3' m -1.28			ν 2.7' d -0.0' m -2.72			ν 2.3' d -0.1' m 1.06					

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	104°50.5	130°28.5	S11°48.6	341°09.3	N23°59.2	33°23.8	N21°45.0	118°02.3	S07°47.3					
1	119°53.0	145°28.5	47.5	356°12.7	59.5	48°26.5	45.0	133°04.6	47.2					
2	134°55.5	160°28.6	46.4	11°16.1	23°59.7	63°29.2	44.9	148°06.8	47.1					
3	149°57.9	175°28.6	•• 45.2	26°19.5	24°00.0	78°31.9	•• 44.9	163°09.1	•• 47.0					
4	165°00.4	190°28.6	44.1	41°22.9	00.3	93°34.7	44.9	178°11.4	46.9					
5	180°02.9	205°28.6	43.0	56°26.4	00.5	108°37.4	44.9	193°13.7	46.8					
6	195°05.3	220°28.7	S11°41.9	71°29.8	N24°00.8	123°40.1	N21°44.9	208°16.0	S07°46.8					
7	210°07.8	235°28.7	40.7	86°33.2	01.1	138°42.8	44.8	223°18.2	46.7					
8	225°10.2	250°28.7	39.6	101°36.6	01.4	153°45.5	44.8	238°20.5	46.6					
9	240°12.7	265°28.8	•• 38.5	116°40.0	•• 01.6	168°48.3	•• 44.8	253°22.8	•• 46.5					
10	255°15.2	280°28.8	37.3	131°43.5	01.9	183°51.0	44.8	268°25.1	46.4					
11	270°17.6	295°28.9	36.2	146°46.9	02.2	198°53.7	44.8	283°27.3	46.3					
12	285°20.1	310°28.9	S11°35.1	161°50.3	N24°02.5	213°56.4	N21°44.7	298°29.6	S07°46.3					
13	300°22.6	325°28.9	33.9	176°53.7	02.7	228°59.1	44.7	313°31.9	46.2					
14	315°25.0	340°29.0	32.8	191°57.1	03.0	244°01.9	44.7	328°34.2	46.1					
15	330°27.5	355°29.0	•• 31.7	207°00.6	•• 03.3	259°04.6	•• 44.7	343°36.4	•• 46.0					
16	345°30.0	10°29.1	30.5	222°04.0	03.6	274°07.3	44.6	358°38.7	45.9					
17	0°32.4	25°29.1	29.4	237°07.4	03.8	289°10.0	44.6	13°41.0	45.8					
18	15°34.9	40°29.2	S11°28.3	252°10.8	N24°04.1	304°12.7	N21°44.6	28°43.3	S07°45.8					
19	30°37.4	55°29.2	27.1	267°14.3	04.4	319°15.5	44.6	43°45.6	45.7					
20	45°39.8	70°29.2	26.0	282°17.7	04.7	334°18.2	44.6	58°47.8	45.6					
21	60°42.3	85°29.3	•• 24.9	297°21.1	•• 04.9	349°20.9	•• 44.5	73°50.1	•• 45.5					
22	75°44.7	100°29.3	23.7	312°24.6	05.2	4°23.6	44.5	88°52.4	45.4					
23	90°47.2	115°29.4	22.6	327°28.0	05.5	19°26.3	44.5	103°54.7	45.3					
Mer.pass. 16:58			ν 0.0' d -1.1' m -4.48			ν 3.4' d 0.3' m -1.30			ν 2.7' d -0.0' m -2.72			ν 2.3' d -0.1' m 1.06		

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	105°49.7	130°29.4	S11°21.5	342°31.4	N24°05.8	34°29.0	N21°44.5	118°56.9	S07°45.2					
1	120°52.1	145°29.5	20.3	357°34.9	06.0	49°31.8	44.4	133°59.2	45.2					
2	135°54.6	160°29.5	19.2	12°38.3	06.3	64°34.5	44.4	149°01.5	45.1					
3	150°57.1	175°29.6	•• 18.1	27°41.7	•• 06.6	79°37.2	•• 44.4	164°03.8	•• 45.0					
4	165°59.5	190°29.7	16.9	42°45.2	06.8	94°39.9	44.4	179°06.0	44.9					
5	181°02.0	205°29.7	15.8	57°48.6	07.1	109°42.6	44.4	194°08.3	44.8					
6	196°04.5	220°29.8	S11°14.6	72°52.0	N24°07.4	124°45.3	N21°44.3	209°10.6	S07°44.7					
7	211°06.9	235°29.8	13.5	87°55.5	07.7	139°48.0	44.3	224°12.9	44.7					
8	226°09.4	250°29.9	12.4	102°58.9	07.9	154°50.8	44.3	239°15.1	44.6					
9	241°11.8	265°29.9	•• 11.2	118°02.4	•• 08.2	169°53.5	•• 44.3	254°17.4	•• 44.5					
10	256°14.3	280°30.0	10.1	133°05.8	08.5	184°56.2	44.3	269°19.7	44.4					
11	271°16.8	295°30.1	08.9	148°09.2	08.7	199°58.9	44.2	284°22.0	44.3					
12	286°19.2	310°30.1	S11°07.8	163°12.7	N24°09.0	215°01.6	N21°44.2	299°24.2	S07°44.2					
13	301°21.7	325°30.2	06.7	178°16.1	09.3	230°04.3	44.2	314°26.5	44.1					
14	316°24.2	340°30.3	05.5	193°19.6	09.6	245°07.0	44.2	329°28.8	44.1					
15	331°26.6	355°30.3	•• 04.4	208°23.0	•• 09.8	260°09.7	•• 44.2	344°31.0	•• 44.0					
16	346°29.1	10°30.4	03.2	223°26.5	10.1	275°12.5	44.1	359°33.3	43.9					
17	1°31.6	25°30.5	02.1	238°29.9	10.4	290°15.2	44.1	14°35.6	43.8					
18	16°34.0	40°30.5	S11°01.0	253°33.4	N24°10.6	305°17.9	N21°44.1	29°37.9	S07°43.7					
19	31°36.5	55°30.6	10°59.8	268°36.8	10.9	320°20.6	44.1	44°40.1	43.6					
20	46°39.0	70°30.7	58.7	283°40.2	11.2	335°23.3	44.0	59°42.4	43.6					
21	61°41.4	85°30.7	•• 57.5	298°43.7	•• 11.5	350°26.0	•• 44.0	74°44.7	•• 43.5					
22	76°43.9	100°30.8	56.4	313°47.1	11.7	5°28.7	44.0	89°47.0	43.4					
23	91°46.3	115°30.9	55.3	328°50.6	12.0	20°31.4	44.0	104°49.2	43.3					
Mer.pass. 16:54			ν 0.1' d -1.1' m -4.49			ν 3.4' d 0.3' m -1.32			ν 2.7' d -0.0' m -2.71			ν 2.		

DUT1 = UT1-UTC = +0.0462 sec ΔT = TT-UT1 = +69.1378 sec

2025 January 04 to Jan. 06 UT

h Sun Moon						
Sat	GHA	Dec	GHA	ν	Dec	d HP
0	178°47.5	S22°43.0	126°12.8	11.6'	S11°42.0	-15.4' 58.6'
1	193°47.2	42.7	140°43.4	11.6'	11°26.6	-15.4' 58.6'
2	208°46.9	42.4	155°14.0	11.7'	11°11.2	-15.5' 58.6'
3	223°46.6	· · 42.2	169°44.7	11.7'	10°55.7	-15.5' 58.6'
4	238°46.3	41.9	184°15.4	11.8'	10°40.2	-15.6' 58.6'
5	253°46.0	41.6	198°46.2	11.8'	10°24.6	-15.7' 58.6'
6	268°45.8	S22°41.4	213°17.0	11.9'	S10°08.9	-15.7' 58.7'
7	283°45.5	41.1	227°47.9	11.9'	09°53.2	-15.8' 58.7'
8	298°45.2	40.8	242°18.8	11.9'	09°37.4	-15.8' 58.7'
9	313°44.9	· · 40.6	256°49.8	12.0'	09°21.6	-15.9' 58.7'
10	328°44.6	40.3	271°20.7	12.0'	09°05.7	-15.9' 58.7'
11	343°44.4	40.0	285°51.8	12.1'	08°49.8	-16.0' 58.7'
12	358°44.1	S22°39.7	300°22.8	12.1'	S08°33.8	-16.0' 58.7'
13	13°43.8	39.5	314°53.9	12.1'	08°17.8	-16.1' 58.7'
14	28°43.5	39.2	329°25.0	12.2'	08°01.8	-16.1' 58.7'
15	43°43.2	· · 38.9	343°56.2	12.2'	07°45.6	-16.2' 58.7'
16	58°42.9	38.6	358°27.4	12.2'	07°29.5	-16.2' 58.8'
17	73°42.7	38.4	12°58.6	12.3'	07°13.3	-16.2' 58.8'
18	88°42.4	S22°38.1	27°29.9	12.3'	S06°57.1	-16.3' 58.8'
19	103°42.1	37.8	42°01.2	12.3'	06°40.8	-16.3' 58.8'
20	118°41.8	37.5	56°32.5	12.3'	06°24.5	-16.3' 58.8'
21	133°41.5	· · 37.3	71°03.8	12.4'	06°08.1	-16.4' 58.8'
22	148°41.3	37.0	85°35.2	12.4'	05°51.8	-16.4' 58.8'
23	163°41.0	36.7	100°06.5	12.4'	05°35.3	-16.4' 58.8'
SD = 16.3' d = -0.3' SD = 16.0'						
Sun	GHA	Dec	GHA	ν	Dec	d HP
0	178°40.7	S22°36.4	114°37.9	12.4'	S05°18.9	-16.5' 58.8'
1	193°40.4	36.1	129°09.4	12.4'	05°02.4	-16.5' 58.8'
2	208°40.1	35.8	143°40.8	12.5'	04°45.9	-16.5' 58.8'
3	223°39.9	· · 35.6	158°12.3	12.5'	04°29.4	-16.6' 58.9'
4	238°39.6	35.3	172°43.7	12.5'	04°12.8	-16.6' 58.9'
5	253°39.3	35.0	187°15.2	12.5'	03°56.3	-16.6' 58.9'
6	268°39.0	S22°34.7	201°46.7	12.5'	S03°39.7	-16.6' 58.9'
7	283°38.8	34.4	216°18.2	12.5'	03°23.0	-16.6' 58.9'
8	298°38.5	34.1	230°49.8	12.5'	03°06.4	-16.7' 58.9'
9	313°38.2	· · 33.8	245°21.3	12.5'	02°49.7	-16.7' 58.9'
10	328°37.9	33.5	259°52.9	12.6'	02°33.0	-16.7' 58.9'
11	343°37.6	33.3	274°24.4	12.6'	02°16.3	-16.7' 58.9'
12	358°37.4	S22°33.0	288°56.0	12.6'	S01°59.6	-16.7' 58.9'
13	13°37.1	32.7	303°27.5	12.6'	01°42.9	-16.7' 58.9'
14	28°36.8	32.4	317°59.1	12.6'	01°26.2	-16.7' 58.9'
15	43°36.5	· · 32.1	332°30.7	12.6'	01°09.4	-16.8' 58.9'
16	58°36.3	31.8	347°02.2	12.6'	00°52.7	-16.8' 59.0'
17	73°36.0	31.5	1°33.8	12.6'	00°35.9	-16.8' 59.0'
18	88°35.7	S22°31.2	16°05.4	12.6'	S00°19.1	-16.8' 59.0'
19	103°35.4	30.9	30°36.9	12.6'	S00°02.4	-16.8' 59.0'
20	118°35.2	30.6	45°08.5	12.5'	N00°14.4	16.8' 59.0'
21	133°34.9	· · 30.3	59°40.0	12.5'	00°31.2	16.8' 59.0'
22	148°34.6	30.0	74°11.5	12.5'	00°48.0	16.8' 59.0'
23	163°34.3	29.7	88°43.1	12.5'	01°04.7	16.8' 59.0'
SD = 16.3' d = -0.3' SD = 16.0'						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	178°34.1	S22°29.4	103°14.6	12.5'	N01°21.5	16.8' 59.0'
1	193°33.8	29.1	117°46.1	12.5'	01°38.3	16.8' 59.0'
2	208°33.5	28.8	132°17.6	12.5'	01°55.1	16.8' 59.0'
3	223°33.2	· · 28.5	146°49.1	12.5'	02°11.8	16.8' 59.0'
4	238°33.0	28.2	161°20.5	12.4'	02°28.6	16.7' 59.0'
5	253°32.7	27.9	175°52.0	12.4'	02°45.3	16.7' 59.0'
6	268°32.4	S22°27.6	190°23.4	12.4'	N03°02.1	16.7' 59.0'
7	283°32.1	27.3	204°54.8	12.4'	03°18.8	16.7' 59.0'
8	298°31.9	27.0	219°26.2	12.4'	03°35.5	16.7' 59.1'
9	313°31.6	· · 26.7	233°57.5	12.3'	03°52.2	16.7' 59.1'
10	328°31.3	26.4	248°28.9	12.3'	04°08.9	16.7' 59.1'
11	343°31.0	26.1	263°00.2	12.3'	04°25.5	16.6' 59.1'
12	358°30.8	S22°25.7	277°31.5	12.2'	N04°42.1	16.6' 59.1'
13	13°30.5	25.4	292°02.7	12.2'	04°58.8	16.6' 59.1'
14	28°30.2	25.1	306°33.9	12.2'	05°15.4	16.6' 59.1'
15	43°30.0	· · 24.8	321°05.1	12.2'	05°31.9	16.5' 59.1'
16	58°29.7	24.5	335°36.3	12.1'	05°48.5	16.5' 59.1'
17	73°29.4	24.2	350°07.4	12.1'	06°05.0	16.5' 59.1'
18	88°29.1	S22°23.9	4°38.5	12.0'	N06°21.5	16.5' 59.1'
19	103°28.9	23.6	19°09.5	12.0'	06°38.0	16.4' 59.1'
20	118°28.6	23.2	33°40.5	12.0'	06°54.4	16.4' 59.1'
21	133°28.3	· · 22.9	48°11.5	11.9'	07°10.8	16.4' 59.1'
22	148°28.1	22.6	62°42.4	11.9'	07°27.1	16.3' 59.1'
23	163°27.8	22.3	77°13.3	11.8'	07°43.5	16.3' 59.1'
SD = 16.3' d = -0.3' SD = 16.1'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:19	10:29	■	■	13:43	15:53
N 70°	08:01	09:42	■	■	14:29	16:10
68°	07:47	09:12	11:27	12:44	15:00	16:25
66°	07:35	08:49	10:19	13:52	15:22	16:37
64°	07:25	08:31	09:44	14:27	15:40	16:47
62°	07:16	08:16	09:19	14:52	15:55	16:56
60°	07:08	08:04	08:59	15:12	16:08	17:03
N 58°	07:01	07:53	08:43	15:28	16:19	17:10
56°	06:55	07:43	08:30	15:42	16:28	17:17
54°	06:49	07:35	08:18	15:54	16:37	17:22
52°	06:44	07:27	08:07	16:04	16:44	17:28
50°	06:39	07:20	07:58	16:13	16:51	17:32
45°	06:28	07:05	07:38	16:33	17:07	17:43
N 40°	06:18	06:52	07:22	16:49	17:19	17:53
35°	06:09	06:41	07:09	17:03	17:31	18:02
30°	06:01	06:31	06:57	17:14	17:41	18:10
20°	05:45	06:12	06:36	17:35	17:59	18:26
N 10°	05:30	05:56	06:19	17:53	18:15	18:41
0°	05:13	05:39	06:02	18:09	18:32	18:58
S 10°	04:55	05:22	05:45	18:26	18:49	19:16
20°	04:33	05:02	05:27	18:44	19:09	19:38
30°	04:05	04:38	05:06	19:05	19:32	20:05
35°	03:47	04:24	04:53	19:18	19:47	20:23
40°	03:25	04:06	04:39	19:32	20:04	20:45
45°	02:56	03:45	04:21	19:49	20:26	21:14
S 50°	02:14	03:16	04:00	20:11	20:54	21:56
52°	01:49	03:02	03:50	20:21	21:08	22:20
54°	01:14	02:46	03:38	20:32	21:24	22:55
56°	—:—	02:25	03:25	20:46	21:45	—:—
58°	—:—	01:59	03:09	21:01	22:10	—:—
S 60°	—:—	01:20	02:50	21:20	22:48	—:—

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	11:51	11:12	10:36	20:56	23:18	· · ·
N 70°	11:39	11:09	10:41	21:06	23:17	· · ·
68°	11:28	11:06	10:46	21:13	23:15	· · ·
66°	11:20	11:04	10:50	21:20	23:14	· · ·
64°	11:12	11:02	10:53	21:25	23:14	· · ·
62°	11:06	11:01	10:56	21:30	23:13	· · ·
60°	11:00	10:59	10:58	21:34	23:12	· · ·
N 58°	10:55	10:58	11:01	21:37	23:12	· · ·
56°	10:51	10:57	11:03	21:40	23:11	· · ·
54°	10:47	10:56	11:05	21:43	23:11	· · ·
52°	10:43	10:55	11:06	21:46	23:10	· · ·
50°	10:40	10:54	11:08	21:48	23:10	· · ·
45°	10:33	10:52	11:11	21:53	23:09	· · ·
N 40°	10:27	10:51	11:14	21:57	23:08	· · ·
35°	10:22	10:49	11:17	22:00	23:08	· · ·
30°	10:17	10:48	11:19	22:04	23:07	· · ·
20°	10:09	10:46	11:23	22:09	23:06	· · ·
N 10°	10:02	10:44	11:26	22:13	23:05	23:58
0°	09:55	10:42	11:30	22:18	23:05	23:52
S 10°	09:48	10:41	11:33	22:22	23:04	23:46
20°	09:41	10:39	11:36	22:26	23:03	23:40
30°	09:33	10:37	11:41	22:31	23:02	23:33
35°	09:28	10:36	11:43	22:34	23:01	23:29
40°	09:23	10:34	11:46	22:37	23:01	23:25
45°	09:16	10:33	11:49	22:41	23:00	23:19
S 50°	09:09	10:31	11:53	22:45	22:59	23:13
52°	09:05	10:30	11:55	22:47	22:59	23:10
54°	09:01	10:29	11:57	22:49	22:58	23:07
56°	08:57	10:28	11:59	22:52	22:58	23:04
58°	08:52	10:27	12:01	22:54	22:57	23:00
S 60°	08:47	10:26	12:04	22:57	22:56	22:56

Day	Sun		Mer. Pass	Moon		Age 5-7 19-39%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
04	04:50	05:04	12:05	16:06	03:42	
05	05:17	05:31	12:06	16:54	04:30	
06	05:44	05:57	12:06	17:41	05:17	

January 07, 08, 09 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	106°48.8	130°31.0	S10°54.1	343°54.1	N24°12.3	35°34.1	N21°44.0	119°51.5	S07°43.2	Alpheratz	357°34.9 29°13.8
1	121°51.3	145°31.0	53.0	358°57.5	12.5	50°36.8	43.9	134°53.8	43.1	Ankaa	353°07.2 -42°10.4
2	136°53.7	160°31.1	51.8	14°01.0	12.8	65°39.6	43.9	149°56.0	43.0	Schedar	349°31.1 56°40.7
3	151°56.2	175°31.2	· · 50.7	29°04.4	· · 13.1	80°42.3	· · 43.9	164°58.3	· · 43.0	Diphda	348°47.3 -17°51.1
4	166°58.7	190°31.3	49.5	44°07.9	13.4	95°45.0	43.9	180°00.6	42.9	Achernar	335°20.1 -57°06.9
5	182°01.1	205°31.3	48.4	59°11.3	13.6	110°47.7	43.9	195°02.9	42.8	Hamal	327°51.1 23°34.9
6	197°03.6	220°31.4	S10°47.3	74°14.8	N24°13.9	125°50.4	N21°43.8	210°05.1	S07°42.7	Polaris	313°46.5 89°22.5
7	212°06.1	235°31.5	46.1	89°18.2	14.2	140°53.1	43.8	225°07.4	42.6	Acamar	315°11.5 -40°12.5
8	227°08.5	250°31.6	45.0	104°21.7	14.4	155°55.8	43.8	240°09.7	42.5	Menkar	314°06.0 4°11.3
9	242°11.0	265°31.7	· · 43.8	119°25.2	· · 14.7	170°58.5	· · 43.8	255°11.9	· · 42.4	Mirfak	308°28.0 49°57.2
10	257°13.5	280°31.7	42.7	134°28.6	15.0	186°01.2	43.8	270°14.2	42.4	Aldebaran	290°39.3 16°33.6
11	272°15.9	295°31.8	41.5	149°32.1	15.2	201°03.9	43.7	285°16.5	42.3	Rigel	281°03.6 -8°10.4
12	287°18.4	310°31.9	S10°40.4	164°35.5	N24°15.5	216°06.6	N21°43.7	300°18.8	S07°42.2	Capella	280°21.5 46°01.5
13	302°20.8	325°32.0	39.2	179°39.0	15.8	231°09.3	43.7	315°21.0	42.1	Bellatrix	278°22.6 6°22.3
14	317°23.3	340°32.1	38.1	194°42.5	16.1	246°12.0	43.7	330°23.3	42.0	Elnath	278°01.5 28°37.7
15	332°25.8	355°32.2	· · 36.9	209°45.9	· · 16.3	261°14.7	· · 43.7	345°25.6	· · 41.9	Alnilam	275°37.4 -1°11.2
16	347°28.2	10°32.3	35.8	224°49.4	16.6	276°17.4	43.6	0°27.8	41.8	Betelgeuse	270°51.8 7°24.7
17	2°30.7	25°32.4	34.6	239°52.8	16.9	291°20.1	43.6	15°30.1	41.8	Canopus	263°51.8 -52°42.5
18	17°33.2	40°32.5	S10°33.5	254°56.3	N24°17.1	306°22.8	N21°43.6	30°32.4	S07°41.7	Sirius	258°25.9 -16°45.0
19	32°35.6	55°32.6	32.3	269°59.8	17.4	321°25.5	43.6	45°34.7	41.6	Adhara	255°05.5 -29°00.4
20	47°38.1	70°32.6	31.2	285°03.2	17.7	336°28.3	43.6	60°36.9	41.5	Procyon	244°50.5 5°09.6
21	62°40.6	85°32.7	· · 30.0	300°06.7	· · 17.9	351°31.0	· · 43.5	75°39.2	· · 41.4	Pollux	243°16.9 27°57.9
22	77°43.0	100°32.8	28.9	315°10.2	18.2	6°33.7	43.5	90°41.5	41.3	Avior	234°14.1 -59°35.2
23	92°45.5	115°32.9	27.7	330°13.6	18.5	21°36.4	43.5	105°43.7	41.2	Suhail	222°45.9 -43°31.9
Mer.pass. 16:50		ν 0.1' d -1.1' m -4.50		ν 3.5' d 0.3' m -1.34		ν 2.7' d -0.0' m -2.71		ν 2.3' d -0.1' m 1.07			

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	107°47.9	130°33.0	S10°26.6	345°17.1	N24°18.7	36°39.1	N21°43.5	120°46.0	S07°41.1
1	122°50.4	145°33.1	25.4	0°20.6	19.0	51°41.8	43.5	135°48.3	41.1
2	137°52.9	160°33.2	24.3	15°24.1	19.3	66°44.5	43.4	150°50.5	41.0
3	152°55.3	175°33.3	· · 23.1	30°27.5	· · 19.5	81°47.2	· · 43.4	165°52.8	· · 40.9
4	167°57.8	190°33.4	22.0	45°31.0	19.8	96°49.9	43.4	180°55.1	40.8
5	183°00.3	205°33.5	20.8	60°34.5	20.1	111°52.6	43.4	195°57.4	40.7
6	198°02.7	220°33.6	S10°19.7	75°37.9	N24°20.3	126°55.3	N21°43.4	210°59.6	S07°40.6
7	213°05.2	235°33.7	18.5	90°41.4	20.6	141°58.0	43.3	226°01.9	40.5
8	228°07.7	250°33.9	17.4	105°44.9	20.9	157°00.7	43.3	241°04.2	40.5
9	243°10.1	265°34.0	· · 16.2	120°48.4	· · 21.1	172°03.4	· · 43.3	256°06.4	· · 40.4
10	258°12.6	280°34.1	15.1	135°51.8	21.4	187°06.1	43.3	271°08.7	40.3
11	273°15.1	295°34.2	13.9	150°55.3	21.7	202°08.8	43.3	286°11.0	40.2
12	288°17.5	310°34.3	S10°12.8	165°58.8	N24°21.9	217°11.5	N21°43.2	301°13.2	S07°40.1
13	303°20.0	325°34.4	11.6	181°02.3	22.2	232°14.2	43.2	316°15.5	40.0
14	318°22.4	340°34.5	10.5	196°05.8	22.5	247°16.8	43.2	331°17.8	39.9
15	333°24.9	355°34.6	· · 09.3	211°09.2	· · 22.7	262°19.5	· · 43.2	346°20.0	· · 39.8
16	348°27.4	10°34.7	08.2	226°12.7	23.0	277°22.2	43.2	1°22.3	39.8
17	3°29.8	25°34.9	07.0	241°16.2	23.3	292°24.9	43.1	16°24.6	39.7
18	18°32.3	40°35.0	S10°05.9	256°19.7	N24°23.5	307°27.6	N21°43.1	31°26.8	S07°39.6
19	33°34.8	55°35.1	04.7	271°23.2	23.8	322°30.3	43.1	46°29.1	39.5
20	48°37.2	70°35.2	03.6	286°26.6	24.1	337°33.0	43.1	61°31.4	39.4
21	63°39.7	85°35.3	· · 02.4	301°30.1	· · 24.3	352°35.7	· · 43.1	76°33.6	· · 39.3
22	78°42.2	100°35.4	01.2	316°33.6	24.6	7°38.4	43.1	91°35.9	39.2
23	93°44.6	115°35.6	00.1	331°37.1	24.9	22°41.1	43.0	106°38.2	39.1
Mer.pass. 16:46		ν 0.1' d -1.1' m -4.51		ν 3.5' d 0.3' m -1.36		ν 2.7' d -0.0' m -2.70		ν 2.3' d -0.1' m 1.07	

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	108°47.1	130°35.7	S09°58.9	346°40.6	N24°25.1	37°43.8	N21°43.0	121°40.4	S07°39.1
1	123°49.6	145°35.8	57.8	1°44.1	25.4	52°46.5	43.0	136°42.7	39.0
2	138°52.0	160°35.9	56.6	16°47.5	25.6	67°49.2	43.0	151°45.0	38.9
3	153°54.5	175°36.1	· · 55.5	31°51.0	· · 25.9	82°51.9	· · 43.0	166°47.2	· · 38.8
4	168°56.9	190°36.2	54.3	46°54.5	26.2	97°54.6	42.9	181°49.5	38.7
5	183°59.4	205°36.3	53.2	61°58.0	26.4	112°57.3	42.9	196°51.8	38.6
6	199°01.9	220°36.4	S09°52.0	77°01.5	N24°26.7	128°00.0	N21°42.9	211°54.0	S07°38.5
7	214°04.3	235°36.6	50.8	92°05.0	27.0	143°02.7	42.9	226°56.3	38.4
8	229°06.8	250°36.7	49.7	107°08.5	27.2	158°05.3	42.9	241°58.6	38.4
9	244°09.3	265°36.8	· · 48.5	122°12.0	· · 27.5	173°08.0	· · 42.8	257°00.8	· · 38.3
10	259°11.7	280°37.0	47.4	137°15.5	27.7	188°10.7	42.8	272°03.1	38.2
11	274°14.2	295°37.1	46.2	152°18.9	28.0	203°13.4	42.8	287°05.4	38.1
12	289°16.7	310°37.2	S09°45.0	167°22.4	N24°28.3	218°16.1	N21°42.8	302°07.6	S07°38.0
13	304°19.1	325°37.4	43.9	182°25.9	28.5	233°18.8	42.8	317°09.9	37.9
14	319°21.6	340°37.5	42.7	197°29.4	28.8	248°21.5	42.7	332°12.2	37.8
15	334°24.1	355°37.6	· · 41.6	212°32.9	· · 29.1	263°24.2	· · 42.7	347°14.4	· · 37.7
16	349°26.5	10°37.8	40.4	227°36.4	29.3	278°26.9	42.7	2°16.7	37.6
17	4°29.0	25°37.9	39.3	242°39.9	29.6	293°29.6	42.7	17°19.0	37.6
18	19°31.4	40°38.1	S09°38.1	257°43.4	N24°29.8	308°32.2	N21°42.7	32°21.2	S07°37.5
19	34°33.9	55°38.2	36.9	272°46.9	30.1	323°34.9	42.7	47°23.5	37.4
20	49°36.4	70°38.4	35.8	287°50.4	30.4	338°37.6	42.6	62°25.8	37.3
21	64°38.8	85°38.5	· · 34.6	302°53.9	· · 30.6	353°40.3	· · 42.6	77°28.0	· · 37.2
22	79°41.3	100°38.6	33.5	317°57.4	30.9	8°43.0	42.6	92°30.3	37.1
23	94°43.8	115°38.8	32.3	333°00.9	31.1	23°45.7	42.6	107°32.5	37.0
Mer.pass. 16:42		ν 0.1' d -1.2' m -4.52		ν 3.5' d 0.3' m -1.37		ν 2.7' d -0.0' m -2.69		ν 2.3' d -0.1' m 1.07	

Jan 07 Tue			SHA	Mer.pass
Venus			23°42.1	15:18
Mars			237°05.2	01:04
Jupiter			288°45.3	21:34
Saturn			13°02.7	15:58

Jan 08 Wed			SHA	Mer.pass
Venus			22°45.1	15:18
Mars			237°29.2	00:59
Jupiter			288°51.1	21:30
Saturn			12°58.1	15:55

Jan 09 Thu			SHA	Mer.pass
Venus			21°48.6	15:17
Mars			237°53.5	00:53
Jupiter			288°56.7	21:25
Saturn			12°53.4	15:51

Horizontal parallax		
Venus: 0.2		
Mars: 0.2		

Sun							Moon						
h	GHA	Dec	GHA	ν	Dec	d	HP						
Tue													
0	178°27.5	S22°22.0	91°44.1	11.8'	N07°59.8	16.3'	59.1'						
1	193°27.3	21.6	106°14.9	11.7'	08°16.0	16.2'	59.1'						
2	208°27.0	21.3	120°45.7	11.7'	08°32.2	16.2'	59.1'						
3	223°26.7	· · 21.0	135°16.4	11.6'	08°48.4	16.1'	59.1'						
4	238°26.4	20.7	149°47.0	11.6'	09°04.5	16.1'	59.1'						
5	253°26.2	20.4	164°17.6	11.5'	09°20.6	16.0'	59.1'						
6	268°25.9	S22°20.0	178°48.2	11.5'	N09°36.7	16.0'	59.1'						
7	283°25.6	19.7	193°18.7	11.4'	09°52.7	15.9'	59.1'						
8	298°25.4	19.4	207°49.1	11.4'	10°08.6	15.9'	59.1'						
9	313°25.1	· · 19.1	222°19.5	11.3'	10°24.5	15.8'	59.1'						
10	328°24.8	18.7	236°49.8	11.3'	10°40.3	15.8'	59.1'						
11	343°24.6	18.4	251°20.1	11.2'	10°56.1	15.7'	59.1'						
12	358°24.3	S22°18.1	265°50.3	11.1'	N11°11.8	15.7'	59.1'						
13	13°24.0	17.8	280°20.4	11.1'	11°27.5	15.6'	59.1'						
14	28°23.8	17.4	294°50.5	11.0'	11°43.1	15.6'	59.1'						
15	43°23.5	· · 17.1	309°20.5	11.0'	11°58.7	15.5'	59.1'						
16	58°23.2	16.8	323°50.5	10.9'	12°14.2	15.4'	59.2'						
17	73°23.0	16.4	338°20.4	10.8'	12°29.6	15.4'	59.2'						
18	88°22.7	S22°16.1	352°50.2	10.8'	N12°45.0	15.3'	59.2'						
19	103°22.4	15.8	7°20.0	10.7'	13°00.3	15.2'	59.2'						
20	118°22.2	15.4	21°49.7	10.6'	13°15.5	15.2'	59.2'						
21	133°21.9	· · 15.1	36°19.3	10.5'	13°30.7	15.1'	59.2'						
22	148°21.7	14.8	50°48.8	10.5'	13°45.8	15.0'	59.2'						
23	163°21.4	14.4	65°18.3	10.4'	14°00.8	14.9'	59.2'						
	SD = 16.3'	$d = -0.3'$			SD = 16.1'								
Wed	GHA	Dec	GHA	ν	Dec	d	HP						
0	178°21.1	S22°14.1	79°47.7	10.3'	N14°15.8	14.9'	59.2'						
1	193°20.9	13.8	94°17.0	10.3'	14°30.6	14.8'	59.2'						
2	208°20.6	13.4	108°46.3	10.2'	14°45.4	14.7'	59.2'						
3	223°20.3	· · 13.1	123°15.5	10.1'	15°00.1	14.6'	59.2'						
4	238°20.1	12.7	137°44.6	10.0'	15°14.8	14.6'	59.2'						
5	253°19.8	12.4	152°13.6	9.9'	15°29.3	14.5'	59.2'						
6	268°19.5	S22°12.0	166°42.5	9.9'	N15°43.8	14.4'	59.2'						
7	283°19.3	11.7	181°11.4	9.8'	15°58.2	14.3'	59.2'						
8	298°19.0	11.4	195°40.2	9.7'	16°12.5	14.2'	59.2'						
9	313°18.8	· · 11.0	210°08.9	9.6'	16°26.7	14.1'	59.1'						
10	328°18.5	10.7	224°37.5	9.5'	16°40.8	14.0'	59.1'						
11	343°18.2	10.3	239°06.0	9.5'	16°54.8	13.9'	59.1'						
12	358°18.0	S22°10.0	253°34.5	9.4'	N17°08.8	13.8'	59.1'						
13	13°17.7	09.6	268°02.9	9.3'	17°22.6	13.7'	59.1'						
14	28°17.5	09.3	282°31.1	9.2'	17°36.3	13.6'	59.1'						
15	43°17.2	· · 08.9	296°59.3	9.1'	17°50.0	13.5'	59.1'						
16	58°16.9	08.6	311°27.4	9.0'	18°03.5	13.4'	59.1'						
17	73°16.7	08.2	325°55.5	8.9'	18°17.0	13.3'	59.1'						
18	88°16.4	S22°07.9	340°23.4	8.8'	N18°30.3	13.2'	59.1'						
19	103°16.2	07.5	354°51.2	8.8'	18°43.5	13.1'	59.1'						
20	118°15.9	07.2	9°19.0	8.7'	18°56.6	13.0'	59.1'						
21	133°15.6	· · 06.8	23°46.7	8.6'	19°09.7	12.9'	59.1'						
22	148°15.4	06.5	38°14.2	8.5'	19°22.6	12.8'	59.1'						
23	163°15.1	06.1	52°41.7	8.4'	19°35.4	12.7'	59.1'						
	SD = 16.3'	$d = -0.3'$			SD = 16.1'								
Thu	GHA	Dec	GHA	ν	Dec	d	HP						
0	178°14.9	S22°05.8	67°09.1	8.3'	N19°48.0	12.6'	59.1'						
1	193°14.6	05.4	81°36.4	8.2'	20°00.6	12.4'	59.1'						
2	208°14.3	05.1	96°03.7	8.1'	20°13.1	12.3'	59.1'						
3	223°14.1	· · 04.7	110°30.8	8.0'	20°25.4	12.2'	59.1'						
4	238°13.8	04.3	124°57.8	7.9'	20°37.6	12.1'	59.1'						
5	253°13.6	04.0	139°24.8	7.8'	20°49.7	12.0'	59.1'						
6	268°13.3	S22°03.6	153°51.6	7.8'	N21°01.6	11.8'	59.1'						
7	283°13.1	03.3	168°18.4	7.7'	21°13.5	11.7'	59.1'						
8	298°12.8	02.9	182°45.0	7.6'	21°25.2	11.6'	59.1'						
9	313°12.5	· · 02.5	197°11.6	7.5'	21°36.8	11.5'	59.1'						
10	328°12.3	02.2	211°38.1	7.4'	21°48.2	11.3'	59.1'						
11	343°12.0	01.8	226°04.5	7.3'	21°59.5	11.2'	59.1'						
12	358°11.8	S22°01.5	240°30.8	7.2'	N22°10.7	11.1'	59.1'						
13	13°11.5	01.1	254°57.0	7.1'	22°21.8	10.9'	59.1'						
14	28°11.3	00.7	269°23.1	7.0'	22°32.7	10.8'	59.0'						
15	43°11.0	· · 00.4	283°49.1	6.9'	22°43.5	10.6'	59.0'						
16	58°10.8	22°00.0	298°15.0	6.8'	22°54.1	10.5'	59.0'						
17	73°10.5	21°59.6	312°40.9	6.8'	23°04.6	10.4'	59.0'						
18	88°10.2	S21°59.2	327°06.6	6.7'	N23°15.0	10.2'	59.0'						
19	103°10.0	58.9	341°32.3	6.6'	23°25.2	10.1'	59.0'						
20	118°09.7	58.5	355°57.9	6.5'	23°35.2	9.9'	59.0'						
21	133°09.5	· · 58.1	10°23.4	6.4'	23°45.2	9.8'	59.0'						
22	148°09.2	57.8	24°48.8	6.3'	23°54.9	9.6'	59.0'						
23	163°09.0	57.4	39°14.1	6.2'	24°04.5	9.5'	59.0'						
	SD = 16.3'	$d = -0.4'$			SD = 16.1'								

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:14	10:18	■	■	13:56	16:00
N 70°	07:57	09:36	■	■	14:39	16:17
68°	07:44	09:07	11:08	13:07	15:07	16:31
66°	07:32	08:45	10:12	14:02	15:29	16:42
64°	07:22	08:28	09:39	14:35	15:46	16:52
62°	07:14	08:14	09:15	14:59	16:00	17:00
60°	07:06	08:02	08:57	15:17	16:12	17:08
N 58°	07:00	07:51	08:41	15:33	16:23	17:14
56°	06:54	07:42	08:28	15:46	16:32	17:20
54°	06:48	07:33	08:16	15:58	16:40	17:26
52°	06:43	07:26	08:06	16:08	16:48	17:31
50°	06:38	07:19	07:57	16:17	16:55	17:36
45°	06:27	07:04	07:37	16:36	17:10	17:46
N 40°	06:18	06:52	07:22	16:52	17:22	17:56
35°	06:09	06:41	07:09	17:05	17:33	18:04
30°	06:01	06:31	06:57	17:17	17:43	18:12
20°	05:46	06:13	06:37	17:37	18:00	18:28
N 10°	05:31	05:57	06:20	17:54	18:17	18:43
0°	05:15	05:41	06:03	18:10	18:33	18:59
S 10°	04:57	05:24	05:47	18:27	18:50	19:17
20°	04:35	05:04	05:29	18:45	19:09	19:38
30°	04:08	04:41	05:08	19:05	19:33	20:05
35°	03:50	04:26	04:56	19:18	19:47	20:23
40°	03:29	04:09	04:42	19:32	20:04	20:44
45°	03:00	03:48	04:25	19:48	20:25	21:12
S 50°	02:20	03:21	04:04	20:09	20:52	21:53
52°	01:56	03:07	03:54	20:19	21:06	22:16
54°	01:23	02:51	03:42	20:30	21:22	22:48
56°	—	02:31	03:29	20:43	21:41	—
58°	—	02:06	03:14	20:58	22:06	—
S 60°	—	01:31	02:56	21:16	22:40	—

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	09:54	08:44	☐	01:43	04:39	☐
N 70°	10:10	09:25	☐	01:30	04:00	☐
68°	10:23	09:53	08:46	01:19	03:34	06:33
66°	10:34	10:14	09:42	01:10	03:14	05:38
64°	10:43	10:31	10:15	01:03	02:58	05:06
62°	10:51	10:46	10:40	00:57	02:45	04:42
60°	10:58	10:58	11:00	00:52	02:34	04:23
N 58°	11:04	11:08	11:16	00:47	02:25	04:07
56°	11:09	11:18	11:30	00:43	02:17	03:54
54°	11:14	11:26	11:42	00:39	02:09	03:43
52°	11:19	11:33	11:53	00:35	02:03	03:33
50°	11:23	11:40	12:03	00:32	01:57	03:24
45°	11:31	11:55	12:23	00:26	01:44	03:05
N 40°	11:39	12:07	12:40	00:20	01:34	02:49
35°	11:45	12:17	12:54	00:16	01:25	02:36
30°	11:51	12:26	13:06	00:11	01:17	02:25
20°	12:01	12:42	13:27	00:04	01:04	02:06
N 10°	12:10	12:56	13:46		00:52	01:50
0°	12:18	13:09	14:03		00:42	01:34
S 10°	12:26	13:22	14:21		00:31	01:19
20°	12:35	13:36	14:40		00:19	01:02
30°	12:46	13:53	15:02		00:06	00:44
35°	12:52	14:02	15:14	23:59		00:33
40°	12:59	14:13	15:29	23:51		00:21
45°	13:07	14:26	15:47	23:41		00:06
S 50°	13:16	14:42	16:10	23:29	23:48	
52°	13:21	14:50	16:20	23:23	23:40	
54°	13:26	14:58	16:33	23:17	23:31	23:51
56°	13:32	15:08	16:47	23:11	23:21	23:36
58°	13:38	15:19	17:03	23:03	23:09	23:19
S 60°	13:45	15:31	17:23	22:55	22:55	22:58

January 10, 11, 12 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	109°46.2	130°38.9	S09°31.1	348°04.4	N24°31.4	38°48.4	N21°42.6	122°34.8	S07°36.9					Alpheratz	357°34.9	29°13.8	
1	124°48.7	145°39.1	30.0	3°07.9	31.7	53°51.1	42.5	137°37.1	36.8					Ankaa	353°07.2	-42°10.4	
2	139°51.2	160°39.2	28.8	18°11.4	31.9	68°53.7	42.5	152°39.3	36.8					Schedar	349°31.1	56°40.7	
3	154°53.6	175°39.4	•• 27.6	33°14.9	•• 32.2	83°56.4	•• 42.5	167°41.6	•• 36.7					Diphda	348°47.3	-17°51.1	
4	169°56.1	190°39.5	26.5	48°18.4	32.4	98°59.1	42.5	182°43.9	36.6					Achernar	335°20.1	-57°06.9	
5	184°58.5	205°39.7	25.3	63°21.9	32.7	114°01.8	42.5	197°46.1	36.5					Hamal	327°51.1	23°34.9	
6	200°01.0	220°39.8	S09°24.2	78°25.4	N24°33.0	129°04.5	N21°42.5	212°48.4	S07°36.4					Polaris	313°47.5	89°22.5	
7	215°03.5	235°40.0	23.0	93°28.9	33.2	144°07.2	42.4	227°50.7	36.3					Acamar	315°11.6	-40°12.5	
8	230°05.9	250°40.1	21.8	108°32.4	33.5	159°09.8	42.4	242°52.9	36.2					Menkar	314°06.0	4°11.3	
9	245°08.4	265°40.3	•• 20.7	123°35.9	•• 33.7	174°12.5	•• 42.4	257°55.2	•• 36.1					Mirfak	308°28.0	49°57.2	
10	260°10.9	280°40.5	19.5	138°39.4	34.0	189°15.2	42.4	272°57.4	36.0					Aldebaran	290°39.3	16°33.6	
11	275°13.3	295°40.6	18.3	153°42.9	34.2	204°17.9	42.4	287°59.7	36.0					Rigel	281°03.6	-8°10.4	
12	290°15.8	310°40.8	S09°17.2	168°46.4	N24°34.5	219°20.6	N21°42.3	303°02.0	S07°35.9					Capella	280°21.4	46°01.5	
13	305°18.3	325°40.9	16.0	183°49.9	34.8	234°23.3	42.3	318°04.2	35.8					Bellatrix	278°22.6	6°22.3	
14	320°20.7	340°41.1	14.9	198°53.4	35.0	249°25.9	42.3	333°06.5	35.7					Elnath	278°01.5	28°37.7	
15	335°23.2	355°41.3	•• 13.7	213°56.9	•• 35.3	264°28.6	•• 42.3	348°08.8	•• 35.6					Alnilam	275°37.4	-1°11.2	
16	350°25.7	10°41.4	12.5	229°00.4	35.5	279°31.3	42.3	3°11.0	35.5					Betelgeuse	270°51.8	7°24.7	
17	5°28.1	25°41.6	11.4	244°03.9	35.8	294°34.0	42.3	18°13.3	35.4					Canopus	263°51.8	-52°42.6	
18	20°30.6	40°41.8	S09°10.2	259°07.4	N24°36.0	309°36.7	N21°42.2	33°15.5	S07°35.3					Sirius	258°25.9	-16°45.1	
19	35°33.0	55°41.9	09.0	274°11.0	36.3	324°39.3	42.2	48°17.8	35.2					Adhara	255°05.5	-29°00.4	
20	50°35.5	70°42.1	07.9	289°14.5	36.6	339°42.0	42.2	63°20.1	35.2					Procyon	244°50.5	5°09.6	
21	65°38.0	85°42.3	•• 06.7	304°18.0	•• 36.8	354°44.7	•• 42.2	78°22.3	•• 35.1					Pollux	243°16.9	27°57.9	
22	80°40.4	100°42.4	05.5	319°21.5	37.1	9°47.4	42.2	93°24.6	35.0					Avior	234°14.1	-59°35.3	
23	95°42.9	115°42.6	04.4	334°25.0	37.3	24°50.1	42.2	108°26.9	34.9					Suhail	222°45.9	-43°31.9	
Mer.pass. 16:38			ν 0.1' d -1.2' m -4.53	ν 3.5' d 0.3' m -1.39	ν 2.7' d -0.0' m -2.69	ν 2.3' d -0.1' m 1.08											

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	110°45.4	130°42.8	S09°03.2	349°28.5	N24°37.6	39°52.7	N21°42.1	123°29.1	S07°34.8
1	125°47.8	145°42.9	02.0	4°32.0	37.8	54°55.4	42.1	138°31.4	34.7
2	140°50.3	160°43.1	09°00.9	19°35.5	38.1	69°58.1	42.1	153°33.6	34.6
3	155°52.8	175°43.3	08°59.7	34°39.0	•• 38.3	85°00.8	•• 42.1	168°35.9	•• 34.5
4	170°55.2	190°43.5	58.5	49°42.5	38.6	100°03.5	42.1	183°38.2	34.4
5	185°57.7	205°43.6	57.4	64°46.1	38.9	115°06.1	42.1	198°40.4	34.3
6	201°00.2	220°43.8	S08°56.2	79°49.6	N24°39.1	130°08.8	N21°42.0	213°42.7	S07°34.3
7	216°02.6	235°44.0	55.0	94°53.1	39.4	145°11.5	42.0	228°44.9	34.2
8	231°05.1	250°44.2	53.9	109°56.6	39.6	160°14.2	42.0	243°47.2	34.1
9	246°07.5	265°44.4	•• 52.7	125°00.1	•• 39.9	175°16.8	•• 42.0	258°49.5	•• 34.0
10	261°10.0	280°44.5	51.5	140°03.6	40.1	190°19.5	42.0	273°51.7	33.9
11	276°12.5	295°44.7	50.4	155°07.1	40.4	205°22.2	42.0	288°54.0	33.8
12	291°14.9	310°44.9	S08°49.2	170°10.7	N24°40.6	220°24.9	N21°41.9	303°56.2	S07°33.7
13	306°17.4	325°45.1	48.0	185°14.2	40.9	235°27.5	41.9	318°58.5	33.6
14	321°19.9	340°45.3	46.9	200°17.7	41.1	250°30.2	41.9	334°00.8	33.5
15	336°22.3	355°45.5	•• 45.7	215°21.2	•• 41.4	265°32.9	•• 41.9	349°03.0	•• 33.4
16	351°24.8	10°45.6	44.5	230°24.7	41.6	280°35.6	41.9	4°05.3	33.3
17	6°27.3	25°45.8	43.4	245°28.2	41.9	295°38.2	41.9	19°07.5	33.3
18	21°29.7	40°46.0	S08°42.2	260°31.8	N24°42.1	310°40.9	N21°41.8	34°09.8	S07°33.2
19	36°32.2	55°46.2	41.0	275°35.3	42.4	325°43.6	41.8	49°12.1	33.1
20	51°34.7	70°46.4	39.8	290°38.8	42.6	340°46.2	41.8	64°14.3	33.0
21	66°37.1	85°46.6	•• 38.7	305°42.3	•• 42.9	355°48.9	•• 41.8	79°16.6	•• 32.9
22	81°39.6	100°46.8	37.5	320°45.8	43.1	10°51.6	41.8	94°18.8	32.8
23	96°42.0	115°47.0	36.3	335°49.3	43.4	25°54.3	41.8	109°21.1	32.7
Mer.pass. 16:34			ν 0.2' d -1.2' m -4.54	ν 3.5' d 0.3' m -1.41	ν 2.7' d -0.0' m -2.68	ν 2.3' d -0.1' m 1.08			

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	111°44.5	130°47.2	S08°35.2	350°52.9	N24°43.6	40°56.9	N21°41.7	124°23.4	S07°32.6
1	126°47.0	145°47.4	34.0	5°56.4	43.9	55°59.6	41.7	139°25.6	32.5
2	141°49.4	160°47.6	32.8	20°59.9	44.1	71°02.3	41.7	154°27.9	32.4
3	156°51.9	175°47.8	•• 31.7	36°03.4	•• 44.4	86°04.9	•• 41.7	169°30.1	•• 32.3
4	171°54.4	190°48.0	30.5	51°06.9	44.6	101°07.6	41.7	184°32.4	32.3
5	186°56.8	205°48.2	29.3	66°10.5	44.9	116°10.3	41.7	199°34.6	32.2
6	201°59.3	220°48.4	S08°28.1	81°14.0	N24°45.1	131°12.9	N21°41.6	214°36.9	S07°32.1
7	217°01.8	235°48.6	27.0	96°17.5	45.4	146°15.6	41.6	229°39.2	32.0
8	232°04.2	250°48.8	25.8	111°21.0	45.6	161°18.3	41.6	244°41.4	31.9
9	247°06.7	265°49.0	•• 24.6	126°24.6	•• 45.9	176°21.0	•• 41.6	259°43.7	•• 31.8
10	262°09.2	280°49.2	23.5	141°28.1	46.1	191°23.6	41.6	274°45.9	31.7
11	277°11.6	295°49.4	22.3	156°31.6	46.4	206°26.3	41.6	289°48.2	31.6
12	292°14.1	310°49.6	S08°21.1	171°35.1	N24°46.6	221°29.0	N21°41.5	304°50.4	S07°31.5
13	307°16.5	325°49.8	19.9	186°38.7	46.9	236°31.6	41.5	319°52.7	31.4
14	322°19.0	340°50.0	18.8	201°42.2	47.1	251°34.3	41.5	334°55.0	31.3
15	337°21.5	355°50.3	•• 17.6	216°45.7	•• 47.4	266°37.0	•• 41.5	349°57.2	•• 31.3
16	352°23.9	10°50.5	16.4	231°49.2	47.6	281°39.6	41.5	4°59.5	31.2
17	7°26.4	25°50.7	15.3	246°52.8	47.8	296°42.3	41.5	20°01.7	31.1
18	22°28.9	40°50.9	S08°14.1	261°56.3	N24°48.1	311°44.9	N21°41.4	35°04.0	S07°31.0
19	37°31.3	55°51.1	12.9	276°59.8	48.3	326°47.6	41.4	50°06.2	30.9
20	52°33.8	70°51.3	11.7	292°03.3	48.6	341°50.3	41.4	65°08.5	30.8
21	67°36.3	85°51.6	•• 10.6	307°06.9	•• 48.8	356°52.9	•• 41.4	80°10.8	•• 30.7
22	82°38.7	100°51.8	09.4	322°10.4	49.1	11°55.6	41.4	95°13.0	30.6
23	97°41.2	115°52.0	08.2	337°13.9	49.3	26°58.3	41.4	110°15.3	30.5
Mer.pass. 16:30			ν 0.2' d -1.2' m -4.55	ν 3.5' d 0.2' m -1.42	ν 2.7' d -0.0' m -2.67	ν 2.3' d -0.1' m 1.08			

Jan 10 Fri	SHA	Mer.pass
Venus	20°52.7	15:17
Mars	238°18.2	00:48
Jupiter	289°02.1	21:21
Saturn	12°48.6	15:47

Jan 11 Sat	SHA	Mer.pass
Venus	19°57.4	15:17
Mars	238°43.1	00:42
Jupiter	289°07.4	21:17
Saturn	12°43.7	15:44

Jan 12 Sun	SHA	Mer.pass
Venus	19°02.7	15:17
Mars	239°08.4	00:36
Jupiter	289°12.4	21:12
Saturn	12°38.8	15:40

Horizontal parallax	
Venus:	0.2
Mars:	0.2

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°08.7	S21°57.0	53°39.3	6.1'	N24°14.0	9.3'	59.0'
1	193°08.5	56.6	68°04.5	6.1'	24°23.3	9.2'	59.0'
2	208°08.2	56.3	82°29.5	6.0'	24°32.5	9.0'	59.0'
3	223°08.0	· · 55.9	96°54.5	5.9'	24°41.5	8.8'	59.0'
4	238°07.7	55.5	111°19.4	5.8'	24°50.3	8.7'	58.9'
5	253°07.5	55.1	125°44.2	5.7'	24°59.0	8.5'	58.9'
6	268°07.2	S21°54.8	140°09.0	5.7'	N25°07.5	8.4'	58.9'
7	283°07.0	54.4	154°33.6	5.6'	25°15.9	8.2'	58.9'
8	298°06.7	54.0	168°58.2	5.5'	25°24.1	8.0'	58.9'
9	313°06.5	· · 53.6	183°22.7	5.4'	25°32.1	7.9'	58.9'
10	328°06.2	53.3	197°47.1	5.4'	25°40.0	7.7'	58.9'
11	343°06.0	52.9	212°11.5	5.3'	25°47.7	7.5'	58.9'
12	358°05.7	S21°52.5	226°35.8	5.2'	N25°55.2	7.4'	58.9'
13	13°05.5	52.1	241°00.0	5.1'	26°02.6	7.2'	58.9'
14	28°05.2	51.7	255°24.1	5.1'	26°09.8	7.0'	58.9'
15	43°05.0	· · 51.3	269°48.2	5.0'	26°16.8	6.9'	58.8'
16	58°04.7	51.0	284°12.2	4.9'	26°23.7	6.7'	58.8'
17	73°04.5	50.6	298°36.1	4.9'	26°30.4	6.5'	58.8'
18	88°04.2	S21°50.2	313°00.0	4.8'	N26°36.9	6.3'	58.8'
19	103°04.0	49.8	327°23.8	4.8'	26°43.2	6.1'	58.8'
20	118°03.7	49.4	341°47.6	4.7'	26°49.3	6.0'	58.8'
21	133°03.5	· · 49.0	356°11.3	4.7'	26°55.3	5.8'	58.8'
22	148°03.2	48.6	10°35.0	4.6'	27°01.1	5.6'	58.8'
23	163°03.0	48.2	24°58.6	4.5'	27°06.7	5.4'	58.8'
SD = 16.3'			d = -0.4'				
			SD = 16.1'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°02.8	S21°47.8	39°22.1	4.5'	N27°12.1	5.3'	58.7'
1	193°02.5	47.5	53°45.6	4.5'	27°17.4	5.1'	58.7'
2	208°02.3	47.1	68°09.1	4.4'	27°22.5	4.9'	58.7'
3	223°02.0	· · 46.7	82°32.5	4.4'	27°27.3	4.7'	58.7'
4	238°01.8	46.3	96°55.8	4.3'	27°32.0	4.5'	58.7'
5	253°01.5	45.9	111°19.2	4.3'	27°36.6	4.3'	58.7'
6	268°01.3	S21°45.5	125°42.4	4.3'	N27°40.9	4.1'	58.7'
7	283°01.0	45.1	140°05.7	4.2'	27°45.0	4.0'	58.7'
8	298°00.8	44.7	154°28.9	4.2'	27°49.0	3.8'	58.6'
9	313°00.5	· · 44.3	168°52.1	4.2'	27°52.8	3.6'	58.6'
10	328°00.3	43.9	183°15.3	4.1'	27°56.4	3.4'	58.6'
11	343°00.1	43.5	197°38.4	4.1'	27°59.8	3.2'	58.6'
12	357°59.8	S21°43.1	212°01.5	4.1'	N28°03.0	3.0'	58.6'
13	12°59.6	42.7	226°24.6	4.1'	28°06.0	2.8'	58.6'
14	27°59.3	42.3	240°47.7	4.1'	28°08.8	2.6'	58.6'
15	42°59.1	· · 41.9	255°10.7	4.0'	28°11.5	2.5'	58.5'
16	57°58.8	41.5	269°33.8	4.0'	28°13.9	2.3'	58.5'
17	72°58.6	41.1	283°56.8	4.0'	28°16.2	2.1'	58.5'
18	87°58.4	S21°40.7	298°19.8	4.0'	N28°18.2	1.9'	58.5'
19	102°58.1	40.3	312°42.8	4.0'	28°20.1	1.7'	58.5'
20	117°57.9	39.9	327°05.8	4.0'	28°21.8	1.5'	58.5'
21	132°57.6	· · 39.5	341°28.8	4.0'	28°23.3	1.3'	58.4'
22	147°57.4	39.1	355°51.9	4.0'	28°24.6	1.1'	58.4'
23	162°57.2	38.7	10°14.9	4.0'	28°25.7	0.9'	58.4'
SD = 16.3'			d = -0.4'				
			SD = 16.0'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	177°56.9	S21°38.2	24°37.9	4.0'	N28°26.7	0.7'	58.4'
1	192°56.7	37.8	39°00.9	4.1'	28°27.4	0.5'	58.4'
2	207°56.4	37.4	53°24.0	4.1'	28°27.9	0.4'	58.4'
3	222°56.2	· · 37.0	67°47.1	4.1'	28°28.3	0.2'	58.3'
4	237°56.0	36.6	82°10.1	4.1'	28°28.5	-0.0'	58.3'
5	252°55.7	36.2	96°33.3	4.1'	28°28.5	-0.2'	58.3'
6	267°55.5	S21°35.8	110°56.4	4.2'	N28°28.2	-0.4'	58.3'
7	282°55.3	35.4	125°19.5	4.2'	28°27.8	-0.6'	58.3'
8	297°55.0	35.0	139°42.7	4.2'	28°27.3	-0.8'	58.2'
9	312°54.8	· · 34.5	154°05.9	4.2'	28°26.5	-1.0'	58.2'
10	327°54.5	34.1	168°29.2	4.3'	28°25.5	-1.1'	58.2'
11	342°54.3	33.7	182°52.5	4.3'	28°24.4	-1.3'	58.2'
12	357°54.1	S21°33.3	197°15.8	4.4'	N28°23.0	-1.5'	58.2'
13	12°53.8	32.9	211°39.2	4.4'	28°21.5	-1.7'	58.2'
14	27°53.6	32.5	226°02.6	4.5'	28°19.8	-1.9'	58.1'
15	42°53.4	· · 32.0	240°26.0	4.5'	28°17.9	-2.1'	58.1'
16	57°53.1	31.6	254°49.5	4.6'	28°15.8	-2.3'	58.1'
17	72°52.9	31.2	269°13.1	4.6'	28°13.6	-2.4'	58.1'
18	87°52.7	S21°30.8	283°36.7	4.7'	N28°11.1	-2.6'	58.1'
19	102°52.4	30.4	298°00.4	4.7'	28°08.5	-2.8'	58.0'
20	117°52.2	29.9	312°24.1	4.8'	28°05.7	-3.0'	58.0'
21	132°51.9	· · 29.5	326°47.9	4.9'	28°02.7	-3.2'	58.0'
22	147°51.7	29.1	341°11.7	4.9'	27°59.6	-3.3'	58.0'
23	162°51.5	28.7	355°35.7	5.0'	27°56.2	-3.5'	58.0'
SD = 16.3'			d = -0.4'				
			SD = 15.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:09	10:06	■	■	14:10	16:08
N 70°	07:53	09:28	■	■	14:49	16:24
68°	07:40	09:01	10:51	13:25	15:15	16:37
66°	07:29	08:41	10:04	14:13	15:36	16:48
64°	07:20	08:24	09:34	14:43	15:52	16:57
62°	07:11	08:11	09:11	15:05	16:06	17:05
60°	07:04	07:59	08:53	15:24	16:17	17:12
N 58°	06:58	07:49	08:38	15:39	16:28	17:19
56°	06:52	07:40	08:25	15:51	16:37	17:24
54°	06:47	07:32	08:14	16:02	16:45	17:30
52°	06:42	07:25	08:04	16:12	16:52	17:35
50°	06:37	07:18	07:55	16:21	16:59	17:39
45°	06:27	07:03	07:36	16:40	17:13	17:49
N 40°	06:18	06:51	07:21	16:55	17:25	17:59
35°	06:09	06:41	07:08	17:08	17:36	18:07
30°	06:01	06:31	06:57	17:19	17:45	18:15
20°	05:47	06:14	06:37	17:39	18:02	18:30
N 10°	05:32	05:58	06:20	17:56	18:18	18:44
0°	05:16	05:42	06:04	18:12	18:34	19:00
S 10°	04:59	05:25	05:48	18:28	18:51	19:17
20°	04:38	05:06	05:31	18:45	19:09	19:38
30°	04:11	04:43	05:10	19:05	19:32	20:05
35°	03:54	04:29	04:58	19:17	19:46	20:22
40°	03:32	04:13	04:45	19:31	20:03	20:43
45°	03:05	03:52	04:28	19:47	20:23	21:10
S 50°	02:26	03:25	04:08	20:08	20:50	21:49
52°	02:04	03:12	03:58	20:17	21:03	22:11
54°	01:33	02:56	03:47	20:28	21:18	22:40
56°	00:35	02:38	03:35	20:40	21:37	23:33
58°	—:—	02:14	03:20	20:55	22:00	—:—
S 60°	—:—	01:42	03:03	21:12	22:31	—:—

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	□	□	□	□
66°	□	□	□	□	□	□
64°	09:38	□	□	07:43	□	□
62°	10:33	10:16	□	06:48	09:12	□
60°	11:06	11:25	12:18	06:16	08:03	09:20
N 58°	11:31	12:00	12:57	05:52	07:28	08:41
56°	11:51	12:26	13:24	05:32	07:03	08:14
54°	12:07	12:46	13:45	05:16	06:43	07:52
52°	12:22	13:03	14:03	05:02	06:26	07:34
50°	12:34	13:18	14:18	04:50	06:11	07:19
45°	13:00	13:48	14:48	04:25	05:42	06:48
N 40°	13:21	14:11	15:12	04:05	05:19	06:25
35°	13:38	14:31	15:32	03:49	05:00	06:05
30°	13:53	14:47	15:49	03:35	04:43	05:48
20°	14:19	15:16	16:17	03:11	04:16	05:19
N 10°	14:41	15:40	16:41	02:50	03:52	04:55
0°	15:02	16:03	17:04	02:31	03:30	04:32
S 10°	15:22	16:25	17:27	02:11	03:08	04:09
20°	15:45	16:50	17:51	01:51	02:45	03:44
30°	16:11	17:18	18:19	01:27	02:18	03:16
35°	16:27	17:35	18:36	01:13	02:02	02:59
40°	16:45	17:55	18:55	00:57	01:43	02:39
45°	17:07	18:19	19:19	00:38	01:20	02:15
S 50°	17:35	18:50	19:49	00:15	00:52	01:43
52°	17:48	19:05	20:04	00:03	00:38	01:28
54°	18:04	19:24	20:22	· · ·	00:21	01:10
56°	18:23	19:46	20:43	· · ·	00:02	00:48
58°	18:46	20:14	21:09	23:39	· · ·	00:19
S 60°	19:17	20:55	21:46	23:07	23:38	· · ·

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12h		Mer. Pass.		
	00h			Upper	Lower	11-13
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	82-95%
10	07:25	07:37	12:08	21:16	08:46	
11	07:49	08:01	12:08	22:17	09:46	
12	08:12	08:24	12:08	23:18	10:48	

January 13, 14, 15 UT (Mon., Tue., Wed.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Mon	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec			
0	112°43.7	130°52.2	S08°07.0	352°17.4	N24°49.6	42°00.9	N21°41.4	125°17.5	S07°30.4				Alpheratz	357°34.9	29°13.8		
1	127°46.1	145°52.4	05.9	7°21.0	49.8	57°03.6	41.3	140°19.8	30.3				Ankaa	353°07.2	-42°10.4		
2	142°48.6	160°52.7	04.7	22°24.5	50.0	72°06.3	41.3	155°22.0	30.2				Schedar	349°31.2	56°40.7		
3	157°51.0	175°52.9	· · 03.5	37°28.0	· · 50.3	87°08.9	· · 41.3	170°24.3	· · 30.1				Diphda	348°47.3	-17°51.1		
4	172°53.5	190°53.1	02.3	52°31.6	50.5	102°11.6	41.3	185°26.6	30.1				Achernar	335°20.1	-57°06.9		
5	187°56.0	205°53.3	01.2	67°35.1	50.8	117°14.2	41.3	200°28.8	30.0				Hamal	327°51.1	23°34.9		
6	202°58.4	220°53.6	S08°00.0	82°38.6	N24°51.0	132°16.9	N21°41.3	215°31.1	S07°29.9				Polaris	313°48.5	89°22.5		
7	218°00.9	235°53.8	07°58.8	97°42.1	51.3	147°19.6	41.2	230°33.3	29.8				Acamar	315°11.6	-40°12.5		
8	233°03.4	250°54.0	57.7	112°45.7	51.5	162°22.2	41.2	245°35.6	29.7				Menkar	314°06.0	4°11.3		
9	248°05.8	265°54.3	· · 56.5	127°49.2	· · 51.7	177°24.9	· · 41.2	260°37.8	· · 29.6				Mirfak	308°28.0	49°57.2		
10	263°08.3	280°54.5	55.3	142°52.7	52.0	192°27.5	41.2	275°40.1	29.5				Aldebaran	290°39.3	16°33.6		
11	278°10.8	295°54.7	54.1	157°56.3	52.2	207°30.2	41.2	290°42.3	29.4				Rigel	281°03.6	-8°10.4		
12	293°13.2	310°55.0	S07°53.0	172°59.8	N24°52.5	222°32.9	N21°41.2	305°44.6	S07°29.3				Capella	280°21.4	46°01.5		
13	308°15.7	325°55.2	51.8	188°03.3	52.7	237°35.5	41.2	320°46.8	29.2				Bellatrix	278°22.6	6°22.3		
14	323°18.2	340°55.4	50.6	203°06.8	52.9	252°38.2	41.1	335°49.1	29.1				Elnath	278°01.5	28°37.7		
15	338°20.6	355°55.7	· · 49.4	218°10.4	· · 53.2	267°40.8	· · 41.1	350°51.4	· · 29.0				Anlilam	275°37.4	-1°11.2		
16	353°23.1	10°55.9	48.2	233°13.9	53.4	282°43.5	41.1	5°53.6	28.9				Betelgeuse	270°51.7	7°24.7		
17	8°25.5	25°56.1	47.1	248°17.4	53.7	297°46.1	41.1	20°55.9	28.9				Canopus	263°51.8	-52°42.6		
18	23°28.0	40°56.4	S07°45.9	263°21.0	N24°53.9	312°48.8	N21°41.1	35°58.1	S07°28.8				Sirius	258°25.9	-16°45.1		
19	38°30.5	55°56.6	44.7	278°24.5	54.1	327°51.5	41.1	51°00.4	28.7				Adhara	255°05.5	-29°00.4		
20	53°32.9	70°56.9	43.5	293°28.0	54.4	342°54.1	41.0	66°02.6	28.6				Procyon	244°50.5	5°09.6		
21	68°35.4	85°57.1	· · 42.4	308°31.6	· · 54.6	357°56.8	· · 41.0	81°04.9	· · 28.5				Pollux	243°16.9	27°57.9		
22	83°37.9	100°57.4	41.2	323°35.1	54.9	12°59.4	41.0	96°07.1	28.4				Avior	234°14.1	-59°35.3		
23	98°40.3	115°57.6	40.0	338°38.6	55.1	28°02.1	41.0	111°09.4	28.3				Suhail	222°45.9	-43°31.9		
Mer.pass. 16:26			ν 0.2' d -1.2' m -4.56			ν 3.5' d 0.2' m -1.43			ν 2.7' d -0.0' m -2.67			ν 2.3' d -0.1' m 1.08					
Tue	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	113°42.8	130°57.8	S07°38.8	353°42.2	N24°55.3	43°04.7	N21°41.0	126°11.6	S07°28.2				Dubhe	193°40.5	61°36.7		
1	128°45.3	145°58.1	37.7	8°45.7	55.6	58°07.4	41.0	141°13.9	28.1				Denebola	182°24.8	15°25.8		
2	143°47.7	160°58.3	36.5	23°49.2	55.8	73°10.0	41.0	156°16.1	28.0				Gienah	175°43.5	-17°40.8		
3	158°50.2	175°58.6	· · 35.3	38°52.8	· · 56.0	88°12.7	· · 40.9	171°18.4	· · 27.9				Acrux	172°59.9	-63°14.0		
4	173°52.7	190°58.8	34.1	53°56.3	56.3	103°15.4	40.9	186°20.7	27.8				Gacrux	171°51.5	-57°15.0		
5	188°55.1	205°59.1	33.0	68°59.8	56.5	118°18.0	40.9	201°22.9	27.7				Alioth	166°12.8	55°49.1		
6	203°57.6	220°59.4	S07°31.8	84°03.4	N24°56.8	133°20.7	N21°40.9	216°25.2	S07°27.6				Spica	158°22.3	-11°17.5		
7	219°00.0	235°59.6	30.6	99°06.9	57.0	148°23.3	40.9	231°27.4	27.6				Alkaid	152°52.0	49°11.0		
8	234°02.5	250°59.9	29.4	114°10.4	57.2	163°26.0	40.9	246°29.7	27.5				Hadar	148°36.1	-60°29.4		
9	249°05.0	266°00.1	· · 28.2	129°14.0	· · 57.5	178°28.6	· · 40.9	261°31.9	· · 27.4				Menkent	147°57.7	-36°29.4		
10	264°07.4	281°00.4	27.1	144°17.5	57.7	193°31.3	40.8	276°34.2	27.3				Arcturus	145°48.0	19°02.9		
11	279°09.9	296°00.6	25.9	159°21.0	57.9	208°33.9	40.8	291°36.4	27.2				Rigel Kent.	139°40.5	-60°56.1		
12	294°12.4	311°00.9	S07°24.7	174°24.6	N24°58.2	223°36.6	N21°40.8	306°38.7	S07°27.1				Kochab	137°20.2	74°02.8		
13	309°14.8	326°01.2	23.5	189°28.1	58.4	238°39.2	40.8	321°40.9	27.0				Zuben'ubi	136°56.1	-16°08.7		
14	324°17.3	341°01.4	22.4	204°31.6	58.6	253°41.9	40.8	336°43.2	26.9				Alphecca	126°03.9	26°37.6		
15	339°19.8	356°01.7	· · 21.2	219°35.2	· · 58.9	268°44.5	· · 40.8	351°45.4	· · 26.8				Antares	112°16.1	-26°29.2		
16	354°22.2	11°01.9	20.0	234°38.7	59.1	283°47.2	40.8	6°47.7	26.7				Atria	107°10.7	-69°04.1		
17	9°24.7	26°02.2	18.8	249°42.3	59.3	298°49.8	40.7	21°49.9	26.6				Sabik	102°03.0	-15°45.4		
18	24°27.1	41°02.5	S07°17.6	264°45.8	N24°59.6	313°52.5	N21°40.7	36°52.2	S07°26.5				Shaula	96°10.7	-37°07.3		
19	39°29.6	56°02.7	16.5	279°49.3	24°59.8	328°55.1	40.7	51°54.4	26.4				Rasalhague	95°58.8	12°32.4		
20	54°32.1	71°03.0	15.3	294°52.9	25°00.0	343°57.8	40.7	66°56.7	26.3				Eltanin	90°42.7	51°28.9		
21	69°34.5	86°03.3	· · 14.1	309°56.4	· · 00.3	359°00.4	· · 40.7	81°58.9	· · 26.2				Kaus Aust.	83°32.9	-34°22.4		
22	84°37.0	101°03.5	12.9	324°59.9	00.5	14°03.1	40.7	97°01.2	26.1				Vega	80°33.6	38°48.3		
23	99°39.5	116°03.8	11.8	340°03.5	00.7	29°05.7	40.7	112°03.4	26.1				Nunki	75°48.1	-26°16.0		
Mer.pass. 16:22			ν 0.2' d -1.2' m -4.57			ν 3.5' d 0.2' m -1.44			ν 2.7' d -0.0' m -2.66			ν 2.3' d -0.1' m 1.09					
Wed	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	114°41.9	131°04.1	S07°10.6	355°07.0	N25°01.0	44°08.3	N21°40.7	127°05.7	S07°26.0				Fomalhaut	15°14.7	-29°29.6		
1	129°44.4	146°04.4	09.4	10°10.5	01.2	59°11.0	40.6	142°07.9	25.9				Scheat	13°45.4	28°13.2		
2	144°46.9	161°04.6	08.2	25°14.1	01.4	74°13.6	40.6	157°10.2	25.8				Markab	13°30.1	15°20.4		
3	159°49.3	176°04.9	· · 07.0	40°17.6	· · 01.6	89°16.3	· · 40.6	172°12.4	· · 25.7				Jan 13 Mon	SHA	Mer.pass		
4	174°51.8	191°05.2	05.9	55°21.1	01.9	104°18.9	40.6	187°14.7	25.6				Venus	18°08.6	15:16		
5	189°54.3	206°05.5	04.7	70°24.7	02.1	119°21.6	40.6	202°16.9	25.5				Mars	239°33.8	00:31		
6	204°56.7	221°05.8	S07°03.5	85°28.2	N25°02.3	134°24.2	N21°40.6	217°19.2	S07°25.4				Jupiter	289°17.3	21:08		
7	219°59.2	236°06.0	02.3	100°31.8	02.6	149°26.9	40.6	232°21.4	25.3				Saturn	12°33.9	15:36		
8	235°01.6	251°06.3	01.1	115°35.3	02.8	164°29.5	40.5	247°23.7	25.2				Jan 14 Tue	SHA	Mer.pass		
9	250°04.1	266°06.6	07°00.0	130°38.8	· · 03.0	179°32.2	· · 40.5	262°25.9	· · 25.1				Venus	17°15.1	15:16		

h Sun Moon						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	177°51.2	S21°28.2	9°59.6	5.1'	N27°52.7	-3.7' 57.9'
1	192°51.0	27.8	24°23.7	5.1'	27°49.0	-3.9' 57.9'
2	207°50.8	27.4	38°47.8	5.2'	27°45.2	-4.0' 57.9'
3	222°50.5	· · 27.0	53°12.0	5.3'	27°41.1	-4.2' 57.9'
4	237°50.3	26.5	67°36.3	5.4'	27°36.9	-4.4' 57.8'
5	252°50.1	26.1	82°00.6	5.4'	27°32.5	-4.6' 57.8'
6	267°49.9	S21°25.7	96°25.1	5.5'	N27°28.0	-4.7' 57.8'
7	282°49.6	25.2	110°49.6	5.6'	27°23.3	-4.9' 57.8'
8	297°49.4	24.8	125°14.2	5.7'	27°18.4	-5.1' 57.8'
9	312°49.2	· · 24.4	139°38.9	5.8'	27°13.3	-5.2' 57.7'
10	327°48.9	23.9	154°03.6	5.9'	27°08.1	-5.4' 57.7'
11	342°48.7	23.5	168°28.5	5.9'	27°02.7	-5.5' 57.7'
12	357°48.5	S21°23.1	182°53.4	6.0'	N26°57.2	-5.7' 57.7'
13	12°48.2	22.6	197°18.5	6.1'	26°51.5	-5.9' 57.6'
14	27°48.0	22.2	211°43.6	6.2'	26°45.6	-6.0' 57.6'
15	42°47.8	· · 21.8	226°08.8	6.3'	26°39.6	-6.2' 57.6'
16	57°47.6	21.3	240°34.2	6.4'	26°33.4	-6.3' 57.6'
17	72°47.3	20.9	254°59.6	6.5'	26°27.1	-6.5' 57.5'
18	87°47.1	S21°20.5	269°25.1	6.6'	N26°20.6	-6.6' 57.5'
19	102°46.9	20.0	283°50.7	6.7'	26°13.9	-6.8' 57.5'
20	117°46.6	19.6	298°16.4	6.8'	26°07.2	-6.9' 57.5'
21	132°46.4	· · 19.1	312°42.3	6.9'	26°00.2	-7.1' 57.5'
22	147°46.2	18.7	327°08.2	7.0'	25°53.1	-7.2' 57.4'
23	162°46.0	18.2	341°34.2	7.1'	25°45.9	-7.4' 57.4'
SD = 16.3' d = -0.4' SD = 15.8'						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	177°45.7	S21°17.8	356°00.4	7.2'	N25°38.5	-7.5' 57.4'
1	192°45.5	17.4	10°26.6	7.4'	25°31.0	-7.7' 57.4'
2	207°45.3	16.9	24°53.0	7.5'	25°23.3	-7.8' 57.3'
3	222°45.1	· · 16.5	39°19.4	7.6'	25°15.5	-7.9' 57.3'
4	237°44.8	16.0	53°46.0	7.7'	25°07.6	-8.1' 57.3'
5	252°44.6	15.6	68°12.7	7.8'	24°59.5	-8.2' 57.3'
6	267°44.4	S21°15.1	82°39.5	7.9'	N24°51.3	-8.4' 57.2'
7	282°44.2	14.7	97°06.4	8.0'	24°42.9	-8.5' 57.2'
8	297°43.9	14.2	111°33.4	8.1'	24°34.4	-8.6' 57.2'
9	312°43.7	· · 13.8	126°00.5	8.2'	24°25.8	-8.7' 57.2'
10	327°43.5	13.3	140°27.7	8.4'	24°17.1	-8.9' 57.1'
11	342°43.3	12.9	154°55.1	8.5'	24°08.2	-9.0' 57.1'
12	357°43.0	S21°12.4	169°22.6	8.6'	N23°59.2	-9.1' 57.1'
13	12°42.8	12.0	183°50.1	8.7'	23°50.1	-9.2' 57.1'
14	27°42.6	11.5	198°17.8	8.8'	23°40.8	-9.4' 57.0'
15	42°42.4	· · 11.1	212°45.7	8.9'	23°31.5	-9.5' 57.0'
16	57°42.1	10.6	227°13.6	9.0'	23°22.0	-9.6' 57.0'
17	72°41.9	10.2	241°41.6	9.2'	23°12.4	-9.7' 57.0'
18	87°41.7	S21°09.7	256°09.8	9.3'	N23°02.7	-9.8' 56.9'
19	102°41.5	09.3	270°38.0	9.4'	22°52.8	-9.9' 56.9'
20	117°41.3	08.8	285°06.4	9.5'	22°42.9	-10.1' 56.9'
21	132°41.0	· · 08.3	299°34.9	9.6'	22°32.8	-10.2' 56.8'
22	147°40.8	07.9	314°03.5	9.7'	22°22.7	-10.3' 56.8'
23	162°40.6	07.4	328°32.3	9.8'	22°12.4	-10.4' 56.8'
SD = 16.3' d = -0.4' SD = 15.6'						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	177°40.4	S21°07.0	343°01.1	10.0'	N22°02.0	-10.5' 56.8'
1	192°40.2	06.5	357°30.1	10.1'	21°51.5	-10.6' 56.7'
2	207°39.9	06.0	11°59.1	10.2'	21°40.9	-10.7' 56.7'
3	222°39.7	· · 05.6	26°28.3	10.3'	21°30.3	-10.8' 56.7'
4	237°39.5	05.1	40°57.6	10.4'	21°19.5	-10.9' 56.7'
5	252°39.3	04.7	55°27.0	10.5'	21°08.6	-11.0' 56.6'
6	267°39.1	S21°04.2	69°56.6	10.6'	N20°57.6	-11.1' 56.6'
7	282°38.8	03.7	84°26.2	10.8'	20°46.5	-11.2' 56.6'
8	297°38.6	03.3	98°56.0	10.9'	20°35.4	-11.3' 56.6'
9	312°38.4	· · 02.8	113°25.8	11.0'	20°24.1	-11.3' 56.5'
10	327°38.2	02.3	127°55.8	11.1'	20°12.8	-11.4' 56.5'
11	342°38.0	01.9	142°25.9	11.2'	20°01.3	-11.5' 56.5'
12	357°37.8	S21°01.4	156°56.1	11.3'	N19°49.8	-11.6' 56.5'
13	12°37.5	00.9	171°26.4	11.4'	19°38.2	-11.7' 56.4'
14	27°37.3	00.5	185°56.8	11.5'	19°26.5	-11.8' 56.4'
15	42°37.1	21°00.0	200°27.4	11.6'	19°14.7	-11.9' 56.4'
16	57°36.9	20°59.5	214°58.0	11.7'	19°02.9	-11.9' 56.3'
17	72°36.7	59.0	229°28.7	11.9'	18°50.9	-12.0' 56.3'
18	87°36.5	S20°58.6	243°59.6	12.0'	N18°38.9	-12.1' 56.3'
19	102°36.3	58.1	258°30.6	12.1'	18°26.8	-12.2' 56.3'
20	117°36.0	57.6	273°01.6	12.2'	18°14.7	-12.2' 56.2'
21	132°35.8	· · 57.2	287°32.8	12.3'	18°02.4	-12.3' 56.2'
22	147°35.6	56.7	302°04.1	12.4'	17°50.1	-12.4' 56.2'
23	162°35.4	56.2	316°35.4	12.5'	17°37.7	-12.4' 56.2'
SD = 16.3' d = -0.5' SD = 15.5'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:02	09:55	■	■	14:25	16:17
N 70°	07:47	09:20	■	■	14:59	16:32
68°	07:35	08:55	10:36	13:42	15:24	16:44
66°	07:25	08:36	09:55	14:24	15:43	16:54
64°	07:16	08:20	09:27	14:52	15:59	17:03
62°	07:08	08:07	09:06	15:13	16:12	17:10
60°	07:02	07:56	08:49	15:30	16:23	17:17
N 58°	06:56	07:46	08:34	15:44	16:33	17:23
56°	06:50	07:37	08:22	15:57	16:41	17:29
54°	06:45	07:30	08:11	16:07	16:49	17:34
52°	06:40	07:23	08:02	16:17	16:56	17:39
50°	06:36	07:16	07:53	16:25	17:02	17:43
45°	06:26	07:02	07:35	16:43	17:16	17:53
N 40°	06:17	06:51	07:20	16:58	17:28	18:01
35°	06:09	06:40	07:08	17:11	17:38	18:09
30°	06:01	06:31	06:57	17:22	17:48	18:17
20°	05:47	06:14	06:38	17:41	18:04	18:31
N 10°	05:33	05:59	06:21	17:57	18:20	18:46
0°	05:17	05:43	06:05	18:13	18:35	19:01
S 10°	05:00	05:27	05:50	18:29	18:51	19:18
20°	04:40	05:09	05:33	18:45	19:10	19:38
30°	04:14	04:46	05:13	19:05	19:32	20:04
35°	03:57	04:32	05:01	19:17	19:45	20:21
40°	03:36	04:16	04:48	19:30	20:02	20:41
45°	03:10	03:56	04:32	19:46	20:21	21:08
S 50°	02:32	03:30	04:12	20:05	20:47	21:45
52°	02:11	03:17	04:03	20:15	21:00	22:05
54°	01:44	03:03	03:52	20:25	21:14	22:32
56°	00:59	02:45	03:40	20:37	21:32	23:14
58°	—:—	02:23	03:26	20:51	21:54	—:—
S 60°	—:—	01:53	03:10	21:07	22:22	—:—

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	□	□	□	□	□	□
N 70°	□	□	14:46	□	□	12:44
68°	□	□	15:46	□	□	11:43
66°	□	13:28	16:20	□	12:14	11:08
64°	□	14:37	16:45	□	11:04	10:42
62°	13:06	15:12	17:04	10:37	10:28	10:22
60°	13:51	15:37	17:20	09:51	10:02	10:05
N 58°	14:21	15:57	17:33	09:21	09:41	09:51
56°	14:44	16:14	17:44	08:59	09:24	09:39
54°	15:02	16:28	17:54	08:40	09:10	09:29
52°	15:18	16:40	18:03	08:24	08:57	09:19
50°	15:31	16:51	18:11	08:10	08:46	09:11
45°	15:59	17:13	18:27	07:42	08:22	08:53
N 40°	16:20	17:31	18:41	07:20	08:03	08:38
35°	16:38	17:46	18:52	07:01	07:47	08:25
30°	16:54	17:59	19:02	06:45	07:34	08:14
20°	17:20	18:21	19:19	06:18	07:10	07:55
N 10°	17:42	18:40	19:34	05:55	06:49	07:38
0°	18:03	18:58	19:48	05:33	06:30	07:22
S 10°	18:24	19:16	20:02	05:11	06:10	07:07
20°	18:46	19:34	20:16	04:47	05:50	06:50
30°	19:12	19:56	20:33	04:19	05:25	06:30
35°	19:27	20:09	20:43	04:03	05:11	06:18
40°	19:44	20:23	20:54	03:44	04:54	06:05
45°	20:05	20:40	21:07	03:21	04:34	05:49
S 50°	20:32	21:01	21:22	02:51	04:08	05:29
52°	20:45	21:11	21:30	02:36	03:56	05:20
54°	20:59	21:22	21:38	02:19	03:42	05:09
56°	21:16	21:35	21:47	01:58	03:25	04:57
58°	21:37	21:50	21:57	01:32	03:05	04:43
S 60°	22:02	22:07	22:08	00:55	02:40	04:26

Day	Sun		Mer. Pass hh:mm	Moon		Age 14-16 99-99%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
13	08:35	08:46	12:09	—:—	11:48	
14	08:57	09:08	12:09	00:17	12:44	
15	09:18	09:29	12:09	01:10	13:35	

January 16, 17, 18 UT (Thu., Fri., Sat.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	115°41.1	131°11.0	S06°42.3	356°31.9	N25°06.4	45°11.8	N21°40.3	127°59.7	S07°23.7	Alpheratz	357°34.9	29°13.8											
1	130°43.5	146°11.3	41.1	11°35.4	06.6	60°14.4	40.3	143°01.9	23.6	Ankaa	353°07.2	-42°10.4											
2	145°46.0	161°11.6	39.9	26°39.0	06.8	75°17.0	40.3	158°04.2	23.5	Schedar	349°31.2	56°40.7											
3	160°48.5	176°11.9	· · 38.7	41°42.5	· · 07.1	90°19.7	· · 40.3	173°06.4	· · 23.4	Diphda	348°47.3	-17°51.1											
4	175°50.9	191°12.2	37.5	56°46.0	07.3	105°22.3	40.3	188°08.7	23.3	Achernar	335°20.1	-57°06.9											
5	190°53.4	206°12.5	36.3	71°49.6	07.5	120°24.9	40.3	203°10.9	23.2	Hamal	327°51.1	23°34.9											
6	205°55.9	221°12.8	S06°35.2	86°53.1	N25°07.7	135°27.6	N21°40.3	218°13.2	S07°23.1	Polaris	313°50.0	89°22.5											
7	220°58.3	236°13.1	34.0	101°56.7	08.0	150°30.2	40.3	233°15.4	23.0	Acamar	315°11.6	-40°12.5											
8	236°00.8	251°13.4	32.8	117°00.2	08.2	165°32.9	40.2	248°17.7	22.9	Menkar	314°06.0	4°11.3											
9	251°03.3	266°13.7	· · 31.6	132°03.7	· · 08.4	180°35.5	· · 40.2	263°19.9	· · 22.8	Mirfak	308°28.0	49°57.2											
10	266°05.7	281°14.0	30.4	147°07.3	08.6	195°38.1	40.2	278°22.2	22.7	Aldebaran	290°39.3	16°33.6											
11	281°08.2	296°14.3	29.3	162°10.8	08.8	210°40.8	40.2	293°24.4	22.6	Rigel	281°03.6	-8°10.4											
12	296°10.6	311°14.6	S06°28.1	177°14.3	N25°09.1	225°43.4	N21°40.2	308°26.7	S07°22.5	Capella	280°21.4	46°01.5											
13	311°13.1	326°14.9	26.9	192°17.9	09.3	240°46.0	40.2	323°28.9	22.5	Bellatrix	278°22.6	6°22.3											
14	326°15.6	341°15.3	25.7	207°21.4	09.5	255°48.7	40.2	338°31.1	22.4	Elnath	278°01.5	28°37.8											
15	341°18.0	356°15.6	· · 24.5	222°25.0	· · 09.7	270°51.3	· · 40.2	353°33.4	· · 22.3	Alnilam	275°37.4	-1°11.2											
16	356°20.5	11°15.9	23.4	237°28.5	09.9	285°53.9	40.1	8°35.6	22.2	Betelgeuse	270°51.7	7°24.7											
17	11°23.0	26°16.2	22.2	252°32.0	10.2	300°56.6	40.1	23°37.9	22.1	Canopus	263°51.9	-52°42.6											
18	26°25.4	41°16.5	S06°21.0	267°35.6	N25°10.4	315°59.2	N21°40.1	38°40.1	S07°22.0	Sirius	258°25.9	-16°45.1											
19	41°27.9	56°16.8	19.8	282°39.1	10.6	331°01.8	40.1	53°42.4	21.9	Adhara	255°05.5	-29°00.4											
20	56°30.4	71°17.2	18.6	297°42.6	10.8	346°04.5	40.1	68°44.6	21.8	Procyon	244°50.5	5°09.6											
21	71°32.8	86°17.5	· · 17.4	312°46.2	· · 11.0	1°07.1	· · 40.1	83°46.9	· · 21.7	Pollux	243°16.9	27°57.9											
22	86°35.3	101°17.8	16.3	327°49.7	11.2	16°09.7	40.1	98°49.1	21.6	Avior	234°14.1	-59°35.3											
23	101°37.7	116°18.1	15.1	342°53.3	11.5	31°12.3	40.1	113°51.4	21.5	Suhail	222°45.9	-43°31.9											
Mer.pass. 16:15				ν 0.3' d -1.2' m -4.59				ν 3.5' d 0.2' m -1.45				ν 2.6' d -0.0' m -2.65				ν 2.2' d -0.1' m 1.09							

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	116°40.2	131°18.5	S06°13.9	357°56.8	N25°11.7	46°15.0	N21°40.0	128°53.6	S07°21.4										
1	131°42.7	146°18.8	12.7	13°00.3	11.9	61°17.6	40.0	143°55.9	21.3										
2	146°45.1	161°19.1	11.5	28°03.9	12.1	76°20.2	40.0	158°58.1	21.2										
3	161°47.6	176°19.4	· · 10.4	43°07.4	· · 12.3	91°22.9	· · 40.0	174°00.3	· · 21.1										
4	176°50.1	191°19.8	09.2	58°10.9	12.5	106°25.5	40.0	189°02.6	21.0										
5	191°52.5	206°20.1	08.0	73°14.5	12.7	121°28.1	40.0	204°04.8	20.9										
6	206°55.0	221°20.4	S06°06.8	88°18.0	N25°13.0	136°30.7	N21°40.0	219°07.1	S07°20.8										
7	221°57.5	236°20.8	05.6	103°21.5	13.2	151°33.4	40.0	234°09.3	20.7										
8	236°59.9	251°21.1	04.4	118°25.1	13.4	166°36.0	40.0	249°11.6	20.6										
9	252°02.4	266°21.4	· · 03.3	133°28.6	· · 13.6	181°38.6	· · 39.9	264°13.8	· · 20.5										
10	267°04.9	281°21.8	02.1	148°32.2	13.8	196°41.3	39.9	279°16.1	20.4										
11	282°07.3	296°22.1	06°00.9	163°35.7	14.0	211°43.9	39.9	294°18.3	20.3										
12	297°09.8	311°22.4	S05°59.7	178°39.2	N25°14.2	226°46.5	N21°39.9	309°20.6	S07°20.2										
13	312°12.2	326°22.8	58.5	193°42.8	14.5	241°49.1	39.9	324°22.8	20.2										
14	327°14.7	341°23.1	57.3	208°46.3	14.7	256°51.8	39.9	339°25.0	20.1										
15	342°17.2	356°23.5	· · 56.2	223°49.8	· · 14.9	271°54.4	· · 39.9	354°27.3	· · 20.0										
16	357°19.6	11°23.8	55.0	238°53.4	15.1	286°57.0	39.9	9°29.5	19.9										
17	12°22.1	26°24.1	53.8	253°56.9	15.3	301°59.6	39.9	24°31.8	19.8										
18	27°24.6	41°24.5	S05°52.6	269°00.4	N25°15.5	317°02.3	N21°39.8	39°34.0	S07°19.7										
19	42°27.0	56°24.8	51.4	284°04.0	15.7	332°04.9	39.8	54°36.3	19.6										
20	57°29.5	71°25.2	50.2	299°07.5	15.9	347°07.5	39.8	69°38.5	19.5										
21	72°32.0	86°25.5	· · 49.1	314°11.0	· · 16.1	2°10.1	· · 39.8	84°40.7	· · 19.4										
22	87°34.4	101°25.9	47.9	329°14.6	16.3	17°12.7	39.8	99°43.0	19.3										
23	102°36.9	116°26.2	46.7	344°18.1	16.5	32°15.4	39.8	114°45.2	19.2										
Mer.pass. 16:11				ν 0.3' d -1.2' m -4.60				ν 3.5' d 0.2' m -1.44				ν 2.6' d -0.0' m -2.64				ν 2.2' d -0.1' m 1.09			

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	117°39.4	131°26.6	S05°45.5	359°21.6	N25°16.8	47°18.0	N21°39.8	129°47.5	S07°19.1										
1	132°41.8	146°26.9	44.3	14°25.2	17.0	62°20.6	39.8	144°49.7	19.0										
2	147°44.3	161°27.3	43.2	29°28.7	17.2	77°23.2	39.8	159°52.0	18.9										
3	162°46.7	176°27.6	· · 42.0	44°32.2	· · 17.4	92°25.9	· · 39.7	174°54.2	· · 18.8										
4	177°49.2	191°28.0	40.8	59°35.8	17.6	107°28.5	39.7	189°56.5	18.7										
5	192°51.7	206°28.4	39.6	74°39.3	17.8	122°31.1	39.7	204°58.7	18.6										
6	207°54.1	221°28.7	S05°38.4	89°42.8	N25°18.0	137°33.7	N21°39.7	220°00.9	S07°18.5										
7	222°56.6	236°29.1	37.2	104°46.4	18.2	152°36.3	39.7	235°03.2	18.4										
8	237°59.1	251°29.4	36.1	119°49.9	18.4	167°38.9	39.7	250°05.4	18.3										
9	253°01.5	266°29.8	· · 34.9	134°53.4	· · 18.6	182°41.6	· · 39.7	265°07.7	· · 18.2										
10	268°04.0	281°30.2	33.7	149°57.0	18.8	197°44.2	39.7	280°09.9	18.1										
11	283°06.5	296°30.5	32.5	165°00.5	19.0	212°46.8	39.7	295°12.2	18.0										
12	298°08.9	311°30.9	S05°31.3	180°04.0	N25°19.2	227°49.4	N21°39.7	310°14.4	S07°17.9										
13	313°11.4	326°31.3	30.1	195°07.6	19.4	242°52.0	39.6	325°16.6	17.8										
14	328°13.8	341°31.6	29.0	210°11.1	19.6	257°54.7	39.6	340°18.9	17.7										
15	343°16.3	356°32.0	· · 27.8	225°14.6	· · 19.8	272°57.3	· · 39.6	355°21.1	· · 17.6										
16	358°18.8	11°32.4	26.6	240°18.2	20.0	287°59.9	39.6	10°23.4	17.5										
17	13°21.2	26°32.7	25.4	255°21.7	20.2	303°02.5	39.6	25°25.6	17.4										
18	28°23.7	41°33.1	S05°24.2	270°25.2	N25°20.4	318°05.1	N21°39.6	40°27.8	S07°17.3										
19	43°26.2	56°33.5	23.0	285°28.7	20.6	333°07.7	39.6	55°30.1	17.2										
20	58°28.6	71°33.9	21.9	300°32.3	20.8	348°10.3	39.6	70°32.3	17.1										
21	73°31.1	86°34.2	· · 20.7	315°35.8	· · 21.0	3°13.0	· · 39.6	85°34.6	· · 17.0										
22	88°33.6	101°34.6	19.5	330°39.3	21.2	18°15.6	39.6	100°36.8	16.9										
23	103°36.0	116°35.0	18.3	345°42.9	21.4	33°18.2	39.5	115°39.1	16.8										
Mer.pass. 16:07				ν 0.4' d -1.2' m -4.61				ν 3.5' d 0.2' m -1.42				ν 2.6' d -0.0' m -2.63				ν 2.2' d -0.1' m 1.09			

Jan 16 Thu	SHA	Mer.pass
Venus	15°29.9	15:15
Mars	240°50.8	00:14
Jupiter	289°30.7	20:56
Saturn	12°18.6	15:26

Jan 17 Fri	SHA	Mer.pass
Venus	14°38.2	15:14
Mars	241°16.6	00:08
Jupiter	289°34.8	20:51
Saturn	12°13.4	15:22

Jan 18 Sat	SHA	Mer.pass
Venus	13°47.2	15:14
Mars	241°42.3	00:03
Jupiter	289°38.6	20:47
Saturn	12°08.1	15:19

Horizontal parallax		
Venus:	0.2	
Mars:	0.2	

Sun				Moon				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	177°35.2	S20°55.7		331°06.9	12.6'	N17°25.3	-12.5'	56.1'
1	192°35.0	55.3		345°38.5	12.7'	17°12.8	-12.6'	56.1'
2	207°34.8	54.8		0°10.2	12.8'	17°00.2	-12.6'	56.1'
3	222°34.6	· · 54.3		14°41.9	12.9'	16°47.6	-12.7'	56.1'
4	237°34.3	53.8		29°13.8	13.0'	16°34.8	-12.8'	56.0'
5	252°34.1	53.3		43°45.8	13.1'	16°22.1	-12.8'	56.0'
6	267°33.9	S20°52.9		58°17.9	13.2'	N16°09.2	-12.9'	56.0'
7	282°33.7	52.4		72°50.0	13.3'	15°56.4	-12.9'	56.0'
8	297°33.5	51.9		87°22.3	13.4'	15°43.4	-13.0'	55.9'
9	312°33.3	· · 51.4		101°54.6	13.4'	15°30.4	-13.1'	55.9'
10	327°33.1	50.9		116°27.1	13.5'	15°17.3	-13.1'	55.9'
11	342°32.9	50.4		130°59.6	13.6'	15°04.2	-13.2'	55.9'
12	357°32.7	S20°50.0		145°32.3	13.7'	N14°51.1	-13.2'	55.8'
13	12°32.5	49.5		160°05.0	13.8'	14°37.8	-13.3'	55.8'
14	27°32.2	49.0		174°37.8	13.9'	14°24.6	-13.3'	55.8'
15	42°32.0	· · 48.5		189°10.7	14.0'	14°11.2	-13.4'	55.8'
16	57°31.8	48.0		203°43.7	14.1'	13°57.9	-13.4'	55.7'
17	72°31.6	47.5		218°16.8	14.2'	13°44.5	-13.5'	55.7'
18	87°31.4	S20°47.0		232°49.9	14.2'	N13°31.0	-13.5'	55.7'
19	102°31.2	46.5		247°23.1	14.3'	13°17.5	-13.5'	55.7'
20	117°31.0	46.1		261°56.5	14.4'	13°04.0	-13.6'	55.6'
21	132°30.8	· · 45.6		276°29.9	14.5'	12°50.4	-13.6'	55.6'
22	147°30.6	45.1		291°03.4	14.6'	12°36.8	-13.7'	55.6'
23	162°30.4	44.6		305°36.9	14.6'	12°23.1	-13.7'	55.6'
SD = 16.3' d = -0.5'				SD = 15.3'				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	177°30.2	S20°44.1		320°10.6	14.7'	N12°09.4	-13.7'	55.5'
1	192°30.0	43.6		334°44.3	14.8'	11°55.7	-13.8'	55.5'
2	207°29.8	43.1		349°18.1	14.9'	11°41.9	-13.8'	55.5'
3	222°29.6	· · 42.6		3°51.9	14.9'	11°28.1	-13.8'	55.5'
4	237°29.4	42.1		18°25.9	15.0'	11°14.2	-13.9'	55.4'
5	252°29.1	41.6		32°59.9	15.1'	11°00.3	-13.9'	55.4'
6	267°28.9	S20°41.1		47°34.0	15.2'	N10°46.4	-13.9'	55.4'
7	282°28.7	40.6		62°08.1	15.2'	10°32.5	-14.0'	55.4'
8	297°28.5	40.1		76°42.4	15.3'	10°18.5	-14.0'	55.4'
9	312°28.3	· · 39.6		91°16.7	15.4'	10°04.5	-14.0'	55.3'
10	327°28.1	39.1		105°51.0	15.4'	09°50.5	-14.0'	55.3'
11	342°27.9	38.6		120°25.4	15.5'	09°36.5	-14.1'	55.3'
12	357°27.7	S20°38.1		134°59.9	15.6'	N09°22.4	-14.1'	55.3'
13	12°27.5	37.6		149°34.5	15.6'	09°08.3	-14.1'	55.2'
14	27°27.3	37.1		164°09.1	15.7'	08°54.2	-14.1'	55.2'
15	42°27.1	· · 36.6		178°43.8	15.7'	08°40.0	-14.2'	55.2'
16	57°26.9	36.1		193°18.5	15.8'	08°25.9	-14.2'	55.2'
17	72°26.7	35.6		207°53.3	15.8'	08°11.7	-14.2'	55.2'
18	87°26.5	S20°35.1		222°28.1	15.9'	N07°57.5	-14.2'	55.1'
19	102°26.3	34.6		237°03.0	16.0'	07°43.3	-14.2'	55.1'
20	117°26.1	34.1		251°38.0	16.0'	07°29.0	-14.3'	55.1'
21	132°25.9	· · 33.6		266°13.0	16.1'	07°14.8	-14.3'	55.1'
22	147°25.7	33.1		280°48.1	16.1'	07°00.5	-14.3'	55.0'
23	162°25.5	32.6		295°23.2	16.2'	06°46.2	-14.3'	55.0'
SD = 16.3' d = -0.5'				SD = 15.1'				
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	177°25.3	S20°32.1		309°58.4	16.2'	N06°31.9	-14.3'	55.0'
1	192°25.1	31.5		324°33.6	16.3'	06°17.6	-14.3'	55.0'
2	207°24.9	31.0		339°08.8	16.3'	06°03.3	-14.3'	55.0'
3	222°24.7	· · 30.5		353°44.1	16.4'	05°48.9	-14.3'	54.9'
4	237°24.5	30.0		8°19.5	16.4'	05°34.6	-14.4'	54.9'
5	252°24.3	29.5		22°54.9	16.4'	05°20.2	-14.4'	54.9'
6	267°24.1	S20°29.0		37°30.3	16.5'	N05°05.8	-14.4'	54.9'
7	282°23.9	28.5		52°05.8	16.5'	04°51.5	-14.4'	54.9'
8	297°23.8	28.0		66°41.3	16.6'	04°37.1	-14.4'	54.9'
9	312°23.6	· · 27.4		81°16.9	16.6'	04°22.7	-14.4'	54.8'
10	327°23.4	26.9		95°52.5	16.6'	04°08.3	-14.4'	54.8'
11	342°23.2	26.4		110°28.1	16.7'	03°53.9	-14.4'	54.8'
12	357°23.0	S20°25.9		125°03.8	16.7'	N03°39.5	-14.4'	54.8'
13	12°22.8	25.4		139°39.5	16.7'	03°25.1	-14.4'	54.8'
14	27°22.6	24.9		154°15.2	16.8'	03°10.7	-14.4'	54.7'
15	42°22.4	· · 24.3		168°51.0	16.8'	02°56.3	-14.4'	54.7'
16	57°22.2	23.8		183°26.8	16.8'	02°41.9	-14.4'	54.7'
17	72°22.0	23.3		198°02.6	16.9'	02°27.5	-14.4'	54.7'
18	87°21.8	S20°22.8		212°38.5	16.9'	N02°13.1	-14.4'	54.7'
19	102°21.6	22.3		227°14.4	16.9'	01°58.7	-14.4'	54.7'
20	117°21.4	21.7		241°50.3	16.9'	01°44.3	-14.4'	54.6'
21	132°21.2	· · 21.2		256°26.2	17.0'	01°29.9	-14.4'	54.6'
22	147°21.0	20.7		271°02.1	17.0'	01°15.5	-14.4'	54.6'
23	162°20.9	20.2		285°38.1	17.0'	01°01.1	-14.4'	54.6'
SD = 16.2' d = -0.5'				SD = 15.0'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:55	09:42	■	■	14:39	16:26
N 70°	07:41	09:11	11:37	12:44	15:10	16:40
68°	07:30	08:48	10:22	13:59	15:33	16:51
66°	07:20	08:30	09:46	14:35	15:51	17:01
64°	07:12	08:15	09:20	15:01	16:06	17:09
62°	07:05	08:03	09:00	15:21	16:18	17:16
60°	06:58	07:52	08:44	15:37	16:29	17:23
N 58°	06:53	07:43	08:30	15:51	16:38	17:28
56°	06:47	07:34	08:19	16:02	16:46	17:33
54°	06:43	07:27	08:08	16:13	16:54	17:38
52°	06:38	07:20	07:59	16:22	17:00	17:43
50°	06:34	07:14	07:51	16:30	17:07	17:47
45°	06:25	07:01	07:33	16:47	17:20	17:56
N 40°	06:16	06:49	07:19	17:02	17:31	18:04
35°	06:08	06:39	07:07	17:14	17:41	18:12
30°	06:01	06:30	06:56	17:24	17:50	18:19
20°	05:47	06:14	06:38	17:43	18:06	18:33
N 10°	05:33	05:59	06:22	17:59	18:21	18:47
0°	05:19	05:44	06:07	18:14	18:36	19:02
S 10°	05:02	05:29	05:51	18:29	18:52	19:18
20°	04:42	05:11	05:35	18:45	19:09	19:38
30°	04:17	04:49	05:15	19:05	19:31	20:03
35°	04:00	04:36	05:04	19:16	19:44	20:19
40°	03:40	04:20	04:51	19:29	20:00	20:39
45°	03:15	04:00	04:36	19:44	20:19	21:04
S 50°	02:39	03:36	04:17	20:03	20:44	21:40
52°	02:19	03:23	04:08	20:12	20:56	21:59
54°	01:54	03:09	03:58	20:22	21:10	22:24
56°	01:16	02:52	03:46	20:33	21:27	22:59
58°	—:—	02:31	03:33	20:46	21:47	—:—
S 60°	—:—	02:04	03:17	21:02	22:13	—:—

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	17:15	19:38	21:40	11:54	11:02	10:26
N 70°	17:44	19:49	21:42	11:24	10:48	10:21
68°	18:04	19:59	21:43	11:02	10:37	10:17
66°	18:21	20:06	21:44	10:44	10:27	10:13
64°	18:34	20:13	21:46	10:29	10:19	10:10
62°	18:45	20:18	21:46	10:17	10:12	10:07
60°	18:54	20:23	21:47	10:06	10:06	10:05
N 58°	19:03	20:27	21:48	09:57	10:00	10:02
56°	19:10	20:31	21:48	09:49	09:55	10:00
54°	19:16	20:34	21:49	09:42	09:51	09:59
52°	19:22	20:37	21:50	09:35	09:47	09:57
50°	19:27	20:40	21:50	09:29	09:44	09:56
45°	19:38	20:46	21:51	09:16	09:36	09:52
N 40°	19:47	20:51	21:52	09:06	09:29	09:50
35°	19:55	20:55	21:53	08:57	09:23	09:47
30°	20:02	20:59	21:53	08:48	09:18	09:45
20°	20:14	21:05	21:54	08:34	09:09	09:42
N 10°	20:24	21:11	21:55	08:22	09:02	09:39
0°	20:34	2				

January 19, 20, 21 UT (Sun., Mon., Tue.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	118°38.5	131°35.4	S05°17.1	0°46.4	N25°21.6	48°20.8	N21°39.5	130°41.3	S07°16.7					Alpheratz	357°34.9	29°13.8	
1	133°41.0	146°35.7	16.0	15°49.9	21.8	63°23.4	39.5	145°43.5	16.7					Ankaa	353°07.3	-42°10.4	
2	148°43.4	161°36.1	14.8	30°53.4	22.0	78°26.0	39.5	160°45.8	16.6					Schedar	349°31.2	56°40.7	
3	163°45.9	176°36.5	•• 13.6	45°57.0	•• 22.2	93°28.6	•• 39.5	175°48.0	•• 16.5					Diphda	348°47.4	-17°51.1	
4	178°48.3	191°36.9	12.4	61°00.5	22.4	108°31.2	39.5	190°50.3	16.4					Achernar	335°20.2	-57°06.9	
5	193°50.8	206°37.3	11.2	76°04.0	22.6	123°33.9	39.5	205°52.5	16.3					Hamal	327°51.1	23°34.9	
6	208°53.3	221°37.7	S05°10.0	91°07.6	N25°22.8	138°36.5	N21°39.5	220°54.7	S07°16.2					Polaris	313°51.5	89°22.5	
7	223°55.7	236°38.0	08.9	106°11.1	23.0	153°39.1	39.5	235°57.0	16.1					Acamar	315°11.6	-40°12.5	
8	238°58.2	251°38.4	07.7	121°14.6	23.2	168°41.7	39.5	250°59.2	16.0					Menkar	314°06.0	4°11.3	
9	254°00.7	266°38.8	•• 06.5	136°18.1	•• 23.4	183°44.3	•• 39.5	266°01.5	•• 15.9					Mirfak	308°28.0	49°57.2	
10	269°03.1	281°39.2	05.3	151°21.7	23.6	198°46.9	39.4	281°03.7	15.8					Aldebaran	290°39.3	16°33.6	
11	284°05.6	296°39.6	04.1	166°25.2	23.8	213°49.5	39.4	296°05.9	15.7					Rigel	281°03.6	-8°10.4	
12	299°08.1	311°40.0	S05°02.9	181°28.7	N25°24.0	228°52.1	N21°39.4	311°08.2	S07°15.6					Capella	280°21.5	46°01.5	
13	314°10.5	326°40.4	01.8	196°32.2	24.2	243°54.7	39.4	326°10.4	15.5					Bellatrix	278°22.6	6°22.3	
14	329°13.0	341°40.8	05°00.6	211°35.8	24.4	258°57.3	39.4	341°12.7	15.4					Elnath	278°01.5	28°37.8	
15	344°15.4	356°41.2	04°59.4	226°39.3	•• 24.6	274°00.0	•• 39.4	356°14.9	•• 15.3					Alnilam	275°37.4	-1°11.2	
16	359°17.9	11°41.6	58.2	241°42.8	24.8	289°02.6	39.4	11°17.1	15.2					Betelgeuse	270°51.7	7°24.7	
17	14°20.4	26°42.0	57.0	256°46.3	25.0	304°05.2	39.4	26°19.4	15.1					Canopus	263°51.9	-52°42.6	
18	29°22.8	41°42.4	S04°55.8	271°49.9	N25°25.2	319°07.8	N21°39.4	41°21.6	S07°15.0					Sirius	258°25.9	-16°45.1	
19	44°25.3	56°42.8	54.7	286°53.4	25.4	334°10.4	39.4	56°23.9	14.9					Adhara	255°05.5	-29°00.4	
20	59°27.8	71°43.2	53.5	301°56.9	25.6	349°13.0	39.4	71°26.1	14.8					Procyon	244°50.5	5°09.6	
21	74°30.2	86°43.6	•• 52.3	317°00.4	•• 25.8	4°15.6	•• 39.3	86°28.3	•• 14.7					Pollux	243°16.9	27°57.9	
22	89°32.7	101°44.0	51.1	332°03.9	25.9	19°18.2	39.3	101°30.6	14.6					Avior	234°14.1	-59°35.3	
23	104°35.2	116°44.4	49.9	347°07.5	26.1	34°20.8	39.3	116°32.8	14.5					Suhail	222°45.9	-43°32.0	
Mer.pass. 16:03			ν 0.4' d -1.2' m -4.62			ν 3.5' d 0.2' m -1.41			ν 2.6' d -0.0' m -2.62			ν 2.2' d -0.1' m 1.09					
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	119°37.6	131°44.8	S04°48.8	2°11.0	N25°26.3	49°23.4	N21°39.3	131°35.1	S07°14.4					Alphard	217°47.4	-8°46.1	
1	134°40.1	146°45.2	47.6	17°14.5	26.5	64°26.0	39.3	146°37.3	14.3					Regulus	207°34.1	11°50.6	
2	149°42.6	161°45.6	46.4	32°18.0	26.7	79°28.6	39.3	161°39.5	14.2					Dubhe	193°40.5	61°36.7	
3	164°45.0	176°46.0	•• 45.2	47°21.6	•• 26.9	94°31.2	•• 39.3	176°41.8	•• 14.1					Denebola	182°24.7	14°25.8	
4	179°47.5	191°46.4	44.0	62°25.1	27.1	109°33.8	39.3	191°44.0	14.0					Gienah	175°43.4	-17°40.8	
5	194°49.9	206°46.8	42.8	77°28.6	27.3	124°36.4	39.3	206°46.2	13.9					Acrux	172°59.8	-63°14.0	
6	209°52.4	221°47.3	S04°41.7	92°32.1	N25°27.5	139°39.0	N21°39.3	221°48.5	S07°13.8					Gacrux	171°51.4	-57°15.0	
7	224°54.9	236°47.7	40.5	107°35.6	27.7	154°41.6	39.3	236°50.7	13.7					Alioth	166°12.8	55°49.1	
8	239°57.3	251°48.1	39.3	122°39.1	27.8	169°44.2	39.2	251°53.0	13.6					Spica	158°22.2	-11°17.5	
9	254°59.8	266°48.5	•• 38.1	137°42.7	•• 28.0	184°46.8	•• 39.2	266°55.2	•• 13.5					Alkaid	152°51.9	49°11.0	
10	270°02.3	281°48.9	36.9	152°46.2	28.2	199°49.4	39.2	281°57.4	13.4					Hadar	148°36.0	-60°29.4	
11	285°04.7	296°49.3	35.8	167°49.7	28.4	214°52.0	39.2	296°59.7	13.3					Menkent	147°57.6	-36°29.5	
12	300°07.2	311°49.8	S04°34.6	182°53.2	N25°28.6	229°54.6	N21°39.2	312°01.9	S07°13.2					Arcturus	145°47.9	19°02.9	
13	315°09.7	326°50.2	33.4	197°56.7	28.8	244°57.2	39.2	327°04.1	13.1					Rigel Kent.	139°40.4	-60°56.1	
14	330°12.1	341°50.6	32.2	213°00.3	29.0	259°59.8	39.2	342°06.4	13.0					Kochab	137°20.1	74°02.7	
15	345°14.6	356°51.0	•• 31.0	228°03.8	•• 29.1	275°02.4	•• 39.2	357°08.6	•• 12.9					Zuben'ubi	136°56.1	-16°08.7	
16	0°17.1	11°51.5	29.8	243°07.3	29.3	290°05.0	39.2	12°10.9	12.8					Alphecca	126°03.9	26°37.6	
17	15°19.5	26°51.9	28.7	258°10.8	29.5	305°07.6	39.2	27°13.1	12.7					Antares	112°16.0	-26°29.2	
18	30°22.0	41°52.3	S04°27.5	273°14.3	N25°29.7	320°10.2	N21°39.2	42°15.3	S07°12.6					Atria	107°10.7	-69°04.1	
19	45°24.4	56°52.7	26.3	288°17.8	29.9	335°12.8	39.2	57°17.6	12.5					Sabik	102°03.0	-15°45.4	
20	60°26.9	71°53.2	25.1	303°21.3	30.1	350°15.4	39.2	72°19.8	12.4					Shaula	96°10.7	-37°07.3	
21	75°29.4	86°53.6	•• 23.9	318°24.9	•• 30.3	5°18.0	•• 39.1	87°22.0	•• 12.3					Rasalhague	95°58.8	12°32.4	
22	90°31.8	101°54.0	22.8	333°28.4	30.4	20°20.6	39.1	102°24.3	12.2					Eltanin	90°42.6	51°28.9	
23	105°34.3	116°54.5	21.6	348°31.9	30.6	35°23.2	39.1	117°26.5	12.1					Kaus Aust.	83°32.8	-34°22.4	
Mer.pass. 15:59			ν 0.4' d -1.2' m -4.63			ν 3.5' d 0.2' m -1.39			ν 2.6' d -0.0' m -2.62			ν 2.2' d -0.1' m 1.10					
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	120°36.8	131°54.9	S04°20.4	3°35.4	N25°30.8	50°25.8	N21°39.1	132°28.8	S07°12.0					Venus	12°56.9	15:13	
1	135°39.2	146°55.3	19.2	18°38.9	31.0	65°28.4	39.1	147°31.0	11.9					Mars	242°07.9	23:51	
2	150°41.7	161°55.8	18.0	33°42.4	31.2	80°31.0	39.1	162°33.2	11.8					Jupiter	289°42.3	20:43	
3	165°44.2	176°56.2	•• 16.9	48°45.9	•• 31.3	95°33.6	•• 39.1	177°35.5	•• 11.7					Saturn	12°02.8	15:15	
4	180°46.6	191°56.6	15.7	63°49.4	31.5	110°36.2	39.1	192°37.7	11.6								
5	195°49.1	206°57.1	14.5	78°52.9	31.7	125°38.8	39.1	207°39.9	11.5								
6	210°51.5	221°57.5	S04°13.3	93°56.5	N25°31.9	140°41.4	N21°39.1	222°42.2	S07°11.4								
7	225°54.0	236°58.0	12.1	109°00.0	32.1	155°44.0	39.1	237°44.4	11.3								
8	240°56.5	251°58.4	10.9	124°03.5	32.2	170°46.6	39.1	252°46.6	11.2								
9	255°58.9	266°58.9	•• 09.8	139°07.0	•• 32.4	185°49.1	•• 39.1	267°48.9	•• 11.1								
1																	

Sun				Moon				
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	177°20.7	S20°19.6		300°14.1	17.0'	N00°46.7	-14.4'	54.6'
1	192°20.5	19.1		314°50.1	17.0'	00°32.3	-14.4'	54.6'
2	207°20.3	18.6		329°26.2	17.1'	00°18.0	-14.4'	54.6'
3	222°20.1	18.1		344°02.2	17.1'	N00°03.6	-14.4'	54.5'
4	237°19.9	17.5		358°38.3	17.1'	S00°10.8	14.3'	54.5'
5	252°19.7	17.0		13°14.4	17.1'	00°25.1	14.3'	54.5'
6	267°19.5	S20°16.5		27°50.5	17.1'	S00°39.4	14.3'	54.5'
7	282°19.3	15.9		42°26.6	17.1'	00°53.8	14.3'	54.5'
8	297°19.1	15.4		57°02.7	17.1'	01°08.1	14.3'	54.5'
9	312°19.0	14.9		71°38.9	17.1'	01°22.4	14.3'	54.5'
10	327°18.8	14.3		86°15.0	17.2'	01°36.7	14.3'	54.4'
11	342°18.6	13.8		100°51.2	17.2'	01°51.0	14.3'	54.4'
12	357°18.4	S20°13.3		115°27.3	17.2'	S02°05.2	14.3'	54.4'
13	12°18.2	12.7		130°03.5	17.2'	02°19.5	14.2'	54.4'
14	27°18.0	12.2		144°39.6	17.2'	02°33.7	14.2'	54.4'
15	42°17.8	11.7		159°15.8	17.2'	02°47.9	14.2'	54.4'
16	57°17.6	11.1		173°52.0	17.2'	03°02.1	14.2'	54.4'
17	72°17.5	10.6		188°28.2	17.2'	03°16.3	14.2'	54.4'
18	87°17.3	S20°10.1		203°04.3	17.2'	S03°30.5	14.2'	54.4'
19	102°17.1	09.5		217°40.5	17.2'	03°44.6	14.1'	54.3'
20	117°16.9	09.0		232°16.7	17.2'	03°58.8	14.1'	54.3'
21	132°16.7	08.4		246°52.8	17.2'	04°12.9	14.1'	54.3'
22	147°16.5	07.9		261°29.0	17.2'	04°27.0	14.1'	54.3'
23	162°16.4	07.4		276°05.1	17.1'	04°41.1	14.0'	54.3'
SD = 16.2' d = -0.5'				SD = 14.9'				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	177°16.2	S20°06.8		290°41.3	17.1'	S04°55.1	14.0'	54.3'
1	192°16.0	06.3		305°17.4	17.1'	05°09.1	14.0'	54.3'
2	207°15.8	05.7		319°53.6	17.1'	05°23.1	14.0'	54.3'
3	222°15.6	05.2		334°29.7	17.1'	05°37.1	14.0'	54.3'
4	237°15.4	04.7		349°05.8	17.1'	05°51.1	13.9'	54.3'
5	252°15.3	04.1		3°41.9	17.1'	06°05.0	13.9'	54.3'
6	267°15.1	S20°03.6		18°17.9	17.1'	S06°18.9	13.9'	54.3'
7	282°14.9	03.0		32°54.0	17.0'	06°32.8	13.9'	54.2'
8	297°14.7	02.5		47°30.0	17.0'	06°46.6	13.8'	54.2'
9	312°14.5	01.9		62°06.1	17.0'	07°00.5	13.8'	54.2'
10	327°14.4	01.4		76°42.1	17.0'	07°14.2	13.8'	54.2'
11	342°14.2	00.8		91°18.1	17.0'	07°28.0	13.7'	54.2'
12	357°14.0	S20°00.3		105°54.0	16.9'	S07°41.8	13.7'	54.2'
13	12°13.8	19°59.7		120°30.0	16.9'	07°55.5	13.7'	54.2'
14	27°13.6	59.2		135°05.9	16.9'	08°09.1	13.6'	54.2'
15	42°13.5	58.6		149°41.8	16.9'	08°22.8	13.6'	54.2'
16	57°13.3	58.1		164°17.7	16.8'	08°36.4	13.6'	54.2'
17	72°13.1	57.5		178°53.5	16.8'	08°50.0	13.5'	54.2'
18	87°12.9	S19°57.0		193°29.3	16.8'	S09°03.5	13.5'	54.2'
19	102°12.7	56.4		208°05.1	16.8'	09°17.0	13.5'	54.2'
20	117°12.6	55.9		222°40.9	16.7'	09°30.5	13.4'	54.2'
21	132°12.4	55.3		237°16.6	16.7'	09°43.9	13.4'	54.2'
22	147°12.2	54.8		251°52.3	16.7'	09°57.4	13.4'	54.2'
23	162°12.0	54.2		266°27.9	16.6'	10°10.7	13.3'	54.2'
SD = 16.2' d = -0.5'				SD = 14.8'				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	177°11.9	S19°53.6		281°03.6	16.6'	S10°24.0	13.3'	54.2'
1	192°11.7	53.1		295°39.2	16.6'	10°37.3	13.3'	54.2'
2	207°11.5	52.5		310°14.7	16.5'	10°50.6	13.2'	54.2'
3	222°11.3	52.0		324°50.2	16.5'	11°03.8	13.2'	54.2'
4	237°11.2	51.4		339°25.7	16.4'	11°17.0	13.1'	54.2'
5	252°11.0	50.8		354°01.1	16.4'	11°30.1	13.1'	54.2'
6	267°10.8	S19°50.3		8°36.5	16.3'	S11°43.2	13.0'	54.2'
7	282°10.6	49.7		23°11.9	16.3'	11°56.2	13.0'	54.2'
8	297°10.5	49.2		37°47.2	16.3'	12°09.3	13.0'	54.2'
9	312°10.3	48.6		52°22.4	16.2'	12°22.2	12.9'	54.2'
10	327°10.1	48.0		66°57.7	16.2'	12°35.1	12.9'	54.2'
11	342°10.0	47.5		81°32.8	16.1'	12°48.0	12.8'	54.2'
12	357°09.8	S19°46.9		96°07.9	16.1'	S13°00.8	12.8'	54.2'
13	12°09.6	46.3		110°43.0	16.0'	13°13.6	12.7'	54.2'
14	27°09.4	45.8		125°18.0	16.0'	13°26.3	12.7'	54.2'
15	42°09.3	45.2		139°53.0	15.9'	13°39.0	12.6'	54.2'
16	57°09.1	44.6		154°27.9	15.9'	13°51.6	12.6'	54.2'
17	72°08.9	44.1		169°02.8	15.8'	14°04.2	12.5'	54.2'
18	87°08.8	S19°43.5		183°37.6	15.8'	S14°16.7	12.5'	54.2'
19	102°08.6	42.9		198°12.4	15.7'	14°29.2	12.4'	54.2'
20	117°08.4	42.4		212°47.1	15.6'	14°41.7	12.4'	54.2'
21	132°08.3	41.8		227°21.7	15.6'	14°54.0	12.3'	54.2'
22	147°08.1	41.2		241°56.3	15.5'	15°06.4	12.3'	54.2'
23	162°07.9	40.7		256°30.8	15.5'	15°18.6	12.2'	54.2'
SD = 16.2' d = -0.6'				SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:47	09:30	■	■	14:54	16:36
N 70°	07:34	09:01	11:03	13:21	15:22	16:49
68°	07:24	08:40	10:09	14:14	15:43	16:59
66°	07:15	08:23	09:36	14:47	16:00	17:08
64°	07:07	08:09	09:13	15:10	16:13	17:16
62°	07:01	07:58	08:54	15:29	16:25	17:22
60°	06:55	07:48	08:39	15:44	16:35	17:28
N 58°	06:49	07:39	08:26	15:57	16:44	17:33
56°	06:44	07:31	08:15	16:08	16:52	17:38
54°	06:40	07:24	08:05	16:18	16:59	17:43
52°	06:36	07:18	07:56	16:27	17:05	17:47
50°	06:32	07:12	07:48	16:35	17:11	17:51
45°	06:23	06:59	07:31	16:51	17:24	18:00
N 40°	06:15	06:48	07:17	17:05	17:34	18:08
35°	06:08	06:39	07:06	17:17	17:44	18:15
30°	06:01	06:30	06:55	17:27	17:53	18:22
20°	05:47	06:14	06:38	17:45	18:08	18:35
N 10°	05:34	06:00	06:22	18:00	18:22	18:48
0°	05:20	05:45	06:07	18:15	18:37	19:02
S 10°	05:04	05:30	05:53	18:29	18:52	19:18
20°	04:44	05:13	05:37	18:45	19:09	19:37
30°	04:20	04:52	05:18	19:04	19:30	20:02
35°	04:04	04:39	05:07	19:14	19:43	20:18
40°	03:45	04:24	04:55	19:27	19:58	20:37
45°	03:20	04:05	04:40	19:42	20:16	21:01
S 50°	02:46	03:41	04:22	20:00	20:40	21:35
52°	02:28	03:29	04:13	20:08	20:52	21:53
54°	02:04	03:16	04:03	20:18	21:05	22:15
56°	01:32	03:00	03:52	20:29	21:21	22:46
58°	—:—	02:40	03:39	20:41	21:40	23:52
S 60°	—:—	02:15	03:24	20:56	22:04	—:—

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	23:38		01:44	09:54	09:21	08:40
N 70°	23:31		01:25	09:56	09:31	09:01
68°	23:25		01:10	09:58	09:39	09:18
66°	23:20		00:58	10:00	09:46	09:31
64°	23:16		00:48	10:01	09:52	09:42
62°	23:13		00:40	10:02	09:57	09:52
60°	23:10		00:32	10:03	10:02	10:00
N 58°	23:07		00:26	10:04	10:06	10:08
56°	23:04		00:21	10:05	10:09	10:14
54°	23:02		00:16	10:06	10:12	10:20
52°	23:00		00:11	10:06	10:15	10:25
50°	22:59		00:07	10:07	10:18	10:30
45°	22:55	23:59		10:08	10:24	10:41
N 40°	22:52	23:51		10:09	10:29	10:49
35°	22:49	23:45		10:10	10:33	10:57
30°	22:46	23:40		10:11	10:37	11:03
20°	22:42	23:31		10:13	10:43	11:15
N 10°	22:39	23:22		10:14	1	

Aries		Venus		Mars		Jupiter		Saturn	
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	121°35.9	132°05.7	S03°52.1	4°59.5	N25°35.1	51°28.0	N21°39.0	133°22.4	S07°09.6
1	136°38.4	147°06.1	50.9	20°03.0	35.2	66°30.6	38.9	148°24.6	09.5
2	151°40.8	162°06.6	49.7	35°06.5	35.4	81°33.2	38.9	163°26.9	09.4
3	166°43.3	177°07.1	· · 48.5	50°10.0	· · 35.6	96°35.7	· · 38.9	178°29.1	· · 09.3
4	181°45.8	192°07.5	47.4	65°13.5	35.7	111°38.3	38.9	193°31.3	09.2
5	196°48.2	207°08.0	46.2	80°17.0	35.9	126°40.9	38.9	208°33.6	09.1
6	211°50.7	222°08.5	S03°45.0	95°20.5	N25°36.1	141°43.5	N21°38.9	223°35.8	S07°09.0
7	226°53.2	237°08.9	43.8	110°24.0	36.3	156°46.1	38.9	238°38.0	08.9
8	241°55.6	252°09.4	42.6	125°27.5	36.4	171°48.7	38.9	253°40.3	08.8
9	256°58.1	267°09.9	· · 41.5	140°31.0	· · 36.6	186°51.2	· · 38.9	268°42.5	· · 08.7
10	272°00.5	282°10.4	40.3	155°34.5	36.8	201°53.8	38.9	283°44.7	08.6
11	287°03.0	297°10.8	39.1	170°38.0	36.9	216°56.4	38.9	298°47.0	08.5
12	302°05.5	312°11.3	S03°37.9	185°41.5	N25°37.1	231°59.0	N21°38.9	313°49.2	S07°08.4
13	317°07.9	327°11.8	36.7	200°45.0	37.3	247°01.6	38.9	328°51.4	08.3
14	332°10.4	342°12.3	35.6	215°48.5	37.4	262°04.2	38.9	343°53.7	08.2
15	347°12.9	357°12.8	· · 34.4	230°52.0	· · 37.6	277°06.7	· · 38.9	358°55.9	· · 08.1
16	2°15.3	12°13.2	33.2	245°55.5	37.8	292°09.3	38.9	13°58.1	08.0
17	17°17.8	27°13.7	32.0	260°59.0	37.9	307°11.9	38.9	29°00.4	07.9
18	32°20.3	42°14.2	S03°30.9	276°02.5	N25°38.1	322°14.5	N21°38.8	44°02.6	S07°07.8
19	47°22.7	57°14.7	29.7	291°05.9	38.3	337°17.1	38.8	59°04.8	07.7
20	62°25.2	72°15.2	28.5	306°09.4	38.4	352°19.6	38.8	74°07.1	07.6
21	77°27.6	87°15.7	· · 27.3	321°12.9	· · 38.6	7°22.2	· · 38.8	89°09.3	· · 07.5
22	92°30.1	102°16.2	26.2	336°16.4	38.8	22°24.8	38.8	104°11.5	07.4
23	107°32.6	117°16.6	25.0	351°19.9	38.9	37°27.4	38.8	119°13.8	07.3
Mer.pass. 15:51		ν 0.5' d -1.2' m -4.65		ν 3.5' d 0.2' m -1.34		ν 2.6' d -0.0' m -2.60		ν 2.2' d -0.1' m 1.10	

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	122°35.0	132°17.1	S03°23.8	6°23.4	N25°39.1	52°30.0	N21°38.8	134°16.0	S07°07.2
1	137°37.5	147°17.6	22.6	21°26.9	39.3	67°32.5	38.8	149°18.2	07.1
2	152°40.0	162°18.1	21.4	36°30.4	39.4	82°35.1	38.8	164°20.5	07.0
3	167°42.4	177°18.6	· · 20.3	51°33.8	· · 39.6	97°37.7	· · 38.8	179°22.7	· · 06.9
4	182°44.9	192°19.1	19.1	66°37.3	39.8	112°40.3	38.8	194°24.9	06.8
5	197°47.4	207°19.6	17.9	81°40.8	39.9	127°42.8	38.8	209°27.1	06.7
6	212°49.8	222°20.1	S03°16.7	96°44.3	N25°40.1	142°45.4	N21°38.8	224°29.4	S07°06.6
7	227°52.3	237°20.6	15.6	111°47.8	40.2	157°48.0	38.8	239°31.6	06.5
8	242°54.8	252°21.1	14.4	126°51.3	40.4	172°50.6	38.8	254°33.8	06.4
9	257°57.2	267°21.6	· · 13.2	141°54.7	· · 40.6	187°53.1	· · 38.8	269°36.1	· · 06.3
10	272°59.7	282°22.1	12.0	156°58.2	40.7	202°55.7	38.8	284°38.3	06.2
11	288°02.1	297°22.6	10.9	172°01.7	40.9	217°58.3	38.8	299°40.5	06.1
12	303°04.6	312°23.1	S03°09.7	187°05.2	N25°41.0	233°00.9	N21°38.8	314°42.8	S07°06.0
13	318°07.1	327°23.6	08.5	202°08.7	41.2	248°03.4	38.7	329°45.0	05.9
14	333°09.5	342°24.2	07.3	217°12.1	41.4	263°06.0	38.7	344°47.2	05.8
15	348°12.0	357°24.7	· · 06.2	232°15.6	· · 41.5	278°08.6	· · 38.7	359°49.5	· · 05.7
16	3°14.5	12°25.2	05.0	247°19.1	41.7	293°11.2	38.7	14°51.7	05.5
17	18°16.9	27°25.7	03.8	262°22.6	41.8	308°13.7	38.7	29°53.9	05.4
18	33°19.4	42°26.2	S03°02.6	277°26.0	N25°42.0	323°16.3	N21°38.7	44°56.1	S07°05.3
19	48°21.9	57°26.7	01.5	292°29.5	42.1	338°18.9	38.7	59°58.4	05.2
20	63°24.3	72°27.2	03°00.3	307°33.0	42.3	353°21.4	38.7	75°00.6	05.1
21	78°26.8	87°27.8	02°59.1	322°36.5	· · 42.5	8°24.0	· · 38.7	90°02.8	· · 05.0
22	93°29.3	102°28.3	57.9	337°39.9	42.6	23°26.6	38.7	105°05.1	04.9
23	108°31.7	117°28.8	56.8	352°43.4	42.8	38°29.2	38.7	120°07.3	04.8
Mer.pass. 15:47		ν 0.5' d -1.2' m -4.66		ν 3.5' d 0.2' m -1.32		ν 2.6' d -0.0' m -2.59		ν 2.2' d -0.1' m 1.10	

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	123°34.2	132°29.3	S02°55.6	7°46.9	N25°42.9	53°31.7	N21°38.7	135°09.5	S07°04.7
1	138°36.6	147°29.8	54.4	22°50.3	43.1	68°34.3	38.7	150°11.8	04.6
2	153°39.1	162°30.4	53.2	37°53.8	43.2	83°36.9	38.7	165°14.0	04.5
3	168°41.6	177°30.9	· · 52.1	52°57.3	· · 43.4	98°39.4	· · 38.7	180°16.2	· · 04.4
4	183°44.0	192°31.4	50.9	68°00.8	43.5	113°42.0	38.7	195°18.4	04.3
5	198°46.5	207°31.9	49.7	83°04.2	43.7	128°44.6	38.7	210°20.7	04.2
6	213°49.0	222°32.5	S02°48.5	98°07.7	N25°43.8	143°47.1	N21°38.7	225°22.9	S07°04.1
7	228°51.4	237°33.0	47.4	113°11.2	44.0	158°49.7	38.7	240°25.1	04.0
8	243°53.9	252°33.5	46.2	128°14.6	44.1	173°52.3	38.7	255°27.4	03.9
9	258°56.4	267°34.1	· · 45.0	143°18.1	· · 44.3	188°54.8	· · 38.7	270°29.6	· · 03.8
10	273°58.8	282°34.6	43.9	158°21.6	44.4	203°57.4	38.7	285°31.8	03.7
11	289°01.3	297°35.1	42.7	173°25.0	44.6	219°00.0	38.7	300°34.1	03.6
12	304°03.8	312°35.7	S02°41.5	188°28.5	N25°44.7	234°02.5	N21°38.6	315°36.3	S07°03.5
13	319°06.2	327°36.2	40.3	203°31.9	44.9	249°05.1	38.6	330°38.5	03.4
14	334°08.7	342°36.7	39.2	218°35.4	45.0	264°07.7	38.6	345°40.7	03.3
15	349°11.1	357°37.3	· · 38.0	233°38.9	· · 45.2	279°10.2	· · 38.6	0°43.0	· · 03.2
16	4°13.6	12°37.8	36.8	248°42.3	45.3	294°12.8	38.6	15°45.2	03.1
17	19°16.1	27°38.4	35.7	263°45.8	45.5	309°15.3	38.6	30°47.4	03.0
18	34°18.5	42°38.9	S02°34.5	278°49.2	N25°45.6	324°17.9	N21°38.6	45°49.6	S07°02.9
19	49°21.0	57°39.5	33.3	293°52.7	45.8	339°20.5	38.6	60°51.9	02.8
20	64°23.5	72°40.0	32.1	308°56.2	45.9	354°23.0	38.6	75°54.1	02.7
21	79°25.9	87°40.6	· · 31.0	323°59.6	· · 46.1	9°25.6	· · 38.6	90°56.3	· · 02.6
22	94°28.4	102°41.1	29.8	339°03.1	46.2	24°28.2	38.6	105°58.6	02.5
23	109°30.9	117°41.7	28.6	354°06.5	46.4	39°30.7	38.6	121°00.8	02.4
Mer.pass. 15:43		ν 0.5' d -1.2' m -4.67		ν 3.5' d 0.2' m -1.29		ν 2.6' d -0.0' m -2.58		ν 2.2' d -0.1' m 1.10	

Stars		
	SHA	Dec
Alpheratz	357° 34.9	29° 13.8
Ankaa	353° 07.3	-42° 10.4
Schedar	349° 31.2	-56° 40.7
Diphda	348° 47.4	-17° 51.1
Achernar	335° 20.2	-57° 06.9
Hamal	327° 51.1	23° 34.9
Polaris	313° 52.8	89° 22.5
Acamar	315° 11.6	-40° 12.5
Menkar	314° 06.0	4° 11.3
Mirfak	308° 28.0	49° 57.2
Aldebaran	290° 39.4	16° 33.6
Rigel	281° 03.6	-8° 10.4
Capella	280° 21.5	46° 01.5
Bellatrix	278° 22.6	6° 22.3
Elnath	278° 01.5	28° 37.8
Anilam	275° 37.4	-1° 11.2
Betelgeuse	270° 51.8	7° 24.7
Canopus	263° 51.9	-52° 42.6
Sirius	258° 25.9	-16° 45.1
Adhara	255° 05.5	-29° 00.4
Procyon	244° 50.5	5° 09.6
Pollux	243° 16.9	27° 57.9
Avior	234° 14.1	-59° 35.3
Suhail	222° 45.9	-43° 32.0
Miaplacidus	221° 37.4	-69° 49.1
Alphard	217° 47.4	-8° 46.1
Regulus	207° 34.1	11° 50.6
Dubhe	193° 40.4	61° 36.7
Denebola	182° 24.7	14° 25.8
Gienah	175° 43.4	-17° 40.9
Acrux	172° 59.8	-63° 14.0
Gacrux	171° 51.4	-57° 15.0
Alioth	166° 12.7	55° 49.1
Spica	158° 22.2	-11° 17.6
Alkaid	152° 51.9	49° 11.0
Hadar	148° 36.0	-60° 29.4
Menkent	147° 57.6	-36° 29.5
Arcturus	145° 47.9	19° 02.9
Rigel Kent.	139° 40.4	-60° 56.1
Kochab	137° 20.0	74° 02.7
Zuben 'ubi	136° 56.0	-16° 08.8
Alphecca	126° 03.8	26° 37.6
Antares	112° 16.0	-26° 29.2
Atria	107° 10.6	-69° 04.1
Sabik	102° 03.0	-15° 45.4
Shaula	96° 10.6	-37° 07.3
Rasalhague	95° 58.8	12° 32.4
Eltanin	90° 42.6	51° 28.9
Kaus Aust.	83° 32.8	-34° 22.3
Vega	80° 33.6	38° 48.2
Nunki	75° 48.1	-26° 16.0
Altair	62° 00.3	8° 56.0
Peacock	53° 06.3	-56° 39.3
Deneb	49° 26.3	45° 22.1
Enif	33° 39.1	9° 59.3
Al Na'ir	27° 33.3	-46° 50.6
Fomalhaut	15° 14.7	-29° 29.5
Scheat	13° 45.4	-28° 13.2
Markab	13° 30.1	15° 20.4

Jan 22 Wed	SHA	Mer.pass
Venus	10°29.8	15:11
Mars	243°23.6	23:35
Jupiter	289°52.1	20:31
Saturn	11°46.5	15:04

Jan 23 Thu	SHA	Mer.pass
Venus	9°42.1	15:10
Mars	243°48.3	23:29
Jupiter	289°54.9	20:26
Saturn	11°40.9	15:01

Jan 24 Fri	SHA	Mer.pass
Venus	8°55.1	15:10
Mars	244°12.7	23:23
Jupiter	289°57.5	20:22
Saturn	11°35.3	14:57

Horizontal parallax	
Venus:	0.3
Mars:	0.2

h	Sun			Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	177°07.7	S19°40.1	271°05.3	15.4'	S15°30.8	12.2'	54.2'
1	192°07.6	39.5	285°39.7	15.3'	15°43.0	12.1'	54.2'
2	207°07.4	38.9	300°14.0	15.3'	15°55.1	12.0'	54.2'
3	222°07.2	· · 38.4	314°48.3	15.2'	16°07.1	12.0'	54.2'
4	237°07.1	37.8	329°22.5	15.1'	16°19.1	11.9'	54.3'
5	252°06.9	37.2	343°56.7	15.1'	16°31.0	11.9'	54.3'
6	267°06.7	S19°36.6	358°30.8	15.0'	S16°42.9	11.8'	54.3'
7	282°06.6	36.1	13°04.8	14.9'	16°54.7	11.7'	54.3'
8	297°06.4	35.5	27°38.7	14.9'	17°06.4	11.7'	54.3'
9	312°06.2	· · 34.9	42°12.6	14.8'	17°18.1	11.6'	54.3'
10	327°06.1	34.3	56°46.4	14.7'	17°29.7	11.5'	54.3'
11	342°05.9	33.7	71°20.1	14.7'	17°41.2	11.5'	54.3'
12	357°05.8	S19°33.2	85°53.8	14.6'	S17°52.7	11.4'	54.3'
13	12°05.6	32.6	100°27.4	14.5'	18°04.1	11.3'	54.3'
14	27°05.4	32.0	115°00.9	14.4'	18°15.5	11.3'	54.3'
15	42°05.3	· · 31.4	129°34.4	14.4'	18°26.8	11.2'	54.4'
16	57°05.1	30.8	144°07.7	14.3'	18°38.0	11.1'	54.4'
17	72°04.9	30.3	158°41.0	14.2'	18°49.1	11.1'	54.4'
18	87°04.8	S19°29.7	173°14.2	14.1'	S19°00.2	11.0'	54.4'
19	102°04.6	29.1	187°47.4	14.1'	19°11.2	10.9'	54.4'
20	117°04.5	28.5	202°20.4	14.0'	19°22.1	10.8'	54.4'
21	132°04.3	· · 27.9	216°53.4	13.9'	19°33.0	10.8'	54.4'
22	147°04.1	27.3	231°26.3	13.8'	19°43.7	10.7'	54.4'
23	162°04.0	26.7	245°59.1	13.7'	19°54.4	10.6'	54.4'
SD = 16.2'			SD = 14.8'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	177°03.8	S19°26.2	260°31.9	13.7'	S20°05.0	10.5'	54.5'
1	192°03.7	25.6	275°04.5	13.6'	20°15.6	10.5'	54.5'
2	207°03.5	25.0	289°37.1	13.5'	20°26.1	10.4'	54.5'
3	222°03.3	· · 24.4	304°09.6	13.4'	20°36.4	10.3'	54.5'
4	237°03.2	23.8	318°42.0	13.3'	20°46.7	10.2'	54.5'
5	252°03.0	23.2	333°14.3	13.2'	20°57.0	10.1'	54.5'
6	267°02.9	S19°22.6	347°46.5	13.1'	S21°07.1	10.1'	54.5'
7	282°02.7	22.0	2°18.7	13.1'	21°17.2	10.0'	54.6'
8	297°02.5	21.4	16°50.7	13.0'	21°27.1	9.9'	54.6'
9	312°02.4	· · 20.8	31°22.7	12.9'	21°37.0	9.8'	54.6'
10	327°02.2	20.3	45°54.6	12.8'	21°46.8	9.7'	54.6'
11	342°02.1	19.7	60°26.4	12.7'	21°56.5	9.6'	54.6'
12	357°01.9	S19°19.1	74°58.1	12.6'	S22°06.2	9.5'	54.6'
13	12°01.8	18.5	89°29.7	12.5'	22°15.7	9.4'	54.7'
14	27°01.6	17.9	104°01.2	12.4'	22°25.1	9.3'	54.7'
15	42°01.5	· · 17.3	118°32.6	12.3'	22°34.5	9.3'	54.7'
16	57°01.3	16.7	133°04.0	12.2'	22°43.7	9.2'	54.7'
17	72°01.1	16.1	147°35.2	12.2'	22°52.9	9.1'	54.7'
18	87°01.0	S19°15.5	162°06.4	12.1'	S23°02.0	9.0'	54.7'
19	102°00.8	14.9	176°37.4	12.0'	23°10.9	8.9'	54.8'
20	117°00.7	14.3	191°08.4	11.9'	23°19.8	8.8'	54.8'
21	132°00.5	· · 13.7	205°39.3	11.8'	23°28.6	8.7'	54.8'
22	147°00.4	13.1	220°10.1	11.7'	23°37.3	8.6'	54.8'
23	162°00.2	12.5	234°40.8	11.6'	23°45.8	8.5'	54.8'
SD = 16.2'			SD = 14.9'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	177°00.1	S19°11.9	249°11.4	11.5'	S23°54.3	8.4'	54.9'
1	191°59.9	11.3	263°41.9	11.4'	24°02.7	8.3'	54.9'
2	206°59.8	10.7	278°12.3	11.3'	24°10.9	8.2'	54.9'
3	221°59.6	· · 10.1	292°42.6	11.2'	24°19.1	8.0'	54.9'
4	236°59.5	09.5	307°12.8	11.1'	24°27.1	7.9'	54.9'
5	251°59.3	08.9	321°42.9	11.0'	24°35.1	7.8'	55.0'
6	266°59.2	S19°08.2	336°12.9	10.9'	S24°42.9	7.7'	55.0'
7	281°59.0	07.6	350°42.9	10.8'	24°50.6	7.6'	55.0'
8	296°58.9	07.0	5°12.7	10.7'	24°58.2	7.5'	55.0'
9	311°58.7	· · 06.4	19°42.5	10.6'	25°05.7	7.4'	55.0'
10	326°58.6	05.8	34°12.1	10.6'	25°13.1	7.3'	55.1'
11	341°58.4	05.2	48°41.7	10.5'	25°20.4	7.2'	55.1'
12	356°58.3	S19°04.6	63°11.1	10.4'	S25°27.5	7.0'	55.1'
13	11°58.1	04.0	77°40.5	10.3'	25°34.6	6.9'	55.1'
14	26°58.0	03.4	92°09.7	10.2'	25°41.5	6.8'	55.2'
15	41°57.8	· · 02.8	106°38.9	10.1'	25°48.3	6.7'	55.2'
16	56°57.7	02.2	121°08.0	10.0'	25°55.0	6.6'	55.2'
17	71°57.5	01.5	135°37.0	9.9'	26°01.5	6.4'	55.2'
18	86°57.4	S19°00.9	150°05.9	9.8'	S26°08.0	6.3'	55.2'
19	101°57.2	19°00.3	164°34.7	9.7'	26°14.3	6.2'	55.3'
20	116°57.1	18°59.7	179°03.4	9.6'	26°20.5	6.1'	55.3'
21	131°57.0	· · 59.1	193°32.0	9.5'	26°26.5	5.9'	55.3'
22	146°56.8	58.5	208°00.5	9.4'	26°32.4	5.8'	55.3'
23	161°56.7	57.9	222°29.0	9.3'	26°38.2	5.7'	55.4'
SD = 16.2'			SD = 15.0'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:38	09:17	■	■	15:08	16:47
N 70°	07:27	08:51	10:39	13:46	15:33	16:58
68°	07:17	08:32	09:55	14:29	15:53	17:08
66°	07:09	08:16	09:26	14:58	16:08	17:16
64°	07:02	08:03	09:05	15:20	16:21	17:22
62°	06:56	07:52	08:47	15:37	16:32	17:29
60°	06:51	07:43	08:33	15:51	16:42	17:34
N 58°	06:46	07:35	08:21	16:04	16:50	17:39
56°	06:41	07:27	08:10	16:14	16:57	17:43
54°	06:37	07:21	08:01	16:24	17:04	17:48
52°	06:33	07:14	07:52	16:32	17:10	17:51
50°	06:29	07:09	07:45	16:39	17:15	17:55
45°	06:21	06:57	07:29	16:55	17:27	18:03
N 40°	06:14	06:46	07:16	17:08	17:38	18:11
35°	06:07	06:37	07:04	17:20	17:47	18:17
30°	06:00	06:29	06:55	17:30	17:55	18:24
20°	05:47	06:14	06:37	17:46	18:10	18:37
N 10°	05:35	06:00	06:22	18:01	18:24	18:49
0°	05:21	05:46	06:08	18:15	18:37	19:03
S 10°	05:05	05:32	05:54	18:30	18:52	19:18
20°	04:47	05:15	05:39	18:45	19:09	19:37
30°	04:23	04:54	05:21	19:03	19:29	20:00
35°	04:08	04:42	05:10	19:13	19:41	20:15
40°	03:49	04:27	04:58	19:25	19:56	20:34
45°	03:25	04:09	04:44	19:39	20:13	20:57
S 50°	02:53	03:47	04:27	19:56	20:36	21:29
52°	02:36	03:35	04:18	20:05	20:47	21:46
54°	02:14	03:23	04:09	20:14	21:00	22:07
56°	01:46	03:07	03:58	20:24	21:15	22:34
58°	00:57	02:49	03:46	20:36	21:32	23:18
S 60°	—:—	02:27	03:32	20:50	21:54	—:—

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	04:26	■	■	07:24	■	■
N 70°	03:36	■	■	08:16	■	■
68°	03:05	05:42	■	08:48	07:44	■
66°	02:43	04:45	■	09:12	08:41	■
64°	02:25	04:12	06:23	09:31	09:15	08:43
62°	02:10	03:48	05:35	09:47	09:40	09:32
60°	01:58	03:29	05:04	10:00	10:00	10:03
N 58°	01:48	03:13	04:42	10:11	10:16	10:27
56°	01:39	03:00	04:23	10:21	10:30	10:46
54°	01:31	02:48	04:07	10:30	10:43	11:02
52°	01:24	02:38	03:54	10:37	10:53	11:15
50°	01:17	02:29	03:42	10:44	11:03	11:27
45°	01:03	02:10	03:18	11:00	11:23	11:53
N 40°	00:52	01:55	02:59	11:12	11:39	12:13
35°	00:43	01:42	02:43	11:23	11:53	12:30
30°	00:34	01:31	02:29	11:32	12:06	12:44
20°	00:20	01:11	02:05	11:49	12:26	13:09
N 10°	00:07	00:55	01:45	12:03	12:45	13:30
0°	· · ·	00:39	01:27	12:17	13:02	13:51
S 10°	· · ·	00:24	01:08	12:30	13:19	14:11
20°	· · ·	00:08	00:48	12:45	13:37	14:32
30°	23:49	· · ·	00:25	13:01	13:58	14:58
35°	23:39	· · ·	00:12	13:11	14:11	15:12
40°	23:26	23:56	· · ·	13:22	14:25	15:30
45°	23:12	23:38	· · ·	13:35	14:43	15:51
S 50°	22:54	23:16	23:45	13:51	15:04	16:17
52°	22:46	23:05	23:32	13:59	15:14	16:30
54°	22:37	22:53	23:17	14:08	15:26	16:45
56°	22:27	22:39	22:59	14:17	15:39	17:02
58°	22:15	22:23	22:38	14:28	15:55	17:23
S 60°	22:02	22:04	22:10	14:41	16:13	17:50

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h			Upper	Lower	23-25
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	49-30%
22	11:29	11:37	12:12	06:06	18:28	
23	11:45	11:52	12:12	06:50	19:14	
24	12:00	12:07	12:12	07:38	20:04	

January 25, 26, 27 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	124°33.3	132°42.2	S02°27.5	9°10.0	N25°46.5	54°33.3	N21°38.6	136°03.0	S07°02.3	Alpheratz	357°34.9	29°13.8
1	139°35.8	147°42.8	26.3	24°13.4	46.6	69°35.8	38.6	151°05.2	02.2	Ankaa	353°07.3	-42°10.4
2	154°38.2	162°43.3	25.1	39°16.9	46.8	84°38.4	38.6	166°07.5	02.1	Schedar	349°31.2	56°40.7
3	169°40.7	177°43.9	· · 23.9	54°20.3	· · 46.9	99°41.0	· · 38.6	181°09.7	· · 02.0	Diphda	348°47.4	-17°51.1
4	184°43.2	192°44.4	22.8	69°23.8	47.1	114°43.5	38.6	196°11.9	01.9	Achernar	335°20.2	-57°06.9
5	199°45.6	207°45.0	21.6	84°27.2	47.2	129°46.1	38.6	211°14.2	01.7	Hamal	327°51.2	23°34.9
6	214°48.1	222°45.5	S02°20.4	99°30.7	N25°47.4	144°48.6	N21°38.6	226°16.4	S07°01.6	Polaris	313°54.0	89°22.5
7	229°50.6	237°46.1	19.3	114°34.1	47.5	159°51.2	38.6	241°18.6	01.5	Acamar	315°11.6	-40°12.5
8	244°53.0	252°46.7	18.1	129°37.6	47.6	174°53.7	38.6	256°20.8	01.4	Menkar	314°06.0	4°11.2
9	259°55.5	267°47.2	· · 16.9	144°41.0	· · 47.8	189°56.3	· · 38.6	271°23.1	· · 01.3	Mirfak	308°28.0	49°57.2
10	274°58.0	282°47.8	15.8	159°44.5	47.9	204°58.9	38.6	286°25.3	01.2	Aldebaran	290°39.4	16°33.6
11	290°00.4	297°48.4	14.6	174°47.9	48.1	220°01.4	38.6	301°27.5	01.1	Rigel	281°03.6	-8°10.4
12	305°02.9	312°48.9	S02°13.4	189°51.4	N25°48.2	235°04.0	N21°38.6	316°29.7	S07°01.0	Capella	280°21.5	46°01.5
13	320°05.4	327°49.5	12.3	204°54.8	48.4	250°06.5	38.6	331°32.0	00.9	Bellatrix	278°22.6	6°22.3
14	335°07.8	342°50.1	11.1	219°58.2	48.5	265°09.1	38.6	346°34.2	00.8	Elnath	278°01.5	28°37.8
15	350°10.3	357°50.6	· · 09.9	235°01.7	· · 48.6	280°11.6	· · 38.6	1°36.4	· · 00.7	Alnilam	275°37.4	-1°11.2
16	5°12.7	12°51.2	08.8	250°05.1	48.8	295°14.2	38.6	16°38.6	00.6	Betelgeuse	270°51.8	7°24.7
17	20°15.2	27°51.8	07.6	265°08.6	48.9	310°16.7	38.6	31°40.9	00.5	Canopus	263°51.9	-52°42.6
18	35°17.7	42°52.4	S02°06.4	280°12.0	N25°49.0	325°19.3	N21°38.6	46°43.1	S07°00.4	Sirius	258°25.9	-16°45.1
19	50°20.1	57°52.9	05.3	295°15.4	49.2	340°21.8	38.6	61°45.3	00.3	Adhara	255°05.5	-29°00.4
20	65°22.6	72°53.5	04.1	310°18.9	49.3	355°24.4	38.6	76°47.6	00.2	Procyon	244°50.5	5°09.6
21	80°25.1	87°54.1	· · 02.9	325°22.3	· · 49.5	10°27.0	· · 38.6	91°49.8	· · 00.1	Pollux	243°16.9	27°57.9
22	95°27.5	102°54.7	01.8	340°25.8	49.6	25°29.5	38.5	106°52.0	07°00.0	Avior	234°14.1	-59°35.4
23	110°30.0	117°55.3	00.6	355°29.2	49.7	40°32.1	38.5	121°54.2	06°59.9	Suhail	222°45.8	-43°32.0
Mer.pass. 15:39		ν 0.6' d -1.2' m -4.68		ν 3.5' d 0.1' m -1.27		ν 2.6' d -0.0' m -2.58		ν 2.2' d -0.1' m 1.10				

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	125°32.5	132°55.8	S01°59.4	10°32.6	N25°49.9	55°34.6	N21°38.5	136°56.5	S06°59.8
1	140°34.9	147°56.4	58.3	25°36.1	50.0	70°37.2	38.5	151°58.7	59.7
2	155°37.4	162°57.0	57.1	40°39.5	50.1	85°39.7	38.5	167°00.9	59.6
3	170°39.9	177°57.6	· · 55.9	55°42.9	· · 50.3	100°42.3	· · 38.5	182°03.1	· · 59.5
4	185°42.3	192°58.2	54.8	70°46.4	50.4	115°44.8	38.5	197°05.4	59.4
5	200°44.8	207°58.8	53.6	85°49.8	50.5	130°47.4	38.5	212°07.6	59.3
6	215°47.2	222°59.4	S01°52.4	100°53.2	N25°50.7	145°49.9	N21°38.5	227°09.8	S06°59.2
7	230°49.7	238°00.0	51.3	115°56.6	50.8	160°52.5	38.5	242°12.0	59.0
8	245°52.2	253°00.6	50.1	131°00.1	50.9	175°55.0	38.5	257°14.3	58.9
9	260°54.6	268°01.2	· · 48.9	146°03.5	· · 51.1	190°57.6	· · 38.5	272°16.5	· · 58.8
10	275°57.1	283°01.7	47.8	161°06.9	51.2	206°00.1	38.5	287°18.7	58.7
11	290°59.6	298°02.3	46.6	176°10.3	51.3	221°02.6	38.5	302°20.9	58.6
12	306°02.0	313°02.9	S01°45.4	191°13.8	N25°51.5	236°05.2	N21°38.5	317°23.2	S06°58.5
13	321°04.5	328°03.5	44.3	206°17.2	51.6	251°07.7	38.5	332°25.4	58.4
14	336°07.0	343°04.1	43.1	221°20.6	51.7	266°10.3	38.5	347°27.6	58.3
15	351°09.4	358°04.7	· · 42.0	236°24.0	· · 51.9	281°12.8	· · 38.5	2°29.8	· · 58.2
16	6°11.9	13°05.4	40.8	251°27.5	52.0	296°15.4	38.5	17°32.0	58.1
17	21°14.4	28°06.0	39.6	266°30.9	52.1	311°17.9	38.5	32°34.3	58.0
18	36°16.8	43°06.6	S01°38.5	281°34.3	N25°52.2	326°20.5	N21°38.5	47°36.5	S06°57.9
19	51°19.3	58°07.2	37.3	296°37.7	52.4	341°23.0	38.5	62°38.7	57.8
20	66°21.7	73°07.8	36.1	311°41.1	52.5	356°25.6	38.5	77°40.9	57.7
21	81°24.2	88°08.4	· · 35.0	326°44.5	· · 52.6	11°28.1	· · 38.5	92°43.2	· · 57.6
22	96°26.7	103°09.0	33.8	341°48.0	52.7	26°30.6	38.5	107°45.4	57.5
23	111°29.1	118°09.6	32.7	356°51.4	52.9	41°33.2	38.5	122°47.6	57.4
Mer.pass. 15:35		ν 0.6' d -1.2' m -4.69		ν 3.4' d 0.1' m -1.24		ν 2.6' d -0.0' m -2.57		ν 2.2' d -0.1' m 1.10	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	126°31.6	133°10.2	S01°31.5	11°54.8	N25°53.0	56°35.7	N21°38.5	137°49.8	S06°57.3
1	141°34.1	148°10.9	30.3	26°58.2	53.1	71°38.3	38.5	152°52.1	57.2
2	156°36.5	163°11.5	29.2	42°01.6	53.2	86°40.8	38.5	167°54.3	57.1
3	171°39.0	178°12.1	· · 28.0	57°05.0	· · 53.4	101°43.3	· · 38.5	182°56.5	· · 57.0
4	186°41.5	193°12.7	26.9	72°08.4	53.5	116°45.9	38.5	197°58.7	56.9
5	201°43.9	208°13.3	25.7	87°11.8	53.6	131°48.4	38.5	213°01.0	56.7
6	216°46.4	223°14.0	S01°24.5	102°15.2	N25°53.7	146°51.0	N21°38.5	228°03.2	S06°56.6
7	231°48.9	238°14.6	23.4	117°18.6	53.9	161°53.5	38.5	243°05.4	56.5
8	246°51.3	253°15.2	22.2	132°22.1	54.0	176°56.0	38.5	258°07.6	56.4
9	261°53.8	268°15.8	· · 21.1	147°25.5	· · 54.1	191°58.6	· · 38.5	273°09.8	· · 56.3
10	276°56.2	283°16.5	19.9	162°28.9	54.2	207°01.1	38.5	288°12.1	56.2
11	291°58.7	298°17.1	18.7	177°32.3	54.4	222°03.7	38.5	303°14.3	56.1
12	307°01.2	313°17.7	S01°17.6	192°35.7	N25°54.5	237°06.2	N21°38.5	318°16.5	S06°56.0
13	322°03.6	328°18.4	16.4	207°39.1	54.6	252°08.7	38.5	333°18.7	55.9
14	337°06.1	343°19.0	15.3	222°42.5	54.7	267°11.3	38.5	348°21.0	55.8
15	352°08.6	358°19.6	· · 14.1	237°45.9	· · 54.8	282°13.8	· · 38.5	3°23.2	· · 55.7
16	7°11.0	13°20.3	13.0	252°49.3	55.0	297°16.3	38.5	18°25.4	55.6
17	22°13.5	28°20.9	11.8	267°52.7	55.1	312°18.9	38.5	33°27.6	55.5
18	37°16.0	43°21.5	S01°10.6	282°56.1	N25°55.2	327°21.4	N21°38.5	48°29.8	S06°55.4
19	52°18.4	58°22.2	09.5	297°59.5	55.3	342°24.0	38.5	63°32.1	55.3
20	67°20.9	73°22.8	08.3	313°02.8	55.4	357°26.5	38.5	78°34.3	55.2
21	82°23.4	88°23.5	· · 07.2	328°06.2	· · 55.6	12°29.0	· · 38.5	93°36.5	· · 55.1
22	97°25.8	103°24.1	06.0	343°09.6	55.7	27°31.6	38.5	108°38.7	55.0
23	112°28.3	118°24.8	04.9	358°13.0	55.8	42°34.1	38.5	123°41.0	54.9
Mer.pass. 15:31		ν 0.6' d -1.2' m -4.70		ν 3.4' d 0.1' m -1.21		ν 2.5' d -0.0' m -2.56		ν 2.2' d -0.1' m 1.11	

Jan 25 Sat	SHA	Mer.pass
Venus	8°08.9	15:09
Mars	244°36.7	23:18
Jupiter	290°00.0	20:18
Saturn	11°29.7	14:54

Jan 26 Sun	SHA	Mer.pass
Venus	7°23.4	15:08
Mars	245°00.2	23:13
Jupiter	290°02.1	20:14
Saturn	11°24.0	14:50

Jan 27 Mon	SHA	Mer.pass
Venus	6°38.6	15:07
Mars	245°23.2	23:07
Jupiter	290°04.1	20:10
Saturn	11°18.2	14:46

Horizontal parallax		
Venus:	0.3	
Mars:	0.2	

h								Moon							
Sun															
Sat	GHA	Dec		GHA	ν	Dec	d	HP							
0	176°56.5	S18°57.2		236°57.3	9.3'	S26°43.9	5.5'	55.4'							
1	191°56.4	56.6		251°25.5	9.2'	26°49.5	5.4'	55.4'							
2	206°56.2	56.0		265°53.7	9.1'	26°54.9	5.3'	55.4'							
3	221°56.1	· · 55.4		280°21.8	9.0'	27°00.1	5.1'	55.5'							
4	236°55.9	54.8		294°49.8	8.9'	27°05.3	5.0'	55.5'							
5	251°55.8	54.1		309°17.7	8.8'	27°10.3	4.9'	55.5'							
6	266°55.7	S18°53.5		323°45.5	8.7'	S27°15.1	4.7'	55.5'							
7	281°55.5	52.9		338°13.2	8.6'	27°19.9	4.6'	55.6'							
8	296°55.4	52.3		352°40.8	8.6'	27°24.5	4.5'	55.6'							
9	311°55.2	· · 51.7		7°08.4	8.5'	27°28.9	4.3'	55.6'							
10	326°55.1	51.0		21°35.9	8.4'	27°33.2	4.2'	55.7'							
11	341°55.0	50.4		36°03.3	8.3'	27°37.4	4.0'	55.7'							
12	356°54.8	S18°49.8		50°30.6	8.2'	S27°41.4	3.9'	55.7'							
13	11°54.7	49.2		64°57.8	8.2'	27°45.3	3.7'	55.7'							
14	26°54.5	48.5		79°25.0	8.1'	27°49.0	3.6'	55.8'							
15	41°54.4	· · 47.9		93°52.0	8.0'	27°52.6	3.4'	55.8'							
16	56°54.3	47.3		108°19.0	7.9'	27°56.1	3.3'	55.8'							
17	71°54.1	46.7		122°45.9	7.8'	27°59.4	3.1'	55.8'							
18	86°54.0	S18°46.0		137°12.8	7.8'	S28°02.5	3.0'	55.9'							
19	101°53.8	45.4		151°39.6	7.7'	28°05.5	2.8'	55.9'							
20	116°53.7	44.8		166°06.3	7.6'	28°08.3	2.7'	55.9'							
21	131°53.6	· · 44.1		180°32.9	7.6'	28°11.0	2.5'	56.0'							
22	146°53.4	43.5		194°59.5	7.5'	28°13.6	2.4'	56.0'							
23	161°53.3	42.9		209°26.0	7.4'	28°15.9	2.2'	56.0'							
SD = 16.2'				$d = -0.6'$				SD = 15.1'							
Sun	GHA	Dec		GHA	ν	Dec	d	HP							
0	176°53.2	S18°42.2		223°52.4	7.4'	S28°18.2	2.1'	56.0'							
1	191°53.0	41.6		238°18.7	7.3'	28°20.2	1.9'	56.1'							
2	206°52.9	41.0		252°45.0	7.2'	28°22.2	1.8'	56.1'							
3	221°52.8	· · 40.4		267°11.3	7.2'	28°23.9	1.6'	56.1'							
4	236°52.6	39.7		281°37.5	7.1'	28°25.5	1.4'	56.2'							
5	251°52.5	39.1		296°03.6	7.1'	28°26.9	1.3'	56.2'							
6	266°52.4	S18°38.4		310°29.6	7.0'	S28°28.2	1.1'	56.2'							
7	281°52.2	37.8		324°55.7	7.0'	28°29.3	1.0'	56.3'							
8	296°52.1	37.2		339°21.6	6.9'	28°30.3	0.8'	56.3'							
9	311°52.0	· · 36.5		353°47.5	6.9'	28°31.1	0.6'	56.3'							
10	326°51.8	35.9		8°13.4	6.8'	28°31.7	0.5'	56.3'							
11	341°51.7	35.3		22°39.2	6.8'	28°32.2	0.3'	56.4'							
12	356°51.6	S18°34.6		37°04.9	6.7'	S28°32.5	0.1'	56.4'							
13	11°51.4	34.0		51°30.6	6.7'	28°32.6	-0.0'	56.4'							
14	26°51.3	33.3		65°56.3	6.6'	28°32.6	-0.2'	56.5'							
15	41°51.2	· · 32.7		80°21.9	6.6'	28°32.4	-0.4'	56.5'							
16	56°51.0	32.1		94°47.5	6.5'	28°32.1	-0.5'	56.5'							
17	71°50.9	31.4		109°13.1	6.5'	28°31.5	-0.7'	56.6'							
18	86°50.8	S18°30.8		123°38.6	6.5'	S28°30.8	-0.9'	56.6'							
19	101°50.6	30.1		138°04.1	6.4'	28°30.0	-1.0'	56.6'							
20	116°50.5	29.5		152°29.5	6.4'	28°29.0	-1.2'	56.6'							
21	131°50.4	· · 28.9		166°54.9	6.4'	28°27.8	-1.4'	56.7'							
22	146°50.3	28.2		181°20.3	6.4'	28°26.4	-1.5'	56.7'							
23	161°50.1	27.6		195°45.7	6.3'	28°24.9	-1.7'	56.7'							
SD = 16.2'				$d = -0.6'$				SD = 15.3'							
Mon	GHA	Dec		GHA	ν	Dec	d	HP							
0	176°50.0	S18°26.9		210°11.0	6.3'	S28°23.2	-1.9'	56.8'							
1	191°49.9	26.3		224°36.3	6.3'	28°21.3	-2.0'	56.8'							
2	206°49.8	25.6		239°01.6	6.3'	28°19.2	-2.2'	56.8'							
3	221°49.6	· · 25.0		253°26.9	6.3'	28°17.0	-2.4'	56.9'							
4	236°49.5	24.3		267°52.2	6.2'	28°14.6	-2.6'	56.9'							
5	251°49.4	23.7		282°17.4	6.2'	28°12.1	-2.7'	56.9'							
6	266°49.2	S18°23.0		296°42.7	6.2'	S28°09.3	-2.9'	57.0'							
7	281°49.1	22.4		311°07.9	6.2'	28°06.4	-3.1'	57.0'							
8	296°49.0	21.7		325°33.1	6.2'	28°03.4	-3.2'	57.0'							
9	311°48.9	· · 21.1		339°58.3	6.2'	28°00.1	-3.4'	57.0'							
10	326°48.8	20.4		354°23.5	6.2'	27°56.7	-3.6'	57.1'							
11	341°48.6	19.8		8°48.7	6.2'	27°53.1	-3.8'	57.1'							
12	356°48.5	S18°19.1		23°13.9	6.2'	S27°49.3	-3.9'	57.1'							
13	11°48.4	18.5		37°39.2	6.2'	27°45.4	-4.1'	57.2'							
14	26°48.3	17.8		52°04.4	6.2'	27°41.3	-4.3'	57.2'							
15	41°48.1	· · 17.2		66°29.6	6.2'	27°37.0	-4.4'	57.2'							
16	56°48.0	16.5		80°54.8	6.2'	27°32.6	-4.6'	57.3'							
17	71°47.9	15.9		95°20.0	6.2'	27°28.0	-4.8'	57.3'							
18	86°47.8	S18°15.2		109°45.3	6.3'	S27°23.2	-5.0'	57.3'							
19	101°47.6	14.5		124°10.5	6.3'	27°18.2	-5.1'	57.3'							
20	116°47.5	13.9		138°35.8	6.3'	27°13.1	-5.3'	57.4'							
21	131°47.4	· · 13.2		153°01.1	6.3'	27°07.8	-5.5'	57.4'							
22	146°47.3	12.6		167°26.4	6.3'	27°02.3	-5.6'	57.4'							
23	161°47.2	11.9		181°51.7	6.3'	26°56.7	-5.8'	57.5'							
SD = 16.2'				$d = -0.6'$				SD = 15.5'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:29	09:04	11:33	12:53	15:23	16:58
N 70°	07:19	08:41	10:19	14:07	15:45	17:08
68°	07:10	08:23	09:42	14:44	16:03	17:16
66°	07:03	08:09	09:16	15:10	16:17	17:23
64°	06:57	07:57	08:56	15:30	16:29	17:30
62°	06:51	07:47	08:40	15:46	16:39	17:35
60°	06:46	07:38	08:27	15:59	16:48	17:40
N 58°	06:41	07:30	08:15	16:11	16:56	17:45
56°	06:37	07:23	08:05	16:21	17:03	17:49
54°	06:33	07:17	07:56	16:29	17:09	17:52
52°	06:30	07:11	07:48	16:37	17:15	17:56
50°	06:26	07:06	07:41	16:44	17:20	17:59
45°	06:19	06:54	07:26	17:00	17:31	18:07
N 40°	06:12	06:44	07:13	17:12	17:41	18:14
35°	06:05	06:36	07:03	17:23	17:50	18:20
30°	05:59	06:28	06:53	17:32	17:57	18:26
20°	05:47	06:14	06:37	17:48	18:12	18:38
N 10°	05:35	06:00	06:23	18:03	18:25	18:50
0°	05:22	05:47	06:09	18:16	18:38	19:03
S 10°	05:07	05:33	05:55	18:30	18:52	19:18
20°	04:49	05:17	05:41	18:44	19:08	19:36
30°	04:26	04:57	05:23	19:01	19:27	19:59
35°	04:11	04:45	05:13	19:11		

January 28, 29, 30 UT (Tue., Wed., Thu.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Tue	GHA	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	SHA Dec	SHA Dec	SHA Dec		
0	127° 30.7	133° 25.4	S01° 03.7	13° 16.4	N25° 55.9	57° 36.6	N21° 38.5	138° 43.2	S06° 54.7				Alpheratz	357° 34.9	29° 13.8		
1	142° 33.2	148° 26.1	02.6	28° 19.8	56.0	72° 39.2	38.5	153° 45.4	54.6				Ankaa	353° 07.3	-42° 10.4		
2	157° 35.7	163° 26.7	01.4	43° 23.2	56.1	87° 41.7	38.5	168° 47.6	54.5				Schedar	349° 31.3	56° 40.7		
3	172° 38.1	178° 27.4	01° 00.2	58° 26.6	56.2	102° 44.2	38.5	183° 49.8	54.4				Diphda	348° 47.4	-17° 51.1		
4	187° 40.6	193° 28.0	00° 59.1	73° 30.0	56.4	117° 46.8	38.5	198° 52.1	54.3				Achernar	335° 20.2	-57° 06.9		
5	202° 43.1	208° 28.7	57.9	88° 33.3	56.5	132° 49.3	38.5	213° 54.3	54.2				Hamal	327° 51.2	23° 34.9		
6	217° 45.5	223° 29.3	S00° 56.8	103° 36.7	N25° 56.6	147° 51.8	N21° 38.5	228° 56.5	S06° 54.1				Polaris	313° 55.3	89° 22.5		
7	232° 48.0	238° 30.0	55.6	118° 40.1	56.7	162° 54.3	38.5	243° 58.7	54.0				Acamar	315° 11.6	-40° 12.5		
8	247° 50.5	253° 30.6	54.5	133° 43.5	56.8	177° 56.9	38.5	259° 00.9	53.9				Menkar	314° 06.0	4° 11.2		
9	262° 52.9	268° 31.3	53.3	148° 46.9	56.9	192° 59.4	38.5	274° 03.2	53.8				Mirfak	308° 28.0	49° 57.2		
10	277° 55.4	283° 32.0	52.2	163° 50.3	57.0	208° 01.9	38.5	289° 05.4	53.7				Aldebaran	290° 39.4	16° 33.6		
11	292° 57.9	298° 32.6	51.0	178° 53.6	57.2	223° 04.5	38.5	304° 07.6	53.6				Rigel	281° 03.6	-8° 10.4		
12	308° 00.3	313° 33.3	S00° 49.9	193° 57.0	N25° 57.3	238° 07.0	N21° 38.5	319° 09.8	S06° 53.5				Capella	280° 21.5	46° 01.5		
13	323° 02.8	328° 33.9	48.7	209° 00.4	57.4	253° 09.5	38.5	334° 12.0	53.4				Bellatrix	278° 22.6	6° 22.3		
14	338° 05.2	343° 34.6	47.6	224° 03.8	57.5	268° 12.1	38.5	349° 14.3	53.3				Elnath	278° 01.5	28° 37.8		
15	353° 07.7	358° 35.3	46.4	239° 07.1	57.6	283° 14.6	38.5	4° 16.5	53.2				Alnilam	275° 37.4	-1° 11.2		
16	8° 10.2	13° 36.0	45.3	254° 10.5	57.7	298° 17.1	38.5	19° 18.7	53.1				Betelgeuse	270° 51.8	7° 24.7		
17	23° 12.6	28° 36.6	44.1	269° 13.9	57.8	313° 19.6	38.5	34° 20.9	52.9				Canopus	263° 51.9	-52° 42.7		
18	38° 15.1	43° 37.3	S00° 43.0	284° 17.3	N25° 57.9	328° 22.2	N21° 38.5	49° 23.2	S06° 52.8				Sirius	258° 25.9	-16° 45.1		
19	53° 17.6	58° 38.0	41.8	299° 20.6	58.0	343° 24.7	38.5	64° 25.4	52.7				Adhara	255° 05.5	-29° 00.5		
20	68° 20.0	73° 38.7	40.7	314° 24.0	58.1	358° 27.2	38.5	79° 27.6	52.6				Procyon	244° 50.4	5° 09.6		
21	83° 22.5	88° 39.3	39.5	329° 27.4	58.3	373° 29.7	38.5	94° 29.8	52.5				Pollux	243° 16.9	27° 57.9		
22	98° 25.0	103° 40.0	38.4	344° 30.7	58.4	388° 32.3	38.5	109° 32.0	52.4				Avior	234° 14.1	-59° 35.4		
23	113° 27.4	118° 40.7	37.2	359° 34.1	58.5	403° 34.8	38.5	124° 34.2	52.3				Suhail	222° 45.8	-43° 32.0		
Mer.pass. 15:27			ν 0.6' d -1.2' m -4.71			ν 3.4' d 0.1' m -1.19			ν 2.5' d 0.0' m -2.55			ν 2.2' d -0.1' m 1.11					
Wed	GHA	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	SHA Dec	SHA Dec	SHA Dec		
0	128° 29.9	133° 41.4	S00° 36.1	14° 37.5	N25° 58.6	58° 37.3	N21° 38.5	139° 36.5	S06° 52.2				Denebola	182° 24.7	14° 25.8		
1	143° 32.3	148° 42.1	34.9	29° 40.8	58.7	73° 39.8	38.5	154° 38.7	52.1				Gienah	175° 43.3	-17° 40.9		
2	158° 34.8	163° 42.7	33.8	44° 44.2	58.8	88° 42.4	38.5	169° 40.9	52.0				Acrux	172° 59.7	-63° 14.1		
3	173° 37.3	178° 43.4	32.6	59° 47.6	58.9	103° 44.9	38.5	184° 43.1	51.9				Gacrux	171° 51.3	-57° 15.0		
4	188° 39.7	193° 44.1	31.5	74° 50.9	59.0	118° 47.4	38.5	199° 45.3	51.8				Alioth	166° 12.7	55° 49.1		
5	203° 42.2	208° 44.8	30.3	89° 54.3	59.1	133° 49.9	38.5	214° 47.6	51.7				Spica	158° 22.2	-11° 17.6		
6	218° 44.7	223° 45.5	S00° 29.2	104° 57.6	N25° 59.2	148° 52.4	N21° 38.5	229° 49.8	S06° 51.6				Alkaid	152° 51.8	49° 10.9		
7	233° 47.1	238° 46.2	28.0	120° 01.0	59.3	163° 55.0	38.5	244° 52.0	51.5				Hadar	148° 35.9	-60° 29.4		
8	248° 49.6	253° 46.9	26.9	135° 04.3	59.4	178° 57.5	38.5	259° 54.2	51.3				Menkent	147° 57.5	-36° 29.5		
9	263° 52.1	268° 47.6	25.7	150° 07.7	59.5	194° 00.0	38.5	274° 56.4	51.2				Arcturus	145° 47.8	19° 02.9		
10	278° 54.5	283° 48.3	24.6	165° 11.1	59.6	209° 02.5	38.5	289° 58.7	51.1				Rigel Kent.	139° 40.3	-60° 56.1		
11	293° 57.0	298° 49.0	23.4	180° 14.4	59.7	224° 05.1	38.5	305° 00.9	51.0				Kochab	137° 19.9	74° 02.7		
12	308° 59.5	313° 49.7	S00° 22.3	195° 17.8	N25° 59.8	239° 07.6	N21° 38.5	320° 03.1	S06° 50.9				Zuben'ubi	136° 56.0	-16° 08.8		
13	324° 01.9	328° 50.4	21.1	210° 21.1	59.9	254° 10.1	38.6	335° 05.3	50.8				Alphecca	126° 03.8	26° 37.6		
14	339° 04.4	343° 51.1	20.0	225° 24.5	60.0	269° 12.6	38.6	350° 07.5	50.7				Antares	112° 16.0	-26° 29.2		
15	354° 06.8	358° 51.8	18.9	240° 27.8	60.1	284° 15.1	38.6	5° 09.8	50.6				Atria	107° 10.5	-69° 04.1		
16	9° 09.3	13° 52.5	17.7	255° 31.2	60.2	299° 17.6	38.6	20° 12.0	50.5				Sabik	102° 02.9	-15° 45.4		
17	24° 11.8	28° 53.2	16.6	270° 34.5	60.3	314° 20.2	38.6	35° 14.2	50.4				Shaula	96° 10.6	-37° 07.3		
18	39° 14.2	43° 53.9	S00° 15.4	285° 37.9	N26° 00.4	329° 22.7	N21° 38.6	50° 16.4	S06° 50.3				Rasalhague	95° 58.7	12° 32.3		
19	54° 16.7	58° 54.6	14.3	300° 41.2	60.5	344° 25.2	38.6	65° 18.6	50.2				Eltanin	90° 42.6	51° 28.8		
20	69° 19.2	73° 55.3	13.1	315° 44.5	60.6	359° 27.7	38.6	80° 20.8	50.1				Kaus Aust.	83° 32.8	-34° 22.3		
21	84° 21.6	88° 56.0	12.0	330° 47.9	60.7	374° 30.2	38.6	95° 23.1	50.0				Vega	80° 33.6	38° 48.2		
22	99° 24.1	103° 56.7	10.9	345° 51.2	60.8	389° 32.7	38.6	110° 25.3	49.9				Nunki	75° 48.0	-26° 16.0		
23	114° 26.6	118° 57.4	09.7	360° 54.6	60.9	404° 35.3	38.6	125° 27.5	49.7				Altair	62° 00.2	8° 56.0		
Mer.pass. 15:23			ν 0.7' d -1.1' m -4.72			ν 3.4' d 0.1' m -1.16			ν 2.5' d 0.0' m -2.54			ν 2.2' d -0.1' m 1.11					
Thu	GHA	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	SHA Dec	SHA Dec	SHA Dec		
0	129° 29.0	133° 58.2	S00° 08.6	15° 57.9	N26° 01.0	59° 37.8	N21° 38.6	140° 29.7	S06° 49.6				Jan 28 Tue	SHA	Mer.pass		
1	144° 31.5	148° 58.9	07.4	31° 01.2	01.1	74° 40.3	38.6	155° 31.9	49.5				Venus	5° 54.7	15:06		
2	159° 34.0	163° 59.6	06.3	46° 04.6	01.2	89° 42.8	38.6	170° 34.1	49.4				Mars	245° 45.7	23:02		
3	174° 36.4	179° 00.3	05.1	61° 07.9	01.3	104° 45.3	38.6	185° 36.4	49.3				Jupiter	290° 05.9	20:06		
4	189° 38.9	194° 01.0	04.0	76° 11.2	01.4	119° 47.8	38.6	200° 38.6	49.2				Saturn	11° 12.4	14:43		
5	204° 41.3	209° 01.8	02.9	91° 14.6	01.5	134° 50.3	38.6	215° 40.8	49.1								
6	219° 43.8	224° 02.5	S00° 01.7	106° 17.9	N26° 01.6	149° 52.9	N21° 38.6	230° 43.0	S06° 49.0								
7	234° 46.3	239° 03.2	S00° 00.6	121° 21.2	01.7	164° 55.4	38.6	245° 45.2	48.9								
8	249° 48.7	254° 03.9	N00° 00.5	136° 24.6	01.8	179° 57.9	38.6	260° 47.5	48.8								
9	2>&																

h		Sun			Moon		
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	176°47.0	S18°11.3	196°17.1	6.4'	S26°50.9	-6.0'	57.5'
1	191°46.9	10.6	210°42.4	6.4'	26°44.9	-6.1'	57.5'
2	206°46.8	09.9	225°07.8	6.4'	26°38.8	-6.3'	57.6'
3	221°46.7	· · 09.3	239°33.3	6.5'	26°32.5	-6.5'	57.6'
4	236°46.6	08.6	253°58.7	6.5'	26°26.0	-6.6'	57.6'
5	251°46.5	08.0	268°24.2	6.5'	26°19.4	-6.8'	57.6'
6	266°46.3	S18°07.3	282°49.7	6.5'	S26°12.6	-7.0'	57.7'
7	281°46.2	06.6	297°15.2	6.6'	26°05.6	-7.1'	57.7'
8	296°46.1	06.0	311°40.8	6.6'	25°58.5	-7.3'	57.7'
9	311°46.0	· · 05.3	326°06.4	6.7'	25°51.2	-7.5'	57.8'
10	326°45.9	04.6	340°32.1	6.7'	25°43.7	-7.6'	57.8'
11	341°45.8	04.0	354°57.8	6.7'	25°36.1	-7.8'	57.8'
12	356°45.6	S18°03.3	9°23.5	6.8'	S25°28.3	-7.9'	57.9'
13	11°45.5	02.6	23°49.3	6.8'	25°20.4	-8.1'	57.9'
14	26°45.4	02.0	38°15.1	6.9'	25°12.3	-8.3'	57.9'
15	41°45.3	· · 01.3	52°41.0	6.9'	25°04.0	-8.4'	57.9'
16	56°45.2	00.6	67°06.9	7.0'	24°55.6	-8.6'	58.0'
17	71°45.1	18°00.0	81°32.9	7.0'	24°47.0	-8.7'	58.0'
18	86°45.0	S17°59.3	95°58.9	7.1'	S24°38.3	-8.9'	58.0'
19	101°44.9	58.6	110°24.9	7.1'	24°29.4	-9.0'	58.1'
20	116°44.7	58.0	124°51.0	7.2'	24°20.3	-9.2'	58.1'
21	131°44.6	· · 57.3	139°17.2	7.2'	24°11.1	-9.4'	58.1'
22	146°44.5	56.6	153°43.4	7.3'	24°01.8	-9.5'	58.1'
23	161°44.4	55.9	168°09.7	7.3'	23°52.3	-9.7'	58.2'
SD = 16.2'		$d = -0.7'$		SD = 15.7'			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	176°44.3	S17°55.3	182°36.0	7.4'	S23°42.6	-9.8'	58.2'
1	191°44.2	54.6	197°02.4	7.4'	23°32.8	-10.0'	58.2'
2	206°44.1	53.9	211°28.8	7.5'	23°22.9	-10.1'	58.2'
3	221°44.0	· · 53.2	225°55.3	7.5'	23°12.8	-10.2'	58.3'
4	236°43.9	52.6	240°21.9	7.6'	23°02.5	-10.4'	58.3'
5	251°43.7	51.9	254°48.5	7.7'	22°52.1	-10.5'	58.3'
6	266°43.6	S17°51.2	269°15.1	7.7'	S22°41.6	-10.7'	58.3'
7	281°43.5	50.5	283°41.9	7.8'	22°30.9	-10.8'	58.4'
8	296°43.4	49.9	298°08.6	7.8'	22°20.1	-11.0'	58.4'
9	311°43.3	· · 49.2	312°35.5	7.9'	22°09.1	-11.1'	58.4'
10	326°43.2	48.5	327°02.4	8.0'	21°58.0	-11.2'	58.4'
11	341°43.1	47.8	341°29.4	8.0'	21°46.8	-11.4'	58.5'
12	356°43.0	S17°47.1	355°56.4	8.1'	S21°35.5	-11.5'	58.5'
13	11°42.9	46.5	10°23.5	8.2'	21°23.9	-11.6'	58.5'
14	26°42.8	45.8	24°50.7	8.2'	21°12.3	-11.8'	58.5'
15	41°42.7	· · 45.1	39°17.9	8.3'	21°00.5	-11.9'	58.6'
16	56°42.6	44.4	53°45.2	8.4'	20°48.6	-12.0'	58.6'
17	71°42.5	43.7	68°12.6	8.4'	20°36.6	-12.2'	58.6'
18	86°42.4	S17°43.1	82°40.0	8.5'	S20°24.5	-12.3'	58.6'
19	101°42.3	42.4	97°07.5	8.6'	20°12.2	-12.4'	58.7'
20	116°42.2	41.7	111°35.0	8.6'	19°59.8	-12.5'	58.7'
21	131°42.0	· · 41.0	126°02.6	8.7'	19°47.2	-12.7'	58.7'
22	146°41.9	40.3	140°30.3	8.8'	19°34.6	-12.8'	58.7'
23	161°41.8	39.6	154°58.1	8.8'	19°21.8	-12.9'	58.8'
SD = 16.2'		$d = -0.7'$		SD = 15.9'			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	176°41.7	S17°38.9	169°25.9	8.9'	S19°08.9	-13.0'	58.8'
1	191°41.6	38.3	183°53.8	8.9'	18°55.9	-13.1'	58.8'
2	206°41.5	37.6	198°21.7	9.0'	18°42.8	-13.2'	58.8'
3	221°41.4	· · 36.9	212°49.7	9.1'	18°29.6	-13.4'	58.8'
4	236°41.3	36.2	227°17.8	9.1'	18°16.2	-13.5'	58.9'
5	251°41.2	35.5	241°46.0	9.2'	18°02.7	-13.6'	58.9'
6	266°41.1	S17°34.8	256°14.2	9.3'	S17°49.2	-13.7'	58.9'
7	281°41.0	34.1	270°42.4	9.3'	17°35.5	-13.8'	58.9'
8	296°40.9	33.4	285°10.8	9.4'	17°21.7	-13.9'	58.9'
9	311°40.8	· · 32.8	299°39.2	9.5'	17°07.8	-14.0'	59.0'
10	326°40.7	32.1	314°07.6	9.5'	16°53.8	-14.1'	59.0'
11	341°40.6	31.4	328°36.2	9.6'	16°39.7	-14.2'	59.0'
12	356°40.5	S17°30.7	343°04.8	9.7'	S16°25.5	-14.3'	59.0'
13	11°40.4	30.0	357°33.4	9.7'	16°11.2	-14.4'	59.0'
14	26°40.3	29.3	12°02.1	9.8'	15°56.8	-14.5'	59.1'
15	41°40.2	· · 28.6	26°30.9	9.8'	15°42.3	-14.6'	59.1'
16	56°40.1	27.9	40°59.8	9.9'	15°27.7	-14.7'	59.1'
17	71°40.1	27.2	55°28.7	10.0'	15°13.0	-14.8'	59.1'
18	86°40.0	S17°26.5	69°57.6	10.0'	S14°58.2	-14.9'	59.1'
19	101°39.9	25.8	84°26.6	10.1'	14°43.4	-15.0'	59.1'
20	116°39.8	25.1	98°55.7	10.1'	14°28.4	-15.0'	59.2'
21	131°39.7	· · 24.4	113°24.9	10.2'	14°13.4	-15.1'	59.2'
22	146°39.6	23.7	127°54.1	10.3'	13°58.2	-15.2'	59.2'
23	161°39.5	23.0	142°23.3	10.3'	13°43.0	-15.3'	59.2'
SD = 16.2'		$d = -0.7'$		SD = 16.0'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:19	08:51	10:53	13:35	15:37	17:09
N 70°	07:10	08:30	10:01	14:27	15:57	17:18
68°	07:02	08:14	09:29	14:58	16:13	17:25
66°	06:56	08:01	09:06	15:21	16:27	17:31
64°	06:50	07:50	08:48	15:39	16:37	17:37
62°	06:45	07:40	08:33	15:54	16:47	17:42
60°	06:41	07:32	08:20	16:07	16:55	17:46
N 58°	06:37	07:25	08:09	16:18	17:02	17:50
56°	06:33	07:18	08:00	16:27	17:09	17:54
54°	06:30	07:12	07:52	16:35	17:14	17:57
52°	06:26	07:07	07:44	16:43	17:20	18:01
50°	06:23	07:02	07:37	16:49	17:25	18:04
45°	06:16	06:51	07:23	17:04	17:35	18:11
N 40°	06:10	06:42	07:11	17:16	17:44	18:17
35°	06:04	06:34	07:01	17:26	17:53	18:23
30°	05:58	06:27	06:52	17:35	18:00	18:29
20°	05:47	06:13	06:36	17:50	18:13	18:40
N 10°	05:35	06:01	06:23	18:04	18:26	18:51
0°	05:23	05:48	06:10	18:17	18:38	19:04
S 10°	05:08	05:34	05:57	18:30	18:52	19:18
20°	04:51	05:19	05:42	18:44	19:07	19:35
30°	04:29	05:00	05:26	19:00	19:26	19:57
35°	04:15	04:49	05:17	19:09	19:37	20:10
40°	03:58	04:35	05:06	19:20	19:50	20:27
45°	03:36	04:19	04:53	19:33	20:06	20:49
S 50°	03:07	03:58	04:37	19:48	20:27	21:17
52°	02:52	03:48	04:29	19:56	20:37	21:32
54°	02:34	03:37	04:21	20:04	20:48	21:49
56°	02:12	03:23	04:12	20:13	21:01	22:12
58°	01:40	03:08	04:01	20:24	21:16	22:42
S 60°	00:31	02:49	03:49	20:36	21:35	23:37

Lat.	Moonrise		Moonset	
	Tue	Wed	Tue	Thu
N 72°	■	■	11:35	■
N 70°	■	■	10:50	■
68°	■	11:42	10:20	■
66°	■	10:34	09:58	■
64°	10:46	09:58	09:40	12:06
62°	09:41	09:31	09:25	13:11
60°	09:06	09:11	09:12	13:45
N 58°	08:40	08:54	09:01	14:10
56°	08:20	08:39	08:51	14:30
54°	08:03	08:27	08:43	14:47
52°	07:49	08:16	08:35	15:01
50°	07:36	08:06	08:28	15:13
45°	07:09	07:45	08:13	15:39
N 40°	06:49	07:28	08:01	15:59
35°	06:31	07:14	07:50	16:16
30°	06:16	07:01	07:41	16:30
20°	05:50	06:40	07:25	16:54
N 10°	05:28	06:21	07:11	17:15
0°	05:07	06:04	06:58	17:35
S 10°	04:46	05:46	06:44	17:54
20°	04:24	05:27	06:30	18:14
30°	03:58	05:05	06:14	18:38
35°	03:43	04:52	06:04	18:52
40°	03:25	04:38	05:53	19:08
45°	03:03	04:20	05:40	19:26
S 50°	02:36	03:58	05:24	19:50
52°	02:22	03:47	05:16	20:01
54°	02:06	03:35	05:08	20:14
56°	01:48	03:21	04:59	20:28
58°	01:25	03:05	04:48	20:45
S 60°	00:56	02:45	04:35	21:05

Day	Sun		Mer. Pass	Moon		Age 29-1 3-0%
	Eqn.of Time 00 ^h	12 ^h		Mer.Pass. Upper	Lower	
28	12:52	12:57	12:13	11:21	23:49	
29	13:03	13:08	12:13	12:12:		

January 31, 01, 02 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	130°28.2	134°15.8	N00°18.7	17°17.7	N26°03.2	60°38.0	N21°38.7	141°22.9	S06°47.1					Alpheratz	357°34.9	29°13.8	
1	145°30.6	149°16.5	19.9	32°21.0	03.3	75°40.5	38.7	156°25.1	47.0					Ankaa	353°07.3	-42°10.4	
2	160°33.1	164°17.3	21.0	47°24.3	03.4	90°43.0	38.7	171°27.3	46.8					Schedar	349°31.3	56°40.7	
3	175°35.6	179°18.0	· · 22.1	62°27.6	· · 03.5	105°45.5	· · 38.7	186°29.6	· · 46.7					Diphda	348°47.4	-17°51.1	
4	190°38.0	194°18.8	23.3	77°30.9	03.6	120°48.0	38.7	201°31.8	46.6					Achernar	335°20.3	-57°06.8	
5	205°40.5	209°19.6	24.4	92°34.2	03.7	135°50.5	38.7	216°34.0	46.5					Hamal	327°51.2	23°34.9	
6	220°42.9	224°20.3	N00°25.5	107°37.5	N26°03.8	150°53.0	N21°38.7	231°36.2	S06°46.4					Polaris	313°57.0	89°22.5	
7	235°45.4	239°21.1	26.7	122°40.8	03.9	165°55.6	38.7	246°38.4	46.3					Acamar	315°11.7	-40°12.5	
8	250°47.9	254°21.9	27.8	137°44.1	03.9	180°58.1	38.7	261°40.6	46.2					Menkar	314°06.0	4°11.2	
9	265°50.3	269°22.6	· · 28.9	152°47.4	· · 04.0	196°00.6	· · 38.7	276°42.9	· · 46.1					Mirfak	308°28.1	49°57.2	
10	280°52.8	284°23.4	30.1	167°50.7	04.1	211°03.1	38.7	291°45.1	46.0					Aldebaran	290°39.4	16°33.6	
11	295°55.3	299°24.2	31.2	182°54.0	04.2	226°05.6	38.7	306°47.3	45.9					Rigel	281°03.6	-8°10.4	
12	310°57.7	314°24.9	N00°32.3	197°57.3	N26°04.3	241°08.1	N21°38.7	321°49.5	S06°45.8					Capella	280°21.5	46°01.5	
13	326°00.2	329°25.7	33.4	213°00.6	04.4	256°10.6	38.7	336°51.7	45.7					Bellatrix	278°22.6	6°22.3	
14	341°02.7	344°26.5	34.6	228°03.9	04.4	271°13.1	38.7	351°53.9	45.5					Elnath	278°01.5	28°37.8	
15	356°05.1	359°27.2	· · 35.7	243°07.2	· · 04.5	286°15.6	· · 38.7	6°56.1	· · 45.4					Alnilam	275°37.4	-1°11.2	
16	11°07.6	14°28.0	36.8	258°10.5	04.6	301°18.1	38.7	21°58.4	45.3					Betelgeuse	270°51.8	7°24.7	
17	26°10.1	29°28.8	38.0	273°13.8	04.7	316°20.6	38.7	37°00.6	45.2					Canopus	263°51.9	-52°42.7	
18	41°12.5	44°29.6	N00°39.1	288°17.1	N26°04.8	331°23.1	N21°38.7	52°02.8	S06°45.1					Sirius	258°25.9	-16°45.1	
19	56°15.0	59°30.4	40.2	303°20.4	04.9	346°25.6	38.7	67°05.0	45.0					Adhara	255°05.5	-29°00.5	
20	71°17.4	74°31.1	41.3	318°23.7	04.9	1°28.1	38.7	82°07.2	44.9					Procyon	244°50.4	5°09.6	
21	86°19.9	89°31.9	· · 42.5	333°27.0	· · 05.0	16°30.6	· · 38.7	97°09.4	· · 44.8					Pollux	243°16.9	27°57.9	
22	101°22.4	104°32.7	43.6	348°30.2	05.1	31°33.1	38.7	112°11.6	44.7					Avior	234°14.1	-59°35.4	
23	116°24.8	119°33.5	44.7	3°33.5	05.2	46°35.6	38.8	127°13.9	44.6					Suhail	222°45.8	-43°32.0	
Mer.pass. 15:16			ν 0.8' d 1.1' m -4.74			ν 3.3' d 0.1' m -1.10			ν 2.5' d 0.0' m -2.53			ν 2.2' d -0.1' m 1.11					

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	131°27.3	134°34.3	N00°45.8	18°36.8	N26°05.2	61°38.0	N21°38.8	142°16.1	S06°44.5					
1	146°29.8	149°35.1	47.0	33°40.1	05.3	76°40.5	38.8	157°18.3	44.4					
2	161°32.2	164°35.9	48.1	48°43.4	05.4	91°43.0	38.8	172°20.5	44.2					
3	176°34.7	179°36.7	· · 49.2	63°46.6	· · 05.5	106°45.5	· · 38.8	187°22.7	· · 44.1					
4	191°37.2	194°37.5	50.3	78°49.9	05.6	121°48.0	38.8	202°24.9	44.0					
5	206°39.6	209°38.3	51.5	93°53.2	05.6	136°50.5	38.8	217°27.1	43.9					
6	221°42.1	224°39.1	N00°52.6	108°56.5	N26°05.7	151°53.0	N21°38.8	232°29.4	S06°43.8					
7	236°44.5	239°39.9	53.7	123°59.7	05.8	166°55.5	38.8	247°31.6	43.7					
8	251°47.0	254°40.7	54.8	139°03.0	05.9	181°58.0	38.8	262°33.8	43.6					
9	266°49.5	269°41.5	· · 55.9	154°06.3	· · 05.9	197°00.5	· · 38.8	277°36.0	· · 43.5					
10	281°51.9	284°42.3	57.1	169°09.6	06.0	212°03.0	38.8	292°38.2	43.4					
11	296°54.4	299°43.1	58.2	184°12.8	06.1	227°05.5	38.8	307°40.4	43.3					
12	311°56.9	314°43.9	N00°59.3	199°16.1	N26°06.2	242°08.0	N21°38.8	322°42.6	S06°43.2					
13	326°59.3	329°44.7	01°00.4	214°19.4	06.2	257°10.5	38.8	337°44.8	43.1					
14	342°01.8	344°45.5	01.6	229°22.6	06.3	272°13.0	38.8	352°47.1	42.9					
15	357°04.3	359°46.3	· · 02.7	244°25.9	· · 06.4	287°15.5	· · 38.8	7°49.3	· · 42.8					
16	12°06.7	14°47.1	03.8	259°29.1	06.5	302°17.9	38.8	22°51.5	42.7					
17	27°09.2	29°47.9	04.9	274°32.4	06.5	317°20.4	38.8	37°53.7	42.6					
18	42°11.7	44°48.8	N01°06.0	289°35.7	N26°06.6	332°22.9	N21°38.9	52°55.9	S06°42.5					
19	57°14.1	59°49.6	07.1	304°38.9	06.7	347°25.4	38.9	67°58.1	42.4					
20	72°16.6	74°50.4	08.3	319°42.2	06.7	2°27.9	38.9	83°00.3	42.3					
21	87°19.0	89°51.2	· · 09.4	334°45.4	· · 06.8	17°30.4	· · 38.9	98°02.5	· · 42.2					
22	102°21.5	104°52.0	10.5	349°48.7	06.9	32°32.9	38.9	113°04.8	42.1					
23	117°24.0	119°52.9	11.6	4°51.9	07.0	47°35.4	38.9	128°07.0	42.0					
Mer.pass. 15:12			ν 0.8' d 1.1' m -4.75			ν 3.3' d 0.1' m -1.07			ν 2.5' d 0.0' m -2.52			ν 2.2' d -0.1' m 1.11		

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	132°26.4	134°53.7	N01°12.7	19°55.2	N26°07.0	62°37.9	N21°38.9	143°09.2	S06°41.9					
1	147°28.9	149°54.5	13.8	34°58.4	07.1	77°40.3	38.9	158°11.4	41.7					
2	162°31.4	164°55.4	15.0	50°01.7	07.2	92°42.8	38.9	173°13.6	41.6					
3	177°33.8	179°56.2	· · 16.1	65°04.9	· · 07.2	107°45.3	· · 38.9	188°15.8	· · 41.5					
4	192°36.3	194°57.0	17.2	80°08.2	07.3	122°47.8	38.9	203°18.0	41.4					
5	207°38.8	209°57.9	18.3	95°11.4	07.4	137°50.3	38.9	218°20.2	41.3					
6	222°41.2	224°58.7	N01°19.4	110°14.7	N26°07.4	152°52.8	N21°38.9	233°22.5	S06°41.2					
7	237°43.7	239°59.5	20.5	125°17.9	07.5	167°55.3	38.9	248°24.7	41.1					
8	252°46.2	255°00.4	21.6	140°21.2	07.6	182°57.7	38.9	263°26.9	41.0					
9	267°48.6	270°01.2	· · 22.8	155°24.4	· · 07.6	198°00.2	· · 38.9	278°29.1	· · 40.9					
10	282°51.1	285°02.1	23.9	170°27.6	07.7	213°02.7	39.0	293°31.3	40.8					
11	297°53.5	300°02.9	25.0	185°30.9	07.8	228°05.2	39.0	308°33.5	40.6					
12	312°56.0	315°03.7	N01°26.1	200°34.1	N26°07.8	243°07.7	N21°39.0	323°35.7	S06°40.5					
13	327°58.5	330°04.6	27.2	215°37.3	07.9	258°10.2	39.0	338°37.9	40.4					
14	343°00.9	345°05.4	28.3	230°40.6	08.0	273°12.6	39.0	353°40.1	40.3					
15	358°03.4	0°06.3	· · 29.4	245°43.8	· · 08.0	288°15.1	· · 39.0	8°42.4	· · 40.2					
16	13°05.9	15°07.2	30.5	260°47.0	08.1	303°17.6	39.0	23°44.6	40.1					
17	28°08.3	30°08.0	31.6	275°50.3	08.2	318°20.1	39.0	38°46.8	40.0					
18	43°10.8	45°08.9	N01°32.7	290°53.5	N26°08.2	333°22.6	N21°39.0	53°49.0	S06°39.9					
19	58°13.3	60°09.7	33.8	305°56.7	08.3	348°25.0	39.0	68°51.2	39.8					
20	73°15.7	75°10.6	35.0	321°00.0	08.3	3°27.5	39.0	83°53.4	39.7					
21	88°18.2	90°11.4	· · 36.1	336°03.2	· · 08.4	18°30.0	· · 39.0	98°55.6	· · 39.6					
22	103°20.6	105°12.3	37.2	351°06.4	08.5	33°32.5	39.0	113°57.8	39.4					
23	118°23.1	120°13.2	38.3	6°09.6	08.5	48°35.0	39.0	129°00.0	39.3					
Mer.pass. 15:08			ν 0.8' d 1.1' m -4.76			ν 3.3' d 0.1' m -1.05			ν 2.5' d 0.0' m -2.51			ν 2.2' d -0.1' m 1.11		

Jan 31 Fri	SHA	Mer.pass
Venus	3°47.6	15:02
Mars	246°49.5	22:46
Jupiter	290°09.9	19:54
Saturn	10°54.8	14:32

Feb 01 Sat	SHA	Mer.pass
Venus	3°07.0	15:01
Mars	247°09.5	22:41
Jupiter	290°10.7	19:50
Saturn	10°48.8	14:29

Feb 02 Sun	SHA	Mer.pass
Venus	2°27.3	15:00
Mars	247°28.8	22:35
Jupiter	290°11.4	19:46
Saturn	10°42.7	14:25

Horizontal parallax		
Venus:	0.3	
Mars:	0.2	

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	176°39.4	S17°22.3	156°52.6	10.4'	S13°27.7	-15.4'	59.2'
1	191°39.3	21.6	171°22.0	10.4'	13°12.4	-15.4'	59.2'
2	206°39.2	20.9	185°51.4	10.5'	12°56.9	-15.5'	59.3'
3	221°39.1	· · 20.2	200°20.9	10.5'	12°41.4	-15.6'	59.3'
4	236°39.0	19.5	214°50.4	10.6'	12°25.8	-15.7'	59.3'
5	251°38.9	18.8	229°20.0	10.6'	12°10.1	-15.7'	59.3'
6	266°38.8	S17°18.1	243°49.6	10.7'	S11°54.4	-15.8'	59.3'
7	281°38.7	17.4	258°19.3	10.7'	11°38.6	-15.9'	59.3'
8	296°38.7	16.7	272°49.0	10.8'	11°22.7	-15.9'	59.3'
9	311°38.6	· · 16.0	287°18.8	10.8'	11°06.8	-16.0'	59.3'
10	326°38.5	15.3	301°48.6	10.9'	10°50.8	-16.1'	59.4'
11	341°38.4	14.6	316°18.5	10.9'	10°34.7	-16.1'	59.4'
12	356°38.3	S17°13.9	330°48.4	11.0'	S10°18.6	-16.2'	59.4'
13	11°38.2	13.2	345°18.4	11.0'	10°02.4	-16.3'	59.4'
14	26°38.1	12.5	359°48.4	11.1'	09°46.1	-16.3'	59.4'
15	41°38.0	· · 11.8	14°18.5	11.1'	09°29.8	-16.4'	59.4'
16	56°37.9	11.1	28°48.6	11.1'	09°13.4	-16.4'	59.4'
17	71°37.9	10.4	43°18.7	11.2'	08°57.0	-16.5'	59.4'
18	86°37.8	S17°09.6	57°48.9	11.2'	S08°40.6	-16.5'	59.4'
19	101°37.7	08.9	72°19.1	11.3'	08°24.0	-16.6'	59.5'
20	116°37.6	08.2	86°49.4	11.3'	08°07.5	-16.6'	59.5'
21	131°37.5	· · 07.5	101°19.7	11.3'	07°50.9	-16.7'	59.5'
22	146°37.4	06.8	115°50.0	11.4'	07°34.2	-16.7'	59.5'
23	161°37.3	06.1	130°20.3	11.4'	07°17.5	-16.7'	59.5'
SD = 16.2' $d = -0.7'$ SD = 16.2'							

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	176°37.2	S17°05.4	144°50.7	11.4'	S07°00.8	-16.8'	59.5'
1	191°37.2	04.7	159°21.2	11.5'	06°44.0	-16.8'	59.5'
2	206°37.1	04.0	173°51.6	11.5'	06°27.2	-16.9'	59.5'
3	221°37.0	· · 03.2	188°22.1	11.5'	06°10.3	-16.9'	59.5'
4	236°36.9	02.5	202°52.6	11.5'	05°53.4	-16.9'	59.5'
5	251°36.8	01.8	217°23.2	11.6'	05°36.5	-17.0'	59.5'
6	266°36.7	S17°01.1	231°53.8	11.6'	S05°19.6	-17.0'	59.5'
7	281°36.7	17°00.4	246°24.4	11.6'	05°02.6	-17.0'	59.5'
8	296°36.6	16°59.7	260°55.0	11.6'	04°45.6	-17.0'	59.5'
9	311°36.5	· · 59.0	275°25.6	11.7'	04°28.5	-17.1'	59.6'
10	326°36.4	58.2	289°56.3	11.7'	04°11.5	-17.1'	59.6'
11	341°36.3	57.5	304°27.0	11.7'	03°54.4	-17.1'	59.6'
12	356°36.3	S16°56.8	318°57.7	11.7'	S03°37.3	-17.1'	59.6'
13	11°36.2	56.1	333°28.4	11.7'	03°20.1	-17.1'	59.6'
14	26°36.1	55.4	347°59.2	11.8'	03°03.0	-17.2'	59.6'
15	41°36.0	· · 54.6	2°29.9	11.8'	02°45.8	-17.2'	59.6'
16	56°35.9	53.9	17°00.7	11.8'	02°28.7	-17.2'	59.6'
17	71°35.9	53.2	31°31.5	11.8'	02°11.5	-17.2'	59.6'
18	86°35.8	S16°52.5	46°02.3	11.8'	S01°54.3	-17.2'	59.6'
19	101°35.7	51.8	60°33.1	11.8'	01°37.1	-17.2'	59.6'
20	116°35.6	51.0	75°03.9	11.8'	01°19.8	-17.2'	59.6'
21	131°35.5	· · 50.3	89°34.7	11.8'	01°02.6	-17.2'	59.6'
22	146°35.5	49.6	104°05.6	11.8'	00°45.4	-17.2'	59.6'
23	161°35.4	48.9	118°36.4	11.8'	00°28.2	-17.2'	59.6'
SD = 16.2' $d = -0.7'$ SD = 16.2'							

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	176°35.3	S16°48.2	133°07.3	11.8'	S00°10.9	-17.2'	59.6'
1	191°35.2	47.4	147°38.1	11.8'	N00°06.3	17.2'	59.6'
2	206°35.2	46.7	162°08.9	11.8'	00°23.6	17.2'	59.6'
3	221°35.1	· · 46.0	176°39.8	11.8'	00°40.8	17.2'	59.6'
4	236°35.0	45.2	191°10.6	11.8'	00°58.0	17.2'	59.6'
5	251°34.9	44.5	205°41.5	11.8'	01°15.2	17.2'	59.6'
6	266°34.9	S16°43.8	220°12.3	11.8'	N01°32.4	17.2'	59.6'
7	281°34.8	43.1	234°43.2	11.8'	01°49.7	17.2'	59.6'
8	296°34.7	42.3	249°14.0	11.8'	02°06.8	17.2'	59.6'
9	311°34.6	· · 41.6	263°44.8	11.8'	02°24.0	17.2'	59.6'
10	326°34.6	40.9	278°15.6	11.8'	02°41.2	17.2'	59.6'
11	341°34.5	40.2	292°46.4	11.8'	02°58.3	17.1'	59.6'
12	356°34.4	S16°39.4	307°17.2	11.8'	N03°15.5	17.1'	59.6'
13	11°34.3	38.7	321°48.0	11.8'	03°32.6	17.1'	59.6'
14	26°34.3	38.0	336°18.7	11.7'	03°49.7	17.1'	59.6'
15	41°34.2	· · 37.2	350°49.5	11.7'	04°06.8	17.0'	59.6'
16	56°34.1	36.5	5°20.2	11.7'	04°23.8	17.0'	59.6'
17	71°34.1	35.8	19°50.9	11.7'	04°40.8	17.0'	59.6'
18	86°34.0	S16°35.0	34°21.6	11.7'	N04°57.8	17.0'	59.6'
19	101°33.9	34.3	48°52.3	11.6'	05°14.8	16.9'	59.6'
20	116°33.9	33.6	63°22.9	11.6'	05°31.7	16.9'	59.6'
21	131°33.8	· · 32.8	77°53.6	11.6'	05°48.6	16.9'	59.6'
22	146°33.7	32.1	92°24.2	11.6'	06°05.5	16.8'	59.5'
23	161°33.7	31.4	106°54.7	11.5'	06°22.3	16.8'	59.5'
SD = 16.2' $d = -0.7'$ SD = 16.3'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:09	08:37	10:25	14:03	15:51	17:20
N 70°	07:01	08:19	09:44	14:44	16:10	17:28
68°	06:54	08:04	09:16	15:12	16:24	17:34
66°	06:49	07:52	08:55	15:33	16:36	17:40
64°	06:44	07:42	08:39	15:49	16:46	17:45
62°	06:39	07:34	08:25	16:03	16:54	17:49
60°	06:35	07:26	08:13	16:15	17:02	17:53
N 58°	06:32	07:19	08:03	16:25	17:09	17:56
56°	06:28	07:13	07:54	16:33	17:15	18:00
54°	06:25	07:08	07:47	16:41	17:20	18:03
52°	06:22	07:03	07:40	16:48	17:25	18:05
50°	06:20	06:58	07:33	16:55	17:29	18:08
45°	06:13	06:48	07:20	17:08	17:39	18:15
N 40°	06:07	06:40	07:08	17:19	17:48	18:20
35°	06:02	06:32	06:59	17:29	17:55	18:26
30°	05:57	06:25	06:50	17:37	18:02	18:31
20°	05:46	06:12	06:35	17:52	18:15	18:41
N 10°	05:35	06:00	06:22	18:05	18:27	18:52
0°	05:23	05:48	06:10	18:17	18:39	19:04
S 10°	05:10	05:36	05:58	18:29	18:51	19:17
20°	04:53	05:21	05:44	18:43	19:06	19:33
30°	04:32	05:03	05:29	18:58	19:24	19:54
35°	04:19	04:52	05:20	19:07	19:34	20:08
40°	04:03	04:39	05:09	19:17	19:47	20:24
45°	03:42	04:24	04:57	19:29	20:02	20:44
S 50°	03:15	04:04	04:42	19:44	20:22	21:11
52°	03:01	03:55	04:35	19:51	20:31	21:25
54°	02:44	03:44	04:27	19:59	20:42	21:41
56°	02:23	03:31	04:18	20:07	20:54	22:01
58°	01:56	03:17	04:08	20:17	21:08	22:27
S 60°	01:13	02:59	03:57	20:28	21:25	23:07

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	10:19	09:35	08:57	18:16	20:46	23:12
N 70°	10:02	09:29	09:00	18:30	20:47	23:02
68°	09:48	09:24	09:03	18:42	20:49	22:54
66°	09:37	09:20	09:05	18:51	20:50	22:48
64°	09:27	09:17	09:07	18:59	20:51	22:43
62°	09:19	09:13	09:08	19:05	20:52	22:38
60°	09:12	09:11	09:10	19:11	20:53	22:34
N 58°	09:05	09:08	09:11	19:16	20:53	22:30
56°	09:00	09:06	09:12	19:20	20:54	22:27
54°	08:55	09:04	09:13	19:24	20:55	22:25
52°	08:50	09:02	09:14	19:28	20:55	22:22
50°	08:46	09:01	09:15	19:31	20:55	22:20
45°	08:37	08:57	09:17	19:38	20:56	22:15
N 40°	08:29	08:54	09:18	19:44	20:57	22:11
35°	08:22	08:51	09:19	19:48	20:58	22:07
30°	08:17	08:49	09:21	19:53	20:58	22:04
20°	08:06	08:45	09:23	20:00	20:59	21:59
N 10°	07:57	08:42	09:25	20:07	21:00	21:54
0°	07:49	08:38	09:26	20:12	21:01	21:49
S 10°	07:41	08:35	09:28	20:18	21:02	21:45
20°	07:31	08:31	09:30	20:25	21:03	21:40
30°	07:21	08:27	09:33	20:32	21:03	21:35
35°	07:15	08:25	09:34	20:36	21:04	21:32
40°	07:08	08:22	09:36	20:40	21:04	21:28
45°	07:00	08:19	09:37	20:45	21:05	21:25
S 50°	06:50	08:15	09:40	20:51	21:06	21:20
52°	06:46	08:14	09:41	20:54	21:06	21:18
54°	06:41	08:12	09:42	20:57	21:06	21:15
56°	06:35	08:10	09:43	21:01	21:07	21:13
58°	06:29	08:07	09:44	21:04	21:07	21:10
S 60°	06:22	08:05	09:46	21:09	21:08	21:07

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass		
	00 ^h	mm:ss	hh:mm	Upper	Lower	2-4
				hh:mm	hh:mm	3-15%
31	13:22	13:27	12:13	14:01	01:36	
01	13:31	13:35	12:14	14:50	02:25	
02	13:39	13:42	12:14	15:38	03:14	

February 03, 04, 05 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	133°25.6	135°14.0	N01°39.4	21°12.8	N26°08.6	63°37.4	N21°39.1	144°02.3	S06°39.2	Alpheratz	357°35.0 29°13.8
1	148°28.0	150°14.9	40.5	36°16.1	08.7	78°39.9	39.1	159°04.5	39.1	Ankaa	353°07.3 -42°10.4
2	163°30.5	165°15.8	41.6	51°19.3	08.7	93°42.4	39.1	174°06.7	39.0	Schedar	349°31.3 56°40.7
3	178°33.0	180°16.6	· · 42.7	66°22.5	· · 08.8	108°44.9	· · 39.1	189°08.9	· · 38.9	Diphda	348°47.4 -17°51.1
4	193°35.4	195°17.5	43.8	81°25.7	08.8	123°47.4	39.1	204°11.1	38.8	Achernar	335°20.3 -57°06.8
5	208°37.9	210°18.4	44.9	96°28.9	08.9	138°49.8	39.1	219°13.3	38.7	Hamal	327°51.2 23°34.9
6	223°40.4	225°19.3	N01°46.0	111°32.1	N26°09.0	153°52.3	N21°39.1	234°15.5	S06°38.6	Polaris	313°58.5 89°22.5
7	238°42.8	240°20.2	47.1	126°35.3	09.0	168°54.8	39.1	249°17.7	38.5	Acamar	315°11.7 -40°12.5
8	253°45.3	255°21.0	48.2	141°38.6	09.1	183°57.3	39.1	264°19.9	38.3	Menkar	314°06.1 4°11.2
9	268°47.8	270°21.9	· · 49.3	156°41.8	· · 09.1	198°59.7	· · 39.1	279°22.1	· · 38.2	Mirfak	308°28.1 49°57.2
10	283°50.2	285°22.8	50.4	171°45.0	09.2	214°02.2	39.1	294°24.4	38.1	Aldebaran	290°39.4 16°33.6
11	298°52.7	300°23.7	51.5	186°48.2	09.2	229°04.7	39.1	309°26.6	38.0	Rigel	281°03.6 -8°10.5
12	313°55.1	315°24.6	N01°52.6	201°51.4	N26°09.3	244°07.2	N21°39.1	324°28.8	S06°37.9	Capella	280°21.5 46°01.5
13	328°57.6	330°25.5	53.7	216°54.6	09.4	259°09.6	39.1	339°31.0	37.8	Bellatrix	278°22.6 6°22.3
14	344°00.1	345°26.3	54.8	231°57.8	09.4	274°12.1	39.2	354°33.2	37.7	Elnath	278°01.5 28°37.8
15	359°02.5	0°27.2	· · 55.9	247°01.0	· · 09.5	289°14.6	· · 39.2	0°35.4	· · 37.6	Alnilam	275°37.4 -1°11.2
16	14°05.0	15°28.1	57.0	262°04.2	09.5	304°17.0	39.2	24°37.6	37.5	Betelgeuse	270°51.8 7°24.7
17	29°07.5	30°29.0	58.1	277°07.4	09.6	319°19.5	39.2	39°39.8	37.4	Canopus	263°51.9 -52°42.7
18	44°09.9	45°29.9	N01°59.2	292°10.6	N26°09.6	334°22.0	N21°39.2	54°42.0	S06°37.2	Sirius	258°25.9 -16°45.1
19	59°12.4	60°30.8	02°00.3	307°13.8	09.7	349°24.5	39.2	69°44.2	37.1	Adhara	255°05.5 -29°00.5
20	74°14.9	75°31.7	01.4	322°17.0	09.7	4°26.9	39.2	84°46.4	37.0	Procyon	244°50.4 5°09.6
21	89°17.3	90°32.6	· · 02.5	337°20.2	· · 09.8	19°29.4	· · 39.2	99°48.7	· · 36.9	Pollux	243°16.9 27°57.9
22	104°19.8	105°33.5	03.6	352°23.3	09.8	34°31.9	39.2	114°50.9	36.8	Avior	234°14.1 -59°35.4
23	119°22.2	120°34.4	04.7	7°26.5	09.9	49°34.3	39.2	129°53.1	36.7	Suhail	222°45.8 -43°32.0
Mer.pass. 15:04		ν 0.9' d 1.1' m -4.77		ν 3.2' d 0.1' m -1.02		ν 2.5' d 0.0' m -2.50		ν 2.2' d -0.1' m 1.11			

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	134°24.7	135°35.3	N02°05.8	22°29.7	N26°09.9	64°36.8	N21°39.2	144°55.3	S06°36.6		
1	149°27.2	150°36.3	06.9	37°32.9	10.0	79°39.3	39.2	159°57.5	36.5		
2	164°29.6	165°37.2	08.0	52°36.1	10.1	94°41.7	39.3	174°59.7	36.4		
3	179°32.1	180°38.1	· · 09.1	67°39.3	· · 10.1	109°44.2	· · 39.3	190°01.9	· · 36.2		
4	194°34.6	195°39.0	10.1	82°42.5	10.2	124°46.7	39.3	205°04.1	36.1		
5	209°37.0	210°39.9	11.2	97°45.6	10.2	139°49.2	39.3	220°06.3	36.0		
6	224°39.5	225°40.8	N02°12.3	112°48.8	N26°10.3	154°51.6	N21°39.3	235°08.5	S06°35.9		
7	239°42.0	240°41.7	13.4	127°52.0	10.3	169°54.1	39.3	250°10.7	35.8		
8	254°44.4	255°42.7	14.5	142°55.2	10.4	184°56.6	39.3	265°13.0	35.7		
9	269°46.9	270°43.6	· · 15.6	157°58.3	· · 10.4	199°59.0	· · 39.3	280°15.2	· · 35.6		
10	284°49.4	285°44.5	16.7	173°01.5	10.4	215°01.5	39.3	295°17.4	35.5		
11	299°51.8	300°45.4	17.8	188°04.7	10.5	230°03.9	39.3	310°19.6	35.4		
12	314°54.3	315°46.4	N02°18.9	203°07.9	N26°10.5	245°06.4	N21°39.3	325°21.8	S06°35.3		
13	329°56.7	330°47.3	20.0	218°11.0	10.6	260°08.9	39.3	340°24.0	35.1		
14	344°59.2	345°48.2	21.0	233°14.2	10.6	275°11.3	39.4	355°26.2	35.0		
15	0°01.7	0°49.2	· · 22.1	248°17.4	· · 10.7	290°13.8	· · 39.4	10°28.4	· · 34.9		
16	15°04.1	15°50.1	23.2	263°20.5	10.7	305°16.3	39.4	25°30.6	34.8		
17	30°06.6	30°51.0	24.3	278°23.7	10.8	320°18.7	39.4	40°32.8	34.7		
18	45°09.1	45°52.0	N02°25.4	293°26.9	N26°10.8	335°21.2	N21°39.4	55°35.0	S06°34.6		
19	60°11.5	60°52.9	26.5	308°30.0	10.9	350°23.7	39.4	70°37.2	34.5		
20	75°14.0	75°53.9	27.6	323°33.2	10.9	5°26.1	39.4	85°39.4	34.4		
21	90°16.5	90°54.8	· · 28.6	338°36.3	· · 11.0	20°28.6	· · 39.4	100°41.7	· · 34.3		
22	105°18.9	105°55.8	29.7	353°39.5	11.0	35°31.0	39.4	115°43.9	34.1		
23	120°21.4	120°56.7	30.8	8°42.7	11.0	50°33.5	39.4	130°46.1	34.0		
Mer.pass. 15:00		ν 0.9' d 1.1' m -4.78		ν 3.2' d 0.1' m -0.99		ν 2.5' d 0.0' m -2.50		ν 2.2' d -0.1' m 1.11			

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	135°23.9	135°57.7	N02°31.9	23°45.8	N26°11.1	65°36.0	N21°39.5	145°48.3	S06°33.9		
1	150°26.3	150°58.6	33.0	38°49.0	11.1	80°38.4	39.5	160°50.5	33.8		
2	165°28.8	165°59.6	34.0	53°52.1	11.2	95°40.9	39.5	175°52.7	33.7		
3	180°31.2	181°00.5	· · 35.1	68°55.3	· · 11.2	110°43.3	· · 39.5	190°54.9	· · 33.6		
4	195°33.7	196°01.5	36.2	83°58.4	11.3	125°45.8	39.5	205°57.1	33.5		
5	210°36.2	211°02.4	37.3	99°01.6	11.3	140°48.3	39.5	220°59.3	33.4		
6	225°38.6	226°03.4	N02°38.4	114°04.7	N26°11.3	155°50.7	N21°39.5	236°01.5	S06°33.3		
7	240°41.1	241°04.4	39.4	129°07.9	11.4	170°53.2	39.5	251°03.7	33.1		
8	255°43.6	256°05.3	40.5	144°11.0	11.4	185°55.6	39.5	266°05.9	33.0		
9	270°46.0	271°06.3	· · 41.6	159°14.1	· · 11.5	200°58.1	· · 39.5	281°08.1	· · 32.9		
10	285°48.5	286°07.3	42.7	174°17.3	11.5	216°00.5	39.5	296°10.3	32.8		
11	300°51.0	301°08.2	43.8	189°20.4	11.6	231°03.0	39.6	311°12.6	32.7		
12	315°53.4	316°09.2	N02°44.8	204°23.6	N26°11.6	246°05.5	N21°39.6	326°14.8	S06°32.6		
13	330°55.9	331°10.2	45.9	219°26.7	11.6	261°07.9	39.6	341°17.0	32.5		
14	345°58.4	346°11.1	47.0	234°29.8	11.7	276°10.4	39.6	356°19.2	32.4		
15	1°00.8	1°12.1	· · 48.1	249°33.0	· · 11.7	291°12.8	· · 39.6	11°21.4	· · 32.3		
16	16°03.3	16°13.1	49.1	264°36.1	11.7	306°15.3	39.6	26°23.6	32.1		
17	31°05.7	31°14.1	50.2	279°39.2	11.8	321°17.7	39.6	41°25.8	32.0		
18	46°08.2	46°15.1	N02°51.3	294°42.3	N26°11.8	336°20.2	N21°39.6	56°28.0	S06°31.9		
19	61°10.7	61°16.1	52.3	309°45.5	11.9	351°22.6	39.6	71°30.2	31.8		
20	76°13.1	76°17.0	53.4	324°48.6	11.9	6°25.1	39.7	86°32.4	31.7		
21	91°15.6	91°18.0	· · 54.5	339°51.7	· · 11.9	21°27.5	· · 39.7	101°34.6	· · 31.6		
22	106°18.1	106°19.0	55.6	354°54.8	12.0	36°30.0	39.7	116°36.8	31.5		
23	121°20.5	121°20.0	56.6	9°58.0	12.0	51°32.4	39.7	131°39.0	31.4		
Mer.pass. 14:56		ν 1.0' d 1.1' m -4.79		ν 3.2' d 0.0' m -0.96		ν 2.5' d 0.0' m -2.49		ν 2.2' d -0.1' m 1.11			

Feb 03 Mon		SHA	Mer.pass
Venus		1°48.5	14:58
Mars		247°47.3	22:30
Jupiter		290°11.9	19:42
Saturn		10°36.7	14:22
Feb 04 Tue		SHA	Mer.pass
Venus		1°10.6	14:57
Mars		248°05.0	22:25
Jupiter		290°12.1	19:38
Saturn		10°30.6	14:18
Feb 05 Wed		SHA	Mer.pass
Venus		0°33.8	14:55
Mars		248°22.0	22:20
Jupiter		290°12.1	19:34
Saturn		10°24.4	14:15
Horizontal parallax			
Venus:		0.3	
Mars:		0.2	

DUT1 = UT1-UTC = +0.0448 sec $\Delta T = TT-UT1 = +69.1392$ sec

2025 February 03 to Feb. 05 UT

h Sun Moon						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	176°33.6	S16°30.6	121°25.3	11.5'	N06°39.1	16.8' 59.5'
1	191°33.5	29.9	135°55.8	11.5'	06°55.9	16.7' 59.5'
2	206°33.4	29.1	150°26.3	11.5'	07°12.6	16.7' 59.5'
3	221°33.4	· · 28.4	164°56.7	11.4'	07°29.3	16.6' 59.5'
4	236°33.3	27.7	179°27.2	11.4'	07°45.9	16.6' 59.5'
5	251°33.3	26.9	193°57.6	11.4'	08°02.5	16.5' 59.5'
6	266°33.2	S16°26.2	208°27.9	11.3'	N08°19.1	16.5' 59.5'
7	281°33.1	25.5	222°58.2	11.3'	08°35.6	16.5' 59.5'
8	296°33.1	24.7	237°28.5	11.2'	08°52.0	16.4' 59.5'
9	311°33.0	· · 24.0	251°58.8	11.2'	09°08.4	16.3' 59.5'
10	326°32.9	23.2	266°29.0	11.2'	09°24.8	16.3' 59.5'
11	341°32.9	22.5	280°59.1	11.1'	09°41.1	16.2' 59.5'
12	356°32.8	S16°21.7	295°29.3	11.1'	N09°57.3	16.2' 59.5'
13	11°32.7	21.0	309°59.3	11.0'	10°13.5	16.1' 59.4'
14	26°32.7	20.3	324°29.4	11.0'	10°29.6	16.1' 59.4'
15	41°32.6	· · 19.5	338°59.4	10.9'	10°45.7	16.0' 59.4'
16	56°32.5	18.8	353°29.3	10.9'	11°01.7	15.9' 59.4'
17	71°32.5	18.0	7°59.2	10.8'	11°17.6	15.9' 59.4'
18	86°32.4	S16°17.3	22°29.1	10.8'	N11°33.5	15.8' 59.4'
19	101°32.4	16.5	36°58.8	10.7'	11°49.3	15.7' 59.4'
20	116°32.3	15.8	51°28.6	10.7'	12°05.1	15.7' 59.4'
21	131°32.2	· · 15.0	65°58.3	10.6'	12°20.7	15.6' 59.4'
22	146°32.2	14.3	80°27.9	10.6'	12°36.3	15.5' 59.4'
23	161°32.1	13.6	94°57.5	10.5'	12°51.9	15.5' 59.4'
SD = 16.2' $d = -0.7'$			SD = 16.2'			
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	176°32.1	S16°12.8	109°27.0	10.5'	N13°07.3	15.4' 59.4'
1	191°32.0	12.1	123°56.5	10.4'	13°22.7	15.3' 59.3'
2	206°31.9	11.3	138°25.9	10.4'	13°38.0	15.2' 59.3'
3	221°31.9	· · 10.6	152°55.3	10.3'	13°53.2	15.1' 59.3'
4	236°31.8	09.8	167°24.6	10.2'	14°08.4	15.1' 59.3'
5	251°31.8	09.1	181°53.8	10.2'	14°23.4	15.0' 59.3'
6	266°31.7	S16°08.3	196°23.0	10.1'	N14°38.4	14.9' 59.3'
7	281°31.7	07.6	210°52.1	10.0'	14°53.3	14.8' 59.3'
8	296°31.6	06.8	225°21.1	10.0'	15°08.1	14.7' 59.3'
9	311°31.5	· · 06.1	239°50.1	9.9'	15°22.8	14.6' 59.3'
10	326°31.5	05.3	254°19.1	9.9'	15°37.4	14.5' 59.2'
11	341°31.4	04.5	268°47.9	9.8'	15°52.0	14.4' 59.2'
12	356°31.4	S16°03.8	283°16.7	9.7'	N16°06.4	14.4' 59.2'
13	11°31.3	03.0	297°45.4	9.7'	16°20.8	14.3' 59.2'
14	26°31.3	02.3	312°14.1	9.6'	16°35.0	14.2' 59.2'
15	41°31.2	· · 01.5	326°42.7	9.5'	16°49.2	14.1' 59.2'
16	56°31.2	00.8	341°11.2	9.4'	17°03.3	14.0' 59.2'
17	71°31.1	16°00.0	355°39.6	9.4'	17°17.2	13.9' 59.2'
18	86°31.1	S15°59.3	10°08.0	9.3'	N17°31.1	13.8' 59.2'
19	101°31.0	58.5	24°36.3	9.2'	17°44.8	13.6' 59.1'
20	116°31.0	57.7	39°04.5	9.2'	17°58.5	13.5' 59.1'
21	131°30.9	· · 57.0	53°32.7	9.1'	18°12.0	13.4' 59.1'
22	146°30.9	56.2	68°00.8	9.0'	18°25.5	13.3' 59.1'
23	161°30.8	55.5	82°28.8	8.9'	18°38.8	13.2' 59.1'
SD = 16.2' $d = -0.7'$			SD = 16.2'			
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	176°30.8	S15°54.7	96°56.7	8.9'	N18°52.0	13.1' 59.1'
1	191°30.7	54.0	111°24.6	8.8'	19°05.1	13.0' 59.1'
2	206°30.7	53.2	125°52.3	8.7'	19°18.1	12.9' 59.0'
3	221°30.6	· · 52.4	140°20.1	8.6'	19°31.0	12.8' 59.0'
4	236°30.6	51.7	154°47.7	8.6'	19°43.7	12.6' 59.0'
5	251°30.5	50.9	169°15.2	8.5'	19°56.4	12.5' 59.0'
6	266°30.5	S15°50.1	183°42.7	8.4'	N20°08.9	12.4' 59.0'
7	281°30.4	49.4	198°10.1	8.3'	20°21.3	12.3' 59.0'
8	296°30.4	48.6	212°37.5	8.3'	20°33.6	12.2' 59.0'
9	311°30.3	· · 47.9	227°04.7	8.2'	20°45.7	12.0' 59.0'
10	326°30.3	47.1	241°31.9	8.1'	20°57.7	11.9' 58.9'
11	341°30.2	46.3	255°59.0	8.0'	21°09.6	11.8' 58.9'
12	356°30.2	S15°45.6	270°26.0	7.9'	N21°21.4	11.6' 58.9'
13	11°30.1	44.8	284°52.9	7.9'	21°33.0	11.5' 58.9'
14	26°30.1	44.0	299°19.8	7.8'	21°44.5	11.4' 58.9'
15	41°30.0	· · 43.3	313°46.6	7.7'	21°55.9	11.2' 58.9'
16	56°30.0	42.5	328°13.3	7.6'	22°07.2	11.1' 58.9'
17	71°29.9	41.7	342°39.9	7.6'	22°18.3	11.0' 58.8'
18	86°29.9	S15°41.0	357°06.5	7.5'	N22°29.2	10.8' 58.8'
19	101°29.9	40.2	11°33.0	7.4'	22°40.1	10.7' 58.8'
20	116°29.8	39.4	25°59.4	7.3'	22°50.8	10.6' 58.8'
21	131°29.8	· · 38.7	40°25.7	7.2'	23°01.3	10.4' 58.8'
22	146°29.7	37.9	54°51.9	7.2'	23°11.7	10.3' 58.8'
23	161°29.7	37.1	69°18.1	7.1'	23°22.0	10.1' 58.7'
SD = 16.2' $d = -0.8'$			SD = 16.1'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:58	08:24	10:02	14:27	16:06	17:32
N 70°	06:51	08:07	09:28	15:01	16:22	17:38
68°	06:46	07:54	09:03	15:26	16:35	17:44
66°	06:41	07:44	08:45	15:44	16:45	17:48
64°	06:37	07:34	08:29	15:59	16:54	17:52
62°	06:33	07:27	08:17	16:12	17:02	17:56
60°	06:30	07:20	08:06	16:23	17:09	17:59
N 58°	06:26	07:14	07:57	16:32	17:15	18:02
56°	06:23	07:08	07:49	16:40	17:21	18:05
54°	06:21	07:03	07:41	16:47	17:26	18:08
52°	06:18	06:58	07:35	16:54	17:30	18:10
50°	06:16	06:54	07:29	17:00	17:34	18:13
45°	06:10	06:45	07:16	17:12	17:43	18:18
N 40°	06:05	06:37	07:05	17:23	17:51	18:24
35°	06:00	06:30	06:56	17:32	17:58	18:28
30°	05:55	06:23	06:48	17:40	18:05	18:33
20°	05:45	06:11	06:34	17:54	18:17	18:43
N 10°	05:35	06:00	06:22	18:06	18:28	18:53
0°	05:24	05:49	06:10	18:17	18:39	19:04
S 10°	05:11	05:37	05:59	18:29	18:51	19:17
20°	04:55	05:23	05:46	18:42	19:05	19:32
30°	04:35	05:06	05:31	18:56	19:22	19:52
35°	04:22	04:55	05:23	19:04	19:32	20:05
40°	04:07	04:43	05:13	19:14	19:44	20:20
45°	03:48	04:29	05:01	19:26	19:58	20:39
S 50°	03:22	04:10	04:47	19:39	20:17	21:04
52°	03:09	04:01	04:41	19:46	20:25	21:17
54°	02:53	03:51	04:33	19:53	20:35	21:32
56°	02:35	03:39	04:25	20:01	20:47	21:50
58°	02:11	03:26	04:16	20:10	21:00	22:13
S 60°	01:36	03:10	04:05	20:21	21:16	22:46

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	08:17	07:19	□	· · ·	01:56	□
N 70°	08:30	07:50	□	· · ·	01:28	□
68°	08:40	08:12	07:23	· · ·	01:07	03:46
66°	08:49	08:30	08:03	· · ·	00:51	03:08
64°	08:57	08:45	08:30	· · ·	00:38	02:41
62°	09:03	08:58	08:52	· · ·	00:27	02:21
60°	09:09	09:08	09:09	· · ·	00:17	02:05
N 58°	09:14	09:18	09:24	· · ·	00:09	01:51
56°	09:18	09:26	09:37	· · ·	00:02	01:39
54°	09:22	09:33	09:48	23:56	· · ·	01:29
52°	09:26	09:40	09:58	23:50	· · ·	01:20
50°	09:29	09:46	10:06	23:45	· · ·	01:11
45°	09:36	09:59	10:25	23:34	· · ·	00:54
N 40°	09:43	10:09	10:41	23:25	· · ·	00:40
35°	09:48	10:19	10:54	23:17	· · ·	00:28
30°	09:53	10:27	11:05	23:10	· · ·	00:18
20°	10:01	10:41	11:25	22:59	· · ·	00:00
N 10°	10:08	10:54	11:43	22:48	23:45	· · ·
0°	10:15	11:06	11:59	22:39	23:31	· · ·
S 10°	10:22	11:18	12:15	22:29	23:16	· · ·
20°	10:30	11:30	12:33	22:19	23:01	23:47
30°	10:38	11:45	12:54	22:08	22:44	23:25
35°	10:44	11:54	13:06	22:01	22:34	23:12
40°	10:49	12:04	13:20	21:54	22:23	22:57
45°	10:56	12:16	13:36	21:45	22:09	22:39
S 50°	11:04	12:30	13:57	21:35	21:53	22:17
52°	11:08	12:37	14:07	21:30	21:46	22:06
54°	11:12	12:44	14:18	21:25	21:37	21:55
56°	11:17	12:53	14:31	21:19	21:28	21:41
58°	11:22	13:02	14:46	21:13	21:18	21:25
S 60°	11:28	13:13	15:04	21:06	21:05	21:07

Day	Sun		Mer. Pass hh:mm	Moon		Age 5-7 25-46%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
03	13:46	13:49	12:14	16:27	04:02	
04	13:52	13:54	12:14	17:18	04:52	
05	13:57	13:59	12:14	18:12	05:45	

February 06, 07, 08 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	136°23.0	136°21.0	N02°57.7	25°01.1	N26°12.0	66°34.9	N21°39.7	146°41.2	S06°31.2					Alpheratz	357°35.0	29°13.8	
1	151°25.5	151°22.0	58.8	40°04.2	12.1	81°37.3	39.7	161°43.4	31.1					Ankaa	353°07.3	-42°10.4	
2	166°27.9	166°23.0	02°59.8	55°07.3	12.1	96°39.8	39.7	176°45.6	31.0					Schedar	349°31.3	56°40.7	
3	181°30.4	181°24.0	03°00.9	70°10.4	•• 12.1	111°42.2	•• 39.7	191°47.8	•• 30.9					Diphda	348°47.4	-17°51.1	
4	196°32.8	196°25.0	02.0	85°13.6	12.2	126°44.7	39.7	206°50.1	30.8					Achernar	335°20.3	-57°06.8	
5	211°35.3	211°26.0	03.0	100°16.7	12.2	141°47.1	39.7	221°52.3	30.7					Hamal	327°51.2	23°34.9	
6	226°37.8	226°27.0	N03°04.1	115°19.8	N26°12.2	156°49.6	N21°39.8	236°54.5	S06°30.6					Polaris	313°59.8	89°22.5	
7	241°40.2	241°28.0	05.2	130°22.9	12.3	171°52.0	39.8	251°56.7	30.5					Acamar	315°11.7	-40°12.5	
8	256°42.7	256°29.0	06.2	145°26.0	12.3	186°54.5	39.8	266°58.9	30.4					Menkar	314°06.1	4°11.2	
9	271°45.2	271°30.0	•• 07.3	160°29.1	•• 12.3	201°56.9	•• 39.8	282°01.1	•• 30.2					Mirfak	308°28.1	49°57.2	
10	286°47.6	286°31.0	08.4	175°32.2	12.4	216°59.4	39.8	297°03.3	30.1					Aldebaran	290°39.4	16°33.6	
11	301°50.1	301°32.1	09.4	190°35.3	12.4	232°01.8	39.8	312°05.5	30.0					Rigel	281°03.6	-8°10.5	
12	316°52.6	316°33.1	N03°10.5	205°38.4	N26°12.4	247°04.3	N21°39.8	327°07.7	S06°29.9					Capella	280°21.5	46°01.5	
13	331°55.0	331°34.1	11.5	220°41.5	12.5	262°06.7	39.8	342°09.9	29.8					Bellatrix	278°22.6	6°22.3	
14	346°57.5	346°35.1	12.6	235°44.6	12.5	277°09.2	39.8	357°12.1	29.7					Elnath	278°01.5	28°37.8	
15	2°00.0	1°36.1	•• 13.7	250°47.7	•• 12.5	292°11.6	•• 39.9	12°14.3	•• 29.6					Alnilam	275°37.4	-1°11.2	
16	17°02.4	16°37.2	14.7	265°50.8	12.6	307°14.1	39.9	27°16.5	29.5					Betelgeuse	270°51.8	7°24.7	
17	32°04.9	31°38.2	15.8	280°53.9	12.6	322°16.5	39.9	42°18.7	29.3					Canopus	263°51.9	-52°42.7	
18	47°07.3	46°39.2	N03°16.8	295°57.0	N26°12.6	337°18.9	N21°39.9	57°20.9	S06°29.2					Sirius	258°25.9	-16°45.1	
19	62°09.8	61°40.2	17.9	311°00.1	12.6	352°21.4	39.9	72°23.1	29.1					Adhara	255°05.5	-29°00.5	
20	77°12.3	76°41.3	19.0	326°03.2	12.7	7°23.8	39.9	87°25.3	29.0					Procyon	244°50.4	5°09.6	
21	92°14.7	91°42.3	•• 20.0	341°06.3	•• 12.7	22°26.3	•• 39.9	102°27.5	•• 28.9					Pollux	243°16.9	27°57.9	
22	107°17.2	106°43.3	21.1	356°09.4	12.7	37°28.7	39.9	117°29.7	28.8					Avior	234°14.1	-59°35.4	
23	122°19.7	121°44.4	22.1	11°12.5	12.8	52°31.2	40.0	132°31.9	28.7					Suhail	222°45.8	-43°32.1	
Mer.pass. 14:52			ν 1.0' d 1.1' m -4.79	ν 3.1' d 0.0' m -0.93	ν 2.5' d 0.0' m -2.48	ν 2.2' d -0.1' m 1.11											

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	137°22.1	136°45.4	N03°23.2	26°15.5	N26°12.8	67°33.6	N21°40.0	147°34.2	S06°28.6
1	152°24.6	151°46.5	24.2	41°18.6	12.8	82°36.0	40.0	162°36.4	28.4
2	167°27.1	166°47.5	25.3	56°21.7	12.8	97°38.5	40.0	177°38.6	28.3
3	182°29.5	181°48.6	•• 26.3	71°24.8	•• 12.9	112°40.9	•• 40.0	192°40.8	•• 28.2
4	197°32.0	196°49.6	27.4	86°27.9	12.9	127°43.4	40.0	207°43.0	28.1
5	212°34.5	211°50.7	28.5	101°31.0	12.9	142°45.8	40.0	222°45.2	28.0
6	227°36.9	226°51.7	N03°29.5	116°34.0	N26°12.9	157°48.2	N21°40.0	237°47.4	S06°27.9
7	242°39.4	241°52.8	30.6	131°37.1	13.0	172°50.7	40.0	252°49.6	27.8
8	257°41.8	256°53.8	31.6	146°40.2	13.0	187°53.1	40.1	267°51.8	27.7
9	272°44.3	271°54.9	•• 32.7	161°43.3	•• 13.0	202°55.6	•• 40.1	282°54.0	•• 27.5
10	287°46.8	286°55.9	33.7	176°46.3	13.0	217°58.0	40.1	297°56.2	27.4
11	302°49.2	301°57.0	34.8	191°49.4	13.1	233°00.4	40.1	312°58.4	27.3
12	317°51.7	316°58.1	N03°35.8	206°52.5	N26°13.1	248°02.9	N21°40.1	328°00.6	S06°27.2
13	332°54.2	331°59.1	36.8	221°55.5	13.1	263°05.3	40.1	343°02.8	27.1
14	347°56.6	347°00.2	37.9	236°58.6	13.1	278°07.8	40.1	358°05.0	27.0
15	2°59.1	2°01.3	•• 38.9	252°01.7	•• 13.2	293°10.2	•• 40.1	13°07.2	•• 26.9
16	18°01.6	17°02.3	40.0	267°04.7	13.2	308°12.6	40.2	28°09.4	26.8
17	33°04.0	32°03.4	41.0	282°07.8	13.2	323°15.1	40.2	43°11.6	26.6
18	48°06.5	47°04.5	N03°42.1	297°10.8	N26°13.2	338°17.5	N21°40.2	58°13.8	S06°26.5
19	63°09.0	62°05.5	43.1	312°13.9	13.2	353°19.9	40.2	73°16.0	26.4
20	78°11.4	77°06.6	44.2	327°17.0	13.3	8°22.4	40.2	88°18.2	26.3
21	93°13.9	92°07.7	•• 45.2	342°20.0	•• 13.3	23°24.8	•• 40.2	103°20.4	•• 26.2
22	108°16.3	107°08.8	46.2	357°23.1	13.3	38°27.2	40.2	118°22.6	26.1
23	123°18.8	122°09.9	47.3	12°26.1	13.3	53°29.7	40.2	133°24.8	26.0
Mer.pass. 14:48			ν 1.0' d 1.1' m -4.80	ν 3.1' d 0.0' m -0.90	ν 2.4' d 0.0' m -2.47	ν 2.2' d -0.1' m 1.12			

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	138°21.3	137°11.0	N03°48.3	27°29.2	N26°13.3	68°32.1	N21°40.3	148°27.0	S06°25.9
1	153°23.7	152°12.1	49.4	42°32.2	13.4	83°34.5	40.3	163°29.2	25.7
2	168°26.2	167°13.1	50.4	57°35.3	13.4	98°37.0	40.3	178°31.4	25.6
3	183°28.7	182°14.2	•• 51.4	72°38.3	•• 13.4	113°39.4	•• 40.3	193°33.6	•• 25.5
4	198°31.1	197°15.3	52.5	87°41.3	13.4	128°41.8	40.3	208°35.8	25.4
5	213°33.6	212°16.4	53.5	102°44.4	13.4	143°44.3	40.3	223°38.0	25.3
6	228°36.1	227°17.5	N03°54.6	117°47.4	N26°13.5	158°46.7	N21°40.3	238°40.2	S06°25.2
7	243°38.5	242°18.6	55.6	132°50.5	13.5	173°49.1	40.4	253°42.5	25.1
8	258°41.0	257°19.7	56.6	147°53.5	13.5	188°51.5	40.4	268°44.7	25.0
9	273°43.5	272°20.8	•• 57.7	162°56.6	•• 13.5	203°54.0	•• 40.4	283°46.9	•• 24.8
10	288°45.9	287°21.9	58.7	177°59.6	13.5	218°56.4	40.4	298°49.1	24.7
11	303°48.4	302°23.1	03°59.7	193°02.6	13.5	233°58.8	40.4	313°51.3	24.6
12	318°50.8	317°24.2	N04°00.8	208°05.7	N26°13.6	249°01.3	N21°40.4	328°53.5	S06°24.5
13	333°53.3	332°25.3	01.8	223°08.7	13.6	264°03.7	40.4	343°55.7	24.4
14	348°55.8	347°26.4	02.8	238°11.7	13.6	279°06.1	40.4	358°57.9	24.3
15	3°58.2	2°27.5	•• 03.9	253°14.7	•• 13.6	294°08.5	•• 40.5	14°00.1	•• 24.2
16	19°00.7	17°28.6	04.9	268°17.8	13.6	309°11.0	40.5	29°02.3	24.0
17	34°03.2	32°29.7	05.9	283°20.8	13.6	324°13.4	40.5	44°04.5	23.9
18	49°05.6	47°30.9	N04°06.9	298°23.8	N26°13.6	339°15.8	N21°40.5	59°06.7	S06°23.8
19	64°08.1	62°32.0	08.0	313°26.8	13.7	354°18.3	40.5	74°08.9	23.7
20	79°10.6	77°33.1	09.0	328°29.9	13.7	9°20.7	40.5	89°11.1	23.6
21	94°13.0	92°34.3	•• 10.0	343°32.9	•• 13.7	24°23.1	•• 40.5	104°13.3	•• 23.5
22	109°15.5	107°35.4	11.0	358°35.9	13.7	39°25.5	40.6	119°15.5	23.4
23	124°17.9	122°36.5	12.1	13°38.9	13.7	54°28.0	40.6	134°17.7	23.3
Mer.pass. 14:44			ν 1.1' d 1.0' m -4.81	ν 3.0' d 0.0' m -0.87	ν 2.4' d 0.0' m -2.46	ν 2.2' d -0.1' m 1.12			

Feb 06 Thu	SHA	Mer.pass
Venus	359°58.0	14:54
Mars	248°38.1	22:15
Jupiter	290°11.9	19:30
Saturn	10°18.2	14:11

Feb 07 Fri	SHA	Mer.pass
Venus	359°23.3	14:52
Mars	248°53.4	22:10
Jupiter	290°11.5	19:27
Saturn	10°12.0	14:08

Feb 08 Sat	SHA	Mer.pass
Venus	358°49.7	14:50
Mars	249°07.9	22:06
Jupiter	290°10.8	19:23
Saturn	10°05.8	14:04

Horizontal parallax		
Venus:	0.3	
Mars:	0.2	

Sun							Moon								
Thu	GHA	Dec	GHA	ν	Dec	d	HP								
0	176°29.6	S15°36.3	83°44.2	7.0'	N23°32.1	10.0'	58.7'								
1	191°29.6	35.6	98°10.2	6.9'	23°42.1	9.8'	58.7'								
2	206°29.6	34.8	112°36.2	6.9'	23°51.9	9.7'	58.7'								
3	221°29.5	· · 34.0	127°02.1	6.8'	24°01.6	9.5'	58.7'								
4	236°29.5	33.3	141°27.9	6.7'	24°11.1	9.4'	58.7'								
5	251°29.4	32.5	155°53.6	6.7'	24°20.5	9.2'	58.7'								
6	266°29.4	S15°31.7	170°19.2	6.6'	N24°29.7	9.1'	58.6'								
7	281°29.4	30.9	184°44.8	6.5'	24°38.8	8.9'	58.6'								
8	296°29.3	30.2	199°10.4	6.4'	24°47.7	8.8'	58.6'								
9	311°29.3	· · 29.4	213°35.8	6.4'	24°56.5	8.6'	58.6'								
10	326°29.2	28.6	228°01.2	6.3'	25°05.1	8.4'	58.6'								
11	341°29.2	27.8	242°26.5	6.2'	25°13.5	8.3'	58.6'								
12	356°29.2	S15°27.1	256°51.7	6.2'	N25°21.8	8.1'	58.5'								
13	11°29.1	26.3	271°16.9	6.1'	25°29.9	8.0'	58.5'								
14	26°29.1	25.5	285°42.0	6.1'	25°37.9	7.8'	58.5'								
15	41°29.1	· · 24.7	300°07.1	6.0'	25°45.7	7.6'	58.5'								
16	56°29.0	24.0	314°32.1	5.9'	25°53.3	7.5'	58.5'								
17	71°29.0	23.2	328°57.0	5.9'	26°00.8	7.3'	58.5'								
18	86°29.0	S15°22.4	343°21.9	5.8'	N26°08.1	7.1'	58.4'								
19	101°28.9	21.6	357°46.7	5.8'	26°15.3	7.0'	58.4'								
20	116°28.9	20.8	12°11.5	5.7'	26°22.2	6.8'	58.4'								
21	131°28.9	· · 20.1	26°36.2	5.6'	26°29.0	6.6'	58.4'								
22	146°28.8	19.3	41°00.8	5.6'	26°35.7	6.5'	58.4'								
23	161°28.8	18.5	55°25.4	5.5'	26°42.1	6.3'	58.4'								
SD = 16.2' d = -0.8'								SD = 16.0'							
Fri	GHA	Dec	GHA	ν	Dec	d	HP								
0	176°28.7	S15°17.7	69°49.9	5.5'	N26°48.4	6.1'	58.3'								
1	191°28.7	16.9	84°14.4	5.4'	26°54.6	5.9'	58.3'								
2	206°28.7	16.2	98°38.9	5.4'	27°00.5	5.8'	58.3'								
3	221°28.7	· · 15.4	113°03.3	5.4'	27°06.3	5.6'	58.3'								
4	236°28.6	14.6	127°27.6	5.3'	27°11.9	5.4'	58.3'								
5	251°28.6	13.8	141°52.0	5.3'	27°17.3	5.3'	58.3'								
6	266°28.6	S15°13.0	156°16.2	5.2'	N27°22.6	5.1'	58.2'								
7	281°28.5	12.2	170°40.5	5.2'	27°27.6	4.9'	58.2'								
8	296°28.5	11.4	185°04.6	5.2'	27°32.5	4.7'	58.2'								
9	311°28.5	· · 10.7	199°28.8	5.1'	27°37.3	4.5'	58.2'								
10	326°28.4	09.9	213°52.9	5.1'	27°41.8	4.4'	58.2'								
11	341°28.4	09.1	228°17.0	5.1'	27°46.2	4.2'	58.2'								
12	356°28.4	S15°08.3	242°41.1	5.0'	N27°50.3	4.0'	58.1'								
13	11°28.3	07.5	257°05.1	5.0'	27°54.3	3.8'	58.1'								
14	26°28.3	06.7	271°29.1	5.0'	27°58.2	3.6'	58.1'								
15	41°28.3	· · 05.9	285°53.1	5.0'	28°01.8	3.5'	58.1'								
16	56°28.3	05.2	300°17.1	4.9'	28°05.3	3.3'	58.1'								
17	71°28.2	04.4	314°41.0	4.9'	28°08.6	3.1'	58.1'								
18	86°28.2	S15°03.6	329°04.9	4.9'	N28°11.7	2.9'	58.0'								
19	101°28.2	02.8	343°28.9	4.9'	28°14.6	2.7'	58.0'								
20	116°28.2	02.0	357°52.7	4.9'	28°17.3	2.6'	58.0'								
21	131°28.1	· · 01.2	12°16.6	4.9'	28°19.9	2.4'	58.0'								
22	146°28.1	15°00.4	26°40.5	4.9'	28°22.2	2.2'	58.0'								
23	161°28.1	14°59.6	41°04.4	4.9'	28°24.4	2.0'	58.0'								
SD = 16.2' d = -0.8'								SD = 15.9'							
Sat	GHA	Dec	GHA	ν	Dec	d	HP								
0	176°28.1	S14°58.8	55°28.2	4.9'	N28°26.4	1.8'	57.9'								
1	191°28.0	58.0	69°52.1	4.9'	28°28.3	1.6'	57.9'								
2	206°28.0	57.2	84°16.0	4.9'	28°29.9	1.5'	57.9'								
3	221°28.0	· · 56.4	98°39.8	4.9'	28°31.4	1.3'	57.9'								
4	236°28.0	55.7	113°03.7	4.9'	28°32.6	1.1'	57.9'								
5	251°27.9	54.9	127°27.6	4.9'	28°33.7	0.9'	57.8'								
6	266°27.9	S14°54.1	141°51.5	4.9'	N28°34.7	0.7'	57.8'								
7	281°27.9	53.3	156°15.4	4.9'	28°35.4	0.5'	57.8'								
8	296°27.9	52.5	170°39.3	4.9'	28°35.9	0.4'	57.8'								
9	311°27.8	· · 51.7	185°03.2	4.9'	28°36.3	0.2'	57.8'								
10	326°27.8	50.9	199°27.1	5.0'	28°36.5	0.0'	57.8'								
11	341°27.8	50.1	213°51.1	5.0'	28°36.5	-0.2'	57.7'								
12	356°27.8	S14°49.3	228°15.1	5.0'	N28°36.3	-0.4'	57.7'								
13	11°27.8	48.5	242°39.1	5.0'	28°35.9	-0.5'	57.7'								
14	26°27.7	47.7	257°03.1	5.1'	28°35.4	-0.7'	57.7'								
15	41°27.7	· · 46.9	271°27.2	5.1'	28°34.7	-0.9'	57.7'								
16	56°27.7	46.1	285°51.3	5.1'	28°33.8	-1.1'	57.6'								
17	71°27.7	45.3	300°15.4	5.2'	28°32.7	-1.3'	57.6'								
18	86°27.7	S14°44.5	314°39.6	5.2'	N28°31.4	-1.4'	57.6'								
19	101°27.6	43.7	329°03.8	5.2'	28°30.0	-1.6'	57.6'								
20	116°27.6	42.9	343°28.0	5.3'	28°28.4	-1.8'	57.6'								
21	131°27.6	· · 42.1	357°52.3	5.3'	28°26.6	-2.0'	57.5'								
22	146°27.6	41.3	12°16.7	5.4'	28°24.6	-2.1'	57.5'								
23	161°27.6	40.5	26°41.0	5.4'	28°22.5	-2.3'	57.5'								
SD = 16.2' d = -0.8'								SD = 15.8'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:46	08:10	09:41	14:48	16:20	17:44
N 70°	06:41	07:56	09:12	15:17	16:34	17:49
68°	06:37	07:44	08:51	15:39	16:45	17:53
66°	06:33	07:35	08:34	15:56	16:55	17:57
64°	06:29	07:26	08:20	16:09	17:03	18:00
62°	06:26	07:19	08:08	16:21	17:10	18:03
60°	06:23	07:13	07:59	16:31	17:16	18:06
N 58°	06:21	07:07	07:50	16:39	17:22	18:09
56°	06:18	07:02	07:42	16:47	17:27	18:11
54°	06:16	06:58	07:36	16:53	17:31	18:13
52°	06:14	06:54	07:30	16:59	17:35	18:15
50°	06:12	06:50	07:24	17:05	17:39	18:17
45°	06:07	06:41	07:12	17:17	17:47	18:22
N 40°	06:02	06:34	07:02	17:26	17:55	18:27
35°	05:57	06:27	06:54	17:35	18:01	18:31
30°	05:53	06:21	06:46	17:42	18:07	18:36
20°	05:44	06:10	06:33	17:55	18:18	18:44
N 10°	05:35	06:00	06:22	18:07	18:28	18:54
0°	05:24	05:49	06:11	18:18	18:39	19:04
S 10°	05:12	05:38	06:00	18:28	18:50	19:16
20°	04:57	05:25	05:48	18:40	19:03	19:30
30°	04:38	05:09	05:34	18:54	19:19	19:49
35°	04:26	04:59	05:26	19:02	19:29	20:01
40°	04:11	04:47	05:17	19:11	19:40	20:16
45°						

February 09, 10, 11 UT (Sun., Mon., Tue.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	139°20.4	137°37.7	N04°13.1	28°41.9	N26°13.7	69°30.4	N21°40.6	149°19.9	S06°23.1					Alpheratz	357°35.0	29°13.8	
1	154°22.9	152°38.8	14.1	43°44.9	13.7	84°32.8	40.6	164°22.1	23.0					Ankaa	353°07.3	-42°10.4	
2	169°25.3	167°39.9	15.1	58°48.0	13.7	99°35.2	40.6	179°24.3	22.9					Schedar	349°31.3	56°40.7	
3	184°27.8	182°41.1	•• 16.2	73°51.0	•• 13.8	114°37.6	•• 40.6	194°26.5	•• 22.8					Diphda	348°47.4	-17°51.1	
4	199°30.3	197°42.2	17.2	88°54.0	13.8	129°40.1	40.6	209°28.7	22.7					Achernar	335°20.3	-57°06.8	
5	214°32.7	212°43.4	18.2	103°57.0	13.8	144°42.5	40.7	224°30.9	22.6					Hamal	327°51.2	23°34.9	
6	229°35.2	227°44.5	N04°19.2	119°00.0	N26°13.8	159°44.9	N21°40.7	239°33.1	S06°22.5					Polaris	314°01.1	89°22.5	
7	244°37.7	242°45.7	20.3	134°03.0	13.8	174°47.3	40.7	254°35.3	22.3					Acamar	315°11.7	-40°12.5	
8	259°40.1	257°46.8	21.3	149°06.0	13.8	189°49.8	40.7	269°37.5	22.2					Menkar	314°06.1	4°11.2	
9	274°42.6	272°48.0	•• 22.3	164°09.0	•• 13.8	204°52.2	•• 40.7	284°39.7	•• 22.1					Mirfak	308°28.1	49°57.2	
10	289°45.1	287°49.1	23.3	179°12.0	13.8	219°54.6	40.7	299°41.9	22.0					Aldebaran	290°39.4	16°33.6	
11	304°47.5	302°50.3	24.3	194°15.0	13.8	234°57.0	40.7	314°44.1	21.9					Rigel	281°03.6	-8°10.5	
12	319°50.0	317°51.4	N04°25.3	209°18.0	N26°13.8	249°59.4	N21°40.8	329°46.3	S06°21.8					Capella	280°21.5	46°01.5	
13	334°52.4	332°52.6	26.4	224°21.0	13.9	265°01.9	40.8	344°48.5	21.7					Bellatrix	278°22.6	6°22.3	
14	349°54.9	347°53.8	27.4	239°24.0	13.9	280°04.3	40.8	359°50.7	21.6					Elnath	278°01.5	28°37.8	
15	4°57.4	2°54.9	•• 28.4	254°27.0	•• 13.9	295°06.7	•• 40.8	14°52.9	•• 21.4					Anilam	275°37.4	-1°11.2	
16	19°59.8	17°56.1	29.4	269°30.0	13.9	310°09.1	40.8	29°55.1	21.3					Betelgeuse	270°51.8	7°24.7	
17	35°02.3	32°57.3	30.4	284°33.0	13.9	325°11.5	40.8	44°57.3	21.2					Canopus	263°52.0	-52°42.7	
18	50°04.8	47°58.5	N04°31.4	299°35.9	N26°13.9	340°13.9	N21°40.8	59°59.5	S06°21.1					Sirius	258°25.9	-16°45.2	
19	65°07.2	62°59.6	32.4	314°38.9	13.9	355°16.4	40.9	75°01.7	21.0					Adhara	255°05.5	-29°00.5	
20	80°09.7	78°00.8	33.4	329°41.9	13.9	10°18.8	40.9	90°03.9	20.9					Procyon	244°50.4	5°09.6	
21	95°12.2	93°02.0	•• 34.5	344°44.9	•• 13.9	25°21.2	•• 40.9	105°06.1	•• 20.8					Pollux	243°16.9	27°57.9	
22	110°14.6	108°03.2	35.5	359°47.9	13.9	40°23.6	40.9	120°08.3	20.6					Avior	234°14.1	-59°35.5	
23	125°17.1	123°04.4	36.5	14°50.9	13.9	55°26.0	40.9	135°10.5	20.5					Suhail	222°45.8	-43°32.1	
Mer.pass. 14:40			ν 1.1' d 1.0' m -4.82	ν 3.0' d 0.0' m -0.84	ν 2.4' d 0.0' m -2.46	ν 2.2' d -0.1' m 1.12											

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	140°19.6	138°05.5	N04°37.5	29°53.8	N26°13.9	70°28.4	N21°40.9	150°12.7	S06°20.4
1	155°22.0	153°06.7	38.5	44°56.8	13.9	85°30.9	40.9	165°14.9	20.3
2	170°24.5	168°07.9	39.5	59°59.8	13.9	100°33.3	41.0	180°17.1	20.2
3	185°26.9	183°09.1	•• 40.5	75°02.8	•• 13.9	115°35.7	•• 41.0	195°19.3	•• 20.1
4	200°29.4	198°10.3	41.5	90°05.7	13.9	130°38.1	41.0	210°21.5	20.0
5	215°31.9	213°11.5	42.5	105°08.7	13.9	145°40.5	41.0	225°23.7	19.8
6	230°34.3	228°12.7	N04°43.5	120°11.7	N26°13.9	160°42.9	N21°41.0	240°25.9	S06°19.7
7	245°36.8	243°13.9	44.5	135°14.7	14.0	175°45.3	41.0	255°28.1	19.6
8	260°39.3	258°15.1	45.5	150°17.6	14.0	190°47.8	41.1	270°30.3	19.5
9	275°41.7	273°16.3	•• 46.5	165°20.6	•• 14.0	205°50.2	•• 41.1	285°32.5	•• 19.4
10	290°44.2	288°17.5	47.5	180°23.6	14.0	220°52.6	41.1	300°34.7	19.3
11	305°46.7	303°18.7	48.5	195°26.5	14.0	235°55.0	41.1	315°36.9	19.2
12	320°49.1	318°19.9	N04°49.5	210°29.5	N26°14.0	250°57.4	N21°41.1	330°39.1	S06°19.0
13	335°51.6	333°21.2	50.5	225°32.4	14.0	265°59.8	41.1	345°41.3	18.9
14	350°54.1	348°22.4	51.5	240°35.4	14.0	281°02.2	41.2	0°43.5	18.8
15	5°56.5	3°23.6	•• 52.5	255°38.3	•• 14.0	296°04.6	•• 41.2	15°45.7	•• 18.7
16	20°59.0	18°24.8	53.5	270°41.3	14.0	311°07.0	41.2	30°47.9	18.6
17	36°01.4	33°26.0	54.5	285°44.3	14.0	326°09.4	41.2	45°50.1	18.5
18	51°03.9	48°27.3	N04°55.5	300°47.2	N26°14.0	341°11.9	N21°41.2	60°52.3	S06°18.4
19	66°06.4	63°28.5	56.5	315°50.2	14.0	356°14.3	41.2	75°54.5	18.2
20	81°08.8	78°29.7	57.5	330°53.1	14.0	11°16.7	41.2	90°56.7	18.1
21	96°11.3	93°31.0	•• 58.5	345°56.1	•• 14.0	26°19.1	•• 41.3	105°58.9	•• 18.0
22	111°13.8	108°32.2	04°59.5	0°59.0	14.0	41°21.5	41.3	121°01.1	17.9
23	126°16.2	123°33.4	05°00.4	16°01.9	14.0	56°23.9	41.3	136°03.3	17.8
Mer.pass. 14:36			ν 1.2' d 1.0' m -4.82	ν 3.0' d 0.0' m -0.81	ν 2.4' d 0.0' m -2.45	ν 2.2' d -0.1' m 1.12			

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	141°18.7	138°34.7	N05°01.4	31°04.9	N26°14.0	71°26.3	N21°41.3	151°05.5	S06°17.7
1	156°21.2	153°35.9	02.4	46°07.8	14.0	86°28.7	41.3	166°07.7	17.6
2	171°23.6	168°37.2	03.4	61°10.8	14.0	101°31.1	41.3	181°09.9	17.4
3	186°26.1	183°38.4	•• 04.4	76°13.7	•• 13.9	116°33.5	•• 41.4	196°12.1	•• 17.3
4	201°28.6	198°39.7	05.4	91°16.6	13.9	131°35.9	41.4	211°14.3	17.2
5	216°31.0	213°40.9	06.4	106°19.6	13.9	146°38.3	41.4	226°16.5	17.1
6	231°33.5	228°42.2	N05°07.4	121°22.5	N26°13.9	161°40.7	N21°41.4	241°18.7	S06°17.0
7	246°35.9	243°43.4	08.3	136°25.5	13.9	176°43.1	41.4	256°20.9	16.9
8	261°38.4	258°44.7	09.3	151°28.4	13.9	191°45.5	41.4	271°23.1	16.8
9	276°40.9	273°45.9	•• 10.3	166°31.3	•• 13.9	206°47.9	•• 41.5	286°25.3	•• 16.6
10	291°43.3	288°47.2	11.3	181°34.2	13.9	221°50.3	41.5	301°27.5	16.5
11	306°45.8	303°48.4	12.3	196°37.2	13.9	236°52.7	41.5	316°29.7	16.4
12	321°48.3	318°49.7	N05°13.3	211°40.1	N26°13.9	251°55.1	N21°41.5	331°31.9	S06°16.3
13	336°50.7	333°51.0	14.2	226°43.0	13.9	266°57.5	41.5	346°34.1	16.2
14	351°53.2	348°52.3	15.2	241°45.9	13.9	281°59.9	41.5	1°36.3	16.1
15	6°55.7	3°53.5	•• 16.2	256°48.9	•• 13.9	297°02.3	•• 41.6	16°38.5	•• 16.0
16	21°58.1	18°54.8	17.2	271°51.8	13.9	312°04.7	41.6	31°40.6	15.8
17	37°00.6	33°56.1	18.1	286°54.7	13.9	327°07.1	41.6	46°42.8	15.7
18	52°03.0	48°57.4	N05°19.1	301°57.6	N26°13.9	342°09.5	N21°41.6	61°45.0	S06°15.6
19	67°05.5	63°58.6	20.1	317°00.5	13.9	357°11.9	41.6	76°47.2	15.5
20	82°08.0	78°59.9	21.1	332°03.4	13.9	12°14.3	41.6	91°49.4	15.4
21	97°10.4	94°01.2	•• 22.0	347°06.3	•• 13.8	27°16.7	•• 41.7	106°51.6	•• 15.3
22	112°12.9	109°02.5	23.0	2°09.3	13.8	42°19.1	41.7	121°53.8	15.2
23	127°15.4	124°03.8	24.0	17°12.2	13.8	57°21.5	41.7	136°56.0	15.0
Mer.pass. 14:32			ν 1.2' d 1.0' m -4.83	ν 2.9' d -0.0' m -0.79	ν 2.4' d 0.0' m -2.44	ν 2.2' d -0.1' m 1.12			

Feb 09 Sun	SHA	Mer.pass
Venus	358°17.2	14:48
Mars	249°21.5	22:01
Jupiter	290°10.0	19:19
Saturn	9°59.5	14:01

Feb 10 Mon	SHA	Mer.pass
Venus	357°46.0	14:46
Mars	249°34.3	21:56
Jupiter	290°08.9	19:15
Saturn	9°53.1	13:57

Feb 11 Tue	SHA	Mer.pass
Venus	357°16.0	14:44
Mars	249°46.2	21:51
Jupiter	290°07.6	19:11
Saturn	9°46.8	13:54

Horizontal parallax		
Venus:	0.3	
Mars:	0.2	

h	Sun			Moon			
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	176°27.6	S14°39.7	41°05.5	5.5'	N28°20.1	-2.5'	57.5'
1	191°27.5	38.9	55°29.9	5.5'	28°17.6	-2.7'	57.5'
2	206°27.5	38.1	69°54.5	5.6'	28°15.0	-2.8'	57.5'
3	221°27.5	· · 37.3	84°19.0	5.6'	28°12.1	-3.0'	57.4'
4	236°27.5	36.5	98°43.7	5.7'	28°09.1	-3.2'	57.4'
5	251°27.5	35.7	113°08.4	5.8'	28°05.9	-3.4'	57.4'
6	266°27.5	S14°34.9	127°33.1	5.8'	N28°02.6	-3.5'	57.4'
7	281°27.5	34.1	141°58.0	5.9'	27°59.1	-3.7'	57.4'
8	296°27.4	33.3	156°22.9	6.0'	27°55.4	-3.9'	57.3'
9	311°27.4	· · 32.4	170°47.8	6.0'	27°51.5	-4.0'	57.3'
10	326°27.4	31.6	185°12.8	6.1'	27°47.5	-4.2'	57.3'
11	341°27.4	30.8	199°37.9	6.2'	27°43.3	-4.4'	57.3'
12	356°27.4	S14°30.0	214°03.1	6.2'	N27°38.9	-4.5'	57.3'
13	11°27.4	29.2	228°28.3	6.3'	27°34.4	-4.7'	57.2'
14	26°27.4	28.4	242°53.6	6.4'	27°29.7	-4.8'	57.2'
15	41°27.4	· · 27.6	257°19.0	6.5'	27°24.9	-5.0'	57.2'
16	56°27.3	26.8	271°44.5	6.5'	27°19.9	-5.2'	57.2'
17	71°27.3	26.0	286°10.0	6.6'	27°14.7	-5.3'	57.2'
18	86°27.3	S14°25.2	300°35.6	6.7'	N27°09.4	-5.5'	57.1'
19	101°27.3	24.4	315°01.3	6.8'	27°03.9	-5.6'	57.1'
20	116°27.3	23.6	329°27.1	6.9'	26°58.2	-5.8'	57.1'
21	131°27.3	· · 22.7	343°53.0	7.0'	26°52.5	-5.9'	57.1'
22	146°27.3	21.9	358°18.9	7.0'	26°46.5	-6.1'	57.1'
23	161°27.3	21.1	12°45.0	7.1'	26°40.4	-6.2'	57.0'
SD = 16.2' $d = -0.8'$			SD = 15.7'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	176°27.3	S14°20.3	27°11.1	7.2'	N26°34.2	-6.4'	57.0'
1	191°27.3	19.5	41°37.3	7.3'	26°27.8	-6.5'	57.0'
2	206°27.2	18.7	56°03.6	7.4'	26°21.2	-6.7'	57.0'
3	221°27.2	· · 17.9	70°30.0	7.5'	26°14.5	-6.8'	57.0'
4	236°27.2	17.1	84°56.5	7.6'	26°07.7	-7.0'	57.0'
5	251°27.2	16.2	99°23.1	7.7'	26°00.7	-7.1'	56.9'
6	266°27.2	S14°15.4	113°49.8	7.8'	N25°53.6	-7.3'	56.9'
7	281°27.2	14.6	128°16.5	7.9'	25°46.3	-7.4'	56.9'
8	296°27.2	13.8	142°43.4	8.0'	25°38.9	-7.5'	56.9'
9	311°27.2	· · 13.0	157°10.4	8.1'	25°31.4	-7.7'	56.9'
10	326°27.2	12.2	171°37.5	8.2'	25°23.7	-7.8'	56.8'
11	341°27.2	11.3	186°04.6	8.3'	25°15.9	-8.0'	56.8'
12	356°27.2	S14°10.5	200°31.9	8.4'	N25°07.9	-8.1'	56.8'
13	11°27.2	09.7	214°59.2	8.5'	24°59.8	-8.2'	56.8'
14	26°27.2	08.9	229°26.7	8.6'	24°51.6	-8.4'	56.8'
15	41°27.2	· · 08.1	243°54.3	8.7'	24°43.2	-8.5'	56.7'
16	56°27.2	07.3	258°21.9	8.8'	24°34.8	-8.6'	56.7'
17	71°27.2	06.4	272°49.7	8.9'	24°26.2	-8.7'	56.7'
18	86°27.2	S14°05.6	287°17.6	9.0'	N24°17.4	-8.9'	56.7'
19	101°27.2	04.8	301°45.5	9.1'	24°08.6	-9.0'	56.7'
20	116°27.2	04.0	316°13.6	9.2'	23°59.6	-9.1'	56.6'
21	131°27.2	· · 03.1	330°41.8	9.3'	23°50.5	-9.2'	56.6'
22	146°27.2	02.3	345°10.1	9.4'	23°41.3	-9.3'	56.6'
23	161°27.2	01.5	359°38.5	9.5'	23°31.9	-9.5'	56.6'
SD = 16.2' $d = -0.8'$			SD = 15.6'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	176°27.2	S14°00.7	14°06.9	9.6'	N23°22.5	-9.6'	56.5'
1	191°27.2	13°59.9	28°35.5	9.7'	23°12.9	-9.7'	56.5'
2	206°27.2	59.0	43°04.2	9.8'	23°03.2	-9.8'	56.5'
3	221°27.2	· · 58.2	57°33.0	9.9'	22°53.4	-9.9'	56.5'
4	236°27.2	57.4	72°01.9	10.0'	22°43.5	-10.0'	56.5'
5	251°27.2	56.6	86°31.0	10.1'	22°33.5	-10.1'	56.4'
6	266°27.2	S13°55.7	101°00.1	10.2'	N22°23.3	-10.2'	56.4'
7	281°27.2	54.9	115°29.3	10.3'	22°13.1	-10.3'	56.4'
8	296°27.2	54.1	129°58.6	10.4'	22°02.7	-10.4'	56.4'
9	311°27.2	· · 53.3	144°28.0	10.5'	21°52.3	-10.6'	56.4'
10	326°27.2	52.4	158°57.6	10.6'	21°41.7	-10.7'	56.3'
11	341°27.2	51.6	173°27.2	10.7'	21°31.1	-10.8'	56.3'
12	356°27.2	S13°50.8	187°56.9	10.8'	N21°20.3	-10.8'	56.3'
13	11°27.2	50.0	202°26.8	10.9'	21°09.5	-10.9'	56.3'
14	26°27.2	49.1	216°56.7	11.0'	20°58.5	-11.0'	56.3'
15	41°27.2	· · 48.3	231°26.7	11.1'	20°47.5	-11.1'	56.2'
16	56°27.2	47.5	245°56.9	11.2'	20°36.4	-11.2'	56.2'
17	71°27.2	46.6	260°27.1	11.3'	20°25.1	-11.3'	56.2'
18	86°27.2	S13°45.8	274°57.5	11.4'	N20°13.8	-11.4'	56.2'
19	101°27.2	45.0	289°27.9	11.5'	20°02.4	-11.5'	56.2'
20	116°27.2	44.2	303°58.5	11.6'	19°50.9	-11.6'	56.1'
21	131°27.2	· · 43.3	318°29.1	11.7'	19°39.3	-11.7'	56.1'
22	146°27.2	42.5	332°59.9	11.8'	19°27.7	-11.8'	56.1'
23	161°27.2	41.7	347°30.7	11.9'	19°15.9	-11.8'	56.1'
SD = 16.2' $d = -0.8'$			SD = 15.4'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:34	07:56	09:22	15:08	16:34	17:56
N 70°	06:30	07:44	08:57	15:33	16:46	18:00
68°	06:27	07:34	08:38	15:52	16:56	18:03
66°	06:24	07:25	08:23	16:07	17:04	18:06
64°	06:21	07:18	08:10	16:19	17:12	18:08
62°	06:19	07:12	08:00	16:30	17:18	18:11
60°	06:17	07:06	07:51	16:38	17:23	18:13
N 58°	06:15	07:01	07:43	16:46	17:28	18:15
56°	06:13	06:56	07:36	16:53	17:33	18:17
54°	06:11	06:52	07:30	16:59	17:37	18:19
52°	06:09	06:49	07:24	17:05	17:40	18:20
50°	06:07	06:45	07:19	17:10	17:44	18:22
45°	06:03	06:37	07:08	17:21	17:52	18:26
N 40°	05:59	06:31	06:59	17:30	17:58	18:30
35°	05:55	06:25	06:51	17:38	18:04	18:34
30°	05:51	06:19	06:44	17:45	18:09	18:38
20°	05:43	06:09	06:32	17:57	18:20	18:46
N 10°	05:34	05:59	06:21	18:08	18:29	18:54
0°	05:25	05:49	06:11	18:18	18:39	19:04
S 10°	05:13	05:39	06:00	18:28	18:50	19:15
20°	04:59	05:26	05:49	18:39	19:02	19:29
30°	04:41	05:11	05:36	18:52	19:17	19:47
35°	04:30	05:02	05:29	18:59	19:26	19:58
40°	04:16	04:51	05:20	19:07	19:36	20:12
45°	03:59	04:38	05:10	19:17	19:49	20:29
S 50°	03:36	04:22	04:58	19:29	20:05	20:51
52°	03:25	04:14	04:52	19:35	20:13	21:02
54°	03:11	04:05	04:46	19:41	20:22	21:15
56°	02:56	03:55	04:39	19:48	20:31	21:30
58°	02:37	03:44	04:31	19:56	20:43	21:48
S 60°	02:12	03:30	04:22	20:05	20:56	22:12

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	12:38	□	□	10:33
66°	□	□	13:34	□	□	09:36
64°	□	11:44	14:07	□	09:37	09:02
62°	10:20	12:35	14:31	09:02	08:44	08:37
60°	11:24	13:07	14:50	07:58	08:12	08:17
N 58°	11:58	13:30	15:06	07:24	07:48	08:00
56°	12:23	13:49	15:19	06:58	07:29	07:46
54°	12:43	14:05	15:31	06:38	07:12	07:34
52°	13:00	14:19	15:41	06:21	06:58	07:24
50°	13:14	14:31	15:50	06:06	06:46	07:14
45°	13:43	14:55	16:09	05:37	06:20	06:54
N 40°	14:06	15:15	16:24	05:14	06:00	06:37
35°	14:25	15:31	16:37	04:54	05:43	06:23
30°	14:41	15:45	16:49	04:38	05:28	06:11
20°	15:08	16:09	17:08	04:10	05:03	05:50
N 10°	15:32	16:30	17:25	03:46	04:41	05:31
0°	15:53	16:49	17:40	03:23	04:20	05:14
S 10°	16:15	17:08	17:55	03:00	04:00	04:56
20°	16:38	17:28	18:12	02:36	03:38	04:38
30°	17:05	17:52	18:31	02:08	03:12	04:16
35°	17:21	18:05	18:42	01:51	02:56	04:03
40°	17:40	18:21	18:54	01:31	02:38	03:48
45°	18:02	18:40	19:09	01:07	02:17	03:31
S 50°	18:30	19:03	19:26	00:36	01:49	03:08
52°	18:44	19:14	19:35	00:21	01:36	02:58
54°	19:00	19:27	19:44	00:03	01:20	02:46
56°	19:19	19:41	19:54	· · ·	01:02	02:32
58°	19:42	19:58	20:06	· · ·	00:39	02:16
S 60°	20:12	20:19	20:20	· · ·	00:10	01:56

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss	hh:mm	Mer. Pass. Upper hh:mm	Lower hh:mm	11-13 86-97%
09	14:10	14:10	12:14	22:07	09:38	
10	14:11	14:11	12:14	23:01	10:35	
11	14:11	14:11	12:14	23:52	11:27	

February 12, 13, 14 UT (Wed., Thu., Fri.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	142°17.8	139°05.1	N05°25.0	32°15.1	N26°13.8	72°23.9	N21°41.7	151°58.2	S06°14.9	Alpheratz	357°35.0	29°13.8											
1	157°20.3	154°06.4	25.9	47°18.0	13.8	87°26.3	41.7	167°00.4	14.8	Ankaa	353°07.3	-42°10.4											
2	172°22.8	169°07.7	26.9	62°20.9	13.8	102°28.7	41.8	182°02.6	14.7	Schedar	349°31.4	56°40.7											
3	187°25.2	184°09.0	•• 27.9	77°23.8	•• 13.8	117°31.1	•• 41.8	197°04.8	•• 14.6	Diphda	348°47.4	-17°51.1											
4	202°27.7	199°10.3	28.8	92°26.7	13.8	132°33.5	41.8	212°07.0	14.5	Achernar	335°20.3	-57°06.8											
5	217°30.2	214°11.6	29.8	107°29.6	13.8	147°35.9	41.8	227°09.2	14.3	Hamal	327°51.2	23°34.9											
6	232°32.6	229°12.9	N05°30.8	122°32.5	N26°13.8	162°38.3	N21°41.8	242°11.4	S06°14.2	Polaris	314°02.6	89°22.5											
7	247°35.1	244°14.2	31.7	137°35.4	13.8	177°40.7	41.8	257°13.6	14.1	Acamar	315°11.7	-40°12.5											
8	262°37.5	259°15.5	32.7	152°38.3	13.7	192°43.1	41.9	272°15.8	14.0	Menkar	314°06.1	4°11.2											
9	277°40.0	274°16.8	•• 33.7	167°41.2	•• 13.7	207°45.5	•• 41.9	287°18.0	•• 13.9	Mirfak	308°28.1	49°57.2											
10	292°42.5	289°18.1	34.6	182°44.1	13.7	222°47.9	41.9	302°20.2	13.8	Aldebaran	290°39.4	16°33.6											
11	307°44.9	304°19.5	35.6	197°46.9	13.7	237°50.3	41.9	317°22.4	13.7	Rigel	281°03.6	-8°10.5											
12	322°47.4	319°20.8	N05°36.5	212°49.8	N26°13.7	252°52.7	N21°41.9	332°24.6	S06°13.5	Capella	280°21.5	46°01.5											
13	337°49.9	334°22.1	37.5	227°52.7	13.7	267°55.1	41.9	347°26.8	13.4	Bellatrix	278°22.6	6°22.3											
14	352°52.3	349°23.4	38.5	242°55.6	13.7	282°57.5	42.0	2°29.0	13.3	Elnath	278°01.6	28°37.8											
15	7°54.8	4°24.8	•• 39.4	257°58.5	•• 13.7	297°59.9	•• 42.0	17°31.2	•• 13.2	Alnilam	275°37.5	-1°11.2											
16	22°57.3	19°26.1	40.4	273°01.4	13.6	313°02.2	42.0	32°33.4	13.1	Betelgeuse	270°51.8	7°24.7											
17	37°59.7	34°27.4	41.3	288°04.3	13.6	328°04.6	42.0	47°35.6	13.0	Canopus	263°52.0	-52°42.7											
18	53°02.2	49°28.8	N05°42.3	303°07.1	N26°13.6	343°07.0	N21°42.0	62°37.8	S06°12.9	Sirius	258°25.9	-16°45.2											
19	68°04.7	64°30.1	43.2	318°10.0	13.6	358°09.4	42.1	77°40.0	12.7	Adhara	255°05.5	-29°00.5											
20	83°07.1	79°31.4	44.2	333°12.9	13.6	13°11.8	42.1	92°42.2	12.6	Procyon	244°50.4	5°09.6											
21	98°09.6	94°32.8	•• 45.1	348°15.8	•• 13.6	28°14.2	•• 42.1	107°44.4	•• 12.5	Pollux	243°16.9	27°57.9											
22	113°12.0	109°34.1	46.1	3°18.6	13.6	43°16.6	42.1	122°46.6	12.4	Avior	234°14.1	-59°35.5											
23	128°14.5	124°35.5	47.0	18°21.5	13.5	58°19.0	42.1	137°48.8	12.3	Suhail	222°45.8	-43°32.1											
Mer.pass. 14:28				ν 1.3' d 1.0' m -4.84				ν 2.9' d -0.0' m -0.76				ν 2.4' d 0.0' m -2.43				ν 2.2' d -0.1' m 1.12							

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	143°17.0	139°36.8	N05°48.0	33°24.4	N26°13.5	73°21.4	N21°42.1	152°50.9	S06°12.2										
1	158°19.4	154°38.2	48.9	48°27.3	13.5	88°23.7	42.2	167°53.1	12.0										
2	173°21.9	169°39.5	49.9	63°30.1	13.5	103°26.1	42.2	182°55.3	11.9										
3	188°24.4	184°40.9	•• 50.8	78°33.0	•• 13.5	118°28.5	•• 42.2	197°57.5	•• 11.8										
4	203°26.8	199°42.3	51.8	93°35.8	13.5	133°30.9	42.2	212°59.7	11.7										
5	218°29.3	214°43.6	52.7	108°38.7	13.4	148°33.3	42.2	228°01.9	11.6										
6	233°31.8	229°45.0	N05°53.7	123°41.6	N26°13.4	163°35.7	N21°42.3	243°04.1	S06°11.5										
7	248°34.2	244°46.3	54.6	138°44.4	13.4	178°38.1	42.3	258°06.3	11.4										
8	263°36.7	259°47.7	55.6	153°47.3	13.4	193°40.5	42.3	273°08.5	11.2										
9	278°39.1	274°49.1	•• 56.5	168°50.1	•• 13.4	208°42.8	•• 42.3	288°10.7	•• 11.1										
10	293°41.6	289°50.5	57.5	183°53.0	13.4	223°45.2	42.3	303°12.9	11.0										
11	308°44.1	304°51.8	58.4	198°55.9	13.3	238°47.6	42.4	318°15.1	10.9										
12	323°46.5	319°53.2	N05°59.3	213°58.7	N26°13.3	253°50.0	N21°42.4	333°17.3	S06°10.8										
13	338°49.0	334°54.6	06°00.3	229°01.6	13.3	268°52.4	42.4	348°19.5	10.7										
14	353°51.5	349°56.0	01.2	244°04.4	13.3	283°54.8	42.4	3°21.7	10.5										
15	8°53.9	4°57.4	•• 02.2	259°07.3	•• 13.3	298°57.1	•• 42.4	18°23.9	•• 10.4										
16	23°56.4	19°58.8	03.1	274°10.1	13.2	313°59.5	42.4	33°26.1	10.3										
17	38°58.9	35°00.1	04.0	289°12.9	13.2	329°01.9	42.5	48°28.3	10.2										
18	54°01.3	50°01.5	N06°05.0	304°15.8	N26°13.2	344°04.3	N21°42.5	63°30.5	S06°10.1										
19	69°03.8	65°02.9	05.9	319°18.6	13.2	359°06.7	42.5	78°32.7	10.0										
20	84°06.3	80°04.3	06.8	334°21.5	13.2	14°09.1	42.5	93°34.9	09.9										
21	99°08.7	95°05.7	•• 07.8	349°24.3	•• 13.1	29°11.4	•• 42.5	108°37.1	•• 09.7										
22	114°11.2	110°07.1	08.7	4°27.1	13.1	44°13.8	42.6	123°39.2	09.6										
23	129°13.6	125°08.5	09.6	19°30.0	13.1	59°16.2	42.6	138°41.4	09.5										
Mer.pass. 14:25				ν 1.4' d 0.9' m -4.84				ν 2.9' d -0.0' m -0.73				ν 2.4' d 0.0' m -2.42				ν 2.2' d -0.1' m 1.12			

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	144°16.1	140°09.9	N06°10.5	34°32.8	N26°13.1	74°18.6	N21°42.6	153°43.6	S06°09.4										
1	159°18.6	155°11.4	11.5	49°35.7	13.1	89°21.0	42.6	168°45.8	09.3										
2	174°21.0	170°12.8	12.4	64°38.5	13.0	104°23.3	42.6	183°48.0	09.2										
3	189°23.5	185°14.2	•• 13.3	79°41.3	•• 13.0	119°25.7	•• 42.7	198°50.2	•• 09.0										
4	204°26.0	200°15.6	14.3	94°44.1	13.0	134°28.1	42.7	213°52.4	08.9										
5	219°28.4	215°17.0	15.2	109°47.0	13.0	149°30.5	42.7	228°54.6	08.8										
6	234°30.9	230°18.4	N06°16.1	124°49.8	N26°12.9	164°32.8	N21°42.7	243°56.8	S06°08.7										
7	249°33.4	245°19.9	17.0	139°52.6	12.9	179°35.2	42.7	258°59.0	08.6										
8	264°35.8	260°21.3	17.9	154°55.4	12.9	194°37.6	42.8	274°01.2	08.5										
9	279°38.3	275°22.7	•• 18.9	169°58.3	•• 12.9	209°40.0	•• 42.8	289°03.4	•• 08.3										
10	294°40.8	290°24.2	19.8	185°01.1	12.8	224°42.4	42.8	304°05.6	08.2										
11	309°43.2	305°25.6	20.7	200°03.9	12.8	239°44.7	42.8	319°07.8	08.1										
12	324°45.7	320°27.0	N06°21.6	215°06.7	N26°12.8	254°47.1	N21°42.8	334°10.0	S06°08.0										
13	339°48.1	335°28.5	22.5	230°09.5	12.8	269°49.5	42.9	349°12.2	07.9										
14	354°50.6	350°29.9	23.5	245°12.3	12.7	284°51.9	42.9	4°14.4	07.8										
15	9°53.1	5°31.4	•• 24.4	260°15.2	•• 12.7	299°54.2	•• 42.9	19°16.6	•• 07.7										
16	24°55.5	20°32.8	25.3	275°18.0	12.7	314°56.6	42.9	34°18.8	07.5										
17	39°58.0	35°34.3	26.2	290°20.8	12.7	329°59.0	42.9	49°20.9	07.4										
18	55°00.5	50°35.7	N06°27.1	305°23.6	N26°12.6	345°01.4	N21°43.0	64°23.1	S06°07.3										
19	70°02.9	65°37.2	28.0	320°26.4	12.6	0°03.7	43.0	79°25.3	07.2										
20	85°05.4	80°38.6	28.9	335°29.2	12.6	15°06.1	43.0	94°27.5	07.1										
21	100°07.9	95°40.1	•• 29.8	350°32.0	•• 12.6	30°08.5	•• 43.0	109°29.7	•• 07.0										
22	115°10.3	110°41.6	30.8	5°34.8	12.5	45°10.8	43.0	124°31.9	06.8										
23	130°12.8	125°43.0	31.7	20°37.6	12.5	60°13.2	43.1	139°34.1	06.7										
Mer.pass. 14:21				ν 1.4' d 0.9' m -4.85				ν 2.8' d -0.0' m -0.70				ν 2.4' d 0.0' m -2.42				ν 2.2' d -0.1' m 1.12			

Feb 12 Wed	SHA	Mer.pass
Venus	356°47.3	14:42
Mars	249°57.2	21:47
Jupiter	290°06.1	19:07
Saturn	9°40.4	13:50

Feb 13 Thu	SHA	Mer.pass
Venus	356°19.9	14:40
Mars	250°07.4	21:42
Jupiter	290°04.4	19:04
Saturn	9°34.0	13:47

Feb 14 Fri	SHA	Mer.pass
Venus	355°53.8	14:38
Mars	250°16.7	21:38
Jupiter	290°02.5	19:00
Saturn	9°27.5	13:43

Horizontal parallax		
Venus:	0.3	
Mars:	0.2	

h	Sun			Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	176°27.3	S13°40.8	2°01.7	12.0'	N19°04.1	-11.9'	56.1'
1	191°27.3	40.0	16°32.7	12.1'	18°52.2	-12.0'	56.0'
2	206°27.3	39.2	31°03.8	12.2'	18°40.2	-12.1'	56.0'
3	221°27.3	· · 38.3	45°35.1	12.3'	18°28.1	-12.1'	56.0'
4	236°27.3	37.5	60°06.4	12.4'	18°16.0	-12.2'	56.0'
5	251°27.3	36.7	74°37.8	12.5'	18°03.7	-12.3'	56.0'
6	266°27.3	S13°35.8	89°09.3	12.6'	N17°51.4	-12.4'	55.9'
7	281°27.3	35.0	103°41.0	12.7'	17°39.1	-12.4'	55.9'
8	296°27.3	34.2	118°12.7	12.8'	17°26.6	-12.5'	55.9'
9	311°27.3	· · 33.3	132°44.5	12.9'	17°14.1	-12.6'	55.9'
10	326°27.3	32.5	147°16.3	13.0'	17°01.5	-12.6'	55.9'
11	341°27.4	31.7	161°48.3	13.1'	16°48.9	-12.7'	55.8'
12	356°27.4	S13°30.8	176°20.4	13.2'	N16°36.2	-12.8'	55.8'
13	11°27.4	30.0	190°52.6	13.2'	16°23.4	-12.8'	55.8'
14	26°27.4	29.2	205°24.8	13.3'	16°10.6	-12.9'	55.8'
15	41°27.4	· · 28.3	219°57.1	13.4'	15°57.7	-13.0'	55.8'
16	56°27.4	27.5	234°29.6	13.5'	15°44.7	-13.0'	55.7'
17	71°27.4	26.6	249°02.1	13.6'	15°31.7	-13.1'	55.7'
18	86°27.4	S13°25.8	263°34.7	13.7'	N15°18.6	-13.1'	55.7'
19	101°27.5	25.0	278°07.3	13.8'	15°05.5	-13.2'	55.7'
20	116°27.5	24.1	292°40.1	13.8'	14°52.3	-13.2'	55.7'
21	131°27.5	· · 23.3	307°12.9	13.9'	14°39.0	-13.3'	55.6'
22	146°27.5	22.4	321°45.9	14.0'	14°25.8	-13.3'	55.6'
23	161°27.5	21.6	336°18.9	14.1'	14°12.4	-13.4'	55.6'
SD = 16.2'			SD = 15.3'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	176°27.5	S13°20.8	350°52.0	14.2'	N13°59.0	-13.4'	55.6'
1	191°27.6	19.9	5°25.1	14.2'	13°45.6	-13.5'	55.6'
2	206°27.6	19.1	19°58.4	14.3'	13°32.1	-13.5'	55.5'
3	221°27.6	· · 18.2	34°31.7	14.4'	13°18.5	-13.6'	55.5'
4	236°27.6	17.4	49°05.1	14.5'	13°04.9	-13.6'	55.5'
5	251°27.6	16.6	63°38.5	14.5'	12°51.3	-13.7'	55.5'
6	266°27.6	S13°15.7	78°12.1	14.6'	N12°37.6	-13.7'	55.5'
7	281°27.7	14.9	92°45.7	14.7'	12°23.9	-13.8'	55.4'
8	296°27.7	14.0	107°19.4	14.8'	12°10.1	-13.8'	55.4'
9	311°27.7	· · 13.2	121°53.1	14.8'	11°56.3	-13.8'	55.4'
10	326°27.7	12.3	136°26.9	14.9'	11°42.5	-13.9'	55.4'
11	341°27.7	11.5	151°00.8	15.0'	11°28.6	-13.9'	55.4'
12	356°27.7	S13°10.6	165°34.8	15.0'	N11°14.7	-13.9'	55.3'
13	11°27.8	09.8	180°08.8	15.1'	11°00.8	-14.0'	55.3'
14	26°27.8	08.9	194°42.9	15.2'	10°46.8	-14.0'	55.3'
15	41°27.8	· · 08.1	209°17.1	15.2'	10°32.8	-14.0'	55.3'
16	56°27.8	07.3	223°51.3	15.3'	10°18.7	-14.1'	55.3'
17	71°27.8	06.4	238°25.6	15.3'	10°04.7	-14.1'	55.2'
18	86°27.9	S13°05.6	252°59.9	15.4'	N09°50.6	-14.1'	55.2'
19	101°27.9	04.7	267°34.3	15.5'	09°36.4	-14.2'	55.2'
20	116°27.9	03.9	282°08.8	15.5'	09°22.3	-14.2'	55.2'
21	131°27.9	· · 03.0	296°43.3	15.6'	09°08.1	-14.2'	55.2'
22	146°28.0	02.2	311°17.9	15.6'	08°53.8	-14.2'	55.2'
23	161°28.0	01.3	325°52.5	15.7'	08°39.6	-14.3'	55.1'
SD = 16.2'			SD = 15.2'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	176°28.0	S13°00.5	340°27.2	15.7'	N08°25.3	-14.3'	55.1'
1	191°28.0	12°59.6	355°02.0	15.8'	08°11.1	-14.3'	55.1'
2	206°28.1	58.8	9°36.8	15.8'	07°56.8	-14.3'	55.1'
3	221°28.1	· · 57.9	24°11.6	15.9'	07°42.4	-14.3'	55.1'
4	236°28.1	57.1	38°46.5	15.9'	07°28.1	-14.4'	55.0'
5	251°28.1	56.2	53°21.5	16.0'	07°13.7	-14.4'	55.0'
6	266°28.1	S12°55.4	67°56.5	16.0'	N06°59.3	-14.4'	55.0'
7	281°28.2	54.5	82°31.5	16.1'	06°44.9	-14.4'	55.0'
8	296°28.2	53.7	97°06.6	16.1'	06°30.5	-14.4'	55.0'
9	311°28.2	· · 52.8	111°41.7	16.2'	06°16.1	-14.4'	55.0'
10	326°28.3	51.9	126°16.9	16.2'	06°01.6	-14.5'	54.9'
11	341°28.3	51.1	140°52.1	16.3'	05°47.2	-14.5'	54.9'
12	356°28.3	S12°50.2	155°27.4	16.3'	N05°32.7	-14.5'	54.9'
13	11°28.3	49.4	170°02.7	16.3'	05°18.2	-14.5'	54.9'
14	26°28.4	48.5	184°38.1	16.4'	05°03.7	-14.5'	54.9'
15	41°28.4	· · 47.7	199°13.4	16.4'	04°49.2	-14.5'	54.9'
16	56°28.4	46.8	213°48.9	16.5'	04°34.7	-14.5'	54.8'
17	71°28.4	46.0	228°24.3	16.5'	04°20.2	-14.5'	54.8'
18	86°28.5	S12°45.1	242°59.8	16.5'	N04°05.7	-14.5'	54.8'
19	101°28.5	44.3	257°35.3	16.6'	03°51.2	-14.5'	54.8'
20	116°28.5	43.4	272°10.9	16.6'	03°36.6	-14.5'	54.8'
21	131°28.6	· · 42.5	286°46.5	16.6'	03°22.1	-14.5'	54.8'
22	146°28.6	41.7	301°22.1	16.7'	03°07.5	-14.5'	54.7'
23	161°28.6	40.8	315°57.8	16.7'	02°53.0	-14.5'	54.7'
SD = 16.2'			SD = 15.0'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:22	07:43	09:04	15:26	16:47	18:08
N 70°	06:19	07:32	08:42	15:48	16:58	18:11
68°	06:17	07:23	08:25	16:04	17:07	18:13
66°	06:15	07:16	08:12	16:18	17:14	18:15
64°	06:13	07:09	08:01	16:29	17:20	18:16
62°	06:11	07:04	07:51	16:38	17:26	18:18
60°	06:10	06:59	07:43	16:46	17:31	18:20
N 58°	06:08	06:54	07:36	16:54	17:35	18:21
56°	06:07	06:50	07:29	17:00	17:39	18:23
54°	06:05	06:47	07:24	17:05	17:42	18:24
52°	06:04	06:43	07:18	17:11	17:46	18:25
50°	06:02	06:40	07:14	17:15	17:49	18:27
45°	05:59	06:33	07:04	17:25	17:56	18:30
N 40°	05:56	06:27	06:55	17:34	18:02	18:33
35°	05:52	06:22	06:48	17:41	18:07	18:37
30°	05:49	06:17	06:41	17:47	18:12	18:40
20°	05:42	06:08	06:30	17:58	18:21	18:47
N 10°	05:34	05:59	06:20	18:08	18:30	18:55
0°	05:25	05:50	06:11	18:18	18:39	19:03
S 10°	05:14	05:40	06:01	18:27	18:49	19:14
20°	05:01	05:28	05:51	18:37	19:00	19:27
30°	04:44	05:14	05:39	18:49	19:14	19:44
35°	04:33	05:05	05:32	18:56	19:22	19:54
40°	04:20	04:55	05:24	19:04	19:32	20:07
45°	04:04	04:43	05:15	19:13	19:44	20:23
S 50°	03:43	04:28	05:03	19:24	19:59	20:44
52°	03:32	04:20	04:58	19:29	20:07	20:54
54°	03:20	04:12	04:52	19:35	20:15	21:06
56°	03:06	04:03	04:46	19:41	20:24	21:20
58°	02:49	03:52	04:38	19:48	20:34	21:37
S 60°	02:27	03:40	04:30	19:56	20:46	21:58

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	14:10	16:55	19:03	10:44	09:32	08:51
N 70°	14:55	17:12	19:09	09:58	09:13	08:43
68°	15:24	17:25	19:13	09:27	08:57	08:36
66°	15:45	17:36	19:17	09:04	08:45	08:30
64°	16:02	17:45	19:20	08:46	08:34	08:25
62°	16:16	17:52	19:23	08:31	08:25	08:20
60°	16:28	17:59	19:25	08:18	08:18	08:16
N 58°	16:38	18:05	19:27	08:07	08:11	08:13
56°	16:47	18:10	19:29	07:57	08:05	08:10
54°	16:54	18:14	19:31	07:49	07:59	08:07
52°	17:01	18:18	19:32	07:41	07:54	08:05
50°	17:08	18:22	19:33	07:34	07:49	08:02
45°	17:21	18:30	19:36	07:19	07:40	07:57
N 40°	17:32	18:37	19:39	07:07	07:31	07:53
35°	17:41	18:42	19:41	06:56	07:24	07:49
30°	17:49	18:47	19:43	06:47	07:18	07:46
20°	18:03	18:56	19:46	06:31	07:07	07:40
N 10°	18:16	19:03	19:49	06:16	06:57	07:35
0°	18:27	19:10	19:51	06:03	06:48	07:30
S 10°	18:38	19:17	19:54	05:50	06:39	07:25
20°	18:50	19:25	19:57	05:35	06:29	07:20
30°	19:04	19:33	20:00	05:18	06:18	07:14
35°	19:12	19:38	20:02	05:08	06:11	07:11
40°	19:21	19:44	20:04	04:57	06:03	07:07
45°	19:31	19:50	20:06	04:44	05:55	07:02
S 50°	19:44	19:57	20:09	04:28	05:44	06:57
52°	19:49	20:01	20:10	04:20	05:39	06:54
54°	19:56	20:05	20:12	04:11	05:33	06:51
56°	20:03	20:09	20:13	04:02	05:27	06:48
58°	20:11	20:13	20:15	03:50	05:20	06:45
S 60°	20:20	20:19	20:17	03:38	05:12	06:41

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	14-16
				hh:mm	hh:mm	100-98%
12	14:11	14:11	12:14	—	12:15	
13	14:10	14:09	12:14	00:38	12:59	
14	14:08	14:07	12:14	01:20	13:41	

Aries		Venus		Mars		Jupiter		Saturn	
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	145°15.2	140°44.5	N06°32.6	35°40.4	N26°12.5	75°15.6	N21°43.1	154°36.3	S06°06.6
1	160°17.7	155°46.0	33.5	50°43.2	12.5	90°18.0	43.1	169°38.5	06.5
2	175°20.2	170°47.4	34.4	65°46.0	12.4	105°20.3	43.1	184°40.7	06.4
3	190°22.6	185°48.9	•• 35.3	80°48.8	•• 12.4	120°22.7	•• 43.2	199°42.9	•• 06.3
4	205°25.1	200°50.4	36.2	95°51.6	12.4	135°25.1	43.2	214°45.1	06.1
5	220°27.6	215°51.9	37.1	110°54.4	12.3	150°27.4	43.2	229°47.3	06.0
6	235°30.0	230°53.4	N06°38.0	125°57.1	N26°12.3	165°29.8	N21°43.2	244°49.5	S06°05.9
7	250°32.5	245°54.8	38.9	140°59.9	12.3	180°32.2	43.2	259°51.7	05.8
8	265°35.0	260°56.3	39.8	156°02.7	12.2	195°34.5	43.2	274°53.9	05.7
9	280°37.4	275°57.8	•• 40.7	171°05.5	•• 12.2	210°36.9	•• 43.3	289°56.0	•• 05.6
10	295°39.9	290°59.3	41.6	186°08.3	12.2	225°39.3	43.3	304°58.2	05.4
11	310°42.4	306°00.8	42.5	201°11.1	12.2	240°41.6	43.3	320°00.4	05.3
12	325°44.8	321°02.3	N06°43.4	216°13.8	N26°12.1	255°44.0	N21°43.3	335°02.6	S06°05.2
13	340°47.3	336°03.8	44.3	231°16.6	12.1	270°46.4	43.4	350°04.8	05.1
14	355°49.7	351°05.3	45.2	246°19.4	12.1	285°48.7	43.4	5°07.0	05.0
15	10°52.2	6°06.8	•• 46.1	261°22.2	•• 12.0	300°51.1	•• 43.4	20°09.2	•• 04.9
16	25°54.7	21°08.3	46.9	276°24.9	12.0	315°53.5	43.4	35°11.4	04.7
17	40°57.1	36°09.9	47.8	291°27.7	12.0	330°55.8	43.4	50°13.6	04.6
18	55°59.6	51°11.4	N06°48.7	306°30.5	N26°11.9	345°58.2	N21°43.5	65°15.8	S06°04.5
19	71°02.1	66°12.9	49.6	321°33.3	11.9	1°00.6	43.5	80°18.0	04.4
20	86°04.5	81°14.4	50.5	336°36.0	11.9	16°02.9	43.5	95°20.2	04.3
21	101°07.0	96°15.9	•• 51.4	351°38.8	•• 11.8	31°05.3	•• 43.5	110°22.4	•• 04.2
22	116°09.5	111°17.5	52.3	6°41.6	11.8	46°07.7	43.5	125°24.6	04.0
23	131°11.9	126°19.0	53.2	21°44.3	11.8	61°10.0	43.6	140°26.7	03.9
Mer.pass. 14:17		ν 1.5' d 0.9' m -4.85		ν 2.8' d -0.0' m -0.67		ν 2.4' d 0.0' m -2.41		ν 2.2' d -0.1' m 1.12	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	147°13.5	141°58.1	N07°14.9	37°52.9	N26°10.9	77°09.0	N21°44.1	156°21.5	S06°01.0
1	162°16.0	156°59.7	15.8	52°55.6	10.8	92°11.3	44.1	171°23.7	00.9
2	177°18.4	172°01.3	16.6	67°58.4	10.8	107°13.7	44.2	186°25.9	00.8
3	192°20.9	187°02.9	· · 17.5	83°01.1	· · 10.7	122°16.0	· · 44.2	201°28.1	· · 00.7
4	207°23.4	202°04.5	18.3	98°03.8	10.7	137°18.4	44.2	216°30.3	00.5
5	222°25.8	217°06.1	19.2	113°06.5	10.7	152°20.7	44.2	231°32.5	00.4
6	237°28.3	232°07.7	N07°20.0	128°09.2	N26°10.6	167°23.1	N21°44.3	246°34.7	S06°00.3
7	252°30.8	247°09.3	20.9	143°11.9	10.6	182°25.4	44.3	261°36.9	00.2
8	267°33.2	262°10.9	21.7	158°14.7	10.5	197°27.8	44.3	276°39.1	00.1
9	282°35.7	277°12.6	· · 22.6	173°17.4	· · 10.5	212°30.1	· · 44.3	291°41.3	06°00.0
10	297°38.2	292°14.2	23.4	188°20.1	10.5	227°32.5	44.3	306°43.5	05°59.8
11	312°40.6	307°15.8	24.3	203°22.8	10.4	242°34.8	44.4	321°45.7	59.7
12	327°43.1	322°17.4	N07°25.1	218°25.5	N26°10.4	257°37.2	N21°44.4	336°47.8	S05°59.6
13	342°45.6	337°19.1	26.0	233°28.2	10.3	272°39.5	44.4	351°50.0	59.5
14	357°48.0	352°20.7	26.8	248°30.9	10.3	287°41.9	44.4	6°52.2	59.4
15	12°50.5	7°22.3	· · 27.7	263°33.6	· · 10.2	302°44.2	· · 44.5	21°54.4	· · 59.3
16	27°52.9	22°24.0	28.5	278°36.3	10.2	317°46.6	44.5	36°56.6	59.1
17	42°55.4	37°25.6	29.3	293°39.0	10.2	332°48.9	44.5	51°58.8	59.0
18	57°57.9	52°27.3	N07°30.2	308°41.7	N26°10.1	347°51.3	N21°44.5	67°01.0	S05°58.9
19	73°00.3	67°28.9	31.0	323°44.4	10.1	2°53.6	44.6	82°03.2	58.8
20	88°02.8	82°30.6	31.8	338°47.1	10.0	17°56.0	44.6	97°05.4	58.7
21	103°05.3	97°32.2	· · 32.7	353°49.8	· · 10.0	32°58.3	· · 44.6	112°07.6	· · 58.6
22	118°07.7	112°33.9	33.5	8°52.5	09.9	48°00.7	44.6	127°09.7	58.4
23	133°10.2	127°35.5	34.3	23°55.2	09.9	63°03.0	44.6	142°11.9	58.3
Mer.pass. 14:09		ν 1.6' d 0.9' m -4.86		ν 2.7' d -0.0' m -0.61		ν 2.4' d 0.0' m -2.39		ν 2.2' d -0.1' m 1.12	

Feb 15 Sat	SHA	Mer.pass
Venus	355° 29.2	14:36
Mars	250° 25.1	21:33
Jupiter	290° 00.3	18:56
Saturn	9° 21.1	13:40

Feb 16 Sun	SHA	Mer.pass
Venus	355° 06.1	14:33
Mars	250° 32.7	21:29
Jupiter	289° 58.0	18:52
Saturn	9° 14.6	13:36

Feb 17 Mon	SHA	Mer.pass
Venus	354° 44.5	14:31
Mars	250° 39.4	21:25
Jupiter	289° 55.5	18:48
Saturn	9° 08.0	13:33

Horizontal parallax	
Venus:	0.4
Mars:	0.2

DUT1 = UT1-UTC = +0.0445 sec ΔT = TT-UT1 = +69.1395 sec

2025 February 15 to Feb. 17 UT

h	Sun			Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	176°28.7	S12°40.0	330°33.5	16.7'	N02°38.5	-14.5'	54.7'
1	191°28.7	39.1	345°09.2	16.7'	02°23.9	-14.5'	54.7'
2	206°28.7	38.2	359°44.9	16.8'	02°09.4	-14.5'	54.7'
3	221°28.7	· · 37.4	14°20.7	16.8'	01°54.8	-14.5'	54.7'
4	236°28.8	36.5	28°56.4	16.8'	01°40.3	-14.5'	54.7'
5	251°28.8	35.7	43°32.3	16.8'	01°25.7	-14.5'	54.6'
6	266°28.8	S12°34.8	58°08.1	16.9'	N01°11.2	-14.5'	54.6'
7	281°28.9	33.9	72°43.9	16.9'	00°56.7	-14.5'	54.6'
8	296°28.9	33.1	87°19.8	16.9'	00°42.2	-14.5'	54.6'
9	311°28.9	· · 32.2	101°55.7	16.9'	00°27.6	-14.5'	54.6'
10	326°29.0	31.4	116°31.6	16.9'	N00°13.1	-14.5'	54.6'
11	341°29.0	30.5	131°07.5	16.9'	S00°01.4	14.5'	54.6'
12	356°29.0	S12°29.6	145°43.5	17.0'	S00°15.9	14.5'	54.5'
13	11°29.1	28.8	160°19.4	17.0'	00°30.4	14.5'	54.5'
14	26°29.1	27.9	174°55.4	17.0'	00°44.8	14.5'	54.5'
15	41°29.2	· · 27.0	189°31.4	17.0'	00°59.3	14.5'	54.5'
16	56°29.2	26.2	204°07.4	17.0'	01°13.8	14.4'	54.5'
17	71°29.2	25.3	218°43.4	17.0'	01°28.2	14.4'	54.5'
18	86°29.3	S12°24.5	233°19.4	17.0'	S01°42.6	14.4'	54.5'
19	101°29.3	23.6	247°55.4	17.0'	01°57.0	14.4'	54.5'
20	116°29.3	22.7	262°31.4	17.0'	02°11.4	14.4'	54.4'
21	131°29.4	· · 21.9	277°07.5	17.0'	02°25.8	14.4'	54.4'
22	146°29.4	21.0	291°43.5	17.0'	02°40.2	14.4'	54.4'
23	161°29.4	20.1	306°19.5	17.0'	02°54.6	14.3'	54.4'
	SD = 16.2'	$d = -0.9'$			SD = 14.9'		

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	176°29.5	S12°19.3	320°55.6	17.0'	S03°08.9	14.3'	54.4'
1	191°29.5	18.4	335°31.6	17.0'	03°23.2	14.3'	54.4'
2	206°29.6	17.5	350°07.7	17.0'	03°37.5	14.3'	54.4'
3	221°29.6	· · 16.7	4°43.7	17.0'	03°51.8	14.3'	54.4'
4	236°29.6	15.8	19°19.8	17.0'	04°06.0	14.2'	54.3'
5	251°29.7	14.9	33°55.8	17.0'	04°20.3	14.2'	54.3'
6	266°29.7	S12°14.0	48°31.8	17.0'	S04°34.5	14.2'	54.3'
7	281°29.8	13.2	63°07.9	17.0'	04°48.7	14.2'	54.3'
8	296°29.8	12.3	77°43.9	17.0'	05°02.9	14.1'	54.3'
9	311°29.8	· · 11.4	92°19.9	17.0'	05°17.0	14.1'	54.3'
10	326°29.9	10.6	106°55.9	17.0'	05°31.1	14.1'	54.3'
11	341°29.9	09.7	121°31.9	17.0'	05°45.2	14.1'	54.3'
12	356°30.0	S12°08.8	136°07.9	17.0'	S05°59.3	14.0'	54.3'
13	11°30.0	08.0	150°43.9	17.0'	06°13.3	14.0'	54.3'
14	26°30.0	07.1	165°19.8	17.0'	06°27.3	14.0'	54.3'
15	41°30.1	· · 06.2	179°55.8	16.9'	06°41.3	14.0'	54.2'
16	56°30.1	05.3	194°31.7	16.9'	06°55.3	13.9'	54.2'
17	71°30.2	04.5	209°07.6	16.9'	07°09.2	13.9'	54.2'
18	86°30.2	S12°03.6	223°43.6	16.9'	S07°23.1	13.9'	54.2'
19	101°30.3	02.7	238°19.4	16.9'	07°37.0	13.8'	54.2'
20	116°30.3	01.8	252°55.3	16.8'	07°50.8	13.8'	54.2'
21	131°30.4	· · 01.0	267°31.2	16.8'	08°04.6	13.8'	54.2'
22	146°30.4	12°00.1	282°07.0	16.8'	08°18.4	13.7'	54.2'
23	161°30.4	11°59.2	296°42.8	16.8'	08°32.1	13.7'	54.2'
	SD = 16.2'	$d = -0.9'$			SD = 14.8'		

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	176°30.5	S11°58.3	311°18.6	16.8'	S08°45.8	13.7'	54.2'
1	191°30.5	57.5	325°54.3	16.7'	08°59.5	13.6'	54.2'
2	206°30.6	56.6	340°30.1	16.7'	09°13.1	13.6'	54.2'
3	221°30.6	· · 55.7	355°05.8	16.7'	09°26.7	13.6'	54.2'
4	236°30.7	54.8	9°41.4	16.7'	09°40.2	13.5'	54.2'
5	251°30.7	54.0	24°17.1	16.6'	09°53.7	13.5'	54.1'
6	266°30.8	S11°53.1	38°52.7	16.6'	S10°07.2	13.4'	54.1'
7	281°30.8	52.2	53°28.3	16.6'	10°20.6	13.4'	54.1'
8	296°30.9	51.3	68°03.9	16.5'	10°34.0	13.4'	54.1'
9	311°30.9	· · 50.5	82°39.4	16.5'	10°47.4	13.3'	54.1'
10	326°31.0	49.6	97°14.9	16.5'	11°00.7	13.3'	54.1'
11	341°31.0	48.7	111°50.4	16.4'	11°14.0	13.2'	54.1'
12	356°31.1	S11°47.8	126°25.8	16.4'	S11°27.2	13.2'	54.1'
13	11°31.1	46.9	141°01.2	16.4'	11°40.4	13.1'	54.1'
14	26°31.2	46.1	155°36.5	16.3'	11°53.5	13.1'	54.1'
15	41°31.2	· · 45.2	170°11.9	16.3'	12°06.6	13.0'	54.1'
16	56°31.3	44.3	184°47.1	16.2'	12°19.6	13.0'	54.1'
17	71°31.3	43.4	199°22.4	16.2'	12°32.6	12.9'	54.1'
18	86°31.4	S11°42.5	213°57.6	16.2'	S12°45.6	12.9'	54.1'
19	101°31.4	41.7	228°32.7	16.1'	12°58.5	12.9'	54.1'
20	116°31.5	40.8	243°07.8	16.1'	13°11.3	12.8'	54.1'
21	131°31.5	· · 39.9	257°42.9	16.0'	13°24.1	12.7'	54.1'
22	146°31.6	39.0	272°17.9	16.0'	13°36.9	12.7'	54.1'
23	161°31.6	38.1	286°52.9	15.9'	13°49.6	12.6'	54.1'
	SD = 16.2'	$d = -0.9'$			SD = 14.8'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:09	07:29	08:46	15:43	17:01	18:21
N 70°	06:08	07:20	08:28	16:02	17:10	18:22
68°	06:07	07:12	08:13	16:17	17:17	18:23
66°	06:06	07:06	08:01	16:28	17:24	18:24
64°	06:05	07:00	07:51	16:38	17:29	18:25
62°	06:04	06:55	07:42	16:47	17:34	18:26
60°	06:03	06:51	07:35	16:54	17:38	18:27
N 58°	06:02	06:47	07:28	17:01	17:42	18:28
56°	06:01	06:44	07:22	17:06	17:45	18:29
54°	05:59	06:41	07:17	17:12	17:48	18:30
52°	05:58	06:38	07:13	17:16	17:51	18:30
50°	05:57	06:35	07:08	17:20	17:54	18:31
45°	05:55	06:29	06:59	17:30	18:00	18:34
N 40°	05:52	06:24	06:51	17:37	18:05	18:37
35°	05:49	06:19	06:45	17:44	18:10	18:39
30°	05:46	06:14	06:39	17:50	18:14	18:42
20°	05:40	06:06	06:28	18:00	18:22	18:48
N 10°	05:33	05:58	06:19	18:09	18:30	18:55
0°	05:25	05:49	06:11	18:17	18:39	19:03
S 10°	05:15	05:40	06:02	18:26	18:48	19:13
20°	05:03	05:30	05:52	18:35	18:58	19:25
30°	04:47	05:16	05:41	18:46	19:11	19:40
35°	04:37	05:08	05:35	18:53	19:19	19:50
40°	04:24	04:59	05:28	19:00	19:28	20:03
45°	04:09	04:48	05:19	19:08	19:39	20:18
S 50°	03:49	04:33	05:09	19:18	19:53	20:37
52°	03:40	04:27	05:04	19:23	20:00	20:47
54°	03:28	04:19	04:58	19:28	20:07	20:58
56°	03:15	04:10	04:52	19:34	20:16	21:10
58°	03:00	04:01	04:46	19:41	20:25	21:25
S 60°	02:41	03:49	04:38	19:48	20:36	21:44

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	21:03	23:05	· · ·	08:18	07:45	07:08
N 70°	20:59	22:51	· · ·	08:17	07:52	07:24
68°	20:56	22:40	· · ·	08:16	07:58	07:37
66°	20:54	22:31	· · ·	08:16	08:02	07:48
64°	20:52	22:23	23:58	08:15	08:06	07:57
62°	20:50	22:17	23:46	08:15	08:10	08:05
60°	20:48	22:11	23:36	08:15	08:13	08:12
N 58°	20:47	22:07	23:27	08:14	08:16	08:18
56°	20:46	22:02	23:20	08:14	08:18	08:23
54°	20:45	21:58	23:13	08:14	08:21	08:28
52°	20:44	21:55	23:07	08:14	08:23	08:32
50°	20:43	21:52	23:01	08:14	08:25	08:36
45°	20:41	21:45	22:50	08:13	08:29	08:45
N 40°	20:39	21:39	22:40	08:13	08:32	08:52
35°	20:38	21:34	22:32	08:12	08:35	08:58
30°	20:37	21:30	22:24	08:12	08:38	09:04
20°	20:35	21:23	22:12	08:12	08:42	09:13
N 10°	20:33	21:17	22:01	08:11	08:46	09:22
0°	20:31	21:11	21:51	08:11	08:50	09:30
S 10°	20:29	21:05	21:41	08:10	08:54	09:38
20°	20:28	20:58	21:30	08:10	08:58	09:46
30°	20:26	20:51	21:18	08:09	09:03	09:56
35°	20:25	20:47	21:11	08:09	09:05	10:02
40°	20:23	20:43	21:04	08:08	09:08	10:08
45°	20:22	20:38	20:54	08:08	09:12	10:16
S 50°	20:20	20:31	20:44	08:07	09:16	10:25
52°	20:19	20:28	20:39	08:07	09:18	10:29
54°	20:19	20:25	20:33	08:06	09:20	10:34
56°	20:18	20:22	20:27	08:06	09:23	10:39
58°	20:17	20:18	20:20	08:06	09:25	10:45
S 60°	20:15	20:14	20:12	08:05	09:28	10:52

Day	Sun		Mer. Pass	Moon		Age
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss	hh:mm	Mer.Pass. Upper hh:mm	Lower hh:mm	17-19 94-83%
15	14:05	14:04	12:14	02:01	14:21	
16	14:02	14:00	12:14	02:41	15:00	
17	13:58	13:56	12:14	03:20	15:40	

February 18, 19, 20 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	148° 12.7	142° 37.2	N07° 35.2	38° 57.9	N26° 09.9	78° 05.4	N21° 44.7	157° 14.1	S05° 58.2	Alpheratz	357° 35.0 29° 13.7
1	163° 15.1	157° 38.9	36.0	54° 00.6	09.8	93° 07.7	44.7	172° 16.3	58.1	Ankaa	353° 07.4 -42° 10.4
2	178° 17.6	172° 40.5	36.8	69° 03.2	09.8	108° 10.0	44.7	187° 18.5	58.0	Schedar	349° 31.4 56° 40.6
3	193° 20.1	187° 42.2	· · 37.7	84° 05.9	· · 09.7	123° 12.4	· · 44.7	202° 20.7	· · 57.8	Diphda	348° 47.4 -17° 51.1
4	208° 22.5	202° 43.9	38.5	99° 08.6	09.7	138° 14.7	44.8	217° 22.9	57.7	Achernar	335° 20.4 -57° 06.8
5	223° 25.0	217° 45.6	39.3	114° 11.3	09.6	153° 17.1	44.8	232° 25.1	57.6	Hamal	327° 51.2 23° 34.9
6	238° 27.4	232° 47.2	N07° 40.1	129° 14.0	N26° 09.6	168° 19.4	N21° 44.8	247° 27.3	S05° 57.5	Polaris	314° 05.7 89° 22.5
7	253° 29.9	247° 48.9	41.0	144° 16.6	09.5	183° 21.8	44.8	262° 29.5	57.4	Acamar	315° 11.8 -40° 12.5
8	268° 32.4	262° 50.6	41.8	159° 19.3	09.5	198° 24.1	44.9	277° 31.7	57.3	Menkar	314° 06.1 4° 11.2
9	283° 34.8	277° 52.3	· · 42.6	174° 22.0	· · 09.4	213° 26.4	· · 44.9	292° 33.8	· · 57.1	Mirfak	308° 28.2 49° 57.2
10	298° 37.3	292° 54.0	43.4	189° 24.7	09.4	228° 28.8	44.9	307° 36.0	57.0	Aldebaran	290° 39.4 16° 33.6
11	313° 39.8	307° 55.7	44.3	204° 27.3	09.3	243° 31.1	44.9	322° 38.2	56.9	Rigel	281° 03.7 -8° 10.5
12	328° 42.2	322° 57.4	N07° 45.1	219° 30.0	N26° 09.3	258° 33.5	N21° 45.0	337° 40.4	S05° 56.8	Capella	280° 21.6 46° 01.5
13	343° 44.7	337° 59.1	45.9	234° 32.7	09.2	273° 35.8	45.0	352° 42.6	56.7	Bellatrix	278° 22.6 6° 22.3
14	358° 47.2	353° 00.8	46.7	249° 35.4	09.2	288° 38.2	45.0	7° 44.8	56.6	Elnath	278° 01.6 28° 37.8
15	13° 49.6	8° 02.5	· · 47.5	264° 38.0	· · 09.2	303° 40.5	· · 45.0	22° 47.0	· · 56.4	Alnilam	275° 37.5 -1° 11.2
16	28° 52.1	23° 04.2	48.3	279° 40.7	09.1	318° 42.8	45.1	37° 49.2	56.3	Betelgeuse	270° 51.8 7° 24.7
17	43° 54.5	38° 05.9	49.1	294° 43.4	09.1	333° 45.2	45.1	52° 51.4	56.2	Canopus	263° 52.0 -52° 42.7
18	58° 57.0	53° 07.6	N07° 50.0	309° 46.0	N26° 09.0	348° 47.5	N21° 45.1	67° 53.6	S05° 56.1	Sirius	258° 25.9 -16° 45.2
19	73° 59.5	68° 09.3	50.8	324° 48.7	09.0	3° 49.8	45.1	82° 55.7	56.0	Adhara	255° 05.5 -29° 00.5
20	89° 01.9	83° 11.0	51.6	339° 51.3	08.9	18° 52.2	45.2	97° 57.9	55.9	Procyon	244° 50.5 5° 09.6
21	104° 04.4	98° 12.8	· · 52.4	354° 54.0	· · 08.9	33° 54.5	· · 45.2	113° 00.1	· · 55.7	Pollux	243° 16.9 27° 57.9
22	119° 06.9	113° 14.5	53.2	9° 56.7	08.8	48° 56.9	45.2	128° 02.3	55.6	Avior	234° 14.1 -59° 35.5
23	134° 09.3	128° 16.2	54.0	24° 59.3	08.8	63° 59.2	45.2	143° 04.5	55.5	Suhail	222° 45.8 -43° 32.1
Mer.pass. 14:05		ν 1.7' d 0.8' m -4.86		ν 2.7' d -0.0' m -0.58		ν 2.3' d 0.0' m -2.38		ν 2.2' d -0.1' m 1.12			

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	149° 11.8	143° 18.0	N07° 54.8	40° 02.0	N26° 08.7	79° 01.5	N21° 45.2	158° 06.7	S05° 55.4		
1	164° 14.3	158° 19.7	55.6	55° 04.6	08.7	94° 03.9	45.3	173° 08.9	55.3		
2	179° 16.7	173° 21.4	56.4	70° 07.3	08.6	109° 06.2	45.3	188° 11.1	55.1		
3	194° 19.2	188° 23.2	· · 57.2	85° 09.9	· · 08.6	124° 08.5	· · 45.3	203° 13.3	· · 55.0		
4	209° 21.7	203° 24.9	58.0	100° 12.6	08.5	139° 10.9	45.3	218° 15.4	54.9		
5	224° 24.1	218° 26.7	58.8	115° 15.2	08.5	154° 13.2	45.4	233° 17.6	54.8		
6	239° 26.6	233° 28.4	N07° 59.6	130° 17.8	N26° 08.4	169° 15.5	N21° 45.4	248° 19.8	S05° 54.7		
7	254° 29.0	248° 30.2	08° 00.4	145° 20.5	08.4	184° 17.9	45.4	263° 22.0	54.6		
8	269° 31.5	263° 31.9	01.2	160° 23.1	08.3	199° 20.2	45.4	278° 24.2	54.4		
9	284° 34.0	278° 33.7	· · 02.0	175° 25.8	· · 08.3	214° 22.5	· · 45.5	293° 26.4	· · 54.3		
10	299° 36.4	293° 35.4	02.8	190° 28.4	08.2	229° 24.9	45.5	308° 28.6	54.2		
11	314° 38.9	308° 37.2	03.6	205° 31.0	08.1	244° 27.2	45.5	323° 30.8	54.1		
12	329° 41.4	323° 39.0	N08° 04.3	220° 33.7	N26° 08.1	259° 29.5	N21° 45.5	338° 33.0	S05° 54.0		
13	344° 43.8	338° 40.7	05.1	235° 36.3	08.0	274° 31.9	45.6	353° 35.1	53.8		
14	359° 46.3	353° 42.5	05.9	250° 39.0	08.0	289° 34.2	45.6	8° 37.3	53.7		
15	14° 48.8	8° 44.3	· · 06.7	265° 41.6	· · 07.9	304° 36.5	· · 45.6	23° 39.5	· · 53.6		
16	29° 51.2	23° 46.1	07.5	280° 44.2	07.9	319° 38.9	45.6	38° 41.7	53.5		
17	44° 53.7	38° 47.8	08.3	295° 46.8	07.8	334° 41.2	45.7	53° 43.9	53.4		
18	59° 56.2	53° 49.6	N08° 09.1	310° 49.5	N26° 07.8	349° 43.5	N21° 45.7	68° 46.1	S05° 53.3		
19	74° 58.6	68° 51.4	09.8	325° 52.1	07.7	4° 45.9	45.7	83° 48.3	53.1		
20	90° 01.1	83° 53.2	10.6	340° 54.7	07.7	19° 48.2	45.7	98° 50.5	53.0		
21	105° 03.5	98° 55.0	· · 11.4	355° 57.3	· · 07.6	34° 50.5	· · 45.8	113° 52.7	· · 52.9		
22	120° 06.0	113° 56.8	12.2	11° 00.0	07.6	49° 52.9	45.8	128° 54.8	52.8		
23	135° 08.5	128° 58.6	12.9	26° 02.6	07.5	64° 55.2	45.8	143° 57.0	52.7		
Mer.pass. 14:01		ν 1.7' d 0.8' m -4.86		ν 2.7' d -0.1' m -0.56		ν 2.3' d 0.0' m -2.38		ν 2.2' d -0.1' m 1.12			

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	150° 10.9	144° 00.4	N08° 13.7	41° 05.2	N26° 07.4	79° 57.5	N21° 45.9	158° 59.2	S05° 52.6		
1	165° 13.4	159° 02.2	14.5	56° 07.8	07.4	94° 59.8	45.9	174° 01.4	52.4		
2	180° 15.9	174° 04.0	15.3	71° 10.4	07.3	110° 02.2	45.9	189° 03.6	52.3		
3	195° 18.3	189° 05.8	· · 16.0	86° 13.0	· · 07.3	125° 04.5	· · 45.9	204° 05.8	· · 52.2		
4	210° 20.8	204° 07.6	16.8	101° 15.7	07.2	140° 06.8	46.0	219° 08.0	52.1		
5	225° 23.3	219° 09.5	17.6	116° 18.3	07.2	155° 09.1	46.0	234° 10.2	52.0		
6	240° 25.7	234° 11.3	N08° 18.3	131° 20.9	N26° 07.1	170° 11.5	N21° 46.0	249° 12.4	S05° 51.8		
7	255° 28.2	249° 13.1	19.1	146° 23.5	07.1	185° 13.8	46.0	264° 14.5	51.7		
8	270° 30.6	264° 14.9	19.9	161° 26.1	07.0	200° 16.1	46.1	279° 16.7	51.6		
9	285° 33.1	279° 16.8	· · 20.6	176° 28.7	· · 06.9	215° 18.4	· · 46.1	294° 18.9	· · 51.5		
10	300° 35.6	294° 18.6	21.4	191° 31.3	06.9	230° 20.8	46.1	309° 21.1	51.4		
11	315° 38.0	309° 20.4	22.2	206° 33.9	06.8	245° 23.1	46.1	324° 23.3	51.3		
12	330° 40.5	324° 22.3	N08° 22.9	221° 36.5	N26° 06.8	260° 25.4	N21° 46.2	339° 25.5	S05° 51.1		
13	345° 43.0	339° 24.1	23.7	236° 39.1	06.7	275° 27.7	46.2	354° 27.7	51.0		
14	0° 45.4	354° 26.0	24.4	251° 41.7	06.7	290° 30.1	46.2	9° 29.9	50.9		
15	15° 47.9	9° 27.8	· · 25.2	266° 44.3	· · 06.6	305° 32.4	· · 46.2	24° 32.0	· · 50.8		
16	30° 50.4	24° 29.7	25.9	281° 46.9	06.5	320° 34.7	46.3	39° 34.2	50.7		
17	45° 52.8	39° 31.5	26.7	296° 49.5	06.5	335° 37.0	46.3	54° 36.4	50.5		
18	60° 55.3	54° 33.4	N08° 27.4	311° 52.1	N26° 06.4	350° 39.4	N21° 46.3	69° 38.6	S05° 50.4		
19	75° 57.8	69° 35.2	28.2	326° 54.7	06.4	5° 41.7	46.3	84° 40.8	50.3		
20	91° 00.2	84° 37.1	28.9	341° 57.2	06.3	20° 44.0	46.4	99° 43.0	50.2		
21	106° 02.7	99° 39.0	· · 29.7	356° 59.8	· · 06.2	35° 46.3	· · 46.4	114° 45.2	· · 50.1		
22	121° 05.1	114° 40.8	30.4	12° 02.4	06.2	50° 48.6	46.4	129° 47.4	50.0		
23	136° 07.6	129° 42.7	31.2	27° 05.0	06.1	65° 51.0	46.5	144° 49.6	49.8		
Mer.pass. 13:57		ν 1.8' d 0.8' m -4.86		ν 2.6' d -0.1' m -0.53		ν 2.3' d 0.0' m -2.37		ν 2.2' d -0.1' m 1.12			

Feb 18 Tue		SHA	Mer.pass
Venus	354° 24.5	14:28	
Mars	250° 45.2	21:20	
Jupiter	289° 52.7	18:45	
Saturn	9° 01.5	13:29	
Feb 19 Wed		SHA	Mer.pass
Venus	354° 06.2	14:25	
Mars	250° 50.2	21:16	
Jupiter	289° 49.7	18:41	
Saturn	8° 54.9	13:26	
Feb 20 Thu		SHA	Mer.pass
Venus	353° 49.5	14:22	
Mars	250° 54.3	21:12	
Jupiter	289° 46.6	18:37	
Saturn	8° 48.3	13:22	
Horizontal parallax			
Venus:		0.4	
Mars:		0.2	

h	Sun			Moon			
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	176°31.7	S11°37.3	301°27.8	15.9'	S14°02.2	12.6'	54.1'
1	191°31.7	36.4	316°02.7	15.8'	14°14.8	12.5'	54.1'
2	206°31.8	35.5	330°37.5	15.8'	14°27.4	12.5'	54.1'
3	221°31.8	· · 34.6	345°12.3	15.7'	14°39.9	12.4'	54.1'
4	236°31.9	33.7	359°47.0	15.7'	14°52.3	12.4'	54.1'
5	251°31.9	32.8	14°21.7	15.6'	15°04.7	12.3'	54.1'
6	266°32.0	S11°31.9	28°56.3	15.6'	S15°17.0	12.3'	54.1'
7	281°32.0	31.1	43°30.9	15.5'	15°29.2	12.2'	54.1'
8	296°32.1	30.2	58°05.4	15.5'	15°41.4	12.1'	54.1'
9	311°32.1	· · 29.3	72°39.9	15.4'	15°53.6	12.1'	54.1'
10	326°32.2	28.4	87°14.3	15.3'	16°05.7	12.0'	54.1'
11	341°32.3	27.5	101°48.6	15.3'	16°17.7	12.0'	54.1'
12	356°32.3	S11°26.6	116°22.9	15.2'	S16°29.6	11.9'	54.1'
13	11°32.4	25.7	130°57.1	15.2'	16°41.5	11.8'	54.1'
14	26°32.4	24.9	145°31.3	15.1'	16°53.4	11.8'	54.1'
15	41°32.5	· · 24.0	160°05.4	15.0'	17°05.1	11.7'	54.1'
16	56°32.5	23.1	174°39.5	15.0'	17°16.8	11.6'	54.1'
17	71°32.6	22.2	189°13.5	14.9'	17°28.5	11.6'	54.1'
18	86°32.7	S11°21.3	203°47.4	14.9'	S17°40.0	11.5'	54.1'
19	101°32.7	20.4	218°21.2	14.8'	17°51.5	11.4'	54.1'
20	116°32.8	19.5	232°55.0	14.7'	18°03.0	11.4'	54.1'
21	131°32.8	· · 18.6	247°28.7	14.7'	18°14.3	11.3'	54.1'
22	146°32.9	17.7	262°02.4	14.6'	18°25.6	11.2'	54.2'
23	161°32.9	16.9	276°36.0	14.5'	18°36.9	11.2'	54.2'
SD = 16.2' $d = -0.9'$			SD = 14.8'				

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	176°33.0	S11°16.0	291°09.5	14.5'	S18°48.0	11.1'	54.2'
1	191°33.1	15.1	305°43.0	14.4'	18°59.1	11.0'	54.2'
2	206°33.1	14.2	320°16.4	14.3'	19°10.1	10.9'	54.2'
3	221°33.2	· · 13.3	334°49.7	14.2'	19°21.1	10.9'	54.2'
4	236°33.2	12.4	349°22.9	14.2'	19°31.9	10.8'	54.2'
5	251°33.3	11.5	3°56.1	14.1'	19°42.7	10.7'	54.2'
6	266°33.4	S11°10.6	18°29.2	14.0'	S19°53.4	10.6'	54.2'
7	281°33.4	09.7	33°02.2	14.0'	20°04.0	10.6'	54.2'
8	296°33.5	08.8	47°35.2	13.9'	20°14.6	10.5'	54.2'
9	311°33.5	· · 07.9	62°08.0	13.8'	20°25.1	10.4'	54.2'
10	326°33.6	07.0	76°40.8	13.7'	20°35.4	10.3'	54.2'
11	341°33.7	06.2	91°13.6	13.6'	20°45.8	10.2'	54.3'
12	356°33.7	S11°05.3	105°46.2	13.6'	S20°56.0	10.1'	54.3'
13	11°33.8	04.4	120°18.8	13.5'	21°06.1	10.1'	54.3'
14	26°33.9	03.5	134°51.3	13.4'	21°16.2	10.0'	54.3'
15	41°33.9	· · 02.6	149°23.7	13.3'	21°26.2	9.9'	54.3'
16	56°34.0	01.7	163°56.0	13.3'	21°36.1	9.8'	54.3'
17	71°34.1	11°00.8	178°28.3	13.2'	21°45.9	9.7'	54.3'
18	86°34.1	S10°59.9	193°00.5	13.1'	S21°55.6	9.6'	54.3'
19	101°34.2	59.0	207°32.6	13.0'	22°05.2	9.5'	54.3'
20	116°34.2	58.1	222°04.6	12.9'	22°14.7	9.4'	54.4'
21	131°34.3	· · 57.2	236°36.5	12.8'	22°24.2	9.4'	54.4'
22	146°34.4	56.3	251°08.4	12.8'	22°33.5	9.3'	54.4'
23	161°34.4	55.4	265°40.1	12.7'	22°42.8	9.2'	54.4'
SD = 16.2' $d = -0.9'$			SD = 14.8'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	176°34.5	S10°54.5	280°11.8	12.6'	S22°52.0	9.1'	54.4'
1	191°34.6	53.6	294°43.4	12.5'	23°01.0	9.0'	54.4'
2	206°34.6	52.7	309°14.9	12.4'	23°10.0	8.9'	54.4'
3	221°34.7	· · 51.8	323°46.4	12.3'	23°18.9	8.8'	54.4'
4	236°34.8	50.9	338°17.7	12.3'	23°27.7	8.7'	54.5'
5	251°34.8	50.0	352°49.0	12.2'	23°36.4	8.6'	54.5'
6	266°34.9	S10°49.1	7°20.1	12.1'	S23°44.9	8.5'	54.5'
7	281°35.0	48.2	21°51.2	12.0'	23°53.4	8.4'	54.5'
8	296°35.0	47.3	36°22.2	11.9'	24°01.8	8.3'	54.5'
9	311°35.1	· · 46.4	50°53.2	11.8'	24°10.1	8.2'	54.5'
10	326°35.2	45.5	65°24.0	11.7'	24°18.3	8.1'	54.6'
11	341°35.3	44.6	79°54.7	11.7'	24°26.3	8.0'	54.6'
12	356°35.3	S10°43.7	94°25.4	11.6'	S24°34.3	7.9'	54.6'
13	11°35.4	42.8	108°56.0	11.5'	24°42.2	7.8'	54.6'
14	26°35.5	41.9	123°26.5	11.4'	24°49.9	7.6'	54.6'
15	41°35.5	· · 41.0	137°56.9	11.3'	24°57.6	7.5'	54.6'
16	56°35.6	40.1	152°27.2	11.2'	25°05.1	7.4'	54.7'
17	71°35.7	39.2	166°57.4	11.1'	25°12.5	7.3'	54.7'
18	86°35.7	S10°38.3	181°27.5	11.1'	S25°19.8	7.2'	54.7'
19	101°35.8	37.4	195°57.6	11.0'	25°27.0	7.1'	54.7'
20	116°35.9	36.5	210°27.5	10.9'	25°34.1	7.0'	54.7'
21	131°36.0	· · 35.6	224°57.4	10.8'	25°41.1	6.9'	54.8'
22	146°36.0	34.7	239°27.2	10.7'	25°47.9	6.7'	54.8'
23	161°36.1	33.8	253°56.9	10.6'	25°54.7	6.6'	54.8'
SD = 16.2' $d = -0.9'$			SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:56	07:15	08:29	16:00	17:15	18:34
N 70°	05:56	07:07	08:13	16:16	17:22	18:33
68°	05:56	07:01	08:00	16:29	17:28	18:33
66°	05:56	06:56	07:50	16:39	17:33	18:33
64°	05:56	06:51	07:41	16:48	17:38	18:33
62°	05:55	06:47	07:33	16:56	17:42	18:33
60°	05:55	06:43	07:26	17:02	17:45	18:34
N 58°	05:55	06:40	07:21	17:08	17:48	18:34
56°	05:54	06:37	07:15	17:13	17:51	18:35
54°	05:53	06:35	07:11	17:18	17:54	18:35
52°	05:53	06:32	07:07	17:22	17:56	18:36
50°	05:52	06:30	07:03	17:26	17:59	18:36
45°	05:50	06:25	06:54	17:34	18:04	18:38
N 40°	05:48	06:20	06:47	17:41	18:08	18:40
35°	05:46	06:16	06:41	17:47	18:12	18:42
30°	05:44	06:12	06:36	17:52	18:16	18:44
20°	05:38	06:04	06:27	18:01	18:23	18:49
N 10°	05:32	05:57	06:18	18:09	18:31	18:55
0°	05:25	05:49	06:10	18:17	18:38	19:03
S 10°	05:16	05:41	06:02	18:25	18:46	19:11
20°	05:05	05:31	05:54	18:34	18:56	19:23
30°	04:50	05:19	05:44	18:43	19:08	19:37
35°	04:40	05:11	05:38	18:49	19:15	19:47
40°	04:29	05:03	05:31	18:56	19:24	19:58
45°	04:14	04:52	05:23	19:03	19:34	20:12
S 50°	03:56	04:39	05:14	19:13	19:47	20:30
52°	03:47	04:33	05:09	19:17	19:53	20:39
54°	03:37	04:26	05:05	19:22	20:00	20:49
56°	03:25	04:18	04:59	19:27	20:08	21:01
58°	03:10	04:09	04:53	19:33	20:17	21:14
S 60°	02:53	03:59	04:46	19:39	20:27	21:31

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	01:26	■	■	06:12	■	■
N 70°	00:54	03:55	■	06:46	05:15	■
68°	00:31	02:42	■	07:11	06:28	■
66°	00:13	02:06	04:48	07:30	07:06	05:59
64°	· · ·	01:41	03:38	07:46	07:32	07:09
62°	· · ·	01:21	03:03	07:59	07:53	07:45
60°	· · ·	01:05	02:38	08:10	08:10	08:11
N 58°	· · ·	00:51	02:18	08:20	08:24	08:32
56°	· · ·	00:39	02:02	08:29	08:37	08:49
54°	· · ·	00:29	01:48	08:36	08:47	09:03
52°	· · ·	00:20	01:35	08:43	08:57	09:16
50°	· · ·	00:12	01:25	08:49	09:06	09:27
45°	23:55	· · ·	01:02	09:03	09:24	09:50
N 40°	23:42	· · ·	00:45	09:14	09:39	10:09
35°	23:30	· · ·	00:30	09:23	09:52	10:25
30°	23:20	· · ·	00:17	09:32	10:03	10:38
20°	23:02	23:55	· · ·	09:46	10:22	11:02
N 10°	22:47	23:36	· · ·	09:59	10:39	11:22
0°	22:33	23:18	· · ·	10:11	10:55	11:41
S 10°	22:19	23:01	23:46	10:23	11:10	12:00
20°	22:04	22:42	23:25	10:36	11:27	12:21
30°	21:48	22:21	23:00	10:51	11:47	12:45
35°	21:38	22:09	22:45	11:00	11:59	12:59
40°	21:27	21:54	22:28	11:09	12:12	13:15
45°	21:14	21:37	22:07	11:21	12:28	13:35
S 50°	20:58	21:17	21:42	11:35	12:47	13:59
52°	20:51	21:07	21:30	11:42	12:56	14:11
54°	20:43	20:56	21:16	11:49	13:07	14:25
56°	20:34	20:44	20:59	11:58	13:18	14:41
58°	20:23	20:29	20:40	12:07	13:32	15:00
S 60°	20:12	20:12	20:16	12:18	13:48	15:23

Day	Sun		Mer. Pass	Moon		Age 20-22 75-57%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
18	13:53	13:51	12:14	04:01	16:22	
19	13:48	13:45	12:14	04:44	17:06	
20	13:42	13:39	12:14	05:30	17:54	

February 21, 22, 23 UT (Fri., Sat., Sun.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	151°10.1	144°44.6	N08°31.9	42°07.6	N26°06.1	80°53.3	N21°46.5	159°51.7	S05°49.7							Alpheratz	357°35.0	29°13.7					
1	166°12.5	159°46.5	32.7	57°10.2	06.0	95°55.6	46.5	174°53.9	49.6							Ankaa	353°07.4	-42°10.4					
2	181°15.0	174°48.3	33.4	72°12.7	05.9	110°57.9	46.5	189°56.1	49.5							Schedar	349°31.4	56°40.6					
3	196°17.5	189°50.2	· · 34.2	87°15.3	· · 05.9	126°00.2	· · 46.6	204°58.3	· · 49.4							Diphda	348°47.4	-17°51.1					
4	211°19.9	204°52.1	34.9	102°17.9	05.8	141°02.6	46.6	220°00.5	49.2							Achernar	335°20.4	-57°06.8					
5	226°22.4	219°54.0	35.6	117°20.5	05.8	156°04.9	46.6	235°02.7	49.1							Hamal	327°51.3	23°34.9					
6	241°24.9	234°55.9	N08°36.4	132°23.1	N26°05.7	171°07.2	N21°46.6	250°04.9	S05°49.0							Polaris	314°07.0	89°22.5					
7	256°27.3	249°57.8	37.1	147°25.6	05.6	186°09.5	46.7	265°07.1	48.9							Acamar	315°11.8	-40°12.5					
8	271°29.8	264°59.7	37.8	162°28.2	05.6	201°11.8	46.7	280°09.2	48.8							Menkar	314°06.1	4°11.2					
9	286°32.3	280°01.6	· · 38.6	177°30.8	· · 05.5	216°14.1	· · 46.7	295°11.4	· · 48.6							Mirfak	308°28.2	49°57.2					
10	301°34.7	295°03.5	39.3	192°33.3	05.5	231°16.5	46.7	310°13.6	48.5							Aldebaran	290°39.4	16°33.6					
11	316°37.2	310°05.4	40.0	207°35.9	05.4	246°18.8	46.8	325°15.8	48.4							Rigel	281°03.7	-8°10.5					
12	331°39.6	325°07.3	N08°40.8	222°38.5	N26°05.3	261°21.1	N21°46.8	340°18.0	S05°48.3							Capella	280°21.6	46°01.5					
13	346°42.1	340°09.3	41.5	237°41.0	05.3	276°23.4	46.8	355°20.2	48.2							Bellatrix	278°22.6	6°22.3					
14	1°44.6	355°11.2	42.2	252°43.6	05.2	291°25.7	46.9	10°22.4	48.1							Elnath	278°01.6	28°37.8					
15	16°47.0	10°13.1	· · 42.9	267°46.2	· · 05.1	306°28.0	· · 46.9	25°24.5	· · 47.9							Alnilam	275°37.5	-1°11.2					
16	31°49.5	25°15.0	43.6	282°48.7	05.1	321°30.3	46.9	40°26.7	47.8							Betelgeuse	270°51.8	7°24.7					
17	46°52.0	40°17.0	44.4	297°51.3	05.0	336°32.7	46.9	55°28.9	47.7							Canopus	263°52.0	-52°42.7					
18	61°54.4	55°18.9	N08°45.1	312°53.8	N26°05.0	351°35.0	N21°47.0	70°31.1	S05°47.6							Sirius	258°25.9	-16°45.2					
19	76°56.9	70°20.8	45.8	327°56.4	04.9	6°37.3	47.0	85°33.3	47.5							Adhara	255°05.5	-29°00.5					
20	91°59.4	85°22.8	46.5	342°58.9	04.8	21°39.6	47.0	100°35.5	47.3							Procyon	244°50.5	5°09.6					
21	107°01.8	100°24.7	· · 47.2	358°01.5	· · 04.8	36°41.9	· · 47.0	115°37.7	· · 47.2							Pollux	243°16.9	27°57.9					
22	122°04.3	115°26.7	48.0	13°04.0	04.7	51°44.2	47.1	130°39.9	47.1							Avior	234°14.2	-59°35.5					
23	137°06.7	130°28.6	48.7	28°06.6	04.6	66°46.5	47.1	145°42.0	47.0							Suhail	222°45.8	-43°32.1					
Mer.pass. 13:53				ν 1.9' d 0.7' m -4.85				ν 2.6' d -0.1' m -0.50				ν 2.3' d 0.0' m -2.36				ν 2.2' d -0.1' m 1.12							

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec												
0	152°09.2	145°30.6	N08°49.4	43°09.1	N26°04.6	81°48.8	N21°47.1	160°44.2	S05°46.9														
1	167°11.7	160°32.5	50.1	58°11.7	04.5	96°51.2	47.2	175°46.4	46.7														
2	182°14.1	175°34.5	50.8	73°14.2	04.4	111°53.5	47.2	190°48.6	46.6														
3	197°16.6	190°36.4	· · 51.5	88°16.8	· · 04.4	126°55.8	· · 47.2	205°50.8	· · 46.5														
4	212°19.1	205°38.4	52.2	103°19.3	04.3	141°58.1	47.2	220°53.0	46.4														
5	227°21.5	220°40.4	52.9	118°21.8	04.2	157°00.4	47.3	235°55.2	46.3														
6	242°24.0	235°42.3	N08°53.6	133°24.4	N26°04.2	172°02.7	N21°47.3	250°57.4	S05°46.2														
7	257°26.5	250°44.3	54.3	148°26.9	04.1	187°05.0	47.3	265°59.5	46.0														
8	272°28.9	265°46.3	55.0	163°29.5	04.0	202°07.3	47.3	281°01.7	45.9														
9	287°31.4	280°48.3	· · 55.7	178°32.0	· · 04.0	217°09.6	· · 47.4	296°03.9	· · 45.8														
10	302°33.9	295°50.3	56.4	193°34.5	03.9	232°11.9	47.4	311°06.1	45.7														
11	317°36.3	310°52.2	57.1	208°37.1	03.8	247°14.3	47.4	326°08.3	45.6														
12	332°38.8	325°54.2	N08°57.8	223°39.6	N26°03.8	262°16.6	N21°47.5	341°10.5	S05°45.4														
13	347°41.2	340°56.2	58.5	238°42.1	03.7	277°18.9	47.5	356°12.7	45.3														
14	2°43.7	355°58.2	59.2	253°44.6	03.6	292°21.2	47.5	11°14.8	45.2														
15	17°46.2	11°00.2	08°59.9	268°47.2	· · 03.6	307°23.5	· · 47.5	26°17.0	· · 45.1														
16	32°48.6	26°02.2	09°00.6	283°49.7	03.5	322°25.8	47.6	41°19.2	45.0														
17	47°51.1	41°04.2	01.3	298°52.2	03.4	337°28.1	47.6	56°21.4	44.8														
18	62°53.6	56°06.2	N09°01.9	313°54.7	N26°03.4	352°30.4	N21°47.6	71°23.6	S05°44.7														
19	77°56.0	71°08.3	02.6	328°57.3	03.3	7°32.7	47.7	86°25.8	44.6														
20	92°58.5	86°10.3	03.3	343°59.8	03.2	22°35.0	47.7	101°28.0	44.5														
21	108°01.0	101°12.3	· · 04.0	359°02.3	· · 03.2	37°37.3	· · 47.7	116°30.1	· · 44.4														
22	123°03.4	116°14.3	04.7	14°04.8	03.1	52°39.6	47.7	131°32.3	44.3														
23	138°05.9	131°16.3	05.3	29°07.3	03.0	67°41.9	47.8	146°34.5	44.1														
Mer.pass. 13:49				ν 2.0' d 0.7' m -4.85				ν 2.5' d -0.1' m -0.47				ν 2.3' d 0.0' m -2.35				ν 2.2' d -0.1' m 1.12							

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	153°08.4	146°18.4	N09°06.0	44°09.8	N26°03.0	82°44.2	N21°47.8	161°36.7	S05°44.0		
1	168°10.8	161°20.4	06.7	59°12.3	02.9	97°46.5	47.8	176°38.9	43.9		
2	183°13.3	176°22.4	07.4	74°14.9	02.8	112°48.8	47.9	191°41.1	43.8		
3	198°15.7	191°24.5	· · 08.0	89°17.4	· · 02.8	127°51.1	· · 47.9	206°43.3	· · 43.7		
4	213°18.2	206°26.5	08.7	104°19.9	02.7	142°53.4	47.9	221°45.4	43.5		
5	228°20.7	221°28.6	09.4	119°22.4	02.6	157°55.7	47.9	236°47.6	43.4		
6	243°23.1	236°30.6	N09°10.1	134°24.9	N26°02.5	172°58.0	N21°48.0	251°49.8	S05°43.3		
7	258°25.6	251°32.7	10.7	149°27.4	02.5	188°00.3	48.0	266°52.0	43.2		
8	273°28.1	266°34.7	11.4	164°29.9	02.4	203°02.6	48.0	281°54.2	43.1		
9	288°30.5	281°36.8	· · 12.1	179°32.4	· · 02.3	218°04.9	· · 48.1	296°56.4	· · 42.9		
10	303°33.0	296°38.9	12.7	194°34.9	02.3	233°07.2	48.1	311°58.6	42.8		
11	318°35.5	311°40.9	13.4	209°37.4	02.2	248°09.5	48.1	327°00.7	42.7		
12	333°37.9	326°43.0	N09°14.0	224°39.9	N26°02.1	263°11.8	N21°48.1	342°02.9	S05°42.6		
13	348°40.4	341°45.1	14.7	239°42.4	02.0	278°14.1	48.2	357°05.1	42.5		
14	3°42.9	356°47.1	15.3	254°44.9	02.0	293°16.4	48.2	12°07.3	42.3		
15	18°45.3	11°49.2	· · 16.0	269°47.4	· · 01.9	308°18.7	· · 48.2	27°09.5	· · 42.2		
16	33°47.8	26°51.3	16.7	284°49.8	01.8	323°21.0	48.3	42°11.7	42.1		
17	48°50.2	41°53.4	17.3	299°52.3	01.8	338°23.3	48.3	57°13.9	42.0		
18	63°52.7	56°55.5	N09°18.0	314°54.8	N26°01.7	353°25.6	N21°48.3	72°16.0	S05°41.9		
19	78°55.2	71°57.6	18.6	329°57.3	01.6	8°27.9	48.3	87°18.2	41.8		
20	93°57.6	86°59.7	19.3	344°59.8	01.5	23°30.2	4				

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	176°36.2	S10°32.9	268°26.5	10.5'	S26°01.3	6.5'	54.8'
1	191°36.2	32.0	282°56.1	10.4'	26°07.8	6.4'	54.8'
2	206°36.3	31.1	297°25.5	10.4'	26°14.2	6.3'	54.9'
3	221°36.4	· · 30.2	311°54.9	10.3'	26°20.4	6.1'	54.9'
4	236°36.5	29.2	326°24.1	10.2'	26°26.6	6.0'	54.9'
5	251°36.5	28.3	340°53.3	10.1'	26°32.6	5.9'	54.9'
6	266°36.6	S10°27.4	355°22.4	10.0'	S26°38.5	5.8'	54.9'
7	281°36.7	26.5	9°51.5	9.9'	26°44.2	5.6'	55.0'
8	296°36.8	25.6	24°20.4	9.9'	26°49.9	5.5'	55.0'
9	311°36.8	· · 24.7	38°49.2	9.8'	26°55.4	5.4'	55.0'
10	326°36.9	23.8	53°18.0	9.7'	27°00.8	5.3'	55.0'
11	341°37.0	22.9	67°46.7	9.6'	27°06.0	5.1'	55.1'
12	356°37.1	S10°22.0	82°15.3	9.5'	S27°11.1	5.0'	55.1'
13	11°37.1	21.1	96°43.8	9.4'	27°16.1	4.9'	55.1'
14	26°37.2	20.2	111°12.3	9.4'	27°21.0	4.7'	55.1'
15	41°37.3	· · 19.3	125°40.6	9.3'	27°25.7	4.6'	55.2'
16	56°37.4	18.4	140°08.9	9.2'	27°30.3	4.5'	55.2'
17	71°37.4	17.4	154°37.1	9.1'	27°34.7	4.3'	55.2'
18	86°37.5	S10°16.5	169°05.2	9.0'	S27°39.1	4.2'	55.2'
19	101°37.6	15.6	183°33.3	9.0'	27°43.2	4.0'	55.3'
20	116°37.7	14.7	198°01.2	8.9'	27°47.3	3.9'	55.3'
21	131°37.8	· · 13.8	212°29.1	8.8'	27°51.2	3.8'	55.3'
22	146°37.8	12.9	226°57.0	8.7'	27°55.0	3.6'	55.3'
23	161°37.9	12.0	241°24.7	8.7'	27°58.6	3.5'	55.4'
SD = 16.2' $d = -0.9'$ SD = 15.0'							
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	176°38.0	S10°11.1	255°52.4	8.6'	S28°02.1	3.3'	55.4'
1	191°38.1	10.2	270°20.0	8.5'	28°05.4	3.2'	55.4'
2	206°38.2	09.2	284°47.5	8.5'	28°08.6	3.0'	55.4'
3	221°38.2	· · 08.3	299°14.9	8.4'	28°11.6	2.9'	55.5'
4	236°38.3	07.4	313°42.3	8.3'	28°14.5	2.8'	55.5'
5	251°38.4	06.5	328°09.7	8.3'	28°17.3	2.6'	55.5'
6	266°38.5	S10°05.6	342°36.9	8.2'	S28°19.9	2.5'	55.6'
7	281°38.6	04.7	357°04.1	8.1'	28°22.4	2.3'	55.6'
8	296°38.6	03.8	11°31.2	8.1'	28°24.7	2.2'	55.6'
9	311°38.7	· · 02.9	25°58.3	8.0'	28°26.8	2.0'	55.6'
10	326°38.8	01.9	40°25.3	7.9'	28°28.8	1.9'	55.7'
11	341°38.9	01.0	54°52.2	7.9'	28°30.7	1.7'	55.7'
12	356°39.0	S10°00.1	69°19.1	7.8'	S28°32.4	1.6'	55.7'
13	11°39.0	09°59.2	83°45.9	7.8'	28°34.0	1.4'	55.8'
14	26°39.1	58.3	98°12.6	7.7'	28°35.4	1.2'	55.8'
15	41°39.2	· · 57.4	112°39.3	7.6'	28°36.6	1.1'	55.8'
16	56°39.3	56.5	127°06.0	7.6'	28°37.7	0.9'	55.9'
17	71°39.4	55.5	141°32.6	7.5'	28°38.6	0.8'	55.9'
18	86°39.5	S09°54.6	155°59.1	7.5'	S28°39.4	0.6'	55.9'
19	101°39.5	53.7	170°25.6	7.4'	28°40.0	0.5'	56.0'
20	116°39.6	52.8	184°52.0	7.4'	28°40.5	0.3'	56.0'
21	131°39.7	· · 51.9	199°18.4	7.3'	28°40.8	0.1'	56.0'
22	146°39.8	51.0	213°44.8	7.3'	28°40.9	-0.0'	56.0'
23	161°39.9	50.0	228°11.1	7.3'	28°40.9	-0.2'	56.1'
SD = 16.2' $d = -0.9'$ SD = 15.1'							
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	176°40.0	S09°49.1	242°37.3	7.2'	S28°40.7	-0.3'	56.1'
1	191°40.1	48.2	257°03.6	7.2'	28°40.4	-0.5'	56.1'
2	206°40.1	47.3	271°29.7	7.1'	28°39.9	-0.7'	56.2'
3	221°40.2	· · 46.4	285°55.9	7.1'	28°39.2	-0.8'	56.2'
4	236°40.3	45.4	300°22.0	7.1'	28°38.4	-1.0'	56.2'
5	251°40.4	44.5	314°48.0	7.0'	28°37.4	-1.2'	56.3'
6	266°40.5	S09°43.6	329°14.1	7.0'	S28°36.3	-1.3'	56.3'
7	281°40.6	42.7	343°40.1	7.0'	28°35.0	-1.5'	56.3'
8	296°40.7	41.8	358°06.1	6.9'	28°33.5	-1.6'	56.4'
9	311°40.7	· · 40.8	12°32.0	6.9'	28°31.8	-1.8'	56.4'
10	326°40.8	39.9	26°57.9	6.9'	28°30.0	-2.0'	56.4'
11	341°40.9	39.0	41°23.8	6.9'	28°28.1	-2.1'	56.5'
12	356°41.0	S09°38.1	55°49.7	6.8'	S28°25.9	-2.3'	56.5'
13	11°41.1	37.2	70°15.5	6.8'	28°23.6	-2.5'	56.5'
14	26°41.2	36.2	84°41.4	6.8'	28°21.1	-2.6'	56.6'
15	41°41.3	· · 35.3	99°07.2	6.8'	28°18.5	-2.8'	56.6'
16	56°41.4	34.4	113°33.0	6.8'	28°15.7	-3.0'	56.7'
17	71°41.5	33.5	127°58.7	6.8'	28°12.7	-3.1'	56.7'
18	86°41.5	S09°32.5	142°24.5	6.8'	S28°09.6	-3.3'	56.7'
19	101°41.6	31.6	156°50.2	6.7'	28°06.3	-3.5'	56.8'
20	116°41.7	30.7	171°16.0	6.7'	28°02.8	-3.6'	56.8'
21	131°41.8	· · 29.8	185°41.7	6.7'	27°59.1	-3.8'	56.8'
22	146°41.9	28.9	200°07.5	6.7'	27°55.3	-4.0'	56.9'
23	161°42.0	27.9	214°33.2	6.7'	27°51.3	-4.1'	56.9'
SD = 16.2' $d = -0.9'$ SD = 15.3'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:42	07:00	08:13	16:16	17:28	18:47
N 70°	05:44	06:54	07:59	16:29	17:34	18:45
68°	05:45	06:49	07:48	16:41	17:39	18:44
66°	05:46	06:45	07:38	16:50	17:43	18:43
64°	05:47	06:42	07:31	16:57	17:47	18:42
62°	05:47	06:38	07:24	17:04	17:50	18:41
60°	05:47	06:35	07:18	17:10	17:53	18:41
N 58°	05:47	06:33	07:13	17:15	17:55	18:41
56°	05:47	06:30	07:08	17:19	17:57	18:41
54°	05:47	06:28	07:04	17:24	18:00	18:41
52°	05:47	06:26	07:00	17:27	18:02	18:41
50°	05:47	06:24	06:57	17:31	18:03	18:41
45°	05:46	06:20	06:50	17:38	18:08	18:42
N 40°	05:44	06:16	06:43	17:44	18:12	18:43
35°	05:43	06:12	06:38	17:49	18:15	18:44
30°	05:41	06:09	06:33	17:54	18:18	18:46
20°	05:37	06:02	06:25	18:02	18:25	18:50
N 10°	05:31	05:56	06:17	18:10	18:31	18:56
0°	05:25	05:49	06:10	18:17	18:38	19:02
S 10°	05:17	05:41	06:03	18:24	18:45	19:10
20°	05:06	05:32	05:55	18:32	18:54	19:20
30°	04:52	05:21	05:46	18:40	19:05	19:34
35°	04:43	05:14	05:41	18:46	19:12	19:43
40°	04:33	05:06	05:35	18:51	19:20	19:53
45°	04:19	04:57	05:28	18:58	19:29	20:06
S 50°	04:02	04:45	05:19	19:07	19:41	20:23
52°	03:54	04:39	05:15	19:11	19:46	20:31
54°	03:45	04:33	05:11	19:15	19:53	20:40
56°	03:34	04:25	05:06	19:20	20:00	20:51
58°	03:21	04:17	05:00	19:25	20:08	21:04
S 60°	03:05	04:08	04:54	19:31	20:17	21:19

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	■	■	■	■	■	■
64°	■	■	■	■	■	■
62°	04:59	■	■	07:32	■	■
60°	04:15	05:48	06:51	08:16	08:34	09:27
N 58°	03:46	05:07	06:08	08:46	09:14	10:09
56°	03:24	04:40	05:40	09:08	09:42	10:38
54°	03:05	04:18	05:18	09:27	10:04	11:00
52°	02:50	04:00	04:59	09:43	10:22	11:18
50°	02:37	03:45	04:44	09:56	10:37	11:34
45°	02:10	03:14	04:12	10:24	11:08	12:05
N 40°	01:48	02:50	03:48	10:46	11:32	12:29
35°	01:30	02:31	03:28	11:04	11:52	12:49
30°	01:15	02:14	03:11	11:20	12:09	13:06
20°	00:49	01:46	02:42	11:47	12:38	13:34
N 10°	00:27	01:22	02:17	12:10	13:02	13:59
0°	00:07	00:59	01:54	12:32	13:25	14:22
S 10°	· · ·	00:37	01:31	12:53	13:48	14:44
20°	· · ·	00:13	01:07	13:16	14:13	15:08
30°	23:45	· · ·	00:38	13:43	14:41	15:37
35°	23:29	· · ·	00:21	13:59	14:58	15:53
40°	23:09	· · ·	00:01	14:18	15:18	16:13
45°	22:46	23:37	· · ·	14:41	15:42	16:36
S 50°	22:17	23:06	· · ·	15:10	16:13	17:06
52°	22:03	22:50	23:55	15:24	16:29	17:21
54°	21:46	22:32	23:38	15:40	16:47	17:38
56°	21:26	22:10	23:18	16:00	17:09	17:59
58°	21:01	21:42	22:52	16:25	17:37	18:25
S 60°	20:27	21:01	22:17	16:58	18:19	19:00

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Upper	Lower	23-25
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	48-28%
21	13:35	13:32	12:14	06:19	18:45	
22	13:28	13:24	12:13	07:12	19:40	
23	13:20	13:16	12:13	08:08	20:36	

February 24, 25, 26 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	154°07.5	147°08.1	N09°21.8	45°09.7	N26°01.2	83°39.4	N21°48.5	162°29.2	S05°41.2	Alpheratz	357°35.0	29°13.7
1	169°10.0	162°10.2	22.5	60°12.2	01.2	98°41.7	48.5	177°31.3	41.0	Ankaa	353°07.4	-42°10.3
2	184°12.4	177°12.3	23.1	75°14.7	01.1	113°44.0	48.6	192°33.5	40.9	Schedar	349°31.4	56°40.6
3	199°14.9	192°14.4	•• 23.7	90°17.1	•• 01.0	128°46.3	•• 48.6	207°35.7	•• 40.8	Diphda	348°47.4	-17°51.1
4	214°17.4	207°16.6	24.4	105°19.6	00.9	143°48.6	48.6	222°37.9	40.7	Achernar	335°20.4	-57°06.8
5	229°19.8	222°18.7	25.0	120°22.1	00.9	158°50.9	48.6	237°40.1	40.6	Hamal	327°51.3	23°34.9
6	244°22.3	237°20.8	N09°25.6	135°24.6	N26°00.8	173°53.1	N21°48.7	252°42.3	S05°40.4	Polaris	314°08.3	89°22.5
7	259°24.7	252°23.0	26.3	150°27.0	00.7	188°55.4	48.7	267°44.5	40.3	Acamar	315°11.8	-40°12.5
8	274°27.2	267°25.1	26.9	165°29.5	00.6	203°57.7	48.7	282°46.6	40.2	Menkar	314°06.1	4°11.2
9	289°29.7	282°27.2	•• 27.5	180°32.0	•• 00.6	219°00.0	•• 48.8	297°48.8	•• 40.1	Mirfak	308°28.2	49°57.2
10	304°32.1	297°29.4	28.2	195°34.4	00.5	234°02.3	48.8	312°51.0	40.0	Aldebaran	290°39.5	16°33.6
11	319°34.6	312°31.5	28.8	210°36.9	00.4	249°04.6	48.8	327°53.2	39.8	Rigel	281°03.7	-8°10.5
12	334°37.1	327°33.7	N09°29.4	225°39.3	N26°00.3	264°06.9	N21°48.8	342°55.4	S05°39.7	Capella	280°21.6	46°01.5
13	349°39.5	342°35.8	30.0	240°41.8	00.3	279°09.2	48.9	357°57.6	39.6	Bellatrix	278°22.7	6°22.3
14	4°42.0	357°38.0	30.6	255°44.3	00.2	294°11.5	48.9	12°59.7	39.5	Elnath	278°01.6	28°37.8
15	19°44.5	12°40.1	•• 31.3	270°46.7	•• 00.1	309°13.8	•• 48.9	28°01.9	•• 39.4	Alnilam	275°37.5	-1°11.2
16	34°46.9	27°42.3	31.9	285°49.2	00.0	324°16.1	49.0	43°04.1	39.2	Betelgeuse	270°51.8	7°24.7
17	49°49.4	42°44.5	32.5	300°51.6	26°00.0	339°18.3	49.0	58°06.3	39.1	Canopus	263°52.1	-52°42.8
18	64°51.8	57°46.7	N09°33.1	315°54.1	N25°59.9	354°20.6	N21°49.0	73°08.5	S05°39.0	Sirius	258°25.9	-16°45.2
19	79°54.3	72°48.8	33.7	330°56.5	59.8	9°22.9	49.1	88°10.7	38.9	Adhara	255°05.5	-29°00.5
20	94°56.8	87°51.0	34.3	345°59.0	59.7	24°25.2	49.1	103°12.9	38.8	Procyon	244°50.5	5°09.5
21	109°59.2	102°53.2	•• 34.9	1°01.4	•• 59.7	39°27.5	•• 49.1	118°15.0	•• 38.7	Pollux	243°16.9	27°57.9
22	125°01.7	117°55.4	35.5	16°03.9	59.6	54°29.8	49.2	133°17.2	38.5	Avior	234°14.2	-59°35.5
23	140°04.2	132°57.6	36.1	31°06.3	59.5	69°32.1	49.2	148°19.4	38.4	Suhail	222°45.8	-43°32.2
Mer.pass. 13:41		ν 2.1' d 0.6' m -4.84		ν 2.5' d -0.1' m -0.42		ν 2.3' d 0.0' m -2.34		ν 2.2' d -0.1' m 1.12				

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	155°06.6	147°59.8	N09°36.8	46°08.8	N25°59.4	84°34.4	N21°49.2	163°21.6	S05°38.3
1	170°09.1	163°01.9	37.4	61°11.2	59.3	99°36.6	49.2	178°23.8	38.2
2	185°11.6	178°04.1	38.0	76°13.7	59.3	114°38.9	49.3	193°26.0	38.1
3	200°14.0	193°06.3	•• 38.6	91°16.1	•• 59.2	129°41.2	•• 49.3	208°28.1	•• 37.9
4	215°16.5	208°08.6	39.1	106°18.5	59.1	144°43.5	49.3	223°30.3	37.8
5	230°19.0	223°10.8	39.7	121°21.0	59.0	159°45.8	49.4	238°32.5	37.7
6	245°21.4	238°13.0	N09°40.3	136°23.4	N25°58.9	174°48.1	N21°49.4	253°34.7	S05°37.6
7	260°23.9	253°15.2	40.9	151°25.8	58.9	189°50.4	49.4	268°36.9	37.5
8	275°26.3	268°17.4	41.5	166°28.3	58.8	204°52.6	49.5	283°39.1	37.3
9	290°28.8	283°19.6	•• 42.1	181°30.7	•• 58.7	219°54.9	•• 49.5	298°41.2	•• 37.2
10	305°31.3	298°21.9	42.7	196°33.1	58.6	234°57.2	49.5	313°43.4	37.1
11	320°33.7	313°24.1	43.3	211°35.6	58.6	249°59.5	49.5	328°45.6	37.0
12	335°36.2	328°26.3	N09°43.9	226°38.0	N25°58.5	265°01.8	N21°49.6	343°47.8	S05°36.9
13	350°38.7	343°28.6	44.1	241°40.4	58.4	280°04.1	49.6	358°50.0	36.7
14	5°41.1	358°30.8	45.0	256°42.9	58.3	295°06.3	49.6	13°52.2	36.6
15	20°43.6	13°33.0	•• 45.6	271°45.3	•• 58.2	310°08.6	•• 49.7	28°54.4	•• 36.5
16	35°46.1	28°35.3	46.2	286°47.7	58.1	325°10.9	49.7	43°56.5	36.4
17	50°48.5	43°37.5	46.8	301°50.1	58.1	340°13.2	49.7	58°58.7	36.3
18	65°51.0	58°39.8	N09°47.3	316°52.5	N25°58.0	355°15.5	N21°49.8	74°00.9	S05°36.1
19	80°53.5	73°42.1	47.9	331°55.0	57.9	10°17.7	49.8	89°03.1	36.0
20	95°55.9	88°44.3	48.5	346°57.4	57.8	25°20.0	49.8	104°05.3	35.9
21	110°58.4	103°46.6	•• 49.0	1°59.8	•• 57.7	40°22.3	•• 49.9	119°07.5	•• 35.8
22	126°00.8	118°48.9	49.6	17°02.2	57.7	55°24.6	49.9	134°09.6	35.7
23	141°03.3	133°51.1	50.2	32°04.6	57.6	70°26.9	49.9	149°11.8	35.5
Mer.pass. 13:37		ν 2.2' d 0.6' m -4.83		ν 2.4' d -0.1' m -0.39		ν 2.3' d 0.0' m -2.33		ν 2.2' d -0.1' m 1.11	

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	156°05.8	148°53.4	N09°50.7	47°07.0	N25°57.5	85°29.1	N21°50.0	164°14.0	S05°35.4
1	171°08.2	163°55.7	51.3	62°09.4	57.4	100°31.4	50.0	179°16.2	35.3
2	186°10.7	178°58.0	51.9	77°11.8	57.3	115°33.7	50.0	194°18.4	35.2
3	201°13.2	194°00.3	•• 52.4	92°14.2	•• 57.2	130°36.0	•• 50.0	209°20.6	•• 35.1
4	216°15.6	209°02.5	53.0	107°16.7	57.2	145°38.2	50.1	224°22.7	34.9
5	231°18.1	224°04.8	53.5	122°19.1	57.1	160°40.5	50.1	239°24.9	34.8
6	246°20.6	239°07.1	N09°54.1	137°21.5	N25°57.0	175°42.8	N21°50.1	254°27.1	S05°34.7
7	261°23.0	254°09.4	54.6	152°23.9	56.9	190°45.1	50.2	269°29.3	34.6
8	276°25.5	269°11.7	55.2	167°26.3	56.8	205°47.3	50.2	284°31.5	34.5
9	291°28.0	284°14.1	•• 55.7	182°28.7	•• 56.7	220°49.6	•• 50.2	299°33.7	•• 34.3
10	306°30.4	299°16.4	56.3	197°31.1	56.7	235°51.9	50.3	314°35.8	34.2
11	321°32.9	314°18.7	56.8	212°33.5	56.6	250°54.2	50.3	329°38.0	34.1
12	336°35.3	329°21.0	N09°57.4	227°35.8	N25°56.5	265°56.4	N21°50.3	344°40.2	S05°34.0
13	351°37.8	344°23.3	57.9	242°38.2	56.4	280°58.7	50.4	359°42.4	33.9
14	6°40.3	359°25.6	58.5	257°40.6	56.3	296°01.0	50.4	14°44.6	33.7
15	21°42.7	14°28.0	•• 59.0	272°43.0	•• 56.2	311°03.3	•• 50.4	29°46.8	•• 33.6
16	36°45.2	29°30.3	09°59.5	287°45.4	56.2	326°05.5	50.5	44°48.9	33.5
17	51°47.7	44°32.6	10°00.1	302°47.8	56.1	341°07.8	50.5	59°51.1	33.4
18	66°50.1	59°35.0	N10°00.6	317°50.2	N25°56.0	356°10.1	N21°50.5	74°53.3	S05°33.3
19	81°52.6	74°37.3	01.1	332°52.6	55.9	11°12.4	50.6	89°55.5	33.1
20	96°55.1	89°39.7	01.7	347°55.0	55.8	26°14.6	50.6	104°57.7	33.0
21	111°57.5	104°42.0	•• 02.2	2°57.3	•• 55.7	41°16.9	•• 50.6	119°59.9	•• 32.9
22	127°00.0	119°44.4	02.7	17°59.7	55.6	56°19.2	50.6	135°02.0	32.8
23	142°02.4	134°46.7	03.2	33°02.1	55.6	71°21.4	50.7	150°04.2	32.7
Mer.pass. 13:33		ν 2.3' d 0.6' m -4.82		ν 2.4' d -0.1' m -0.36		ν 2.3' d 0.0' m -2.32		ν 2.2' d -0.1' m 1.11	

Feb 24 Mon	SHA	Mer.pass
Venus	353°00.6	14:09
Mars	251°02.2	20:56
Jupiter	289°31.9	18:23
Saturn	8°21.7	13:08

Feb 25 Tue	SHA	Mer.pass
Venus	352°53.1	14:06
Mars	251°02.1	20:52
Jupiter	289°27.7	18:19
Saturn	8°15.0	13:05

Feb 26 Wed	SHA	Mer.pass
Venus	352°47.6	14:02
Mars	251°01.2	20:48
Jupiter	289°23.4	18:15
Saturn	8°08.2	13:01

Horizontal parallax		
Venus:	0.4	
Mars:	0.2	

DUT1 = UT1-UTC = +0.0464 sec ΔT = TT-UT1 = +69.1376 sec

2025 February 24 to Feb. 26 UT

h	Sun				Moon			
Mon	GHA	Dec	GHA	ν	Dec	d	HP	
0	176°42.1	S09°27.0	228°58.9	6.7'	S27°47.2	-4.3'	56.9'	
1	191°42.2	26.1	243°24.6	6.7'	27°42.9	-4.5'	57.0'	
2	206°42.3	25.2	257°50.3	6.7'	27°38.4	-4.7'	57.0'	
3	221°42.4	· · 24.2	272°16.0	6.7'	27°33.7	-4.8'	57.0'	
4	236°42.5	23.3	286°41.8	6.7'	27°28.9	-5.0'	57.1'	
5	251°42.6	22.4	301°07.5	6.7'	27°23.9	-5.2'	57.1'	
6	266°42.6	S09°21.5	315°33.2	6.7'	S27°18.8	-5.3'	57.2'	
7	281°42.7	20.5	329°59.0	6.8'	27°13.5	-5.5'	57.2'	
8	296°42.8	19.6	344°24.7	6.8'	27°08.0	-5.7'	57.2'	
9	311°42.9	· · 18.7	358°50.5	6.8'	27°02.3	-5.8'	57.3'	
10	326°43.0	17.8	13°16.2	6.8'	26°56.5	-6.0'	57.3'	
11	341°43.1	16.8	27°42.0	6.8'	26°50.5	-6.2'	57.3'	
12	356°43.2	S09°15.9	42°07.8	6.8'	S26°44.3	-6.3'	57.4'	
13	11°43.3	15.0	56°33.7	6.8'	26°38.0	-6.5'	57.4'	
14	26°43.4	14.0	70°59.5	6.9'	26°31.5	-6.6'	57.4'	
15	41°43.5	· · 13.1	85°25.4	6.9'	26°24.9	-6.8'	57.5'	
16	56°43.6	12.2	99°51.2	6.9'	26°18.1	-7.0'	57.5'	
17	71°43.7	11.3	114°17.1	6.9'	26°11.1	-7.1'	57.6'	
18	86°43.8	S09°10.3	128°43.1	7.0'	S26°04.0	-7.3'	57.6'	
19	101°43.9	09.4	143°09.0	7.0'	25°56.7	-7.5'	57.6'	
20	116°44.0	08.5	157°35.0	7.0'	25°49.2	-7.6'	57.7'	
21	131°44.1	· · 07.5	172°01.0	7.0'	25°41.6	-7.8'	57.7'	
22	146°44.2	06.6	186°27.0	7.1'	25°33.8	-8.0'	57.7'	
23	161°44.3	05.7	200°53.1	7.1'	25°25.8	-8.1'	57.8'	
	SD = 16.2'	$d = -0.9'$	SD = 15.5'					

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	176° 44.4	S09° 04.8	215° 19.2	7.1'	S25° 17.7	-8.3'	57.8'
1	191° 44.5	03.8	229° 45.3	7.2'	25° 09.4	-8.4'	57.9'
2	206° 44.6	02.9	244° 11.5	7.2'	25° 01.0	-8.6'	57.9'
3	221° 44.7	· · 02.0	258° 37.7	7.2'	24° 52.4	-8.7'	57.9'
4	236° 44.8	01.0	273° 03.9	7.3'	24° 43.7	-8.9'	58.0'
5	251° 44.9	09° 00.1	287° 30.2	7.3'	24° 34.8	-9.1'	58.0'
6	266° 45.0	S08° 59.2	301° 56.5	7.3'	S24° 25.7	-9.2'	58.0'
7	281° 45.0	58.2	316° 22.9	7.4'	24° 16.5	-9.4'	58.1'
8	296° 45.1	57.3	330° 49.3	7.4'	24° 07.1	-9.5'	58.1'
9	311° 45.2	· · 56.4	345° 15.7	7.5'	23° 57.6	-9.7'	58.1'
10	326° 45.3	55.4	359° 42.2	7.5'	23° 47.9	-9.8'	58.2'
11	341° 45.4	54.5	14° 08.7	7.6'	23° 38.1	-10.0'	58.2'
12	356° 45.5	S08° 53.6	28° 35.2	7.6'	S23° 28.1	-10.1'	58.3'
13	11° 45.6	52.6	43° 01.8	7.6'	23° 18.0	-10.3'	58.3'
14	26° 45.7	51.7	57° 28.5	7.7'	23° 07.7	-10.4'	58.3'
15	41° 45.9	· · 50.8	71° 55.2	7.7'	22° 57.3	-10.6'	58.4'
16	56° 46.0	49.8	86° 21.9	7.8'	22° 46.7	-10.7'	58.4'
17	71° 46.1	48.9	100° 48.7	7.8'	22° 36.0	-10.9'	58.4'
18	86° 46.2	S08° 48.0	115° 15.5	7.9'	S22° 25.1	-11.0'	58.5'
19	101° 46.3	47.0	129° 42.4	7.9'	22° 14.1	-11.2'	58.5'
20	116° 46.4	46.1	144° 09.3	8.0'	22° 02.9	-11.3'	58.5'
21	131° 46.5	· · 45.2	158° 36.3	8.0'	21° 51.6	-11.4'	58.6'
22	146° 46.6	44.2	173° 03.3	8.1'	21° 40.2	-11.6'	58.6'
23	161° 46.7	43.3	187° 30.4	8.1'	21° 28.6	-11.7'	58.6'
	SD = 16.2'	$d = -0.9'$			SD = 15.8'		

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	176°46.8	S08°42.4	201°57.6	8.2'	S21°16.9	-11.9'	58.7'
1	191°46.9	41.4	216°24.7	8.2'	21°05.1	-12.0'	58.7'
2	206°47.0	40.5	230°52.0	8.3'	20°53.1	-12.1'	58.7'
3	221°47.1	· · 39.6	245°19.2	8.3'	20°40.9	-12.3'	58.8'
4	236°47.2	38.6	259°46.6	8.4'	20°28.7	-12.4'	58.8'
5	251°47.3	37.7	274°14.0	8.4'	20°16.3	-12.5'	58.8'
6	266°47.4	S08°36.7	288°41.4	8.5'	S20°03.8	-12.7'	58.9'
7	281°47.5	35.8	303°08.9	8.5'	19°51.1	-12.8'	58.9'
8	296°47.6	34.9	317°36.4	8.6'	19°38.3	-12.9'	58.9'
9	311°47.7	· · 33.9	332°04.0	8.6'	19°25.4	-13.0'	59.0'
10	326°47.8	33.0	346°31.6	8.7'	19°12.4	-13.2'	59.0'
11	341°47.9	32.1	0°59.3	8.7'	18°59.2	-13.3'	59.0'
12	356°48.0	S08°31.1	15°27.1	8.8'	S18°46.0	-13.4'	59.1'
13	11°48.1	30.2	29°54.9	8.9'	18°32.6	-13.5'	59.1'
14	26°48.2	29.2	44°22.7	8.9'	18°19.0	-13.6'	59.1'
15	41°48.3	· · 28.3	58°50.6	9.0'	18°05.4	-13.8'	59.2'
16	56°48.5	27.4	73°18.6	9.0'	17°51.6	-13.9'	59.2'
17	71°48.6	26.4	87°46.6	9.1'	17°37.8	-14.0'	59.2'
18	86°48.7	S08°25.5	102°14.7	9.1'	S17°23.8	-14.1'	59.3'
19	101°48.8	24.5	116°42.8	9.2'	17°09.7	-14.2'	59.3'
20	116°48.9	23.6	131°10.9	9.2'	16°55.4	-14.3'	59.3'
21	131°49.0	· · 22.7	145°39.1	9.3'	16°41.1	-14.4'	59.4'
22	146°49.1	21.7	160°07.4	9.3'	16°26.7	-14.5'	59.4'
23	161°49.2	20.8	174°35.7	9.4'	16°12.1	-14.6'	59.4'
	SD = 16.1'	$d = -0.9'$			SD = 16.0'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:28	06:46	07:57	16:31	17:42	19:00
N 70°	05:31	06:42	07:45	16:43	17:46	18:57
68°	05:34	06:38	07:35	16:52	17:50	18:54
66°	05:36	06:35	07:27	17:00	17:53	18:52
64°	05:37	06:32	07:20	17:07	17:55	18:50
62°	05:38	06:29	07:15	17:13	17:58	18:49
60°	05:39	06:27	07:09	17:18	18:00	18:48
N 58°	05:40	06:25	07:05	17:22	18:02	18:47
56°	05:40	06:23	07:01	17:26	18:04	18:47
54°	05:41	06:22	06:57	17:30	18:05	18:46
52°	05:41	06:20	06:54	17:33	18:07	18:46
50°	05:41	06:18	06:51	17:36	18:08	18:46
45°	05:41	06:15	06:45	17:42	18:12	18:46
N 40°	05:40	06:12	06:39	17:47	18:15	18:46
35°	05:39	06:09	06:34	17:52	18:18	18:47
30°	05:38	06:06	06:30	17:56	18:20	18:48
20°	05:35	06:00	06:23	18:04	18:26	18:52
N 10°	05:30	05:55	06:16	18:10	18:31	18:56
0°	05:24	05:49	06:10	18:16	18:37	19:01
S 10°	05:17	05:42	06:03	18:23	18:44	19:09
20°	05:08	05:34	05:56	18:29	18:52	19:18
30°	04:55	05:24	05:48	18:37	19:02	19:30
35°	04:47	05:17	05:43	18:42	19:08	19:39
40°	04:37	05:10	05:38	18:47	19:15	19:48
45°	04:24	05:01	05:32	18:53	19:24	20:00
S 50°	04:09	04:50	05:24	19:01	19:34	20:16
52°	04:01	04:45	05:21	19:04	19:40	20:23
54°	03:52	04:39	05:17	19:08	19:45	20:32
56°	03:42	04:33	05:12	19:12	19:52	20:42
58°	03:30	04:25	05:08	19:17	19:59	20:53
S 60°	03:17	04:17	05:02	19:22	20:07	21:07

February 27, 28, 01 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	157°04.9	149°49.1	N10°03.8	48°04.5	N25°55.5	86°23.7	N21°50.7	165°06.4	S05°32.5	Alpheratz	357°35.0	29°13.7					
1	172°07.4	164°51.5	04.3	63°06.8	55.4	101°26.0	50.7	180°08.6	32.4	Ankaa	353°07.4	-42°10.3					
2	187°09.8	179°53.8	04.8	78°09.2	55.3	116°28.3	50.8	195°10.8	32.3	Schedar	349°31.4	56°40.6					
3	202°12.3	194°56.2	· · 05.3	93°11.6	· · 55.2	131°30.5	· · 50.8	210°13.0	· · 32.2	Diphda	348°47.4	-17°51.1					
4	217°14.8	209°58.6	05.8	108°14.0	55.1	146°32.8	50.8	225°15.1	32.1	Achernar	335°20.4	-57°06.8					
5	232°17.2	225°01.0	06.3	123°16.3	55.0	161°35.1	50.9	240°17.3	31.9	Hamal	327°51.3	23°34.9					
6	247°19.7	240°03.3	N10°06.9	138°18.7	N25°55.0	176°37.3	N21°50.9	255°19.5	S05°31.8	Polaris	314°09.8	89°22.5					
7	262°22.2	255°05.7	07.4	153°21.1	54.9	191°39.6	50.9	270°21.7	31.7	Acamar	315°11.8	-40°12.5					
8	277°24.6	270°08.1	07.9	168°23.4	54.8	206°41.9	51.0	285°23.9	31.6	Menkar	314°06.1	4°11.2					
9	292°27.1	285°10.5	· · 08.4	183°25.8	· · 54.7	221°44.1	· · 51.0	300°26.1	· · 31.5	Mirfak	308°28.2	49°57.2					
10	307°29.6	300°12.9	08.9	198°28.2	54.6	236°46.4	51.0	315°28.2	31.3	Aldebaran	290°39.5	16°33.6					
11	322°32.0	315°15.3	09.4	213°30.5	54.5	251°48.7	51.1	330°30.4	31.2	Rigel	281°03.7	-8°10.5					
12	337°34.5	330°17.7	N10°09.9	228°32.9	N25°54.4	266°50.9	N21°51.1	345°32.6	S05°31.1	Capella	280°21.6	46°01.5					
13	352°36.9	345°20.1	10.4	243°35.3	54.3	281°53.2	51.1	0°34.8	31.0	Bellatrix	278°22.7	6°22.3					
14	7°39.4	0°22.6	10.9	258°37.6	54.2	296°55.5	51.2	15°37.0	30.9	Elnath	278°01.6	28°37.8					
15	22°41.9	15°25.0	· · 11.4	273°40.0	· · 54.2	311°57.7	· · 51.2	30°39.2	· · 30.7	Alnilam	275°37.5	-1°11.2					
16	37°44.3	30°27.4	11.9	288°42.3	54.1	327°00.0	51.2	45°41.3	30.6	Betelgeuse	270°51.8	7°24.7					
17	52°46.8	45°29.8	12.4	303°44.7	54.0	342°02.3	51.3	60°43.5	30.5	Canopus	263°52.1	-52°42.8					
18	67°49.3	60°32.2	N10°12.9	318°47.0	N25°53.9	357°04.5	N21°51.3	75°45.7	S05°30.4	Sirius	258°26.0	-16°45.2					
19	82°51.7	75°34.7	13.3	333°49.4	53.8	12°06.8	51.3	90°47.9	30.3	Adhara	255°05.6	-29°00.6					
20	97°54.2	90°37.1	13.8	348°51.7	53.7	27°09.0	51.4	105°50.1	30.1	Procyon	244°50.5	5°09.5					
21	112°56.7	105°39.5	· · 14.3	3°54.1	· · 53.6	42°11.3	· · 51.4	120°52.2	· · 30.0	Pollux	243°16.9	27°57.9					
22	127°59.1	120°42.0	14.8	18°56.4	53.5	57°13.6	51.4	135°54.4	29.9	Avior	234°14.2	-59°35.5					
23	143°01.6	135°44.4	15.3	33°58.8	53.4	72°15.8	51.5	150°56.6	29.8	Suhail	222°45.8	-43°32.2					
Mer.pass. 13:29		ν 2.4' d 0.5' m -4.81			ν 2.4' d -0.1' m -0.33			ν 2.3' d 0.0' m -2.31			ν 2.2' d -0.1' m 1.11						

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	158°04.1	150°46.9	N10°15.8	49°01.1	N25°53.3	87°18.1	N21°51.5	165°58.8	S05°29.7				
1	173°06.5	165°49.3	16.2	64°03.5	53.3	102°20.4	51.5	181°01.0	29.5				
2	188°09.0	180°51.8	16.7	79°05.8	53.2	117°22.6	51.6	196°03.2	29.4				
3	203°11.4	195°54.3	· · 17.2	94°08.2	· · 53.1	132°24.9	· · 51.6	211°05.3	· · 29.3				
4	218°13.9	210°56.7	17.7	109°10.5	53.0	147°27.1	51.6	226°07.5	29.2				
5	233°16.4	225°59.2	18.1	124°12.8	52.9	162°29.4	51.7	241°09.7	29.1				
6	248°18.8	241°01.7	N10°18.6	139°15.2	N25°52.8	177°31.7	N21°51.7	256°11.9	S05°28.9				
7	263°21.3	256°04.1	19.1	154°17.5	52.7	192°33.9	51.7	271°14.1	28.8				
8	278°23.8	271°06.6	19.5	169°19.8	52.6	207°36.2	51.8	286°16.3	28.7				
9	293°26.2	286°09.1	· · 20.0	184°22.2	· · 52.5	222°38.4	· · 51.8	301°18.4	· · 28.6				
10	308°28.7	301°11.6	20.4	199°24.5	52.4	237°40.7	51.8	316°20.6	28.5				
11	323°31.2	316°14.1	20.9	214°26.8	52.3	252°43.0	51.9	331°22.8	28.3				
12	338°33.6	331°16.6	N10°21.4	229°29.2	N25°52.3	267°45.2	N21°51.9	346°25.0	S05°28.2				
13	353°36.1	346°19.1	21.8	244°31.5	52.2	282°47.5	51.9	1°27.2	28.1				
14	8°38.5	1°21.6	22.3	259°33.8	52.1	297°49.7	52.0	16°29.3	28.0				
15	23°41.0	16°24.1	· · 22.7	274°36.1	· · 52.0	312°52.0	· · 52.0	31°31.5	· · 27.9				
16	38°43.5	31°26.6	23.2	289°38.5	51.9	327°54.2	52.0	46°33.7	27.7				
17	53°45.9	46°29.1	23.6	304°40.8	51.8	342°56.5	52.1	61°35.9	27.6				
18	68°48.4	61°31.6	N10°24.1	319°43.1	N25°51.7	357°58.8	N21°52.1	76°38.1	S05°27.5				
19	83°50.9	76°34.1	24.5	334°45.4	51.6	13°01.0	52.1	91°40.3	27.4				
20	98°53.3	91°36.6	24.9	349°47.7	51.5	28°03.3	52.2	106°42.4	27.3				
21	113°55.8	106°39.2	· · 25.4	4°50.1	· · 51.4	43°05.5	· · 52.2	121°44.6	· · 27.1				
22	128°58.3	121°41.7	25.8	19°52.4	51.3	58°07.8	52.2	136°46.8	27.0				
23	144°00.7	136°44.2	26.3	34°54.7	51.2	73°10.0	52.3	151°49.0	26.9				
Mer.pass. 13:26		ν 2.5' d 0.5' m -4.80			ν 2.3' d -0.1' m -0.31			ν 2.3' d 0.0' m -2.31			ν 2.2' d -0.1' m 1.11		

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	159°03.2	151°46.8	N10°26.7	49°57.0	N25°51.1	88°12.3	N21°52.3	166°51.2	S05°26.8				
1	174°05.7	166°49.3	27.1	64°59.3	51.0	103°14.5	52.3	181°53.4	26.7				
2	189°08.1	181°51.9	27.5	80°01.6	50.9	118°16.8	52.4	196°55.5	26.5				
3	204°10.6	196°54.4	· · 28.0	95°03.9	· · 50.8	133°19.1	· · 52.4	211°57.7	· · 26.4				
4	219°13.0	211°57.0	28.4	110°06.2	50.8	148°21.3	52.4	226°59.9	26.3				
5	234°15.5	226°59.5	28.8	125°08.5	50.7	163°23.6	52.5	242°02.1	26.2				
6	249°18.0	242°02.1	N10°29.2	140°10.9	N25°50.6	178°25.8	N21°52.5	257°04.3	S05°26.1				
7	264°20.4	257°04.7	29.7	155°13.2	50.5	193°28.1	52.5	272°06.4	25.9				
8	279°22.9	272°07.2	30.1	170°15.5	50.4	208°30.3	52.6	287°08.6	25.8				
9	294°25.4	287°09.8	· · 30.5	185°17.8	· · 50.3	223°32.6	· · 52.6	302°10.8	· · 25.7				
10	309°27.8	302°12.4	30.9	200°20.1	50.2	238°34.8	52.6	317°13.0	25.6				
11	324°30.3	317°14.9	31.3	215°22.4	50.1	253°37.1	52.7	332°15.2	25.5				
12	339°32.8	332°17.5	N10°31.7	230°24.7	N25°50.0	268°39.3	N21°52.7	347°17.4	S05°25.3				
13	354°35.2	347°20.1	32.1	245°26.9	49.9	283°41.6	52.7	2°19.5	25.2				
14	9°37.7	2°22.7	32.5	260°29.2	49.8	298°43.8	52.8	17°21.7	25.1				
15	24°40.1	17°25.3	· · 32.9	275°31.5	· · 49.7	313°46.1	· · 52.8	32°23.9	· · 25.0				
16	39°42.6	32°27.9	33.3	290°33.8	49.6	328°48.3	52.8	47°26.1	24.9				
17	54°45.1	47°30.5	33.7	305°36.1	49.5	343°50.6	52.9	62°28.3	24.7				
18	69°47.5	62°33.1	N10°34.1	320°38.4	N25°49.4	358°52.8	N21°52.9	77°30.4	S05°24.6				
19	84°50.0	77°35.7	34.5	335°40.7	49.3	13°55.1	52.9	92°32.6	24.5				
20	99°52.5	92°38.3	34.9	350°43.0	49.2	28°57.3	53.0	107°34.8	24.4				
21	114°54.9	107°41.0	· · 35.3	5°45.3	· · 49.1	43°59.6	· · 53.0	122°37.0	· · 24.3				
22	129°57.4	122°43.6	35.7	20°47.5	49.0	59°01.8	53.0	137°39.2	24.1				
23	144°59.9	137°46.2	36.1	35°49.8	48.9	74°04.1	53.1	152°41.4	24.0				
Mer.pass. 13:22		ν 2.5' d 0.4' m -4.78			ν 2.3' d -0.1' m -0.28			ν 2.3' d 0.0' m -2.30			ν 2.2' d -0.1' m 1.11		

Feb 27 Thu	SHA	Mer.pass
Venus	352°44.2	13:58
Mars	250°59.6	20:44
Jupiter	289°18.8	18:12
Saturn	8°01.5	12:58

Feb 28 Fri	SHA	Mer.pass
Venus	352°42.8	13:55
Mars	250°57.1	20:41
Jupiter	289°14.0	18:08
Saturn	7°54.7	12:54

Mar 01 Sat	SHA	Mer.pass
Venus	352°43.6	13:51
Mars	250°53.8	20:37
Jupiter	289°09.1	18:04
Saturn	7°48.0	12:51

Horizontal parallax		
Venus:	0.4	
Mars:	0.2	

h	Sun				Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP	
0	176°49.3	S08°19.8	189°04.1	9.4'	S15°57.5	-14.8'	59.4'	
1	191°49.4	18.9	203°32.5	9.5'	15°42.7	-14.9'	59.5'	
2	206°49.5	18.0	218°00.9	9.5'	15°27.9	-15.0'	59.5'	
3	221°49.6	· · 17.0	232°29.4	9.6'	15°12.9	-15.1'	59.5'	
4	236°49.8	16.1	246°58.0	9.6'	14°57.9	-15.2'	59.6'	
5	251°49.9	15.1	261°26.6	9.6'	14°42.7	-15.3'	59.6'	
6	266°50.0	S08°14.2	275°55.2	9.7'	S14°27.5	-15.3'	59.6'	
7	281°50.1	13.3	290°23.9	9.7'	14°12.1	-15.4'	59.6'	
8	296°50.2	12.3	304°52.7	9.8'	13°56.7	-15.5'	59.7'	
9	311°50.3	· · 11.4	319°21.5	9.8'	13°41.1	-15.6'	59.7'	
10	326°50.4	10.4	333°50.3	9.9'	13°25.5	-15.7'	59.7'	
11	341°50.5	09.5	348°19.2	9.9'	13°09.8	-15.8'	59.7'	
12	356°50.6	S08°08.5	2°48.1	10.0'	S12°54.0	-15.9'	59.8'	
13	11°50.8	07.6	17°17.0	10.0'	12°38.1	-16.0'	59.8'	
14	26°50.9	06.7	31°46.0	10.0'	12°22.2	-16.0'	59.8'	
15	41°51.0	· · 05.7	46°15.1	10.1'	12°06.1	-16.1'	59.8'	
16	56°51.1	04.8	60°44.1	10.1'	11°50.0	-16.2'	59.9'	
17	71°51.2	03.8	75°13.3	10.2'	11°33.8	-16.3'	59.9'	
18	86°51.3	S08°02.9	89°42.4	10.2'	S11°17.5	-16.4'	59.9'	
19	101°51.4	01.9	104°11.6	10.2'	11°01.2	-16.4'	59.9'	
20	116°51.5	01.0	118°40.8	10.3'	10°44.7	-16.5'	60.0'	
21	131°51.7	08°00.0	133°10.1	10.3'	10°28.2	-16.6'	60.0'	
22	146°51.8	07°59.1	147°39.4	10.3'	10°11.7	-16.6'	60.0'	
23	161°51.9	58.1	162°08.8	10.4'	09°55.0	-16.7'	60.0'	
	SD = 16.1'	$d = -0.9'$	SD = 16.2'					
Fri	GHA	Dec	GHA	ν	Dec	d	HP	
0	176°52.0	S07°57.2	176°38.1	10.4'	S09°38.3	-16.8'	60.0'	
1	191°52.1	56.3	191°07.5	10.4'	09°21.6	-16.8'	60.1'	
2	206°52.2	55.3	205°37.0	10.5'	09°04.8	-16.9'	60.1'	
3	221°52.3	· · 54.4	220°06.4	10.5'	08°47.9	-16.9'	60.1'	
4	236°52.5	53.4	234°35.9	10.5'	08°30.9	-17.0'	60.1'	
5	251°52.6	52.5	249°05.4	10.5'	08°13.9	-17.1'	60.1'	
6	266°52.7	S07°51.5	263°35.0	10.6'	S07°56.9	-17.1'	60.1'	
7	281°52.8	50.6	278°04.6	10.6'	07°39.8	-17.2'	60.2'	
8	296°52.9	49.6	292°34.2	10.6'	07°22.6	-17.2'	60.2'	
9	311°53.0	· · 48.7	307°03.8	10.6'	07°05.4	-17.3'	60.2'	
10	326°53.2	47.7	321°33.4	10.7'	06°48.1	-17.3'	60.2'	
11	341°53.3	46.8	336°03.1	10.7'	06°30.8	-17.3'	60.2'	
12	356°53.4	S07°45.8	350°32.8	10.7'	S06°13.5	-17.4'	60.2'	
13	11°53.5	44.9	5°02.5	10.7'	05°56.1	-17.4'	60.3'	
14	26°53.6	43.9	19°32.2	10.7'	05°38.6	-17.5'	60.3'	
15	41°53.7	· · 43.0	34°02.0	10.8'	05°21.2	-17.5'	60.3'	
16	56°53.9	42.0	48°31.8	10.8'	05°03.7	-17.5'	60.3'	
17	71°54.0	41.1	63°01.5	10.8'	04°46.1	-17.6'	60.3'	
18	86°54.1	S07°40.1	77°31.3	10.8'	S04°28.6	-17.6'	60.3'	
19	101°54.2	39.2	92°01.2	10.8'	04°10.9	-17.6'	60.3'	
20	116°54.3	38.2	106°31.0	10.8'	03°53.3	-17.7'	60.4'	
21	131°54.5	· · 37.3	121°00.8	10.8'	03°35.6	-17.7'	60.4'	
22	146°54.6	36.3	135°30.7	10.9'	03°18.0	-17.7'	60.4'	
23	161°54.7	35.4	150°00.5	10.9'	03°00.2	-17.7'	60.4'	
	SD = 16.1'	$d = -0.9'$	SD = 16.4'					
Sat	GHA	Dec	GHA	ν	Dec	d	HP	
0	176°54.8	S07°34.4	164°30.4	10.9'	S02°42.5	-17.8'	60.4'	
1	191°54.9	33.5	179°00.2	10.9'	02°24.8	-17.8'	60.4'	
2	206°55.1	32.5	193°30.1	10.9'	02°07.0	-17.8'	60.4'	
3	221°55.2	· · 31.6	208°00.0	10.9'	01°49.2	-17.8'	60.4'	
4	236°55.3	30.6	222°29.9	10.9'	01°31.4	-17.8'	60.4'	
5	251°55.4	29.7	236°59.8	10.9'	01°13.6	-17.8'	60.4'	
6	266°55.5	S07°28.7	251°29.6	10.9'	S00°55.8	-17.8'	60.4'	
7	281°55.7	27.8	265°59.5	10.9'	00°37.9	-17.8'	60.5'	
8	296°55.8	26.8	280°29.4	10.9'	00°20.1	-17.8'	60.5'	
9	311°55.9	· · 25.9	294°59.3	10.9'	S00°02.2	-17.8'	60.5'	
10	326°56.0	24.9	309°29.1	10.9'	N00°15.6	17.8'	60.5'	
11	341°56.1	24.0	323°59.0	10.9'	00°33.5	17.8'	60.5'	
12	356°56.3	S07°23.0	338°28.9	10.9'	N00°51.3	17.8'	60.5'	
13	11°56.4	22.1	352°58.7	10.8'	01°09.1	17.8'	60.5'	
14	26°56.5	21.1	7°28.6	10.8'	01°27.0	17.8'	60.5'	
15	41°56.6	· · 20.2	21°58.4	10.8'	01°44.8	17.8'	60.5'	
16	56°56.8	19.2	36°28.2	10.8'	02°02.6	17.8'	60.5'	
17	71°56.9	18.3	50°58.0	10.8'	02°20.5	17.8'	60.5'	
18	86°57.0	S07°17.3	65°27.8	10.8'	N02°38.3	17.8'	60.5'	
19	101°57.1	16.3	79°57.6	10.8'	02°56.0	17.8'	60.5'	
20	116°57.3	15.4	94°27.3	10.7'	03°13.8	17.8'	60.5'	
21	131°57.4	· · 14.4	108°57.1	10.7'	03°31.6	17.7'	60.5'	
22	146°57.5	13.5	123°26.8	10.7'	03°49.3	17.7'	60.5'	
23	161°57.6	12.5	137°56.5	10.7'	04°07.0	17.7'	60.5'	
	SD = 16.1'	$d = -1.0'$	SD = 16.5'					

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:14	06:32	07:41	16:46	17:56	19:14
N 70°	05:18	06:29	07:31	16:56	17:58	19:09
68°	05:22	06:26	07:23	17:04	18:01	19:05
66°	05:25	06:24	07:16	17:10	18:02	19:02
64°	05:27	06:22	07:10	17:16	18:04	18:59
62°	05:29	06:20	07:05	17:21	18:06	18:57
60°	05:31	06:19	07:01	17:25	18:07	18:55
N 58°	05:32	06:17	06:57	17:29	18:09	18:54
56°	05:33	06:16	06:54	17:32	18:10	18:53
54°	05:34	06:15	06:50	17:35	18:11	18:52
52°	05:35	06:14	06:48	17:38	18:12	18:51
50°	05:35	06:13	06:45	17:41	18:13	18:51
45°	05:36	06:10	06:39	17:46	18:16	18:50
N 40°	05:36	06:07	06:35	17:51	18:18	18:49
35°	05:36	06:05	06:31	17:55	18:20	18:50
30°	05:35	06:03	06:27	17:58	18:22	18:50
20°	05:33	05:58	06:20	18:05	18:27	18:53
N 10°	05:29	05:53	06:15	18:10	18:32	18:56
0°	05:24	05:48	06:09	18:16	18:37	19:01
S 10°	05:18	05:42	06:04	18:21	18:42	19:07
20°	05:09	05:35	05:57	18:27	18:49	19:16
30°	04:57	05:26	05:50	18:34	18:58	19:27
35°	04:50	05:20	05:46	18:38	19:04	19:34
40°	04:41	05:14	05:41	18:43	19:10	19:43
45°	04:29	05:06	05:36	18:48	19:18	19:54
S 50°	04:15	04:56	05:29	18:55	19:28	20:09
52°	04:08	04:51	05:26	18:58	19:33	20:16
54°	04:00	04:46	05:23	19:01	19:38	20:23
56°	03:50	04:40	05:19	19:04	19:44	20:32
58°	03:40	04:33	05:15	19:09	19:50	20:43
S 60°	03:27	04:25	05:10	19:13	19:58	20:55

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	09:00	08:06	07:25	15:08	17:51	20:22
N 70°	08:34	07:55	07:24	15:31	17:58	20:17
68°	08:14	07:47	07:24	15:49	18:04	20:14
66°	07:58	07:39	07:23	16:03	18:08	20:10
64°	07:45	07:33	07:23	16:15	18:12	20:08
62°	07:33	07:28	07:22	16:25	18:16	20:06
60°	07:24	07:23	07:22	16:33	18:19	20:04
N 58°	07:15	07:19	07:21	16:40	18:21	20:02
56°	07:08	07:15	07:21	16:46	18:24	20:00
54°	07:01	07:12	07:21	16:52	18:26	19:59
52°	06:55	07:09	07:21	16:57	18:27	19:58
50°	06:50	07:06	07:20	17:02	18:29	19:56
45°	06:38	07:00	07:20	17:12	18:33	19:54
N 40°	06:28	06:54	07:19	17:20	18:36	19:52
35°	06:19	06:50	07:19	17:27	18:38	19:50
30°	06:12	06:46	07:19	17:33	18:41	19:48
20°	05:59	06:39	07:18	17:43	18:44	19:46
N 10°	05:47	06:33	07:18	17:52	18:48	19:43
0°	05:36	06:27	07:17	18:01	18:51	19:41
S 10°	05:25	06:21	07:17	18:09	18:54	19:39
20°	05:14	06:15	07:16	18:18	18:57	19:36
30°	05:00	06:08	07:16	18:28	19:01	19:33
35°	04:52	06:04	07:16	18:33	19:03	19:32
40°	04:43	06:00	07:16	18:40	19:05	19:30
45°	04:33	05:54	07:15	18:47	19:08	19:28
S 50°	04:20	05:48	07:15	18:56	19:11	19:25
52°	04:14	05:45	07:15	19:00	19:12	19:24
54°	04:07	05:41	07:15	19:04	19:14	19:23
56°	04:00	05:38	07:14	19:09	19:16	19:22
58°	03:52	05:34	07:14	19:15	19:18	19:20
S 60°	03:42	05:29	07:14	19:21	19:20	19:18

March 02, 03, 04 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	160°02.3	152°48.8	N10°36.5	50°52.1	N25°48.8	89°06.3	N21°53.1	167°43.5	S05°23.9	Alpheratz	357°35.0	29°13.7
1	175°04.8	167°51.5	36.9	65°54.4	48.7	104°08.5	53.1	182°45.7	23.8	Ankaa	353°07.4	-42°10.3
2	190°07.3	182°54.1	37.3	80°56.7	48.6	119°10.8	53.2	197°47.9	23.7	Schedar	349°31.4	56°40.6
3	205°09.7	197°56.7	•• 37.6	95°58.9	•• 48.5	134°13.0	•• 53.2	212°50.1	•• 23.5	Diphda	348°47.4	-17°51.1
4	220°12.2	212°59.4	38.0	111°01.2	48.4	149°15.3	53.2	227°52.3	23.4	Achernar	335°20.5	-57°06.8
5	235°14.6	228°02.0	38.4	126°03.5	48.3	164°17.5	53.3	242°54.4	23.3	Hamal	327°51.3	23°34.9
6	250°17.1	243°04.7	N10°38.8	141°05.8	N25°48.2	179°19.8	N21°53.3	257°56.6	S05°23.2	Polaris	314°11.3	89°22.5
7	265°19.6	258°07.3	39.1	156°08.0	48.1	194°22.0	53.3	272°58.8	23.1	Acamar	315°11.8	-40°12.5
8	280°22.0	273°10.0	39.5	171°10.3	48.0	209°24.3	53.4	288°01.0	22.9	Menkar	314°06.2	4°11.2
9	295°24.5	288°12.7	•• 39.9	186°12.6	•• 47.9	224°26.5	•• 53.4	303°03.2	•• 22.8	Mirfak	308°28.3	49°57.2
10	310°27.0	303°15.3	40.2	201°14.9	47.8	239°28.7	53.5	318°05.3	22.7	Aldebaran	290°39.5	16°33.6
11	325°29.4	318°18.0	40.6	216°17.1	47.7	254°31.0	53.5	333°07.5	22.6	Rigel	281°03.7	-8°10.5
12	340°31.9	333°20.7	N10°41.0	231°19.4	N25°47.6	269°33.2	N21°53.5	348°09.7	S05°22.5	Capella	280°21.6	46°01.5
13	355°34.4	348°23.3	41.3	246°21.6	47.5	284°35.5	53.6	3°11.9	22.3	Bellatrix	278°22.7	6°22.3
14	10°36.8	3°26.0	41.7	261°23.9	47.4	299°37.7	53.6	18°14.1	22.2	Elnath	278°01.6	28°37.8
15	25°39.3	18°28.7	•• 42.0	276°26.2	•• 47.3	314°40.0	•• 53.6	33°16.3	•• 22.1	Alnilam	275°37.5	-1°11.2
16	40°41.7	33°31.4	42.4	291°28.4	47.2	329°42.2	53.7	48°18.4	22.0	Betelgeuse	270°51.8	7°24.7
17	55°44.2	48°34.1	42.7	306°30.7	47.1	344°44.4	53.7	63°20.6	21.9	Canopus	263°52.1	-52°42.8
18	70°46.7	63°36.8	N10°43.1	321°32.9	N25°47.0	359°46.7	N21°53.7	78°22.8	S05°21.7	Sirius	258°26.0	-16°45.2
19	85°49.1	78°39.5	43.4	336°35.2	46.9	14°48.9	53.8	93°25.0	21.6	Adhara	255°05.6	-29°00.6
20	100°51.6	93°42.2	43.8	351°37.5	46.8	29°51.2	53.8	108°27.2	21.5	Procyon	244°50.5	5°09.5
21	115°54.1	108°44.9	•• 44.1	6°39.7	•• 46.7	44°53.4	•• 53.8	123°29.3	•• 21.4	Pollux	243°16.9	27°57.9
22	130°56.5	123°47.6	44.5	21°42.0	46.6	59°55.6	53.9	138°31.5	21.3	Avior	234°14.2	-59°35.6
23	145°59.0	138°50.3	44.8	36°44.2	46.5	74°57.9	53.9	153°33.7	21.1	Suhail	222°45.8	-43°32.2
Mer.pass. 13:18		ν 2.6' d 0.4' m -4.76		ν 2.3' d -0.1' m -0.25		ν 2.2' d 0.0' m -2.29		ν 2.2' d -0.1' m 1.11				

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	161°01.5	153°53.1	N10°45.1	51°46.5	N25°46.4	90°00.1	N21°53.9	168°35.9	S05°21.0
1	176°03.9	168°55.8	45.5	66°48.7	46.3	105°02.4	54.0	183°38.1	20.9
2	191°06.4	183°58.5	45.8	81°51.0	46.2	120°04.6	54.0	198°40.2	20.8
3	206°08.9	199°01.2	•• 46.1	96°53.2	•• 46.1	135°06.8	•• 54.0	213°42.4	•• 20.6
4	221°11.3	214°04.0	46.4	111°55.5	46.0	150°09.1	54.1	228°44.6	20.5
5	236°13.8	229°06.7	46.8	126°57.7	45.9	165°11.3	54.1	243°46.8	20.4
6	251°16.2	244°09.5	N10°47.1	141°59.9	N25°45.8	180°13.5	N21°54.2	258°49.0	S05°20.3
7	266°18.7	259°12.2	47.4	157°02.2	45.7	195°15.8	54.2	273°51.1	20.2
8	281°21.2	274°15.0	47.7	172°04.4	45.6	210°18.0	54.2	288°53.3	20.0
9	296°23.6	289°17.7	•• 48.1	187°06.7	•• 45.5	225°20.2	•• 54.3	303°55.5	•• 19.9
10	311°26.1	304°20.5	48.4	202°08.9	45.4	240°22.5	54.3	318°57.7	19.8
11	326°28.6	319°23.2	48.7	217°11.1	45.3	255°24.7	54.3	333°59.9	19.7
12	341°31.0	334°26.0	N10°49.0	232°13.4	N25°45.2	270°27.0	N21°54.4	349°02.1	S05°19.6
13	356°33.5	349°28.8	49.3	247°15.6	45.1	285°29.2	54.4	4°04.2	19.4
14	11°36.0	4°31.5	49.6	262°17.8	45.0	300°31.4	54.4	19°06.4	19.3
15	26°38.4	19°34.3	•• 49.9	277°20.1	•• 44.9	315°33.7	•• 54.5	34°08.6	•• 19.2
16	41°40.9	34°37.1	50.2	292°22.3	44.8	330°35.9	54.5	49°10.8	19.1
17	56°43.4	49°39.9	50.5	307°24.5	44.7	345°38.1	54.5	64°13.0	19.0
18	71°45.8	64°42.7	N10°50.8	322°26.7	N25°44.6	0°40.4	N21°54.6	79°15.1	S05°18.8
19	86°48.3	79°45.5	51.1	337°29.0	44.4	15°42.6	54.6	94°17.3	18.7
20	101°50.7	94°48.3	51.4	352°31.2	44.3	30°44.8	54.7	109°19.5	18.6
21	116°53.2	109°51.1	•• 51.7	7°33.4	•• 44.2	45°47.0	•• 54.7	124°21.7	•• 18.5
22	131°55.7	124°53.9	52.0	22°35.6	44.1	60°49.3	54.7	139°23.9	18.4
23	146°58.1	139°56.7	52.3	37°37.9	44.0	75°51.5	54.8	154°26.0	18.2
Mer.pass. 13:14		ν 2.7' d 0.3' m -4.74		ν 2.2' d -0.1' m -0.23		ν 2.2' d 0.0' m -2.29		ν 2.2' d -0.1' m 1.11	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	162°00.6	154°59.5	N10°52.5	52°40.1	N25°43.9	90°53.7	N21°54.8	169°28.2	S05°18.1
1	177°03.1	170°02.3	52.8	67°42.3	43.8	105°56.0	54.8	184°30.4	18.0
2	192°05.5	185°05.1	53.1	82°44.5	43.7	120°58.2	54.9	199°32.6	17.9
3	207°08.0	200°07.9	•• 53.4	97°46.7	•• 43.6	136°00.4	•• 54.9	214°34.8	•• 17.8
4	222°10.5	215°10.8	53.7	112°48.9	43.5	151°02.7	54.9	229°36.9	17.6
5	237°12.9	230°13.6	53.9	127°51.2	43.4	166°04.9	55.0	244°39.1	17.5
6	252°15.4	245°16.4	N10°54.2	142°53.4	N25°43.3	181°07.1	N21°55.0	259°41.3	S05°17.4
7	267°17.8	260°19.3	54.5	157°55.6	43.2	196°09.4	55.1	274°43.5	17.3
8	282°20.3	275°22.1	54.7	172°57.8	43.1	211°11.6	55.1	289°45.7	17.2
9	297°22.8	290°25.0	•• 55.0	188°00.0	•• 43.0	226°13.8	•• 55.1	304°47.8	•• 17.0
10	312°25.2	305°27.8	55.3	203°02.2	42.9	241°16.0	55.2	319°50.0	16.9
11	327°27.7	320°30.7	55.5	218°04.4	42.7	256°18.3	55.2	334°52.2	16.8
12	342°30.2	335°33.5	N10°55.8	233°06.6	N25°42.6	271°20.5	N21°55.2	349°54.4	S05°16.7
13	357°32.6	350°36.4	56.0	248°08.8	42.5	286°22.7	55.3	4°56.6	16.6
14	12°35.1	5°39.2	56.3	263°11.0	42.4	301°24.9	55.3	19°58.8	16.4
15	27°37.6	20°42.1	•• 56.5	278°13.2	•• 42.3	316°27.2	•• 55.3	35°00.9	•• 16.3
16	42°40.0	35°45.0	56.8	293°15.4	42.2	331°29.4	55.4	50°03.1	16.2
17	57°42.5	50°47.9	57.0	308°17.6	42.1	346°31.6	55.4	65°05.3	16.1
18	72°45.0	65°50.7	N10°57.3	323°19.8	N25°42.0	1°33.8	N21°55.5	80°07.5	S05°15.9
19	87°47.4	80°53.6	57.5	338°22.0	41.9	16°36.1	55.5	95°09.7	15.8
20	102°49.9	95°56.5	57.7	353°24.2	41.8	31°38.3	55.5	110°11.8	15.7
21	117°52.3	110°59.4	•• 58.0	8°26.4	•• 41.7	46°40.5	•• 55.6	125°14.0	•• 15.6
22	132°54.8	126°02.3	58.2	23°28.6	41.6	61°42.7	55.6	140°16.2	15.5
23	147°57.3	141°05.2	58.5	38°30.8	41.4	76°45.0	55.6	155°18.4	15.3
Mer.pass. 13:10		ν 2.8' d 0.3' m -4.72		ν 2.2' d -0.1' m -0.20		ν 2.2' d 0.0' m -2.28		ν 2.2' d -0.1' m 1.11	

Mar 02 Sun	SHA	Mer.pass
Venus	352°46.5	13:46
Mars	250°49.8	20:33
Jupiter	289°04.0	18:01
Saturn	7°41.2	12:47

Mar 03 Mon	SHA	Mer.pass
Venus	352°51.6	13:42
Mars	250°45.0	20:30
Jupiter	288°58.7	17:57
Saturn	7°34.4	12:44

Mar 04 Tue	SHA	Mer.pass
Venus	352°58.9	13:37
Mars	250°39.5	20:26
Jupiter	288°53.1	17:54
Saturn	7°27.6	12:40

Horizontal parallax	
Venus:	0.4
Mars:	0.2

DUT1 = UT1-UTC = +0.0443 sec ΔT = TT-UT1 = +69.1397 sec

2025 March 02 to Mar. 04 UT

h Sun Moon						
Sun	GHA	Dec	GHA	ν	Dec	d HP
0	176°57.8	S07°11.6	152°26.2	10.7'	N04°24.7	17.7' 60.5'
1	191°57.9	10.6	166°55.9	10.6'	04°42.4	17.6' 60.5'
2	206°58.0	09.7	181°25.5	10.6'	05°00.0	17.6' 60.5'
3	221°58.1	· · 08.7	195°55.1	10.6'	05°17.6	17.6' 60.5'
4	236°58.3	07.8	210°24.7	10.6'	05°35.2	17.5' 60.5'
5	251°58.4	06.8	224°54.2	10.5'	05°52.7	17.5' 60.5'
6	266°58.5	S07°05.8	239°23.8	10.5'	N06°10.2	17.5' 60.5'
7	281°58.6	04.9	253°53.3	10.5'	06°27.7	17.4' 60.5'
8	296°58.8	03.9	268°22.7	10.4'	06°45.1	17.4' 60.5'
9	311°58.9	· · 03.0	282°52.2	10.4'	07°02.5	17.3' 60.5'
10	326°59.0	02.0	297°21.6	10.4'	07°19.9	17.3' 60.5'
11	341°59.1	01.1	311°50.9	10.3'	07°37.2	17.3' 60.5'
12	356°59.3	S07°00.1	326°20.3	10.3'	N07°54.4	17.2' 60.5'
13	11°59.4	06°59.1	340°49.6	10.3'	08°11.6	17.2' 60.4'
14	26°59.5	58.2	355°18.8	10.2'	08°28.8	17.1' 60.4'
15	41°59.7	· · 57.2	9°48.0	10.2'	08°45.9	17.1' 60.4'
16	56°59.8	56.3	24°17.2	10.1'	09°02.9	17.0' 60.4'
17	71°59.9	55.3	38°46.3	10.1'	09°19.9	16.9' 60.4'
18	87°00.0	S06°54.4	53°15.4	10.0'	N09°36.9	16.9' 60.4'
19	102°00.2	53.4	67°44.5	10.0'	09°53.8	16.8' 60.4'
20	117°00.3	52.4	82°13.5	10.0'	10°10.6	16.8' 60.4'
21	132°00.4	· · 51.5	96°42.4	9.9'	10°27.3	16.7' 60.4'
22	147°00.6	50.5	111°11.4	9.9'	10°44.0	16.6' 60.4'
23	162°00.7	49.6	125°40.2	9.8'	11°00.6	16.6' 60.4'
SD = 16.1' d = -1.0'			SD = 16.5'			
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	177°00.8	S06°48.6	140°09.0	9.8'	N11°17.2	16.5' 60.3'
1	192°00.9	47.6	154°37.8	9.7'	11°33.7	16.4' 60.3'
2	207°01.1	46.7	169°06.5	9.7'	11°50.1	16.3' 60.3'
3	222°01.2	· · 45.7	183°35.2	9.6'	12°06.4	16.3' 60.3'
4	237°01.3	44.8	198°03.8	9.6'	12°22.7	16.2' 60.3'
5	252°01.5	43.8	212°32.4	9.5'	12°38.9	16.1' 60.3'
6	267°01.6	S06°42.8	227°00.9	9.4'	N12°55.0	16.0' 60.3'
7	282°01.7	41.9	241°29.3	9.4'	13°11.0	15.9' 60.3'
8	297°01.9	40.9	255°57.7	9.3'	13°26.9	15.9' 60.2'
9	312°02.0	· · 40.0	270°26.0	9.3'	13°42.8	15.8' 60.2'
10	327°02.1	39.0	284°54.3	9.2'	13°58.5	15.7' 60.2'
11	342°02.3	38.0	299°22.5	9.2'	14°14.2	15.6' 60.2'
12	357°02.4	S06°37.1	313°50.7	9.1'	N14°29.8	15.5' 60.2'
13	12°02.5	36.1	328°18.8	9.0'	14°45.3	15.4' 60.2'
14	27°02.7	35.2	342°46.8	9.0'	15°00.7	15.3' 60.2'
15	42°02.8	· · 34.2	357°14.8	8.9'	15°16.0	15.2' 60.1'
16	57°02.9	33.2	11°42.7	8.9'	15°31.2	15.1' 60.1'
17	72°03.1	32.3	26°10.6	8.8'	15°46.3	15.0' 60.1'
18	87°03.2	S06°31.3	40°38.4	8.7'	N16°01.3	14.9' 60.1'
19	102°03.3	30.4	55°06.1	8.7'	16°16.3	14.8' 60.1'
20	117°03.5	29.4	69°33.7	8.6'	16°31.1	14.7' 60.1'
21	132°03.6	· · 28.4	84°01.3	8.5'	16°45.8	14.6' 60.0'
22	147°03.7	27.5	98°28.9	8.5'	17°00.3	14.5' 60.0'
23	162°03.9	26.5	112°56.3	8.4'	17°14.8	14.4' 60.0'
SD = 16.1' d = -1.0'			SD = 16.5'			
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	177°04.0	S06°25.5	127°23.7	8.3'	N17°29.2	14.3' 60.0'
1	192°04.1	24.6	141°51.0	8.3'	17°43.4	14.1' 60.0'
2	207°04.3	23.6	156°18.3	8.2'	17°57.6	14.0' 59.9'
3	222°04.4	· · 22.7	170°45.5	8.1'	18°11.6	13.9' 59.9'
4	237°04.5	21.7	185°12.6	8.1'	18°25.5	13.8' 59.9'
5	252°04.7	20.7	199°39.7	8.0'	18°39.3	13.7' 59.9'
6	267°04.8	S06°19.8	214°06.6	7.9'	N18°53.0	13.5' 59.9'
7	282°05.0	18.8	228°33.6	7.8'	19°06.5	13.4' 59.8'
8	297°05.1	17.8	243°00.4	7.8'	19°19.9	13.3' 59.8'
9	312°05.2	· · 16.9	257°27.2	7.7'	19°33.2	13.2' 59.8'
10	327°05.4	15.9	271°53.9	7.6'	19°46.4	13.0' 59.8'
11	342°05.5	14.9	286°20.5	7.6'	19°59.4	12.9' 59.8'
12	357°05.6	S06°14.0	300°47.1	7.5'	N20°12.3	12.8' 59.7'
13	12°05.8	13.0	315°13.6	7.4'	20°25.1	12.6' 59.7'
14	27°05.9	12.0	329°40.0	7.4'	20°37.8	12.5' 59.7'
15	42°06.1	· · 11.1	344°06.4	7.3'	20°50.3	12.4' 59.7'
16	57°06.2	10.1	358°32.6	7.2'	21°02.6	12.2' 59.6'
17	72°06.3	09.2	12°58.9	7.1'	21°14.9	12.1' 59.6'
18	87°06.5	S06°08.2	27°25.0	7.1'	N21°27.0	12.0' 59.6'
19	102°06.6	07.2	41°51.1	7.0'	21°38.9	11.8' 59.6'
20	117°06.8	06.3	56°17.1	6.9'	21°50.7	11.7' 59.6'
21	132°06.9	· · 05.3	70°43.0	6.9'	22°02.4	11.5' 59.5'
22	147°07.0	04.3	85°08.9	6.8'	22°13.9	11.4' 59.5'
23	162°07.2	03.4	99°34.7	6.7'	22°25.3	11.2' 59.5'
SD = 16.1' d = -1.0'			SD = 16.4'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:59	06:17	07:25	17:01	18:09	19:28
N 70°	05:05	06:15	07:17	17:09	18:10	19:21
68°	05:10	06:14	07:10	17:15	18:11	19:16
66°	05:14	06:13	07:05	17:21	18:12	19:12
64°	05:17	06:12	07:00	17:25	18:13	19:08
62°	05:20	06:11	06:56	17:29	18:14	19:05
60°	05:22	06:10	06:52	17:33	18:15	19:03
N 58°	05:24	06:10	06:49	17:36	18:15	19:01
56°	05:26	06:09	06:46	17:39	18:16	18:59
54°	05:27	06:08	06:43	17:41	18:17	18:58
52°	05:28	06:07	06:41	17:44	18:17	18:56
50°	05:29	06:07	06:39	17:46	18:18	18:55
45°	05:31	06:05	06:34	17:50	18:20	18:54
N 40°	05:32	06:03	06:30	17:54	18:21	18:53
35°	05:32	06:01	06:27	17:58	18:23	18:52
30°	05:32	06:00	06:24	18:01	18:25	18:52
20°	05:30	05:56	06:18	18:06	18:28	18:54
N 10°	05:28	05:52	06:13	18:11	18:32	18:56
0°	05:24	05:48	06:09	18:15	18:36	19:00
S 10°	05:18	05:43	06:04	18:20	18:41	19:06
20°	05:10	05:36	05:59	18:25	18:47	19:13
30°	05:00	05:28	05:52	18:31	18:55	19:23
35°	04:53	05:23	05:49	18:34	19:00	19:30
40°	04:44	05:17	05:45	18:38	19:06	19:38
45°	04:34	05:10	05:40	18:43	19:13	19:49
S 50°	04:21	05:01	05:34	18:48	19:21	20:02
52°	04:14	04:57	05:32	18:51	19:26	20:08
54°	04:07	04:52	05:29	18:54	19:30	20:15
56°	03:59	04:47	05:26	18:57	19:35	20:23
58°	03:49	04:41	05:22	19:00	19:41	20:33
S 60°	03:38	04:34	05:18	19:04	19:48	20:44

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	06:46	05:56	☐	23:02	☐	☐
N 70°	06:54	06:18	05:13	22:43	· · ·	01:40
68°	07:01	06:35	05:56	22:28	· · ·	00:59
66°	07:07	06:49	06:25	22:16	· · ·	00:31
64°	07:12	07:01	06:47	22:06	· · ·	00:10
62°	07:17	07:11	07:05	21:57	23:54	· · ·
60°	07:21	07:20	07:20	21:50	23:40	· · ·
N 58°	07:24	07:27	07:33	21:43	23:28	· · ·
56°	07:27	07:34	07:44	21:38	23:18	· · ·
54°	07:30	07:40	07:54	21:33	23:09	· · ·
52°	07:32	07:46	08:02	21:28	23:01	· · ·
50°	07:35	07:51	08:10	21:24	22:54	· · ·
45°	07:40	08:02	08:27	21:16	22:38	· · ·
N 40°	07:44	08:11	08:41	21:08	22:26	23:44
35°	07:48	08:19	08:53	21:02	22:15	23:29
30°	07:51	08:26	09:04	20:57	22:06	23:17
20°	07:57	08:38	09:22	20:47	21:51	22:55
N 10°	08:02	08:49	09:38	20:39	21:37	22:37
0°	08:07	08:59	09:53	20:32	21:24	22:19
S 10°	08:12	09:09	10:08	20:24	21:11	22:02
20°	08:18	09:20	10:24	20:16	20:58	21:44
30°	08:24	09:33	10:43	20:07	20:43	21:23
35°	08:28	09:40	10:54	20:01	20:34	21:11
40°	08:32	09:49	11:07	19:55	20:24	20:57
45°	08:37	09:59	11:22	19:49	20:12	20:40
S 50°	08:42	10:11	11:41	19:40	19:58	20:20
52°	08:45	10:17	11:50	19:37	19:51	20:10
54°	08:48	10:23	12:00	19:33	19:44	20:00
56°	08:51	10:30	12:11	19:28	19:36	19:47
58°	08:55	10:38	12:24	19:23	19:27	19:33
S 60°	08:59	10:47	12:40	19:17	19:17	19:17

Day	Sun		Mer. Pass hh:mm	Moon		Age 2-4 6-21%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
02	12:09	12:03	12:12	14:19	01:54	
03	11:57	11:50	12:12	15:11	02:45	
04	11:44	11:37	12:12	16:06	03:38	

March 05, 06, 07 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	162°59.7	156°08.1	N10°58.7	53°33.0	N25°41.3	91°47.2	N21°55.7	170°20.6	S05°15.2	Alpheratz	357°35.0	29°13.7
1	178°02.2	171°11.0	58.9	68°35.2	41.2	106°49.4	55.7	185°22.7	15.1	Ankaa	353°07.4	-42°10.3
2	193°04.7	186°13.9	59.1	83°37.3	41.1	121°51.6	55.7	200°24.9	15.0	Schedar	349°31.4	56°40.6
3	208°07.1	201°16.8	· · 59.4	98°39.5	· · 41.0	136°53.9	· · 55.8	215°27.1	· · 14.9	Diphda	348°47.4	-17°51.1
4	223°09.6	216°19.8	59.6	113°41.7	40.9	151°56.1	55.8	230°29.3	14.7	Achernar	335°20.5	-57°06.7
5	238°12.1	231°22.7	10°59.8	128°43.9	40.8	166°58.3	55.9	245°31.5	14.6	Hamal	327°51.3	23°34.9
6	253°14.5	246°25.6	N11°00.0	143°46.1	N25°40.7	182°00.5	N21°55.9	260°33.6	S05°14.5	Polaris	314°12.4	89°22.5
7	268°17.0	261°28.5	00.2	158°48.3	40.6	197°02.7	55.9	275°35.8	14.4	Acamar	315°11.9	-40°12.5
8	283°19.5	276°31.5	00.4	173°50.4	40.5	212°05.0	56.0	290°38.0	14.3	Menkar	314°06.2	4°11.2
9	298°21.9	291°34.4	· · 00.6	188°52.6	· · 40.3	227°07.2	· · 56.0	305°40.2	· · 14.1	Mirfak	308°28.3	49°57.2
10	313°24.4	306°37.4	00.9	203°54.8	40.2	242°09.4	56.0	320°42.4	14.0	Aldebaran	290°39.5	16°33.6
11	328°26.8	321°40.3	01.1	218°57.0	40.1	257°11.6	56.1	335°44.5	13.9	Rigel	281°03.7	-8°10.5
12	343°29.3	336°43.2	N11°01.3	233°59.1	N25°40.0	272°13.8	N21°56.1	350°46.7	S05°13.8	Capella	280°21.7	46°01.5
13	358°31.8	351°46.2	01.5	249°01.3	39.9	287°16.1	56.2	5°48.9	13.7	Bellatrix	278°22.7	6°22.3
14	13°34.2	6°49.2	01.7	264°03.5	39.8	302°18.3	56.2	20°51.1	13.5	Elnath	278°01.7	28°37.8
15	28°36.7	21°52.1	· · 01.9	279°05.7	· · 39.7	317°20.5	· · 56.2	35°53.3	· · 13.4	Alnilam	275°37.5	-1°11.2
16	43°39.2	36°55.1	02.0	294°07.8	39.6	332°22.7	56.3	50°55.4	13.3	Betelgeuse	270°51.9	7°24.6
17	58°41.6	51°58.1	02.2	309°10.0	39.5	347°24.9	56.3	65°57.6	13.2	Canopus	263°52.1	-52°42.8
18	73°44.1	67°01.0	N11°02.4	324°12.2	N25°39.3	2°27.2	N21°56.3	80°59.8	S05°13.1	Sirius	258°26.0	-16°45.2
19	88°46.6	82°04.0	02.6	339°14.3	39.2	17°29.4	56.4	96°02.0	12.9	Adhara	255°05.6	-29°00.6
20	103°49.0	97°07.0	02.8	354°16.5	39.1	32°31.6	56.4	111°04.2	12.8	Procyon	244°50.5	5°09.5
21	118°51.5	112°10.0	· · 03.0	9°18.7	· · 39.0	47°33.8	· · 56.4	126°06.3	· · 12.7	Pollux	243°16.9	27°57.9
22	133°53.9	127°12.9	03.2	24°20.8	38.9	62°36.0	56.5	141°08.5	12.6	Avior	234°14.2	-59°35.6
23	148°56.4	142°15.9	03.3	39°23.0	38.8	77°38.2	56.5	156°10.7	12.4	Suhail	222°45.9	-43°32.2
Mer.pass. 13:06		ν 2.9' d 0.2' m -4.69		ν 2.2' d -0.1' m -0.17		ν 2.2' d 0.0' m -2.27		ν 2.2' d -0.1' m 1.11				

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	163°58.9	157°18.9	N11°03.5	54°25.1	N25°38.7	92°40.5	N21°56.6	171°12.9	S05°12.3	Denebola	182°24.5	14°25.7
1	179°01.3	172°21.9	03.7	69°27.3	38.6	107°42.7	56.6	186°15.1	12.2	Gienah	175°43.2	-17°41.0
2	194°03.8	187°24.9	03.8	84°29.5	38.4	122°44.9	56.6	201°17.2	12.1	Acrux	172°59.4	-63°14.3
3	209°06.3	202°27.9	· · 04.0	99°31.6	· · 38.3	137°47.1	· · 56.7	216°19.4	· · 12.0	Gacrux	171°51.0	-57°15.2
4	224°08.7	217°31.0	04.2	114°33.8	38.2	152°49.3	56.7	231°21.6	11.8	Alioth	166°12.3	55°49.2
5	239°11.2	232°34.0	04.3	129°35.9	38.1	167°51.5	56.7	246°23.8	11.7	Spica	158°21.9	-11°17.7
6	254°13.7	247°37.0	N11°04.5	144°38.1	N25°38.0	182°53.7	N21°56.8	261°26.0	S05°11.6	Alkaid	152°51.5	49°11.0
7	269°16.1	262°40.0	04.7	159°40.2	37.9	197°56.0	56.8	276°28.1	11.5	Hadar	148°35.4	-60°29.5
8	284°18.6	277°43.0	04.8	174°42.4	37.8	212°58.2	56.9	291°30.3	11.4	Menkent	147°57.2	-36°29.6
9	299°21.1	292°46.1	· · 05.0	189°44.5	· · 37.6	228°00.4	· · 56.9	306°32.5	· · 11.2	Arcturus	145°47.6	19°02.8
10	314°23.5	307°49.1	05.1	204°46.7	37.5	243°02.6	56.9	321°34.7	11.1	Rigil Kent.	139°39.8	-60°56.2
11	329°26.0	322°52.1	05.3	219°48.8	37.4	258°04.8	57.0	336°36.9	11.0	Kochab	137°19.1	74°02.7
12	344°28.4	337°55.2	N11°05.4	234°51.0	N25°37.3	273°07.0	N21°57.0	351°39.0	S05°10.9	Zuben'ubi	136°55.7	-16°08.9
13	359°30.9	352°58.2	05.6	249°53.1	37.2	288°09.2	57.0	6°41.2	10.8	Alphecca	126°03.5	26°37.5
14	14°33.4	8°01.3	05.7	264°55.2	37.1	303°11.4	57.1	21°43.4	10.6	Antares	112°15.6	-26°29.3
15	29°35.8	23°04.3	· · 05.8	279°57.4	· · 37.0	318°13.6	· · 57.1	36°45.6	· · 10.5	Atria	107°09.8	-69°04.1
16	44°38.3	38°07.4	06.0	294°59.5	36.8	333°15.9	57.2	51°47.8	10.4	Sabik	102°02.7	-15°45.4
17	59°40.8	53°10.4	06.1	310°01.7	36.7	348°18.1	57.2	66°49.9	10.3	Shaula	96°10.3	-37°07.3
18	74°43.2	68°13.5	N11°06.2	325°03.8	N25°36.6	3°20.3	N21°57.2	81°52.1	S05°10.2	Rasalhague	95°58.5	12°32.3
19	89°45.7	83°16.6	06.4	340°05.9	36.5	18°22.5	57.3	96°54.3	10.0	Eltanin	90°42.3	51°28.7
20	104°48.2	98°19.6	06.5	355°08.1	36.4	33°24.7	57.3	111°56.5	09.9	Kaus Aust.	83°32.5	-34°22.3
21	119°50.6	113°22.7	· · 06.6	10°10.2	· · 36.3	48°26.9	· · 57.4	126°58.7	· · 09.8	Vega	80°33.3	38°48.1
22	134°53.1	128°25.8	06.7	25°12.4	36.1	63°29.1	57.4	142°00.8	09.7	Nunki	75°47.8	-26°15.9
23	149°55.6	143°28.9	06.9	40°14.5	36.0	78°31.3	57.4	157°03.0	09.6	Altair	62°00.1	8°55.9
Mer.pass. 13:02		ν 3.0' d 0.2' m -4.66		ν 2.2' d -0.1' m -0.15		ν 2.2' d 0.0' m -2.26		ν 2.2' d -0.1' m 1.11				

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	164°58.0	158°31.9	N11°07.0	55°16.6	N25°35.9	93°33.5	N21°57.5	172°05.2	S05°09.4	0		
1	180°00.5	173°35.0	07.1	70°18.7	35.8	108°35.7	57.5	187°07.4	09.3	1		
2	195°02.9	188°38.1	07.2	85°20.9	35.7	123°38.0	57.5	202°09.6	09.2	2		
3	210°05.4	203°41.2	· · 07.3	100°23.0	· · 35.6	138°40.2	· · 57.6	217°11.7	· · 09.1	3		
4	225°07.9	218°44.3	07.4	115°25.1	35.4	153°42.4	57.6	232°13.9	08.9	4		
5	240°10.3	233°47.4	07.5	130°27.3	35.3	168°44.6	57.7	247°16.1	08.8	5		
6	255°12.8	248°50.5	N11°07.6	145°29.4	N25°35.2	183°46.8	N21°57.7	262°18.3	S05°08.7	6		
7	270°15.3	263°53.6	07.7	160°31.5	35.1	198°49.0	57.7	277°20.5	08.6	7		
8	285°17.7	278°56.8	07.8	175°33.6	35.0	213°51.2	57.8	292°22.6	08.5	8		
9	300°20.2	293°59.9	· · 07.9	190°35.7	· · 34.9	228°53.4	· · 57.8	307°24.8	· · 08.3	9		
10	315°22.7	309°03.0	08.0	205°37.9	34.7	243°55.6	57.8	322°27.0	08.2	10		
11	330°25.1	324°06.1	08.1	220°40.0	34.6	258°57.8	57.9	337°29.2	08.1	11		
12	345°27.6	339°09.3	N11°08.2	235°42.1	N25°34.5	274°00.0	N21°57.9	352°31.4	S05°08.0	12		
13	0°30.1	354°12.4	08.3	250°44.2	34.4	289°02.2	58.0	7°33.5	07.9	13		
14	15°32.5	9°15.5	08.3	265°46.3	34.3	304°04.4	58.0	22°35.7	07.7	14		
15	30°35.0	24°18.7	· · 08.4	280°48.4	· · 34.1	319°06.6	· · 58.0	37°37.9	· · 07.6	15		
16	45°37.4	39°21.8	08.5	295°50.6	34.0	334°08.8	58.1	52°40.1	07.5	16		
17	60°39.9	54°25.0	08.6	310°52.7	33.9	349°11.0	58.1	67°42.2	07.4	17		
18	75°42.4	69°28.1	N11°08.7	325°54.8	N25°33.8	4°13.2	N21°58.2	82°44.4	S05°07.3	18		
19	90°44.8	84°31.3	08.7	340°56.9	33.7	19°15.4	58.2	97°46.6	07.1	19		
20	105°47.3	99°34.4	08.8	355°59.0	33.5	34°17.6	58.2	112°48.8	07.0	20		
21	120°49.8	114°37.6	· · 08.9	11°01.1	· · 33.4	49°19.8	· · 58.3	127°51.0	· · 06.9	21		
22	135°52.2	129°40.8	08.9	26°03.2	33.3	64°22.0	58.3	142°53.1	06.8	22		
23	150°54.7	144°43.9	09.0	41°05.3	33.2	79°24.2	58.3	157°55.3	06.7	23		
Mer.pass. 12:58		ν 3.1' d 0.1' m -4.63		ν 2.1' d -0.1' m -0.12		ν 2.2' d 0.0' m -2.26		ν 2.2' d -0.1' m 1.11				

Mar 05 Wed		SHA	Mer.pass
Venus	353°08.4		13:33
Mars	250°33.2		20:23
Jupiter	288°47.5		17:50
Saturn	7°20.8		12:37

Mar 06 Thu		SHA	Mer.pass
Venus	353°20.1		13:28
Mars	250°26.3		20:19
Jupiter	288°41.6		17:47
Saturn	7°14.0		12:33

Mar 07 Fri		SHA	Mer.pass
Venus	353°33.9		13:23
Mars	250°18.6		20:16
Jupiter	288°35.5		17:43
Saturn	7°07.2		12:30

Horizontal parallax		
Venus:		0.5
Mars:		0.2

DUT1 = UT1-UTC = +0.0421 sec $\Delta T = TT-UT1 = +69.1419$ sec

2025 March 05 to Mar. 07 UT

h		Sun			Moon		
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	177°07.3	S06°02.4	114°00.4	6.7'	N22°36.5	11.1'	59.5'
1	192°07.4	01.4	128°26.1	6.6'	22°47.6	10.9'	59.4'
2	207°07.6	06°00.5	142°51.7	6.5'	22°58.5	10.8'	59.4'
3	222°07.7	05°59.5	157°17.2	6.5'	23°09.3	10.6'	59.4'
4	237°07.9	58.5	171°42.7	6.4'	23°19.9	10.5'	59.4'
5	252°08.0	57.6	186°08.1	6.3'	23°30.4	10.3'	59.3'
6	267°08.2	S05°56.6	200°33.4	6.3'	N23°40.7	10.2'	59.3'
7	282°08.3	55.6	214°58.7	6.2'	23°50.9	10.0'	59.3'
8	297°08.4	54.7	229°23.9	6.1'	24°00.9	9.8'	59.3'
9	312°08.6	· · 53.7	243°49.0	6.1'	24°10.7	9.7'	59.2'
10	327°08.7	52.7	258°14.1	6.0'	24°20.4	9.5'	59.2'
11	342°08.9	51.8	272°39.1	6.0'	24°29.9	9.4'	59.2'
12	357°09.0	S05°50.8	287°04.1	5.9'	N24°39.3	9.2'	59.2'
13	12°09.1	49.8	301°29.0	5.8'	24°48.5	9.0'	59.1'
14	27°09.3	48.8	315°53.8	5.8'	24°57.5	8.9'	59.1'
15	42°09.4	· · 47.9	330°18.6	5.7'	25°06.3	8.7'	59.1'
16	57°09.6	46.9	344°43.4	5.7'	25°15.0	8.5'	59.1'
17	72°09.7	45.9	359°08.0	5.6'	25°23.6	8.4'	59.0'
18	87°09.9	S05°45.0	13°32.7	5.6'	N25°31.9	8.2'	59.0'
19	102°10.0	44.0	27°57.2	5.5'	25°40.1	8.0'	59.0'
20	117°10.1	43.0	42°21.8	5.5'	25°48.1	7.8'	58.9'
21	132°10.3	· · 42.1	56°46.2	5.4'	25°56.0	7.7'	58.9'
22	147°10.4	41.1	71°10.7	5.4'	26°03.6	7.5'	58.9'
23	162°10.6	40.1	85°35.0	5.3'	26°11.1	7.3'	58.9'
SD = 16.1'		$d = -1.0'$	SD = 16.2'				
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	177°10.7	S05°39.2	99°59.4	5.3'	N26°18.5	7.1'	58.8'
1	192°10.9	38.2	114°23.7	5.2'	26°25.6	7.0'	58.8'
2	207°11.0	37.2	128°47.9	5.2'	26°32.6	6.8'	58.8'
3	222°11.2	· · 36.2	143°12.1	5.2'	26°39.4	6.6'	58.8'
4	237°11.3	35.3	157°36.3	5.1'	26°46.0	6.4'	58.7'
5	252°11.4	34.3	172°00.4	5.1'	26°52.4	6.3'	58.7'
6	267°11.6	S05°33.3	186°24.5	5.1'	N26°58.7	6.1'	58.7'
7	282°11.7	32.4	200°48.5	5.0'	27°04.8	5.9'	58.7'
8	297°11.9	31.4	215°12.6	5.0'	27°10.7	5.7'	58.6'
9	312°12.0	· · 30.4	229°36.6	5.0'	27°16.4	5.5'	58.6'
10	327°12.2	29.5	244°00.5	4.9'	27°22.0	5.4'	58.6'
11	342°12.3	28.5	258°24.5	4.9'	27°27.3	5.2'	58.5'
12	357°12.5	S05°27.5	272°48.4	4.9'	N27°32.5	5.0'	58.5'
13	12°12.6	26.5	287°12.3	4.9'	27°37.5	4.8'	58.5'
14	27°12.8	25.6	301°36.2	4.8'	27°42.3	4.6'	58.5'
15	42°12.9	· · 24.6	316°00.0	4.8'	27°47.0	4.5'	58.4'
16	57°13.1	23.6	330°23.8	4.8'	27°51.4	4.3'	58.4'
17	72°13.2	22.7	344°47.6	4.8'	27°55.7	4.1'	58.4'
18	87°13.4	S05°21.7	359°11.5	4.8'	N27°59.8	3.9'	58.4'
19	102°13.5	20.7	13°35.2	4.8'	28°03.7	3.7'	58.3'
20	117°13.7	19.7	27°59.0	4.8'	28°07.4	3.5'	58.3'
21	132°13.8	· · 18.8	42°22.8	4.8'	28°10.9	3.3'	58.3'
22	147°14.0	17.8	56°46.6	4.8'	28°14.3	3.2'	58.2'
23	162°14.1	16.8	71°10.3	4.8'	28°17.4	3.0'	58.2'
SD = 16.1'		$d = -1.0'$	SD = 16.0'				
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	177°14.2	S05°15.9	85°34.1	4.8'	N28°20.4	2.8'	58.2'
1	192°14.4	14.9	99°57.9	4.8'	28°23.2	2.6'	58.2'
2	207°14.5	13.9	114°21.6	4.8'	28°25.8	2.4'	58.1'
3	222°14.7	· · 12.9	128°45.4	4.8'	28°28.2	2.2'	58.1'
4	237°14.8	12.0	143°09.2	4.8'	28°30.5	2.1'	58.1'
5	252°15.0	11.0	157°33.0	4.8'	28°32.5	1.9'	58.1'
6	267°15.1	S05°10.0	171°56.8	4.8'	N28°34.4	1.7'	58.0'
7	282°15.3	09.0	186°20.6	4.8'	28°36.1	1.5'	58.0'
8	297°15.4	08.1	200°44.4	4.8'	28°37.6	1.3'	58.0'
9	312°15.6	· · 07.1	215°08.3	4.9'	28°38.9	1.1'	57.9'
10	327°15.7	06.1	229°32.1	4.9'	28°40.0	1.0'	57.9'
11	342°15.9	05.1	243°56.0	4.9'	28°41.0	0.8'	57.9'
12	357°16.0	S05°04.2	258°19.9	4.9'	N28°41.7	0.6'	57.9'
13	12°16.2	03.2	272°43.9	5.0'	28°42.3	0.4'	57.8'
14	27°16.4	02.2	287°07.9	5.0'	28°42.7	0.2'	57.8'
15	42°16.5	· · 01.2	301°31.9	5.0'	28°43.0	0.0'	57.8'
16	57°16.7	05°00.3	315°55.9	5.1'	28°43.0	-0.1'	57.8'
17	72°16.8	04°59.3	330°20.0	5.1'	28°42.9	-0.3'	57.7'
18	87°17.0	S04°58.3	344°44.1	5.1'	N28°42.5	-0.5'	57.7'
19	102°17.1	57.4	359°08.2	5.2'	28°42.0	-0.7'	57.7'
20	117°17.3	56.4	13°32.4	5.2'	28°41.4	-0.9'	57.6'
21	132°17.4	· · 55.4	27°56.6	5.3'	28°40.5	-1.0'	57.6'
22	147°17.6	54.4	42°20.9	5.3'	28°39.5	-1.2'	57.6'
23	162°17.7	53.5	56°45.2	5.4'	28°38.2	-1.4'	57.6'
SD = 16.1'		$d = -1.0'$	SD = 15.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:43	06:02	07:09	17:15	18:23	19:42
N 70°	04:51	06:02	07:03	17:21	18:22	19:34
68°	04:58	06:02	06:58	17:26	18:22	19:27
66°	05:03	06:02	06:53	17:31	18:22	19:22
64°	05:07	06:02	06:49	17:34	18:22	19:17
62°	05:11	06:02	06:46	17:38	18:22	19:13
60°	05:14	06:02	06:43	17:40	18:22	19:10
N 58°	05:16	06:01	06:41	17:43	18:22	19:08
56°	05:18	06:01	06:38	17:45	18:22	19:05
54°	05:20	06:01	06:36	17:47	18:22	19:03
52°	05:22	06:01	06:34	17:49	18:23	19:02
50°	05:23	06:00	06:33	17:51	18:23	19:00
45°	05:26	05:59	06:29	17:54	18:24	18:58
N 40°	05:27	05:59	06:26	17:57	18:24	18:56
35°	05:28	05:57	06:23	18:00	18:25	18:55
30°	05:29	05:56	06:20	18:03	18:26	18:54
20°	05:28	05:54	06:16	18:07	18:29	18:54
N 10°	05:26	05:51	06:12	18:11	18:32	18:56
0°	05:23	05:47	06:08	18:14	18:35	18:59
S 10°	05:18	05:43	06:04	18:18	18:39	19:04
20°	05:11	05:37	06:00	18:22	18:45	19:10
30°	05:02	05:30	05:54	18:27	18:51	19:20
35°	04:56	05:26	05:52	18:30	18:56	19:26
40°	04:48	05:21	05:48	18:34	19:01	19:33
45°	04:39	05:14	05:44	18:37	19:07	19:43
S 50°	04:26	05:06	05:39	18:42	19:15	19:54
52°	04:21	05:03	05:37	18:44	19:19	20:00
54°	04:14	04:58	05:35	18:46	19:23	20:07
56°	04:06	04:54	05:32	18:49	19:27	20:14
58°	03:58	04:48	05:29	18:52	19:32	20:23
S 60°	03:47	04:42	05:26	18:55	19:38	20:33

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	□	□	□	□
66°	05:29	□	□	03:24	□	□
64°	06:24	□	□	02:30	□	□
62°	06:58	06:46	□	01:58	04:13	□
60°	07:22	07:31	07:57	01:34	03:28	05:08
N 58°	07:42	08:00	08:37	01:15	02:59	04:28
56°	07:59	08:23	09:05	00:59	02:36	04:00
54°	08:13	08:42	09:27	00:45	02:18	03:38
52°	08:25	08:58	09:45	00:34	02:03	03:20
50°	08:36	09:12	10:00	00:23	01:49	03:05
45°	08:59	09:40	10:32	00:01	01:22	02:34
N 40°	09:18	10:02	10:56	· ·	01:00	02:10
35°	09:33	10:21	11:16	· ·	00:42	01:50
30°	09:47	10:36	11:33	· ·	00:27	01:34
20°	10:10	11:04	12:01	· ·	00:01	01:05
N 10°	10:31	11:27	12:26	23:38	· ·	00:41
0°	10:50	11:49	12:49	23:18	· ·	00:18
S 10°	11:09	12:11	13:12	22:57	23:55	· ·
20°	11:30	12:35	13:37	22:35	23:31	· ·
30°	11:54	13:02	14:06	22:09	23:02	· ·
35°	12:08	13:19	14:23	21:54	22:45	23:44
40°	12:25	13:38	14:44	21:37	22:26	23:24
45°	12:44	14:01	15:08	21:16	22:02	23:00
S 50°	13:10	14:31	15:40	20:50	21:32	22:28
52°	13:22	14:46	15:56	20:37	21:16	22:12
54°	13:36	15:04	16:14	20:23	20:59	21:54
56°	13:52	15:25	16:37	20:06	20:38	21:31
58°	14:12	15:51	17:06	19:46	20:11	21:02
S 60°	14:36	16:29	17:51	19:21	19:33	20:17

Day	Sun		Mer. Pass	Moon		Age 5-7 31-54%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
05	11:31	11:24	12:11	17:04	04:34	
06	11:17	11:10	12:11	18:03	05:33	
07	11:03	10:56	12:11	19:04	06:34	

March 08, 09, 10 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	165°57.2	159°47.1	N11°09.0	56°07.4	N25°33.1	94°26.4	N21°58.4	172°57.5	S05°06.5	Alpheratz	357°35.0 29°13.7
1	180°59.6	174°50.3	09.1	71°09.5	32.9	109°28.6	58.4	187°59.7	06.4	Ankaa	353°07.4 -42°10.3
2	196°02.1	189°53.5	09.1	86°11.6	32.8	124°30.8	58.5	203°01.9	06.3	Schedar	349°31.4 56°40.6
3	211°04.6	204°56.6	· · 09.2	101°13.7	· · 32.7	139°33.0	· · 58.5	218°04.0	· · 06.2	Diphda	348°47.4 -17°51.1
4	226°07.0	219°59.8	09.2	116°15.8	32.6	154°35.2	58.5	233°06.2	06.1	Achernar	335°20.5 -57°06.7
5	241°09.5	235°03.0	09.3	131°17.9	32.5	169°37.4	58.6	248°08.4	05.9	Hamal	327°51.3 23°34.9
6	256°11.9	250°06.2	N11°09.3	146°20.0	N25°32.3	184°39.6	N21°58.6	263°10.6	S05°05.8	Polaris	314°13.5 89°22.5
7	271°14.4	265°09.4	09.4	161°22.1	32.2	199°41.8	58.7	278°12.8	05.7	Acamar	315°11.9 -40°12.4
8	286°16.9	280°12.6	09.4	176°24.2	32.1	214°44.0	58.7	293°14.9	05.6	Menkar	314°06.2 4°11.2
9	301°19.3	295°15.8	· · 09.4	191°26.3	· · 32.0	229°46.2	· · 58.7	308°17.1	· · 05.4	Mirfak	308°28.3 49°57.2
10	316°21.8	310°19.0	09.5	206°28.4	31.9	244°48.4	58.8	323°19.3	05.3	Aldebaran	290°39.5 16°33.6
11	331°24.3	325°22.3	09.5	221°30.5	31.7	259°50.6	58.8	338°21.5	05.2	Rigel	281°03.7 -8°10.5
12	346°26.7	340°25.5	N11°09.5	236°32.6	N25°31.6	274°52.8	N21°58.9	353°23.7	S05°05.1	Capella	280°21.7 46°01.5
13	1°29.2	355°28.7	09.6	251°34.6	31.5	289°55.0	58.9	8°25.8	05.0	Bellatrix	278°22.7 6°22.3
14	16°31.7	10°31.9	09.6	266°36.7	31.4	304°57.2	58.9	23°28.0	04.8	Elnath	278°01.7 28°37.8
15	31°34.1	25°35.1	· · 09.6	281°38.8	· · 31.3	319°59.4	· · 59.0	38°30.2	· · 04.7	Alnilam	275°37.5 -1°11.2
16	46°36.6	40°38.4	09.6	296°40.9	31.1	335°01.6	59.0	53°32.4	04.6	Betelgeuse	270°51.9 7°24.6
17	61°39.0	55°41.6	09.6	311°43.0	31.0	350°03.8	59.1	68°34.6	04.5	Canopus	263°52.2 -52°42.8
18	76°41.5	70°44.8	N11°09.6	326°45.1	N25°30.9	5°06.0	N21°59.1	83°36.7	S05°04.4	Sirius	258°26.0 -16°45.2
19	91°44.0	85°48.1	09.6	341°47.1	30.8	20°08.2	59.1	98°38.9	04.2	Adhara	255°05.6 -29°00.6
20	106°46.4	100°51.3	09.7	356°49.2	30.6	35°10.4	59.2	113°41.1	04.1	Procyon	244°50.5 5°09.5
21	121°48.9	115°54.6	· · 09.7	11°51.3	· · 30.5	50°12.6	· · 59.2	128°43.3	· · 04.0	Pollux	243°16.9 27°57.9
22	136°51.4	130°57.8	09.7	26°53.4	30.4	65°14.8	59.2	143°45.5	03.9	Avior	234°14.2 -59°35.6
23	151°53.8	146°01.1	09.7	41°55.5	30.3	80°17.0	59.3	158°47.6	03.8	Suhail	222°45.9 -43°32.2
Mer.pass. 12:54		ν 3.2' d 0.1' m -4.59		ν 2.1' d -0.1' m -0.10		ν 2.2' d 0.0' m -2.25		ν 2.2' d -0.1' m 1.10		Miaplacidus	221°37.5 -69°49.4
Alphard 217°47.3 -8°46.2 Regulus 207°34.0 11°50.5 Dubhe 193°40.1 61°36.8 Denebola 182°24.5 14°25.7 Gienah 175°43.1 -17°41.0 Acrux 172°59.3 -63°14.3 Gacrux 171°51.0 -57°15.2 Alioth 166°12.3 55°49.2 Spica 158°21.9 -11°17.7 Alkaid 152°51.5 49°11.0 Hadar 148°35.4 -60°29.6 Menkent 147°57.2 -36°29.6 Arcturus 145°47.5 19°02.8 Rigil Kent. 139°39.7 -60°56.2 Kochab 137°19.1 74°02.7 Zuben'ubi 136°55.7 -16°08.9 Alphecca 126°03.5 26°37.5 Antares 112°15.6 -26°29.3 Atria 107°09.7 -69°04.1 Sabik 102°02.6 -15°45.4 Shaula 96°10.2 -37°07.3 Rasalhague 95°58.5 12°32.3 Eltanin 90°42.2 51°28.7 Kaus Aust. 83°32.4 -34°22.3 Vega 80°33.3 38°48.1 Nunki 75°47.7 -26°15.9 Altair 62°00.0 8°55.9 Peacock 53°05.9 -56°39.1 Deneb 49°26.1 45°21.9 Enif 33°39.0 9°59.3 Al Na'ir 27°33.2 -46°50.4 Fomalhaut 15°14.7 -29°29.4 Scheat 13°45.4 28°13.0 Markab 13°30.1 15°20.3											
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	166°56.3	161°04.4	N11°09.7	56°57.5	N25°30.1	95°19.2	N21°59.3	173°49.8	S05°03.6		
1	181°58.8	176°07.6	09.7	71°59.6	30.0	110°21.4	59.4	188°52.0	03.5		
2	197°01.2	191°10.9	09.6	87°01.7	29.9	125°23.6	59.4	203°54.2	03.4		
3	212°03.7	206°14.2	· · 09.6	102°03.8	· · 29.8	140°25.7	· · 59.4	218°56.4	· · 03.3		
4	227°06.2	221°17.4	09.6	117°05.8	29.7	155°27.9	59.5	233°58.5	03.2		
5	242°08.6	236°20.7	09.6	132°07.9	29.5	170°30.1	59.5	249°00.7	03.0		
6	257°11.1	251°24.0	N11°09.6	147°10.0	N25°29.4	185°32.3	N21°59.6	264°02.9	S05°02.9		
7	272°13.5	266°27.3	09.6	162°12.0	29.3	200°34.5	59.6	279°05.1	02.8		
8	287°16.0	281°30.6	09.5	177°14.1	29.2	215°36.7	59.6	294°07.2	02.7		
9	302°18.5	296°33.9	· · 09.5	192°16.2	· · 29.0	230°38.9	· · 59.7	309°09.4	· · 02.6		
10	317°20.9	311°37.2	09.5	207°18.2	28.9	245°41.1	59.7	324°11.6	02.4		
11	332°23.4	326°40.4	09.5	222°20.3	28.8	260°43.3	59.8	339°13.8	02.3		
12	347°25.9	341°43.8	N11°09.4	237°22.4	N25°28.7	275°45.5	N21°59.8	354°16.0	S05°02.2		
13	2°28.3	356°47.1	09.4	252°24.4	28.5	290°47.7	59.8	9°18.1	02.1		
14	17°30.8	11°50.4	09.3	267°26.5	28.4	305°49.8	59.9	24°20.3	01.9		
15	32°33.3	26°53.7	· · 09.3	282°28.5	· · 28.3	320°52.0	21°59.9	39°22.5	· · 01.8		
16	47°35.7	41°57.0	09.3	297°30.6	28.2	335°54.2	22°00.0	54°24.7	01.7		
17	62°38.2	57°00.3	09.2	312°32.7	28.0	350°56.4	00.0	69°26.9	01.6		
18	77°40.7	72°03.6	N11°09.2	327°34.7	N25°27.9	5°58.6	N22°00.0	84°29.0	S05°01.5		
19	92°43.1	87°07.0	09.1	342°36.8	27.8	21°00.8	00.1	99°31.2	01.3		
20	107°45.6	102°10.3	09.1	357°38.8	27.6	36°03.0	00.1	114°33.4	01.2		
21	122°48.0	117°13.6	· · 09.0	12°40.9	· · 27.5	51°05.2	· · 00.2	129°35.6	· · 01.1		
22	137°50.5	132°17.0	08.9	27°42.9	27.4	66°07.4	00.2	144°37.8	01.0		
23	152°53.0	147°20.3	08.9	42°45.0	27.3	81°09.5	00.2	159°39.9	00.9		
Mer.pass. 12:50		ν 3.3' d -0.0' m -4.55		ν 2.1' d -0.1' m -0.07		ν 2.2' d 0.0' m -2.24		ν 2.2' d -0.1' m 1.10			
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	167°55.4	162°23.6	N11°08.8	57°47.0	N25°27.1	96°11.7	N22°00.3	174°42.1	S05°00.7		
1	182°57.9	177°27.0	08.7	72°49.1	27.0	111°13.9	00.3	189°44.3	00.6		
2	198°00.4	192°30.3	08.7	87°51.1	26.9	126°16.1	00.4	204°46.5	00.5		
3	213°02.8	207°33.7	· · 08.6	102°53.2	· · 26.8	141°18.3	· · 00.4	219°48.7	· · 00.4		
4	228°05.3	222°37.1	08.5	117°55.2	26.6	156°20.5	00.4	234°50.8	00.3		
5	243°07.8	237°40.4	08.4	132°57.2	26.5	171°22.7	00.5	249°53.0	00.1		
6	258°10.2	252°43.8	N11°08.4	147°59.3	N25°26.4	186°24.8	N22°00.5	264°55.2	S05°00.0		
7	273°12.7	267°47.1	08.3	163°01.3	26.2	201°27.0	00.6	279°57.4	04°59.9		
8	288°15.2	282°50.5	08.2	178°03.4	26.1	216°29.2	00.6	294°59.5	59.8		
9	303°17.6	297°53.9	· · 08.1	193°05.4	· · 26.0	231°31.4	· · 00.6	310°01.7	· · 59.7		
10	318°20.1	312°57.3	08.0	208°07.4	25.9	246°33.6	00.7	325°03.9	59.5		
11	333°22.5	328°00.6	07.9	223°09.5	25.7	261°35.8	00.7	340°06.1	59.4		
12	348°25.0	343°04.0	N11°07.8	238°11.5	N25°25.6	276°37.9	N22°00.8	355°08.3	S04°59.3		
13	3°27.5	358°07.4	07.7	253°13.6	25.5	291°40.1	00.8	10°10.4	59.2		
14	18°29.9	13°10.8	07.6	268°15.6	25.3	306°42.3	00.8	25°12.6	59.1		
15	33°32.4	28°14.2	· · 07.5	283°17.6	· · 25.2	321°44.5	· · 00.9	40°14.8	· · 58.9		
16	48°34.9	43°17.6	07.4	298°19.6	25.1	336°46.7	00.9	55°17.0	58.8		
17	63°37.3	58°21.0	07.3	313°21.7	25.0	351°48.9	01.0	70°19.2	58.7		
18	78°39.8	73°24.4	N11°07.2	328°23.7	N25°24.8	6°51.0	N22°01.0	85°21.3	S04°58.6		
19	93°42.3	88°27.8	07.1	343°25.7	24.7	21°53.2	01.0	100°23.5	58.4		
20	108°44.7	103°31.2	07.0	358°27.8	24.6	36°55.4	01.1	115°25.7	58.3		
21	123°47.2	118°34.6	· · 06.8	13°29.8	· · 24.4	51°57.6	· · 01.1	130°27.9	· · 58.2		
22	138°49.6	133°38.1	06.7	28°31.8	24.3	66°59.8	01.2	145°30.1	58.1		
23	153°52.1	148°41.5	06.6	43°33.8	24.2	82°01.9	01.2	160°32.2	58.0		
Mer.pass. 12:46		ν 3.3' d -0.1' m -4.50		ν 2.0' d -0.1' m -0.05		ν 2.2' d 0.0' m -2.24		ν 2.2' d -0.1' m 1.10			
Mar 08 Sat SHA Mer.pass Venus 353°49.9 13:18 Mars 250°10.3 20:13 Jupiter 288°29.3 17:40 Saturn 7°00.3 12:26 Mar 09 Sun SHA Mer.pass Venus 354°08.1 13:13 Mars 250°01.2 20:09 Jupiter 288°22.9 17:36 Saturn 6°53.5 12:23 Mar 10 Mon SHA Mer.pass Venus 354°28.2 13:07 Mars 249°51.6 20:06 Jupiter 288°16.3 17:33 Saturn 6°46.7 12:19 Horizontal parallax Venus: 0.5 Mars: 0.2											

Sun				Moon				
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	177°17.9	S04°52.5		71°09.6	5.4'	N28°36.9	-1.6'	57.5'
1	192°18.0	51.5		85°34.0	5.5'	28°35.3	-1.7'	57.5'
2	207°18.2	50.5		99°58.5	5.5'	28°33.5	-1.9'	57.5'
3	222°18.3	· · 49.6		114°23.0	5.6'	28°31.6	-2.1'	57.5'
4	237°18.5	48.6		128°47.6	5.6'	28°29.5	-2.3'	57.4'
5	252°18.6	47.6		143°12.2	5.7'	28°27.3	-2.4'	57.4'
6	267°18.8	S04°46.6		157°37.0	5.8'	N28°24.8	-2.6'	57.4'
7	282°19.0	45.6		172°01.7	5.8'	28°22.2	-2.8'	57.4'
8	297°19.1	44.7		186°26.6	5.9'	28°19.5	-2.9'	57.3'
9	312°19.3	· · 43.7		200°51.4	6.0'	28°16.5	-3.1'	57.3'
10	327°19.4	42.7		215°16.4	6.0'	28°13.4	-3.3'	57.3'
11	342°19.6	41.7		229°41.4	6.1'	28°10.1	-3.4'	57.3'
12	357°19.7	S04°40.8		244°06.6	6.2'	N28°06.7	-3.6'	57.2'
13	12°19.9	39.8		258°31.7	6.3'	28°03.1	-3.8'	57.2'
14	27°20.0	38.8		272°57.0	6.3'	27°59.3	-3.9'	57.2'
15	42°20.2	· · 37.8		287°22.3	6.4'	27°55.3	-4.1'	57.2'
16	57°20.3	36.9		301°47.7	6.5'	27°51.2	-4.3'	57.1'
17	72°20.5	35.9		316°13.2	6.6'	27°47.0	-4.4'	57.1'
18	87°20.7	S04°34.9		330°38.7	6.6'	N27°42.5	-4.6'	57.1'
19	102°20.8	33.9		345°04.4	6.7'	27°38.0	-4.7'	57.1'
20	117°21.0	33.0		359°30.1	6.8'	27°33.2	-4.9'	57.0'
21	132°21.1	· · 32.0		13°55.9	6.9'	27°28.3	-5.1'	57.0'
22	147°21.3	31.0		28°21.8	7.0'	27°23.3	-5.2'	57.0'
23	162°21.4	30.0		42°47.8	7.1'	27°18.0	-5.4'	57.0'
SD = 16.1'			$d = -1.0'$	SD = 15.7'				
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	177°21.6	S04°29.0		57°13.8	7.2'	N27°12.7	-5.5'	56.9'
1	192°21.8	28.1		71°40.0	7.2'	27°07.2	-5.7'	56.9'
2	207°21.9	27.1		86°06.2	7.3'	27°01.5	-5.8'	56.9'
3	222°22.1	· · 26.1		100°32.6	7.4'	26°55.7	-6.0'	56.9'
4	237°22.2	25.1		114°59.0	7.5'	26°49.7	-6.1'	56.8'
5	252°22.4	24.2		129°25.5	7.6'	26°43.6	-6.3'	56.8'
6	267°22.5	S04°23.2		143°52.1	7.7'	N26°37.4	-6.4'	56.8'
7	282°22.7	22.2		158°18.8	7.8'	26°31.0	-6.5'	56.8'
8	297°22.9	21.2		172°45.6	7.9'	26°24.4	-6.7'	56.7'
9	312°23.0	· · 20.2		187°12.5	8.0'	26°17.7	-6.8'	56.7'
10	327°23.2	19.3		201°39.5	8.1'	26°10.9	-7.0'	56.7'
11	342°23.3	18.3		216°06.6	8.2'	26°03.9	-7.1'	56.7'
12	357°23.5	S04°17.3		230°33.7	8.3'	N25°56.8	-7.2'	56.6'
13	12°23.7	16.3		245°01.0	8.4'	25°49.6	-7.4'	56.6'
14	27°23.8	15.3		259°28.4	8.5'	25°42.2	-7.5'	56.6'
15	42°24.0	· · 14.4		273°55.9	8.6'	25°34.7	-7.6'	56.6'
16	57°24.1	13.4		288°23.5	8.7'	25°27.1	-7.8'	56.5'
17	72°24.3	12.4		302°51.1	8.8'	25°19.3	-7.9'	56.5'
18	87°24.5	S04°11.4		317°18.9	8.9'	N25°11.4	-8.0'	56.5'
19	102°24.6	10.4		331°46.8	9.0'	25°03.3	-8.2'	56.5'
20	117°24.8	09.5		346°14.8	9.1'	24°55.2	-8.3'	56.5'
21	132°24.9	· · 08.5		0°42.9	9.2'	24°46.9	-8.4'	56.4'
22	147°25.1	07.5		15°11.0	9.3'	24°38.5	-8.5'	56.4'
23	162°25.3	06.5		29°39.3	9.4'	24°30.0	-8.7'	56.4'
SD = 16.1'			$d = -1.0'$	SD = 15.5'				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	177°25.4	S04°05.6		44°07.7	9.5'	N24°21.3	-8.8'	56.4'
1	192°25.6	04.6		58°36.2	9.6'	24°12.5	-8.9'	56.3'
2	207°25.7	03.6		73°04.8	9.7'	24°03.6	-9.0'	56.3'
3	222°25.9	· · 02.6		87°33.5	9.8'	23°54.6	-9.1'	56.3'
4	237°26.1	01.6		102°02.3	9.9'	23°45.5	-9.2'	56.3'
5	252°26.2	04°00.7		116°31.2	10.0'	23°36.2	-9.4'	56.3'
6	267°26.4	S03°59.7		131°00.2	10.1'	N23°26.9	-9.5'	56.2'
7	282°26.5	58.7		145°29.3	10.2'	23°17.4	-9.6'	56.2'
8	297°26.7	57.7		159°58.5	10.3'	23°07.8	-9.7'	56.2'
9	312°26.9	· · 56.7		174°27.8	10.4'	22°58.2	-9.8'	56.2'
10	327°27.0	55.7		188°57.2	10.5'	22°48.4	-9.9'	56.1'
11	342°27.2	54.8		203°26.8	10.6'	22°38.5	-10.0'	56.1'
12	357°27.4	S03°53.8		217°56.4	10.7'	N22°28.4	-10.1'	56.1'
13	12°27.5	52.8		232°26.1	10.8'	22°18.3	-10.2'	56.1'
14	27°27.7	51.8		246°55.9	10.9'	22°08.1	-10.3'	56.1'
15	42°27.8	· · 50.8		261°25.8	11.0'	21°57.8	-10.4'	56.0'
16	57°28.0	49.9		275°55.9	11.1'	21°47.4	-10.5'	56.0'
17	72°28.2	48.9		290°26.0	11.2'	21°36.9	-10.6'	56.0'
18	87°28.3	S03°47.9		304°56.2	11.3'	N21°26.3	-10.7'	56.0'
19	102°28.5	46.9		319°26.5	11.4'	21°15.6	-10.8'	56.0'
20	117°28.7	45.9		333°57.0	11.5'	21°04.8	-10.9'	55.9'
21	132°28.8	· · 45.0		348°27.5	11.6'	20°53.9	-11.0'	55.9'
22	147°29.0	44.0		2°58.1	11.7'	20°42.9	-11.1'	55.9'
23	162°29.2	43.0		17°28.8	11.8'	20°31.8	-11.2'	55.9'
SD = 16.1'			$d = -1.0'$	SD = 15.4'				
Twilight		Sunrise		Sunset		Twilight		
Lat.	Naut.	Civil				Civil	Naut.	
N 72°	04:27	05:47	06:54	17:29	18:37	19:57		
N 70°	04:37	05:49	06:49	17:34	18:35	19:47		
68°	04:45	05:50	06:45	17:38	18:33	19:39		
66°	04:51	05:51	06:42	17:41	18:32	19:32		
64°	04:56	05:52	06:39	17:43	18:31	19:26		
62°	05:01	05:52	06:36	17:46	18:30	19:22		
60°	05:05	05:53	06:34	17:48	18:29	19:18		
N 58°	05:08	05:53	06:32	17:50	18:29	19:15		
56°	05:11	05:54	06:31	17:51	18:28	19:12		
54°	05:13	05:54	06:29	17:53	18:28	19:09		
52°	05:15	05:54	06:28	17:54	18:28	19:07		
50°	05:17	05:54	06:26	17:56	18:28	19:05		
45°	05:20	05:54	06:23	17:58	18:28	19:02		
N 40°	05:23	05:54	06:21	18:01	18:28	18:59		
35°	05:24	05:53	06:19	18:03	18:28	18:57		
30°	05:25	05:53	06:17	18:05	18:28	18:56		
20°	05:26	05:51	06:13	18:08	18:30	18:55		
N 10°	05:25	05:49	06:10	18:11	18:32	18:56		
0°	05:22	05:46	06:07	18:14	18:34	18:58		
S 10°	05:18	05:43	06:04	18:17	18:38	19:02		
20°	05:13	05:38	06:01	18:20	18:42	19:08		
30°	05:04	05:32	05:56	18:24	18:48	19:16		
35°	04:59	05:29	05:54	18:26	18:52	19:21		
40°	04:52	05:24	05:51	18:29	18:56	19:28		
45°	04:43	05:18	05:48	18:32	19:01	19:37		
S 50°	04:32	05:11	05:44	18:36	19:08	19:47		
52°	04:27	05:08	05:42	18:37	19:11	19:53		
54°	04:21	05:04	05:41	18:39	19:15	19:59		
56°	04:14	05:00	05:38	18:41	19:19	20:05		
58°	04:06	04:56	05:36	18:44	19:24	20:13		
S 60°	03:57	04:50	05:33	18:46	19:29	20:22		
Lat.	Moonrise			Moonset				
	Sat	Sun	Mon	Sat	Sun	Mon		
N 72°	☐	☐	☐	☐	☐	☐		
N 70°	☐	☐	☐	☐	☐	☐		
68°	☐	☐	☐	☐	☐	☐		
66°	☐	☐	10:50	☐	☐	08:10		
64°	☐	☐	11:36	☐	☐	07:23		
62°	☐	10:06	12:06	☐	07:03	06:53		
60°	09:05	10:45	12:28	06:05	06:24	06:29		
N 58°	09:43	11:12	12:46	05:26	05:56	06:11		
56°	10:10	11:33	13:02	04:59	05:35	05:55		
54°	10:31	11:50	13:15	04:38	05:17	05:41		
52°	10:49	12:05	13:26	04:20	05:02	05:29		
50°	11:04	12:18	13:36	04:05	04:48	05:19		
45°	11:34	12:44	13:57	03:34	04:21	04:57		
N 40°	11:58	13:05	14:14	03:11	04:00	04:39		
35°	12:17	13:22	14:28	02:51	03:42	04:24		
30°	12:34	13:37	14:40	02:34	03:26	04:10		
20°	13:02	14:02	15:01	02:05	03:00	03:48		
N 10°	13:26	14:						

March 11, 12, 13 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	168°54.6	163°44.9	N11°06.5	58°35.9	N25°24.1	97°04.1	N22°01.2	175°34.4	S04°57.8	Alpheratz	357°35.0 29°13.7
1	183°57.0	178°48.3	06.3	73°37.9	23.9	112°06.3	01.3	190°36.6	57.7	Ankaa	353°07.4 -42°10.3
2	198°59.5	193°51.7	06.2	88°39.9	23.8	127°08.5	01.3	205°38.8	57.6	Schedar	349°31.5 56°40.6
3	214°02.0	208°55.2	•• 06.1	103°41.9	•• 23.7	142°10.6	•• 01.4	220°41.0	•• 57.5	Diphda	348°47.4 -17°51.0
4	229°04.4	223°58.6	05.9	118°43.9	23.5	157°12.8	01.4	235°43.1	57.4	Achernar	335°20.5 -57°06.7
5	244°06.9	239°02.1	05.8	133°46.0	23.4	172°15.0	01.4	250°45.3	57.2	Hamal	327°51.3 23°34.9
6	259°09.4	254°05.5	N11°05.6	148°48.0	N25°23.3	187°17.2	N22°01.5	265°47.5	S04°57.1	Polaris	314°14.8 89°22.5
7	274°11.8	269°08.9	05.5	163°50.0	23.1	202°19.4	01.5	280°49.7	57.0	Acamar	315°11.9 -40°12.4
8	289°14.3	284°12.4	05.4	178°52.0	23.0	217°21.5	01.6	295°51.8	56.9	Menkar	314°06.2 4°11.2
9	304°16.8	299°15.8	•• 05.2	193°54.0	•• 22.9	232°23.7	•• 01.6	310°54.0	•• 56.8	Mirfak	308°28.3 49°57.2
10	319°19.2	314°19.3	05.0	208°56.0	22.7	247°25.9	01.7	325°56.2	56.6	Aldebaran	290°39.5 16°33.6
11	334°21.7	329°22.8	04.9	223°58.0	22.6	262°28.1	01.7	340°58.4	56.5	Rigel	281°03.7 -8°10.5
12	349°24.1	344°26.2	N11°04.7	239°00.0	N25°22.5	277°30.2	N22°01.7	356°00.6	S04°56.4	Capella	280°21.7 46°01.5
13	4°26.6	359°29.7	04.6	254°02.1	22.3	292°32.4	01.8	11°02.7	56.3	Bellatrix	278°22.7 6°22.3
14	19°29.1	14°33.2	04.4	269°04.1	22.2	307°34.6	01.8	26°04.9	56.2	Elnath	278°01.7 28°37.8
15	34°31.5	29°36.6	•• 04.2	284°06.1	•• 22.1	322°36.8	•• 01.9	41°07.1	•• 56.0	Alnilam	275°37.6 -1°11.2
16	49°34.0	44°40.1	04.1	299°08.1	21.9	337°38.9	01.9	56°09.3	55.9	Betelgeuse	270°51.9 7°24.6
17	64°36.5	59°43.6	03.9	314°10.1	21.8	352°41.1	01.9	71°11.5	55.8	Canopus	263°52.2 -52°42.8
18	79°38.9	74°47.1	N11°03.7	329°12.1	N25°21.7	7°43.3	N22°02.0	86°13.6	S04°55.7	Sirius	258°26.0 -16°45.2
19	94°41.4	89°50.5	03.5	344°14.1	21.5	22°45.5	02.0	101°15.8	55.6	Adhara	255°05.6 -29°00.6
20	109°43.9	104°54.0	03.4	359°16.1	21.4	37°47.6	02.1	116°18.0	55.4	Procyon	244°50.5 5°09.5
21	124°46.3	119°57.5	•• 03.2	14°18.1	•• 21.3	52°49.8	•• 02.1	131°20.2	•• 55.3	Pollux	243°16.9 27°57.9
22	139°48.8	135°01.0	03.0	29°20.1	21.1	67°52.0	02.1	146°22.4	55.2	Avior	234°14.3 -59°35.6
23	154°51.3	150°04.5	02.8	44°22.1	21.0	82°54.2	02.2	161°24.5	55.1	Suhail	222°45.9 -43°32.2
Mer.pass. 12:42		ν 3.4' d -0.1' m -4.45		ν 2.0' d -0.1' m -0.02		ν 2.2' d 0.0' m -2.23		ν 2.2' d -0.1' m 1.10		Miaplacidus	221°37.6 -69°49.4
Alphard 217°47.3 -8°46.2 Regulus 207°34.0 11°50.5 Dubhe 193°40.1 61°36.9 Denebola 182°24.5 14°25.7 Gienah 175°43.1 -17°41.0 Acrux 172°59.3 -63°14.3 Gacrux 171°51.0 -57°15.3 Alioth 166°12.3 55°49.2 Spica 158°21.9 -11°17.7 Alkaid 152°51.5 49°11.0 Hadar 148°35.4 -60°29.6 Menkent 147°57.2 -36°29.6 Arcturus 145°47.5 19°02.8 Rigel Kent. 139°39.7 -60°56.2 Kochab 137°19.0 74°02.8 Zuben'ubi 136°55.7 -16°08.9 Alphecca 126°03.4 26°37.5 Antares 112°15.6 -26°29.3 Atria 107°09.6 -69°04.1 Sabik 102°02.6 -15°45.4 Shaula 96°10.2 -37°07.3 Rasalhague 95°58.4 12°32.2 Eltanin 90°42.2 51°28.7 Kaus Aust. 83°32.4 -34°22.3 Vega 80°33.3 38°48.1 Nunki 75°47.7 -26°15.9 Altair 62°00.0 8°55.9 Peacock 53°05.9 -56°39.1 Deneb 49°26.1 45°21.9 Enif 33°38.9 9°59.3 Al Na'ir 27°33.1 -46°50.4 Fomalhaut 15°14.7 -29°29.4 Scheat 13°45.4 28°13.0 Markab 13°30.1 15°20.3											
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	169°53.7	165°08.0	N11°02.6	59°24.1	N25°20.9	97°56.3	N22°02.2	176°26.7	S04°55.0		
1	184°56.2	180°11.5	02.4	74°26.1	20.7	112°58.5	02.3	191°28.9	54.8		
2	199°58.6	195°15.0	02.2	89°28.1	20.6	128°00.7	02.3	206°31.1	54.7		
3	215°01.1	210°18.5	•• 02.0	104°30.1	•• 20.5	143°02.8	•• 02.3	221°33.3	•• 54.6		
4	230°03.6	225°22.0	01.8	119°32.1	20.3	158°05.0	02.4	236°35.4	54.5		
5	245°06.0	240°25.5	01.6	134°34.1	20.2	173°07.2	02.4	251°37.6	54.4		
6	260°08.5	255°29.0	N11°01.4	149°36.0	N25°20.1	188°09.4	N22°02.5	266°39.8	S04°54.2		
7	275°11.0	270°32.6	01.2	164°38.0	19.9	203°11.5	02.5	281°42.0	54.1		
8	290°13.4	285°36.1	01.0	179°40.0	19.8	218°13.7	02.6	296°44.1	54.0		
9	305°15.9	300°39.6	•• 00.8	194°42.0	•• 19.7	233°15.9	•• 02.6	311°46.3	•• 53.9		
10	320°18.4	315°43.1	00.6	209°44.0	19.5	248°18.0	02.6	326°48.5	53.7		
11	335°20.8	330°46.7	00.3	224°46.0	19.4	263°20.2	02.7	341°50.7	53.6		
12	350°23.3	345°50.2	N11°00.1	239°48.0	N25°19.3	278°22.4	N22°02.7	356°52.9	S04°53.5		
13	5°25.7	0°53.7	10°59.9	254°50.0	19.1	293°24.5	02.8	11°55.0	53.4		
14	20°28.2	15°57.3	59.7	269°51.9	19.0	308°26.7	02.8	26°57.2	53.3		
15	35°30.7	31°00.8	•• 59.4	284°53.9	•• 18.9	323°28.9	•• 02.8	41°59.4	•• 53.1		
16	50°33.1	46°04.4	59.2	299°55.9	18.7	338°31.0	02.9	57°01.6	53.0		
17	65°35.6	61°07.9	59.0	314°57.9	18.6	353°33.2	02.9	72°03.8	52.9		
18	80°38.1	76°11.5	N10°58.7	329°59.9	N25°18.4	8°35.4	N22°03.0	87°05.9	S04°52.8		
19	95°40.5	91°15.0	58.5	345°01.8	18.3	23°37.5	03.0	102°08.1	52.7		
20	110°43.0	106°18.6	58.3	0°03.8	18.2	38°39.7	03.0	117°10.3	52.5		
21	125°45.5	121°22.1	•• 58.0	15°05.8	•• 18.0	53°41.9	•• 03.1	132°12.5	•• 52.4		
22	140°47.9	136°25.7	57.8	30°07.8	17.9	68°44.0	03.1	147°14.7	52.3		
23	155°50.4	151°29.3	57.5	45°09.7	17.8	83°46.2	03.2	162°16.8	52.2		
Mer.pass. 12:38		ν 3.5' d -0.2' m -4.40		ν 2.0' d -0.1' m 0.00		ν 2.2' d 0.0' m -2.22		ν 2.2' d -0.1' m 1.10			
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	170°52.9	166°32.8	N10°57.3	60°11.7	N25°17.6	98°48.4	N22°03.2	177°19.0	S04°52.1		
1	185°55.3	181°36.4	57.0	75°13.7	17.5	113°50.5	03.3	192°21.2	51.9		
2	200°57.8	196°40.0	56.7	90°15.6	17.4	128°52.7	03.3	207°23.4	51.8		
3	216°00.2	211°43.5	•• 56.5	105°17.6	•• 17.2	143°54.9	•• 03.3	222°25.6	•• 51.7		
4	231°02.7	226°47.1	56.2	120°19.6	17.1	158°57.0	03.4	237°27.7	51.6		
5	246°05.2	241°50.7	55.9	135°21.6	16.9	173°59.2	03.4	252°29.9	51.5		
6	261°07.6	256°54.3	N10°55.7	150°23.5	N25°16.8	189°01.4	N22°03.5	267°32.1	S04°51.3		
7	276°10.1	271°57.9	55.4	165°25.5	16.7	204°03.5	03.5	282°34.3	51.2		
8	291°12.6	287°01.5	55.1	180°27.4	16.5	219°05.7	03.5	297°36.4	51.1		
9	306°15.0	302°05.1	•• 54.9	195°29.4	•• 16.4	234°07.9	•• 03.6	312°38.6	•• 51.0		
10	321°17.5	317°08.7	54.6	210°31.4	16.2	249°10.0	03.6	327°40.8	50.9		
11	336°20.0	332°12.3	54.3	225°33.3	16.1	264°12.2	03.7	342°43.0	50.7		
12	351°22.4	347°15.9	N10°54.0	240°35.3	N25°16.0	279°14.3	N22°03.7	357°45.2	S04°50.6		
13	6°24.9	2°19.5	53.7	255°37.3	15.8	294°16.5	03.8	12°47.3	50.5		
14	21°27.3	17°23.1	53.4	270°39.2	15.7	309°18.7	03.8	27°49.5	50.4		
15	36°29.8	32°26.7	•• 53.1	285°41.2	•• 15.6	324°20.8	•• 03.8	42°51.7	•• 50.3		
16	51°32.3	47°30.3	52.8	300°43.1	15.4	339°23.0	03.9	57°53.9	50.1		
17	66°34.7	62°33.9	52.5	315°45.1	15.3	354°25.1	03.9	72°56.1	50.0		
18	81°37.2	77°37.5	N10°52.2	330°47.0	N25°15.1	9°27.3	N22°04.0	87°58.2	S04°49.9		
19	96°39.7	92°41.1	51.9	345°49.0	15.0	24°29.5	04.0	103°00.4	49.8		
20	111°42.1	107°44.7	51.6	0°50.9	14.9	39°31.6	04.1	118°02.6	49.7		
21	126°44.6	122°48.4	•• 51.3	15°52.9	•• 14.7	54°33.8	•• 04.1	133°04.8	•• 49.5		
22	141°47.1	137°52.0	51.0	30°54.8	14.6	69°35.9	04.1	148°07.0	49.4		
23	156°49.5	152°55.6	50.7	45°56.8	14.4	84°38.1	04.2	163°09.1	49.3		
Mer.pass. 12:34		ν 3.6' d -0.3' m -4.34		ν 2.0' d -0.1' m 0.03		ν 2.2' d 0.0' m -2.22		ν 2.2' d -0.1' m 1.11			

Mar 11 Tue			SHA	Mer.pass
Venus	354°50.3		50.3	13:02
Mars	249°41.3			20:03
Jupiter	288°09.5			17:29
Saturn	6°39.8			12:16
Mar 12 Wed			SHA	Mer.pass
Venus	355°14.3		14.3	12:56
Mars	249°30.4			20:00
Jupiter	288°02.6			17:26
Saturn	6°33.0			12:12
Mar 13 Thu			SHA	Mer.pass
Venus	355°40.0		40.0	12:51
Mars	249°18.9			19:57
Jupiter	287°55.5			17:22
Saturn	6°26.2			12:09
Horizontal parallax				
Venus:			0.5	
Mars:			0.2	

DUT1 = UT1-UTC = +0.0410 sec $\Delta T = TT-UT1 = +69.1430$ sec

2025 March 11 to Mar. 13 UT

h Sun Moon						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	177°29.3	S03°42.0	31°59.6	11.9'	N20°20.7	-11.3' 55.9'
1	192°29.5	41.0	46°30.5	12.0'	20°09.4	-11.3' 55.8'
2	207°29.6	40.0	61°01.6	12.1'	19°58.1	-11.4' 55.8'
3	222°29.8	· · 39.1	75°32.7	12.2'	19°46.6	-11.5' 55.8'
4	237°30.0	38.1	90°03.9	12.3'	19°35.1	-11.6' 55.8'
5	252°30.1	37.1	104°35.2	12.4'	19°23.6	-11.7' 55.8'
6	267°30.3	S03°36.1	119°06.5	12.5'	N19°11.9	-11.7' 55.7'
7	282°30.5	35.1	133°38.0	12.6'	19°00.1	-11.8' 55.7'
8	297°30.6	34.2	148°09.6	12.7'	18°48.3	-11.9' 55.7'
9	312°30.8	· · 33.2	162°41.3	12.8'	18°36.4	-12.0' 55.7'
10	327°31.0	32.2	177°13.0	12.8'	18°24.4	-12.1' 55.7'
11	342°31.1	31.2	191°44.9	12.9'	18°12.4	-12.1' 55.6'
12	357°31.3	S03°30.2	206°16.8	13.0'	N18°00.3	-12.2' 55.6'
13	12°31.5	29.2	220°48.8	13.1'	17°48.1	-12.3' 55.6'
14	27°31.6	28.3	235°20.9	13.2'	17°35.8	-12.3' 55.6'
15	42°31.8	· · 27.3	249°53.1	13.3'	17°23.4	-12.4' 55.6'
16	57°32.0	26.3	264°25.4	13.4'	17°11.0	-12.5' 55.5'
17	72°32.1	25.3	278°57.8	13.5'	16°58.6	-12.5' 55.5'
18	87°32.3	S03°24.3	293°30.2	13.5'	N16°46.0	-12.6' 55.5'
19	102°32.5	23.3	308°02.8	13.6'	16°33.4	-12.7' 55.5'
20	117°32.6	22.4	322°35.4	13.7'	16°20.8	-12.7' 55.5'
21	132°32.8	· · 21.4	337°08.1	13.8'	16°08.0	-12.8' 55.4'
22	147°33.0	20.4	351°40.9	13.9'	15°55.2	-12.9' 55.4'
23	162°33.1	19.4	6°13.8	13.9'	15°42.4	-12.9' 55.4'
SD = 16.1' $d = -1.0'$			SD = 15.2'			
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	177°33.3	S03°18.4	20°46.7	14.0'	N15°29.5	-13.0' 55.4'
1	192°33.5	17.4	35°19.7	14.1'	15°16.5	-13.0' 55.4'
2	207°33.6	16.5	49°52.8	14.2'	15°03.5	-13.1' 55.4'
3	222°33.8	· · 15.5	64°26.0	14.3'	14°50.4	-13.1' 55.3'
4	237°34.0	14.5	78°59.3	14.3'	14°37.3	-13.2' 55.3'
5	252°34.1	13.5	93°32.6	14.4'	14°24.1	-13.2' 55.3'
6	267°34.3	S03°12.5	108°06.0	14.5'	N14°10.8	-13.3' 55.3'
7	282°34.5	11.5	122°39.5	14.6'	13°57.5	-13.3' 55.3'
8	297°34.6	10.6	137°13.1	14.6'	13°44.2	-13.4' 55.3'
9	312°34.8	· · 09.6	151°46.7	14.7'	13°30.8	-13.4' 55.2'
10	327°35.0	08.6	166°20.4	14.8'	13°17.4	-13.5' 55.2'
11	342°35.1	07.6	180°54.1	14.8'	13°03.9	-13.5' 55.2'
12	357°35.3	S03°06.6	195°28.0	14.9'	N12°50.4	-13.6' 55.2'
13	12°35.5	05.6	210°01.9	15.0'	12°36.8	-13.6' 55.2'
14	27°35.7	04.7	224°35.8	15.0'	12°23.2	-13.7' 55.1'
15	42°35.8	· · 03.7	239°09.9	15.1'	12°09.5	-13.7' 55.1'
16	57°36.0	02.7	253°44.0	15.2'	11°55.9	-13.7' 55.1'
17	72°36.2	01.7	268°18.1	15.2'	11°42.1	-13.8' 55.1'
18	87°36.3	S03°00.7	282°52.3	15.3'	N11°28.3	-13.8' 55.1'
19	102°36.5	02°59.7	297°26.6	15.3'	11°14.5	-13.8' 55.1'
20	117°36.7	58.8	312°01.0	15.4'	11°00.7	-13.9' 55.1'
21	132°36.8	· · 57.8	326°35.4	15.5'	10°46.8	-13.9' 55.0'
22	147°37.0	56.8	341°09.8	15.5'	10°32.9	-14.0' 55.0'
23	162°37.2	55.8	355°44.3	15.6'	10°18.9	-14.0' 55.0'
SD = 16.1' $d = -1.0'$			SD = 15.1'			
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	177°37.4	S02°54.8	10°18.9	15.6'	N10°05.0	-14.0' 55.0'
1	192°37.5	53.8	24°53.5	15.7'	09°50.9	-14.0' 55.0'
2	207°37.7	52.8	39°28.2	15.7'	09°36.9	-14.1' 55.0'
3	222°37.9	· · 51.9	54°03.0	15.8'	09°22.8	-14.1' 54.9'
4	237°38.0	50.9	68°37.8	15.8'	09°08.7	-14.1' 54.9'
5	252°38.2	49.9	83°12.6	15.9'	08°54.6	-14.2' 54.9'
6	267°38.4	S02°48.9	97°47.5	15.9'	N08°40.4	-14.2' 54.9'
7	282°38.6	47.9	112°22.4	16.0'	08°26.2	-14.2' 54.9'
8	297°38.7	46.9	126°57.4	16.0'	08°12.0	-14.2' 54.9'
9	312°38.9	· · 45.9	141°32.4	16.1'	07°57.8	-14.3' 54.8'
10	327°39.1	45.0	156°07.5	16.1'	07°43.5	-14.3' 54.8'
11	342°39.2	44.0	170°42.6	16.2'	07°29.3	-14.3' 54.8'
12	357°39.4	S02°43.0	185°17.8	16.2'	N07°15.0	-14.3' 54.8'
13	12°39.6	42.0	199°53.0	16.3'	07°00.7	-14.3' 54.8'
14	27°39.8	41.0	214°28.3	16.3'	06°46.3	-14.4' 54.8'
15	42°39.9	· · 40.0	229°03.6	16.3'	06°32.0	-14.4' 54.8'
16	57°40.1	39.0	243°38.9	16.4'	06°17.6	-14.4' 54.7'
17	72°40.3	38.1	258°14.3	16.4'	06°03.2	-14.4' 54.7'
18	87°40.4	S02°37.1	272°49.7	16.4'	N05°48.8	-14.4' 54.7'
19	102°40.6	36.1	287°25.1	16.5'	05°34.4	-14.4' 54.7'
20	117°40.8	35.1	302°00.6	16.5'	05°20.0	-14.4' 54.7'
21	132°41.0	· · 34.1	316°36.1	16.5'	05°05.5	-14.5' 54.7'
22	147°41.1	33.1	331°11.7	16.6'	04°51.1	-14.5' 54.7'
23	162°41.3	32.1	345°47.3	16.6'	04°36.6	-14.5' 54.6'
SD = 16.1' $d = -1.0'$			SD = 15.0'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:10	05:32	06:38	17:43	18:50	20:13
N 70°	04:22	05:35	06:35	17:46	18:47	20:01
68°	04:31	05:37	06:33	17:49	18:44	19:51
66°	04:39	05:40	06:30	17:51	18:42	19:43
64°	04:46	05:41	06:28	17:52	18:40	19:36
62°	04:51	05:43	06:27	17:54	18:38	19:30
60°	04:55	05:44	06:25	17:55	18:37	19:26
N 58°	04:59	05:45	06:24	17:57	18:36	19:22
56°	05:03	05:46	06:23	17:58	18:35	19:18
54°	05:05	05:47	06:22	17:59	18:34	19:15
52°	05:08	05:47	06:21	18:00	18:33	19:13
50°	05:10	05:48	06:20	18:00	18:33	19:10
45°	05:15	05:49	06:18	18:02	18:31	19:06
N 40°	05:18	05:49	06:16	18:04	18:31	19:02
35°	05:20	05:49	06:15	18:05	18:30	19:00
30°	05:22	05:49	06:13	18:06	18:30	18:58
20°	05:23	05:49	06:11	18:09	18:31	18:56
N 10°	05:23	05:48	06:09	18:11	18:32	18:56
0°	05:22	05:46	06:06	18:13	18:34	18:58
S 10°	05:19	05:43	06:04	18:15	18:36	19:01
20°	05:14	05:39	06:01	18:17	18:40	19:05
30°	05:06	05:34	05:58	18:20	18:44	19:12
35°	05:01	05:31	05:57	18:22	18:47	19:17
40°	04:55	05:27	05:55	18:24	18:51	19:23
45°	04:48	05:23	05:52	18:26	18:56	19:31
S 50°	04:38	05:17	05:49	18:29	19:02	19:40
52°	04:33	05:14	05:48	18:30	19:04	19:45
54°	04:27	05:11	05:46	18:32	19:07	19:50
56°	04:21	05:07	05:45	18:33	19:11	19:56
58°	04:14	05:03	05:43	18:35	19:15	20:03
S 60°	04:06	04:58	05:41	18:37	19:19	20:11

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	☐	14:15	16:30	☐	08:04	07:17
N 70°	12:05	14:38	16:39	08:38	07:39	07:05
68°	12:47	14:55	16:46	07:55	07:19	06:56
66°	13:15	15:09	16:52	07:26	07:04	06:47
64°	13:36	15:21	16:58	07:04	06:51	06:41
62°	13:53	15:31	17:02	06:46	06:40	06:35
60°	14:07	15:39	17:06	06:31	06:30	06:29
N 58°	14:19	15:46	17:09	06:18	06:22	06:25
56°	14:29	15:52	17:12	06:07	06:15	06:20
54°	14:38	15:58	17:15	05:57	06:08	06:17
52°	14:46	16:03	17:17	05:48	06:02	06:13
50°	14:53	16:08	17:20	05:41	05:57	06:10
45°	15:09	16:18	17:24	05:24	05:45	06:03
N 40°	15:21	16:26	17:28	05:10	05:36	05:58
35°	15:32	16:33	17:32	04:58	05:27	05:53
30°	15:41	16:39	17:35	04:48	05:20	05:48
20°	15:57	16:50	17:40	04:30	05:07	05:41
N 10°	16:11	16:59	17:45	04:14	04:56	05:34
0°	16:23	17:07	17:49	03:59	04:45	05:28
S 10°	16:36	17:16	17:53	03:44	04:34	05:21
20°	16:50	17:25	17:58	03:28	04:23	05:14
30°	17:05	17:35	18:03	03:10	04:09	05:06
35°	17:14	17:41	18:06	02:59	04:01	05:02
40°	17:25	17:48	18:09	02:46	03:53	04:56
45°	17:36	17:56	18:13	02:31	03:42	04:50
S 50°	17:51	18:05	18:17	02:13	03:29	04:43
52°	17:57	18:09	18:19	02:04	03:23	04:39
54°	18:05	18:14	18:22	01:55	03:17	04:35
56°	18:13	18:19	18:24	01:44	03:09	04:31
58°	18:22	18:25	18:27	01:31	03:01	04:26
S 60°	18:33	18:32	18:30	01:16	02:52	04:21

Day	Sun		Mer. Pass hh:mm	Moon		Age 11-13 90-98%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
11	10:03	09:55	12:10	22:34	10:11	
12	09:47	09:39	12:10	23:18	10:56	
13	09:31	09:22	12:09	23:58	11:38	

Aries Venus

<https://pypi.org/project/sfalmanac/>

Sun				Moon			
h	GHA	Dec		GHA	ν	Dec	d
Fri							HP
0	177°41.5	S02°31.2		0°22.9	16.6'	N04°22.1	-14.5'
1	192°41.7	30.2		14°58.5	16.7'	04°07.7	-14.5'
2	207°41.8	29.2		29°34.2	16.7'	03°53.2	-14.5'
3	222°42.0	· · 28.2		44°09.9	16.7'	03°38.7	-14.5'
4	237°42.2	27.2		58°45.6	16.8'	03°24.2	-14.5'
5	252°42.4	26.2		73°21.4	16.8'	03°09.6	-14.5'
6	267°42.5	S02°25.2		87°57.1	16.8'	N02°55.1	-14.5'
7	282°42.7	24.3		102°32.9	16.8'	02°40.6	-14.5'
8	297°42.9	23.3		117°08.8	16.8'	02°26.1	-14.5'
9	312°43.0	· · 22.3		131°44.6	16.9'	02°11.6	-14.5'
10	327°43.2	21.3		146°20.5	16.9'	01°57.0	-14.5'
11	342°43.4	20.3		160°56.4	16.9'	01°42.5	-14.5'
12	357°43.6	S02°19.3		175°32.3	16.9'	N01°28.0	-14.5'
13	12°43.7	18.3		190°08.2	16.9'	01°13.5	-14.5'
14	27°43.9	17.4		204°44.1	17.0'	00°59.0	-14.5'
15	42°44.1	· · 16.4		219°20.1	17.0'	00°44.4	-14.5'
16	57°44.3	15.4		233°56.1	17.0'	00°29.9	-14.5'
17	72°44.4	14.4		248°32.1	17.0'	00°15.4	-14.5'
18	87°44.6	S02°13.4		263°08.1	17.0'	N00°00.9	-14.5'
19	102°44.8	12.4		277°44.1	17.0'	S00°13.6	14.5'
20	117°45.0	11.4		292°20.1	17.0'	00°28.1	14.5'
21	132°45.1	· · 10.4		306°56.1	17.0'	00°42.6	14.5'
22	147°45.3	09.5		321°32.2	17.0'	00°57.0	14.5'
23	162°45.5	08.5		336°08.2	17.1'	01°11.5	14.5'
	SD = 16.1'	$d = -1.0'$		SD = 14.9'			

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	177°45.7	S02°07.5	350°44.3	17.1'	S01°26.0	14.4'	54.3'
1	192°45.9	06.5	5°20.3	17.1'	01°40.4	14.4'	54.3'
2	207°46.0	05.5	19°56.4	17.1'	01°54.8	14.4'	54.3'
3	222°46.2	· · 04.5	34°32.4	17.1'	02°09.3	14.4'	54.3'
4	237°46.4	03.5	49°08.5	17.1'	02°23.7	14.4'	54.3'
5	252°46.6	02.6	63°44.6	17.1'	02°38.1	14.4'	54.3'
6	267°46.7	S02°01.6	78°20.6	17.1'	S02°52.4	14.4'	54.3'
7	282°46.9	02°00.6	92°56.7	17.1'	03°06.8	14.3'	54.3'
8	297°47.1	01°59.6	107°32.8	17.1'	03°21.1	14.3'	54.3'
9	312°47.3	· · 58.6	122°08.8	17.1'	03°35.5	14.3'	54.3'
10	327°47.4	57.6	136°44.9	17.1'	03°49.8	14.3'	54.2'
11	342°47.6	56.6	151°21.0	17.1'	04°04.1	14.3'	54.2'
12	357°47.8	S01°55.6	165°57.0	17.0'	S04°18.3	14.2'	54.2'
13	12°48.0	54.7	180°33.1	17.0'	04°32.6	14.2'	54.2'
14	27°48.1	53.7	195°09.1	17.0'	04°46.8	14.2'	54.2'
15	42°48.3	· · 52.7	209°45.1	17.0'	05°01.0	14.2'	54.2'
16	57°48.5	51.7	224°21.1	17.0'	05°15.2	14.2'	54.2'
17	72°48.7	50.7	238°57.1	17.0'	05°29.4	14.1'	54.2'
18	87°48.9	S01°49.7	253°33.1	17.0'	S05°43.5	14.1'	54.2'
19	102°49.0	48.7	268°09.1	17.0'	05°57.6	14.1'	54.2'
20	117°49.2	47.7	282°45.1	17.0'	06°11.7	14.1'	54.2'
21	132°49.4	· · 46.8	297°21.0	16.9'	06°25.7	14.0'	54.2'
22	147°49.6	45.8	311°57.0	16.9'	06°39.8	14.0'	54.1'
23	162°49.7	44.8	326°32.9	16.9'	06°53.8	14.0'	54.1'
	SD = 16.1'	$d = -1.0'$	SD = 14.8'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	177°49.9	S01°43.8	341°08.8	16.9'	S07°07.8	13.9'	54.1'
1	192°50.1	42.8	355°44.7	16.9'	07°21.7	13.9'	54.1'
2	207°50.3	41.8	10°20.6	16.8'	07°35.6	13.9'	54.1'
3	222°50.5	· · 40.8	24°56.4	16.8'	07°49.5	13.8'	54.1'
4	237°50.6	39.8	39°32.2	16.8'	08°03.3	13.8'	54.1'
5	252°50.8	38.9	54°08.0	16.8'	08°17.2	13.8'	54.1'
6	267°51.0	S01°37.9	68°43.8	16.8'	S08°30.9	13.7'	54.1'
7	282°51.2	36.9	83°19.6	16.7'	08°44.7	13.7'	54.1'
8	297°51.4	35.9	97°55.3	16.7'	08°58.4	13.7'	54.1'
9	312°51.5	· · 34.9	112°31.0	16.7'	09°12.1	13.6'	54.1'
10	327°51.7	33.9	127°06.7	16.7'	09°25.7	13.6'	54.1'
11	342°51.9	32.9	141°42.4	16.6'	09°39.3	13.6'	54.1'
12	357°52.1	S01°31.9	156°18.0	16.6'	S09°52.9	13.5'	54.1'
13	12°52.2	31.0	170°53.6	16.6'	10°06.4	13.5'	54.1'
14	27°52.4	30.0	185°29.1	16.5'	10°19.9	13.4'	54.0'
15	42°52.6	· · 29.0	200°04.7	16.5'	10°33.3	13.4'	54.0'
16	57°52.8	28.0	214°40.2	16.5'	10°46.7	13.4'	54.0'
17	72°53.0	27.0	229°15.6	16.4'	11°00.1	13.3'	54.0'
18	87°53.1	S01°26.0	243°51.1	16.4'	S11°13.4	13.3'	54.0'
19	102°53.3	25.0	258°26.5	16.4'	11°26.7	13.2'	54.0'
20	117°53.5	24.0	273°01.8	16.3'	11°39.9	13.2'	54.0'
21	132°53.7	· · 23.0	287°37.1	16.3'	11°53.1	13.1'	54.0'
22	147°53.9	22.1	302°12.4	16.2'	12°06.2	13.1'	54.0'
23	162°54.0	21.1	316°47.7	16.2'	12°19.3	13.0'	54.0'
	SD = 16.1'	$d = -1.0'$	SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:52	05:16	06:23	17:57	19:05	20:30
N 70°	04:06	05:21	06:21	17:58	18:59	20:15
68°	04:18	05:25	06:20	18:00	18:55	20:03
66°	04:27	05:28	06:19	18:00	18:52	19:53
64°	04:34	05:31	06:18	18:01	18:49	19:46
62°	04:41	05:33	06:17	18:02	18:46	19:39
60°	04:46	05:35	06:16	18:03	18:44	19:33
N 58°	04:51	05:37	06:16	18:03	18:42	19:29
56°	04:54	05:38	06:15	18:04	18:41	19:25
54°	04:58	05:39	06:14	18:04	18:40	19:21
52°	05:01	05:40	06:14	18:05	18:38	19:18
50°	05:04	05:41	06:13	18:05	18:37	19:15
45°	05:09	05:43	06:12	18:06	18:35	19:10
N 40°	05:13	05:44	06:11	18:07	18:34	19:05
35°	05:16	05:45	06:10	18:08	18:33	19:02
30°	05:18	05:46	06:10	18:08	18:32	19:00
20°	05:21	05:46	06:08	18:10	18:32	18:57
N 10°	05:22	05:46	06:07	18:11	18:32	18:56
0°	05:21	05:45	06:06	18:12	18:33	18:57
S 10°	05:19	05:43	06:04	18:13	18:34	18:59
20°	05:15	05:40	06:02	18:15	18:37	19:03
30°	05:08	05:36	06:00	18:17	18:41	19:09
35°	05:04	05:34	05:59	18:18	18:43	19:13
40°	04:59	05:31	05:58	18:19	18:46	19:18
45°	04:52	05:27	05:56	18:21	18:50	19:25
S 50°	04:43	05:22	05:54	18:23	18:55	19:33
52°	04:39	05:19	05:53	18:23	18:57	19:38
54°	04:34	05:16	05:52	18:24	19:00	19:42
56°	04:28	05:13	05:51	18:26	19:03	19:48
58°	04:22	05:10	05:50	18:27	19:06	19:54
S 60°	04:15	05:06	05:48	18:28	19:10	20:01

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	18:31	20:31	22:42	06:42	06:10	05:35
N 70°	18:31	20:21	22:18	06:39	06:14	05:47
68°	18:30	20:13	22:01	06:36	06:17	05:57
66°	18:30	20:07	21:46	06:33	06:20	06:05
64°	18:30	20:01	21:35	06:31	06:22	06:13
62°	18:30	19:57	21:25	06:29	06:24	06:19
60°	18:30	19:53	21:17	06:28	06:26	06:24
N 58°	18:30	19:49	21:09	06:26	06:28	06:29
56°	18:30	19:46	21:03	06:25	06:29	06:33
54°	18:29	19:43	20:57	06:24	06:30	06:37
52°	18:29	19:41	20:52	06:23	06:31	06:41
50°	18:29	19:38	20:48	06:22	06:33	06:44
45°	18:29	19:33	20:38	06:20	06:35	06:51
N 40°	18:29	19:29	20:29	06:18	06:37	06:57
35°	18:29	19:26	20:22	06:16	06:39	07:02
30°	18:29	19:22	20:16	06:15	06:40	07:06
20°	18:29	19:17	20:06	06:12	06:43	07:14
N 10°	18:29	19:12	19:56	06:10	06:45	07:21
0°	18:29	19:08	19:48	06:08	06:48	07:27
S 10°	18:29	19:04	19:39	06:06	06:50	07:34
20°	18:28	18:59	19:30	06:04	06:52	07:40
30°	18:28	18:54	19:20	06:01	06:55	07:48
35°	18:28	18:51	19:14	06:00	06:56	07:53
40°	18:28	18:48	19:08	05:58	06:58	07:58
45°	18:28	18:44	19:00	05:56	07:00	08:04
S 50°	18:28	18:39	18:51	05:53	07:03	08:12
52°	18:28	18:37	18:47	05:52	07:04	08:15
54°	18:28	18:35	18:42	05:51	07:05	08:19
56°	18:28	18:32	18:37	05:50	07:06	08:23
58°	18:28	18:30	18:31	05:48	07:08	08:28
S 60°	18:28	18:26	18:25	05:46	07:10	08:33

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
14	09:14	09:06	12:09	—:—	12:18	
15	08:57	08:49	12:09	00:38	12:58	
16	08:40	08:32	12:09	01:17	13:37	

March 17, 18, 19 UT (Mon., Tue., Wed.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Mon	GHA	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	SHA Dec	SHA Dec		
0	174°49.4	172°26.5	N10°20.9	63°16.4	N25°03.8	102°14.9	N22°07.3	180°48.2	S04°40.5					Alpheratz	357°35.0 29°13.7		
1	189°51.9	187°30.3	20.4	78°18.3	03.7	117°17.1	07.3	195°50.4	40.4					Ankaa	353°07.4 -42°10.3		
2	204°54.3	202°34.1	19.9	93°20.2	03.5	132°19.2	07.4	210°52.6	40.3					Schedar	349°31.5 56°40.5		
3	219°56.8	217°37.9	•• 19.4	108°22.1	•• 03.4	147°21.4	•• 07.4	225°54.8	•• 40.2					Diphda	348°47.5 -17°51.0		
4	234°59.3	232°41.7	18.9	123°23.9	03.2	162°23.5	07.5	240°56.9	40.0					Achernar	335°20.5 -57°06.7		
5	250°01.7	247°45.4	18.4	138°25.8	03.1	177°25.6	07.5	255°59.1	39.9					Hamal	327°51.3 23°34.9		
6	265°04.2	262°49.2	N10°17.9	153°27.7	N25°02.9	192°27.8	N22°07.6	271°01.3	S04°39.8					Polaris	314°17.2 89°22.5		
7	280°06.6	277°53.0	17.3	168°29.5	02.8	207°29.9	07.6	286°03.5	39.7					Acamar	315°11.9 -40°12.4		
8	295°09.1	292°56.8	16.8	183°31.4	02.6	222°32.0	07.6	301°05.7	39.6					Menkar	314°06.2 4°11.2		
9	310°11.6	308°00.6	•• 16.3	198°33.3	•• 02.4	237°34.2	•• 07.7	316°07.8	•• 39.4					Mirfak	308°28.3 49°57.2		
10	325°14.0	323°04.4	15.8	213°35.2	02.3	252°36.3	07.7	331°10.0	39.3					Aldebaran	290°39.5 16°33.6		
11	340°16.5	338°08.2	15.3	228°37.0	02.1	267°38.4	07.8	346°12.2	39.2					Rigel	281°03.8 -8°10.5		
12	355°19.0	353°12.0	N10°14.7	243°38.9	N25°02.0	282°40.6	N22°07.8	1°14.4	S04°39.1					Capella	280°21.7 46°01.5		
13	10°21.4	8°15.8	14.2	258°40.8	01.8	297°42.7	07.9	16°16.6	39.0					Bellatrix	278°22.7 6°22.3		
14	25°23.9	23°19.6	13.7	273°42.6	01.7	312°44.8	07.9	31°18.7	38.8					Elnath	278°01.7 28°37.8		
15	40°26.4	38°23.4	•• 13.1	288°44.5	•• 01.5	327°47.0	•• 07.9	46°20.9	•• 38.7					Alnilam	275°37.6 -1°11.2		
16	55°28.8	53°27.2	12.6	303°46.4	01.4	342°49.1	08.0	61°23.1	38.6					Betelgeuse	270°51.9 7°24.6		
17	70°31.3	68°31.0	12.1	318°48.2	01.2	357°51.3	08.0	76°25.3	38.5					Canopus	263°52.3 -52°42.8		
18	85°33.8	83°34.8	N10°11.5	333°50.1	N25°01.1	12°53.4	N22°08.1	91°27.5	S04°38.4					Sirius	258°26.0 -16°45.2		
19	100°36.2	98°38.6	11.0	348°51.9	00.9	27°55.5	08.1	106°29.6	38.2					Adhara	255°05.6 -29°00.6		
20	115°38.7	113°42.4	10.4	3°53.8	00.8	42°57.7	08.2	121°31.8	38.1					Procyon	244°50.5 5°09.5		
21	130°41.1	128°46.2	•• 09.9	18°55.7	•• 00.6	57°59.8	•• 08.2	136°34.0	•• 38.0					Pollux	243°16.9 27°57.9		
22	145°43.6	143°50.0	09.3	33°57.5	00.5	73°01.9	08.3	151°36.2	37.9					Avior	234°14.3 -59°35.6		
23	160°46.1	158°53.8	08.8	48°59.4	00.3	88°04.0	08.3	166°38.4	37.8					Suhail	222°45.9 -43°32.2		
Mer.pass. 12:19			ν 3.8' d -0.5' m -4.09			ν 1.9' d -0.2' m 0.12			ν 2.1' d 0.0' m -2.19			ν 2.2' d -0.1' m 1.12					
Tue	GHA	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec				
0	175°48.5	173°57.6	N10°08.2	64°01.2	N25°00.2	103°06.2	N22°08.3	181°40.5	S04°37.6					Denebola	182°24.5 14°25.7		
1	190°51.0	189°01.5	07.7	79°03.1	25°00.0	118°08.3	08.4	196°42.7	37.5					Gienah	175°43.1 -17°41.0		
2	205°53.5	204°05.3	07.1	94°04.9	24°59.8	133°10.4	08.4	211°44.9	37.4					Acrux	172°59.3 -63°14.3		
3	220°55.9	219°09.1	•• 06.6	109°06.8	•• 59.7	148°12.6	•• 08.5	226°47.1	•• 37.3					Gacrux	171°51.0 -57°15.3		
4	235°58.4	234°12.9	06.0	124°08.6	59.5	163°14.7	08.5	241°49.3	37.2					Alioth	166°12.3 55°49.2		
5	251°00.9	249°16.7	05.4	139°10.5	59.4	178°16.8	08.6	256°51.4	37.0					Spica	158°21.9 -11°17.7		
6	266°03.3	264°20.5	N10°04.9	154°12.3	N24°59.2	193°19.0	N22°08.6	271°53.6	S04°36.9					Alkaid	152°51.4 49°11.0		
7	281°05.8	279°24.4	04.3	169°14.2	59.1	208°21.1	08.6	286°55.8	36.8					Hadar	148°35.3 -60°29.6		
8	296°08.2	294°28.2	03.7	184°16.0	58.9	223°23.2	08.7	301°58.0	36.7					Menkent	147°57.1 -36°29.7		
9	311°10.7	309°32.0	•• 03.1	199°17.9	•• 58.8	238°25.4	•• 08.7	317°00.2	•• 36.6					Arcturus	145°47.5 19°02.8		
10	326°13.2	324°35.8	02.6	214°19.7	58.6	253°27.5	08.8	332°02.3	36.4					Rigil Kent.	139°39.7 -60°56.3		
11	341°15.6	339°39.7	02.0	229°21.6	58.4	268°29.6	08.8	347°04.5	36.3					Kochab	137°18.9 74°02.8		
12	356°18.1	354°43.5	N10°01.4	244°23.4	N24°58.3	283°31.7	N22°08.9	2°06.7	S04°36.2					Zuben'ubi	136°55.6 -16°08.9		
13	11°20.6	9°47.3	00.8	259°25.3	58.1	298°33.9	08.9	17°08.9	36.1					Alphecca	126°03.4 26°37.5		
14	26°23.0	24°51.1	10°00.2	274°27.1	58.0	313°36.0	09.0	32°11.1	36.0					Antares	112°15.5 -26°29.3		
15	41°25.5	39°55.0	09°59.7	289°29.0	•• 57.8	328°38.1	•• 09.0	47°13.2	•• 35.8					Atria	107°09.5 -69°04.1		
16	56°28.0	54°58.8	59.1	304°30.8	57.7	343°40.3	09.0	62°15.4	35.7					Sabik	102°02.6 -15°45.4		
17	71°30.4	70°02.6	58.5	319°32.6	57.5	358°42.4	09.1	77°17.6	35.6					Shaula	96°10.1 -37°07.3		
18	86°32.9	85°06.5	N09°57.9	334°34.5	N24°57.4	13°44.5	N22°09.1	92°19.8	S04°35.5					Rasalhague	95°58.4 12°32.2		
19	101°35.4	100°10.3	57.3	349°36.3	57.2	28°46.6	09.2	107°21.9	35.7					Eltanin	90°42.1 51°28.7		
20	116°37.8	115°14.1	56.7	4°38.2	57.0	43°48.8	09.2	122°24.1	35.2					Kaus Aust.	83°32.4 -34°22.3		
21	131°40.3	130°17.9	•• 56.1	19°40.0	•• 56.9	58°50.9	•• 09.3	137°26.3	•• 35.1					Vega	80°33.2 38°48.1		
22	146°42.7	145°21.8	55.5	34°41.8	56.7	73°53.0	09.3	152°28.5	35.0					Nunki	75°47.7 -26°15.9		
23	161°45.2	160°25.6	54.9	49°43.7	56.6	88°55.1	09.4	167°30.7	34.9					Altair	62°00.0 8°55.9		
Mer.pass. 12:15			ν 3.8' d -0.6' m -4.04			ν 1.9' d -0.2' m 0.14			ν 2.1' d 0.0' m -2.19			ν 2.2' d -0.1' m 1.12					
Wed	GHA	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec				
0	176°47.7	175°29.5	N09°54.3	64°45.5	N24°56.4	103°57.3	N22°09.4	182°32.8	S04°34.8					Mar 17 Mon	SHA Mer.pass		
1	191°50.1	190°33.3	53.7	79°47.3	56.3	118°59.4	09.4	197°35.0	34.6					Venus	357°37.1 12:27		
2	206°52.6	205°37.1	53.0	94°49.2	56.1	134°01.5	09.5	212°37.2	34.5					Mars	248°27.0 19:44		
3	221°55.1	220°41.0	•• 52.4	109°51.0	•• 55.9	149°03.6	•• 09.5	227°39.4	•• 34.4					Jupiter	287°25.5 17:09		
4	236°57.5	235°44.8	51.8	124°52.8	55.8	164°05.8	09.6	242°41.6	34.3					Saturn	5°58.8 11:55		
5	252°00.0	250°48.6	51.2	139°54.7	55.6	179°07.9	09.6	257°43.7	34.2								
6	267°02.5	265°52.5	N09°50.6	154°56.5	N24°55.5	194°10.0	N22°09.7	272°45.9	S04°34.0								
7	282°04.9	280°56.3	50.0	169°58.3	55.3	209°12.1	09.7	287°48.1	33.9								
8	297°07.4	296°00.2	49.3	185°00.2	55.1	224°14.3	09.8	302°50.3	33.8								
9	312°09.9	311°04.0	•• 48.7	200°02.0	•• 55.0	239°16.4	•• 09.8	317°52.5	•• 33.7								
10	327°12.3	326°07.9	48.1	215°03.8	54.8	254°18.5	09.8	332°54.6	33.6								
11	342°14.8	341°11.7	47.4	230°05.6	54.7	269°20.6	09.9	347°56.8	33.4								
12	357°17.2	356°15.5	N09°46.8	245°07.5	N24°54.5	284°22.8	N22°09.9	2°59.0	S04°33.3								
13	12°19.7	11°19.4	46.2	260°09.3	54.4	299°24.9	10.0	18°01.2	33.2								
14	27°22.2	26°23.2	45.5	275°11.1	54.2	314°27.0	10.0	33°03.4	33.1								
15	42°24.6	41°27.1	•• 44.9	290°12.9	•• 54.0	329°29.1	•• 10.1	48°05.5	•• 33.0								
16	57°27.1	56°30.9	44.3	305°14.7	53.9	344°31.2	10.1	63°07.7	32.8								
17	72°29.6	71°34.8	43.6	320°16.6	53.7	359°33.4	10.2	78°09.9	32.7								
18	87°32.0	86°38.6	N09°43.0	335°18.4	N24°53.6	14°35.5	N22°10.2	93°12.1	S04°32.6								
19	102°34.5	101°42.5	42.3	350°20.2	53.4	29°37.6	10.2	108°14.3	32.5								
20	117°37.0	116°46.3	41.7	5°22.0	53.2	44°39.7	10.3	123°16.4	32.4								
21	132°39.4	131°50.2	•• 41.0	20°23.8	•• 53.1	59°41.8	•• 10.3	138°18.6	•• 32.3								
22	147°41.9	146°54.0	40.4	35°25.7	52.9	74°44.0	10.4	153°20.8	32.1								
23	162°44.3	161°57.9	39.7	50°27.5	52.8	89°46.1	10.4	168°23.0	32.0								
Mer.pass. 12:11			ν 3.8' d -0.6' m -4.13			ν 1.8' d -0.2' m 0.17			ν 2.1' d 0.0' m -2.18			ν 2.2' d -0.1' m 1.13					
															Horizontal parallax		
															Venus:	0.5	
															Mars:	0.1	

h	Sun			Moon				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	177°54.2	S01°20.1		331°22.9	16.2'	S12°32.4	13.0'	54.0'
1	192°54.4	19.1		345°58.0	16.1'	12°45.4	12.9'	54.0'
2	207°54.6	18.1		0°33.2	16.1'	12°58.3	12.9'	54.0'
3	222°54.8	· · 17.1		15°08.3	16.0'	13°11.2	12.8'	54.0'
4	237°54.9	16.1		29°43.3	16.0'	13°24.1	12.8'	54.0'
5	252°55.1	15.1		44°18.3	15.9'	13°36.9	12.7'	54.0'
6	267°55.3	S01°14.2		58°53.2	15.9'	S13°49.6	12.7'	54.0'
7	282°55.5	13.2		73°28.1	15.9'	14°02.3	12.6'	54.0'
8	297°55.7	12.2		88°03.0	15.8'	14°14.9	12.6'	54.0'
9	312°55.8	· · 11.2		102°37.8	15.8'	14°27.5	12.5'	54.0'
10	327°56.0	10.2		117°12.6	15.7'	14°40.0	12.5'	54.0'
11	342°56.2	09.2		131°47.3	15.7'	14°52.5	12.4'	54.0'
12	357°56.4	S01°08.2		146°21.9	15.6'	S15°04.9	12.4'	54.0'
13	12°56.6	07.2		160°56.6	15.6'	15°17.3	12.3'	54.0'
14	27°56.8	06.2		175°31.1	15.5'	15°29.5	12.2'	54.0'
15	42°56.9	· · 05.3		190°05.6	15.5'	15°41.8	12.2'	54.0'
16	57°57.1	04.3		204°40.1	15.4'	15°54.0	12.1'	54.0'
17	72°57.3	03.3		219°14.5	15.3'	16°06.1	12.0'	54.0'
18	87°57.5	S01°02.3		233°48.8	15.3'	S16°18.1	12.0'	54.0'
19	102°57.7	01.3		248°23.1	15.2'	16°30.1	11.9'	54.0'
20	117°57.8	01°00.3		262°57.4	15.2'	16°42.0	11.9'	54.0'
21	132°58.0	00°59.3		277°31.5	15.1'	16°53.9	11.8'	54.0'
22	147°58.2	58.3		292°05.7	15.1'	17°05.7	11.7'	54.0'
23	162°58.4	57.4		306°39.7	15.0'	17°17.4	11.7'	54.0'
	SD = 16.1'	$d = -1.0'$				SD = 14.7'		

Tue		GHA	Dec		GHA	ν	Dec	d	HP
0		177°58.6	S00°56.4		321°13.7	14.9'	S17°29.1	11.6'	54.0'
1		192°58.8	55.4		335°47.7	14.9'	17°40.6	11.5'	54.0'
2		207°58.9	54.4		350°21.6	14.8'	17°52.2	11.5'	54.0'
3		222°59.1	· · 53.4		4°55.4	14.8'	18°03.6	11.4'	54.0'
4		237°59.3	52.4		19°29.1	14.7'	18°15.0	11.3'	54.0'
5		252°59.5	51.4		34°02.8	14.6'	18°26.3	11.2'	54.0'
6		267°59.7	S00°50.4		48°36.5	14.6'	S18°37.5	11.2'	54.0'
7		282°59.8	49.4		63°10.0	14.5'	18°48.7	11.1'	54.0'
8		298°00.0	48.5		77°43.5	14.4'	18°59.8	11.0'	54.0'
9		313°00.2	· · 47.5		92°17.0	14.4'	19°10.8	10.9'	54.0'
10		328°00.4	46.5		106°50.4	14.3'	19°21.8	10.9'	54.0'
11		343°00.6	45.5		121°23.7	14.2'	19°32.6	10.8'	54.0'
12		358°00.8	S00°44.5		135°56.9	14.2'	S19°43.4	10.7'	54.0'
13		13°00.9	43.5		150°30.1	14.1'	19°54.1	10.6'	54.0'
14		28°01.1	42.5		165°03.2	14.0'	20°04.7	10.5'	54.0'
15		43°01.3	· · 41.5		179°36.2	14.0'	20°15.3	10.5'	54.0'
16		58°01.5	40.6		194°09.2	13.9'	20°25.8	10.4'	54.0'
17		73°01.7	39.6		208°42.1	13.8'	20°36.1	10.3'	54.0'
18		88°01.9	S00°38.6		223°14.9	13.8'	S20°46.4	10.2'	54.0'
19		103°02.0	37.6		237°47.7	13.7'	20°56.7	10.1'	54.1'
20		118°02.2	36.6		252°20.4	13.6'	21°06.8	10.0'	54.1'
21		133°02.4	· · 35.6		266°53.0	13.5'	21°16.8	10.0'	54.1'
22		148°02.6	34.6		281°25.5	13.5'	21°26.8	9.9'	54.1'
23		163°02.8	33.6		295°58.0	13.4'	21°36.7	9.8'	54.1'
		SD = 16.1' $d = -1.0'$			SD = 14.7'				

Wed		GHA	Dec		GHA	ν	Dec	d	HP
0		178°03.0	S00°32.6		310°30.4	13.3'	S21°46.5	9.7'	54.1'
1		193°03.1	31.7		325°02.7	13.2'	21°56.2	9.6'	54.1'
2		208°03.3	30.7		339°35.0	13.2'	22°05.8	9.5'	54.1'
3		223°03.5	· · 29.7		354°07.1	13.1'	22°15.3	9.4'	54.1'
4		238°03.7	28.7		8°39.3	13.0'	22°24.7	9.3'	54.1'
5		253°03.9	27.7		23°11.3	13.0'	22°34.1	9.2'	54.1'
6		268°04.1	S00°26.7		37°43.2	12.9'	S22°43.3	9.1'	54.1'
7		283°04.2	25.7		52°15.1	12.8'	22°52.4	9.0'	54.1'
8		298°04.4	24.7		66°46.9	12.7'	23°01.5	9.0'	54.2'
9		313°04.6	· · 23.8		81°18.6	12.6'	23°10.4	8.9'	54.2'
10		328°04.8	22.8		95°50.3	12.6'	23°19.3	8.8'	54.2'
11		343°05.0	21.8		110°21.8	12.5'	23°28.0	8.7'	54.2'
12		358°05.2	S00°20.8		124°53.3	12.4'	S23°36.7	8.6'	54.2'
13		13°05.3	19.8		139°24.8	12.3'	23°45.3	8.5'	54.2'
14		28°05.5	18.8		153°56.1	12.3'	23°53.7	8.4'	54.2'
15		43°05.7	· · 17.8		168°27.4	12.2'	24°02.1	8.3'	54.2'
16		58°05.9	16.8		182°58.5	12.1'	24°10.3	8.1'	54.2'
17		73°06.1	15.8		197°29.7	12.0'	24°18.5	8.0'	54.2'
18		88°06.3	S00°14.9		212°00.7	12.0'	S24°26.5	7.9'	54.3'
19		103°06.5	13.9		226°31.6	11.9'	24°34.4	7.8'	54.3'
20		118°06.6	12.9		241°02.5	11.8'	24°42.3	7.7'	54.3'
21		133°06.8	· · 11.9		255°33.3	11.7'	24°50.0	7.6'	54.3'
22		148°07.0	10.9		270°04.0	11.6'	24°57.6	7.5'	54.3'
23		163°07.2	09.9		284°34.7	11.6'	25°05.1	7.4'	54.3'
		SD = 16.1' $d = -1.0'$			SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:33	05:00	06:08	18:11	19:19	20:47
N 70°	03:50	05:07	06:08	18:11	19:12	20:29
68°	04:03	05:12	06:07	18:10	19:06	20:16
66°	04:14	05:16	06:07	18:10	19:02	20:04
64°	04:23	05:20	06:07	18:10	18:58	19:55
62°	04:30	05:23	06:07	18:10	18:54	19:48
60°	04:36	05:26	06:07	18:10	18:52	19:41
N 58°	04:42	05:28	06:07	18:10	18:49	19:36
56°	04:46	05:30	06:07	18:10	18:47	19:31
54°	04:50	05:32	06:07	18:10	18:45	19:27
52°	04:54	05:33	06:07	18:10	18:44	19:23
50°	04:57	05:35	06:07	18:10	18:42	19:20
45°	05:03	05:37	06:07	18:10	18:39	19:14
N 40°	05:08	05:40	06:06	18:10	18:37	19:09
35°	05:12	05:41	06:06	18:10	18:35	19:05
30°	05:14	05:42	06:06	18:10	18:34	19:02
20°	05:18	05:44	06:06	18:10	18:32	18:58
N 10°	05:20	05:44	06:05	18:11	18:32	18:56
0°	05:20	05:44	06:05	18:11	18:32	18:56
S 10°	05:19	05:43	06:04	18:12	18:33	18:57
20°	05:16	05:41	06:03	18:12	18:34	19:00
30°	05:10	05:38	06:02	18:13	18:37	19:05
35°	05:07	05:36	06:02	18:14	18:39	19:08
40°	05:02	05:34	06:01	18:14	18:41	19:13
45°	04:56	05:31	06:00	18:15	18:44	19:19
S 50°	04:48	05:26	05:59	18:16	18:48	19:26
52°	04:44	05:25	05:58	18:17	18:50	19:30
54°	04:40	05:22	05:58	18:17	18:52	19:34
56°	04:35	05:20	05:57	18:18	18:55	19:39
58°	04:30	05:17	05:56	18:18	18:57	19:45
S 60°	04:23	05:14	05:56	18:19	19:01	19:51

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	■	■	■	04:49	■	■
N 70°		00:41	■	05:14	04:19	■
68°		00:02	■	05:34	05:00	■
66°	23:35		01:46	05:49	05:28	04:49
64°	23:14		01:04	06:02	05:50	05:31
62°	22:57		00:36	06:13	06:07	06:00
60°	22:44		00:15	06:23	06:22	06:22
N 58°	22:32	23:58		06:31	06:34	06:40
56°	22:22	23:43		06:38	06:45	06:55
54°	22:13	23:31		06:45	06:55	07:08
52°	22:05	23:20		06:51	07:03	07:20
50°	21:58	23:10		06:56	07:11	07:30
45°	21:43	22:50	23:56	07:08	07:28	07:52
N 40°	21:31	22:33	23:36	07:17	07:41	08:09
35°	21:20	22:19	23:20	07:26	07:53	08:24
30°	21:11	22:07	23:05	07:33	08:03	08:36
20°	20:56	21:47	22:41	07:46	08:20	08:58
N 10°	20:42	21:30	22:20	07:57	08:36	09:18
0°	20:29	21:13	22:00	08:08	08:50	09:35
S 10°	20:17	20:57	21:41	08:18	09:05	09:53
20°	20:04	20:40	21:20	08:30	09:20	10:13
30°	19:48	20:20	20:56	08:43	09:38	10:35
35°	19:40	20:09	20:42	08:50	09:49	10:48
40°	19:30	19:55	20:26	08:59	10:01	11:04
45°	19:18	19:40	20:07	09:09	10:15	11:22
S 50°	19:04	19:21	19:43	09:21	10:33	11:45
52°	18:58	19:12	19:32	09:27	10:41	11:56
54°	18:51	19:02	19:19	09:34	10:50	12:08
56°	18:43	18:51	19:04	09:41	11:01	12:22
58°	18:34	18:39	18:47	09:49	11:13	12:39
S 60°	18:24	18:24	18:26	09:58	11:27	13:00

March 20, 21, 22 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec		GHA	Dec		SHA	Dec	
0	177°46.8	177°01.7	N09°39.1	65°29.3	N24°52.6	104°48.2	N22°10.5	183°25.2	S04°31.9			183°25.2	S04°31.9		Alpheratz	357°35.0	29°13.7
1	192°49.3	192°05.6	38.4	80°31.1	52.4	119°50.3	10.5	198°27.3	31.8			198°27.3	31.8		Ankaa	353°07.4	-42°10.2
2	207°51.7	207°09.4	37.7	95°32.9	52.3	134°52.4	10.6	213°29.5	31.7			213°29.5	31.7		Schedar	349°31.5	56°40.5
3	222°54.2	222°13.3	· · 37.1	110°34.7	· · 52.1	149°54.6	· · 10.6	228°31.7	· · 31.5			228°31.7	· · 31.5		Diphda	348°47.5	-17°51.0
4	237°56.7	237°17.1	36.4	125°36.5	51.9	164°56.7	10.6	243°33.9	31.4			243°33.9	31.4		Achernar	335°20.6	-57°06.7
5	252°59.1	252°21.0	35.7	140°38.3	51.8	179°58.8	10.7	258°36.1	31.3			258°36.1	31.3		Hamal	327°51.3	23°34.9
6	268°01.6	267°24.8	N09°35.1	155°40.1	N24°51.6	195°00.9	N22°10.7	273°38.2	S04°31.2			273°38.2	S04°31.2		Polaris	314°18.1	89°22.5
7	283°04.1	282°28.7	34.4	170°42.0	51.5	210°03.0	10.8	288°40.4	31.1			288°40.4	31.1		Acamar	315°11.9	-40°12.4
8	298°06.5	297°32.5	33.7	185°43.8	51.3	225°05.1	10.8	303°42.6	30.9			303°42.6	30.9		Menkar	314°06.2	4°11.2
9	313°09.0	312°36.4	· · 33.0	200°45.6	· · 51.1	240°07.3	· · 10.9	318°44.8	· · 30.8			318°44.8	· · 30.8		Mirfak	308°28.4	49°57.2
10	328°11.5	327°40.2	32.4	215°47.4	51.0	255°09.4	10.9	333°47.0	30.7			333°47.0	30.7		Aldebaran	290°39.6	16°33.6
11	343°13.9	342°44.1	31.7	230°49.2	50.8	270°11.5	11.0	348°49.2	30.6			348°49.2	30.6		Rigel	281°03.8	-8°10.5
12	358°16.4	357°48.0	N09°31.0	245°51.0	N24°50.7	285°13.6	N22°11.0	3°51.3	S04°30.5			3°51.3	S04°30.5		Capella	280°21.7	46°01.5
13	13°18.8	12°51.8	30.3	260°52.8	50.5	300°15.7	11.0	18°53.5	30.3			18°53.5	30.3		Bellatrix	278°22.8	6°22.3
14	28°21.3	27°55.7	29.6	275°54.6	50.3	315°17.8	11.1	33°55.7	30.2			33°55.7	30.2		Elnath	278°01.7	28°37.8
15	43°23.8	42°59.5	· · 29.0	290°56.4	· · 50.2	330°20.0	· · 11.1	48°57.9	· · 30.1			48°57.9	· · 30.1		Alnilam	275°37.6	-1°11.3
16	58°26.2	58°03.4	28.3	305°58.2	50.0	345°22.1	11.2	64°00.1	30.0			64°00.1	30.0		Betelgeuse	270°51.9	7°24.6
17	73°28.7	73°07.2	27.6	321°00.0	49.8	0°24.2	11.2	79°02.2	29.9			79°02.2	29.9		Canopus	263°52.3	-52°42.8
18	88°31.2	88°11.1	N09°26.9	336°01.8	N24°49.7	15°26.3	N22°11.3	94°04.4	S04°29.7			94°04.4	S04°29.7		Sirius	258°26.0	-16°45.2
19	103°33.6	103°14.9	26.2	351°03.6	49.5	30°28.4	11.3	109°06.6	29.6			109°06.6	29.6		Adhara	255°05.7	-29°00.6
20	118°36.1	118°18.8	25.5	6°05.4	49.3	45°30.5	11.4	124°08.8	29.5			124°08.8	29.5		Procyon	244°50.5	5°09.5
21	133°38.6	133°22.7	· · 24.8	21°07.2	· · 49.2	60°32.6	· · 11.4	139°11.0	· · 29.4			139°11.0	· · 29.4		Pollux	243°17.0	27°57.9
22	148°41.0	148°26.5	24.1	36°09.0	49.0	75°34.8	11.4	154°13.1	29.3			154°13.1	29.3		Avior	234°14.4	-59°35.3
23	163°43.5	163°30.4	23.4	51°10.8	48.9	90°36.9	11.5	169°15.3	29.1			169°15.3	29.1		Suhail	222°45.9	-43°32.3
Mer.pass. 12:07			ν 3.9' d -0.7' m -4.18	ν 1.8' d -0.2' m 0.19	ν 2.1' d 0.0' m -2.17	ν 2.2' d -0.1' m 1.13											

Fri	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec		GHA	Dec		GHA	Dec	
0	178°46.0	178°34.2	N09°22.7	66°12.6	N24°48.7	105°39.0	N22°11.5	184°17.5	S04°29.0			184°17.5	S04°29.0		Regulus	207°34.0	11°50.5
1	193°48.4	193°38.1	22.0	81°14.3	48.5	120°41.1	11.6	199°19.7	28.9			199°19.7	28.9		Dubhe	193°40.1	61°36.9
2	208°50.9	208°41.9	21.3	96°16.1	48.4	135°43.2	11.6	214°21.9	28.8			214°21.9	28.8		Denebola	182°24.5	14°25.7
3	223°53.3	223°45.8	· · 20.6	111°17.9	· · 48.2	150°45.3	· · 11.7	229°24.0	· · 28.7			229°24.0	· · 28.7		Gienah	175°43.1	-17°41.0
4	238°55.8	238°49.6	19.9	126°19.7	48.0	165°47.4	11.7	244°26.2	28.6			244°26.2	28.6		Acrux	172°59.3	-63°14.4
5	253°58.3	253°53.5	19.1	141°21.5	47.9	180°49.5	11.8	259°28.4	28.4			259°28.4	28.4		Gacrux	171°51.0	-57°15.3
6	269°00.7	268°57.4	N09°18.4	156°23.3	N24°47.7	195°51.6	N22°11.8	274°30.6	S04°28.3			274°30.6	S04°28.3		Alioth	166°12.2	55°49.2
7	284°03.2	284°01.2	17.7	171°25.1	47.5	210°53.8	11.8	289°32.8	28.2			289°32.8	28.2		Spica	158°21.8	-11°17.7
8	299°05.7	299°05.1	17.0	186°26.9	47.4	225°55.9	11.9	304°34.9	28.1			304°34.9	28.1		Alkaid	152°51.4	49°11.0
9	314°08.1	314°08.9	· · 16.3	201°28.7	· · 47.2	240°58.0	· · 11.9	319°37.1	· · 28.0			319°37.1	· · 28.0		Hadar	148°35.3	-60°29.6
10	329°10.6	329°12.8	15.5	216°30.4	47.0	256°00.1	12.0	334°39.3	27.8			334°39.3	27.8		Menkent	147°57.1	-36°29.7
11	344°13.1	344°16.6	14.8	231°32.2	46.9	271°02.2	12.0	349°41.5	27.7			349°41.5	27.7		Arcturus	145°47.5	19°02.8
12	359°15.5	359°20.5	N09°14.1	246°34.0	N24°46.7	286°04.3	N22°12.1	4°43.7	S04°27.6			4°43.7	S04°27.6		Rigel Kent.	139°39.6	-60°56.3
13	14°18.0	14°24.3	13.4	261°35.8	46.5	301°06.4	12.1	19°45.8	27.5			19°45.8	27.5		Kochab	137°18.9	74°02.8
14	29°20.5	29°28.2	12.6	276°37.6	46.4	316°08.5	12.2	34°48.0	27.4			34°48.0	27.4		Zuben'ubi	136°55.6	-16°08.9
15	44°22.9	44°32.1	· · 11.9	291°39.3	· · 46.2	331°10.6	· · 12.2	49°50.2	· · 27.2			49°50.2	· · 27.2		Alphecca	126°03.4	26°37.5
16	59°25.4	59°35.9	11.2	306°41.1	46.0	346°12.7	12.3	64°52.4	27.1			64°52.4	27.1		Antares	112°15.5	-26°29.3
17	74°27.8	74°39.8	10.4	321°42.9	45.9	1°14.9	12.3	79°54.6	27.0			79°54.6	27.0		Atria	107°09.5	-69°04.1
18	89°30.3	89°43.6	N09°09.7	336°44.7	N24°45.7	16°17.0	N22°12.3	94°56.7	S04°26.9			94°56.7	S04°26.9		Sabik	102°02.5	-15°45.4
19	104°32.8	104°47.5	09.0	351°46.5	45.5	31°19.1	12.4	109°58.9	26.8			109°58.9	26.8		Shaula	96°10.1	-37°07.3
20	119°35.2	119°51.3	08.2	6°48.2	45.4	46°21.2	12.4	125°01.1	26.6			125°01.1	26.6		Rasalhague	95°58.4	12°32.3
21	134°37.7	134°55.2	· · 07.5	21°50.0	· · 45.2	61°23.3	· · 12.5	140°03.3	· · 26.5			140°03.3	· · 26.5		Eltanin	90°42.1	51°28.7
22	149°40.2	149°59.0	06.8	36°51.8	45.0	76°25.4	12.5	155°05.5	26.4			155°05.5	26.4		Kaus Aust.	83°32.3	-34°22.3
23	164°42.6	165°02.9	06.0	51°53.6	44.9	91°27.5	12.6	170°07.7	26.3			170°07.7	26.3		Vega	80°33.2	38°48.1
Mer.pass. 12:03			ν 3.9' d -0.7' m -4.20	ν 1.8' d -0.2' m 0.21	ν 2.1' d 0.0' m -2.17	ν 2.2' d -0.1' m 1.13											

Sat	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec		GHA	Dec		GHA	Dec	
0	179°45.1	180°06.7	N09°05.3	66°55.3	N24°44.7	106°29.6	N22°12.6	185°09.8	S04°26.2			185°09.8	S04°26.2		Mer 20 Thu	SHA	Mer.pass
1	194°47.6	195°10.6	04.5	81°57.1	44.5	121°31.7	12.7	200°12.0	26.1			200°12.0	26.1		Venus	359°14.9	12:09
2	209°50.0	210°14.4	03.8	96°58.9	44.4	136°33.8	12.7	215°14.2	25.9			215°14.2	25.9		Mars	247°42.5	19:36
3	224°52.5	225°18.3	· · 03.0	112°00.7	· · 44.2	151°35.9	· · 12.7	230°16.4	· · 25.8			230°16.4	· · 25.8		Jupiter	287°01.4	16:58
4	239°54.9	240°22.1	02.3	127°02.4	44.0	166°38.0	12.8	245°18.6	25.7			245°18.6	25.7		Saturn	5°38.4	11:45
5	254°57.4	255°26.0	01.5	142°04.2	43.9	181°40.1	12.8	260°20.7	25.6			260°20.7	25.6				
6	269°59.9	270°29.8	N09°00.8	157°06.0	N24°43.7	196°42.2	N22°12.9	275°22.9	S04°25.5			275°22.9	S04°25.5		Mer 21 Fri	SHA	Mer.pass
7	285°02.3	285°33.7	09°00.0	172°07.7	43.5	211°44.3	12.9	290°25.1	25.3			290°25.1	25.3		Venus	359°48.3	12:03
8	300°04.8	300°37.5	08°59.2	187°09.5	43.4	226°46.4	13.0	305°27.3	25.2			305°27.3	25.2		Mars	247°26.6	19:33
9	315°07.3	315°41.4	· · 58.5	202°11.3	· · 43.2	241°48.6	· · 13.0	320°29.5	· · 25.1			320°29.5	· · 25.1		Jupiter	286°53.0	16:55
10	330°09.7	330°45.2	57.7	217°13.0	43.0	256°50.7	13.1	335°31.6	25.0			335°31.6	25.0		Saturn	5°31.5	11:41
11	345°12.2	345°49.1	57.0	232°14.8	42.8	271°52.8	13.1	350°33.8	24.9			350°33.8	24.9				
12	0°14.7	0°52.9	N08°56.2	247°16.5	N24°42.7	286°54.9	N22°13.2	5°36.0	S04°24.7			5°36.0	S04°24.7		Mer 22 Sat	SHA	Mer.pass
13	15°17.1	15°56.8	55.4	262°18.3	42.5	301°57.0	13.2	20°38.2	24.6			20°38.2	24.6		Venus	0°21.6	11:56
14	30°19.6	31°00.6	54.7	277°20.1	42.3	316°59.1	13.2	35°40.4	24.5			35°40.4	24.5		Mars	247°10.2	19:30

Sun				Moon				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	178°07.4	S00°08.9		299°05.2	11.5'	S25°12.5	7.3'	54.3'
1	193°07.6	07.9		313°35.7	11.4'	25°19.8	7.2'	54.3'
2	208°07.7	07.0		328°06.1	11.3'	25°27.0	7.1'	54.4'
3	223°07.9	· · 06.0		342°36.5	11.3'	25°34.0	6.9'	54.4'
4	238°08.1	05.0		357°06.7	11.2'	25°41.0	6.8'	54.4'
5	253°08.3	04.0		11°36.9	11.1'	25°47.8	6.7'	54.4'
6	268°08.5	S00°03.0		26°07.0	11.0'	S25°54.5	6.6'	54.4'
7	283°08.7	02.0		40°37.0	10.9'	26°01.1	6.5'	54.4'
8	298°08.9	01.0		55°07.0	10.9'	26°07.6	6.4'	54.4'
9	313°09.0	S00°00.0		69°36.9	10.8'	26°13.9	6.2'	54.5'
10	328°09.2	N00°01.0		84°06.7	10.7'	26°20.2	6.1'	54.5'
11	343°09.4	01.9		98°36.4	10.6'	26°26.3	6.0'	54.5'
12	358°09.6	N00°02.9		113°06.0	10.6'	S26°32.3	5.9'	54.5'
13	13°09.8	03.9		127°35.6	10.5'	26°38.2	5.8'	54.5'
14	28°10.0	04.9		142°05.1	10.4'	26°43.9	5.6'	54.5'
15	43°10.2	· · 05.9		156°34.5	10.4'	26°49.5	5.5'	54.6'
16	58°10.3	06.9		171°03.9	10.3'	26°55.0	5.4'	54.6'
17	73°10.5	07.9		185°33.2	10.2'	27°00.4	5.3'	54.6'
18	88°10.7	N00°08.9		200°02.4	10.1'	S27°05.7	5.1'	54.6'
19	103°10.9	09.8		214°31.5	10.1'	27°10.8	5.0'	54.6'
20	118°11.1	10.8		229°00.6	10.0'	27°15.8	4.9'	54.7'
21	133°11.3	· · 11.8		243°29.6	9.9'	27°20.7	4.7'	54.7'
22	148°11.5	12.8		257°58.5	9.9'	27°25.4	4.6'	54.7'
23	163°11.6	13.8		272°27.3	9.8'	27°30.0	4.5'	54.7'
SD = 16.1'		$d = -1.0'$		SD = 14.8'				

Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	178°11.8	N00°14.8		286°56.1	9.7'	S27°34.5	4.3'	54.7'
1	193°12.0	15.8		301°24.8	9.7'	27°38.8	4.2'	54.7'
2	208°12.2	16.8		315°53.5	9.6'	27°43.0	4.1'	54.8'
3	223°12.4	· · 17.7		330°22.1	9.5'	27°47.1	3.9'	54.8'
4	238°12.6	18.7		344°50.6	9.5'	27°51.0	3.8'	54.8'
5	253°12.8	19.7		359°19.0	9.4'	27°54.8	3.7'	54.8'
6	268°12.9	N00°20.7		13°47.4	9.3'	S27°58.5	3.5'	54.9'
7	283°13.1	21.7		28°15.8	9.3'	28°02.0	3.4'	54.9'
8	298°13.3	22.7		42°44.0	9.2'	28°05.4	3.2'	54.9'
9	313°13.5	· · 23.7		57°12.2	9.1'	28°08.6	3.1'	54.9'
10	328°13.7	24.7		71°40.4	9.1'	28°11.8	3.0'	54.9'
11	343°13.9	25.6		86°08.5	9.0'	28°14.7	2.8'	55.0'
12	358°14.1	N00°26.6		100°36.5	9.0'	S28°17.5	2.7'	55.0'
13	13°14.2	27.6		115°04.4	8.9'	28°20.2	2.5'	55.0'
14	28°14.4	28.6		129°32.4	8.9'	28°22.8	2.4'	55.0'
15	43°14.6	· · 29.6		144°00.2	8.8'	28°25.2	2.3'	55.1'
16	58°14.8	30.6		158°28.0	8.7'	28°27.4	2.1'	55.1'
17	73°15.0	31.6		172°55.8	8.7'	28°29.5	2.0'	55.1'
18	88°15.2	N00°32.6		187°23.4	8.6'	S28°31.5	1.8'	55.1'
19	103°15.4	33.5		201°51.1	8.6'	28°33.3	1.7'	55.2'
20	118°15.6	34.5		216°18.7	8.5'	28°35.0	1.5'	55.2'
21	133°15.7	· · 35.5		230°46.2	8.5'	28°36.5	1.4'	55.2'
22	148°15.9	36.5		245°13.7	8.4'	28°37.9	1.2'	55.2'
23	163°16.1	37.5		259°41.2	8.4'	28°39.1	1.1'	55.3'
SD = 16.1'		$d = 1.0'$		SD = 14.9'				

Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	178°16.3	N00°38.5		274°08.6	8.4'	S28°40.2	0.9'	55.3'
1	193°16.5	39.5		288°35.9	8.3'	28°41.1	0.8'	55.3'
2	208°16.7	40.4		303°03.3	8.3'	28°41.9	0.6'	55.3'
3	223°16.9	· · 41.4		317°30.5	8.2'	28°42.5	0.5'	55.4'
4	238°17.0	42.4		331°57.8	8.2'	28°43.0	0.3'	55.4'
5	253°17.2	43.4		346°25.0	8.2'	28°43.3	0.2'	55.4'
6	268°17.4	N00°44.4		0°52.1	8.1'	S28°43.5	0.0'	55.4'
7	283°17.6	45.4		15°19.2	8.1'	28°43.5	-0.1'	55.5'
8	298°17.8	46.4		29°46.3	8.1'	28°43.4	-0.3'	55.5'
9	313°18.0	· · 47.4		44°13.4	8.0'	28°43.1	-0.4'	55.5'
10	328°18.2	48.3		58°40.4	8.0'	28°42.7	-0.6'	55.6'
11	343°18.4	49.3		73°07.4	8.0'	28°42.1	-0.8'	55.6'
12	358°18.5	N00°50.3		87°34.3	7.9'	S28°41.3	-0.9'	55.6'
13	13°18.7	51.3		102°01.3	7.9'	28°40.4	-1.1'	55.6'
14	28°18.9	52.3		116°28.2	7.9'	28°39.3	-1.2'	55.7'
15	43°19.1	· · 53.3		130°55.1	7.9'	28°38.1	-1.4'	55.7'
16	58°19.3	54.3		145°21.9	7.8'	28°36.8	-1.5'	55.7'
17	73°19.5	55.2		159°48.8	7.8'	28°35.2	-1.7'	55.8'
18	88°19.7	N00°56.2		174°15.6	7.8'	S28°33.5	-1.8'	55.8'
19	103°19.9	57.2		188°42.4	7.8'	28°31.7	-2.0'	55.8'
20	118°20.0	58.2		203°09.1	7.8'	28°29.7	-2.2'	55.9'
21	133°20.2	00°59.2		217°35.9	7.7'	28°27.5	-2.3'	55.9'
22	148°20.4	01°00.2		232°02.6	7.7'	28°25.2	-2.5'	55.9'
23	163°20.6	01.2		246°29.3	7.7'	28°22.7	-2.6'	56.0'
SD = 16.0'		$d = 1.0'$		SD = 15.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:13	04:44	05:52	18:25	19:34	21:06
N 70°	03:33	04:52	05:54	18:23	19:25	20:45
68°	03:49	04:59	05:55	18:21	19:18	20:29
66°	04:01	05:05	05:56	18:20	19:12	20:16
64°	04:11	05:09	05:57	18:19	19:07	20:06
62°	04:19	05:13	05:57	18:18	19:03	19:57
60°	04:26	05:17	05:58	18:17	18:59	19:50
N 58°	04:33	05:19	05:59	18:17	18:56	19:43
56°	04:38	05:22	05:59	18:16	18:53	19:38
54°	04:42	05:24	06:00	18:16	18:51	19:33
52°	04:46	05:26	06:00	18:15	18:49	19:29
50°	04:50	05:28	06:00	18:15	18:47	19:25
45°	04:57	05:32	06:01	18:14	18:43	19:18
N 40°	05:03	05:35	06:02	18:13	18:40	19:12
35°	05:07	05:37	06:02	18:13	18:38	19:07
30°	05:11	05:39	06:02	18:12	18:36	19:04
20°	05:16	05:41	06:03	18:11	18:33	18:59
N 10°	05:18	05:43	06:04	18:11	18:32	18:56
0°	05:19	05:43	06:04	18:10	18:31	18:55
S 10°	05:19	05:43	06:04	18:10	18:31	18:55
20°	05:17	05:42	06:04	18:10	18:32	18:57
30°	05:12	05:40	06:04	18:10	18:33	19:01
35°	05:09	05:39	06:04	18:10	18:35	19:04
40°	05:05	05:37	06:04	18:09	18:36	19:08
45°	05:00	05:35	06:04	18:10	18:39	19:13
S 50°	04:53	05:31	06:04	18:10	18:42	19:20
52°	04:50	05:30	06:04	18:10	18:43	19:23
54°	04:46	05:28	06:03	18:10	18:45	19:26
56°	04:42	05:26	06:03	18:10	18:47	19:31
58°	04:37	05:24	06:03	18:10	18:49	19:35
S 60°	04:31	05:21	06:03	18:10	18:51	19:41

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	■	■	■	■	■	■
64°	03:35	■	■	04:39	■	■
62°	02:25	■	■	05:49	■	■
60°	01:50	03:26	04:43	06:25	06:35	07:09
N 58°	01:25	02:49	03:59	06:51	07:12	07:54
56°	01:05	02:23	03:30	07:11	07:38	08:23
54°	00:49	02:03	03:07	07:28	07:59	08:46
52°	00:34	01:46	02:49	07:43	08:16	09:04
50°	00:22	01:31	02:33	07:56	08:31	09:20
45°	· · ·	01:01	02:01	08:22	09:01	09:51
N 40°	· · ·	00:38	01:37	08:43	09:25	10:16
35°	· · ·	00:19	01:17	09:00	09:44	10:36
30°	· · ·	00:03	01:00	09:15	10:01	10:53
20°	23:36	· · ·	00:31	09:41	10:29	11:22
N 10°	23:12	· · ·	00:06	10:03	10:53	11:47
0°	22:50	23:43	· · ·	10:24	11:16	12:10
S 10°	22:29	23:20	· · ·	10:45	11:38	12:33
20°	22:05	22:56	23:51	11:07	12:02	12:57
30°	21:38	22:27	23:23	11:33	12:31	13:26
35°	21:22	22:10	23:06	11:48	12:47	13:43
40°	21:04	21:50	22:46	12:06	13:07	14:03
45°	20:42	21:26	22:23	12:28	13:31	14:27
S 50°	20:14	20:56	21:52	12:55	14:01	14:58
52°	20:00	20:40	21:37	13:09	14:17	15:13
54°	19:44	20:22	21:19	13:24	14:34	15:31
56°	19:25	20:01	20:57	13:43	14:56	15:53
58°	19:02	19:33	20:29	14:05	15:23	16:20
S 60°	18:32	18:53	19:49	14:35	16:03	17:01

Day	Sun		Mer. Pass hh:mm	Moon		Age 73-55%
	Eqn. of Time	Mer. Pass		Mer. Pass.	Age	
	00 ^h mm:ss			12 ^h mm:ss		
20	07:30	07:22	12:07	04:12	16:37	
21	07:13	07:04	12:07	05:03	17:29	
22	06:55	06:46	12:07	05:56	18:24	

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:13	04:44	05:52	18:25	19:34	21:06
N 70°	03:33	04:52	05:54	18:23	19:25	20:45
68°	03:49	04:59	05:55	18:21	19:18	20:29
66°	04:01	05:05	05:56	18:20	19:12	20:16
64°	04:11	05:09	05:57	18:19	19:07	20:06
62°	04:19	05:13	05:57	18:18	19:03	19:57
60°	04:26	05:17	05:58	18:17	18:59	19:50
N 58°	04:33	05:19	05:59	18:17	18:56	19:43
56°	04:38	05:22	05:59	18:16	18:53	19:38
54°	04:42	05:24	06:00	18:16	18:51	19:33
52°	04:46	05:26	06:00	18:15	18:49	19:29
50°	04:50	05:28	06:00	18:15	18:47	19:25
45°	04:57	05:32	06:01	18:14	18:43	19:18
N 40°	05:03	05:35	06:02	18:13	18:40	19:12
35°	05:07	05:37	06:02	18:13	18:38	19:07
30°	05:11	05:39	06:02	18:12	18:36	19:04
20°	05:16	05:41	06:03	18:11	18:33	18:59
N 10°	05:18	05:43	06:04	18:11	18:32	18:56
0°	05:19	05:43	06:04	18:10	18:31	18:55
S 10°	05:19	05:43	06:04	18:10	18:31	18:55
20°	05:17	05:42	06:04	18:10	18:32	18:57
30°	05:12	05:40	06:04	18:10	18:33	19:01
35°	05:09	05:39	06:04	18:10	18:35	19:04
40°	05:05	05:37	06:04	18:09	18:36	19:08
45°	05:00	05:35	06:04	18:10	18:39	19:13
S 50°	04:53	05:31	06:04	18:10	18:42	19:20
52°	04:50	05:30	06:04	18:10	18:43	19:23
54°	04:46	05:28	06:03	18:10	18:45	19:26
56°	04:42	05:26	06:03	18:10	18:47	19:31
58°	04:37	05:24	06:03	18:10	18:49	19:35
S 60°	04:31	05:21	06:03	18:10	18:51	19:41

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	■	■	■	■	■	■
64°	03:35	■	■	04:39	■	■
62°	02:25	■	■	05:49	■	■
60°	01:50	03:26	04:43	06:25	06:35	07:09
N 58°	01:25	02:49	03:59	06:51	07:12	07:54
56°	01:05	02:23	03:30	07:11	07:38	08:23
54°	00:49	02:03	03:07	07:28	07:59	08:46
52°	00:34	01:46	02:49	07:43	08:16	09:04
50°	00:22	01:31	02:33	07:56	08:31	09:20
45°	· ·	01:01	02:01	08:22	09:01	09:51
N 40°	· ·	00:38	01:37	08:43	09:25	10:16
35°	· ·	00:19	01:17	09:00	09:44	10:36
30°	· ·	00:03	01:00	09:15	10:01	10:53
20°	23:36	· ·	00:31	09:41	10:29	11:22
N 10°	23:12	· ·	00:06	10:03	10:53	11:47
0°	22:50	23:43	· ·	10:24	11:16	12:10
S 10°	22:29	23:20	· ·	10:45	11:38	12:33
20°	22:05	22:56	23:51	11:07	12:02	12:57
30°	21:38	22:27	23:23	11:33	12:31	13:26
35°	21:22	22:10	23:06	11:48	12:47	13:43
40°	21:04	21:50	22:46	12:06	13:07	14:03
45°	20:42	21:26	22:23	12:28	13:31	14:27
S 50°	20:14	20:56	21:52	12:55	14:01	14:58
52°	20:00	20:40	21:37	13:09	14:17	15:13
54°	19:44	20:22	21:19	13:24	14:34	15:31
56°	19:25	20:01	20:57	13:43	14:56	15:53
58°	19:02	19:33	20:29	14:05	15:23	16:20
S 60°	18:32	18:53	19:49	14:35	16:03	17:01

Day	Sun		Mer. Pass	Moon		Age
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March 23, 24, 25 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	180°44.2	181°39.0	N08°46.9	67°37.6	N24°40.6	107°20.1	N22°13.7	186°02.2	S04°23.3	Alpheratz	357°35.0	29°13.7
1	195°46.7	196°42.9	46.1	82°39.4	40.5	122°22.2	13.7	201°04.4	23.2	Ankaa	353°07.4	-42°10.2
2	210°49.2	211°46.7	45.3	97°41.1	40.3	137°24.3	13.8	216°06.5	23.1	Schedar	349°31.5	56°40.5
3	225°51.6	226°50.5	· · 44.5	112°42.9	· · 40.1	152°26.4	· · 13.8	231°08.7	· · 23.0	Diphda	348°47.4	-17°51.0
4	240°54.1	241°54.4	43.7	127°44.6	39.9	167°28.5	13.9	246°10.9	22.8	Achernar	335°20.6	-57°06.7
5	255°56.6	256°58.2	43.0	142°46.4	39.8	182°30.6	13.9	261°13.1	22.7	Hamal	327°51.3	23°34.9
6	270°59.0	272°02.0	N08°42.2	157°48.1	N24°39.6	197°32.7	N22°14.0	276°15.3	S04°22.6	Polaris	314°19.0	89°22.4
7	286°01.5	287°05.9	41.4	172°49.9	39.4	212°34.8	14.0	291°17.4	22.5	Acamar	315°11.9	-40°12.4
8	301°03.9	302°09.7	40.6	187°51.6	39.3	227°36.9	14.1	306°19.6	22.4	Menkar	314°06.2	4°11.2
9	316°06.4	317°13.5	· · 39.8	202°53.4	· · 39.1	242°39.0	· · 14.1	321°21.8	· · 22.2	Mirfak	308°28.4	49°57.1
10	331°08.9	332°17.3	39.0	217°55.1	38.9	257°41.1	14.2	336°24.0	22.1	Aldebaran	290°39.6	16°33.6
11	346°11.3	347°21.2	38.2	232°56.9	38.7	272°43.2	14.2	351°26.2	22.0	Rigel	281°03.8	-8°10.5
12	1°13.8	2°25.0	N08°37.4	247°58.6	N24°38.6	287°45.3	N22°14.2	6°28.4	S04°21.9	Capella	280°21.8	46°01.5
13	16°16.3	17°28.8	36.6	263°00.4	38.4	302°47.4	14.3	21°30.5	21.8	Bellatrix	278°22.8	6°22.3
14	31°18.7	32°32.7	35.8	278°02.1	38.2	317°49.5	14.3	36°32.7	21.7	Elnath	278°01.7	28°37.8
15	46°21.2	47°36.5	· · 35.0	293°03.8	· · 38.0	332°51.6	· · 14.4	51°34.9	· · 21.5	Alnilam	275°37.6	-1°11.2
16	61°23.7	62°40.3	34.2	308°05.6	37.9	347°53.7	14.4	66°37.1	21.4	Betelgeuse	270°51.9	7°24.6
17	76°26.1	77°44.1	33.4	323°07.3	37.7	2°55.8	14.5	81°39.3	21.3	Canopus	263°52.3	-52°42.8
18	91°28.6	92°47.9	N08°32.6	338°09.1	N24°37.5	17°57.8	N22°14.5	96°41.4	S04°21.2	Sirius	258°26.1	-16°45.2
19	106°31.1	107°51.8	31.7	353°10.8	37.3	32°59.9	14.6	111°43.6	21.1	Adhara	255°05.7	-29°00.6
20	121°33.5	122°55.6	30.9	8°12.5	37.2	48°02.0	14.6	126°45.8	20.9	Procyon	244°50.6	5°09.5
21	136°36.0	137°59.4	· · 30.1	23°14.3	· · 37.0	63°04.1	· · 14.7	141°48.0	· · 20.8	Pollux	243°17.0	27°57.9
22	151°38.4	153°03.2	29.3	38°16.0	36.8	78°06.2	14.7	156°50.2	20.7	Avior	234°14.4	-59°35.6
23	166°40.9	168°07.0	28.5	53°17.8	36.7	93°08.3	14.7	171°52.3	20.6	Suhail	222°45.9	-43°32.3
Mer.pass. 11:55		ν 3.8' d -0.8' m -4.20		ν 1.8' d -0.2' m 0.25		ν 2.1' d 0.0' m -2.16		ν 2.2' d -0.1' m 1.14				

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	181°43.4	183°10.9	N08°27.7	68°19.5	N24°36.5	108°10.4	N22°14.8	186°54.5	S04°20.5	Alphard	217°47.4	-8°46.2
1	196°45.8	198°14.7	26.9	83°21.2	36.3	123°12.5	14.8	201°56.7	20.3	Regulus	207°34.0	11°50.5
2	211°48.3	213°18.5	26.0	98°23.0	36.1	138°14.6	14.9	216°58.9	20.2	Dubhe	193°40.1	61°36.9
3	226°50.8	228°22.3	· · 25.2	113°24.7	· · 36.0	153°16.7	· · 14.9	232°01.1	· · 20.1	Denebola	182°24.5	14°25.7
4	241°53.2	243°26.1	24.4	128°26.4	35.8	168°18.8	15.0	247°03.3	20.0	Gienah	175°43.1	-17°41.1
5	256°55.7	258°29.9	23.6	143°28.1	35.6	183°20.9	15.0	262°05.4	19.9	Acrux	172°59.3	-63°14.4
6	271°58.2	273°33.7	N08°22.8	158°29.9	N24°35.4	198°23.0	N22°15.1	277°07.6	S04°19.8	Gacrux	171°50.9	-57°15.3
7	287°00.6	288°37.5	21.9	173°31.6	35.2	213°25.1	15.1	292°09.8	19.6	Alioth	166°12.2	55°49.2
8	302°03.1	303°41.3	21.1	188°33.3	35.1	228°27.2	15.2	307°12.0	19.5	Spica	158°21.8	-11°17.7
9	317°05.5	318°45.1	· · 20.3	203°35.1	· · 34.9	243°29.3	· · 15.2	322°14.2	· · 19.4	Alkaid	152°51.4	49°11.0
10	332°08.0	333°48.9	19.5	218°36.8	34.7	258°31.3	15.3	337°16.3	19.3	Hadar	148°35.3	-60°29.6
11	347°10.5	348°52.7	18.6	233°38.5	34.5	273°33.4	15.3	352°18.5	19.2	Menkent	147°57.1	-36°29.7
12	2°12.9	3°56.5	N08°17.8	248°40.2	N24°34.4	288°35.5	N22°15.3	7°20.7	S04°19.0	Arcturus	145°47.5	19°02.8
13	17°15.4	19°00.3	17.0	263°42.0	34.2	303°37.6	15.4	22°22.9	18.9	Rigil Kent.	139°39.6	-60°56.3
14	32°17.9	34°04.1	16.1	278°43.7	34.0	318°39.7	15.4	37°25.1	18.8	Kochab	137°18.8	74°02.8
15	47°20.3	49°07.9	· · 15.3	293°45.4	· · 33.8	333°41.8	· · 15.5	52°27.3	· · 18.7	Zuben'ubi	136°55.6	-16°08.9
16	62°22.8	64°11.7	14.5	308°47.1	33.7	348°43.9	15.5	67°29.4	18.6	Alphecca	126°03.4	26°37.5
17	77°25.3	79°15.5	13.6	323°48.9	33.5	3°46.0	15.6	82°31.6	18.5	Antares	112°15.5	-26°29.3
18	92°27.7	94°19.3	N08°12.8	338°50.6	N24°33.3	18°48.1	N22°15.6	97°33.8	S04°18.3	Atria	107°09.4	-69°04.1
19	107°30.2	109°23.1	12.0	353°52.3	33.1	33°50.2	15.7	112°36.0	18.2	Sabik	102°02.5	-15°45.4
20	122°32.7	124°26.9	11.1	8°54.0	32.9	48°52.2	15.7	127°38.2	18.1	Shaula	96°10.1	-37°07.3
21	137°35.1	139°30.7	· · 10.3	23°55.7	· · 32.8	63°54.3	· · 15.8	142°40.3	· · 18.0	Rasalhague	95°58.3	12°32.3
22	152°37.6	154°34.5	09.4	38°57.4	32.6	78°56.4	15.8	157°42.5	17.9	Eltanin	90°42.1	51°28.7
23	167°40.0	169°38.2	08.6	53°59.2	32.4	93°58.5	15.8	172°44.7	17.7	Kaus Aust.	83°32.3	-34°22.3
Mer.pass. 11:51		ν 3.8' d -0.8' m -4.20		ν 1.7' d -0.2' m 0.28		ν 2.1' d 0.0' m -2.15		ν 2.2' d -0.1' m 1.15		Vega	80°33.2	38°48.1

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	182°42.5	184°42.0	N08°07.8	69°00.9	N24°32.2	109°00.6	N22°15.9	187°46.9	S04°17.6	Nunki	75°47.6	-26°15.9
1	197°45.0	199°45.8	06.9	84°02.6	32.1	124°02.7	15.9	202°49.1	17.5	Altair	61°59.9	8°55.9
2	212°47.4	214°49.6	06.1	99°04.3	31.9	139°04.8	16.0	217°51.3	17.4	Peacock	53°05.8	-56°39.1
3	227°49.9	229°53.4	· · 05.2	114°06.0	· · 31.7	154°06.9	· · 16.0	232°53.4	· · 17.3	Deneb	49°26.0	45°21.9
4	242°52.4	244°57.1	04.4	129°07.7	31.5	169°09.0	16.1	247°55.6	17.2	Enif	33°38.9	9°59.3
5	257°54.8	260°00.9	03.5	144°09.4	31.3	184°11.0	16.1	262°57.8	17.0	Al Na'ir	27°33.1	-46°50.3
6	272°57.3	275°04.7	N08°02.7	159°11.2	N24°31.2	199°13.1	N22°16.2	278°00.0	S04°16.9	Fomalhaut	15°14.7	-29°29.4
7	287°59.8	290°08.5	01.8	174°12.9	31.0	214°15.2	16.2	293°02.2	16.8	Scheat	13°45.4	28°13.0
8	303°02.2	305°12.2	01.0	189°14.6	30.8	229°17.3	16.3	308°04.4	16.7	Markab	13°30.1	25°20.3
9	318°04.7	320°16.0	08°00.1	204°16.3	· · 30.6	244°19.4	· · 16.3	323°06.5	· · 16.6	Mar 23 Sun		
10	333°07.2	335°19.8	07°59.3	219°18.0	30.4	259°21.5	16.4	338°08.7	16.4	Venus	0°54.8	11:50
11	348°09.6	350°23.5	58.4	234°19.7	30.3	274°23.6	16.4	353°10.9	16.3	Mars	246°53.4	19:27
12	3°12.1	5°27.3	N07°57.6	249°21.4	N24°30.1	289°25.6	N22°16.4	8°13.1	S04°16.2	Jupiter	286°35.9	16:48
13	18°14.5	20°31.1	56.7	264°23.1	29.9	304°27.7	16.5	23°15.3	16.1	Saturn	5°17.9	11:34
14	33°17.0	35°34.8	55.9	279°24.8	29.7	319°29.8	16.5	38°17.4	16.0	Mar 24 Mon		
15	48°19.5	50°38.6	· · 55.0	294°26.5	· · 29.5	334°31.9	· · 16.6	53°19.6	· · 15.9	Venus	1°27.5	11:44
16	63°21.9	65°42.3	54.2	309°28.2	29.4	349°34.0	16.6	68°21.8	15.7	Mars	246°36.1	19:24
17	78°24.4	80°46.1	53.3	324°29.9	29.2	4°36.1	16.7	83°24.0	15.6	Jupiter	286°27.0	16:45
18	93°26.9	95°49.8	N07°52.4	339°31.6	N24°29.0	19°38.2	N22°16.7	98°26.2	S04°15.5	Saturn	5°11.2	11:31
19	108°29.3	110°53.6	51.6	354°33.3	28.8	34°40.2	16.8	113°28.4	15.4	Mar 25 Tue		
20	123°31.8	125°57.3	50.7	9°35.0	28.6	49°42.3	16.8	128°30.5	15.3	Venus	1°59.5	11:38
21	138°34.3	141°01.										

h	Sun			Moon			
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°20.8	N01°02.1	260°56.1	7.7'	S28°20.1	-2.8'	56.0'
1	193°21.0	03.1	275°22.8	7.7'	28°17.3	-3.0'	56.0'
2	208°21.2	04.1	289°49.5	7.7'	28°14.3	-3.1'	56.1'
3	223°21.4	· · 05.1	304°16.1	7.7'	28°11.2	-3.3'	56.1'
4	238°21.5	06.1	318°42.8	7.7'	28°07.9	-3.4'	56.1'
5	253°21.7	07.1	333°09.5	7.7'	28°04.5	-3.6'	56.2'
6	268°21.9	N01°08.1	347°36.1	7.7'	S28°00.9	-3.7'	56.2'
7	283°22.1	09.0	2°02.8	7.7'	27°57.2	-3.9'	56.2'
8	298°22.3	10.0	16°29.5	7.7'	27°53.3	-4.1'	56.3'
9	313°22.5	· · 11.0	30°56.1	7.7'	27°49.2	-4.2'	56.3'
10	328°22.7	12.0	45°22.8	7.7'	27°45.0	-4.4'	56.3'
11	343°22.9	13.0	59°49.4	7.7'	27°40.6	-4.5'	56.4'
12	358°23.0	N01°14.0	74°16.1	7.7'	S27°36.0	-4.7'	56.4'
13	13°23.2	15.0	88°42.8	7.7'	27°31.3	-4.9'	56.4'
14	28°23.4	15.9	103°09.4	7.7'	27°26.5	-5.0'	56.5'
15	43°23.6	· · 16.9	117°36.1	7.7'	27°21.5	-5.2'	56.5'
16	58°23.8	17.9	132°02.8	7.7'	27°16.3	-5.3'	56.5'
17	73°24.0	18.9	146°29.5	7.7'	27°10.9	-5.5'	56.6'
18	88°24.2	N01°19.9	160°56.2	7.7'	S27°05.4	-5.7'	56.6'
19	103°24.4	20.9	175°23.0	7.7'	26°59.8	-5.8'	56.6'
20	118°24.5	21.9	189°49.7	7.8'	26°54.0	-6.0'	56.7'
21	133°24.7	· · 22.8	204°16.5	7.8'	26°48.0	-6.1'	56.7'
22	148°24.9	23.8	218°43.2	7.8'	26°41.9	-6.3'	56.8'
23	163°25.1	24.8	233°10.0	7.8'	26°35.6	-6.4'	56.8'
SD = 16.0'			SD = 15.3'				
$d = 1.0'$							

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	178°25.3	N01°25.8	247°36.8	7.8'	S26°29.2	-6.6'	56.8'
1	193°25.5	26.8	262°03.6	7.8'	26°22.6	-6.8'	56.9'
2	208°25.7	27.8	276°30.5	7.9'	26°15.8	-6.9'	56.9'
3	223°25.9	· · 28.7	290°57.3	7.9'	26°08.9	-7.1'	56.9'
4	238°26.1	29.7	305°24.2	7.9'	26°01.8	-7.2'	57.0'
5	253°26.2	30.7	319°51.1	7.9'	25°54.6	-7.4'	57.0'
6	268°26.4	N01°31.7	334°18.1	8.0'	S25°47.2	-7.5'	57.1'
7	283°26.6	32.7	348°45.0	8.0'	25°39.7	-7.7'	57.1'
8	298°26.8	33.7	3°12.0	8.0'	25°32.0	-7.8'	57.1'
9	313°27.0	· · 34.7	17°39.0	8.0'	25°24.2	-8.0'	57.2'
10	328°27.2	35.6	32°06.1	8.1'	25°16.2	-8.1'	57.2'
11	343°27.4	36.6	46°33.1	8.1'	25°08.1	-8.3'	57.2'
12	358°27.6	N01°37.6	61°00.2	8.1'	S24°59.8	-8.4'	57.3'
13	13°27.7	38.6	75°27.4	8.2'	24°51.3	-8.6'	57.3'
14	28°27.9	39.6	89°54.5	8.2'	24°42.7	-8.7'	57.4'
15	43°28.1	· · 40.6	104°21.7	8.2'	24°34.0	-8.9'	57.4'
16	58°28.3	41.5	118°48.9	8.3'	24°25.1	-9.0'	57.4'
17	73°28.5	42.5	133°16.2	8.3'	24°16.1	-9.2'	57.5'
18	88°28.7	N01°43.5	147°43.5	8.3'	S24°06.9	-9.3'	57.5'
19	103°28.9	44.5	162°10.8	8.4'	23°57.6	-9.5'	57.6'
20	118°29.1	45.5	176°38.2	8.4'	23°48.1	-9.6'	57.6'
21	133°29.2	· · 46.5	191°05.5	8.4'	23°38.4	-9.8'	57.6'
22	148°29.4	47.4	205°33.0	8.5'	23°28.7	-9.9'	57.7'
23	163°29.6	48.4	220°00.4	8.5'	23°18.8	-10.1'	57.7'
SD = 16.0'			SD = 15.5'				
$d = 1.0'$							

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	178°29.8	N01°49.4	234°27.9	8.5'	S23°08.7	-10.2'	57.8'
1	193°30.0	50.4	248°55.5	8.6'	22°58.5	-10.3'	57.8'
2	208°30.2	51.4	263°23.1	8.6'	22°48.1	-10.5'	57.8'
3	223°30.4	· · 52.4	277°50.7	8.7'	22°37.7	-10.6'	57.9'
4	238°30.6	53.3	292°18.3	8.7'	22°27.0	-10.8'	57.9'
5	253°30.8	54.3	306°46.0	8.7'	22°16.3	-10.9'	58.0'
6	268°30.9	N01°55.3	321°13.8	8.8'	S22°05.3	-11.0'	58.0'
7	283°31.1	56.3	335°41.5	8.8'	21°54.3	-11.2'	58.0'
8	298°31.3	57.3	350°09.4	8.9'	21°43.1	-11.3'	58.1'
9	313°31.5	· · 58.3	4°37.2	8.9'	21°31.8	-11.5'	58.1'
10	328°31.7	01°59.2	19°05.1	8.9'	21°20.4	-11.6'	58.2'
11	343°31.9	02°00.2	33°33.0	9.0'	21°08.8	-11.7'	58.2'
12	358°32.1	N02°01.2	48°01.0	9.0'	S20°57.0	-11.9'	58.2'
13	13°32.3	02.2	62°29.0	9.1'	20°45.2	-12.0'	58.3'
14	28°32.4	03.2	76°57.1	9.1'	20°33.2	-12.1'	58.3'
15	43°32.6	· · 04.1	91°25.2	9.1'	20°21.1	-12.2'	58.4'
16	58°32.8	05.1	105°53.3	9.2'	20°08.8	-12.4'	58.4'
17	73°33.0	06.1	120°21.5	9.2'	19°56.5	-12.5'	58.4'
18	88°33.2	N02°07.1	134°49.7	9.3'	S19°44.0	-12.6'	58.5'
19	103°33.4	08.1	149°18.0	9.3'	19°31.3	-12.8'	58.5'
20	118°33.6	09.1	163°46.3	9.3'	19°18.6	-12.9'	58.6'
21	133°33.8	· · 10.0	178°14.6	9.4'	19°05.7	-13.0'	58.6'
22	148°34.0	11.0	192°43.0	9.4'	18°52.7	-13.1'	58.6'
23	163°34.1	12.0	207°11.4	9.5'	18°39.6	-13.2'	58.7'
SD = 16.0'			SD = 15.8'				
$d = 1.0'$							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	02:51	04:27	05:37	18:38	19:49	21:27
N 70°	03:15	04:37	05:40	18:35	19:38	21:02
68°	03:33	04:46	05:42	18:32	19:29	20:43
66°	03:47	04:53	05:44	18:30	19:22	20:28
64°	03:59	04:58	05:46	18:28	19:16	20:16
62°	04:08	05:03	05:48	18:26	19:11	20:06
60°	04:16	05:07	05:49	18:25	19:07	19:58
N 58°	04:23	05:11	05:50	18:23	19:03	19:51
56°	04:29	05:14	05:51	18:22	19:00	19:45
54°	04:34	05:17	05:52	18:21	18:57	19:39
52°	04:39	05:19	05:53	18:20	18:54	19:35
50°	04:43	05:21	05:54	18:20	18:52	19:30
45°	04:51	05:26	05:55	18:18	18:47	19:22
N 40°	04:58	05:30	05:57	18:16	18:43	19:15
35°	05:03	05:33	05:58	18:15	18:40	19:10
30°	05:07	05:35	05:59	18:14	18:38	19:06
20°	05:13	05:38	06:00	18:12	18:34	19:00
N 10°	05:16	05:41	06:02	18:11	18:32	18:56
0°	05:18	05:42	06:03	18:09	18:30	18:54
S 10°	05:19	05:43	06:04	18:08	18:29	18:54
20°	05:17	05:43	06:05	18:07	18:29	18:55
30°	05:14	05:42	06:06	18:06	18:30	18:58
35°	05:12	05:41	06:06	18:05	18:31	19:00
40°	05:09	05:40	06:07	18:05	18:32	19:03
45°	05:04	05:38	06:08	18:04	18:33	19:07
S 50°	04:58	05:36	06:08	18:03	18:35	19:13
52°	04:55	05:35	06:09	18:03	18:36	19:16
54°	04:52	05:34	06:09	18:02	18:37	19:19
56°	04:48	05:32	06:09	18:02	18:39	19:22
58°	04:44	05:31	06:10	18:01	18:40	19:26
S 60°	04:39	05:29	06:10	18:01	18:42	19:31

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	08:09	■	■	09:30
66°	■	■	06:59	■	■	10:39
64°	■	07:12	06:22	■	08:32	11:15
62°	■	06:07	05:56	■	09:37	11:40
60°	05:20	05:32	05:35	08:27	10:11	12:00
N 58°	04:43	05:06	05:18	09:04	10:36	12:16
56°	04:17	04:46	05:04	09:30	10:56	12:30
54°	03:56	04:29	04:51	09:51	11:12	12:42
52°	03:38	04:15	04:40	10:08	11:26	12:52
50°	03:23	04:02	04:30	10:23	11:39	13:02
45°	02:53	03:36	04:09	10:53	12:04	13:21
N 40°	02:30	03:15	03:53	11:16	12:24	13:37
35°	02:10	02:57	03:38	11:35	12:41	13:50
30°	01:53	02:42	03:26	11:52	12:55	14:02
20°	01:25	02:16	03:04	12:19	13:20	14:21
N 10°	01:01	01:54	02:46	12:43	13:40	14:38
0°	00:38	01:34	02:28	13:05	14:00	14:53
S 10°	00:16	01:13	02:10	13:27	14:19	15:09
20°	· · ·	00:50	01:52	13:50	14:40	15:25
30°	· · ·	00:24	01:30	14:17	15:03	15:44
35°	· · ·	00:09	01:17	14:33	15:17	15:55
40°	23:51	· · ·	01:02	14:51	15:33	16:07
45°	23:29	· · ·	00:44	15:14	15:52	16:22
S 50°	23:02	· · ·	00:22	15:42	16:15	16:40
52°	22:49	· · ·	00:11	15:56	16:26	16:48
54°	22:33	23:59	· · ·	16:12	16:39	16:57
56°	22:15	23:45	· · ·	16:30	16:53	17:08
58°	21:52	23:29	· · ·	16:53	17:10	17:19
S 60°	21:22	23:09	· · ·	17:23	17:31	17:33

Day	Sun		Mer.	Moon		Age
	Eqn. of Time	12 ^h	Pass	Upper	Lower	23-25
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	45-25%
23	06:37	06:28	12:06	06:51	19:19	
24	06:19	06:10	12:06	07:47	20:14	
25	06:01	05:52	12:06	08:41	21:07	

March 26, 27, 28 UT (Wed., Thu., Fri.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	183°41.7	186°12.3	N07°47.3	69°41.8	N24°27.9	109°50.6	N22°17.0	188°39.3	S04°14.8	Alpheratz	357°35.0	29°13.7											
1	198°44.1	201°16.1	46.4	84°43.5	27.7	124°52.7	17.0	203°41.5	14.7	Ankaa	353°07.4	-42°10.2											
2	213°46.6	216°19.8	45.5	99°45.2	27.5	139°54.8	17.1	218°43.6	14.6	Schedar	349°31.5	56°40.5											
3	228°49.0	231°23.5	· · 44.7	114°46.9	· · 27.4	154°56.9	· · 17.1	233°45.8	· · 14.4	Diphda	348°47.4	-17°51.0											
4	243°51.5	246°27.3	43.8	129°48.6	27.2	169°59.0	17.2	248°48.0	14.3	Achernar	335°20.6	-57°06.6											
5	258°54.0	261°31.0	42.9	144°50.3	27.0	185°01.1	17.2	263°50.2	14.2	Hamal	327°51.3	23°34.9											
6	273°56.4	276°34.7	N07°42.1	159°52.0	N24°26.8	200°03.1	N22°17.3	278°52.4	S04°14.1	Polaris	314°20.0	89°22.4											
7	288°58.9	291°38.5	41.2	174°53.7	26.6	215°05.2	17.3	293°54.5	14.0	Acamar	315°11.9	-40°12.4											
8	304°01.4	306°42.2	40.3	189°55.4	26.4	230°07.3	17.4	308°56.7	13.9	Menkar	314°06.2	4°11.2											
9	319°03.8	321°45.9	· · 39.4	204°57.1	· · 26.3	245°09.4	· · 17.4	323°58.9	· · 13.7	Mirfak	308°28.4	49°57.1											
10	334°06.3	336°49.6	38.6	219°58.8	26.1	260°11.5	17.5	339°01.1	13.6	Aldebaran	290°39.6	16°33.6											
11	349°08.8	351°53.3	37.7	235°00.4	25.9	275°13.5	17.5	354°03.3	13.5	Rigel	281°03.8	-8°10.5											
12	4°11.2	6°57.1	N07°36.8	250°02.1	N24°25.7	290°15.6	N22°17.6	9°05.5	S04°13.4	Capella	280°21.8	46°01.5											
13	19°13.7	22°00.8	36.0	265°03.8	25.5	305°17.7	17.6	24°07.6	13.3	Bellatrix	278°22.8	6°22.3											
14	34°16.1	37°04.5	35.1	280°05.5	25.3	320°19.8	17.6	39°09.8	13.1	Elnath	278°01.7	28°37.8											
15	49°18.6	52°08.2	· · 34.2	295°07.2	· · 25.2	335°21.9	· · 17.7	54°12.0	· · 13.0	Alnilam	275°37.6	-1°11.2											
16	64°21.1	67°11.9	33.3	310°08.9	25.0	350°23.9	17.7	69°14.2	12.9	Betelgeuse	270°51.9	7°24.6											
17	79°23.5	82°15.6	32.5	325°10.6	24.8	5°26.0	17.8	84°16.4	12.8	Canopus	263°52.3	-52°42.8											
18	94°26.0	97°19.3	N07°31.6	340°12.2	N24°24.6	20°28.1	N22°17.8	99°18.6	S04°12.7	Sirius	258°26.1	-16°45.2											
19	109°28.5	112°23.0	30.7	355°13.9	24.4	35°30.2	17.9	114°20.7	12.6	Adhara	255°05.7	-29°00.6											
20	124°30.9	127°26.7	29.8	10°15.6	24.2	50°32.2	17.9	129°22.9	12.4	Procyon	244°50.6	5°09.5											
21	139°33.4	142°30.4	· · 28.9	25°17.3	· · 24.1	65°34.3	· · 18.0	144°25.1	· · 12.3	Pollux	243°17.0	27°58.0											
22	154°35.9	157°34.1	28.1	40°19.0	23.9	80°36.4	18.0	159°27.3	12.2	Avior	234°14.4	-59°35.7											
23	169°38.3	172°37.8	27.2	55°20.7	23.7	95°38.5	18.1	174°29.5	12.1	Suhail	222°45.9	-43°32.3											
Mer.pass. 11:43				ν 3.7' d -0.9' m -4.18				ν 1.7' d -0.2' m 0.32				ν 2.1' d 0.0' m -2.14				ν 2.2' d -0.1' m 1.15							

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	184°40.8	187°41.5	N07°26.3	70°22.3	N24°23.5	110°40.6	N22°18.1	189°31.7	S04°12.0										
1	199°43.3	202°45.2	25.4	85°24.0	23.3	125°42.6	18.2	204°33.8	11.9										
2	214°45.7	217°48.9	24.5	100°25.7	23.1	140°44.7	18.2	219°36.0	11.7										
3	229°48.2	232°52.6	· · 23.7	115°27.4	· · 22.9	155°46.8	· · 18.2	234°38.2	· · 11.6										
4	244°50.6	247°56.3	22.8	130°29.0	22.8	170°48.9	18.3	249°40.4	11.5										
5	259°53.1	263°00.0	21.9	145°30.7	22.6	185°50.9	18.3	264°42.6	11.4										
6	274°55.6	278°03.6	N07°21.0	160°32.4	N24°22.4	200°53.0	N22°18.4	279°44.8	S04°11.3										
7	289°58.0	293°07.3	20.1	175°34.1	22.2	215°55.1	18.4	294°46.9	11.1										
8	305°00.5	308°11.0	19.2	190°35.7	22.0	230°57.2	18.5	309°49.1	11.0										
9	320°03.0	323°14.7	· · 18.3	205°37.4	· · 21.8	245°59.2	· · 18.5	324°51.3	· · 10.9										
10	335°05.4	338°18.3	17.5	220°39.1	21.6	261°01.3	18.6	339°53.5	10.8										
11	350°07.9	353°22.0	16.6	235°40.8	21.4	276°03.4	18.6	354°55.7	10.7										
12	5°10.4	8°25.7	N07°15.7	250°42.4	N24°21.3	291°05.4	N22°18.7	9°57.9	S04°10.6										
13	20°12.8	23°29.3	14.8	265°44.1	21.1	306°07.5	18.7	25°00.0	10.4										
14	35°15.3	38°33.0	13.9	280°45.8	20.9	321°09.6	18.8	40°02.2	10.3										
15	50°17.7	53°36.6	· · 13.0	295°47.4	· · 20.7	336°11.7	· · 18.8	55°04.4	· · 10.2										
16	65°20.2	68°40.3	12.1	310°49.1	20.5	351°13.7	18.9	70°06.6	10.1										
17	80°22.7	83°44.0	11.2	325°50.8	20.3	6°15.8	18.9	85°08.8	10.0										
18	95°25.1	98°47.6	N07°10.4	340°52.4	N24°20.1	21°17.9	N22°18.9	100°11.0	S04°09.9										
19	110°27.6	113°51.3	09.5	355°54.1	19.9	36°20.0	19.0	115°13.1	09.7										
20	125°30.1	128°54.9	08.6	10°55.8	19.8	51°22.0	19.0	130°15.3	09.6										
21	140°32.5	143°58.6	· · 07.7	25°57.4	· · 19.6	66°24.1	· · 19.1	145°17.5	· · 09.5										
22	155°35.0	159°02.2	06.8	40°59.1	19.4	81°26.2	19.1	160°19.7	09.4										
23	170°37.5	174°05.8	05.9	56°00.8	19.2	96°28.2	19.2	175°21.9	09.3										
Mer.pass. 11:39				ν 3.7' d -0.9' m -4.13				ν 1.7' d -0.2' m 0.34				ν 2.1' d 0.0' m -2.13				ν 2.2' d -0.1' m 1.15			

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	185°39.9	189°09.5	N07°05.0	71°02.4	N24°19.0	111°30.3	N22°19.2	190°24.1	S04°09.2										
1	200°42.4	204°13.1	04.1	86°04.1	18.8	126°32.4	19.3	205°26.2	09.0										
2	215°44.9	219°16.7	03.2	101°05.7	18.6	141°34.5	19.3	220°28.4	08.9										
3	230°47.3	234°20.4	· · 02.3	116°07.4	· · 18.4	156°36.5	· · 19.4	235°30.6	· · 08.8										
4	245°49.8	249°24.0	01.4	131°09.1	18.2	171°38.6	19.4	250°32.8	08.7										
5	260°52.2	264°27.6	07°00.5	146°10.7	18.0	186°40.7	19.5	265°35.0	08.6										
6	275°54.7	279°31.2	N06°59.7	161°12.4	N24°17.9	201°42.7	N22°19.5	280°37.2	S04°08.5										
7	290°57.2	294°34.9	58.8	176°14.0	17.7	216°44.8	19.5	295°39.3	08.3										
8	305°59.6	309°38.5	57.9	191°15.7	17.5	231°46.9	19.6	310°41.5	08.2										
9	321°02.1	324°42.1	· · 57.0	206°17.3	· · 17.3	246°48.9	· · 19.6	325°43.7	· · 08.1										
10	336°04.6	339°45.7	56.1	221°19.0	17.1	261°51.0	19.7	340°45.9	08.0										
11	351°07.0	354°49.3	55.2	236°20.7	16.9	276°53.1	19.7	355°48.1	07.9										
12	6°09.5	9°52.9	N06°54.3	251°22.3	N24°16.7	291°55.1	N22°19.8	10°50.3	S04°07.7										
13	21°12.0	24°56.5	53.4	266°24.0	16.5	306°57.2	19.8	25°52.5	07.6										
14	36°14.4	40°00.1	52.5	281°25.6	16.3	321°59.3	19.9	40°54.6	07.5										
15	51°16.9	55°03.7	· · 51.6	296°27.3	· · 16.1	337°01.3	· · 19.9	55°56.8	· · 07.4										
16	66°19.3	70°07.3	50.7	311°28.9	15.9	352°03.4	20.0	70°59.0	07.3										
17	81°21.8	85°10.9	49.8	326°30.6	15.8	7°05.5	20.0	86°01.2	07.2										
18	96°24.3	100°14.5	N06°48.9	341°32.2	N24°15.6	22°07.5	N22°20.1	101°03.4	S04°07.0										
19	111°26.7	115°18.1	48.0	356°33.9	15.4	37°09.6	20.1	116°05.6	06.9										
20	126°29.2	130°21.7	47.1	11°35.5	15.2	52°11.7	20.2	131°07.7	06.8										
21	141°31.7	145°25.2	· · 46.2	26°37.2	· · 15.0	67°13.7	· · 20.2	146°09.9	· · 06.7										
22	156°34.1	160°28.8	45.3	41°38.8	14.8	82°15.8	20.2	161°12.1	06.6										
23	171°36.6	175°32.4	44.4	56°40.4	14.6	97°17.9	20.3	176°14.3	06.5										
Mer.pass. 11:35				ν 3.6' d -0.9' m -4.05				ν 1.7' d -0.2' m 0.36				ν 2.1' d 0.0' m -2.13				ν 2.2' d -0.1' m 1.16			

Mar 26 Wed	SHA	Mer.pass
Venus	2°30.7	11:32
Mars	246°00.2	19:19
Jupiter	286°09.0	16:38
Saturn	4°57.6	11:24

Mar 27 Thu	SHA	Mer.pass
Venus	3°00.7	11:26
Mars	245°41.5	19:16
Jupiter	285°59.8	16:35
Saturn	4°50.9	11:20

Mar 28 Fri	SHA	Mer.pass
Venus	3°29.5	11:21
Mars	245°22.5	19:14
Jupiter	285°50.4	16:32
Saturn	4°44.1	11:17

Horizontal parallax		
Venus:	0.5	
Mars:	0.1	

DUT1 = UT1-UTC = +0.0321 sec ΔT = TT-UT1 = +69.1519 sec

2025 March 26 to Mar. 28 UT

h Sun Moon						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	178°34.3	N02°13.0	221°39.9	9.5'	S18°26.3	-13.4' 58.7'
1	193°34.5	14.0	236°08.4	9.5'	18°13.0	-13.5' 58.8'
2	208°34.7	14.9	250°36.9	9.6'	17°59.5	-13.6' 58.8'
3	223°34.9	· · 15.9	265°05.5	9.6'	17°45.9	-13.7' 58.8'
4	238°35.1	16.9	279°34.1	9.7'	17°32.2	-13.8' 58.9'
5	253°35.3	17.9	294°02.8	9.7'	17°18.4	-13.9' 58.9'
6	268°35.5	N02°18.9	308°31.4	9.7'	S17°04.4	-14.1' 58.9'
7	283°35.6	19.9	323°00.2	9.8'	16°50.4	-14.2' 59.0'
8	298°35.8	20.8	337°28.9	9.8'	16°36.2	-14.3' 59.0'
9	313°36.0	· · 21.8	351°57.7	9.8'	16°21.9	-14.4' 59.1'
10	328°36.2	22.8	6°26.6	9.9'	16°07.5	-14.5' 59.1'
11	343°36.4	23.8	20°55.5	9.9'	15°53.0	-14.6' 59.1'
12	358°36.6	N02°24.8	35°24.4	9.9'	S15°38.4	-14.7' 59.2'
13	13°36.8	25.7	49°53.3	10.0'	15°23.7	-14.8' 59.2'
14	28°37.0	26.7	64°22.3	10.0'	15°08.9	-14.9' 59.3'
15	43°37.2	· · 27.7	78°51.3	10.0'	14°54.0	-15.0' 59.3'
16	58°37.3	28.7	93°20.3	10.1'	14°39.0	-15.1' 59.3'
17	73°37.5	29.7	107°49.4	10.1'	14°23.9	-15.2' 59.4'
18	88°37.7	N02°30.6	122°18.5	10.1'	S14°08.7	-15.3' 59.4'
19	103°37.9	31.6	136°47.7	10.2'	13°53.4	-15.4' 59.4'
20	118°38.1	32.6	151°16.8	10.2'	13°38.0	-15.5' 59.5'
21	133°38.3	· · 33.6	165°46.0	10.2'	13°22.5	-15.6' 59.5'
22	148°38.5	34.6	180°15.2	10.3'	13°06.9	-15.7' 59.5'
23	163°38.7	35.5	194°44.5	10.3'	12°51.2	-15.8' 59.6'
SD = 16.0' d = 1.0'			SD = 16.0'			
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	178°38.8	N02°36.5	209°13.8	10.3'	S12°35.5	-15.9' 59.6'
1	193°39.0	37.5	223°43.1	10.3'	12°19.6	-15.9' 59.7'
2	208°39.2	38.5	238°12.4	10.4'	12°03.7	-16.0' 59.7'
3	223°39.4	· · 39.4	252°41.8	10.4'	11°47.7	-16.1' 59.7'
4	238°39.6	40.4	267°11.2	10.4'	11°31.5	-16.2' 59.8'
5	253°39.8	41.4	281°40.6	10.4'	11°15.4	-16.3' 59.8'
6	268°40.0	N02°42.4	296°10.0	10.4'	S10°59.1	-16.3' 59.8'
7	283°40.2	43.4	310°39.4	10.5'	10°42.7	-16.4' 59.9'
8	298°40.3	44.3	325°08.9	10.5'	10°26.3	-16.5' 59.9'
9	313°40.5	· · 45.3	339°38.4	10.5'	10°09.8	-16.6' 59.9'
10	328°40.7	46.3	354°07.9	10.5'	09°53.2	-16.6' 60.0'
11	343°40.9	47.3	8°37.4	10.5'	09°36.6	-16.7' 60.0'
12	358°41.1	N02°48.3	23°07.0	10.6'	S09°19.9	-16.8' 60.0'
13	13°41.3	49.2	37°36.5	10.6'	09°03.1	-16.9' 60.1'
14	28°41.5	50.2	52°06.1	10.6'	08°46.2	-16.9' 60.1'
15	43°41.7	· · 51.2	66°35.7	10.6'	08°29.3	-17.0' 60.1'
16	58°41.9	52.2	81°05.3	10.6'	08°12.3	-17.0' 60.1'
17	73°42.0	53.1	95°34.9	10.6'	07°55.3	-17.1' 60.2'
18	88°42.2	N02°54.1	110°04.5	10.6'	S07°38.2	-17.2' 60.2'
19	103°42.4	55.1	124°34.1	10.6'	07°21.0	-17.2' 60.2'
20	118°42.6	56.1	139°03.8	10.6'	07°03.8	-17.3' 60.3'
21	133°42.8	· · 57.1	153°33.4	10.7'	06°46.5	-17.3' 60.3'
22	148°43.0	58.0	168°03.1	10.7'	06°29.1	-17.4' 60.3'
23	163°43.2	59.0	182°32.7	10.7'	06°11.8	-17.4' 60.4'
SD = 16.0' d = 1.0'			SD = 16.3'			
Fri	GHA	Dec	GHA	ν	Dec	d HP
0	178°43.4	N03°00.0	197°02.4	10.7'	S05°54.3	-17.5' 60.4'
1	193°43.5	01.0	211°32.1	10.7'	05°36.8	-17.5' 60.4'
2	208°43.7	01.9	226°01.7	10.7'	05°19.3	-17.6' 60.4'
3	223°43.9	· · 02.9	240°31.4	10.7'	05°01.7	-17.6' 60.5'
4	238°44.1	03.9	255°01.0	10.7'	04°44.1	-17.7' 60.5'
5	253°44.3	04.9	269°30.7	10.7'	04°26.4	-17.7' 60.5'
6	268°44.5	N03°05.8	284°00.4	10.7'	S04°08.7	-17.7' 60.5'
7	283°44.7	06.8	298°30.0	10.7'	03°51.0	-17.8' 60.6'
8	298°44.9	07.8	312°59.7	10.6'	03°33.2	-17.8' 60.6'
9	313°45.0	· · 08.8	327°29.3	10.6'	03°15.4	-17.8' 60.6'
10	328°45.2	09.8	341°59.0	10.6'	02°57.6	-17.9' 60.6'
11	343°45.4	10.7	356°28.6	10.6'	02°39.7	-17.9' 60.7'
12	358°45.6	N03°11.7	10°58.2	10.6'	S02°21.8	-17.9' 60.7'
13	13°45.8	12.7	25°27.8	10.6'	02°03.9	-18.0' 60.7'
14	28°46.0	13.7	39°57.4	10.6'	01°45.9	-18.0' 60.7'
15	43°46.2	· · 14.6	54°27.0	10.6'	01°27.9	-18.0' 60.7'
16	58°46.4	15.6	68°56.6	10.6'	01°09.9	-18.0' 60.8'
17	73°46.5	16.6	83°26.1	10.5'	00°51.9	-18.0' 60.8'
18	88°46.7	N03°17.6	97°55.7	10.5'	S00°33.9	-18.0' 60.8'
19	103°46.9	18.5	112°25.2	10.5'	S00°15.8	-18.1' 60.8'
20	118°47.1	19.5	126°54.7	10.5'	N00°02.2	18.1' 60.8'
21	133°47.3	· · 20.5	141°24.2	10.5'	00°20.3	18.1' 60.9'
22	148°47.5	21.5	155°53.6	10.4'	00°38.4	18.1' 60.9'
23	163°47.7	22.4	170°23.0	10.4'	00°56.5	18.1' 60.9'
SD = 16.0' d = 1.0'			SD = 16.5'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	02:27	04:10	05:21	18:52	20:04	21:51
N 70°	02:56	04:22	05:26	18:47	19:51	21:20
68°	03:17	04:32	05:30	18:43	19:41	20:58
66°	03:33	04:40	05:33	18:40	19:33	20:41
64°	03:46	04:47	05:36	18:37	19:26	20:27
62°	03:57	04:53	05:38	18:34	19:20	20:16
60°	04:06	04:58	05:40	18:32	19:14	20:07
N 58°	04:14	05:02	05:42	18:30	19:10	19:59
56°	04:21	05:06	05:43	18:28	19:06	19:52
54°	04:26	05:09	05:45	18:27	19:03	19:46
52°	04:32	05:12	05:46	18:26	19:00	19:40
50°	04:36	05:15	05:47	18:24	18:57	19:36
45°	04:46	05:20	05:50	18:22	18:51	19:26
N 40°	04:53	05:25	05:52	18:19	18:46	19:19
35°	04:59	05:28	05:54	18:17	18:43	19:13
30°	05:03	05:31	05:55	18:16	18:40	19:08
20°	05:10	05:36	05:58	18:13	18:35	19:01
N 10°	05:15	05:39	06:00	18:11	18:32	18:56
0°	05:17	05:41	06:02	18:09	18:29	18:53
S 10°	05:18	05:43	06:04	18:07	18:28	18:52
20°	05:18	05:44	06:06	18:04	18:26	18:52
30°	05:16	05:44	06:08	18:02	18:26	18:54
35°	05:14	05:44	06:09	18:01	18:26	18:56
40°	05:12	05:43	06:10	18:00	18:27	18:58
45°	05:08	05:42	06:11	17:58	18:27	19:02
S 50°	05:03	05:41	06:13	17:57	18:29	19:06
52°	05:01	05:40	06:14	17:56	18:29	19:08
54°	04:58	05:39	06:15	17:55	18:30	19:11
56°	04:55	05:38	06:15	17:54	18:31	19:14
58°	04:51	05:37	06:16	17:53	18:32	19:18
S 60°	04:47	05:36	06:17	17:52	18:33	19:22

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	08:01	06:43	05:58	11:32	14:40	17:15
N 70°	07:15	06:26	05:52	12:16	14:54	17:16
68°	06:45	06:12	05:47	12:45	15:06	17:17
66°	06:22	06:00	05:43	13:06	15:15	17:18
64°	06:04	05:51	05:40	13:23	15:22	17:19
62°	05:49	05:43	05:37	13:37	15:29	17:20
60°	05:36	05:35	05:34	13:48	15:35	17:20
N 58°	05:25	05:29	05:32	13:58	15:40	17:21
56°	05:15	05:23	05:30	14:07	15:44	17:21
54°	05:07	05:18	05:28	14:15	15:48	17:22
52°	04:59	05:14	05:26	14:22	15:52	17:22
50°	04:52	05:09	05:24	14:28	15:55	17:22
45°	04:37	05:00	05:21	14:41	16:02	17:23
N 40°	04:25	04:52	05:18	14:52	16:07	17:24
35°	04:14	04:46	05:16	15:01	16:12	17:24
30°	04:05	04:40	05:13	15:09	16:17	17:25
20°	03:48	04:30	05:09	15:23	16:24	17:25
N 10°	03:34	04:21	05:06	15:34	16:30	17:26
0°	03:21	04:12	05:03	15:46	16:36	17:26
S 10°	03:08	04:04	04:59	15:56	16:42	17:27
20°	02:53	03:55	04:56	16:08	16:48	17:28
30°	02:37	03:44	04:52	16:21	16:55	17:28
35°	02:27	03:38	04:50	16:29	16:59	17:28
40°	02:16	03:31	04:47	16:37	17:04	17:29
45°	02:03	03:23	04:44	16:47	17:09	17:29
S 50°	01:47	03:13	04:41	16:59	17:15	17:30
52°	01:39	03:09	04:39	17:04	17:18	17:30
54°	01:31	03:04	04:37	17:10	17:21	17:30
56°	01:21	02:58	04:35	17:17	17:24	17:30
58°	01:10	02:52	04:33	17:25	17:28	17:31
S 60°	00:58	02:45	04:30	17:33	17:32	17:31

Day	Sun		Mer. Pass hh:mm	Moon		Age 26-28 16-3%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
26	05:43	05:34	12:06	09:33	21:59	
27	05:25	05:16	12:05	10:24	22:49	
28	05:07	04:58	12:05	11:15	23:40	

March 29, 30, 31 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	186°39.1	190°36.0	N06°43.5	71°42.1	N24°14.4	112°19.9	N22°20.3	191°16.5	S04°06.3	Alpheratz	357°35.0	29°13.7
1	201°41.5	205°39.5	42.6	86°43.7	14.2	127°22.0	20.4	206°18.7	06.2	Ankaa	353°07.4	-42°10.2
2	216°44.0	220°43.1	41.7	101°45.4	14.0	142°24.1	20.4	221°20.9	06.1	Schedar	349°31.5	56°40.5
3	231°46.5	235°46.7	· · 40.8	116°47.0	· · 13.8	157°26.1	· · 20.5	236°23.0	· · 06.0	Diphda	348°47.4	-17°51.0
4	246°48.9	250°50.2	39.9	131°48.7	13.6	172°28.2	20.5	251°25.2	05.9	Achernar	335°20.6	-57°06.6
5	261°51.4	265°53.8	39.0	146°50.3	13.4	187°30.3	20.6	266°27.4	05.8	Hamal	327°51.3	23°34.9
6	276°53.8	280°57.4	N06°38.1	161°51.9	N24°13.2	202°32.3	N22°20.6	281°29.6	S04°05.6	Polaris	314°21.1	89°22.4
7	291°56.3	296°00.9	37.2	176°53.6	13.0	217°34.4	20.7	296°31.8	05.5	Acamar	315°12.0	-40°12.4
8	306°58.8	311°04.5	36.3	191°55.2	12.9	232°36.4	20.7	311°34.0	05.4	Menkar	314°06.2	4°11.2
9	322°01.2	326°08.0	· · 35.4	206°56.9	· · 12.7	247°38.5	· · 20.8	326°36.1	· · 05.3	Mirfak	308°28.4	49°57.1
10	337°03.7	341°11.6	34.5	221°58.5	12.5	262°40.6	20.8	341°38.3	05.2	Aldebaran	290°39.6	16°33.6
11	352°06.2	356°15.1	33.6	237°00.1	12.3	277°42.6	20.9	356°40.5	05.1	Rigel	281°03.8	-8°10.5
12	7°08.6	11°18.6	N06°32.8	252°01.8	N24°12.1	292°44.7	N22°20.9	11°42.7	S04°04.9	Capella	280°21.8	46°01.5
13	22°11.1	26°22.2	31.9	267°03.4	11.9	307°46.8	20.9	26°44.9	04.8	Bellatrix	278°22.8	6°22.3
14	37°13.6	41°25.7	31.0	282°05.1	11.7	322°48.8	21.0	41°47.1	04.7	Elnath	278°01.8	28°37.8
15	52°16.0	56°29.2	· · 30.1	297°06.7	· · 11.5	337°50.9	· · 21.0	56°49.3	· · 04.6	Alnilam	275°37.6	-1°11.2
16	67°18.5	71°32.8	29.2	312°08.3	11.3	352°52.9	21.1	71°51.4	04.5	Betelgeuse	270°52.0	7°24.6
17	82°21.0	86°36.3	28.3	327°10.0	11.1	7°55.0	21.1	86°53.6	04.4	Canopus	263°52.4	-52°42.8
18	97°23.4	101°39.8	N06°27.4	342°11.6	N24°10.9	22°57.1	N22°21.2	101°55.8	S04°04.2	Sirius	258°26.1	-16°45.2
19	112°25.9	116°43.3	26.5	357°13.2	10.7	37°59.1	21.2	116°58.0	04.1	Adhara	255°05.7	-29°00.6
20	127°28.3	131°46.9	25.6	12°14.8	10.5	53°01.2	21.3	132°00.2	04.0	Procyon	244°50.6	5°09.5
21	142°30.8	146°50.4	· · 24.7	27°16.5	· · 10.3	68°03.2	· · 21.3	147°02.4	· · 03.9	Pollux	243°17.0	27°58.0
22	157°33.3	161°53.9	23.8	42°18.1	10.1	83°05.3	21.4	162°04.5	03.8	Avior	234°14.4	-59°35.7
23	172°35.7	176°57.4	22.9	57°19.7	09.9	98°07.4	21.4	177°06.7	03.7	Suhail	222°46.0	-43°32.3
Mer.pass. 11:32		ν 3.6' d -0.9' m -4.06		ν 1.6' d -0.2' m 0.38		ν 2.1' d 0.0' m -2.12		ν 2.2' d -0.1' m 1.16		Miaplacidus	221°37.8	-69°49.5
Alphard217°47.4-8°46.2												
Regulus207°34.011°50.6												
Dubhe193°40.161°36.9												
Denebola182°24.514°25.7												
Gienah175°43.1-17°41.1												
Acrux172°59.3-63°14.4												
Gacrux171°50.9-57°15.4												
Alioth166°12.255°49.3												
Spica158°21.8-11°17.7												
Alkaid152°51.449°11.0												
Hadar148°35.2-60°29.7												
Menkent147°57.1-36°29.7												
Arcturus145°47.419°02.9												
Rigil Kent.139°39.5-60°56.3												
Kochab137°18.874°02.8												
Zuben'ubi136°55.6-16°08.9												
Alphecca126°03.326°37.5												
Antares112°15.4-26°29.3												
Atria107°09.3-69°04.2												
Sabik102°02.5-15°45.4												
Shaula96°10.0-37°07.3												
Rasalhague95°58.312°32.3												
Eltanin90°42.051°28.7												
Kaus Aust.83°32.3-34°22.3												
Vega80°33.138°48.0												
Nunki75°47.6-26°15.9												
Altair61°59.98°55.9												
Peacock53°05.7-56°39.1												
Deneb49°25.945°21.9												
Enif33°38.99°59.3												
Al Na'ir27°33.0-46°50.3												
Fomalhaut15°14.6-29°29.4												
Scheat13°45.428°13.0												
Markab13°30.015°20.3												
Mar 29 Sat			SHA	Mer.pass								
Venus			3°56.9	11:15								
Mars			245°03.0	19:11								
Jupiter			285°40.9	16:28								
Saturn			4°37.4	11:13								
Mar 30 Sun			SHA	Mer.pass								
Venus			4°22.7	11:09								
Mars			244°43.2	19:09								
Jupiter			285°31.2	16:25								
Saturn			4°30.7	11:10								
Mar 31 Mon			SHA	Mer.pass								
Venus			4°46.7	11:04								
Mars			244°22.9	19:06								
Jupiter			285°21.4	16:22								
Saturn			4°24.0	11:06								
Horizontal parallax												
Venus: 0.5												
Mars: 0.1												
Mer.pass. 11:24		ν 3.4' d -0.9' m -4.18		ν 1.6' d -0.2' m 0.42		ν 2.1' d 0.0' m -2.11		ν 2.2' d -0.1' m 1.16				

DUT1 = UT1-UTC = +0.0288 sec ΔT = TT-UT1 = +69.1552 sec

2025 March 29 to Mar. 31 UT

h Sun				Moon			
Sat	GHA	Dec		GHA	ν	Dec	d HP
0	178°47.9	N03°23.4		184°52.5	10.4'	N01°14.6	18.1' 60.9'
1	193°48.0	24.4		199°21.8	10.4'	01°32.7	18.1' 60.9'
2	208°48.2	25.4		213°51.2	10.3'	01°50.8	18.1' 60.9'
3	223°48.4	· · 26.3		228°20.5	10.3'	02°08.9	18.1' 61.0'
4	238°48.6	27.3		242°49.8	10.3'	02°26.9	18.1' 61.0'
5	253°48.8	28.3		257°19.1	10.2'	02°45.0	18.1' 61.0'
6	268°49.0	N03°29.3		271°48.3	10.2'	N03°03.1	18.1' 61.0'
7	283°49.2	30.2		286°17.5	10.2'	03°21.2	18.1' 61.0'
8	298°49.4	31.2		300°46.7	10.1'	03°39.2	18.0' 61.0'
9	313°49.5	· · 32.2		315°15.8	10.1'	03°57.3	18.0' 61.0'
10	328°49.7	33.1		329°44.9	10.1'	04°15.3	18.0' 61.0'
11	343°49.9	34.1		344°14.0	10.0'	04°33.3	18.0' 61.1'
12	358°50.1	N03°35.1		358°43.0	10.0'	N04°51.3	18.0' 61.1'
13	13°50.3	36.1		13°12.0	9.9'	05°09.3	17.9' 61.1'
14	28°50.5	37.0		27°40.9	9.9'	05°27.2	17.9' 61.1'
15	43°50.7	· · 38.0		42°09.8	9.8'	05°45.2	17.9' 61.1'
16	58°50.8	39.0		56°38.6	9.8'	06°03.1	17.9' 61.1'
17	73°51.0	40.0		71°07.4	9.8'	06°20.9	17.8' 61.1'
18	88°51.2	N03°40.9		85°36.2	9.7'	N06°38.8	17.8' 61.1'
19	103°51.4	41.9		100°04.9	9.7'	06°56.6	17.8' 61.1'
20	118°51.6	42.9		114°33.6	9.6'	07°14.3	17.7' 61.1'
21	133°51.8	· · 43.8		129°02.2	9.6'	07°32.0	17.7' 61.1'
22	148°52.0	44.8		143°30.7	9.5'	07°49.7	17.6' 61.1'
23	163°52.2	45.8		157°59.2	9.5'	08°07.3	17.6' 61.1'
SD = 16.0'			$d = 1.0'$	SD = 16.6'			

Sun	GHA	Dec		GHA	ν	Dec	d HP
0	178°52.3	N03°46.8		172°27.7	9.4'	N08°24.9	17.5' 61.1'
1	193°52.5	47.7		186°56.1	9.3'	08°42.5	17.5' 61.1'
2	208°52.7	48.7		201°24.4	9.3'	09°00.0	17.4' 61.1'
3	223°52.9	· · 49.7		215°52.7	9.2'	09°17.4	17.4' 61.1'
4	238°53.1	50.6		230°20.9	9.2'	09°34.8	17.3' 61.1'
5	253°53.3	51.6		244°49.1	9.1'	09°52.1	17.3' 61.1'
6	268°53.5	N03°52.6		259°17.2	9.1'	N10°09.4	17.2' 61.1'
7	283°53.6	53.6		273°45.3	9.0'	10°26.6	17.2' 61.1'
8	298°53.8	54.5		288°13.3	8.9'	10°43.8	17.1' 61.1'
9	313°54.0	· · 55.5		302°41.2	8.9'	11°00.9	17.0' 61.1'
10	328°54.2	56.5		317°09.1	8.8'	11°17.9	16.9' 61.1'
11	343°54.4	57.4		331°36.9	8.7'	11°34.8	16.9' 61.1'
12	358°54.6	N03°58.4		346°04.6	8.7'	N11°51.7	16.8' 61.1'
13	13°54.8	03°59.4		0°32.3	8.6'	12°08.5	16.7' 61.1'
14	28°55.0	04°00.4		14°59.9	8.5'	12°25.2	16.7' 61.1'
15	43°55.1	· · 01.3		29°27.4	8.5'	12°41.9	16.6' 61.1'
16	58°55.3	02.3		43°54.9	8.4'	12°58.5	16.5' 61.1'
17	73°55.5	03.3		58°22.3	8.3'	13°15.0	16.4' 61.1'
18	88°55.7	N04°04.2		72°49.6	8.3'	N13°31.4	16.3' 61.1'
19	103°55.9	05.2		87°16.8	8.2'	13°47.7	16.2' 61.1'
20	118°56.1	06.2		101°44.0	8.1'	14°03.9	16.1' 61.1'
21	133°56.3	· · 07.1		116°11.1	8.0'	14°20.1	16.0' 61.1'
22	148°56.4	08.1		130°38.2	8.0'	14°36.1	16.0' 61.1'
23	163°56.6	09.1		145°05.1	7.9'	14°52.1	15.9' 61.1'
SD = 16.0'			$d = 1.0'$	SD = 16.7'			

Mon	GHA	Dec		GHA	ν	Dec	d HP
0	178°56.8	N04°10.0		159°32.0	7.8'	N15°07.9	15.8' 61.1'
1	193°57.0	11.0		173°58.8	7.7'	15°23.7	15.7' 61.0'
2	208°57.2	12.0		188°25.6	7.7'	15°39.3	15.6' 61.0'
3	223°57.4	· · 13.0		202°52.2	7.6'	15°54.9	15.4' 61.0'
4	238°57.6	13.9		217°18.8	7.5'	16°10.3	15.3' 61.0'
5	253°57.7	14.9		231°45.3	7.4'	16°25.7	15.2' 61.0'
6	268°57.9	N04°15.9		246°11.8	7.4'	N16°40.9	15.1' 61.0'
7	283°58.1	16.8		260°38.1	7.3'	16°56.0	15.0' 61.0'
8	298°58.3	17.8		275°04.4	7.2'	17°11.0	14.9' 61.0'
9	313°58.5	· · 18.8		289°30.6	7.1'	17°25.9	14.8' 60.9'
10	328°58.7	19.7		303°56.7	7.0'	17°40.7	14.7' 60.9'
11	343°58.9	20.7		318°22.7	7.0'	17°55.3	14.5' 60.9'
12	358°59.0	N04°21.7		332°48.7	6.9'	N18°09.8	14.4' 60.9'
13	13°59.2	22.6		347°14.6	6.8'	18°24.3	14.3' 60.9'
14	28°59.4	23.6		1°40.4	6.7'	18°38.5	14.2' 60.9'
15	43°59.6	· · 24.6		16°06.1	6.6'	18°52.7	14.0' 60.8'
16	58°59.8	25.5		30°31.7	6.6'	19°06.7	13.9' 60.8'
17	74°00.0	26.5		44°57.3	6.5'	19°20.6	13.8' 60.8'
18	89°00.2	N04°27.5		59°22.8	6.4'	N19°34.4	13.6' 60.8'
19	104°00.3	28.4		73°48.2	6.3'	19°48.0	13.5' 60.8'
20	119°00.5	29.4		88°13.5	6.2'	20°01.5	13.3' 60.7'
21	134°00.7	· · 30.4		102°38.7	6.2'	20°14.8	13.2' 60.7'
22	149°00.9	31.3		117°03.9	6.1'	20°28.0	13.1' 60.7'
23	164°01.1	32.3		131°29.0	6.0'	20°41.1	12.9' 60.7'
SD = 16.0'			$d = 1.0'$	SD = 16.7'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	01:58	03:52	05:05	19:06	20:20	22:19
N 70°	02:35	04:07	05:12	19:00	20:05	21:40
68°	03:00	04:18	05:17	18:54	19:53	21:13
66°	03:18	04:28	05:21	18:50	19:43	20:54
64°	03:33	04:36	05:25	18:46	19:35	20:38
62°	03:45	04:43	05:28	18:42	19:28	20:26
60°	03:56	04:48	05:31	18:39	19:22	20:15
N 58°	04:04	04:53	05:33	18:37	19:17	20:06
56°	04:12	04:58	05:35	18:35	19:13	19:59
54°	04:18	05:02	05:37	18:33	19:09	19:52
52°	04:24	05:05	05:39	18:31	19:05	19:46
50°	04:29	05:08	05:41	18:29	19:02	19:41
45°	04:40	05:15	05:44	18:25	18:55	19:30
N 40°	04:48	05:20	05:47	18:22	18:50	19:22
35°	04:54	05:24	05:49	18:20	18:45	19:15
30°	05:00	05:28	05:52	18:18	18:42	19:10
20°	05:07	05:33	05:55	18:14	18:36	19:02
N 10°	05:13	05:37	05:58	18:11	18:32	18:56
0°	05:16	05:40	06:01	18:08	18:28	18:52
S 10°	05:18	05:43	06:04	18:05	18:26	18:50
20°	05:19	05:45	06:07	18:02	18:24	18:49
30°	05:18	05:46	06:10	17:59	18:23	18:50
35°	05:17	05:46	06:11	17:57	18:22	18:51
40°	05:15	05:46	06:13	17:55	18:22	18:53
45°	05:12	05:46	06:15	17:53	18:22	18:56
S 50°	05:08	05:46	06:18	17:50	18:22	19:00
52°	05:06	05:45	06:19	17:49	18:22	19:01
54°	05:04	05:45	06:20	17:48	18:23	19:04
56°	05:01	05:44	06:22	17:46	18:23	19:06
58°	04:58	05:44	06:23	17:44	18:23	19:09
S 60°	04:55	05:43	06:25	17:43	18:24	19:12

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	05:18	04:35	03:26	19:50	22:54	
N 70°	05:22	04:49	04:03	19:39	22:19	
68°	05:25	05:01	04:29	19:31	21:55	· · ·
66°	05:27	05:10	04:50	19:23	21:36	· · ·
64°	05:29	05:19	05:06	19:17	21:22	23:38
62°	05:31	05:26	05:20	19:12	21:09	23:14
60°	05:33	05:32	05:32	19:07	20:59	22:55
N 58°	05:34	05:38	05:42	19:04	20:50	22:40
56°	05:36	05:42	05:51	19:00	20:42	22:26
54°	05:37	05:47	05:59	18:57	20:35	22:15
52°	05:38	05:51	06:06	18:54	20:28	22:05
50°	05:39	05:55	06:13	18:51	20:23	21:56
45°	05:41	06:03	06:27	18:46	20:11	21:37
N 40°	05:43	06:09	06:39	18:41	20:01	21:22
35°	05:45	06:15	06:49	18:37	19:52	21:09
30°	05:46	06:20	06:58	18:34	19:45	20:58
20°	05:49	06:29	07:13	18:28	19:32	20:39
N 10°	05:51	06:38	07:27	18:22	19:21	20:22
0°	05:53	06:45	07:40	18:17	19:11	20:07
S 10°	05:55	06:53	07:53	18:13	19:00	19:51
20°	05:58	07:01	08:07	18:07	18:49	19:35
30°	06:01	07:11	08:23	18:01	18:37	19:16
35°	06:02	07:16	08:32	17:58	18:30	19:06
40°	06:04	07:23	08:43	17:54	18:22	18:53
45°	06:06	07:30	08:56	17:50	18:12	18:39
S 50°	06:09	07:39	09:12	17:44	18:01	18:21
52°	06:10	07:43	09:19	17:42	17:56	18:13
54°	06:11	07:48	09:28	17:39	17:50	18:04
56°	06:13	07:53	09:37	17:36	17:44	17:54
58°	06:15	07:59	09:48	17:33	17:37	17:42
S 60°	06:16	08:06	10:00	17:30	17:29	17:29

Day	Sun		Mer. Pass hh:mm	Moon		Age 0-2 0-4%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
29	04:49	04:40	12:05	12:05	—:—	
30	04:31	04:22	12:04	12:58	00:31	
31	04:13	04:04	12:04	13:53	01:25	

April 01, 02, 03 UT (Tue., Wed., Thu.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	189°36.5	194°45.4	N05°39.1	73°38.7	N24°00.1	114°48.0	N22°23.7	193°53.8	S03°58.0					Alpheratz	357°35.0	29°13.7	
1	204°38.9	209°48.8	38.2	88°40.3	23°59.9	129°50.1	23.7	208°56.0	57.9					Ankaa	353°07.4	-42°10.2	
2	219°41.4	224°52.1	37.4	103°41.9	59.7	144°52.1	23.8	223°58.2	57.8					Schedar	349°31.5	56°40.5	
3	234°43.9	239°55.4	•• 36.5	118°43.5	•• 59.5	159°54.1	•• 23.8	239°00.4	•• 57.6					Diphda	348°47.4	-17°51.0	
4	249°46.3	254°58.8	35.6	133°45.1	59.3	174°56.2	23.9	254°02.6	57.5					Achernar	335°20.6	-57°06.6	
5	264°48.8	270°02.1	34.7	148°46.7	59.1	189°58.2	23.9	269°04.8	57.4					Hamal	327°51.3	23°34.8	
6	279°51.3	285°05.4	N05°33.8	163°48.3	N23°58.8	205°00.3	N22°24.0	284°07.0	S03°57.3					Polaris	314°21.6	89°22.4	
7	294°53.7	300°08.8	32.9	178°49.9	58.6	220°02.3	24.0	299°09.1	57.2					Acamar	315°12.0	-40°12.4	
8	309°56.2	315°12.1	32.1	193°51.5	58.4	235°04.4	24.1	314°11.3	57.1					Menkar	314°06.2	4°11.2	
9	324°58.7	330°15.4	•• 31.2	208°53.1	•• 58.2	250°06.4	•• 24.1	329°13.5	•• 56.9					Mirfak	308°28.4	49°57.1	
10	340°01.1	345°18.7	30.3	223°54.7	58.0	265°08.5	24.2	344°15.7	56.8					Aldebaran	290°39.6	16°33.6	
11	355°03.6	0°22.0	29.4	238°56.3	57.8	280°10.5	24.2	359°17.9	56.7					Rigel	281°03.8	-8°10.5	
12	10°06.0	15°25.3	N05°28.5	253°57.8	N23°57.6	295°12.6	N22°24.3	14°20.1	S03°56.6					Capella	280°21.8	46°01.5	
13	25°08.5	30°28.6	27.7	268°59.4	57.4	310°14.6	24.3	29°22.3	56.5					Bellatrix	278°22.8	6°22.3	
14	40°11.0	45°32.0	26.8	284°01.0	57.2	325°16.7	24.4	44°24.5	56.4					Elnath	278°01.8	28°37.8	
15	55°13.4	60°35.2	•• 25.9	299°02.6	•• 57.0	340°18.7	•• 24.4	59°26.6	•• 56.2					Alnilam	275°37.7	-1°11.2	
16	70°15.9	75°38.5	25.0	314°04.2	56.8	355°20.7	24.4	74°28.8	56.1					Betelgeuse	270°52.0	7°24.6	
17	85°18.4	90°41.8	24.2	329°05.8	56.6	10°22.8	24.5	89°31.0	56.0					Canopus	263°52.4	-52°42.8	
18	100°20.8	105°45.1	N05°23.3	344°07.4	N23°56.4	25°24.8	N22°24.5	104°33.2	S03°55.9					Sirius	258°26.1	-16°45.2	
19	115°23.3	120°48.4	22.4	359°09.0	56.2	40°26.9	24.6	119°35.4	55.8					Adhara	255°05.7	-29°00.6	
20	130°25.8	135°51.7	21.5	14°10.5	55.9	55°28.9	24.6	134°37.6	55.7					Procyon	244°50.6	5°09.5	
21	145°28.2	150°55.0	•• 20.7	29°12.1	•• 55.7	70°31.0	•• 24.7	149°39.8	•• 55.6					Pollux	243°17.0	27°58.0	
22	160°30.7	165°58.2	19.8	44°13.7	55.5	85°33.0	24.7	164°42.0	55.4					Avior	234°14.5	-59°35.7	
23	175°33.1	181°01.5	18.9	59°15.3	55.3	100°35.1	24.8	179°44.1	55.3					Suhail	222°46.0	-43°32.3	
Mer.pass. 11:20			ν 3.3' d -0.9' m -4.24			ν 1.6' d -0.2' m 0.44			ν 2.0' d 0.0' m -2.11			ν 2.2' d -0.1' m 1.16					
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	190°35.6	196°04.8	N05°18.1	74°16.9	N23°55.1	115°37.1	N22°24.8	194°46.3	S03°55.2					Denebola	182°24.5	14°25.7	
1	205°38.1	211°08.0	17.2	89°18.4	54.9	130°39.1	24.9	209°48.5	55.1					Gienah	175°43.1	-17°41.1	
2	220°40.5	226°11.3	16.3	104°20.0	54.7	145°41.2	24.9	224°50.7	55.0					Acrux	172°59.3	-63°14.4	
3	235°43.0	241°14.6	•• 15.5	119°21.6	•• 54.5	160°43.2	•• 25.0	239°52.9	•• 54.9					Gacrux	171°50.9	-57°15.4	
4	250°45.5	256°17.8	14.6	134°23.2	54.3	175°45.3	25.0	254°55.1	54.8					Alioth	166°12.2	55°49.3	
5	265°47.9	271°21.1	13.7	149°24.8	54.1	190°47.3	25.1	269°57.3	54.6					Spica	158°21.8	-11°17.7	
6	280°50.4	286°24.3	N05°12.8	164°26.3	N23°53.9	205°49.4	N22°25.1	284°59.5	S03°54.5					Alkaid	152°51.4	49°11.1	
7	295°52.9	301°27.5	12.0	179°27.9	53.7	220°51.4	25.1	300°01.6	54.4					Hadar	148°35.2	-60°29.7	
8	310°55.3	316°30.8	11.1	194°29.5	53.4	235°53.4	25.2	315°03.8	54.3					Menkent	147°57.1	-36°29.7	
9	325°57.8	331°34.0	•• 10.3	209°31.1	•• 53.2	250°55.5	•• 25.2	330°06.0	•• 54.2					Arcturus	145°47.4	19°02.9	
10	341°00.3	346°37.2	09.4	224°32.6	53.0	265°57.5	25.3	345°08.2	54.1					Rigil Kent.	139°39.5	-60°56.3	
11	356°02.7	1°40.5	08.5	239°34.2	52.8	280°59.6	25.3	0°10.4	53.9					Kochab	137°18.7	74°02.8	
12	11°05.2	16°43.7	N05°07.7	254°35.8	N23°52.6	296°01.6	N22°25.4	15°12.6	S03°53.8					Zuben'ubi	136°55.5	-16°08.9	
13	26°07.6	31°46.9	06.8	269°37.4	52.4	311°03.6	25.4	30°14.8	53.7					Alphecca	126°03.3	26°37.5	
14	41°10.1	46°50.1	05.9	284°38.9	52.2	326°05.7	25.5	45°17.0	53.6					Antares	112°15.4	-26°29.3	
15	56°12.6	61°53.3	•• 05.1	299°40.5	•• 52.0	341°07.7	•• 25.5	60°19.1	•• 53.5					Atria	107°09.3	-69°04.2	
16	71°15.0	76°56.6	04.2	314°42.1	51.8	356°09.8	25.6	75°21.3	53.4					Sabik	102°02.4	-15°45.4	
17	86°17.5	91°59.8	03.4	329°43.7	51.5	11°11.8	25.6	90°23.5	53.3					Shaula	96°10.0	-37°07.3	
18	101°20.0	107°03.0	N05°02.5	344°45.2	N23°51.3	26°13.8	N22°25.7	105°25.7	S03°53.1					Rasalhague	95°58.3	12°32.3	
19	116°22.4	122°06.2	01.7	359°46.8	51.1	41°15.9	25.7	120°27.9	53.0					Eltanin	90°42.0	51°28.7	
20	131°24.9	137°09.4	05°00.8	14°48.4	50.9	56°17.9	25.8	135°30.1	52.9					Kaus Aust.	83°32.2	-34°22.3	
21	146°27.4	152°12.5	04°59.9	29°49.9	•• 50.7	71°19.9	•• 25.8	150°32.3	•• 52.8					Vega	80°33.1	38°48.1	
22	161°29.8	167°15.7	59.1	44°51.5	50.5	86°22.0	25.8	165°34.5	52.7					Nunki	75°47.5	-26°15.9	
23	176°32.3	182°18.9	58.2	59°53.1	50.3	101°24.0	25.9	180°36.7	52.6					Altair	61°59.9	8°55.9	
Mer.pass. 11:16			ν 3.3' d -0.9' m -4.30			ν 1.6' d -0.2' m 0.46			ν 2.0' d 0.0' m -2.10			ν 2.2' d -0.1' m 1.16					
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	191°34.8	197°22.1	N04°57.4	74°54.6	N23°50.1	116°26.1	N22°25.9	195°38.8	S03°52.5								
1	206°37.2	212°25.3	56.5	89°56.2	49.8	131°28.1	26.0	210°41.0	52.3								
2	221°39.7	227°28.4	55.7	104°57.8	49.6	146°30.1	26.0	225°43.2	52.2								
3	236°42.1	242°31.6	•• 54.8	119°59.3	•• 49.4	161°32.2	•• 26.1	240°45.4	•• 52.1								
4	251°44.6	257°34.8	54.0	135°00.9	49.2	176°34.2	26.1	255°47.6	52.0								
5	266°47.1	272°37.9	53.1	150°02.5	49.0	191°36.2	26.2	270°49.8	51.9								
6	281°49.5	287°41.1	N04°52.3	165°04.0	N23°48.8	206°38.3	N22°26.2	285°52.0	S03°51.8								
7	296°52.0	302°44.2	51.4	180°05.6	48.6	221°40.3	26.3	300°54.2	51.7								
8	311°54.5	317°47.4	50.6	195°07.1	48.4	236°42.4	26.3	315°56.4	51.5								
9	326°56.9	332°50.5	•• 49.7	210°08.7	•• 48.1	251°44.4	•• 26.4	330°58.5	•• 51.4								
10	341°59.4	347°53.7	48.9	225°10.3													

Sun			Moon				
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	179°01.3	N04°33.3	145°54.0	5.9'	N20°54.0	12.8'	60.7'
1	194°01.4	34.2	160°18.9	5.9'	21°06.8	12.6'	60.6'
2	209°01.6	35.2	174°43.8	5.8'	21°19.4	12.5'	60.6'
3	224°01.8	· · 36.1	189°08.6	5.7'	21°31.9	12.3'	60.6'
4	239°02.0	37.1	203°33.3	5.6'	21°44.2	12.2'	60.6'
5	254°02.2	38.1	217°57.9	5.5'	21°56.4	12.0'	60.6'
6	269°02.4	N04°39.0	232°22.4	5.5'	N22°08.4	11.9'	60.5'
7	284°02.6	40.0	246°46.9	5.4'	22°20.2	11.7'	60.5'
8	299°02.7	41.0	261°11.3	5.3'	22°31.9	11.5'	60.5'
9	314°02.9	· · 41.9	275°35.7	5.3'	22°43.5	11.4'	60.5'
10	329°03.1	42.9	289°59.9	5.2'	22°54.8	11.2'	60.4'
11	344°03.3	43.9	304°24.1	5.1'	23°06.1	11.0'	60.4'
12	359°03.5	N04°44.8	318°48.2	5.0'	N23°17.1	10.9'	60.4'
13	14°03.7	45.8	333°12.3	5.0'	23°28.0	10.7'	60.4'
14	29°03.8	46.8	347°36.2	4.9'	23°38.7	10.5'	60.3'
15	44°04.0	· · 47.7	2°00.1	4.8'	23°49.2	10.4'	60.3'
16	59°04.2	48.7	16°24.0	4.8'	23°59.6	10.2'	60.3'
17	74°04.4	49.6	30°47.8	4.7'	24°09.8	10.0'	60.2'
18	89°04.6	N04°50.6	45°11.5	4.7'	N24°19.9	9.9'	60.2'
19	104°04.8	51.6	59°35.1	4.6'	24°29.7	9.7'	60.2'
20	119°05.0	52.5	73°58.7	4.5'	24°39.4	9.5'	60.2'
21	134°05.1	· · 53.5	88°22.2	4.5'	24°48.9	9.3'	60.1'
22	149°05.3	54.5	102°45.7	4.4'	24°58.2	9.1'	60.1'
23	164°05.5	55.4	117°09.1	4.4'	25°07.4	9.0'	60.1'
SD = 16.0'			$d = 1.0'$		SD = 16.5'		
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	179°05.7	N04°56.4	131°32.5	4.3'	N25°16.4	8.8'	60.0'
1	194°05.9	57.3	145°55.8	4.3'	25°25.1	8.6'	60.0'
2	209°06.1	58.3	160°19.0	4.2'	25°33.7	8.4'	60.0'
3	224°06.2	04°59.3	174°42.2	4.2'	25°42.2	8.2'	60.0'
4	239°06.4	05°00.2	189°05.4	4.1'	25°50.4	8.1'	59.9'
5	254°06.6	01.2	203°28.5	4.1'	25°58.5	7.9'	59.9'
6	269°06.8	N05°02.1	217°51.6	4.0'	N26°06.3	7.7'	59.9'
7	284°07.0	03.1	232°14.6	4.0'	26°14.0	7.5'	59.8'
8	299°07.2	04.1	246°37.6	3.9'	26°21.5	7.3'	59.8'
9	314°07.3	· · 05.0	261°00.5	3.9'	26°28.8	7.1'	59.8'
10	329°07.5	06.0	275°23.4	3.9'	26°35.9	6.9'	59.7'
11	344°07.7	06.9	289°46.3	3.8'	26°42.8	6.7'	59.7'
12	359°07.9	N05°07.9	304°09.1	3.8'	N26°49.5	6.5'	59.7'
13	14°08.1	08.9	318°31.9	3.8'	26°56.1	6.3'	59.6'
14	29°08.3	09.8	332°54.7	3.7'	27°02.4	6.1'	59.6'
15	44°08.4	· · 10.8	347°17.4	3.7'	27°08.6	6.0'	59.6'
16	59°08.6	11.7	1°40.1	3.7'	27°14.5	5.8'	59.5'
17	74°08.8	12.7	16°02.8	3.7'	27°20.3	5.6'	59.5'
18	89°09.0	N05°13.7	30°25.5	3.7'	N27°25.9	5.4'	59.5'
19	104°09.2	14.6	44°48.2	3.6'	27°31.2	5.2'	59.4'
20	119°09.4	15.6	59°10.8	3.6'	27°36.4	5.0'	59.4'
21	134°09.5	· · 16.5	73°33.4	3.6'	27°41.4	4.8'	59.4'
22	149°09.7	17.5	87°56.0	3.6'	27°46.2	4.6'	59.3'
23	164°09.9	18.5	102°18.6	3.6'	27°50.8	4.4'	59.3'
SD = 16.0'			$d = 1.0'$		SD = 16.4'		
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	179°10.1	N05°19.4	116°41.2	3.6'	N27°55.2	4.2'	59.3'
1	194°10.3	20.4	131°03.8	3.6'	27°59.3	4.0'	59.2'
2	209°10.4	21.3	145°26.4	3.6'	28°03.3	3.8'	59.2'
3	224°10.6	· · 22.3	159°49.0	3.6'	28°07.1	3.6'	59.2'
4	239°10.8	23.2	174°11.6	3.6'	28°10.8	3.4'	59.1'
5	254°11.0	24.2	188°34.2	3.6'	28°14.2	3.2'	59.1'
6	269°11.2	N05°25.2	202°56.8	3.6'	N28°17.4	3.0'	59.1'
7	284°11.4	26.1	217°19.5	3.6'	28°20.4	2.8'	59.0'
8	299°11.5	27.1	231°42.1	3.7'	28°23.2	2.6'	59.0'
9	314°11.7	· · 28.0	246°04.8	3.7'	28°25.8	2.4'	58.9'
10	329°11.9	29.0	260°27.4	3.7'	28°28.3	2.2'	58.9'
11	344°12.1	29.9	274°50.1	3.7'	28°30.5	2.0'	58.9'
12	359°12.3	N05°30.9	289°12.8	3.7'	N28°32.5	1.8'	58.8'
13	14°12.5	31.8	303°35.6	3.8'	28°34.4	1.6'	58.8'
14	29°12.6	32.8	317°58.3	3.8'	28°36.0	1.5'	58.8'
15	44°12.8	· · 33.8	332°21.1	3.8'	28°37.5	1.3'	58.7'
16	59°13.0	34.7	346°44.0	3.9'	28°38.7	1.1'	58.7'
17	74°13.2	35.7	1°06.8	3.9'	28°39.8	0.9'	58.7'
18	89°13.4	N05°36.6	15°29.7	3.9'	N28°40.7	0.7'	58.6'
19	104°13.5	37.6	29°52.7	4.0'	28°41.3	0.5'	58.6'
20	119°13.7	38.5	44°15.7	4.0'	28°41.8	0.3'	58.6'
21	134°13.9	· · 39.5	58°38.7	4.1'	28°42.1	0.1'	58.5'
22	149°14.1	40.4	73°01.8	4.1'	28°42.2	-0.1'	58.5'
23	164°14.3	41.4	87°24.9	4.2'	28°42.1	-0.3'	58.5'
SD = 16.0'			$d = 1.0'$		SD = 16.2'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	01:22	03:34	04:50	19:20	20:38	23:00
N 70°	02:11	03:51	04:58	19:12	20:20	22:03
68°	02:41	04:04	05:04	19:05	20:06	21:31
66°	03:03	04:15	05:10	18:59	19:54	21:08
64°	03:20	04:24	05:14	18:55	19:45	20:50
62°	03:34	04:32	05:18	18:50	19:37	20:36
60°	03:45	04:39	05:22	18:47	19:30	20:24
N 58°	03:55	04:44	05:25	18:44	19:24	20:14
56°	04:03	04:49	05:28	18:41	19:19	20:06
54°	04:10	04:54	05:30	18:38	19:14	19:59
52°	04:16	04:58	05:32	18:36	19:10	19:52
50°	04:22	05:01	05:34	18:34	19:07	19:46
45°	04:34	05:09	05:39	18:29	18:59	19:34
N 40°	04:43	05:15	05:42	18:25	18:53	19:25
35°	04:50	05:20	05:45	18:22	18:48	19:18
30°	04:56	05:24	05:48	18:19	18:43	19:12
20°	05:05	05:31	05:53	18:15	18:37	19:03
N 10°	05:11	05:36	05:57	18:10	18:32	18:56
0°	05:15	05:39	06:00	18:07	18:27	18:52
S 10°	05:18	05:43	06:04	18:03	18:24	18:49
20°	05:20	05:45	06:07	17:59	18:21	18:47
30°	05:20	05:47	06:11	17:55	18:19	18:47
35°	05:19	05:48	06:14	17:53	18:18	18:47
40°	05:18	05:49	06:16	17:50	18:17	18:48
45°	05:16	05:50	06:19	17:47	18:16	18:50
S 50°	05:13	05:50	06:22	17:44	18:16	18:53
52°	05:11	05:50	06:24	17:42	18:16	18:55
54°	05:09	05:50	06:26	17:40	18:15	18:56
56°	05:07	05:50	06:28	17:38	18:15	18:58
58°	05:05	05:50	06:30	17:36	18:15	19:01
S 60°	05:02	05:50	06:32	17:34	18:15	19:03

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°						
N 70°						
68°	03:19			01:05		
66°	04:15			00:10		
64°	04:49	04:02		· · ·	02:30	
62°	05:13	05:04		· · ·	01:29	
60°	05:33	05:39	05:57	· · ·	00:55	02:47
N 58°	05:50	06:04	06:34	· · ·	00:30	02:11
56°	06:04	06:24	07:00	· · ·	00:10	01:44
54°	06:16	06:41	07:21	23:54	· · ·	01:23
52°	06:27	06:56	07:39	23:40	· · ·	01:06
50°	06:36	07:09	07:54	23:28	· · ·	00:51
45°	06:57	07:35	08:24	23:02	· · ·	00:21
N 40°	07:14	07:56	08:48	22:42	23:58	· · ·
35°	07:28	08:14	09:08	22:25	23:38	· · ·
30°	07:40	08:29	09:25	22:11	23:22	· · ·
20°	08:01	08:55	09:53	21:47	22:54	23:58
N 10°	08:20	09:17	10:18	21:26	22:30	23:33
0°	08:37	09:38	10:41	21:06	22:08	23:10
S 10°	08:55	09:59	11:03	20:47	21:46	22:47
20°	09:14	10:22	11:28	20:26	21:22	22:22
30°	09:36	10:49	11:57	20:02	20:54	21:53

April 04, 05, 06 UT (Fri., Sat., Sun.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	192°33.9	198°37.3	N04°37.2	75°32.0	N23°44.9	117°14.9	N22°27.1	196°31.4	S03°49.7	Alpheratz	357°34.9	29°13.6											
1	207°36.4	213°40.4	36.3	90°33.6	44.7	132°16.9	27.1	211°33.6	49.6	Ankaa	353°07.4	-42°10.2											
2	222°38.8	228°43.5	35.5	105°35.1	44.5	147°19.0	27.2	226°35.8	49.5	Schedar	349°31.4	56°40.5											
3	237°41.3	243°46.5	· · 34.7	120°36.7	· · 44.3	162°21.0	· · 27.2	241°37.9	· · 49.4	Diphda	348°47.4	-17°51.0											
4	252°43.8	258°49.6	33.9	135°38.2	44.0	177°23.0	27.3	256°40.1	49.3	Achernar	335°20.6	-57°06.6											
5	267°46.2	273°52.7	33.0	150°39.8	43.8	192°25.1	27.3	271°42.3	49.1	Hamal	327°51.3	23°34.8											
6	282°48.7	288°55.7	N04°32.2	165°41.3	N23°43.6	207°27.1	N22°27.3	286°44.5	S03°49.0	Polaris	314°22.1	89°22.4											
7	297°51.1	303°58.8	31.4	180°42.9	43.4	222°29.1	27.4	301°46.7	48.9	Acamar	315°12.0	-40°12.4											
8	312°53.6	319°01.9	30.6	195°44.4	43.2	237°31.2	27.4	316°48.9	48.8	Menkar	314°06.2	4°11.2											
9	327°56.1	334°04.9	· · 29.7	210°46.0	· · 43.0	252°33.2	· · 27.5	331°51.1	· · 48.7	Mirfak	308°28.4	49°57.1											
10	342°58.5	349°08.0	28.9	225°47.5	42.7	267°35.2	27.5	346°53.3	48.6	Aldebaran	290°39.6	16°33.6											
11	358°01.0	4°11.0	28.1	240°49.1	42.5	282°37.2	27.6	1°55.5	48.5	Rigel	281°03.8	-8°10.5											
12	13°03.5	19°14.1	N04°27.3	255°50.6	N23°42.3	297°39.3	N22°27.6	16°57.7	S03°48.3	Capella	280°21.8	46°01.5											
13	28°05.9	34°17.1	26.5	270°52.2	42.1	312°41.3	27.7	31°59.8	48.2	Bellatrix	278°22.8	6°22.3											
14	43°08.4	49°20.1	25.6	285°53.7	41.9	327°43.3	27.7	47°02.0	48.1	Elnath	278°01.8	28°37.8											
15	58°10.9	64°23.2	· · 24.8	300°55.3	· · 41.6	342°45.4	· · 27.8	62°04.2	· · 48.0	Alnilam	275°37.7	-1°11.2											
16	73°13.3	79°26.2	24.0	315°56.8	41.4	357°47.4	27.8	77°06.4	47.9	Betelgeuse	270°52.0	7°24.6											
17	88°15.8	94°29.2	23.2	330°58.3	41.2	12°49.4	27.9	92°08.6	47.8	Canopus	263°52.4	-52°42.8											
18	103°18.3	109°32.2	N04°22.4	345°59.9	N23°41.0	27°51.5	N22°27.9	107°10.8	S03°47.7	Sirius	258°26.1	-16°45.2											
19	118°20.7	124°35.3	21.6	1°01.4	40.8	42°53.5	28.0	122°13.0	47.5	Adhara	255°05.7	-29°00.6											
20	133°23.2	139°38.3	20.8	16°03.0	40.6	57°55.5	28.0	137°15.2	47.4	Procyon	244°50.6	5°09.5											
21	148°25.6	154°41.3	· · 20.0	31°04.5	· · 40.3	72°57.5	· · 28.0	152°17.4	· · 47.3	Pollux	243°17.0	27°58.0											
22	163°28.1	169°44.3	19.2	46°06.0	40.1	87°59.6	28.1	167°19.6	47.2	Avior	234°14.5	-59°35.7											
23	178°30.6	184°47.3	18.3	61°07.6	39.9	103°01.6	28.1	182°21.7	47.1	Suhail	222°46.0	-43°32.3											
Mer.pass. 11:08				ν 3.1' d -0.8' m -4.40				ν 1.6' d -0.2' m 0.50				ν 2.0' d 0.0' m -2.09				ν 2.2' d -0.1' m 1.16							

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	193°33.0	199°50.3	N04°17.5	76°09.1	N23°39.7	118°03.6	N22°28.2	197°23.9	S03°47.0										
1	208°35.5	214°53.3	16.7	91°10.7	39.5	133°05.6	28.2	212°26.1	46.9										
2	223°38.0	229°56.3	15.9	106°12.2	39.2	148°07.7	28.3	227°28.3	46.7										
3	238°40.4	244°59.3	· · 15.1	121°13.7	· · 39.0	163°09.7	· · 28.3	242°30.5	· · 46.6										
4	253°42.9	260°02.2	14.3	136°15.3	38.8	178°11.7	28.4	257°32.7	46.5										
5	268°45.4	275°05.2	13.5	151°16.8	38.6	193°13.8	28.4	272°34.9	46.4										
6	283°47.8	290°08.2	N04°12.7	166°18.3	N23°38.3	208°15.8	N22°28.5	287°37.1	S03°46.3										
7	298°50.3	305°11.2	11.9	181°19.9	38.1	223°17.8	28.5	302°39.3	46.2										
8	313°52.7	320°14.1	11.1	196°21.4	37.9	238°19.8	28.6	317°41.5	46.1										
9	328°55.2	335°17.1	· · 10.3	211°22.9	· · 37.7	253°21.9	· · 28.6	332°43.6	· · 46.0										
10	343°57.7	350°20.1	09.5	226°24.5	37.5	268°23.9	28.7	347°45.8	45.8										
11	359°00.1	5°23.0	08.8	241°26.0	37.2	283°25.9	28.7	2°48.0	45.7										
12	14°02.6	20°26.0	N04°08.0	256°27.5	N23°37.0	298°27.9	N22°28.7	17°50.2	S03°45.6										
13	29°05.1	35°28.9	07.2	271°29.1	36.8	313°30.0	28.8	32°52.4	45.5										
14	44°07.5	50°31.8	06.4	286°30.6	36.6	328°32.0	28.8	47°54.6	45.4										
15	59°10.0	65°34.8	· · 05.6	301°32.1	· · 36.3	343°34.0	· · 28.9	62°56.8	· · 45.3										
16	74°12.5	80°37.7	04.8	316°33.7	36.1	358°36.0	28.9	77°59.0	45.2										
17	89°14.9	95°40.7	04.0	331°35.2	35.9	13°38.1	29.0	93°01.2	45.0										
18	104°17.4	110°43.6	N04°03.2	346°36.7	N23°35.7	28°40.1	N22°29.0	108°03.4	S03°44.9										
19	119°19.9	125°46.5	02.4	1°38.2	35.5	43°42.1	29.1	123°05.6	44.8										
20	134°22.3	140°49.4	01.7	16°39.8	35.2	58°44.1	29.1	138°07.7	44.7										
21	149°24.8	155°52.3	· · 00.9	31°41.3	· · 35.0	73°46.2	· · 29.2	153°09.9	· · 44.6										
22	164°27.2	170°55.3	04°00.1	46°42.8	34.8	88°48.2	29.2	168°12.1	44.5										
23	179°29.7	185°58.2	03°59.3	61°44.3	34.6	103°50.2	29.3	183°14.3	44.4										
Mer.pass. 11:04				ν 3.0' d -0.8' m -4.45				ν 1.5' d -0.2' m 0.52				ν 2.0' d 0.0' m -2.09				ν 2.2' d -0.1' m 1.16			

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	194°32.2	201°01.1	N03°58.6	76°45.9	N23°34.3	118°52.2	N22°29.3	198°16.5	S03°44.3										
1	209°34.6	216°04.0	57.8	91°47.4	34.1	133°54.2	29.4	213°18.7	44.1										
2	224°37.1	231°06.9	57.0	106°48.9	33.9	148°56.3	29.4	228°20.9	44.0										
3	239°39.6	246°09.8	· · 56.2	121°50.4	· · 33.7	163°58.3	· · 29.4	243°23.1	· · 43.9										
4	254°42.0	261°12.6	55.5	136°52.0	33.4	179°00.3	29.5	258°25.3	43.8										
5	269°44.5	276°15.5	54.7	151°53.5	33.2	194°02.3	29.5	273°27.5	43.7										
6	284°47.0	291°18.4	N03°53.9	166°55.0	N23°33.0	209°04.3	N22°29.6	288°29.7	S03°43.6										
7	299°49.4	306°21.3	53.1	181°56.5	32.8	224°06.4	29.6	303°31.9	43.5										
8	314°51.9	321°24.2	52.4	196°58.0	32.5	239°08.4	29.7	318°34.0	43.4										
9	329°54.4	336°27.0	· · 51.6	211°59.6	· · 32.3	254°10.4	· · 29.7	333°36.2	· · 43.2										
10	344°56.8	351°29.9	50.8	227°01.1	32.1	269°12.4	29.8	348°38.4	43.1										
11	359°59.3	6°32.8	50.1	242°02.6	31.9	284°14.5	29.8	3°40.6	43.0										
12	15°01.7	21°35.6	N03°49.3	257°04.1	N23°31.6	299°16.5	N22°29.9	18°42.8	S03°42.9										
13	30°04.2	36°38.5	48.6	272°05.6	31.4	314°18.5	29.9	33°45.0	42.8										
14	45°06.7	51°41.3	47.8	287°07.2	31.2	329°20.5	30.0	48°47.2	42.7										
15	60°09.1	66°44.2	· · 47.0	302°08.7	· · 31.0	344°22.5	· · 30.0	63°49.4	· · 42.6										
16	75°11.6	81°47.0	46.3	317°10.2	30.7	359°24.6	30.1	78°51.6	42.4										
17	90°14.1	96°49.8	45.5	332°11.7	30.5	14°26.6	30.1	93°53.8	42.3										
18	105°16.5	111°52.7	N03°44.8	347°13.2	N23°30.3	29°28.6	N22°30.1	108°56.0	S03°42.2										
19	120°19.0	126°55.5	44.0	2°14.7	30.0	44°30.6	30.2	123°58.2	42.1										
20	135°21.5	141°58.3	43.3	17°16.2	29.8	59°32.6	30.2	139°00.3	42.0										
21	150°23.9	157°01.1	· · 42.5	32°17.8	· · 29.6	74°34.6	· · 30.3	154°02.5	· · 41.9										
22	165°26.4	172°04.0	41.8	47°19.3	29.4	89°36.7	30.3	169°04.7	41.8										
23	180°28.9	187°06.8	41.0	62°20.8	29.1	104°38.7	30.4	184°06.9	41.7										
Mer.pass. 11:00				ν 2.9' d -0.8' m -4.49				ν 1.5' d -0.2' m 0.54				ν 2.0' d 0.0' m -2.08				ν 2.2' d -0.1' m 1.17			

Apr 04 Fri	SHA	Mer.pass
Venus	6°03.4	10:43
Mars	242°58.1	18:56
Jupiter	284°41.0	16:09
Saturn	3°57.5	10:52

Apr 05 Sat	SHA	Mer.pass
Venus	6°17.3	10:39
Mars	242°36.1	18:53
Jupiter	284°30.6	16:06
Saturn	3°50.9	10:49

Apr 06 Sun	SHA	Mer.pass
Venus	6°28.9	10:34
Mars	242°13.7	18:51
Jupiter	284°20.0	16:02
Saturn	3°44.3	10:45

Horizontal parallax		
Venus:		0.5
Mars:		0.1

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	179°14.4	N05°42.3	101°48.1	4.2'	N28°41.9	-0.5'	58.4'
1	194°14.6	43.3	116°11.3	4.3'	28°41.4	-0.7'	58.4'
2	209°14.8	44.3	130°34.6	4.3'	28°40.7	-0.8'	58.3'
3	224°15.0	· · 45.2	144°57.9	4.4'	28°39.9	-1.0'	58.3'
4	239°15.2	46.2	159°21.3	4.5'	28°38.9	-1.2'	58.3'
5	254°15.4	47.1	173°44.8	4.5'	28°37.7	-1.4'	58.2'
6	269°15.5	N05°48.1	188°08.3	4.6'	N28°36.3	-1.6'	58.2'
7	284°15.7	49.0	202°32.0	4.7'	28°34.7	-1.8'	58.2'
8	299°15.9	50.0	216°55.6	4.7'	28°32.9	-1.9'	58.1'
9	314°16.1	· · 50.9	231°19.4	4.8'	28°31.0	-2.1'	58.1'
10	329°16.3	51.9	245°43.2	4.9'	28°28.9	-2.3'	58.1'
11	344°16.4	52.8	260°07.1	5.0'	28°26.6	-2.5'	58.0'
12	359°16.6	N05°53.8	274°31.0	5.0'	N28°24.1	-2.7'	58.0'
13	14°16.8	54.7	288°55.1	5.1'	28°21.4	-2.8'	58.0'
14	29°17.0	55.7	303°19.2	5.2'	28°18.6	-3.0'	57.9'
15	44°17.2	· · 56.6	317°43.4	5.3'	28°15.6	-3.2'	57.9'
16	59°17.3	57.6	332°07.7	5.4'	28°12.4	-3.3'	57.8'
17	74°17.5	58.5	346°32.1	5.5'	28°09.1	-3.5'	57.8'
18	89°17.7	N05°59.5	0°56.6	5.6'	N28°05.6	-3.7'	57.8'
19	104°17.9	06°00.4	15°21.1	5.6'	28°01.9	-3.9'	57.7'
20	119°18.1	01.4	29°45.8	5.7'	27°58.0	-4.0'	57.7'
21	134°18.2	· · 02.3	44°10.5	5.8'	27°54.0	-4.2'	57.7'
22	149°18.4	03.3	58°35.3	5.9'	27°49.8	-4.4'	57.6'
23	164°18.6	04.2	73°00.3	6.0'	27°45.4	-4.5'	57.6'
SD = 16.0'			SD = 15.9'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	179°18.8	N06°05.2	87°25.3	6.1'	N27°40.9	-4.7'	57.6'
1	194°19.0	06.1	101°50.4	6.2'	27°36.2	-4.8'	57.5'
2	209°19.1	07.1	116°15.6	6.3'	27°31.4	-5.0'	57.5'
3	224°19.3	· · 08.0	130°40.9	6.4'	27°26.4	-5.2'	57.5'
4	239°19.5	09.0	145°06.3	6.5'	27°21.2	-5.3'	57.4'
5	254°19.7	09.9	159°31.9	6.6'	27°15.9	-5.5'	57.4'
6	269°19.8	N06°10.9	173°57.5	6.7'	N27°10.5	-5.6'	57.4'
7	284°20.0	11.8	188°23.2	6.8'	27°04.8	-5.8'	57.3'
8	299°20.2	12.8	202°49.0	6.9'	26°59.1	-5.9'	57.3'
9	314°20.4	· · 13.7	217°15.0	7.0'	26°53.1	-6.1'	57.3'
10	329°20.6	14.7	231°41.0	7.2'	26°47.1	-6.2'	57.2'
11	344°20.7	15.6	246°07.2	7.3'	26°40.8	-6.4'	57.2'
12	359°20.9	N06°16.6	260°33.4	7.4'	N26°34.5	-6.5'	57.2'
13	14°21.1	17.5	274°59.8	7.5'	26°28.0	-6.7'	57.1'
14	29°21.3	18.5	289°26.3	7.6'	26°21.3	-6.8'	57.1'
15	44°21.5	· · 19.4	303°52.9	7.7'	26°14.5	-6.9'	57.1'
16	59°21.6	20.3	318°19.6	7.8'	26°07.6	-7.1'	57.0'
17	74°21.8	21.3	332°46.4	7.9'	26°00.5	-7.2'	57.0'
18	89°22.0	N06°22.2	347°13.3	8.0'	N25°53.3	-7.3'	57.0'
19	104°22.2	23.2	1°40.3	8.1'	25°45.9	-7.5'	56.9'
20	119°22.3	24.1	16°07.5	8.3'	25°38.4	-7.6'	56.9'
21	134°22.5	· · 25.1	30°34.8	8.4'	25°30.8	-7.7'	56.9'
22	149°22.7	26.0	45°02.1	8.5'	25°23.1	-7.9'	56.8'
23	164°22.9	27.0	59°29.6	8.6'	25°15.2	-8.0'	56.8'
SD = 16.0'			SD = 15.7'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	179°23.1	N06°27.9	73°57.2	8.7'	N25°07.2	-8.1'	56.8'
1	194°23.2	28.9	88°24.9	8.8'	24°59.1	-8.3'	56.7'
2	209°23.4	29.8	102°52.8	8.9'	24°50.8	-8.4'	56.7'
3	224°23.6	· · 30.7	117°20.7	9.1'	24°42.4	-8.5'	56.7'
4	239°23.8	31.7	131°48.8	9.2'	24°33.9	-8.6'	56.7'
5	254°23.9	32.6	146°17.0	9.3'	24°25.3	-8.7'	56.6'
6	269°24.1	N06°33.6	160°45.2	9.4'	N24°16.6	-8.9'	56.6'
7	284°24.3	34.5	175°13.6	9.5'	24°07.7	-9.0'	56.6'
8	299°24.5	35.5	189°42.2	9.6'	23°58.7	-9.1'	56.5'
9	314°24.7	· · 36.4	204°10.8	9.7'	23°49.6	-9.2'	56.5'
10	329°24.8	37.3	218°39.5	9.9'	23°40.4	-9.3'	56.5'
11	344°25.0	38.3	233°08.4	10.0'	23°31.1	-9.4'	56.4'
12	359°25.2	N06°39.2	247°37.4	10.1'	N23°21.7	-9.5'	56.4'
13	14°25.4	40.2	262°06.5	10.2'	23°12.1	-9.6'	56.4'
14	29°25.5	41.1	276°35.7	10.3'	23°02.5	-9.8'	56.4'
15	44°25.7	· · 42.1	291°05.0	10.4'	22°52.7	-9.9'	56.3'
16	59°25.9	43.0	305°34.4	10.5'	22°42.9	-10.0'	56.3'
17	74°26.1	43.9	320°03.9	10.6'	22°32.9	-10.1'	56.3'
18	89°26.2	N06°44.9	334°33.6	10.8'	N22°22.9	-10.2'	56.2'
19	104°26.4	45.8	349°03.3	10.9'	22°12.7	-10.3'	56.2'
20	119°26.6	46.8	3°33.2	11.0'	22°02.4	-10.4'	56.2'
21	134°26.8	· · 47.7	18°03.2	11.1'	21°52.1	-10.5'	56.2'
22	149°26.9	48.6	32°33.2	11.2'	21°41.6	-10.5'	56.1'
23	164°27.1	49.6	47°03.4	11.3'	21°31.1	-10.6'	56.1'
SD = 16.0'			SD = 15.5'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	03:14	04:34	19:35	20:56	—:—
N 70°	01:43	03:34	04:43	19:25	20:35	22:31
68°	02:21	03:50	04:51	19:16	20:18	21:50
66°	02:47	04:03	04:58	19:09	20:05	21:23
64°	03:06	04:13	05:04	19:03	19:55	21:03
62°	03:21	04:22	05:08	18:58	19:46	20:47
60°	03:34	04:29	05:13	18:54	19:38	20:34
N 58°	03:45	04:36	05:16	18:50	19:31	20:23
56°	03:54	04:41	05:20	18:47	19:25	20:13
54°	04:02	04:46	05:23	18:44	19:20	20:05
52°	04:09	04:51	05:25	18:41	19:16	19:58
50°	04:15	04:55	05:28	18:38	19:12	19:52
45°	04:27	05:03	05:33	18:33	19:03	19:39
N 40°	04:37	05:10	05:37	18:28	18:56	19:29
35°	04:45	05:16	05:41	18:25	18:50	19:21
30°	04:52	05:20	05:44	18:21	18:45	19:14
20°	05:02	05:28	05:50	18:15	18:38	19:04
N 10°	05:09	05:34	05:55	18:10	18:32	18:56
0°	05:14	05:39	05:59	18:06	18:27	18:51
S 10°	05:18	05:43	06:04	18:01	18:23	18:47
20°	05:20	05:46	06:08	17:57	18:19	18:44
30°	05:21	05:49	06:13	17:52	18:16	18:43
35°	05:21	05:51	06:16	17:49	18:14	18:43
40°	05:21	05:52	06:19	17:45	18:13	18:44
45°	05:19	05:53	06:23	17:42	18:11	18:45
S 50°	05:17	05:55	06:27	17:37	18:10	18:47
52°	05:16	05:55	06:29	17:35	18:09	18:48
54°	05:15	05:56	06:31	17:33	18:08	18:49
56°	05:13	05:56	06:34	17:31	18:08	18:51
58°	05:11	05:57	06:36	17:28	18:07	18:52
S 60°	05:09	05:57	06:39	17:25	18:06	18:54

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	□	□	□	□	□	□
66°	□	□	07:42	□	□	07:11
64°	□	□	09:07	□	□	05:45
62°	□	07:38	09:44	□	05:20	05:07
60°	06:50	08:25	10:10	04:03	04:32	04:41
N 58°	07:31	08:56	10:30	03:23	04:01	04:20
56°	07:59	09:19	10:47	02:55	03:38	04:02
54°	08:21	09:37	11:01	02:33	03:19	03:47
52°	08:39	09:53	11:14	02:15	03:03	03:34
50°	08:54	10:07	11:25	01:59	02:49	03:23
45°	09:25	10:35	11:47	01:28	02:20	02:59
N 40°	09:49	10:56	12:05	01:04	01:58	02:40
35°	10:09	11:14	12:21	00:44	01:39	02:24
30°	10:26	11:30	12:34	00:27	01:23	02:10
20°	10:54	11:56	12:56	· ·	00:56	01:46
N 10°	11:19	12:19	13:15	· ·	00:32	01:25
0°	11:42	12:40	13:33	· ·	00:10	01:06
S 10°	12:05	13:01	13:51	23:48	· ·	00:46
20°	12:29	13:23	14:10	23:24	· ·	00:25
30°	12:58	13:49	14:32	22:56	· ·	00:01
35°	13:14	14:04	14:45	22:40	23:46	· ·
40°	13:34	14:22	14:59	22:21	23:29	· ·
45°	13:58	14:43	15:17	21:57	23:09	· ·
S 50°	14:29	15:10	15:38	21:27	22:43	· ·
52°	14:44	15:23	15:48	21:12	22:31	23:52
54°	15:02	15:37	15:59	20:54	22:16	23:42
56°	15:23	15:55	16:12	20:33	22:00	23:29
58°	15:50	16:15	16:27	20:06	21:39	23:15
S 60°	16:28	16:42	16:45	19:28	21:14	22:58

Day	Sun		Mer. Pass hh:mm	Moon		Age 6-8 38-60%
	Eqn.of Time 00 ^h mm:ss	Time 12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
04	03:02	02:54	12:03	17:56	05:26	
05	02:45	02:36	12:03	18:53	06:25	
06	02:28	02:19	12:02	19:45	07:20	

April 07, 08, 09 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	195°31.3	202°09.6	N03°40.3	77°22.3	N23°28.9	119°40.7	N22°30.4	199°09.1	S03°41.5	Alpheratz	357°34.9	29°13.6
1	210°33.8	217°12.4		92°23.8	28.7	134°42.7	30.5	214°11.3	41.4	Ankaa	353°07.3	-42°10.2
2	225°36.2	232°15.2		107°25.3	28.4	149°44.7	30.5	229°13.5	41.3	Schedar	349°31.4	56°40.5
3	240°38.7	247°18.0	•• 38.1	122°26.8	•• 28.2	164°46.7	•• 30.6	244°15.7	•• 41.2	Diphda	348°47.4	-17°51.0
4	255°41.2	262°20.8		137°28.3	28.0	179°48.8	30.6	259°17.9	41.1	Achernar	335°20.6	-57°06.6
5	270°43.6	277°23.6		152°29.8	27.8	194°50.8	30.7	274°20.1	41.0	Hamal	327°51.3	23°34.8
6	285°46.1	292°26.4	N03°35.8	167°31.4	N23°27.5	209°52.8	N22°30.7	289°22.3	S03°40.9	Polaris	314°22.8	89°22.4
7	300°48.6	307°29.1		182°32.9	27.3	224°54.8	30.8	304°24.5	40.8	Acamar	315°12.0	-40°12.3
8	315°51.0	322°31.9		197°34.4	27.1	239°56.8	30.8	319°26.7	40.6	Menkar	314°06.2	4°11.2
9	330°53.5	337°34.7	•• 33.6	212°35.9	•• 26.8	254°58.8	•• 30.8	334°28.8	•• 40.5	Mirfak	308°28.4	49°57.1
10	345°56.0	352°37.5		227°37.4	26.6	270°00.9	30.9	349°31.0	40.4	Aldebaran	290°39.6	16°33.6
11	0°58.4	7°40.2		242°38.9	26.4	285°02.9	30.9	4°33.2	40.3	Rigel	281°03.8	-8°10.5
12	16°00.9	22°43.0	N03°31.4	257°40.4	N23°26.2	300°04.9	N22°31.0	19°35.4	S03°40.2	Capella	280°21.8	46°01.5
13	31°03.3	37°45.7		272°41.9	25.9	315°06.9	31.0	34°37.6	40.1	Bellatrix	278°22.8	6°22.3
14	46°05.8	52°48.5		300°28.7	25.7	330°08.9	31.1	49°39.8	40.0	Elnath	278°01.8	28°37.8
15	61°08.3	67°51.2	•• 29.3	302°44.9	•• 25.5	345°10.9	•• 31.1	64°42.0	•• 39.9	Alnilam	275°37.7	-1°11.2
16	76°10.7	82°54.0		317°46.4	25.2	0°13.0	31.2	79°44.2	39.8	Betelgeuse	270°52.0	7°24.7
17	91°13.2	97°56.7		332°47.9	25.0	15°15.0	31.2	94°46.4	39.6	Canopus	263°52.4	-52°42.8
18	106°15.7	112°59.5	N03°27.1	347°49.4	N23°24.8	30°17.0	N22°31.3	109°48.6	S03°39.5	Sirius	258°26.1	-16°45.2
19	121°18.1	128°02.2		2°50.9	24.5	45°19.0	31.3	124°50.8	39.4	Adhara	255°05.7	-29°00.6
20	136°20.6	143°04.9		17°52.4	24.3	60°21.0	31.4	139°53.0	39.3	Procyon	244°50.6	5°09.5
21	151°23.1	158°07.7	•• 24.9	32°53.9	•• 24.1	75°23.0	•• 31.4	154°55.2	•• 39.2	Pollux	243°17.0	27°58.0
22	166°25.5	173°10.4		47°55.4	23.8	90°25.0	31.4	169°57.4	39.1	Avior	234°14.5	-59°35.7
23	181°28.0	188°13.1		62°56.9	23.6	105°27.0	31.5	184°59.6	39.0	Suhail	222°46.0	-43°32.3
Mer.pass. 10:56		ν 2.8' d -0.7' m -4.53		ν 1.5' d -0.2' m 0.55		ν 2.0' d 0.0' m -2.08		ν 2.2' d -0.1' m 1.17				

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	196°30.5	203°15.8	N03°22.8	77°58.4	N23°23.4	120°29.1	N22°31.5	200°01.7	S03°38.9	Denebola	182°24.4	14°25.8
1	211°32.9	218°18.5		92°59.9	23.1	135°31.1	31.6	215°03.9	38.7	Gienah	175°43.1	-17°41.1
2	226°35.4	233°21.2		108°01.4	22.9	150°33.1	31.6	230°06.1	38.6	Acrux	172°59.3	-63°14.5
3	241°37.8	248°23.9	•• 20.7	123°02.9	•• 22.7	165°35.1	•• 31.7	245°08.3	•• 38.5	Gacrux	171°50.9	-57°15.4
4	256°40.3	263°26.6		138°04.4	22.4	180°37.1	31.7	260°10.5	38.4	Alioth	166°12.2	55°49.3
5	271°42.8	278°29.3		153°05.9	22.2	195°39.1	31.8	275°12.7	38.3	Spica	158°21.8	-11°17.7
6	286°45.2	293°32.0	N03°18.6	168°07.4	N23°22.0	210°41.1	N22°31.8	280°14.9	S03°38.2	Alkaid	152°51.3	49°11.1
7	301°47.7	308°34.7		183°08.9	21.7	225°43.1	31.9	305°17.1	38.1	Hadar	148°35.2	-60°29.7
8	316°50.2	323°37.4		198°10.4	21.5	240°45.2	31.9	320°19.3	38.0	Menkent	147°57.0	-36°29.7
9	331°52.6	338°40.1	•• 16.5	213°11.9	•• 21.3	255°47.2	•• 32.0	335°21.5	•• 37.8	Arcturus	145°47.4	19°02.9
10	346°55.1	353°42.7		228°13.4	21.0	270°49.2	32.0	350°23.7	37.7	Rigel Kent.	139°39.5	-60°56.4
11	1°57.6	8°45.4		243°14.9	20.8	285°51.2	32.0	5°25.9	37.6	Kochab	137°18.7	74°02.9
12	17°00.0	23°48.1	N03°14.4	258°16.4	N23°20.6	300°53.2	N22°32.1	20°28.1	S03°37.5	Zuben'ubi	136°55.5	-16°08.9
13	32°02.5	38°50.7		273°17.9	20.3	315°55.2	32.1	35°30.3	37.4	Alphecca	126°03.3	26°37.5
14	47°04.9	53°53.4		288°19.3	20.1	330°57.2	32.2	50°32.5	37.3	Antares	112°15.4	-26°29.3
15	62°07.4	68°56.1	•• 12.3	303°20.8	•• 19.9	345°59.2	•• 32.2	65°34.7	•• 37.2	Atria	107°09.1	-69°04.2
16	77°09.9	83°58.7		318°22.3	19.6	1°01.2	32.3	80°36.8	37.1	Sabik	102°02.4	-15°45.4
17	92°12.3	99°01.4		333°23.8	19.4	16°03.2	32.3	95°39.0	37.0	Shaula	96°09.9	-37°07.3
18	107°14.8	114°04.0	N03°10.2	348°25.3	N23°19.2	31°05.3	N22°32.4	110°41.2	S03°36.8	Rasalhague	95°58.2	12°32.3
19	122°17.3	129°06.6		3°26.8	18.9	46°07.3	32.4	125°43.4	36.7	Eltanin	90°41.9	51°28.7
20	137°19.7	144°09.3		18°28.3	18.7	61°09.3	32.5	140°45.6	36.6	Kaus Aust.	83°32.2	-34°22.3
21	152°22.2	159°11.9	•• 08.2	33°29.8	•• 18.5	76°11.3	•• 32.5	155°47.8	•• 36.5	Vega	80°33.0	38°48.1
22	167°24.7	174°14.5		48°31.3	18.2	91°13.3	32.6	170°50.0	36.4	Nunki	75°47.5	-26°15.9
23	182°27.1	189°17.1		63°32.7	18.0	106°15.3	32.6	185°52.2	36.3	Altair	61°59.8	8°55.9
Mer.pass. 10:52		ν 2.7' d -0.7' m -4.56		ν 1.5' d -0.2' m 0.57		ν 2.0' d 0.0' m -2.07		ν 2.2' d -0.1' m 1.17				

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	197°29.6	204°19.8	N03°06.2	78°34.2	N23°17.7	121°17.3	N22°32.7	200°54.4	S03°36.2	Peacock	53°05.6	-56°39.0	
1	212°32.1	219°22.4		93°35.7	17.5	136°19.3	32.7	215°56.6	36.1	Deneb	49°25.9	45°21.9	
2	227°34.5	234°25.0		108°37.2	17.3	151°21.3	32.7	230°58.8	36.0	Enif	33°38.8	9°59.3	
3	242°37.0	249°27.6	•• 04.8	123°38.7	•• 17.0	166°23.3	•• 32.8	246°01.0	•• 35.8	Al Na'ir	27°33.0	-46°50.2	
4	257°39.4	264°30.2		138°40.2	16.8	181°25.3	32.8	261°03.2	35.7	Fomalhaut	15°14.6	-29°29.3	
5	272°41.9	279°32.8		153°41.7	16.6	196°27.3	32.9	276°05.4	35.6	Scheat	13°45.3	28°13.0	
6	287°44.4	294°35.4	N03°02.1	168°43.1	N23°16.3	211°29.4	N22°32.9	291°07.6	S03°35.5	Markab	13°30.0	15°20.3	
7	302°46.8	309°38.0		183°44.6	16.1	226°31.4	33.0	306°09.8	35.4	Apr 07 Mon		SHA	Mer.pass
8	317°49.3	324°40.6		198°46.1	15.9	241°33.4	33.0	321°12.0	35.3	Venus	6°38.3	10:29	
9	332°51.8	339°43.2	03°00.1	213°47.6	•• 15.6	256°35.4	•• 33.1	336°14.2	•• 35.2	Mars	241°51.0	18:49	
10	347°54.2	354°45.7	02°59.5	228°49.1	15.4	271°37.4	33.1	351°16.3	35.1	Jupiter	284°09.4	15:59	
11	2°56.7	9°48.3		243°50.6	15.1	286°39.4	33.2	6°18.5	35.0	Saturn	3°37.8	10:42	
12	17°59.2	24°50.9	N02°58.2	258°52.0	N23°14.9	301°41.4	N22°33.2	21°20.7	S03°34.8	Apr 08 Tue		SHA	Mer.pass
13	33°01.6	39°53.5		273°53.5	14.7	316°43.4	33.3	36°22.9	34.7	Venus	6°45.4	10:25	
14	48°04.1	54°56.0		288°55.0	14.4	331°45.4	33.3	51°25.1	34.6	Mars	241°28.0	18:46	
15	63°06.6	69°58.6	•• 56.2	303°56.5	•• 14.2	346°47.4	•• 33.3	66°27.3	•• 34.5	Jupiter	283°58.6	15:56	
16	78°09.0	85°01.1		318°58.0	13.9	1°49.4	33.4	81°29.5	34.4	Saturn	3°31.3	10:38	
17	93°11.5	100°03.7		333°59.4	13.7	16°51.4	33.4	96°31.7	34.3	Apr 09 Wed		SHA	Mer.pass
18	108°13.9	115°06.2	N02°54.3	349°00.9	N23°13.5	31°53.4	N22°33.5	111°33.9	S03°34.2	Venus	6°50.2	10:21	
19	123°16.4	130°08.8		4°02.4	13.2	46°55.4	33.5	126°36.1	34.1	Mars	241°04.6	18:44	
20	138°18.9	145°11.3		19°03.9	13.0	61°57.4	33.6	141°38.3	34.0	Jupiter	283°47.7	15:53	
21	153°21.3	160°13.9	•• 52.3	34°05.3	•• 12.7	76°59.4	•• 33.6	156°40.5	•• 33.8	Saturn	3°24.8	10:35	
22	168°23.8	175°16.4		49°06.8	12.5	92°01.4	33.7	171°42.7	33.7	Horizontal parallax			
23	183°26.3	190°18.9		64°08.3	12.3	107°03.4	33.7	186°44.9	33.6	Venus:		0.5	
Mer.pass. 10:48		ν 2.6' d -0.7' m -4.59		ν 1.5' d -0.2' m 0.59		ν 2.0' d 0.0' m -2.07		ν 2.2' d -0.1' m 1.17		Mars:		0.1	

h Sun Moon						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	179°27.3	N06°50.5	61°33.7	11.4'	N21°20.4	-10.7' 56.1'
1	194°27.5	51.5	76°04.1	11.5'	21°09.7	-10.8' 56.0'
2	209°27.6	52.4	90°34.7	11.6'	20°58.9	-10.9' 56.0'
3	224°27.8	· · 53.3	105°05.3	11.7'	20°48.0	-11.0' 56.0'
4	239°28.0	54.3	119°36.0	11.8'	20°37.0	-11.1' 56.0'
5	254°28.2	55.2	134°06.8	11.9'	20°25.9	-11.2' 55.9'
6	269°28.3	N06°56.2	148°37.8	12.0'	N20°14.7	-11.3' 55.9'
7	284°28.5	57.1	163°08.8	12.1'	20°03.5	-11.3' 55.9'
8	299°28.7	58.0	177°39.9	12.2'	19°52.1	-11.4' 55.9'
9	314°28.9	· · 59.0	192°11.2	12.3'	19°40.7	-11.5' 55.8'
10	329°29.0	06°59.9	206°42.5	12.4'	19°29.2	-11.6' 55.8'
11	344°29.2	07°00.9	221°14.0	12.5'	19°17.7	-11.7' 55.8'
12	359°29.4	N07°01.8	235°45.5	12.6'	N19°06.0	-11.7' 55.8'
13	14°29.6	02.7	250°17.2	12.7'	18°54.3	-11.8' 55.7'
14	29°29.7	03.7	264°48.9	12.8'	18°42.5	-11.9' 55.7'
15	44°29.9	· · 04.6	279°20.7	12.9'	18°30.6	-11.9' 55.7'
16	59°30.1	05.5	293°52.7	13.0'	18°18.6	-12.0' 55.7'
17	74°30.3	06.5	308°24.7	13.1'	18°06.6	-12.1' 55.6'
18	89°30.4	N07°07.4	322°56.8	13.2'	N17°54.5	-12.2' 55.6'
19	104°30.6	08.3	337°29.0	13.3'	17°42.4	-12.2' 55.6'
20	119°30.8	09.3	352°01.3	13.4'	17°30.1	-12.3' 55.6'
21	134°31.0	· · 10.2	6°33.7	13.5'	17°17.9	-12.4' 55.5'
22	149°31.1	11.2	21°06.2	13.6'	17°05.5	-12.4' 55.5'
23	164°31.3	12.1	35°38.8	13.7'	16°53.1	-12.5' 55.5'
SD = 16.0' $d = 0.9'$ SD = 15.3'						

Tue	GHA	Dec	GHA	ν	Dec	d HP
0	179°31.5	N07°13.0	50°11.4	13.8'	N16°40.6	-12.5' 55.5'
1	194°31.7	14.0	64°44.2	13.8'	16°28.1	-12.6' 55.4'
2	209°31.8	14.9	79°17.0	13.9'	16°15.5	-12.7' 55.4'
3	224°32.0	· · 15.8	93°50.0	14.0'	16°02.8	-12.7' 55.4'
4	239°32.2	16.8	108°23.0	14.1'	15°50.1	-12.8' 55.4'
5	254°32.3	17.7	122°56.0	14.2'	15°37.3	-12.8' 55.4'
6	269°32.5	N07°18.6	137°29.2	14.3'	N15°24.5	-12.9' 55.3'
7	284°32.7	19.6	152°02.5	14.3'	15°11.6	-12.9' 55.3'
8	299°32.9	20.5	166°35.8	14.4'	14°58.6	-13.0' 55.3'
9	314°33.0	· · 21.4	181°09.2	14.5'	14°45.6	-13.0' 55.3'
10	329°33.2	22.4	195°42.7	14.6'	14°32.6	-13.1' 55.2'
11	344°33.4	23.3	210°16.3	14.6'	14°19.5	-13.1' 55.2'
12	359°33.6	N07°24.2	224°49.9	14.7'	N14°06.4	-13.2' 55.2'
13	14°33.7	25.2	239°23.6	14.8'	13°53.2	-13.2' 55.2'
14	29°33.9	26.1	253°57.4	14.9'	13°39.9	-13.3' 55.2'
15	44°34.1	· · 27.0	268°31.3	14.9'	13°26.6	-13.3' 55.1'
16	59°34.2	27.9	283°05.2	15.0'	13°13.3	-13.4' 55.1'
17	74°34.4	28.9	297°39.2	15.1'	12°59.9	-13.4' 55.1'
18	89°34.6	N07°29.8	312°13.3	15.1'	N12°46.5	-13.5' 55.1'
19	104°34.8	30.7	326°47.4	15.2'	12°33.1	-13.5' 55.1'
20	119°34.9	31.7	341°21.6	15.3'	12°19.6	-13.5' 55.0'
21	134°35.1	· · 32.6	355°55.9	15.3'	12°06.0	-13.6' 55.0'
22	149°35.3	33.5	10°30.2	15.4'	11°52.4	-13.6' 55.0'
23	164°35.4	34.5	25°04.6	15.5'	11°38.8	-13.7' 55.0'
SD = 16.0' $d = 0.9'$ SD = 15.1'						

Wed	GHA	Dec	GHA	ν	Dec	d HP
0	179°35.6	N07°35.4	39°39.1	15.5'	N11°25.2	-13.7' 55.0'
1	194°35.8	36.3	54°13.6	15.6'	11°11.5	-13.7' 55.0'
2	209°36.0	37.3	68°48.2	15.6'	10°57.8	-13.8' 54.9'
3	224°36.1	· · 38.2	83°22.9	15.7'	10°44.0	-13.8' 54.9'
4	239°36.3	39.1	97°57.6	15.8'	10°30.2	-13.8' 54.9'
5	254°36.5	40.0	112°32.3	15.8'	10°16.4	-13.9' 54.9'
6	269°36.6	N07°41.0	127°07.1	15.9'	N10°02.5	-13.9' 54.9'
7	284°36.8	41.9	141°42.0	15.9'	09°48.6	-13.9' 54.8'
8	299°37.0	42.8	156°16.9	16.0'	09°34.7	-13.9' 54.8'
9	314°37.1	· · 43.7	170°51.9	16.0'	09°20.8	-14.0' 54.8'
10	329°37.3	44.7	185°26.9	16.1'	09°06.8	-14.0' 54.8'
11	344°37.5	45.6	200°02.0	16.1'	08°52.8	-14.0' 54.8'
12	359°37.7	N07°46.5	214°37.1	16.2'	N08°38.8	-14.0' 54.8'
13	14°37.8	47.5	229°12.3	16.2'	08°24.7	-14.1' 54.7'
14	29°38.0	48.4	243°47.5	16.3'	08°10.7	-14.1' 54.7'
15	44°38.2	· · 49.3	258°22.8	16.3'	07°56.6	-14.1' 54.7'
16	59°38.3	50.2	272°58.1	16.4'	07°42.5	-14.1' 54.7'
17	74°38.5	51.2	287°33.5	16.4'	07°28.3	-14.2' 54.7'
18	89°38.7	N07°52.1	302°08.9	16.4'	N07°14.2	-14.2' 54.7'
19	104°38.8	53.0	316°44.3	16.5'	07°00.0	-14.2' 54.6'
20	119°39.0	53.9	331°19.8	16.5'	06°45.8	-14.2' 54.6'
21	134°39.2	· · 54.9	345°55.3	16.6'	06°31.6	-14.2' 54.6'
22	149°39.3	55.8	0°30.9	16.6'	06°17.3	-14.2' 54.6'
23	164°39.5	56.7	15°06.4	16.6'	06°03.1	-14.3' 54.6'
SD = 16.0' $d = 0.9'$ SD = 15.0'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	02:53	04:17	19:50	21:16	—:—
N 70°	01:06	03:17	04:29	19:37	20:51	23:14
68°	01:58	03:35	04:39	19:27	20:32	22:12
66°	02:29	03:50	04:46	19:19	20:17	21:39
64°	02:51	04:01	04:53	19:12	20:05	21:16
62°	03:09	04:11	04:59	19:07	19:55	20:58
60°	03:23	04:19	05:04	19:01	19:46	20:43
N 58°	03:34	04:27	05:08	18:57	19:39	20:31
56°	03:45	04:33	05:12	18:53	19:32	20:21
54°	03:53	04:39	05:15	18:49	19:26	20:12
52°	04:01	04:44	05:19	18:46	19:21	20:04
50°	04:08	04:48	05:21	18:43	19:17	19:57
45°	04:21	04:58	05:28	18:37	19:07	19:43
N 40°	04:32	05:05	05:33	18:31	18:59	19:32
35°	04:41	05:11	05:37	18:27	18:53	19:23
30°	04:48	05:17	05:41	18:23	18:47	19:16
20°	04:59	05:25	05:48	18:16	18:39	19:05
N 10°	05:07	05:32	05:53	18:10	18:32	18:56
0°	05:13	05:38	05:58	18:05	18:26	18:50
S 10°	05:18	05:42	06:04	18:00	18:21	18:45
20°	05:21	05:47	06:09	17:54	18:16	18:42
30°	05:23	05:51	06:15	17:48	18:12	18:40
35°	05:24	05:53	06:18	17:45	18:10	18:39
40°	05:24	05:55	06:22	17:41	18:08	18:39
45°	05:23	05:57	06:27	17:36	18:06	18:40
S 50°	05:22	05:59	06:32	17:31	18:03	18:41
52°	05:21	06:00	06:34	17:28	18:02	18:41
54°	05:20	06:01	06:37	17:26	18:01	18:42
56°	05:19	06:02	06:40	17:23	18:00	18:43
58°	05:18	06:03	06:43	17:20	17:59	18:44
S 60°	05:16	06:04	06:46	17:16	17:58	18:46

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	☐	11:38	14:01	☐	06:37	05:43
N 70°	09:01	12:08	14:14	07:38	06:05	05:27
68°	10:12	12:31	14:24	06:25	05:41	05:15
66°	10:49	12:48	14:33	05:47	05:22	05:05
64°	11:14	13:02	14:40	05:21	05:07	04:56
62°	11:34	13:13	14:46	05:00	04:54	04:49
60°	11:50	13:23	14:51	04:43	04:43	04:42
N 58°	12:04	13:32	14:55	04:29	04:34	04:36
56°	12:15	13:39	14:59	04:16	04:25	04:31
54°	12:25	13:46	15:03	04:05	04:18	04:27
52°	12:34	13:52	15:06	03:56	04:11	04:23
50°	12:42	13:57	15:09	03:47	04:05	04:19
45°	12:59	14:09	15:16	03:29	03:52	04:10
N 40°	13:13	14:18	15:21	03:14	03:41	04:03
35°	13:25	14:27	15:26	03:01	03:31	03:57
30°	13:35	14:34	15:30	02:49	03:23	03:52
20°	13:53	14:46	15:36	02:30	03:08	03:43
N 10°	14:08	14:57	15:43	02:13	02:56	03:35
0°	14:22	15:07	15:48	01:57	02:44	03:27
S 10°	14:36	15:16	15:54	01:41	02:32	03:19
20°	14:51	15:27	16:00	01:24	02:19	03:11
30°	15:08	15:39	16:07	01:04	02:04	03:01
35°	15:18	15:46	16:11	00:52	01:55	02:55
40°	15:29	15:54	16:15	00:38	01:45	02:49
45°	15:42	16:03	16:20	00:22	01:33	02:41
S 50°	15:58	16:13	16:26	00:02	01:19	02:32
52°	16:06	16:19	16:29	· · ·	01:12	02:28
54°	16:14	16:24	16:32	· · ·	01:04	02:23
56°	16:23	16:30	16:35	· · ·	00:56	02:18
58°	16:34	16:37	16:39	· · ·	00:46	02:12
S 60°	16:45	16:45	16:43	· · ·	00:35	02:06

Day	Sun		Mer. Pass	Moon		Age 9-11 70-86%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
07	02:11	02:02	12:02	20:33	08:10	
08	01:54	01:46	12:02	21:17	08:55	
09	01:38	01:29	12:01	21:58	09:38	

April 10, 11, 12 UT (Thu., Fri., Sat.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	198°28.7	205°21.4	N02°50.4	79°09.8	N23°12.0	122°05.4	N22°33.8	201°47.1	S03°33.5	Alpheratz	357°34.9	29°13.6
1	213°31.2	220°24.0	49.8	94°11.2	11.8	137°07.4	33.8	216°49.3	33.4	Ankaa	353°07.3	-42°10.1
2	228°33.7	235°26.5	49.1	109°12.7	11.5	152°09.4	33.9	231°51.5	33.3	Schedar	349°31.4	56°40.4
3	243°36.1	250°29.0	· · 48.5	124°14.2	· · 11.3	167°11.5	· · 33.9	246°53.7	· · 33.2	Diphda	348°47.4	-17°51.0
4	258°38.6	265°31.5	47.9	139°15.7	11.1	182°13.5	33.9	261°55.9	33.1	Achernar	335°20.6	-57°06.6
5	273°41.0	280°34.0	47.2	154°17.1	10.8	197°15.5	34.0	276°58.1	33.0	Hamal	327°51.3	23°34.8
6	288°43.5	295°36.5	N02°46.6	169°18.6	N23°10.6	212°17.5	N22°34.0	292°00.3	S03°32.8	Polaris	314°23.5	89°22.4
7	303°46.0	310°39.0	46.0	184°20.1	10.3	227°19.5	34.1	307°02.5	32.7	Acamar	315°12.0	-40°12.3
8	318°48.4	325°41.5	45.3	199°21.5	10.1	242°21.5	34.1	322°04.7	32.6	Menkar	314°06.2	4°11.2
9	333°50.9	340°44.0	· · 44.7	214°23.0	· · 09.9	257°23.5	· · 34.2	337°06.8	· · 32.5	Mirfak	308°28.4	49°57.1
10	348°53.4	355°46.5	44.1	229°24.5	09.6	272°25.5	34.2	352°09.0	32.4	Aldebaran	290°39.6	16°33.6
11	3°55.8	10°49.0	43.5	244°26.0	09.4	287°27.5	34.3	7°11.2	32.3	Rigel	281°03.9	-8°10.5
12	18°58.3	25°51.4	N02°42.9	259°27.4	N23°09.1	302°29.5	N22°34.3	22°13.4	S03°32.2	Capella	280°21.9	46°01.5
13	34°00.8	40°53.9	42.2	274°28.9	08.9	317°31.5	34.4	37°15.6	32.1	Bellatrix	278°22.8	6°22.3
14	49°03.2	55°56.4	41.6	289°30.4	08.6	332°33.5	34.4	52°17.8	32.0	Elnath	278°01.8	28°37.8
15	64°05.7	70°58.8	· · 41.0	304°31.8	· · 08.4	347°35.5	· · 34.4	67°20.0	· · 31.9	Alnilam	275°37.7	-1°11.2
16	79°08.2	86°01.3	40.4	319°33.3	08.2	2°37.5	34.5	82°22.2	31.7	Betelgeuse	270°52.0	7°24.7
17	94°10.6	101°03.8	39.8	334°34.8	07.9	17°39.5	34.5	97°24.4	31.6	Canopus	263°52.5	-52°42.8
18	109°13.1	116°06.2	N02°39.2	349°36.2	N23°07.7	32°41.5	N22°34.6	112°26.6	S03°31.5	Sirius	258°26.1	-16°45.2
19	124°15.5	131°08.7	38.6	4°37.7	07.4	47°43.5	34.6	127°28.8	31.4	Adhara	255°05.8	-29°00.6
20	139°18.0	146°11.1	38.0	19°39.2	07.2	62°45.5	34.7	142°31.0	31.3	Procyon	244°50.6	5°09.5
21	154°20.5	161°13.6	· · 37.4	34°40.6	· · 06.9	77°47.5	· · 34.7	157°33.2	· · 31.2	Pollux	243°17.0	27°58.0
22	169°22.9	176°16.0	36.8	49°42.1	06.7	92°49.5	34.8	172°35.4	31.1	Avior	234°14.6	-59°35.7
23	184°25.4	191°18.4	36.2	64°43.6	06.5	107°51.5	34.8	187°37.6	31.0	Suhail	222°46.0	-43°32.3
Mer.pass. 10:44		ν 2.5' d -0.6' m -4.62		ν 1.5' d -0.2' m 0.61		ν 2.0' d 0.0' m -2.06		ν 2.2' d -0.1' m 1.17				

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	199°27.9	206°20.9	N02°35.6	79°45.0	N23°06.2	122°53.5	N22°34.9	202°39.8	S03°30.9
1	214°30.3	221°23.3	35.0	94°46.5	06.0	137°55.5	34.9	217°42.0	30.7
2	229°32.8	236°25.7	34.4	109°47.9	05.7	152°57.5	35.0	232°44.2	30.6
3	244°35.3	251°28.1	· · 33.8	124°49.4	· · 05.5	167°59.5	· · 35.0	247°46.4	· · 30.5
4	259°37.7	266°30.5	33.2	139°50.9	05.2	183°01.5	35.0	262°48.6	30.4
5	274°40.2	281°33.0	32.6	154°52.3	05.0	198°03.5	35.1	277°50.8	30.3
6	289°42.6	296°35.4	N02°32.0	169°53.8	N23°04.7	213°05.5	N22°35.1	292°53.0	S03°30.2
7	304°45.1	311°37.8	31.4	184°55.2	04.5	228°07.5	35.2	307°55.2	30.1
8	319°47.6	326°40.2	30.8	199°56.7	04.2	243°09.4	35.2	322°57.4	30.0
9	334°50.0	341°42.6	· · 30.2	214°58.2	· · 04.0	258°11.4	· · 35.3	337°59.6	· · 29.9
10	349°52.5	356°45.0	29.7	229°59.6	03.8	273°13.4	35.3	353°01.8	29.8
11	4°55.0	11°47.3	29.1	245°01.1	03.5	288°15.4	35.4	8°04.0	29.6
12	19°57.4	26°49.7	N02°28.5	260°02.5	N23°03.3	303°17.4	N22°35.4	23°06.2	S03°29.5
13	34°59.9	41°52.1	27.9	275°04.0	03.0	318°19.4	35.5	38°08.4	29.4
14	50°02.4	56°54.5	27.4	290°05.5	02.8	333°21.4	35.5	53°10.6	29.3
15	65°04.8	71°56.9	· · 26.8	305°06.9	· · 02.5	348°23.4	· · 35.5	68°12.8	· · 29.2
16	80°07.3	86°59.2	26.2	320°08.4	02.3	3°25.4	35.6	83°15.0	29.1
17	95°09.8	102°01.6	25.6	335°09.8	02.0	18°27.4	35.6	98°17.2	29.0
18	110°12.2	117°04.0	N02°25.1	350°11.3	N23°01.8	33°29.4	N22°35.7	113°19.3	S03°28.9
19	125°14.7	132°06.3	24.5	5°12.7	01.5	48°31.4	35.7	128°21.5	28.8
20	140°17.1	147°08.7	23.9	20°14.2	01.3	63°33.4	35.8	143°23.7	28.7
21	155°19.6	162°11.0	· · 23.4	35°15.6	· · 01.0	78°35.4	· · 35.8	158°25.9	· · 28.6
22	170°22.1	177°13.4	22.8	50°17.1	00.8	93°37.4	35.9	173°28.1	28.4
23	185°24.5	192°15.7	22.2	65°18.5	00.5	108°39.4	35.9	188°30.3	28.3
Mer.pass. 10:40		ν 2.4' d -0.6' m -4.65		ν 1.5' d -0.2' m 0.63		ν 2.0' d 0.0' m -2.06		ν 2.2' d -0.1' m 1.17	

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	200°27.0	207°18.0	N02°21.7	80°20.0	N23°00.3	123°41.4	N22°36.0	203°32.5	S03°28.2
1	215°29.5	222°20.4	21.1	95°21.4	23°00.0	138°43.4	36.0	218°34.7	28.1
2	230°31.9	237°22.7	20.6	110°22.9	22°59.8	153°45.4	36.1	233°36.9	28.0
3	245°34.4	252°25.0	· · 20.0	125°24.4	· · 59.5	168°47.4	· · 36.1	248°39.1	· · 27.9
4	260°36.9	267°27.4	19.5	140°25.8	59.3	183°49.4	36.1	263°41.3	27.8
5	275°39.3	282°29.7	18.9	155°27.3	59.0	198°51.3	36.2	278°43.5	27.7
6	290°41.8	297°32.0	N02°18.4	170°28.7	N22°58.8	213°53.3	N22°36.2	293°45.7	S03°27.6
7	305°44.2	312°34.3	17.8	185°30.1	58.5	228°55.3	36.3	308°47.9	27.5
8	320°46.7	327°36.6	17.3	200°31.6	58.3	243°57.3	36.3	323°50.1	27.3
9	335°49.2	342°38.9	· · 16.7	215°33.0	· · 58.0	258°59.3	· · 36.4	338°52.3	· · 27.2
10	350°51.6	357°41.2	16.2	230°34.5	57.8	274°01.3	36.4	353°54.5	27.1
11	5°54.1	12°43.5	15.7	245°35.9	57.5	289°03.3	36.5	8°56.7	27.0
12	20°56.6	27°45.8	N02°15.1	260°37.4	N22°57.3	304°05.3	N22°36.5	23°58.9	S03°26.9
13	35°59.0	42°48.1	14.6	275°38.8	57.0	319°07.3	36.6	39°01.1	26.8
14	51°01.5	57°50.4	14.0	290°40.3	56.8	334°09.3	36.6	54°03.3	26.7
15	66°04.0	72°52.7	· · 13.5	305°41.7	· · 56.5	349°11.3	· · 36.6	69°05.5	· · 26.6
16	81°06.4	87°54.9	13.0	320°43.2	56.3	4°13.3	36.7	84°07.7	26.5
17	96°08.9	102°57.2	12.5	335°44.6	56.0	19°15.3	36.7	99°09.9	26.4
18	111°11.4	117°59.5	N02°11.9	350°46.1	N22°55.8	34°17.2	N22°36.8	114°12.1	S03°26.3
19	126°13.8	133°01.8	11.4	5°47.5	55.5	49°19.2	36.8	129°14.3	26.1
20	141°16.3	148°04.0	10.9	20°48.9	55.3	64°21.2	36.9	144°16.5	26.0
21	156°18.7	163°06.3	· · 10.4	35°50.4	· · 55.0	79°23.2	· · 36.9	159°18.7	· · 25.9
22	171°21.2	178°08.5	09.8	50°51.8	54.8	94°25.2	37.0	174°20.9	25.8
23	186°23.7	193°10.8	09.3	65°53.3	54.5	109°27.2	37.0	189°23.1	25.7
Mer.pass. 10:36		ν 2.3' d -0.6' m -4.67		ν 1.5' d -0.2' m 0.64		ν 2.0' d 0.0' m -2.05		ν 2.2' d -0.1' m 1.17	

Apr 10 Thu		SHA		Mer.pass	
Venus		6°52.7		10:17	
Mars		240°41.0		18:42	
Jupiter		283°36.7		15:50	
Saturn		3°18.4		10:31	
Apr 11 Fri		SHA		Mer.pass	
Venus		6°53.0		10:13	
Mars		240°17.2		18:39	
Jupiter		283°25.6		15:46	
Saturn		3°11.9		10:28	
Apr 12 Sat		SHA		Mer.pass	
Venus		6°51.0		10:09	
Mars		239°53.0		18:37	
Jupiter		283°14.4		15:43	
Saturn		3°05.5		10:24	
Horizontal parallax					
Venus:		0.4			
Mars:		0.1			

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	179°39.7	N07°57.6	29°42.1	16.7'	N05°48.8	-14.3'	54.6'
1	194°39.9	58.6	44°17.7	16.7'	05°34.5	-14.3'	54.6'
2	209°40.0	07°59.5	58°53.4	16.7'	05°20.3	-14.3'	54.5'
3	224°40.2	08°00.4	73°29.2	16.8'	05°05.9	-14.3'	54.5'
4	239°40.4	01.3	88°04.9	16.8'	04°51.6	-14.3'	54.5'
5	254°40.5	02.2	102°40.7	16.8'	04°37.3	-14.3'	54.5'
6	269°40.7	N08°03.2	117°16.5	16.8'	N04°23.0	-14.3'	54.5'
7	284°40.9	04.1	131°52.4	16.9'	04°08.6	-14.4'	54.5'
8	299°41.0	05.0	146°28.3	16.9'	03°54.3	-14.4'	54.5'
9	314°41.2	· · 05.9	161°04.2	16.9'	03°39.9	-14.4'	54.5'
10	329°41.4	06.9	175°40.1	17.0'	03°25.5	-14.4'	54.4'
11	344°41.5	07.8	190°16.0	17.0'	03°11.1	-14.4'	54.4'
12	359°41.7	N08°08.7	204°52.0	17.0'	N02°56.8	-14.4'	54.4'
13	14°41.9	09.6	219°28.0	17.0'	02°42.4	-14.4'	54.4'
14	29°42.0	10.5	234°04.0	17.0'	02°28.0	-14.4'	54.4'
15	44°42.2	· · 11.5	248°40.1	17.1'	02°13.6	-14.4'	54.4'
16	59°42.4	12.4	263°16.1	17.1'	01°59.2	-14.4'	54.4'
17	74°42.5	13.3	277°52.2	17.1'	01°44.8	-14.4'	54.4'
18	89°42.7	N08°14.2	292°28.3	17.1'	N01°30.4	-14.4'	54.3'
19	104°42.9	15.1	307°04.4	17.1'	01°16.0	-14.4'	54.3'
20	119°43.0	16.1	321°40.5	17.1'	01°01.6	-14.4'	54.3'
21	134°43.2	· · 17.0	336°16.6	17.1'	00°47.2	-14.4'	54.3'
22	149°43.4	17.9	350°52.8	17.2'	00°32.8	-14.4'	54.3'
23	164°43.5	18.8	5°28.9	17.2'	00°18.4	-14.4'	54.3'
SD = 16.0'			SD = 14.9'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	179°43.7	N08°19.7	20°05.1	17.2'	N00°04.0	-14.4'	54.3'
1	194°43.9	20.7	34°41.2	17.2'	S00°10.4	14.4'	54.3'
2	209°44.0	21.6	49°17.4	17.2'	00°24.8	14.4'	54.3'
3	224°44.2	· · 22.5	63°53.6	17.2'	00°39.1	14.4'	54.2'
4	239°44.4	23.4	78°29.8	17.2'	00°53.5	14.4'	54.2'
5	254°44.5	24.3	93°06.0	17.2'	01°07.8	14.3'	54.2'
6	269°44.7	N08°25.2	107°42.2	17.2'	S01°22.2	14.3'	54.2'
7	284°44.8	26.2	122°18.4	17.2'	01°36.5	14.3'	54.2'
8	299°45.0	27.1	136°54.6	17.2'	01°50.9	14.3'	54.2'
9	314°45.2	· · 28.0	151°30.8	17.2'	02°05.2	14.3'	54.2'
10	329°45.3	28.9	166°07.0	17.2'	02°19.5	14.3'	54.2'
11	344°45.5	29.8	180°43.2	17.2'	02°33.8	14.3'	54.2'
12	359°45.7	N08°30.7	195°19.4	17.2'	S02°48.1	14.3'	54.2'
13	14°45.8	31.7	209°55.6	17.2'	03°02.3	14.3'	54.2'
14	29°46.0	32.6	224°31.8	17.2'	03°16.6	14.2'	54.1'
15	44°46.2	· · 33.5	239°08.0	17.2'	03°30.8	14.2'	54.1'
16	59°46.3	34.4	253°44.2	17.2'	03°45.1	14.2'	54.1'
17	74°46.5	35.3	268°20.3	17.2'	03°59.3	14.2'	54.1'
18	89°46.6	N08°36.2	282°56.5	17.2'	S04°13.4	14.2'	54.1'
19	104°46.8	37.1	297°32.6	17.1'	04°27.6	14.1'	54.1'
20	119°47.0	38.1	312°08.8	17.1'	04°41.8	14.1'	54.1'
21	134°47.1	· · 39.0	326°44.9	17.1'	04°55.9	14.1'	54.1'
22	149°47.3	39.9	341°21.0	17.1'	05°10.0	14.1'	54.1'
23	164°47.5	40.8	355°57.1	17.1'	05°24.1	14.1'	54.1'
SD = 16.0'			SD = 14.8'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	179°47.6	N08°41.7	10°33.2	17.1'	S05°38.1	14.0'	54.1'
1	194°47.8	42.6	25°09.3	17.1'	05°52.2	14.0'	54.1'
2	209°47.9	43.5	39°45.4	17.0'	06°06.2	14.0'	54.1'
3	224°48.1	· · 44.4	54°21.4	17.0'	06°20.2	14.0'	54.0'
4	239°48.4	45.4	68°57.4	17.0'	06°34.2	13.9'	54.0'
5	254°48.4	46.3	83°33.4	17.0'	06°48.1	13.9'	54.0'
6	269°48.6	N08°47.2	98°09.4	17.0'	S07°02.0	13.9'	54.0'
7	284°48.8	48.1	112°45.4	16.9'	07°15.9	13.9'	54.0'
8	299°48.9	49.0	127°21.3	16.9'	07°29.8	13.8'	54.0'
9	314°49.1	· · 49.9	141°57.3	16.9'	07°43.6	13.8'	54.0'
10	329°49.2	50.8	156°33.1	16.9'	07°57.4	13.8'	54.0'
11	344°49.4	51.7	171°09.0	16.8'	08°11.2	13.7'	54.0'
12	359°49.6	N08°52.6	185°44.9	16.8'	S08°24.9	13.7'	54.0'
13	14°49.7	53.5	200°20.7	16.8'	08°38.6	13.7'	54.0'
14	29°49.9	54.5	214°56.5	16.8'	08°52.3	13.6'	54.0'
15	44°50.0	· · 55.4	229°32.2	16.7'	09°05.9	13.6'	54.0'
16	59°50.2	56.3	244°08.0	16.7'	09°19.5	13.6'	54.0'
17	74°50.4	57.2	258°43.7	16.7'	09°33.1	13.5'	54.0'
18	89°50.5	N08°58.1	273°19.3	16.6'	S09°46.6	13.5'	54.0'
19	104°50.7	59.0	287°55.0	16.6'	10°00.1	13.5'	54.0'
20	119°50.8	08°59.9	302°30.6	16.6'	10°13.5	13.4'	54.0'
21	134°51.0	09°00.8	317°06.1	16.5'	10°27.0	13.4'	54.0'
22	149°51.2	01.7	331°41.7	16.5'	10°40.3	13.3'	53.9'
23	164°51.3	02.6	346°17.2	16.5'	10°53.7	13.3'	53.9'
SD = 16.0'			SD = 14.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	02:31	04:01	20:05	21:38	—:—
N 70°	—:—	02:59	04:15	19:50	21:07	—:—
68°	01:31	03:20	04:26	19:39	20:46	22:40
66°	02:10	03:36	04:35	19:29	20:29	21:58
64°	02:36	03:49	04:42	19:21	20:15	21:30
62°	02:55	04:00	04:49	19:15	20:04	21:10
60°	03:11	04:10	04:55	19:09	19:54	20:53
N 58°	03:24	04:18	05:00	19:04	19:46	20:40
56°	03:35	04:25	05:04	18:59	19:39	20:29
54°	03:45	04:31	05:08	18:55	19:32	20:19
52°	03:53	04:37	05:12	18:51	19:27	20:10
50°	04:00	04:41	05:15	18:48	19:22	20:03
45°	04:15	04:52	05:22	18:41	19:11	19:48
N 40°	04:27	05:00	05:28	18:35	19:02	19:36
35°	04:37	05:07	05:33	18:29	18:55	19:26
30°	04:44	05:13	05:38	18:25	18:49	19:18
20°	04:57	05:23	05:45	18:17	18:39	19:06
N 10°	05:06	05:30	05:52	18:10	18:32	18:56
0°	05:13	05:37	05:58	18:04	18:25	18:49
S 10°	05:18	05:42	06:04	17:58	18:19	18:44
20°	05:22	05:48	06:10	17:52	18:14	18:40
30°	05:25	05:53	06:17	17:45	18:09	18:37
35°	05:26	05:55	06:21	17:41	18:06	18:36
40°	05:26	05:58	06:25	17:36	18:03	18:35
45°	05:27	06:01	06:30	17:31	18:00	18:34
S 50°	05:26	06:04	06:36	17:25	17:57	18:35
52°	05:26	06:05	06:39	17:22	17:56	18:35
54°	05:26	06:07	06:42	17:19	17:54	18:35
56°	05:25	06:08	06:46	17:15	17:53	18:36
58°	05:24	06:10	06:49	17:11	17:51	18:36
S 60°	05:23	06:11	06:54	17:07	17:49	18:37

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	16:04	18:02	20:07	05:06	04:33	04:00
N 70°	16:07	17:56	19:49	05:00	04:35	04:09
68°	16:09	17:51	19:35	04:55	04:36	04:17
66°	16:11	17:46	19:24	04:50	04:37	04:23
64°	16:12	17:43	19:15	04:47	04:38	04:29
62°	16:14	17:40	19:07	04:44	04:38	04:33
60°	16:15	17:37	19:00	04:41	04:39	04:37
N 58°	16:16	17:35	18:54	04:38	04:40	04:41
56°	16:17	17:33	18:49	04:36	04:40	04:44
54°	16:17	17:31	18:44	04:34	04:41	04:47
52°	16:18	17:29	18:40	04:32	04:41	04:50
50°	16:19	17:28	18:36	04:31	04:42	04:52
45°	16:20	17:24	18:28	04:27	04:42	04:58
N 40°	16:22	17:21	18:21	04:24	04:43	05:02
35°	16:23	17:19	18:15	04:21	04:44	05:06
30°	16:24	17:17	18:10	04:19	04:44	05:10
20°	16:25	17:13	18:02	04:15	04:45	05:16
N 10°	16:27	17:10	17:54	04:11	04:46	05:21
0°	16:28	17:07	17:47	04:07	04:47	05:26
S 10°	16:29	17:04	17:40	04:04	04:48	05:31
20°	16:31	17:01	17:32	04:00	04:48	05:37
30°	16:33	16:58	17:24	03:56	04:49	05:43
35°	16:33	16:56	17:19	03:53	04:50	05:46
40°	16:35	16:54	17:13	03:50	04:51	05:50
45°	16:36	16:51	17:07	03:47	04:51	05:55
S 50°	16:37	16:48	16:59	03:43	04:52	06:01
52°	16:38	16:47	16:56	03:41	04:52	06:03
54°	16:39	16:45	16:52	03:39	04:53	06:06
56°	16:39	16:44	16:48	03:37	04:53	06:09
58°	16:40	16:42	16:43	03:34	04:54	06:13
S 60°	16:41	16:40	16:38	03:31	04:54	06:17

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	12-14 92-99%
10	01:21	01:13	12:01	22:37	10:18	
11	01:05	00:57	12:01	23:17	10:57	
12	00:50	00:42	12:01	23:56	11:36	

April 13, 14, 15 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	201°26.1	208°13.0	N02°08.8	80°54.7	N22°54.3	124°29.2	N22°37.1	204°25.3	S03°25.6	Alpheratz	357°34.9	29°13.6
1	216°28.6	223°15.3	08.3	95°56.2	54.0	139°31.2	37.1	219°27.5	25.5	Ankaa	353°07.3	-42°10.1
2	231°31.1	238°17.5	07.8	110°57.6	53.8	154°33.2	37.1	234°29.7	25.4	Schedar	349°31.4	56°40.4
3	246°33.5	253°19.7	· · 07.3	125°59.0	· · 53.5	169°35.2	· · 37.2	249°31.9	· · 25.3	Diphda	348°47.4	-17°51.0
4	261°36.0	268°22.0	06.7	141°00.5	53.3	184°37.1	37.2	264°34.1	25.2	Achernar	335°20.6	-57°06.5
5	276°38.5	283°24.2	06.2	156°01.9	53.0	199°39.1	37.3	279°36.3	25.1	Hamal	327°51.3	23°34.8
6	291°40.9	298°26.4	N02°05.7	171°03.4	N22°52.7	214°41.1	N22°37.3	294°38.5	S03°25.0	Polaris	314°24.0	89°22.3
7	306°43.4	313°28.7	05.2	186°04.8	52.5	229°43.1	37.4	309°40.7	24.8	Acamar	315°12.0	-40°12.3
8	321°45.9	328°30.9	04.7	201°06.2	52.2	244°45.1	37.4	324°42.9	24.7	Menkar	314°06.2	4°11.2
9	336°48.3	343°33.1	· · 04.2	216°07.7	· · 52.0	259°47.1	· · 37.5	339°45.1	· · 24.6	Mirfak	308°28.4	49°57.1
10	351°50.8	358°35.3	03.7	231°09.1	51.7	274°49.1	37.5	354°47.3	24.5	Aldebaran	290°39.6	16°33.6
11	6°53.2	13°37.5	03.2	246°10.5	51.5	289°51.1	37.5	9°49.5	24.4	Rigel	281°03.9	-8°10.5
12	21°55.7	28°39.7	N02°02.7	261°12.0	N22°51.2	304°53.1	N22°37.6	24°51.7	S03°24.3	Capella	280°21.9	46°01.5
13	36°58.2	43°41.9	02.2	276°13.4	51.0	319°55.0	37.6	39°53.9	24.2	Bellatrix	278°22.8	6°22.3
14	52°00.6	58°44.1	01.7	291°14.8	50.7	334°57.0	37.7	54°56.1	24.1	Elnath	278°01.8	28°37.8
15	67°03.1	73°46.3	· · 01.2	306°16.3	· · 50.5	349°59.0	· · 37.7	69°58.3	· · 24.0	Alnilam	275°37.7	-1°11.2
16	82°05.6	88°48.5	00.8	321°17.7	50.2	5°01.0	37.8	85°00.5	23.9	Betelgeuse	270°52.0	7°24.7
17	97°08.0	103°50.7	02°00.3	336°19.1	49.9	20°03.0	37.8	100°02.7	23.8	Canopus	263°52.5	-52°42.8
18	112°10.5	118°52.8	N01°59.8	351°20.6	N22°49.7	35°05.0	N22°37.9	115°04.9	S03°23.6	Sirius	258°26.2	-16°45.2
19	127°13.0	133°55.0	59.3	6°22.0	49.4	50°07.0	37.9	130°07.1	23.5	Adhara	255°05.8	-29°00.6
20	142°15.4	148°57.2	58.8	21°23.4	49.2	65°08.9	38.0	145°09.3	23.4	Procyon	244°50.6	5°09.5
21	157°17.9	163°59.4	· · 58.3	36°24.9	· · 48.9	80°10.9	· · 38.0	160°11.5	· · 23.3	Pollux	243°17.1	27°58.0
22	172°20.3	179°01.5	57.9	51°26.3	48.7	95°12.9	38.0	175°13.7	23.2	Avior	234°14.6	-59°35.7
23	187°22.8	194°03.7	57.4	66°27.7	48.4	110°14.9	38.1	190°15.9	23.1	Suhail	222°46.0	-43°32.3
Mer.pass. 10:33		ν 2.2' d -0.5' m -4.69		ν 1.4' d -0.3' m 0.66		ν 2.0' d 0.0' m -2.05		ν 2.2' d -0.1' m 1.17				

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	202°25.3	209°05.9	N01°56.9	81°29.2	N22°48.2	125°16.9	N22°38.1	205°18.1	S03°23.0
1	217°27.7	224°08.0	56.4	96°30.6	47.9	140°18.9	38.2	220°20.3	22.9
2	232°30.2	239°10.2	56.0	111°32.0	47.6	155°20.9	38.2	235°22.5	22.8
3	247°32.7	254°12.3	· · 55.5	126°33.5	· · 47.4	170°22.8	· · 38.3	250°24.7	· · 22.7
4	262°35.1	269°14.4	55.0	141°34.9	47.1	185°24.8	38.3	265°26.9	22.6
5	277°37.6	284°16.6	54.6	156°36.3	46.9	200°26.8	38.4	280°29.1	22.5
6	292°40.1	299°18.7	N01°54.1	171°37.7	N22°46.6	215°28.8	N22°38.4	295°31.3	S03°22.3
7	307°42.5	314°20.9	53.6	186°39.2	46.3	230°30.8	38.5	310°33.5	22.2
8	322°45.0	329°23.0	53.2	201°40.6	46.1	245°32.8	38.5	325°35.7	22.1
9	337°47.5	344°25.1	· · 52.7	216°42.0	· · 45.8	260°34.8	· · 38.5	340°37.9	· · 22.0
10	352°49.9	359°27.2	52.2	231°43.5	45.6	275°36.7	38.6	355°40.1	21.9
11	7°52.4	14°29.4	51.8	246°44.9	45.3	290°38.7	38.6	10°42.3	21.8
12	22°54.8	29°31.5	N01°51.3	261°46.3	N22°45.1	305°40.7	N22°38.7	25°44.5	S03°21.7
13	37°57.3	44°33.6	50.9	276°47.7	44.8	320°42.7	38.7	40°46.7	21.6
14	52°59.8	59°35.7	50.4	291°49.2	44.5	335°44.7	38.8	55°48.9	21.5
15	68°02.2	74°37.8	· · 50.0	306°50.6	· · 44.3	350°46.6	· · 38.8	70°51.1	· · 21.4
16	83°04.7	89°39.9	49.5	321°52.0	44.0	5°48.6	38.9	85°53.3	21.3
17	98°07.2	104°42.0	49.1	336°53.4	43.8	20°50.6	38.9	100°55.5	21.2
18	113°09.6	119°44.1	N01°48.6	351°54.8	N22°43.5	35°52.6	N22°38.9	115°57.7	S03°21.1
19	128°12.1	134°46.2	48.2	6°56.3	43.2	50°54.6	39.0	130°59.9	20.9
20	143°14.6	149°48.3	47.8	21°57.7	43.0	65°56.6	39.0	146°02.1	20.8
21	158°17.0	164°50.3	· · 47.3	36°59.1	· · 42.7	80°58.5	· · 39.1	161°04.3	· · 20.7
22	173°19.5	179°52.4	46.9	52°00.5	42.5	96°00.5	39.1	176°06.5	20.6
23	188°21.9	194°54.5	46.5	67°02.0	42.2	111°02.5	39.2	191°08.7	20.5
Mer.pass. 10:29		ν 2.2' d -0.5' m -4.70		ν 1.4' d -0.3' m 0.68		ν 2.0' d 0.0' m -2.05		ν 2.2' d -0.1' m 1.17	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	203°24.4	209°56.6	N01°46.0	82°03.4	N22°41.9	126°04.5	N22°39.2	206°10.9	S03°20.4
1	218°26.9	224°58.6	45.6	97°04.8	41.7	141°06.5	39.3	221°13.1	20.3
2	233°29.3	240°00.7	45.2	112°06.2	41.4	156°08.4	39.3	236°15.3	20.2
3	248°31.8	255°02.8	· · 44.7	127°07.6	· · 41.1	171°10.4	· · 39.3	251°17.5	· · 20.1
4	263°34.3	270°04.8	44.3	142°09.1	40.9	186°12.4	39.4	266°19.7	20.0
5	278°36.7	285°06.9	43.9	157°10.5	40.6	201°14.4	39.4	281°21.9	19.9
6	293°39.2	300°08.9	N01°43.5	172°11.9	N22°40.4	216°16.4	N22°39.5	296°24.2	S03°19.8
7	308°41.7	315°11.0	43.0	187°13.3	40.1	231°18.3	39.5	311°26.4	19.7
8	323°44.1	330°13.0	42.6	202°14.7	39.8	246°20.3	39.6	326°28.6	19.5
9	338°46.6	345°15.0	· · 42.2	217°16.1	· · 39.6	261°22.3	· · 39.6	341°30.8	· · 19.4
10	353°49.1	0°17.1	41.8	232°17.6	39.3	276°24.3	39.7	356°33.0	19.3
11	8°51.5	15°19.1	41.4	247°19.0	39.0	291°26.3	39.7	11°35.2	19.2
12	23°54.0	30°21.1	N01°41.0	262°20.4	N22°38.8	306°28.2	N22°39.8	26°37.4	S03°19.1
13	38°56.4	45°23.2	40.6	277°21.8	38.5	321°30.2	39.8	41°39.6	19.0
14	53°58.9	60°25.2	40.1	292°23.2	38.3	336°32.2	39.8	56°41.8	18.9
15	69°01.4	75°27.2	· · 39.7	307°24.6	· · 38.0	351°34.2	· · 39.9	71°44.0	· · 18.8
16	84°03.8	90°29.2	39.3	322°26.0	37.7	6°36.2	39.9	86°46.2	18.7
17	99°06.3	105°31.2	38.9	337°27.5	37.5	21°38.1	40.0	101°48.4	18.6
18	114°08.8	120°33.2	N01°38.5	352°28.9	N22°37.2	36°40.1	N22°40.0	116°50.6	S03°18.5
19	129°11.2	135°35.2	38.1	7°30.3	36.9	51°42.1	40.1	131°52.8	18.4
20	144°13.7	150°37.2	37.7	22°31.7	36.7	66°44.1	40.1	146°55.0	18.3
21	159°16.2	165°39.2	· · 37.3	37°33.1	· · 36.4	81°46.1	· · 40.2	161°57.2	· · 18.2
22	174°18.6	180°41.2	36.9	52°34.5	36.1	96°48.0	40.2	176°59.4	18.1
23	189°21.1	195°43.2	36.5	67°35.9	35.9	111°50.0	40.2	192°01.6	17.9
Mer.pass. 10:25		ν 2.1' d -0.4' m -4.72		ν 1.4' d -0.3' m 0.70		ν 2.0' d 0.0' m -2.04		ν 2.2' d -0.1' m 1.17	

Apr 13 Sun		SHA	Mer.pass
Venus		6°46.9	10:06
Mars		239°28.6	18:35
Jupiter		283°03.0	15:40
Saturn		2°59.2	10:21

Apr 14 Mon		SHA	Mer.pass
Venus		6°40.6	10:02
Mars		239°03.9	18:32
Jupiter		282°51.6	15:37
Saturn		2°52.8	10:17

Apr 15 Tue		SHA	Mer.pass
Venus		6°32.1	09:59
Mars		238°39.0	18:30
Jupiter		282°40.1	15:34
Saturn		2°46.5	10:14

Horizontal parallax		
Venus:		0.4
Mars:		0.1

h Sun			Moon				
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	179°51.5	N09°03.5	0°52.6	16.4'	S11°07.0	13.3'	53.9'
1	194°51.6	04.4	15°28.0	16.4'	11°20.2	13.2'	53.9'
2	209°51.8	05.3	30°03.4	16.3'	11°33.4	13.2'	53.9'
3	224°52.0	· · 06.2	44°38.8	16.3'	11°46.6	13.1'	53.9'
4	239°52.1	07.1	59°14.1	16.3'	11°59.7	13.1'	53.9'
5	254°52.3	08.1	73°49.3	16.2'	12°12.8	13.0'	53.9'
6	269°52.4	N09°09.0	88°24.6	16.2'	S12°25.8	13.0'	53.9'
7	284°52.6	09.9	102°59.7	16.1'	12°38.8	12.9'	53.9'
8	299°52.7	10.8	117°34.9	16.1'	12°51.7	12.9'	53.9'
9	314°52.9	· · 11.7	132°10.0	16.0'	13°04.6	12.8'	53.9'
10	329°53.1	12.6	146°45.0	16.0'	13°17.4	12.8'	53.9'
11	344°53.2	13.5	161°20.0	16.0'	13°30.2	12.7'	53.9'
12	359°53.4	N09°14.4	175°55.0	15.9'	S13°43.0	12.7'	53.9'
13	14°53.5	15.3	190°29.9	15.9'	13°55.6	12.6'	53.9'
14	29°53.7	16.2	205°04.7	15.8'	14°08.3	12.6'	53.9'
15	44°53.9	· · 17.1	219°39.5	15.8'	14°20.8	12.5'	53.9'
16	59°54.0	18.0	234°14.3	15.7'	14°33.4	12.5'	53.9'
17	74°54.2	18.9	248°49.0	15.7'	14°45.8	12.4'	53.9'
18	89°54.3	N09°19.8	263°23.7	15.6'	S14°58.2	12.4'	53.9'
19	104°54.5	20.7	277°58.3	15.6'	15°10.6	12.3'	53.9'
20	119°54.6	21.6	292°32.8	15.5'	15°22.9	12.2'	53.9'
21	134°54.8	· · 22.5	307°07.3	15.4'	15°35.1	12.2'	53.9'
22	149°54.9	23.4	321°41.8	15.4'	15°47.3	12.1'	53.9'
23	164°55.1	24.3	336°16.1	15.3'	15°59.4	12.1'	53.9'
SD = 15.9'			SD = 14.7'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	179°55.3	N09°25.2	350°50.5	15.3'	S16°11.5	12.0'	53.9'
1	194°55.4	26.1	5°24.7	15.2'	16°23.5	11.9'	53.9'
2	209°55.6	27.0	19°59.0	15.2'	16°35.4	11.9'	53.9'
3	224°55.7	· · 27.9	34°33.1	15.1'	16°47.3	11.8'	53.9'
4	239°55.9	28.8	49°07.2	15.0'	16°59.1	11.7'	53.9'
5	254°56.0	29.7	63°41.3	15.0'	17°10.8	11.7'	53.9'
6	269°56.2	N09°30.6	78°15.3	14.9'	S17°22.5	11.6'	53.9'
7	284°56.3	31.5	92°49.2	14.9'	17°34.1	11.5'	53.9'
8	299°56.5	32.4	107°23.0	14.8'	17°45.6	11.5'	53.9'
9	314°56.7	· · 33.3	121°56.8	14.7'	17°57.1	11.4'	53.9'
10	329°56.8	34.2	136°30.6	14.7'	18°08.4	11.3'	53.9'
11	344°57.0	35.1	151°04.2	14.6'	18°19.8	11.2'	53.9'
12	359°57.1	N09°36.0	165°37.9	14.5'	S18°31.0	11.2'	53.9'
13	14°57.3	36.9	180°11.4	14.5'	18°42.2	11.1'	53.9'
14	29°57.4	37.8	194°44.9	14.4'	18°53.3	11.0'	53.9'
15	44°57.6	· · 38.7	209°18.3	14.4'	19°04.3	11.0'	53.9'
16	59°57.7	39.6	223°51.6	14.3'	19°15.3	10.9'	53.9'
17	74°57.9	40.5	238°24.9	14.2'	19°26.1	10.8'	53.9'
18	89°58.0	N09°41.3	252°58.1	14.2'	S19°36.9	10.7'	53.9'
19	104°58.2	42.2	267°31.3	14.1'	19°47.7	10.6'	53.9'
20	119°58.3	43.1	282°04.4	14.0'	19°58.3	10.6'	53.9'
21	134°58.5	· · 44.0	296°37.4	13.9'	20°08.9	10.5'	53.9'
22	149°58.6	44.9	311°10.3	13.9'	20°19.3	10.4'	53.9'
23	164°58.8	45.8	325°43.2	13.8'	20°29.7	10.3'	54.0'
SD = 15.9'			SD = 14.7'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	179°59.0	N09°46.7	340°16.0	13.7'	S20°40.0	10.2'	54.0'
1	194°59.1	47.6	354°48.8	13.7'	20°50.3	10.1'	54.0'
2	209°59.3	48.5	9°21.4	13.6'	21°00.4	10.1'	54.0'
3	224°59.4	· · 49.4	23°54.0	13.5'	21°10.5	10.0'	54.0'
4	239°59.6	50.3	38°26.6	13.5'	21°20.4	9.9'	54.0'
5	254°59.7	51.2	52°59.0	13.4'	21°30.3	9.8'	54.0'
6	269°59.9	N09°52.1	67°31.4	13.3'	S21°40.1	9.7'	54.0'
7	285°00.0	53.0	82°03.7	13.2'	21°49.8	9.6'	54.0'
8	300°00.2	53.8	96°36.0	13.2'	21°59.4	9.5'	54.0'
9	315°00.3	· · 54.7	111°08.1	13.1'	22°09.0	9.4'	54.0'
10	330°00.5	55.6	125°40.2	13.0'	22°18.4	9.3'	54.0'
11	345°00.6	56.5	140°12.2	13.0'	22°27.7	9.2'	54.0'
12	0°00.8	N09°57.4	154°44.2	12.9'	S22°37.0	9.1'	54.0'
13	15°00.9	58.3	169°16.1	12.8'	22°46.1	9.1'	54.0'
14	30°01.1	09°59.2	183°47.9	12.7'	22°55.2	9.0'	54.0'
15	45°01.2	10°00.1	198°19.6	12.7'	23°04.1	8.9'	54.0'
16	60°01.4	01.0	212°51.3	12.6'	23°13.0	8.8'	54.0'
17	75°01.5	01.9	227°22.8	12.5'	23°21.8	8.7'	54.1'
18	90°01.7	N10°02.7	241°54.4	12.4'	S23°30.4	8.6'	54.1'
19	105°01.8	03.6	256°25.8	12.4'	23°39.0	8.5'	54.1'
20	120°02.0	04.5	270°57.2	12.3'	23°47.4	8.4'	54.1'
21	135°02.1	· · 05.4	285°28.4	12.2'	23°55.8	8.3'	54.1'
22	150°02.3	06.3	299°59.7	12.1'	24°04.0	8.1'	54.1'
23	165°02.4	07.2	314°30.8	12.1'	24°12.2	8.0'	54.1'
SD = 15.9'			SD = 14.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	02:05	03:44	20:21	22:03	—:—
N 70°	—:—	02:40	04:00	20:04	21:26	—:—
68°	00:53	03:04	04:13	19:50	21:00	23:26
66°	01:48	03:23	04:23	19:40	20:41	22:19
64°	02:19	03:37	04:32	19:31	20:26	21:46
62°	02:42	03:50	04:39	19:23	20:13	21:22
60°	02:59	04:00	04:46	19:16	20:02	21:04
N 58°	03:14	04:09	04:52	19:10	19:53	20:49
56°	03:26	04:17	04:57	19:05	19:45	20:37
54°	03:36	04:23	05:01	19:00	19:38	20:26
52°	03:45	04:29	05:05	18:56	19:32	20:17
50°	03:53	04:35	05:09	18:53	19:27	20:09
45°	04:09	04:46	05:17	18:44	19:15	19:52
N 40°	04:22	04:56	05:24	18:38	19:06	19:39
35°	04:32	05:03	05:29	18:32	18:58	19:29
30°	04:41	05:10	05:34	18:27	18:51	19:20
20°	04:54	05:20	05:43	18:18	18:40	19:07
N 10°	05:04	05:29	05:50	18:10	18:32	18:57
0°	05:12	05:36	05:57	18:04	18:24	18:49
S 10°	05:18	05:42	06:04	17:57	18:18	18:43
20°	05:23	05:48	06:11	17:49	18:12	18:37
30°	05:27	05:54	06:19	17:41	18:06	18:33
35°	05:28	05:57	06:23	17:37	18:02	18:32
40°	05:29	06:01	06:28	17:32	17:59	18:30
45°	05:30	06:04	06:34	17:26	17:55	18:29
S 50°	05:31	06:08	06:41	17:19	17:51	18:29
52°	05:31	06:10	06:44	17:15	17:50	18:29
54°	05:31	06:12	06:48	17:12	17:48	18:29
56°	05:31	06:14	06:52	17:08	17:46	18:29
58°	05:30	06:16	06:56	17:03	17:43	18:29
S 60°	05:30	06:18	07:01	16:58	17:41	18:29

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	22:43	■	■	03:20	02:10	■
N 70°	21:59	■	■	03:39	02:56	■
68°	21:30	23:57	■	03:55	03:26	02:29
66°	21:08	23:08	■	04:08	03:49	03:20
64°	20:51	22:37	· ·	04:19	04:07	03:52
62°	20:38	22:14	23:58	04:28	04:22	04:16
60°	20:26	21:56	23:30	04:36	04:35	04:35
N 58°	20:16	21:40	23:07	04:43	04:46	04:51
56°	20:07	21:28	22:49	04:49	04:55	05:04
54°	19:59	21:16	22:34	04:55	05:04	05:16
52°	19:53	21:07	22:21	05:00	05:11	05:27
50°	19:46	20:58	22:10	05:04	05:18	05:36
45°	19:33	20:39	21:46	05:14	05:33	05:56
N 40°	19:22	20:24	21:27	05:23	05:45	06:12
35°	19:13	20:12	21:11	05:30	05:56	06:25
30°	19:05	20:01	20:58	05:36	06:05	06:37
20°	18:51	19:42	20:35	05:47	06:21	06:58
N 10°	18:39	19:26	20:15	05:57	06:35	07:16
0°	18:28	19:11	19:56	06:06	06:48	07:32
S 10°	18:16	18:56	19:38	06:15	07:01	07:49
20°	18:05	18:40	19:19	06:25	07:15	08:07
30°	17:51	18:22	18:56	06:37	07:32	08:28
35°	17:43	18:11	18:43	06:43	07:41	08:40
40°	17:35	17:59	18:28	06:51	07:52	08:55
45°	17:24	17:45	18:10	06:59	08:05	09:11
S 50°	17:12	17:28	17:48	07:10	08:21	09:32
52°	17:07	17:20	17:37	07:15	08:28	09:42
54°	17:00	17:11	17:26	07:20	08:36	09:54
56°	16:53	17:01	17:12	07:26	08:46	10:07
58°	16:46	16:50	16:57	07:33	08:56	10:22
S 60°	16:37	16:37	16:38	07:41	09:08	10:40

Day	Sun		Mer. Pass hh:mm	Moon		Age 15-17 100-96%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Upper Mer.Pass. hh:mm	Lower Mer.Pass. hh:mm	
13	00:34	00:26	12:00	—:—	12:17	
14	00:19	00:12	12:00	00:38	12:59	
15	00:04	00:03	12:00	01:21	13:44	

April 16, 17, 18 UT (Wed., Thu., Fri.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	204°23.6	210°45.2	N01°36.2	82°37.3	N22°35.6	126°52.0	N22°40.3	207°03.8	S03°17.8	Alpheratz	357°34.9	29°13.6											
1	219°26.0	225°47.2	35.8	97°38.8	35.3	141°54.0	40.3	222°06.0	17.7	Ankaa	353°07.3	-42°10.1											
2	234°28.5	240°49.2	35.4	112°40.2	35.1	156°55.9	40.4	237°08.2	17.6	Schedar	349°31.4	56°40.4											
3	249°30.9	255°51.1	•• 35.0	127°41.6	•• 34.8	171°57.9	•• 40.4	252°10.4	•• 17.5	Diphda	348°47.4	-17°50.9											
4	264°33.4	270°53.1	34.6	142°43.0	34.5	186°59.9	40.5	267°12.6	17.4	Achernar	335°20.6	-57°06.5											
5	279°35.9	285°55.1	34.2	157°44.4	34.3	202°01.9	40.5	282°14.8	17.3	Hamal	327°51.3	23°34.8											
6	294°38.3	300°57.0	N01°33.9	172°45.8	N22°34.0	217°03.8	N22°40.6	297°17.0	S03°17.2	Polaris	314°24.2	89°22.3											
7	309°40.8	315°59.0	33.5	187°47.2	33.8	232°05.8	40.6	312°19.2	17.1	Acamar	315°12.0	-40°12.3											
8	324°43.3	331°01.0	33.1	202°48.6	33.5	247°07.8	40.6	327°21.4	17.0	Menkar	314°06.2	4°11.2											
9	339°45.7	346°02.9	•• 32.7	217°50.0	•• 33.2	262°09.8	•• 40.7	342°23.6	•• 16.9	Mirfak	308°28.4	49°57.1											
10	354°48.2	1°04.9	32.3	232°51.4	33.0	277°11.7	40.7	357°25.8	16.8	Aldebaran	290°39.6	16°33.6											
11	9°50.7	16°06.8	32.0	247°52.8	32.7	292°13.7	40.8	12°28.0	16.7	Rigel	281°03.9	-8°10.5											
12	24°53.1	31°08.8	N01°31.6	262°54.2	N22°32.4	307°15.7	N22°40.8	27°30.2	S03°16.6	Capella	280°21.9	46°01.5											
13	39°55.6	46°10.7	31.2	277°55.6	32.1	322°17.7	40.9	42°32.5	16.4	Bellatrix	278°22.9	6°22.3											
14	54°58.1	61°12.6	30.9	292°57.0	31.9	337°19.6	40.9	57°34.7	16.3	Elnath	278°01.8	28°37.8											
15	70°00.5	76°14.6	•• 30.5	307°58.5	•• 31.6	352°21.6	•• 41.0	72°36.9	•• 16.2	Alnilam	275°37.7	-1°11.2											
16	85°03.0	91°16.5	30.1	322°59.9	31.3	7°23.6	41.0	87°39.1	16.1	Betelgeuse	270°52.0	7°24.7											
17	100°05.4	106°18.4	29.8	338°01.3	31.1	22°25.6	41.0	102°41.3	16.0	Canopus	263°52.5	-52°42.8											
18	115°07.9	121°20.4	N01°29.4	353°02.7	N22°30.8	37°27.5	N22°41.1	117°43.5	S03°15.9	Sirius	258°26.2	-16°45.2											
19	130°10.4	136°22.3	29.1	8°04.1	30.5	52°29.5	41.1	132°45.7	15.8	Adhara	255°05.8	-29°00.6											
20	145°12.8	151°24.2	28.7	23°05.5	30.3	67°31.5	41.2	147°47.9	15.7	Procyon	244°50.7	5°09.5											
21	160°15.3	166°26.1	•• 28.4	38°06.9	•• 30.0	82°33.5	•• 41.2	162°50.1	•• 15.6	Pollux	243°17.1	27°58.0											
22	175°17.8	181°28.0	28.0	53°08.3	29.7	97°35.4	41.3	177°52.3	15.5	Avior	234°14.6	-59°35.7											
23	190°20.2	196°29.9	27.7	68°09.7	29.5	112°37.4	41.3	192°54.5	15.4	Suhail	222°46.0	-43°32.3											
Mer.pass. 10:21				ν 2.0' d -0.4' m -4.73				ν 1.4' d -0.3' m 0.71				ν 2.0' d 0.0' m -2.04				ν 2.2' d -0.1' m 1.17							

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	205°22.7	211°31.8	N01°27.3	83°11.1	N22°29.2	127°39.4	N22°41.4	207°56.7	S03°15.3						
1	220°25.2	226°33.7	27.0	98°12.5	28.9	142°41.3	41.4	222°58.9	15.2						
2	235°27.6	241°35.6	26.6	113°13.9	28.7	157°43.3	41.4	238°01.1	15.1						
3	250°30.1	256°37.5	•• 26.3	128°15.3	•• 28.4	172°45.3	•• 41.5	253°03.3	•• 15.0						
4	265°32.5	271°39.4	25.9	143°16.7	28.1	187°47.3	41.5	268°05.5	14.9						
5	280°35.0	286°41.3	25.6	158°18.1	27.8	202°49.2	41.6	283°07.7	14.7						
6	295°37.5	301°43.2	N01°25.3	173°19.5	N22°27.6	217°51.2	N22°41.6	298°09.9	S03°14.6						
7	310°39.9	316°45.1	24.9	188°20.9	27.3	232°53.2	41.7	313°12.1	14.5						
8	325°42.4	331°46.9	24.6	203°22.3	27.0	247°55.2	41.7	328°14.3	14.4						
9	340°44.9	346°48.8	•• 24.3	218°23.7	•• 26.8	262°57.1	•• 41.7	343°16.5	•• 14.3						
10	355°47.3	1°50.7	23.9	233°25.1	26.5	277°59.1	41.8	358°18.8	14.2						
11	10°49.8	16°52.6	23.6	248°26.5	26.2	293°01.1	41.8	13°21.0	14.1						
12	25°52.3	31°54.4	N01°23.3	263°27.9	N22°25.9	308°03.0	N22°41.9	28°23.2	S03°14.0						
13	40°54.7	46°56.3	23.0	278°29.2	25.7	323°05.0	41.9	43°25.4	13.9						
14	55°57.2	61°58.1	22.6	293°30.6	25.4	338°07.0	42.0	58°27.6	13.8						
15	70°59.7	77°00.0	•• 22.3	308°32.0	•• 25.1	353°08.9	•• 42.0	73°29.8	•• 13.7						
16	86°02.1	92°01.8	22.0	323°33.4	24.9	8°10.9	42.1	88°32.0	13.6						
17	101°04.6	107°03.7	21.7	338°34.8	24.6	23°12.9	42.1	103°34.2	13.5						
18	116°07.0	122°05.5	N01°21.4	353°36.2	N22°24.3	38°14.9	N22°42.1	118°36.4	S03°13.4						
19	131°09.5	137°07.4	21.0	8°37.6	24.0	53°16.8	42.2	133°38.6	13.3						
20	146°12.0	152°09.2	20.7	23°39.0	23.8	68°18.8	42.2	148°40.8	13.2						
21	161°14.4	167°11.0	•• 20.4	38°40.4	•• 23.5	83°20.8	•• 42.3	163°43.0	•• 13.1						
22	176°16.9	182°12.9	20.1	53°41.8	23.2	98°22.7	42.3	178°45.2	13.0						
23	191°19.4	197°14.7	19.8	68°43.2	22.9	113°24.7	42.4	193°47.4	12.8						
Mer.pass. 10:17				ν 1.9' d -0.3' m -4.74				ν 2.0' d 0.0' m -2.03				ν 2.2' d -0.1' m 1.17			

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	206°21.8	212°16.5	N01°19.5	83°44.6	N22°22.7	128°26.7	N22°42.4	208°49.6	S03°12.7						
1	221°24.3	227°18.3	19.2	98°46.0	22.4	143°28.6	42.5	223°51.8	12.6						
2	236°26.8	242°20.2	18.9	113°47.4	22.1	158°30.6	42.5	238°54.0	12.5						
3	251°29.2	257°22.0	•• 18.6	128°48.7	•• 21.8	173°32.6	•• 42.5	253°56.3	•• 12.4						
4	266°31.7	272°23.8	18.3	143°50.1	21.6	188°34.5	42.6	268°58.5	12.3						
5	281°34.2	287°25.6	18.0	158°51.5	21.3	203°36.5	42.6	284°00.7	12.2						
6	296°36.6	302°27.4	N01°17.7	173°52.9	N22°21.0	218°38.5	N22°42.7	299°02.9	S03°12.1						
7	311°39.1	317°29.2	17.4	188°54.3	20.7	233°40.4	42.7	314°05.1	12.0						
8	326°41.5	332°31.0	17.1	203°55.7	20.5	248°42.4	42.8	329°07.3	11.9						
9	341°44.0	347°32.8	•• 16.8	218°57.1	•• 20.2	263°44.4	•• 42.8	344°09.5	•• 11.8						
10	356°46.5	2°34.6	16.5	233°58.5	19.9	278°46.3	42.8	359°11.7	11.7						
11	11°48.9	17°36.4	16.2	248°59.9	19.6	293°48.3	42.9	14°13.9	11.6						
12	26°51.4	32°38.2	N01°16.0	264°01.2	N22°19.4	308°50.3	N22°42.9	29°16.1	S03°11.5						
13	41°53.9	47°39.9	15.7	279°02.6	19.1	323°52.2	43.0	44°18.3	11.4						
14	56°56.3	62°41.7	15.4	294°04.0	18.8	338°54.2	43.0	59°20.5	11.3						
15	71°58.8	77°43.5	•• 15.1	309°05.4	•• 18.5	353°56.2	•• 43.1	74°22.7	•• 11.2						
16	87°01.3	92°45.3	14.8	324°06.8	18.3	8°58.1	43.1	89°24.9	11.1						
17	102°03.7	107°47.0	14.6	339°08.2	18.0	24°00.1	43.2	104°27.1	11.0						
18	117°06.2	122°48.8	N01°14.3	354°09.6	N22°17.7	39°02.1	N22°43.2	119°29.4	S03°10.8						
19	132°08.7	137°50.6	14.0	9°10.9	17.4	54°04.0	43.2	134°31.6	10.7						
20	147°11.1	152°52.3	13.7	24°12.3	17.2	69°06.0	43.3	149°33.8	10.6						
21	162°13.6	167°54.1	•• 13.5	39°13.7	•• 16.9	84°08.0	•• 43.3	164°36.0	•• 10.5						
22	177°16.0	182°55.8	13.2	54°15.1	16.6	99°09.9	43.4	179°38.2	10.4						
23	192°18.5	197°57.6	12.9	69°16.5	16.3	114°11.9	43.4	194°40.4	10.3						
Mer.pass. 10:13				ν 1.8' d -0.3' m -4.74				ν 2.0' d 0.0' m -2.03				ν 2.2' d -0.1' m 1.17			

Apr 16 Wed	SHA	Mer.pass
Venus	6°21.7	09:56
Mars	238°13.8	18:28
Jupiter	282°28.4	15:30
Saturn	2°40.3	10:10

Apr 17 Thu	SHA	Mer.pass
Venus	6°09.1	09:53
Mars	237°48.4	18:26
Jupiter	282°16.7	15:27
Saturn	2°34.0	10:07

Apr 18 Fri	SHA	Mer.pass
Venus	5°54.7	09:50
Mars	237°22.7	18:23
Jupiter	282°04.8	15:24
Saturn	2°27.8	10:03

Horizontal parallax		
Venus:	0.4	
Mars:	0.1	

h Sun							Moon						
Wed	GHA	Dec	GHA	ν	Dec	d	HP						
0	180°02.6	N10°08.1	329°01.9	12.0'	S24°20.2	7.9'	54.1'						
1	195°02.7	09.0	343°32.9	11.9'	24°28.2	7.8'	54.1'						
2	210°02.8	09.8	358°03.8	11.8'	24°36.0	7.7'	54.1'						
3	225°03.0	· · 10.7	12°34.6	11.8'	24°43.7	7.6'	54.1'						
4	240°03.1	11.6	27°05.4	11.7'	24°51.3	7.5'	54.1'						
5	255°03.3	12.5	41°36.1	11.6'	24°58.8	7.4'	54.1'						
6	270°03.4	N10°13.4	56°06.8	11.6'	S25°06.2	7.3'	54.2'						
7	285°03.6	14.3	70°37.3	11.5'	25°13.5	7.2'	54.2'						
8	300°03.7	15.1	85°07.8	11.4'	25°20.7	7.1'	54.2'						
9	315°03.9	· · 16.0	99°38.2	11.3'	25°27.7	6.9'	54.2'						
10	330°04.0	16.9	114°08.6	11.3'	25°34.7	6.8'	54.2'						
11	345°04.2	17.8	128°38.8	11.2'	25°41.5	6.7'	54.2'						
12	0°04.3	N10°18.7	143°09.0	11.1'	S25°48.2	6.6'	54.2'						
13	15°04.5	19.6	157°39.2	11.1'	25°54.8	6.5'	54.2'						
14	30°04.6	20.4	172°09.2	11.0'	26°01.3	6.4'	54.2'						
15	45°04.8	· · 21.3	186°39.2	10.9'	26°07.6	6.2'	54.3'						
16	60°04.9	22.2	201°09.1	10.9'	26°13.8	6.1'	54.3'						
17	75°05.0	23.1	215°39.0	10.8'	26°20.0	6.0'	54.3'						
18	90°05.2	N10°24.0	230°08.8	10.7'	S26°25.9	5.9'	54.3'						
19	105°05.3	24.9	244°38.5	10.6'	26°31.8	5.7'	54.3'						
20	120°05.5	25.7	259°08.1	10.6'	26°37.6	5.6'	54.3'						
21	135°05.6	· · 26.6	273°37.7	10.5'	26°43.2	5.5'	54.3'						
22	150°05.8	27.5	288°07.2	10.5'	26°48.7	5.4'	54.3'						
23	165°05.9	28.4	302°36.7	10.4'	26°54.1	5.2'	54.4'						
SD = 15.9'							d = 0.9'	SD = 14.8'					
Thu	GHA	Dec	GHA	ν	Dec	d	HP						
0	180°06.1	N10°29.3	317°06.1	10.3'	S26°59.3	5.1'	54.4'						
1	195°06.2	30.1	331°35.4	10.3'	27°04.4	5.0'	54.4'						
2	210°06.3	31.0	346°04.6	10.2'	27°09.4	4.9'	54.4'						
3	225°06.5	· · 31.9	0°33.8	10.1'	27°14.3	4.7'	54.4'						
4	240°06.6	32.8	15°03.0	10.1'	27°19.0	4.6'	54.4'						
5	255°06.8	33.6	29°32.0	10.0'	27°23.6	4.5'	54.4'						
6	270°06.9	N10°34.5	44°01.0	10.0'	S27°28.1	4.3'	54.4'						
7	285°07.1	35.4	58°30.0	9.9'	27°32.4	4.2'	54.5'						
8	300°07.2	36.3	72°58.9	9.8'	27°36.6	4.1'	54.5'						
9	315°07.4	· · 37.2	87°27.7	9.8'	27°40.7	3.9'	54.5'						
10	330°07.5	38.0	101°56.5	9.7'	27°44.7	3.8'	54.5'						
11	345°07.6	38.9	116°25.2	9.7'	27°48.5	3.7'	54.5'						
12	0°07.8	N10°39.8	130°53.9	9.6'	S27°52.1	3.5'	54.5'						
13	15°07.9	40.7	145°22.5	9.6'	27°55.7	3.4'	54.6'						
14	30°08.1	41.5	159°51.1	9.5'	27°59.1	3.3'	54.6'						
15	45°08.2	· · 42.4	174°19.6	9.5'	28°02.3	3.1'	54.6'						
16	60°08.3	43.3	188°48.0	9.4'	28°05.5	3.0'	54.6'						
17	75°08.5	44.2	203°16.4	9.4'	28°08.4	2.8'	54.6'						
18	90°08.6	N10°45.0	217°44.8	9.3'	S28°11.3	2.7'	54.6'						
19	105°08.8	45.9	232°13.1	9.3'	28°14.0	2.6'	54.7'						
20	120°08.9	46.8	246°41.4	9.2'	28°16.6	2.4'	54.7'						
21	135°09.1	· · 47.7	261°09.6	9.2'	28°19.0	2.3'	54.7'						
22	150°09.2	48.5	275°37.7	9.1'	28°21.3	2.1'	54.7'						
23	165°09.3	49.4	290°05.9	9.1'	28°23.4	2.0'	54.7'						
SD = 15.9'							d = 0.9'	SD = 14.8'					
Fri	GHA	Dec	GHA	ν	Dec	d	HP						
0	180°09.5	N10°50.3	304°34.0	9.0'	S28°25.4	1.9'	54.7'						
1	195°09.6	51.1	319°02.0	9.0'	28°27.3	1.7'	54.8'						
2	210°09.8	52.0	333°30.0	9.0'	28°29.0	1.6'	54.8'						
3	225°09.9	· · 52.9	347°58.0	8.9'	28°30.5	1.4'	54.8'						
4	240°10.0	53.8	2°25.9	8.9'	28°32.0	1.3'	54.8'						
5	255°10.2	54.6	16°53.8	8.9'	28°33.3	1.1'	54.8'						
6	270°10.3	N10°55.5	31°21.6	8.8'	S28°34.4	1.0'	54.9'						
7	285°10.4	56.4	45°49.5	8.8'	28°35.4	0.8'	54.9'						
8	300°10.6	57.2	60°17.2	8.8'	28°36.2	0.7'	54.9'						
9	315°10.7	· · 58.1	74°45.0	8.7'	28°36.9	0.5'	54.9'						
10	330°10.9	59.0	89°12.7	8.7'	28°37.5	0.4'	54.9'						
11	345°11.0	10°59.8	103°40.4	8.7'	28°37.9	0.3'	55.0'						
12	0°11.1	N11°00.7	118°08.1	8.6'	S28°38.1	0.1'	55.0'						
13	15°11.3	01.6	132°35.7	8.6'	28°38.2	-0.0'	55.0'						
14	30°11.4	02.5	147°03.3	8.6'	28°38.2	-0.2'	55.0'						
15	45°11.6	· · 03.3	161°30.9	8.6'	28°38.0	-0.3'	55.0'						
16	60°11.7	04.2	175°58.5	8.6'	28°37.6	-0.5'	55.1'						
17	75°11.8	05.1	190°26.1	8.5'	28°37.1	-0.6'	55.1'						
18	90°12.0	N11°05.9	204°53.6	8.5'	S28°36.5	-0.8'	55.1'						
19	105°12.1	06.8	219°21.1	8.5'	28°35.7	-0.9'	55.1'						
20	120°12.2	07.7	233°48.6	8.5'	28°34.8	-1.1'	55.2'						
21	135°12.4	· · 08.5	248°16.1	8.5'	28°33.7	-1.2'	55.2'						
22	150°12.5	09.4	262°43.5	8.5'	28°32.4	-1.4'	55.2'						
23	165°12.6	10.2	277°11.0	8.4'	28°31.0	-1.5'	55.2'						
SD = 15.9'							d = 0.9'	SD = 14.9'					

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	01:35	03:26	20:37	22:35	—:—
N 70°	—:—	02:19	03:45	20:17	21:46	—:—
68°	—:—	02:47	04:00	20:02	21:16	—:—
66°	01:22	03:09	04:11	19:50	20:54	22:46
64°	02:01	03:25	04:21	19:40	20:37	22:03
62°	02:27	03:39	04:30	19:31	20:23	21:36
60°	02:47	03:50	04:37	19:24	20:11	21:15
N 58°	03:03	04:00	04:43	19:17	20:01	20:59
56°	03:16	04:08	04:49	19:11	19:52	20:45
54°	03:28	04:16	04:54	19:06	19:45	20:33
52°	03:37	04:22	04:59	19:01	19:38	20:23
50°	03:46	04:28	05:03	18:57	19:32	20:15
45°	04:03	04:41	05:12	18:48	19:19	19:57
N 40°	04:17	04:51	05:19	18:41	19:09	19:43
35°	04:28	04:59	05:25	18:34	19:00	19:32
30°	04:37	05:06	05:31	18:29	18:53	19:22
20°	04:51	05:18	05:40	18:19	18:41	19:08
N 10°	05:02	05:27	05:49	18:11	18:32	18:57
0°	05:11	05:35	05:56	18:03	18:24	18:48
S 10°	05:18	05:42	06:04	17:55	18:17	18:41
20°	05:23	05:49	06:12	17:47	18:10	18:35
30°	05:28	05:56	06:20	17:38	18:02	18:30
35°	05:30	06:00	06:25	17:33	17:59	18:28
40°	05:32	06:04	06:31	17:27	17:55	18:26
45°	05:34	06:08	06:38	17:21	17:50	18:25
S 50°	05:35	06:13	06:46	17:13	17:45	18:23
52°	05:36	06:15	06:49	17:09	17:43	18:23
54°	05:36	06:17	06:53	17:05	17:41	18:22
56°	05:36	06:19	06:58	17:00	17:39	18:22
58°	05:36	06:22	07:03	16:55	17:36	18:21
S 60°	05:37	06:25	07:08	16:50	17:33	18:21

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	■	■	■	■	■	■
64°	00:42	■	■	03:23	■	■
62°		01:58	■	04:07	03:51	■
60°		01:05	02:30	04:37	04:44	05:08
N 58°		00:33	01:48	05:00	05:17	05:50
56°		00:09	01:20	05:18	05:41	06:18
54°	23:50		00:58	05:34	06:00	06:41
52°	23:34		00:40	05:47	06:17	06:59
50°	23:20		00:24	05:59	06:31	07:14
45°	22:52	23:53		06:24	07:00	07:45
N 40°	22:30	23:29		06:43	07:22	08:10
35°	22:11	23:10		07:00	07:41	08:30
30°	21:56	22:53	23:47	07:14	07:57	08:47
20°	21:29	22:24	23:18	07:39	08:25	09:15
N 10°	21:07	22:00	22:54	08:00	08:48	09:40
0°	20:45	21:37	22:31	08:20	09:10	10:03
S 10°	20:24	21:15	22:08	08:40	09:32	10:26
20°	20:02	20:50	21:43	09:01	09:56	10:50
30°	19:36	20:22	21:15	09:26	10:23	11:19
35°	19:21	20:05	20:58	09:40	10:40	11:36
40°	19:03	19:46	20:38	09:57	10:59	11:56
45°	18:42	19:23	20:14	10:18	11:22	12:20
S 50°	18:15	18:53	19:43	10:44	11:51	12:51
52°	18:02	18:38	19:28	10:56	12:06	13:06
54°	17:47	18:21	19:10	11:11	12:23	13:24
56°	17:30	18:00	18:48	11:28	12:44	13:46
58°	17:09	17:34	18:20	11:48	13:10	14:14
S 60°	16:43	16:58	17:39	12:14	13:46	14:55

April 19, 20, 21 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	207°21.0	212°59.3	N01°12.7	84°17.9	N22°16.0	129°13.9	N22°43.5	209°42.6	S03°10.2	Alpheratz	357°34.9	29°13.6
1	222°23.4	228°01.1	12.4	99°19.2	15.8	144°15.8	43.5	224°44.8	10.1	Ankaa	353°07.3	-42°10.1
2	237°25.9	243°02.8	12.2	114°20.6	15.5	159°17.8	43.5	239°47.0	10.0	Schedar	349°31.4	56°40.4
3	252°28.4	258°04.5	•• 11.9	129°22.0	•• 15.2	174°19.8	•• 43.6	254°49.2	•• 09.9	Diphda	348°47.4	-17°50.9
4	267°30.8	273°06.3	11.6	144°23.4	14.9	189°21.7	43.6	269°51.4	09.8	Achernar	335°20.6	-57°06.5
5	282°33.3	288°08.0	11.4	159°24.8	14.7	204°23.7	43.7	284°53.6	09.7	Hamal	327°51.3	23°34.8
6	297°35.8	303°09.7	N01°11.1	174°26.1	N22°14.4	219°25.7	N22°43.7	299°55.8	S03°09.6	Polaris	314°24.4	89°22.3
7	312°38.2	318°11.5	10.9	189°27.5	14.1	234°27.6	43.8	314°58.1	09.5	Acamar	315°12.0	-40°12.3
8	327°40.7	333°13.2	10.6	204°28.9	13.8	249°29.6	43.8	330°00.3	09.4	Menkar	314°06.2	4°11.2
9	342°43.1	348°14.9	•• 10.4	219°30.3	•• 13.5	264°31.5	•• 43.8	345°02.5	•• 09.3	Mirfak	308°28.4	49°57.1
10	357°45.6	3°16.6	10.1	234°31.7	13.3	279°33.5	43.9	0°04.7	09.2	Aldebaran	290°39.6	16°33.6
11	12°48.1	18°18.3	09.9	249°33.0	13.0	294°35.5	43.9	15°06.9	09.1	Rigel	281°03.9	-8°10.5
12	27°50.5	33°20.0	N01°09.6	264°34.4	N22°12.7	309°37.4	N22°44.0	30°09.1	S03°09.0	Capella	280°21.9	46°01.5
13	42°53.0	48°21.7	09.4	279°35.8	12.4	324°39.4	44.0	45°11.3	08.9	Bellatrix	278°22.9	6°22.3
14	57°55.5	63°23.4	09.2	294°37.2	12.1	339°41.4	44.1	60°13.5	08.8	Elnath	278°01.8	28°37.8
15	72°57.9	78°25.1	•• 08.9	309°38.5	•• 11.9	354°43.3	•• 44.1	75°15.7	•• 08.7	Alnilam	275°37.7	-1°11.2
16	88°00.4	93°26.8	08.7	324°39.9	11.6	9°45.3	44.2	90°17.9	08.5	Betelgeuse	270°52.0	7°24.7
17	103°02.9	108°28.5	08.5	339°41.3	11.3	24°47.2	44.2	105°20.1	08.4	Canopus	263°52.5	-52°42.8
18	118°05.3	123°30.2	N01°08.2	354°42.7	N22°11.0	39°49.2	N22°44.2	120°22.4	S03°08.3	Sirius	258°26.2	-16°45.2
19	133°07.8	138°31.9	08.0	9°44.0	10.7	54°51.2	44.3	135°24.6	08.2	Adhara	255°05.8	-29°00.6
20	148°10.3	153°33.6	07.8	24°45.4	10.4	69°53.1	44.3	150°26.8	08.1	Procyon	244°50.7	5°09.5
21	163°12.7	168°35.3	•• 07.5	39°46.8	•• 10.2	84°55.1	•• 44.4	165°29.0	•• 08.0	Pollux	243°17.1	27°58.0
22	178°15.2	183°36.9	07.3	54°48.2	09.9	99°57.0	44.4	180°31.2	07.9	Avior	234°14.6	-59°35.7
23	193°17.6	198°38.6	07.1	69°49.5	09.6	114°59.0	44.5	195°33.4	07.8	Suhail	222°46.1	-43°32.3
Mer.pass. 10:09		ν 1.7' d -0.3' m -4.75		ν 1.4' d -0.3' m 0.76		ν 2.0' d 0.0' m -2.03		ν 2.2' d -0.1' m 1.17				

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	208°20.1	213°40.3	N01°06.9	84°50.9	N22°09.3	130°01.0	N22°44.5	210°35.6	S03°07.7		
1	223°22.6	228°42.0	06.6	99°52.3	09.0	145°02.9	44.5	225°37.8	07.6		
2	238°25.0	243°43.6	06.4	114°53.7	08.8	160°04.9	44.6	240°40.0	07.5		
3	253°27.5	258°45.3	•• 06.2	129°55.0	•• 08.5	175°06.9	•• 44.6	255°42.2	•• 07.4		
4	268°30.0	273°46.9	06.0	144°56.4	08.2	190°08.8	44.7	270°44.4	07.3		
5	283°32.4	288°48.6	05.8	159°57.8	07.9	205°10.8	44.7	285°46.7	07.2		
6	298°34.9	303°50.3	N01°05.6	174°59.1	N22°07.6	220°12.7	N22°44.8	300°48.9	S03°07.1		
7	313°37.4	318°51.9	05.4	190°00.5	07.3	235°14.7	44.8	315°51.1	07.0		
8	328°39.8	333°53.6	05.1	205°01.9	07.1	250°16.7	44.8	330°53.3	06.9		
9	343°42.3	348°55.2	•• 04.9	220°03.3	•• 06.8	265°18.6	•• 44.9	345°55.5	•• 06.8		
10	358°44.8	3°56.8	04.7	235°04.6	06.5	280°20.6	44.9	0°57.7	06.7		
11	13°47.2	18°58.5	04.5	250°06.0	06.2	295°22.5	45.0	15°59.9	06.6		
12	28°49.7	34°00.1	N01°04.3	265°07.4	N22°05.9	310°24.5	N22°45.0	31°02.1	S03°06.5		
13	43°52.1	49°01.7	04.1	280°08.7	05.6	325°26.4	45.1	46°04.3	06.4		
14	58°54.6	64°03.4	03.9	295°10.1	05.3	340°28.4	45.1	61°06.5	06.3		
15	73°57.1	79°05.0	•• 03.7	310°11.5	•• 05.1	355°30.4	•• 45.1	76°08.8	•• 06.2		
16	88°59.5	94°06.6	03.5	325°12.8	04.8	10°32.3	45.2	91°11.0	06.1		
17	104°02.0	109°08.2	03.3	340°14.2	04.5	25°34.3	45.2	106°13.2	06.0		
18	119°04.5	124°09.9	N01°03.2	355°15.6	N22°04.2	40°36.2	N22°45.3	121°15.4	S03°05.8		
19	134°06.9	139°11.5	03.0	10°16.9	03.9	55°38.2	45.3	136°17.6	05.7		
20	149°09.4	154°13.1	02.8	25°18.3	03.6	70°40.2	45.4	151°19.8	05.6		
21	164°11.9	169°14.7	•• 02.6	40°19.7	•• 03.3	85°42.1	•• 45.4	166°22.0	•• 05.5		
22	179°14.3	184°16.3	02.4	55°21.0	03.1	100°44.1	45.4	181°24.2	05.4		
23	194°16.8	199°17.9	02.2	70°22.4	02.8	115°46.0	45.5	196°26.4	05.3		
Mer.pass. 10:05		ν 1.7' d -0.2' m -4.75		ν 1.4' d -0.3' m 0.78		ν 2.0' d 0.0' m -2.02		ν 2.2' d -0.1' m 1.17			

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	209°19.3	214°19.5	N01°02.0	85°23.8	N22°02.5	130°48.0	N22°45.5	211°28.6	S03°05.2		
1	224°21.7	229°21.1	01.9	100°25.1	02.2	145°49.9	45.6	226°30.9	05.1		
2	239°24.2	244°22.7	01.7	115°26.5	01.9	160°51.9	45.6	241°33.1	05.0		
3	254°26.6	259°24.3	•• 01.5	130°27.9	•• 01.6	175°53.8	•• 45.7	256°35.3	•• 04.9		
4	269°29.1	274°25.9	01.3	145°29.2	01.3	190°55.8	45.7	271°37.5	04.8		
5	284°31.6	289°27.4	01.2	160°30.6	01.1	205°57.8	45.7	286°39.7	04.7		
6	299°34.0	304°29.0	N01°01.0	175°31.9	N22°00.8	220°59.7	N22°45.8	301°41.9	S03°04.6		
7	314°36.5	319°30.6	00.8	190°33.3	00.5	236°01.7	45.8	316°44.1	04.5		
8	329°39.0	334°32.2	00.7	205°34.7	22°00.2	251°03.6	45.9	331°46.3	04.4		
9	344°41.4	349°33.8	•• 00.5	220°36.0	21°59.9	266°05.6	•• 45.9	346°48.5	•• 04.3		
10	359°43.9	4°35.3	00.3	235°37.4	59.6	281°07.5	46.0	1°50.8	04.2		
11	14°46.4	19°36.9	00.2	250°38.8	59.3	296°09.5	46.0	16°53.0	04.1		
12	29°48.8	34°38.5	N01°00.0	265°40.1	N21°59.0	311°11.4	N22°46.0	31°55.2	S03°04.0		
13	44°51.3	49°40.0	00°59.8	280°41.5	58.7	326°13.4	46.1	46°57.4	03.9		
14	59°53.7	64°41.6	59.7	295°42.8	58.5	341°15.4	46.1	61°59.6	03.8		
15	74°56.2	79°43.1	•• 59.5	310°44.2	•• 58.2	356°17.3	•• 46.2	77°01.8	•• 03.7		
16	89°58.7	94°44.7	59.4	325°45.5	57.9	11°19.3	46.2	92°04.0	03.6		
17	105°01.1	109°46.2	59.2	340°46.9	57.6	26°21.2	46.3	107°06.2	03.5		
18	120°03.6	124°47.8	N00°59.1	355°48.3	N21°57.3	41°23.2	N22°46.3	122°08.5	S03°03.4		
19	135°06.1	139°49.3	58.9	10°49.6	57.0	56°25.1	46.3	137°10.7	03.3		
20	150°08.5	154°50.9	58.8	25°51.0	56.7	71°27.1	46.4	152°12.9	03.2		
21	165°11.0	169°52.4	•• 58.6	40°52.3	•• 56.4	86°29.0	•• 46.4	167°15.1	•• 03.1		
22	180°13.5	184°53.9	58.5	55°53.7	56.1	101°31.0	46.5	182°17.3	03.0		
23	195°15.9	199°55.5	58.3	70°55.1	55.8	116°32.9	46.5	197°19.5	02.9		
Mer.pass. 10:01		ν 1.6' d -0.2' m -4.76		ν 1.4' d -0.3' m 0.79		ν 2.0' d 0.0' m -2.02		ν 2.2' d -0.1' m 1.17			

Apr 19 Sat			SHA	Mer.pass
Venus			5°38.4	09:47
Mars			236°56.9	18:21
Jupiter			281°52.9	15:21
Saturn			2°21.6	10:00
Apr 20 Sun			SHA	Mer.pass
Venus			5°20.2	09:44
Mars			236°30.8	18:19
Jupiter			281°40.9	15:18
Saturn			2°15.5	09:56
Apr 21 Mon			SHA	Mer.pass
Venus			5°00.2	09:42
Mars			236°04.5	18:17
Jupiter			281°28.7	15:15
Saturn			2°09.4	09:53
Horizontal parallax				
Venus:			0.4	
Mars:			0.1	

h	Sun			Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°12.8	N11°11.1	291°38.4	8.4'	S28°29.5	-1.7'	55.3'
1	195°12.9	12.0	306°05.9	8.4'	28°27.8	-1.8'	55.3'
2	210°13.1	12.8	320°33.3	8.4'	28°25.9	-2.0'	55.3'
3	225°13.2	· · 13.7	335°00.7	8.4'	28°23.9	-2.1'	55.3'
4	240°13.3	14.6	349°28.1	8.4'	28°21.8	-2.3'	55.4'
5	255°13.5	15.4	3°55.5	8.4'	28°19.5	-2.5'	55.4'
6	270°13.6	N11°16.3	18°22.9	8.4'	S28°17.0	-2.6'	55.4'
7	285°13.7	17.2	32°50.3	8.4'	28°14.4	-2.8'	55.4'
8	300°13.9	18.0	47°17.7	8.4'	28°11.7	-2.9'	55.5'
9	315°14.0	· · 18.9	61°45.1	8.4'	28°08.8	-3.1'	55.5'
10	330°14.1	19.7	76°12.5	8.4'	28°05.7	-3.2'	55.5'
11	345°14.3	20.6	90°39.9	8.4'	28°02.5	-3.4'	55.5'
12	0°14.4	N11°21.5	105°07.3	8.4'	S27°59.1	-3.5'	55.6'
13	15°14.5	22.3	119°34.7	8.4'	27°55.6	-3.7'	55.6'
14	30°14.7	23.2	134°02.1	8.4'	27°52.0	-3.8'	55.6'
15	45°14.8	· · 24.0	148°29.5	8.4'	27°48.1	-4.0'	55.6'
16	60°14.9	24.9	162°57.0	8.4'	27°44.2	-4.1'	55.7'
17	75°15.1	25.8	177°24.4	8.5'	27°40.1	-4.3'	55.7'
18	90°15.2	N11°26.6	191°51.9	8.5'	S27°35.8	-4.4'	55.7'
19	105°15.3	27.5	206°19.3	8.5'	27°31.4	-4.6'	55.8'
20	120°15.5	28.3	220°46.8	8.5'	27°26.8	-4.7'	55.8'
21	135°15.6	· · 29.2	235°14.3	8.5'	27°22.1	-4.9'	55.8'
22	150°15.7	30.1	249°41.8	8.5'	27°17.2	-5.0'	55.8'
23	165°15.8	30.9	264°09.3	8.5'	27°12.2	-5.2'	55.9'
SD = 15.9'			SD = 15.1'				
$d = 0.9'$							

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	180°16.0	N11°31.8	278°36.9	8.6'	S27°07.0	-5.3'	55.9'
1	195°16.1	32.6	293°04.4	8.6'	27°01.7	-5.5'	55.9'
2	210°16.2	33.5	307°32.0	8.6'	26°56.2	-5.6'	56.0'
3	225°16.4	· · 34.3	321°59.6	8.6'	26°50.6	-5.8'	56.0'
4	240°16.5	35.2	336°27.3	8.6'	26°44.9	-5.9'	56.0'
5	255°16.6	36.1	350°54.9	8.7'	26°39.0	-6.1'	56.0'
6	270°16.8	N11°36.9	5°22.6	8.7'	S26°32.9	-6.2'	56.1'
7	285°16.9	37.8	19°50.3	8.7'	26°26.7	-6.4'	56.1'
8	300°17.0	38.6	34°18.0	8.7'	26°20.3	-6.5'	56.1'
9	315°17.2	· · 39.5	48°45.7	8.8'	26°13.8	-6.6'	56.2'
10	330°17.3	40.3	63°13.5	8.8'	26°07.2	-6.8'	56.2'
11	345°17.4	41.2	77°41.3	8.8'	26°00.4	-6.9'	56.2'
12	0°17.5	N11°42.0	92°09.2	8.9'	S25°53.5	-7.1'	56.3'
13	15°17.7	42.9	106°37.0	8.9'	25°46.4	-7.2'	56.3'
14	30°17.8	43.7	121°04.9	8.9'	25°39.1	-7.4'	56.3'
15	45°17.9	· · 44.6	135°32.8	9.0'	25°31.8	-7.5'	56.4'
16	60°18.1	45.4	150°00.8	9.0'	25°24.3	-7.7'	56.4'
17	75°18.2	46.3	164°28.8	9.0'	25°16.6	-7.8'	56.4'
18	90°18.3	N11°47.1	178°56.8	9.1'	S25°08.8	-7.9'	56.5'
19	105°18.4	48.0	193°24.9	9.1'	25°00.9	-8.1'	56.5'
20	120°18.6	48.8	207°53.0	9.1'	24°52.8	-8.2'	56.5'
21	135°18.7	· · 49.7	222°21.1	9.2'	24°44.6	-8.4'	56.6'
22	150°18.8	50.5	236°49.3	9.2'	24°36.2	-8.5'	56.6'
23	165°18.9	51.4	251°17.5	9.2'	24°27.7	-8.6'	56.6'
SD = 15.9'			SD = 15.2'				
$d = 0.9'$							

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	180°19.1	N11°52.2	265°45.7	9.3'	S24°19.1	-8.8'	56.7'
1	195°19.2	53.1	280°14.0	9.3'	24°10.3	-8.9'	56.7'
2	210°19.3	53.9	294°42.3	9.3'	24°01.4	-9.1'	56.7'
3	225°19.5	· · 54.8	309°10.6	9.4'	23°52.3	-9.2'	56.8'
4	240°19.6	55.6	323°39.0	9.4'	23°43.1	-9.3'	56.8'
5	255°19.7	56.5	338°07.4	9.5'	23°33.8	-9.5'	56.8'
6	270°19.8	N11°57.3	352°35.9	9.5'	S23°24.3	-9.6'	56.9'
7	285°20.0	58.2	7°04.4	9.5'	23°14.7	-9.7'	56.9'
8	300°20.1	59.0	21°32.9	9.6'	23°05.0	-9.9'	56.9'
9	315°20.2	11°59.9	36°01.5	9.6'	22°55.1	-10.0'	57.0'
10	330°20.3	12°00.7	50°30.1	9.7'	22°45.2	-10.1'	57.0'
11	345°20.5	01.6	64°58.8	9.7'	22°35.0	-10.3'	57.1'
12	0°20.6	N12°02.4	79°27.5	9.7'	S22°24.8	-10.4'	57.1'
13	15°20.7	03.3	93°56.2	9.8'	22°14.4	-10.5'	57.1'
14	30°20.8	04.1	108°25.0	9.8'	22°03.9	-10.6'	57.2'
15	45°20.9	· · 04.9	122°53.9	9.9'	21°53.2	-10.8'	57.2'
16	60°21.1	05.8	137°22.7	9.9'	21°42.5	-10.9'	57.2'
17	75°21.2	06.6	151°51.6	9.9'	21°31.6	-11.0'	57.3'
18	90°21.3	N12°07.5	166°20.6	10.0'	S21°20.5	-11.1'	57.3'
19	105°21.4	08.3	180°49.5	10.0'	21°09.4	-11.3'	57.3'
20	120°21.6	09.2	195°18.6	10.1'	20°58.1	-11.4'	57.4'
21	135°21.7	· · 10.0	209°47.6	10.1'	20°46.7	-11.5'	57.4'
22	150°21.8	10.8	224°16.7	10.1'	20°35.2	-11.6'	57.5'
23	165°21.9	11.7	238°45.9	10.2'	20°23.5	-11.8'	57.5'
SD = 15.9'			SD = 15.5'				
$d = 0.8'$							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	00:52	03:08	20:54	23:30	—:—
N 70°	—:—	01:55	03:30	20:32	22:09	—:—
68°	—:—	02:30	03:46	20:14	21:33	—:—
66°	00:45	02:54	04:00	20:00	21:07	23:34
64°	01:41	03:13	04:11	19:49	20:48	22:23
62°	02:12	03:28	04:20	19:39	20:32	21:50
60°	02:34	03:40	04:28	19:31	20:19	21:27
N 58°	02:52	03:51	04:35	19:24	20:08	21:08
56°	03:07	04:00	04:42	19:17	19:59	20:53
54°	03:19	04:08	04:47	19:12	19:51	20:41
52°	03:29	04:16	04:52	19:07	19:43	20:30
50°	03:39	04:22	04:57	19:02	19:37	20:20
45°	03:58	04:35	05:07	18:52	19:23	20:01
N 40°	04:12	04:46	05:15	18:44	19:12	19:46
35°	04:24	04:55	05:22	18:37	19:03	19:34
30°	04:34	05:03	05:28	18:30	18:55	19:25
20°	04:49	05:15	05:38	18:20	18:42	19:09
N 10°	05:01	05:26	05:47	18:11	18:32	18:57
0°	05:10	05:34	05:55	18:02	18:23	18:48
S 10°	05:18	05:42	06:04	17:54	18:15	18:40
20°	05:24	05:50	06:12	17:45	18:07	18:33
30°	05:30	05:58	06:22	17:35	17:59	18:27
35°	05:32	06:02	06:28	17:29	17:55	18:25
40°	05:35	06:07	06:34	17:23	17:51	18:22
45°	05:37	06:11	06:41	17:16	17:46	18:20
S 50°	05:39	06:17	06:50	17:07	17:40	18:18
52°	05:40	06:20	06:54	17:03	17:37	18:17
54°	05:41	06:22	06:59	16:58	17:35	18:16
56°	05:42	06:25	07:04	16:53	17:32	18:15
58°	05:42	06:28	07:09	16:48	17:28	18:14
S 60°	05:43	06:32	07:15	16:41	17:25	18:13

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	■	■	05:43	■	■	07:32
64°	■	■	04:44	■	■	08:30
62°	■	04:23	04:10	■	07:00	09:04
60°	03:21	03:39	03:45	06:09	07:44	09:28
N 58°	02:41	03:10	03:25	06:49	08:12	09:47
56°	02:13	02:48	03:09	07:17	08:35	10:03
54°	01:51	02:29	02:54	07:38	08:53	10:17
52°	01:33	02:14	02:42	07:56	09:08	10:29
50°	01:18	02:00	02:31	08:12	09:21	10:39
45°	00:47	01:32	02:08	08:42	09:48	11:01
N 40°	00:23	01:10	01:50	09:06	10:10	11:18
35°	00:04	00:52	01:34	09:25	10:27	11:33
30°	· · ·	00:36	01:20	09:42	10:42	11:46
20°	· · ·	00:09	00:57	10:10	11:08	12:07
N 10°	23:46	· · ·	00:37	10:34	11:30	12:26
0°	23:25	· · ·	00:18	10:57	11:51	12:43
S 10°	23:03	23:59	· · ·	11:19	12:11	13:00
20°	22:40	23:39	· · ·	11:43	12:33	13:18
30°	22:13	23:15	· · ·	12:11	12:58	13:39
35°	21:57	23:01	· · ·	12:27	13:12	13:51
40°	21:38	22:45	23:56	12:46	13:29	14:05
45°	21:16	22:26	23:41	13:09	13:49	14:22
S 50°	20:47	22:01	23:22	13:39	14:15	14:42
52°	20:33	21:50	23:13	13:53	14:27	14:51
54°	20:16	21:36	23:03	14:10	14:41	15:02
56°	19:57	21:21	22:52	14:30	14:57	15:14
58°	19:32	21:02	22:39	14:54	15:16	15:27
S 60°	18:59	20:39	22:24	15:28	15:39	15:43

Day	Sun		Mer.	Moon		Age
	Eqn.of Time	12 ^h	Pass	Mer.Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	21-23
				hh:mm	hh:mm	70-51%
19	00:51	00:58	11:59	04:44	17:11	
20	01:04	01:10	11:59	05:38	18:04	
21	01:16	01:22	11:59	06:31	18:57	

April 22, 23, 24 UT (Tue., Wed., Thu.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Tue	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec			
0	210°18.4	214°57.0	N00°58.2	85°56.4	N21°55.6	131°34.9	N22°46.6	212°21.7	S03°02.8				Alpheratz	357°34.9	29°13.6		
1	225°20.9	229°58.5	58.1	100°57.8	55.3	146°36.8	46.6	227°23.9	02.7				Ankaa	353°07.3	-42°10.1		
2	240°23.3	245°00.0	57.9	115°59.1	55.0	161°38.8	46.6	242°26.1	02.6				Schedar	349°31.4	56°40.4		
3	255°25.8	260°01.6	•• 57.8	131°00.5	•• 54.7	176°40.8	•• 46.7	257°28.4	•• 02.5				Diphda	348°47.4	-17°50.9		
4	270°28.2	275°03.1	57.6	146°01.8	54.4	191°42.7	46.7	272°30.6	02.4				Achernar	335°20.6	-57°06.5		
5	285°30.7	290°04.6	57.5	161°03.2	54.1	206°44.7	46.8	287°32.8	02.3				Hamal	327°51.3	23°34.8		
6	300°33.2	305°06.1	N00°57.4	176°04.5	N21°53.8	221°46.6	N22°46.8	302°35.0	S03°02.2				Polaris	314°24.7	89°22.3		
7	315°35.6	320°07.6	57.3	191°05.9	53.5	236°48.6	46.8	317°37.2	02.0				Acamar	315°12.0	-40°12.3		
8	330°38.1	335°09.1	57.1	206°07.2	53.2	251°50.5	46.9	332°39.4	01.9				Menkar	314°06.2	4°11.3		
9	345°40.6	350°10.6	•• 57.0	221°08.6	•• 52.9	266°52.5	•• 46.9	347°41.6	•• 01.8				Mirfak	308°28.4	49°57.1		
10	0°43.0	5°12.1	56.9	236°10.0	52.6	281°54.4	47.0	2°43.9	01.7				Aldebaran	290°39.6	16°33.6		
11	15°45.5	20°13.6	56.7	251°11.3	52.3	296°56.4	47.0	17°46.1	01.6				Rigel	281°03.9	-8°10.4		
12	30°48.0	35°15.1	N00°56.6	266°12.7	N21°52.0	311°58.3	N22°47.1	32°48.3	S03°01.5				Capella	280°21.9	46°01.5		
13	45°50.4	50°16.6	56.5	281°14.0	51.8	327°00.3	47.1	47°50.5	01.4				Bellatrix	278°22.9	6°22.3		
14	60°52.9	65°18.1	56.4	296°15.4	51.5	342°02.2	47.1	62°52.7	01.3				Elnath	278°01.8	28°37.8		
15	75°55.4	80°19.6	•• 56.3	311°16.7	•• 51.2	357°04.2	•• 47.2	77°54.9	•• 01.2				Alnilam	275°37.7	-1°11.2		
16	90°57.8	95°21.1	56.2	326°18.1	50.9	12°06.1	47.2	92°57.1	01.1				Betelgeuse	290°52.0	7°24.7		
17	106°00.3	110°22.6	56.0	341°19.4	50.6	27°08.1	47.3	107°59.3	01.0				Canopus	263°52.6	-52°42.8		
18	121°02.7	125°24.0	N00°55.9	356°20.8	N21°50.3	42°10.0	N22°47.3	123°01.6	S03°00.9				Sirius	258°26.2	-16°45.2		
19	136°05.2	140°25.5	55.8	11°22.1	50.0	57°12.0	47.4	138°03.8	00.8				Adhara	255°05.8	-29°00.6		
20	151°07.7	155°27.0	55.7	26°23.5	49.7	72°13.9	47.4	153°06.0	00.7				Procyon	244°50.7	5°09.5		
21	166°10.1	170°28.5	•• 55.6	41°24.8	•• 49.4	87°15.9	•• 47.4	168°08.2	•• 00.6				Pollux	243°17.1	27°58.0		
22	181°12.6	185°29.9	55.5	56°26.2	49.1	102°17.8	47.5	183°10.4	00.5				Avior	234°14.7	-59°35.7		
23	196°15.1	200°31.4	55.4	71°27.5	48.8	117°19.8	47.5	198°12.6	00.4				Suhail	222°46.1	-43°32.3		
Mer.pass. 09:57			ν 1.5' d -0.1' m -4.76	ν 1.4' d -0.3' m 0.81	ν 2.0' d 0.0' m -2.01	ν 2.2' d -0.1' m 1.17											
Wed	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec					
0	211°17.5	215°32.9	N00°55.3	86°28.9	N21°48.5	132°21.7	N22°47.6	213°14.8	S03°00.3				Denebola	182°24.5	14°25.8		
1	226°20.0	230°34.3	55.2	101°30.2	48.2	147°23.7	47.6	228°17.1	00.2				Gienah	175°43.1	-17°41.1		
2	241°22.5	245°35.8	55.1	116°31.5	47.9	162°25.6	47.6	243°19.3	00.1				Acrux	172°59.3	-63°14.5		
3	256°24.9	260°37.2	•• 55.0	131°32.9	•• 47.6	177°27.6	•• 47.7	258°21.5	03°00.0				Gacrux	171°50.9	-57°15.5		
4	271°27.4	275°38.7	54.9	146°34.2	47.3	192°29.5	47.7	273°23.7	02°59.9				Alioth	166°12.2	55°49.4		
5	286°29.9	290°40.1	54.8	161°35.6	47.0	207°31.5	47.8	288°25.9	59.8				Spica	158°21.8	-11°17.7		
6	301°32.3	305°41.6	N00°54.7	176°36.9	N21°46.7	222°33.4	N22°47.8	303°28.1	S02°59.7				Alkaid	152°51.3	49°11.1		
7	316°34.8	320°43.0	54.6	191°38.3	46.4	237°35.4	47.9	318°30.3	59.6				Hadar	148°35.1	-60°29.8		
8	331°37.2	335°44.5	54.5	206°39.6	46.1	252°37.3	47.9	333°32.6	59.5				Menkent	147°57.0	-36°29.8		
9	346°39.7	350°45.9	•• 54.5	221°41.0	•• 45.8	267°39.3	•• 47.9	348°34.8	•• 59.4				Arcturus	145°47.4	19°02.9		
10	1°42.2	5°47.3	54.4	236°42.3	45.6	282°41.2	48.0	3°37.0	59.3				Rigel Kent.	139°39.4	-60°56.4		
11	16°44.6	20°48.8	54.3	251°43.7	45.3	297°43.1	48.0	18°39.2	59.2				Kochab	137°18.6	74°02.9		
12	31°47.1	35°50.2	N00°54.2	266°45.0	N21°45.0	312°45.1	N22°48.1	33°41.4	S02°59.1				Zuben'ubi	136°55.4	-16°08.9		
13	46°49.6	50°51.6	54.1	281°46.3	44.7	327°47.0	48.1	48°43.6	59.0				Alphecca	126°03.2	26°37.6		
14	61°52.0	65°53.0	54.0	296°47.7	44.4	342°49.0	48.1	63°45.8	58.9				Antares	112°15.3	-26°29.3		
15	76°54.5	80°54.5	•• 54.0	311°49.0	•• 44.1	357°50.9	•• 48.2	78°48.1	•• 58.8				Atria	107°08.9	-69°04.2		
16	91°57.0	95°55.9	53.9	326°50.4	43.8	12°52.9	48.2	93°50.3	58.7				Sabik	102°02.3	-15°45.4		
17	106°59.4	110°57.3	53.8	341°51.7	43.5	27°54.8	48.3	108°52.5	58.6				Shaula	96°09.8	-37°07.3		
18	122°01.9	125°58.7	N00°53.7	356°53.1	N21°43.2	42°56.8	N22°48.3	123°54.7	S02°58.5				Rasalhague	95°58.1	12°32.3		
19	137°04.3	141°00.1	53.7	11°54.4	42.9	57°58.7	48.4	138°56.9	58.4				Eltanin	90°41.8	51°28.8		
20	152°06.8	156°01.5	53.6	26°55.7	42.6	73°00.7	48.4	153°59.1	58.3				Kaus Aust.	83°32.0	-34°22.3		
21	167°09.3	171°02.9	•• 53.5	41°57.1	•• 42.3	88°02.6	•• 48.4	169°01.4	•• 58.2				Vega	80°32.9	38°48.1		
22	182°11.7	186°04.3	53.5	56°58.4	42.0	103°04.6	48.5	184°03.6	58.1				Nunki	75°47.4	-26°15.9		
23	197°14.2	201°05.7	53.4	71°59.8	41.7	118°06.5	48.5	199°05.8	58.0				Altair	61°59.7	8°55.9		
Mer.pass. 09:53			ν 1.5' d -0.1' m -4.76	ν 1.3' d -0.3' m 0.82	ν 1.9' d 0.0' m -2.01	ν 2.2' d -0.1' m 1.17											
Thu	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec					
0	212°16.7	216°07.1	N00°53.3	87°01.1	N21°41.4	133°08.5	N22°48.6	214°08.0	S02°57.9								
1	227°19.1	231°08.5	53.3	102°02.4	41.1	148°10.4	48.6	229°10.2	57.8								
2	242°21.6	246°09.9	53.2	117°03.8	40.8	163°12.3	48.6	244°12.4	57.7								
3	257°24.1	261°11.3	•• 53.2	132°05.1	•• 40.5	178°14.3	•• 48.7	259°14.6	•• 57.6								
4	272°26.5	276°12.7	53.1	147°06.5	40.2	193°16.2	48.7	274°16.9	57.5								
5	287°29.0	291°14.1	53.1	162°07.8	39.9	208°18.2	48.8	289°19.1	57.4								
6	302°31.5	306°15.5	N00°53.0	177°09.1	N21°39.6	223°20.1	N22°48.8	304°21.3	S02°57.3								
7	317°33.9	321°16.9	52.9	192°10.5	39.3	238°22.1	48.9	319°23.5	57.2								
8	332°36.4	336°18.2	52.9	207°11.8	39.0	253°24.0	48.9	334°25.7	57.1								
9	347°38.8	351°19.6	•• 52.8	222°13.1	•• 38.7	268°26.0	•• 48.9	349°27.9	•• 57.0								
10	2°41.3	6°21.0	52.8	237°14.5	38.4	283°27.9	49.0	4°30.2	56.9								
11	17°43.8	21°22.3	52.8	252°15.8	38.1	298°29.8	49.0	19°32.4	56.8								
12	32°46.2	36°23.7	N00°52.7	267°17.2	N21°37.8	313°31.8	N22°49.1	34°34.6	S02°56.7								
13	47°48.7	51°25.1	52.7	282°18.5	37.5	328°33.7	49.1	49°36.8	56.6								
14	62°51.2	66°26.4	52.6	297°19.8	37.2	343°35.7	49.1	64°39.0	56.5								
15	77°53.6	81°27.8	•• 52.6	312°21.2	•• 36.9	358°37.6	•• 49.2	79°41.2	•• 56.4								
16	92°56.1	96°29.2	52.6	327°22.5	36.5	13°39.6	49.2	94°43.5	56.3								
17	107°58.6	111°30.5	52.5	342°23.8	36.2	28°41.5	49.3	109°45.7	56.2								
18	123°01.0	126°31.9	N00°52.5	357°25.2	N21°35.9	43°43.4	N22°49.3	124°47.9	S02°56.1								
19	138°03.5	141°33.2	52.4	12°26.5	35.6	58°45.4	49.3	139°50.1	56.0								
20	153°05.9	156°34.6	52.4	27°27.8	35.3	73°47.3	49.4	154°52.3	55.9								
21	168°08.4	171°35.9	•• 52.4	42°29.2	•• 35.0	88°49.3	•• 49.4	169°54.5	•• 55.8								
22	183°10.9	186°37.2	52.4	57°30.5	34.7	103°51.2	49.5	184°56.8	55.7								

h	Sun			Moon			
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	180°22.0	N12°12.5	253°15.1	10.2'	S20°11.8	-11.9'	57.5'
1	195°22.2	13.4	267°44.3	10.3'	19°59.9	-12.0'	57.6'
2	210°22.3	14.2	282°13.5	10.3'	19°47.9	-12.1'	57.6'
3	225°22.4	15.0	296°42.8	10.3'	19°35.8	-12.2'	57.6'
4	240°22.5	15.9	311°12.2	10.4'	19°23.5	-12.4'	57.7'
5	255°22.7	16.7	325°41.6	10.4'	19°11.2	-12.5'	57.7'
6	270°22.8	N12°17.6	340°11.0	10.5'	S18°58.7	-12.6'	57.8'
7	285°22.9	18.4	354°40.4	10.5'	18°46.1	-12.7'	57.8'
8	300°23.0	19.2	9°09.9	10.5'	18°33.4	-12.8'	57.8'
9	315°23.1	20.1	23°39.5	10.6'	18°20.6	-12.9'	57.9'
10	330°23.3	20.9	38°09.0	10.6'	18°07.7	-13.0'	57.9'
11	345°23.4	21.7	52°38.6	10.6'	17°54.7	-13.1'	58.0'
12	0°23.5	N12°22.6	67°08.3	10.7'	S17°41.5	-13.3'	58.0'
13	15°23.6	23.4	81°37.9	10.7'	17°28.3	-13.4'	58.0'
14	30°23.7	24.3	96°07.6	10.7'	17°14.9	-13.5'	58.1'
15	45°23.9	25.1	110°37.4	10.8'	17°01.4	-13.6'	58.1'
16	60°24.0	25.9	125°07.1	10.8'	16°47.9	-13.7'	58.1'
17	75°24.1	26.8	139°37.0	10.8'	16°34.2	-13.8'	58.2'
18	90°24.2	N12°27.6	154°06.8	10.9'	S16°20.4	-13.9'	58.2'
19	105°24.3	28.4	168°36.7	10.9'	16°06.5	-14.0'	58.3'
20	120°24.4	29.3	183°06.6	10.9'	15°52.5	-14.1'	58.3'
21	135°24.6	30.1	197°36.5	11.0'	15°38.5	-14.2'	58.3'
22	150°24.7	30.9	212°06.5	11.0'	15°24.3	-14.3'	58.4'
23	165°24.8	31.8	226°36.4	11.0'	15°10.0	-14.4'	58.4'
SD = 15.9' $d = 0.8'$			SD = 15.7'				
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	180°24.9	N12°32.6	241°06.5	11.0'	S14°55.6	-14.5'	58.5'
1	195°25.0	33.4	255°36.5	11.1'	14°41.1	-14.6'	58.5'
2	210°25.1	34.3	270°06.6	11.1'	14°26.6	-14.7'	58.5'
3	225°25.3	35.1	284°36.7	11.1'	14°11.9	-14.8'	58.6'
4	240°25.4	35.9	299°06.8	11.1'	13°57.1	-14.9'	58.6'
5	255°25.5	36.8	313°36.9	11.2'	13°42.3	-14.9'	58.7'
6	270°25.6	N12°37.6	328°07.1	11.2'	S13°27.3	-15.0'	58.7'
7	285°25.7	38.4	342°37.3	11.2'	13°12.3	-15.1'	58.7'
8	300°25.8	39.2	357°07.5	11.2'	12°57.2	-15.2'	58.8'
9	315°26.0	40.1	11°37.8	11.3'	12°42.0	-15.3'	58.8'
10	330°26.1	40.9	26°08.0	11.3'	12°26.7	-15.4'	58.9'
11	345°26.2	41.7	40°38.3	11.3'	12°11.3	-15.5'	58.9'
12	0°26.3	N12°42.6	55°08.6	11.3'	S11°55.9	-15.5'	58.9'
13	15°26.4	43.4	69°38.9	11.3'	11°40.3	-15.6'	59.0'
14	30°26.5	44.2	84°09.2	11.3'	11°24.7	-15.7'	59.0'
15	45°26.6	45.0	98°39.6	11.4'	11°09.0	-15.8'	59.0'
16	60°26.7	45.9	113°09.9	11.4'	10°53.2	-15.9'	59.1'
17	75°26.9	46.7	127°40.3	11.4'	10°37.4	-15.9'	59.1'
18	90°27.0	N12°47.5	142°10.7	11.4'	S10°21.4	-16.0'	59.2'
19	105°27.1	48.3	156°41.0	11.4'	10°05.4	-16.1'	59.2'
20	120°27.2	49.2	171°11.4	11.4'	09°49.4	-16.2'	59.2'
21	135°27.3	50.0	185°41.9	11.4'	09°33.2	-16.2'	59.3'
22	150°27.4	50.8	200°12.3	11.4'	09°17.0	-16.3'	59.3'
23	165°27.5	51.6	214°42.7	11.4'	09°00.7	-16.4'	59.3'
SD = 15.9' $d = 0.8'$			SD = 15.9'				
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	180°27.6	N12°52.5	229°13.1	11.4'	S08°44.3	-16.4'	59.4'
1	195°27.8	53.3	243°43.6	11.4'	08°27.9	-16.5'	59.4'
2	210°27.9	54.1	258°14.0	11.4'	08°11.4	-16.6'	59.5'
3	225°28.0	54.9	272°44.4	11.4'	07°54.9	-16.6'	59.5'
4	240°28.1	55.8	287°14.9	11.4'	07°38.2	-16.7'	59.5'
5	255°28.2	56.6	301°45.3	11.4'	07°21.6	-16.7'	59.6'
6	270°28.3	N12°57.4	316°15.8	11.4'	S07°04.8	-16.8'	59.6'
7	285°28.4	58.2	330°46.2	11.4'	06°48.0	-16.8'	59.6'
8	300°28.5	59.0	345°16.6	11.4'	06°31.2	-16.9'	59.7'
9	315°28.6	12°59.9	359°47.1	11.4'	06°14.3	-17.0'	59.7'
10	330°28.8	13°00.7	14°17.5	11.4'	05°57.3	-17.0'	59.7'
11	345°28.9	01.5	28°47.9	11.4'	05°40.3	-17.1'	59.8'
12	0°29.0	N13°02.3	43°18.3	11.4'	S05°23.3	-17.1'	59.8'
13	15°29.1	03.1	57°48.7	11.4'	05°06.2	-17.2'	59.9'
14	30°29.2	04.0	72°19.1	11.4'	04°49.0	-17.2'	59.9'
15	45°29.3	04.8	86°49.5	11.4'	04°31.8	-17.2'	59.9'
16	60°29.4	05.6	101°19.8	11.3'	04°14.5	-17.3'	60.0'
17	75°29.5	06.4	115°50.2	11.3'	03°57.3	-17.3'	60.0'
18	90°29.6	N13°07.2	130°20.5	11.3'	S03°39.9	-17.4'	60.0'
19	105°29.7	08.1	144°50.8	11.3'	03°22.6	-17.4'	60.1'
20	120°29.8	08.9	159°21.1	11.3'	03°05.2	-17.4'	60.1'
21	135°29.9	09.7	173°51.4	11.3'	02°47.7	-17.5'	60.1'
22	150°30.1	10.5	188°21.6	11.2'	02°30.2	-17.5'	60.2'
23	165°30.2	11.3	202°51.9	11.2'	02°12.7	-17.5'	60.2'
SD = 15.9' $d = 0.8'$			SD = 16.2'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	02:49	21:12	—:—	—:—
N 70°	—:—	01:27	03:14	20:46	22:39	—:—
68°	—:—	02:11	03:33	20:27	21:51	—:—
66°	—:—	02:39	03:48	20:11	21:22	—:—
64°	01:16	03:00	04:00	19:58	21:00	22:49
62°	01:55	03:17	04:11	19:48	20:42	22:06
60°	02:21	03:31	04:20	19:38	20:28	21:39
N 58°	02:41	03:42	04:28	19:31	20:16	21:19
56°	02:57	03:52	04:34	19:24	20:06	21:02
54°	03:10	04:01	04:40	19:17	19:57	20:48
52°	03:22	04:09	04:46	19:12	19:49	20:37
50°	03:32	04:16	04:51	19:07	19:42	20:27
45°	03:52	04:30	05:02	18:56	19:27	20:06
N 40°	04:07	04:42	05:10	18:47	19:15	19:50
35°	04:20	04:52	05:18	18:39	19:06	19:37
30°	04:30	05:00	05:25	18:32	18:57	19:27
20°	04:47	05:13	05:36	18:21	18:44	19:10
N 10°	04:59	05:24	05:46	18:11	18:32	18:58
0°	05:09	05:34	05:55	18:02	18:23	18:47
S 10°	05:18	05:42	06:04	17:52	18:14	18:39
20°	05:25	05:51	06:13	17:43	18:05	18:31
30°	05:31	06:00	06:24	17:32	17:57	18:25
35°	05:35	06:04	06:30	17:26	17:52	18:21
40°	05:38	06:09	06:37	17:19	17:47	18:18
45°	05:41	06:15	06:45	17:11	17:41	18:15
S 50°	05:43	06:21	06:55	17:01	17:34	18:12
52°	05:45	06:24	06:59	16:56	17:31	18:11
54°	05:46	06:27	07:04	16:52	17:28	18:10
56°	05:47	06:31	07:10	16:46	17:25	18:08
58°	05:48	06:34	07:16	16:40	17:21	18:07
S 60°	05:50	06:38	07:22	16:33	17:17	18:06

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	■	05:22	04:30	■	11:32	14:09
N 70°	06:07	04:56	04:19	08:59	11:55	14:17
68°	05:17	04:37	04:10	09:47	12:12	14:23
66°	04:45	04:21	04:03	10:18	12:27	14:28
64°	04:22	04:08	03:56	10:40	12:38	14:32
62°	04:03	03:56	03:51	10:58	12:48	14:35
60°	03:47	03:47	03:46	11:13	12:56	14:38
N 58°	03:33	03:38	03:42	11:25	13:03	14:41
56°	03:22	03:31	03:38	11:36	13:10	14:44
54°	03:12	03:24	03:34	11:45	13:15	14:46
52°	03:03	03:18	03:31	11:54	13:20	14:48
50°	02:54	03:13	03:28	12:01	13:25	14:49
45°	02:37	03:01	03:22	12:17	13:35	14:53
N 40°	02:23	02:51	03:17	12:30	13:43	14:56
35°	02:10	02:43	03:12	12:41	13:50	14:59
30°	02:00	02:35	03:08	12:51	13:56	15:02
20°	01:41	02:22	03:01	13:07	14:06	15:06
N 10°	01:25	02:11	02:55	13:21	14:15	15:09
0°	01:10	02:00	02:49	13:34	14:24	15:12
S 10°	00:55	01:49	02:43	13:47	14:32	15:16
20°	00:38	01:38	02:37	14:01	14:41	15:19
30°	00:19	01:24	02:30	14:17	14:51	15:23
35°	00:08	01:17	02:26	14:26	14:56	15:25
40°	...	01:08	02:21	14:36	15:03	15:28
45°	...	00:58	02:16	14:48	15:10	15:30
S 50°	...	00:45	02:09	15:02	15:19	15:34
52°	...	00:39	02:06	15:09	15:23	15:35
54°	...	00:32	02:03	15:16	15:27	15:37
56°	...	00:25	01:59	15:25	15:32	15:39
58°	...	00:17	01:55	15:34	15:38	15:41
S 60°	...	00:07	01:50	15:44	15:44	15:43

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss	hh:mm	Upper hh:mm	Lower hh:mm	24-26 40-20%
22	01:28	01:34	11:58	07:22	19:47	
23	01:40	01:45	11:58	08:12	20:36	
24	01:51	01:56	11:58	09:01	21:25	

April 25, 26, 27 UT (Fri., Sat., Sun.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	213°15.8	216°39.9	N00°52.3	87°33.2	N21°34.1	133°55.1	N22°49.6	215°01.2	S02°55.5	Alpheratz	357°34.9 29°13.6
1	228°18.3	231°41.3	52.3	102°34.5	33.8	148°57.0	49.6	230°03.4	55.4	Ankaa	353°07.3 -42°10.1
2	243°20.7	246°42.6	52.3	117°35.8	33.5	163°59.0	49.6	245°05.6	55.3	Schedar	349°31.3 56°40.4
3	258°23.2	261°43.9	· · 52.2	132°37.2	· · 33.2	179°00.9	· · 49.7	260°07.9	· · 55.2	Diphda	348°47.4 -17°50.9
4	273°25.7	276°45.2	52.2	147°38.5	32.9	194°02.9	49.7	275°10.1	55.1	Achernar	335°20.6 -57°06.5
5	288°28.1	291°46.6	52.2	162°39.8	32.6	209°04.8	49.8	290°12.3	55.0	Hamal	327°51.3 23°34.8
6	303°30.6	306°47.9	N00°52.2	177°41.1	N21°32.3	224°06.7	N22°49.8	305°14.5	S02°54.9	Polaris	314°25.0 89°22.3
7	318°33.1	321°49.2	52.2	192°42.5	32.0	239°08.7	49.8	320°16.7	54.8	Acamar	315°12.0 -40°12.3
8	333°35.5	336°50.5	52.2	207°43.8	31.7	254°10.6	49.9	335°18.9	54.7	Menkar	314°06.2 4°11.3
9	348°38.0	351°51.8	· · 52.1	222°45.1	· · 31.4	269°12.6	· · 49.9	350°21.2	· · 54.6	Mirfak	308°28.4 49°57.1
10	3°40.4	6°53.1	52.1	237°46.5	31.1	284°14.5	50.0	5°23.4	54.5	Aldebaran	290°39.7 16°33.6
11	18°42.9	21°54.5	52.1	252°47.8	30.8	299°16.5	50.0	20°25.6	54.4	Rigel	281°03.9 -8°10.4
12	33°45.4	36°55.8	N00°52.1	267°49.1	N21°30.5	314°18.4	N22°50.0	35°27.8	S02°54.3	Capella	280°21.9 46°01.5
13	48°47.8	51°57.1	52.1	282°50.4	30.1	329°20.3	50.1	50°30.0	54.2	Bellatrix	278°22.9 6°22.3
14	63°50.3	66°58.4	52.1	297°51.8	29.8	344°22.3	50.1	65°32.3	54.1	Elnath	278°01.9 28°37.7
15	78°52.8	81°59.7	· · 52.1	312°53.1	· · 29.5	359°24.2	· · 50.2	80°34.5	· · 54.0	Alnilam	275°37.7 -1°11.2
16	93°55.2	97°01.0	52.1	327°54.4	29.2	14°26.1	50.2	95°36.7	53.9	Betelgeuse	270°52.1 7°24.7
17	108°57.7	112°02.3	52.1	342°55.8	28.9	29°28.1	50.2	110°38.9	53.8	Canopus	263°52.6 -52°42.8
18	124°00.2	127°03.5	N00°52.1	357°57.1	N21°28.6	44°30.0	N22°50.3	125°41.1	S02°53.7	Sirius	258°26.2 -16°45.2
19	139°02.6	142°04.8	52.1	12°58.4	28.3	59°32.0	50.3	140°43.3	53.6	Adhara	255°05.8 -29°00.6
20	154°05.1	157°06.1	52.1	27°59.7	28.0	74°33.9	50.4	155°45.6	53.5	Procyon	244°50.7 5°09.6
21	169°07.5	172°07.4	· · 52.1	43°01.1	· · 27.7	89°35.8	· · 50.4	170°47.8	· · 53.4	Pollux	243°17.1 27°58.0
22	184°10.0	187°08.7	52.1	58°02.4	27.4	104°37.8	50.4	185°50.0	53.3	Avior	234°14.7 -59°35.7
23	199°12.5	202°10.0	52.2	73°03.7	27.1	119°39.7	50.5	200°52.2	53.2	Suhail	222°46.1 -43°32.3
Mer.pass. 09:45		ν 1.3' d -0.0' m -4.75		ν 1.3' d -0.3' m 0.85		ν 1.9' d 0.0' m -2.00		ν 2.2' d -0.1' m 1.17			

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	214°14.9	217°11.2	N00°52.2	88°05.0	N21°26.8	134°41.7	N22°50.5	215°54.4	S02°53.1
1	229°17.4	232°12.5	52.2	103°06.4	26.5	149°43.6	50.6	230°56.7	53.0
2	244°19.9	247°13.8	52.2	118°07.7	26.1	164°45.5	50.6	245°58.9	52.9
3	259°22.3	262°15.1	· · 52.2	133°09.0	· · 25.8	179°47.5	· · 50.6	261°01.1	· · 52.8
4	274°24.8	277°16.3	52.2	148°10.3	25.5	194°49.4	50.7	276°03.3	52.7
5	289°27.3	292°17.6	52.2	163°11.7	25.2	209°51.3	50.7	291°05.5	52.6
6	304°29.7	307°18.9	N00°52.3	178°13.0	N21°24.9	224°53.3	N22°50.8	306°07.8	S02°52.5
7	319°32.2	322°20.1	52.3	193°14.3	24.6	239°55.2	50.8	321°10.0	52.4
8	334°34.7	337°21.4	52.3	208°15.6	24.3	254°57.2	50.9	336°12.2	52.3
9	349°37.1	352°22.6	· · 52.3	223°16.9	· · 24.0	269°59.1	· · 50.9	351°14.4	· · 52.2
10	4°39.6	7°23.9	52.4	238°18.3	23.7	285°01.0	50.9	6°16.6	52.1
11	19°42.0	22°25.1	52.4	253°19.6	23.4	300°03.0	51.0	21°18.9	52.0
12	34°44.5	37°26.4	N00°52.4	268°20.9	N21°23.0	315°04.9	N22°51.0	36°21.1	S02°51.9
13	49°47.0	52°27.6	52.5	283°22.2	22.7	330°06.8	51.1	51°23.3	51.8
14	64°49.4	67°28.9	52.5	298°23.5	22.4	345°08.8	51.1	66°25.5	51.7
15	79°51.9	82°30.1	· · 52.5	313°24.9	· · 22.1	0°10.7	· · 51.1	81°27.7	· · 51.6
16	94°54.4	97°31.4	52.6	328°26.2	21.8	15°12.7	51.2	96°30.0	51.5
17	109°56.8	112°32.6	52.6	343°27.5	21.5	30°14.6	51.2	111°32.2	51.4
18	124°59.3	127°33.8	N00°52.6	358°28.8	N21°21.2	45°16.5	N22°51.3	126°34.4	S02°51.3
19	140°01.8	142°35.1	52.7	13°30.1	20.9	60°18.5	51.3	141°36.6	51.2
20	155°04.2	157°36.3	52.7	28°31.5	20.5	75°20.4	51.3	156°38.8	51.1
21	170°06.7	172°37.5	· · 52.8	43°32.8	· · 20.2	90°22.3	· · 51.4	171°41.1	· · 51.0
22	185°09.2	187°38.7	52.8	58°34.1	19.9	105°24.3	51.4	186°43.3	50.9
23	200°11.6	202°40.0	52.8	73°35.4	19.6	120°26.2	51.5	201°45.5	50.8
Mer.pass. 09:41		ν 1.3' d 0.0' m -4.75		ν 1.3' d -0.3' m 0.87		ν 1.9' d 0.0' m -2.00		ν 2.2' d -0.1' m 1.17	

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	215°14.1	217°41.2	N00°52.9	88°36.7	N21°19.3	135°28.1	N22°51.5	216°47.7	S02°50.7
1	230°16.5	232°42.4	52.9	103°38.0	19.0	150°30.1	51.5	231°49.9	50.6
2	245°19.0	247°43.6	53.0	118°39.4	18.7	165°32.0	51.6	246°52.2	50.5
3	260°21.5	262°44.8	· · 53.0	133°40.7	· · 18.4	180°33.9	· · 51.6	261°54.4	· · 50.4
4	275°23.9	277°46.0	53.1	148°42.0	18.0	195°35.9	51.7	276°56.6	50.3
5	290°26.4	292°47.3	53.2	163°43.3	17.7	210°37.8	51.7	291°58.8	50.2
6	305°28.9	307°48.5	N00°53.2	178°44.6	N21°17.4	225°39.7	N22°51.7	307°01.1	S02°50.2
7	320°31.3	322°49.7	53.3	193°45.9	17.1	240°41.7	51.8	322°03.3	50.1
8	335°33.8	337°50.9	53.3	208°47.3	16.8	255°43.6	51.8	337°05.5	50.0
9	350°36.3	352°52.1	· · 53.4	223°48.6	· · 16.5	270°45.5	· · 51.9	352°07.7	· · 49.9
10	5°38.7	7°53.3	53.4	238°49.9	16.2	285°47.5	51.9	7°09.9	49.8
11	20°41.2	22°54.5	53.5	253°51.2	15.8	300°49.4	51.9	22°12.2	49.7
12	35°43.6	37°55.7	N00°53.6	268°52.5	N21°15.5	315°51.3	N22°52.0	37°14.4	S02°49.6
13	50°46.1	52°56.8	53.6	283°53.8	15.2	330°53.3	52.0	52°16.6	49.5
14	65°48.6	67°58.0	53.7	298°55.1	14.9	345°55.2	52.0	67°18.8	49.4
15	80°51.0	82°59.2	· · 53.8	313°56.4	· · 14.6	0°57.1	· · 52.1	82°21.0	· · 49.3
16	95°53.5	98°00.4	53.9	328°57.8	14.3	15°59.1	52.1	97°23.3	49.2
17	110°56.0	113°01.6	53.9	343°59.1	13.9	31°01.0	52.2	112°25.5	49.1
18	125°58.4	128°02.8	N00°54.0	359°00.4	N21°13.6	46°02.9	N22°52.2	127°27.7	S02°49.0
19	141°00.9	143°03.9	54.1	14°01.7	13.3	61°04.9	52.2	142°29.9	48.9
20	156°03.4	158°05.1	54.2	29°03.0	13.0	76°06.8	52.3	157°32.2	48.8
21	171°05.8	173°06.3	· · 54.2	44°04.3	· · 12.7	91°08.7	· · 52.3	172°34.4	· · 48.7
22	186°08.3	188°07.5	54.3	59°05.6	12.4	106°10.7	52.4	187°36.6	48.6
23	201°10.8	203°08.6	54.4	74°06.9	12.0	121°12.6	52.4	202°38.8	48.5
Mer.pass. 09:37		ν 1.2' d 0.0' m -4.74		ν 1.3' d -0.3' m 0.88		ν 1.9' d 0.0' m -2.00		ν 2.2' d -0.1' m 1.17	

Apr 25 Fri	SHA	Mer.pass
Venus	3°24.1	09:32
Mars	234°17.4	18:08
Jupiter	280°39.3	15:02
Saturn	1°45.4	09:38

Apr 26 Sat	SHA	Mer.pass
Venus	2°56.3	09:30
Mars	233°50.1	18:06
Jupiter	280°26.7	14:59
Saturn	1°39.5	09:35

Apr 27 Sun	SHA	Mer.pass
Venus	2°27.1	09:28
Mars	233°22.6	18:04
Jupiter	280°14.1	14:56
Saturn	1°33.6	09:31

Horizontal parallax	
Venus:	0.3
Mars:	0.1

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	180°30.3	N13°12.1	217°22.1	11.2'	S01°55.2	-17.6'	60.2'
1	195°30.4	12.9	231°52.2	11.2'	01°37.6	-17.6'	60.2'
2	210°30.5	13.8	246°22.4	11.1'	01°20.0	-17.6'	60.3'
3	225°30.6	· · 14.6	260°52.5	11.1'	01°02.4	-17.7'	60.3'
4	240°30.7	15.4	275°22.6	11.1'	00°44.7	-17.7'	60.3'
5	255°30.8	16.2	289°52.7	11.0'	00°27.1	-17.7'	60.4'
6	270°30.9	N13°17.0	304°22.7	11.0'	S00°09.4	-17.7'	60.4'
7	285°31.0	17.8	318°52.7	11.0'	N00°08.4	17.7'	60.4'
8	300°31.1	18.6	333°22.7	10.9'	00°26.1	17.7'	60.5'
9	315°31.2	· · 19.5	347°52.6	10.9'	00°43.8	17.8'	60.5'
10	330°31.3	20.3	2°22.5	10.9'	01°01.6	17.8'	60.5'
11	345°31.4	21.1	16°52.4	10.8'	01°19.4	17.8'	60.5'
12	0°31.5	N13°21.9	31°22.2	10.8'	N01°37.1	17.8'	60.6'
13	15°31.6	22.7	45°52.0	10.7'	01°54.9	17.8'	60.6'
14	30°31.7	23.5	60°21.7	10.7'	02°12.7	17.8'	60.6'
15	45°31.8	· · 24.3	74°51.4	10.6'	02°30.5	17.8'	60.6'
16	60°31.9	25.1	89°21.0	10.6'	02°48.3	17.8'	60.7'
17	75°32.0	25.9	103°50.6	10.5'	03°06.1	17.8'	60.7'
18	90°32.1	N13°26.7	118°20.2	10.5'	N03°23.9	17.8'	60.7'
19	105°32.3	27.5	132°49.7	10.4'	03°41.7	17.8'	60.7'
20	120°32.4	28.4	147°19.1	10.4'	03°59.5	17.8'	60.8'
21	135°32.5	· · 29.2	161°48.5	10.3'	04°17.3	17.8'	60.8'
22	150°32.6	30.0	176°17.9	10.3'	04°35.1	17.8'	60.8'
23	165°32.7	30.8	190°47.2	10.2'	04°52.9	17.8'	60.8'
SD = 15.9'			SD = 16.4'				
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°32.8	N13°31.6	205°16.4	10.2'	N05°10.6	17.7'	60.9'
1	195°32.9	32.4	219°45.6	10.1'	05°28.4	17.7'	60.9'
2	210°33.0	33.2	234°14.7	10.1'	05°46.1	17.7'	60.9'
3	225°33.1	· · 34.0	248°43.8	10.0'	06°03.8	17.7'	60.9'
4	240°33.2	34.8	263°12.7	9.9'	06°21.5	17.7'	60.9'
5	255°33.3	35.6	277°41.7	9.9'	06°39.1	17.6'	61.0'
6	270°33.4	N13°36.4	292°10.6	9.8'	N06°56.7	17.6'	61.0'
7	285°33.5	37.2	306°39.4	9.7'	07°14.3	17.6'	61.0'
8	300°33.6	38.0	321°08.1	9.7'	07°31.9	17.5'	61.0'
9	315°33.7	· · 38.8	335°36.8	9.6'	07°49.5	17.5'	61.0'
10	330°33.8	39.6	350°05.4	9.5'	08°07.0	17.5'	61.1'
11	345°33.9	40.4	4°33.9	9.5'	08°24.4	17.4'	61.1'
12	0°34.0	N13°41.2	19°02.4	9.4'	N08°41.9	17.4'	61.1'
13	15°34.1	42.0	33°30.8	9.3'	08°59.2	17.3'	61.1'
14	30°34.2	42.8	47°59.1	9.2'	09°16.6	17.3'	61.1'
15	45°34.3	· · 43.6	62°27.3	9.2'	09°33.9	17.3'	61.1'
16	60°34.4	44.4	76°55.5	9.1'	09°51.2	17.2'	61.2'
17	75°34.5	45.2	91°23.6	9.0'	10°08.4	17.2'	61.2'
18	90°34.5	N13°46.0	105°51.6	8.9'	N10°25.5	17.1'	61.2'
19	105°34.6	46.8	120°19.5	8.9'	10°42.6	17.0'	61.2'
20	120°34.7	47.6	134°47.4	8.8'	10°59.7	17.0'	61.2'
21	135°34.8	· · 48.4	149°15.2	8.7'	11°16.6	16.9'	61.2'
22	150°34.9	49.2	163°42.9	8.6'	11°33.6	16.9'	61.2'
23	165°35.0	50.0	178°10.5	8.5'	11°50.4	16.8'	61.2'
SD = 15.9'			SD = 16.6'				
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	180°35.1	N13°50.8	192°38.0	8.4'	N12°07.2	16.7'	61.2'
1	195°35.2	51.6	207°05.4	8.4'	12°23.9	16.7'	61.3'
2	210°35.3	52.4	221°32.8	8.3'	12°40.6	16.6'	61.3'
3	225°35.4	· · 53.2	236°00.1	8.2'	12°57.2	16.5'	61.3'
4	240°35.5	54.0	250°27.2	8.1'	13°13.7	16.4'	61.3'
5	255°35.6	54.8	264°54.3	8.0'	13°30.1	16.4'	61.3'
6	270°35.7	N13°55.6	279°21.3	7.9'	N13°46.5	16.3'	61.3'
7	285°35.8	56.4	293°48.2	7.8'	14°02.8	16.2'	61.3'
8	300°35.9	57.2	308°15.0	7.7'	14°18.9	16.1'	61.3'
9	315°36.0	· · 58.0	322°41.8	7.6'	14°35.0	16.0'	61.3'
10	330°36.1	58.7	337°08.4	7.5'	14°51.1	15.9'	61.3'
11	345°36.2	13°59.5	351°34.9	7.4'	15°07.0	15.8'	61.3'
12	0°36.3	N14°00.3	6°01.4	7.4'	N15°22.8	15.7'	61.3'
13	15°36.4	01.1	20°27.7	7.3'	15°38.5	15.6'	61.3'
14	30°36.4	01.9	34°54.0	7.2'	15°54.2	15.5'	61.3'
15	45°36.5	· · 02.7	49°20.2	7.1'	16°09.7	15.4'	61.3'
16	60°36.6	03.5	63°46.2	7.0'	16°25.2	15.3'	61.3'
17	75°36.7	04.3	78°12.2	6.9'	16°40.5	15.2'	61.3'
18	90°36.8	N14°05.1	92°38.1	6.8'	N16°55.7	15.1'	61.3'
19	105°36.9	05.9	107°03.8	6.7'	17°10.8	15.0'	61.3'
20	120°37.0	06.6	121°29.5	6.6'	17°25.8	14.9'	61.3'
21	135°37.1	· · 07.4	135°55.1	6.5'	17°40.7	14.8'	61.3'
22	150°37.2	08.2	150°20.6	6.4'	17°55.5	14.7'	61.3'
23	165°37.3	09.0	164°45.9	6.3'	18°10.1	14.5'	61.3'
SD = 15.9'			SD = 16.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	02:29	21:32	—:—	—:—
N 70°	—:—	00:48	02:58	21:02	23:28	—:—
68°	—:—	01:50	03:19	20:39	22:12	—:—
66°	—:—	02:23	03:36	20:22	21:37	—:—
64°	00:42	02:47	03:50	20:08	21:12	23:32
62°	01:37	03:06	04:01	19:56	20:53	22:25
60°	02:07	03:21	04:11	19:46	20:37	21:52
N 58°	02:29	03:33	04:20	19:37	20:24	21:29
56°	02:47	03:44	04:27	19:30	20:13	21:11
54°	03:01	03:54	04:34	19:23	20:03	20:56
52°	03:14	04:02	04:40	19:17	19:55	20:44
50°	03:24	04:09	04:45	19:11	19:47	20:33
45°	03:46	04:25	04:57	18:59	19:31	20:11
N 40°	04:03	04:38	05:06	18:50	19:19	19:54
35°	04:16	04:48	05:15	18:41	19:08	19:40
30°	04:27	04:57	05:22	18:34	18:59	19:29
20°	04:44	05:11	05:34	18:22	18:45	19:12
N 10°	04:58	05:23	05:45	18:11	18:33	18:58
0°	05:08	05:33	05:54	18:01	18:22	18:47
S 10°	05:18	05:43	06:04	17:51	18:13	18:38
20°	05:26	05:52	06:14	17:41	18:03	18:30
30°	05:33	06:01	06:26	17:29	17:54	18:22
35°	05:37	06:07	06:33	17:22	17:48	18:18
40°	05:40	06:12	06:40	17:15	17:43	18:15
45°	05:44	06:18	06:49	17:06	17:36	18:11
S 50°	05:48	06:26	06:59	16:55	17:29	18:07
52°	05:49	06:29	07:04	16:51	17:26	18:05
54°	05:51	06:32	07:10	16:45	17:22	18:04
56°	05:52	06:36	07:15	16:39	17:18	18:02
58°	05:54	06:40	07:22	16:32	17:14	18:00
S 60°	05:56	06:45	07:30	16:25	17:09	17:59

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	03:50	03:11	02:21	16:39	19:21	□
N 70°	03:49	03:19	02:42	16:35	19:02	22:06
68°	03:48	03:26	03:00	16:31	18:47	21:24
66°	03:47	03:31	03:13	16:28	18:35	20:56
64°	03:46	03:36	03:25	16:26	18:25	20:35
62°	03:46	03:40	03:35	16:24	18:17	20:18
60°	03:45	03:44	03:44	16:22	18:10	20:04
N 58°	03:44	03:47	03:51	16:21	18:04	19:53
56°	03:44	03:50	03:58	16:19	17:58	19:42
54°	03:44	03:53	04:04	16:18	17:54	19:33
52°	03:43	03:55	04:09	16:17	17:49	19:25
50°	03:43	03:58	04:14	16:16	17:45	19:18
45°	03:42	04:03	04:25	16:14	17:37	19:03
N 40°	03:42	04:07	04:34	16:12	17:30	18:50
35°	03:41	04:10	04:42	16:10	17:23	18:40
30°	03:41	04:14	04:49	16:09	17:18	18:31
20°	03:40	04:19	05:01	16:06	17:09	18:15
N 10°	03:39	04:24	05:12	16:04	17:01	18:01
0°	03:38	04:29	05:22	16:02	16:53	17:48
S 10°	03:38	04:34	05:32	16:00	16:46	17:36
20°	03:37	04:39	05:43	15:58	16:38	17:22
30°	03:36	04:45	05:56	15:55	16:29	17:07
35°	03:36	04:48	06:03	15:54	16:24	16:58
40°	03:36	04:52	06:12	15:52	16:18	16:48
45°	03:35	04:57	06:22	15:50	16:11	16:36
S 50°	03:35	05:02	06:34	15:48	16:03	16:22
52°	03:34	05:05	06:39	15:47	16:00	16:15
54°	03:34	05:08	06:46	15:46	15:56	16:08
56°	03:34	05:11	06:53	15:45	15:51	16:00
58°	03:33	05:15	07:01	15:43	15:46	15:51
S 60°	03:33	05:19	07:10	15:42	15:41	15:40

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss	hh:mm	Mer. Pass. Upper hh:mm	Lower hh:mm	27-0 12-1%
25	02:01	02:06	11:58	09:50	22:15	
26	02:11	02:16	11:58	10:41	23:08	
27	02:21	02:25	11:58	11:35	—:—	

April 28, 29, 30 UT (Mon., Tue., Wed.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Mon	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec		GHA	Dec		SHA	Dec	
0	216°13.2	218°09.8	N00°54.5	89°08.2	N21°11.7	136°14.5	N22°52.4	217°41.1	S02°48.4						Alpheratz	357°34.8	
1	231°15.7	233°11.0	54.6	104°09.6	11.4	151°16.5	52.5	232°43.3	48.3						Ankaa	353°07.3	
2	246°18.1	248°12.1	54.6	116°10.9	11.1	166°18.4	52.5	247°45.5	48.2						Schedar	349°31.3	
3	261°20.6	263°13.3	· · 54.7	134°12.2	· · 10.8	181°20.3	· · 52.6	262°47.7	· · 48.1						Diphda	348°47.4	
4	276°23.1	278°14.4	54.8	149°13.5	10.5	196°22.3	52.6	277°49.9	48.0						Achernar	335°20.6	
5	291°25.5	293°15.6	54.9	164°14.8	10.1	211°24.2	52.6	292°52.2	47.9						Hamal	327°51.3	
6	306°28.0	308°16.7	N00°55.0	179°16.1	N21°09.8	226°26.1	N22°52.7	307°54.4	S02°47.8						Polaris	314°24.9	
7	321°30.5	323°17.9	55.1	194°17.4	09.5	241°28.0	52.7	322°56.6	47.7						Acamar	315°12.0	
8	336°32.9	338°19.0	55.2	209°18.7	09.2	256°30.0	52.8	337°58.8	47.6						Menkar	314°06.2	
9	351°35.4	353°20.2	· · 55.3	224°20.0	· · 08.9	271°31.9	· · 52.8	353°01.1	· · 47.5						Mirfak	308°28.4	
10	6°37.9	8°21.3	55.4	239°21.3	08.5	286°33.8	52.8	8°03.3	47.4						Aldebaran	290°39.7	
11	21°40.3	23°22.5	55.5	254°22.6	08.2	301°35.8	52.9	23°05.5	47.3						Rigel	281°03.9	
12	36°42.8	38°23.6	N00°55.6	269°23.9	N21°07.9	316°37.7	N22°52.9	38°07.7	S02°47.2						Capella	280°21.9	
13	51°45.3	53°24.8	55.7	284°25.2	07.6	331°39.6	53.0	53°10.0	47.1						Bellatrix	278°22.9	
14	66°47.7	68°25.9	55.8	299°26.6	07.3	346°41.6	53.0	68°12.2	47.0						Elnath	278°01.9	
15	81°50.2	83°27.0	· · 55.9	314°27.9	· · 06.9	1°43.5	· · 53.0	83°14.4	· · 46.9						Alnilam	275°37.7	
16	96°52.6	98°28.2	56.0	329°29.2	06.6	16°45.4	53.1	98°16.6	46.8						Betelgeuse	270°52.1	
17	111°55.1	113°29.3	56.1	344°30.5	06.3	31°47.3	53.1	113°18.9	46.7						Canopus	263°52.6	
18	126°57.6	128°30.4	N00°56.2	359°31.8	N21°06.0	46°49.3	N22°53.1	128°21.1	S02°46.7						Sirius	258°26.2	
19	142°00.0	143°31.5	56.3	14°33.1	05.7	61°51.2	53.2	143°23.3	46.6						Adhara	255°05.9	
20	157°02.5	158°32.7	56.4	29°34.4	05.3	76°53.1	53.2	158°25.5	46.5						Procyon	244°50.7	
21	172°05.0	173°33.8	· · 56.5	44°35.7	· · 05.0	91°55.1	· · 53.3	173°27.8	· · 46.4						Pollux	243°17.1	
22	187°07.4	188°34.9	56.6	59°37.0	04.7	106°57.0	53.3	188°30.0	46.3						Avior	234°14.7	
23	202°09.9	203°36.0	56.8	74°38.3	04.4	121°58.9	53.3	203°32.2	46.2						Suhail	222°46.1	
Mer.pass. 09:34			ν 1.2' d 0.1' m -4.74			ν 1.3' d -0.3' m 0.90			ν 1.9' d 0.0' m -1.99			ν 2.2' d -0.1' m 1.17					

Tue	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec		GHA	Dec				
0	217°12.4	218°37.1	N00°56.9	89°39.6	N21°04.0	137°00.8	N22°53.4	218°34.4	S02°46.1						Denebola	182°24.5	
1	232°14.8	233°38.2	57.0	104°40.9	03.7	152°02.8	53.4	233°36.7	46.0						Gienah	175°43.1	
2	247°17.3	248°39.3	57.1	119°42.2	03.4	167°04.7	53.5	248°38.9	45.9						Acrux	172°59.3	
3	262°19.7	263°40.5	· · 57.2	134°43.5	· · 03.1	182°06.6	· · 53.5	263°41.1	· · 45.8						Gacrux	171°50.9	
4	277°22.2	278°41.6	57.4	149°44.8	02.8	197°08.5	53.5	278°43.3	45.7						Alioth	166°12.2	
5	292°24.7	293°42.7	57.5	164°46.1	02.4	212°10.5	53.6	293°45.6	45.6						Spica	158°21.8	
6	307°27.1	308°43.8	N00°57.6	179°47.4	N21°02.1	227°12.4	N22°53.6	308°47.8	S02°45.5						Alkaid	152°51.3	
7	322°29.6	323°44.9	57.7	194°48.7	01.8	242°14.3	53.7	323°50.0	45.4						Hadar	148°35.1	
8	337°32.1	338°46.0	57.9	209°50.0	01.5	257°16.3	53.7	338°52.2	45.3						Menkent	147°57.0	
9	352°34.5	353°47.1	· · 58.0	224°51.3	· · 01.1	272°18.2	· · 53.7	353°54.5	· · 45.2						Arcturus	145°47.4	
10	7°37.0	8°48.2	58.1	239°52.6	00.8	287°20.1	53.8	8°56.7	45.1						Rigel Kent.	139°39.4	
11	22°39.5	23°49.2	58.2	254°53.9	00.5	302°22.0	53.8	23°58.9	45.0						Kochab	137°18.6	
12	37°41.9	38°50.3	N00°58.4	269°55.2	N21°00.2	317°24.0	N22°53.8	39°01.1	S02°44.9						Zuben'ubi	136°55.4	
13	52°44.4	53°51.4	58.5	284°56.5	20°59.8	332°25.9	53.9	54°03.4	44.8						Alphecca	126°03.2	
14	67°46.9	68°52.5	58.7	299°57.8	59.5	347°27.8	53.9	69°05.6	44.7						Antares	112°15.2	
15	82°49.3	83°53.6	· · 58.8	314°59.1	· · 59.2	2°29.7	· · 54.0	84°07.8	· · 44.6						Atria	107°08.8	
16	97°51.8	98°54.7	58.9	330°00.4	58.9	17°31.7	54.0	99°10.0	44.5						Sabik	102°02.3	
17	112°54.2	113°55.7	59.1	345°01.7	58.5	32°33.6	54.0	114°12.3	44.4						Shaula	96°09.8	
18	127°56.7	128°56.8	N00°59.2	0°03.0	N20°58.2	47°35.5	N22°54.1	129°14.5	S02°44.3						Rasalhague	95°58.1	
19	142°59.2	143°57.9	59.4	15°04.3	57.9	62°37.4	54.1	144°16.7	44.3						Eltanin	90°41.7	
20	158°01.6	158°59.0	59.5	30°05.6	57.6	77°39.4	54.2	159°18.9	44.2						Kaus Aust.	83°32.0	
21	173°04.1	174°00.0	· · 59.6	45°06.9	· · 57.2	92°41.3	· · 54.2	174°21.2	· · 44.1						Vega	80°32.9	
22	188°06.6	189°01.1	59.8	60°08.2	56.9	107°43.2	54.2	189°23.4	44.0						Nunki	75°47.3	
23	203°09.0	204°02.2	59.9	75°09.5	56.6	122°45.1	54.3	204°25.6	43.9						Altair	61°59.7	
Mer.pass. 09:30			ν 1.1' d 0.1' m -4.73			ν 1.3' d -0.3' m 0.91			ν 1.9' d 0.0' m -1.99			ν 2.2' d -0.1' m 1.17					

Wed	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec		GHA	Dec				
0	218°11.5	219°03.2	N01°00.1	90°10.8	N20°56.3	137°47.1	N22°54.3	219°27.8	S02°43.8						Venus	1°56.6	
1	233°14.0	234°04.3	00.2	105°12.1	55.9	152°49.0	54.3	234°30.1	43.7						Mars	232°55.0	
2	248°16.4	249°05.4	00.4	120°13.4	55.6	167°50.9	54.4	249°32.3	43.6						Jupiter	280°01.3	
3	263°18.9	264°06.4	· · 00.5	135°14.7	· · 55.3	182°52.8	· · 54.4	264°34.5	· · 43.5						Saturn	1°27.8	
4	278°21.4	279°07.5	00.7	150°16.0	54.9	197°54.8	54.5	279°36.8	43.4								
5	293°23.8	294°08.5	00.9	165°17.3	54.6	212°56.7	54.5	294°39.0	43.3								
6	308°26.3	309°09.6	N01°01.0	180°18.5	N20°54.3	227°58.6	N22°54.5	309°41.2	S02°43.2								
7	323°28.7	324°10.6	01.2	195°19.8	54.0	243°00.5	54.6	324°43.4	43.1								
8	338°31.2	339°11.7	01.3	210°21.1	53.6	258°02.5	54.6	339°45.7	43.0								
9	353°33.7	354°12.7	· · 01.5	225°22.4	· · 53.3	273°04.4	· · 54.6	354°47.9	· · 42.9								
10	8°36.1	9°13.8	01.7	240°23.7	53.0	288°06.3	54.7	9°50.1	42.8								
11	23°38.6	24°14.8	01.8	255°25.0	52.6	303°08.2	54.7	24°52.3	42.7								
12	38°41.1	39°15.9	N01°02.0	270°26.3	N20°52.3	318°10.2	N22°54.8	39°54.6	S02°42.6								
13	53°43.5	54°16.9	02.1	285°27.6	52.0	333°12.1	54.8	54°56.8	42.5								
14	68°46.0	69°17.9	02.3	300°28.9	51.7	348°14.0	54.8	69°59.0	42.4								
15	83°48.5	84°19.0	· · 02.5	315°30.2	· · 51.3	3°15.9	· · 54.9	85°01.3	· · 42.4								
16	98°50.9	99°20.0															

h Sun Moon						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	180°37.4	N14°09.8	179°11.2	6.2'	N18°24.6	14.4' 61.3'
1	195°37.5	10.6	193°36.4	6.1'	18°39.1	14.3' 61.3'
2	210°37.5	11.4	208°01.5	6.0'	18°53.3	14.2' 61.3'
3	225°37.6	· · 12.2	222°26.5	5.9'	19°07.5	14.0' 61.3'
4	240°37.7	12.9	236°51.4	5.8'	19°21.5	13.9' 61.3'
5	255°37.8	13.7	251°16.2	5.7'	19°35.4	13.8' 61.3'
6	270°37.9	N14°14.5	265°40.8	5.6'	N19°49.2	13.6' 61.3'
7	285°38.0	15.3	280°05.4	5.5'	20°02.8	13.5' 61.3'
8	300°38.1	16.1	294°29.9	5.4'	20°16.3	13.3' 61.2'
9	315°38.2	· · 16.9	308°54.3	5.3'	20°29.6	13.2' 61.2'
10	330°38.3	17.6	323°18.6	5.2'	20°42.8	13.0' 61.2'
11	345°38.4	18.4	337°42.8	5.1'	20°55.8	12.9' 61.2'
12	0°38.4	N14°19.2	352°06.9	5.0'	N21°08.7	12.8' 61.2'
13	15°38.5	20.0	6°31.0	4.9'	21°21.5	12.6' 61.2'
14	30°38.6	20.8	20°54.9	4.8'	21°34.1	12.4' 61.2'
15	45°38.7	· · 21.6	35°18.7	4.7'	21°46.5	12.3' 61.2'
16	60°38.8	22.3	49°42.4	4.6'	21°58.8	12.1' 61.2'
17	75°38.9	23.1	64°06.1	4.5'	22°11.0	12.0' 61.1'
18	90°39.0	N14°23.9	78°29.6	4.5'	N22°22.9	11.8' 61.1'
19	105°39.0	24.7	92°53.0	4.4'	22°34.7	11.6' 61.1'
20	120°39.1	25.4	107°16.4	4.3'	22°46.4	11.5' 61.1'
21	135°39.2	· · 26.2	121°39.7	4.2'	22°57.9	11.3' 61.1'
22	150°39.3	27.0	136°02.9	4.1'	23°09.2	11.1' 61.1'
23	165°39.4	27.8	150°26.0	4.0'	23°20.3	11.0' 61.0'
SD = 15.9' $d = 0.8'$ SD = 16.7'						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	180°39.5	N14°28.6	164°49.0	3.9'	N23°31.3	10.8' 61.0'
1	195°39.6	29.3	179°11.9	3.8'	23°42.1	10.6' 61.0'
2	210°39.6	30.1	193°34.8	3.8'	23°52.7	10.4' 61.0'
3	225°39.7	· · 30.9	207°57.5	3.7'	24°03.2	10.3' 61.0'
4	240°39.8	31.7	222°20.2	3.6'	24°13.4	10.1' 61.0'
5	255°39.9	32.4	236°42.8	3.5'	24°23.5	9.9' 60.9'
6	270°40.0	N14°33.2	251°33.5	3.5'	N24°33.4	9.7' 60.9'
7	285°40.1	34.0	265°27.8	3.4'	24°43.2	9.5' 60.9'
8	300°40.2	34.8	279°50.2	3.3'	24°52.7	9.4' 60.9'
9	315°40.2	· · 35.5	294°12.5	3.2'	25°02.1	9.2' 60.8'
10	330°40.3	36.3	308°34.7	3.2'	25°11.2	9.0' 60.8'
11	345°40.4	37.1	322°56.9	3.1'	25°20.2	8.8' 60.8'
12	0°40.5	N14°37.9	337°19.0	3.0'	N25°29.0	8.6' 60.8'
13	15°40.6	38.6	351°41.1	3.0'	25°37.6	8.4' 60.8'
14	30°40.7	39.4	6°03.0	2.9'	25°46.0	8.2' 60.7'
15	45°40.7	· · 40.2	20°25.0	2.9'	25°54.2	8.0' 60.7'
16	60°40.8	40.9	34°46.8	2.8'	26°02.2	7.8' 60.7'
17	75°40.9	41.7	49°08.6	2.8'	26°10.0	7.6' 60.7'
18	90°41.0	N14°42.5	63°30.4	2.7'	N26°17.6	7.4' 60.6'
19	105°41.1	43.2	77°52.1	2.7'	26°25.0	7.2' 60.6'
20	120°41.1	44.0	92°13.8	2.6'	26°32.3	7.0' 60.6'
21	135°41.2	· · 44.8	106°35.4	2.6'	26°39.3	6.8' 60.5'
22	150°41.3	45.5	120°56.9	2.5'	26°46.1	6.6' 60.5'
23	165°41.4	46.3	135°18.5	2.5'	26°52.7	6.4' 60.5'
SD = 15.9' $d = 0.8'$ SD = 16.6'						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	180°41.5	N14°47.1	149°40.0	2.5'	N26°59.1	6.2' 60.5'
1	195°41.5	47.8	164°01.4	2.4'	27°05.3	6.0' 60.4'
2	210°41.6	48.6	178°22.9	2.4'	27°11.3	5.8' 60.4'
3	225°41.7	· · 49.4	192°44.3	2.4'	27°17.1	5.6' 60.4'
4	240°41.8	50.1	207°05.6	2.3'	27°22.7	5.4' 60.3'
5	255°41.9	50.9	221°27.0	2.3'	27°28.1	5.2' 60.3'
6	270°41.9	N14°51.7	235°48.3	2.3'	N27°33.3	5.0' 60.3'
7	285°42.0	52.4	250°09.6	2.3'	27°38.2	4.8' 60.2'
8	300°42.1	53.2	264°30.9	2.3'	27°43.0	4.5' 60.2'
9	315°42.2	· · 54.0	278°52.2	2.3'	27°47.5	4.3' 60.2'
10	330°42.2	54.7	293°13.4	2.3'	27°51.9	4.1' 60.2'
11	345°42.3	55.5	307°34.7	2.3'	27°56.0	3.9' 60.1'
12	0°42.4	N14°56.3	321°56.0	2.3'	N27°59.9	3.7' 60.1'
13	15°42.5	57.0	336°17.2	2.3'	28°03.6	3.5' 60.1'
14	30°42.6	57.8	350°38.5	2.3'	28°07.1	3.3' 60.0'
15	45°42.6	· · 58.5	4°59.8	2.3'	28°10.4	3.1' 60.0'
16	60°42.7	14°59.3	19°21.0	2.3'	28°13.5	2.9' 60.0'
17	75°42.8	15°00.1	33°42.3	2.3'	28°16.4	2.7' 59.9'
18	90°42.9	N15°00.8	48°03.6	2.3'	N28°19.1	2.5' 59.9'
19	105°42.9	01.6	62°24.9	2.3'	28°21.5	2.2' 59.8'
20	120°43.0	02.3	76°46.3	2.4'	28°23.8	2.0' 59.8'
21	135°43.1	· · 03.1	91°07.6	2.4'	28°25.8	1.8' 59.8'
22	150°43.2	03.8	105°29.0	2.4'	28°27.6	1.6' 59.7'
23	165°43.2	04.6	119°50.4	2.5'	28°29.2	1.4' 59.7'
SD = 15.9' $d = 0.8'$ SD = 16.5'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	02:07	21:54	—:—	—:—
N 70°	—:—	—:—	02:41	21:18	—:—	—:—
68°	—:—	01:25	03:06	20:52	22:38	—:—
66°	—:—	02:06	03:25	20:33	21:53	—:—
64°	—:—	02:33	03:40	20:17	21:25	—:—
62°	01:15	02:54	03:52	20:04	21:03	22:47
60°	01:52	03:11	04:03	19:53	20:46	22:07
N 58°	02:17	03:25	04:12	19:44	20:32	21:40
56°	02:37	03:36	04:20	19:36	20:20	21:20
54°	02:53	03:47	04:27	19:28	20:10	21:04
52°	03:06	03:55	04:34	19:22	20:00	20:51
50°	03:17	04:03	04:40	19:16	19:52	20:39
45°	03:40	04:20	04:52	19:03	19:36	20:15
N 40°	03:58	04:33	05:02	18:53	19:22	19:58
35°	04:12	04:44	05:11	18:44	19:11	19:43
30°	04:24	04:54	05:19	18:36	19:01	19:32
20°	04:42	05:09	05:32	18:23	18:46	19:13
N 10°	04:56	05:22	05:43	18:11	18:33	18:59
0°	05:08	05:33	05:54	18:01	18:22	18:47
S 10°	05:18	05:43	06:04	17:50	18:12	18:37
20°	05:27	05:53	06:15	17:39	18:02	18:28
30°	05:35	06:03	06:28	17:26	17:51	18:19
35°	05:39	06:09	06:35	17:19	17:45	18:15
40°	05:43	06:15	06:43	17:11	17:39	18:11
45°	05:47	06:22	06:53	17:01	17:32	18:07
S 50°	05:52	06:30	07:04	16:50	17:24	18:02
52°	05:54	06:33	07:09	16:45	17:20	18:00
54°	05:55	06:37	07:15	16:39	17:16	17:58
56°	05:58	06:42	07:21	16:32	17:12	17:56
58°	06:00	06:46	07:29	16:25	17:07	17:54
S 60°	06:02	06:51	07:37	16:17	17:02	17:52

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	□	□	□	□	□	□
N 70°	01:36	□	□	□	□	□
68°	02:19	□	□	□	□	□
66°	02:49	01:46	□	· ·	00:05	□
64°	03:11	02:48	□	23:04	□	□
62°	03:29	03:23	03:11	22:30	· ·	00:54
60°	03:44	03:48	04:00	22:05	· ·	00:06
N 58°	03:57	04:09	04:31	21:46	23:36	· ·
56°	04:09	04:25	04:54	21:29	23:12	· ·
54°	04:19	04:40	05:14	21:16	22:54	· ·
52°	04:27	04:53	05:30	21:04	22:38	23:58
50°	04:35	05:04	05:44	20:53	22:24	23:42
45°	04:53	05:27	06:13	20:31	21:56	23:11
N 40°	05:07	05:46	06:35	20:13	21:34	22:47
35°	05:19	06:02	06:54	19:58	21:16	22:28
30°	05:29	06:16	07:10	19:45	21:00	22:11
20°	05:48	06:40	07:38	19:24	20:34	21:42
N 10°	06:04	07:00	08:02	19:05	20:11	21:17
0°	06:19	07:20	08:24	18:47	19:50	20:55
S 10°	06:34	07:39	08:46	18:30	19:29	20:32
20°	06:51	08:00	09:10	18:11	19:06	20:07
30°	07:10	08:25	09:38	17:50	18:40	19:38
35°	07:21	08:39	09:55	17:37	18:25	19:21
40°	07:34	08:56	10:14	17:23	18:07	19:02
45°	07:49	09:17	10:38	17:06	17:46	18:38
S 50°	08:08	09:42	11:08	16:46	17:19	18:07
52°	08:17	09:55	11:24	16:36	17:06	17:51
54°	08:28	10:09	11:41	16:25	16:51	17:34
56°	08:39	10:26	12:03	16:13	16:34	17:12
58°	08:53	10:46	12:30	15:58	16:13	16:44
S 60°	09:08	11:12	13:09	15:42	15:47	16:05

Day	Sun		Mer. Pass hh:mm	Moon		Age 1-3 0-7%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
28	02:29	02:34	11:57	12:33	00:03	
29	02:38	02:42	11:57	13:35	01:03	
30	02:46	02:50	11:57	14:39	02:07	

May 01, 02, 03 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	219°10.6	219°28.2	N01°04.1	90°41.8	N20°48.4	138°33.2	N22°55.2	220°21.3	S02°41.5					Alpheratz	357°34.8	29°13.6	
1	234°13.1	234°29.2		105°43.1	48.0	153°35.1	55.3	235°23.5	41.4					Ankaa	353°07.2	-42°10.0	
2	249°15.6	249°30.2	04.4	120°44.4	47.7	168°37.1	55.3	250°25.8	41.3					Schedar	349°31.3	56°40.4	
3	264°18.0	264°31.2	· · 04.6	135°45.7	· · 47.4	183°39.0	· · 55.3	265°28.0	· · 41.2					Diphda	348°47.3	-17°50.9	
4	279°20.5	279°32.2	04.8	150°47.0	47.0	198°40.9	55.4	280°30.2	41.1					Achernar	335°20.5	-57°06.4	
5	294°23.0	294°33.2	05.0	165°48.2	46.7	213°42.8	55.4	295°32.5	41.0					Hamal	327°51.3	23°34.8	
6	309°25.4	309°34.3	N01°05.2	180°49.5	N20°46.4	228°44.8	N22°55.4	310°34.7	S02°40.9					Polaris	314°24.6	89°22.3	
7	324°27.9	324°35.3	05.4	195°50.8	46.0	243°46.7	55.5	325°36.9	40.8					Acamar	315°12.0	-40°12.2	
8	339°30.4	339°36.3	05.6	210°52.1	45.7	258°48.6	55.5	340°39.1	40.7					Menkar	314°06.2	4°11.3	
9	354°32.8	354°37.3	· · 05.8	225°53.4	· · 45.4	273°50.5	· · 55.6	355°41.4	· · 40.7					Mirfak	308°28.4	49°57.0	
10	9°35.3	9°38.3	06.0	240°54.7	45.0	288°52.4	55.6	10°43.6	40.6					Aldebaran	290°39.7	16°33.6	
11	24°37.7	24°39.2	06.2	255°56.0	44.7	303°54.4	55.6	25°45.8	40.5					Rigel	281°03.9	-8°10.4	
12	39°40.2	39°40.2	N01°06.3	270°57.3	N20°44.4	318°56.3	N22°55.7	40°48.1	S02°40.4					Capella	280°21.9	46°01.5	
13	54°42.7	54°41.2	06.5	285°58.5	44.0	333°58.2	55.7	55°50.3	40.3					Bellatrix	278°22.9	6°22.3	
14	69°45.1	69°42.2	06.7	300°59.8	43.7	349°00.1	55.7	70°52.5	40.2					Elnath	278°01.9	28°37.7	
15	84°47.6	84°43.2	· · 06.9	316°01.1	· · 43.4	4°02.0	· · 55.8	85°54.8	· · 40.1					Alnilam	275°37.7	-1°11.2	
16	99°50.1	99°44.2	07.1	331°02.4	43.0	19°04.0	55.8	100°57.0	40.0					Betelgeuse	270°52.1	7°24.7	
17	114°52.5	114°45.2	07.4	346°03.7	42.7	34°05.9	55.9	115°59.2	39.9					Canopus	263°52.6	-52°42.7	
18	129°55.0	129°46.2	N01°07.6	1°05.0	N20°42.4	49°07.8	N22°55.9	131°01.4	S02°39.8					Sirius	258°26.2	-16°45.2	
19	144°57.5	144°47.1	07.8	16°06.3	42.0	64°09.7	55.9	146°03.7	39.7					Adhara	255°05.9	-29°00.6	
20	159°59.9	159°48.1	08.0	31°07.5	41.7	79°11.6	56.0	161°05.9	39.6					Procyon	244°50.7	5°09.6	
21	175°02.4	174°49.1	· · 08.2	46°08.8	· · 41.4	94°13.5	· · 56.0	176°08.1	· · 39.5					Pollux	243°17.1	27°58.0	
22	190°04.9	189°50.1	08.4	61°10.1	41.0	109°15.5	56.0	191°10.4	39.4					Avior	234°14.8	-59°35.7	
23	205°07.3	204°51.0	08.6	76°11.4	40.7	124°17.4	56.1	206°12.6	39.3					Suhail	222°46.1	-43°32.3	
Mer.pass. 09:22			ν 1.0' d 0.2' m -4.72			ν 1.3' d -0.3' m 0.94			ν 1.9' d 0.0' m -1.98			ν 2.2' d -0.1' m 1.17					

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	220°09.8	219°52.0	N01°08.8	91°12.7	N20°40.4	139°19.3	N22°56.1	221°14.8	S02°39.2					
1	235°12.2	234°53.0	09.0	106°14.0	40.0	154°21.2	56.1	236°17.1	39.2					
2	250°14.7	249°54.0	09.2	121°15.2	39.7	169°23.1	56.2	251°19.3	39.1					
3	265°17.2	264°54.9	· · 09.5	136°16.5	· · 39.3	184°25.1	· · 56.2	266°21.5	· · 39.0					
4	280°19.6	279°55.9	09.7	151°17.8	39.0	199°27.0	56.3	281°23.8	38.9					
5	295°22.1	294°56.8	09.9	166°19.1	38.7	214°28.9	56.3	296°26.0	38.8					
6	310°24.6	309°57.8	N01°10.1	181°20.4	N20°38.3	229°30.8	N22°56.3	311°28.2	S02°38.7					
7	325°27.0	324°58.8	10.3	196°21.6	38.0	244°32.7	56.4	326°30.4	38.6					
8	340°29.5	339°59.7	10.6	211°22.9	37.7	259°34.6	56.4	341°32.7	38.5					
9	355°32.0	355°00.7	· · 10.8	226°24.2	· · 37.3	274°36.6	· · 56.4	356°34.9	· · 38.4					
10	10°34.4	10°01.6	11.0	241°25.5	37.0	289°38.5	56.5	11°37.1	38.3					
11	25°36.9	25°02.6	11.2	256°26.8	36.6	304°40.4	56.5	26°39.4	38.2					
12	40°39.4	40°03.5	N01°11.5	271°28.0	N20°36.3	319°42.3	N22°56.6	41°41.6	S02°38.1					
13	55°41.8	55°04.5	11.7	286°29.3	36.0	334°44.2	56.6	56°43.8	38.0					
14	70°44.3	70°05.4	11.9	301°30.6	35.6	349°46.2	56.6	71°46.1	37.9					
15	85°46.7	85°06.4	· · 12.2	316°31.9	· · 35.3	4°48.1	· · 56.7	86°48.3	· · 37.9					
16	100°49.2	100°07.3	12.4	331°33.2	35.0	19°50.0	56.7	101°50.5	37.8					
17	115°51.7	115°08.2	12.6	346°34.4	34.6	34°51.9	56.7	116°52.8	37.7					
18	130°54.1	130°09.2	N01°12.9	1°35.7	N20°34.3	49°53.8	N22°56.8	131°55.0	S02°37.6					
19	145°56.6	145°10.1	13.1	16°37.0	33.9	64°55.7	56.8	146°57.2	37.5					
20	160°59.1	160°11.1	13.3	31°38.3	33.6	79°57.6	56.8	161°59.5	37.4					
21	176°01.5	175°12.0	· · 13.6	46°39.6	· · 33.3	94°59.6	· · 56.9	177°01.7	· · 37.3					
22	191°04.0	190°12.9	13.8	61°40.8	32.9	110°01.5	56.9	192°03.9	37.2					
23	206°06.5	205°13.8	14.0	76°42.1	32.6	125°03.4	57.0	207°06.2	37.1					
Mer.pass. 09:18			ν 1.0' d 0.2' m -4.71			ν 1.3' d -0.3' m 0.95			ν 1.9' d 0.0' m -1.98			ν 2.2' d -0.1' m 1.17		

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	221°08.9	220°14.8	N01°14.3	91°43.4	N20°32.2	140°05.3	N22°57.0	222°08.4	S02°37.0					
1	236°11.4	235°15.7	14.5	106°44.7	31.9	155°07.2	57.0	237°10.6	36.9					
2	251°13.8	250°16.6	14.8	121°45.9	31.6	170°09.1	57.1	252°12.9	36.8					
3	266°16.3	265°17.5	· · 15.0	136°47.2	· · 31.2	185°11.1	· · 57.1	267°15.1	· · 36.7					
4	281°18.8	280°18.5	15.3	151°48.5	30.9	200°13.0	57.1	282°17.3	36.7					
5	296°21.2	295°19.4	15.5	166°49.8	30.5	215°14.9	57.2	297°19.6	36.6					
6	311°23.7	310°20.3	N01°15.8	181°51.1	N20°30.2	230°16.8	N22°57.2	312°21.8	S02°36.5					
7	326°26.2	325°21.2	16.0	196°52.3	29.9	245°18.7	57.2	327°24.0	36.4					
8	341°28.6	340°22.1	16.3	211°53.6	29.5	260°20.6	57.3	342°26.3	36.3					
9	356°31.1	355°23.0	· · 16.5	226°54.9	· · 29.2	275°22.5	· · 57.3	357°28.5	· · 36.2					
10	11°33.6	10°24.0	16.8	241°56.2	28.8	290°24.5	57.4	12°30.7	36.1					
11	26°36.0	25°24.9	17.0	256°57.4	28.5	305°26.4	57.4	27°33.0	36.0					
12	41°38.5	40°25.8	N01°17.3	271°58.7	N20°28.1	320°28.3	N22°57.4	42°35.2	S02°35.9					
13	56°41.0	55°26.7	17.6	287°00.0	27.8	335°30.2	57.5	57°37.4	35.8					
14	71°43.4	70°27.6	17.8	302°01.2	27.5	350°32.1	57.5	72°39.7	35.7					
15	86°45.9	85°28.5	· · 18.1	317°02.5	· · 27.1	5°34.0	· · 57.5	87°41.9	· · 35.6					
16	101°48.3	100°29.4	18.3	332°03.8	26.8	20°35.9	57.6	102°44.1	35.6					
17	116°50.8	115°30.3	18.6	347°05.1	26.4	35°37.9	57.6	117°46.4	35.5					
18	131°53.3	130°31.2	N01°18.9	2°06.3	N20°26.1	50°39.8	N22°57.6	132°48.6	S02°35.4					
19	146°55.7	145°32.1	19.1	17°07.6	25.7	65°41.7	57.7	147°50.8	35.3					
20	161°58.2	160°33.0	19.4	32°08.9	25.4	80°43.6	57.7	162°53.1	35.2					
21	177°00.7	175°33.8	· · 19.7	47°10.2	· · 25.1	95°45.5	· · 57.8	177°55.3	· · 35.1					
22	192°03.1	190°34.7	19.9	62°11.4	24.7	110°47.4	57.8	192°57.5	35.0					
23	207°05.6	205°35.6	20.2	77°12.7	24.4	125°49.3	57.8	207°59.8	34.9					
Mer.pass. 09:14			ν 0.9' d 0.2' m -4.70			ν 1.3' d -0.3' m 0.97			ν 1.9' d 0.0' m -1.98			ν 2.2' d -0.1' m 1.17		

May 01 Thu			SHA	Mer.pass
Venus			0°17.6	09

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	180°43.3	N15°05.4	134°11.9	2.5'	N28°30.7	1.2'	59.7'
1	195°43.4	06.1	148°33.4	2.5'	28°31.9	1.0'	59.6'
2	210°43.5	06.9	162°54.9	2.6'	28°32.9	0.8'	59.6'
3	225°43.5	· · 07.6	177°16.5	2.6'	28°33.7	0.6'	59.6'
4	240°43.6	08.4	191°38.1	2.7'	28°34.3	0.4'	59.5'
5	255°43.7	09.1	205°59.7	2.7'	28°34.6	0.2'	59.5'
6	270°43.8	N15°09.9	220°21.5	2.8'	N28°34.8	-0.0'	59.5'
7	285°43.8	10.6	234°43.2	2.8'	28°34.8	-0.2'	59.4'
8	300°43.9	11.4	249°05.0	2.9'	28°34.6	-0.4'	59.4'
9	315°44.0	· · 12.1	263°26.9	2.9'	28°34.2	-0.6'	59.3'
10	330°44.0	12.9	277°48.9	3.0'	28°33.5	-0.8'	59.3'
11	345°44.1	13.7	292°10.9	3.1'	28°32.7	-1.0'	59.3'
12	0°44.2	N15°14.4	306°32.9	3.1'	N28°31.7	-1.2'	59.2'
13	15°44.3	15.2	320°55.1	3.2'	28°30.5	-1.4'	59.2'
14	30°44.3	15.9	335°17.3	3.3'	28°29.0	-1.6'	59.1'
15	45°44.4	· · 16.7	349°39.6	3.4'	28°27.4	-1.8'	59.1'
16	60°44.5	17.4	4°01.9	3.4'	28°25.6	-2.0'	59.1'
17	75°44.5	18.2	18°24.4	3.5'	28°23.6	-2.2'	59.0'
18	90°44.6	N15°18.9	32°46.9	3.6'	N28°21.4	-2.4'	59.0'
19	105°44.7	19.6	47°09.5	3.7'	28°19.1	-2.6'	59.0'
20	120°44.8	20.4	61°32.2	3.8'	28°16.5	-2.8'	58.9'
21	135°44.8	· · 21.1	75°55.0	3.9'	28°13.7	-2.9'	58.9'
22	150°44.9	21.9	90°17.8	4.0'	28°10.8	-3.1'	58.8'
23	165°45.0	22.6	104°40.8	4.1'	28°07.6	-3.3'	58.8'
SD = 15.9'			SD = 16.3'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	180°45.0	N15°23.4	119°03.9	4.2'	N28°04.3	-3.5'	58.8'
1	195°45.1	24.1	133°27.0	4.3'	28°00.8	-3.7'	58.7'
2	210°45.2	24.9	147°50.3	4.4'	27°57.2	-3.9'	58.7'
3	225°45.2	· · 25.6	162°13.6	4.5'	27°53.3	-4.0'	58.6'
4	240°45.3	26.4	176°37.1	4.6'	27°49.3	-4.2'	58.6'
5	255°45.4	27.1	191°00.6	4.7'	27°45.0	-4.4'	58.6'
6	270°45.4	N15°27.8	205°24.3	4.8'	N27°40.7	-4.6'	58.5'
7	285°45.5	28.6	219°48.1	4.9'	27°36.1	-4.7'	58.5'
8	300°45.6	29.3	234°12.0	5.0'	27°31.4	-4.9'	58.4'
9	315°45.6	· · 30.1	248°36.0	5.1'	27°26.5	-5.1'	58.4'
10	330°45.7	30.8	263°00.1	5.2'	27°21.4	-5.2'	58.4'
11	345°45.8	31.6	277°24.3	5.3'	27°16.1	-5.4'	58.3'
12	0°45.8	N15°32.3	291°48.7	5.5'	N27°10.7	-5.6'	58.3'
13	15°45.9	33.0	306°13.1	5.6'	27°05.2	-5.7'	58.2'
14	30°46.0	33.8	320°37.7	5.7'	26°59.4	-5.9'	58.2'
15	45°46.0	· · 34.5	335°02.4	5.8'	26°53.6	-6.0'	58.2'
16	60°46.1	35.3	349°27.2	5.9'	26°47.5	-6.2'	58.1'
17	75°46.2	36.0	3°52.1	6.1'	26°41.3	-6.4'	58.1'
18	90°46.2	N15°36.7	18°17.2	6.2'	N26°34.9	-6.5'	58.0'
19	105°46.3	37.5	32°42.4	6.3'	26°28.4	-6.7'	58.0'
20	120°46.4	38.2	47°07.7	6.4'	26°21.8	-6.8'	58.0'
21	135°46.4	· · 38.9	61°33.1	6.6'	26°15.0	-7.0'	57.9'
22	150°46.5	39.7	75°58.6	6.7'	26°08.0	-7.1'	57.9'
23	165°46.6	40.4	90°24.3	6.8'	26°00.9	-7.3'	57.8'
SD = 15.9'			SD = 16.0'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°46.6	N15°41.2	104°50.1	6.9'	N25°53.6	-7.4'	57.8'
1	195°46.7	41.9	119°16.1	7.1'	25°46.2	-7.5'	57.8'
2	210°46.7	42.6	133°42.1	7.2'	25°38.7	-7.7'	57.7'
3	225°46.8	· · 43.4	148°08.3	7.3'	25°31.0	-7.8'	57.7'
4	240°46.9	44.1	162°34.6	7.4'	25°23.2	-8.0'	57.7'
5	255°46.9	44.8	177°01.1	7.6'	25°15.2	-8.1'	57.6'
6	270°47.0	N15°45.6	191°27.6	7.7'	N25°07.2	-8.2'	57.6'
7	285°47.1	46.3	205°54.3	7.8'	24°58.9	-8.3'	57.5'
8	300°47.1	47.0	220°21.2	8.0'	24°50.6	-8.5'	57.5'
9	315°47.2	· · 47.7	234°48.1	8.1'	24°42.1	-8.6'	57.5'
10	330°47.2	48.5	249°15.2	8.2'	24°33.5	-8.7'	57.4'
11	345°47.3	49.2	263°42.4	8.4'	24°24.8	-8.9'	57.4'
12	0°47.4	N15°49.9	278°09.8	8.5'	N24°15.9	-9.0'	57.3'
13	15°47.4	50.7	292°37.3	8.6'	24°07.0	-9.1'	57.3'
14	30°47.5	51.4	307°04.9	8.7'	23°57.9	-9.2'	57.3'
15	45°47.5	· · 52.1	321°32.6	8.9'	23°48.6	-9.3'	57.2'
16	60°47.6	52.9	336°00.5	9.0'	23°39.3	-9.4'	57.2'
17	75°47.7	53.6	350°28.5	9.1'	23°29.9	-9.6'	57.2'
18	90°47.7	N15°54.3	4°56.6	9.3'	N23°20.3	-9.7'	57.1'
19	105°47.8	55.0	19°24.9	9.4'	23°10.6	-9.8'	57.1'
20	120°47.8	55.8	33°53.3	9.5'	23°00.9	-9.9'	57.0'
21	135°47.9	· · 56.5	48°21.8	9.6'	22°51.0	-10.0'	57.0'
22	150°48.0	57.2	62°50.4	9.8'	22°41.0	-10.1'	57.0'
23	165°48.0	57.9	77°19.2	9.9'	22°30.9	-10.2'	56.9'
SD = 15.9'			SD = 15.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	01:42	22:20	—:—	—:—
N 70°	—:—	—:—	02:24	21:35	—:—	—:—
68°	—:—	00:52	02:52	21:06	23:16	—:—
66°	—:—	01:48	03:13	20:44	22:12	—:—
64°	—:—	02:20	03:29	20:27	21:38	—:—
62°	00:46	02:43	03:43	20:13	21:14	23:21
60°	01:36	03:01	03:55	20:01	20:55	22:23
N 58°	02:05	03:16	04:05	19:51	20:40	21:52
56°	02:27	03:29	04:13	19:42	20:27	21:30
54°	02:44	03:40	04:21	19:34	20:16	21:12
52°	02:58	03:49	04:28	19:27	20:06	20:58
50°	03:10	03:58	04:34	19:21	19:58	20:45
45°	03:35	04:15	04:48	19:07	19:40	20:20
N 40°	03:53	04:29	04:59	18:56	19:25	20:01
35°	04:08	04:41	05:08	18:46	19:14	19:46
30°	04:21	04:51	05:16	18:38	19:04	19:34
20°	04:40	05:07	05:30	18:24	18:47	19:14
N 10°	04:55	05:20	05:42	18:12	18:34	18:59
0°	05:07	05:32	05:54	18:00	18:22	18:47
S 10°	05:18	05:43	06:05	17:49	18:11	18:36
20°	05:27	05:54	06:16	17:37	18:00	18:26
30°	05:36	06:05	06:30	17:24	17:49	18:17
35°	05:41	06:11	06:37	17:16	17:42	18:12
40°	05:46	06:18	06:46	17:07	17:36	18:08
45°	05:50	06:25	06:56	16:57	17:28	18:03
S 50°	05:56	06:34	07:08	16:45	17:19	17:58
52°	05:58	06:38	07:14	16:39	17:15	17:55
54°	06:00	06:42	07:20	16:33	17:11	17:53
56°	06:03	06:47	07:27	16:26	17:06	17:50
58°	06:05	06:52	07:35	16:18	17:01	17:48
S 60°	06:08	06:58	07:44	16:09	16:55	17:45

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	☐	☐	☐	☐	☐	☐
64°	☐	☐	06:26	☐	☐	04:10
62°	☐	04:57	07:15	☐	03:36	03:20
60°	04:37	06:00	07:46	01:45	02:33	02:48
N 58°	05:17	06:34	08:10	01:05	01:59	02:25
56°	05:45	06:59	08:28	00:37	01:34	02:05
54°	06:06	07:19	08:44	00:16	01:13	01:49
52°	06:24	07:36	08:58	· ·	00:56	01:35
50°	06:40	07:50	09:09	· ·	00:42	01:23
45°	07:11	08:20	09:34	· ·	00:12	00:57
N 40°	07:35	08:42	09:53	23:49	· ·	00:37
35°	07:55	09:01	10:10	23:29	· ·	00:20
30°	08:12	09:17	10:23	23:13	· ·	00:05
20°	08:40	09:45	10:47	22:45	23:40	· ·
N 10°	09:05	10:08	11:08	22:21	23:18	· ·
0°	09:28	10:30	11:27	21:58	22:58	23:52
S 10°	09:51	10:52	11:46	21:35	22:37	23:35
20°	10:16	11:15	12:06	21:11	22:15	23:16
30°	10:45	11:42	12:29	20:43	21:49	22:54
35°	11:02	11:58	12:43	20:26	21:34	22:42
40°	11:22	12:16	12:59	20:06	21:16	22:27
45°	11:46	12:39	13:17	19:42	20:54	22:09
S 50°	12:17	13:07	13:40	19:11	20:27	21:48
52°	12:33	13:21	13:51	18:55	20:13	21:37
54°	12:51	13:36	14:04	18:37	19:58	21:25
56°	13:13	13:55	14:18	18:15	19:40	21:11
58°	13:42	14:18	14:35	17:47	19:17	20:55
S 60°	14:24	14:48	14:55	17:05	18:47	20:36

Day	Sun			Moon		Age 4-6 15-34%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss	Mer. Pass hh:mm	Mer.Pass. Upper hh:mm	Lower hh:mm	
01	02:53	02:57	11:57	15:43	03:11	
02	03:00	03:03	11:57	16:44	04:14	
03	03:06	03:09	11:57	17:39	05:12	

May 04, 05, 06 UT (Sun., Mon., Tue.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Sun	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec			
0	222°08.1	220°36.5	N01°20.5	92°14.0	N20°24.0	140°51.2	N22°57.9	223°02.0	S02°34.8				Alpheratz	357°34.8	29°13.6		
1	237°10.5	235°37.4		107°15.2	23.7	155°53.2	57.9	238°04.2	34.7				Ankaa	353°07.2	-42°10.0		
2	252°13.0	250°38.3		122°16.5	23.3	170°55.1	57.9	253°06.5	34.6				Schedar	349°31.3	56°40.4		
3	267°15.5	265°39.1	•• 21.3	137°17.8	•• 23.0	185°57.0	•• 58.0	268°08.7	•• 34.5				Diphda	348°47.3	-17°50.9		
4	282°17.9	280°40.0		152°19.1	22.6	200°58.9	58.0	283°11.0	34.5				Achernar	335°20.5	-57°06.4		
5	297°20.4	295°40.9		167°20.3	22.3	216°00.8	58.0	298°13.2	34.4				Hamal	327°51.3	23°34.8		
6	312°22.8	310°41.8	N01°22.1	182°21.6	N20°21.9	231°02.7	N22°58.1	313°15.4	S02°34.3				Polaris	314°24.5	89°22.2		
7	327°25.3	325°42.7		197°22.9	21.6	246°04.6	58.1	328°17.7	34.2				Acamar	315°12.0	-40°12.2		
8	342°27.8	340°43.5		212°24.1	21.3	261°06.5	58.1	343°19.9	34.1				Menkar	314°06.2	4°11.3		
9	357°30.2	355°44.4	•• 23.0	227°25.4	•• 20.9	276°08.5	•• 58.2	358°22.1	•• 34.0				Mirfak	308°28.4	49°57.0		
10	12°32.7	10°45.3		242°26.7	20.6	291°10.4	58.2	13°24.4	33.9				Aldebaran	290°39.7	16°33.6		
11	27°35.2	25°46.1		257°27.9	20.2	306°12.3	58.3	28°26.6	33.8				Rigel	281°03.9	-8°10.4		
12	42°37.6	40°47.0	N01°23.8	272°29.2	N20°19.9	321°14.2	N22°58.3	43°28.8	S02°33.7				Capella	280°21.9	46°01.5		
13	57°40.1	55°47.9		287°30.5	19.5	336°16.1	58.3	58°31.1	33.6				Bellatrix	278°22.9	6°22.3		
14	72°42.6	70°48.7		302°31.8	19.2	351°18.0	58.4	73°33.3	33.5				Elnath	278°01.9	28°37.7		
15	87°45.0	85°49.6	•• 24.7	317°33.0	•• 18.8	6°19.9	•• 58.4	88°35.5	•• 33.5				Alnilam	275°37.7	-1°11.2		
16	102°47.5	100°50.4		332°34.3	18.5	21°21.8	58.4	103°37.8	33.4				Betelgeuse	270°52.1	7°24.7		
17	117°50.0	115°51.3		347°35.6	18.1	36°23.7	58.5	118°40.0	33.3				Canopus	263°52.6	-52°42.7		
18	132°52.4	130°52.1	N01°25.6	2°36.8	N20°17.8	51°25.7	N22°58.5	133°42.3	S02°33.2				Sirius	258°26.2	-16°45.2		
19	147°54.9	145°53.0		17°38.1	17.4	66°27.6	58.5	148°44.5	33.1				Adhara	255°05.9	-29°00.6		
20	162°57.3	160°53.9		32°39.4	17.1	81°29.5	58.6	163°46.7	33.0				Procyon	244°50.7	5°09.6		
21	177°59.8	175°54.7	•• 26.5	47°40.6	•• 16.7	96°31.4	•• 58.6	178°49.0	•• 32.9				Pollux	243°17.1	27°58.0		
22	193°02.3	190°55.5		62°41.9	16.4	111°33.3	58.6	193°51.2	32.8				Avior	234°14.8	-59°35.7		
23	208°04.7	205°56.4		77°43.2	16.0	126°35.2	58.7	208°53.4	32.7				Suhail	222°46.1	-43°32.3		
Mer.pass. 09:10			ν 0.9' d 0.3' m -4.69			ν 1.3' d -0.3' m 0.98			ν 1.9' d 0.0' m -1.97			ν 2.2' d -0.1' m 1.17					

Mon	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	223°07.2	220°57.2	N01°27.4	92°44.4	N20°15.7	141°37.1	N22°58.7	223°55.7	S02°32.6					
1	238°09.7	235°58.1		107°45.7	15.3	156°39.0	58.7	238°57.9	32.6					
2	253°12.1	250°58.9		122°47.0	15.0	171°40.9	58.8	254°00.2	32.5					
3	268°14.6	265°59.8	•• 28.3	137°48.2	•• 14.6	186°42.8	•• 58.8	269°02.4	•• 32.4					
4	283°17.1	281°00.6		152°49.5	14.3	201°44.8	58.9	284°04.6	32.3					
5	298°19.5	296°01.4		167°50.7	13.9	216°46.7	58.9	299°06.9	32.2					
6	313°22.0	311°02.3	N01°29.2	182°52.0	N20°13.6	231°48.6	N22°58.9	314°09.1	S02°32.1					
7	328°24.4	326°03.1		197°53.3	13.2	246°50.5	59.0	329°11.3	32.0					
8	343°26.9	341°03.9		212°54.5	12.9	261°52.4	59.0	344°13.6	31.9					
9	358°29.4	356°04.8	•• 30.1	227°55.8	•• 12.5	276°54.3	•• 59.0	359°15.8	•• 31.8					
10	13°31.8	11°05.6		242°57.1	12.2	291°56.2	59.1	14°18.1	31.7					
11	28°34.3	26°06.4		257°58.3	11.8	306°58.1	59.1	29°20.3	31.7					
12	43°36.8	41°07.3	N01°31.0	272°59.6	N20°11.5	322°00.0	N22°59.1	44°22.5	S02°31.6					
13	58°39.2	56°08.1		288°00.9	11.1	337°01.9	59.2	59°24.8	31.5					
14	73°41.7	71°08.9		303°02.1	10.8	352°03.8	59.2	74°27.0	31.4					
15	88°44.2	86°09.7	•• 32.0	318°03.4	•• 10.4	7°05.7	•• 59.2	89°29.2	•• 31.3					
16	103°46.6	101°10.5		333°04.6	10.1	22°07.7	59.3	104°31.5	31.2					
17	118°49.1	116°11.4		348°05.9	09.7	37°09.6	59.3	119°33.7	31.1					
18	133°51.6	131°12.2	N01°33.0	3°07.2	N20°09.4	52°11.5	N22°59.3	134°36.0	S02°31.0					
19	148°54.0	146°13.0		18°08.4	09.0	67°13.4	59.4	149°38.2	30.9					
20	163°56.5	161°13.8		33°09.7	08.7	82°15.3	59.4	164°40.4	30.8					
21	178°58.9	176°14.6	•• 33.9	48°11.0	•• 08.3	97°17.2	•• 59.4	179°42.7	•• 30.8					
22	194°01.4	191°15.4		63°12.2	08.0	112°19.1	59.5	194°44.9	30.7					
23	209°03.9	206°16.2		78°13.5	07.6	127°21.0	59.5	209°47.2	30.6					
Mer.pass. 09:06			ν 0.8' d 0.3' m -4.68			ν 1.3' d -0.3' m 0.99			ν 1.9' d 0.0' m -1.97			ν 2.2' d -0.1' m 1.16		

Tue	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	224°06.3	221°17.0	N01°34.9	93°14.7	N20°07.3	142°22.9	N22°59.5	224°49.4	S02°30.5					
1	239°08.8	236°17.8		108°16.0	06.9	157°24.8	59.6	239°51.6	30.4					
2	254°11.3	251°18.6		123°17.3	06.6	172°26.7	59.6	254°53.9	30.3					
3	269°13.7	266°19.4	•• 35.9	138°18.5	•• 06.2	187°28.6	•• 59.7	269°56.1	•• 30.2					
4	284°16.2	281°20.2		153°19.8	05.8	202°30.5	59.7	284°58.4	30.1					
5	299°18.7	296°21.0		168°21.0	05.5	217°32.4	59.7	300°00.6	30.0					
6	314°21.1	311°21.8	N01°36.9	183°22.3	N20°05.1	232°34.4	N22°59.8	315°02.8	S02°30.0					
7	329°23.6	326°22.6		198°23.6	04.8	247°36.3	59.8	330°05.1	29.9					
8	344°26.1	341°23.4		213°24.8	04.4	262°38.2	59.8	345°07.3	29.8					
9	359°28.5	356°24.2	•• 37.9	228°26.1	•• 04.1	277°40.1	•• 59.9	0°09.6	•• 29.7					
10	14°31.0	11°25.0		243°27.3	03.7	292°42.0	59.9	15°11.8	29.6					
11	29°33.4	26°25.8		258°28.6	03.4	307°43.9	22°59.9	30°14.0	29.5					
12	44°35.9	41°26.6	N01°38.9	273°29.8	N20°03.0	322°45.8	N23°00.0	45°16.3	S02°29.4					
13	59°38.4	56°27.4		288°31.1	02.6	337°47.7	00.0	60°18.5	29.3					
14	74°40.8	71°28.2		303°32.4	02.3	352°49.6	00.0	75°20.8	29.2					
15	89°43.3	86°28.9	•• 39.9	318°33.6	•• 01.9	7°51.5	•• 00.1	90°23.0	•• 29.2					
16	104°45.8	101°29.7		333°34.9	01.6	22°53.4	00.1	105°25.2	29.1					
17	119°48.2	116°30.5		348°36.1	01.2	37°55.3	00.1	120°27.5	29.0					
18	134°50.7	131°31.3	N01°41.0	3°37.4	N20°00.9	52°57.2	N23°00.2	135°29.7	S02°28.9					
19	149°53.2	146°32.0		18°38.6	00.5	67°59.1	00.2	150°32.0	28.8					
20	164°55.6	161°32.8		33°39.9	20°00.2	83°01.0	00.2	165°34.2	28.7					
21	179°58.1	176°33.6	•• 42.0	48°41.2	19°59.8	98°02.9	•• 00.3	180°36.4	•• 28.6					
22	195°00.5	191°34.4		63°42.4	59.4	113°04.8	00.3	195°38.7	28.5					
23	210°03.0	206°35.1		78°43.7	59.1	128°06.7	00.3	210°40.9	28.4					
Mer.pass. 09:02			ν 0.8' d 0.3' m -4.67			ν 1.3' d -0.4' m 1.00			ν 1.9' d 0.0' m -1.97			ν 2.2' d -0.1' m 1.16		

May 04 Sun	SHA	Mer.pass
Venus	358°28.4	09:17
Mars	230°05.9	17:50
Jupiter	278°43.2	14:35
Saturn	0°54.0	09:07

May 05 Mon	SHA	Mer.pass
Venus	357°50.0	09:16
Mars	229°37.2	17:48
Jupiter	278°29.9	14:32
Saturn	0°48.5	09:03

May 06 Tue	SHA	Mer.pass
Venus	357°10.7	09:14
Mars	229°08.4	17:46
Jupiter	278°16.6	14:29
Saturn	0°43.1	08:59

Horizontal parallax		
Venus:		0.3
Mars:		0.1

h Sun Moon						
Sun	GHA	Dec	GHA	ν	Dec	d HP
0	180°48.1	N15°58.7	91°48.1	10.0'	N22°20.7	-10.3' 56.9'
1	195°48.1	15°59.4	106°17.1	10.1'	22°10.4	-10.4' 56.9'
2	210°48.2	16°00.1	120°46.2	10.3'	22°00.0	-10.5' 56.8'
3	225°48.3	· · 00.8	135°15.5	10.4'	21°49.5	-10.6' 56.8'
4	240°48.3	01.6	149°44.9	10.5'	21°38.9	-10.7' 56.8'
5	255°48.4	02.3	164°14.4	10.6'	21°28.2	-10.8' 56.7'
6	270°48.4	N16°03.0	178°44.0	10.8'	N21°17.4	-10.9' 56.7'
7	285°48.5	03.7	193°13.8	10.9'	21°06.5	-11.0' 56.6'
8	300°48.5	04.4	207°43.7	11.0'	20°55.5	-11.1' 56.6'
9	315°48.6	· · 05.2	222°13.7	11.1'	20°44.5	-11.1' 56.6'
10	330°48.6	05.9	236°43.8	11.2'	20°33.3	-11.2' 56.5'
11	345°48.7	06.6	251°14.0	11.4'	20°22.1	-11.3' 56.5'
12	0°48.8	N16°07.3	265°44.4	11.5'	N20°10.8	-11.4' 56.5'
13	15°48.8	08.0	280°14.8	11.6'	19°59.4	-11.5' 56.4'
14	30°48.9	08.7	294°45.4	11.7'	19°47.9	-11.6' 56.4'
15	45°48.9	· · 09.5	309°16.1	11.8'	19°36.3	-11.6' 56.4'
16	60°49.0	10.2	323°46.9	11.9'	19°24.7	-11.7' 56.3'
17	75°49.0	10.9	338°17.9	12.0'	19°13.0	-11.8' 56.3'
18	90°49.1	N16°11.6	352°48.9	12.2'	N19°01.2	-11.9' 56.3'
19	105°49.1	12.3	7°20.1	12.3'	18°49.3	-11.9' 56.2'
20	120°49.2	13.0	21°51.3	12.4'	18°37.4	-12.0' 56.2'
21	135°49.2	· · 13.8	36°22.7	12.5'	18°25.4	-12.1' 56.2'
22	150°49.3	14.5	50°54.2	12.6'	18°13.3	-12.2' 56.1'
23	165°49.4	15.2	65°25.7	12.7'	18°01.1	-12.2' 56.1'
SD = 15.9' $d = 0.7'$ SD = 15.5'						
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	180°49.4	N16°15.9	79°57.4	12.8'	N17°48.9	-12.3' 56.1'
1	195°49.5	16.6	94°29.2	12.9'	17°36.6	-12.4' 56.0'
2	210°49.5	17.3	109°01.1	13.0'	17°24.3	-12.4' 56.0'
3	225°49.6	· · 18.0	123°33.1	13.1'	17°11.9	-12.5' 56.0'
4	240°49.6	18.7	138°05.2	13.2'	16°59.4	-12.5' 56.0'
5	255°49.7	19.5	152°37.4	13.3'	16°46.9	-12.6' 55.9'
6	270°49.7	N16°20.2	167°09.7	13.4'	N16°34.3	-12.7' 55.9'
7	285°49.8	20.9	181°42.1	13.5'	16°21.6	-12.7' 55.9'
8	300°49.8	21.6	196°14.6	13.6'	16°08.9	-12.8' 55.8'
9	315°49.9	· · 22.3	210°47.2	13.7'	15°56.1	-12.8' 55.8'
10	330°49.9	23.0	225°19.9	13.8'	15°43.3	-12.9' 55.8'
11	345°50.0	23.7	239°52.7	13.9'	15°30.4	-12.9' 55.7'
12	0°50.0	N16°24.4	254°25.6	14.0'	N15°17.5	-13.0' 55.7'
13	15°50.1	25.1	268°58.5	14.1'	15°04.5	-13.0' 55.7'
14	30°50.1	25.8	283°31.6	14.1'	14°51.5	-13.1' 55.7'
15	45°50.2	· · 26.5	298°04.7	14.2'	14°38.4	-13.1' 55.6'
16	60°50.2	27.2	312°37.9	14.3'	14°25.3	-13.2' 55.6'
17	75°50.3	27.9	327°11.2	14.4'	14°12.1	-13.2' 55.6'
18	90°50.3	N16°28.6	341°44.6	14.5'	N13°58.9	-13.3' 55.5'
19	105°50.4	29.4	356°18.1	14.6'	13°45.6	-13.3' 55.5'
20	120°50.4	30.1	10°51.7	14.6'	13°32.3	-13.4' 55.5'
21	135°50.4	· · 30.8	25°25.3	14.7'	13°18.9	-13.4' 55.5'
22	150°50.5	31.5	39°59.1	14.8'	13°05.5	-13.4' 55.4'
23	165°50.5	32.2	54°32.9	14.9'	12°52.1	-13.5' 55.4'
SD = 15.9' $d = 0.7'$ SD = 15.3'						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	180°50.6	N16°32.9	69°06.7	15.0'	N12°38.6	-13.5' 55.4'
1	195°50.6	33.6	83°40.7	15.0'	12°25.1	-13.6' 55.4'
2	210°50.7	34.3	98°14.7	15.1'	12°11.6	-13.6' 55.3'
3	225°50.7	· · 35.0	112°48.8	15.2'	11°58.0	-13.6' 55.3'
4	240°50.8	35.7	127°23.0	15.2'	11°44.4	-13.7' 55.3'
5	255°50.8	36.4	141°57.2	15.3'	11°30.7	-13.7' 55.3'
6	270°50.9	N16°37.1	156°31.6	15.4'	N11°17.0	-13.7' 55.2'
7	285°50.9	37.8	171°05.9	15.5'	11°03.3	-13.8' 55.2'
8	300°51.0	38.5	185°40.4	15.5'	10°49.5	-13.8' 55.2'
9	315°51.0	· · 39.2	200°14.9	15.6'	10°35.8	-13.8' 55.2'
10	330°51.0	39.9	214°49.5	15.6'	10°21.9	-13.8' 55.1'
11	345°51.1	40.6	229°24.1	15.7'	10°08.1	-13.9' 55.1'
12	0°51.1	N16°41.2	243°58.9	15.8'	N09°54.2	-13.9' 55.1'
13	15°51.2	41.9	258°33.6	15.8'	09°40.3	-13.9' 55.1'
14	30°51.2	42.6	273°08.4	15.9'	09°26.4	-13.9' 55.0'
15	45°51.3	· · 43.3	287°43.3	15.9'	09°12.5	-14.0' 55.0'
16	60°51.3	44.0	302°18.3	16.0'	08°58.5	-14.0' 55.0'
17	75°51.3	44.7	316°53.3	16.1'	08°44.5	-14.0' 55.0'
18	90°51.4	N16°45.4	331°28.3	16.1'	N08°30.5	-14.0' 54.9'
19	105°51.4	46.1	346°03.4	16.2'	08°16.4	-14.1' 54.9'
20	120°51.5	46.8	0°38.6	16.2'	08°02.4	-14.1' 54.9'
21	135°51.5	· · 47.5	15°13.8	16.3'	07°48.3	-14.1' 54.9'
22	150°51.6	48.2	29°49.1	16.3'	07°34.2	-14.1' 54.9'
23	165°51.6	48.9	44°24.4	16.4'	07°20.1	-14.1' 54.8'
SD = 15.8' $d = 0.7'$ SD = 15.1'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	01:11	22:54	—:—	—:—
N 70°	—:—	—:—	02:05	21:54	—:—	—:—
68°	—:—	—:—	02:37	21:20	—:—	—:—
66°	—:—	01:27	03:01	20:55	22:33	—:—
64°	—:—	02:05	03:19	20:36	21:52	—:—
62°	—:—	02:31	03:34	20:21	21:25	—:—
60°	01:17	02:51	03:47	20:08	21:05	22:42
N 58°	01:52	03:07	03:58	19:57	20:48	22:05
56°	02:16	03:21	04:07	19:48	20:34	21:40
54°	02:35	03:33	04:15	19:39	20:22	21:21
52°	02:50	03:43	04:23	19:32	20:12	21:05
50°	03:04	03:52	04:29	19:25	20:03	20:51
45°	03:30	04:11	04:43	19:11	19:44	20:25
N 40°	03:49	04:25	04:55	18:59	19:29	20:05
35°	04:05	04:38	05:05	18:49	19:16	19:49
30°	04:18	04:48	05:14	18:40	19:06	19:36
20°	04:38	05:05	05:28	18:25	18:48	19:16
N 10°	04:54	05:19	05:41	18:12	18:34	19:00
0°	05:07	05:32	05:53	18:00	18:22	18:47
S 10°	05:18	05:43	06:05	17:48	18:10	18:35
20°	05:28	05:55	06:17	17:36	17:59	18:25
30°	05:38	06:07	06:32	17:21	17:46	18:15
35°	05:43	06:13	06:40	17:13	17:40	18:10
40°	05:48	06:21	06:49	17:04	17:32	18:05
45°	05:54	06:29	07:00	16:53	17:24	17:59
S 50°	06:00	06:38	07:13	16:40	17:15	17:53
52°	06:02	06:42	07:19	16:34	17:10	17:51
54°	06:05	06:47	07:26	16:27	17:05	17:48
56°	06:08	06:52	07:33	16:20	17:00	17:45
58°	06:11	06:58	07:41	16:11	16:55	17:42
S 60°	06:14	07:04	07:51	16:02	16:48	17:39

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	☐	08:53	11:33	☐	05:17	04:07
N 70°	☐	09:36	11:50	☐	04:32	03:49
68°	07:24	10:04	12:03	05:04	04:02	03:33
66°	08:16	10:25	12:14	04:10	03:40	03:21
64°	08:48	10:42	12:23	03:37	03:22	03:11
62°	09:12	10:55	12:30	03:13	03:07	03:02
60°	09:30	11:07	12:37	02:53	02:54	02:54
N 58°	09:46	11:17	12:42	02:37	02:43	02:47
56°	09:59	11:26	12:47	02:23	02:34	02:41
54°	10:10	11:33	12:52	02:11	02:25	02:36
52°	10:20	11:40	12:56	02:00	02:18	02:31
50°	10:29	11:46	13:00	01:51	02:11	02:26
45°	10:48	11:59	13:07	01:31	01:56	02:17
N 40°	11:03	12:10	13:14	01:14	01:44	02:08
35°	11:16	12:20	13:20	01:01	01:34	02:01
30°	11:27	12:28	13:25	00:48	01:24	01:55
20°	11:46	12:41	13:33	00:27	01:08	01:44
N 10°	12:03	12:53	13:41	00:09	00:54	01:35
0°	12:18	13:05	13:47	· · ·	00:41	01:26
S 10°	12:33	13:16	13:54	· · ·	00:28	01:16
20°	12:50	13:28	14:02	· · ·	00:13	01:07
30°	13:08	13:41	14:10	23:57	· · ·	00:55
35°	13:19	13:49	14:15	23:47	· · ·	00:49
40°	13:31	13:58	14:20	23:36	· · ·	00:41
45°	13:46	14:08	14:27	23:23	· · ·	00:33
S 50°	14:03	14:21	14:34	23:07	· · ·	00:22
52°	14:12	14:26	14:38	22:59	· · ·	00:17
54°	14:21	14:33	14:41	22:50	· · ·	00:11
56°	14:31	14:40	14:45	22:41	· · ·	00:05
58°	14:43	14:47	14:50	22:30	23:58	· · ·
S 60°	14:57	14:56	14:55	22:17	23:51	· · ·

Day	Sun		Mer. Pass	Moon		Age 7-9 44-64%
	Eqn. of Time 00 ^h mm:ss	Time 12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
04	03:12	03:15	11:57	18:30	06:05	
05	03:18	03:20	11:57	19:15	06:53	
06	03:22	03:25	11:57	19:57	07:37	

May 07, 08, 09 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	225°05.5	221°35.9	N01°43.1	93°44.9	N19°58.7	143°08.6	N23°00.4	225°43.2	S02°28.4	Alpheratz	357°34.8 29°13.6
1	240°07.9	236°36.7	43.4	108°46.2	58.4	158°10.5	00.4	240°45.4	28.3	Ankaa	353°07.2 -42°10.0
2	255°10.4	251°37.4	43.8	123°47.4	58.0	173°12.5	00.4	255°47.6	28.2	Schedar	349°31.2 56°40.4
3	270°12.9	266°38.2	· · 44.2	138°48.7	· · 57.6	188°14.4	· · 00.5	270°49.9	· · 28.1	Diphda	348°47.3 -17°50.9
4	285°15.3	281°39.0	44.5	153°49.9	57.3	203°16.3	00.5	285°52.1	28.0	Achernar	335°20.5 -57°06.4
5	300°17.8	296°39.7	44.9	168°51.2	56.9	218°18.2	00.5	300°54.4	27.9	Hamal	327°51.3 23°34.8
6	315°20.3	311°40.5	N01°45.2	183°52.4	N19°56.6	233°20.1	N23°00.6	315°56.6	S02°27.8	Polaris	314°24.5 89°22.2
7	330°22.7	326°41.2	45.6	198°53.7	56.2	248°22.0	00.6	330°58.9	27.7	Acamar	315°12.0 -40°12.2
8	345°25.2	341°42.0	46.0	213°55.0	55.9	263°23.9	00.6	346°01.1	27.7	Menkar	314°06.2 4°11.3
9	0°27.7	356°42.8	· · 46.3	228°56.2	· · 55.5	278°25.8	· · 00.7	1°03.3	· · 27.6	Mirfak	308°28.4 49°57.0
10	15°30.1	11°43.5	46.7	243°57.5	55.1	293°27.7	00.7	16°05.6	27.5	Aldebaran	290°39.7 16°33.6
11	30°32.6	26°44.3	47.1	258°58.7	54.8	308°29.6	00.7	31°07.8	27.4	Rigel	281°03.9 -8°10.4
12	45°35.0	41°45.0	N01°47.4	274°00.0	N19°54.4	323°31.5	N23°00.8	46°10.1	S02°27.3	Capella	280°21.9 46°01.5
13	60°37.5	56°45.8	47.8	289°01.2	54.1	338°33.4	00.8	61°12.3	27.2	Bellatrix	278°22.9 6°22.3
14	75°40.0	71°46.5	48.2	304°02.5	53.7	353°35.3	00.8	76°14.6	27.1	Elnath	278°01.9 28°37.7
15	90°42.4	86°47.3	· · 48.5	319°03.7	· · 53.3	8°37.2	· · 00.9	91°16.8	· · 27.0	Alnilam	275°37.7 -1°11.2
16	105°44.9	101°48.0	48.9	334°05.0	53.0	23°39.1	00.9	106°19.0	27.0	Betelgeuse	270°52.1 7°24.7
17	120°47.4	116°48.7	49.3	349°06.2	52.6	38°41.0	00.9	121°21.3	26.9	Canopus	263°52.7 -52°42.7
18	135°49.8	131°49.5	N01°49.6	4°07.5	N19°52.2	53°42.9	N23°01.0	136°23.5	S02°26.8	Sirius	258°26.2 -16°45.2
19	150°52.3	146°50.2	50.0	19°08.7	51.9	68°44.8	01.0	151°25.8	26.7	Adhara	255°05.9 -29°00.5
20	165°54.8	161°51.0	50.4	34°10.0	51.5	83°46.7	01.0	166°28.0	26.6	Procyon	244°50.7 5°09.6
21	180°57.2	176°51.7	· · 50.8	49°11.2	· · 51.2	98°48.6	· · 01.1	181°30.3	· · 26.5	Pollux	243°17.1 27°58.0
22	195°59.7	191°52.4	51.2	64°12.5	50.8	113°50.5	01.1	196°32.5	26.4	Avior	234°14.8 -59°35.7
23	211°02.1	206°53.2	51.5	79°13.7	50.4	128°52.4	01.1	211°34.7	26.3	Suhail	222°46.2 -43°32.3
Mer.pass. 08:58		ν 0.8' d 0.4' m -4.66		ν 1.3' d -0.4' m 1.02		ν 1.9' d 0.0' m -1.97		ν 2.2' d -0.1' m 1.16			

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	226°04.6	221°53.9	N01°51.9	94°15.0	N19°50.1	143°54.3	N23°01.2	226°37.0	S02°26.3	Denebola	182°24.5 14°25.8
1	241°07.1	236°54.6	52.3	109°16.2	49.7	158°56.2	01.2	241°39.2	26.2	Gienah	175°43.1 -17°41.1
2	256°09.5	251°55.4	52.7	124°17.5	49.4	173°58.1	01.2	256°41.5	26.1	Acrux	172°59.3 -63°14.6
3	271°12.0	266°56.1	· · 53.1	139°18.7	· · 49.0	189°00.0	· · 01.3	271°43.7	· · 26.0	Gacrux	171°51.0 -57°15.5
4	286°14.5	281°56.8	53.4	154°20.0	48.6	204°01.9	01.3	286°46.0	25.9	Alioth	166°12.2 55°49.4
5	301°16.9	296°57.6	53.8	169°21.2	48.3	219°03.8	01.3	301°48.2	25.8	Spica	158°21.7 -11°17.8
6	316°19.4	311°58.3	N01°54.2	184°22.5	N19°47.9	234°05.7	N23°01.4	316°50.5	S02°25.7	Alkaid	152°51.3 49°11.2
7	331°21.9	326°59.0	54.6	199°23.7	47.5	249°07.6	01.4	331°52.7	25.6	Hadar	148°35.1 -60°29.9
8	346°24.3	341°59.7	55.0	214°25.0	47.2	264°09.5	01.4	346°54.9	25.6	Menkent	147°57.0 -36°29.8
9	1°26.8	357°00.4	· · 55.4	229°26.2	· · 46.8	279°11.4	· · 01.5	1°57.2	· · 25.5	Arcturus	145°47.3 19°02.9
10	16°29.3	12°01.2	55.8	244°27.5	46.4	294°13.3	01.5	16°59.4	25.4	Rigel Kent.	139°39.3 -60°56.5
11	31°31.7	27°01.9	56.2	259°28.7	46.1	309°15.2	01.5	32°01.7	25.3	Kochab	137°18.6 74°03.0
12	46°34.2	42°02.6	N01°56.5	274°30.0	N19°45.7	324°17.1	N23°01.6	47°03.9	S02°25.2	Zuben'ubi	136°55.4 -16°08.9
13	61°36.6	57°03.3	56.9	289°31.2	45.3	339°19.0	01.6	62°06.2	25.1	Alphecca	126°03.2 26°37.6
14	76°39.1	72°04.0	57.3	304°32.5	45.0	354°20.9	01.6	77°08.4	25.0	Antares	112°15.2 -26°29.3
15	91°41.6	87°04.7	· · 57.7	319°33.7	· · 44.6	9°22.8	· · 01.7	92°10.7	· · 25.0	Atria	107°08.7 -69°04.3
16	106°44.0	102°05.4	58.1	334°35.0	44.3	24°24.7	01.7	107°12.9	24.9	Sabik	102°02.2 -15°45.4
17	121°46.5	117°06.2	58.5	349°36.2	43.9	39°26.6	01.7	122°15.1	24.8	Shaula	96°09.7 -37°07.3
18	136°49.0	132°06.9	N01°58.9	4°37.5	N19°43.5	54°28.5	N23°01.8	137°17.4	S02°24.7	Rasalhague	95°58.0 12°32.3
19	151°51.4	147°07.6	59.3	19°38.7	43.2	69°30.4	01.8	152°19.6	24.6	Eltanin	90°41.7 51°28.8
20	166°53.9	162°08.3	01°59.7	34°40.0	42.8	84°32.3	01.8	167°21.9	24.5	Kaus Aust.	83°31.9 -34°22.3
21	181°56.4	177°09.0	02°00.1	49°41.2	· · 42.4	99°34.2	· · 01.9	182°24.1	· · 24.4	Vega	80°32.8 38°48.1
22	196°58.8	192°09.7	00.5	64°42.4	42.1	114°36.1	01.9	197°26.4	24.4	Nunki	75°47.2 -26°15.9
23	212°01.3	207°10.4	00.9	79°43.7	41.7	129°38.0	01.9	212°28.6	24.3	Altair	61°59.6 8°55.9
Mer.pass. 08:54		ν 0.7' d 0.4' m -4.65		ν 1.3' d -0.4' m 1.03		ν 1.9' d 0.0' m -1.96		ν 2.2' d -0.1' m 1.16		Peacock	53°05.2 -56°39.0

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	227°03.8	222°11.1	N02°01.3	94°44.9	N19°41.3	144°39.9	N23°02.0	227°30.9	S02°24.2	Deneb	49°25.6 45°21.9
1	242°06.2	237°11.8	01.7	109°46.2	41.0	159°41.8	02.0	242°33.1	24.1	Enif	33°38.6 9°59.3
2	257°08.7	252°12.5	02.1	124°47.4	40.6	174°43.7	02.0	257°35.4	24.0	Al Na'ir	27°32.7 -46°50.1
3	272°11.1	267°13.2	· · 02.6	139°48.7	· · 40.2	189°45.6	· · 02.1	272°37.6	· · 23.9	Fomalhaut	15°14.4 -29°29.2
4	287°13.6	282°13.9	03.0	154°49.9	39.9	204°47.5	02.1	287°39.9	23.8	Scheat	13°45.1 28°13.0
5	302°16.1	297°14.5	03.4	169°51.2	39.5	219°49.4	02.1	302°42.1	23.8	Markab	13°29.8 15°20.3
6	317°18.5	312°15.2	N02°03.8	184°52.4	N19°39.1	234°51.3	N23°02.1	317°44.3	S02°23.7	May 07 Wed	
7	332°21.0	327°15.9	04.2	199°53.6	38.8	249°53.2	02.2	332°46.6	23.6	SHA	Mer.pass
8	347°23.5	342°16.6	04.6	214°54.9	38.4	264°55.1	02.2	347°48.8	23.5	Venus	356°30.4 09:13
9	2°25.9	357°17.3	· · 05.0	229°56.1	· · 38.0	279°57.0	· · 02.2	2°51.1	· · 23.4	Mars	228°39.4 17:44
10	17°28.4	12°18.0	05.4	244°57.4	37.6	294°58.9	02.3	17°53.3	23.3	Jupiter	278°03.2 14:26
11	32°30.9	27°18.7	05.8	259°58.6	37.3	310°00.8	02.3	32°55.6	23.2	Saturn	0°37.7 08:56
12	47°33.3	42°19.3	N02°06.3	274°59.9	N19°36.9	325°02.7	N23°02.3	47°57.8	S02°23.2	May 08 Thu	
13	62°35.8	57°20.0	06.7	290°01.1	36.5	340°04.6	02.4	63°00.1	23.1	SHA	Mer.pass
14	77°38.2	72°20.7	07.1	305°02.4	36.2	355°06.5	02.4	78°02.3	23.0	Venus	355°49.3 09:12
15	92°40.7	87°21.4	· · 07.5	320°03.6	· · 35.8	10°08.4	· · 02.4	93°04.6	· · 22.9	Mars	228°10.4 17:42
16	107°43.2	102°22.1	07.9	335°04.8	35.4	25°10.3	02.5	108°06.8	22.8	Jupiter	277°49.7 14:23
17	122°45.6	117°22.7	08.4	350°06.1	35.1	40°12.2	02.5	123°09.1	22.7	Saturn	0°32.4 08:52
18	137°48.1	132°23.4	N02°08.8	5°07.3	N19°34.7	55°14.1	N23°02.5	138°11.3	S02°22.6	May 09 Fri	
19	152°50.6	147°24.1	09.2	20°08.6	34.3	70°16.0	02.6	153°13.6	22.6	SHA	Mer.pass
20	167°53.0	162°24.8	09.6	35°09.8	33.9	85°17.9	02.6	168°15.8	22.5	Venus	355°07.3 09:11
21	182°55.5	177°25.4	· · 10.1	50°11.0	· · 33.6	100°19.8	· · 02.6	183°18.1	· · 22.4	Mars	227°41.2 17:40
22	197°58.0	192°26.1	10.5	65°12.3	33.2	115°21.7	02.7	198°20.3	22.3	Jupiter	277°36.2 14:20
23	213°00.4	207°26.8	10.9	80°13.5	32.8	130°23.6	02.7	213°22.6	22.2	Saturn	0°27.1 08:49
Mer.pass. 08:50		ν 0.7' d 0.4' m -4.64		ν 1.2' d -0.4' m 1.04		ν 1.9' d 0.0' m -1.96		ν 2.2' d -0.1' m 1.15		Horizontal parallax	

Venus:		0.3
Mars:		0.1

h Sun Moon						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	180°51.6	N16°49.6	58°59.7	16.4'	N07°05.9	-14.2' 54.8'
1	195°51.7	50.2	73°35.1	16.4'	06°51.8	-14.2' 54.8'
2	210°51.7	50.9	88°10.6	16.5'	06°37.6	-14.2' 54.8'
3	225°51.8	51.6	102°46.1	16.5'	06°23.4	-14.2' 54.8'
4	240°51.8	52.3	117°21.6	16.6'	06°09.2	-14.2' 54.7'
5	255°51.8	53.0	131°57.2	16.6'	05°55.0	-14.2' 54.7'
6	270°51.9	N16°53.7	146°32.8	16.6'	N05°40.8	-14.2' 54.7'
7	285°51.9	54.4	161°08.4	16.7'	05°26.6	-14.2' 54.7'
8	300°52.0	55.1	175°44.1	16.7'	05°12.3	-14.3' 54.7'
9	315°52.0	55.7	190°19.8	16.8'	04°58.1	-14.3' 54.7'
10	330°52.0	56.4	204°55.6	16.8'	04°43.8	-14.3' 54.6'
11	345°52.1	57.1	219°31.4	16.8'	04°29.5	-14.3' 54.6'
12	0°52.1	N16°57.8	234°07.2	16.9'	N04°15.3	-14.3' 54.6'
13	15°52.2	58.5	248°43.1	16.9'	04°01.0	-14.3' 54.6'
14	30°52.2	59.2	263°18.9	16.9'	03°46.7	-14.3' 54.6'
15	45°52.2	16°59.8	277°54.9	16.9'	03°32.4	-14.3' 54.5'
16	60°52.3	17°00.5	292°30.8	17.0'	03°18.1	-14.3' 54.5'
17	75°52.3	01.2	307°06.8	17.0'	03°03.8	-14.3' 54.5'
18	90°52.3	N17°01.9	321°42.8	17.0'	N02°49.5	-14.3' 54.5'
19	105°52.4	02.6	336°18.8	17.0'	02°35.1	-14.3' 54.5'
20	120°52.4	03.2	350°54.8	17.1'	02°20.8	-14.3' 54.5'
21	135°52.4	03.9	5°30.9	17.1'	02°06.5	-14.3' 54.5'
22	150°52.5	04.6	20°07.0	17.1'	01°52.2	-14.3' 54.4'
23	165°52.5	05.3	34°43.1	17.1'	01°37.9	-14.3' 54.4'
SD = 15.8' d = 0.7' SD = 15.0'						
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	180°52.6	N17°06.0	49°19.2	17.1'	N01°23.6	-14.3' 54.4'
1	195°52.6	06.6	63°55.3	17.2'	01°09.2	-14.3' 54.4'
2	210°52.6	07.3	78°31.5	17.2'	00°54.9	-14.3' 54.4'
3	225°52.7	08.0	93°07.7	17.2'	00°40.6	-14.3' 54.4'
4	240°52.7	08.7	107°43.8	17.2'	00°26.3	-14.3' 54.3'
5	255°52.7	09.3	122°20.0	17.2'	N00°12.0	-14.3' 54.3'
6	270°52.8	N17°10.0	136°56.2	17.2'	S00°02.3	14.3' 54.3'
7	285°52.8	10.7	151°32.5	17.2'	00°16.6	14.3' 54.3'
8	300°52.8	11.4	166°08.7	17.2'	00°30.9	14.3' 54.3'
9	315°52.9	12.0	180°44.9	17.2'	00°45.1	14.3' 54.3'
10	330°52.9	12.7	195°21.2	17.2'	00°59.4	14.3' 54.3'
11	345°52.9	13.4	209°57.4	17.3'	01°13.7	14.2' 54.3'
12	0°53.0	N17°14.1	224°33.7	17.3'	S01°27.9	14.2' 54.2'
13	15°53.0	14.7	239°09.9	17.3'	01°42.1	14.2' 54.2'
14	30°53.0	15.4	253°46.2	17.3'	01°56.4	14.2' 54.2'
15	45°53.1	16.1	268°22.4	17.3'	02°10.6	14.2' 54.2'
16	60°53.1	16.7	282°58.7	17.3'	02°24.8	14.2' 54.2'
17	75°53.1	17.4	297°34.9	17.3'	02°39.0	14.2' 54.2'
18	90°53.1	N17°18.1	312°11.2	17.3'	S02°53.1	14.2' 54.2'
19	105°53.2	18.8	326°47.5	17.2'	03°07.3	14.1' 54.2'
20	120°53.2	19.4	341°23.7	17.2'	03°21.5	14.1' 54.2'
21	135°53.2	20.1	355°59.9	17.2'	03°35.6	14.1' 54.2'
22	150°53.3	20.8	10°36.2	17.2'	03°49.7	14.1' 54.1'
23	165°53.3	21.4	25°12.4	17.2'	04°03.8	14.1' 54.1'
SD = 15.8' d = 0.7' SD = 14.8'						
Fri	GHA	Dec	GHA	ν	Dec	d HP
0	180°53.3	N17°22.1	39°48.6	17.2'	S04°17.9	14.1' 54.1'
1	195°53.4	22.8	54°24.8	17.2'	04°31.9	14.0' 54.1'
2	210°53.4	23.4	69°01.1	17.2'	04°46.0	14.0' 54.1'
3	225°53.4	24.1	83°37.2	17.2'	05°00.0	14.0' 54.1'
4	240°53.4	24.7	98°13.4	17.2'	05°14.0	14.0' 54.1'
5	255°53.5	25.4	112°49.6	17.1'	05°28.0	14.0' 54.1'
6	270°53.5	N17°26.1	127°25.7	17.1'	S05°41.9	13.9' 54.1'
7	285°53.5	26.7	142°01.9	17.1'	05°55.8	13.9' 54.1'
8	300°53.6	27.4	156°38.0	17.1'	06°09.7	13.9' 54.1'
9	315°53.6	28.1	171°14.1	17.1'	06°23.6	13.9' 54.0'
10	330°53.6	28.7	185°50.2	17.1'	06°37.5	13.8' 54.0'
11	345°53.6	29.4	200°26.2	17.0'	06°51.3	13.8' 54.0'
12	0°53.7	N17°30.0	215°02.3	17.0'	S07°05.1	13.8' 54.0'
13	15°53.7	30.7	229°38.3	17.0'	07°18.9	13.7' 54.0'
14	30°53.7	31.4	244°14.3	17.0'	07°32.6	13.7' 54.0'
15	45°53.7	32.0	258°50.2	16.9'	07°46.4	13.7' 54.0'
16	60°53.8	32.7	273°26.2	16.9'	08°00.1	13.7' 54.0'
17	75°53.8	33.3	288°02.1	16.9'	08°13.7	13.6' 54.0'
18	90°53.8	N17°34.0	302°38.0	16.9'	S08°27.3	13.6' 54.0'
19	105°53.8	34.6	317°13.8	16.8'	08°40.9	13.6' 54.0'
20	120°53.9	35.3	331°49.7	16.8'	08°54.5	13.5' 54.0'
21	135°53.9	36.0	346°25.5	16.8'	09°08.0	13.5' 54.0'
22	150°53.9	36.6	1°01.3	16.7'	09°21.5	13.5' 54.0'
23	165°53.9	37.3	15°37.0	16.7'	09°35.0	13.4' 54.0'
SD = 15.8' d = 0.7' SD = 14.8'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	00:17	—:—	—:—	—:—
N 70°	—:—	—:—	01:45	22:15	—:—	—:—
68°	—:—	—:—	02:23	21:34	—:—	—:—
66°	—:—	01:01	02:49	21:07	23:01	—:—
64°	—:—	01:49	03:09	20:46	22:08	—:—
62°	—:—	02:19	03:25	20:29	21:37	—:—
60°	00:54	02:41	03:39	20:16	21:14	23:07
N 58°	01:38	02:59	03:51	20:04	20:56	22:19
56°	02:05	03:14	04:01	19:54	20:41	21:50
54°	02:26	03:26	04:09	19:45	20:28	21:29
52°	02:43	03:37	04:17	19:37	20:17	21:12
50°	02:57	03:46	04:24	19:30	20:08	20:58
45°	03:24	04:06	04:39	19:14	19:48	20:30
N 40°	03:45	04:22	04:52	19:02	19:32	20:09
35°	04:01	04:35	05:02	18:51	19:19	19:52
30°	04:15	04:46	05:11	18:42	19:08	19:39
20°	04:36	05:04	05:27	18:26	18:50	19:17
N 10°	04:53	05:18	05:40	18:13	18:35	19:01
0°	05:06	05:31	05:53	18:00	18:22	18:47
S 10°	05:18	05:44	06:05	17:47	18:09	18:35
20°	05:29	05:56	06:19	17:34	17:57	18:24
30°	05:40	06:08	06:34	17:19	17:44	18:13
35°	05:45	06:15	06:42	17:10	17:37	18:07
40°	05:51	06:23	06:52	17:01	17:29	18:02
45°	05:57	06:32	07:03	16:49	17:20	17:56
S 50°	06:03	06:42	07:17	16:35	17:10	17:49
52°	06:06	06:47	07:24	16:29	17:05	17:46
54°	06:09	06:52	07:31	16:22	17:00	17:43
56°	06:12	06:57	07:39	16:14	16:55	17:40
58°	06:16	07:04	07:48	16:05	16:49	17:36
S 60°	06:19	07:10	07:58	15:54	16:42	17:33

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	13:39	15:37	17:38	03:28	02:55	02:23
N 70°	13:45	15:34	17:24	03:19	02:54	02:29
68°	13:49	15:31	17:14	03:12	02:53	02:35
66°	13:53	15:29	17:05	03:06	02:53	02:39
64°	13:56	15:27	16:58	03:01	02:52	02:43
62°	13:59	15:25	16:51	02:57	02:52	02:47
60°	14:01	15:24	16:46	02:53	02:51	02:50
N 58°	14:03	15:22	16:41	02:49	02:51	02:53
56°	14:05	15:21	16:37	02:46	02:51	02:55
54°	14:07	15:20	16:33	02:44	02:51	02:57
52°	14:08	15:19	16:30	02:41	02:50	02:59
50°	14:10	15:18	16:27	02:39	02:50	03:01
45°	14:13	15:16	16:20	02:34	02:50	03:05
N 40°	14:15	15:15	16:14	02:30	02:49	03:08
35°	14:17	15:14	16:10	02:26	02:49	03:11
30°	14:19	15:12	16:05	02:23	02:48	03:14
20°	14:22	15:10	15:58	02:17	02:48	03:18
N 10°	14:25	15:09	15:52	02:12	02:47	03:22
0°	14:28	15:07	15:46	02:07	02:47	03:26
S 10°	14:31	15:05	15:40	02:02	02:46	03:30
20°	14:33	15:04	15:34	01:57	02:46	03:34
30°	14:37	15:02	15:27	01:51	02:45	03:38
35°	14:38	15:01	15:24	01:48	02:45	03:41
40°	14:40	15:00	15:19	01:44	02:44	03:44
45°	14:43	14:58	15:14	01:39	02:44	03:47
S 50°	14:46	14:57	15:08	01:34	02:43	03:51
52°	14:47	14:56	15:05	01:31	02:43	03:53
54°	14:48	14:55	15:02	01:28	02:42	03:55
56°	14:50	14:54	14:59	01:25	02:42	03:58
58°	14:52	14:53	14:55	01:22	02:42	04:00
S 60°	14:54	14:52	14:51	01:18	02:41	04:03

Day	Sun		Mer. Pass	Moon		Age 10-12 73-88%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
07	03:27	03:28	11:57	20:37	08:18	
08	03:30	03:32	11:56	21:16	08:57	
09	03:33	03:35	11:56	21:56	09:36	

May 10, 11, 12 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	228°02.9	222°27.4	N02°11.3	95°14.8	N19°32.5	145°25.5	N23°02.7	228°24.8	S02°22.1	Alpheratz	357°34.8 29°13.6	
1	243°05.4	237°28.1	11.8	110°16.0	32.1	160°27.4	02.8	243°27.0	22.0	Ankaa	353°07.2 -42°10.0	
2	258°07.8	252°28.8	12.2	125°17.3	31.7	175°29.2	02.8	258°29.3	22.0	Schedar	349°31.2 56°40.4	
3	273°10.3	267°29.4	•• 12.6	140°18.5	•• 31.3	190°31.1	•• 02.8	273°31.5	•• 21.9	Diphda	348°47.3 -17°50.9	
4	288°12.7	282°30.1	13.1	155°19.7	31.0	205°33.0	02.8	288°33.8	21.8	Achernar	335°20.5 -57°06.4	
5	303°15.2	297°30.7	13.5	170°21.0	30.6	220°34.9	02.9	303°36.0	21.7	Hamal	327°51.3 23°34.8	
6	318°17.7	312°31.4	N02°13.9	185°22.2	N19°30.2	235°36.8	N23°02.9	318°38.3	S02°21.6	Polaris	314°24.3 89°22.2	
7	333°20.1	327°32.0	14.4	200°23.4	29.9	250°38.7	02.9	333°40.5	21.5	Acamar	315°12.0 -40°12.2	
8	348°22.6	342°32.7	14.8	215°24.7	29.5	265°40.6	03.0	348°42.8	21.5	Menkar	314°06.2 4°11.3	
9	3°25.1	357°33.4	•• 15.2	230°25.9	•• 29.1	280°42.5	•• 03.0	3°45.0	•• 21.4	Mirfak	308°28.4 49°57.0	
10	18°27.5	12°34.0	15.7	245°27.2	28.7	295°44.4	03.0	18°47.3	21.3	Aldebaran	290°39.7 16°33.6	
11	33°30.0	27°34.7	16.1	260°28.4	28.4	310°46.3	03.1	33°49.5	21.2	Rigel	281°03.9 -8°10.4	
12	48°32.5	42°35.3	N02°16.6	275°29.6	N19°28.0	325°48.2	N23°03.1	48°51.8	S02°21.1	Capella	280°21.9 46°01.5	
13	63°34.9	57°36.0	17.0	290°30.9	27.6	340°50.1	03.1	63°54.0	21.0	Bellatrix	278°22.9 6°22.3	
14	78°37.4	72°36.6	17.4	305°32.1	27.2	355°52.0	03.2	78°56.3	20.9	Elnath	278°01.9 28°37.7	
15	93°39.8	87°37.2	•• 17.9	320°33.4	•• 26.9	10°53.9	•• 03.2	93°58.5	•• 20.9	Alnilam	275°37.8 -1°11.2	
16	108°42.3	102°37.9	18.3	335°34.6	26.5	25°55.8	03.2	109°00.8	20.8	Betelgeuse	270°52.1 7°24.7	
17	123°44.8	117°38.5	18.8	350°35.8	26.1	40°57.7	03.3	124°03.0	20.7	Canopus	263°52.7 -52°42.7	
18	138°47.2	132°39.2	N02°19.2	5°37.1	N19°25.8	55°59.6	N23°03.3	139°05.3	S02°20.6	Sirius	258°26.2 -16°45.2	
19	153°49.7	147°39.8	19.7	20°38.3	25.4	71°01.5	03.3	154°07.5	20.5	Adhara	255°05.9 -29°00.5	
20	168°52.2	162°40.5	20.1	35°39.5	25.0	86°03.4	03.3	169°09.8	20.4	Procyon	244°50.7 5°09.6	
21	183°54.6	177°41.1	•• 20.6	50°40.8	•• 24.6	101°05.3	•• 03.4	184°12.0	•• 20.4	Pollux	243°17.2 27°58.0	
22	198°57.1	192°41.7	21.0	65°42.0	24.3	116°07.2	03.4	199°14.3	20.3	Avior	234°14.9 -59°35.7	
23	213°59.6	207°42.4	21.5	80°43.3	23.9	131°09.0	03.4	214°16.5	20.2	Suhail	222°46.2 -43°32.3	
Mer.pass. 08:46		ν 0.7' d 0.4' m -4.63		ν 1.2' d -0.4' m 1.05		ν 1.9' d 0.0' m -1.96		ν 2.2' d -0.1' m 1.15		Miaplacidus	221°38.4 -69°49.6	
											Alphard	217°47.5 -8°46.2
											Regulus	207°34.1 11°50.6
											Dubhe	193°40.4 61°37.1
											Denebola	182°24.5 14°25.8
											Gienah	175°43.1 -17°41.1
											Acrux	172°59.4 -63°14.6
											Gacrux	171°51.0 -57°15.6
											Alioth	166°12.3 55°49.4
											Spica	158°21.8 -11°17.7
											Alkaid	152°51.3 49°11.2
											Hadar	148°35.1 -60°29.9
											Menkent	147°57.0 -36°29.8
											Arcturus	145°47.3 19°02.9
											Rigel Kent.	139°39.3 -60°56.5
											Kochab	137°18.6 74°03.0
											Zuben'ubi	136°55.4 -16°08.9
											Alphecca	126°03.2 26°37.6
											Antares	112°15.2 -26°29.3
											Atria	107°08.7 -69°04.3
											Sabik	102°02.2 -15°45.4
											Shaula	96°09.7 -37°07.3
											Rasalhague	95°58.0 12°32.3
											Eltanin	90°41.6 51°28.9
											Kaus Aust.	83°31.9 -34°22.3
											Vega	80°32.8 38°48.2
											Nunki	75°47.2 -26°15.9
											Altair	61°59.6 8°56.0
											Peacock	53°05.1 -56°39.0
											Deneb	49°25.5 45°21.9
											Enif	33°38.6 9°59.3
											Al Na'ir	27°32.7 -46°50.1
											Fomalhaut	15°14.4 -29°29.2
											Scheat	13°45.1 28°13.0
											Markab	13°29.8 15°20.3
											May 10 Sat	SHA Mer.pass
											Venus	354°24.5 09:10
											Mars	227°11.9 17:38
											Jupiter	277°22.6 14:16
											Saturn	0°21.9 08:45
											May 11 Sun	SHA Mer.pass
											Venus	353°41.0 09:09
											Mars	226°42.5 17:36
											Jupiter	277°08.9 14:13
											Saturn	0°16.8 08:41
											May 12 Mon	SHA Mer.pass
											Venus	352°56.7 09:08
											Mars	226°12.9 17:34
											Jupiter	276°55.2 14:10
											Saturn	0°11.7 08:38
											Horizontal parallax	
											Venus:	0.3
											Mars:	0.1
Mer.pass. 08:39		ν 0.6' d 0.5' m -4.61		ν 1.2' d -0.4' m 1.08		ν 1.9' d 0.0' m -1.95		ν 2.3' d -0.1' m 1.14				

Sun							Moon						
Sat	GHA	Dec	GHA	ν	Dec	d	HP						
0	180°54.0	N17°37.9	30°12.7	16.7'	S09°48.4	13.4'	54.0'						
1	195°54.0	38.6	44°48.4	16.6'	10°01.8	13.4'	54.0'						
2	210°54.0	39.2	59°24.0	16.6'	10°15.2	13.3'	54.0'						
3	225°54.0	· · 39.9	73°59.6	16.6'	10°28.5	13.3'	53.9'						
4	240°54.1	40.5	88°35.2	16.5'	10°41.7	13.2'	53.9'						
5	255°54.1	41.2	103°10.7	16.5'	10°55.0	13.2'	53.9'						
6	270°54.1	N17°41.8	117°46.2	16.5'	S11°08.2	13.2'	53.9'						
7	285°54.1	42.5	132°21.7	16.4'	11°21.3	13.1'	53.9'						
8	300°54.1	43.1	146°57.1	16.4'	11°34.4	13.1'	53.9'						
9	315°54.2	· · 43.8	161°32.4	16.3'	11°47.5	13.0'	53.9'						
10	330°54.2	44.4	176°07.8	16.3'	12°00.5	13.0'	53.9'						
11	345°54.2	45.1	190°43.1	16.2'	12°13.5	12.9'	53.9'						
12	0°54.2	N17°45.7	205°18.3	16.2'	S12°26.4	12.9'	53.9'						
13	15°54.2	46.4	219°53.5	16.2'	12°39.3	12.8'	53.9'						
14	30°54.3	47.0	234°28.6	16.1'	12°52.2	12.8'	53.9'						
15	45°54.3	· · 47.7	249°03.7	16.1'	13°05.0	12.7'	53.9'						
16	60°54.3	48.3	263°38.8	16.0'	13°17.7	12.7'	53.9'						
17	75°54.3	49.0	278°13.8	16.0'	13°30.4	12.6'	53.9'						
18	90°54.3	N17°49.6	292°48.8	15.9'	S13°43.1	12.6'	53.9'						
19	105°54.4	50.2	307°23.7	15.9'	13°55.7	12.5'	53.9'						
20	120°54.4	50.9	321°58.5	15.8'	14°08.2	12.5'	53.9'						
21	135°54.4	· · 51.5	336°33.3	15.8'	14°20.7	12.4'	53.9'						
22	150°54.4	52.2	351°08.1	15.7'	14°33.1	12.4'	53.9'						
23	165°54.4	52.8	5°42.8	15.6'	14°45.5	12.3'	53.9'						
SD = 15.8'			d = 0.7'	SD = 14.7'									
Sun	GHA	Dec	GHA	ν	Dec	d	HP						
0	180°54.5	N17°53.5	20°17.5	15.6'	S14°57.8	12.3'	53.9'						
1	195°54.5	54.1	34°52.0	15.5'	15°10.1	12.2'	53.9'						
2	210°54.5	54.7	49°26.6	15.5'	15°22.3	12.2'	53.9'						
3	225°54.5	· · 55.4	64°01.1	15.4'	15°34.5	12.1'	53.9'						
4	240°54.5	56.0	78°35.5	15.4'	15°46.6	12.0'	53.9'						
5	255°54.5	56.6	93°09.9	15.3'	15°58.6	12.0'	53.9'						
6	270°54.6	N17°57.3	107°44.2	15.2'	S16°10.6	11.9'	53.9'						
7	285°54.6	57.9	122°18.4	15.2'	16°22.5	11.9'	53.9'						
8	300°54.6	58.6	136°52.6	15.1'	16°34.4	11.8'	53.9'						
9	315°54.6	· · 59.2	151°26.7	15.1'	16°46.2	11.7'	53.9'						
10	330°54.6	17°59.8	166°00.8	15.0'	16°57.9	11.7'	53.9'						
11	345°54.6	18°00.5	180°34.8	14.9'	17°09.6	11.6'	53.9'						
12	0°54.6	N18°01.1	195°08.7	14.9'	S17°21.2	11.5'	53.9'						
13	15°54.7	01.7	209°42.6	14.8'	17°32.7	11.5'	53.9'						
14	30°54.7	02.4	224°16.4	14.8'	17°44.2	11.4'	53.9'						
15	45°54.7	· · 03.0	238°50.2	14.7'	17°55.6	11.3'	53.9'						
16	60°54.7	03.6	253°23.9	14.6'	18°06.9	11.3'	53.9'						
17	75°54.7	04.3	267°57.5	14.6'	18°18.2	11.2'	53.9'						
18	90°54.7	N18°04.9	282°31.1	14.5'	S18°29.3	11.1'	53.9'						
19	105°54.7	05.5	297°04.5	14.4'	18°40.5	11.0'	53.9'						
20	120°54.8	06.2	311°38.0	14.4'	18°51.5	11.0'	53.9'						
21	135°54.8	· · 06.8	326°11.3	14.3'	19°02.5	10.9'	53.9'						
22	150°54.8	07.4	340°44.6	14.2'	19°13.4	10.8'	53.9'						
23	165°54.8	08.1	355°17.8	14.1'	19°24.2	10.7'	53.9'						
SD = 15.8'			d = 0.6'	SD = 14.7'									
Mon	GHA	Dec	GHA	ν	Dec	d	HP						
0	180°54.8	N18°08.7	9°51.0	14.1'	S19°34.9	10.7'	54.0'						
1	195°54.8	09.3	24°24.0	14.0'	19°45.6	10.6'	54.0'						
2	210°54.8	09.9	38°57.0	13.9'	19°56.2	10.5'	54.0'						
3	225°54.8	· · 10.6	53°30.0	13.9'	20°06.7	10.4'	54.0'						
4	240°54.8	11.2	68°02.8	13.8'	20°17.1	10.3'	54.0'						
5	255°54.9	11.8	82°35.6	13.7'	20°27.4	10.3'	54.0'						
6	270°54.9	N18°12.4	97°08.4	13.6'	S20°37.7	10.2'	54.0'						
7	285°54.9	13.1	111°41.0	13.6'	20°47.8	10.1'	54.0'						
8	300°54.9	13.7	126°13.6	13.5'	20°57.9	10.0'	54.0'						
9	315°54.9	· · 14.3	140°46.1	13.4'	21°07.9	9.9'	54.0'						
10	330°54.9	14.9	155°18.5	13.4'	21°17.9	9.8'	54.0'						
11	345°54.9	15.6	169°50.9	13.3'	21°27.7	9.7'	54.0'						
12	0°54.9	N18°16.2	184°23.1	13.2'	S21°37.4	9.7'	54.0'						
13	15°54.9	16.8	198°55.4	13.1'	21°47.1	9.6'	54.0'						
14	30°54.9	17.4	213°27.5	13.1'	21°56.7	9.5'	54.0'						
15	45°55.0	· · 18.1	227°59.5	13.0'	22°06.1	9.4'	54.0'						
16	60°55.0	18.7	242°31.5	12.9'	22°15.5	9.3'	54.0'						
17	75°55.0	19.3	257°03.4	12.8'	22°24.8	9.2'	54.0'						
18	90°55.0	N18°19.9	271°35.3	12.8'	S22°34.0	9.1'	54.0'						
19	105°55.0	20.5	286°07.0	12.7'	22°43.1	9.0'	54.1'						
20	120°55.0	21.1	300°38.7	12.6'	22°52.1	8.9'	54.1'						
21	135°55.0	· · 21.8	315°10.3	12.5'	23°01.0	8.8'	54.1'						
22	150°55.0	22.4	329°41.9	12.5'	23°09.8	8.7'	54.1'						
23	165°55.0	23.0	344°13.3	12.4'	23°18.5	8.6'	54.1'						
SD = 15.8'			d = 0.6'	SD = 14.7'									

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	—:—	—:—	01:21	22:39	—:—	—:—
68°	—:—	—:—	02:07	21:50	—:—	—:—
66°	—:—	00:19	02:37	21:19	—:—	—:—
64°	—:—	01:33	02:59	20:56	22:25	—:—
62°	—:—	02:07	03:17	20:38	21:49	—:—
60°	00:17	02:31	03:31	20:23	21:24	—:—
N 58°	01:23	02:51	03:44	20:10	21:04	22:34
56°	01:55	03:06	03:54	20:00	20:48	22:01
54°	02:17	03:20	04:04	19:50	20:35	21:38
52°	02:35	03:31	04:12	19:42	20:23	21:19
50°	02:50	03:41	04:20	19:34	20:13	21:04
45°	03:19	04:02	04:36	19:18	19:52	20:34
N 40°	03:41	04:18	04:49	19:05	19:35	20:13
35°	03:58	04:32	05:00	18:54	19:21	19:55
30°	04:12	04:43	05:09	18:44	19:10	19:41
20°	04:34	05:02	05:26	18:27	18:51	19:19
N 10°	04:52	05:18	05:40	18:13	18:35	19:01
0°	05:06	05:31	05:53	18:00	18:22	18:47
S 10°	05:18	05:44	06:06	17:47	18:09	18:34
20°	05:30	05:57	06:20	17:33	17:56	18:22
30°	05:41	06:10	06:35	17:17	17:42	18:11
35°	05:47	06:18	06:44	17:08	17:35	18:05
40°	05:53	06:26	06:55	16:58	17:26	17:59
45°	06:00	06:35	07:07	16:45	17:17	17:52
S 50°	06:07	06:46	07:21	16:31	17:06	17:45
52°	06:10	06:51	07:28	16:24	17:01	17:42
54°	06:13	06:57	07:36	16:16	16:56	17:39
56°	06:17	07:02	07:44	16:08	16:50	17:35
58°	06:21	07:09	07:54	15:58	16:43	17:31
S 60°	06:25	07:16	08:05	15:47	16:36	17:27

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	19:57	■	■	01:46	00:53	■
N 70°	19:26	22:17	■	02:02	01:26	00:04
68°	19:03	21:13	■	02:15	01:50	01:09
66°	18:46	20:38	23:10	02:25	02:08	01:45
64°	18:32	20:13	22:09	02:34	02:24	02:11
62°	18:20	19:54	21:35	02:42	02:37	02:31
60°	18:10	19:38	21:11	02:49	02:48	02:48
N 58°	18:02	19:25	20:51	02:54	02:57	03:02
56°	17:54	19:13	20:35	03:00	03:06	03:14
54°	17:47	19:03	20:21	03:04	03:13	03:24
52°	17:41	18:55	20:09	03:09	03:20	03:34
50°	17:36	18:47	19:59	03:13	03:26	03:42
45°	17:24	18:30	19:37	03:21	03:39	04:00
N 40°	17:15	18:16	19:19	03:28	03:50	04:15
35°	17:07	18:05	19:04	03:34	04:00	04:28
30°	16:59	17:55	18:52	03:40	04:08	04:39
20°	16:47	17:38	18:30	03:49	04:22	04:58
N 10°	16:36	17:23	18:11	03:58	04:35	05:15
0°	16:27	17:09	17:54	04:05	04:47	05:30
S 10°	16:17	16:55	17			

May 13, 14, 15 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	231°00.3	223°11.9	N02°44.7	96°43.6	N19°05.3	147°41.7	N23°04.9	231°06.9	S02°16.1	Alpheratz	357°34.7	29°13.6
1	246°02.8	238°12.5	45.2	111°44.9	04.9	162°43.6	04.9	246°09.2	16.1	Ankaa	353°07.2	-42°10.0
2	261°05.2	253°13.1	45.7	126°46.1	04.5	177°45.5	05.0	261°11.4	16.0	Schedar	349°31.2	56°40.4
3	276°07.7	268°13.6	· · 46.1	141°47.3	· · 04.1	192°47.4	· · 05.0	276°13.7	· · 15.9	Diphda	348°47.3	-17°50.8
4	291°10.2	283°14.2	46.6	156°48.5	03.7	207°49.3	05.0	291°16.0	15.8	Achernar	335°20.5	-57°06.4
5	306°12.6	298°14.8	47.1	171°49.8	03.3	222°51.2	05.1	306°18.2	15.7	Hamal	327°51.2	23°34.8
6	321°15.1	313°15.3	N02°47.6	186°51.0	N19°02.9	237°53.1	N23°05.1	321°20.5	S02°15.7	Polaris	314°23.7	89°22.2
7	336°17.6	328°15.9	48.1	201°52.2	02.6	252°54.9	05.1	336°22.7	15.6	Acamar	315°12.0	-40°12.2
8	351°20.0	343°16.5	48.6	216°53.4	02.2	267°56.8	05.1	351°25.0	15.5	Menkar	314°06.2	4°11.3
9	6°22.5	358°17.0	· · 49.2	231°54.7	· · 01.8	282°58.7	· · 05.2	6°27.2	· · 15.4	Mirfak	308°28.4	49°57.0
10	21°24.9	13°17.6	49.7	246°55.9	01.4	298°00.6	05.2	21°29.5	15.3	Aldebaran	290°39.7	16°33.6
11	36°27.4	28°18.2	50.2	261°57.1	01.0	313°02.5	05.2	36°31.8	15.2	Rigel	281°03.9	-8°10.4
12	51°29.9	43°18.7	N02°50.7	276°58.3	N19°00.6	328°04.4	N23°05.3	51°34.0	S02°15.2	Capella	280°21.9	46°01.4
13	66°32.3	58°19.3	51.2	291°59.6	19°00.2	343°06.3	05.3	66°36.3	15.1	Bellatrix	278°22.9	6°22.3
14	81°34.8	73°19.8	51.7	307°00.8	18°59.9	358°08.2	05.3	81°38.5	15.0	Elnath	278°01.9	28°37.7
15	96°37.3	88°20.4	· · 52.2	322°02.0	· · 59.5	13°10.1	· · 05.4	96°40.8	· · 14.9	Alnilam	275°37.8	-1°11.2
16	111°39.7	103°20.9	52.7	337°03.2	59.1	28°11.9	05.4	111°43.0	14.8	Betelgeuse	270°52.1	7°24.7
17	126°42.2	118°21.5	53.2	352°04.5	58.7	43°13.8	05.4	126°45.3	14.8	Canopus	263°52.7	-52°42.7
18	141°44.7	133°22.0	N02°53.7	7°05.7	N18°58.3	58°15.7	N23°05.4	141°47.5	S02°14.7	Sirius	258°26.3	-16°45.2
19	156°47.1	148°22.6	54.2	22°06.9	57.9	73°17.6	05.5	156°49.8	14.6	Adhara	255°05.9	-29°00.5
20	171°49.6	163°23.1	54.7	37°08.1	57.5	88°19.5	05.5	171°52.1	14.5	Procyon	244°50.7	5°09.6
21	186°52.1	178°23.7	· · 55.3	52°09.4	· · 57.1	103°21.4	· · 05.5	186°54.3	· · 14.4	Pollux	243°17.2	27°58.0
22	201°54.5	193°24.2	55.8	67°10.6	56.8	118°23.3	05.6	201°56.6	14.4	Avior	234°14.9	-59°35.7
23	216°57.0	208°24.8	56.3	82°11.8	56.4	133°25.2	05.6	216°58.8	14.3	Suhail	222°46.2	-43°32.3
Mer.pass. 08:35		ν 0.6' d 0.5' m -4.60		ν 1.2' d -0.4' m 1.09		ν 1.9' d 0.0' m -1.95		ν 2.3' d -0.1' m 1.14				

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	231°59.4	223°25.3	N02°56.8	97°13.0	N18°56.0	148°27.0	N23°05.6	232°01.1	S02°14.2
1	247°01.9	238°25.9	57.3	112°14.3	55.6	163°28.9	05.6	247°03.4	14.1
2	262°04.4	253°26.4	57.8	127°15.5	55.2	178°30.8	05.7	262°05.6	14.0
3	277°06.8	268°27.0	· · 58.3	142°16.7	· · 54.8	193°32.7	· · 05.7	277°07.9	· · 14.0
4	292°09.3	283°27.5	58.9	157°17.9	54.4	208°34.6	05.7	292°10.1	13.9
5	307°11.8	298°28.0	59.4	172°19.2	54.0	223°36.5	05.8	307°12.4	13.8
6	322°14.2	313°28.6	N02°59.9	187°20.4	N18°53.6	238°38.4	N23°05.8	322°14.6	S02°13.7
7	337°16.7	328°29.1	03°00.4	202°21.6	53.2	253°40.2	05.8	337°16.9	13.6
8	352°19.2	343°29.6	00.9	217°22.8	52.9	268°42.1	05.8	352°19.2	13.6
9	7°21.6	358°30.2	· · 01.5	232°24.0	· · 52.5	283°44.0	· · 05.9	7°21.4	· · 13.5
10	22°24.1	13°30.7	02.0	247°25.3	52.1	298°45.9	05.9	22°23.7	13.4
11	37°26.5	28°31.2	02.5	262°26.5	51.7	313°47.8	05.9	37°25.9	13.3
12	52°29.0	43°31.8	N03°03.0	277°27.7	N18°51.3	328°49.7	N23°06.0	52°28.2	S02°13.2
13	67°31.5	58°32.3	03.6	292°28.9	50.9	343°51.6	06.0	67°30.5	13.2
14	82°33.9	73°32.8	04.1	307°30.2	50.5	358°53.4	06.0	82°32.7	13.1
15	97°36.4	88°33.4	· · 04.6	322°31.4	· · 50.1	13°55.3	· · 06.0	97°35.0	· · 13.0
16	112°38.9	103°33.9	05.2	337°32.6	49.7	28°57.2	06.1	112°37.2	12.9
17	127°41.3	118°34.4	05.7	352°33.8	49.3	43°59.1	06.1	127°39.5	12.8
18	142°43.8	133°34.9	N03°06.2	7°35.0	N18°48.9	59°01.0	N23°06.1	142°41.8	S02°12.8
19	157°46.3	148°35.5	06.7	22°36.3	48.6	74°02.9	06.1	157°44.0	12.7
20	172°48.7	163°36.0	07.3	37°37.5	48.2	89°04.8	06.2	172°46.3	12.6
21	187°51.2	178°36.5	· · 07.8	52°38.7	· · 47.8	104°06.6	· · 06.2	187°48.5	· · 12.5
22	202°53.7	193°37.0	08.3	67°39.9	47.4	119°08.5	06.2	202°50.8	12.4
23	217°56.1	208°37.5	08.9	82°41.1	47.0	134°10.4	06.3	217°53.1	12.4
Mer.pass. 08:31		ν 0.5' d 0.5' m -4.59		ν 1.2' d -0.4' m 1.10		ν 1.9' d 0.0' m -1.95		ν 2.3' d -0.1' m 1.14	

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	232°58.6	223°38.1	N03°09.4	97°42.4	N18°46.6	149°12.3	N23°06.3	232°55.3	S02°12.3
1	248°01.0	238°38.6	10.0	112°43.6	46.2	164°14.2	06.3	247°57.6	12.2
2	263°03.5	253°39.1	10.5	127°44.8	45.8	179°16.1	06.3	262°59.8	12.1
3	278°06.0	268°39.6	· · 11.0	142°46.0	· · 45.4	194°17.9	· · 06.4	278°02.1	· · 12.0
4	293°08.4	283°40.1	11.6	157°47.2	45.0	209°19.8	06.4	293°04.4	12.0
5	308°10.9	298°40.6	12.1	172°48.5	44.6	224°21.7	06.4	308°06.6	11.9
6	323°13.4	313°41.1	N03°12.6	187°49.7	N18°44.2	239°23.6	N23°06.5	323°08.9	S02°11.8
7	338°15.8	328°41.7	13.2	202°50.9	43.8	254°25.5	06.5	338°11.1	11.7
8	353°18.3	343°42.2	13.7	217°52.1	43.4	269°27.4	06.5	353°13.4	11.7
9	8°20.8	358°42.7	· · 14.3	232°53.3	· · 43.0	284°29.3	· · 06.5	8°15.7	· · 11.6
10	23°23.2	13°43.2	14.8	247°54.5	42.6	299°31.1	06.6	23°17.9	11.5
11	38°25.7	28°43.7	15.4	262°55.8	42.3	314°33.0	06.6	38°20.2	11.4
12	53°28.2	43°44.2	N03°15.9	277°57.0	N18°41.9	329°34.9	N23°06.6	53°22.4	S02°11.3
13	68°30.6	58°44.7	16.4	292°58.2	41.5	344°36.8	06.6	68°24.7	11.3
14	83°33.1	73°45.2	17.0	307°59.4	41.1	359°38.7	06.7	83°27.0	11.2
15	98°35.5	88°45.7	· · 17.5	323°00.6	· · 40.7	14°40.6	· · 06.7	98°29.2	· · 11.1
16	113°38.0	103°46.2	18.1	338°01.9	40.3	29°42.4	06.7	113°31.5	11.0
17	128°40.5	118°46.7	18.6	353°03.1	39.9	44°44.3	06.8	128°33.8	10.9
18	143°42.9	133°47.2	N03°19.2	8°04.3	N18°39.5	59°46.2	N23°06.8	143°36.0	S02°10.9
19	158°45.4	148°47.7	19.7	23°05.5	39.1	74°48.1	06.8	158°38.3	10.8
20	173°47.9	163°48.2	20.3	38°06.7	38.7	89°50.0	06.8	173°40.5	10.7
21	188°50.3	178°48.7	· · 20.8	53°07.9	· · 38.3	104°51.8	· · 06.9	188°42.8	· · 10.6
22	203°52.8	193°49.2	21.4	68°09.2	37.9	119°53.7	06.9	203°45.1	10.6
23	218°55.3	208°49.7	21.9	83°10.4	37.5	134°55.6	06.9	218°47.3	10.5
Mer.pass. 08:27		ν 0.5' d 0.5' m -4.58		ν 1.2' d -0.4' m 1.11		ν 1.9' d 0.0' m -1.95		ν 2.3' d -0.1' m 1.14	

May 13 Tue		SHA	Mer.pass
Venus	352°11.6		09:07
Mars	225°43.3		17:32
Jupiter	276°41.4		14:07
Saturn	0°06.6		08:34

May 14 Wed		SHA	Mer.pass
Venus	351°25.9		09:06
Mars	225°13.6		17:30
Jupiter	276°27.6		14:04
Saturn	0°01.7		08:31

May 15 Thu		SHA	Mer.pass
Venus	350°39.5		09:05
Mars	224°43.8		17:28
Jupiter	276°13.7		14:01
Saturn	359°56.7		08:27

Horizontal parallax		
Venus:		0.3
Mars:		0.1

Sun				Moon				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	180°55.0	N18°23.6		358°44.7	12.3'	S23°27.1	8.5'	54.1'
1	195°55.0	24.2		13°16.0	12.2'	23°35.6	8.4'	54.1'
2	210°55.0	24.8		27°47.2	12.2'	23°44.0	8.3'	54.1'
3	225°55.0	· · 25.5		42°18.4	12.1'	23°52.3	8.2'	54.1'
4	240°55.0	26.1		56°49.5	12.0'	24°00.5	8.1'	54.1'
5	255°55.0	26.7		71°20.5	11.9'	24°08.6	8.0'	54.1'
6	270°55.0	N18°27.3		85°51.4	11.9'	S24°16.6	7.9'	54.1'
7	285°55.0	27.9		100°22.3	11.8'	24°24.5	7.8'	54.1'
8	300°55.1	28.5		114°53.1	11.7'	24°32.2	7.7'	54.1'
9	315°55.1	· · 29.1		129°23.8	11.6'	24°39.9	7.6'	54.2'
10	330°55.1	29.7		143°54.4	11.6'	24°47.5	7.4'	54.2'
11	345°55.1	30.4		158°25.0	11.5'	24°54.9	7.3'	54.2'
12	0°55.1	N18°31.0		172°55.4	11.4'	S25°02.2	7.2'	54.2'
13	15°55.1	31.6		187°25.9	11.3'	25°09.4	7.1'	54.2'
14	30°55.1	32.2		201°56.2	11.3'	25°16.5	7.0'	54.2'
15	45°55.1	· · 32.8		216°26.5	11.2'	25°23.5	6.9'	54.2'
16	60°55.1	33.4		230°56.7	11.1'	25°30.4	6.8'	54.2'
17	75°55.1	34.0		245°26.8	11.1'	25°37.2	6.6'	54.2'
18	90°55.1	N18°34.6		259°56.8	11.0'	S25°43.8	6.5'	54.2'
19	105°55.1	35.2		274°26.8	10.9'	25°50.3	6.4'	54.2'
20	120°55.1	35.8		288°56.7	10.8'	25°56.7	6.3'	54.3'
21	135°55.1	· · 36.4		303°26.6	10.8'	26°03.0	6.2'	54.3'
22	150°55.1	37.0		317°56.4	10.7'	26°09.2	6.0'	54.3'
23	165°55.1	37.6		332°26.1	10.6'	26°15.2	5.9'	54.3'
SD = 15.8'			d = 0.6'	SD = 14.8'				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	180°55.1	N18°38.2		346°55.7	10.6'	S26°21.2	5.8'	54.3'
1	195°55.1	38.8		1°25.3	10.5'	26°27.0	5.7'	54.3'
2	210°55.1	39.4		15°54.8	10.4'	26°32.6	5.5'	54.3'
3	225°55.1	· · 40.0		30°24.2	10.4'	26°38.2	5.4'	54.3'
4	240°55.1	40.6		44°53.6	10.3'	26°43.6	5.3'	54.3'
5	255°55.1	41.2		59°22.9	10.2'	26°48.9	5.2'	54.4'
6	270°55.1	N18°41.8		73°52.1	10.2'	S26°54.1	5.0'	54.4'
7	285°55.1	42.4		88°21.3	10.1'	26°59.1	4.9'	54.4'
8	300°55.1	43.0		102°50.4	10.1'	27°04.0	4.8'	54.4'
9	315°55.1	· · 43.6		117°19.5	10.0'	27°08.8	4.7'	54.4'
10	330°55.1	44.2		131°48.5	9.9'	27°13.5	4.5'	54.4'
11	345°55.1	44.8		146°17.4	9.9'	27°18.0	4.4'	54.4'
12	0°55.1	N18°45.4		160°46.3	9.8'	S27°22.4	4.3'	54.4'
13	15°55.1	46.0		175°15.1	9.8'	27°26.6	4.1'	54.5'
14	30°55.0	46.6		189°43.9	9.7'	27°30.7	4.0'	54.5'
15	45°55.0	· · 47.2		204°12.6	9.7'	27°34.7	3.9'	54.5'
16	60°55.0	47.8		218°41.3	9.6'	27°38.6	3.7'	54.5'
17	75°55.0	48.4		233°09.9	9.6'	27°42.3	3.6'	54.5'
18	90°55.0	N18°49.0		247°38.4	9.5'	S27°45.9	3.4'	54.5'
19	105°55.0	49.6		262°06.9	9.5'	27°49.3	3.3'	54.5'
20	120°55.0	50.2		276°35.4	9.4'	27°52.6	3.2'	54.5'
21	135°55.0	· · 50.8		291°03.8	9.4'	27°55.8	3.0'	54.6'
22	150°55.0	51.4		305°32.2	9.3'	27°58.8	2.9'	54.6'
23	165°55.0	52.0		320°00.5	9.3'	28°01.7	2.8'	54.6'
SD = 15.8'			d = 0.6'	SD = 14.8'				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	180°55.0	N18°52.5		334°28.7	9.2'	S28°04.5	2.6'	54.6'
1	195°55.0	53.1		348°57.0	9.2'	28°07.1	2.5'	54.6'
2	210°55.0	53.7		3°25.1	9.1'	28°09.6	2.3'	54.6'
3	225°55.0	· · 54.3		17°53.3	9.1'	28°11.9	2.2'	54.6'
4	240°55.0	54.9		32°21.4	9.1'	28°14.1	2.0'	54.7'
5	255°55.0	55.5		46°49.4	9.0'	28°16.1	1.9'	54.7'
6	270°55.0	N18°56.1		61°17.4	9.0'	S28°18.0	1.8'	54.7'
7	285°55.0	56.7		75°45.4	8.9'	28°19.8	1.6'	54.7'
8	300°54.9	57.2		90°13.4	8.9'	28°21.4	1.5'	54.7'
9	315°54.9	· · 57.8		104°41.3	8.9'	28°22.8	1.3'	54.7'
10	330°54.9	58.4		119°09.2	8.9'	28°24.2	1.2'	54.8'
11	345°54.9	59.0		133°37.0	8.8'	28°25.3	1.0'	54.8'
12	0°54.9	N18°59.6		148°04.8	8.8'	S28°26.4	0.9'	54.8'
13	15°54.9	19°00.2		162°32.6	8.8'	28°27.3	0.7'	54.8'
14	30°54.9	00.7		177°00.4	8.7'	28°28.0	0.6'	54.8'
15	45°54.9	· · 01.3		191°28.1	8.7'	28°28.6	0.4'	54.8'
16	60°54.9	01.9		205°55.9	8.7'	28°29.0	0.3'	54.8'
17	75°54.9	02.5		220°23.5	8.7'	28°29.3	0.1'	54.9'
18	90°54.8	N19°03.1		234°51.2	8.7'	S28°29.5	0.0'	54.9'
19	105°54.8	03.6		249°18.9	8.6'	28°29.5	-0.1'	54.9'
20	120°54.8	04.2		263°46.5	8.6'	28°29.3	-0.3'	54.9'
21	135°54.8	· · 04.8		278°14.1	8.6'	28°29.0	-0.4'	54.9'
22	150°54.8	05.4		292°41.7	8.6'	28°28.6	-0.6'	54.9'
23	165°54.8	06.0		307°09.3	8.6'	28°28.0	-0.7'	55.0'
SD = 15.8'			d = 0.6'	SD = 14.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	00:51	23:15	☐	☐
68°	☐	☐	01:51	22:06	☐	☐
66°	☐	☐	02:25	21:31	☐	☐
64°	☐	01:14	02:50	21:05	22:45	☐
62°	☐	01:55	03:09	20:46	22:02	☐
60°	☐	02:22	03:24	20:30	21:34	☐
N 58°	01:06	02:42	03:37	20:17	21:12	22:52
56°	01:43	02:59	03:49	20:05	20:55	22:13
54°	02:09	03:13	03:59	19:55	20:41	21:46
52°	02:28	03:25	04:07	19:46	20:28	21:26
50°	02:44	03:36	04:15	19:38	20:18	21:10
45°	03:15	03:58	04:32	19:21	19:56	20:39
N 40°	03:37	04:15	04:46	19:08	19:38	20:16
35°	03:55	04:29	04:57	18:56	19:24	19:58
30°	04:10	04:41	05:07	18:46	19:12	19:43
20°	04:33	05:01	05:24	18:29	18:52	19:20
N 10°	04:51	05:17	05:39	18:14	18:36	19:02
0°	05:06	05:31	05:53	18:00	18:22	18:47
S 10°	05:19	05:44	06:06	17:46	18:08	18:34
20°	05:31	05:58	06:21	17:32	17:55	18:22
30°	05:43	06:12	06:37	17:15	17:40	18:09
35°	05:49	06:20	06:47	17:06	17:33	18:03
40°	05:56	06:29	06:58	16:55	17:24	17:57
45°	06:03	06:38	07:10	16:42	17:14	17:49
S 50°	06:11	06:50	07:26	16:27	17:02	17:42
52°	06:14	06:55	07:33	16:19	16:57	17:38
54°	06:18	07:01	07:41	16:11	16:51	17:34
56°	06:21	07:07	07:50	16:02	16:45	17:31
58°	06:26	07:14	08:00	15:52	16:38	17:26
S 60°	06:30	07:22	08:12	15:40	16:30	17:22

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	☐	☐	☐	00:48	☐	☐
64°	☐	☐	☐	01:50	☐	☐
62°	23:27	☐	☐	02:24	02:14	☐
60°	22:46	· ·	00:15	02:49	02:55	03:13
N 58°	22:18	23:37	· ·	03:09	03:24	03:52
56°	21:56	23:10	· ·	03:26	03:46	04:19
54°	21:38	22:49	23:47	03:40	04:04	04:40
52°	21:23	22:32	23:29	03:53	04:19	04:58
50°	21:10	22:17	23:14	04:04	04:33	05:13
45°	20:43	21:46	22:43	04:27	05:00	05:43
N 40°	20:22	21:23	22:19	04:45	05:22	06:07
35°	20:05	21:04	21:59	05:01	05:40	06:27
30°	19:50	20:47	21:42	05:15	05:56	06:43
20°	19:24	20:19	21:14	05:38	06:22	07:12
N 10°	19:02	19:55	20:49	0		

May 16, 17, 18 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	233°57.7	223°50.2	N03°22.5	98°11.6	N18°37.1	149°57.5	N23°06.9	233°49.6	S02°10.4	Alpheratz	357°34.7	29°13.6					
1	249°00.2	238°50.6	23.1	113°12.8	36.7	164°59.4	07.0	248°51.9	10.3	Ankaa	353°07.1	-42°10.0					
2	264°02.7	253°51.1	23.6	128°14.0	36.3	180°01.3	07.0	263°54.1	10.2	Schedar	349°31.2	56°40.4					
3	279°05.1	268°51.6	· · 24.2	143°15.2	· · 35.9	195°03.1	· · 07.0	278°56.4	· · 10.2	Diphda	348°47.3	-17°50.8					
4	294°07.6	283°52.1	24.7	158°16.4	35.5	210°05.0	07.1	293°58.6	10.1	Achernar	335°20.5	-57°06.3					
5	309°10.0	298°52.6	25.3	173°17.7	35.1	225°06.9	07.1	309°00.9	10.0	Hamal	327°51.2	23°34.8					
6	324°12.5	313°53.1	N03°25.8	188°18.9	N18°34.7	240°08.8	N23°07.1	324°03.2	S02°09.9	Polaris	314°23.1	89°22.2					
7	339°15.0	328°53.6	26.4	203°20.1	34.3	255°10.7	07.1	339°05.4	09.9	Acamar	315°12.0	-40°12.1					
8	354°17.4	343°54.0	27.0	218°21.3	33.9	270°12.6	07.2	354°07.7	09.8	Menkar	314°06.2	4°11.3					
9	9°19.9	358°54.5	· · 27.5	233°22.5	· · 33.5	285°14.4	· · 07.2	9°10.0	· · 09.7	Mirfak	308°28.4	49°57.0					
10	24°22.4	13°55.0	28.1	248°23.7	33.1	300°16.3	07.2	24°12.2	09.6	Aldebaran	290°39.6	16°33.6					
11	39°24.8	28°55.5	28.6	263°25.0	32.7	315°18.2	07.2	39°14.5	09.5	Rigel	281°03.9	-8°10.4					
12	54°27.3	43°56.0	N03°29.2	278°26.2	N18°32.3	330°20.1	N23°07.3	54°16.8	S02°09.5	Capella	280°21.9	46°01.4					
13	69°29.8	58°56.4	29.8	293°27.4	31.9	345°22.0	07.3	69°19.0	09.4	Bellatrix	278°22.9	6°22.3					
14	84°32.2	73°56.9	30.3	308°28.6	31.5	0°23.8	07.3	84°21.3	09.3	Elnath	278°01.9	28°37.7					
15	99°34.7	88°57.4	· · 30.9	323°29.8	· · 31.1	15°25.7	· · 07.3	99°23.5	· · 09.2	Alnilam	275°37.8	-1°11.2					
16	114°37.2	103°57.9	31.5	338°31.0	30.7	30°27.6	07.4	114°25.8	09.2	Betelgeuse	270°52.1	7°24.7					
17	129°39.6	118°58.3	32.0	353°32.2	30.3	45°29.5	07.4	129°28.1	09.1	Canopus	263°52.7	-52°42.7					
18	144°42.1	133°58.8	N03°32.6	8°33.4	N18°29.9	60°31.4	N23°07.4	144°30.3	S02°09.0	Sirius	258°26.3	-16°45.2					
19	159°44.5	148°59.3	33.2	23°34.7	29.5	75°33.2	07.5	159°32.6	08.9	Adhara	255°05.9	-29°00.5					
20	174°47.0	163°59.7	33.7	38°35.9	29.1	90°35.1	07.5	174°34.9	08.9	Procyon	244°50.7	5°09.6					
21	189°49.5	179°00.2	· · 34.3	53°37.1	· · 28.7	105°37.0	· · 07.5	189°37.1	· · 08.8	Pollux	243°17.2	27°58.0					
22	204°51.9	194°00.7	34.9	68°38.3	28.3	120°38.9	07.5	204°39.4	08.7	Avior	234°14.9	-59°35.7					
23	219°54.4	209°01.2	35.5	83°39.5	27.9	135°40.8	07.6	219°41.7	08.6	Suhail	222°46.2	-43°32.3					
Mer.pass. 08:23			ν 0.5' d 0.6' m -4.57			ν 1.2' d -0.4' m 1.12			ν 1.9' d 0.0' m -1.94			ν 2.3' d -0.1' m 1.13					

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	234°56.9	224°01.6	N03°36.0	98°40.7	N18°27.5	150°42.6	N23°07.6	234°43.9	S02°08.5					
1	249°59.3	239°02.1	36.6	113°41.9	27.1	165°44.5	07.6	249°46.2	08.5					
2	265°01.8	254°02.5	37.2	128°43.1	26.7	180°46.4	07.6	264°48.5	08.4					
3	280°04.3	269°03.0	· · 37.7	143°44.4	· · 26.3	195°48.3	· · 07.7	279°50.7	· · 08.3					
4	295°06.7	284°03.5	38.3	158°45.6	25.9	210°50.2	07.7	294°53.0	08.2					
5	310°09.2	299°03.9	38.9	173°46.8	25.5	225°52.0	07.7	309°55.3	08.2					
6	325°11.7	314°04.4	N03°39.5	188°48.0	N18°25.1	240°53.9	N23°07.7	324°57.5	S02°08.1					
7	340°14.1	329°04.8	40.0	203°49.2	24.7	255°55.8	07.8	339°59.8	08.0					
8	355°16.6	344°05.3	40.6	218°50.4	24.3	270°57.7	07.8	355°02.1	07.9					
9	10°19.0	359°05.8	· · 41.2	233°51.6	· · 23.9	285°59.6	· · 07.8	10°04.3	· · 07.9					
10	25°21.5	14°06.2	41.8	248°52.8	23.5	301°01.4	07.8	25°06.6	07.8					
11	40°24.0	29°06.7	42.4	263°54.0	23.1	316°03.3	07.9	40°08.9	07.7					
12	55°26.4	44°07.1	N03°42.9	278°55.3	N18°22.7	331°05.2	N23°07.9	55°11.1	S02°07.6					
13	70°28.9	59°07.6	43.5	293°56.5	22.3	346°07.1	07.9	70°13.4	07.5					
14	85°31.4	74°08.0	44.1	308°57.7	21.9	1°09.0	07.9	85°15.7	07.5					
15	100°33.8	89°08.5	· · 44.7	323°58.9	· · 21.5	16°10.8	· · 08.0	100°17.9	· · 07.4					
16	115°36.3	104°08.9	45.3	339°00.1	21.1	31°12.7	08.0	115°20.2	07.3					
17	130°38.8	119°09.4	45.9	354°01.3	20.6	46°14.6	08.0	130°22.5	07.2					
18	145°41.2	134°09.8	N03°46.4	9°02.5	N18°20.2	61°16.5	N23°08.0	145°24.7	S02°07.2					
19	160°43.7	149°10.3	47.0	24°03.7	19.8	76°18.4	08.1	160°27.0	07.1					
20	175°46.1	164°10.7	47.6	39°04.9	19.4	91°20.2	08.1	175°29.3	07.0					
21	190°48.6	179°11.1	· · 48.2	54°06.1	· · 19.0	106°22.1	· · 08.1	190°31.5	· · 06.9					
22	205°51.1	194°11.6	48.8	69°07.4	18.6	121°24.0	08.2	205°33.8	06.9					
23	220°53.5	209°12.0	49.4	84°08.6	18.2	136°25.9	08.2	220°36.1	06.8					
Mer.pass. 08:19			ν 0.5' d 0.6' m -4.55			ν 1.2' d -0.4' m 1.14			ν 1.9' d 0.0' m -1.94			ν 2.3' d -0.1' m 1.13		

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	235°56.0	224°12.5	N03°50.0	99°09.8	N18°17.8	151°27.7	N23°08.2	235°38.3	S02°06.7					
1	250°58.5	239°12.9	50.6	114°11.0	17.4	166°29.6	08.2	250°40.6	06.6					
2	266°00.9	254°13.4	51.2	129°12.2	17.0	181°31.5	08.3	265°42.9	06.6					
3	281°03.4	269°13.8	· · 51.7	144°13.4	· · 16.6	196°33.4	· · 08.3	280°45.1	· · 06.5					
4	296°05.9	284°14.2	52.3	159°14.6	16.2	211°35.3	08.3	295°47.4	06.4					
5	311°08.3	299°14.7	52.9	174°15.8	15.8	226°37.1	08.3	310°49.7	06.3					
6	326°10.8	314°15.1	N03°53.5	189°17.0	N18°15.4	241°39.0	N23°08.4	325°51.9	S02°06.3					
7	341°13.3	329°15.5	54.1	204°18.2	15.0	256°40.9	08.4	340°54.2	06.2					
8	356°15.7	344°16.0	54.7	219°19.4	14.6	271°42.8	08.4	355°56.5	06.1					
9	11°18.2	359°16.4	· · 55.3	234°20.6	· · 14.1	286°44.6	· · 08.4	10°58.8	· · 06.0					
10	26°20.6	14°16.8	55.9	249°21.8	13.7	301°46.5	08.5	26°01.0	06.0					
11	41°23.1	29°17.3	56.5	264°23.1	13.3	316°48.4	08.5	41°03.3	05.9					
12	56°25.6	44°17.7	N03°57.1	279°24.3	N18°12.9	331°50.3	N23°08.5	56°05.6	S02°05.8					
13	71°28.0	59°18.1	57.7	294°25.5	12.5	346°52.2	08.5	71°07.8	05.7					
14	86°30.5	74°18.5	58.3	309°26.7	12.1	1°54.0	08.6	86°10.1	05.7					
15	101°33.0	89°19.0	· · 58.9	324°27.9	· · 11.7	16°55.9	· · 08.6	101°12.4	· · 05.6					
16	116°35.4	104°19.4	03°59.5	339°29.1	11.3	31°57.8	08.6	116°14.6	05.5					
17	131°37.9	119°19.8	04°00.1	354°30.3	10.9	46°59.7	08.6	131°16.9	05.4					
18	146°40.4	134°20.2	N04°00.7	9°31.5	N18°10.5	62°01.5	N23°08.7	146°19.2	S02°05.4					
19	161°42.8	149°20.7	01.3	24°32.7	10.1	77°03.4	08.7	161°21.4	05.3					
20	176°45.3	164°21.1	01.9	39°33.9	09.6	92°05.3	08.7	176°23.7	05.2					
21	191°47.8	179°21.5	· · 02.5	54°35.1	· · 09.2	107°07.2	· · 08.7	191°26.0	· · 05.1					
22	206°50.2	194°21.9	03.1	69°36.3	08.8	122°09.0	08.8	206°28.3	05.1					
23	221°52.7	209°22.3	03.7	84°37.5	08.4	137°10.9	08.8	221°30.5	05.0					
Mer.pass. 08:15			ν 0.4' d 0.6' m -4.54			ν 1.2' d -0.4' m 1.15			ν 1.9' d 0.0' m -1.94			ν 2.3' d -0.1' m 1.13		

May 16 Fri	SHA	Mer.pass
Venus	349°52.4	09:04
Mars	224°13.9	17:26
Jupiter	275°59.8	13:58
Saturn	359°51.9	08:23
May 17 Sat	SHA	Mer.pass
Venus	349°04.8	09:04
Mars	223°43.9	17:24
Jupiter	275°45.8	13:55
Saturn	359°47.1	08:20
May 18 Sun	SHA	Mer.pass
Venus	348°16.5	09:03
Mars	223°13.8	17:22
Jupiter	275°31.7	13:52
Saturn	359°42.3	08:16
Horizontal parallax		
Venus:	0.2	
Mars:	0.1	

Sun				Moon				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	180°54.8	N19°06.5		321°36.9	8.6'	S28°27.2	-0.9'	55.0'
1	195°54.8	07.1		336°04.5	8.6'	28°26.3	-1.0'	55.0'
2	210°54.8	07.7		350°32.0	8.6'	28°25.3	-1.2'	55.0'
3	225°54.7	· · 08.3		4°59.6	8.5'	28°24.1	-1.3'	55.0'
4	240°54.7	08.8		19°27.1	8.5'	28°22.8	-1.5'	55.1'
5	255°54.7	09.4		33°54.7	8.5'	28°21.3	-1.6'	55.1'
6	270°54.7	N19°10.0		48°22.2	8.5'	S28°19.6	-1.8'	55.1'
7	285°54.7	10.6		62°49.7	8.5'	28°17.9	-1.9'	55.1'
8	300°54.7	11.1		77°17.3	8.5'	28°15.9	-2.1'	55.1'
9	315°54.7	· · 11.7		91°44.8	8.5'	28°13.8	-2.2'	55.2'
10	330°54.6	12.3		106°12.3	8.5'	28°11.6	-2.4'	55.2'
11	345°54.6	12.8		120°39.9	8.5'	28°09.2	-2.5'	55.2'
12	0°54.6	N19°13.4		135°07.4	8.6'	S28°06.7	-2.7'	55.2'
13	15°54.6	14.0		149°35.0	8.6'	28°04.0	-2.8'	55.2'
14	30°54.6	14.5		164°02.5	8.6'	28°01.2	-3.0'	55.2'
15	45°54.6	· · 15.1		178°30.1	8.6'	27°58.2	-3.1'	55.3'
16	60°54.6	15.7		192°57.7	8.6'	27°55.0	-3.3'	55.3'
17	75°54.5	16.2		207°25.3	8.6'	27°51.8	-3.4'	55.3'
18	90°54.5	N19°16.8		221°52.9	8.6'	S27°48.3	-3.6'	55.3'
19	105°54.5	17.4		236°20.5	8.6'	27°44.7	-3.7'	55.4'
20	120°54.5	17.9		250°48.1	8.6'	27°41.0	-3.9'	55.4'
21	135°54.5	· · 18.5		265°15.8	8.7'	27°37.1	-4.0'	55.4'
22	150°54.5	19.1		279°43.4	8.7'	27°33.1	-4.2'	55.4'
23	165°54.4	19.6		294°11.1	8.7'	27°29.0	-4.3'	55.4'
SD = 15.8'			d = 0.6'	SD = 15.0'				

Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	180°54.4	N19°20.2		308°38.8	8.7'	S27°24.6	-4.5'	55.5'
1	195°54.4	20.8		323°06.5	8.8'	27°20.2	-4.6'	55.5'
2	210°54.4	21.3		337°34.3	8.8'	27°15.6	-4.8'	55.5'
3	225°54.4	· · 21.9		352°02.1	8.8'	27°10.8	-4.9'	55.5'
4	240°54.3	22.4		6°29.9	8.8'	27°05.9	-5.0'	55.5'
5	255°54.3	23.0		20°57.7	8.9'	27°00.9	-5.2'	55.6'
6	270°54.3	N19°23.6		35°25.6	8.9'	S26°55.7	-5.3'	55.6'
7	285°54.3	24.1		49°53.4	8.9'	26°50.3	-5.5'	55.6'
8	300°54.3	24.7		64°21.4	8.9'	26°44.8	-5.6'	55.6'
9	315°54.2	· · 25.2		78°49.3	9.0'	26°39.2	-5.8'	55.7'
10	330°54.2	25.8		93°17.3	9.0'	26°33.4	-5.9'	55.7'
11	345°54.2	26.4		107°45.3	9.0'	26°27.5	-6.1'	55.7'
12	0°54.2	N19°26.9		122°13.3	9.1'	S26°21.5	-6.2'	55.7'
13	15°54.2	27.5		136°41.4	9.1'	26°15.3	-6.3'	55.8'
14	30°54.1	28.0		151°09.5	9.1'	26°08.9	-6.5'	55.8'
15	45°54.1	· · 28.6		165°37.7	9.2'	26°02.5	-6.6'	55.8'
16	60°54.1	29.1		180°05.8	9.2'	25°55.9	-6.8'	55.8'
17	75°54.1	29.7		194°34.1	9.3'	25°49.1	-6.9'	55.8'
18	90°54.0	N19°30.2		209°02.3	9.3'	S25°42.2	-7.0'	55.9'
19	105°54.0	30.8		223°30.6	9.3'	25°35.2	-7.2'	55.9'
20	120°54.0	31.3		237°59.0	9.4'	25°28.0	-7.3'	55.9'
21	135°54.0	· · 31.9		252°27.3	9.4'	25°20.7	-7.4'	55.9'
22	150°54.0	32.4		266°55.8	9.5'	25°13.2	-7.6'	56.0'
23	165°53.9	33.0		281°24.2	9.5'	25°05.6	-7.7'	56.0'
SD = 15.8'			d = 0.6'	SD = 15.1'				

Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	180°53.9	N19°33.5		295°52.7	9.5'	S24°57.9	-7.9'	56.0'
1	195°53.9	34.1		310°21.3	9.6'	24°50.1	-8.0'	56.0'
2	210°53.9	34.6		324°49.9	9.6'	24°42.1	-8.1'	56.1'
3	225°53.8	· · 35.2		339°18.5	9.7'	24°33.9	-8.3'	56.1'
4	240°53.8	35.7		353°47.2	9.7'	24°25.7	-8.4'	56.1'
5	255°53.8	36.3		8°15.9	9.8'	24°17.3	-8.5'	56.2'
6	270°53.8	N19°36.8		22°44.7	9.8'	S24°08.8	-8.7'	56.2'
7	285°53.7	37.4		37°13.5	9.9'	24°00.1	-8.8'	56.2'
8	300°53.7	37.9		51°42.4	9.9'	23°51.4	-8.9'	56.2'
9	315°53.7	· · 38.5		66°11.3	10.0'	23°42.4	-9.0'	56.3'
10	330°53.7	39.0		80°40.3	10.0'	23°33.4	-9.2'	56.3'
11	345°53.6	39.5		95°09.3	10.1'	23°24.2	-9.3'	56.3'
12	0°53.6	N19°40.1		109°38.3	10.1'	S23°14.9	-9.4'	56.3'
13	15°53.6	40.6		124°07.4	10.1'	23°05.5	-9.5'	56.4'
14	30°53.5	41.2		138°36.6	10.2'	22°56.0	-9.7'	56.4'
15	45°53.5	· · 41.7		153°05.8	10.2'	22°46.3	-9.8'	56.4'
16	60°53.5	42.3		167°35.0	10.3'	22°36.5	-9.9'	56.5'
17	75°53.5	42.8		182°04.3	10.3'	22°26.6	-10.0'	56.5'
18	90°53.4	N19°43.3		196°33.6	10.4'	S22°16.6	-10.2'	56.5'
19	105°53.4	43.9		211°03.0	10.4'	22°06.4	-10.3'	56.5'
20	120°53.4	44.4		225°32.5	10.5'	21°56.1	-10.4'	56.6'
21	135°53.3	· · 44.9		240°02.0	10.5'	21°45.7	-10.5'	56.6'
22	150°53.3	45.5		254°31.5	10.6'	21°35.2	-10.6'	56.6'
23	165°53.3	46.0		269°01.1	10.6'	21°24.6	-10.8'	56.7'
SD = 15.8'			d = 0.5'	SD = 15.3'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°	--	--	01:34	22:24	--	--
66°	--	--	02:13	21:43	--	--
64°	--	00:50	02:40	21:15	23:12	--
62°	--	01:42	03:01	20:54	22:15	--
60°	--	02:12	03:17	20:37	21:43	--
N 58°	00:45	02:34	03:31	20:23	21:20	23:16
56°	01:32	02:52	03:43	20:11	21:02	22:25
54°	02:00	03:07	03:54	20:00	20:47	21:55
52°	02:21	03:20	04:03	19:51	20:34	21:34
50°	02:38	03:31	04:11	19:42	20:22	21:16
45°	03:10	03:54	04:29	19:25	19:59	20:44
N 40°	03:34	04:12	04:43	19:10	19:41	20:20
35°	03:52	04:27	04:55	18:58	19:27	20:01
30°	04:07	04:39	05:05	18:48	19:14	19:46
20°	04:31	04:59	05:23	18:30	18:54	19:22
N 10°	04:50	05:16	05:39	18:14	18:37	19:03
0°	05:05	05:31	05:53	18:00	18:22	18:47
S 10°	05:19	05:45	06:07	17:46	18:08	18:34
20°	05:32	05:59	06:22	17:31	17:54	18:21
30°	05:45	06:14	06:39	17:13	17:39	18:08
35°	05:51	06:22	06:49	17:03	17:31	18:01
40°	05:58	06:31	07:00	16:52	17:21	17:54
45°	06:06	06:41	07:14	16:39	17:11	17:47
S 50°	06:14	06:54	07:30	16:23	16:59	17:38
52°	06:18	06:59	07:37	16:15	16:53	17:35
54°	06:22	07:05	07:46	16:07	16:47	17:31
56°	06:26	07:12	07:55	15:57	16:40	17:26
58°	06:30	07:19	08:06	15:46	16:33	17:22
S 60°	06:35	07:28	08:18	15:34	16:24	17:17

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°						
N 70°						
68°						
66°						
64°			03:04			05:59
62°		02:36	02:22		04:36	06:40
60°	01:18	01:44	01:54	04:02	05:28	07:08
N 58°	00:37	01:13	01:31	04:43	05:59	07:30
56°	00:09	00:49	01:13	05:11	06:23	07:47
54°	· · ·	00:29	00:58	05:33	06:42	08:02
52°	· · ·	00:13	00:45	05:51	06:58	08:15
50°	23:59	· · ·	00:33	06:06	07:12	08:26
45°	23:30	· · ·	00:08	06:37	07:40	08:50
N 40°	23:08	23:49	· · ·	07:01	08:02	09:08
35°	22:49	23:32	· · ·	07:20	08:20	09:24
30°	22:33	23:18	23:58	07:37	08:36	09:37
20°	22:06	22:54	23:38	08:05	09:02	10:00
N 10°	21:42	22:33	23:21	08:30	09:25	10:20
0°	21:20	22:13	23:05	08:52	09:46	10:38
S 10°	20:58	21:53	22:48	09:15	10:07	10:56
20°	20:35	21:32	22:31	09:39	10:29	11:16
30°	20:07	21:08	22:10	10:07	10:55	11:38
35°	19:51	20:53	21:58	10:24	11:10	11:51
40°	19:32	20:36	21:44	10:43	11:28	12:05
45°	19:09	20:16	21:28	11:06	11:49	12:23
S 50°	18:39	19:50	21:08	11:36	12:15	12:44
52°	18:25	19:38	20:58	11:51	12:28	12:55
54°	18:08	19:23	20:47	12:08	12:43	13:06
56°	17:47	19:07	20:34	12:28	13:00	13:19
58°	17:22	18:47	20:20	12:54	13:21	13:35
S 60°	16:47	18:21	20:03	13:30	13:46	13:53

Day	Sun		Mer.	Moon		Age
	Eqn.of Time	Time	Pass	Mer.Pass.		
	00 ^h	12 ^h	hh:mm	Upper	Lower	19-21
	mm:ss	mm:ss		hh:mm	hh:mm	90-75%
16	03:39	03:38	11:56	02:39	15:06	
17	03:38	03:37	11:56	03:33	16:00	
18	03:36	03:34	11:56	04:26	16:51	

May 19, 20, 21 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	236°55.1	224°22.8	N04°04.3	99°38.7	N18°08.0	152°12.8	N23°08.8	236°32.8	S02°04.9	Alpheratz	357°34.7	29°13.6
1	251°57.6	239°23.2	04.9	114°39.9	07.6	167°14.7	08.8	251°35.1	04.8	Ankaa	353°07.1	-42°09.9
2	267°00.1	254°23.6	05.5	129°41.1	07.2	182°16.5	08.9	266°37.3	04.8	Schedar	349°31.1	56°40.3
3	282°02.5	269°24.0	•• 06.1	144°42.3	•• 06.8	197°18.4	•• 08.9	281°39.6	•• 04.7	Diphda	348°47.2	-17°50.8
4	297°05.0	284°24.4	06.8	159°43.5	06.4	212°20.3	08.9	296°41.9	04.6	Achernar	335°20.4	-57°06.3
5	312°07.5	299°24.8	07.4	174°44.7	06.0	227°22.2	08.9	311°44.2	04.5	Hamal	327°51.2	23°34.8
6	327°09.9	314°25.2	N04°08.0	189°46.0	N18°05.5	242°24.1	N23°08.9	326°46.4	S02°04.5	Polaris	314°22.7	89°22.2
7	342°12.4	329°25.6	08.6	204°47.2	05.1	257°25.9	09.0	341°48.7	04.4	Acamar	315°12.0	-40°12.1
8	357°14.9	344°26.0	09.2	219°48.4	04.7	272°27.8	09.0	356°51.0	04.3	Menkar	314°06.2	4°11.3
9	12°17.3	359°26.5	•• 09.8	234°49.6	•• 04.3	287°29.7	•• 09.0	11°53.2	•• 04.2	Mirfak	308°28.4	49°57.0
10	27°19.8	14°26.9	10.4	249°50.8	03.9	302°31.6	09.0	26°55.5	04.2	Aldebaran	290°39.6	16°33.6
11	42°22.3	29°27.3	11.0	264°52.0	03.5	317°33.4	09.1	41°57.8	04.1	Rigel	281°03.9	-8°10.4
12	57°24.7	44°27.7	N04°11.6	279°53.2	N18°03.1	332°35.3	N23°09.1	57°00.1	S02°04.0	Capella	280°21.9	46°01.4
13	72°27.2	59°28.1	12.3	294°54.4	02.7	347°37.2	09.1	72°02.3	04.0	Bellatrix	278°22.9	6°22.3
14	87°29.6	74°28.5	12.9	309°55.6	02.2	2°39.1	09.1	87°04.6	03.9	Elnath	278°01.9	28°37.7
15	102°32.1	89°28.9	•• 13.5	324°56.8	•• 01.8	17°40.9	•• 09.2	102°06.9	•• 03.8	Alnilam	275°37.7	-1°11.2
16	117°34.6	104°29.3	14.1	339°58.0	01.4	32°42.8	09.2	117°09.1	03.7	Betelgeuse	270°52.1	7°24.7
17	132°37.0	119°29.7	14.7	354°59.2	01.0	47°44.7	09.2	132°11.4	03.7	Canopus	263°52.7	-52°42.7
18	147°39.5	134°30.1	N04°15.3	10°00.4	N18°00.6	62°46.6	N23°09.2	147°13.7	S02°03.6	Sirius	258°26.3	-16°45.1
19	162°42.0	149°30.5	16.0	25°01.6	18°00.2	77°48.4	09.3	162°16.0	03.5	Adhara	255°05.9	-29°00.5
20	177°44.4	164°30.9	16.6	40°02.8	17°59.8	92°50.3	09.3	177°18.2	03.4	Procyon	244°50.7	5°09.6
21	192°46.9	179°31.3	•• 17.2	55°04.0	•• 59.3	107°52.2	•• 09.3	192°20.5	•• 03.4	Pollux	243°17.2	27°58.0
22	207°49.4	194°31.7	17.8	70°05.2	58.9	122°54.1	09.3	207°22.8	03.3	Avior	234°14.9	-59°35.7
23	222°51.8	209°32.1	18.4	85°06.4	58.5	137°55.9	09.4	222°25.1	03.2	Suhail	222°46.2	-43°32.3
Mer.pass. 08:11		ν 0.4' d 0.6' m -4.53		ν 1.2' d -0.4' m 1.16		ν 1.9' d 0.0' m -1.94		ν 2.3' d -0.1' m 1.12				

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	237°54.3	224°32.5	N04°19.1	100°07.6	N17°58.1	152°57.8	N23°09.4	237°27.3	S02°03.1
1	252°56.7	239°32.9	19.7	115°08.8	57.7	167°59.7	09.4	252°29.6	03.1
2	267°59.2	254°33.2	20.3	130°10.0	57.3	183°01.5	09.4	267°31.9	03.0
3	283°01.7	269°33.6	•• 20.9	145°11.2	•• 56.9	198°03.4	•• 09.5	282°34.1	•• 02.9
4	298°04.1	284°34.0	21.5	160°12.4	56.4	213°05.3	09.5	297°36.4	02.9
5	313°06.6	299°34.4	22.2	175°13.6	56.0	228°07.2	09.5	312°38.7	02.8
6	328°09.1	314°34.8	N04°22.8	190°14.8	N17°55.6	243°09.0	N23°09.5	327°41.0	S02°02.7
7	343°11.5	329°35.2	23.4	205°16.0	55.2	258°10.9	09.5	342°43.2	02.6
8	358°14.0	344°35.6	24.0	220°17.2	54.8	273°12.8	09.6	357°45.5	02.6
9	13°16.5	359°36.0	•• 24.7	235°18.4	•• 54.4	288°14.7	•• 09.6	12°47.8	•• 02.5
10	28°18.9	14°36.3	25.3	250°19.6	53.9	303°16.5	09.6	27°50.1	02.4
11	43°21.4	29°36.7	25.9	265°20.8	53.5	318°18.4	09.6	42°52.3	02.3
12	58°23.9	44°37.1	N04°26.6	280°22.0	N17°53.1	333°20.3	N23°09.7	57°54.6	S02°02.3
13	73°26.3	59°37.5	27.2	295°23.2	52.7	348°22.2	09.7	72°56.9	02.2
14	88°28.8	74°37.9	27.8	310°24.4	52.3	3°24.0	09.7	87°59.2	02.1
15	103°31.2	89°38.3	•• 28.5	325°25.6	•• 51.9	18°25.9	•• 09.7	103°01.4	•• 02.1
16	118°33.7	104°38.6	29.1	340°26.8	51.4	33°27.8	09.8	118°03.7	02.0
17	133°36.2	119°39.0	29.7	355°28.0	51.0	48°29.6	09.8	133°06.0	01.9
18	148°38.6	134°39.4	N04°30.3	10°29.2	N17°50.6	63°31.5	N23°09.8	148°08.3	S02°01.8
19	163°41.1	149°39.8	31.0	25°30.4	50.2	78°33.4	09.8	163°10.5	01.8
20	178°43.6	164°40.1	31.6	40°31.6	49.8	93°35.3	09.8	178°12.8	01.7
21	193°46.0	179°40.5	•• 32.2	55°32.8	•• 49.3	108°37.1	•• 09.9	193°15.1	•• 01.6
22	208°48.5	194°40.9	32.9	70°34.0	48.9	123°39.0	09.9	208°17.4	01.6
23	223°51.0	209°41.3	33.5	85°35.2	48.5	138°40.9	09.9	223°19.6	01.5
Mer.pass. 08:07		ν 0.4' d 0.6' m -4.52		ν 1.2' d -0.4' m 1.17		ν 1.9' d 0.0' m -1.94		ν 2.3' d -0.1' m 1.12	

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	238°53.4	224°41.6	N04°34.2	100°36.4	N17°48.1	153°42.8	N23°09.9	238°21.9	S02°01.4
1	253°55.9	239°42.0	34.8	115°37.6	47.7	168°44.6	10.0	253°24.2	01.3
2	268°58.4	254°42.4	35.4	130°38.8	47.3	183°46.5	10.0	268°26.5	01.3
3	284°00.8	269°42.7	•• 36.1	145°40.0	•• 46.8	198°48.4	•• 10.0	283°28.7	•• 01.2
4	299°03.3	284°43.1	36.7	160°41.2	46.4	213°50.2	10.0	298°31.0	01.1
5	314°05.7	299°43.5	37.3	175°42.4	46.0	228°52.1	10.1	313°33.3	01.0
6	329°08.2	314°43.8	N04°38.0	190°43.6	N17°45.6	243°54.0	N23°10.1	328°35.6	S02°01.0
7	344°10.7	329°44.2	38.6	205°44.8	45.2	258°55.9	10.1	343°37.8	00.9
8	359°13.1	344°44.6	39.3	220°46.0	44.7	273°57.7	10.1	358°40.1	00.8
9	14°15.6	359°44.9	•• 39.9	235°47.2	•• 44.3	288°59.6	•• 10.1	13°42.4	•• 00.8
10	29°18.1	14°45.3	40.5	250°48.4	43.9	304°01.5	10.2	28°44.7	00.7
11	44°20.5	29°45.6	41.2	265°49.6	43.5	319°03.3	10.2	43°47.0	00.6
12	59°23.0	44°46.0	N04°41.8	280°50.8	N17°43.0	334°05.2	N23°10.2	58°49.2	S02°00.5
13	74°25.5	59°46.4	42.5	295°52.0	42.6	349°07.1	10.2	73°51.5	00.5
14	89°27.9	74°46.7	43.1	310°53.2	42.2	4°09.0	10.3	88°53.8	00.4
15	104°30.4	89°47.1	•• 43.8	325°54.4	•• 41.8	19°10.8	•• 10.3	103°56.1	•• 00.3
16	119°32.8	104°47.4	44.4	340°55.5	41.4	34°12.7	10.3	118°58.3	00.3
17	134°35.3	119°47.8	45.1	355°56.7	40.9	49°14.6	10.3	134°00.6	00.2
18	149°37.8	134°48.2	N04°45.7	10°57.9	N17°40.5	64°16.4	N23°10.3	149°02.9	S02°00.1
19	164°40.2	149°48.5	46.4	25°59.1	40.1	79°18.3	10.4	164°05.2	00.1
20	179°42.7	164°48.9	47.0	41°00.3	39.7	94°20.2	10.4	179°07.5	02°00.0
21	194°45.2	179°49.2	•• 47.7	56°01.5	•• 39.2	109°22.1	•• 10.4	194°09.7	01°59.9
22	209°47.6	194°49.6	48.3	71°02.7	38.8	124°23.9	10.4	209°12.0	59.8
23	224°50.1	209°49.9	48.9	86°03.9	38.4	139°25.8	10.5	224°14.3	59.8
Mer.pass. 08:03		ν 0.4' d 0.6' m -4.51		ν 1.2' d -0.4' m 1.18		ν 1.9' d 0.0' m -1.93		ν 2.3' d -0.1' m 1.12	

May 19 Mon	SHA	Mer.pass
Venus	347°27.6	09:02
Mars	222°43.6	17:20
Jupiter	275°17.7	13:49
Saturn	359°37.7	08:13

May 20 Tue	SHA	Mer.pass
Venus	346°38.2	09:02
Mars	222°13.3	17:18
Jupiter	275°03.5	13:46
Saturn	359°33.0	08:09

May 21 Wed	SHA	Mer.pass
Venus	345°48.2	09:01
Mars	221°43.0	17:16
Jupiter	274°49.3	13:43
Saturn	359°28.5	08:05

Horizontal parallax	
Venus:	0.2
Mars:	0.1

Sun								Moon									
Mon	GHA	Dec		GHA	ν	Dec	d	HP									
0	180°53.3	N19°46.5		283°30.7	10.7'	S21°13.8	-10.9'	56.7'									
1	195°53.2	47.1		298°00.4	10.7'	21°02.9	-11.0'	56.7'									
2	210°53.2	47.6		312°30.1	10.8'	20°51.9	-11.1'	56.7'									
3	225°53.2	· · 48.2		326°59.9	10.8'	20°40.8	-11.2'	56.8'									
4	240°53.1	48.7		341°29.7	10.9'	20°29.6	-11.3'	56.8'									
5	255°53.1	49.2		355°59.6	10.9'	20°18.3	-11.4'	56.8'									
6	270°53.1	N19°49.7		10°29.5	11.0'	S20°06.9	-11.6'	56.9'									
7	285°53.0	50.3		24°59.5	11.0'	19°55.3	-11.7'	56.9'									
8	300°53.0	50.8		39°29.5	11.1'	19°43.6	-11.8'	56.9'									
9	315°53.0	· · 51.3		53°59.5	11.1'	19°31.9	-11.9'	56.9'									
10	330°52.9	51.9		68°29.6	11.1'	19°20.0	-12.0'	57.0'									
11	345°52.9	52.4		82°59.8	11.2'	19°08.0	-12.1'	57.0'									
12	0°52.9	N19°52.9		97°30.0	11.2'	S18°55.9	-12.2'	57.0'									
13	15°52.8	53.5		112°00.2	11.3'	18°43.7	-12.3'	57.1'									
14	30°52.8	54.0		126°30.5	11.3'	18°31.4	-12.4'	57.1'									
15	45°52.8	· · 54.5		141°00.8	11.4'	18°19.0	-12.5'	57.1'									
16	60°52.7	55.0		155°31.2	11.4'	18°06.5	-12.6'	57.2'									
17	75°52.7	55.6		170°01.6	11.5'	17°53.9	-12.7'	57.2'									
18	90°52.7	N19°56.1		184°32.1	11.5'	S17°41.2	-12.8'	57.2'									
19	105°52.6	56.6		199°02.6	11.5'	17°28.3	-12.9'	57.3'									
20	120°52.6	57.1		213°33.1	11.6'	17°15.4	-13.0'	57.3'									
21	135°52.6	· · 57.7		228°03.7	11.6'	17°02.4	-13.1'	57.3'									
22	150°52.5	58.2		242°34.3	11.7'	16°49.3	-13.2'	57.4'									
23	165°52.5	58.7		257°04.9	11.7'	16°36.1	-13.3'	57.4'									
SD = 15.8'			$d = 0.5'$			SD = 15.5'											
Tue	GHA	Dec		GHA	ν	Dec	d	HP									
0	180°52.5	N19°59.2		271°35.6	11.7'	S16°22.8	-13.4'	57.4'									
1	195°52.4	19°59.7		286°06.4	11.8'	16°09.4	-13.5'	57.4'									
2	210°52.4	20°00.3		300°37.1	11.8'	15°55.9	-13.6'	57.5'									
3	225°52.4	· · 00.8		315°07.9	11.8'	15°42.4	-13.7'	57.5'									
4	240°52.3	01.3		329°38.8	11.9'	15°28.7	-13.8'	57.5'									
5	255°52.3	01.8		344°09.6	11.9'	15°14.9	-13.8'	57.6'									
6	270°52.2	N20°02.3		358°40.5	11.9'	S15°01.1	-13.9'	57.6'									
7	285°52.2	02.9		13°11.5	12.0'	14°47.1	-14.0'	57.6'									
8	300°52.2	03.4		27°42.5	12.0'	14°33.1	-14.1'	57.7'									
9	315°52.1	· · 03.9		42°13.5	12.0'	14°19.0	-14.2'	57.7'									
10	330°52.1	04.4		56°44.5	12.1'	14°04.8	-14.3'	57.7'									
11	345°52.1	04.9		71°15.6	12.1'	13°50.5	-14.4'	57.8'									
12	0°52.0	N20°05.4		85°46.7	12.1'	S13°36.2	-14.4'	57.8'									
13	15°52.0	05.9		100°17.8	12.1'	13°21.7	-14.5'	57.8'									
14	30°51.9	06.5		114°48.9	12.2'	13°07.2	-14.6'	57.9'									
15	45°51.9	· · 07.0		129°20.1	12.2'	12°52.6	-14.7'	57.9'									
16	60°51.9	07.5		143°51.3	12.2'	12°37.9	-14.8'	57.9'									
17	75°51.8	08.0		158°22.5	12.2'	12°23.2	-14.8'	58.0'									
18	90°51.8	N20°08.5		172°53.8	12.3'	S12°08.3	-14.9'	58.0'									
19	105°51.7	09.0		187°25.0	12.3'	11°53.4	-15.0'	58.0'									
20	120°51.7	09.5		201°56.3	12.3'	11°38.4	-15.1'	58.1'									
21	135°51.7	· · 10.0		216°27.6	12.3'	11°23.4	-15.1'	58.1'									
22	150°51.6	10.5		230°58.9	12.3'	11°08.3	-15.2'	58.1'									
23	165°51.6	11.0		245°30.3	12.4'	10°53.1	-15.3'	58.2'									
SD = 15.8'			$d = 0.5'$			SD = 15.7'											
Wed	GHA	Dec		GHA	ν	Dec	d	HP									
0	180°51.5	N20°11.6		260°01.6	12.4'	S10°37.8	-15.3'	58.2'									
1	195°51.5	12.1		274°33.0	12.4'	10°22.4	-15.4'	58.2'									
2	210°51.5	12.6		289°04.4	12.4'	10°07.0	-15.5'	58.3'									
3	225°51.4	· · 13.1		303°35.8	12.4'	09°51.6	-15.5'	58.3'									
4	240°51.4	13.6		318°07.2	12.4'	09°36.0	-15.6'	58.3'									
5	255°51.3	14.1		332°38.7	12.4'	09°20.4	-15.7'	58.4'									
6	270°51.3	N20°14.6		347°10.1	12.4'	S09°04.7	-15.7'	58.4'									
7	285°51.2	15.1		1°41.5	12.5'	08°49.0	-15.8'	58.4'									
8	300°51.2	15.6		16°13.0	12.5'	08°33.2	-15.9'	58.5'									
9	315°51.2	· · 16.1		30°44.5	12.5'	08°17.4	-15.9'	58.5'									
10	330°51.1	16.6		45°15.9	12.5'	08°01.5	-16.0'	58.5'									
11	345°51.1	17.1		59°47.4	12.5'	07°45.5	-16.0'	58.6'									
12	0°51.0	N20°17.6		74°18.9	12.5'	S07°29.5	-16.1'	58.6'									
13	15°51.0	18.1		88°50.3	12.5'	07°13.4	-16.1'	58.6'									
14	30°50.9	18.6		103°21.8	12.5'	06°57.3	-16.2'	58.7'									
15	45°50.9	· · 19.1		117°53.3	12.5'	06°41.1	-16.2'	58.7'									
16	60°50.8	19.6		132°24.7	12.5'	06°24.8	-16.3'	58.7'									
17	75°50.8	20.1		146°56.2	12.5'	06°08.6	-16.3'	58.8'									
18	90°50.7	N20°20.6		161°27.7	12.5'	S05°52.2	-16.4'	58.8'									
19	105°50.7	21.1		175°59.1	12.4'	05°35.8	-16.4'	58.8'									
20	120°50.7	21.6		190°30.6	12.4'	05°19.4	-16.5'	58.9'									
21	135°50.6	· · 22.1		205°02.0	12.4'	05°02.9	-16.5'	58.9'									
22	150°50.6	22.6		219°33.4	12.4'	04°46.4	-16.6'	58.9'									
23	165°50.5	23.1		234°04.9	12.4'	04°29.9	-16.6'	59.0'									
SD = 15.8'			$d = 0.5'$			SD = 15.9'											

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°	—	—	01:15	22:45	—	—
66°	—	—	02:01	21:55	—	—
64°	—	00:05	02:31	21:25	—	—
62°	—	01:28	02:53	21:02	22:29	—
60°	—	02:02	03:11	20:44	21:53	—
N 58°	00:04	02:27	03:26	20:29	21:28	—
56°	01:19	02:46	03:38	20:16	21:09	22:37
54°	01:51	03:02	03:49	20:05	20:53	22:04
52°	02:14	03:15	03:59	19:55	20:39	21:41
50°	02:32	03:27	04:07	19:46	20:27	21:22
45°	03:06	03:51	04:26	19:28	20:03	20:48
N 40°	03:31	04:09	04:40	19:13	19:44	20:23
35°	03:50	04:24	04:53	19:01	19:29	20:04
30°	04:05	04:37	05:04	18:50	19:16	19:48
20°	04:30	04:58	05:22	18:31	18:55	19:23
N 10°	04:49	05:16	05:38	18:15	18:	

May 22, 23, 24 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec		GHA	Dec		SHA	Dec	
0	239°52.6	224°50.3	N04°49.6	101°05.1	N17°38.0	154°27.7	N23°10.5	239°16.6	S01°59.7			239°16.6	S01°59.7		Alpheratz	357°34.7	29°13.6
1	254°55.0	239°50.6	50.3	116°06.3	37.6	169°29.5	10.5	254°18.9	59.6			254°18.9	59.6		Ankaa	353°07.1	-42°09.9
2	269°57.5	254°51.0	50.9	131°07.5	37.1	184°31.4	10.5	269°21.1	59.6			269°21.1	59.6		Schedar	349°31.1	56°40.3
3	285°00.0	269°51.3	· · 51.6	146°08.7	· · 36.7	199°33.3	· · 10.5	284°23.4	· · 59.5			284°23.4	· · 59.5		Diphda	348°47.2	-17°50.8
4	300°02.4	284°51.6	52.2	161°09.9	36.3	214°35.1	10.6	299°25.7	59.4			299°25.7	59.4		Achernar	335°20.4	-57°06.3
5	315°04.9	299°52.0	52.9	176°11.1	35.9	229°37.0	10.6	314°28.0	59.3			314°28.0	59.3		Hamal	327°51.2	23°34.8
6	330°07.3	314°52.3	N04°53.5	191°12.3	N17°35.4	244°38.9	N23°10.6	329°30.2	S01°59.3			329°30.2	S01°59.3		Polaris	314°22.3	89°22.2
7	345°09.8	329°52.7	54.2	206°13.5	35.0	259°40.8	10.6	344°32.5	59.2			344°32.5	59.2		Acamar	315°12.0	-40°12.1
8	0°12.3	344°53.0	54.8	221°14.7	34.6	274°42.6	10.7	359°34.8	59.1			359°34.8	59.1		Menkar	314°06.2	4°11.3
9	15°14.7	359°53.4	· · 55.5	236°15.9	· · 34.2	289°44.5	· · 10.7	14°37.1	· · 59.1			14°37.1	· · 59.1		Mirfak	308°28.4	49°57.0
10	30°17.2	14°53.7	56.1	251°17.1	33.7	304°46.4	10.7	29°39.4	59.0			29°39.4	59.0		Aldebaran	290°39.6	16°33.6
11	45°19.7	29°54.0	56.8	266°18.2	33.3	319°48.2	10.7	44°41.6	58.9			44°41.6	58.9		Rigel	281°03.9	-8°10.4
12	60°22.1	44°54.4	N04°57.4	281°19.4	N17°32.9	334°50.1	N23°10.7	59°43.9	S01°58.9			59°43.9	S01°58.9		Capella	280°21.9	46°01.4
13	75°24.6	59°54.7	58.1	296°20.6	32.5	349°52.0	10.8	74°46.2	58.8			74°46.2	58.8		Bellatrix	278°22.9	6°22.3
14	90°27.1	74°55.1	58.8	311°21.8	32.0	4°53.8	10.8	89°48.5	58.7			89°48.5	58.7		Elnath	278°01.9	28°37.7
15	105°29.5	89°55.4	04°59.4	326°23.0	· · 31.6	19°55.7	· · 10.8	104°50.8	· · 58.6			104°50.8	· · 58.6		Alnilam	275°37.8	-1°11.2
16	120°32.0	104°55.7	05°00.1	341°24.2	31.2	34°57.6	10.8	119°53.0	58.6			119°53.0	58.6		Betelgeuse	270°52.1	7°24.7
17	135°34.5	119°56.1	00.7	356°25.4	30.8	49°59.5	10.8	134°55.3	58.5			134°55.3	58.5		Canopus	263°52.7	-52°42.7
18	150°36.9	134°56.4	N05°01.4	11°26.6	N17°30.3	65°01.3	N23°10.9	149°57.6	S01°58.4			149°57.6	S01°58.4		Sirius	258°26.3	-16°45.1
19	165°39.4	149°56.7	02.1	26°27.8	29.9	80°03.2	10.9	164°59.9	58.4			164°59.9	58.4		Adhara	255°05.9	-29°00.5
20	180°41.8	164°57.1	02.7	41°29.0	29.5	95°05.1	10.9	180°02.2	58.3			180°02.2	58.3		Procyon	244°50.8	5°09.6
21	195°44.3	179°57.4	· · 03.4	56°30.2	· · 29.0	110°06.9	· · 10.9	195°04.4	· · 58.2			195°04.4	· · 58.2		Pollux	243°17.2	27°58.0
22	210°46.8	194°57.7	04.0	71°31.4	28.6	125°08.8	11.0	210°06.7	58.2			210°06.7	58.2		Avior	234°15.0	-59°35.7
23	225°49.2	209°58.1	04.7	86°32.6	28.2	140°10.7	11.0	225°09.0	58.1			225°09.0	58.1		Suhail	222°46.3	-43°32.3
Mer.pass. 07:59			ν 0.3' d 0.7' m -4.50	ν 1.2' d -0.4' m 1.19	ν 1.9' d 0.0' m -1.93	ν 2.3' d -0.1' m 1.11											

Fri	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec	
0	240°51.7	224°58.4	N05°05.4	101°33.7	N17°27.8	155°12.5	N23°11.0	240°11.3	S01°58.0		
1	255°54.2	239°58.7	06.0	116°34.9	27.3	170°14.4	11.0	255°13.6	58.0		
2	270°56.6	254°59.0	06.7	131°36.1	26.9	185°16.3	11.0	270°15.9	57.9		
3	285°59.1	269°59.4	· · 07.4	146°37.3	· · 26.5	200°18.1	· · 11.1	285°18.1	· · 57.8		
4	301°01.6	284°59.7	08.0	161°38.5	26.1	215°20.0	11.1	300°20.4	57.7		
5	316°04.0	300°00.0	08.7	176°39.7	25.6	230°21.9	11.1	315°22.7	57.7		
6	331°06.5	315°00.3	N05°09.4	191°40.9	N17°25.2	245°23.7	N23°11.1	330°25.0	S01°57.6		
7	346°08.9	330°00.7	10.0	206°42.1	24.8	260°25.6	11.1	345°27.3	57.5		
8	1°11.4	345°01.0	10.7	221°43.3	24.3	275°27.5	11.2	0°29.5	57.5		
9	16°13.9	0°01.3	· · 11.4	236°44.5	· · 23.9	290°29.3	· · 11.2	15°31.8	· · 57.4		
10	31°16.3	15°01.6	12.0	251°45.7	23.5	305°31.2	11.2	30°34.1	57.3		
11	46°18.8	30°01.9	12.7	266°46.8	23.0	320°33.1	11.2	45°36.4	57.3		
12	61°21.3	45°02.3	N05°13.4	281°48.0	N17°22.6	335°35.0	N23°11.2	60°38.7	S01°57.2		
13	76°23.7	60°02.6	14.0	296°49.2	22.2	350°36.8	11.3	75°41.0	57.1		
14	91°26.2	75°02.9	14.7	311°50.4	21.8	5°38.7	11.3	90°43.2	57.1		
15	106°28.7	90°03.2	· · 15.4	326°51.6	· · 21.3	20°40.6	· · 11.3	105°45.5	· · 57.0		
16	121°31.1	105°03.5	16.1	341°52.8	20.9	35°42.4	11.3	120°47.8	56.9		
17	136°33.6	120°03.8	16.7	356°54.0	20.5	50°44.3	11.3	135°50.1	56.9		
18	151°36.1	135°04.1	N05°17.4	11°55.2	N17°20.0	65°46.2	N23°11.4	150°52.4	S01°56.8		
19	166°38.5	150°04.5	18.1	26°56.4	19.6	80°48.0	11.4	165°54.7	56.7		
20	181°41.0	165°04.8	18.7	41°57.5	19.2	95°49.9	11.4	180°56.9	56.7		
21	196°43.4	180°05.1	· · 19.4	56°58.7	· · 18.7	110°51.8	· · 11.4	195°59.2	· · 56.6		
22	211°45.9	195°05.4	20.1	71°59.9	18.3	125°53.6	11.5	211°01.5	56.5		
23	226°48.4	210°05.7	20.8	87°01.1	17.9	140°55.5	11.5	226°03.8	56.4		
Mer.pass. 07:55			ν 0.3' d 0.7' m -4.49	ν 1.2' d -0.4' m 1.20	ν 1.9' d 0.0' m -1.93	ν 2.3' d -0.1' m 1.11					

Sat	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec	
0	241°50.8	225°06.0	N05°21.4	102°02.3	N17°17.4	155°57.4	N23°11.5	241°06.1	S01°56.4		
1	256°53.3	240°06.3	22.1	117°03.5	17.0	170°59.2	11.5	256°08.4	56.3		
2	271°55.8	255°06.6	22.8	132°04.7	16.6	186°01.1	11.5	271°10.7	56.2		
3	286°58.2	270°06.9	· · 23.5	147°05.9	· · 16.1	201°03.0	· · 11.6	286°12.9	· · 56.2		
4	302°00.7	285°07.2	24.2	162°07.1	15.7	216°04.8	11.6	301°15.2	56.1		
5	317°03.2	300°07.5	24.8	177°08.2	15.3	231°06.7	11.6	316°17.5	56.0		
6	332°05.6	315°07.8	N05°25.5	192°09.4	N17°14.8	246°08.6	N23°11.6	331°19.8	S01°56.0		
7	347°08.1	330°08.1	26.2	207°10.6	14.4	261°10.4	11.6	346°22.1	55.9		
8	2°10.5	345°08.4	26.9	222°11.8	14.0	276°12.3	11.7	1°24.4	55.8		
9	17°13.0	0°08.7	· · 27.6	237°13.0	· · 13.5	291°14.2	· · 11.7	16°26.6	· · 55.8		
10	32°15.5	15°09.0	28.2	252°14.2	13.1	306°16.0	11.7	31°28.9	55.7		
11	47°17.9	30°09.3	28.9	267°15.4	12.7	321°17.9	11.7	46°31.2	55.6		
12	62°20.4	45°09.6	N05°29.6	282°16.6	N17°12.2	336°19.8	N23°11.7	61°33.5	S01°55.6		
13	77°22.9	60°09.9	30.3	297°17.7	11.8	351°21.6	11.7	76°35.8	55.5		
14	92°25.3	75°10.2	31.0	312°18.9	11.4	6°23.5	11.8	91°38.1	55.4		
15	107°27.8	90°10.5	· · 31.6	327°20.1	· · 10.9	21°25.4	· · 11.8	106°40.4	· · 55.4		
16	122°30.3	105°10.8	32.3	342°21.3	10.5	36°27.2	11.8	121°42.6	55.3		
17	137°32.7	120°11.1	33.0	357°22.5	10.1	51°29.1	11.8	136°44.9	55.2		
18	152°35.2	135°11.4	N05°33.7	12°23.7	N17°09.6	66°30.9	N23°11.8	151°47.2	S01°55.2		
19	167°37.7	150°11.7	34.4	27°24.9	09.2	81°32.8	11.9	166°49.5	55.1		
20	182°40.1	165°12.0	35.1	42°26.0	08.8	96°34.7	11.9	181°51.8	55.0		
21	197°42.6	180°12.3	· · 35.8	57°27.2	· · 08.3	111°36.5	· · 11.9	196°54.1	· · 55.0		
22	212°45.0	195°12.6	36.4	72°28.4	07.9	126°38.4	11.9	211°56.4	54.9		
23	227°47.5	210°12.9	37.1	87°29.6	07.5	141°40.3	11.9	226°58.7	54.8		
Mer.pass. 07:51			ν 0.3' d 0.7' m -4.48	ν 1.2' d -0.4' m 1.21	ν 1.9' d 0.0' m -1.93	ν 2.3' d -0.1' m 1.11					

May 22 Thu	SHA	Mer.pass
Venus	344°57.7	09:00
Mars	221°12.5	17:14
Jupiter	274°35.1	13:40
Saturn	359°24.0	08:02

May 23 Fri	SHA	Mer.pass
Venus	344°06.7	09:00
Mars	220°42.0	17:12
Jupiter	274°20.8	13:37
Saturn	359°19.6	07:58

May 24 Sat	SHA	Mer.pass
Venus	343°15.2	08:59
Mars	220°11.5	17:10
Jupiter	274°06.5	13:34
Saturn	359°15.2	07:54

Horizontal parallax		
Venus:	0.2	
Mars:	0.1	

h Sun Moon						
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	180°50.5	N20°23.5	248°36.3	12.4'	S04°13.3	-16.6' 59.0'
1	195°50.4	24.0	263°07.7	12.4'	03°56.6	-16.7' 59.0'
2	210°50.4	24.5	277°39.0	12.4'	03°40.0	-16.7' 59.1'
3	225°50.3	· · 25.0	292°10.4	12.3'	03°23.2	-16.8' 59.1'
4	240°50.3	25.5	306°41.7	12.3'	03°06.5	-16.8' 59.1'
5	255°50.2	26.0	321°13.1	12.3'	02°49.7	-16.8' 59.2'
6	270°50.2	N20°26.5	335°44.4	12.3'	S02°32.9	-16.9' 59.2'
7	285°50.1	27.0	350°15.6	12.3'	02°16.0	-16.9' 59.2'
8	300°50.1	27.5	4°46.9	12.2'	01°59.1	-16.9' 59.3'
9	315°50.0	· · 28.0	19°18.1	12.2'	01°42.2	-16.9' 59.3'
10	330°50.0	28.4	33°49.3	12.2'	01°25.3	-17.0' 59.3'
11	345°49.9	28.9	48°20.5	12.1'	01°08.3	-17.0' 59.4'
12	0°49.9	N20°29.4	62°51.7	12.1'	S00°51.3	-17.0' 59.4'
13	15°49.8	29.9	77°22.8	12.1'	00°34.3	-17.0' 59.4'
14	30°49.8	30.4	91°53.9	12.1'	00°17.3	-17.1' 59.5'
15	45°49.7	· · 30.9	106°24.9	12.0'	S00°00.3	-17.1' 59.5'
16	60°49.7	31.3	120°55.9	12.0'	N00°16.8	17.1' 59.5'
17	75°49.6	31.8	135°26.9	11.9'	00°33.9	17.1' 59.6'
18	90°49.6	N20°32.3	149°57.8	11.9'	N00°51.0	17.1' 59.6'
19	105°49.5	32.8	164°28.7	11.9'	01°08.1	17.1' 59.6'
20	120°49.5	33.3	178°59.6	11.8'	01°25.3	17.1' 59.6'
21	135°49.4	· · 33.7	193°30.4	11.8'	01°42.4	17.2' 59.7'
22	150°49.4	34.2	208°01.2	11.7'	01°59.6	17.2' 59.7'
23	165°49.3	34.7	222°31.9	11.7'	02°16.7	17.2' 59.7'
SD = 15.8' $d = 0.5'$			SD = 16.1'			
Fri	GHA	Dec	GHA	ν	Dec	d HP
0	180°49.3	N20°35.2	237°02.6	11.6'	N02°33.9	17.2' 59.8'
1	195°49.2	35.7	251°33.3	11.6'	02°51.1	17.2' 59.8'
2	210°49.2	36.1	266°03.9	11.5'	03°08.2	17.2' 59.8'
3	225°49.1	· · 36.6	280°34.4	11.5'	03°25.4	17.2' 59.9'
4	240°49.0	37.1	295°04.9	11.4'	03°42.6	17.2' 59.9'
5	255°49.0	37.6	309°35.3	11.4'	03°59.8	17.2' 59.9'
6	270°48.9	N20°38.0	324°05.7	11.3'	N04°16.9	17.2' 59.9'
7	285°48.9	38.5	338°36.0	11.3'	04°34.1	17.2' 60.0'
8	300°48.8	39.0	353°06.3	11.2'	04°51.2	17.1' 60.0'
9	315°48.8	· · 39.5	7°36.5	11.1'	05°08.4	17.1' 60.0'
10	330°48.7	39.9	22°06.6	11.1'	05°25.5	17.1' 60.0'
11	345°48.7	40.4	36°36.7	11.0'	05°42.7	17.1' 60.1'
12	0°48.6	N20°40.9	51°06.7	11.0'	N05°59.8	17.1' 60.1'
13	15°48.6	41.3	65°36.7	10.9'	06°16.9	17.1' 60.1'
14	30°48.5	41.8	80°06.5	10.8'	06°33.9	17.1' 60.2'
15	45°48.4	· · 42.3	94°36.4	10.7'	06°51.0	17.0' 60.2'
16	60°48.4	42.7	109°06.1	10.7'	07°08.0	17.0' 60.2'
17	75°48.3	43.2	123°35.8	10.6'	07°25.1	17.0' 60.2'
18	90°48.3	N20°43.7	138°05.4	10.5'	N07°42.1	17.0' 60.3'
19	105°48.2	44.1	152°34.9	10.5'	07°59.0	16.9' 60.3'
20	120°48.2	44.6	167°04.4	10.4'	08°15.9	16.9' 60.3'
21	135°48.1	· · 45.1	181°33.8	10.3'	08°32.9	16.9' 60.3'
22	150°48.0	45.5	196°03.1	10.2'	08°49.7	16.8' 60.4'
23	165°48.0	46.0	210°32.3	10.1'	09°06.6	16.8' 60.4'
SD = 15.8' $d = 0.5'$			SD = 16.3'			
Sat	GHA	Dec	GHA	ν	Dec	d HP
0	180°47.9	N20°46.5	225°01.4	10.1'	N09°23.4	16.8' 60.4'
1	195°47.9	46.9	239°30.5	10.0'	09°40.1	16.7' 60.4'
2	210°47.8	47.4	253°59.5	9.9'	09°56.8	16.7' 60.4'
3	225°47.7	· · 47.9	268°28.4	9.8'	10°13.5	16.6' 60.5'
4	240°47.7	48.3	282°57.2	9.7'	10°30.2	16.6' 60.5'
5	255°47.6	48.8	297°25.9	9.6'	10°46.7	16.5' 60.5'
6	270°47.6	N20°49.2	311°54.6	9.5'	N11°03.3	16.5' 60.5'
7	285°47.5	49.7	326°23.1	9.5'	11°19.8	16.4' 60.5'
8	300°47.4	50.2	340°51.6	9.4'	11°36.2	16.4' 60.6'
9	315°47.4	· · 50.6	355°19.9	9.3'	11°52.6	16.3' 60.6'
10	330°47.3	51.1	9°48.2	9.2'	12°08.9	16.3' 60.6'
11	345°47.3	51.5	24°16.4	9.1'	12°25.2	16.2' 60.6'
12	0°47.2	N20°52.0	38°44.5	9.0'	N12°41.4	16.1' 60.6'
13	15°47.1	52.4	53°12.5	8.9'	12°57.5	16.1' 60.7'
14	30°47.1	52.9	67°40.3	8.8'	13°13.6	16.0' 60.7'
15	45°47.0	· · 53.3	82°08.1	8.7'	13°29.6	15.9' 60.7'
16	60°47.0	53.8	96°35.8	8.6'	13°45.5	15.9' 60.7'
17	75°46.9	54.2	111°03.4	8.5'	14°01.4	15.8' 60.7'
18	90°46.8	N20°54.7	125°30.9	8.4'	N14°17.2	15.7' 60.7'
19	105°46.8	55.2	139°58.3	8.3'	14°32.9	15.6' 60.8'
20	120°46.7	55.6	154°25.6	8.2'	14°48.5	15.6' 60.8'
21	135°46.6	· · 56.1	168°52.8	8.1'	15°04.1	15.5' 60.8'
22	150°46.6	56.5	183°19.9	8.0'	15°19.5	15.4' 60.8'
23	165°46.5	57.0	197°46.8	7.9'	15°34.9	15.3' 60.8'
SD = 15.8' $d = 0.5'$			SD = 16.5'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°	--	--	00:52	23:11	--	--
66°	--	--	01:49	22:08	--	--
64°	--	--	02:22	21:34	--	--
62°	--	01:13	02:46	21:10	22:45	--
60°	--	01:53	03:05	20:50	22:03	--
N 58°	--	02:19	03:20	20:35	21:36	--
56°	01:06	02:40	03:33	20:21	21:15	22:52
54°	01:43	02:57	03:45	20:10	20:58	22:13
52°	02:07	03:11	03:55	19:59	20:44	21:48
50°	02:27	03:23	04:04	19:50	20:32	21:28
45°	03:02	03:48	04:23	19:31	20:07	20:52
N 40°	03:28	04:07	04:38	19:16	19:47	20:27
35°	03:47	04:22	04:51	19:03	19:31	20:07
30°	04:04	04:36	05:02	18:51	19:18	19:50
20°	04:29	04:57	05:21	18:32	18:56	19:25
N 10°	04:49	05:15	05:38	18:16	18:38	19:05
0°	05:05	05:31	05:53	18:00	18:22	18:48
S 10°	05:20	05:46	06:08	17:45	18:07	18:33
20°	05:34	06:01	06:24	17:29	17:52	18:19
30°	05:48	06:17	06:43	17:11	17:36	18:06
35°	05:55	06:26	06:53	17:00	17:27	17:58
40°	06:03	06:36	07:06	16:48	17:17	17:51
45°	06:11	06:47	07:20	16:33	17:06	17:42
S 50°	06:20	07:01	07:37	16:16	16:52	17:33
52°	06:25	07:07	07:46	16:08	16:46	17:28
54°	06:29	07:13	07:55	15:58	16:40	17:24
56°	06:34	07:21	08:05	15:48	16:32	17:19
58°	06:39	07:29	08:17	15:36	16:24	17:14
S 60°	06:45	07:38	08:31	15:22	16:15	17:08

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	02:16	01:39	00:58 23:53	13:46	16:13	19:10
N 70°	02:11	01:43	01:12	13:46	16:02	18:37
68°	02:07	01:46	01:24	13:47	15:53	18:13
66°	02:04	01:49	01:33	13:47	15:46	17:55
64°	02:01	01:51	01:41	13:48	15:40	17:40
62°	01:58	01:53	01:49	13:48	15:35	17:28
60°	01:56	01:55	01:55	13:48	15:30	17:18
N 58°	01:54	01:57	02:00	13:49	15:26	17:09
56°	01:52	01:58	02:05	13:49	15:23	17:01
54°	01:51	02:00	02:09	13:49	15:20	16:55
52°	01:49	02:01	02:13	13:49	15:17	16:48
50°	01:48	02:02	02:17	13:50	15:14	16:43
45°	01:45	02:04	02:25	13:50	15:09	16:31
N 40°	01:42	02:06	02:32	13:50	15:04	16:21
35°	01:40	02:08	02:37	13:50	15:00	16:13
30°	01:38	02:09	02:43	13:51	14:57	16:06
20°	01:35	02:12	02:52	13:51	14:51	15:53
N 10°	01:32	02:15	02:59	13:51	14:45	15:42
0°	01:29	02:17	03:07	13:52	14:40	15:32
S 10°	01:26	02:19	03:15	13:52	14:35	15:22
20°	01:23	02:22	03:23	13:52	14:30	15:11
30°	01:20	02:25	03:32	13:52	14:24	14:59
35°	01:18	02:27	03:38	13:52	14:21	14:52
40°	01:16	02:29	03:44	13:52	14:17	14:44
45°	01:13	02:31	03:51	13:53	14:12	14:34
S 50°	01:10	02:34	04:00	13:53	14:07	14:23
52°	01:09	02:35	04:04	13:53	14:05	14:18
54°	01:07	02:36	04:09	13:53	14:02	14:13
56°	01:05	02:38	04:14	13:53	13:59	14:06
58°	01:03	02:40	04:20	13:53	13:56	14:00
S 60°	01:01	02:42	04:26	13:53	13:52	13:52

Day	Sun		Mer. Pass hh:mm	Moon		Age 25-27 34-14%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
22	03:22	03:20	11:57	07:40	20:04	
23	03:17	03:14	11:57	08:29	20:54	
24	03:12	03:09	11:57	09:19	21:46	

May 25, 26, 27 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	242°50.0	225°13.2	N05°37.8	102°30.8	N17°07.0	156°42.1	N23°12.0	242°00.9	S01°54.8	Alpheratz	357°34.6 29°13.6
1	257°52.4	240°13.4	38.5	117°32.0	06.6	171°44.0	12.0	257°03.2	54.7	Ankaa	353°07.1 -42°09.9
2	272°54.9	255°13.7	39.2	132°33.2	06.1	186°45.9	12.0	272°05.5	54.6	Schedar	349°31.1 56°40.3
3	287°57.4	270°14.0	•• 39.9	147°34.3	•• 05.7	201°47.7	•• 12.0	287°07.8	•• 54.6	Diphda	348°47.2 -17°50.8
4	302°59.8	285°14.3	40.6	162°35.5	05.3	216°49.6	12.0	302°10.1	54.5	Achernar	335°20.4 -57°06.3
5	318°02.3	300°14.6	41.3	177°36.7	04.8	231°51.5	12.1	317°12.4	54.4	Hamal	327°51.2 23°34.8
6	333°04.8	315°14.9	N05°41.9	192°37.9	N17°04.4	246°53.3	N23°12.1	332°14.7	S01°54.4	Polaris	314°21.6 89°22.1
7	348°07.2	330°15.1	42.6	207°39.1	04.0	261°55.2	12.1	347°17.0	54.3	Acamar	315°11.9 -40°12.1
8	3°09.7	345°15.4	43.3	222°40.3	03.5	276°57.1	12.1	2°19.2	54.2	Menkar	314°06.2 4°11.3
9	18°12.2	0°15.7	•• 44.0	237°41.5	•• 03.1	291°58.9	•• 12.1	17°21.5	•• 54.2	Mirfak	308°28.3 49°57.0
10	33°14.6	15°16.0	44.7	252°42.6	02.6	307°00.8	12.2	32°23.8	54.1	Aldebaran	290°39.6 16°33.6
11	48°17.1	30°16.3	45.4	267°43.8	02.2	322°02.7	12.2	47°26.1	54.0	Rigel	281°03.9 -8°10.4
12	63°19.5	45°16.5	N05°46.1	282°45.0	N17°01.8	337°04.5	N23°12.2	62°28.4	S01°54.0	Capella	280°21.9 46°01.4
13	78°22.0	60°16.8	46.8	297°46.2	01.3	352°06.4	12.2	77°30.7	53.9	Bellatrix	278°22.9 6°22.3
14	93°24.5	75°17.1	47.5	312°47.4	00.9	7°08.2	12.2	92°33.0	53.8	Elnath	278°01.9 28°37.7
15	108°26.9	90°17.4	•• 48.2	327°48.6	•• 00.5	22°10.1	•• 12.2	107°35.3	•• 53.8	Alnilam	275°37.8 -1°11.2
16	123°29.4	105°17.7	48.9	342°49.7	17°00.0	37°12.0	12.3	122°37.6	53.7	Betelgeuse	270°52.1 7°24.7
17	138°31.9	120°17.9	49.6	357°50.9	16°59.6	52°13.8	12.3	137°39.8	53.6	Canopus	263°52.8 -52°42.6
18	153°34.3	135°18.2	N05°50.3	12°52.1	N16°59.1	67°15.7	N23°12.3	152°42.1	S01°53.6	Sirius	258°26.3 -16°45.1
19	168°36.8	150°18.5	51.0	27°53.3	58.7	82°17.6	12.3	167°44.4	53.5	Adhara	255°05.9 -29°00.5
20	183°39.3	165°18.7	51.7	42°54.5	58.3	97°19.4	12.3	182°46.7	53.4	Procyon	244°50.8 5°09.6
21	198°41.7	180°19.0	•• 52.4	57°55.7	•• 57.8	112°21.3	•• 12.4	197°49.0	•• 53.4	Pollux	243°17.2 27°58.0
22	213°44.2	195°19.3	53.1	72°56.8	57.4	127°23.2	12.4	212°51.3	53.3	Avior	234°15.0 -59°35.7
23	228°46.7	210°19.6	53.8	87°58.0	56.9	142°25.0	12.4	227°53.6	53.2	Suhail	222°46.3 -43°32.3
Mer.pass. 07:47		ν 0.3' d 0.7' m -4.47		ν 1.2' d -0.4' m 1.22		ν 1.9' d 0.0' m -1.93		ν 2.3' d -0.1' m 1.11			

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	243°49.1	225°19.8	N05°54.5	102°59.2	N16°56.5	157°26.9	N23°12.4	242°55.9	S01°53.2
1	258°51.6	240°20.1	55.2	118°00.4	56.1	172°28.7	12.4	257°58.2	53.1
2	273°54.0	255°20.4	55.9	133°01.6	55.6	187°30.6	12.5	273°00.4	53.1
3	288°56.5	270°20.6	•• 56.6	148°02.7	•• 55.2	202°32.5	•• 12.5	288°02.7	•• 53.0
4	303°59.0	285°20.9	57.3	163°03.9	54.7	217°34.3	12.5	303°05.0	52.9
5	319°01.4	300°21.2	58.0	178°05.1	54.3	232°36.2	12.5	318°07.3	52.9
6	334°03.9	315°21.4	N05°58.7	193°06.3	N16°53.8	247°38.1	N23°12.5	333°09.6	S01°52.8
7	349°06.4	330°21.7	05°59.4	208°07.5	53.4	262°39.9	12.5	348°11.9	52.7
8	4°08.8	345°22.0	06°00.1	223°08.7	53.0	277°41.8	12.6	3°14.2	52.7
9	19°11.3	0°22.2	•• 00.8	238°09.8	•• 52.5	292°43.7	•• 12.6	18°16.5	•• 52.6
10	34°13.8	15°22.5	01.5	253°11.0	52.1	307°45.5	12.6	33°18.8	52.5
11	49°16.2	30°22.7	02.2	268°12.2	51.6	322°47.4	12.6	48°21.1	52.5
12	64°18.7	45°23.0	N06°02.9	283°13.4	N16°51.2	337°49.2	N23°12.6	63°23.4	S01°52.4
13	79°21.1	60°23.3	03.6	298°14.6	50.7	352°51.1	12.7	78°25.6	52.3
14	94°23.6	75°23.5	04.3	313°15.7	50.3	7°53.0	12.7	93°27.9	52.3
15	109°26.1	90°23.8	•• 05.0	328°16.9	•• 49.9	22°54.8	•• 12.7	108°30.2	•• 52.2
16	124°28.5	105°24.0	05.7	343°18.1	49.4	37°56.7	12.7	123°32.5	52.1
17	139°31.0	120°24.3	06.4	358°19.3	49.0	52°58.6	12.7	138°34.8	52.1
18	154°33.5	135°24.5	N06°07.1	13°20.5	N16°48.5	68°00.4	N23°12.7	153°37.1	S01°52.0
19	169°35.9	150°24.8	07.8	28°21.6	48.1	83°02.3	12.8	168°39.4	52.0
20	184°38.4	165°25.0	08.5	43°22.8	47.6	98°04.1	12.8	183°41.7	51.9
21	199°40.9	180°25.3	•• 09.2	58°24.0	•• 47.2	113°06.0	•• 12.8	198°44.0	•• 51.8
22	214°43.3	195°25.6	09.9	73°25.2	46.8	128°07.9	12.8	213°46.3	51.8
23	229°45.8	210°25.8	10.6	88°26.4	46.3	143°09.7	12.8	228°48.6	51.7
Mer.pass. 07:43		ν 0.3' d 0.7' m -4.45		ν 1.2' d -0.4' m 1.23		ν 1.9' d 0.0' m -1.92		ν 2.3' d -0.1' m 1.10	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	244°48.3	225°26.1	N06°11.3	103°27.5	N16°45.9	158°11.6	N23°12.8	243°50.9	S01°51.6
1	259°50.7	240°26.3	12.1	118°28.7	45.4	173°13.4	12.9	258°53.2	51.6
2	274°53.2	255°26.6	12.8	133°29.9	45.0	188°15.3	12.9	273°55.4	51.5
3	289°55.6	270°26.8	•• 13.5	148°31.1	•• 44.5	203°17.2	•• 12.9	288°57.7	•• 51.4
4	304°58.1	285°27.0	14.2	163°32.3	44.1	218°19.0	12.9	304°00.0	51.4
5	320°00.6	300°27.3	14.9	178°33.4	43.6	233°20.9	12.9	319°02.3	51.3
6	335°03.0	315°27.5	N06°15.6	193°34.6	N16°43.2	248°22.8	N23°12.9	334°04.6	S01°51.2
7	350°05.5	330°27.8	16.3	208°35.8	42.7	263°24.6	13.0	349°06.9	51.2
8	5°08.0	345°28.0	17.0	223°37.0	42.3	278°26.5	13.0	4°09.2	51.1
9	20°10.4	0°28.3	•• 17.7	238°38.2	•• 41.9	293°28.3	•• 13.0	19°11.5	•• 51.1
10	35°12.9	15°28.5	18.5	253°39.3	41.4	308°30.2	13.0	34°13.8	51.0
11	50°15.4	30°28.8	19.2	268°40.5	41.0	323°32.1	13.0	49°16.1	50.9
12	65°17.8	45°29.0	N06°19.9	283°41.7	N16°40.5	338°33.9	N23°13.1	64°18.4	S01°50.9
13	80°20.3	60°29.2	20.6	298°42.9	40.1	353°35.8	13.1	79°20.7	50.8
14	95°22.8	75°29.5	21.3	313°44.0	39.6	8°37.6	13.1	94°23.0	50.7
15	110°25.2	90°29.7	•• 22.0	328°45.2	•• 39.2	23°39.5	•• 13.1	109°25.3	•• 50.7
16	125°27.7	105°30.0	22.7	343°46.4	38.7	38°41.4	13.1	124°27.6	50.6
17	140°30.1	120°30.2	23.5	358°47.6	38.3	53°43.2	13.1	139°29.9	50.6
18	155°32.6	135°30.4	N06°24.2	13°48.8	N16°37.8	68°45.1	N23°13.2	154°32.2	S01°50.5
19	170°35.1	150°30.7	24.9	28°49.9	37.4	83°47.0	13.2	169°34.5	50.4
20	185°37.5	165°30.9	25.6	43°51.1	36.9	98°48.8	13.2	184°36.7	50.4
21	200°40.0	180°31.1	•• 26.3	58°52.3	•• 36.5	113°50.7	•• 13.2	199°39.0	•• 50.3
22	215°42.5	195°31.4	27.0	73°53.5	36.0	128°52.5	13.2	214°41.3	50.2
23	230°44.9	210°31.6	27.8	88°54.6	35.6	143°54.4	13.2	229°43.6	50.2
Mer.pass. 07:40		ν 0.2' d 0.7' m -4.44		ν 1.2' d -0.4' m 1.24		ν 1.9' d 0.0' m -1.92		ν 2.3' d -0.1' m 1.10	

May 25 Sun		SHA	Mer.pass
Venus	342°23.2	08:59	
Mars	219°40.8	17:09	
Jupiter	273°52.2	13:32	
Saturn	359°11.0	07:51	

May 26 Mon		SHA	Mer.pass
Venus	341°30.7	08:59	
Mars	219°10.1	17:07	
Jupiter	273°37.8	13:29	
Saturn	359°06.8	07:47	

May 27 Tue		SHA	Mer.pass
Venus	340°37.8	08:58	
Mars	218°39.3	17:05	
Jupiter	273°23.3	13:26	
Saturn	359°02.6	07:43	

Horizontal parallax		
Venus:	0.2	
Mars:	0.1	

h Sun Moon						
Sun	GHA	Dec	GHA	ν	Dec	d HP
0	180°46.5	N20°57.4	212°13.7	7.8'	N15°50.2	15.2' 60.8'
1	195°46.4	57.8	226°40.5	7.7'	16°05.4	15.1' 60.8'
2	210°46.3	58.3	241°07.1	7.5'	16°20.5	15.0' 60.9'
3	225°46.3	· · 58.7	255°33.7	7.4'	16°35.6	14.9' 60.9'
4	240°46.2	59.2	270°00.1	7.3'	16°50.5	14.8' 60.9'
5	255°46.1	20°59.6	284°26.4	7.2'	17°05.3	14.7' 60.9'
6	270°46.1	N21°00.1	298°52.6	7.1'	N17°20.0	14.6' 60.9'
7	285°46.0	00.5	313°18.7	7.0'	17°34.6	14.5' 60.9'
8	300°45.9	01.0	327°44.7	6.9'	17°49.2	14.4' 60.9'
9	315°45.9	· · 01.4	342°10.6	6.8'	18°03.6	14.3' 60.9'
10	330°45.8	01.8	356°36.4	6.7'	18°17.9	14.2' 60.9'
11	345°45.7	02.3	11°02.0	6.5'	18°32.0	14.1' 60.9'
12	0°45.7	N21°02.7	25°27.6	6.4'	N18°46.1	13.9' 61.0'
13	15°45.6	03.2	39°53.0	6.3'	19°00.0	13.8' 61.0'
14	30°45.5	03.6	54°18.3	6.2'	19°13.9	13.7' 61.0'
15	45°45.5	· · 04.0	68°43.5	6.1'	19°27.6	13.6' 61.0'
16	60°45.4	04.5	83°08.6	6.0'	19°41.2	13.5' 61.0'
17	75°45.3	04.9	97°33.6	5.9'	19°54.6	13.3' 61.0'
18	90°45.3	N21°05.4	111°58.5	5.8'	N20°07.9	13.2' 61.0'
19	105°45.2	05.8	126°23.2	5.6'	20°21.1	13.1' 61.0'
20	120°45.1	06.2	140°47.9	5.5'	20°34.2	12.9' 61.0'
21	135°45.1	· · 06.7	155°12.4	5.4'	20°47.1	12.8' 61.0'
22	150°45.0	07.1	169°36.8	5.3'	20°59.9	12.6' 61.0'
23	165°44.9	07.5	184°01.1	5.2'	21°12.5	12.5' 61.0'
SD = 15.8'			d = 0.4'			
SD = 15.8'			d = 0.4'			
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	180°44.9	N21°08.0	198°25.3	5.1'	N21°25.1	12.4' 61.0'
1	195°44.8	08.4	212°49.4	5.0'	21°37.4	12.2' 61.0'
2	210°44.7	08.8	227°13.3	4.9'	21°49.6	12.1' 61.0'
3	225°44.6	· · 09.3	241°37.2	4.7'	22°01.7	11.9' 61.0'
4	240°44.6	09.7	256°00.9	4.6'	22°13.6	11.8' 61.0'
5	255°44.5	10.1	270°24.6	4.5'	22°25.3	11.6' 61.0'
6	270°44.4	N21°10.6	284°48.1	4.4'	N22°36.9	11.4' 61.0'
7	285°44.4	11.0	299°11.5	4.3'	22°48.4	11.3' 61.0'
8	300°44.3	11.4	313°34.8	4.2'	22°59.7	11.1' 61.0'
9	315°44.2	· · 11.8	327°58.1	4.1'	23°10.8	11.0' 61.0'
10	330°44.2	12.3	342°21.2	4.0'	23°21.7	10.8' 61.0'
11	345°44.1	12.7	356°44.2	3.9'	23°32.5	10.6' 61.0'
12	0°44.0	N21°13.1	11°07.1	3.8'	N23°43.2	10.5' 61.0'
13	15°43.9	13.5	25°29.9	3.7'	23°53.6	10.3' 61.0'
14	30°43.9	14.0	39°52.6	3.6'	24°03.9	10.1' 61.0'
15	45°43.8	· · 14.4	54°15.2	3.5'	24°14.0	9.9' 61.0'
16	60°43.7	14.8	68°37.7	3.4'	24°23.9	9.8' 60.9'
17	75°43.7	15.2	83°00.1	3.3'	24°33.7	9.6' 60.9'
18	90°43.6	N21°15.7	97°22.5	3.2'	N24°43.2	9.4' 60.9'
19	105°43.5	16.1	111°44.7	3.1'	24°52.6	9.2' 60.9'
20	120°43.4	16.5	126°06.8	3.1'	25°01.8	9.0' 60.9'
21	135°43.4	· · 16.9	140°28.9	3.0'	25°10.9	8.8' 60.9'
22	150°43.3	17.3	154°50.9	2.9'	25°19.7	8.6' 60.9'
23	165°43.2	17.8	169°12.7	2.8'	25°28.4	8.5' 60.9'
SD = 15.8'			d = 0.4'			
SD = 15.8'			d = 0.4'			
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	180°43.1	N21°18.2	183°34.5	2.7'	N25°36.8	8.3' 60.9'
1	195°43.1	18.6	197°56.3	2.6'	25°45.1	8.1' 60.9'
2	210°43.0	19.0	212°17.9	2.6'	25°53.1	7.9' 60.8'
3	225°42.9	· · 19.4	226°39.5	2.5'	26°01.0	7.7' 60.8'
4	240°42.8	19.8	241°01.0	2.4'	26°08.7	7.5' 60.8'
5	255°42.8	20.2	255°22.4	2.4'	26°16.2	7.3' 60.8'
6	270°42.7	N21°20.7	269°43.8	2.3'	N26°23.5	7.1' 60.8'
7	285°42.6	21.1	284°05.1	2.2'	26°30.6	6.9' 60.8'
8	300°42.5	21.5	298°26.3	2.2'	26°37.5	6.7' 60.8'
9	315°42.5	· · 21.9	312°47.5	2.1'	26°44.1	6.5' 60.7'
10	330°42.4	22.3	327°08.6	2.1'	26°50.6	6.3' 60.7'
11	345°42.3	22.7	341°29.7	2.0'	26°56.9	6.1' 60.7'
12	0°42.2	N21°23.1	355°50.7	2.0'	N27°03.0	5.9' 60.7'
13	15°42.2	23.5	10°11.7	1.9'	27°08.8	5.7' 60.7'
14	30°42.1	24.0	24°32.6	1.9'	27°14.5	5.4' 60.6'
15	45°42.0	· · 24.4	38°53.5	1.8'	27°19.9	5.2' 60.6'
16	60°41.9	24.8	53°14.3	1.8'	27°25.2	5.0' 60.6'
17	75°41.8	25.2	67°35.1	1.8'	27°30.2	4.8' 60.6'
18	90°41.8	N21°25.6	81°55.9	1.7'	N27°35.0	4.6' 60.6'
19	105°41.7	26.0	96°16.6	1.7'	27°39.6	4.4' 60.5'
20	120°41.6	26.4	110°37.3	1.7'	27°44.0	4.2' 60.5'
21	135°41.5	· · 26.8	124°58.0	1.7'	27°48.2	4.0' 60.5'
22	150°41.5	27.2	139°18.7	1.6'	27°52.1	3.8' 60.5'
23	165°41.4	27.6	153°39.4	1.6'	27°55.9	3.5' 60.5'
SD = 15.8'			d = 0.4'			
SD = 15.8'			d = 0.4'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	—:—	—:—	00:11	—:—	—:—	—:—
66°	—:—	—:—	01:37	22:21	—:—	—:—
64°	—:—	—:—	02:13	21:43	—:—	—:—
62°	—:—	00:57	02:39	21:17	23:03	—:—
60°	—:—	01:44	02:59	20:57	22:13	—:—
N 58°	—:—	02:12	03:15	20:40	21:44	—:—
56°	00:52	02:34	03:29	20:26	21:21	23:08
54°	01:34	02:52	03:41	20:14	21:04	22:22
52°	02:01	03:06	03:52	20:03	20:49	21:55
50°	02:21	03:19	04:01	19:54	20:36	21:34
45°	02:59	03:45	04:20	19:34	20:10	20:56
N 40°	03:25	04:05	04:36	19:18	19:50	20:30
35°	03:45	04:21	04:50	19:05	19:34	20:09
30°	04:02	04:34	05:01	18:53	19:20	19:52
20°	04:28	04:57	05:21	18:34	18:58	19:26
N 10°	04:48	05:15	05:38	18:16	18:39	19:06
0°	05:06	05:31	05:53	18:01	18:23	18:49
S 10°	05:21	05:47	06:09	17:45	18:07	18:33
20°	05:35	06:02	06:26	17:28	17:52	18:19
30°	05:49	06:19	06:45	17:09	17:35	18:05
35°	05:57	06:28	06:56	16:58	17:26	17:57
40°	06:05	06:38	07:08	16:46	17:16	17:49
45°	06:14	06:50	07:23	16:31	17:04	17:40
S 50°	06:23	07:04	07:41	16:13	16:50	17:30
52°	06:28	07:10	07:49	16:04	16:43	17:26
54°	06:32	07:17	07:59	15:55	16:36	17:21
56°	06:38	07:25	08:10	15:44	16:29	17:16
58°	06:43	07:34	08:22	15:32	16:20	17:11
S 60°	06:49	07:43	08:36	15:17	16:10	17:04

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	□	□	□	□	□	□
N 70°	00:28	□	□	□	□	□
68°	00:54 23:50	□	□	21:16	□	□
66°	01:14	00:42	□	20:26	□	□
64°	01:30	01:14	00:34	19:55	22:44	□
62°	01:44	01:38	01:31	19:32	21:48	· ·
60°	01:55	01:57	02:05	19:13	21:15	23:09
N 58°	02:05	02:14	02:30	18:58	20:50	22:33
56°	02:14	02:27	02:50	18:45	20:31	22:08
54°	02:22	02:39	03:06	18:34	20:15	21:47
52°	02:29	02:50	03:21	18:24	20:01	21:30
50°	02:35	03:00	03:33	18:15	19:49	21:15
45°	02:49	03:20	04:00	17:57	19:24	20:45
N 40°	03:01	03:36	04:20	17:42	19:04	20:22
35°	03:11	03:50	04:38	17:29	18:47	20:03
30°	03:19	04:02	04:53	17:18	18:33	19:47
20°	03:35	04:23	05:19	16:59	18:09	19:19
N 10°	03:48	04:42	05:41	16:43	17:48	18:55
0°	04:01	04:59	06:02	16:28	17:28	18:33
S 10°	04:14	05:17	06:23	16:13	17:09	18:11
20°	04:27	05:35	06:46	15:57	16:48	17:47
30°	04:43	05:57	07:12	15:38	16:25	17:20
35°	04:52	06:10	07:28	15:28	16:11	17:03
40°	05:03	06:25	07:46	15:15	15:55	16:44
45°	05:16	06:43	08:08	15:01	15:36	16:22
S 50°	05:31	07:05	08:37	14:44	15:12	15:53
52°	05:39	07:16	08:50	14:36	15:01	15:38
54°	05:47	07:28	09:07	14:27	14:48	15:22
56°	05:56	07:42	09:26	14:17	14:33	15:02
58°	06:06	07:59	09:50	14:05	14:16	14:38
S 60°	06:18	08:19	10:21	13:52	13:55	14:06

Day	Sun		Mer. Pass hh:mm	Moon		Age 28-0 7-0%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
25	03:06	03:03	11:57	10:14	22:43	
26	02:59	02:56	11:57	11:14	23:45	
27	02:53	02:49	11:57	12:17	—:—	

May 28, 29, 30 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	245°47.4	225°31.8	N06°28.5	103°55.8	N16°35.1	158°56.3	N23°13.3	244°45.9	S01°50.1	Alpheratz	357°34.6	29°13.7
1	260°49.9	240°32.1	29.2	118°57.0	34.7	173°58.1	13.3	259°48.2	50.1	Ankaa	353°07.0	-42°09.9
2	275°52.3	255°32.3	29.9	133°58.2	34.2	189°00.0	13.3	274°50.5	50.0	Schedar	349°31.0	56°40.3
3	290°54.8	270°32.5	•• 30.6	148°59.3	•• 33.8	204°01.8	•• 13.3	289°52.8	•• 49.9	Diphda	348°47.2	-17°50.8
4	305°57.3	285°32.8	31.3	164°00.5	33.3	219°03.7	13.3	304°55.1	49.9	Achernar	335°20.4	-57°06.3
5	320°59.7	300°33.0	32.1	179°01.7	32.9	234°05.6	13.3	319°57.4	49.8	Hamal	327°51.1	23°34.8
6	336°02.2	315°33.2	N06°32.8	194°02.9	N16°32.4	249°07.4	N23°13.4	334°59.7	S01°49.7	Polaris	314°20.5	89°22.1
7	351°04.6	330°33.4	33.5	209°04.0	32.0	264°09.3	13.4	350°02.0	49.7	Acamar	315°11.9	-40°12.1
8	6°07.1	345°33.7	34.2	224°05.2	31.5	279°11.1	13.4	5°04.3	49.6	Menkar	314°06.1	4°11.3
9	21°09.6	0°33.9	•• 35.0	239°06.4	•• 31.1	294°13.0	•• 13.4	20°06.6	•• 49.6	Mirfak	308°28.3	49°57.0
10	36°12.0	15°34.1	35.7	254°07.6	30.6	309°14.9	13.4	35°08.9	49.5	Aldebaran	290°39.6	16°33.6
11	51°14.5	30°34.3	36.4	269°08.7	30.2	324°16.7	13.4	50°11.2	49.4	Rigel	281°03.9	-8°10.4
12	66°17.0	45°34.6	N06°37.1	284°09.9	N16°29.7	339°18.6	N23°13.4	65°13.5	S01°49.4	Capella	280°21.9	46°01.4
13	81°19.4	60°34.8	37.8	299°11.1	29.3	354°20.4	13.5	80°15.8	49.3	Bellatrix	278°22.9	6°22.3
14	96°21.9	75°35.0	38.6	314°12.3	28.8	9°22.3	13.5	95°18.1	49.2	Elnath	278°01.9	28°37.7
15	111°24.4	90°35.2	•• 39.3	329°13.4	•• 28.4	24°24.2	•• 13.5	110°20.4	•• 49.2	Alnilam	275°37.7	-1°11.2
16	126°26.8	105°35.4	40.0	344°14.6	27.9	39°26.0	13.5	125°22.7	49.1	Betelgeuse	270°52.1	7°24.7
17	141°29.3	120°35.7	40.7	359°15.8	27.5	54°27.9	13.5	140°25.0	49.1	Canopus	263°52.8	-52°42.6
18	156°31.8	135°35.9	N06°41.5	14°17.0	N16°27.0	69°29.7	N23°13.5	155°27.3	S01°49.0	Sirius	258°26.3	-16°45.1
19	171°34.2	150°36.1	42.2	29°18.1	26.6	84°31.6	13.6	170°29.6	48.9	Adhara	255°05.9	-29°00.5
20	186°36.7	165°36.3	42.9	44°19.3	26.1	99°33.5	13.6	185°31.9	48.9	Procyon	244°50.8	5°09.6
21	201°39.1	180°36.5	•• 43.6	59°20.5	•• 25.7	114°35.3	•• 13.6	200°34.2	•• 48.8	Pollux	243°17.2	27°58.0
22	216°41.6	195°36.7	44.4	74°21.7	25.2	129°37.2	13.6	215°36.5	48.8	Avior	234°15.0	-59°35.7
23	231°44.1	210°37.0	45.1	89°22.8	24.8	144°39.0	13.6	230°38.8	48.7	Suhail	222°46.3	-43°32.3
Mer.pass. 07:36		ν 0.2' d 0.7' m -4.43		ν 1.2' d -0.4' m 1.25		ν 1.9' d 0.0' m -1.92		ν 2.3' d -0.1' m 1.10		Miaplacidus	221°38.6	-69°49.5
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	Alphard	217°47.6	-8°46.2
0	246°46.5	225°37.2	N06°45.8	104°24.0	N16°24.3	159°40.9	N23°13.6	245°41.1	S01°48.6	Regulus	207°34.2	11°50.6
1	261°49.0	240°37.4	46.5	119°25.2	23.8	174°42.7	13.7	260°43.4	48.6	Dubhe	193°40.5	61°37.1
2	276°51.5	255°37.6	47.3	134°26.4	23.4	189°44.6	13.7	275°45.7	48.5	Denebola	182°24.5	14°25.8
3	291°53.9	270°37.8	•• 48.0	149°27.5	•• 22.9	204°46.5	•• 13.7	290°48.0	•• 48.4	Gienah	175°43.1	-17°41.1
4	306°56.4	285°38.0	48.7	164°28.7	22.5	219°48.3	13.7	305°50.3	48.4	Acrux	172°59.4	-63°14.7
5	321°58.9	300°38.2	49.5	179°29.9	22.0	234°50.2	13.7	320°52.6	48.3	Gacrux	171°51.0	-57°15.6
6	337°01.3	315°38.4	N06°50.2	194°31.1	N16°21.6	249°52.0	N23°13.7	335°54.9	S01°48.3	Alioth	166°12.3	55°49.5
7	352°03.8	330°38.7	50.9	209°32.2	21.1	264°53.9	13.7	350°57.2	48.2	Spica	158°21.8	-11°17.7
8	7°06.3	345°38.9	51.6	224°33.4	20.7	279°55.8	13.8	5°59.5	48.1	Alkaid	152°51.4	49°11.3
9	22°08.7	0°39.1	•• 52.4	239°34.6	•• 20.2	294°57.6	•• 13.8	21°01.8	•• 48.1	Hadar	148°35.1	-60°29.9
10	37°11.2	15°39.3	53.1	254°35.8	19.8	309°59.5	13.8	36°04.1	48.0	Menkent	147°56.9	-36°29.9
11	52°13.6	30°39.5	53.8	269°36.9	19.3	325°01.3	13.8	51°06.4	48.0	Arcturus	145°47.3	19°03.0
12	67°16.1	45°39.7	N06°54.6	284°38.1	N16°18.8	340°03.2	N23°13.8	66°08.7	S01°47.9	Rigil Kent.	139°39.3	-60°56.6
13	82°18.6	60°39.9	55.3	299°39.3	18.4	355°05.1	13.8	81°11.0	47.8	Kochab	137°18.7	74°03.1
14	97°21.0	75°40.1	56.0	314°40.4	17.9	10°06.9	13.9	96°13.3	47.8	Zuben'ubi	136°55.4	-16°08.9
15	112°23.5	90°40.3	•• 56.8	329°41.6	•• 17.5	25°08.8	•• 13.9	111°15.6	•• 47.7	Alphecca	126°03.1	26°37.7
16	127°26.0	105°40.5	57.5	344°42.8	17.0	40°10.6	13.9	126°17.9	47.7	Antares	112°15.1	-26°29.4
17	142°28.4	120°40.7	58.2	359°44.0	16.6	55°12.5	13.9	141°20.2	47.6	Atria	107°08.5	-69°04.4
18	157°30.9	135°40.9	N06°59.0	14°45.1	N16°16.1	70°14.3	N23°13.9	156°22.5	S01°47.5	Sabik	102°02.1	-15°45.4
19	172°33.4	150°41.1	06°59.7	29°46.3	15.7	85°16.2	13.9	171°24.8	47.5	Shaula	96°09.5	-37°07.3
20	187°35.8	165°41.3	07°00.4	44°47.5	15.2	100°18.1	13.9	186°27.1	47.4	Rasalhague	95°57.9	12°32.4
21	202°38.3	180°41.5	•• 01.2	59°48.6	•• 14.7	115°19.9	•• 14.0	201°29.4	•• 47.4	Eltanin	90°41.5	51°28.9
22	217°40.8	195°41.7	01.9	74°49.8	14.3	130°21.8	14.0	216°31.7	47.3	Kaus Aust.	83°31.7	-34°22.3
23	232°43.2	210°41.9	02.6	89°51.0	13.8	145°23.6	14.0	231°34.0	47.2	Vega	80°32.6	38°48.2
Mer.pass. 07:32		ν 0.2' d 0.7' m -4.42		ν 1.2' d -0.5' m 1.26		ν 1.9' d 0.0' m -1.92		ν 2.3' d -0.1' m 1.09		Nunki	75°47.1	-26°15.9
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	Altair	61°59.4	8°56.0
0	247°45.7	225°42.1	N07°03.4	104°52.2	N16°13.4	160°25.5	N23°14.0	246°36.3	S01°47.2	Peacock	53°04.9	-56°39.0
1	262°48.1	240°42.3	04.1	119°53.3	12.9	175°27.4	14.0	261°38.6	47.1	Deneb	49°25.4	45°22.0
2	277°50.6	255°42.5	04.8	134°54.5	12.5	190°29.2	14.0	276°40.9	47.1	Enif	33°38.4	9°59.4
3	292°53.1	270°42.7	•• 05.6	149°55.7	•• 12.0	205°31.1	•• 14.0	291°43.2	•• 47.0	Al Na'ir	27°32.5	-46°50.1
4	307°55.5	285°42.9	06.3	164°56.8	11.5	220°32.9	14.1	306°45.5	46.9	Fomalhaut	15°14.2	-29°29.1
5	322°58.0	300°43.1	07.0	179°58.0	11.1	235°34.8	14.1	321°47.8	46.9	Scheat	13°45.0	28°13.0
6	338°00.5	315°43.3	N07°07.8	194°59.2	N16°10.6	250°36.6	N23°14.1	336°50.1	S01°46.8	Markab	13°29.6	15°20.4
7	353°02.9	330°43.5	08.5	210°00.4	10.2	265°38.5	14.1	351°52.4	46.8	May 28 Wed SHA Mer.pass		
8	8°05.4	345°43.6	09.3	225°01.5	09.7	280°40.4	14.1	6°54.7	46.7	Venus	339°44.4	08:58
9	23°07.9	0°43.8	•• 10.0	240°02.7	•• 09.2	295°42.2	•• 14.1	21°57.0	•• 46.6	Mars	218°08.4	17:03
10	38°10.3	15°44.0	10.7	255°03.9	08.8	310°44.1	14.1	36°59.3	46.6	Jupiter	273°08.9	13:23
11	53°12.8	30°44.2	11.5	270°05.0	08.3	325°45.9	14.2	52°01.6	46.5	Saturn	358°58.5	07:40
12	68°15.3	45°44.4	N07°12.2	285°06.2	N16°07.9	340°47.8	N23°14.2	67°03.9	S01°46.5	May 29 Thu SHA Mer.pass		
13	83°17.7	60°44.6	12.9	300°07.4	07.4	355°49.6	14.2	82°06.2	46.4	Venus	338°50.6	08:57
14	98°20.2	75°44.8	13.7	315°08.5	06.9	10°51.5	14.2	97°08.5	46.3	Mars	217°37.5	17:01
15	113°22.6	90°45.0	•• 14.4	330°09.7	•• 06.5	25°53.4	•• 14.2	112°10.8	•• 46.3	Jupiter	272°54.4	13:20
16	128°25.1	105°45.2	15.2	345°10.9	06.0	40°55.2	14.2	127°13.1	46.2	Saturn	358°54.5	07:36
17	143°27.6	120°45.3	15.9	0°12.1	05.6	55°57.1	14.2	142°15.4	46.2	May 30 Fri SHA Mer.pass		
18	158°30.0	135°45.5	N07°16.6	15°13.2	N16°05.1	70°58.9	N23°14.3	157°17.7	S01°46.1	Venus	337°56.4	08:57
19	173°32.5	150°45.7	17.4	30°14.4</								

h	Sun			Moon			
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	180° 41.3	N21° 28.0	168° 00.0	1.6'	N27° 59.4	3.3'	60.4'
1	195° 41.2	28.4	182° 20.6	1.6'	28° 02.8	3.1'	60.4'
2	210° 41.1	28.8	196° 41.2	1.6'	28° 05.9	2.9'	60.4'
3	225° 41.1	· · 29.2	211° 01.8	1.6'	28° 08.8	2.7'	60.4'
4	240° 41.0	29.6	225° 22.5	1.6'	28° 11.5	2.5'	60.3'
5	255° 40.9	30.0	239° 43.1	1.6'	28° 13.9	2.3'	60.3'
6	270° 40.8	N21° 30.4	254° 03.7	1.6'	N28° 16.2	2.0'	60.3'
7	285° 40.7	30.8	268° 24.3	1.6'	28° 18.2	1.8'	60.3'
8	300° 40.7	31.2	282° 45.0	1.7'	28° 20.0	1.6'	60.2'
9	315° 40.6	· · 31.6	297° 05.6	1.7'	28° 21.6	1.4'	60.2'
10	330° 40.5	32.0	311° 26.3	1.7'	28° 23.0	1.2'	60.2'
11	345° 40.4	32.4	325° 47.0	1.7'	28° 24.2	1.0'	60.2'
12	0° 40.3	N21° 32.8	340° 07.7	1.8'	N28° 25.2	0.8'	60.1'
13	15° 40.3	33.2	354° 28.5	1.8'	28° 25.9	0.5'	60.1'
14	30° 40.2	33.6	8° 49.3	1.8'	28° 26.5	0.3'	60.1'
15	45° 40.1	· · 34.0	23° 10.1	1.9'	28° 26.8	0.1'	60.0'
16	60° 40.0	34.4	37° 31.0	1.9'	28° 26.9	-0.1'	60.0'
17	75° 39.9	34.7	51° 51.9	2.0'	28° 26.8	-0.3'	60.0'
18	90° 39.9	N21° 35.1	66° 12.9	2.0'	N28° 26.5	-0.5'	59.9'
19	105° 39.8	35.5	80° 33.9	2.1'	28° 26.0	-0.7'	59.9'
20	120° 39.7	35.9	94° 54.9	2.1'	28° 25.3	-0.9'	59.9'
21	135° 39.6	· · 36.3	109° 16.0	2.2'	28° 24.4	-1.1'	59.9'
22	150° 39.5	36.7	123° 37.2	2.2'	28° 23.2	-1.3'	59.8'
23	165° 39.4	37.1	137° 58.5	2.3'	28° 21.9	-1.5'	59.8'
SD = 15.8'			SD = 16.5'				

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	180° 39.4	N21° 37.5	152° 19.8	2.4'	N28° 20.3	-1.8'	59.8'
1	195° 39.3	37.9	166° 41.1	2.4'	28° 18.6	-2.0'	59.7'
2	210° 39.2	38.2	181° 02.6	2.5'	28° 16.6	-2.2'	59.7'
3	225° 39.1	· · 38.6	195° 24.1	2.6'	28° 14.5	-2.4'	59.7'
4	240° 39.0	39.0	209° 45.7	2.7'	28° 12.1	-2.6'	59.6'
5	255° 38.9	39.4	224° 07.3	2.8'	28° 09.6	-2.8'	59.6'
6	270° 38.8	N21° 39.8	238° 29.1	2.8'	N28° 06.8	-3.0'	59.6'
7	285° 38.8	40.2	252° 50.9	2.9'	28° 03.8	-3.1'	59.5'
8	300° 38.7	40.5	267° 12.9	3.0'	28° 00.7	-3.3'	59.5'
9	315° 38.6	· · 40.9	281° 34.9	3.1'	27° 57.4	-3.5'	59.5'
10	330° 38.5	41.3	295° 57.0	3.2'	27° 53.8	-3.7'	59.4'
11	345° 38.4	41.7	310° 19.2	3.3'	27° 50.1	-3.9'	59.4'
12	0° 38.3	N21° 42.1	324° 41.5	3.4'	N27° 46.2	-4.1'	59.3'
13	15° 38.3	42.4	339° 03.9	3.5'	27° 42.1	-4.3'	59.3'
14	30° 38.2	42.8	353° 26.4	3.6'	27° 37.8	-4.5'	59.3'
15	45° 38.1	· · 43.2	7° 49.0	3.7'	27° 33.3	-4.7'	59.2'
16	60° 38.0	43.6	22° 11.7	3.8'	27° 28.6	-4.8'	59.2'
17	75° 37.9	43.9	36° 34.5	3.9'	27° 23.8	-5.0'	59.2'
18	90° 37.8	N21° 44.3	50° 57.5	4.0'	N27° 18.8	-5.2'	59.1'
19	105° 37.7	44.7	65° 20.5	4.2'	27° 13.6	-5.4'	59.1'
20	120° 37.6	45.1	79° 43.7	4.3'	27° 08.2	-5.6'	59.1'
21	135° 37.6	· · 45.4	94° 06.9	4.4'	27° 02.6	-5.7'	59.0'
22	150° 37.5	45.8	108° 30.3	4.5'	26° 56.9	-5.9'	59.0'
23	165° 37.4	46.2	122° 53.8	4.6'	26° 51.0	-6.1'	58.9'
SD = 15.8'			SD = 16.3'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	180° 37.3	N21° 46.6	137° 17.5	4.8'	N26° 45.0	-6.2'	58.9'
1	195° 37.2	46.9	151° 41.2	4.9'	26° 38.7	-6.4'	58.9'
2	210° 37.1	47.3	166° 05.1	5.0'	26° 32.3	-6.6'	58.8'
3	225° 37.0	· · 47.7	180° 29.1	5.1'	26° 25.8	-6.7'	58.8'
4	240° 36.9	48.0	194° 53.2	5.3'	26° 19.0	-6.9'	58.8'
5	255° 36.9	48.4	209° 17.5	5.4'	26° 12.1	-7.0'	58.7'
6	270° 36.8	N21° 48.8	223° 41.9	5.5'	N26° 05.1	-7.2'	58.7'
7	285° 36.7	49.1	238° 06.4	5.6'	25° 57.9	-7.4'	58.6'
8	300° 36.6	49.5	252° 31.0	5.8'	25° 50.5	-7.5'	58.6'
9	315° 36.5	· · 49.9	266° 55.8	5.9'	25° 43.0	-7.7'	58.6'
10	330° 36.4	50.2	281° 20.7	6.0'	25° 35.4	-7.8'	58.5'
11	345° 36.3	50.6	295° 45.8	6.2'	25° 27.6	-8.0'	58.5'
12	0° 36.2	N21° 51.0	310° 10.9	6.3'	N25° 19.6	-8.1'	58.5'
13	15° 36.1	51.3	324° 36.2	6.4'	25° 11.5	-8.2'	58.4'
14	30° 36.0	51.7	339° 01.7	6.6'	25° 03.3	-8.4'	58.4'
15	45° 36.0	· · 52.0	353° 27.3	6.7'	24° 54.9	-8.5'	58.3'
16	60° 35.9	52.4	7° 53.0	6.9'	24° 46.3	-8.7'	58.3'
17	75° 35.8	52.8	22° 18.8	7.0'	24° 37.7	-8.8'	58.3'
18	90° 35.7	N21° 53.1	36° 44.8	7.1'	N24° 28.9	-8.9'	58.2'
19	105° 35.6	53.5	51° 10.9	7.3'	24° 20.0	-9.1'	58.2'
20	120° 35.5	53.8	65° 37.2	7.4'	24° 10.9	-9.2'	58.1'
21	135° 35.4	· · 54.2	80° 03.6	7.5'	24° 01.7	-9.3'	58.1'
22	150° 35.3	54.5	94° 30.2	7.7'	23° 52.4	-9.4'	58.1'
23	165° 35.2	54.9	108° 56.8	7.8'	23° 43.0	-9.6'	58.0'
SD = 15.8'			SD = 16.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	—:—	—:—	01:24	22:35	—:—	—:—
64°	—:—	—:—	02:05	21:52	—:—	—:—
62°	—:—	00:37	02:33	21:24	23:26	—:—
60°	—:—	01:34	02:54	21:03	22:23	—:—
N 58°	—:—	02:06	03:11	20:45	21:51	—:—
56°	00:33	02:29	03:25	20:31	21:27	23:29
54°	01:26	02:47	03:38	20:18	21:09	22:31
52°	01:55	03:03	03:48	20:07	20:53	22:01
50°	02:17	03:16	03:58	19:57	20:40	21:39
45°	02:55	03:42	04:18	19:37	20:13	21:00
N 40°	03:23	04:03	04:35	19:21	19:53	20:33
35°	03:44	04:19	04:48	19:07	19:36	20:12
30°	04:01	04:33	05:00	18:55	19:22	19:55
20°	04:27	04:56	05:20	18:35	18:59	19:28
N 10°	04:48	05:15	05:38	18:17	18:40	19:07
0°	05:06	05:32	05:54	18:01	18:23	18:49
S 10°	05:21	05:47	06:10	17:45	18:08	18:34
20°	05:36	06:03	06:27	17:28	17:52	18:19
30°	05:51	06:20	06:46	17:08	17:34	18:04
35°	05:59	06:30	06:57	16:57	17:25	17:56
40°	06:07	06:40	07:10	16:44	17:14	17:48
45°	06:16	06:53	07:26	16:29	17:02	17:39
S 50°	06:26	07:07	07:44	16:10	16:48	17:28
52°	06:31	07:14	07:53	16:02	16:41	17:24
54°	06:36	07:21	08:03	15:52	16:34	17:19
56°	06:41	07:29	08:14	15:40	16:26	17:13
58°	06:47	07:38	08:27	15:28	16:17	17:08
S 60°	06:53	07:48	08:42	15:13	16:07	17:01

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	☐	☐	☐	☐	☐	☐
64°	☐	☐	☐	☐	☐	☐
62°	01:13	☐	04:33	00:23	☐	01:32
60°	02:27	03:30	05:12	· ·	00:24	00:52
N 58°	03:03	04:07	05:40	23:46	· ·	00:24
56°	03:29	04:34	06:01	23:19	· ·	00:03
54°	03:50	04:55	06:18	22:58	23:45	· ·
52°	04:07	05:13	06:33	22:41	23:29	· ·
50°	04:22	05:28	06:46	22:25	23:16	23:50
45°	04:52	05:58	07:13	21:55	22:49	23:28
N 40°	05:16	06:22	07:34	21:31	22:27	23:10
35°	05:35	06:41	07:51	21:11	22:09	22:55
30°	05:52	06:58	08:06	20:55	21:53	22:42
20°	06:20	07:26	08:31	20:26	21:27	22:19
N 10°	06:45	07:50	08:53	20:02	21:04	21:59
0°	07:07	08:12	09:13	19:39	20:42	21:41
S 10°	07:30	08:35	09:34	19:16	20:21	21:22
20°	07:55	08:59	09:55	18:51	19:57	21:02
30°	08:24	09:27	10:20	18:22	19:30	20:38
35°	08:41	09:44	10:35	18:05	19:14	20:25
40°	09:00	10:03	10:52	17:45	18:55	20:09
45°	09:25	10:26	11:12	17:21	18:32	19:49
S 50°	09:56	10:56	11:37	16:50	18:03	19:25
52°	10:11	11:11	11:49	16:34	17:49	19:13
54°	10:30	11:28	12:03	16:16	17:32	19:00
56°	10:52	11:48	12:19	15:54	17:12	18:44
58°	11:20	12:14	12:38	15:25	16:46	18:26
S 60°	12:02	12:49	13:02	14:43	16:12	18:03

Day	Sun		Mer. Pass hh:mm	Moon		Age 1-3 1-12%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
28	02:45	02:41	11:57	13:23	00:50	
29	02:37	02:33	11:57	14:27	01:56	
30	02:29	02:25	11:58	15:27	02:58	

May 31, 01, 02 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	248°44.8	225°46.6	N07°21.1	105°20.2	N16°02.3	161°10.1	N23°14.3	247°31.6	S01°45.8	Alpheratz	357°34.6 29°13.7	
1	263°47.3	240°46.8	21.8	120°21.4	01.9	176°11.9	14.4	262°33.9	45.7	Ankaa	353°07.0 -42°09.9	
2	278°49.8	255°47.0	22.6	135°22.6	01.4	191°13.8	14.4	277°36.2	45.6	Schedar	349°31.0 56°40.3	
3	293°52.2	270°47.1	· · 23.3	150°23.7	· · 01.0	206°15.6	· · 14.4	292°38.5	· · 45.6	Diphda	348°47.1 -17°50.8	
4	308°54.7	285°47.3	24.1	165°24.9	00.5	221°17.5	14.4	307°40.8	45.5	Achernar	335°20.3 -57°06.3	
5	323°57.1	300°47.5	24.8	180°26.1	16°00.0	236°19.3	14.4	322°43.1	45.5	Hamal	327°51.1 23°34.8	
6	338°59.6	315°47.7	N07°25.6	195°27.3	N15°59.6	251°21.2	N23°14.4	337°45.4	S01°45.4	Polaris	314°19.7 89°22.1	
7	354°02.1	330°47.8	26.3	210°28.4	59.1	266°23.1	14.4	352°47.7	45.4	Acamar	315°11.9 -40°12.1	
8	9°04.5	345°48.0	27.0	225°29.6	58.6	281°24.9	14.5	7°50.0	45.3	Menkar	314°06.1 4°11.3	
9	24°07.0	0°48.2	· · 27.8	240°30.8	· · 58.2	296°26.8	· · 14.5	22°52.3	· · 45.2	Mirfak	308°28.3 49°57.0	
10	39°09.5	15°48.4	28.5	255°31.9	57.7	311°28.6	14.5	37°54.6	45.2	Aldebaran	290°39.6 16°33.6	
11	54°11.9	30°48.5	29.3	270°33.1	57.3	326°30.5	14.5	52°56.9	45.1	Rigel	281°03.9 -8°10.4	
12	69°14.4	45°48.7	N07°30.0	285°34.3	N15°56.8	341°32.3	N23°14.5	67°59.2	S01°45.1	Capella	280°21.9 46°01.4	
13	84°16.9	60°48.9	30.8	300°35.4	56.3	356°34.2	14.5	83°01.5	45.0	Bellatrix	278°22.9 6°22.3	
14	99°19.3	75°49.0	31.5	315°36.6	55.9	11°36.0	14.5	98°03.8	45.0	Elnath	278°01.8 28°37.7	
15	114°21.8	90°49.2	· · 32.3	330°37.8	· · 55.4	26°37.9	· · 14.5	113°06.2	· · 44.9	Alnilam	275°37.7 -1°11.2	
16	129°24.2	105°49.4	33.0	345°38.9	54.9	41°39.8	14.6	128°08.5	44.8	Betelgeuse	270°52.1 7°24.7	
17	144°26.7	120°49.6	33.8	0°40.1	54.5	56°41.6	14.6	143°10.8	44.8	Canopus	263°52.8 -52°42.6	
18	159°29.2	135°49.7	N07°34.5	15°41.3	N15°54.0	71°43.5	N23°14.6	158°13.1	S01°44.7	Sirius	258°26.3 -16°45.1	
19	174°31.6	150°49.9	35.3	30°42.4	53.5	86°45.3	14.6	173°15.4	44.7	Adhara	255°05.9 -29°00.5	
20	189°34.1	165°50.1	36.0	45°43.6	53.1	101°47.2	14.6	188°17.7	44.6	Procyon	244°50.8 5°09.6	
21	204°36.6	180°50.2	· · 36.8	60°44.8	· · 52.6	116°49.0	· · 14.6	203°20.0	· · 44.6	Pollux	243°17.2 27°58.0	
22	219°39.0	195°50.4	37.5	75°45.9	52.1	131°50.9	14.6	218°22.3	44.5	Avior	234°15.0 -59°35.7	
23	234°41.5	210°50.5	38.2	90°47.1	51.7	146°52.7	14.7	233°24.6	44.4	Suhail	222°46.3 -43°32.3	
Mer.pass. 07:24		ν 0.2' d 0.7' m -4.40		ν 1.2' d -0.5' m 1.28		ν 1.9' d 0.0' m -1.92		ν 2.3' d -0.1' m 1.09		Miaplacidus	221°38.6 -69°49.5	
											Alphard	217°47.6 -8°46.2
											Regulus	207°34.2 11°50.6
											Dubhe	193°40.5 61°37.1
											Denebola	182°24.5 14°25.8
											Gienah	175°43.1 -17°41.1
											Acrux	172°59.5 -63°14.7
											Gacrux	171°51.0 -57°15.6
											Alioth	166°12.4 55°49.5
											Spica	158°21.8 -11°17.7
											Alkaid	152°51.4 49°11.3
											Hadar	148°35.1 -60°29.9
											Menkent	147°56.9 -36°29.9
											Arcturus	145°47.3 19°03.0
											Rigel Kent.	139°39.3 -60°56.6
											Kochab	137°18.7 74°03.1
											Zuben'ubi	136°55.4 -16°08.9
											Alphecca	126°03.1 26°37.7
											Antares	112°15.1 -26°29.4
											Atria	107°08.4 -69°04.4
											Sabik	102°02.1 -15°45.4
											Shaula	96°09.5 -37°07.3
											Rasalhague	95°57.9 12°32.4
											Eltanin	90°41.5 51°29.0
											Kaus Aust.	83°31.7 -34°22.3
											Vega	80°32.6 38°48.3
											Nunki	75°47.0 -26°15.9
											Altair	61°59.4 8°56.0
											Peacock	53°04.9 -56°39.0
											Deneb	49°25.3 45°22.0
											Enif	33°38.4 9°59.4
											Al Na'ir	27°32.4 -46°50.1
											Fomalhaut	15°14.2 -29°29.1
											Scheat	13°44.9 28°13.0
											Markab	13°29.6 15°20.4
											May 31 Sat	SHA Mer.pass
											Venus	337°01.8 08:57
											Mars	216°35.4 16:57
											Jupiter	272°25.2 13:14
											Saturn	358°46.7 07:29
											Jun 01 Sun	SHA Mer.pass
											Venus	336°06.7 08:57
											Mars	216°04.3 16:55
											Jupiter	272°10.6 13:11
											Saturn	358°43.0 07:25
											Jun 02 Mon	SHA Mer.pass
											Venus	335°11.3 08:56
											Mars	215°33.1 16:54
											Jupiter	271°56.0 13:08
											Saturn	358°39.3 07:21
											Horizontal parallax	
											Venus:	0.2
											Mars:	0.1
Mer.pass. 07:16		ν 0.1' d 0.8' m -4.38		ν 1.2' d -0.5' m 1.29		ν 1.9' d 0.0' m -1.91		ν 2.3' d -0.1' m 1.08				

Sun				Moon				
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	180°35.1	N21°55.3		123°23.6	8.0'	N23°33.4	-9.7'	58.0'
1	195°35.0	55.6		137°50.6	8.1'	23°23.7	-9.8'	57.9'
2	210°34.9	56.0		152°17.7	8.2'	23°13.9	-9.9'	57.9'
3	225°34.9	· · 56.3		166°44.9	8.4'	23°04.0	-10.0'	57.9'
4	240°34.8	56.7		181°12.3	8.5'	22°53.9	-10.2'	57.8'
5	255°34.7	57.0		195°39.8	8.6'	22°43.8	-10.3'	57.8'
6	270°34.6	N21°57.4		210°07.4	8.8'	N22°33.5	-10.4'	57.7'
7	285°34.5	57.7		224°35.2	8.9'	22°23.1	-10.5'	57.7'
8	300°34.4	58.1		239°03.1	9.0'	22°12.7	-10.6'	57.7'
9	315°34.3	· · 58.4		253°31.1	9.2'	22°02.1	-10.7'	57.6'
10	330°34.2	58.8		267°59.3	9.3'	21°51.4	-10.8'	57.6'
11	345°34.1	59.1		282°27.6	9.4'	21°40.6	-10.9'	57.6'
12	0°34.0	N21°59.5		296°56.1	9.6'	N21°29.7	-11.0'	57.5'
13	15°33.9	21°59.8		311°24.7	9.7'	21°18.7	-11.1'	57.5'
14	30°33.8	22°00.2		325°53.4	9.8'	21°07.6	-11.2'	57.4'
15	45°33.7	· · 00.5		340°22.2	10.0'	20°56.4	-11.3'	57.4'
16	60°33.6	00.8		354°51.2	10.1'	20°45.1	-11.4'	57.4'
17	75°33.5	01.2		9°20.3	10.2'	20°33.7	-11.5'	57.3'
18	90°33.4	N22°01.5		23°49.5	10.4'	N20°22.3	-11.6'	57.3'
19	105°33.4	01.9		38°18.9	10.5'	20°10.7	-11.6'	57.2'
20	120°33.3	02.2		52°48.4	10.6'	19°59.1	-11.7'	57.2'
21	135°33.2	· · 02.6		67°18.0	10.7'	19°47.4	-11.8'	57.2'
22	150°33.1	02.9		81°47.8	10.9'	19°35.5	-11.9'	57.1'
23	165°33.0	03.2		96°17.6	11.0'	19°23.7	-12.0'	57.1'
SD = 15.8'		$d = 0.4'$		SD = 15.8'				

Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	180°32.9	N22°03.6		110°47.6	11.1'	N19°11.7	-12.0'	57.1'
1	195°32.8	03.9		125°17.8	11.2'	18°59.6	-12.1'	57.0'
2	210°32.7	04.3		139°48.0	11.4'	18°47.5	-12.2'	57.0'
3	225°32.6	· · 04.6		154°18.4	11.5'	18°35.3	-12.3'	56.9'
4	240°32.5	04.9		168°48.8	11.6'	18°23.0	-12.3'	56.9'
5	255°32.4	05.3		183°19.5	11.7'	18°10.7	-12.4'	56.9'
6	270°32.3	N22°05.6		197°50.2	11.8'	N17°58.3	-12.5'	56.8'
7	285°32.2	05.9		212°21.0	12.0'	17°45.8	-12.6'	56.8'
8	300°32.1	06.3		226°52.0	12.1'	17°33.2	-12.6'	56.8'
9	315°32.0	· · 06.6		241°23.0	12.2'	17°20.6	-12.7'	56.7'
10	330°31.9	06.9		255°54.2	12.3'	17°07.9	-12.7'	56.7'
11	345°31.8	07.3		270°25.5	12.4'	16°55.2	-12.8'	56.6'
12	0°31.7	N22°07.6		284°56.9	12.5'	N16°42.4	-12.9'	56.6'
13	15°31.6	07.9		299°28.5	12.6'	16°29.5	-12.9'	56.6'
14	30°31.5	08.3		314°00.1	12.7'	16°16.6	-13.0'	56.5'
15	45°31.4	· · 08.6		328°31.8	12.8'	16°03.6	-13.0'	56.5'
16	60°31.3	08.9		343°03.7	12.9'	15°50.5	-13.1'	56.5'
17	75°31.2	09.2		357°35.6	13.1'	15°37.4	-13.2'	56.4'
18	90°31.1	N22°09.6		12°07.7	13.2'	N15°24.3	-13.2'	56.4'
19	105°31.0	09.9		26°39.8	13.3'	15°11.1	-13.3'	56.4'
20	120°30.9	10.2		41°12.1	13.4'	14°57.8	-13.3'	56.3'
21	135°30.8	· · 10.5		55°44.4	13.5'	14°44.5	-13.4'	56.3'
22	150°30.7	10.9		70°16.9	13.6'	14°31.2	-13.4'	56.3'
23	165°30.6	11.2		84°49.4	13.7'	14°17.8	-13.4'	56.2'
SD = 15.8'		$d = 0.3'$		SD = 15.6'				

Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	180°30.5	N22°11.5		99°22.1	13.7'	N14°04.3	-13.5'	56.2'
1	195°30.4	11.8		113°54.8	13.8'	13°50.8	-13.5'	56.2'
2	210°30.3	12.2		128°27.7	13.9'	13°37.3	-13.6'	56.1'
3	225°30.2	· · 12.5		143°00.6	14.0'	13°23.7	-13.6'	56.1'
4	240°30.1	12.8		157°33.6	14.1'	13°10.1	-13.7'	56.1'
5	255°30.0	13.1		172°06.8	14.2'	12°56.5	-13.7'	56.0'
6	270°29.9	N22°13.4		186°40.0	14.3'	N12°42.8	-13.7'	56.0'
7	285°29.8	13.8		201°13.2	14.4'	12°29.0	-13.8'	56.0'
8	300°29.7	14.1		215°46.6	14.5'	12°15.3	-13.8'	55.9'
9	315°29.6	· · 14.4		230°20.1	14.5'	12°01.5	-13.8'	55.9'
10	330°29.5	14.7		244°53.6	14.6'	11°47.7	-13.9'	55.9'
11	345°29.4	15.0		259°27.2	14.7'	11°33.8	-13.9'	55.8'
12	0°29.3	N22°15.3		274°01.0	14.8'	N11°19.9	-13.9'	55.8'
13	15°29.2	15.7		288°34.7	14.9'	11°06.0	-14.0'	55.8'
14	30°29.1	16.0		303°08.6	14.9'	10°52.0	-14.0'	55.7'
15	45°29.0	· · 16.3		317°42.5	15.0'	10°38.0	-14.0'	55.7'
16	60°28.9	16.6		332°16.5	15.1'	10°24.0	-14.0'	55.7'
17	75°28.8	16.9		346°50.6	15.2'	10°10.0	-14.1'	55.7'
18	90°28.7	N22°17.2		1°24.8	15.2'	N09°55.9	-14.1'	55.6'
19	105°28.6	17.5		15°59.0	15.3'	09°41.8	-14.1'	55.6'
20	120°28.5	17.8		30°33.3	15.4'	09°27.7	-14.1'	55.6'
21	135°28.4	· · 18.1		45°07.7	15.4'	09°13.6	-14.2'	55.5'
22	150°28.3	18.5		59°42.1	15.5'	08°59.4	-14.2'	55.5'
23	165°28.2	18.8		74°16.6	15.6'	08°45.2	-14.2'	55.5'
SD = 15.8'		$d = 0.3'$		SD = 15.3'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°	—:—	—:—	01:11	22:49	—:—	—:—
64°	—:—	—:—	01:57	22:01	—:—	—:—
62°	—:—	—:—	02:27	21:30	—:—	—:—
60°	—:—	01:26	02:49	21:08	22:33	—:—
N 58°	—:—	02:00	03:07	20:50	21:58	—:—
56°	—:—	02:24	03:22	20:35	21:33	—:—
54°	01:18	02:43	03:35	20:22	21:13	22:40
52°	01:50	02:59	03:46	20:11	20:57	22:08
50°	02:13	03:13	03:56	20:01	20:44	21:44
45°	02:53	03:40	04:17	19:40	20:16	21:04
N 40°	03:21	04:01	04:33	19:23	19:55	20:36
35°	03:42	04:18	04:47	19:09	19:38	20:14
30°	04:00	04:32	04:59	18:57	19:24	19:56
20°	04:27	04:56	05:20	18:36	19:00	19:29
N 10°	04:48	05:15	05:38	18:18	18:41	19:08
0°	05:06	05:32	05:54	18:02	18:24	18:50
S 10°	05:22	05:48	06:11	17:45	18:08	18:34
20°	05:37	06:04	06:28	17:28	17:51	18:19
30°	05:52	06:22	06:48	17:08	17:34	18:03
35°	06:00	06:32	06:59	16:56	17:24	17:55
40°	06:09	06:42	07:13	16:43	17:13	17:47
45°	06:18	06:55	07:28	16:27	17:01	17:37
S 50°	06:29	07:10	07:47	16:08	16:46	17:27
52°	06:34	07:17	07:56	15:59	16:39	17:22
54°	06:39	07:24	08:07	15:49	16:31	17:17
56°	06:44	07:32	08:18	15:37	16:23	17:11
58°	06:50	07:42	08:31	15:24	16:14	17:05
S 60°	06:57	07:52	08:47	15:09	16:03	16:59

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°			08:56			02:35
N 70°		06:45	09:19		03:08	02:11
68°		07:25	09:35		02:27	01:52
66°	05:23	07:53	09:49	02:41	01:58	01:37
64°	06:09	08:13	10:00	01:55	01:36	01:24
62°	06:39	08:30	10:10	01:24	01:19	01:14
60°	07:01	08:44	10:18	01:01	01:04	01:04
N 58°	07:19	08:55	10:25	00:42	00:51	00:56
56°	07:35	09:05	10:31	00:26	00:40	00:49
54°	07:47	09:14	10:36	00:13	00:31	00:43
52°	07:59	09:22	10:41	00:01	00:22	00:37
50°	08:09	09:29	10:46	· · ·	00:14	00:32
45°	08:30	09:45	10:55	23:57	· · ·	00:20
N 40°	08:47	09:57	11:03	23:44	· · ·	00:11
35°	09:01	10:07	11:10	23:32	· · ·	00:02
30°	09:13	10:17	11:16	23:22	23:55	· · ·
20°	09:34	10:32	11:27	23:04	23:43	· · ·
N 10°	09:52	10:46	11:36	22:48	23:31	· · ·
0°	10:09	10:59	11:44	22:33	23:21	· · ·

June 03, 04, 05 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	251°42.2	225°57.7	N08°15.2	106°44.1	N15°28.7	163°23.6	N23°15.2	250°17.9	S01°41.7	Alpheratz	357°34.6	29°13.7
1	266°44.7	240°57.8	16.0	121°45.3	28.2	178°25.5	15.2	265°20.2	41.7	Ankaa	353°07.0	-42°09.9
2	281°47.2	255°58.0	16.8	136°46.5	27.7	193°27.3	15.3	280°22.5	41.6	Schedar	349°31.0	56°40.3
3	296°49.6	270°58.1	•• 17.5	151°47.6	•• 27.2	208°29.2	•• 15.3	295°24.8	•• 41.6	Diphda	348°47.1	-17°50.8
4	311°52.1	285°58.2	18.3	166°48.8	26.8	223°31.0	15.3	310°27.1	41.5	Achernar	335°20.3	-57°06.2
5	326°54.6	300°58.3	19.1	181°50.0	26.3	238°32.9	15.3	325°29.4	41.5	Hamal	327°51.1	23°34.8
6	341°57.0	315°58.5	N08°19.8	196°51.1	N15°25.8	253°34.7	N23°15.3	340°31.7	S01°41.4	Polaris	314°19.0	89°22.1
7	356°59.5	330°58.6	20.6	211°52.3	25.3	268°36.6	15.3	355°34.1	41.3	Acamar	315°11.9	-40°12.0
8	12°02.0	345°58.7	21.3	226°53.4	24.9	283°38.4	15.3	10°36.4	41.3	Menkar	314°06.1	4°11.3
9	27°04.4	0°58.8	•• 22.1	241°54.6	•• 24.4	298°40.3	•• 15.3	25°38.7	•• 41.2	Mirfak	308°28.3	49°57.0
10	42°06.9	15°59.0	22.9	256°55.8	23.9	313°42.1	15.3	40°41.0	41.2	Aldebaran	290°39.6	16°33.6
11	57°09.3	30°59.1	23.6	271°56.9	23.4	328°44.0	15.4	55°43.3	41.1	Rigel	281°03.9	-8°10.3
12	72°11.8	45°59.2	N08°24.4	286°58.1	N15°23.0	343°45.8	N23°15.4	70°45.6	S01°41.1	Capella	280°21.9	46°01.4
13	87°14.3	60°59.3	25.2	301°59.2	22.5	358°47.7	15.4	85°48.0	41.0	Bellatrix	278°22.9	6°22.3
14	102°16.7	75°59.4	25.9	317°00.4	22.0	13°49.5	15.4	100°50.3	41.0	Elnath	278°01.8	28°37.7
15	117°19.2	90°59.6	•• 26.7	332°01.6	•• 21.5	28°51.4	•• 15.4	115°52.6	•• 40.9	Alnilam	275°37.7	-1°11.2
16	132°21.7	105°59.7	27.4	347°02.7	21.1	43°53.2	15.4	130°54.9	40.9	Betelgeuse	270°52.1	7°24.7
17	147°24.1	120°59.8	28.2	2°03.9	20.6	58°55.1	15.4	145°57.2	40.8	Canopus	263°52.8	-52°42.6
18	162°26.6	135°59.9	N08°29.0	17°05.1	N15°20.1	73°57.0	N23°15.4	160°59.5	S01°40.8	Sirius	258°26.3	-16°45.1
19	177°29.1	151°00.0	29.7	32°06.2	19.6	88°58.8	15.4	176°01.9	40.7	Adhara	255°06.0	-29°00.5
20	192°31.5	166°00.2	30.5	47°07.4	19.2	104°00.7	15.4	191°04.2	40.7	Procyon	244°50.8	5°09.6
21	207°34.0	181°00.3	•• 31.3	62°08.5	•• 18.7	119°02.5	•• 15.5	206°06.5	•• 40.6	Pollux	243°17.2	27°58.0
22	222°36.4	196°00.4	32.0	77°09.7	18.2	134°04.4	15.5	221°08.8	40.5	Avior	234°15.0	-59°35.6
23	237°38.9	211°00.5	32.8	92°10.9	17.7	149°06.2	15.5	236°11.1	40.5	Suhail	222°46.3	-43°32.3
Mer.pass. 07:12		ν 0.1' d 0.8' m -4.37		ν 1.2' d -0.5' m 1.30		ν 1.9' d 0.0' m -1.91		ν 2.3' d -0.1' m 1.08				

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	252°41.4	226°00.6	N08°33.6	107°12.0	N15°17.3	164°08.1	N23°15.5	251°13.4	S01°40.4	Alphard	217°47.6	-8°46.2
1	267°43.8	241°00.7	34.3	122°13.2	16.8	179°09.9	15.5	266°15.8	40.4	Regulus	207°34.2	11°50.6
2	282°46.3	256°00.8	35.1	137°14.3	16.3	194°11.8	15.5	281°18.1	40.3	Dubhe	193°40.6	61°37.1
3	297°48.8	271°00.9	•• 35.9	152°15.5	•• 15.8	209°13.6	•• 15.5	296°20.4	•• 40.3	Denebola	182°24.5	14°25.8
4	312°51.2	286°01.1	36.6	167°16.7	15.3	224°15.5	15.5	311°22.7	40.2	Gienah	175°43.1	-17°41.1
5	327°53.7	301°01.2	37.4	182°17.8	14.9	239°17.3	15.5	326°25.0	40.2	Acrux	172°59.5	-63°14.7
6	342°56.2	316°01.3	N08°38.2	197°19.0	N15°14.4	254°19.2	N23°15.5	341°27.4	S01°40.1	Gacrux	171°51.1	-57°15.6
7	357°58.6	331°01.4	38.9	212°20.1	13.9	269°21.0	15.6	356°29.7	40.1	Alioth	166°12.4	55°49.5
8	13°01.1	346°01.5	39.7	227°21.3	13.4	284°22.9	15.6	11°32.0	40.0	Spica	158°21.8	-11°17.7
9	28°03.6	1°01.6	•• 40.5	242°22.4	•• 12.9	299°24.7	•• 15.6	26°34.3	•• 40.0	Alkaid	152°51.4	49°11.3
10	43°06.0	16°01.7	41.2	257°23.6	12.5	314°26.6	15.6	41°36.6	39.9	Hadar	148°35.1	-60°30.0
11	58°08.5	31°01.8	42.0	272°24.8	12.0	329°28.4	15.6	56°38.9	39.9	Menkent	147°56.9	-36°29.9
12	73°10.9	46°01.9	N08°42.8	287°25.9	N15°11.5	344°30.3	N23°15.6	71°41.3	S01°39.8	Arcturus	145°47.3	19°03.0
13	88°13.4	61°02.0	43.5	302°27.1	11.0	359°32.1	15.6	86°43.6	39.8	Rigil Kent.	139°39.3	-60°56.6
14	103°15.9	76°02.1	44.3	317°28.2	10.5	14°34.0	15.6	101°45.9	39.7	Kochab	137°18.7	74°03.1
15	118°18.3	91°02.2	•• 45.1	332°29.4	•• 10.1	29°35.8	•• 15.6	116°48.2	•• 39.7	Zuben'ubi	136°55.4	-16°08.9
16	133°20.8	106°02.3	45.8	347°30.6	09.6	44°37.7	15.6	131°50.5	39.6	Alphecca	126°03.1	26°37.7
17	148°23.3	121°02.4	46.6	2°31.7	09.1	59°39.5	15.6	146°52.9	39.6	Antares	112°15.1	-26°29.4
18	163°25.7	136°02.5	N08°47.4	17°32.9	N15°08.6	74°41.4	N23°15.7	161°55.2	S01°39.5	Atria	107°08.4	-69°04.4
19	178°28.2	151°02.6	48.1	32°34.0	08.1	89°43.2	15.7	176°57.5	39.5	Sabik	102°02.1	-15°45.4
20	193°30.7	166°02.7	48.9	47°35.2	07.7	104°45.1	15.7	191°59.8	39.4	Shaula	96°09.5	-37°07.3
21	208°33.1	181°02.8	•• 49.7	62°36.4	•• 07.2	119°46.9	•• 15.7	207°02.1	•• 39.4	Rasalhague	95°57.9	12°32.4
22	223°35.6	196°02.9	50.5	77°37.5	06.7	134°48.8	15.7	222°04.5	39.3	Eltanin	90°41.5	51°29.0
23	238°38.1	211°03.0	51.2	92°38.7	06.2	149°50.7	15.7	237°06.8	39.3	Kaus Aust.	83°31.7	-34°22.3
Mer.pass. 07:08		ν 0.1' d 0.8' m -4.36		ν 1.2' d -0.5' m 1.31		ν 1.9' d 0.0' m -1.91		ν 2.3' d -0.1' m 1.07		Vega	80°32.6	38°48.3

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	253°40.5	226°03.1	N08°52.0	107°39.8	N15°05.7	164°52.5	N23°15.7	252°09.1	S01°39.2	Nunki	75°47.0	-26°15.9
1	268°43.0	241°03.2	52.8	122°41.0	05.3	179°54.4	15.7	267°11.4	39.2	Altair	61°59.4	8°56.0
2	283°45.4	256°03.3	53.5	137°42.1	04.8	194°56.2	15.7	282°13.7	39.1	Peacock	53°04.8	-56°39.0
3	298°47.9	271°03.4	•• 54.3	152°43.3	•• 04.3	209°58.1	•• 15.7	297°16.1	•• 39.0	Deneb	49°25.3	45°22.0
4	313°50.4	286°03.5	55.1	167°44.5	03.8	224°59.9	15.7	312°18.4	39.0	Enif	33°38.4	9°59.4
5	328°52.8	301°03.6	55.8	182°45.6	03.3	240°01.8	15.8	327°20.7	38.9	Al Na'ir	27°32.4	-46°50.1
6	343°55.3	316°03.7	N08°56.6	197°46.8	N15°02.8	255°03.6	N23°15.8	342°23.0	S01°38.9	Fomalhaut	15°14.2	-29°29.1
7	358°57.8	331°03.8	57.4	212°47.9	02.4	270°05.5	15.8	357°25.4	38.8	Scheat	13°44.9	28°13.1
8	14°00.2	346°03.9	58.2	227°49.1	01.9	285°07.3	15.8	12°27.7	38.8	Markab	13°29.6	15°20.4
9	29°02.7	1°04.0	•• 58.9	242°50.2	•• 01.4	300°09.2	•• 15.8	27°30.0	•• 38.7			
10	44°05.2	16°04.1	08°59.7	257°51.4	00.9	315°11.0	15.8	42°32.3	38.7	Jun 03 Tue	SHA	Mer.pass
11	59°07.6	31°04.2	09°00.5	272°52.6	15°00.4	330°12.9	15.8	57°34.6	38.6	Venus	334°15.5	08:56
12	74°10.1	46°04.2	N09°01.2	287°53.7	N14°59.9	345°14.7	N23°15.8	72°37.0	S01°38.6	Mars	215°01.9	16:52
13	89°12.5	61°04.3	02.0	302°54.9	59.5	0°16.6	15.8	87°39.3	38.5	Jupiter	271°41.4	13:05
14	104°15.0	76°04.4	02.8	317°56.0	59.0	15°18.4	15.8	102°41.6	38.5	Saturn	358°35.6	07:18
15	119°17.5	91°04.5	•• 03.6	332°57.2	•• 58.5	30°20.3	•• 15.8	117°43.9	•• 38.4			
16	134°19.9	106°04.6	04.3	347°58.3	58.0	45°22.1	15.9	132°46.3	38.4	Jun 04 Wed	SHA	Mer.pass
17	149°22.4	121°04.7	05.1	2°59.5	57.5	60°24.0	15.9	147°48.6	38.3	Venus	333°19.2	08:56
18	164°24.9	136°04.8	N09°05.9	18°00.7	N14°57.0	75°25.8	N23°15.9	162°50.9	S01°38.3	Mars	214°30.6	16:50
19	179°27.3	151°04.8	06.6	33°01.8	56.6	90°27.7	15.9	177°53.2	38.2	Jupiter	271°26.7	13:02
20	194°29.8	166°04.9	07.4	48°03.								

h	Sun			Moon			
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	180°28.1	N22°19.1	88°51.2	15.6'	N08°31.0	-14.2'	55.5'
1	195°28.0	19.4	103°25.8	15.7'	08°16.8	-14.2'	55.4'
2	210°27.9	19.7	118°00.5	15.7'	08°02.6	-14.2'	55.4'
3	225°27.8	· · 20.0	132°35.2	15.8'	07°48.3	-14.3'	55.4'
4	240°27.7	20.3	147°10.1	15.9'	07°34.1	-14.3'	55.3'
5	255°27.6	20.6	161°44.9	15.9'	07°19.8	-14.3'	55.3'
6	270°27.4	N22°20.9	176°19.8	16.0'	N07°05.5	-14.3'	55.3'
7	285°27.3	21.2	190°54.8	16.0'	06°51.2	-14.3'	55.3'
8	300°27.2	21.5	205°29.8	16.1'	06°36.9	-14.3'	55.2'
9	315°27.1	· · 21.8	220°04.9	16.1'	06°22.5	-14.3'	55.2'
10	330°27.0	22.1	234°40.1	16.2'	06°08.2	-14.4'	55.2'
11	345°26.9	22.4	249°15.2	16.2'	05°53.8	-14.4'	55.2'
12	0°26.8	N22°22.7	263°50.5	16.3'	N05°39.5	-14.4'	55.1'
13	15°26.7	23.0	278°25.7	16.3'	05°25.1	-14.4'	55.1'
14	30°26.6	23.3	293°01.1	16.4'	05°10.7	-14.4'	55.1'
15	45°26.5	· · 23.6	307°36.4	16.4'	04°56.3	-14.4'	55.1'
16	60°26.4	23.9	322°11.8	16.4'	04°42.0	-14.4'	55.0'
17	75°26.3	24.2	336°47.3	16.5'	04°27.6	-14.4'	55.0'
18	90°26.2	N22°24.5	351°22.8	16.5'	N04°13.2	-14.4'	55.0'
19	105°26.1	24.8	5°58.3	16.6'	03°58.8	-14.4'	55.0'
20	120°26.0	25.1	20°33.9	16.6'	03°44.3	-14.4'	54.9'
21	135°25.9	· · 25.3	35°09.5	16.6'	03°29.9	-14.4'	54.9'
22	150°25.8	25.6	49°45.1	16.7'	03°15.5	-14.4'	54.9'
23	165°25.6	25.9	64°20.8	16.7'	03°01.1	-14.4'	54.9'
SD = 15.8' $d = 0.3'$			SD = 15.1'				
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	180°25.5	N22°26.2	78°56.5	16.7'	N02°46.7	-14.4'	54.9'
1	195°25.4	26.5	93°32.2	16.8'	02°32.3	-14.4'	54.8'
2	210°25.3	26.8	108°08.0	16.8'	02°17.9	-14.4'	54.8'
3	225°25.2	· · 27.1	122°43.8	16.8'	02°03.5	-14.4'	54.8'
4	240°25.1	27.4	137°19.6	16.8'	01°49.1	-14.4'	54.8'
5	255°25.0	27.7	151°55.5	16.9'	01°34.7	-14.4'	54.8'
6	270°24.9	N22°27.9	166°31.3	16.9'	N01°20.3	-14.4'	54.7'
7	285°24.8	28.2	181°07.2	16.9'	01°05.9	-14.4'	54.7'
8	300°24.7	28.5	195°43.1	16.9'	00°51.5	-14.4'	54.7'
9	315°24.6	· · 28.8	210°19.1	17.0'	00°37.1	-14.4'	54.7'
10	330°24.5	29.1	224°55.0	17.0'	00°22.8	-14.4'	54.7'
11	345°24.4	29.4	239°31.0	17.0'	N00°08.4	-14.4'	54.6'
12	0°24.2	N22°29.7	254°07.0	17.0'	S00°05.9	14.3'	54.6'
13	15°24.1	29.9	268°43.0	17.0'	00°20.3	14.3'	54.6'
14	30°24.0	30.2	283°19.1	17.0'	00°34.6	14.3'	54.6'
15	45°23.9	· · 30.5	297°55.1	17.1'	00°48.9	14.3'	54.6'
16	60°23.8	30.8	312°31.1	17.1'	01°03.2	14.3'	54.5'
17	75°23.7	31.1	327°07.2	17.1'	01°17.5	14.3'	54.5'
18	90°23.6	N22°31.3	341°43.3	17.1'	S01°31.8	14.3'	54.5'
19	105°23.5	31.6	356°19.4	17.1'	01°46.1	14.3'	54.5'
20	120°23.4	31.9	10°55.5	17.1'	02°00.3	14.2'	54.5'
21	135°23.3	· · 32.2	25°31.6	17.1'	02°14.6	14.2'	54.5'
22	150°23.2	32.4	40°07.7	17.1'	02°28.8	14.2'	54.4'
23	165°23.0	32.7	54°43.8	17.1'	02°43.0	14.2'	54.4'
SD = 15.8' $d = 0.3'$			SD = 15.0'				
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	180°22.9	N22°33.0	69°19.9	17.1'	S02°57.2	14.2'	54.4'
1	195°22.8	33.3	83°56.0	17.1'	03°11.4	14.2'	54.4'
2	210°22.7	33.5	98°32.1	17.1'	03°25.5	14.1'	54.4'
3	225°22.6	· · 33.8	113°08.2	17.1'	03°39.7	14.1'	54.4'
4	240°22.5	34.1	127°44.3	17.1'	03°53.8	14.1'	54.4'
5	255°22.4	34.3	142°20.4	17.1'	04°07.9	14.1'	54.3'
6	270°22.3	N22°34.6	156°56.5	17.1'	S04°22.0	14.1'	54.3'
7	285°22.2	34.9	171°32.6	17.1'	04°36.0	14.0'	54.3'
8	300°22.0	35.2	186°08.7	17.1'	04°50.1	14.0'	54.3'
9	315°21.9	· · 35.4	200°44.8	17.1'	05°04.1	14.0'	54.3'
10	330°21.8	35.7	215°20.9	17.1'	05°18.1	14.0'	54.3'
11	345°21.7	36.0	229°56.9	17.1'	05°32.0	13.9'	54.3'
12	0°21.6	N22°36.2	244°33.0	17.0'	S05°46.0	13.9'	54.3'
13	15°21.5	36.5	259°09.0	17.0'	05°59.9	13.9'	54.2'
14	30°21.4	36.7	273°45.1	17.0'	06°13.8	13.9'	54.2'
15	45°21.3	· · 37.0	288°21.1	17.0'	06°27.6	13.8'	54.2'
16	60°21.1	37.3	302°57.1	17.0'	06°41.5	13.8'	54.2'
17	75°21.0	37.5	317°33.1	17.0'	06°55.3	13.8'	54.2'
18	90°20.9	N22°37.8	332°09.0	16.9'	S07°09.0	13.7'	54.2'
19	105°20.8	38.1	346°45.0	16.9'	07°22.8	13.7'	54.2'
20	120°20.7	38.3	1°20.9	16.9'	07°36.5	13.7'	54.2'
21	135°20.6	· · 38.6	15°56.8	16.9'	07°50.2	13.7'	54.2'
22	150°20.5	38.8	30°32.7	16.9'	08°03.8	13.6'	54.2'
23	165°20.4	39.1	45°08.5	16.8'	08°17.5	13.6'	54.1'
SD = 15.8' $d = 0.3'$			SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	--	--	00:58	23:03	--	--
64°	--	--	01:50	22:08	--	--
62°	--	--	02:22	21:36	--	--
60°	--	01:17	02:45	21:13	22:42	--
N 58°	--	01:54	03:04	20:54	22:04	--
56°	--	02:20	03:19	20:39	21:38	--
54°	01:10	02:40	03:32	20:25	21:18	22:49
52°	01:45	02:57	03:44	20:14	21:01	22:13
50°	02:09	03:11	03:54	20:03	20:47	21:49
45°	02:50	03:38	04:15	19:42	20:19	21:07
N 40°	03:19	04:00	04:32	19:25	19:57	20:38
35°	03:41	04:17	04:46	19:11	19:40	20:16
30°	03:59	04:32	04:59	18:58	19:25	19:58
20°	04:27	04:55	05:20	18:37	19:01	19:30
N 10°	04:48	05:15	05:38	18:19	18:42	19:09
0°	05:06	05:32	05:55	18:02	18:24	18:50
S 10°	05:23	05:49	06:11	17:45	18:08	18:34
20°	05:38	06:05	06:29	17:28	17:51	18:19
30°	05:54	06:23	06:49	17:07	17:33	18:03
35°	06:02	06:33	07:01	16:55	17:23	17:55
40°	06:11	06:44	07:15	16:42	17:12	17:46
45°	06:20	06:57	07:31	16:26	16:59	17:36
S 50°	06:31	07:12	07:50	16:06	16:44	17:25
52°	06:36	07:19	07:59	15:57	16:37	17:20
54°	06:41	07:27	08:10	15:47	16:30	17:15
56°	06:47	07:35	08:22	15:35	16:21	17:09
58°	06:53	07:45	08:35	15:21	16:11	17:03
S 60°	07:00	07:56	08:51	15:05	16:01	16:56

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	11:11	13:11	15:10	01:50	01:16	00:44
N 70°	11:20	13:11	15:00	01:38	01:12	00:48
68°	11:27	13:10	14:52	01:29	01:10	00:52
66°	11:32	13:10	14:46	01:21	01:08	00:55
64°	11:37	13:09	14:40	01:15	01:06	00:57
62°	11:41	13:09	14:35	01:09	01:04	00:59
60°	11:45	13:09	14:31	01:04	01:03	01:01
N 58°	11:48	13:08	14:27	00:59	01:01	01:03
56°	11:51	13:08	14:24	00:55	01:00	01:05
54°	11:54	13:08	14:21	00:52	00:59	01:06
52°	11:56	13:08	14:19	00:48	00:58	01:07
50°	11:58	13:08	14:16	00:45	00:57	01:09
45°	12:03	13:07	14:11	00:39	00:55	01:11
N 40°	12:06	13:07	14:07	00:33	00:54	01:13
35°	12:10	13:07	14:03	00:29	00:52	01:15
30°	12:12	13:07	14:00	00:24	00:51	01:17
20°	12:17	13:06	13:54	00:17	00:49	01:20
N 10°	12:22	13:06	13:49	00:10	00:47	01:22
0°	12:26	13:06	13:45	00:04	00:45	01:25
S 10°	12:30	13:05	13:40	· · ·	00:43	01:27
20°	12:34	13:05	13:36	· · ·	00:41	01:29
30°	12:39	13:05	13:30	· · ·	00:39	01:32
35°	12:42	13:05	13:27	· · ·	00:37	01:34
40°	12:45	13:05	13:24	· · ·	00:36	01:36
45°	12:49	13:04	13:20	· · ·	00:34	01:38
S 50°	12:53	13:04	13:15	· · ·	00:32	01:41
52°	12:55	13:04	13:13	· · ·	00:31	01:42
54°	12:57	13:04	13:11	· · ·	00:30	01:43
56°	12:59	13:04	13:08	· · ·	00:28	01:45
58°	13:02	13:04	13:05	· · ·	00:27	01:46
S 60°	13:05	13:04	13:02	· · ·	00:26	01:48

Day	Sun		Mer. Pass hh:mm	Moon		Age 7-9 49-68%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
03	01:52	01:47	11:58	18:35	06:15	
04	01:42	01:37	11:58	19:15	06:55	
05	01:32	01:26	11:59	19:54	07:35	

June 06, 07, 08 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	254°39.7	226°05.3	N09°10.5	108°07.6	N14°54.1	165°36.9	N23°15.9	253°04.8	S01°38.0					Alpheratz	357°34.5	29°13.7	
1	260°42.1	241°05.3		123°08.7	53.6	180°38.8	15.9	268°07.2	37.9					Ankaa	353°07.0	-42°09.9	
2	284°44.6	256°05.4		138°09.9	53.2	195°40.6	15.9	283°09.5	37.9					Schedar	349°30.9	56°40.3	
3	299°47.0	271°05.5	· · 12.8	153°11.1	· · 52.7	210°42.5	· · 15.9	298°11.8	· · 37.8					Diphda	348°47.1	-17°50.7	
4	314°49.5	286°05.6		168°12.2	52.2	225°44.3	15.9	313°14.1	37.8					Achernar	335°20.3	-57°06.2	
5	329°52.0	301°05.7		183°13.4	51.7	240°46.2	16.0	328°16.5	37.7					Hamal	327°51.1	23°34.8	
6	344°54.4	316°05.7	N09°15.2	198°14.5	N14°51.2	255°48.0	N23°16.0	343°18.8	S01°37.7					Polaris	314°18.1	89°22.1	
7	359°56.9	331°05.8		213°15.7	50.7	270°49.9	16.0	358°21.1	37.6					Acamar	315°11.9	-40°12.0	
8	14°59.4	346°05.9		228°16.8	50.2	285°51.7	16.0	13°23.4	37.6					Menkar	314°06.1	4°11.3	
9	30°01.8	1°06.0	· · 17.5	243°18.0	· · 49.8	300°53.6	· · 16.0	28°25.8	· · 37.6					Mirfak	308°28.3	49°57.0	
10	45°04.3	16°06.0		258°19.1	49.3	315°55.4	16.0	43°28.1	37.5					Aldebaran	290°39.6	16°33.6	
11	60°06.8	31°06.1		273°20.3	48.8	330°57.3	16.0	58°30.4	37.5					Rigel	281°03.9	-8°10.3	
12	75°09.2	46°06.2	N09°19.8	288°21.5	N14°48.3	345°59.1	N23°16.0	73°32.7	S01°37.4					Capella	280°21.9	46°01.4	
13	90°11.7	61°06.3		303°22.6	47.8	1°01.0	16.0	88°35.1	37.4					Bellatrix	278°22.9	6°22.4	
14	105°14.2	76°06.3		318°23.8	47.3	16°02.8	16.0	103°37.4	37.3					Elnath	278°01.8	28°37.7	
15	120°16.6	91°06.4	· · 22.1	333°24.9	· · 46.8	31°04.7	· · 16.0	118°39.7	· · 37.3					Alnilam	275°37.7	-1°11.1	
16	135°19.1	106°06.5		348°26.1	46.3	46°06.5	16.0	133°42.0	37.2					Betelgeuse	270°52.1	7°24.7	
17	150°21.5	121°06.5		3°27.2	45.9	61°08.4	16.0	148°44.4	37.2					Canopus	263°52.8	-52°42.6	
18	165°24.0	136°06.6	N09°24.5	18°28.4	N14°45.4	76°10.2	N23°16.1	163°46.7	S01°37.1					Sirius	258°26.3	-16°45.1	
19	180°26.5	151°06.7		33°29.5	44.9	91°12.1	16.1	178°49.0	37.1					Adhara	255°06.0	-29°00.5	
20	195°28.9	166°06.7		48°30.7	44.4	106°13.9	16.1	193°51.3	37.0					Procyon	244°50.8	5°09.6	
21	210°31.4	181°06.8	· · 26.8	63°31.8	· · 43.9	121°15.8	· · 16.1	208°53.7	· · 37.0					Pollux	243°17.2	27°58.0	
22	225°33.9	196°06.9		78°33.0	43.4	136°17.6	16.1	223°56.0	36.9					Avior	234°15.1	-59°35.6	
23	240°36.3	211°06.9		93°34.2	42.9	151°19.5	16.1	238°58.3	36.9					Suhail	222°46.3	-43°32.3	
Mer.pass. 07:00			ν 0.1' d 0.8' m -4.34	ν 1.2' d -0.5' m 1.33	ν 1.9' d 0.0' m -1.91	ν 2.3' d -0.0' m 1.06											

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	255°38.8	226°07.0	N09°29.1	108°35.3	N14°42.4	166°21.3	N23°16.1	254°00.7	S01°36.8
1	270°41.3	241°07.1		123°36.5	41.9	181°23.2	16.1	269°03.0	36.8
2	285°43.7	256°07.1		138°37.6	41.5	196°25.0	16.1	284°05.3	36.7
3	300°46.2	271°07.2	· · 31.4	153°38.8	· · 41.0	211°26.9	· · 16.1	299°07.6	· · 36.7
4	315°48.6	286°07.3		168°39.9	40.5	226°28.7	16.1	314°10.0	36.6
5	330°51.1	301°07.3		183°41.1	40.0	241°30.6	16.1	329°12.3	36.6
6	345°53.6	316°07.4	N09°33.8	198°42.2	N14°39.5	256°32.4	N23°16.1	344°14.6	S01°36.5
7	0°56.0	331°07.5		213°43.4	39.0	271°34.3	16.1	359°16.9	36.5
8	15°58.5	346°07.5		228°44.5	38.5	286°36.1	16.1	14°19.3	36.4
9	31°01.0	1°07.6	· · 36.1	243°45.7	· · 38.0	301°38.0	· · 16.2	29°21.6	· · 36.4
10	46°03.4	16°07.6		258°46.8	37.5	316°39.8	16.2	44°23.9	36.3
11	61°05.9	31°07.7		273°48.0	37.0	331°41.7	16.2	59°26.3	36.3
12	76°08.4	46°07.7	N09°38.4	288°49.2	N14°36.5	346°43.5	N23°16.2	74°28.6	S01°36.2
13	91°10.8	61°07.8		303°50.3	36.1	1°45.4	16.2	89°30.9	36.2
14	106°13.3	76°07.9		318°51.5	35.6	16°47.2	16.2	104°33.3	36.2
15	121°15.8	91°07.9	· · 40.8	333°52.6	· · 35.1	31°49.0	· · 16.2	119°35.6	· · 36.1
16	136°18.2	106°08.0		348°53.8	34.6	46°50.9	16.2	134°37.9	36.1
17	151°20.7	121°08.0		3°54.9	34.1	61°52.7	16.2	149°40.2	36.0
18	166°23.1	136°08.1	N09°43.1	18°56.1	N14°33.6	76°54.6	N23°16.2	164°42.6	S01°36.0
19	181°25.6	151°08.1		33°57.2	33.1	91°56.4	16.2	179°44.9	35.9
20	196°28.1	166°08.2		48°58.4	32.6	106°58.3	16.2	194°47.2	35.9
21	211°30.5	181°08.2	· · 45.4	63°59.5	· · 32.1	122°00.1	· · 16.2	209°49.6	· · 35.8
22	226°33.0	196°08.3		79°00.7	31.6	137°02.0	16.2	224°51.9	35.8
23	241°35.5	211°08.3		94°01.8	31.1	152°03.8	16.2	239°54.2	35.7
Mer.pass. 06:56			ν 0.1' d 0.8' m -4.33	ν 1.2' d -0.5' m 1.34	ν 1.8' d 0.0' m -1.91	ν 2.3' d -0.0' m 1.06			

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	256°37.9	226°08.4	N09°47.8	109°03.0	N14°30.6	167°05.7	N23°16.3	254°56.5	S01°35.7
1	271°40.4	241°08.4		124°04.1	30.1	182°07.5	16.3	269°58.9	35.6
2	286°42.9	256°08.5		139°05.3	29.7	197°09.4	16.3	285°01.2	35.6
3	301°45.3	271°08.5	· · 50.1	154°06.4	· · 29.2	212°11.2	· · 16.3	300°03.5	· · 35.5
4	316°47.8	286°08.6		169°07.6	28.7	227°13.1	16.3	315°05.9	35.5
5	331°50.3	301°08.6		184°08.7	28.2	242°14.9	16.3	330°08.2	35.5
6	346°52.7	316°08.7	N09°52.5	199°09.9	N14°27.7	257°16.8	N23°16.3	345°10.5	S01°35.4
7	1°55.2	331°08.7		214°11.0	27.2	272°18.6	16.3	0°12.9	35.4
8	16°57.6	346°08.8		229°12.2	26.7	287°20.5	16.3	15°15.2	35.3
9	32°00.1	1°08.8	· · 54.8	244°13.3	· · 26.2	302°22.3	· · 16.3	30°17.5	· · 35.3
10	47°02.6	16°08.8		259°14.5	25.7	317°24.2	16.3	45°19.9	35.2
11	62°05.0	31°08.9		274°15.7	25.2	332°26.0	16.3	60°22.2	35.2
12	77°07.5	46°08.9	N09°57.1	289°16.8	N14°24.7	347°27.9	N23°16.3	75°24.5	S01°35.1
13	92°10.0	61°09.0		304°18.0	24.2	2°29.7	16.3	90°26.9	35.1
14	107°12.4	76°09.0		319°19.1	23.7	17°31.6	16.3	105°29.2	35.0
15	122°14.9	91°09.1	09°59.5	334°20.3	· · 23.2	32°33.4	· · 16.3	120°31.5	· · 35.0
16	137°17.4	106°09.1	10°00.3	349°21.4	22.7	47°35.3	16.3	135°33.9	34.9
17	152°19.8	121°09.1		4°22.6	22.2	62°37.1	16.3	150°36.2	34.9
18	167°22.3	136°09.2	N10°01.8	19°23.7	N14°21.7	77°39.0	N23°16.4	165°38.5	S01°34.9
19	182°24.7	151°09.2		34°24.9	21.2	92°40.8	16.4	180°40.9	34.8
20	197°27.2	166°09.2		49°26.0	20.7	107°42.7	16.4	195°43.2	34.8
21	212°29.7	181°09.3	· · 04.2	64°27.2	· · 20.3	122°44.5	· · 16.4	210°45.5	· · 34.7
22	227°32.1	196°09.3		79°28.3	19.8	137°46.3	16.4	225°47.9	34.7
23	242°34.6	211°09.3		94°29.5	19.3	152°48.2	16.4	240°50.2	34.6
Mer.pass. 06:52			ν 0.0' d 0.8' m -4.32	ν 1.2' d -0.5' m 1.34	ν 1.8' d 0.0' m -1.91	ν 2.3' d -0.0' m 1.06			

Stars		
Jun 06 Fri	SHA	Mer.pass
Venus	331°25.6	08:56
Mars	213°27.9	16:46
Jupiter	270°57.3	12:56
Saturn	358°25.2	07:07
Jun 07 Sat	SHA	Mer.pass
Venus	330°28.2	08:55
Mars	212°56.5	16:44
Jupiter	270°42.5	12:53
Saturn	358°21.9	07:03
Jun 08 Sun	SHA	Mer.pass
Venus	329°30.5	08:55
Mars	212°25.1	16:43
Jupiter	270°27.8	12:50
Saturn	358°18.6	06:59
Horizontal parallax		
Venus:		0.2
Mars:		0.1

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	180°20.2	N22°39.4	59°44.4	16.8'	S08°31.1	13.6'	54.1'
1	195°20.1	39.6	74°20.2	16.8'	08°44.6	13.5'	54.1'
2	210°20.0	39.9	88°56.0	16.8'	08°58.1	13.5'	54.1'
3	225°19.9	· · 40.1	103°31.7	16.7'	09°11.6	13.4'	54.1'
4	240°19.8	40.4	118°07.5	16.7'	09°25.1	13.4'	54.1'
5	255°19.7	40.6	132°43.1	16.7'	09°38.5	13.4'	54.1'
6	270°19.6	N22°40.9	147°18.8	16.6'	S09°51.8	13.3'	54.1'
7	285°19.4	41.1	161°54.5	16.6'	10°05.2	13.3'	54.1'
8	300°19.3	41.4	176°30.1	16.6'	10°18.5	13.3'	54.1'
9	315°19.2	· · 41.6	191°05.6	16.5'	10°31.7	13.2'	54.1'
10	330°19.1	41.9	205°41.2	16.5'	10°45.0	13.2'	54.1'
11	345°19.0	42.1	220°16.7	16.5'	10°58.1	13.1'	54.1'
12	0°18.9	N22°42.4	234°52.1	16.4'	S11°11.3	13.1'	54.1'
13	15°18.8	42.6	249°27.5	16.4'	11°24.4	13.1'	54.1'
14	30°18.6	42.9	264°02.9	16.3'	11°37.4	13.0'	54.0'
15	45°18.5	· · 43.1	278°38.3	16.3'	11°50.4	13.0'	54.0'
16	60°18.4	43.4	293°13.6	16.3'	12°03.4	12.9'	54.0'
17	75°18.3	43.6	307°48.8	16.2'	12°16.3	12.9'	54.0'
18	90°18.2	N22°43.9	322°24.1	16.2'	S12°29.2	12.8'	54.0'
19	105°18.1	44.1	336°59.2	16.1'	12°42.0	12.8'	54.0'
20	120°17.9	44.4	351°34.4	16.1'	12°54.8	12.7'	54.0'
21	135°17.8	· · 44.6	6°09.5	16.0'	13°07.5	12.7'	54.0'
22	150°17.7	44.8	20°44.5	16.0'	13°20.2	12.6'	54.0'
23	165°17.6	45.1	35°19.5	15.9'	13°32.8	12.6'	54.0'
SD = 15.8'			$d = 0.3'$ SD = 14.8'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	180°17.5	N22°45.3	49°54.4	15.9'	S13°45.4	12.5'	54.0'
1	195°17.4	45.6	64°29.3	15.8'	13°58.0	12.5'	54.0'
2	210°17.2	45.8	79°04.2	15.8'	14°10.4	12.4'	54.0'
3	225°17.1	· · 46.0	93°39.0	15.7'	14°22.9	12.4'	54.0'
4	240°17.0	46.3	108°13.7	15.7'	14°35.2	12.3'	54.0'
5	255°16.9	46.5	122°48.4	15.6'	14°47.5	12.3'	54.0'
6	270°16.8	N22°46.8	137°23.1	15.6'	S14°59.8	12.2'	54.0'
7	285°16.7	47.0	151°57.6	15.5'	15°12.0	12.1'	54.0'
8	300°16.5	47.2	166°32.2	15.5'	15°24.2	12.1'	54.0'
9	315°16.4	· · 47.5	181°06.6	15.4'	15°36.3	12.0'	54.0'
10	330°16.3	47.7	195°41.1	15.4'	15°48.3	12.0'	54.0'
11	345°16.2	47.9	210°15.4	15.3'	16°00.3	11.9'	54.0'
12	0°16.1	N22°48.2	224°49.7	15.2'	S16°12.2	11.9'	54.0'
13	15°16.0	48.4	239°23.9	15.2'	16°24.0	11.8'	54.0'
14	30°15.8	48.6	253°58.1	15.1'	16°35.8	11.7'	54.0'
15	45°15.7	· · 48.9	268°32.2	15.1'	16°47.5	11.7'	54.0'
16	60°15.6	49.1	283°06.3	15.0'	16°59.2	11.6'	54.0'
17	75°15.5	49.3	297°40.3	14.9'	17°10.8	11.5'	54.0'
18	90°15.4	N22°49.5	312°14.2	14.9'	S17°22.3	11.5'	54.0'
19	105°15.2	49.8	326°48.1	14.8'	17°33.8	11.4'	54.0'
20	120°15.1	50.0	341°21.9	14.7'	17°45.2	11.3'	54.0'
21	135°15.0	· · 50.2	355°55.6	14.7'	17°56.5	11.3'	54.0'
22	150°14.9	50.4	10°29.3	14.6'	18°07.8	11.2'	54.0'
23	165°14.8	50.7	25°02.9	14.5'	18°19.0	11.1'	54.0'
SD = 15.8'			$d = 0.2'$ SD = 14.7'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	180°14.7	N22°50.9	39°36.4	14.5'	S18°30.1	11.0'	54.0'
1	195°14.5	51.1	54°09.9	14.4'	18°41.1	11.0'	54.0'
2	210°14.4	51.3	68°43.3	14.3'	18°52.1	10.9'	54.0'
3	225°14.3	· · 51.6	83°16.6	14.3'	19°03.0	10.8'	54.0'
4	240°14.2	51.8	97°49.9	14.2'	19°13.9	10.8'	54.0'
5	255°14.1	52.0	112°23.1	14.1'	19°24.6	10.7'	54.0'
6	270°13.9	N22°52.2	126°56.2	14.0'	S19°35.3	10.6'	54.0'
7	285°13.8	52.4	141°29.2	14.0'	19°45.9	10.5'	54.0'
8	300°13.7	52.7	156°02.2	13.9'	19°56.4	10.4'	54.0'
9	315°13.6	· · 52.9	170°35.1	13.8'	20°06.9	10.4'	54.1'
10	330°13.5	53.1	185°07.9	13.8'	20°17.2	10.3'	54.1'
11	345°13.3	53.3	199°40.7	13.7'	20°27.5	10.2'	54.1'
12	0°13.2	N22°53.5	214°13.4	13.6'	S20°37.7	10.1'	54.1'
13	15°13.1	53.7	228°46.0	13.5'	20°47.8	10.0'	54.1'
14	30°13.0	54.0	243°18.5	13.5'	20°57.9	9.9'	54.1'
15	45°12.9	· · 54.2	257°51.0	13.4'	21°07.8	9.9'	54.1'
16	60°12.7	54.4	272°23.3	13.3'	21°17.7	9.8'	54.1'
17	75°12.6	54.6	286°55.6	13.2'	21°27.4	9.7'	54.1'
18	90°12.5	N22°54.8	301°27.9	13.2'	S21°37.1	9.6'	54.1'
19	105°12.4	55.0	316°00.0	13.1'	21°46.7	9.5'	54.1'
20	120°12.2	55.2	330°32.1	13.0'	21°56.2	9.4'	54.1'
21	135°12.1	· · 55.4	345°04.1	12.9'	22°05.7	9.3'	54.1'
22	150°12.0	55.6	359°36.0	12.8'	22°15.0	9.2'	54.1'
23	165°11.9	55.9	14°07.9	12.8'	22°24.2	9.1'	54.1'
SD = 15.8'			$d = 0.2'$ SD = 14.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°	--	--	00:44	23:19	--	--
64°	--	--	01:44	22:15	--	--
62°	--	--	02:17	21:42	--	--
60°	--	01:09	02:42	21:17	22:51	--
N 58°	--	01:50	03:01	20:58	22:10	--
56°	--	02:17	03:17	20:42	21:42	--
54°	01:03	02:37	03:30	20:28	21:21	22:57
52°	01:41	02:54	03:42	20:16	21:04	22:19
50°	02:06	03:09	03:52	20:06	20:50	21:53
45°	02:48	03:37	04:14	19:44	20:21	21:10
N 40°	03:18	03:59	04:31	19:27	19:59	20:40
35°	03:40	04:16	04:46	19:12	19:42	20:18
30°	03:58	04:31	04:58	19:00	19:27	20:00
20°	04:26	04:55	05:20	18:38	19:03	19:32
N 10°	04:48	05:15	05:38	18:20	18:43	19:10
0°	05:07	05:33	05:55	18:03	18:25	18:51
S 10°	05:23	05:49	06:12	17:46	18:08	18:35
20°	05:39	06:06	06:30	17:28	17:52	18:19
30°	05:55	06:25	06:51	17:07	17:33	18:03
35°	06:03	06:35	07:03	16:55	17:23	17:55
40°	06:12	06:46	07:16	16:41	17:12	17:46
45°	06:22	06:59	07:33	16:25	16:59	17:36
S 50°	06:33	07:15	07:53	16:05	16:43	17:24
52°	06:38	07:22	08:02	15:56	16:36	17:19
54°	06:44	07:30	08:13	15:45	16:28	17:14
56°	06:50	07:38	08:25	15:33	16:19	17:08
58°	06:56	07:48	08:39	15:19	16:10	17:02
S 60°	07:03	07:59	08:55	15:02	15:58	16:54

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	17:20			00:10 23:25		
N 70°	16:57	19:19		00:22 23:50	22:56	
68°	16:39	18:39		00:33	00:10 23:37	
66°	16:24	18:12	20:21	00:41	00:26	00:05 23:29
64°	16:13	17:51	19:41	00:48	00:39	00:27
62°	16:03	17:35	19:13	00:55	00:50	00:44
60°	15:54	17:21	18:52	01:00	00:59	00:59
N 58°	15:47	17:09	18:34	01:05	01:08	01:11
56°	15:41	16:59	18:20	01:09	01:15	01:22
54°	15:35	16:50	18:08	01:13	01:21	01:32
52°	15:30	16:42	17:57	01:17	01:27	01:40
50°	15:25	16:35	17:47	01:20	01:33	01:48
45°	15:15	16:20	17:27	01:27	01:44	02:05
N 40°	15:07	16:08	17:10	01:33	01:54	02:18
35°	15:00	15:57	16:57	01:38	02:03	02:30
30°	14:53	15:48	16:45	01:43	02:10	02:40
20°	14:43	15:33	16:24	01:51	02:23	02:58
N 10°	14:33	15:19	16:07	01:57	02:34	03:13
0°	14:25	15:06	15:51	02:04	02:45	03:27
S 10°	14:16	14:54	15:34	02:11	02:55	03:42
20°	14:07	14:41	15:17	02:18	03:07	03:57
30°	13:57	14:25	14:57	02:26	03:20	04:15
35°	13:51	14:17	14:46	02:30	03:27	04:26
40°	13:44	14:07	14:33	02:36	03:36	04:38
45°	13:37	13:55	14:18	02:42	03:46	04:52
S 50°	13:27	13:41	13:59	02:49	03:59	05:10
52°	13:23	13:35	13:50	02:53	04:05	05:18
54°	13:18	13:28	13:40	02:57	04:11	05:27
56°	13:13	13:20	13:29	03:01	04:18	05:38
58°	13:08	13:11	13:16	03:06	04:26	05:50
S 60°	13:01	13:01	13:01	03:11	04:35	06:04

Day	Sun		Mer. Pass	Moon		Age
	Eqn.of Time 00 ^h mm:ss	Time 12 ^h mm:ss		Upper Mer.Pass. hh:mm	Lower Mer.Pass. hh:mm	
06	01:21	01:15	11:59	20:35	08:14	
07	01:10	01:04	11:59	21:17	08:55	
08	00:59	00:53	11:59	22:02	09:39	

June 09, 10, 11 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	257°37.1	226°09.4	N10°06.5	109°30.6	N14°18.8	167°50.0	N23°16.4	255°52.5	S01°34.6	Alpheratz	357°34.5	29°13.7
1	272°39.5	241°09.4	07.3	124°31.8	18.3	182°51.9	16.4	270°54.9	34.5	Ankaa	353°06.9	-42°09.8
2	287°42.0	256°09.4	08.1	139°32.9	17.8	197°53.7	16.4	285°57.2	34.5	Schedar	349°30.9	56°40.3
3	302°44.5	271°09.5	· · 08.8	154°34.1	· · 17.3	212°55.6	· · 16.4	300°59.5	· · 34.5	Diphda	348°47.1	-17°50.7
4	317°46.9	286°09.5	09.6	169°35.2	16.8	227°57.4	16.4	316°01.9	34.4	Achernar	335°20.3	-57°06.2
5	332°49.4	301°09.5	10.4	184°36.4	16.3	242°59.3	16.4	331°04.2	34.4	Hamal	327°51.1	23°34.9
6	347°51.9	316°09.6	N10°11.2	199°37.5	N14°15.8	258°01.1	N23°16.4	346°06.5	S01°34.3	Polaris	314°17.0	89°22.1
7	2°54.3	331°09.6	12.0	214°38.7	15.3	273°03.0	16.4	1°08.9	34.3	Acamar	315°11.9	-40°12.0
8	17°56.8	346°09.6	12.7	229°39.8	14.8	288°04.8	16.4	16°11.2	34.2	Menkar	314°06.1	4°11.3
9	32°59.2	1°09.7	· · 13.5	244°41.0	· · 14.3	303°06.7	· · 16.4	31°13.5	· · 34.2	Mirfak	308°28.2	49°56.9
10	48°01.7	16°09.7	14.3	259°42.1	13.8	318°08.5	16.4	46°15.9	34.1	Aldebaran	290°39.6	16°33.6
11	63°04.2	31°09.7	15.1	274°43.3	13.3	333°10.4	16.4	61°18.2	34.1	Rigel	281°03.9	-8°10.3
12	78°06.6	46°09.7	N10°15.9	289°44.4	N14°12.8	348°12.2	N23°16.4	76°20.5	S01°34.0	Capella	280°21.9	46°01.4
13	93°09.1	61°09.8	16.7	304°45.6	12.3	3°14.1	16.4	91°22.9	34.0	Bellatrix	278°22.9	6°22.4
14	108°11.6	76°09.8	17.4	319°46.7	11.8	18°15.9	16.4	106°25.2	34.0	Elnath	278°01.8	28°37.7
15	123°14.0	91°09.8	· · 18.2	334°47.9	· · 11.3	33°17.8	· · 16.5	121°27.5	· · 33.9	Alnilam	275°37.7	-1°11.1
16	138°16.5	106°09.8	19.0	349°49.0	10.8	48°19.6	16.5	136°29.9	33.9	Betelgeuse	270°52.1	7°24.7
17	153°19.0	121°09.9	19.8	4°50.2	10.3	63°21.5	16.5	151°32.2	33.8	Canopus	263°52.8	-52°42.6
18	168°21.4	136°09.9	N10°20.6	19°51.3	N14°09.8	78°23.3	N23°16.5	166°34.6	S01°33.8	Sirius	258°26.3	-16°45.1
19	183°23.9	151°09.9	21.3	34°52.5	09.3	93°25.1	16.5	181°36.9	33.7	Adhara	255°06.0	-29°00.4
20	198°26.4	166°09.9	22.1	49°53.6	08.8	108°27.0	16.5	196°39.2	33.7	Procyon	244°50.8	5°09.6
21	213°28.8	181°09.9	· · 22.9	64°54.8	· · 08.3	123°28.8	· · 16.5	211°41.6	· · 33.7	Pollux	243°17.2	27°58.0
22	228°31.3	196°10.0	23.7	79°55.9	07.8	138°30.7	16.5	226°43.9	33.6	Avior	234°15.1	-59°35.6
23	243°33.7	211°10.0	24.5	94°57.1	07.3	153°32.5	16.5	241°46.2	33.6	Suhail	222°46.3	-43°32.3
Mer.pass. 06:48		ν 0.0' d 0.8' m -4.31		ν 1.2' d -0.5' m 1.35		ν 1.8' d 0.0' m -1.90		ν 2.3' d -0.0' m 1.05				

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	258°36.2	226°10.0	N10°25.2	109°58.2	N14°06.8	168°34.4	N23°16.5	256°48.6	S01°33.5		
1	273°38.7	241°10.0	26.0	124°59.3	06.3	183°36.2	16.5	271°50.9	33.5		
2	288°41.1	256°10.0	26.8	140°00.5	05.8	198°38.1	16.5	286°53.2	33.4		
3	303°43.6	271°10.1	· · 27.6	155°01.6	· · 05.3	213°39.9	· · 16.5	301°55.6	· · 33.4		
4	318°46.1	286°10.1	28.4	170°02.8	04.8	228°41.8	16.5	316°57.9	33.3		
5	333°48.5	301°10.1	29.2	185°03.9	04.3	243°43.6	16.5	332°00.3	33.3		
6	348°51.0	316°10.1	N10°29.9	200°05.1	N14°03.8	258°45.5	N23°16.5	347°02.6	S01°33.3		
7	3°53.5	331°10.1	30.7	215°06.2	03.3	273°47.3	16.5	2°04.9	33.2		
8	18°55.9	346°10.1	31.5	230°07.4	02.8	288°49.2	16.5	17°07.3	33.2		
9	33°58.4	1°10.1	· · 32.3	245°08.5	· · 02.3	303°51.0	· · 16.5	32°09.6	· · 33.1		
10	49°00.9	16°10.1	33.1	260°09.7	01.8	318°52.9	16.5	47°12.0	33.1		
11	64°03.3	31°10.2	33.9	275°10.8	01.3	333°54.7	16.5	62°14.3	33.0		
12	79°05.8	46°10.2	N10°34.6	290°12.0	N14°00.8	348°56.5	N23°16.5	77°16.6	S01°33.0		
13	94°08.2	61°10.2	35.4	305°13.1	14°00.3	3°58.4	16.5	92°19.0	33.0		
14	109°10.7	76°10.2	36.2	320°14.3	13°59.8	19°00.2	16.5	107°21.3	32.9		
15	124°13.2	91°10.2	· · 37.0	335°15.4	· · 59.3	34°02.1	· · 16.5	122°23.6	· · 32.9		
16	139°15.6	106°10.2	37.8	350°16.6	58.8	49°03.9	16.6	137°26.0	32.8		
17	154°18.1	121°10.2	38.5	5°17.7	58.3	64°05.8	16.6	152°28.3	32.8		
18	169°20.6	136°10.2	N10°39.3	20°18.9	N13°57.7	79°07.6	N23°16.6	167°30.7	S01°32.7		
19	184°23.0	151°10.2	40.1	35°20.0	57.2	94°09.5	16.6	182°33.0	32.7		
20	199°25.5	166°10.2	40.9	50°21.2	56.7	109°11.3	16.6	197°35.3	32.7		
21	214°28.0	181°10.2	· · 41.7	65°22.3	· · 56.2	124°13.2	· · 16.6	212°37.7	· · 32.6		
22	229°30.4	196°10.2	42.5	80°23.5	55.7	139°15.0	16.6	227°40.0	32.6		
23	244°32.9	211°10.2	43.2	95°24.6	55.2	154°16.9	16.6	242°42.4	32.5		
Mer.pass. 06:44		ν 0.0' d 0.8' m -4.30		ν 1.1' d -0.5' m 1.36		ν 1.8' d 0.0' m -1.90		ν 2.3' d -0.0' m 1.05			

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	259°35.4	226°10.2	N10°44.0	110°25.7	N13°54.7	169°18.7	N23°16.6	257°44.7	S01°32.5		
1	274°37.8	241°10.2	44.8	125°26.9	54.2	184°20.5	16.6	272°47.0	32.4		
2	289°40.3	256°10.3	45.6	140°28.0	53.7	199°22.4	16.6	287°49.4	32.4		
3	304°42.7	271°10.3	· · 46.4	155°29.2	· · 53.2	214°24.2	· · 16.6	302°51.7	· · 32.4		
4	319°45.2	286°10.3	47.2	170°30.3	52.7	229°26.1	16.6	317°54.1	32.3		
5	334°47.7	301°10.3	47.9	185°31.5	52.2	244°27.9	16.6	332°56.4	32.3		
6	349°50.1	316°10.3	N10°48.7	200°32.6	N13°51.7	259°29.8	N23°16.6	347°58.8	S01°32.2		
7	4°52.6	331°10.2	49.5	215°33.8	51.2	274°31.6	16.6	3°01.1	32.2		
8	19°55.1	346°10.2	50.3	230°34.9	50.7	289°33.5	16.6	18°03.4	32.2		
9	34°57.5	1°10.2	· · 51.1	245°36.1	· · 50.2	304°35.3	· · 16.6	33°05.8	· · 32.1		
10	50°00.0	16°10.2	51.9	260°37.2	49.7	319°37.2	16.6	48°08.1	32.1		
11	65°02.5	31°10.2	52.6	275°38.4	49.2	334°39.0	16.6	63°10.5	32.0		
12	80°04.9	46°10.2	N10°53.4	290°39.5	N13°48.7	349°40.9	N23°16.6	78°12.8	S01°32.0		
13	95°07.4	61°10.2	54.2	305°40.7	48.2	4°42.7	16.6	93°15.1	31.9		
14	110°09.8	76°10.2	55.0	320°41.8	47.6	19°44.6	16.6	108°17.5	31.9		
15	125°12.3	91°10.2	· · 55.8	335°42.9	· · 47.1	34°46.4	· · 16.6	123°19.8	· · 31.9		
16	140°14.8	106°10.2	56.5	350°44.1	46.6	49°48.2	16.6	138°22.2	31.8		
17	155°17.2	121°10.2	57.3	5°45.2	46.1	64°50.1	16.6	153°24.5	31.8		
18	170°19.7	136°10.2	N10°58.1	20°46.4	N13°45.6	79°51.9	N23°16.6	168°26.9	S01°31.7		
19	185°22.2	151°10.2	58.9	35°47.5	45.1	94°53.8	16.6	183°29.2	31.7		
20	200°24.6	166°10.2	10°59.7	50°48.7	44.6	109°55.6	16.6	198°31.5	31.7		
21	215°27.1	181°10.2	11°00.5	65°49.8	· · 44.1	124°57.5	· · 16.6	213°33.9	· · 31.6		
22	230°29.6	196°10.1	01.2	80°51.0	43.6	139°59.3	16.6	228°36.2	31.6		
23	245°32.0	211°10.1	02.0	95°52.1	43.1	155°01.2	16.6	243°38.6	31.5		
Mer.pass. 06:41		ν 0.0' d 0.8' m -4.29		ν 1.1' d -0.5' m 1.37		ν 1.8' d 0.0' m -1.90		ν 2.3' d -0.0' m 1.04			

Jun 09 Mon			SHA	Mer.pass
Venus			328°32.3	08:55
Mars			211°53.5	16:41
Jupiter			270°13.0	12:47
Saturn			358°15.5	06:55

Jun 10 Tue			SHA	Mer.pass
Venus			327°33.8	08:55
Mars			211°22.0	16:39
Jupiter			269°58.2	12:44
Saturn			358°12.4	06:52

Jun 11 Wed			SHA	Mer.pass
Venus			326°34.9	08:55
Mars			210°50.4	16:37
Jupiter			269°43.4	12:41
Saturn			358°09.4	06:48

Horizontal parallax			
Venus:			0.2
Mars:			0.1

Sun				Moon			
h	GHA	Dec		GHA	ν	Dec	HP
0	180°11.8	N22°56.1		28°39.6	12.7'	S22°33.4	9.0'
1	195°11.6	56.3		43°11.3	12.6'	22°42.4	9.0'
2	210°11.5	56.5		57°42.9	12.5'	22°51.3	8.9'
3	225°11.4	56.7		72°14.4	12.5'	23°00.2	8.8'
4	240°11.3	56.9		86°45.9	12.4'	23°09.0	8.7'
5	255°11.2	57.1		101°17.3	12.3'	23°17.6	8.6'
6	270°11.0	N22°57.3		115°48.6	12.2'	S23°26.2	8.5'
7	285°10.9	57.5		130°19.8	12.1'	23°34.6	8.4'
8	300°10.8	57.7		144°50.9	12.1'	23°43.0	8.3'
9	315°10.7	57.9		159°22.0	12.0'	23°51.2	8.1'
10	330°10.5	58.1		173°53.0	11.9'	23°59.4	8.0'
11	345°10.4	58.3		188°23.9	11.8'	24°07.4	7.9'
12	0°10.3	N22°58.5		202°54.7	11.7'	S24°15.4	7.8'
13	15°10.2	58.7		217°25.4	11.7'	24°23.2	7.7'
14	30°10.0	58.9		231°56.1	11.6'	24°30.9	7.6'
15	45°09.9	59.1		246°26.7	11.5'	24°38.5	7.5'
16	60°09.8	59.3		260°57.2	11.4'	24°46.1	7.4'
17	75°09.7	59.5		275°27.6	11.4'	24°53.4	7.3'
18	90°09.6	N22°59.7		289°58.0	11.3'	S25°00.7	7.2'
19	105°09.4	22°59.9		304°28.3	11.2'	25°07.9	7.1'
20	120°09.3	23°00.1		318°58.5	11.1'	25°15.0	6.9'
21	135°09.2	00.3		333°28.6	11.1'	25°21.9	6.8'
22	150°09.1	00.4		347°58.6	11.0'	25°28.7	6.7'
23	165°08.9	00.6		2°28.6	10.9'	25°35.5	6.6'
SD = 15.8'			d = 0.2'	SD = 14.8'			

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	180°08.8	N23°00.8	16°58.5	10.8'	S25°42.0	6.5'	54.4'
1	195°08.7	01.0	31°28.3	10.8'	25°48.5	6.4'	54.4'
2	210°08.6	01.2	45°58.1	10.7'	25°54.9	6.2'	54.4'
3	225°08.4	01.4	60°27.8	10.6'	26°01.1	6.1'	54.4'
4	240°08.3	01.6	74°57.4	10.5'	26°07.2	6.0'	54.4'
5	255°08.2	01.8	89°26.9	10.5'	26°13.2	5.9'	54.4'
6	270°08.1	N23°02.0	103°56.4	10.4'	S26°19.1	5.7'	54.4'
7	285°07.9	02.1	118°25.8	10.3'	26°24.8	5.6'	54.4'
8	300°07.8	02.3	132°55.1	10.3'	26°30.5	5.5'	54.5'
9	315°07.7	02.5	147°24.3	10.2'	26°36.0	5.4'	54.5'
10	330°07.6	02.7	161°53.5	10.1'	26°41.3	5.2'	54.5'
11	345°07.4	02.9	176°22.6	10.0'	26°46.6	5.1'	54.5'
12	0°07.3	N23°03.1	190°51.7	10.0'	S26°51.7	5.0'	54.5'
13	15°07.2	03.2	205°20.7	9.9'	26°56.7	4.9'	54.5'
14	30°07.1	03.4	219°49.6	9.9'	27°01.5	4.7'	54.5'
15	45°06.9	03.6	234°18.4	9.8'	27°06.3	4.6'	54.5'
16	60°06.8	03.8	248°47.2	9.7'	27°10.9	4.5'	54.6'
17	75°06.7	04.0	263°15.9	9.7'	27°15.3	4.3'	54.6'
18	90°06.6	N23°04.1	277°44.6	9.6'	S27°19.6	4.2'	54.6'
19	105°06.4	04.3	292°13.2	9.5'	27°23.8	4.1'	54.6'
20	120°06.3	04.5	306°41.7	9.5'	27°27.9	3.9'	54.6'
21	135°06.2	04.7	321°10.2	9.4'	27°31.8	3.8'	54.6'
22	150°06.1	04.8	335°38.6	9.4'	27°35.6	3.6'	54.6'
23	165°05.9	05.0	350°07.0	9.3'	27°39.3	3.5'	54.7'
SD = 15.7'			d = 0.2'	SD = 14.8'			

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	180°05.8	N23°05.2	4°35.3	9.3'	S27°42.8	3.4'	54.7'
1	195°05.7	05.4	19°03.6	9.2'	27°46.1	3.2'	54.7'
2	210°05.6	05.5	33°31.8	9.2'	27°49.4	3.1'	54.7'
3	225°05.4	05.7	47°59.9	9.1'	27°52.5	3.0'	54.7'
4	240°05.3	05.9	62°28.1	9.1'	27°55.4	2.8'	54.7'
5	255°05.2	06.0	76°56.1	9.0'	27°58.2	2.7'	54.7'
6	270°05.0	N23°06.2	91°24.1	9.0'	S28°00.9	2.5'	54.8'
7	285°04.9	06.4	105°52.1	8.9'	28°03.4	2.4'	54.8'
8	300°04.8	06.6	120°20.0	8.9'	28°05.8	2.2'	54.8'
9	315°04.7	06.7	134°47.9	8.8'	28°08.1	2.1'	54.8'
10	330°04.5	06.9	149°15.7	8.8'	28°10.2	2.0'	54.8'
11	345°04.4	07.1	163°43.5	8.7'	28°12.1	1.8'	54.8'
12	0°04.3	N23°07.2	178°11.2	8.7'	S28°13.9	1.7'	54.9'
13	15°04.2	07.4	192°38.9	8.7'	28°15.6	1.5'	54.9'
14	30°04.0	07.5	207°06.6	8.6'	28°17.1	1.4'	54.9'
15	45°03.9	07.7	221°34.3	8.6'	28°18.5	1.2'	54.9'
16	60°03.8	07.9	236°01.9	8.6'	28°19.7	1.1'	54.9'
17	75°03.6	08.0	250°29.4	8.5'	28°20.8	0.9'	54.9'
18	90°03.5	N23°08.2	264°57.0	8.5'	S28°21.7	0.8'	54.9'
19	105°03.4	08.4	279°24.5	8.5'	28°22.5	0.6'	55.0'
20	120°03.3	08.5	293°52.0	8.5'	28°23.1	0.5'	55.0'
21	135°03.1	08.7	308°19.4	8.4'	28°23.6	0.3'	55.0'
22	150°03.0	08.8	322°46.9	8.4'	28°23.9	0.2'	55.0'
23	165°02.9	09.0	337°14.3	8.4'	28°24.1	0.0'	55.0'
SD = 15.7'			d = 0.2'	SD = 14.9'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°	—:—	—:—	00:27	23:39	—:—	—:—
64°	—:—	—:—	01:39	22:21	—:—	—:—
62°	—:—	—:—	02:14	21:46	—:—	—:—
60°	—:—	01:02	02:39	21:21	22:59	—:—
N 58°	—:—	01:46	02:59	21:01	22:14	—:—
56°	—:—	02:14	03:15	20:45	21:46	—:—
54°	00:57	02:35	03:29	20:31	21:25	23:04
52°	01:37	02:52	03:41	20:19	21:07	22:23
50°	02:03	03:07	03:51	20:08	20:52	21:57
45°	02:47	03:36	04:13	19:46	20:23	21:12
N 40°	03:17	03:58	04:31	19:28	20:01	20:42
35°	03:40	04:16	04:46	19:14	19:43	20:20
30°	03:58	04:31	04:58	19:01	19:28	20:01
20°	04:26	04:55	05:20	18:39	19:04	19:33
N 10°	04:49	05:16	05:39	18:20	18:43	19:10
0°	05:07	05:33	05:56	18:03	18:26	18:52
S 10°	05:24	05:50	06:13	17:46	18:09	18:35
20°	05:40	06:07	06:31	17:28	17:52	18:19
30°	05:56	06:26	06:52	17:07	17:33	18:03
35°	06:04	06:36	07:04	16:55	17:23	17:54
40°	06:14	06:48	07:18	16:41	17:11	17:45
45°	06:24	07:01	07:35	16:24	16:58	17:35
S 50°	06:35	07:17	07:55	16:04	16:42	17:24
52°	06:40	07:24	08:04	15:54	16:35	17:19
54°	06:46	07:32	08:15	15:44	16:27	17:13
56°	06:52	07:41	08:28	15:31	16:18	17:07
58°	06:58	07:51	08:42	15:17	16:08	17:00
S 60°	07:06	08:02	08:59	15:00	15:57	16:53

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°						
N 70°						
68°						
66°						
64°	22:04			00:10 23:26		
62°	21:00	23:07		00:38	00:30	00:10
60°	20:26	21:59	23:12	01:00	01:05	01:19
N 58°	20:01	21:24	22:31	01:18	01:30	01:54
56°	19:42	20:59	22:03	01:33	01:51	02:19
54°	19:25	20:39	21:42	01:46	02:07	02:40
52°	19:11	20:22	21:24	01:58	02:22	02:57
50°	18:59	20:08	21:08	02:08	02:35	03:11
45°	18:34	19:38	20:38	02:29	03:01	03:41
N 40°	18:13	19:15	20:14	02:47	03:21	04:04
35°	17:57	18:57	19:54	03:01	03:39	04:23
30°	17:42	18:41	19:37	03:14	03:54	04:40
20°	17:18	18:13	19:09	03:36	04:19	05:07
N 10°	16:57	17:50	18:44	03:55	04:41	05:31
0°	16:38	17:28	18:21	04:13	05:02	05:54
S 10°	16:19	17:07	17:59	04:31	05:23	06:16
20°	15:58	16:44	17:34	04:50	05:45	06:40
30°	15:34	16:17	17:06	05:12	06:11	07:08
35°	15:20	16:01	16:49	05:26	06:26	07:25
40°	15:04	15:43	16:30	05:41	06:44	07:44
45°	14:45	15:21	16:06	05:59	07:05	08:07
S 50°	14:22	14:53	15:36	06:21	07:32	08:37
52°	14:10	14:39	15:21	06:32	07:45	08:52
54°	13:58	14:24	15:04	06:45	08:01	09:10
56°	13:43	14:05	14:43	06:59	08:19	09:31
58°	13:26	13:43	14:16	07:16	08:41	09:57
S 60°	13:05	13:14	13:38	07:36	09:10	10:35

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
09	00:47	00:41	11:59	22:50	10:25	
10	00:35	00:29	12:00	23:41	11:15	
11	00:23	00:17	12:00	—:—	12:08	

June 12, 13, 14 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	260°34.5	226°10.1	N11°02.8	110°53.3	N13°42.6	170°03.0	N23°16.6	258°40.9	S01°31.5					Alpheratz	357°34.5	29°13.7	
1	275°37.0	241°10.1	03.6	125°54.4	42.1	185°04.9	16.6	273°43.3	31.5					Ankaa	353°06.9	-42°09.8	
2	290°39.4	256°10.1	04.4	140°55.5	41.6	200°06.7	16.6	288°45.6	31.4					Schedar	349°30.8	56°40.3	
3	305°41.9	271°10.1	· · 05.2	155°56.7	· · 41.0	215°08.5	· · 16.6	303°47.9	· · 31.4					Diphda	348°47.1	-17°50.7	
4	320°44.3	286°10.1	05.9	170°57.8	40.5	230°10.4	16.6	318°50.3	31.3					Achernar	335°20.2	-57°06.2	
5	335°46.8	301°10.0	06.7	185°59.0	40.0	245°12.2	16.6	333°52.6	31.3					Hamal	327°51.0	23°34.9	
6	350°49.3	316°10.0	N11°07.5	201°00.1	N13°39.5	260°14.1	N23°16.7	348°55.0	S01°31.3					Polaris	314°15.8	89°22.1	
7	5°51.7	331°10.0	08.3	216°01.3	39.0	275°15.9	16.7	3°57.3	31.2					Acamar	315°11.9	-40°12.0	
8	20°54.2	346°10.0	09.1	231°02.4	38.5	290°17.8	16.7	18°59.7	31.2					Menkar	314°06.1	4°11.4	
9	35°56.7	1°10.0	· · 09.8	246°03.6	· · 38.0	305°19.6	· · 16.7	34°02.0	· · 31.1					Mirfak	308°28.2	49°56.9	
10	50°59.1	16°10.0	10.6	261°04.7	37.5	320°21.5	16.7	49°04.4	31.1					Aldebaran	290°39.6	16°33.6	
11	66°01.6	31°09.9	11.4	276°05.8	37.0	335°23.3	16.7	64°06.7	31.1					Rigel	281°03.9	-8°10.3	
12	81°04.1	46°09.9	N11°12.2	291°07.0	N13°36.5	350°25.2	N23°16.7	79°09.1	S01°31.0					Capella	280°21.9	46°01.4	
13	96°06.5	61°09.9	13.0	306°08.1	36.0	5°27.0	16.7	94°11.4	31.0					Bellatrix	278°22.8	6°22.4	
14	111°09.0	76°09.9	13.8	321°09.3	35.4	20°28.8	16.7	109°13.7	30.9					Elnath	278°01.8	28°37.7	
15	126°11.5	91°09.9	· · 14.5	336°10.4	· · 34.9	35°30.7	· · 16.7	124°16.1	· · 30.9					Alnilam	275°37.7	-1°11.1	
16	141°13.9	106°09.8	15.3	351°11.6	34.4	50°32.5	16.7	139°18.4	30.9					Betelgeuse	270°52.1	7°24.7	
17	156°16.4	121°09.8	16.1	6°12.7	33.9	65°34.4	16.7	154°20.8	30.8					Canopus	263°52.8	-52°42.6	
18	171°18.8	136°09.8	N11°16.9	21°13.9	N13°33.4	80°36.2	N23°16.7	169°23.1	S01°30.8					Sirius	258°26.3	-16°45.1	
19	186°21.3	151°09.8	17.7	36°15.0	32.9	95°38.1	16.7	184°25.5	30.7					Adhara	255°06.0	-29°00.4	
20	201°23.8	166°09.7	18.5	51°16.1	32.4	110°39.9	16.7	199°27.8	30.7					Procyon	244°50.8	5°09.6	
21	216°26.2	181°09.7	· · 19.2	66°17.3	· · 31.9	125°41.8	· · 16.7	214°30.2	· · 30.7					Pollux	243°17.2	27°57.9	
22	231°28.7	196°09.7	20.0	81°18.4	31.4	140°43.6	16.7	229°32.5	30.6					Avior	234°15.1	-59°35.6	
23	246°31.2	211°09.7	20.8	96°19.6	30.8	155°45.5	16.7	244°34.9	30.6					Suhail	222°46.3	-43°32.3	
Mer.pass. 06:37			ν -0.0' d 0.8' m -4.29	ν 1.1' d -0.5' m 1.37	ν 1.8' d 0.0' m -1.90	ν 2.3' d -0.0' m 1.04											

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	261°33.6	226°09.6	N11°21.6	111°20.7	N13°30.3	170°47.3	N23°16.7	259°37.2	S01°30.5
1	276°36.1	241°09.6	22.4	126°21.9	29.8	185°49.1	16.7	274°39.6	30.5
2	291°38.6	256°09.6	23.1	141°23.0	29.3	200°51.0	16.7	289°41.9	30.5
3	306°41.0	271°09.5	· · 23.9	156°24.1	· · 28.8	215°52.8	· · 16.7	304°44.3	· · 30.4
4	321°43.5	286°09.5	24.7	171°25.3	28.3	230°54.7	16.7	319°46.6	30.4
5	336°46.0	301°09.5	25.5	186°26.4	27.8	245°56.5	16.7	334°48.9	30.3
6	351°48.4	316°09.4	N11°26.3	201°27.6	N13°27.3	260°58.4	N23°16.7	349°51.3	S01°30.3
7	6°50.9	331°09.4	27.1	216°28.7	26.7	276°00.2	16.7	4°53.6	30.3
8	21°53.3	346°09.4	27.8	231°29.9	26.2	291°02.1	16.7	19°56.0	30.2
9	36°55.8	1°09.3	· · 28.6	246°31.0	· · 25.7	306°03.9	· · 16.7	34°58.3	· · 30.2
10	51°58.3	16°09.3	29.4	261°32.2	25.2	321°05.8	16.7	50°00.7	30.2
11	67°00.7	31°09.3	30.2	276°33.3	24.7	336°07.6	16.7	65°03.0	30.1
12	82°03.2	46°09.2	N11°31.0	291°34.4	N13°24.2	351°09.4	N23°16.7	80°05.4	S01°30.1
13	97°05.7	61°09.2	31.7	306°35.6	23.7	6°11.3	16.7	95°07.7	30.0
14	112°08.1	76°09.2	32.5	321°36.7	23.2	21°13.1	16.7	110°10.1	30.0
15	127°10.6	91°09.1	· · 33.3	336°37.9	· · 22.6	36°15.0	· · 16.7	125°12.4	· · 30.0
16	142°13.1	106°09.1	34.1	351°39.0	22.1	51°16.8	16.7	140°14.8	29.9
17	157°15.5	121°09.0	34.9	6°40.1	21.6	66°18.7	16.7	155°17.1	29.9
18	172°18.0	136°09.0	N11°35.7	21°41.3	N13°21.1	81°20.5	N23°16.7	170°19.5	S01°29.9
19	187°20.5	151°09.0	36.4	36°42.4	20.6	96°22.4	16.7	185°21.8	29.8
20	202°22.9	166°08.9	37.2	51°43.6	20.1	111°24.2	16.7	200°24.2	29.8
21	217°25.4	181°08.9	· · 38.0	66°44.7	· · 19.6	126°26.0	· · 16.7	215°26.5	· · 29.7
22	232°27.8	196°08.8	38.8	81°45.9	19.0	141°27.9	16.7	230°28.9	29.7
23	247°30.3	211°08.8	39.6	96°47.0	18.5	156°29.7	16.7	245°31.2	29.7
Mer.pass. 06:33			ν -0.0' d 0.8' m -4.28	ν 1.1' d -0.5' m 1.38	ν 1.8' d 0.0' m -1.90	ν 2.3' d -0.0' m 1.04			

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	262°32.8	226°08.8	N11°40.3	111°48.1	N13°18.0	171°31.6	N23°16.7	260°33.6	S01°29.6
1	277°35.2	241°08.7	41.1	126°49.3	17.5	186°33.4	16.7	275°35.9	29.6
2	292°37.7	256°08.7	41.9	141°50.4	17.0	201°35.3	16.7	290°38.3	29.6
3	307°40.2	271°08.6	· · 42.7	156°51.6	· · 16.5	216°37.1	· · 16.7	305°40.6	· · 29.5
4	322°42.6	286°08.6	43.5	171°52.7	15.9	231°39.0	16.7	320°43.0	29.5
5	337°45.1	301°08.5	44.2	186°53.9	15.4	246°40.8	16.7	335°45.3	29.4
6	352°47.6	316°08.5	N11°45.0	201°55.0	N13°14.9	261°42.7	N23°16.7	350°47.7	S01°29.4
7	7°50.0	331°08.4	45.8	216°56.1	14.4	276°44.5	16.7	5°50.0	29.4
8	22°52.5	346°08.4	46.6	231°57.3	13.9	291°46.3	16.7	20°52.4	29.3
9	37°55.0	1°08.3	· · 47.4	246°58.4	· · 13.4	306°48.2	· · 16.7	35°54.7	· · 29.3
10	52°57.4	16°08.3	48.1	261°59.6	12.8	321°50.0	16.7	50°57.1	29.3
11	67°59.9	31°08.2	48.9	277°00.7	12.3	336°51.9	16.7	65°59.5	29.2
12	83°02.3	46°08.2	N11°49.7	292°01.8	N13°11.8	351°53.7	N23°16.7	81°01.8	S01°29.2
13	98°04.8	61°08.1	50.5	307°03.0	11.3	6°55.6	16.7	96°04.2	29.1
14	113°07.3	76°08.1	51.3	322°04.1	10.8	21°57.4	16.7	111°06.5	29.1
15	128°09.7	91°08.0	· · 52.0	337°05.3	· · 10.3	36°59.3	· · 16.7	126°08.9	· · 29.1
16	143°12.2	106°08.0	52.8	352°06.4	09.7	52°01.1	16.7	141°11.2	29.0
17	158°14.7	121°07.9	53.6	7°07.5	09.2	67°02.9	16.7	156°13.6	29.0
18	173°17.1	136°07.9	N11°54.4	22°08.7	N13°08.7	82°04.8	N23°16.7	171°15.9	S01°29.0
19	188°19.6	151°07.8	55.2	37°09.8	08.2	97°06.6	16.7	186°18.3	28.9
20	203°22.1	166°07.8	55.9	52°11.0	07.7	112°08.5	16.7	201°20.6	28.9
21	218°24.5	181°07.7	· · 56.7	67°12.1	· · 07.1	127°10.3	· · 16.7	216°23.0	· · 28.9
22	233°27.0	196°07.6	57.5	82°13.3	06.6	142°12.2	16.7	231°25.3	28.8
23	248°29.4	211°07.6	58.3	97°14.4	06.1	157°14.0	16.7	246°27.7	28.8
Mer.pass. 06:29			ν -0.0' d 0.8' m -4.27	ν 1.1' d -0.5' m 1.39	ν 1.8' d 0.0' m -1.90	ν 2.4' d -0.0' m 1.03			

Jun 12 Thu	SHA	Mer.pass
Venus	325°35.6	08:55
Mars	210°18.8	16:35
Jupiter	269°28.5	12:38
Saturn	358°06.4	06:44

Jun 13 Fri	SHA	Mer.pass
Venus	324°36.0	08:55
Mars	209°47.1	16:33
Jupiter	269°13.7	12:35
Saturn	358°03.6	06:40

Jun 14 Sat	SHA	Mer.pass
Venus	323°36.0	08:55
Mars	209°15.4	16:32
Jupiter	268°58.8	12:32
Saturn	358°00.8	06:37

Horizontal parallax		
Venus:	0.2	
Mars:	0.1	

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	180°02.7	N23°09.1	351°41.7	8.4'	S28°24.1	-0.1'	55.0'
1	195°02.6	09.3	6°09.0	8.4'	28°24.0	-0.3'	55.1'
2	210°02.5	09.5	20°36.4	8.3'	28°23.7	-0.4'	55.1'
3	225°02.4	· · 09.6	35°03.7	8.3'	28°23.3	-0.6'	55.1'
4	240°02.2	09.8	49°31.0	8.3'	28°22.7	-0.7'	55.1'
5	255°02.1	09.9	63°58.4	8.3'	28°22.0	-0.9'	55.1'
6	270°02.0	N23°10.1	78°25.6	8.3'	S28°21.1	-1.0'	55.1'
7	285°01.8	10.2	92°52.9	8.3'	28°20.1	-1.2'	55.2'
8	300°01.7	10.4	107°20.2	8.3'	28°18.9	-1.3'	55.2'
9	315°01.6	· · 10.5	121°47.5	8.3'	28°17.6	-1.5'	55.2'
10	330°01.5	10.7	136°14.7	8.3'	28°16.1	-1.6'	55.2'
11	345°01.3	10.8	150°42.0	8.3'	28°14.5	-1.8'	55.2'
12	0°01.2	N23°11.0	165°09.2	8.3'	S28°12.7	-1.9'	55.3'
13	15°01.1	11.1	179°36.5	8.3'	28°10.8	-2.1'	55.3'
14	30°00.9	11.3	194°03.8	8.3'	28°08.7	-2.2'	55.3'
15	45°00.8	· · 11.4	208°31.0	8.3'	28°06.5	-2.4'	55.3'
16	60°00.7	11.6	222°58.3	8.3'	28°04.1	-2.5'	55.3'
17	75°00.6	11.7	237°25.5	8.3'	28°01.5	-2.7'	55.3'
18	90°00.4	N23°11.8	251°52.8	8.3'	S27°58.8	-2.8'	55.4'
19	105°00.3	12.0	266°20.1	8.3'	27°56.0	-3.0'	55.4'
20	120°00.2	12.1	280°47.4	8.3'	27°53.0	-3.1'	55.4'
21	135°00.0	· · 12.3	295°14.7	8.3'	27°49.8	-3.3'	55.4'
22	149°59.9	12.4	309°42.0	8.3'	27°46.6	-3.4'	55.4'
23	164°59.8	12.6	324°09.3	8.3'	27°43.1	-3.6'	55.5'
SD = 15.7'			SD = 15.0'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	179°59.6	N23°12.7	338°36.7	8.4'	S27°39.5	-3.7'	55.5'
1	194°59.5	12.8	353°04.1	8.4'	27°35.8	-3.9'	55.5'
2	209°59.4	13.0	7°31.4	8.4'	27°31.9	-4.0'	55.5'
3	224°59.3	· · 13.1	21°58.8	8.4'	27°27.8	-4.2'	55.5'
4	239°59.1	13.2	36°26.3	8.4'	27°23.6	-4.3'	55.5'
5	254°59.0	13.4	50°53.7	8.5'	27°19.3	-4.5'	55.6'
6	269°58.9	N23°13.5	65°21.2	8.5'	S27°14.8	-4.6'	55.6'
7	284°58.7	13.7	79°48.7	8.5'	27°10.1	-4.8'	55.6'
8	299°58.6	13.8	94°16.2	8.6'	27°05.3	-4.9'	55.6'
9	314°58.5	· · 13.9	108°43.8	8.6'	27°00.4	-5.1'	55.6'
10	329°58.3	14.1	123°11.4	8.6'	26°55.3	-5.2'	55.7'
11	344°58.2	14.2	137°39.0	8.6'	26°50.1	-5.4'	55.7'
12	359°58.1	N23°14.3	152°06.6	8.7'	S26°44.7	-5.5'	55.7'
13	14°57.9	14.4	166°34.3	8.7'	26°39.2	-5.7'	55.7'
14	29°57.8	14.6	181°02.0	8.8'	26°33.5	-5.8'	55.7'
15	44°57.7	· · 14.7	195°29.8	8.8'	26°27.7	-6.0'	55.8'
16	59°57.6	14.8	209°57.6	8.8'	26°21.7	-6.1'	55.8'
17	74°57.4	15.0	224°25.4	8.9'	26°15.6	-6.2'	55.8'
18	89°57.3	N23°15.1	238°53.3	8.9'	S26°09.4	-6.4'	55.8'
19	104°57.2	15.2	253°21.2	8.9'	26°03.0	-6.5'	55.8'
20	119°57.0	15.3	267°49.1	9.0'	25°56.5	-6.7'	55.9'
21	134°56.9	· · 15.5	282°17.1	9.0'	25°49.8	-6.8'	55.9'
22	149°56.8	15.6	296°45.1	9.1'	25°43.0	-6.9'	55.9'
23	164°56.6	15.7	311°13.2	9.1'	25°36.1	-7.1'	55.9'
SD = 15.7'			SD = 15.1'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	179°56.5	N23°15.8	325°41.3	9.2'	S25°29.0	-7.2'	55.9'
1	194°56.4	16.0	340°09.5	9.2'	25°21.8	-7.4'	56.0'
2	209°56.2	16.1	354°37.7	9.3'	25°14.4	-7.5'	56.0'
3	224°56.1	· · 16.2	9°05.9	9.3'	25°06.9	-7.6'	56.0'
4	239°56.0	16.3	23°34.3	9.4'	24°59.3	-7.8'	56.0'
5	254°55.8	16.4	38°02.6	9.4'	24°51.5	-7.9'	56.0'
6	269°55.7	N23°16.6	52°31.0	9.5'	S24°43.6	-8.0'	56.1'
7	284°55.6	16.7	66°59.5	9.5'	24°35.6	-8.2'	56.1'
8	299°55.4	16.8	81°28.0	9.6'	24°27.4	-8.3'	56.1'
9	314°55.3	· · 16.9	95°56.5	9.6'	24°19.1	-8.4'	56.1'
10	329°55.2	17.0	110°25.1	9.7'	24°10.7	-8.6'	56.1'
11	344°55.1	17.1	124°53.8	9.7'	24°02.1	-8.7'	56.2'
12	359°54.9	N23°17.3	139°22.5	9.8'	S23°53.4	-8.8'	56.2'
13	14°54.8	17.4	153°51.3	9.8'	23°44.6	-9.0'	56.2'
14	29°54.7	17.5	168°20.1	9.9'	23°35.6	-9.1'	56.2'
15	44°54.5	· · 17.6	182°49.0	9.9'	23°26.6	-9.2'	56.3'
16	59°54.4	17.7	197°17.9	10.0'	23°17.3	-9.3'	56.3'
17	74°54.3	17.8	211°46.9	10.0'	23°08.0	-9.5'	56.3'
18	89°54.1	N23°17.9	226°15.9	10.1'	S22°58.6	-9.6'	56.3'
19	104°54.0	18.0	240°45.0	10.1'	22°49.0	-9.7'	56.3'
20	119°53.9	18.1	255°14.1	10.2'	22°39.3	-9.8'	56.4'
21	134°53.7	· · 18.2	269°43.3	10.3'	22°29.5	-9.9'	56.4'
22	149°53.6	18.4	284°12.6	10.3'	22°19.5	-10.1'	56.4'
23	164°53.5	18.5	298°41.9	10.4'	22°09.5	-10.2'	56.4'
SD = 15.7'			SD = 15.3'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	☐	☐	☐	☐	☐	☐
64°	—:—	—:—	01:35	22:26	—:—	—:—
62°	—:—	—:—	02:11	21:50	—:—	—:—
60°	—:—	00:56	02:37	21:24	23:06	—:—
N 58°	—:—	01:43	02:57	21:04	22:18	—:—
56°	—:—	02:12	03:14	20:47	21:49	—:—
54°	00:51	02:34	03:28	20:33	21:27	23:11
52°	01:34	02:51	03:40	20:21	21:09	22:27
50°	02:01	03:06	03:50	20:10	20:54	21:59
45°	02:46	03:35	04:13	19:48	20:25	21:14
N 40°	03:16	03:58	04:31	19:30	20:03	20:44
35°	03:39	04:16	04:45	19:15	19:45	20:21
30°	03:58	04:31	04:58	19:02	19:29	20:03
20°	04:27	04:56	05:20	18:40	19:05	19:34
N 10°	04:49	05:16	05:39	18:21	18:44	19:11
0°	05:08	05:34	05:56	18:04	18:26	18:52
S 10°	05:25	05:51	06:14	17:46	18:09	18:36
20°	05:41	06:08	06:32	17:28	17:52	18:20
30°	05:57	06:27	06:53	17:07	17:33	18:03
35°	06:06	06:37	07:05	16:55	17:23	17:55
40°	06:15	06:49	07:19	16:41	17:11	17:45
45°	06:25	07:02	07:36	16:24	16:58	17:35
S 50°	06:37	07:18	07:57	16:03	16:42	17:23
52°	06:42	07:26	08:06	15:54	16:34	17:18
54°	06:48	07:34	08:17	15:43	16:26	17:13
56°	06:54	07:43	08:30	15:30	16:17	17:06
58°	07:00	07:53	08:44	15:16	16:07	17:00
S 60°	07:08	08:04	09:02	14:58	15:56	16:52

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	■	■	■	■	■	■
64°	■	■	01:28	■	■	03:27
62°	■	00:52	00:34	■	02:11	04:20
60°	23:48	· ·	00:02	01:57	03:14	04:52
N 58°	23:14	23:37	23:49	02:38	03:48	05:16
56°	22:49	23:18	23:35	03:06	04:13	05:35
54°	22:29	23:01	23:23	03:27	04:33	05:51
52°	22:12	22:47	23:12	03:45	04:49	06:05
50°	21:58	22:35	23:02	04:01	05:04	06:17
45°	21:28	22:09	22:42	04:32	05:33	06:41
N 40°	21:05	21:49	22:25	04:56	05:55	07:01
35°	20:46	21:32	22:11	05:15	06:14	07:17
30°	20:30	21:17	21:59	05:32	06:30	07:31
20°	20:02	20:52	21:37	06:00	06:57	07:55
N 10°	19:38	20:30	21:19	06:25	07:20	08:16
0°	19:16	20:10	21:02	06:48	07:42	08:35
S 10°	18:53	19:49	20:44	07:10	08:03	08:54
20°	18:30	19:27	20:26	07:34	08:26	09:14
30°	18:02	19:02	20:04	08:03	08:53	09:37
35°	17:45	18:47	19:51	08:19	09:08	09:51
40°	17:26	18:29	19:37	08:39	09:26	10:06
45°	17:03	18:08	19:19	09:02	09:48	10:25
S 50°	16:33	17:41	18:57	09:33	10:16	10:48
52°	16:18	17:28	18:47	09:48	10:29	10:59
54°	16:00	17:13	18:35	10:05	10:44	11:11
56°	15:40	16:55	18:22	10:26	11:03	11:25
58°	15:13	16:34	18:06	10:52	11:24	11:42
S 60°	14:36	16:06	17:47	11:29	11:53	12:01

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	16-18
				hh:mm	hh:mm	99-93%
12	00:11	00:05	12:00	00:34	13:02	
13	00:01	00:08	12:00	01:29	13:56	
14	00:14	00:20	12:00	02:22	14:48	

June 15, 16, 17 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	263°31.9	226°07.5	N11°59.1	112°15.5	N13°05.6	172°15.9	N23°16.7	261°30.0	S01°28.7	Alpheratz	357°34.5	29°13.7
1	278°34.4	241°07.5	11°59.8	127°16.7	05.1	187°17.7	16.7	276°32.4	28.7	Ankaa	353°06.9	-42°09.8
2	293°36.8	256°07.4	12°00.6	142°17.8	04.6	202°19.5	16.7	291°34.8	28.7	Schedar	349°30.8	56°40.4
3	308°39.3	271°07.3	· · 01.4	157°19.0	· · 04.0	217°21.4	· · 16.7	306°37.1	· · 28.6	Diphda	348°47.0	-17°50.7
4	323°41.8	286°07.3	02.2	172°20.1	03.5	232°23.2	16.7	321°39.5	28.6	Achernar	335°20.2	-57°06.2
5	338°44.2	301°07.2	02.9	187°21.2	03.0	247°25.1	16.7	336°41.8	28.6	Hamal	327°51.0	23°34.9
6	353°46.7	316°07.2	N12°03.7	202°22.4	N13°02.5	262°26.9	N23°16.7	351°44.2	S01°28.5	Polaris	314°14.7	89°22.1
7	8°49.2	331°07.1	04.5	217°23.5	02.0	277°28.8	16.7	6°46.5	28.5	Acamar	315°11.8	-40°12.0
8	23°51.6	346°07.0	05.3	232°24.7	01.4	292°30.6	16.7	21°48.9	28.5	Menkar	314°06.0	4°11.4
9	38°54.1	1°07.0	· · 06.1	247°25.8	· · 00.9	307°32.5	· · 16.7	36°51.2	· · 28.4	Mirfak	308°28.2	49°56.9
10	53°56.6	16°06.9	06.8	262°26.9	13°00.4	322°34.3	16.7	51°53.6	28.4	Aldebaran	290°39.6	16°33.6
11	68°59.0	31°06.8	07.6	277°28.1	12°59.9	337°36.1	16.7	66°56.0	28.4	Rigel	281°03.9	-8°10.3
12	84°01.5	46°06.8	N12°08.4	292°29.2	N12°59.4	352°38.0	N23°16.7	81°58.3	S01°28.3	Capella	280°21.8	46°01.4
13	99°03.9	61°06.7	09.2	307°30.4	58.8	7°39.8	16.7	97°00.7	28.3	Bellatrix	278°22.8	6°22.4
14	114°06.4	76°06.6	10.0	322°31.5	58.3	22°41.7	16.7	112°03.0	28.2	Elnath	278°01.8	28°37.7
15	129°08.9	91°06.6	· · 10.7	337°32.6	· · 57.8	37°43.5	· · 16.7	127°05.4	· · 28.2	Alnilam	275°37.7	-1°11.1
16	144°11.3	106°06.5	11.5	352°33.8	57.3	52°45.4	16.7	142°07.7	28.2	Betelgeuse	270°52.0	7°24.7
17	159°13.8	121°06.4	12.3	7°34.9	56.7	67°47.2	16.7	157°10.1	28.1	Canopus	263°52.8	-52°42.5
18	174°16.3	136°06.4	N12°13.1	22°36.0	N12°56.2	82°49.0	N23°16.7	172°12.4	S01°28.1	Sirius	258°26.3	-16°45.1
19	189°18.7	151°06.3	13.8	37°37.2	55.7	97°50.9	16.7	187°14.8	28.1	Adhara	255°06.0	-29°00.4
20	204°21.2	166°06.2	14.6	52°38.3	55.2	112°52.7	16.7	202°17.2	28.0	Procyon	244°50.8	5°09.6
21	219°23.7	181°06.1	· · 15.4	67°39.5	· · 54.7	127°54.6	· · 16.7	217°19.5	· · 28.0	Pollux	243°17.2	27°57.9
22	234°26.1	196°06.1	16.2	82°40.6	54.1	142°56.4	16.7	232°21.9	28.0	Avior	234°15.1	-59°35.6
23	249°28.6	211°06.0	16.9	97°41.7	53.6	157°58.3	16.7	247°24.2	27.9	Suhail	222°46.4	-43°32.3
Mer.pass. 06:25		ν -0.1' d 0.8' m -4.26		ν 1.1' d -0.5' m 1.40		ν 1.8' d -0.0' m -1.90		ν 2.4' d -0.0' m 1.03				

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	264°31.1	226°05.9	N12°17.7	112°42.9	N12°53.1	173°00.1	N23°16.7	262°26.6	S01°27.9
1	279°33.5	241°05.8	18.5	127°44.0	52.6	188°02.0	16.7	277°28.9	27.9
2	294°36.0	256°05.8	19.3	142°45.2	52.0	203°03.8	16.6	292°31.3	27.8
3	309°38.4	271°05.7	· · 20.1	157°46.3	· · 51.5	218°05.6	· · 16.6	307°33.7	· · 27.8
4	324°40.9	286°05.6	20.8	172°47.4	51.0	233°07.5	16.6	322°36.0	27.8
5	339°43.4	301°05.5	21.6	187°48.6	50.5	248°09.3	16.6	337°38.4	27.7
6	354°45.8	316°05.5	N12°22.4	202°49.7	N12°50.0	263°11.2	N23°16.6	352°40.7	S01°27.7
7	9°48.3	331°05.4	23.2	217°50.9	49.4	278°13.0	16.6	7°43.1	27.7
8	24°50.8	346°05.3	23.9	232°52.0	48.9	293°14.9	16.6	22°45.5	27.6
9	39°53.2	1°05.2	· · 24.7	247°53.1	· · 48.4	308°16.7	· · 16.6	37°47.8	· · 27.6
10	54°55.7	16°05.1	25.5	262°54.3	47.9	323°18.6	16.6	52°50.2	27.6
11	69°58.2	31°05.1	26.3	277°55.4	47.3	338°20.4	16.6	67°52.5	27.5
12	85°00.6	46°05.0	N12°27.0	292°56.5	N12°46.8	353°22.2	N23°16.6	82°54.9	S01°27.5
13	100°03.1	61°04.9	27.8	307°57.7	46.3	8°24.1	16.6	97°57.2	27.5
14	115°05.6	76°04.8	28.6	322°58.8	45.8	23°25.9	16.6	112°59.6	27.4
15	130°08.0	91°04.7	· · 29.4	338°00.0	· · 45.2	38°27.8	· · 16.6	128°02.0	· · 27.4
16	145°10.5	106°04.6	30.1	353°01.1	44.7	53°29.6	16.6	143°04.3	27.4
17	160°12.9	121°04.6	30.9	8°02.2	44.2	68°31.5	16.6	158°06.7	27.3
18	175°15.4	136°04.5	N12°31.7	23°03.4	N12°43.7	83°33.3	N23°16.6	173°09.0	S01°27.3
19	190°17.9	151°04.4	32.5	38°04.5	43.1	98°35.1	16.6	188°11.4	27.3
20	205°20.3	166°04.3	33.2	53°05.6	42.6	113°37.0	16.6	203°13.8	27.2
21	220°22.8	181°04.2	· · 34.0	68°06.8	· · 42.1	128°38.8	· · 16.6	218°16.1	· · 27.2
22	235°25.3	196°04.1	34.8	83°07.9	41.6	143°40.7	16.6	233°18.5	27.2
23	250°27.7	211°04.0	35.6	98°09.1	41.0	158°42.5	16.6	248°20.9	27.1
Mer.pass. 06:21		ν -0.1' d 0.8' m -4.25		ν 1.1' d -0.5' m 1.40		ν 1.8' d -0.0' m -1.90		ν 2.4' d -0.0' m 1.02	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	265°30.2	226°03.9	N12°36.3	113°10.2	N12°40.5	173°44.4	N23°16.6	263°23.2	S01°27.1
1	280°32.7	241°03.9	37.1	128°11.3	40.0	188°46.2	16.6	278°25.6	27.1
2	295°35.1	256°03.8	37.9	143°12.5	39.5	203°48.0	16.6	293°27.9	27.0
3	310°37.6	271°03.7	· · 38.6	158°13.6	· · 38.9	218°49.9	· · 16.6	308°30.3	· · 27.0
4	325°40.0	286°03.6	39.4	173°14.7	38.4	233°51.7	16.6	323°32.7	27.0
5	340°42.5	301°03.5	40.2	188°15.9	37.9	248°53.6	16.6	338°35.0	26.9
6	355°45.0	316°03.4	N12°41.0	203°17.0	N12°37.4	263°55.4	N23°16.6	353°37.4	S01°26.9
7	10°47.4	331°03.3	41.7	218°18.1	36.8	278°57.3	16.6	8°39.7	26.9
8	25°49.9	346°03.2	42.5	233°19.3	36.3	293°59.1	16.6	23°42.1	26.8
9	40°52.4	1°03.1	· · 43.3	248°20.4	· · 35.8	309°01.0	· · 16.6	38°44.5	· · 26.8
10	55°54.8	16°03.0	44.1	263°21.6	35.2	324°02.8	16.6	53°46.8	26.8
11	70°57.3	31°02.9	44.8	278°22.7	34.7	339°04.6	16.6	68°49.2	26.7
12	85°59.8	46°02.8	N12°45.6	293°23.8	N12°34.2	354°06.5	N23°16.6	83°51.6	S01°26.7
13	101°02.2	61°02.7	46.4	308°25.0	33.7	9°08.3	16.6	98°53.9	26.7
14	116°04.7	76°02.6	47.1	323°26.1	33.1	24°10.2	16.6	113°56.3	26.6
15	131°07.2	91°02.5	· · 47.9	338°27.2	· · 32.6	39°12.0	· · 16.5	128°58.6	· · 26.6
16	146°09.6	106°02.4	48.7	353°28.4	32.1	54°13.9	16.5	144°01.0	26.6
17	161°12.1	121°02.3	49.5	8°29.5	31.5	69°15.7	16.5	159°03.4	26.5
18	176°14.5	136°02.2	N12°50.2	23°30.6	N12°31.0	84°17.5	N23°16.5	174°05.7	S01°26.5
19	191°17.0	151°02.1	51.0	38°31.8	30.5	99°19.4	16.5	189°08.1	26.5
20	206°19.5	166°02.0	51.8	53°32.9	30.0	114°21.2	16.5	204°10.5	26.5
21	221°21.9	181°01.9	· · 52.5	68°34.1	· · 29.4	129°23.1	· · 16.5	219°12.8	· · 26.4
22	236°24.4	196°01.8	53.3	83°35.2	28.9	144°24.9	16.5	234°15.2	26.4
23	251°26.9	211°01.7	54.1	98°36.3	28.4	159°26.8	16.5	249°17.6	26.4
Mer.pass. 06:17		ν -0.1' d 0.8' m -4.24		ν 1.1' d -0.5' m 1.41		ν 1.8' d -0.0' m -1.90		ν 2.4' d -0.0' m 1.02	

Jun 15 Sun			SHA	Mer.pass
Venus			322°35.6	08:56
Mars			208°43.6	16:30
Jupiter			268°43.9	12:29
Saturn			357°58.1	06:33

Jun 16 Mon			SHA	Mer.pass
Venus			321°34.9	08:56
Mars			208°11.8	16:28
Jupiter			268°29.1	12:26
Saturn			357°55.5	06:29

Jun 17 Tue			SHA	Mer.pass
Venus			320°33.8	08:56
Mars			207°40.0	16:26
Jupiter			268°14.2	12:24
Saturn			357°53.0	06:25

Horizontal parallax			
Venus:			0.2
Mars:			0.1

Sun				Moon				
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	179°53.3	N23°18.6		313°11.3	10.4'	S21°59.3	-10.3'	56.5'
1	194°53.2	18.7		327°40.7	10.5'	21°49.0	-10.4'	56.5'
2	209°53.1	18.8		342°10.2	10.5'	21°38.6	-10.5'	56.5'
3	224°52.9	•• 18.9		356°39.7	10.6'	21°28.0	-10.6'	56.5'
4	239°52.8	19.0		11°09.3	10.7'	21°17.4	-10.8'	56.5'
5	254°52.7	19.1		25°39.0	10.7'	21°06.6	-10.9'	56.6'
6	269°52.5	N23°19.2		40°08.7	10.8'	S20°55.8	-11.0'	56.6'
7	284°52.4	19.3		54°38.5	10.8'	20°44.8	-11.1'	56.6'
8	299°52.3	19.4		69°08.3	10.9'	20°33.7	-11.2'	56.6'
9	314°52.1	•• 19.5		83°38.2	10.9'	20°22.5	-11.3'	56.7'
10	329°52.0	19.6		98°08.1	11.0'	20°11.1	-11.4'	56.7'
11	344°51.9	19.7		112°38.1	11.0'	19°59.7	-11.5'	56.7'
12	359°51.7	N23°19.8		127°08.1	11.1'	S19°48.2	-11.6'	56.7'
13	14°51.6	19.9		141°38.2	11.2'	19°36.6	-11.7'	56.7'
14	29°51.5	20.0		156°08.4	11.2'	19°24.8	-11.8'	56.8'
15	44°51.3	•• 20.1		170°38.6	11.3'	19°13.0	-11.9'	56.8'
16	59°51.2	20.2		185°08.9	11.3'	19°01.0	-12.1'	56.8'
17	74°51.1	20.3		199°39.2	11.4'	18°49.0	-12.2'	56.8'
18	89°50.9	N23°20.3		214°09.6	11.4'	S18°36.8	-12.3'	56.9'
19	104°50.8	20.4		228°40.0	11.5'	18°24.6	-12.4'	56.9'
20	119°50.7	20.5		243°10.5	11.5'	18°12.2	-12.4'	56.9'
21	134°50.5	•• 20.6		257°41.0	11.6'	17°59.8	-12.5'	56.9'
22	149°50.4	20.7		272°11.6	11.6'	17°47.2	-12.6'	57.0'
23	164°50.3	20.8		286°42.2	11.7'	17°34.6	-12.7'	57.0'
SD = 15.7' d = 0.1'				SD = 15.4'				

Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	179°50.1	N23°20.9		301°12.9	11.7'	S17°21.8	-12.8'	57.0'
1	194°50.0	21.0		315°43.6	11.8'	17°09.0	-12.9'	57.0'
2	209°49.8	21.1		330°14.4	11.8'	16°56.1	-13.0'	57.0'
3	224°49.7	•• 21.1		344°45.2	11.9'	16°43.1	-13.1'	57.1'
4	239°49.6	21.2		359°16.1	11.9'	16°30.0	-13.2'	57.1'
5	254°49.4	21.3		13°47.0	12.0'	16°16.8	-13.3'	57.1'
6	269°49.3	N23°21.4		28°18.0	12.0'	S16°03.5	-13.4'	57.1'
7	284°49.2	21.5		42°49.0	12.1'	15°50.1	-13.5'	57.2'
8	299°49.0	21.6		57°20.1	12.1'	15°36.7	-13.5'	57.2'
9	314°48.9	•• 21.7		71°51.2	12.2'	15°23.2	-13.6'	57.2'
10	329°48.8	21.7		86°22.4	12.2'	15°09.5	-13.7'	57.2'
11	344°48.6	21.8		100°53.6	12.2'	14°55.8	-13.8'	57.3'
12	359°48.5	N23°21.9		115°24.8	12.3'	S14°42.0	-13.9'	57.3'
13	14°48.4	22.0		129°56.1	12.3'	14°28.2	-13.9'	57.3'
14	29°48.2	22.0		144°27.4	12.4'	14°14.2	-14.0'	57.3'
15	44°48.1	•• 22.1		158°58.8	12.4'	14°00.2	-14.1'	57.4'
16	59°48.0	22.2		173°30.2	12.4'	13°46.1	-14.2'	57.4'
17	74°47.8	22.3		188°01.6	12.5'	13°31.9	-14.3'	57.4'
18	89°47.7	N23°22.4		202°33.1	12.5'	S13°17.7	-14.3'	57.4'
19	104°47.6	22.4		217°04.6	12.5'	13°03.4	-14.4'	57.5'
20	119°47.4	22.5		231°36.1	12.6'	12°49.0	-14.5'	57.5'
21	134°47.3	•• 22.6		246°07.7	12.6'	12°34.5	-14.5'	57.5'
22	149°47.2	22.7		260°39.3	12.6'	12°20.0	-14.6'	57.5'
23	164°47.0	22.7		275°11.0	12.7'	12°05.3	-14.7'	57.6'
SD = 15.7' d = 0.1'				SD = 15.5'				

Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	179°46.9	N23°22.8		289°42.7	12.7'	S11°50.7	-14.7'	57.6'
1	194°46.7	22.9		304°14.4	12.7'	11°35.9	-14.8'	57.6'
2	209°46.6	22.9		318°46.1	12.8'	11°21.1	-14.9'	57.6'
3	224°46.5	•• 23.0		333°17.9	12.8'	11°06.2	-14.9'	57.7'
4	239°46.3	23.1		347°49.7	12.8'	10°51.3	-15.0'	57.7'
5	254°46.2	23.1		2°21.5	12.8'	10°36.3	-15.1'	57.7'
6	269°46.1	N23°23.2		16°53.4	12.9'	S10°21.2	-15.1'	57.7'
7	284°45.9	23.3		31°25.2	12.9'	10°06.1	-15.2'	57.8'
8	299°45.8	23.3		45°57.1	12.9'	09°50.9	-15.2'	57.8'
9	314°45.7	•• 23.4		60°29.0	12.9'	09°35.7	-15.3'	57.8'
10	329°45.5	23.5		75°01.0	13.0'	09°20.4	-15.4'	57.8'
11	344°45.4	23.5		89°32.9	13.0'	09°05.0	-15.4'	57.9'
12	359°45.3	N23°23.6		104°04.9	13.0'	S08°49.6	-15.5'	57.9'
13	14°45.1	23.7		118°36.9	13.0'	08°34.1	-15.5'	57.9'
14	29°45.0	23.7		133°08.9	13.0'	08°18.6	-15.6'	57.9'
15	44°44.8	•• 23.8		147°41.0	13.0'	08°03.1	-15.6'	58.0'
16	59°44.7	23.8		162°13.0	13.1'	07°47.4	-15.7'	58.0'
17	74°44.6	23.9		176°45.1	13.1'	07°31.8	-15.7'	58.0'
18	89°44.4	N23°24.0		191°17.1	13.1'	S07°16.0	-15.8'	58.0'
19	104°44.3	24.0		205°49.2	13.1'	07°00.3	-15.8'	58.1'
20	119°44.2	24.1		220°21.3	13.1'	06°44.5	-15.9'	58.1'
21	134°44.0	•• 24.1		234°53.4	13.1'	06°28.6	-15.9'	58.1'
22	149°43.9	24.2		249°25.5	13.1'	06°12.7	-15.9'	58.1'
23	164°43.8	24.2		263°57.6	13.1'	05°56.8	-16.0'	58.2'
SD = 15.7' d = 0.1'				SD = 15.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°						
64°	—:—	—:—	01:32	22:30	—:—	—:—
62°	—:—	—:—	02:10	21:52	—:—	—:—
60°	—:—	00:52	02:36	21:26	23:11	—:—
N 58°	—:—	01:41	02:56	21:06	22:21	—:—
56°	—:—	02:11	03:13	20:49	21:51	—:—
54°	00:47	02:33	03:27	20:35	21:29	23:15
52°	01:33	02:51	03:39	20:22	21:11	22:29
50°	02:00	03:06	03:50	20:11	20:56	22:02
45°	02:46	03:35	04:13	19:49	20:26	21:16
N 40°	03:16	03:58	04:31	19:31	20:04	20:46
35°	03:39	04:16	04:46	19:16	19:46	20:22
30°	03:58	04:31	04:59	19:03	19:30	20:04
20°	04:27	04:56	05:21	18:41	19:05	19:35
N 10°	04:49	05:16	05:40	18:22	18:45	19:12
0°	05:08	05:35	05:57	18:04	18:27	18:53
S 10°	05:25	05:52	06:14	17:47	18:10	18:36
20°	05:41	06:09	06:33	17:29	17:53	18:20
30°	05:58	06:28	06:54	17:07	17:34	18:04
35°	06:07	06:38	07:07	16:55	17:23	17:55
40°	06:16	06:50	07:21	16:41	17:11	17:45
45°	06:26	07:04	07:37	16:24	16:58	17:35
S 50°	06:38	07:20	07:58	16:03	16:42	17:23
52°	06:43	07:27	08:08	15:53	16:34	17:18
54°	06:49	07:35	08:19	15:42	16:26	17:12
56°	06:55	07:44	08:32	15:30	16:17	17:06
58°	07:02	07:55	08:46	15:15	16:07	17:00
S 60°	07:09	08:06	09:04	14:58	15:55	16:52

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°		02:38	01:21		05:53	08:50
N 70°		01:49	01:02		06:39	09:06
68°	02:31	01:18	00:47	04:14	07:09	09:19
66°	01:27	00:54	00:35	05:16	07:31	09:30
64°	00:52	00:36	00:24	05:50	07:49	09:38
62°	00:26	00:20	00:15	06:15	08:03	09:46
60°	00:06	00:08	00:07	06:34	08:15	09:52
N 58°	23:56	•• •	00:01	06:50	08:25	09:58
56°	23:47	23:55	•• •	07:04	08:34	10:03
54°	23:38	23:49	23:58	07:16	08:42	10:07
52°	23:30	23:44	23:56	07:26	08:49	10:11
50°	23:23	23:40	23:54	07:35	08:55	10:15
45°	23:08	23:30	23:50	07:54	09:08	10:22
N 40°	22:55	23:22	23:46	08:10	09:19	10:29
35°	22:45	23:15	23:42	08:23	09:29	10:34
30°	22:35	23:08	23:39	08:34	09:37	10:39
20°	22:19	22:58	23:34	08:54	09:51	10:47
N 10°	22:05	22:48	23:30	09:10	10:03	10:55
0°	21:51	22:39	23:26	09:26	10:14	11:01
S 10°						

June 18, 19, 20 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	266°29.3	226°01.6	N12°54.9	113°37.5	N12°27.8	174°28.6	N23°16.5	264°19.9	S01°26.3	Alpheratz	357°34.4	29°13.7
1	281°31.8	241°01.5	55.6	128°38.6	27.3	189°30.4	16.5	279°22.3	26.3	Ankaa	353°06.8	-42°09.8
2	296°34.3	256°01.4	56.4	143°39.7	26.8	204°32.3	16.5	294°24.7	26.3	Schedar	349°30.8	56°40.4
3	311°36.7	271°01.3	•• 57.2	158°40.9	•• 26.3	219°34.1	•• 16.5	309°27.0	•• 26.2	Diphda	348°47.0	-17°50.7
4	326°39.2	286°01.2	57.9	173°42.0	25.7	234°36.0	16.5	324°29.4	26.2	Achernar	335°20.2	-57°06.2
5	341°41.7	301°01.1	58.7	188°43.1	25.2	249°37.8	16.5	339°31.8	26.2	Hamal	327°51.0	23°34.9
6	356°44.1	316°01.0	N12°59.5	203°44.3	N12°24.7	264°39.7	N23°16.5	354°34.1	S01°26.1	Polaris	314°13.7	89°22.1
7	11°46.6	331°00.9	13°00.2	218°45.4	24.1	279°41.5	16.5	9°36.5	26.1	Acamar	315°11.8	-40°12.0
8	26°49.0	346°00.8	01.0	233°46.5	23.6	294°43.3	16.5	24°38.8	26.1	Menkar	314°06.0	4°11.4
9	41°51.5	1°00.6	•• 01.8	248°47.7	•• 23.1	309°45.2	•• 16.5	39°41.2	•• 26.1	Mirfak	308°28.2	49°56.9
10	56°54.0	16°00.5	02.5	263°48.8	22.5	324°47.0	16.5	54°43.6	26.0	Aldebaran	290°39.5	16°33.6
11	71°56.4	31°00.4	03.3	278°49.9	22.0	339°48.9	16.5	69°45.9	26.0	Rigel	281°03.9	-8°10.3
12	86°58.9	46°00.3	N13°04.1	293°51.1	N12°21.5	354°50.7	N23°16.5	84°48.3	S01°26.0	Capella	280°21.8	46°01.4
13	102°01.4	61°00.2	04.8	308°52.2	20.9	9°52.6	16.5	99°50.7	25.9	Bellatrix	278°22.8	6°22.4
14	117°03.8	76°00.1	05.6	323°53.4	20.4	24°54.4	16.5	114°53.0	25.9	Elnath	278°01.8	28°37.7
15	132°06.3	91°00.0	•• 06.4	338°54.5	•• 19.9	39°56.3	•• 16.5	129°55.4	•• 25.9	Alnilam	275°37.7	-1°11.1
16	147°08.8	105°59.9	07.1	353°55.6	19.4	54°58.1	16.4	144°57.8	25.8	Betelgeuse	270°52.0	7°24.7
17	162°11.2	120°59.7	07.9	8°56.8	18.8	69°59.9	16.4	160°00.1	25.8	Canopus	263°52.8	-52°42.5
18	177°13.7	135°59.6	N13°08.7	23°57.9	N12°18.3	85°01.8	N23°16.4	175°02.5	S01°25.8	Sirius	258°26.3	-16°45.1
19	192°16.1	150°59.5	09.5	38°59.0	17.8	100°03.6	16.4	190°04.9	25.8	Adhara	255°06.0	-29°00.4
20	207°18.6	165°59.4	10.2	54°00.2	17.2	115°05.5	16.4	205°07.2	25.7	Procyon	244°50.8	5°09.6
21	222°21.1	180°59.3	•• 11.0	69°01.3	•• 16.7	130°07.3	•• 16.4	220°09.6	•• 25.7	Pollux	243°17.2	27°57.9
22	237°23.5	195°59.2	11.8	84°02.4	16.2	145°09.2	16.4	235°12.0	25.7	Avior	234°15.1	-59°35.6
23	252°26.0	210°59.0	12.5	99°03.6	15.6	160°11.0	16.4	250°14.4	25.6	Suhail	222°46.4	-43°32.3
Mer.pass. 06:13		ν -0.1' d 0.8' m -4.23		ν 1.1' d -0.5' m 1.42		ν 1.8' d -0.0' m -1.90		ν 2.4' d -0.0' m 1.02				

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	267°28.5	225°58.9	N13°13.3	114°04.7	N12°15.1	175°12.8	N23°16.4	265°16.7	S01°25.6			
1	282°30.9	240°58.8	14.0	129°05.8	14.6	190°14.7	16.4	280°19.1	25.6			
2	297°33.4	255°58.7	14.8	144°07.0	14.0	205°16.5	16.4	295°21.5	25.5			
3	312°35.9	270°58.6	•• 15.6	159°08.1	•• 13.5	220°18.4	•• 16.4	310°23.8	•• 25.5			
4	327°38.3	285°58.4	16.3	174°09.2	13.0	235°20.2	16.4	325°26.2	25.5			
5	342°40.8	300°58.3	17.1	189°10.4	12.4	250°22.1	16.4	340°28.6	25.5			
6	357°43.3	315°58.2	N13°17.9	204°11.5	N12°11.9	265°23.9	N23°16.4	355°30.9	S01°25.4			
7	12°45.7	330°58.1	18.6	219°12.6	11.4	280°25.7	16.4	10°33.3	25.4			
8	27°48.2	345°57.9	19.4	234°13.8	10.8	295°27.6	16.4	25°35.7	25.4			
9	42°50.6	0°57.8	•• 20.2	249°14.9	•• 10.3	310°29.4	•• 16.4	40°38.0	•• 25.3			
10	57°53.1	15°57.7	20.9	264°16.0	09.8	325°31.3	16.4	55°40.4	25.3			
11	72°55.6	30°57.6	21.7	279°17.2	09.2	340°33.1	16.4	70°42.8	25.3			
12	87°58.0	45°57.4	N13°22.5	294°18.3	N12°08.7	355°35.0	N23°16.3	85°45.2	S01°25.3			
13	103°00.5	60°57.3	23.2	309°19.4	08.1	10°36.8	16.3	100°47.5	25.2			
14	118°03.0	75°57.2	24.0	324°20.6	07.6	25°38.6	16.3	115°49.9	25.2			
15	133°05.4	90°57.0	•• 24.7	339°21.7	•• 07.1	40°40.5	•• 16.3	130°52.3	•• 25.2			
16	148°07.9	105°56.9	25.5	354°22.8	06.5	55°42.3	16.3	145°54.6	25.1			
17	163°10.4	120°56.8	26.3	9°24.0	06.0	70°44.2	16.3	160°57.0	25.1			
18	178°12.8	135°56.7	N13°27.0	24°25.1	N12°05.5	85°46.0	N23°16.3	175°59.4	S01°25.1			
19	193°15.3	150°56.5	27.8	39°26.2	04.9	100°47.9	16.3	191°01.7	25.1			
20	208°17.8	165°56.4	28.6	54°27.4	04.4	115°49.7	16.3	206°04.1	25.0			
21	223°20.2	180°56.3	•• 29.3	69°28.5	•• 03.9	130°51.5	•• 16.3	221°06.5	•• 25.0			
22	238°22.7	195°56.1	30.1	84°29.6	03.3	145°53.4	16.3	236°08.9	25.0			
23	253°25.1	210°56.0	30.8	99°30.8	02.8	160°55.2	16.3	251°11.2	24.9			
Mer.pass. 06:09		ν -0.1' d 0.8' m -4.23		ν 1.1' d -0.5' m 1.42		ν 1.8' d -0.0' m -1.90		ν 2.4' d -0.0' m 1.01				

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	268°27.6	225°55.9	N13°31.6	114°31.9	N12°02.3	175°57.1	N23°16.3	266°13.6	S01°24.9			
1	283°30.1	240°55.7	32.4	129°33.0	01.7	190°58.9	16.3	281°16.0	24.9			
2	298°32.5	255°55.6	33.1	144°34.2	01.2	206°00.8	16.3	296°18.3	24.9			
3	313°35.0	270°55.4	•• 33.9	159°35.3	•• 00.6	221°02.6	•• 16.3	311°20.7	•• 24.8			
4	328°37.5	285°55.3	34.6	174°36.4	12°00.1	236°04.4	16.3	326°23.1	24.8			
5	343°39.9	300°55.2	35.4	189°37.6	11°59.6	251°06.3	16.3	341°25.5	24.8			
6	358°42.4	315°55.0	N13°36.2	204°38.7	N11°59.0	266°08.1	N23°16.2	356°27.8	S01°24.7			
7	13°44.9	330°54.9	36.9	219°39.8	58.5	281°10.0	16.2	11°30.2	24.7			
8	28°47.3	345°54.8	37.7	234°40.9	58.0	296°11.8	16.2	26°32.6	24.7			
9	43°49.8	0°54.6	•• 38.4	249°42.1	•• 57.4	311°13.7	•• 16.2	41°35.0	•• 24.7			
10	58°52.2	15°54.5	39.2	264°43.2	56.9	326°15.5	16.2	56°37.3	24.6			
11	73°54.7	30°54.3	40.0	279°44.3	56.3	341°17.3	16.2	71°39.7	24.6			
12	88°57.2	45°54.2	N13°40.7	294°45.5	N11°55.8	356°19.2	N23°16.2	86°42.1	S01°24.6			
13	103°59.6	60°54.0	41.5	309°46.6	55.3	11°21.0	16.2	101°44.5	24.6			
14	119°02.1	75°53.9	42.2	324°47.7	54.7	26°22.9	16.2	116°46.8	24.5			
15	134°04.6	90°53.8	•• 43.0	339°48.9	•• 54.2	41°24.7	•• 16.2	131°49.2	•• 24.5			
16	149°07.0	105°53.6	43.8	354°50.0	53.7	56°26.6	16.2	146°51.6	24.5			
17	164°09.5	120°53.5	44.5	9°51.1	53.1	71°28.4	16.2	161°53.9	24.4			
18	179°12.0	135°53.3	N13°45.3	24°52.3	N11°52.6	86°30.2	N23°16.2	176°56.3	S01°24.4			
19	194°14.4	150°53.2	46.0	39°53.4	52.0	101°32.1	16.2	191°58.7	24.4			
20	209°16.9	165°53.0	46.8	54°54.5	51.5	116°33.9	16.2	207°01.1	24.4			
21	224°19.4	180°52.9	•• 47.5	69°55.7	•• 51.0	131°35.8	•• 16.1	222°03.4	•• 24.3			
22	239°21.8	195°52.7	48.3	84°56.8	50.4	146°37.6	16.1	237°05.8	24.3			
23	254°24.3	210°52.6	49.0	99°57.9	49.9	161°39.5	16.1	252°08.2	24.3			
Mer.pass. 06:05		ν -0.1' d 0.8' m -4.22		ν 1.1' d -0.5' m 1.43		ν 1.8' d -0.0' m -1.90		ν 2.4' d -0.0' m 1.01				

Jun 18 Wed	SHA	Mer.pass
Venus	319°32.3	08:56
Mars	207°08.1	16:24
Jupiter	267°59.3	12:21
Saturn	357°50.6	06:22

Jun 19 Thu	SHA	Mer.pass
Venus	318°30.4	08:56
Mars	206°36.2	16:22
Jupiter	267°44.4	12:18
Saturn	357°48.3	06:18

Jun 20 Fri	SHA	Mer.pass
Venus	317°28.2	08:56
Mars	206°04.3	16:21
Jupiter	267°29.5	12:15
Saturn	357°46.0	06:14

Horizontal parallax		
Venus:	0.2	
Mars:	0.1	

DUT1 = UT1-UTC = +0.0297 sec ΔT = TT-UT1 = +69.1543 sec

2025 June 18 to Jun. 20 UT

Sun				Moon			
h							
Wed	GHA	Dec		GHA	ν	Dec	HP
0	179°43.6	N23°24.3		278°29.7	13.1'	S05°40.8	-16.0'
1	194°43.5	24.3		293°01.8	13.1'	05°24.8	-16.1'
2	209°43.4	24.4		307°33.9	13.1'	05°08.7	-16.1'
3	224°43.2	· · 24.4		322°06.0	13.1'	04°52.6	-16.1'
4	239°43.1	24.5		336°38.1	13.1'	04°36.5	-16.2'
5	254°42.9	24.6		351°10.3	13.1'	04°20.3	-16.2'
6	269°42.8	N23°24.6		5°42.4	13.1'	S04°04.1	-16.2'
7	284°42.7	24.6		20°14.5	13.1'	03°47.9	-16.3'
8	299°42.5	24.7		34°46.6	13.1'	03°31.6	-16.3'
9	314°42.4	· · 24.7		49°18.6	13.1'	03°15.3	-16.3'
10	329°42.3	24.8		63°50.7	13.1'	02°59.0	-16.4'
11	344°42.1	24.8		78°22.8	13.1'	02°42.7	-16.4'
12	359°42.0	N23°24.9		92°54.8	13.0'	S02°26.3	-16.4'
13	14°41.9	24.9		107°26.9	13.0'	02°09.9	-16.4'
14	29°41.7	25.0		121°58.9	13.0'	01°53.4	-16.5'
15	44°41.6	· · 25.0		136°30.9	13.0'	01°37.0	-16.5'
16	59°41.4	25.1		151°02.9	13.0'	01°20.5	-16.5'
17	74°41.3	25.1		165°34.9	13.0'	01°04.0	-16.5'
18	89°41.2	N23°25.1		180°06.9	12.9'	S00°47.5	-16.5'
19	104°41.0	25.2		194°38.8	12.9'	00°31.0	-16.5'
20	119°40.9	25.2		209°10.7	12.9'	S00°14.4	-16.6'
21	134°40.8	· · 25.3		223°42.6	12.9'	N00°02.1	16.6'
22	149°40.6	25.3		238°14.5	12.8'	00°18.7	16.6'
23	164°40.5	25.3		252°46.3	12.8'	00°35.3	16.6'
SD = 15.7'			d = 0.1'	SD = 15.9'			

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	179°40.3	N23°25.4	267°18.1	12.8'	N00°51.9	16.6'	58.8'
1	194°40.2	25.4	281°49.9	12.7'	01°08.5	16.6'	58.8'
2	209°40.1	25.4	296°21.6	12.7'	01°25.1	16.6'	58.8'
3	224°39.9	· · 25.5	310°53.3	12.7'	01°41.7	16.6'	58.8'
4	239°39.8	25.5	325°25.0	12.6'	01°58.3	16.6'	58.9'
5	254°39.7	25.5	339°56.6	12.6'	02°15.0	16.6'	58.9'
6	269°39.5	N23°25.6	354°28.2	12.6'	N02°31.6	16.6'	58.9'
7	284°39.4	25.6	8°59.8	12.5'	02°48.2	16.6'	58.9'
8	299°39.3	25.6	23°31.3	12.5'	03°04.9	16.6'	59.0'
9	314°39.1	· · 25.7	38°02.8	12.4'	03°21.5	16.6'	59.0'
10	329°39.0	25.7	52°34.3	12.4'	03°38.2	16.6'	59.0'
11	344°38.8	25.7	67°05.6	12.3'	03°54.8	16.6'	59.0'
12	359°38.7	N23°25.8	81°37.0	12.3'	N04°11.4	16.6'	59.1'
13	14°38.6	25.8	96°08.3	12.3'	04°28.0	16.6'	59.1'
14	29°38.4	25.8	110°39.6	12.2'	04°44.6	16.6'	59.1'
15	44°38.3	· · 25.8	125°10.8	12.2'	05°01.2	16.6'	59.1'
16	59°38.2	25.9	139°41.9	12.1'	05°17.8	16.6'	59.2'
17	74°38.0	25.9	154°13.0	12.0'	05°34.4	16.6'	59.2'
18	89°37.9	N23°25.9	168°44.0	12.0'	N05°50.9	16.5'	59.2'
19	104°37.8	25.9	183°15.0	11.9'	06°07.5	16.5'	59.2'
20	119°37.6	26.0	197°46.0	11.9'	06°24.0	16.5'	59.2'
21	134°37.5	· · 26.0	212°16.8	11.8'	06°40.5	16.5'	59.3'
22	149°37.3	26.0	226°47.6	11.7'	06°57.0	16.5'	59.3'
23	164°37.2	26.0	241°18.4	11.7'	07°13.5	16.4'	59.3'
	SD = 15.7'	d = 0.0'			SD = 16.0'		

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	179° 37.1	N23° 26.0	255° 49.1	11.6'	N07° 29.9	16.4'	59.3'
1	194° 36.9	26.1	270° 19.7	11.6'	07° 46.3	16.4'	59.4'
2	209° 36.8	26.1	284° 50.3	11.5'	08° 02.7	16.4'	59.4'
3	224° 36.7	· · 26.1	299° 20.7	11.4'	08° 19.1	16.3'	59.4'
4	239° 36.5	26.1	313° 51.2	11.3'	08° 35.4	16.3'	59.4'
5	254° 36.4	26.1	328° 21.5	11.3'	08° 51.7	16.3'	59.4'
6	269° 36.2	N23° 26.1	342° 51.8	11.2'	N09° 07.9	16.2'	59.5'
7	284° 36.1	26.2	357° 22.0	11.1'	09° 24.2	16.2'	59.5'
8	299° 36.0	26.2	11° 52.1	11.1'	09° 40.3	16.2'	59.5'
9	314° 35.8	· · 26.2	26° 22.2	11.0'	09° 56.5	16.1'	59.5'
10	329° 35.7	26.2	40° 52.1	10.9'	10° 12.6	16.1'	59.5'
11	344° 35.6	26.2	55° 22.0	10.8'	10° 28.7	16.0'	59.6'
12	359° 35.4	N23° 26.2	69° 51.8	10.7'	N10° 44.7	16.0'	59.6'
13	14° 35.3	26.2	84° 21.6	10.7'	11° 00.7	15.9'	59.6'
14	29° 35.1	26.2	98° 51.2	10.6'	11° 16.6	15.9'	59.6'
15	44° 35.0	· · 26.3	113° 20.8	10.5'	11° 32.5	15.8'	59.6'
16	59° 34.9	26.3	127° 50.3	10.4'	11° 48.3	15.8'	59.7'
17	74° 34.7	26.3	142° 19.7	10.3'	12° 04.1	15.7'	59.7'
18	89° 34.6	N23° 26.3	156° 49.0	10.2'	N12° 19.8	15.7'	59.7'
19	104° 34.5	26.3	171° 18.2	10.1'	12° 35.5	15.6'	59.7'
20	119° 34.3	26.3	185° 47.3	10.0'	12° 51.1	15.5'	59.7'
21	134° 34.2	· · 26.3	200° 16.4	9.9'	13° 06.7	15.5'	59.8'
22	149° 34.1	26.3	214° 45.3	9.9'	13° 22.1	15.4'	59.8'
23	164° 33.9	26.3	229° 14.2	9.8'	13° 37.6	15.4'	59.8'
	SD = 15.7'	d = 0.0'			SD = 16.2'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	☐	☐	☐	☐	☐	☐
64°	—:—	—:—	01:31	22:32	—:—	—:—
62°	—:—	—:—	02:09	21:54	—:—	—:—
60°	—:—	00:49	02:36	21:27	23:14	—:—
N 58°	—:—	01:40	02:56	21:07	22:23	—:—
56°	—:—	02:10	03:13	20:50	21:53	—:—
54°	00:45	02:33	03:27	20:36	21:30	23:18
52°	01:32	02:51	03:39	20:23	21:12	22:31
50°	02:00	03:06	03:50	20:13	20:57	22:03
45°	02:46	03:35	04:13	19:50	20:27	21:17
N 40°	03:16	03:58	04:31	19:32	20:05	20:47
35°	03:40	04:16	04:46	19:17	19:47	20:23
30°	03:58	04:32	04:59	19:04	19:31	20:05
20°	04:27	04:57	05:21	18:42	19:06	19:36
N 10°	04:50	05:17	05:40	18:23	18:46	19:13
0°	05:09	05:35	05:58	18:05	18:28	18:54
S 10°	05:26	05:52	06:15	17:48	18:10	18:37
20°	05:42	06:10	06:34	17:29	17:53	18:21
30°	05:59	06:29	06:55	17:08	17:34	18:04
35°	06:08	06:39	07:07	16:55	17:24	17:55
40°	06:17	06:51	07:22	16:41	17:12	17:46
45°	06:27	07:05	07:38	16:24	16:58	17:36
S 50°	06:39	07:21	07:59	16:04	16:42	17:24
52°	06:44	07:28	08:09	15:54	16:35	17:18
54°	06:50	07:36	08:20	15:43	16:26	17:13
56°	06:56	07:45	08:33	15:30	16:17	17:07
58°	07:03	07:56	08:48	15:15	16:07	17:00
S 60°	07:11	08:07	09:05	14:58	15:55	16:52

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	00:38	00:02 23:24	22:34	11:12	13:32	16:07
N 70°	00:30	00:03 23:34	22:59	11:16	13:25	15:45
68°	00:24	00:04 23:43	23:17	11:20	13:20	15:29
66°	00:19	00:04 23:50	23:33	11:22	13:16	15:15
64°	00:14	00:05 23:56	23:45	11:25	13:12	15:04
62°	00:10	00:05 23:56	00:01	11:27	13:09	14:55
60°	00:07	00:06	00:05	11:28	13:06	14:47
N 58°	00:04	00:06	00:09	11:30	13:03	14:40
56°	00:01	00:07	00:13	11:31	13:01	14:34
54°	· · ·	00:07	00:16	11:32	12:59	14:29
52°	· · ·	00:07	00:19	11:33	12:57	14:24
50°	· · ·	00:08	00:22	11:34	12:56	14:20
45°	· · ·	00:08	00:28	11:37	12:52	14:10
N 40°	· · ·	00:09	00:33	11:38	12:49	14:02
35°	· · ·	00:09	00:37	11:40	12:47	13:56
30°	· · ·	00:10	00:41	11:41	12:44	13:50
20°	· · ·	00:10	00:48	11:44	12:41	13:40
N 10°	· · ·	00:11	00:54	11:46	12:37	13:31
0°	· · ·	00:12	00:59	11:47	12:34	13:23
S 10°	· · ·	00:12	01:05	11:49	12:31	13:14
20°	· · ·	00:13	01:11	11:51	12:27	13:06
30°	· · ·	00:14	01:18	11:53	12:24	12:56
35°	· · ·	00:15	01:22	11:54	12:21	12:50
40°	· · ·	00:15	01:27	11:56	12:19	12:44
45°	· · ·	00:16	01:33	11:57	12:16	12:36
S 50°	· · ·	00:17	01:39	11:59	12:13	12:28
52°	· · ·	00:17	01:42	12:00	12:11	12:23
54°	· · ·	00:18	01:46	12:01	12:09	12:19
56°	· · ·	00:18	01:50	12:02	12:08	12:14
58°	· · ·	00:19	01:54	12:03	12:05	12:09
S 60°	· · ·	00:19	01:58	12:04	12:03	12:02

Day	Sun			Moon		
	Eqn.of Time		Mer.	Mer.Pass.		Age
	00 ^h mm:ss	12 ^h mm:ss	Pass hh:mm	Upper hh:mm	Lower hh:mm	22-24 59-37%
18	01:06	01:12	12:01	05:36	18:00	
19	01:19	01:25	12:01	06:23	18:47	
20	01:32	01:38	12:02	07:11	19:36	

June 21, 22, 23 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	269°26.7	225°52.4	N13°49.8	114°59.0	N11°49.3	176°41.3	N23°16.1	267°10.6	S01°24.3	Alpheratz	357°34.4	29°13.7
1	284°29.2	240°52.3	50.6	130°00.2	48.8	191°43.1	16.1	282°12.9	24.2	Ankaa	353°06.8	-42°09.8
2	299°31.7	255°52.1	51.3	145°01.3	48.3	206°45.0	16.1	297°15.3	24.2	Schedar	349°30.7	56°40.4
3	314°34.1	270°52.0	•• 52.1	160°02.4	•• 47.7	221°46.8	•• 16.1	312°17.7	•• 24.2	Diphda	348°47.0	-17°50.7
4	329°36.6	285°51.8	52.8	175°03.6	47.2	236°48.7	16.1	327°20.1	24.2	Achernar	335°20.1	-57°06.2
5	344°39.1	300°51.7	53.6	190°04.7	46.6	251°50.5	16.1	342°22.5	24.1	Hamal	327°51.0	23°34.9
6	359°41.5	315°51.5	N13°54.3	205°05.8	N11°46.1	266°52.3	N23°16.1	357°24.8	S01°24.1	Polaris	314°12.5	89°22.0
7	14°44.0	330°51.4	55.1	220°07.0	45.6	281°54.2	16.1	12°27.2	24.1	Acamar	315°11.8	-40°12.0
8	29°46.5	345°51.2	55.8	235°08.1	45.0	296°56.0	16.1	27°29.6	24.1	Menkar	314°06.0	4°11.4
9	44°48.9	0°51.1	•• 56.6	250°09.2	•• 44.5	311°57.9	•• 16.1	42°32.0	•• 24.0	Mirfak	308°28.1	49°56.9
10	59°51.4	15°50.9	57.3	265°10.4	43.9	326°59.7	16.1	57°34.3	24.0	Aldebaran	290°39.5	16°33.6
11	74°53.9	30°50.7	58.1	280°11.5	43.4	342°01.6	16.0	72°36.7	24.0	Rigel	281°03.9	-8°10.3
12	89°56.3	45°50.6	N13°58.9	295°12.6	N11°42.8	357°03.4	N23°16.0	87°39.1	S01°23.9	Capella	280°21.8	46°01.4
13	104°58.8	60°50.4	13°59.6	310°13.7	42.3	12°05.2	16.0	102°41.5	23.9	Bellatrix	278°22.8	6°22.4
14	120°01.2	75°50.3	14°00.4	325°14.9	41.8	27°07.1	16.0	117°43.8	23.9	Elnath	278°01.8	28°37.7
15	135°03.7	90°50.1	•• 01.1	340°16.0	•• 41.2	42°08.9	•• 16.0	132°46.2	•• 23.9	Alnilam	275°37.7	-1°11.1
16	150°06.2	105°49.9	01.9	355°17.1	40.7	57°10.8	16.0	147°48.6	23.8	Betelgeuse	270°52.0	7°24.7
17	165°08.6	120°49.8	02.6	10°18.3	40.1	72°12.6	16.0	162°51.0	23.8	Canopus	263°52.8	-52°42.5
18	180°11.1	135°49.6	N14°03.4	25°19.4	N11°39.6	87°14.5	N23°16.0	177°53.4	S01°23.8	Sirius	258°26.3	-16°45.0
19	195°13.6	150°49.5	04.1	40°20.5	39.1	102°16.3	16.0	192°55.7	23.8	Adhara	255°06.0	-29°00.4
20	210°16.0	165°49.3	04.9	55°21.6	38.5	117°18.1	16.0	207°58.1	23.7	Procyon	244°50.8	5°09.6
21	225°18.5	180°49.1	•• 05.6	70°22.8	•• 38.0	132°20.0	•• 16.0	223°00.5	•• 23.7	Pollux	243°17.2	27°57.9
22	240°21.0	195°49.0	06.4	85°23.9	37.4	147°21.8	16.0	238°02.9	23.7	Avior	234°15.2	-59°35.6
23	255°23.4	210°48.8	07.1	100°25.0	36.9	162°23.7	16.0	253°05.2	23.7	Suhail	222°46.4	-43°32.3
Mer.pass. 06:01		ν -0.2' d 0.8' m -4.21		ν 1.1' d -0.5' m 1.44		ν 1.8' d -0.0' m -1.90		ν 2.4' d -0.0' m 1.00				

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	270°25.9	225°48.6	N14°07.9	115°26.2	N11°36.3	177°25.5	N23°16.0	268°07.6	S01°23.6
1	285°28.3	240°48.5	08.6	130°27.3	35.8	192°27.4	15.9	283°10.0	23.6
2	300°30.8	255°48.3	09.4	145°28.4	35.2	207°29.2	15.9	298°12.4	23.6
3	315°33.3	270°48.1	•• 10.1	160°29.6	•• 34.7	222°31.0	•• 15.9	313°14.8	•• 23.6
4	330°35.7	285°48.0	10.9	175°30.7	34.2	237°32.9	15.9	328°17.1	23.5
5	345°38.2	300°47.8	11.6	190°31.8	33.6	252°34.7	15.9	343°19.5	23.5
6	0°40.7	315°47.6	N14°12.4	205°32.9	N11°33.1	267°36.6	N23°15.9	358°21.9	S01°23.5
7	15°43.1	330°47.5	13.1	220°34.1	32.5	282°38.4	15.9	13°24.3	23.5
8	30°45.6	345°47.3	13.9	235°35.2	32.0	297°40.3	15.9	28°26.7	23.5
9	45°48.1	0°47.1	•• 14.6	250°36.3	•• 31.4	312°42.1	•• 15.9	43°29.0	•• 23.4
10	60°50.5	15°47.0	15.4	265°37.5	30.9	327°43.9	15.9	58°31.4	23.4
11	75°53.0	30°46.8	16.1	280°38.6	30.3	342°45.8	15.9	73°33.8	23.4
12	90°55.5	45°46.6	N14°16.8	295°39.7	N11°29.8	357°47.6	N23°15.9	88°36.2	S01°23.4
13	105°57.9	60°46.4	17.6	310°40.8	29.3	12°49.5	15.8	103°38.6	23.3
14	121°00.4	75°46.3	18.3	325°42.0	28.7	27°51.3	15.8	118°41.0	23.3
15	136°02.8	90°46.1	•• 19.1	340°43.1	•• 28.2	42°53.2	•• 15.8	133°43.3	•• 23.3
16	151°05.3	105°45.9	19.8	355°44.2	27.6	57°55.0	15.8	148°45.7	23.3
17	166°07.8	120°45.7	20.6	10°45.4	27.1	72°56.8	15.8	163°48.1	23.2
18	181°10.2	135°45.6	N14°21.3	25°46.5	N11°26.5	87°58.7	N23°15.8	178°50.5	S01°23.2
19	196°12.7	150°45.4	22.1	40°47.6	26.0	103°00.5	15.8	193°52.9	23.2
20	211°15.2	165°45.2	22.8	55°48.7	25.4	118°02.4	15.8	208°55.2	23.2
21	226°17.6	180°45.0	•• 23.6	70°49.9	•• 24.9	133°04.2	•• 15.8	223°57.6	•• 23.1
22	241°20.1	195°44.9	24.3	85°51.0	24.3	148°06.0	15.8	239°00.0	23.1
23	256°22.6	210°44.7	25.0	100°52.1	23.8	163°07.9	15.8	254°02.4	23.1
Mer.pass. 05:57		ν -0.2' d 0.7' m -4.20		ν 1.1' d -0.5' m 1.44		ν 1.8' d -0.0' m -1.89		ν 2.4' d -0.0' m 1.00	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	271°25.0	225°44.5	N14°25.8	115°53.2	N11°23.3	178°09.7	N23°15.7	269°04.8	S01°23.1
1	286°27.5	240°44.3	26.5	130°54.4	22.7	193°11.6	15.7	284°07.2	23.1
2	301°30.0	255°44.1	27.3	145°55.5	22.2	208°13.4	15.7	299°09.5	23.0
3	316°32.4	270°44.0	•• 28.0	160°56.6	•• 21.6	223°15.3	•• 15.7	314°11.9	•• 23.0
4	331°34.9	285°43.8	28.8	175°57.8	21.1	238°17.1	15.7	329°14.3	23.0
5	346°37.3	300°43.6	29.5	190°58.9	20.5	253°18.9	15.7	344°16.7	23.0
6	1°39.8	315°43.4	N14°30.2	206°00.0	N11°20.0	268°20.8	N23°15.7	359°19.1	S01°22.9
7	16°42.3	330°43.2	31.0	221°01.1	19.4	283°22.6	15.7	14°21.5	22.9
8	31°44.7	345°43.0	31.7	236°02.3	18.9	298°24.5	15.7	29°23.8	22.9
9	46°47.2	0°42.9	•• 32.5	251°03.4	•• 18.3	313°26.3	•• 15.7	44°26.2	•• 22.9
10	61°49.7	15°42.7	33.2	266°04.5	17.8	328°28.2	15.7	59°28.6	22.8
11	76°52.1	30°42.5	33.9	281°05.6	17.2	343°30.0	15.6	74°31.0	22.8
12	91°54.6	45°42.3	N14°34.7	296°06.8	N11°16.7	358°31.8	N23°15.6	89°33.4	S01°22.8
13	106°57.1	60°42.1	35.4	311°07.9	16.1	13°33.7	15.6	104°35.8	22.8
14	121°59.5	75°41.9	36.2	326°09.0	15.6	28°35.5	15.6	119°38.1	22.8
15	137°02.0	90°41.7	•• 36.9	341°10.2	•• 15.0	43°37.4	•• 15.6	134°40.5	•• 22.7
16	152°04.5	105°41.5	37.6	356°11.3	14.5	58°39.2	15.6	149°42.9	22.7
17	167°06.9	120°41.4	38.4	11°12.4	13.9	73°41.1	15.6	164°45.3	22.7
18	182°09.4	135°41.2	N14°39.1	26°13.5	N11°13.4	88°42.9	N23°15.6	179°47.7	S01°22.7
19	197°11.8	150°41.0	39.8	41°14.7	12.8	103°44.7	15.6	194°50.1	22.6
20	212°14.3	165°40.8	40.6	56°15.8	12.3	118°46.6	15.6	209°52.5	22.6
21	227°16.8	180°40.6	•• 41.3	71°16.9	•• 11.7	133°48.4	•• 15.5	224°54.8	•• 22.6
22	242°19.2	195°40.4	42.1	86°18.0	11.2	148°50.3	15.5	239°57.2	22.6
23	257°21.7	210°40.2	42.8	101°19.2	10.6	163°52.1	15.5	254°59.6	22.6
Mer.pass. 05:53		ν -0.2' d 0.7' m -4.20		ν 1.1' d -0.5' m 1.45		ν 1.8' d -0.0' m -1.89		ν 2.4' d -0.0' m 0.99	

Jun 21 Sat	SHA	Mer.pass
Venus	316°25.7	08:57
Mars	205°32.3	16:19
Jupiter	267°14.5	12:12
Saturn	357°43.8	06:10

Jun 22 Sun	SHA	Mer.pass
Venus	315°22.8	08:57
Mars	205°00.3	16:17
Jupiter	266°59.6	12:09
Saturn	357°41.7	06:07

Jun 23 Mon	SHA	Mer.pass
Venus	314°19.5	08:57
Mars	204°28.2	16:15
Jupiter	266°44.7	12:06
Saturn	357°39.7	06:03

Horizontal parallax	
Venus:	0.2
Mars:	0.1

h	Sun			Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	179°33.8	N23°26.3	243°42.9	9.7'	N13°52.9	15.3'	59.8'
1	194°33.6	26.3	258°11.6	9.6'	14°08.2	15.2'	59.8'
2	209°33.5	26.3	272°40.1	9.5'	14°23.4	15.1'	59.8'
3	224°33.4	· · 26.3	287°08.6	9.4'	14°38.6	15.1'	59.9'
4	239°33.2	26.3	301°37.0	9.3'	14°53.7	15.0'	59.9'
5	254°33.1	26.3	316°05.2	9.2'	15°08.7	14.9'	59.9'
6	269°33.0	N23°26.3	330°33.4	9.1'	N15°23.6	14.8'	59.9'
7	284°32.8	26.3	345°01.5	9.0'	15°38.4	14.8'	59.9'
8	299°32.7	26.3	359°29.4	8.9'	15°53.2	14.7'	59.9'
9	314°32.5	· · 26.3	13°57.3	8.7'	16°07.9	14.6'	60.0'
10	329°32.4	26.3	28°25.0	8.6'	16°22.5	14.5'	60.0'
11	344°32.3	26.3	42°52.7	8.5'	16°37.0	14.4'	60.0'
12	359°32.1	N23°26.3	57°20.2	8.4'	N16°51.4	14.3'	60.0'
13	14°32.0	26.3	71°47.6	8.3'	17°05.7	14.2'	60.0'
14	29°31.9	26.3	86°14.9	8.2'	17°20.0	14.1'	60.0'
15	44°31.7	· · 26.2	100°42.1	8.1'	17°34.1	14.0'	60.0'
16	59°31.6	26.2	115°09.2	8.0'	17°48.1	13.9'	60.1'
17	74°31.5	26.2	129°36.2	7.9'	18°02.1	13.8'	60.1'
18	89°31.3	N23°26.2	144°03.1	7.8'	N18°15.9	13.7'	60.1'
19	104°31.2	26.2	158°29.9	7.7'	18°29.7	13.6'	60.1'
20	119°31.0	26.2	172°56.5	7.5'	18°43.3	13.5'	60.1'
21	134°30.9	· · 26.2	187°23.1	7.4'	18°56.8	13.4'	60.1'
22	149°30.8	26.2	201°49.5	7.3'	19°10.2	13.3'	60.1'
23	164°30.6	26.2	216°15.8	7.2'	19°23.5	13.2'	60.1'
SD = 15.7'			SD = 16.3'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	179°30.5	N23°26.1	230°42.0	7.1'	N19°36.7	13.1'	60.1'
1	194°30.4	26.1	245°08.1	7.0'	19°49.8	13.0'	60.2'
2	209°30.2	26.1	259°34.1	6.9'	20°02.7	12.8'	60.2'
3	224°30.1	· · 26.1	273°59.9	6.7'	20°15.6	12.7'	60.2'
4	239°29.9	26.1	288°25.7	6.6'	20°28.3	12.6'	60.2'
5	254°29.8	26.1	302°51.3	6.5'	20°40.9	12.5'	60.2'
6	269°29.7	N23°26.0	317°16.8	6.4'	N20°53.3	12.3'	60.2'
7	284°29.5	26.0	331°42.2	6.3'	21°05.7	12.2'	60.2'
8	299°29.4	26.0	346°07.5	6.2'	21°17.9	12.1'	60.2'
9	314°29.3	· · 26.0	0°32.6	6.1'	21°29.9	11.9'	60.2'
10	329°29.1	25.9	14°57.7	5.9'	21°41.9	11.8'	60.2'
11	344°29.0	25.9	29°22.6	5.8'	21°53.7	11.7'	60.2'
12	359°28.9	N23°25.9	43°47.5	5.7'	N22°05.3	11.5'	60.2'
13	14°28.7	25.9	58°12.2	5.6'	22°16.8	11.4'	60.3'
14	29°28.6	25.9	72°36.8	5.5'	22°28.2	11.2'	60.3'
15	44°28.4	· · 25.8	87°01.2	5.4'	22°39.4	11.1'	60.3'
16	59°28.3	25.8	101°25.6	5.3'	22°50.5	10.9'	60.3'
17	74°28.2	25.8	115°49.9	5.1'	23°01.4	10.8'	60.3'
18	89°28.0	N23°25.7	130°14.0	5.0'	N23°12.2	10.6'	60.3'
19	104°27.9	25.7	144°38.1	4.9'	23°22.8	10.5'	60.3'
20	119°27.8	25.7	159°02.0	4.8'	23°33.3	10.3'	60.3'
21	134°27.6	· · 25.7	173°25.8	4.7'	23°43.6	10.2'	60.3'
22	149°27.5	25.6	187°49.5	4.6'	23°53.8	10.0'	60.3'
23	164°27.4	25.6	202°13.1	4.5'	24°03.8	9.8'	60.3'
SD = 15.7'			SD = 16.4'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	179°27.2	N23°25.6	216°36.6	4.4'	N24°13.6	9.7'	60.3'
1	194°27.1	25.5	231°00.0	4.3'	24°23.3	9.5'	60.3'
2	209°26.9	25.5	245°23.3	4.2'	24°32.8	9.3'	60.3'
3	224°26.8	· · 25.5	259°46.5	4.1'	24°42.1	9.2'	60.3'
4	239°26.7	25.4	274°09.6	4.0'	24°51.3	9.0'	60.3'
5	254°26.5	25.4	288°32.6	3.9'	25°00.3	8.8'	60.3'
6	269°26.4	N23°25.4	302°55.5	3.8'	N25°09.1	8.6'	60.3'
7	284°26.3	25.3	317°18.2	3.7'	25°17.7	8.5'	60.3'
8	299°26.1	25.3	331°40.9	3.6'	25°26.2	8.3'	60.3'
9	314°26.0	· · 25.2	346°03.6	3.5'	25°34.4	8.1'	60.3'
10	329°25.9	25.2	0°26.1	3.4'	25°42.6	7.9'	60.3'
11	344°25.7	25.2	14°48.5	3.3'	25°50.5	7.7'	60.3'
12	359°25.6	N23°25.1	29°10.8	3.3'	N25°58.2	7.5'	60.3'
13	14°25.4	25.1	43°33.1	3.2'	26°05.8	7.4'	60.3'
14	29°25.3	25.0	57°55.3	3.1'	26°13.1	7.2'	60.3'
15	44°25.2	· · 25.0	72°17.3	3.0'	26°20.3	7.0'	60.3'
16	59°25.0	25.0	86°39.4	2.9'	26°27.3	6.8'	60.3'
17	74°24.9	24.9	101°01.3	2.9'	26°34.1	6.6'	60.3'
18	89°24.8	N23°24.9	115°23.2	2.8'	N26°40.7	6.4'	60.3'
19	104°24.6	24.8	129°44.9	2.7'	26°47.1	6.2'	60.3'
20	119°24.5	24.8	144°06.7	2.7'	26°53.3	6.0'	60.3'
21	134°24.4	· · 24.7	158°28.3	2.6'	26°59.3	5.8'	60.2'
22	149°24.2	24.7	172°49.9	2.5'	27°05.1	5.6'	60.2'
23	164°24.1	24.6	187°11.4	2.5'	27°10.7	5.4'	60.2'
SD = 15.7'			SD = 16.4'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	☐	☐	☐	☐	☐	☐
64°	--	--	01:31	22:33	--	--
62°	--	--	02:10	21:54	--	--
60°	--	00:50	02:36	21:28	23:14	--
N 58°	--	01:41	02:57	21:07	22:23	--
56°	--	02:11	03:13	20:51	21:53	--
54°	00:45	02:33	03:28	20:36	21:31	23:18
52°	01:33	02:51	03:40	20:24	21:13	22:31
50°	02:01	03:06	03:51	20:13	20:58	22:03
45°	02:46	03:36	04:14	19:51	20:28	21:18
N 40°	03:17	03:59	04:32	19:33	20:05	20:47
35°	03:40	04:17	04:47	19:17	19:47	20:24
30°	03:59	04:32	05:00	19:04	19:32	20:05
20°	04:28	04:57	05:22	18:42	19:07	19:36
N 10°	04:51	05:18	05:41	18:23	18:46	19:14
0°	05:10	05:36	05:58	18:06	18:28	18:55
S 10°	05:27	05:53	06:16	17:48	18:11	18:37
20°	05:43	06:10	06:34	17:30	17:54	18:21
30°	05:59	06:29	06:56	17:08	17:35	18:05
35°	06:08	06:40	07:08	16:56	17:24	17:56
40°	06:18	06:52	07:22	16:42	17:12	17:47
45°	06:28	07:05	07:39	16:25	16:59	17:36
S 50°	06:40	07:21	08:00	16:04	16:43	17:24
52°	06:45	07:29	08:10	15:54	16:35	17:19
54°	06:51	07:37	08:21	15:43	16:27	17:13
56°	06:57	07:46	08:34	15:31	16:18	17:07
58°	07:04	07:56	08:48	15:16	16:08	17:00
S 60°	07:11	08:08	09:06	14:58	15:56	16:53

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	☐	☐	☐	☐	☐	☐
N 70°	21:49	☐	☐	18:46	☐	☐
68°	22:37	☐	☐	18:00	☐	☐
66°	23:09	22:04	☐	17:30	20:35	☐
64°	23:32	23:10	☐	17:08	19:31	☐
62°	23:51	23:45	23:37	16:50	18:56	21:16
60°	00:05	00:07	00:11	16:35	18:31	20:28
N 58°	00:13	00:20	00:32	16:23	18:11	19:58
56°	00:21	00:31	00:49	16:13	17:55	19:35
54°	00:27	00:42	01:03	16:03	17:41	19:16
52°	00:33	00:51	01:16	15:55	17:29	19:00
50°	00:38	00:59	01:27	15:48	17:18	18:47
45°	00:50	01:16	01:51	15:32	16:56	18:19
N 40°	00:59	01:31	02:10	15:19	16:38	17:57
35°	01:08	01:43	02:26	15:08	16:23	17:39
30°	01:15	01:54	02:39	14:58	16:10	17:23
20°	01:28	02:12	03:03	14:42	15:48	16:57
N 10°	01:39	02:29	03:24	14:28	15:29	16:34
0°	01:50	02:44	03:43	14:15	15:12	16:13
S 10°	02:00	03:00	04:03	14:02	14:54	15:52
20°	02:12	03:16	04:24	13:48	14:35	15:30
30°	02:25	03:36	04:49	13:32	14:14	15:04
35°	02:33	03:47	05:03	13:23	14:01	14:48
40°	02:42	04:00	05:20	13:12	13:47	14:31
45°	02:53	04:16	05:40	13:00	13:30	14:09
S 50°	03:05	04:35	06:06	12:45	13:09	13:43
52°	03:11	04:44	06:19	12:39	12:59	13:30
54°	03:18	04:55	06:33	12:31	12:48	13:15
56°	03:25	05:07	06:50	12:23	12:36	12:58
58°	03:34	05:20	07:10	12:13	12:21	12:37
S 60°	03:44	05:36	07:36	12:02	12:04	12:11

Day	Sun		Mer.	Moon		Age
	Eqn.of Time	12 ^h	Pass	Upper	Lower	25-27
	00 ^h	mm:ss	hh:mm	hh:mm	hh:mm	26-9%
21	01:45	01:51	12:02	08:02	20:29	
22	01:58	02:05	12:02	08:58	21:27	
23	02:11	02:18	12:02	09:58	22:30	

June 24, 25, 26 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	272°24.2	225°40.0	N14°43.5	116°20.3	N11°10.1	178°53.9	N23°15.5	270°02.0	S01°22.5	Alpheratz	357°34.4	29°13.7
1	287°26.6	240°39.8	44.3	131°21.4	09.5	193°55.8	15.5	285°04.4	22.5	Ankaa	353°06.8	-42°09.8
2	302°29.1	255°39.6	45.0	146°22.5	09.0	208°57.6	15.5	300°06.8	22.5	Schedar	349°30.7	56°40.4
3	317°31.6	270°39.4	· · 45.7	161°23.7	· · 08.4	223°59.5	· · 15.5	315°09.2	· · 22.5	Diphda	348°47.0	-17°50.7
4	332°34.0	285°39.2	46.5	176°24.8	07.9	239°01.3	15.5	330°11.6	22.5	Achernar	335°20.1	-57°06.1
5	347°36.5	300°39.0	47.2	191°25.9	07.3	254°03.2	15.5	345°13.9	22.4	Hamal	327°50.9	23°34.9
6	2°39.0	315°38.8	N14°47.9	206°27.0	N11°06.8	269°05.0	N23°15.5	0°16.3	S01°22.4	Polaris	314°10.9	89°22.0
7	17°41.4	330°38.6	48.7	221°28.2	06.2	284°06.8	15.4	15°18.7	22.4	Acamar	315°11.8	-40°11.9
8	32°43.9	345°38.4	49.4	236°29.3	05.7	299°08.7	15.4	30°21.1	22.4	Menkar	314°06.0	4°11.4
9	47°46.3	0°38.2	· · 50.1	251°30.4	· · 05.1	314°10.5	· · 15.4	45°23.5	· · 22.3	Mirfak	308°28.1	49°56.9
10	62°48.8	15°38.0	50.9	266°31.5	04.6	329°12.4	15.4	60°25.9	22.3	Aldebaran	290°39.5	16°33.6
11	77°51.3	30°37.8	51.6	281°32.7	04.0	344°14.2	15.4	75°28.3	22.3	Rigel	281°03.8	-8°10.3
12	92°53.7	45°37.6	N14°52.3	296°33.8	N11°03.5	359°16.0	N23°15.4	90°30.7	S01°22.3	Capella	280°21.8	46°01.4
13	107°56.2	60°37.4	53.1	311°34.9	02.9	14°17.9	15.4	105°33.0	22.3	Bellatrix	278°22.8	6°22.4
14	122°58.7	75°37.2	53.8	326°36.0	02.4	29°19.7	15.3	120°35.4	22.2	Elnath	278°01.8	28°37.7
15	138°01.1	90°37.0	· · 54.5	341°37.2	· · 01.8	44°21.6	· · 15.3	135°37.8	· · 22.2	Alnilam	275°37.7	-1°11.1
16	153°03.6	105°36.8	55.3	356°38.3	01.3	59°23.4	15.3	150°40.2	22.2	Betelgeuse	270°52.0	7°24.7
17	168°06.1	120°36.6	56.0	11°39.4	00.7	74°25.3	15.3	165°42.6	22.2	Canopus	263°52.8	-52°42.5
18	183°08.5	135°36.4	N14°56.7	26°40.5	N11°00.2	89°27.1	N23°15.3	180°45.0	S01°22.2	Sirius	258°26.3	-16°45.0
19	198°11.0	150°36.2	57.5	41°41.7	10°59.6	104°29.0	15.3	195°47.4	22.1	Adhara	255°06.0	-29°00.4
20	213°13.5	165°36.0	58.2	56°42.8	59.1	119°30.8	15.3	210°49.8	22.1	Procyon	244°50.8	5°09.6
21	228°15.9	180°35.8	· · 58.9	71°43.9	· · 58.5	134°32.7	· · 15.3	225°52.2	· · 22.1	Pollux	243°17.2	27°57.9
22	243°18.4	195°35.6	14°59.6	86°45.0	58.0	149°34.5	15.3	240°54.5	22.1	Avior	234°15.2	-59°35.6
23	258°20.8	210°35.4	15°00.4	101°46.2	57.4	164°36.4	15.3	255°56.9	22.1	Suhail	222°46.4	-43°32.3
Mer.pass. 05:49		ν -0.2' d 0.7' m -4.19		ν 1.1' d -0.6' m 1.45		ν 1.8' d -0.0' m -1.89		ν 2.4' d -0.0' m 0.99				

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	273°23.3	225°35.2	N15°01.1	116°47.3	N10°56.8	179°38.2	N23°15.3	270°59.3	S01°22.0	Alphard	217°47.6	-8°46.2
1	288°25.8	240°34.9	01.8	131°48.4	56.3	194°40.0	15.3	286°01.7	22.0	Regulus	207°34.2	11°50.6
2	303°28.2	255°34.7	02.6	146°49.5	55.7	209°41.9	15.2	301°04.1	22.0	Dubhe	193°40.7	61°37.1
3	318°30.7	270°34.5	· · 03.3	161°50.7	· · 55.2	224°43.7	· · 15.2	316°06.5	· · 22.0	Denebola	182°24.6	14°25.9
4	333°33.2	285°34.3	04.0	176°51.8	54.6	239°45.6	15.2	331°08.9	22.0	Gienah	175°43.2	-17°41.1
5	348°35.6	300°34.1	04.7	191°52.9	54.1	254°47.4	15.2	346°11.3	21.9	Acrux	172°59.7	-63°14.7
6	3°38.1	315°33.9	N15°05.5	206°54.0	N10°53.5	269°49.3	N23°15.2	1°13.7	S01°21.9	Gacrux	171°51.2	-57°15.6
7	18°40.6	330°33.7	06.2	221°55.2	53.0	284°51.1	15.2	16°16.1	21.9	Alioth	166°12.5	55°49.5
8	33°43.0	345°33.5	06.9	236°56.3	52.4	299°52.9	15.2	31°18.5	21.9	Spica	158°21.8	-11°17.7
9	48°45.5	0°33.2	· · 07.6	251°57.4	· · 51.9	314°54.8	· · 15.2	46°20.8	· · 21.9	Alkaid	152°51.5	-49°11.4
10	63°48.0	15°33.0	08.4	266°58.5	51.3	329°56.6	15.2	61°23.2	21.8	Hadar	148°35.2	-60°30.0
11	78°50.4	30°32.8	09.1	281°59.7	50.8	344°58.5	15.1	76°25.6	21.8	Menkent	147°57.0	-36°29.9
12	93°52.9	45°32.6	N15°09.8	297°00.8	N10°50.2	0°00.3	N23°15.1	91°28.0	S01°21.8	Arcturus	145°47.4	19°03.1
13	108°55.3	60°32.4	10.5	312°01.9	49.6	15°02.1	15.1	106°30.4	21.8	Rigil Kent.	139°39.4	-60°56.7
14	123°57.8	75°32.2	11.3	327°03.0	49.1	30°04.0	15.1	121°32.8	21.8	Kochab	137°19.0	74°03.2
15	139°00.3	90°31.9	· · 12.0	342°04.1	· · 48.5	45°05.8	· · 15.1	136°35.2	· · 21.8	Zuben'ubi	136°55.4	-16°08.9
16	154°02.7	105°31.7	12.7	357°05.3	48.0	60°07.7	15.1	151°37.6	21.7	Alphecca	126°03.1	26°37.8
17	169°05.2	120°31.5	13.4	12°06.4	47.4	75°09.5	15.1	166°40.0	21.7	Antares	112°15.0	-26°29.4
18	184°07.7	135°31.3	N15°14.2	27°07.5	N10°46.9	90°11.4	N23°15.1	181°42.4	S01°21.7	Atria	107°08.4	-69°04.5
19	199°10.1	150°31.1	14.9	42°08.6	46.3	105°13.2	15.1	196°44.8	21.7	Sabik	102°02.0	-15°45.4
20	214°12.6	165°30.8	15.6	57°09.8	45.8	120°15.0	15.0	211°47.2	21.7	Shaula	96°09.4	-37°07.4
21	229°15.1	180°30.6	· · 16.3	72°10.9	· · 45.2	135°16.9	· · 15.0	226°49.6	· · 21.6	Rasalhague	95°57.9	12°32.5
22	244°17.5	195°30.4	17.0	87°12.0	44.6	150°18.7	15.0	241°51.9	21.6	Eltanin	90°41.5	51°29.1
23	259°20.0	210°30.2	17.8	102°13.1	44.1	165°20.6	15.0	256°54.3	21.6	Kaus Aust.	83°31.6	-34°22.3
Mer.pass. 05:45		ν -0.2' d 0.7' m -4.18		ν 1.1' d -0.6' m 1.46		ν 1.8' d -0.0' m -1.89		ν 2.4' d -0.0' m 0.99		Vega	80°32.5	38°48.4

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	274°22.5	225°30.0	N15°18.5	117°14.3	N10°43.5	180°22.4	N23°15.0	271°56.7	S01°21.6	Nunki	75°46.9	-26°15.9
1	289°24.9	240°29.7	19.2	132°15.4	43.0	195°24.3	15.0	286°59.1	21.6	Altair	61°59.3	8°56.1
2	304°27.4	255°29.5	19.9	147°16.5	42.4	210°26.1	15.0	302°01.5	21.6	Peacock	53°04.6	-56°39.0
3	319°29.8	270°29.3	· · 20.6	162°17.6	· · 41.9	225°27.9	· · 15.0	317°03.9	· · 21.5	Deneb	49°25.2	45°22.1
4	334°32.3	285°29.1	21.4	177°18.7	41.3	240°29.8	15.0	332°06.3	21.5	Enif	33°38.2	9°59.5
5	349°34.8	300°28.8	22.1	192°19.9	40.7	255°31.6	14.9	347°08.7	21.5	Al Na'ir	27°32.2	-46°50.0
6	4°37.2	315°28.6	N15°22.8	207°21.0	N10°40.2	270°33.5	N23°14.9	2°11.1	S01°21.5	Fomalhaut	15°14.0	-29°29.1
7	19°39.7	330°28.4	23.5	222°22.1	39.6	285°35.3	14.9	17°13.5	21.5	Scheat	13°44.7	28°13.1
8	34°42.2	345°28.1	24.2	237°23.2	39.1	300°37.2	14.9	32°15.9	21.4	Markab	13°29.4	15°20.5
9	49°44.6	0°27.9	· · 25.0	252°24.4	· · 38.5	315°39.0	· · 14.9	47°18.3	· · 21.4			
10	64°47.1	15°27.7	25.7	267°25.5	38.0	330°40.8	14.9	62°20.7	21.4	Jun 24 Tue	SHA	Mer.pass
11	79°49.6	30°27.5	26.4	282°26.6	37.4	345°42.7	14.9	77°23.1	21.4	Venus	313°15.8	08:57
12	94°52.0	45°27.2	N15°27.1	297°27.7	N10°36.8	0°44.5	N23°14.9	92°25.5	S01°21.4	Mars	203°56.1	16:13
13	109°54.5	60°27.0	27.8	312°28.8	36.3	15°46.4	14.8	107°27.9	21.4	Jupiter	266°29.8	12:03
14	124°56.9	75°26.8	28.5	327°30.0	35.7	30°48.2	14.8	122°30.3	21.3	Saturn	357°37.8	05:59
15	139°59.4	90°26.5	· · 29.2	342°31.1	· · 35.2	45°50.1	· · 14.8	137°32.7	· · 21.3			
16	155°01.9	105°26.3	30.0	357°32.2	34.6	60°51.9	14.8	152°35.1	21.3	Jun 25 Wed	SHA	Mer.pass
17	170°04.3	120°26.1	30.7	12°33.3	34.0	75°53.7	14.8	167°37.5	21.3	Venus	312°11.8	08:58
18	185°06.8	135°25.8	N15°31.4	27°34.5	N10°33.5	90°55.6	N23°14.8	182°39.9	S01°21.3	Mars	203°24.0	16:12
19	200°09.3	150°25.6	32.1	42°35.6	32.9	105°57.4	14.8	197°42.3	21.3	Jupiter	266°14.9	12:00
20	215°11.7	165°25.4	32.8	57°36.7	32.4	120°59.3	14.8	212°44.6	21.2	Saturn		

Sun				Moon				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	179°24.0	N23°24.6		201°32.9	2.4'	N27°16.2	5.2'	60.2'
1	194°23.8	24.5		215°54.3	2.4'	27°21.4	5.0'	60.2'
2	209°23.7	24.5		230°15.6	2.3'	27°26.4	4.8'	60.2'
3	224°23.6	· · 24.4		244°37.0	2.3'	27°31.2	4.6'	60.2'
4	239°23.4	24.4		258°58.2	2.2'	27°35.8	4.4'	60.2'
5	254°23.3	24.3		273°19.4	2.2'	27°40.2	4.2'	60.2'
6	269°23.1	N23°24.3		287°40.6	2.1'	N27°44.4	4.0'	60.2'
7	284°23.0	24.2		302°01.7	2.1'	27°48.4	3.8'	60.2'
8	299°22.9	24.2		316°22.8	2.1'	27°52.2	3.6'	60.1'
9	314°22.7	· · 24.1		330°43.9	2.0'	27°55.8	3.4'	60.1'
10	329°22.6	24.0		345°04.9	2.0'	27°59.1	3.2'	60.1'
11	344°22.5	24.0		359°25.9	2.0'	28°02.3	3.0'	60.1'
12	359°22.3	N23°23.9		13°46.9	2.0'	N28°05.2	2.7'	60.1'
13	14°22.2	23.9		28°07.9	1.9'	28°08.0	2.5'	60.1'
14	29°22.1	23.8		42°28.8	1.9'	28°10.5	2.3'	60.1'
15	44°21.9	· · 23.8		56°49.7	1.9'	28°12.8	2.1'	60.1'
16	59°21.8	23.7		71°10.7	1.9'	28°15.0	1.9'	60.0'
17	74°21.7	23.6		85°31.6	1.9'	28°16.9	1.7'	60.0'
18	89°21.5	N23°23.6		99°52.5	1.9'	N28°18.5	1.5'	60.0'
19	104°21.4	23.5		114°13.4	1.9'	28°20.0	1.3'	60.0'
20	119°21.3	23.4		128°34.3	1.9'	28°21.3	1.1'	60.0'
21	134°21.1	· · 23.4		142°55.2	1.9'	28°22.3	0.8'	60.0'
22	149°21.0	23.3		157°16.2	1.9'	28°23.2	0.6'	59.9'
23	164°20.9	23.2		171°37.1	2.0'	28°23.8	0.4'	59.9'
SD = 15.7' $d = -0.1'$				SD = 16.4'				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	179°20.7	N23°23.2		185°58.1	2.0'	N28°24.3	0.2'	59.9'
1	194°20.6	23.1		200°19.0	2.0'	28°24.5	0.0'	59.9'
2	209°20.5	23.0		214°40.0	2.0'	28°24.5	-0.2'	59.9'
3	224°20.3	· · 23.0		229°01.1	2.1'	28°24.3	-0.4'	59.9'
4	239°20.2	22.9		243°22.1	2.1'	28°23.9	-0.6'	59.8'
5	254°20.0	22.8		257°43.2	2.1'	28°23.2	-0.8'	59.8'
6	269°19.9	N23°22.8		272°04.3	2.2'	N28°22.4	-1.0'	59.8'
7	284°19.8	22.7		286°25.5	2.2'	28°21.4	-1.2'	59.8'
8	299°19.6	22.6		300°46.7	2.3'	28°20.1	-1.5'	59.8'
9	314°19.5	· · 22.5		315°08.0	2.3'	28°18.7	-1.7'	59.7'
10	329°19.4	22.5		329°29.3	2.4'	28°17.0	-1.9'	59.7'
11	344°19.2	22.4		343°50.6	2.4'	28°15.2	-2.1'	59.7'
12	359°19.1	N23°22.3		358°12.1	2.5'	N28°13.1	-2.3'	59.7'
13	14°19.0	22.2		12°33.5	2.5'	28°10.8	-2.5'	59.7'
14	29°18.8	22.2		26°55.1	2.6'	28°08.4	-2.7'	59.6'
15	44°18.7	· · 22.1		41°16.7	2.7'	28°05.7	-2.9'	59.6'
16	59°18.6	22.0		55°38.3	2.7'	28°02.8	-3.1'	59.6'
17	74°18.4	21.9		70°00.0	2.8'	27°59.7	-3.3'	59.6'
18	89°18.3	N23°21.9		84°21.9	2.9'	N27°56.5	-3.5'	59.5'
19	104°18.2	21.8		98°43.7	3.0'	27°53.0	-3.7'	59.5'
20	119°18.0	21.7		113°05.7	3.0'	27°49.4	-3.9'	59.5'
21	134°17.9	· · 21.6		127°27.7	3.1'	27°45.5	-4.0'	59.5'
22	149°17.8	21.5		141°49.9	3.2'	27°41.5	-4.2'	59.4'
23	164°17.6	21.4		156°12.1	3.3'	27°37.2	-4.4'	59.4'
SD = 15.7' $d = -0.1'$				SD = 16.3'				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	179°17.5	N23°21.4		170°34.4	3.4'	N27°32.8	-4.6'	59.4'
1	194°17.4	21.3		184°56.8	3.5'	27°28.2	-4.8'	59.4'
2	209°17.2	21.2		199°19.2	3.6'	27°23.4	-5.0'	59.3'
3	224°17.1	· · 21.1		213°41.8	3.7'	27°18.4	-5.2'	59.3'
4	239°17.0	21.0		228°04.5	3.8'	27°13.2	-5.4'	59.3'
5	254°16.9	20.9		242°27.3	3.9'	27°07.9	-5.5'	59.2'
6	269°16.7	N23°20.8		256°50.2	4.0'	N27°02.4	-5.7'	59.2'
7	284°16.6	20.8		271°13.1	4.1'	26°56.7	-5.9'	59.2'
8	299°16.5	20.7		285°36.2	4.2'	26°50.8	-6.1'	59.2'
9	314°16.3	· · 20.6		299°59.4	4.3'	26°44.7	-6.2'	59.1'
10	329°16.2	20.5		314°22.7	4.4'	26°38.5	-6.4'	59.1'
11	344°16.1	20.4		328°46.1	4.5'	26°32.1	-6.6'	59.1'
12	359°15.9	N23°20.3		343°09.7	4.6'	N26°25.5	-6.7'	59.1'
13	14°15.8	20.2		357°33.3	4.8'	26°18.7	-6.9'	59.0'
14	29°15.7	20.1		11°57.1	4.9'	26°11.8	-7.1'	59.0'
15	44°15.5	· · 20.0		26°21.0	5.0'	26°04.7	-7.2'	59.0'
16	59°15.4	19.9		40°45.0	5.1'	25°57.5	-7.4'	58.9'
17	74°15.3	19.8		55°09.1	5.2'	25°50.1	-7.6'	58.9'
18	89°15.1	N23°19.7		69°33.3	5.4'	N25°42.5	-7.7'	58.9'
19	104°15.0	19.6		83°57.7	5.5'	25°34.8	-7.9'	58.8'
20	119°14.9	19.5		98°22.2	5.6'	25°26.9	-8.0'	58.8'
21	134°14.7	· · 19.4		112°46.8	5.7'	25°18.9	-8.2'	58.8'
22	149°14.6	19.3		127°11.6	5.9'	25°10.7	-8.3'	58.7'
23	164°14.5	19.2		141°36.4	6.0'	25°02.3	-8.5'	58.7'
SD = 15.7' $d = -0.1'$				SD = 16.2'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°						
64°	—:—	—:—	01:33	22:31	—:—	—:—
62°	—:—	—:—	02:11	21:54	—:—	—:—
60°	—:—	00:52	02:37	21:28	23:12	—:—
N 58°	—:—	01:42	02:58	21:07	22:23	—:—
56°	—:—	02:12	03:15	20:51	21:53	—:—
54°	00:48	02:34	03:29	20:36	21:31	23:16
52°	01:34	02:52	03:41	20:24	21:13	22:31
50°	02:02	03:07	03:52	20:13	20:58	22:03
45°	02:47	03:37	04:14	19:51	20:28	21:18
N 40°	03:18	04:00	04:32	19:33	20:06	20:47
35°	03:41	04:18	04:48	19:18	19:48	20:24
30°	04:00	04:33	05:01	19:05	19:32	20:06
20°	04:29	04:58	05:23	18:43	19:07	19:37
N 10°	04:51	05:18	05:41	18:24	18:47	19:14
0°	05:10	05:37	05:59	18:06	18:29	18:55
S 10°	05:27	05:54	06:16	17:49	18:12	18:38
20°	05:43	06:11	06:35	17:30	17:54	18:22
30°	06:00	06:30	06:56	17:09	17:36	18:06
35°	06:09	06:40	07:09	16:57	17:25	17:57
40°	06:18	06:52	07:23	16:43	17:13	17:47
45°	06:28	07:06	07:39	16:26	17:00	17:37
S 50°	06:40	07:22	08:00	16:05	16:44	17:25
52°	06:45	07:29	08:10	15:55	16:36	17:20
54°	06:51	07:37	08:21	15:44	16:28	17:14
56°	06:57	07:46	08:34	15:32	16:19	17:08
58°	07:04	07:57	08:48	15:17	16:09	17:02
S 60°	07:12	08:08	09:06	15:00	15:57	16:54

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°						
N 70°						
68°						
66°						
64°						
62°						
60°	00:24	01:05	02:33	22:05	22:52	23:08
N 58°	00:55	01:44	03:05	21:26	22:19	22:45
56°	01:19	02:11	03:29	20:59	21:55	22:27
54°	01:38	02:32	03:48	20:38	21:35	22:11
52°	01:54	02:50	04:05	20:20	21:19	21:58
50°	02:08	03:05	04:19	20:05	21:04	21:46
45°	02:36	03:36	04:47	19:34	20:35	21:21
N 40°	02:59	04:00	05:10	19:10	20:12	21:02
35°	03:18	04:19	05:28	18:50	19:53	20:45
30°	03:34	04:36	05:44	18:34	19:37	20:31
20°	04:01	05:05	06:11	18:05	19:09	20:06
N 10°	04:25	05:29	06:34	17:41	18:45	19:45
0°	04:47	05:52	06:56	17:18	18:23	19:24
S 10°	05:09	06:15	07:17	16:55	18:00	19:04
20°	05:3:53					

June 27, 28, 29 UT (Fri., Sat., Sun.)

Aries		Venus		Mars		Jupiter		Saturn	
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	275°21.6	225°24.4	N15°35.7	117°41.2	N10°30.1	181°06.6	N23°14.7	272°54.2	S01°21.2
1	290°24.1	240°24.2	36.4	132°42.3	29.6	196°08.5	14.7	287°56.6	21.2
2	305°26.5	255°23.9	37.1	147°43.4	29.0	211°10.3	14.7	302°59.0	21.1
3	320°29.0	270°23.7	· · 37.8	162°44.5	· · 28.4	226°12.2	· · 14.7	318°01.4	· · 21.1
4	335°31.4	285°23.4	38.5	177°45.7	27.9	241°14.0	14.7	333°03.8	21.1
5	350°33.9	300°23.2	39.2	192°46.8	27.3	256°15.9	14.6	348°06.2	21.1
6	5°36.4	315°23.0	N15°39.9	207°47.9	N10°26.8	271°17.7	N23°14.6	3°08.6	S01°21.1
7	20°38.8	330°22.7	40.6	222°49.0	26.2	286°19.5	14.6	18°11.0	21.1
8	35°41.3	345°22.5	41.3	237°50.2	25.6	301°21.4	14.6	33°13.4	21.0
9	50°43.8	0°22.2	· · 42.0	252°51.3	· · 25.1	316°23.2	· · 14.6	48°15.8	· · 21.0
10	65°46.2	15°22.0	42.8	267°52.4	24.5	331°25.1	14.6	63°18.2	21.0
11	80°48.7	30°21.7	43.5	282°53.5	24.0	346°26.9	14.6	78°20.6	21.0
12	95°51.2	45°21.5	N15°44.2	297°54.6	N10°23.4	1°28.8	N23°14.5	93°23.0	S01°21.0
13	110°53.6	60°21.3	44.9	312°55.8	22.8	16°30.6	14.5	108°25.4	21.0
14	125°56.1	75°21.0	45.6	327°56.9	22.3	31°32.4	14.5	123°27.8	20.9
15	140°58.6	90°20.8	· · 46.3	342°58.0	· · 21.7	46°34.3	· · 14.5	138°30.2	· · 20.9
16	156°01.0	105°20.5	47.0	357°59.1	21.2	61°36.1	14.5	153°32.6	20.9
17	171°03.5	120°20.3	47.7	13°00.2	20.6	76°38.0	14.5	168°35.0	20.9
18	186°05.9	135°20.0	N15°48.4	28°01.4	N10°20.0	91°39.8	N23°14.5	183°37.4	S01°20.9
19	201°08.4	150°19.8	49.1	43°02.5	19.5	106°41.7	14.5	198°39.8	20.9
20	216°10.9	165°19.5	49.8	58°03.6	18.9	121°43.5	14.4	213°42.2	20.9
21	231°13.3	180°19.3	· · 50.5	73°04.7	· · 18.3	136°45.3	· · 14.4	228°44.6	· · 20.8
22	246°15.8	195°19.0	51.2	88°05.8	17.8	151°47.2	14.4	243°47.0	20.8
23	261°18.3	210°18.8	51.9	103°07.0	17.2	166°49.0	14.4	258°49.4	20.8
Mer.pass. 05:38		ν -0.2' d 0.7' m -4.17		ν 1.1' d -0.6' m 1.47		ν 1.8' d -0.0' m -1.89		ν 2.4' d -0.0' m 0.98	

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	276°20.7	225°18.5	N15°52.6	118°08.1	N10°16.7	181°50.9	N23°14.4	273°51.8	S01°20.8
1	291°23.2	240°18.3	53.3	133°09.2	16.1	196°52.7	14.4	288°54.2	20.8
2	306°25.7	255°18.0	54.0	148°10.3	15.5	211°54.6	14.4	303°56.6	20.8
3	321°28.1	270°17.7	· · 54.7	163°11.4	· · 15.0	226°56.4	· · 14.3	318°59.0	· · 20.8
4	336°30.6	285°17.5	55.4	178°12.6	14.4	241°58.3	14.3	334°01.4	20.7
5	351°33.1	300°17.2	56.1	193°13.7	13.8	257°00.1	14.3	349°03.8	20.7
6	6°35.5	315°17.0	N15°56.8	208°14.8	N10°13.3	272°01.9	N23°14.3	4°06.2	S01°20.7
7	21°38.0	330°16.7	57.5	223°15.9	12.7	287°03.8	14.3	19°08.6	20.7
8	36°40.4	345°16.5	58.2	238°17.0	12.1	302°05.6	14.3	34°11.0	20.7
9	51°42.9	0°16.2	· · 58.9	253°18.2	· · 11.6	317°07.5	· · 14.3	49°13.4	· · 20.7
10	66°45.4	15°15.9	15°59.6	268°19.3	11.0	332°09.3	14.2	64°15.9	20.7
11	81°47.8	30°15.7	16°00.3	283°20.4	10.5	347°11.2	14.2	79°18.3	20.6
12	96°50.3	45°15.4	N16°01.0	298°21.5	N10°09.9	2°13.0	N23°14.2	94°20.7	S01°20.6
13	111°52.8	60°15.2	01.7	313°22.6	09.3	17°14.8	14.2	109°23.1	20.6
14	126°55.2	75°14.9	02.4	328°23.7	08.8	32°16.7	14.2	124°25.5	20.6
15	141°57.7	90°14.6	· · 03.1	343°24.9	· · 08.2	47°18.5	· · 14.2	139°27.9	· · 20.6
16	157°00.2	105°14.4	03.8	358°26.0	07.6	62°20.4	14.2	154°30.3	20.6
17	172°02.6	120°14.1	04.5	13°27.1	07.1	77°22.2	14.1	169°32.7	20.6
18	187°05.1	135°13.9	N16°05.2	28°28.2	N10°06.5	92°24.1	N23°14.1	184°35.1	S01°20.5
19	202°07.6	150°13.6	05.9	43°29.3	05.9	107°25.9	14.1	199°37.5	20.5
20	217°10.0	165°13.3	06.6	58°30.5	05.4	122°27.7	14.1	214°39.9	20.5
21	232°12.5	180°13.1	· · 07.3	73°31.6	· · 04.8	137°29.6	· · 14.1	229°42.3	· · 20.5
22	247°14.9	195°12.8	08.0	88°32.7	04.2	152°31.4	14.1	244°44.7	20.5
23	262°17.4	210°12.5	08.7	103°33.8	03.7	167°33.3	14.1	259°47.1	20.5
Mer.pass. 05:34		ν -0.3' d 0.7' m -4.16		ν 1.1' d -0.6' m 1.48		ν 1.8' d -0.0' m -1.89		ν 2.4' d -0.0' m 0.97	

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	277°19.9	225°12.3	N16°09.3	118°34.9	N10°03.1	182°35.1	N23°14.0	274°49.5	S01°20.5
1	292°22.3	240°12.0	10.0	133°36.1	02.5	197°37.0	14.0	289°51.9	20.4
2	307°24.8	255°11.7	10.7	148°37.2	02.0	212°38.8	14.0	304°54.3	20.4
3	322°27.3	270°11.5	· · 11.4	163°38.3	· · 01.4	227°40.7	· · 14.0	319°56.7	· · 20.4
4	337°29.7	285°11.2	12.1	178°39.4	00.8	242°42.5	14.0	334°59.1	20.4
5	352°32.2	300°10.9	12.8	193°40.5	10°00.3	257°44.3	14.0	350°01.5	20.4
6	7°34.7	315°10.6	N16°13.5	208°41.6	N09°59.7	272°46.2	N23°14.0	5°03.9	S01°20.4
7	22°37.1	330°10.4	14.2	223°42.8	59.1	287°48.0	13.9	20°06.4	20.4
8	37°39.6	345°10.1	14.9	238°43.9	58.6	302°49.9	13.9	35°08.8	20.4
9	52°42.0	0°09.8	· · 15.6	253°45.0	· · 58.0	317°51.7	· · 13.9	50°11.2	· · 20.3
10	67°44.5	15°09.6	16.2	268°46.1	57.4	332°53.6	13.9	65°13.6	20.3
11	82°47.0	30°09.3	16.9	283°47.2	56.9	347°55.4	13.9	80°16.0	20.3
12	97°49.4	45°09.0	N16°17.6	298°48.4	N09°56.3	2°57.2	N23°13.9	95°18.4	S01°20.3
13	112°51.9	60°08.7	18.3	313°49.5	55.7	17°59.1	13.8	110°20.8	20.3
14	127°54.4	75°08.5	19.0	328°50.6	55.2	33°00.9	13.8	125°23.2	20.3
15	142°56.8	90°08.2	· · 19.7	343°51.7	· · 54.6	48°02.8	· · 13.8	140°25.6	· · 20.3
16	157°59.3	105°07.9	20.4	358°52.8	54.0	63°04.6	13.8	155°28.0	20.3
17	173°01.8	120°07.6	21.0	13°53.9	53.5	78°06.5	13.8	170°30.4	20.2
18	188°04.2	135°07.4	N16°21.7	28°55.1	N09°52.9	93°08.3	N23°13.8	185°32.8	S01°20.2
19	203°06.7	150°07.1	22.4	43°56.2	52.3	108°10.2	13.8	200°35.2	20.2
20	218°09.2	165°06.8	23.1	58°57.3	51.8	123°12.0	13.7	215°37.6	20.2
21	233°11.6	180°06.5	· · 23.8	73°58.4	· · 51.2	138°13.8	· · 13.7	230°40.1	· · 20.2
22	248°14.1	195°06.2	24.5	88°59.5	50.6	153°15.7	13.7	245°42.5	20.2
23	263°16.5	210°06.0	25.1	104°00.6	50.1	168°17.5	13.7	260°44.9	20.2
Mer.pass. 05:30		ν -0.3' d 0.7' m -4.15		ν 1.1' d -0.6' m 1.48		ν 1.8' d -0.0' m -1.89		ν 2.4' d -0.0' m 0.97	

Stars		
	SHA	Dec
Alpheratz	357°34.3	29°13.7
Ankaa	353°06.8	-42°09.8
Schedar	349°30.7	56°40.4
Diphda	348°46.9	-17°50.7
Achernar	335°20.1	-57°06.1
Hamal	327°50.9	23°34.9
Polaris	314°09.5	89°22.0
Acamar	315°11.7	-40°11.9
Menkar	314°06.0	4°11.4
Mirfak	308°28.1	49°56.9
Aldebaran	290°39.5	16°33.6
Rigel	281°03.8	-8°10.3
Capella	280°21.8	46°01.3
Bellatrix	278°22.8	6°22.4
Elnath	278°01.7	28°37.7
Alnilam	275°37.7	-1°11.1
Betelgeuse	270°52.0	7°24.7
Canopus	263°52.8	-52°42.5
Sirius	258°26.2	-16°45.0
Adhara	255°05.9	-29°00.4
Procyon	244°50.7	5°09.6
Pollux	243°17.2	27°57.9
Avior	234°15.2	-59°35.6
Suhail	222°46.4	-43°32.2
Miaplacidus	221°39.0	-69°49.5
Alphard	217°47.6	-8°46.2
Regulus	207°34.2	11°50.6
Dubhe	193°40.8	61°37.1
Denebola	182°24.6	14°25.9
Gienah	175°43.2	-17°41.1
Acrux	172°59.7	-63°14.7
Gacrux	171°51.2	-57°15.6
Alioth	166°12.5	55°49.5
Spica	158°21.8	-11°17.7
Alkaid	152°51.5	49°11.4
Hadar	148°35.2	-60°30.0
Menkent	147°57.0	-36°29.9
Arcturus	145°47.4	19°03.1
Rigil Kent.	139°39.4	-60°56.7
Kochab	137°19.0	74°03.2
Zuben'ubi	136°55.4	-16°08.9
Alphecca	126°03.1	26°37.8
Antares	112°15.0	-26°29.4
Atria	107°08.4	-69°04.5
Sabik	102°02.0	-15°45.4
Shaula	96°09.4	-37°07.4
Rasalhague	95°57.8	12°32.5
Eltanin	90°41.5	51°29.1
Kaus Aust.	83°31.6	-34°22.3
Vega	80°32.5	38°48.4
Nunki	75°46.9	-26°15.9
Altair	61°59.3	8°56.1
Peacock	53°04.6	-56°39.0
Deneb	49°25.1	45°22.1
Enif	33°38.2	9°59.5
Al Na'ir	27°32.1	-46°50.0
Fomalhaut	15°13.9	-29°29.0
Scheat	13°44.7	28°13.1
Markab	13°29.4	15°20.5

Jun 27 Fri	SHA	Mer.pass
Venus	310°02.8	08:59
Mars	202°19.6	16:08
Jupiter	265°45.0	11:54
Saturn	357°32.6	05:47

Jun 28 Sat	SHA	Mer.pass
Venus	308°57.8	08:59
Mars	201°47.3	16:06
Jupiter	265°30.1	11:51
Saturn	357°31.1	05:44

Jun 29 Sun	SHA	Mer.pass
Venus	307°52.4	08:59
Mars	201°15.1	16:04
Jupiter	265°15.2	11:48
Saturn	357°29.6	05:40

Horizontal parallax		
Venus:	0.2	
Mars:	0.1	

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	179°14.3	N23°19.1	156°01.4	6.1'	N24°53.9	-8.6'	58.7'
1	194°14.2	19.0	170°26.6	6.3'	24°45.2	-8.8'	58.6'
2	209°14.1	18.9	184°51.8	6.4'	24°36.5	-8.9'	58.6'
3	224°14.0	•• 18.8	199°17.2	6.5'	24°27.5	-9.1'	58.6'
4	239°13.8	18.7	213°42.7	6.7'	24°18.5	-9.2'	58.6'
5	254°13.7	18.6	228°08.4	6.8'	24°09.3	-9.3'	58.5'
6	269°13.6	N23°18.5	242°34.2	6.9'	N24°00.0	-9.5'	58.5'
7	284°13.4	18.4	257°00.1	7.0'	23°50.5	-9.6'	58.5'
8	299°13.3	18.3	271°26.1	7.2'	23°40.9	-9.7'	58.4'
9	314°13.2	•• 18.2	285°52.3	7.3'	23°31.2	-9.9'	58.4'
10	329°13.0	18.1	300°18.6	7.4'	23°21.3	-10.0'	58.4'
11	344°12.9	18.0	314°45.1	7.6'	23°11.3	-10.1'	58.3'
12	359°12.8	N23°17.9	329°11.6	7.7'	N23°01.2	-10.2'	58.3'
13	14°12.6	17.8	343°38.4	7.8'	22°51.0	-10.3'	58.3'
14	29°12.5	17.7	358°05.2	8.0'	22°40.7	-10.5'	58.2'
15	44°12.4	•• 17.5	12°32.2	8.1'	22°30.2	-10.6'	58.2'
16	59°12.3	17.4	26°59.3	8.3'	22°19.6	-10.7'	58.1'
17	74°12.1	17.3	41°26.6	8.4'	22°08.9	-10.8'	58.1'
18	89°12.0	N23°17.2	55°53.9	8.5'	N21°58.1	-10.9'	58.1'
19	104°11.9	17.1	70°21.5	8.6'	21°47.2	-11.0'	58.0'
20	119°11.7	17.0	84°49.1	8.8'	21°36.2	-11.1'	58.0'
21	134°11.6	•• 16.9	99°16.9	8.9'	21°25.0	-11.2'	58.0'
22	149°11.5	16.7	113°44.8	9.0'	21°13.8	-11.3'	57.9'
23	164°11.4	16.6	128°12.8	9.2'	21°02.4	-11.4'	57.9'
SD = 15.7'			SD = 16.0'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	179°11.2	N23°16.5	142°41.0	9.3'	N20°51.0	-11.5'	57.9'
1	194°11.1	16.4	157°09.3	9.4'	20°39.5	-11.6'	57.8'
2	209°11.0	16.3	171°37.8	9.6'	20°27.8	-11.7'	57.8'
3	224°10.8	•• 16.2	186°06.3	9.7'	20°16.1	-11.8'	57.8'
4	239°10.7	16.0	200°35.0	9.8'	20°04.3	-11.9'	57.7'
5	254°10.6	15.9	215°03.9	10.0'	19°52.3	-12.0'	57.7'
6	269°10.4	N23°15.8	229°32.8	10.1'	N19°40.3	-12.1'	57.7'
7	284°10.3	15.7	244°01.9	10.2'	19°28.2	-12.2'	57.6'
8	299°10.2	15.5	258°31.1	10.3'	19°16.1	-12.3'	57.6'
9	314°10.1	•• 15.4	273°00.4	10.5'	19°03.8	-12.4'	57.6'
10	329°09.9	15.3	287°29.9	10.6'	18°51.4	-12.4'	57.5'
11	344°09.8	15.2	301°59.5	10.7'	18°39.0	-12.5'	57.5'
12	359°09.7	N23°15.0	316°29.2	10.8'	N18°26.5	-12.6'	57.5'
13	14°09.6	14.9	330°59.0	10.9'	18°13.9	-12.7'	57.4'
14	29°09.4	14.8	345°28.9	11.1'	18°01.2	-12.7'	57.4'
15	44°09.3	•• 14.7	359°59.0	11.2'	17°48.5	-12.8'	57.3'
16	59°09.2	14.5	14°29.2	11.3'	17°35.7	-12.9'	57.3'
17	74°09.0	14.4	28°59.5	11.4'	17°22.8	-12.9'	57.3'
18	89°08.9	N23°14.3	43°29.9	11.5'	N17°09.9	-13.0'	57.2'
19	104°08.8	14.1	58°00.5	11.7'	16°56.8	-13.1'	57.2'
20	119°08.7	14.0	72°31.1	11.8'	16°43.8	-13.1'	57.2'
21	134°08.5	•• 13.9	87°01.9	11.9'	16°30.6	-13.2'	57.1'
22	149°08.4	13.7	101°32.8	12.0'	16°17.4	-13.3'	57.1'
23	164°08.3	13.6	116°03.8	12.1'	16°04.1	-13.3'	57.1'
SD = 15.7'			SD = 15.8'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	179°08.1	N23°13.5	130°34.9	12.2'	N15°50.8	-13.4'	57.0'
1	194°08.0	13.3	145°06.1	12.3'	15°37.4	-13.4'	57.0'
2	209°07.9	13.2	159°37.4	12.4'	15°24.0	-13.5'	57.0'
3	224°07.8	•• 13.1	174°08.8	12.5'	15°10.5	-13.6'	56.9'
4	239°07.6	12.9	188°40.4	12.6'	14°56.9	-13.6'	56.9'
5	254°07.5	12.8	203°12.0	12.7'	14°43.3	-13.7'	56.9'
6	269°07.4	N23°12.6	217°43.8	12.8'	N14°29.7	-13.7'	56.8'
7	284°07.3	12.5	232°15.6	12.9'	14°16.0	-13.8'	56.8'
8	299°07.1	12.4	246°47.6	13.1'	14°02.2	-13.8'	56.8'
9	314°07.0	•• 12.2	261°19.6	13.1'	13°48.4	-13.8'	56.7'
10	329°06.9	12.1	275°51.8	13.2'	13°34.6	-13.9'	56.7'
11	344°06.8	11.9	290°24.0	13.3'	13°20.7	-13.9'	56.7'
12	359°06.6	N23°11.8	304°56.4	13.4'	N13°06.8	-14.0'	56.6'
13	14°06.5	11.6	319°28.8	13.5'	12°52.8	-14.0'	56.6'
14	29°06.4	11.5	334°01.3	13.6'	12°38.8	-14.0'	56.6'
15	44°06.3	•• 11.4	348°34.0	13.7'	12°24.8	-14.1'	56.5'
16	59°06.1	11.2	3°06.7	13.8'	12°10.7	-14.1'	56.5'
17	74°06.0	11.1	17°39.5	13.9'	11°56.6	-14.2'	56.5'
18	89°05.9	N23°10.9	32°12.4	14.0'	N11°42.4	-14.2'	56.4'
19	104°05.8	10.8	46°45.4	14.1'	11°28.2	-14.2'	56.4'
20	119°05.6	10.6	61°18.4	14.2'	11°14.0	-14.2'	56.4'
21	134°05.5	•• 10.5	75°51.6	14.2'	10°59.8	-14.3'	56.3'
22	149°05.4	10.3	90°24.8	14.3'	10°45.5	-14.3'	56.3'
23	164°05.3	10.2	104°58.1	14.4'	10°31.2	-14.3'	56.3'
SD = 15.7'			SD = 15.6'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	☐	☐	☐	☐	☐	☐
64°	—:—	—:—	01:37	22:29	—:—	—:—
62°	—:—	—:—	02:14	21:52	—:—	—:—
60°	—:—	00:57	02:40	21:27	23:08	—:—
N 58°	—:—	01:45	03:00	21:07	22:21	—:—
56°	—:—	02:14	03:16	20:50	21:52	—:—
54°	00:53	02:36	03:30	20:36	21:30	23:13
52°	01:37	02:54	03:43	20:24	21:12	22:29
50°	02:04	03:09	03:53	20:13	20:57	22:02
45°	02:49	03:38	04:16	19:51	20:28	21:17
N 40°	03:19	04:01	04:34	19:33	20:06	20:47
35°	03:42	04:19	04:49	19:18	19:48	20:24
30°	04:01	04:34	05:01	19:05	19:33	20:06
20°	04:30	04:59	05:23	18:43	19:08	19:37
N 10°	04:52	05:19	05:42	18:24	18:47	19:15
0°	05:11	05:37	06:00	18:07	18:30	18:56
S 10°	05:28	05:54	06:17	17:50	18:13	18:39
20°	05:44	06:11	06:35	17:31	17:55	18:23
30°	06:00	06:30	06:56	17:10	17:37	18:06
35°	06:09	06:41	07:09	16:58	17:26	17:58
40°	06:18	06:52	07:23	16:44	17:14	17:48
45°	06:29	07:06	07:40	16:27	17:01	17:38
S 50°	06:40	07:22	08:00	16:07	16:45	17:27
52°	06:45	07:29	08:10	15:57	16:38	17:21
54°	06:51	07:37	08:21	15:46	16:30	17:16
56°	06:57	07:46	08:33	15:33	16:21	17:10
58°	07:04	07:56	08:48	15:19	16:11	17:03
S 60°	07:11	08:08	09:05	15:02	15:59	16:56

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	☐	☐	06:01	☐	☐	01:13
N 70°	☐	☐	06:33	☐	☐	00:39
68°	☐	04:25	06:56	☐	01:03	00:14 23:47
66°	☐	05:05	07:14	☐	00:21 23:55	23:37
64°	03:13	05:33	07:29	00:19 23:53	23:39	23:28
62°	03:55	05:54	07:41	23:31	23:26	23:21
60°	04:23	06:11	07:51	23:13	23:14	23:14
N 58°	04:45	06:25	07:59	22:58	23:04	23:08
56°	05:02	06:37	08:07	22:45	22:56	23:03
54°	05:17	06:48	08:14	22:34	22:48	22:59
52°	05:30	06:57	08:20	22:24	22:41	22:54
50°	05:42	07:05	08:26	22:15	22:35	22:50
45°	06:05	07:23	08:37	21:55	22:21	22:42
N 40°	06:24	07:37	08:47	21:40	22:10	22:35
35°	06:40	07:49	08:55	21:27	22:00	22:29
30°	06:53	08:00	09:03	21:15	21:52	22:23
20°	07:16	08:18	09:15	20:55	21:37	22:14
N 10°	07:36	08:34	09:26	20:37	21:24	22:06
0°	07:55	08:48	09:36	20:21	21:12	21:58
S 10°	08:13	09:03	09:46	20:04	20:59	21:50
20°	08:33	09:18	09:57	19:46	20:46	21:41
30°	08:55	09:36	10:09	19:26	20:31	21:31
35°	09:09	09:46	10:16	19:13	20:22	21:26
40°	09:24	09:57	10:24	18:59	20:11	21:19
45°	09:42	10:11	10:34	18:43	19:59	21:12
S 50°	10:04	10:27	10:45	18:22	19:44	21:02
52°	10:14	10:35	10:50	18:12	19:37	20:58
54°	10:26	10:44	10:55	18:00	19:29	20:53
56°	10:40	10:53	11:02	17:47	19:21	20:48
58°	10:56	11:04	11:09	17:32	19:11	20:42
S 60°	11:14	11:16	11:17	17:14	18:59	20:35

Day	Sun		Mer. Pass hh:mm	Moon		Age 2-4 3-15%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
27	03:03	03:09	12:03	14:08	01:40	
28	03:15	03:21	12:03	15:00	02:35	
29	03:27	03:33	12:04	15:47	03:24	

June 30, 01, 02 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	278°19.0	225°05.7	N16°25.8	119°01.8	N09°49.5	183°19.4	N23°13.7	275°47.3	S01°20.2	Alpheratz	357°34.3	29°13.7
1	293°21.5	240°05.4	26.5	134°02.9	48.9	198°21.2	13.7	290°49.7	20.2	Ankaa	353°06.7	-42°09.8
2	308°23.9	255°05.1	27.2	149°04.0	48.3	213°23.1	13.6	305°52.1	20.1	Schedar	349°30.6	56°40.4
3	323°26.4	270°04.8	· · 27.9	164°05.1	· · 47.8	228°24.9	· · 13.6	320°54.5	· · 20.1	Diphda	348°46.9	-17°50.7
4	338°28.9	285°04.5	28.5	179°06.2	47.2	243°26.8	13.6	335°56.9	20.1	Achernar	335°20.0	-57°06.1
5	353°31.3	300°04.3	29.2	194°07.3	46.6	258°28.6	13.6	350°59.3	20.1	Hamal	327°50.9	23°34.9
6	8°33.8	315°04.0	N16°29.9	209°08.5	N09°46.1	273°30.4	N23°13.6	6°01.7	S01°20.1	Polaris	314°08.3	89°22.0
7	23°36.3	330°03.7	30.6	224°09.6	45.5	288°32.3	13.6	21°04.2	20.1	Acamar	315°11.7	-40°11.9
8	38°38.7	345°03.4	31.3	239°10.7	44.9	303°34.1	13.5	36°06.6	20.1	Menkar	314°05.9	4°11.4
9	53°41.2	0°03.1	· · 31.9	254°11.8	· · 44.4	318°36.0	· · 13.5	51°09.0	· · 20.1	Mirfak	308°28.0	49°56.9
10	68°43.7	15°02.8	32.6	269°12.9	43.8	333°37.8	13.5	66°11.4	20.1	Aldebaran	290°39.5	16°33.6
11	83°46.1	30°02.5	33.3	284°14.0	43.2	348°39.7	13.5	81°13.8	20.0	Rigel	281°03.8	-8°10.3
12	98°48.6	45°02.2	N16°34.0	299°15.2	N09°42.7	3°41.5	N23°13.5	96°16.2	S01°20.0	Capella	280°21.8	46°01.3
13	113°51.0	60°02.0	34.6	314°16.3	42.1	18°43.3	13.5	111°18.6	20.0	Bellatrix	278°22.8	6°22.4
14	128°53.5	75°01.7	35.3	329°17.4	41.5	33°45.2	13.4	126°21.0	20.0	Elnath	278°01.7	28°37.7
15	143°56.0	90°01.4	· · 36.0	344°18.5	· · 40.9	48°47.0	· · 13.4	141°23.4	· · 20.0	Alnilam	275°37.7	-1°11.1
16	158°58.4	105°01.1	36.7	359°19.6	40.4	63°48.9	13.4	156°25.9	20.0	Betelgeuse	270°52.0	7°24.7
17	174°00.9	120°00.8	37.3	14°20.7	39.8	78°50.7	13.4	171°28.3	20.0	Canopus	263°52.8	-52°42.5
18	189°03.4	135°00.5	N16°38.0	29°21.9	N09°39.2	93°52.6	N23°13.4	186°30.7	S01°20.0	Sirius	258°26.2	-16°45.0
19	204°05.8	150°00.2	38.7	44°23.0	38.7	108°54.4	13.4	201°33.1	20.0	Adhara	255°05.9	-29°00.4
20	219°08.3	164°59.9	39.4	59°24.1	38.1	123°56.3	13.4	216°35.5	20.0	Procyon	244°50.7	5°09.6
21	234°10.8	179°59.6	· · 40.0	74°25.2	· · 37.5	138°58.1	· · 13.3	231°37.9	· · 19.9	Pollux	243°17.2	27°57.9
22	249°13.2	194°59.3	40.7	89°26.3	36.9	154°00.0	13.3	246°40.3	19.9	Avior	234°15.2	-59°35.5
23	264°15.7	209°59.0	41.4	104°27.4	36.4	169°01.8	13.3	261°42.7	19.9	Suhail	222°46.4	-43°32.2
Mer.pass. 05:26		ν -0.3' d 0.7' m -4.15		ν 1.1' d -0.6' m 1.49		ν 1.8' d -0.0' m -1.89		ν 2.4' d -0.0' m 0.96				

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	279°18.1	224°58.7	N16°42.1	119°28.5	N09°35.8	184°03.6	N23°13.3	276°45.1	S01°19.9
1	294°20.6	239°58.4	42.7	134°29.7	35.2	199°05.5	13.3	291°47.6	19.9
2	309°23.1	254°58.1	43.4	149°30.8	34.7	214°07.3	13.3	306°50.0	19.9
3	324°25.5	269°57.8	· · 44.1	164°31.9	· · 34.1	229°09.2	· · 13.2	321°52.4	· · 19.9
4	339°28.0	284°57.5	44.7	179°33.0	33.5	244°11.0	13.2	336°54.8	19.9
5	354°30.5	299°57.2	45.4	194°34.1	32.9	259°12.9	13.2	351°57.2	19.9
6	9°32.9	314°56.9	N16°46.1	209°35.2	N09°32.4	274°14.7	N23°13.2	6°59.6	S01°19.9
7	24°35.4	329°56.6	46.7	224°36.4	31.8	289°16.6	13.2	22°02.0	19.8
8	39°37.9	344°56.3	47.4	239°37.5	31.2	304°18.4	13.1	37°04.5	19.8
9	54°40.3	359°56.0	· · 48.1	254°38.6	· · 30.6	319°20.2	· · 13.1	52°06.9	· · 19.8
10	69°42.8	14°55.7	48.7	269°39.7	30.1	334°22.1	13.1	67°09.3	19.8
11	84°45.3	29°55.4	49.4	284°40.8	29.5	349°23.9	13.1	82°11.7	19.8
12	99°47.7	44°55.1	N16°50.1	299°41.9	N09°28.9	4°25.8	N23°13.1	97°14.1	S01°19.8
13	114°50.2	59°54.8	50.7	314°43.0	28.4	19°27.6	13.1	112°16.5	19.8
14	129°52.6	74°54.5	51.4	329°44.2	27.8	34°29.5	13.0	127°18.9	19.8
15	144°55.1	89°54.2	· · 52.1	344°45.3	· · 27.2	49°31.3	· · 13.0	142°21.4	· · 19.8
16	159°57.6	104°53.9	52.7	359°46.4	26.6	64°33.2	13.0	157°23.8	19.8
17	175°00.0	119°53.6	53.4	14°47.5	26.1	79°35.0	13.0	172°26.2	19.8
18	190°02.5	134°53.3	N16°54.0	29°48.6	N09°25.5	94°36.8	N23°13.0	187°28.6	S01°19.7
19	205°05.0	149°53.0	54.7	44°49.7	24.9	109°38.7	13.0	202°31.0	19.7
20	220°07.4	164°52.7	55.4	59°50.8	24.3	124°40.5	12.9	217°33.4	19.7
21	235°09.9	179°52.4	· · 56.0	74°52.0	· · 23.8	139°42.4	· · 12.9	232°35.9	· · 19.7
22	250°12.4	194°52.1	56.7	89°53.1	23.2	154°44.2	12.9	247°38.3	19.7
23	265°14.8	209°51.8	57.4	104°54.2	22.6	169°46.1	12.9	262°40.7	19.7
Mer.pass. 05:22		ν -0.3' d 0.7' m -4.14		ν 1.1' d -0.6' m 1.49		ν 1.8' d -0.0' m -1.89		ν 2.4' d -0.0' m 0.96	

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	280°17.3	224°51.5	N16°58.0	119°55.3	N09°22.0	184°47.9	N23°12.9	277°43.1	S01°19.7
1	295°19.8	239°51.1	58.7	134°56.4	21.5	199°49.8	12.9	292°45.5	19.7
2	310°22.2	254°50.8	16°59.3	149°57.5	20.9	214°51.6	12.8	307°47.9	19.7
3	325°24.7	269°50.5	17°00.0	164°58.6	· · 20.3	229°53.5	· · 12.8	322°50.4	· · 19.7
4	340°27.1	284°50.2	00.6	179°59.8	19.7	244°55.3	12.8	337°52.8	19.7
5	355°29.6	299°49.9	01.3	195°00.9	19.2	259°57.1	12.8	352°55.2	19.7
6	10°32.1	314°49.6	N17°02.0	210°02.0	N09°18.6	274°59.0	N23°12.8	7°57.6	S01°19.7
7	25°34.5	329°49.3	02.6	225°03.1	18.0	290°00.8	12.7	23°00.0	19.6
8	40°37.0	344°48.9	03.3	240°04.2	17.4	305°02.7	12.7	38°02.4	19.6
9	55°39.5	359°48.6	· · 03.9	255°05.3	· · 16.9	320°04.5	· · 12.7	53°04.9	· · 19.6
10	70°41.9	14°48.3	04.6	270°06.4	16.3	335°06.4	12.7	68°07.3	19.6
11	85°44.4	29°48.0	05.2	285°07.6	15.7	350°08.2	12.7	83°09.7	19.6
12	100°46.9	44°47.7	N17°05.9	300°08.7	N09°15.1	5°10.1	N23°12.7	98°12.1	S01°19.6
13	115°49.3	59°47.4	06.5	315°09.8	14.6	20°11.9	12.6	113°14.5	19.6
14	130°51.8	74°47.0	07.2	330°10.9	14.0	35°13.8	12.6	128°17.0	19.6
15	145°54.2	89°46.7	· · 07.8	345°12.0	· · 13.4	50°15.6	· · 12.6	143°19.4	· · 19.6
16	160°56.7	104°46.4	08.5	0°13.1	12.8	65°17.4	12.6	158°21.8	19.6
17	175°59.2	119°46.1	09.1	15°14.2	12.2	80°19.3	12.6	173°24.2	19.6
18	191°01.6	134°45.8	N17°09.8	30°15.3	N09°11.7	95°21.1	N23°12.5	188°26.6	S01°19.6
19	206°04.1	149°45.4	10.4	45°16.5	11.1	110°23.0	12.5	203°29.1	19.6
20	221°06.6	164°45.1	11.1	60°17.6	10.5	125°24.8	12.5	218°31.5	19.5
21	236°09.0	179°44.8	· · 11.7	75°18.7	· · 09.9	140°26.7	· · 12.5	233°33.9	· · 19.5
22	251°11.5	194°44.5	12.4	90°19.8	09.4	155°28.5	12.5	248°36.3	19.5
23	266°14.0	209°44.2	13.0	105°20.9	08.8	170°30.4	12.5	263°38.7	19.5
Mer.pass. 05:18		ν -0.3' d 0.7' m -4.13		ν 1.1' d -0.6' m 1.50		ν 1.8' d -0.0' m -1.89		ν 2.4' d -0.0' m 0.95	

Jun 30 Mon	SHA	Mer.pass
Venus	306°46.7	09:00
Mars	200°42.7	16:03
Jupiter	265°00.4	11:45
Saturn	357°28.3	05:36
Jul 01 Tue	SHA	Mer.pass
Venus	305°40.6	09:00
Mars	200°10.4	16:01
Jupiter	264°45.5	11:42
Saturn	357°27.0	05:32
Jul 02 Wed	SHA	Mer.pass
Venus	304°34.2	09:01
Mars	199°38.0	15:59
Jupiter	264°30.6	11:39
Saturn	357°25.8	05:28
Horizontal parallax		
Venus:	0.2	
Mars:	0.1	

h Sun			Moon				
Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	179°05.1	N23°10.0	119°31.5	14.5'	N10°16.9	-14.4'	56.2'
1	194°05.0	09.9	134°05.0	14.6'	10°02.5	-14.4'	56.2'
2	209°04.9	09.7	148°38.5	14.6'	09°48.1	-14.4'	56.2'
3	224°04.8	· · 09.6	163°12.2	14.7'	09°33.7	-14.4'	56.1'
4	239°04.6	09.4	177°45.9	14.8'	09°19.3	-14.4'	56.1'
5	254°04.5	09.2	192°19.7	14.9'	09°04.8	-14.5'	56.1'
6	269°04.4	N23°09.1	206°53.5	14.9'	N08°50.4	-14.5'	56.0'
7	284°04.3	08.9	221°27.4	15.0'	08°35.9	-14.5'	56.0'
8	299°04.1	08.8	236°01.4	15.1'	08°21.4	-14.5'	56.0'
9	314°04.0	· · 08.6	250°35.5	15.1'	08°06.8	-14.5'	55.9'
10	329°03.9	08.5	265°09.6	15.2'	07°52.3	-14.6'	55.9'
11	344°03.8	08.3	279°43.8	15.3'	07°37.7	-14.6'	55.9'
12	359°03.7	N23°08.1	294°18.1	15.3'	N07°23.2	-14.6'	55.8'
13	14°03.5	08.0	308°52.4	15.4'	07°08.6	-14.6'	55.8'
14	29°03.4	07.8	323°26.8	15.4'	06°54.0	-14.6'	55.8'
15	44°03.3	· · 07.7	338°01.2	15.5'	06°39.4	-14.6'	55.8'
16	59°03.2	07.5	352°35.7	15.6'	06°24.8	-14.6'	55.7'
17	74°03.0	07.3	7°10.3	15.6'	06°10.2	-14.6'	55.7'
18	89°02.9	N23°07.2	21°44.9	15.7'	N05°55.5	-14.6'	55.7'
19	104°02.8	07.0	36°19.6	15.7'	05°40.9	-14.6'	55.6'
20	119°02.7	06.8	50°54.3	15.8'	05°26.2	-14.7'	55.6'
21	134°02.5	· · 06.7	65°29.1	15.8'	05°11.6	-14.7'	55.6'
22	149°02.4	06.5	80°03.9	15.9'	04°56.9	-14.7'	55.6'
23	164°02.3	06.3	94°38.8	15.9'	04°42.3	-14.7'	55.5'
SD = 15.7' $d = -0.2'$			SD = 15.3'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	179°02.2	N23°06.2	109°13.7	16.0'	N04°27.6	-14.7'	55.5'
1	194°02.1	06.0	123°48.7	16.0'	04°12.9	-14.7'	55.5'
2	209°01.9	05.8	138°23.7	16.1'	03°58.3	-14.7'	55.4'
3	224°01.8	· · 05.7	152°58.8	16.1'	03°43.6	-14.7'	55.4'
4	239°01.7	05.5	167°33.9	16.1'	03°29.0	-14.7'	55.4'
5	254°01.6	05.3	182°09.0	16.2'	03°14.3	-14.7'	55.4'
6	269°01.5	N23°05.1	196°44.2	16.2'	N02°59.6	-14.7'	55.3'
7	284°01.3	05.0	211°19.5	16.3'	02°45.0	-14.7'	55.3'
8	299°01.2	04.8	225°54.7	16.3'	02°30.3	-14.6'	55.3'
9	314°01.1	· · 04.6	240°30.0	16.3'	02°15.7	-14.6'	55.3'
10	329°01.0	04.4	255°05.4	16.4'	02°01.0	-14.6'	55.2'
11	344°00.9	04.3	269°40.7	16.4'	01°46.4	-14.6'	55.2'
12	359°00.7	N23°04.1	284°16.1	16.4'	N01°31.7	-14.6'	55.2'
13	14°00.6	03.9	298°51.6	16.5'	01°17.1	-14.6'	55.2'
14	29°00.5	03.7	313°27.1	16.5'	01°02.5	-14.6'	55.1'
15	44°00.4	· · 03.5	328°02.5	16.5'	00°47.9	-14.6'	55.1'
16	59°00.3	03.4	342°38.1	16.5'	00°33.3	-14.6'	55.1'
17	74°00.1	03.2	357°13.6	16.6'	00°18.7	-14.6'	55.1'
18	89°00.0	N23°03.0	11°49.2	16.6'	N00°04.2	-14.6'	55.0'
19	103°59.9	02.8	26°24.8	16.6'	S00°10.4	14.5'	55.0'
20	118°59.8	02.6	41°00.4	16.6'	00°25.0	14.5'	55.0'
21	133°59.7	· · 02.5	55°36.1	16.7'	00°39.5	14.5'	55.0'
22	148°59.5	02.3	70°11.7	16.7'	00°54.0	14.5'	55.0'
23	163°59.4	02.1	84°47.4	16.7'	01°08.5	14.5'	54.9'
SD = 15.7' $d = -0.2'$			SD = 15.1'				

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	178°59.3	N23°01.9	99°23.1	16.7'	S01°23.0	14.5'	54.9'
1	193°59.2	01.7	113°58.8	16.7'	01°37.5	14.5'	54.9'
2	208°59.1	01.5	128°34.5	16.7'	01°51.9	14.4'	54.9'
3	223°58.9	· · 01.3	143°10.3	16.8'	02°06.4	14.4'	54.9'
4	238°58.8	01.2	157°46.0	16.8'	02°20.8	14.4'	54.8'
5	253°58.7	01.0	172°21.8	16.8'	02°35.2	14.4'	54.8'
6	268°58.6	N23°00.8	186°57.6	16.8'	S02°49.6	14.4'	54.8'
7	283°58.5	00.6	201°33.3	16.8'	03°03.9	14.3'	54.8'
8	298°58.4	00.4	216°09.1	16.8'	03°18.2	14.3'	54.8'
9	313°58.2	· · 00.2	230°44.9	16.8'	03°32.6	14.3'	54.7'
10	328°58.1	23°00.0	245°20.7	16.8'	03°46.9	14.3'	54.7'
11	343°58.0	22°59.8	259°56.5	16.8'	04°01.1	14.2'	54.7'
12	358°57.9	N22°59.6	274°32.4	16.8'	S04°15.4	14.2'	54.7'
13	13°57.8	59.4	289°08.2	16.8'	04°29.6	14.2'	54.7'
14	28°57.7	59.2	303°44.0	16.8'	04°43.8	14.2'	54.6'
15	43°57.5	· · 59.0	318°19.8	16.8'	04°57.9	14.1'	54.6'
16	58°57.4	58.8	332°55.6	16.8'	05°12.1	14.1'	54.6'
17	73°57.3	58.6	347°31.4	16.8'	05°26.2	14.1'	54.6'
18	88°57.2	N22°58.4	2°07.2	16.8'	S05°40.3	14.1'	54.6'
19	103°57.1	58.2	16°43.0	16.8'	05°54.3	14.0'	54.6'
20	118°57.0	58.0	31°18.8	16.8'	06°08.4	14.0'	54.5'
21	133°56.8	· · 57.8	45°54.6	16.8'	06°22.4	14.0'	54.5'
22	148°56.7	57.6	60°30.3	16.8'	06°36.3	13.9'	54.5'
23	163°56.6	57.4	75°06.1	16.7'	06°50.3	13.9'	54.5'
SD = 15.7' $d = -0.2'$			SD = 15.0'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°	--	--	00:22	23:38	--	--
64°	--	--	01:42	22:25	--	--
62°	--	--	02:17	21:50	--	--
60°	--	01:04	02:42	21:25	23:02	--
N 58°	--	01:49	03:02	21:05	22:18	--
56°	--	02:17	03:19	20:49	21:50	--
54°	00:59	02:39	03:32	20:35	21:29	23:07
52°	01:40	02:56	03:45	20:23	21:11	22:27
50°	02:07	03:11	03:55	20:12	20:57	22:00
45°	02:51	03:40	04:17	19:50	20:27	21:17
N 40°	03:21	04:02	04:35	19:33	20:05	20:47
35°	03:44	04:20	04:50	19:18	19:48	20:24
30°	04:02	04:35	05:03	19:05	19:33	20:06
20°	04:31	05:00	05:24	18:44	19:08	19:37
N 10°	04:53	05:20	05:43	18:25	18:48	19:15
0°	05:12	05:38	06:00	18:08	18:30	18:56
S 10°	05:28	05:55	06:17	17:50	18:13	18:40
20°	05:44	06:12	06:36	17:32	17:56	18:24
30°	06:01	06:30	06:57	17:11	17:38	18:07
35°	06:09	06:41	07:09	16:59	17:27	17:59
40°	06:18	06:52	07:23	16:45	17:16	17:50
45°	06:28	07:06	07:39	16:29	17:03	17:40
S 50°	06:40	07:21	08:00	16:09	16:47	17:28
52°	06:45	07:29	08:09	15:59	16:40	17:23
54°	06:51	07:37	08:20	15:48	16:32	17:18
56°	06:57	07:45	08:32	15:36	16:23	17:12
58°	07:03	07:55	08:47	15:21	16:13	17:05
S 60°	07:10	08:07	09:04	15:04	16:01	16:58

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	08:33	10:39	12:39	00:15 23:38	23:05	22:32
N 70°	08:46	10:42	12:32	00:00 23:32	23:07	22:42
68°	08:56	10:44	12:27	23:27	23:08	22:50
66°	09:04	10:45	12:23	23:23	23:10	22:56
64°	09:11	10:47	12:19	23:19	23:11	23:02
62°	09:17	10:48	12:16	23:16	23:11	23:07
60°	09:23	10:49	12:13	23:13	23:12	23:11
N 58°	09:27	10:50	12:10	23:11	23:13	23:15
56°	09:31	10:51	12:08	23:09	23:13	23:18
54°	09:35	10:52	12:06	23:07	23:14	23:21
52°	09:38	10:53	12:05	23:05	23:14	23:24
50°	09:41	10:53	12:03	23:03	23:15	23:26
45°	09:48	10:55	11:59	23:00	23:16	23:32
N 40°	09:53	10:56	11:57	22:57	23:17	23:37
35°	09:58	10:57	11:54	22:54	23:17	23:41
30°	10:02	10:58	11:52	22:52	23:18	23:44
20°	10:09	10:59	11:48	22:48	23:19	23:50
N 10°	10:15	11:00	11:45	22:44	23:20	23:56
0°	10:20	11:02	11:42	22:40	23:21	· ·
S 10°	10:26	11:03	11:39	22:37	23:22	· ·
20°	10:32	11:04	11:36	22:33	23:23	· ·
30°	10:39	11:06	11:32	22:29	23:24	· ·
35°	10:43	11:07	11:30	22:26	23:24	· ·
40°	10:47	11:08	11:28	22:23	23:25	· ·
45°	10:52	11:09	11:25	22:20	23:26	· ·
S 50°	10:59	11:10	11:22	22:16	23:27	· ·
52°	11:01	11:11	11:20	22:14	23:27	· ·
54°	11:04	11:12	11:19	22:12	23:28	· ·
56°	11:08	11:13	11:17	22:10	23:28	· ·
58°	11:11	11:13	11:15	22:07	23:29	· ·
S 60°	11:16	11:14	11:13	22:05	23:30	· ·

Day	Sun		Mer. Pass hh:mm	Moon		Age 5-7 24-42%
	Eqn.of 00 ^h mm:ss	Time 12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
30	03:39	03:45	12:04	16:30	04:09	
01	03:51	03:57	12:04	17:11	04:51	
02	04:03	04:08	12:04	17:51	05:31	

July 03, 04, 05 UT (Thu., Fri., Sat.)

Aries		Venus		Mars		Jupiter		Saturn	
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	281°16.4	224°43.8	N17°13.7	120°22.0	N09°08.2	185°32.2	N23°12.4	278°41.2	S01°19.5
1	296°18.9	239°43.5	14.3	135°23.1	07.6	200°34.1	12.4	293°43.6	19.5
2	311°21.4	254°43.2	15.0	150°24.2	07.0	215°35.9	12.4	308°46.0	19.5
3	326°23.8	269°42.8	· · 15.6	165°25.4	· · 06.5	230°37.7	· · 12.4	323°48.4	· · 19.5
4	341°26.3	284°42.5	16.3	180°26.5	05.9	245°39.6	12.4	338°50.8	19.5
5	356°28.7	299°42.2	16.9	195°27.6	05.3	260°41.4	12.3	353°53.3	19.5
6	11°31.2	314°41.9	N17°17.6	210°28.7	N09°04.7	275°43.3	N23°12.3	8°55.7	S01°19.5
7	26°33.7	329°41.5	18.2	225°29.8	04.2	290°45.1	12.3	23°58.1	19.5
8	41°36.1	344°41.2	18.8	240°30.9	03.6	305°47.0	12.3	39°00.5	19.5
9	56°38.6	359°40.9	· · 19.5	255°32.0	· · 03.0	320°48.8	· · 12.3	54°02.9	· · 19.5
10	71°41.1	14°40.5	20.1	270°33.1	02.4	335°50.7	12.2	69°05.4	19.5
11	86°43.5	29°40.2	20.8	285°34.3	01.8	350°52.5	12.2	84°07.8	19.5
12	101°46.0	44°39.9	N17°21.4	300°35.4	N09°01.3	5°54.4	N23°12.2	99°10.2	S01°19.5
13	116°48.5	59°39.6	22.1	315°36.5	00.7	20°56.2	12.2	114°12.6	19.4
14	131°50.9	74°39.2	22.7	330°37.6	09°00.1	35°58.0	12.2	129°15.1	19.4
15	146°53.4	89°38.9	· · 23.3	345°38.7	08°59.5	50°59.9	· · 12.1	144°17.5	· · 19.4
16	161°55.9	104°38.6	24.0	0°39.8	58.9	66°01.7	12.1	159°19.9	19.4
17	176°58.3	119°38.2	24.6	15°40.9	58.4	81°03.6	12.1	174°22.3	19.4
18	192°00.8	134°37.9	N17°25.3	30°42.0	N08°57.8	96°05.4	N23°12.1	189°24.7	S01°19.4
19	207°03.2	149°37.5	25.9	45°43.2	57.2	111°07.3	12.1	204°27.2	19.4
20	222°05.7	164°37.2	26.5	60°44.3	56.6	126°09.1	12.1	219°29.6	19.4
21	237°08.2	179°36.9	· · 27.2	75°45.4	· · 56.0	141°11.0	· · 12.0	234°32.0	· · 19.4
22	252°10.6	194°36.5	27.8	90°46.5	55.5	156°12.8	12.0	249°34.4	19.4
23	267°13.1	209°36.2	28.4	105°47.6	54.9	171°14.7	12.0	264°36.9	19.4
Mer.pass. 05:14		ν -0.3' d 0.6' m -4.13		ν 1.1' d -0.6' m 1.50		ν 1.8' d -0.0' m -1.89		ν 2.4' d -0.0' m 0.95	

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	282°15.6	224°35.9	N17°29.1	120°48.7	N08°54.3	186°16.5	N23°12.0	279°39.3	S01°19.4
1	297°18.0	239°35.5	29.7	135°49.8	53.7	201°18.4	12.0	294°41.7	19.4
2	312°20.5	254°35.2	30.3	150°50.9	53.1	216°20.2	11.9	309°44.1	19.4
3	327°23.0	269°34.8	· · 31.0	165°52.0	· · 52.6	231°22.1	· · 11.9	324°46.6	· · 19.4
4	342°25.4	284°34.5	31.6	180°53.2	52.0	246°23.9	11.9	339°49.0	19.4
5	357°27.9	299°34.2	32.2	195°54.3	51.4	261°25.7	11.9	354°51.4	19.4
6	12°30.3	314°33.8	N17°32.9	210°55.4	N08°50.8	276°27.6	N23°11.9	9°53.8	S01°19.4
7	27°32.8	329°33.5	33.5	225°56.5	50.2	291°29.4	11.8	24°56.3	19.4
8	42°35.3	344°33.1	34.1	240°57.6	49.7	306°31.3	11.8	39°58.7	19.4
9	57°37.7	359°32.8	· · 34.8	255°58.7	· · 49.1	321°33.1	· · 11.8	55°01.1	· · 19.3
10	72°40.2	14°32.4	35.4	270°59.8	48.5	336°35.0	11.8	70°03.5	19.3
11	87°42.7	29°32.1	36.0	286°00.9	47.9	351°36.8	11.8	85°06.0	19.3
12	102°45.1	44°31.7	N17°36.6	301°02.0	N08°47.3	6°38.7	N23°11.7	100°08.4	S01°19.3
13	117°47.6	59°31.4	37.3	316°03.2	46.7	21°40.5	11.7	115°10.8	19.3
14	132°50.1	74°31.1	37.9	331°04.3	46.2	36°42.4	11.7	130°13.2	19.3
15	147°52.5	89°30.7	· · 38.5	346°05.4	· · 45.6	51°44.2	· · 11.7	145°15.7	· · 19.3
16	162°55.0	104°30.4	39.2	1°06.5	45.0	66°46.1	11.7	160°18.1	19.3
17	177°57.5	119°30.0	39.8	16°07.6	44.4	81°47.9	11.6	175°20.5	19.3
18	192°59.9	134°29.7	N17°40.4	31°08.7	N08°43.8	96°49.8	N23°11.6	190°23.0	S01°19.3
19	208°02.4	149°29.3	41.0	46°09.8	43.3	111°51.6	11.6	205°25.4	19.3
20	223°04.8	164°29.0	41.7	61°10.9	42.7	126°53.4	11.6	220°27.8	19.3
21	238°07.3	179°28.6	· · 42.3	76°12.0	· · 42.1	141°55.3	· · 11.5	235°30.2	· · 19.3
22	253°09.8	194°28.3	42.9	91°13.1	41.5	156°57.1	11.5	250°32.7	19.3
23	268°12.2	209°27.9	43.5	106°14.3	40.9	171°59.0	11.5	265°35.1	19.3
Mer.pass. 05:10		ν -0.3' d 0.6' m -4.12		ν 1.1' d -0.6' m 1.51		ν 1.8' d -0.0' m -1.89		ν 2.4' d -0.0' m 0.94	

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	283°14.7	224°27.6	N17°44.1	121°15.4	N08°40.3	187°00.8	N23°11.5	280°37.5	S01°19.3
1	298°17.2	239°27.2	44.8	136°16.5	39.8	202°02.7	11.5	295°39.9	19.3
2	313°19.6	254°26.8	45.4	151°17.6	39.2	217°04.5	11.4	310°42.4	19.3
3	328°22.1	269°26.5	· · 46.0	166°18.7	· · 38.6	232°06.4	· · 11.4	325°44.8	· · 19.3
4	343°24.6	284°26.1	46.6	181°19.8	38.0	247°08.2	11.4	340°47.2	19.3
5	358°27.0	299°25.8	47.2	196°20.9	37.4	262°10.1	11.4	355°49.7	19.3
6	13°29.5	314°25.4	N17°47.9	211°22.0	N08°36.8	277°11.9	N23°11.4	10°52.1	S01°19.3
7	28°32.0	329°25.1	48.5	226°23.1	36.2	292°13.8	11.3	25°54.5	19.3
8	43°34.4	344°24.7	49.1	241°24.2	35.7	307°15.6	11.3	40°57.0	19.3
9	58°36.9	359°24.3	· · 49.7	256°25.4	· · 35.1	322°17.5	· · 11.3	55°59.4	· · 19.3
10	73°39.3	14°24.0	50.3	271°26.5	34.5	337°19.3	11.3	71°01.8	19.3
11	88°41.8	29°23.6	51.0	286°27.6	33.9	352°21.2	11.3	86°04.2	19.3
12	103°44.3	44°23.3	N17°51.6	301°28.7	N08°33.3	7°23.0	N23°11.2	101°06.7	S01°19.3
13	118°46.7	59°22.9	52.2	316°29.8	32.7	22°24.8	11.2	116°09.1	19.3
14	133°49.2	74°22.5	52.8	331°30.9	32.2	37°26.7	11.2	131°11.5	19.3
15	148°51.7	89°22.2	· · 53.4	346°32.0	· · 31.6	52°28.5	· · 11.2	146°14.0	· · 19.3
16	163°54.1	104°21.8	54.0	1°33.1	31.0	67°30.4	11.2	161°16.4	19.3
17	178°56.6	119°21.5	54.6	16°34.2	30.4	82°32.2	11.1	176°18.8	19.3
18	193°59.1	134°21.1	N17°55.2	31°35.3	N08°29.8	97°34.1	N23°11.1	191°21.3	S01°19.2
19	209°01.5	149°20.7	55.9	46°36.4	29.2	112°35.9	11.1	206°23.7	19.2
20	224°04.0	164°20.4	56.5	61°37.6	28.6	127°37.8	11.1	221°26.1	19.2
21	239°06.4	179°20.0	· · 57.1	76°38.7	· · 28.1	142°39.6	· · 11.0	236°28.5	· · 19.2
22	254°08.9	194°19.6	57.7	91°39.8	27.5	157°41.5	11.0	251°31.0	19.2
23	269°11.4	209°19.3	58.3	106°40.9	26.9	172°43.3	11.0	266°33.4	19.2
Mer.pass. 05:06		ν -0.4' d 0.6' m -4.12		ν 1.1' d -0.6' m 1.51		ν 1.8' d -0.0' m -1.89		ν 2.4' d -0.0' m 0.94	

Stars		
	SHA	Dec
Alpheratz	357°34.3	29°13.8
Ankaa	353°06.7	-42°09.8
Schedar	349°30.6	56°40.4
Diphda	348°46.9	-17°50.6
Achernar	335°20.0	-57°06.1
Hamal	327°50.9	23°34.9
Polaris	314°07.0	89°22.0
Acamar	315°11.7	-40°11.9
Menkar	314°05.9	4°11.4
Mirfak	308°28.0	49°56.9
Aldebaran	290°39.5	16°33.6
Rigel	281°03.8	-8°10.3
Capella	280°21.7	46°01.3
Bellatrix	278°22.8	6°22.4
Elnath	278°01.7	28°37.7
Alnilam	275°37.6	-1°11.1
Betelgeuse	270°52.0	7°24.7
Canopus	263°52.8	-52°42.4
Sirius	258°26.2	-16°45.0
Adhara	255°05.9	-29°00.3
Procyon	244°50.7	5°09.6
Pollux	243°17.2	27°57.9
Avior	234°15.2	-59°35.5
Suhail	222°46.4	-43°32.2
Miaplacidus	221°39.0	-69°49.4
Alphard	217°47.6	-8°46.2
Regulus	207°34.2	11°50.6
Dubhe	193°40.8	61°37.1
Denebola	182°24.6	14°25.9
Gienah	175°43.2	-17°41.1
Acrux	172°59.7	-63°14.7
Gacrux	171°51.3	-57°15.6
Alioth	166°12.6	55°49.5
Spica	158°21.8	-11°17.7
Alkaid	152°51.5	49°11.4
Hadar	148°35.2	-60°30.0
Menkent	147°57.0	-36°29.9
Arcturus	145°47.4	19°03.1
Rigil Kent.	139°39.4	-60°56.7
Kochab	137°19.1	74°03.2
Zuben'ubi	136°55.4	-16°08.9
Alphecca	126°03.1	26°37.8
Antares	112°15.0	-26°29.4
Atria	107°08.4	-69°04.5
Sabik	102°02.0	-15°45.4
Shaula	96°09.4	-37°07.4
Rasalhague	95°57.8	12°32.5
Eltanin	90°41.5	51°29.1
Kaus Aust.	83°31.6	-34°22.3
Vega	80°32.5	38°48.4
Nunki	75°46.9	-26°15.9
Altair	61°59.2	8°56.1
Peacock	53°04.5	-56°39.0
Deneb	49°25.1	45°22.2
Enif	33°38.2	9°59.5
Al Na'ir	27°32.1	-46°50.0
Fomalhaut	15°13.9	-29°29.0
Scheat	13°44.7	28°13.2
Markab	13°29.4	15°20.5

Jul 03 Thu	SHA	Mer.pass
Venus	303°27.4	09:01
Mars	199°05.6	15:57
Jupiter	264°15.8	11:36
Saturn	357°24.7	05:24

Jul 04 Fri	SHA	Mer.pass
Venus	302°20.3	09:02
Mars	198°33.1	15:56
Jupiter	264°00.9	11:33
Saturn	357°23.7	05:21

Jul 05 Sat	SHA	Mer.pass
Venus	301°12.9	09:02
Mars	198°00.7	15:54
Jupiter	263°46.1	11:31
Saturn	357°22.8	05:17

Horizontal parallax		
Venus:		0.2
Mars:		0.1

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	178°56.5	N22°57.2	89°41.8	16.7'	S07°04.2	13.9'	54.5'
1	193°56.4	57.0	104°17.6	16.7'	07°18.0	13.8'	54.5'
2	208°56.3	56.8	118°53.3	16.7'	07°31.9	13.8'	54.5'
3	223°56.2	· · 56.6	133°29.0	16.7'	07°45.7	13.8'	54.4'
4	238°56.0	56.4	148°04.7	16.7'	07°59.4	13.7'	54.4'
5	253°55.9	56.2	162°40.4	16.7'	08°13.2	13.7'	54.4'
6	268°55.8	N22°56.0	177°16.0	16.6'	S08°26.9	13.7'	54.4'
7	283°55.7	55.8	191°51.7	16.6'	08°40.5	13.6'	54.4'
8	298°55.6	55.6	206°27.3	16.6'	08°54.1	13.6'	54.4'
9	313°55.5	· · 55.4	221°02.9	16.6'	09°07.7	13.5'	54.4'
10	328°55.4	55.2	235°38.4	16.5'	09°21.3	13.5'	54.4'
11	343°55.2	55.0	250°14.0	16.5'	09°34.8	13.5'	54.3'
12	358°55.1	N22°54.8	264°49.5	16.5'	S09°48.2	13.4'	54.3'
13	13°55.0	54.5	279°25.0	16.5'	10°01.7	13.4'	54.3'
14	28°54.9	54.3	294°00.5	16.4'	10°15.0	13.3'	54.3'
15	43°54.8	· · 54.1	308°35.9	16.4'	10°28.4	13.3'	54.3'
16	58°54.7	53.9	323°11.3	16.4'	10°41.7	13.3'	54.3'
17	73°54.6	53.7	337°46.7	16.3'	10°54.9	13.2'	54.3'
18	88°54.4	N22°53.5	352°22.1	16.3'	S11°08.1	13.2'	54.3'
19	103°54.3	53.3	6°57.4	16.3'	11°21.3	13.1'	54.3'
20	118°54.2	53.0	21°32.7	16.2'	11°34.4	13.1'	54.3'
21	133°54.1	· · 52.8	36°07.9	16.2'	11°47.5	13.0'	54.2'
22	148°54.0	52.6	50°43.1	16.2'	12°00.5	13.0'	54.2'
23	163°53.9	52.4	65°18.3	16.1'	12°13.5	12.9'	54.2'
SD = 15.7' $d = -0.2'$			SD = 14.9'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°53.8	N22°52.2	79°53.4	16.1'	S12°26.5	12.9'	54.2'
1	193°53.7	52.0	94°28.5	16.1'	12°39.3	12.8'	54.2'
2	208°53.5	51.7	109°03.6	16.0'	12°52.2	12.8'	54.2'
3	223°53.4	· · 51.5	123°38.6	16.0'	13°05.0	12.7'	54.2'
4	238°53.3	51.3	138°13.6	15.9'	13°17.7	12.7'	54.2'
5	253°53.2	51.1	152°48.5	15.9'	13°30.4	12.6'	54.2'
6	268°53.1	N22°50.8	167°23.4	15.8'	S13°43.0	12.6'	54.2'
7	283°53.0	50.6	181°58.3	15.8'	13°55.6	12.5'	54.2'
8	298°52.9	50.4	196°33.1	15.8'	14°08.1	12.5'	54.2'
9	313°52.8	· · 50.2	211°07.8	15.7'	14°20.6	12.4'	54.2'
10	328°52.7	50.0	225°42.5	15.7'	14°33.0	12.4'	54.2'
11	343°52.5	49.7	240°17.2	15.6'	14°45.3	12.3'	54.2'
12	358°52.4	N22°49.5	254°51.8	15.6'	S14°57.6	12.2'	54.1'
13	13°52.3	49.3	269°26.3	15.5'	15°09.9	12.2'	54.1'
14	28°52.2	49.0	284°00.9	15.5'	15°22.0	12.1'	54.1'
15	43°52.1	· · 48.8	298°35.3	15.4'	15°34.2	12.1'	54.1'
16	58°52.0	48.6	313°09.7	15.3'	15°46.2	12.0'	54.1'
17	73°51.9	48.4	327°44.1	15.3'	15°58.2	11.9'	54.1'
18	88°51.8	N22°48.1	342°18.3	15.2'	S16°10.2	11.9'	54.1'
19	103°51.7	47.9	356°52.6	15.2'	16°22.0	11.8'	54.1'
20	118°51.6	47.7	11°26.8	15.1'	16°33.8	11.7'	54.1'
21	133°51.5	· · 47.4	26°00.9	15.1'	16°45.6	11.7'	54.1'
22	148°51.3	47.2	40°34.9	15.0'	16°57.3	11.6'	54.1'
23	163°51.2	47.0	55°08.9	14.9'	17°08.9	11.6'	54.1'
SD = 15.7' $d = -0.2'$			SD = 14.8'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°51.1	N22°46.7	69°42.9	14.9'	S17°20.4	11.5'	54.1'
1	193°51.0	46.5	84°16.8	14.8'	17°31.9	11.4'	54.1'
2	208°50.9	46.2	98°50.6	14.8'	17°43.4	11.3'	54.1'
3	223°50.8	· · 46.0	113°24.4	14.7'	17°54.7	11.3'	54.1'
4	238°50.7	45.8	127°58.1	14.6'	18°06.0	11.2'	54.1'
5	253°50.6	45.5	142°31.7	14.6'	18°17.2	11.1'	54.1'
6	268°50.5	N22°45.3	157°05.3	14.5'	S18°28.3	11.1'	54.1'
7	283°50.4	45.1	171°38.8	14.4'	18°39.4	11.0'	54.1'
8	298°50.3	44.8	186°12.2	14.4'	18°50.4	10.9'	54.1'
9	313°50.2	· · 44.6	200°45.6	14.3'	19°01.3	10.8'	54.1'
10	328°50.1	44.3	215°18.9	14.2'	19°12.1	10.8'	54.1'
11	343°49.9	44.1	229°52.1	14.2'	19°22.9	10.7'	54.1'
12	358°49.8	N22°43.8	244°25.3	14.1'	S19°33.6	10.6'	54.1'
13	13°49.7	43.6	258°58.3	14.0'	19°44.2	10.5'	54.1'
14	28°49.6	43.4	273°31.4	14.0'	19°54.8	10.5'	54.1'
15	43°49.5	· · 43.1	288°04.3	13.9'	20°05.2	10.4'	54.1'
16	58°49.4	42.9	302°37.2	13.8'	20°15.6	10.3'	54.1'
17	73°49.3	42.6	317°10.0	13.7'	20°25.9	10.2'	54.1'
18	88°49.2	N22°42.4	331°42.8	13.7'	S20°36.1	10.1'	54.2'
19	103°49.1	42.1	346°15.4	13.6'	20°46.2	10.0'	54.2'
20	118°49.0	41.9	0°48.0	13.5'	20°56.3	10.0'	54.2'
21	133°48.9	· · 41.6	15°20.5	13.4'	21°06.2	9.9'	54.2'
22	148°48.8	41.4	29°53.0	13.4'	21°16.1	9.8'	54.2'
23	163°48.7	41.1	44°25.3	13.3'	21°25.9	9.7'	54.2'
SD = 15.7' $d = -0.2'$			SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	—:—	—:—	00:43	23:21	—:—	—:—
64°	—:—	—:—	01:48	22:20	—:—	—:—
62°	—:—	—:—	02:21	21:46	—:—	—:—
60°	—:—	01:12	02:46	21:22	22:55	—:—
N 58°	—:—	01:54	03:05	21:03	22:14	—:—
56°	—:—	02:21	03:21	20:47	21:47	—:—
54°	01:06	02:42	03:35	20:34	21:26	23:01
52°	01:45	02:59	03:47	20:22	21:09	22:23
50°	02:10	03:13	03:57	20:11	20:55	21:58
45°	02:53	03:42	04:19	19:50	20:27	21:15
N 40°	03:23	04:04	04:37	19:32	20:05	20:46
35°	03:45	04:22	04:51	19:18	19:47	20:23
30°	04:03	04:37	05:04	19:05	19:32	20:05
20°	04:32	05:01	05:25	18:44	19:08	19:37
N 10°	04:54	05:21	05:44	18:25	18:48	19:15
0°	05:12	05:38	06:01	18:08	18:31	18:57
S 10°	05:29	05:55	06:18	17:51	18:14	18:40
20°	05:45	06:12	06:36	17:33	17:57	18:25
30°	06:01	06:30	06:56	17:13	17:39	18:09
35°	06:09	06:40	07:08	17:01	17:29	18:00
40°	06:18	06:52	07:22	16:47	17:17	17:51
45°	06:28	07:05	07:39	16:31	17:04	17:41
S 50°	06:39	07:20	07:59	16:11	16:49	17:30
52°	06:44	07:28	08:08	16:01	16:42	17:25
54°	06:50	07:36	08:19	15:50	16:34	17:20
56°	06:56	07:44	08:31	15:38	16:25	17:14
58°	07:02	07:54	08:45	15:24	16:15	17:07
S 60°	07:09	08:05	09:02	15:08	16:04	17:00

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	14:44	17:21	■	21:52	20:43	■
N 70°	14:26	16:36	■	22:12	21:29	■
68°	14:12	16:07	18:34	22:28	22:00	21:04
66°	14:01	15:45	17:44	22:41	22:23	21:55
64°	13:51	15:28	17:13	22:52	22:41	22:27
62°	13:43	15:14	16:50	23:02	22:57	22:51
60°	13:36	15:02	16:31	23:10	23:09	23:10
N 58°	13:30	14:52	16:16	23:17	23:20	23:26
56°	13:25	14:43	16:03	23:23	23:30	23:39
54°	13:20	14:35	15:52	23:29	23:38	23:51
52°	13:16	14:28	15:42	23:34	23:46	· · ·
50°	13:12	14:22	15:33	23:39	23:53	· · ·
45°	13:04	14:09	15:15	23:49	· · ·	00:08
N 40°	12:57	13:58	15:00	23:57	· · ·	00:20
35°	12:51	13:48	14:47	· · ·	00:05	00:31
30°	12:46	13:40	14:36	· · ·	00:11	00:40
20°	12:37	13:26	14:17	· · ·	00:22	00:56
N 10°	12:29	13:14	14:01	· · ·	00:32	01:10
0°	12:22	13:03	13:46	00:01	00:41	01:23
S 10°	12:14	12:51	13:31	00:06	00:51	01:36
20°	12:07	12:40	13:15	00:12	01:00	01:51
30°	11:58	12:26	12:57	00:18	01:12	02:07
35°	11:53	12:18	12:46	00:21	01:19	02:17
40°	11:48	12:09	12:34	00:26	01:26	02:27
45°	11:41	11:59	12:20	00:30	01:35	02:40
S 50°	11:34	11:47	12:03	00:36	01:46	02:56
52°	11:30	11:41	11:55	00:39	01:51	03:04
54°	11:26	11:35	11:46	00:42	01:56	03:12
56°	11:22	11:28	11:36	00:45	02:02	03:21
58°	11:17	11:20	11:24	00:49	02:09	03:32
S 60°	11:12	11:11	11:11	00:53	02:17	03:44

Day	Sun		Mer. Pass	Moon		Age 8-10 52-70%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
03	04:14	04:20	12:04	18:31	06:11	
04	04:25	04:30	12:05	19:13	06:52	
05	04:35	04:41	12:05	19:57	07:34	

July 06, 07, 08 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	284°13.8	224°18.9	N17°58.9	121°42.0	N08°26.3	187°45.2	N23°11.0	281°35.8	S01°19.2	Alpheratz	357°34.3	29°13.8
1	299°16.3	239°18.5	17°59.5	136°43.1	25.7	202°47.0	11.0	296°38.3	19.2	Ankaa	353°06.7	-42°09.8
2	314°18.8	254°18.2	18°00.1	151°44.2	25.1	217°48.9	10.9	311°40.7	19.2	Schedar	349°30.5	56°40.4
3	329°21.2	269°17.8	· · 00.7	166°45.3	· · 24.5	232°50.7	· · 10.9	326°43.1	· · 19.2	Diphda	348°46.9	-17°50.6
4	344°23.7	284°17.4	01.3	181°46.4	24.0	247°52.6	10.9	341°45.6	19.2	Achernar	335°20.0	-57°06.1
5	359°26.2	299°17.1	01.9	196°47.5	23.4	262°54.4	10.9	356°48.0	19.2	Hamal	327°50.8	23°34.9
6	14°28.6	314°16.7	N18°02.5	211°48.6	N08°22.8	277°56.3	N23°10.9	11°50.4	S01°19.2	Polaris	314°05.5	89°22.0
7	29°31.1	329°16.3	03.2	226°49.7	22.2	292°58.1	10.8	26°52.9	19.2	Acamar	315°11.7	-40°11.9
8	44°33.6	344°15.9	03.8	241°50.9	21.6	308°00.0	10.8	41°55.3	19.2	Menkar	314°05.9	4°11.4
9	59°36.0	359°15.6	· · 04.4	256°52.0	· · 21.0	323°01.8	· · 10.8	56°57.7	· · 19.2	Mirfak	308°28.0	49°56.9
10	74°38.5	14°15.2	05.0	271°53.1	20.4	338°03.6	10.8	72°00.2	19.2	Aldebaran	290°39.4	16°33.6
11	89°40.9	29°14.8	05.6	286°54.2	19.9	353°05.5	10.7	87°02.6	19.2	Rigel	281°03.8	-8°10.2
12	104°43.4	44°14.5	N18°06.2	301°55.3	N08°19.3	8°07.3	N23°10.7	102°05.0	S01°19.2	Capella	280°21.7	46°01.3
13	119°45.9	59°14.1	06.8	316°56.4	18.7	23°09.2	10.7	117°07.5	19.2	Bellatrix	278°22.8	6°22.4
14	134°48.3	74°13.7	07.4	331°57.5	18.1	38°11.0	10.7	132°09.9	19.2	Elnath	278°01.7	28°37.7
15	149°50.8	89°13.3	· · 08.0	346°58.6	· · 17.5	53°12.9	· · 10.7	147°12.3	· · 19.2	Alnilam	275°37.6	-1°11.1
16	164°53.3	104°13.0	08.6	1°59.7	16.9	68°14.7	10.6	162°14.8	19.2	Betelgeuse	270°52.0	7°24.7
17	179°55.7	119°12.6	09.2	17°00.8	16.3	83°16.6	10.6	177°17.2	19.2	Canopus	263°52.8	-52°42.4
18	194°58.2	134°12.2	N18°09.8	32°01.9	N08°15.7	98°18.4	N23°10.6	192°19.6	S01°19.2	Sirius	258°26.2	-16°45.0
19	210°00.7	149°11.8	10.4	47°03.0	15.1	113°20.3	10.6	207°22.1	19.2	Adhara	255°05.9	-29°00.3
20	225°03.1	164°11.4	11.0	62°04.1	14.6	128°22.1	10.5	222°24.5	19.2	Procyon	244°50.7	5°09.6
21	240°05.6	179°11.1	· · 11.6	77°05.3	· · 14.0	143°24.0	· · 10.5	237°26.9	· · 19.2	Pollux	243°17.2	27°57.9
22	255°08.1	194°10.7	12.1	92°06.4	13.4	158°25.8	10.5	252°29.4	19.2	Avior	234°15.2	-59°35.5
23	270°10.5	209°10.3	12.7	107°07.5	12.8	173°27.7	10.5	267°31.8	19.2	Suhail	222°46.4	-43°32.2
Mer.pass. 05:02		ν -0.4' d 0.6' m -4.11		ν 1.1' d -0.6' m 1.52		ν 1.8' d -0.0' m -1.89		ν 2.4' d -0.0' m 0.93		Miaplacidus	221°39.0	-69°49.4
										Alphard	217°47.6	-8°46.1
										Regulus	207°34.2	11°50.6
										Dubhe	193°40.8	61°37.1
										Denebola	182°24.6	14°25.9
										Gienah	175°43.2	-17°41.1
										Acrux	172°59.8	-63°14.7
										Gacrux	171°51.3	-57°15.6
										Alioth	166°12.6	55°49.5
										Spica	158°21.8	-11°17.7
										Alkaid	152°51.6	49°11.4
										Hadar	148°35.2	-60°30.0
										Menkent	147°57.0	-36°29.9
										Arcturus	145°47.4	19°03.1
										Rigel Kent.	139°39.5	-60°56.7
										Kochab	137°19.2	74°03.3
										Zuben'ubi	136°55.4	-16°08.9
										Alphecca	126°03.1	26°37.8
										Antares	112°15.0	-26°29.4
										Atria	107°08.4	-69°04.5
										Sabik	102°02.0	-15°45.4
										Shaula	96°09.4	-37°07.4
										Rasalhague	95°57.8	12°32.5
										Eltanin	90°41.5	51°29.2
										Kaus Aust.	83°31.6	-34°22.3
										Vega	80°32.5	38°48.4
										Nunki	75°46.9	-26°15.9
										Altair	61°59.2	8°56.2
										Peacock	53°04.5	-56°39.0
										Deneb	49°25.1	45°22.2
										Enif	33°38.1	9°59.5
										Al Na'ir	27°32.1	-46°50.0
										Fomalhaut	15°13.9	-29°29.0
										Scheat	13°44.6	28°13.2
										Markab	13°29.3	15°20.5
										Jul 06 Sun	SHA	Mer.pass
										Venus	300°05.1	09:03
										Mars	197°28.1	15:52
										Jupiter	263°31.3	11:28
										Saturn	357°22.0	05:13
										Jul 07 Mon	SHA	Mer.pass
										Venus	298°56.9	09:04
										Mars	196°55.6	15:50
										Jupiter	263°16.5	11:25
										Saturn	357°21.3	05:09
										Jul 08 Tue	SHA	Mer.pass
										Venus	297°48.5	09:04
										Mars	196°23.0	15:48
										Jupiter	263°01.8	11:22
										Saturn	357°20.6	05:05
										Horizontal parallax		
										Venus:	0.1	
										Mars:	0.1	

Sun							Moon						
Sun	GHA	Dec	GHA	ν	Dec	d	HP						
0	178°48.6	N22°40.9	58°57.6	13.2'	S21°35.6	9.6'	54.2'						
1	193°48.5	40.6	73°29.8	13.1'	21°45.2	9.5'	54.2'						
2	208°48.4	40.4	88°02.0	13.1'	21°54.7	9.4'	54.2'						
3	223°48.3	· · 40.1	102°34.0	13.0'	22°04.2	9.3'	54.2'						
4	238°48.2	39.8	117°06.0	12.9'	22°13.5	9.3'	54.2'						
5	253°48.1	39.6	131°37.9	12.8'	22°22.8	9.2'	54.2'						
6	268°48.0	N22°39.3	146°09.8	12.8'	S22°31.9	9.1'	54.2'						
7	283°47.9	39.1	160°41.5	12.7'	22°41.0	9.0'	54.2'						
8	298°47.7	38.8	175°13.2	12.6'	22°50.0	8.9'	54.2'						
9	313°47.6	· · 38.6	189°44.8	12.5'	22°58.9	8.8'	54.2'						
10	328°47.5	38.3	204°16.3	12.4'	23°07.6	8.7'	54.3'						
11	343°47.4	38.0	218°47.7	12.4'	23°16.3	8.6'	54.3'						
12	358°47.3	N22°37.8	233°19.1	12.3'	S23°24.9	8.5'	54.3'						
13	13°47.2	37.5	247°50.4	12.2'	23°33.4	8.4'	54.3'						
14	28°47.1	37.3	262°21.6	12.1'	23°41.8	8.3'	54.3'						
15	43°47.0	· · 37.0	276°52.7	12.0'	23°50.0	8.2'	54.3'						
16	58°46.9	36.7	291°23.7	12.0'	23°58.2	8.1'	54.3'						
17	73°46.8	36.5	305°54.7	11.9'	24°06.3	8.0'	54.3'						
18	88°46.7	N22°36.2	320°25.6	11.8'	S24°14.2	7.9'	54.3'						
19	103°46.6	35.9	334°56.4	11.7'	24°22.1	7.8'	54.3'						
20	118°46.5	35.7	349°27.1	11.6'	24°29.9	7.6'	54.3'						
21	133°46.4	· · 35.4	3°57.7	11.6'	24°37.5	7.5'	54.4'						
22	148°46.3	35.2	18°28.3	11.5'	24°45.1	7.4'	54.4'						
23	163°46.2	34.9	32°58.8	11.4'	24°52.5	7.3'	54.4'						
SD = 15.7'			$d = -0.3'$		SD = 14.8'								
Mon	GHA	Dec	GHA	ν	Dec	d	HP						
0	178°46.1	N22°34.6	47°29.2	11.3'	S24°59.8	7.2'	54.4'						
1	193°46.0	34.3	61°59.5	11.2'	25°07.0	7.1'	54.4'						
2	208°45.9	34.1	76°29.7	11.2'	25°14.1	7.0'	54.4'						
3	223°45.8	· · 33.8	90°59.9	11.1'	25°21.1	6.9'	54.4'						
4	238°45.7	33.5	105°30.0	11.0'	25°28.0	6.8'	54.4'						
5	253°45.6	33.3	120°00.0	10.9'	25°34.7	6.6'	54.4'						
6	268°45.5	N22°33.0	134°29.9	10.9'	S25°41.3	6.5'	54.5'						
7	283°45.4	32.7	148°59.8	10.8'	25°47.9	6.4'	54.5'						
8	298°45.3	32.4	163°29.6	10.7'	25°54.3	6.3'	54.5'						
9	313°45.2	· · 32.2	177°59.3	10.6'	26°00.5	6.2'	54.5'						
10	328°45.1	31.9	192°28.9	10.5'	26°06.7	6.0'	54.5'						
11	343°45.0	31.6	206°58.4	10.5'	26°12.7	5.9'	54.5'						
12	358°44.9	N22°31.3	221°27.9	10.4'	S26°18.7	5.8'	54.5'						
13	13°44.8	31.1	235°57.3	10.3'	26°24.5	5.7'	54.5'						
14	28°44.7	30.8	250°26.6	10.3'	26°30.1	5.5'	54.6'						
15	43°44.6	· · 30.5	264°55.9	10.2'	26°35.7	5.4'	54.6'						
16	58°44.5	30.2	279°25.1	10.1'	26°41.1	5.3'	54.6'						
17	73°44.4	30.0	293°54.2	10.0'	26°46.4	5.2'	54.6'						
18	88°44.3	N22°29.7	308°23.2	10.0'	S26°51.6	5.0'	54.6'						
19	103°44.2	29.4	322°52.2	9.9'	26°56.6	4.9'	54.6'						
20	118°44.1	29.1	337°21.0	9.8'	27°01.5	4.8'	54.6'						
21	133°44.1	· · 28.8	351°49.9	9.8'	27°06.3	4.6'	54.7'						
22	148°44.0	28.5	6°18.6	9.7'	27°10.9	4.5'	54.7'						
23	163°43.9	28.3	20°47.3	9.6'	27°15.4	4.4'	54.7'						
SD = 15.7'			$d = -0.3'$		SD = 14.8'								
Tue	GHA	Dec	GHA	ν	Dec	d	HP						
0	178°43.8	N22°28.0	35°15.9	9.6'	S27°19.8	4.2'	54.7'						
1	193°43.7	27.7	49°44.5	9.5'	27°24.1	4.1'	54.7'						
2	208°43.6	27.4	64°13.0	9.4'	27°28.2	4.0'	54.7'						
3	223°43.5	· · 27.1	78°41.4	9.4'	27°32.2	3.8'	54.8'						
4	238°43.4	26.8	93°09.8	9.3'	27°36.0	3.7'	54.8'						
5	253°43.3	26.5	107°38.1	9.2'	27°39.7	3.6'	54.8'						
6	268°43.2	N22°26.3	122°06.3	9.2'	S27°43.3	3.4'	54.8'						
7	283°43.1	26.0	136°34.5	9.1'	27°46.7	3.3'	54.8'						
8	298°43.0	25.7	151°02.6	9.1'	27°50.0	3.1'	54.8'						
9	313°42.9	· · 25.4	165°30.7	9.0'	27°53.1	3.0'	54.9'						
10	328°42.8	25.1	179°58.7	9.0'	27°56.1	2.9'	54.9'						
11	343°42.7	24.8	194°26.6	8.9'	27°59.0	2.7'	54.9'						
12	358°42.6	N22°24.5	208°54.5	8.8'	S28°01.7	2.6'	54.9'						
13	13°42.5	24.2	223°22.4	8.8'	28°04.3	2.4'	54.9'						
14	28°42.4	23.9	237°50.2	8.7'	28°06.8	2.3'	54.9'						
15	43°42.3	· · 23.6	252°17.9	8.7'	28°09.0	2.1'	55.0'						
16	58°42.2	23.3	266°45.6	8.6'	28°11.2	2.0'	55.0'						
17	73°42.2	23.0	281°13.3	8.6'	28°13.2	1.9'	55.0'						
18	88°42.1	N22°22.7	295°40.9	8.6'	S28°15.1	1.7'	55.0'						
19	103°42.0	22.4	310°08.4	8.5'	28°16.8	1.6'	55.0'						
20	118°41.9	22.2	324°35.9	8.5'	28°18.3	1.4'	55.1'						
21	133°41.8	· · 21.9	339°03.4	8.4'	28°19.7	1.3'	55.1'						
22	148°41.7	21.6	353°30.8	8.4'	28°21.0	1.1'	55.1'						
23	163°41.6	21.3	7°58.2	8.4'	28°22.1	1.0'	55.1'						
SD = 15.7'			$d = -0.3'$		SD = 14.9'								

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°	—:—	—:—	00:59	23:07	—:—	—:—
64°	—:—	—:—	01:54	22:14	—:—	—:—
62°	—:—	—:—	02:27	21:42	—:—	—:—
60°	—:—	01:20	02:50	21:19	22:47	—:—
N 58°	—:—	01:59	03:09	21:00	22:09	—:—
56°	—:—	02:25	03:25	20:45	21:44	—:—
54°	01:14	02:46	03:38	20:31	21:24	22:54
52°	01:50	03:02	03:49	20:20	21:07	22:19
50°	02:14	03:16	04:00	20:10	20:53	21:55
45°	02:56	03:44	04:21	19:48	20:25	21:13
N 40°	03:25	04:06	04:38	19:31	20:04	20:45
35°	03:47	04:23	04:53	19:17	19:46	20:23
30°	04:05	04:38	05:05	19:05	19:32	20:05
20°	04:33	05:02	05:26	18:44	19:08	19:37
N 10°	04:55	05:22	05:45	18:25	18:48	19:15
0°	05:13	05:39	06:01	18:09	18:31	18:57
S 10°	05:29	05:55	06:18	17:52	18:15	18:41
20°	05:45	06:12	06:36	17:34	17:58	18:25
30°	06:00	06:30	06:56	17:14	17:40	18:10
35°	06:09	06:40	07:08	17:02	17:30	18:02
40°	06:17	06:51	07:22	16:49	17:19	17:53
45°	06:27	07:04	07:38	16:33	17:06	17:43
S 50°	06:38	07:19	07:57	16:13	16:51	17:32
52°	06:43	07:26	08:06	16:04	16:44	17:27
54°	06:48	07:34	08:17	15:53	16:36	17:22
56°	06:54	07:43	08:29	15:41	16:28	17:16
58°	07:00	07:52	08:43	15:28	16:18	17:10
S 60°	07:07	08:03	08:59	15:12	16:07	17:03

July 09, 10, 11 UT (Wed., Thu., Fri.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	287° 11.3	223° 50.9	N18° 41.2	123° 01.7	N07° 43.8	189° 58.3	N23° 09.3	284° 31.4	S01° 19.3					Alpheratz	357° 34.2	29° 13.8	
1	302° 13.7	238° 50.5		138° 02.8	43.2	205° 00.1	09.3	299° 33.8	19.3					Ankaa	353° 06.6	-42° 09.8	
2	317° 16.2	253° 50.1	42.3	153° 03.9	42.6	220° 02.0	09.3	314° 36.2	19.3					Schedar	349° 30.5	56° 40.4	
3	332° 18.7	268° 49.7	· · 42.9	168° 05.0	· · 42.0	235° 03.8	· · 09.3	329° 38.7	· · 19.3					Diphda	348° 46.8	-17° 50.6	
4	347° 21.1	283° 49.3	43.4	183° 06.1	41.4	250° 05.7	09.2	344° 41.1	19.3					Achernar	335° 19.9	-57° 06.1	
5	2° 23.6	298° 48.9	44.0	198° 07.2	40.9	265° 07.5	09.2	359° 43.6	19.3					Hamal	327° 50.8	23° 34.9	
6	17° 26.0	313° 48.5	N18° 44.5	213° 08.3	N07° 40.3	280° 09.4	N23° 09.2	14° 46.0	S01° 19.3					Polaris	314° 03.8	89° 22.0	
7	32° 28.5	328° 48.1	45.1	228° 09.4	39.7	295° 11.3	09.2	29° 48.5	19.3					Acamar	315° 11.7	-40° 11.9	
8	47° 31.0	343° 47.7	45.7	243° 10.5	39.1	310° 13.1	09.1	44° 50.9	19.3					Menkar	314° 05.9	4° 11.4	
9	62° 33.4	358° 47.2	· · 46.2	258° 11.6	· · 38.5	325° 15.0	· · 09.1	59° 53.4	· · 19.3					Mirfak	308° 27.9	49° 56.9	
10	77° 35.9	13° 46.8	46.8	273° 12.7	37.9	340° 16.8	09.1	74° 55.8	19.3					Aldebaran	290° 39.4	16° 33.6	
11	92° 38.4	28° 46.4	47.3	288° 13.8	37.3	355° 18.7	09.1	89° 58.3	19.4					Rigel	281° 03.8	-8° 10.2	
12	107° 40.8	43° 46.0	N18° 47.9	303° 14.9	N07° 36.7	10° 20.5	N23° 09.0	105° 00.7	S01° 19.4					Capella	280° 21.7	46° 01.3	
13	122° 43.3	58° 45.6	48.5	318° 16.0	36.1	25° 22.4	09.0	120° 03.1	19.4					Bellatrix	278° 22.7	6° 22.4	
14	137° 45.8	73° 45.2	49.0	333° 17.1	35.5	40° 24.2	09.0	135° 05.6	19.4					Elnath	278° 01.7	28° 37.7	
15	152° 48.2	88° 44.7	· · 49.6	348° 18.2	· · 34.9	55° 26.1	· · 09.0	150° 08.0	· · 19.4					Alnilam	275° 37.6	-1° 11.1	
16	167° 50.7	103° 44.3	50.1	3° 19.3	34.3	70° 27.9	08.9	165° 10.5	19.4					Betelgeuse	270° 52.0	7° 24.8	
17	182° 53.2	118° 43.9	50.7	18° 20.4	33.7	85° 29.8	08.9	180° 12.9	19.4					Canopus	263° 52.8	-52° 42.4	
18	197° 55.6	133° 43.5	N18° 51.2	33° 21.5	N07° 33.1	100° 31.6	N23° 08.9	195° 15.4	S01° 19.4					Sirius	258° 26.2	-16° 45.0	
19	212° 58.1	148° 43.1	51.8	48° 22.6	32.5	115° 33.5	08.9	210° 17.8	19.4					Adhara	255° 05.9	-29° 00.3	
20	228° 00.5	163° 42.7	52.3	63° 23.7	31.9	130° 35.3	08.8	225° 20.3	19.4					Procyon	244° 50.7	5° 09.6	
21	243° 03.0	178° 42.2	· · 52.9	78° 24.8	· · 31.3	145° 37.2	· · 08.8	240° 22.7	· · 19.4					Pollux	243° 17.2	27° 57.9	
22	258° 05.5	193° 41.8	53.4	93° 25.9	30.7	160° 39.0	08.8	255° 25.2	19.4					Avior	234° 15.2	-59° 35.5	
23	273° 07.9	208° 41.4	54.0	108° 27.0	30.1	175° 40.9	08.8	270° 27.6	19.4					Suhail	222° 46.4	-43° 32.2	
Mer.pass. 04:50			ν -0.4' d 0.6' m -4.09			ν 1.1' d -0.6' m 1.53			ν 1.9' d -0.0' m -1.89			ν 2.4' d 0.0' m 0.92					

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	288° 10.4	223° 41.0	N18° 54.5	123° 28.1	N07° 29.5	190° 42.7	N23° 08.7	285° 30.1	S01° 19.4					Denebola	182° 24.6	14° 25.9	
1	303° 12.9	238° 40.5	55.1	138° 29.2	28.9	205° 44.6	08.7	300° 32.5	19.4					Gienah	175° 43.2	-17° 41.1	
2	318° 15.3	253° 40.1	55.6	153° 30.3	28.3	220° 46.4	08.7	315° 34.9	19.4					Acrux	172° 59.8	-63° 14.7	
3	333° 17.8	268° 39.7	· · 56.2	168° 31.5	· · 27.8	235° 48.3	· · 08.7	330° 37.4	· · 19.4					Gacrux	171° 51.3	-57° 15.6	
4	348° 20.3	283° 39.3	56.7	183° 32.6	27.2	250° 50.1	08.6	345° 39.8	19.4					Alioth	166° 12.6	55° 49.5	
5	3° 22.7	298° 38.8	57.3	198° 33.7	26.6	265° 52.0	08.6	0° 42.3	19.4					Spica	158° 21.8	-11° 17.7	
6	18° 25.2	313° 38.4	N18° 57.8	213° 34.8	N07° 26.0	280° 53.8	N23° 08.6	15° 44.7	S01° 19.4					Alkaid	152° 51.6	49° 11.4	
7	33° 27.7	328° 38.0	58.4	228° 35.9	25.4	295° 55.7	08.6	30° 47.2	19.5					Hadar	148° 35.2	-60° 30.0	
8	48° 30.1	343° 37.6	58.9	243° 37.0	24.8	310° 57.5	08.5	45° 49.6	19.5					Menkent	147° 57.0	-36° 29.9	
9	63° 32.6	358° 37.1	18° 59.5	258° 38.1	· · 24.2	325° 59.4	· · 08.5	60° 52.1	· · 19.5					Arcturus	145° 47.4	19° 03.1	
10	78° 35.0	13° 36.7	19° 00.0	273° 39.2	23.6	341° 01.2	08.5	75° 54.5	19.5					Rigel Kent.	139° 39.5	-60° 56.7	
11	93° 37.5	28° 36.3	00.5	288° 40.3	23.0	356° 03.1	08.5	90° 57.0	19.5					Kochab	137° 19.2	74° 03.3	
12	108° 40.0	43° 35.9	N19° 01.1	303° 41.4	N07° 22.4	11° 04.9	N23° 08.4	105° 59.4	S01° 19.5					Zuben'ubi	136° 55.4	-16° 08.9	
13	123° 42.4	58° 35.4	01.6	318° 42.5	21.8	26° 06.8	08.4	121° 01.9	19.5					Alphecca	126° 03.2	26° 37.8	
14	138° 44.9	73° 35.0	02.2	333° 43.6	21.2	41° 08.6	08.4	136° 04.3	19.5					Antares	112° 15.0	-26° 29.4	
15	153° 47.4	88° 34.6	· · 02.7	348° 44.7	· · 20.6	56° 10.5	· · 08.3	151° 06.8	· · 19.5					Atria	107° 08.4	-69° 04.5	
16	168° 49.8	103° 34.1	03.2	3° 45.8	20.0	71° 12.3	08.3	166° 09.2	19.5					Sabik	102° 02.0	-15° 45.4	
17	183° 52.3	118° 33.7	03.8	18° 46.9	19.4	86° 14.2	08.3	181° 11.7	19.5					Shaula	96° 09.4	-37° 07.4	
18	198° 54.8	133° 33.3	N19° 04.3	33° 48.0	N07° 18.8	101° 16.0	N23° 08.3	196° 14.1	S01° 19.5					Rasalhague	95° 57.8	12° 32.5	
19	213° 57.2	148° 32.8	04.9	48° 49.1	18.2	116° 17.9	08.2	211° 16.6	19.5					Eltanin	90° 41.5	51° 29.2	
20	228° 59.7	163° 32.4	05.4	63° 50.2	17.6	131° 19.7	08.2	226° 19.0	19.5					Kaus Aust.	83° 31.6	-34° 22.3	
21	244° 02.2	178° 32.0	· · 05.9	78° 51.3	· · 17.0	146° 21.6	· · 08.2	241° 21.5	· · 19.5					Vega	80° 32.5	38° 48.5	
22	259° 04.6	193° 31.5	06.5	93° 52.4	16.4	161° 23.5	08.2	256° 23.9	19.5					Nunki	75° 46.9	-26° 15.9	
23	274° 07.1	208° 31.1	07.0	108° 53.5	15.8	176° 25.3	08.1	271° 26.4	19.6					Altair	61° 59.2	8° 56.2	
Mer.pass. 04:47			ν -0.4' d 0.5' m -4.09			ν 1.1' d -0.6' m 1.53			ν 1.9' d -0.0' m -1.90			ν 2.4' d 0.0' m 0.91					

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	289° 09.5	223° 30.7	N19° 07.5	123° 54.6	N07° 15.2	191° 27.2	N23° 08.1	286° 28.8	S01° 19.6					Jul 09 Wed	SHA	Mer.pass
1	304° 12.0	238° 30.2	08.1	138° 55.7	14.6	206° 29.0	08.1	301° 31.3	19.6					Venus	296° 39.7	09:05
2	319° 14.5	253° 29.8	08.6	153° 56.8	14.0	221° 30.9	08.1	316° 33.7	19.6					Mars	195° 50.4	15:47
3	334° 16.9	268° 29.3	· · 09.1	168° 57.9	· · 13.4	236° 32.7	· · 08.0	331° 36.2	· · 19.6					Jupiter	262° 47.0	11:19
4	349° 19.4	283° 28.9	09.7	183° 59.0	12.8	251° 34.6	08.0	346° 38.6	19.6					Saturn	357° 20.1	05:01
5	4° 21.9	298° 28.5	10.2	199° 00.1	12.2	266° 36.4	08.0	1° 41.1	19.6							
6	19° 24.3	313° 28.0	N19° 10.7	214° 01.2	N07° 11.6	281° 38.3	N23° 08.0	16° 43.5	S01° 19.6							
7	34° 26.8	328° 27.6	11.3	229° 02.3	11.											

h Sun Moon						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	178°41.5	N22°21.0	22°25.6	8.3'	S28°23.1	0.8' 55.1'
1	193°41.4	20.7	36°52.9	8.3'	28°23.9	0.7' 55.1'
2	208°41.3	20.4	51°20.2	8.3'	28°24.6	0.5' 55.2'
3	223°41.2	· · 20.0	65°47.4	8.2'	28°25.1	0.4' 55.2'
4	238°41.1	19.7	80°14.7	8.2'	28°25.5	0.2' 55.2'
5	253°41.0	19.4	94°41.8	8.2'	28°25.7	0.1' 55.2'
6	268°41.0	N22°19.1	109°09.0	8.1'	S28°25.8	-0.1' 55.2'
7	283°40.9	18.8	123°36.1	8.1'	28°25.7	-0.2' 55.3'
8	298°40.8	18.5	138°03.3	8.1'	28°25.4	-0.4' 55.3'
9	313°40.7	· · 18.2	152°30.3	8.1'	28°25.0	-0.5' 55.3'
10	328°40.6	17.9	166°57.4	8.0'	28°24.5	-0.7' 55.3'
11	343°40.5	17.6	181°24.5	8.0'	28°23.8	-0.9' 55.3'
12	358°40.4	N22°17.3	195°51.5	8.0'	S28°22.9	-1.0' 55.4'
13	13°40.3	17.0	210°18.5	8.0'	28°21.9	-1.2' 55.4'
14	28°40.2	16.7	224°45.5	8.0'	28°20.8	-1.3' 55.4'
15	43°40.1	· · 16.4	239°12.5	8.0'	28°19.4	-1.5' 55.4'
16	58°40.1	16.1	253°39.5	8.0'	28°18.0	-1.6' 55.4'
17	73°40.0	15.7	268°06.4	8.0'	28°16.4	-1.8' 55.5'
18	88°39.9	N22°15.4	282°33.4	7.9'	S28°14.6	-1.9' 55.5'
19	103°39.8	15.1	297°00.3	7.9'	28°12.6	-2.1' 55.5'
20	118°39.7	14.8	311°27.3	7.9'	28°10.5	-2.2' 55.5'
21	133°39.6	· · 14.5	325°54.2	7.9'	28°08.3	-2.4' 55.5'
22	148°39.5	14.2	340°21.1	7.9'	28°05.9	-2.6' 55.6'
23	163°39.4	13.9	354°48.1	7.9'	28°03.3	-2.7' 55.6'
SD = 15.7' d = -0.3' SD = 15.0'						
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	178°39.4	N22°13.5	9°15.0	7.9'	S28°00.6	-2.9' 55.6'
1	193°39.3	13.2	23°41.9	7.9'	27°57.8	-3.0' 55.6'
2	208°39.2	12.9	38°08.9	7.9'	27°54.8	-3.2' 55.6'
3	223°39.1	· · 12.6	52°35.8	8.0'	27°51.6	-3.3' 55.7'
4	238°39.0	12.3	67°02.8	8.0'	27°48.3	-3.5' 55.7'
5	253°38.9	12.0	81°29.8	8.0'	27°44.8	-3.6' 55.7'
6	268°38.8	N22°11.6	95°56.7	8.0'	S27°41.1	-3.8' 55.7'
7	283°38.7	11.3	110°23.7	8.0'	27°37.3	-3.9' 55.8'
8	298°38.7	11.0	124°50.7	8.0'	27°33.4	-4.1' 55.8'
9	313°38.6	· · 10.7	139°17.7	8.0'	27°29.3	-4.3' 55.8'
10	328°38.5	10.3	153°44.8	8.1'	27°25.1	-4.4' 55.8'
11	343°38.4	10.0	168°11.8	8.1'	27°20.7	-4.6' 55.8'
12	358°38.3	N22°09.7	182°38.9	8.1'	S27°16.1	-4.7' 55.9'
13	13°38.2	09.4	197°06.0	8.1'	27°11.4	-4.9' 55.9'
14	28°38.1	09.0	211°33.1	8.1'	27°06.5	-5.0' 55.9'
15	43°38.1	· · 08.7	226°00.3	8.2'	27°01.5	-5.2' 55.9'
16	58°38.0	08.4	240°27.4	8.2'	26°56.4	-5.3' 55.9'
17	73°37.9	08.1	254°54.6	8.2'	26°51.0	-5.5' 56.0'
18	88°37.8	N22°07.7	269°21.9	8.3'	S26°45.6	-5.6' 56.0'
19	103°37.7	07.4	283°49.1	8.3'	26°40.0	-5.8' 56.0'
20	118°37.6	07.1	298°16.4	8.3'	26°34.2	-5.9' 56.0'
21	133°37.6	· · 06.8	312°43.7	8.4'	26°28.3	-6.1' 56.1'
22	148°37.5	06.4	327°11.1	8.4'	26°22.2	-6.2' 56.1'
23	163°37.4	06.1	341°38.5	8.4'	26°16.0	-6.4' 56.1'
SD = 15.7' d = -0.3' SD = 15.2'						
Fri	GHA	Dec	GHA	ν	Dec	d HP
0	178°37.3	N22°05.8	358°05.9	8.5'	S26°09.7	-6.5' 56.1'
1	193°37.2	05.4	10°33.4	8.5'	26°03.2	-6.6' 56.1'
2	208°37.1	05.1	25°00.9	8.5'	25°56.5	-6.8' 56.2'
3	223°37.1	· · 04.8	39°28.4	8.6'	25°49.7	-6.9' 56.2'
4	238°37.0	04.4	53°56.0	8.6'	25°42.8	-7.1' 56.2'
5	253°36.9	04.1	68°23.6	8.7'	25°35.7	-7.2' 56.2'
6	268°36.8	N22°03.7	82°51.3	8.7'	S25°28.5	-7.4' 56.3'
7	283°36.7	03.4	97°19.0	8.8'	25°21.1	-7.5' 56.3'
8	298°36.7	03.1	111°46.7	8.8'	25°13.6	-7.6' 56.3'
9	313°36.6	· · 02.7	126°14.5	8.8'	25°06.0	-7.8' 56.3'
10	328°36.5	02.4	140°42.4	8.9'	24°58.2	-7.9' 56.3'
11	343°36.4	02.1	155°10.3	8.9'	24°50.2	-8.1' 56.4'
12	358°36.3	N22°01.7	169°38.2	9.0'	S24°42.2	-8.2' 56.4'
13	13°36.2	01.4	184°06.2	9.0'	24°34.0	-8.3' 56.4'
14	28°36.2	01.0	198°34.3	9.1'	24°25.6	-8.5' 56.4'
15	43°36.1	· · 00.7	213°02.4	9.1'	24°17.1	-8.6' 56.5'
16	58°36.0	00.4	227°30.5	9.2'	24°08.5	-8.7' 56.5'
17	73°35.9	22°00.0	241°58.7	9.2'	23°59.8	-8.9' 56.5'
18	88°35.8	N21°59.7	256°26.9	9.3'	S23°50.9	-9.0' 56.5'
19	103°35.8	59.3	270°55.3	9.4'	23°41.9	-9.1' 56.5'
20	118°35.7	59.0	285°23.6	9.4'	23°32.7	-9.3' 56.6'
21	133°35.6	· · 58.6	299°52.0	9.5'	23°23.5	-9.4' 56.6'
22	148°35.5	58.3	314°20.5	9.5'	23°14.0	-9.5' 56.6'
23	163°35.5	57.9	328°49.0	9.6'	23°04.5	-9.7' 56.6'
SD = 15.7' d = -0.3' SD = 15.3'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°						
66°	—:—	—:—	01:14	22:53	—:—	—:—
64°	—:—	—:—	02:02	22:07	—:—	—:—
62°	—:—	—:—	02:32	21:37	—:—	—:—
60°	—:—	01:30	02:55	21:15	22:39	—:—
N 58°	—:—	02:05	03:13	20:57	22:04	—:—
56°	—:—	02:30	03:28	20:42	21:40	—:—
54°	01:22	02:50	03:41	20:29	21:20	22:46
52°	01:55	03:06	03:53	20:18	21:04	22:14
50°	02:19	03:20	04:03	20:08	20:51	21:51
45°	02:59	03:47	04:23	19:47	20:23	21:11
N 40°	03:28	04:08	04:40	19:30	20:02	20:43
35°	03:49	04:25	04:54	19:16	19:45	20:21
30°	04:07	04:40	05:07	19:04	19:31	20:04
20°	04:34	05:03	05:27	18:43	19:08	19:36
N 10°	04:56	05:22	05:45	18:26	18:48	19:15
0°	05:14	05:39	06:02	18:09	18:31	18:57
S 10°	05:30	05:56	06:18	17:53	18:15	18:41
20°	05:45	06:12	06:36	17:35	17:59	18:26
30°	06:00	06:30	06:56	17:15	17:41	18:11
35°	06:08	06:39	07:07	17:04	17:32	18:03
40°	06:17	06:50	07:20	16:51	17:21	17:54
45°	06:26	07:03	07:36	16:35	17:08	17:45
S 50°	06:37	07:18	07:55	16:16	16:53	17:34
52°	06:42	07:25	08:04	16:07	16:47	17:30
54°	06:47	07:32	08:15	15:57	16:39	17:24
56°	06:52	07:40	08:26	15:45	16:31	17:19
58°	06:58	07:50	08:40	15:32	16:22	17:13
S 60°	07:05	08:00	08:55	15:16	16:11	17:06

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°						
N 70°						
68°						
66°			23:58			
64°			23:10			
62°		22:49	22:40			01:52
60°	21:52	22:10	22:16	· · ·	00:55	02:30
N 58°	21:14	21:43	21:58	00:30	01:32	02:57
56°	20:47	21:21	21:42	00:58	01:59	03:18
54°	20:26	21:03	21:28	01:20	02:20	03:36
52°	20:09	20:48	21:16	01:38	02:37	03:51
50°	19:54	20:35	21:06	01:53	02:52	04:03
45°	19:23	20:08	20:44	02:24	03:22	04:30
N 40°	19:00	19:46	20:26	02:48	03:45	04:50
35°	18:40	19:29	20:10	03:07	04:05	05:08
30°	18:23	19:13	19:57	03:24	04:21	05:23
20°	17:55	18:47	19:35	03:53	04:49	05:48
N 10°	17:31	18:24	19:15	04:17	05:13	06:09
0°	17:08	18:03	18:57	04:40	05:35	06:29
S 10°	16:46	17:42	18:38	05:03	05:57	06:49
20°	16:21	17:19	18:18	05:27	06:20	07:10
30°	15:53	16:53	17:55	05:56	06:48	07:35
35°	15:37	16:37	17:42	06:12	07:04	07:49
40°	15:17	16:19	17:26	06:32	07:23	08:05
45°	14:53	15:57	17:08	06:56	07:45	08:25
S 50°	14:23	15:29	16:44	07:26	08:14	08:50
52°	14:08	15:15	16:33	07:41	08:28	09:01
54°	13:50	14:59	16:20	07:59	08:44	09:15
56°	13:29	14:40	16:05	08:21	09:03	09:30
58°	13:01	14:16	15:47	08:48	09:27	09:48
S 60°	12:23	13:45	15:26	09:27	09:59	10:11

Day	Sun		Mer. Pass hh:mm	Moon		Age 14-16 96-100%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
09	05:14	05:18	12:05	23:22	10:54	
10	05:23	05:27	12:05	—:—	11:49	
11	05:31	05:35	12:06	00:16	12:43	

July 12, 13, 14 UT (Sat., Sun., Mon.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	290°08.7	223°20.0	N19°20.1	124°21.0	N07°00.8	192°11.6	N23°07.5	287°27.7	S01°19.7					Alpheratz	357°34.2	29°13.8	
1	305°11.1	238°19.6		139°22.1	07°00.2	207°13.5		302°30.2	19.7					Ankaa	353°06.6	-42°09.7	
2	320°13.6	253°19.1		154°23.2	06°59.6	222°15.3		317°32.6	19.8					Schedar	349°30.5	56°40.4	
3	335°16.1	268°18.7	· ·	169°24.3	· · 59.0	237°17.2	· · 07.4	332°35.1	· · 19.8					Diphda	348°46.8	-17°50.6	
4	350°18.5	283°18.2		184°25.4	58.4	252°19.0		347°37.5	19.8					Achernar	335°19.9	-57°06.1	
5	5°21.0	298°17.8		199°26.5	57.8	267°20.9	07.3	2°40.0	19.8					Hamal	327°50.8	23°34.9	
6	20°23.5	313°17.3	N19°23.2	214°27.6	N06°57.2	282°22.7	N23°07.3	17°42.4	S01°19.8					Polaris	314°02.4	89°22.0	
7	35°25.9	328°16.9		229°28.7	56.6	297°24.6		32°44.9	19.8					Acamar	315°11.6	-40°11.9	
8	50°28.4	343°16.4		244°29.8	56.0	312°26.4		47°47.4	19.8					Menkar	314°05.8	4°11.4	
9	65°30.9	358°15.9	· ·	259°30.9	· · 55.4	327°28.3	· · 07.2	62°49.8	· · 19.8					Mirfak	308°27.9	49°56.9	
10	80°33.3	13°15.5		274°32.0	54.8	342°30.2		77°52.3	19.8					Aldebaran	290°39.4	16°33.6	
11	95°35.8	28°15.0		289°33.1	54.2	357°32.0	07.2	92°54.7	19.8					Rigel	281°03.7	-8°10.2	
12	110°38.3	43°14.6	N19°26.3	304°34.2	N06°53.6	12°33.9	N23°07.1	107°57.2	S01°19.8					Capella	280°21.7	46°01.3	
13	125°40.7	58°14.1		319°35.3	53.0	27°35.7		122°59.6	19.9					Bellatrix	278°22.7	6°22.4	
14	140°43.2	73°13.7		334°36.4	52.4	42°37.6		138°02.1	19.9					Elnath	278°01.7	28°37.7	
15	155°45.6	88°13.2	· ·	349°37.5	· · 51.8	57°39.4	· · 07.1	153°04.6	· · 19.9					Alnilam	275°37.6	-1°11.1	
16	170°48.1	103°12.8		4°38.6	51.2	72°41.3		168°07.0	19.9					Betelgeuse	270°51.9	7°24.8	
17	185°50.6	118°12.3		19°39.7	50.6	87°43.1		183°09.5	19.9					Canopus	263°52.7	-52°42.4	
18	200°53.0	133°11.8	N19°29.3	34°40.8	N06°50.0	102°45.0	N23°07.0	198°11.9	S01°19.9					Sirius	258°26.2	-16°45.0	
19	215°55.5	148°11.4		49°41.9	49.4	117°46.8		213°14.4	19.9					Adhara	255°05.9	-29°00.3	
20	230°58.0	163°10.9		64°43.0	48.8	132°48.7		228°16.8	19.9					Procyon	244°50.7	5°09.6	
21	246°00.4	178°10.5	· ·	79°44.1	· · 48.2	147°50.5	· · 06.9	243°19.3	· · 19.9					Pollux	243°17.2	27°57.9	
22	261°02.9	193°10.0		94°45.2	47.5	162°52.4		258°21.8	19.9					Avior	234°15.2	-59°35.5	
23	276°05.4	208°09.5		109°46.3	46.9	177°54.3	06.8	273°24.2	20.0					Suhail	222°46.4	-43°32.2	
Mer.pass. 04:39		ν -0.5' d 0.5' m -4.08			ν 1.1' d -0.6' m 1.54			ν 1.9' d -0.0' m -1.90			ν 2.5' d 0.0' m 0.90						

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	291°07.8	223°09.1	N19°32.4	124°47.4	N06°46.3	192°56.1	N23°06.8	288°26.7	S01°20.0				
1	306°10.3	238°08.6	32.9	139°48.5	45.7	207°58.0	06.8	303°29.1	20.0				
2	321°12.8	253°08.1	33.4	154°49.6	45.1	222°59.8	06.8	318°31.6	20.0				
3	336°15.2	268°07.7	· · 33.9	169°50.7	· · 44.5	238°01.7	· · 06.7	333°34.1	· · 20.0				
4	351°17.7	283°07.2	34.4	184°51.8	43.9	253°03.5	06.7	348°36.5	20.0				
5	6°20.1	298°06.7	34.9	199°52.9	43.3	268°05.4	06.7	3°39.0	20.0				
6	21°22.6	313°06.3	N19°35.4	214°54.0	N06°42.7	283°07.2	N23°06.6	18°41.4	S01°20.0				
7	36°25.1	328°05.8	35.8	229°55.1	42.1	298°09.1	06.6	33°43.9	20.0				
8	51°27.5	343°05.3	36.3	244°56.2	41.5	313°10.9	06.6	48°46.4	20.0				
9	66°30.0	358°04.9	· · 36.8	259°57.3	· · 40.9	328°12.8	· · 06.6	63°48.8	· · 20.1				
10	81°32.5	13°04.4	37.3	274°58.4	40.3	343°14.7	06.5	78°51.3	20.1				
11	96°34.9	28°03.9	37.8	289°59.5	39.7	358°16.5	06.5	93°53.7	20.1				
12	111°37.4	43°03.5	N19°38.3	305°00.6	N06°39.1	13°18.4	N23°06.5	108°56.2	S01°20.1				
13	126°39.9	58°03.0	38.8	320°01.7	38.5	28°20.2	06.5	123°58.7	20.1				
14	141°42.3	73°02.5	39.3	335°02.8	37.9	43°22.1	06.4	139°01.1	20.1				
15	156°44.8	88°02.1	· · 39.8	350°03.9	· · 37.3	58°23.9	· · 06.4	154°03.6	· · 20.1				
16	171°47.3	103°01.6	40.3	5°05.0	36.7	73°25.8	06.4	169°06.0	20.1				
17	186°49.7	118°01.1	40.8	20°06.1	36.1	88°27.6	06.3	184°08.5	20.1				
18	201°52.2	133°00.6	N19°41.3	35°07.2	N06°35.5	103°29.5	N23°06.3	199°11.0	S01°20.2				
19	216°54.6	148°00.2	41.7	50°08.3	34.9	118°31.4	06.3	214°13.4	20.2				
20	231°57.1	162°59.7	42.2	65°09.4	34.2	133°33.2	06.3	229°15.9	20.2				
21	246°59.6	177°59.2	· · 42.7	80°10.4	· · 33.6	148°35.1	· · 06.2	244°18.4	· · 20.2				
22	262°02.0	192°58.7	43.2	95°11.5	33.0	163°36.9	06.2	259°20.8	20.2				
23	277°04.5	207°58.3	43.7	110°12.6	32.4	178°38.8	06.2	274°23.3	20.2				
Mer.pass. 04:35		ν -0.5' d 0.5' m -4.07			ν 1.1' d -0.6' m 1.55			ν 1.9' d -0.0' m -1.90			ν 2.5' d 0.0' m 0.90		

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	292°07.0	222°57.8	N19°44.2	125°13.7	N06°31.8	193°40.6	N23°06.1	289°25.7	S01°20.2				
1	307°09.4	237°57.3	44.6	140°14.8	31.2	208°42.5	06.1	304°28.2	20.2				
2	322°11.9	252°56.8	45.1	155°15.9	30.6	223°44.3	06.1	319°30.7	20.3				
3	337°14.4	267°56.4	· · 45.6	170°17.0	· · 30.0	238°46.2	· · 06.0	334°33.1	· · 20.3				
4	352°16.8	282°55.9	46.1	185°18.1	29.4	253°48.1	06.0	349°35.6	20.3				
5	7°19.3	297°55.4	46.6	200°19.2	28.8	268°49.9	06.0	4°38.1	20.3				
6	22°21.7	312°54.9	N19°47.0	215°20.3	N06°28.2	283°51.8	N23°06.0	19°40.5	S01°20.3				
7	37°24.2	327°54.4	47.5	230°21.4	27.6	298°53.6	05.9	34°43.0	20.3				
8	52°26.7	342°54.0	48.0	245°22.5	27.0	313°55.5	05.9	49°45.5	20.3				
9	67°29.1	357°53.5	· · 48.5	260°23.6	· · 26.4	328°57.3	· · 05.9	64°47.9	· · 20.3				
10	82°31.6	12°53.0	49.0	275°24.7	25.8	343°59.2	05.8	79°50.4	20.3				
11	97°34.1	27°52.5	49.4	290°25.8	25.2	359°01.0	05.8	94°52.8	20.4				
12	112°36.5	42°52.0	N19°49.9	305°26.9	N06°24.5	14°02.9	N23°05.8	109°55.3	S01°20.4				
13	127°39.0	57°51.6	50.4	320°28.0	23.9	29°04.8	05.8	124°57.8	20.4				
14	142°41.5	72°51.1	50.9	335°29.1	23.3	44°06.6	05.7	140°00.2	20.4				
15	157°43.9	87°50.6	· · 51.3	350°30.2	· · 22.7	59°08.5	· · 05.7	155°02.7	· · 20.4				
16	172°46.4	102°50.1	51.8	5°31.3	22.1	74°10.3	05.7	170°05.2	20.4				
17	187°48.9	117°49.6	52.3	20°32.4	21.5	89°12.2	05.6	185°07.6	20.4				
18	202°51.3	132°49.1	N19°52.7	35°33.5	N06°20.9	104°14.0	N23°05.6	200°10.1	S01°20.5				
19	217°53.8	147°48.7	53.2	50°34.6	20.3	119°15.9	05.6	215°12.6	20.5				
20	232°56.2	162°48.2	53.7	65°35.7	19.7	134°17.7	05.6	230°15.0	20.5				
21	247°58.7	177°47.7	· · 54.1	80°36.8	· · 19.1	149°19.6	· · 05.5	245°17.5	· · 20.5				
22	263°01.2	192°47.2	54.6	95°37.9	18.5	164°21.5	05.5	260°20.0	20.5				
23	278°03.6	207°46.7	55.1	110°38.9	17.9	179°23.3	05.5	275°22.4	20.5				
Mer.pass. 04:31		ν -0.5' d 0.5' m -4.07			ν 1.1' d -0.6' m 1.55			ν 1.9' d -0.0' m -1.90			ν 2.5' d 0.0' m 0.89		

Jul 12 Sat	SHA	Mer.pass
Venus	293°11.3	09:07
Mars	194°12.3	15:41
Jupiter	262°02.9	11:10
Saturn	357°19.0	04:49

Jul 13 Sun	SHA	Mer.pass
Venus	292°01.2	09:08
Mars	193°39.6	15:40
Jupiter	261°48.3	11:07
Saturn	357°18.9	04:45

Jul 14 Mon	SHA	Mer.pass
Venus	290°50.8	09:08
Mars	193°06.8	15:38
Jupiter	261°33.7	11:04
Saturn	357°18.8	04:42

Horizontal parallax	
Venus:	0.1
Mars:	0.1

h	Sun			Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°35.4	N21°57.6	343°17.6	9.6'	S22°54.8	-9.8'	56.6'
1	193°35.3	57.2	357°46.2	9.7'	22°45.0	-9.9'	56.7'
2	208°35.2	56.9	12°14.9	9.7'	22°35.1	-10.0'	56.7'
3	223°35.1	56.5	26°43.7	9.8'	22°25.1	-10.2'	56.7'
4	238°35.1	56.2	41°12.5	9.9'	22°14.9	-10.3'	56.7'
5	253°35.0	55.8	55°41.3	9.9'	22°04.6	-10.4'	56.8'
6	268°34.9	N21°55.5	70°10.3	10.0'	S21°54.2	-10.5'	56.8'
7	283°34.8	55.1	84°39.2	10.0'	21°43.7	-10.7'	56.8'
8	298°34.8	54.8	99°08.3	10.1'	21°33.0	-10.8'	56.8'
9	313°34.7	54.4	113°37.4	10.2'	21°22.3	-10.9'	56.8'
10	328°34.6	54.1	128°06.5	10.2'	21°11.4	-11.0'	56.9'
11	343°34.5	53.7	142°35.7	10.3'	21°00.4	-11.1'	56.9'
12	358°34.5	N21°53.4	157°05.0	10.3'	S20°49.2	-11.2'	56.9'
13	13°34.4	53.0	171°34.3	10.4'	20°38.0	-11.3'	56.9'
14	28°34.3	52.6	186°03.7	10.5'	20°26.7	-11.5'	56.9'
15	43°34.2	52.3	200°33.2	10.5'	20°15.2	-11.6'	57.0'
16	58°34.2	51.9	215°02.7	10.6'	20°03.6	-11.7'	57.0'
17	73°34.1	51.6	229°32.3	10.6'	19°51.9	-11.8'	57.0'
18	88°34.0	N21°51.2	244°01.9	10.7'	S19°40.1	-11.9'	57.0'
19	103°33.9	50.9	258°31.6	10.7'	19°28.2	-12.0'	57.1'
20	118°33.9	50.5	273°01.3	10.8'	19°16.2	-12.1'	57.1'
21	133°33.8	50.1	287°31.1	10.9'	19°04.1	-12.2'	57.1'
22	148°33.7	49.8	302°01.0	10.9'	18°51.9	-12.3'	57.1'
23	163°33.6	49.4	316°30.9	11.0'	18°39.6	-12.4'	57.1'
SD = 15.7'			SD = 15.4'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°33.6	N21°49.0	331°00.9	11.0'	S18°27.2	-12.5'	57.2'
1	193°33.5	48.7	345°30.9	11.1'	18°14.6	-12.6'	57.2'
2	208°33.4	48.3	0°01.0	11.1'	18°02.0	-12.7'	57.2'
3	223°33.3	47.9	14°31.1	11.2'	17°49.3	-12.8'	57.2'
4	238°33.3	47.6	29°01.3	11.3'	17°36.5	-12.9'	57.2'
5	253°33.2	47.2	43°31.6	11.3'	17°23.6	-13.0'	57.3'
6	268°33.1	N21°46.8	58°01.9	11.4'	S17°10.6	-13.1'	57.3'
7	283°33.0	46.5	72°32.3	11.4'	16°57.5	-13.2'	57.3'
8	298°33.0	46.1	87°02.7	11.5'	16°44.3	-13.3'	57.3'
9	313°32.9	45.7	101°33.2	11.5'	16°31.0	-13.4'	57.3'
10	328°32.8	45.4	116°03.7	11.6'	16°17.6	-13.5'	57.4'
11	343°32.8	45.0	130°34.3	11.6'	16°04.2	-13.5'	57.4'
12	358°32.7	N21°44.6	145°04.9	11.7'	S15°50.6	-13.6'	57.4'
13	13°32.6	44.3	159°35.6	11.7'	15°37.0	-13.7'	57.4'
14	28°32.6	43.9	174°06.3	11.8'	15°23.3	-13.8'	57.4'
15	43°32.5	43.5	188°37.1	11.8'	15°09.5	-13.9'	57.5'
16	58°32.4	43.1	203°07.9	11.9'	14°55.6	-14.0'	57.5'
17	73°32.3	42.8	217°38.8	11.9'	14°41.6	-14.0'	57.5'
18	88°32.3	N21°42.4	232°09.8	12.0'	S14°27.5	-14.1'	57.5'
19	103°32.2	42.0	246°40.7	12.0'	14°13.4	-14.2'	57.5'
20	118°32.1	41.6	261°11.8	12.1'	13°59.2	-14.3'	57.6'
21	133°32.1	41.3	275°42.8	12.1'	13°44.9	-14.4'	57.6'
22	148°32.0	40.9	290°13.9	12.2'	13°30.6	-14.4'	57.6'
23	163°31.9	40.5	304°45.1	12.2'	13°16.2	-14.5'	57.6'
SD = 15.7'			SD = 15.6'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	178°31.9	N21°40.1	319°16.3	12.2'	S13°01.6	-14.6'	57.6'
1	193°31.8	39.7	333°47.5	12.3'	12°47.1	-14.6'	57.7'
2	208°31.7	39.4	348°18.8	12.3'	12°32.4	-14.7'	57.7'
3	223°31.7	39.0	2°50.2	12.4'	12°17.7	-14.8'	57.7'
4	238°31.6	38.6	17°21.5	12.4'	12°02.9	-14.8'	57.7'
5	253°31.5	38.2	31°52.9	12.4'	11°48.1	-14.9'	57.7'
6	268°31.5	N21°37.8	46°24.4	12.5'	S11°33.2	-15.0'	57.8'
7	283°31.4	37.5	60°55.9	12.5'	11°18.2	-15.0'	57.8'
8	298°31.3	37.1	75°27.4	12.6'	11°03.2	-15.1'	57.8'
9	313°31.3	36.7	89°58.9	12.6'	10°48.1	-15.2'	57.8'
10	328°31.2	36.3	104°30.5	12.6'	10°32.9	-15.2'	57.8'
11	343°31.1	35.9	119°02.1	12.7'	10°17.7	-15.3'	57.8'
12	358°31.1	N21°35.5	133°33.8	12.7'	S10°02.4	-15.3'	57.9'
13	13°31.0	35.1	148°05.5	12.7'	09°47.1	-15.4'	57.9'
14	28°30.9	34.8	162°37.2	12.7'	09°31.7	-15.4'	57.9'
15	43°30.9	34.4	177°08.9	12.8'	09°16.3	-15.5'	57.9'
16	58°30.8	34.0	191°40.7	12.8'	09°00.8	-15.5'	57.9'
17	73°30.7	33.6	206°12.5	12.8'	08°45.2	-15.6'	58.0'
18	88°30.7	N21°33.2	220°44.3	12.8'	S08°29.6	-15.6'	58.0'
19	103°30.6	32.8	235°16.2	12.9'	08°14.0	-15.7'	58.0'
20	118°30.5	32.4	249°48.0	12.9'	07°58.3	-15.7'	58.0'
21	133°30.5	32.0	264°19.9	12.9'	07°42.6	-15.8'	58.0'
22	148°30.4	31.6	278°51.8	12.9'	07°26.8	-15.8'	58.1'
23	163°30.3	31.2	293°23.8	13.0'	07°10.9	-15.9'	58.1'
SD = 15.7'			SD = 15.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	☐	☐	☐	☐	☐	☐
64°	☐	☐	☐	☐	☐	☐
62°	☐	☐	☐	☐	☐	☐
60°	☐	☐	☐	☐	☐	☐
N 58°	☐	☐	☐	☐	☐	☐
56°	☐	☐	☐	☐	☐	☐
54°	☐	☐	☐	☐	☐	☐
52°	☐	☐	☐	☐	☐	☐
50°	☐	☐	☐	☐	☐	☐
45°	☐	☐	☐	☐	☐	☐
N 40°	☐	☐	☐	☐	☐	☐
35°	☐	☐	☐	☐	☐	☐
30°	☐	☐	☐	☐	☐	☐
20°	☐	☐	☐	☐	☐	☐
N 10°	☐	☐	☐	☐	☐	☐
0°	☐	☐	☐	☐	☐	☐
S 10°	☐	☐	☐	☐	☐	☐
20°	☐	☐	☐	☐	☐	☐
30°	☐	☐	☐	☐	☐	☐
35°	☐	☐	☐	☐	☐	☐
40°	☐	☐	☐	☐	☐	☐
45°	☐	☐	☐	☐	☐	☐
S 50°	☐	☐	☐	☐	☐	☐
52°	☐	☐	☐	☐	☐	☐
54°	☐	☐	☐	☐	☐	☐
56°	☐	☐	☐	☐	☐	☐
58°	☐	☐	☐	☐	☐	☐
S 60°	☐	☐	☐	☐	☐	☐

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	☐	☐	☐	☐	☐	☐
64°	☐	☐	☐	☐	☐	☐
62°	☐	☐	☐	☐	☐	☐
60°	☐	☐	☐	☐	☐	☐
N 58°	☐	☐	☐	☐	☐	☐
56°	☐	☐	☐	☐	☐	☐
54°	☐	☐	☐	☐	☐	☐
52°	☐	☐	☐	☐	☐	☐
50°	☐	☐	☐	☐	☐	☐
45°	☐	☐	☐	☐	☐	☐
N 40°	☐	☐	☐	☐	☐	☐
35°	☐	☐	☐	☐	☐	☐
30°	☐	☐	☐	☐	☐	☐
20°	☐	☐	☐	☐	☐	☐
N 10°	☐	☐	☐	☐	☐	☐
0°	☐	☐	☐	☐	☐	☐
S 10°	☐	☐	☐	☐	☐	☐
20°	☐	☐	☐	☐	☐	☐
30°	☐	☐	☐	☐	☐	☐
35°	☐	☐	☐	☐	☐	☐
40°	☐	☐	☐	☐	☐	☐
45°	☐	☐	☐	☐	☐	☐
S 50°	☐	☐	☐	☐	☐	☐
52°	☐	☐	☐	☐	☐	☐
54°	☐	☐	☐	☐	☐	☐
56°	☐	☐	☐	☐	☐	☐
58°	☐	☐	☐	☐	☐	☐
S 60°	☐	☐	☐	☐	☐	☐

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
12	05:38	05:42	12:06	01:09	13:35	
13	05:46	05:49	12:06	02:00	14:24	
14	05:53	05:56	12:06	02:48	15:12	

July 15, 16, 17 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	293°06.1	222°46.2	N19°55.5	125°40.0	N06°17.3	194°25.2	N23°05.4	290°24.9	S01°20.5	Alpheratz	357°34.2	29°13.8
1	308°08.6	237°45.7	56.0	140°41.1	16.6	209°27.0	05.4	305°27.4	20.5	Ankaa	353°06.6	-42°09.7
2	323°11.0	252°45.2	56.5	155°42.2	16.0	224°28.9	05.4	320°29.8	20.6	Schedar	349°30.4	56°40.4
3	338°13.5	267°44.7	· · 56.9	170°43.3	· · 15.4	239°30.7	· · 05.3	335°32.3	· · 20.6	Diphda	348°46.8	-17°50.6
4	353°16.0	282°44.3	57.4	185°44.4	14.8	254°32.6	05.3	350°34.8	20.6	Achernar	335°19.9	-57°06.1
5	8°18.4	297°43.8	57.9	200°45.5	14.2	269°34.5	05.3	5°37.2	20.6	Hamal	327°50.8	23°34.9
6	23°20.9	312°43.3	N19°58.3	215°46.6	N06°13.6	284°36.3	N23°05.3	20°39.7	S01°20.6	Polaris	314°01.1	89°22.0
7	38°23.4	327°42.8	58.8	230°47.7	13.0	299°38.2	05.2	35°42.2	20.6	Acamar	315°11.6	-40°11.8
8	53°25.8	342°42.3	59.2	245°48.8	12.4	314°40.0	05.2	50°44.6	20.6	Menkar	314°05.8	4°11.5
9	68°28.3	357°41.8	19°59.7	260°49.9	· · 11.8	329°41.9	· · 05.2	65°47.1	· · 20.7	Mirfak	308°27.9	49°56.9
10	83°30.7	12°41.3	20°00.2	275°51.0	11.2	344°43.7	05.1	80°49.6	20.7	Aldebaran	290°39.4	16°33.6
11	98°33.2	27°40.8	00.6	290°52.1	10.6	359°45.6	05.1	95°52.0	20.7	Rigel	281°03.7	-8°10.2
12	113°35.7	42°40.3	N20°01.1	305°53.2	N06°10.0	14°47.5	N23°05.1	110°54.5	S01°20.7	Capella	280°21.6	46°01.3
13	128°38.1	57°39.8	01.5	320°54.3	09.3	29°49.3	05.0	125°57.0	20.7	Bellatrix	278°22.7	6°22.4
14	143°40.6	72°39.3	02.0	335°55.4	08.7	44°51.2	05.0	140°59.4	20.7	Elnath	278°01.6	28°37.7
15	158°43.1	87°38.8	· · 02.4	350°56.5	· · 08.1	59°53.0	· · 05.0	156°01.9	· · 20.7	Alnilam	275°37.6	-1°11.1
16	173°45.5	102°38.3	02.9	5°57.6	07.5	74°54.9	05.0	171°04.4	20.8	Betelgeuse	270°51.9	7°24.8
17	188°48.0	117°37.8	03.3	20°58.7	06.9	89°56.7	04.9	186°06.9	20.8	Canopus	263°52.7	-52°42.4
18	203°50.5	132°37.3	N20°03.8	35°59.7	N06°06.3	104°58.6	N23°04.9	201°09.3	S01°20.8	Sirius	258°26.2	-16°45.0
19	218°52.9	147°36.8	04.3	51°00.8	05.7	120°00.5	04.9	216°11.8	20.8	Adhara	255°05.9	-29°00.3
20	233°55.4	162°36.3	04.7	66°01.9	05.1	135°02.3	04.8	231°14.3	20.8	Procyon	244°50.7	5°09.6
21	248°57.8	177°35.8	· · 05.2	81°03.0	· · 04.5	150°04.2	· · 04.8	246°16.7	· · 20.8	Pollux	243°17.2	27°57.9
22	264°00.3	192°35.3	05.6	96°04.1	03.9	165°06.0	04.8	261°19.2	20.8	Avior	234°15.2	-59°35.5
23	279°02.8	207°34.8	06.0	111°05.2	03.2	180°07.9	04.7	276°21.7	20.9	Suhail	222°46.4	-43°32.2
Mer.pass. 04:27		ν -0.5' d 0.5' m -4.07		ν 1.1' d -0.6' m 1.55		ν 1.9' d -0.0' m -1.90		ν 2.5' d 0.0' m 0.89				

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	294°05.2	222°34.3	N20°06.5	126°06.3	N06°02.6	195°09.7	N23°04.7	291°24.1	S01°20.9	Alphard	217°47.6	-8°46.1
1	309°07.7	237°33.8	06.9	141°07.4	02.0	210°11.6	04.7	306°26.6	20.9	Regulus	207°34.2	11°50.6
2	324°10.2	252°33.3	07.4	156°08.5	01.4	225°13.5	04.6	321°29.1	20.9	Dubhe	193°40.9	61°37.0
3	339°12.6	267°32.8	· · 07.8	171°09.6	· · 00.8	240°15.3	· · 04.6	336°31.6	· · 20.9	Denebola	182°24.6	14°25.9
4	354°15.1	282°32.3	08.3	186°10.7	06°00.2	255°17.2	04.6	351°34.0	20.9	Gienah	175°43.3	-17°41.1
5	9°17.6	297°31.8	08.7	201°11.8	05°59.6	270°19.0	04.6	6°36.5	20.9	Acrux	172°59.8	-63°14.7
6	24°20.0	312°31.3	N20°09.2	216°12.9	N05°59.0	285°20.9	N23°04.5	21°39.0	S01°21.0	Gacrux	171°51.3	-57°15.6
7	39°22.5	327°30.8	09.6	231°14.0	58.4	300°22.8	04.5	36°41.4	21.0	Alioth	166°12.6	55°49.5
8	54°25.0	342°30.3	10.0	246°15.1	57.7	315°24.6	04.5	51°43.9	21.0	Spica	158°21.8	-11°17.7
9	69°27.4	357°29.8	· · 10.5	261°16.2	· · 57.1	330°26.5	· · 04.4	66°46.4	· · 21.0	Alkaid	152°51.6	49°11.4
10	84°29.9	12°29.3	10.9	276°17.2	56.5	345°28.3	04.4	81°48.9	21.0	Hadar	148°35.3	-60°30.0
11	99°32.3	27°28.8	11.4	291°18.3	55.9	0°30.2	04.4	96°51.3	21.0	Menkent	147°57.0	-36°29.9
12	114°34.8	42°28.3	N20°11.8	306°19.4	N05°55.3	15°32.0	N23°04.3	111°53.8	S01°21.1	Arcturus	145°47.4	19°03.1
13	129°37.3	57°27.8	12.2	321°20.5	54.7	30°33.9	04.3	126°56.3	21.1	Rigel Kent.	139°39.5	-60°56.7
14	144°39.7	72°27.3	12.7	336°21.6	54.1	45°35.8	04.3	141°58.7	21.1	Kochab	137°19.3	74°03.3
15	159°42.2	87°26.7	· · 13.1	351°22.7	· · 53.5	60°37.6	· · 04.2	157°01.2	· · 21.1	Zuben'ubi	136°55.4	-16°08.9
16	174°44.7	102°26.2	13.5	6°23.8	52.9	75°39.5	04.2	172°03.7	21.1	Alphecca	126°03.2	26°37.9
17	189°47.1	117°25.7	14.0	21°24.9	52.2	90°41.3	04.2	187°06.2	21.1	Antares	112°15.0	-26°29.4
18	204°49.6	132°25.2	N20°14.4	36°26.0	N05°51.6	105°43.2	N23°04.2	202°08.6	S01°21.2	Atria	107°08.4	-69°04.6
19	219°52.1	147°24.7	14.8	51°27.1	51.0	120°45.1	04.1	217°11.1	21.2	Sabik	102°02.0	-15°45.4
20	234°54.5	162°24.2	15.3	66°28.2	50.4	135°46.9	04.1	232°13.6	21.2	Shaula	96°09.4	-37°07.4
21	249°57.0	177°23.7	· · 15.7	81°29.3	· · 49.8	150°48.8	· · 04.1	247°16.1	· · 21.2	Rasalhague	95°57.8	12°32.5
22	264°59.4	192°23.2	16.1	96°30.4	49.2	165°50.6	04.0	262°18.5	21.2	Eltanin	90°41.5	51°29.2
23	280°01.9	207°22.6	16.6	111°31.4	48.6	180°52.5	04.0	277°21.0	21.2	Kaus Aust.	83°31.5	-34°22.3
Mer.pass. 04:23		ν -0.5' d 0.4' m -4.06		ν 1.1' d -0.6' m 1.56		ν 1.9' d -0.0' m -1.90		ν 2.5' d 0.0' m 0.88				

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	295°04.4	222°22.1	N20°17.0	126°32.5	N05°48.0	195°54.3	N23°04.0	292°23.5	S01°21.3	Nunki	75°46.8	-26°15.9
1	310°06.8	237°21.6	17.4	141°33.6	47.3	210°56.2	03.9	307°26.0	21.3	Altair	61°59.2	8°56.2
2	325°09.3	252°21.1	17.9	156°34.7	46.7	225°58.1	03.9	322°28.4	21.3	Peacock	53°04.4	-56°39.1
3	340°11.8	267°20.6	· · 18.3	171°35.8	· · 46.1	240°59.9	· · 03.9	337°30.9	· · 21.3	Deneb	49°25.1	45°22.2
4	355°14.2	282°20.1	18.7	186°36.9	45.5	256°01.8	03.8	352°33.4	21.3	Enif	33°38.1	9°59.5
5	10°16.7	297°19.6	19.1	201°38.0	44.9	271°03.6	03.8	7°35.9	21.3	Al Na'ir	27°32.0	-46°50.0
6	25°19.2	312°19.0	N20°19.6	216°39.1	N05°44.3	286°05.5	N23°03.8	22°38.3	S01°21.4	Fomalhaut	15°13.8	-29°29.0
7	40°21.6	327°18.5	20.0	231°40.2	43.7	301°07.4	03.7	37°40.8	21.4	Scheat	13°44.6	28°13.2
8	55°24.1	342°18.0	20.4	246°41.3	43.0	316°09.2	03.7	52°43.3	21.4	Markab	13°29.3	15°20.6
9	70°26.6	357°17.5	· · 20.8	261°42.4	· · 42.4	331°11.1	· · 03.7	67°45.8	· · 21.4			
10	85°29.0	12°17.0	21.2	276°43.5	41.8	346°12.9	03.6	82°48.2	21.4	Jul 15 Tue	SHA	Mer.pass
11	100°31.5	27°16.5	21.7	291°44.5	41.2	1°14.8	03.6	97°50.7	21.4	Venus	289°40.1	09:09
12	115°33.9	42°15.9	N20°22.1	306°45.6	N05°40.6	16°16.7	N23°03.6	112°53.2	S01°21.5	Mars	192°33.9	15:36
13	130°36.4	57°15.4	22.5	321°46.7	40.0	31°18.5	03.6	127°55.7	21.5	Jupiter	261°19.1	11:01
14	145°38.9	72°14.9	22.9	336°47.8	39.4	46°20.4	03.5	142°58.1	21.5	Saturn	357°18.8	04:38
15	160°41.3	87°14.4	· · 23.3	351°48.9	· · 38.8	61°22.2	· · 03.5	158°00.6	· · 21.5			
16	175°43.8	102°13.8	23.8	6°50.0	38.1	76°24.1	03.5	173°03.1	21.5	Jul 16 Wed	SHA	Mer.pass
17	190°46.3	117°13.3	24.2	21°51.1	37.5	91°26.0	03.4	188°05.6	21.5	Venus	288°29.1	09:10
18	205°48.7	132°12.8	N20°24.6	36°52.2	N05°36.9	106°27.8	N23°03.4	203°08.0	S01°21.6	Mars	192°01.1	15:34
19	220°51.2	147°12.3	25.0	51°53.3	36.3	121°29.7	03.4	218°10.5	21.6	Jupiter	260°50.0	10:55
20	235°53.7	162°11.8	25.4	66°54.4	35.7	136°31.5	03.3	233°13.0	21.6	Saturn	357°19.1	04:30
21	250°56.1	177°11.2	· · 25.8	81°55.5								

h	Sun			Moon			
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	178°30.3	N21°30.8	307°55.7	13.0'	S06°55.1	-15.9'	58.1'
1	193°30.2	30.4	322°27.7	13.0'	06°39.2	-15.9'	58.1'
2	208°30.2	30.0	336°59.7	13.0'	06°23.2	-16.0'	58.1'
3	223°30.1	· · 29.7	351°31.7	13.0'	06°07.2	-16.0'	58.1'
4	238°30.0	29.3	6°03.7	13.0'	05°51.2	-16.1'	58.2'
5	253°30.0	28.9	20°35.8	13.0'	05°35.1	-16.1'	58.2'
6	268°29.9	N21°28.5	35°07.8	13.1'	S05°19.0	-16.1'	58.2'
7	283°29.8	28.1	49°39.9	13.1'	05°02.9	-16.2'	58.2'
8	298°29.8	27.7	64°11.9	13.1'	04°46.8	-16.2'	58.2'
9	313°29.7	· · 27.3	78°44.0	13.1'	04°30.6	-16.2'	58.2'
10	328°29.7	26.9	93°16.1	13.1'	04°14.3	-16.2'	58.3'
11	343°29.6	26.5	107°48.2	13.1'	03°58.1	-16.3'	58.3'
12	358°29.5	N21°26.1	122°20.3	13.1'	S03°41.8	-16.3'	58.3'
13	13°29.5	25.7	136°52.3	13.1'	03°25.5	-16.3'	58.3'
14	28°29.4	25.3	151°24.4	13.1'	03°09.2	-16.3'	58.3'
15	43°29.4	· · 24.8	165°56.5	13.1'	02°52.8	-16.4'	58.3'
16	58°29.3	24.4	180°28.6	13.1'	02°36.5	-16.4'	58.4'
17	73°29.2	24.0	195°00.7	13.1'	02°20.1	-16.4'	58.4'
18	88°29.2	N21°23.6	209°32.8	13.1'	S02°03.7	-16.4'	58.4'
19	103°29.1	23.2	224°04.9	13.1'	01°47.2	-16.4'	58.4'
20	118°29.1	22.8	238°37.0	13.1'	01°30.8	-16.5'	58.4'
21	133°29.0	· · 22.4	253°09.1	13.1'	01°14.3	-16.5'	58.4'
22	148°28.9	22.0	267°41.1	13.1'	00°57.9	-16.5'	58.5'
23	163°28.9	21.6	282°13.2	13.0'	00°41.4	-16.5'	58.5'
SD = 15.7' $d = -0.4'$			SD = 15.8'				
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	178°28.8	N21°21.2	296°45.2	13.0'	S00°24.9	-16.5'	58.5'
1	193°28.8	20.8	311°17.3	13.0'	S00°08.4	-16.5'	58.5'
2	208°28.7	20.4	325°49.3	13.0'	N00°08.2	16.5'	58.5'
3	223°28.6	· · 20.0	340°21.3	13.0'	00°24.7	16.5'	58.5'
4	238°28.6	19.5	354°53.3	13.0'	00°41.2	16.5'	58.6'
5	253°28.5	19.1	9°25.2	12.9'	00°57.8	16.5'	58.6'
6	268°28.5	N21°18.7	23°57.2	12.9'	N01°14.3	16.5'	58.6'
7	283°28.4	18.3	38°29.1	12.9'	01°30.8	16.5'	58.6'
8	298°28.4	17.9	53°01.0	12.9'	01°47.4	16.5'	58.6'
9	313°28.3	· · 17.5	67°32.9	12.9'	02°03.9	16.5'	58.6'
10	328°28.2	17.1	82°04.8	12.8'	02°20.5	16.5'	58.6'
11	343°28.2	16.6	96°36.6	12.8'	02°37.0	16.5'	58.7'
12	358°28.1	N21°16.2	111°08.4	12.8'	N02°53.5	16.5'	58.7'
13	13°28.1	15.8	125°40.2	12.7'	03°10.1	16.5'	58.7'
14	28°28.0	15.4	140°11.9	12.7'	03°26.6	16.5'	58.7'
15	43°28.0	· · 15.0	154°43.6	12.7'	03°43.1	16.5'	58.7'
16	58°27.9	14.5	169°15.3	12.6'	03°59.6	16.5'	58.7'
17	73°27.9	14.1	183°47.0	12.6'	04°16.1	16.5'	58.7'
18	88°27.8	N21°13.7	198°18.6	12.6'	N04°32.5	16.5'	58.8'
19	103°27.8	13.3	212°50.2	12.5'	04°49.0	16.4'	58.8'
20	118°27.7	12.9	227°21.7	12.5'	05°05.4	16.4'	58.8'
21	133°27.6	· · 12.4	241°53.2	12.5'	05°21.9	16.4'	58.8'
22	148°27.6	12.0	256°24.6	12.4'	05°38.3	16.4'	58.8'
23	163°27.5	11.6	270°56.0	12.4'	05°54.7	16.4'	58.8'
SD = 15.7' $d = -0.4'$			SD = 16.0'				
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	178°27.5	N21°11.2	285°27.4	12.3'	N06°11.0	16.3'	58.8'
1	193°27.4	10.7	299°58.7	12.3'	06°27.4	16.3'	58.9'
2	208°27.4	10.3	314°30.0	12.2'	06°43.7	16.3'	58.9'
3	223°27.3	· · 09.9	329°01.2	12.2'	07°00.0	16.3'	58.9'
4	238°27.3	09.5	343°32.4	12.1'	07°16.2	16.2'	58.9'
5	253°27.2	09.0	358°03.5	12.1'	07°32.5	16.2'	58.9'
6	268°27.2	N21°08.6	12°34.6	12.0'	N07°48.7	16.2'	58.9'
7	283°27.1	08.2	27°05.6	12.0'	08°04.9	16.1'	58.9'
8	298°27.1	07.7	41°36.6	11.9'	08°21.0	16.1'	58.9'
9	313°27.0	· · 07.3	56°07.5	11.9'	08°37.1	16.1'	59.0'
10	328°27.0	06.9	70°38.4	11.8'	08°53.2	16.0'	59.0'
11	343°26.9	06.5	85°09.2	11.7'	09°09.2	16.0'	59.0'
12	358°26.9	N21°06.0	99°39.9	11.7'	N09°25.2	16.0'	59.0'
13	13°26.8	05.6	114°10.6	11.6'	09°41.2	15.9'	59.0'
14	28°26.8	05.2	128°41.2	11.5'	09°57.1	15.9'	59.0'
15	43°26.7	· · 04.7	143°11.7	11.5'	10°12.9	15.8'	59.0'
16	58°26.7	04.3	157°42.2	11.4'	10°28.7	15.8'	59.0'
17	73°26.6	03.9	172°12.6	11.3'	10°44.5	15.7'	59.1'
18	88°26.6	N21°03.4	186°43.0	11.3'	N11°00.2	15.7'	59.1'
19	103°26.5	03.0	201°13.2	11.2'	11°15.9	15.6'	59.1'
20	118°26.5	02.5	215°43.4	11.1'	11°31.5	15.6'	59.1'
21	133°26.4	· · 02.1	230°13.6	11.1'	11°47.1	15.5'	59.1'
22	148°26.4	01.7	244°43.6	11.0'	12°02.6	15.5'	59.1'
23	163°26.3	01.2	259°13.6	10.9'	12°18.1	15.4'	59.1'
SD = 15.7' $d = -0.4'$			SD = 16.0'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	--	--	01:41	22:28	--	--
64°	--	--	02:19	21:51	--	--
62°	--	00:58	02:45	21:25	23:08	--
60°	--	01:49	03:06	21:05	22:21	--
N 58°	--	02:19	03:23	20:48	21:51	--
56°	00:54	02:41	03:37	20:35	21:30	23:13
54°	01:40	02:59	03:49	20:23	21:12	22:30
52°	02:08	03:14	03:59	20:12	20:57	22:03
50°	02:29	03:27	04:09	20:03	20:44	21:42
45°	03:06	03:53	04:29	19:43	20:19	21:05
N 40°	03:33	04:13	04:45	19:27	19:59	20:38
35°	03:54	04:29	04:58	19:14	19:43	20:18
30°	04:11	04:43	05:10	19:02	19:29	20:01
20°	04:37	05:05	05:30	18:42	19:07	19:35
N 10°	04:57	05:24	05:47	18:25	18:48	19:15
0°	05:15	05:40	06:03	18:10	18:32	18:58
S 10°	05:30	05:56	06:18	17:54	18:17	18:43
20°	05:44	06:11	06:35	17:37	18:01	18:28
30°	05:59	06:28	06:54	17:19	17:44	18:14
35°	06:06	06:37	07:05	17:08	17:35	18:06
40°	06:14	06:48	07:18	16:55	17:25	17:58
45°	06:23	07:00	07:32	16:40	17:13	17:49
S 50°	06:33	07:14	07:51	16:22	16:59	17:39
52°	06:38	07:20	07:59	16:13	16:53	17:35
54°	06:42	07:27	08:09	16:04	16:46	17:30
56°	06:47	07:35	08:20	15:53	16:38	17:25
58°	06:53	07:44	08:32	15:41	16:29	17:20
S 60°	06:59	07:53	08:46	15:26	16:19	17:14

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	22:22	21:46	21:02	08:46	11:04	13:31
N 70°	22:21	21:53	21:21	08:53	11:01	13:14
68°	22:20	21:59	21:36	08:58	10:58	13:02
66°	22:19	22:05	21:48	09:03	10:55	12:51
64°	22:18	22:09	21:59	09:07	10:53	12:43
62°	22:18	22:13	22:08	09:10	10:51	12:35
60°	22:17	22:16	22:16	09:13	10:50	12:29
N 58°	22:17	22:19	22:23	09:16	10:49	12:23
56°	22:16	22:22	22:29	09:18	10:47	12:18
54°	22:16	22:24	22:34	09:20	10:46	12:14
52°	22:15	22:26	22:39	09:22	10:45	12:10
50°	22:15	22:29	22:44	09:24	10:44	12:06
45°	22:14	22:33	22:54	09:28	10:42	11:59
N 40°	22:13	22:37	23:02	09:31	10:41	11:52
35°	22:13	22:40	23:09	09:33	10:39	11:47
30°	22:12	22:43	23:15	09:35	10:38	11:42
20°	22:12	22:48	23:26	09:39	10:36	11:34
N 10°	22:11	22:52	23:36	09:43	10:34	11:26
0°	22:10	22:57	23:45	09:46	10:32	11:19
S 10°	22:09	23:01	23:55	09:49	10:30	11:13
20°	22:09	23:06	· · ·	09:52	10:28	11:05
30°	22:08	23:11	· · ·	09:56	10:26	10:57
35°	22:08	23:14	· · ·	09:58	10:25	10:53
40°	22:07	23:18	· · ·	10:00	10:23	10:47
45°	22:07	23:22	· · ·	10:03	10:22	10:41
S 50°	22:06	23:27	· · ·	10:06	10:20	10:34
52°	22:06	23:29	· · ·	10:08	10:19	10:31
54°	22:05	23:32	· · ·	10:09	10:18	10:27
56°	22:05	23:35	· · ·	10:11	10:17	10:23
58°	22:04	23:38	· · ·	10:13	10:16	10:18
S 60°	22:04	23:41	· · ·	10:15	10:14	10:13

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss	hh:mm	Mer. Pass. Upper hh:mm	Lower hh:mm	20-22 82-62%
15	05:59	06:02	12:06	03:35	15:58	
16	06:05	06:07	12:06	04:21	16:44	
17	06:10	06:13	12:06	05:08	17:32	

July 18, 19, 20 UT (Fri., Sat., Sun.)

Aries		Venus		Mars		Jupiter		Saturn	
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	296°03.5	222°09.7	N20°27.1	126°58.7	N05°33.2	196°39.0	N23°03.2	293°22.9	S01°21.7
1	311°06.0	237°09.1	27.5	141°59.8	32.6	211°40.8	03.2	308°25.4	21.7
2	326°08.4	252°08.6	27.9	157°00.9	32.0	226°42.7	03.1	323°27.9	21.7
3	341°10.9	267°08.1	· · 28.3	172°02.0	· · 31.4	241°44.6	· · 03.1	338°30.4	· · 21.7
4	356°13.4	282°07.5	28.7	187°03.1	30.8	256°46.4	03.1	353°32.8	21.8
5	11°15.8	297°07.0	29.1	202°04.2	30.1	271°48.3	03.0	8°35.3	21.8
6	26°18.3	312°06.5	N20°29.5	217°05.3	N05°29.5	286°50.1	N23°03.0	23°37.8	S01°21.8
7	41°20.8	327°06.0	29.9	232°06.4	28.9	301°52.0	03.0	38°40.3	21.8
8	56°23.2	342°05.4	30.3	247°07.4	28.3	316°53.9	02.9	53°42.7	21.8
9	71°25.7	357°04.9	· · 30.7	262°08.5	· · 27.7	331°55.7	· · 02.9	68°45.2	· · 21.8
10	86°28.2	12°04.4	31.1	277°09.6	27.1	346°57.6	02.9	83°47.7	21.9
11	101°30.6	27°03.8	31.5	292°10.7	26.5	1°59.4	02.8	98°50.2	21.9
12	116°33.1	42°03.3	N20°31.9	307°11.8	N05°25.8	17°01.3	N23°02.8	113°52.7	S01°21.9
13	131°35.5	57°02.8	32.3	322°12.9	25.2	32°03.2	02.8	128°55.1	21.9
14	146°38.0	72°02.2	32.7	337°14.0	24.6	47°05.0	02.7	143°57.6	21.9
15	161°40.5	87°01.7	· · 33.1	352°15.1	· · 24.0	62°06.9	· · 02.7	159°00.1	· · 22.0
16	176°42.9	102°01.2	33.5	7°16.2	23.4	77°08.8	02.7	174°02.6	22.0
17	191°45.4	117°00.6	33.9	22°17.2	22.8	92°10.6	02.6	189°05.1	22.0
18	206°47.9	132°00.1	N20°34.3	37°18.3	N05°22.1	107°12.5	N23°02.6	204°07.6	S01°22.0
19	221°50.3	146°59.6	34.7	52°19.4	21.5	122°14.3	02.6	219°10.0	22.0
20	236°52.8	161°59.0	35.1	67°20.5	20.9	137°16.2	02.5	234°12.5	22.1
21	251°55.3	176°58.5	· · 35.5	82°21.6	· · 20.3	152°18.1	· · 02.5	249°15.0	· · 22.1
22	266°57.7	191°58.0	35.9	97°22.7	19.7	167°19.9	02.5	264°17.5	22.1
23	282°00.2	206°57.4	36.3	112°23.8	19.1	182°21.8	02.4	279°20.0	22.1
Mer.pass. 04:15		ν -0.5' d 0.4' m -4.05		ν 1.1' d -0.6' m 1.56		ν 1.9' d -0.0' m -1.90		ν 2.5' d 0.0' m 0.87	

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	297°02.7	221°56.9	N20°36.6	127°24.9	N05°18.4	197°23.6	N23°02.4	294°22.4	S01°22.1
1	312°05.1	236°56.3	37.0	142°26.0	17.8	212°25.5	02.4	309°24.9	22.2
2	327°07.6	251°55.8	37.4	157°27.0	17.2	227°27.4	02.3	324°27.4	22.2
3	342°10.0	266°55.3	· · 37.8	172°28.1	· · 16.6	242°29.2	· · 02.3	339°29.9	· · 22.2
4	357°12.5	281°54.7	38.2	187°29.2	16.0	257°31.1	02.3	354°32.4	22.2
5	12°15.0	296°54.2	38.6	202°30.3	15.4	272°33.0	02.2	9°34.9	22.2
6	27°17.4	311°53.7	N20°39.0	217°31.4	N05°14.7	287°34.8	N23°02.2	24°37.3	S01°22.3
7	42°19.9	326°53.1	39.4	232°32.5	14.1	302°36.7	02.2	39°39.8	22.3
8	57°22.4	341°52.6	39.7	247°33.6	13.5	317°38.5	02.1	54°42.3	22.3
9	72°24.8	356°52.0	· · 40.1	262°34.7	· · 12.9	332°40.4	· · 02.1	69°44.8	· · 22.3
10	87°27.3	11°51.5	40.5	277°35.7	12.3	347°42.3	02.1	84°47.3	22.3
11	102°29.8	26°50.9	40.9	292°36.8	11.6	2°44.1	02.0	99°49.8	22.4
12	117°32.2	41°50.4	N20°41.3	307°37.9	N05°11.0	17°46.0	N23°02.0	114°52.2	S01°22.4
13	132°34.7	56°49.9	41.6	322°39.0	10.4	32°47.9	02.0	129°54.7	22.4
14	147°37.2	71°49.3	42.0	337°40.1	09.8	47°49.7	01.9	144°57.2	22.4
15	162°39.6	86°48.8	· · 42.4	352°41.2	· · 09.2	62°51.6	· · 01.9	159°59.7	· · 22.4
16	177°42.1	101°48.2	42.8	7°42.3	08.6	77°53.4	01.9	175°02.2	22.5
17	192°44.5	116°47.7	43.2	22°43.4	07.9	92°55.3	01.8	190°04.7	22.5
18	207°47.0	131°47.1	N20°43.5	37°44.4	N05°07.3	107°57.2	N23°01.8	205°07.1	S01°22.5
19	222°49.5	146°46.6	43.9	52°45.5	06.7	122°59.0	01.8	220°09.6	22.5
20	237°51.9	161°46.0	44.3	67°46.6	06.1	138°00.9	01.7	235°12.1	22.6
21	252°54.4	176°45.5	· · 44.6	82°47.7	· · 05.5	153°02.8	· · 01.7	250°14.6	· · 22.6
22	267°56.9	191°44.9	45.0	97°48.8	04.8	168°04.6	01.7	265°17.1	22.6
23	282°59.3	206°44.4	45.4	112°49.9	04.2	183°06.5	01.6	280°19.6	22.6
Mer.pass. 04:11		ν -0.5' d 0.4' m -4.05		ν 1.1' d -0.6' m 1.57		ν 1.9' d -0.0' m -1.90		ν 2.5' d 0.0' m 0.87	

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	298°01.8	221°43.8	N20°45.8	127°51.0	N05°03.6	198°08.3	N23°01.6	295°22.1	S01°22.6
1	313°04.3	236°43.3	46.1	142°52.1	03.0	213°10.2	01.6	310°24.5	22.7
2	328°06.7	251°42.7	46.5	157°53.1	02.4	228°12.1	01.5	325°27.0	22.7
3	343°09.2	266°42.2	· · 46.9	172°54.2	· · 01.7	243°13.9	· · 01.5	340°29.5	· · 22.7
4	358°11.7	281°41.6	47.2	187°55.3	01.1	258°15.8	01.5	355°32.0	22.7
5	13°14.1	296°41.1	47.6	202°56.4	05°00.5	273°17.7	01.4	10°34.5	22.7
6	28°16.6	311°40.5	N20°48.0	217°57.5	N04°59.9	288°19.5	N23°01.4	25°37.0	S01°22.8
7	43°19.0	326°40.0	48.3	232°58.6	59.3	303°21.4	01.4	40°39.5	22.8
8	58°21.5	341°39.4	48.7	247°59.7	58.6	318°23.3	01.3	55°42.0	22.8
9	73°24.0	356°38.9	· · 49.1	263°00.7	· · 58.0	333°25.1	· · 01.3	70°44.4	· · 22.8
10	88°26.4	11°38.3	49.4	278°01.8	57.4	348°27.0	01.3	85°46.9	22.9
11	103°28.9	26°37.8	49.8	293°02.9	56.8	3°28.8	01.2	100°49.4	22.9
12	118°31.4	41°37.2	N20°50.1	308°04.0	N04°56.2	18°30.7	N23°01.2	115°51.9	S01°22.9
13	133°33.8	56°36.7	50.5	323°05.1	55.5	33°32.6	01.2	130°54.4	22.9
14	148°36.3	71°36.1	50.9	338°06.2	54.9	48°34.4	01.1	145°56.9	22.9
15	163°38.8	86°35.6	· · 51.2	353°07.3	· · 54.3	63°36.3	· · 01.1	160°59.4	· · 23.0
16	178°41.2	101°35.0	51.6	8°08.3	53.7	78°38.2	01.1	176°01.9	23.0
17	193°43.7	116°34.4	51.9	23°09.4	53.1	93°40.0	01.0	191°04.3	23.0
18	208°46.1	131°33.9	N20°52.3	38°10.5	N04°52.4	108°41.9	N23°01.0	206°06.8	S01°23.0
19	223°48.6	146°33.3	52.6	53°11.6	51.8	123°43.8	01.0	221°09.3	23.1
20	238°51.1	161°32.8	53.0	68°12.7	51.2	138°45.6	00.9	236°11.8	23.1
21	253°53.5	176°32.2	· · 53.3	83°13.8	· · 50.6	153°47.5	· · 00.9	251°14.3	· · 23.1
22	268°56.0	191°31.6	53.7	98°14.9	50.0	168°49.4	00.8	266°16.8	23.1
23	283°58.5	206°31.1	54.0	113°15.9	49.3	183°51.2	00.8	281°19.3	23.2
Mer.pass. 04:07		ν -0.5' d 0.4' m -4.04		ν 1.1' d -0.6' m 1.57		ν 1.9' d -0.0' m -1.90		ν 2.5' d 0.0' m 0.86	

Stars		
	SHA	Dec
Alpheratz	357°34.2	29°13.8
Ankaa	353°06.6	-42°09.7
Schedar	349°30.4	56°40.4
Diphda	348°46.8	-17°50.6
Achernar	335°19.8	-57°06.1
Hamal	327°50.7	23°34.9
Polaris	313°59.5	89°22.0
Acamar	315°11.6	-40°11.8
Menkar	314°05.8	4°11.5
Mirfak	308°27.9	49°56.9
Aldebaran	290°39.4	16°33.6
Rigel	281°03.7	-8°10.2
Capella	280°21.6	46°01.3
Bellatrix	278°22.7	6°22.4
Elnath	278°01.6	28°37.7
Alnilam	275°37.6	-1°11.0
Betelgeuse	270°51.9	7°24.8
Canopus	263°52.7	-52°42.4
Sirius	258°26.2	-16°45.0
Adhara	255°05.9	-29°00.3
Procyon	244°50.7	5°09.6
Pollux	243°17.1	27°57.9
Avior	234°15.2	-57°35.5
Suhail	222°46.4	-43°32.2
Miaplacidus	221°39.1	-69°49.4
Alphard	217°47.6	-8°46.1
Regulus	207°34.2	11°50.6
Dubhe	193°40.9	61°37.0
Denebola	182°24.7	14°25.9
Gienah	175°43.3	-17°41.1
Acrux	172°59.9	-63°14.7
Gacrux	171°51.4	-57°15.6
Alioth	166°12.7	55°49.5
Spica	158°21.9	-11°17.7
Alkaid	152°51.6	49°11.4
Hadar	148°35.3	-60°30.0
Menkent	147°57.1	-36°29.9
Arcturus	145°47.4	19°03.1
Rigel Kent.	139°39.5	-60°56.7
Kochab	137°19.4	74°03.3
Zuben'ubi	136°55.4	-16°08.9
Alphecca	126°03.2	26°37.9
Antares	112°15.0	-26°29.4
Atria	107°08.4	-69°04.6
Sabik	102°02.0	-15°45.4
Shaula	96°09.4	-37°07.4
Rasalhague	95°57.8	12°32.5
Eltanin	90°41.5	51°29.2
Kaus Aust.	83°31.5	-34°22.4
Vega	80°32.5	38°48.5
Nunki	75°46.8	-26°15.9
Altair	61°59.2	8°56.2
Peacock	53°04.4	-56°39.1
Deneb	49°25.1	45°22.2
Enif	33°38.1	9°59.6
Al Na'ir	27°32.0	-46°50.0
Fomalhaut	15°13.8	-29°29.0
Scheat	13°44.5	28°13.2
Markab	13°29.2	15°20.6

Jul 18 Fri	SHA	Mer.pass
Venus	286°06.1	09:12
Mars	190°55.2	15:31
Jupiter	260°35.5	10:52
Saturn	357°19.4	04:26

Jul 19 Sat	SHA	Mer.pass
Venus	284°54.2	09:13
Mars	190°22.2	15:29
Jupiter	260°21.0	10:49
Saturn	357°19.8	04:22

Jul 20 Sun	SHA	Mer.pass
Venus	283°42.0	09:13
Mars	189°49.2	15:27
Jupiter	260°06.6	10:46
Saturn	357°20.3	04:18

Horizontal parallax		
Venus:		0.1
Mars:		0.1

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°26.3	N21°00.8	273°43.5	10.8'	N12°33.5	15.3'	59.1'
1	193°26.2	21°00.3	288°13.4	10.8'	12°48.8	15.3'	59.1'
2	208°26.2	20°59.9	302°43.1	10.7'	13°04.1	15.2'	59.2'
3	223°26.1	· · 59.5	317°12.8	10.6'	13°19.3	15.1'	59.2'
4	238°26.1	59.0	331°42.4	10.5'	13°34.4	15.1'	59.2'
5	253°26.0	58.6	346°11.9	10.4'	13°49.5	15.0'	59.2'
6	268°26.0	N20°58.1	0°41.3	10.3'	N14°04.5	14.9'	59.2'
7	283°25.9	57.7	15°10.7	10.3'	14°19.5	14.9'	59.2'
8	298°25.9	57.3	29°39.9	10.2'	14°34.3	14.8'	59.2'
9	313°25.9	· · 56.8	44°09.1	10.1'	14°49.1	14.7'	59.2'
10	328°25.8	56.4	58°38.2	10.0'	15°03.9	14.6'	59.2'
11	343°25.8	55.9	73°07.2	9.9'	15°18.5	14.6'	59.2'
12	358°25.7	N20°55.5	87°36.1	9.8'	N15°33.1	14.5'	59.3'
13	13°25.7	55.0	102°04.9	9.7'	15°47.6	14.4'	59.3'
14	28°25.6	54.6	116°33.6	9.6'	16°02.0	14.3'	59.3'
15	43°25.6	· · 54.1	131°02.3	9.5'	16°16.3	14.2'	59.3'
16	58°25.5	53.7	145°30.8	9.5'	16°30.5	14.2'	59.3'
17	73°25.5	53.2	159°59.3	9.4'	16°44.7	14.1'	59.3'
18	88°25.5	N20°52.8	174°27.6	9.3'	N16°58.7	14.0'	59.3'
19	103°25.4	52.3	188°55.9	9.2'	17°12.7	13.9'	59.3'
20	118°25.4	51.9	203°24.0	9.1'	17°26.6	13.8'	59.3'
21	133°25.3	· · 51.4	217°52.1	9.0'	17°40.4	13.7'	59.3'
22	148°25.3	51.0	232°20.1	8.9'	17°54.1	13.6'	59.3'
23	163°25.2	50.5	246°48.0	8.8'	18°07.7	13.5'	59.4'
SD = 15.7' $d = -0.4'$ SD = 16.1'							

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°25.2	N20°50.1	261°15.7	8.7'	N18°21.1	13.4'	59.4'
1	193°25.2	49.6	275°43.4	8.6'	18°34.5	13.3'	59.4'
2	208°25.1	49.1	290°11.0	8.5'	18°47.8	13.2'	59.4'
3	223°25.1	· · 48.7	304°38.4	8.4'	19°01.0	13.1'	59.4'
4	238°25.0	48.2	319°05.8	8.3'	19°14.1	13.0'	59.4'
5	253°25.0	47.8	333°33.1	8.2'	19°27.1	12.9'	59.4'
6	268°25.0	N20°47.3	348°00.2	8.1'	N19°39.9	12.7'	59.4'
7	283°24.9	46.9	2°27.3	8.0'	19°52.7	12.6'	59.4'
8	298°24.9	46.4	16°54.2	7.9'	20°05.3	12.5'	59.4'
9	313°24.8	· · 45.9	31°21.1	7.7'	20°17.8	12.4'	59.4'
10	328°24.8	45.5	45°47.8	7.6'	20°30.2	12.3'	59.4'
11	343°24.8	45.0	60°14.5	7.5'	20°42.5	12.2'	59.4'
12	358°24.7	N20°44.6	74°41.0	7.4'	N20°54.6	12.0'	59.4'
13	13°24.7	44.1	89°07.5	7.3'	21°06.7	11.9'	59.4'
14	28°24.6	43.6	103°33.8	7.2'	21°18.6	11.8'	59.4'
15	43°24.6	· · 43.2	118°00.0	7.1'	21°30.4	11.7'	59.5'
16	58°24.6	42.7	132°26.1	7.0'	21°42.0	11.5'	59.5'
17	73°24.5	42.2	146°52.1	6.9'	21°53.5	11.4'	59.5'
18	88°24.5	N20°41.8	161°18.0	6.8'	N22°04.9	11.3'	59.5'
19	103°24.4	41.3	175°43.8	6.7'	22°16.2	11.1'	59.5'
20	118°24.4	40.8	190°09.5	6.6'	22°27.3	11.0'	59.5'
21	133°24.4	· · 40.4	204°35.1	6.5'	22°38.3	10.8'	59.5'
22	148°24.3	39.9	219°00.6	6.4'	22°49.2	10.7'	59.5'
23	163°24.3	39.4	233°26.0	6.3'	22°59.9	10.6'	59.5'
SD = 15.7' $d = -0.5'$ SD = 16.2'							

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°24.3	N20°39.0	247°51.3	6.2'	N23°10.4	10.4'	59.5'
1	193°24.2	38.5	262°16.5	6.1'	23°20.8	10.3'	59.5'
2	208°24.2	38.0	276°41.5	6.0'	23°31.1	10.1'	59.5'
3	223°24.1	· · 37.6	291°06.5	5.9'	23°41.2	10.0'	59.5'
4	238°24.1	37.1	305°31.4	5.8'	23°51.2	9.8'	59.5'
5	253°24.1	36.6	319°56.2	5.7'	24°01.0	9.7'	59.5'
6	268°24.0	N20°36.1	334°20.8	5.6'	N24°10.7	9.5'	59.5'
7	283°24.0	35.7	348°45.4	5.5'	24°20.2	9.4'	59.5'
8	298°24.0	35.2	3°09.9	5.4'	24°29.6	9.2'	59.5'
9	313°23.9	· · 34.7	17°34.3	5.3'	24°38.8	9.0'	59.5'
10	328°23.9	34.2	31°58.5	5.2'	24°47.8	8.9'	59.5'
11	343°23.9	33.8	46°22.7	5.1'	24°56.7	8.7'	59.5'
12	358°23.8	N20°33.3	60°46.8	5.0'	N25°05.4	8.6'	59.5'
13	13°23.8	32.8	75°10.8	4.9'	25°14.0	8.4'	59.5'
14	28°23.8	32.3	89°34.7	4.8'	25°22.4	8.2'	59.5'
15	43°23.7	· · 31.9	103°58.5	4.7'	25°30.6	8.1'	59.5'
16	58°23.7	31.4	118°22.2	4.6'	25°38.7	7.9'	59.5'
17	73°23.7	30.9	132°45.9	4.5'	25°46.5	7.7'	59.5'
18	88°23.6	N20°30.4	147°09.4	4.5'	N25°54.3	7.5'	59.5'
19	103°23.6	29.9	161°32.9	4.4'	26°01.8	7.4'	59.5'
20	118°23.6	29.5	175°56.3	4.3'	26°09.2	7.2'	59.5'
21	133°23.5	· · 29.0	190°19.5	4.2'	26°16.3	7.0'	59.5'
22	148°23.5	28.5	204°42.8	4.1'	26°23.3	6.8'	59.5'
23	163°23.5	28.0	219°05.9	4.1'	26°30.2	6.6'	59.5'
SD = 15.7' $d = -0.5'$ SD = 16.2'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	—:—	—:—	00:49	23:14	—:—	—:—
66°	—:—	—:—	01:54	22:15	—:—	—:—
64°	—:—	—:—	02:28	21:42	—:—	—:—
62°	—:—	01:17	02:53	21:18	22:51	—:—
60°	—:—	01:59	03:12	20:59	22:11	—:—
N 58°	—:—	02:26	03:28	20:44	21:44	—:—
56°	01:10	02:47	03:41	20:30	21:24	22:58
54°	01:49	03:04	03:53	20:19	21:07	22:21
52°	02:14	03:19	04:03	20:09	20:53	21:56
50°	02:34	03:31	04:12	20:00	20:41	21:37
45°	03:10	03:56	04:32	19:41	20:16	21:01
N 40°	03:36	04:16	04:47	19:25	19:57	20:36
35°	03:56	04:31	05:00	19:12	19:41	20:16
30°	04:13	04:45	05:11	19:01	19:28	20:00
20°	04:38	05:07	05:31	18:42	19:06	19:34
N 10°	04:58	05:25	05:47	18:25	18:48	19:14
0°	05:15	05:41	06:03	18:10	18:32	18:58
S 10°	05:30	05:56	06:18	17:55	18:17	18:43
20°	05:44	06:11	06:34	17:39	18:02	18:29
30°	05:58	06:27	06:53	17:20	17:46	18:15
35°	06:05	06:36	07:03	17:09	17:37	18:08
40°	06:13	06:46	07:16	16:57	17:27	18:00
45°	06:21	06:58	07:30	16:43	17:16	17:52
S 50°	06:31	07:11	07:48	16:25	17:02	17:42
52°	06:35	07:17	07:56	16:17	16:56	17:38
54°	06:39	07:24	08:05	16:08	16:49	17:34
56°	06:44	07:31	08:16	15:58	16:42	17:29
58°	06:50	07:40	08:27	15:46	16:34	17:24
S 60°	06:55	07:49	08:41	15:32	16:24	17:18

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	19:36	☐	☐	16:43	☐	☐
N 70°	20:30	☐	☐	15:52	☐	☐
68°	21:03	☐	☐	15:20	☐	☐
66°	21:27	20:49	☐	14:57	17:30	☐
64°	21:47	21:29	☐	14:39	16:51	☐
62°	22:03	21:57	21:49	14:25	16:24	18:35
60°	22:16	22:19	22:27	14:13	16:03	17:58
N 58°	22:28	22:37	22:54	14:02	15:46	17:31
56°	22:38	22:52	23:15	13:53	15:32	17:11
54°	22:47	23:05	23:33	13:45	15:19	16:54
52°	22:55	23:16	23:48	13:38	15:08	16:39
50°	23:02	23:26	· · ·	13:31	14:59	16:26
45°	23:18	23:48	· · ·	13:17	14:39	16:00
N 40°	23:31	· · ·	00:05	13:06	14:22	15:40
35°	23:42	· · ·	00:20	12:56	14:09	15:22
30°	23:51	· · ·	00:33	12:48	13:57	15:08
20°	· · ·	00:08	00:55	12:34	13:37	14:43
N 10°	· · ·	00:23	01:15	12:21	13:19	14:21
0°	· · ·	00:37	01:33	12:09	13:03	14:01
S 10°	· · ·	00:51	01:51	11:58	12:47	13:41
20°	00:05	01:06	02:11	11:45	12:30	13:20
30°	00:16	01:24	02:34	11:31	12:10	12:55
35°	00:23	01:34	02:47	11:23	11:58	12:41
40°	00:30	01:46	03:03	11:14	11:45	12:24
45°	00:40	02:00	03:22	11:03	11:30	12:04
S 50°	00:50	02:17	03:46	10:50	11:11	11:39
52°	00:56	02:25	03:57	10:44	11:02	11:28
54°	01:01	02:34	04:10	10:38	10:52	11:14
56°	01:08	02:45	04:25	10:30	10:41	10:58
58°	01:15	02:57	04:43	10:22	10:28	10:40
S 60°	01:23	03:10	05:05	10:13	10:14	10:17

Day	Sun		Mer. Pass hh:mm	Moon		Age 23-25 50-28%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
18	06:15	06:17	12:06	05:57	18:23	
19	06:19	06:21	12:06	06:50	19:18	
20	06:23	06:25	12:06	07:47	20:17	

July 21, 22, 23 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	299°00.9	221°30.5	N20°54.4	128°17.0	N04°48.7	198°53.1	N23°00.8	296°21.8	S01°23.2	Alpheratz	357°34.1 29°13.8
1	314°03.4	236°30.0	54.7	143°18.1	48.1	213°54.9	00.7	311°24.3	23.2	Ankaa	353°06.5 -42°09.7
2	329°05.9	251°29.4	55.1	158°19.2	47.5	228°56.8	00.7	326°26.7	23.2	Schedar	349°30.4 56°40.4
3	344°08.3	266°28.8	· · 55.4	173°20.3	· · 46.9	243°58.7	· · 00.7	341°29.2	· · 23.2	Diphda	348°46.7 -17°50.6
4	359°10.8	281°28.3	55.8	188°21.4	46.2	259°00.5	00.6	356°31.7	23.3	Achernar	335°19.8 -57°06.1
5	14°13.3	296°27.7	56.1	203°22.4	45.6	274°02.4	00.6	11°34.2	23.3	Hamal	327°50.7 23°34.9
6	29°15.7	311°27.2	N20°56.5	218°23.5	N04°45.0	289°04.3	N23°00.6	26°36.7	S01°23.3	Polaris	313°57.7 89°22.0
7	44°18.2	326°26.6	56.8	233°24.6	44.4	304°06.1	00.5	41°39.2	23.3	Acamar	315°11.5 -40°11.8
8	59°20.6	341°26.0	57.2	248°25.7	43.7	319°08.0	00.5	56°41.7	23.4	Menkar	314°05.8 4°11.5
9	74°23.1	356°25.5	· · 57.5	263°26.8	· · 43.1	334°09.9	· · 00.5	71°44.2	· · 23.4	Mirfak	308°27.8 49°56.9
10	89°25.6	11°24.9	57.8	278°27.9	42.5	349°11.7	00.4	86°46.7	23.4	Aldebaran	290°39.3 16°33.6
11	104°28.0	26°24.3	58.2	293°29.0	41.9	4°13.6	00.4	101°49.2	23.4	Rigel	281°03.7 -8°10.2
12	119°30.5	41°23.8	N20°58.5	308°30.0	N04°41.3	19°15.5	N23°00.4	116°51.7	S01°23.5	Capella	280°21.6 46°01.1
13	134°33.0	56°23.2	58.9	323°31.1	40.6	34°17.3	00.3	131°54.2	23.5	Bellatrix	278°22.7 6°22.4
14	149°35.4	71°22.6	59.2	338°32.2	40.0	49°19.2	00.3	146°56.6	23.5	Elnath	278°01.6 28°37.7
15	164°37.9	86°22.1	· · 59.5	353°33.3	· · 39.4	64°21.1	· · 00.2	161°59.1	· · 23.5	Alnilam	275°37.5 -1°11.0
16	179°40.4	101°21.5	20°59.9	8°34.4	38.8	79°22.9	00.2	177°01.6	23.6	Betelgeuse	270°51.9 7°24.8
17	194°42.8	116°20.9	21°00.2	23°35.5	38.1	94°24.8	00.2	192°04.1	23.6	Canopus	263°52.7 -52°42.4
18	209°45.3	131°20.4	N21°00.5	38°36.5	N04°37.5	109°26.7	N23°00.1	207°06.6	S01°23.6	Sirius	258°26.2 -16°45.0
19	224°47.8	146°19.8	00.9	53°37.6	36.9	124°28.5	00.1	222°09.1	23.6	Adhara	255°05.9 -29°00.3
20	239°50.2	161°19.2	01.2	68°38.7	36.3	139°30.4	00.1	237°11.6	23.7	Procyon	244°50.7 5°09.6
21	254°52.7	176°18.7	· · 01.5	83°39.8	· · 35.7	154°32.3	· · 00.0	252°14.1	· · 23.7	Pollux	243°17.1 27°57.9
22	269°55.1	191°18.1	01.9	98°40.9	35.0	169°34.1	00.0	267°16.6	23.7	Avior	234°15.2 -59°35.4
23	284°57.6	206°17.5	02.2	113°42.0	34.4	184°36.0	00.0	282°19.1	23.7	Suhail	222°46.4 -43°32.2
Mer.pass. 04:03		ν -0.6' d 0.3' m -4.04		ν 1.1' d -0.6' m 1.57		ν 1.9' d -0.0' m -1.90		ν 2.5' d 0.0' m 0.86		Miaplacidus	221°39.1 -69°49.4
Alphard 217°47.6 -8°46.1											
Regulus 207°34.2 11°50.6											
Dubhe 193°40.9 61°37.0											
Denebola 182°24.7 14°25.9											
Gienah 175°43.3 -17°41.1											
Acrux 172°59.9 -63°14.7											
Gacrux 171°51.4 -57°15.6											
Alioth 166°12.7 55°49.5											
Spica 158°21.9 -11°17.7											
Alkaid 152°51.6 49°11.4											
Hadar 148°35.3 -60°30.0											
Menkent 147°57.1 -36°29.9											
Arcturus 145°47.5 19°03.1											
Rigel Kent. 139°39.6 -60°56.7											
Kochab 137°19.5 74°03.3											
Zuben'ubi 136°55.4 -16°08.9											
Alphecca 126°03.2 26°37.9											
Antares 112°15.0 -26°29.4											
Atria 107°08.4 -69°04.6											
Sabik 102°02.0 -15°45.4											
Shaula 96°09.4 -37°07.4											
Rasalhague 95°57.8 12°32.6											
Eltanin 90°41.5 51°29.2											
Kaus Aust. 83°31.5 -34°22.4											
Vega 80°32.5 38°48.5											
Nunki 75°46.8 -26°15.9											
Altair 61°59.2 8°56.2											
Peacock 53°04.4 -56°39.1											
Deneb 49°25.0 45°22.2											
Enif 33°38.1 9°59.6											
Al Na'ir 27°31.9 -46°50.0											
Fomalhaut 15°13.7 -29°29.0											
Scheat 13°44.5 28°13.2											
Markab 13°29.2 15°20.6											
Jul 21 Mon SHA Mer.pass											
Venus 282°29.6 09:14											
Mars 189°16.1 15:26											
Jupiter 259°52.1 10:43											
Saturn 357°20.8 04:14											
Jul 22 Tue SHA Mer.pass											
Venus 281°16.9 09:15											
Mars 188°43.0 15:24											
Jupiter 259°37.8 10:40											
Saturn 357°21.5 04:10											
Jul 23 Wed SHA Mer.pass											
Venus 280°03.9 09:16											
Mars 188°09.8 15:22											
Jupiter 259°23.4 10:37											
Saturn 357°22.2 04:06											
Horizontal parallax											
Venus: 0.1											
Mars: 0.1											

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	299°00.9	221°30.5	N20°54.4	128°17.0	N04°48.7	198°53.1	N23°00.8	296°21.8	S01°23.2	Alpheratz	357°34.1 29°13.8
1	314°03.4	236°30.0	54.7	143°18.1	48.1	213°54.9	00.7	311°24.3	23.2	Ankaa	353°06.5 -42°09.7
2	329°05.9	251°29.4	55.1	158°19.2	47.5	228°56.8	00.7	326°26.7	23.2	Schedar	349°30.4 56°40.4
3	344°08.3	266°28.8	· · 55.4	173°20.3	· · 46.9	243°58.7	· · 00.7	341°29.2	· · 23.2	Diphda	348°46.7 -17°50.6
4	359°10.8	281°28.3	55.8	188°21.4	46.2	259°00.5	00.6	356°31.7	23.3	Achernar	335°19.8 -57°06.1
5	14°13.3	296°27.7	56.1	203°22.4	45.6	274°02.4	00.6	11°34.2	23.3	Hamal	327°50.7 23°34.9
6	29°15.7	311°27.2	N20°56.5	218°23.5	N04°45.0	289°04.3	N23°00.6	26°36.7	S01°23.3	Polaris	313°57.7 89°22.0
7	44°18.2	326°26.6	56.8	233°24.6	44.4	304°06.1	00.5	41°39.2	23.3	Acamar	315°11.5 -40°11.8
8	59°20.6	341°26.0	57.2	248°25.7	43.7	319°08.0	00.5	56°41.7	23.4	Menkar	314°05.8 4°11.5
9	74°23.1	356°25.5	· · 57.5	263°26.8	· · 43.1	334°09.9	· · 00.5	71°44.2	· · 23.4	Mirfak	308°27.8 49°56.9
10	89°25.6	11°24.9	57.8	278°27.9	42.5	349°11.7	00.4	86°46.7	23.4	Aldebaran	290°39.3 16°33.6
11	104°28.0	26°24.3	58.2	293°29.0	41.9	4°13.6	00.4	101°49.2	23.4	Rigel	281°03.7 -8°10.2
12	119°30.5	41°23.8	N20°58.5	308°30.0	N04°41.3	19°15.5	N23°00.4	116°51.7	S01°23.5	Capella	280°21.6 46°01.1
13	134°33.0	56°23.2	58.9	323°31.1	40.6	34°17.3	00.3	131°54.2	23.5	Bellatrix	278°22.7 6°22.4
14	149°35.4	71°22.6	59.2	338°32.2	40.0	49°19.2	00.3	146°56.6	23.5	Elnath	278°01.6 28°37.7
15	164°37.9	86°22.1	· · 59.5	353°33.3	· · 39.4	64°21.1	· · 00.2	161°59.1	· · 23.5	Alnilam	275°37.5 -1°11.0
16	179°40.4	101°21.5	20°59.9	8°34.4	38.8	79°22.9	00.2	177°01.6	23.6	Betelgeuse	270°51.9 7°24.8
17	194°42.8	116°20.9	21°00.2	23°35.5	38.1	94°24.8	00.2	192°04.1	23.6	Canopus	263°52.7 -52°42.4
18	209°45.3	131°20.4	N21°00.5	38°36.5	N04°37.5	109°26.7	N23°00.1	207°06.6	S01°23.6	Sirius	258°26.2 -16°45.0
19	224°47.8	146°19.8	00.9	53°37.6	36.9	124°28.5	00.1	222°09.1	23.6	Adhara	255°05.9 -29°00.3
20	239°50.2	161°19.2	01.2	68°38.7	36.3	139°30.4	00.1	237°11.6	23.7	Procyon	244°50.7 5°09.6
21	254°52.7	176°18.7	· · 01.5	83°39.8	· · 35.7	154°32.3	· · 00.0	252°14.1	· · 23.7	Pollux	243°17.1 27°57.9
22	269°55.1	191°18.1	01.9	98°40.9	35.0	169°34.1	00.0	267°16.6	23.7	Avior	234°15.2 -59°35.4
23	284°57.6	206°17.5	02.2	113°42.0	34.4	184°36.0	00.0	282°19.1	23.7	Suhail	222°46.4 -43°32.2
Mer.pass. 04:03		ν -0.6' d 0.3' m -4.04		ν 1.1' d -0.6' m 1.57		ν 1.9' d -0.0' m -1.90		ν 2.5' d 0.0' m 0.86		Miaplacidus	221°39.1 -69°49.4
Alphard 217°47.6 -8°46.1											
Regulus 207°34.2 11°50.6											
Dubhe 193°40.9 61°37.0											
Denebola 182°24.7 14°25.9											
Gienah 175°43.3 -17°41.1											
Acrux 172°59.9 -63°14.7											
Gacrux 171°51.4 -57°15.6											
Alioth 166°12.7 55°49.5											
Spica 158°21.9 -11°17.7											
Alkaid 152°51.6 49°11.4											
Hadar 148°35.3 -60°30.0											
Menkent 147°57.1 -36°29.9											
Arcturus 145°47.5 19°03.1											
Rigel Kent. 139°39.6 -60°56.7											
Kochab 137°19.5 74°03.3											
Zuben'ubi 136°55.4 -16°08.9											
Alphecca 126°03.2 26°37.9											
Antares 112°15.0 -26°29.4											
Atria 107°08.4 -69°04.6											
Sabik 102°02.0 -15°45.4											
Shaula 96°09.4 -37°07.4											
Rasalhague 95°57.8 12°32.6											
Eltanin 90°41.5 51°29.2											
Kaus Aust. 83°31.5 -34°22.4											
Vega 80°32.5 38°48.5											
Nunki 75°46.8 -26°15.9											
Altair 61°59.2 8°56.2											
Peacock 53°04.4 -56°39.1											
Deneb 49°25.0 45°22.2											
Enif 33°38.1 9°59.6											
Al Na'ir 27°31.9 -46°50.0											
Fomalhaut 15°13.7 -29°29.0											
Scheat 13°44.5 28°13.2											
Markab 13°29.2 15°20.6											
Jul 21 Mon SHA Mer.pass											
Venus 282°29.6 09:14											
Mars 189°16.1 15:26											
Jupiter 259°52.1 10:43											
Saturn 357°20.8 04:14											
Jul 22 Tue SHA Mer.pass											
Venus 281°16.9 09:15											
Mars 188°43.0 15:24											
Jupiter 259°37.8 10:40											
Saturn 357°21.5 04:10											
Jul 23 Wed SHA Mer.pass											
Venus 280°03.9 09:16											
Mars 188°09.8 15:22											
Jupiter 259°23.4 10:37											
Saturn 357°22.2 04:06											
Horizontal parallax											
Venus: 0.1											
Mars: 0.1											

h	Sun			Moon			
Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	178°23.4	N20°27.5	233°29.0	4.0'	N26°36.8	6.5'	59.5'
1	193°23.4	27.0	247°51.9	3.9'	26°43.3	6.3'	59.5'
2	208°23.4	26.6	262°14.9	3.8'	26°49.6	6.1'	59.5'
3	223°23.4	· · 26.1	276°37.7	3.8'	26°55.7	5.9'	59.5'
4	238°23.3	25.6	291°00.5	3.7'	27°01.6	5.7'	59.5'
5	253°23.3	25.1	305°23.2	3.6'	27°07.3	5.5'	59.5'
6	268°23.3	N20°24.6	319°45.8	3.6'	N27°12.9	5.4'	59.5'
7	283°23.2	24.1	334°08.4	3.5'	27°18.2	5.2'	59.5'
8	298°23.2	23.6	348°30.9	3.5'	27°23.4	5.0'	59.5'
9	313°23.2	· · 23.2	2°53.4	3.4'	27°28.3	4.8'	59.5'
10	328°23.1	22.7	17°15.8	3.3'	27°33.1	4.6'	59.5'
11	343°23.1	22.2	31°38.1	3.3'	27°37.7	4.4'	59.5'
12	358°23.1	N20°21.7	46°00.4	3.3'	N27°42.1	4.2'	59.4'
13	13°23.1	21.2	60°22.7	3.2'	27°46.3	4.0'	59.4'
14	28°23.0	20.7	74°44.9	3.2'	27°50.3	3.8'	59.4'
15	43°23.0	· · 20.2	89°07.0	3.1'	27°54.1	3.6'	59.4'
16	58°23.0	19.7	103°29.2	3.1'	27°57.7	3.4'	59.4'
17	73°23.0	19.2	117°51.2	3.0'	28°01.2	3.2'	59.4'
18	88°22.9	N20°18.7	132°13.3	3.0'	N28°04.4	3.0'	59.4'
19	103°22.9	18.2	146°35.3	3.0'	28°07.4	2.8'	59.4'
20	118°22.9	17.7	160°57.3	3.0'	28°10.2	2.6'	59.4'
21	133°22.8	· · 17.2	175°19.2	2.9'	28°12.8	2.4'	59.4'
22	148°22.8	16.7	189°41.2	2.9'	28°15.3	2.2'	59.4'
23	163°22.8	16.2	204°03.1	2.9'	28°17.5	2.0'	59.4'
SD = 15.7'			SD = 16.2'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	178°22.8	N20°15.8	218°25.0	2.9'	N28°19.5	1.8'	59.4'
1	193°22.7	15.3	232°46.9	2.9'	28°21.3	1.6'	59.4'
2	208°22.7	14.8	247°08.7	2.9'	28°23.0	1.4'	59.3'
3	223°22.7	· · 14.3	261°30.6	2.9'	28°24.4	1.2'	59.3'
4	238°22.7	13.8	275°52.4	2.8'	28°25.6	1.0'	59.3'
5	253°22.6	13.3	290°14.3	2.8'	28°26.6	0.8'	59.3'
6	268°22.6	N20°12.8	304°36.1	2.8'	N28°27.4	0.6'	59.3'
7	283°22.6	12.3	318°58.0	2.9'	28°28.0	0.4'	59.3'
8	298°22.6	11.7	333°19.8	2.9'	28°28.5	0.2'	59.3'
9	313°22.6	· · 11.2	347°41.7	2.9'	28°28.7	0.0'	59.3'
10	328°22.5	10.7	2°03.6	2.9'	28°28.7	-0.2'	59.3'
11	343°22.5	10.2	16°25.5	2.9'	28°28.5	-0.4'	59.3'
12	358°22.5	N20°09.7	30°47.4	2.9'	N28°28.1	-0.6'	59.2'
13	13°22.5	09.2	45°09.3	2.9'	28°27.5	-0.8'	59.2'
14	28°22.4	08.7	59°31.2	3.0'	28°26.7	-1.0'	59.2'
15	43°22.4	· · 08.2	73°53.2	3.0'	28°25.7	-1.2'	59.2'
16	58°22.4	07.7	88°15.2	3.0'	28°24.5	-1.4'	59.2'
17	73°22.4	07.2	102°37.3	3.1'	28°23.1	-1.6'	59.2'
18	88°22.4	N20°06.7	116°59.3	3.1'	N28°21.5	-1.8'	59.2'
19	103°22.3	06.2	131°21.4	3.1'	28°19.8	-2.0'	59.2'
20	118°22.3	05.7	145°43.6	3.2'	28°17.8	-2.2'	59.1'
21	133°22.3	· · 05.2	160°05.8	3.2'	28°15.6	-2.4'	59.1'
22	148°22.3	04.7	174°28.0	3.3'	28°13.2	-2.6'	59.1'
23	163°22.3	04.1	188°50.3	3.3'	28°10.6	-2.8'	59.1'
SD = 15.7'			SD = 16.2'				

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	178°22.2	N20°03.6	203°12.6	3.4'	N28°07.9	-3.0'	59.1'
1	193°22.2	03.1	217°35.0	3.4'	28°04.9	-3.2'	59.1'
2	208°22.2	02.6	231°57.5	3.5'	28°01.8	-3.3'	59.0'
3	223°22.2	· · 02.1	246°20.0	3.6'	27°58.4	-3.5'	59.0'
4	238°22.2	01.6	260°42.5	3.6'	27°54.9	-3.7'	59.0'
5	253°22.1	01.1	275°05.2	3.7'	27°51.2	-3.9'	59.0'
6	268°22.1	N20°00.5	289°27.9	3.8'	N27°47.2	-4.1'	59.0'
7	283°22.1	20°00.0	303°50.6	3.8'	27°43.1	-4.3'	59.0'
8	298°22.1	19°59.5	318°13.5	3.9'	27°38.9	-4.5'	58.9'
9	313°22.1	· · 59.0	332°36.4	4.0'	27°34.4	-4.7'	58.9'
10	328°22.1	58.5	346°59.4	4.1'	27°29.7	-4.8'	58.9'
11	343°22.0	58.0	1°22.5	4.2'	27°24.9	-5.0'	58.9'
12	358°22.0	N19°57.4	15°45.6	4.2'	N27°19.9	-5.2'	58.9'
13	13°22.0	56.9	30°08.8	4.3'	27°14.7	-5.4'	58.9'
14	28°22.0	56.4	44°32.2	4.4'	27°09.3	-5.6'	58.8'
15	43°22.0	· · 55.9	58°55.6	4.5'	27°03.7	-5.7'	58.8'
16	58°22.0	55.4	73°19.1	4.6'	26°58.0	-5.9'	58.8'
17	73°21.9	54.8	87°42.6	4.7'	26°52.1	-6.1'	58.8'
18	88°21.9	N19°54.3	102°06.3	4.8'	N26°46.0	-6.3'	58.8'
19	103°21.9	53.8	116°30.1	4.9'	26°39.7	-6.4'	58.7'
20	118°21.9	53.3	130°54.0	5.0'	26°33.3	-6.6'	58.7'
21	133°21.9	· · 52.7	145°17.9	5.1'	26°26.7	-6.8'	58.7'
22	148°21.9	52.2	159°42.0	5.2'	26°20.0	-6.9'	58.7'
23	163°21.9	51.7	174°06.2	5.3'	26°13.0	-7.1'	58.7'
SD = 15.7'			SD = 16.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°	—:—	—:—	01:16	22:51	—:—	—:—
66°	—:—	—:—	02:06	22:03	—:—	—:—
64°	—:—	—:—	02:37	21:33	—:—	—:—
62°	—:—	01:32	03:00	21:11	22:37	—:—
60°	—:—	02:09	03:18	20:53	22:02	—:—
N 58°	—:—	02:34	03:34	20:38	21:37	—:—
56°	01:24	02:54	03:46	20:25	21:18	22:45
54°	01:58	03:10	03:58	20:14	21:02	22:13
52°	02:21	03:24	04:07	20:05	20:48	21:50
50°	02:40	03:35	04:16	19:56	20:37	21:32
45°	03:15	04:00	04:35	19:38	20:13	20:57
N 40°	03:39	04:18	04:50	19:23	19:54	20:33
35°	03:59	04:34	05:02	19:10	19:39	20:14
30°	04:15	04:47	05:13	18:59	19:26	19:58
20°	04:40	05:08	05:32	18:41	19:05	19:33
N 10°	04:59	05:25	05:48	18:25	18:47	19:14
0°	05:15	05:41	06:03	18:10	18:32	18:58
S 10°	05:30	05:55	06:18	17:55	18:18	18:43
20°	05:43	06:10	06:33	17:40	18:03	18:30
30°	05:57	06:26	06:51	17:22	17:47	18:17
35°	06:04	06:34	07:02	17:12	17:39	18:10
40°	06:11	06:44	07:14	17:00	17:29	18:02
45°	06:19	06:55	07:27	16:46	17:18	17:54
S 50°	06:28	07:08	07:44	16:29	17:05	17:45
52°	06:32	07:14	07:52	16:21	17:00	17:41
54°	06:36	07:20	08:01	16:12	16:53	17:37
56°	06:41	07:28	08:11	16:02	16:46	17:33
58°	06:46	07:36	08:23	15:51	16:38	17:28
S 60°	06:51	07:44	08:36	15:38	16:29	17:22

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°						
N 70°						
68°						
66°						
64°						
62°						21:52
60°	22:52	23:59	· · ·	19:44	20:50	21:14
N 58°	23:29	· · ·	00:36	19:07	20:13	20:48
56°	23:56	· · ·	01:03	18:40	19:46	20:26
54°	· · ·	00:17	01:24	18:19	19:25	20:09
52°	· · ·	00:35	01:41	18:02	19:08	19:54
50°	00:01	00:50	01:56	17:47	18:53	19:41
45°	00:28	01:20	02:26	17:17	18:22	19:14
N 40°	00:49	01:44	02:49	16:53	17:59	18:52
35°	01:07	02:04	03:09	16:34	17:39	18:35
30°	01:23	02:20	03:25	16:17	17:22	18:19
20°	01:49	02:49	03:53	15:49	16:54	17:53
N 10°	02:12	03:13	04:17	15:25	16:29	17:30
0°	02:33	03:36	04:39	15:03	16:07	17:09
S 10°	02:55	03:59	05:02	14:41	15:44	16:47
20°	03:18	04:24	05:26	14:17	15:19	16:24
30°	03:45	04:52	05:54	13:49	14:50	15:57
35°	04:01	05:10	06:10	13:32	14:33	15:42
40°	04:19	05:29	06:29	13:13	14:13	15:23
45°	04:42	05:54	06:52	12:50	13:49	15:00
S 50°	05:11	06:25	07:22	12:20	13:18	14:31
52°	05:25	06:41	07:36	12:06	13:02	14:17
54°	05:42	06:59	07:53	11:49	12:44	14:01
56°	06:02	07:21	08:13	11:29	12:22	13:41
58°	06:26	07:50	08:38	11:04	11:53	13:16
S 60°	07:00	08:32	09:12	10:30	11:11	12:43

Day	Sun		Mer. Pass hh:mm	Moon		Age 26-28 18-4%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
21	06:26	06:28	12:06	08:48	21:20	
22	06:29	06:30	12:07	09:51	22:23	
23	06:31	06:32	12:07	10:54	23:25	

July 24, 25, 26 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	301°58.4	220°49.1	N21°17.3	129°34.9	N04°03.8	201°07.5	N22°58.2	299°21.5	S01°25.0	Alpheratz 357°34.1 29°13.8 Ankaa 353°06.5 -42°09.7 Schedar 349°30.3 56°40.5 Diphda 348°46.7 -17°50.6 Achernar 335°19.8 -57°06.1 Hamal 327°50.7 23°35.0 Polaris 313°56.0 89°22.0 Acamar 315°11.5 -40°11.8 Menkar 314°05.7 4°11.5 Mirfak 308°27.8 49°56.9 Aldebaran 290°39.3 16°33.7 Rigel 281°03.7 -8°10.2 Capella 280°21.6 46°01.3 Bellatrix 278°22.6 6°22.4 Elnath 278°01.6 28°37.7 Alnilam 275°37.5 -1°11.0 Betelgeuse 270°51.9 7°24.8 Canopus 263°52.7 -52°42.3 Sirius 258°26.2 -16°44.9 Adhara 255°05.9 -29°00.3 Procyon 244°50.7 5°09.6 Pollux 243°17.1 27°57.9 Avior 234°15.2 -59°35.4 Suhail 222°46.4 -43°32.1 Miaplacidus 221°39.1 -69°49.4 Alphard 217°47.6 -8°46.1 Regulus 207°34.2 11°50.6 Dubhe 193°40.9 61°37.0 Denebola 182°24.7 14°25.9 Gienah 175°43.3 -17°41.1 Acrux 172°59.9 -63°14.7 Gacrux 171°51.4 -57°15.6 Alioth 166°12.7 55°49.5 Spica 158°21.9 -11°17.7 Alkaid 152°51.6 49°11.4 Hadar 148°35.4 -60°30.0 Menkent 147°57.1 -36°29.9 Arcturus 145°47.5 19°03.1 Rigel Kent. 139°39.6 -60°56.7 Kochab 137°19.5 74°03.3 Zuben'ubi 136°55.4 -16°08.9 Alphecca 126°03.2 26°37.9 Antares 112°15.0 -26°29.4 Atria 107°08.4 -69°04.6 Sabik 102°02.0 -15°45.4 Shaula 96°09.4 -37°07.4 Rasalhague 95°57.8 12°32.6 Eltanin 90°41.5 51°29.2 Kaus Aust. 83°31.5 -34°22.4 Vega 80°32.5 38°48.5 Nunki 75°46.8 -26°15.9 Altair 61°59.2 8°56.2 Peacock 53°04.4 -56°39.1 Deneb 49°25.0 45°22.3 Enif 33°38.0 9°59.6 Al Na'ir 27°31.9 -46°50.0 Fomalhaut 15°13.7 -29°29.0 Scheat 13°44.5 28°13.3 Markab 13°29.2 15°20.6 Jul 24 Thu 24 Thu 25 Fri 25 Fri 26 Sat 26 Sat Horizontal parallax Venus: 0.1 Mars: 0.1							
1	317°00.8	235°48.5	17.6	144°36.0	03.2	216°09.4	58.1	314°24.0	25.0								
2	332°03.3	250°47.9	17.9	159°37.1	02.5	231°11.3	58.1	329°26.5	25.1								
3	347°05.8	265°47.3	•• 18.2	174°38.1	•• 01.9	246°13.1	•• 58.1	344°29.0	•• 25.1								
4	2°08.2	280°46.7	18.5	189°39.2	01.3	261°15.0	58.0	359°31.5	25.1								
5	17°10.7	295°46.1	18.7	204°40.3	00.6	276°16.9	58.0	14°34.0	25.2								
6	32°13.1	310°45.5	N21°19.0	219°41.4	N04°00.0	291°18.7	N22°58.0	29°36.5	S01°25.2								
7	47°15.6	325°44.9	19.3	234°42.5	03°59.4	306°20.6	57.9	44°39.0	25.2								
8	62°18.1	340°44.3	19.6	249°43.5	58.8	321°22.5	57.9	59°41.5	25.2								
9	77°20.5	355°43.7	•• 19.9	264°44.6	•• 58.1	336°24.3	•• 57.8	74°44.0	•• 25.3								
10	92°23.0	10°43.1	20.1	279°45.7	57.5	351°26.2	57.8	89°46.5	25.3								
11	107°25.5	25°42.5	20.4	294°46.8	56.9	6°28.1	57.8	104°49.0	25.3								
12	122°27.9	40°41.9	N21°20.7	309°47.9	N03°56.3	21°30.0	N22°57.7	119°51.5	S01°25.4								
13	137°30.4	55°41.3	21.0	324°48.9	55.6	36°31.8	57.7	134°54.0	25.4								
14	152°32.9	70°40.7	21.2	339°50.0	55.0	51°33.7	57.7	149°56.5	25.4								
15	167°35.3	85°40.1	•• 21.5	354°51.1	•• 54.4	66°35.6	•• 57.6	164°59.0	•• 25.4								
16	182°37.8	100°39.5	21.8	9°52.2	53.7	81°37.4	57.6	180°01.5	25.5								
17	197°40.3	115°38.9	22.1	24°53.2	53.1	96°39.3	57.5	195°04.0	25.5								
18	212°42.7	130°38.3	N21°22.3	39°54.3	N03°52.5	111°41.2	N22°57.5	210°06.5	S01°25.5								
19	227°45.2	145°37.7	22.6	54°55.4	51.9	126°43.1	57.5	225°09.0	25.6								
20	242°47.6	160°37.1	22.9	69°56.5	51.2	141°44.9	57.4	240°11.5	25.6								
21	257°50.1	175°36.6	•• 23.1	84°57.5	•• 50.6	156°46.8	•• 57.4	255°14.0	•• 25.6								
22	272°52.6	190°36.0	23.4	99°58.6	50.0	171°48.7	57.4	270°16.5	25.7								
23	287°55.0	205°35.4	23.7	114°59.7	49.3	186°50.5	57.3	285°19.0	25.7								
Mer.pass. 03:51		ν -0.6' d 0.3' m -4.03			ν 1.1' d -0.6' m 1.58			ν 1.9' d -0.0' m -1.91			ν 2.5' d 0.0' m 0.84						
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec								
0	302°57.5	220°34.7	N21°23.9	130°00.8	N03°48.7	201°52.4	N22°57.3	300°21.5	S01°25.7								
1	318°00.0	235°34.1	24.2	145°01.9	48.1	216°54.3	57.2	315°24.0	25.7								
2	333°02.4	250°33.5	24.5	160°02.9	47.5	231°56.1	57.2	330°26.5	25.8								
3	348°04.9	265°32.9	•• 24.7	175°04.0	•• 46.8	246°58.0	•• 57.2	345°29.0	•• 25.8								
4	3°07.4	280°32.3	25.0	190°05.1	46.2	261°59.9	57.1	0°31.6	25.8								
5	18°09.8	295°31.7	25.2	205°06.2	45.6	277°01.8	57.1	15°34.1	25.9								
6	33°12.3	310°31.1	N21°25.5	220°07.2	N03°44.9	292°03.6	N22°57.0	30°36.6	S01°25.9								
7	48°14.7	325°30.5	25.8	235°08.3	44.3	307°05.5	57.0	45°39.1	25.9								
8	63°17.2	340°29.9	26.0	250°09.4	43.7	322°07.4	57.0	60°41.6	26.0								
9	78°19.7	355°29.3	•• 26.3	265°10.5	•• 43.0	337°09.3	•• 56.9	75°44.1	•• 26.0								
10	93°22.1	10°28.7	26.5	280°11.5	42.4	352°11.1	56.9	90°46.6	26.0								
11	108°24.6	25°28.1	26.8	295°12.6	41.8	7°13.0	56.9	105°49.1	26.0								
12	123°27.1	40°27.5	N21°27.0	310°13.7	N03°41.2	22°14.9	N22°56.8	120°51.6	S01°26.1								
13	138°29.5	55°26.9	27.3	325°14.8	40.5	37°16.7	56.8	135°54.1	26.1								
14	153°32.0	70°26.3	27.5	340°15.8	39.9	52°18.6	56.7	150°56.6	26.1								
15	168°34.5	85°25.7	•• 27.8	355°16.9	•• 39.3	67°20.5	•• 56.7	165°59.1	•• 26.2								
16	183°36.9	100°25.1	28.0	10°18.0	38.6	82°22.4	56.7	181°01.6	26.2								
17	198°39.4	115°24.5	28.3	25°19.1	38.0	97°24.2	56.6	196°04.1	26.2								
18	213°41.9	130°23.9	N21°28.5	40°20.2	N03°37.4	112°26.1	N22°56.6	211°06.7	S01°26.3								
19	228°44.3	145°23.3	28.8	55°21.2	36.7	127°28.0	56.6	226°09.2	26.3								
20	243°46.8	160°22.7	29.0	70°22.3	36.1	142°29.8	56.5	241°11.7	26.3								
21	258°49.2	175°22.1	•• 29.3	85°23.4	•• 35.5	157°31.7	•• 56.5	256°14.2	•• 26.3								
22	273°51.7	190°21.4	29.5	100°24.5	34.9	172°33.6	56.4	271°16.7	26.4								
23	288°54.2	205°20.8	29.8	115°25.5	34.2	187°35.5	56.4	286°19.2	26.4								
Mer.pass. 03:48		ν -0.6' d 0.3' m -4.03			ν 1.1' d -0.6' m 1.59			ν 1.9' d -0.0' m -1.91			ν 2.5' d 0.0' m 0.84						
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec								
0	303°56.6	220°20.2	N21°30.0	130°26.6	N03°33.6	202°37.3	N22°56.4	301°21.7	S01°26.4								
1	318°59.1	235°19.6	30.3	145°27.7	33.0	217°39.2	56.3	316°24.2	26.5								
2	334°01.6	250°19.0	30.5	160°28.8	32.3	232°41.1	56.3	331°26.7	26.5								
3	349°04.0	265°18.4	•• 30.7	175°29.8	•• 31.7	247°43.0	•• 56.2	346°29.2	•• 26.5								
4	4°06.5	280°17.8	31.0	190°30.9	31.1	262°44.8	56.2	1°31.7	26.6								
5	19°09.0	295°17.2	31.2	205°32.0	30.4	277°46.7	56.2	16°34.2	26.6								
6	34°11.4	310°16.6	N21°31.5	220°33.0	N03°29.8	292°48.6	N22°56.1	31°36.8	S01°26.6								
7	49°13.9	325°16.0	31.7	235°34.1	29.2	307°50.5	56.1	46°39.3	26.7								
8	64°16.4	340°15.3	31.9	250°35.2	28.6	322°52.3	56.0	61°41.8	26.7								
9	79°18.8	355°14.7	•• 32.2	265°36.3	•• 27.9	337°54.2	•• 56.0	76°44.3	•• 26.7								
10	94°21.3	10°14.1	32.4	280°37.3	27.3	352°56.1	56.0	91°46.8	26.8								
11	109°23.7	25°13.5	32.6	295°38.4	26.7	7°58.0	55.9	106°49.3	26.8								
12	124°26.2	40°12.9	N21°32.9	310°39.5	N03°26.0	22°59.8	N22°55.9	121°51.8	S01°26.8								
13	139°28.7	55°12.3	33.1	325°40.6	25.4	38°01.7	55.9	136°54.3	26.9								
14	154°31.1	70°11.7	33.3	340°41.6	24.8	53°03.6	55.8	151°56.8	26.9								
15	169°33.6	85°11.0	•• 33.6	355°42.7	•• 24.1	68°05.4	•• 55.8	166°59.4	•• 26.9								
16	184°36.1	100°10.4	33.8	10°43.8	23.5	83°07.3	55.7	182°01.9	27.0								
17	199°38.5	115°09.8	34.0	25°44.9	22.9	98°09.2	55.7	197°04.4	27.0								
18	214°41.0	130°09.2	N21°34.2	40°45.9	N03°22.2	113°11.1	N22°55.7	212°06.9	S01°27.0								
19	229°43.5	145°08.6	34.5	55°47.0	21.6	128°12.9	55.6	227°09.4	27.0								
20	244°45.9	160°08.0	34.7	70°48.1	21.0	143°14.8	55.6	242°11.9	27.1								
21	259°48.4	175°07.3	•• 34.9	85°49.2	•• 20.3	158°16.7	•• 55.5	257°14.4	•• 27.1								
22	274°50.8	190°06.7	35.1	100°50.2	19.7	173°18.6	55.5	272°16.9	27.1								
23	289°53.3	205°06.1	35.4	115°51.3	19.1	188°20.4	55.5	287°19.4	27.2								
Mer.pass. 03:44		ν -0.6' d 0.2' m -4.02			ν 1.1' d -0.6' m 1.59			ν 1.9' d -0.0' m -1.91			ν 2.5' d 0.0' m 0.83						

Sun			Moon				
h							
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	178°21.9	N19°51.2	188°30.5	5.4'	N26°05.9	-7.3'	58.6'
1	193°21.8	50.6	202°54.8	5.5'	25°58.7	-7.4'	58.6'
2	208°21.8	50.1	217°19.3	5.6'	25°51.3	-7.6'	58.6'
3	223°21.8	· · 49.6	231°43.9	5.7'	25°43.7	-7.7'	58.6'
4	238°21.8	49.1	246°08.6	5.8'	25°36.0	-7.9'	58.6'
5	253°21.8	48.5	260°33.4	5.9'	25°28.1	-8.0'	58.5'
6	268°21.8	N19°48.0	274°58.3	6.0'	N25°20.0	-8.2'	58.5'
7	283°21.8	47.5	289°23.4	6.1'	25°11.8	-8.3'	58.5'
8	298°21.8	46.9	303°48.5	6.3'	25°03.5	-8.5'	58.5'
9	313°21.7	· · 46.4	318°13.8	6.4'	24°55.0	-8.6'	58.4'
10	328°21.7	45.9	332°39.1	6.5'	24°46.4	-8.8'	58.4'
11	343°21.7	45.4	347°04.6	6.6'	24°37.6	-8.9'	58.4'
12	358°21.7	N19°44.8	1°30.2	6.7'	N24°28.7	-9.1'	58.4'
13	13°21.7	44.3	15°56.0	6.8'	24°19.6	-9.2'	58.3'
14	28°21.7	43.8	30°21.8	7.0'	24°10.4	-9.3'	58.3'
15	43°21.7	· · 43.2	44°47.8	7.1'	24°01.0	-9.5'	58.3'
16	58°21.7	42.7	59°13.8	7.2'	23°51.5	-9.6'	58.3'
17	73°21.7	42.2	73°40.0	7.3'	23°41.9	-9.8'	58.3'
18	88°21.7	N19°41.6	88°06.3	7.4'	N23°32.2	-9.9'	58.2'
19	103°21.6	41.1	102°32.8	7.6'	23°22.3	-10.0'	58.2'
20	118°21.6	40.5	116°59.3	7.7'	23°12.3	-10.1'	58.2'
21	133°21.6	· · 40.0	131°26.0	7.8'	23°02.1	-10.3'	58.2'
22	148°21.6	39.5	145°52.8	7.9'	22°51.9	-10.4'	58.1'
23	163°21.6	38.9	160°19.7	8.0'	22°41.5	-10.5'	58.1'
SD = 15.7'			SD = 16.0'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°21.6	N19°38.4	174°46.8	8.2'	N22°31.0	-10.6'	58.1'
1	193°21.6	37.8	189°14.0	8.3'	22°20.3	-10.7'	58.1'
2	208°21.6	37.3	203°41.2	8.4'	22°09.6	-10.9'	58.0'
3	223°21.6	· · 36.8	218°08.6	8.5'	21°58.7	-11.0'	58.0'
4	238°21.6	36.2	232°36.2	8.7'	21°47.8	-11.1'	58.0'
5	253°21.6	35.7	247°03.8	8.8'	21°36.7	-11.2'	57.9'
6	268°21.6	N19°35.1	261°31.6	8.9'	N21°25.5	-11.3'	57.9'
7	283°21.6	34.6	275°59.5	9.0'	21°14.2	-11.4'	57.9'
8	298°21.6	34.1	290°27.5	9.1'	21°02.8	-11.5'	57.9'
9	313°21.6	· · 33.5	304°55.7	9.3'	20°51.2	-11.6'	57.8'
10	328°21.6	33.0	319°23.9	9.4'	20°39.6	-11.7'	57.8'
11	343°21.5	32.4	333°52.3	9.5'	20°27.9	-11.8'	57.8'
12	358°21.5	N19°31.9	348°20.8	9.6'	N20°16.1	-11.9'	57.8'
13	13°21.5	31.3	2°49.4	9.7'	20°04.2	-12.0'	57.7'
14	28°21.5	30.8	17°18.2	9.9'	19°52.2	-12.1'	57.7'
15	43°21.5	· · 30.2	31°47.0	10.0'	19°40.0	-12.2'	57.7'
16	58°21.5	29.7	46°16.0	10.1'	19°27.8	-12.3'	57.6'
17	73°21.5	29.1	60°45.1	10.2'	19°15.6	-12.4'	57.6'
18	88°21.5	N19°28.6	75°14.3	10.3'	N19°03.2	-12.5'	57.6'
19	103°21.5	28.0	89°43.6	10.4'	18°50.7	-12.5'	57.6'
20	118°21.5	27.5	104°13.1	10.6'	18°38.2	-12.6'	57.5'
21	133°21.5	· · 26.9	118°42.6	10.7'	18°25.5	-12.7'	57.5'
22	148°21.5	26.4	133°12.3	10.8'	18°12.8	-12.8'	57.5'
23	163°21.5	25.8	147°42.1	10.9'	18°00.0	-12.9'	57.4'
SD = 15.7'			SD = 15.8'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°21.5	N19°25.3	162°12.0	11.0'	N17°47.1	-12.9'	57.4'
1	193°21.5	24.7	176°42.0	11.1'	17°34.2	-13.0'	57.4'
2	208°21.5	24.2	191°12.1	11.2'	17°21.2	-13.1'	57.4'
3	223°21.5	· · 23.6	205°42.3	11.3'	17°08.1	-13.2'	57.3'
4	238°21.5	23.1	220°12.7	11.4'	16°54.9	-13.2'	57.3'
5	253°21.5	22.5	234°43.1	11.6'	16°41.7	-13.3'	57.3'
6	268°21.5	N19°21.9	249°13.7	11.7'	N16°28.4	-13.4'	57.2'
7	283°21.5	21.4	263°44.3	11.8'	16°15.0	-13.4'	57.2'
8	298°21.5	20.8	278°15.1	11.9'	16°01.6	-13.5'	57.2'
9	313°21.5	· · 20.3	292°46.0	12.0'	15°48.1	-13.6'	57.2'
10	328°21.5	19.7	307°17.0	12.1'	15°34.5	-13.6'	57.1'
11	343°21.5	19.2	321°48.0	12.2'	15°20.9	-13.7'	57.1'
12	358°21.5	N19°18.6	336°19.2	12.3'	N15°07.2	-13.7'	57.1'
13	13°21.5	18.0	350°50.5	12.4'	14°53.5	-13.8'	57.0'
14	28°21.5	17.5	5°21.9	12.5'	14°39.7	-13.8'	57.0'
15	43°21.5	· · 16.9	19°53.4	12.6'	14°25.8	-13.9'	57.0'
16	58°21.5	16.4	34°25.0	12.7'	14°11.9	-13.9'	57.0'
17	73°21.5	15.8	48°56.6	12.8'	13°58.0	-14.0'	56.9'
18	88°21.5	N19°15.2	63°28.4	12.9'	N13°44.0	-14.0'	56.9'
19	103°21.6	14.7	78°00.3	13.0'	13°29.9	-14.1'	56.9'
20	118°21.6	14.1	92°32.3	13.1'	13°15.8	-14.1'	56.8'
21	133°21.6	· · 13.5	107°04.3	13.1'	13°01.7	-14.2'	56.8'
22	148°21.6	13.0	121°36.5	13.2'	12°47.5	-14.2'	56.8'
23	163°21.6	12.4	136°08.7	13.3'	12°33.3	-14.3'	56.7'
SD = 15.7'			SD = 15.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°						
68°	—:—	—:—	01:37	22:31	—:—	—:—
66°	—:—	—:—	02:19	21:51	—:—	—:—
64°	—:—	00:47	02:47	21:24	23:16	—:—
62°	—:—	01:46	03:08	21:03	22:23	—:—
60°	—:—	02:18	03:25	20:46	21:52	—:—
N 58°	00:43	02:42	03:39	20:32	21:29	23:21
56°	01:37	03:00	03:52	20:20	21:11	22:33
54°	02:07	03:15	04:02	20:10	20:56	22:04
52°	02:28	03:29	04:12	20:01	20:43	21:43
50°	02:46	03:40	04:20	19:52	20:32	21:26
45°	03:19	04:03	04:38	19:35	20:09	20:53
N 40°	03:43	04:21	04:52	19:20	19:51	20:30
35°	04:02	04:36	05:04	19:08	19:37	20:11
30°	04:17	04:49	05:15	18:58	19:24	19:56
20°	04:41	05:09	05:33	18:40	19:04	19:32
N 10°	05:00	05:26	05:49	18:24	18:47	19:13
0°	05:16	05:41	06:03	18:10	18:32	18:58
S 10°	05:29	05:55	06:17	17:56	18:18	18:44
20°	05:42	06:09	06:33	17:41	18:04	18:31
30°	05:55	06:24	06:50	17:24	17:49	18:18
35°	06:02	06:33	07:00	17:14	17:41	18:12
40°	06:09	06:42	07:11	17:02	17:32	18:05
45°	06:17	06:52	07:25	16:49	17:21	17:57
S 50°	06:25	07:05	07:41	16:33	17:09	17:49
52°	06:29	07:10	07:48	16:25	17:03	17:45
54°	06:33	07:17	07:57	16:17	16:57	17:41
56°	06:37	07:23	08:06	16:07	16:50	17:37
58°	06:42	07:31	08:17	15:57	16:43	17:32
S 60°	06:47	07:39	08:30	15:44	16:34	17:27

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°			02:31			00:21 22:47
N 70°			03:32		23:18	22:25
68°			04:07		22:42	22:09
66°		02:05	04:31	22:54	22:16	21:55
64°		02:45	04:50	22:13	21:55	21:43
62°	01:07	03:13	05:06	21:44	21:39	21:34
60°	01:44	03:34	05:18	21:22	21:25	21:25
N 58°	02:10	03:51	05:29	21:04	21:13	21:18
56°	02:31	04:06	05:39	20:49	21:02	21:11
54°	02:48	04:18	05:47	20:36	20:53	21:05
52°	03:02	04:29	05:55	20:24	20:45	21:00
50°	03:15	04:39	06:02	20:14	20:38	20:55
45°	03:41	04:59	06:16	19:52	20:22	20:44
N 40°	04:02	05:16	06:28	19:35	20:08	20:35
35°	04:19	05:29	06:38	19:20	19:57	20:28
30°	04:33	05:41	06:46	19:07	19:47	20:21
20°	04:58	06:02	07:01	18:45	19:30	20:09
N 10°	05:20	06:19	07:14	18:25	19:15	19:59
0°	05:40	06:36	07:26	18:07	19:01	19:49
S 10°	06:00	06:52	07:39	17:49	18:46	19:39
20°	06:21	07:09	07:51	17:29	18:31	19:28
30°	06:46	07:29	08:06	17:06	18:13	19:16
35°	07:00	07:41	08:14	16:53	18:02	19:09
40°	07:17	07:54	08:24	16:37	17:50	19:01
45°	07:37	08:10	08:35	16:18	17:36	18:51
S 50°	08:01	08:29	08:49	15:54	17:19	18:40
52°	08:13	08:38	08:55	15:43	17:10	18:34
54°	08:27	08:48	09:02	15:30	17:01	18:28
56°	08:42	08:59	09:09	15:15	16:50	18:21
58°	09:01	09:12	09:18	14:57	16:38	18:14
S 60°	09:24	09:27	09:28	14:35	16:24	18:05

Day	Sun		Mer. Pass hh:mm	Moon		Age 0-2 1-2%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
24	06:33	06:33	12:07	11:54	—:—	
25	06:34	06:34	12:07	12:48	00:22	
26	06:34	06:34	12:07	13:38	01:14	

July 27, 28, 29 UT (Sun., Mon., Tue.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	304°55.8	220°05.5	N21°35.6	130°52.4	N03°18.4	203°22.3	N22°55.4	302°22.0	S01°27.2	Alpheratz	357°34.1	29°13.9					
1	319°58.2	235°04.9	35.8	145°53.4	17.8	218°24.2	55.4	317°24.5	27.2	Ankaa	353°06.5	-42°09.7					
2	335°00.7	250°04.3	36.0	160°54.5	17.2	233°26.1	55.3	332°27.0	27.3	Schedar	349°30.3	56°40.5					
3	350°03.2	265°03.6	· · 36.2	175°55.6	· · 16.5	248°27.9	· · 55.3	347°29.5	· · 27.3	Diphda	348°46.7	-17°50.6					
4	5°05'06.1	280°03.0	36.5	190°56.7	15.9	263°29.8	55.3	2°32.0	27.3	Achernar	335°19.7	-57°06.1					
5	20°08.1	295°02.4	36.7	205°57.7	15.3	278°31.7	55.2	17°34.5	27.4	Hamal	327°50.7	23°35.0					
6	35°10.6	310°01.8	N21°36.9	220°58.8	N03°14.6	293°33.6	N22°55.2	32°37.0	S01°27.4	Polaris	313°54.6	89°22.0					
7	50°13.0	325°01.2	37.1	235°59.9	14.0	308°35.5	55.1	47°39.6	27.4	Acamar	315°11.5	-40°11.8					
8	65°15.5	340°00.5	37.3	251°01.0	13.4	323°37.3	55.1	62°42.1	27.5	Menkar	314°05.7	4°11.5					
9	80°18.0	354°59.9	· · 37.5	266°02.0	· · 12.8	338°39.2	· · 55.1	77°44.6	· · 27.5	Mirfak	308°27.7	49°56.9					
10	95°20.4	9°59.3	37.7	281°03.1	12.1	353°41.1	55.0	92°47.1	27.5	Aldebaran	290°39.3	16°33.7					
11	110°22.9	24°58.7	37.9	296°04.2	11.5	8°43.0	55.0	107°49.6	27.6	Rigel	281°03.7	-8°10.2					
12	125°25.3	39°58.0	N21°38.2	311°05.2	N03°10.9	23°44.8	N22°54.9	122°52.1	S01°27.6	Capella	280°21.5	46°01.3					
13	140°27.8	54°57.4	38.4	326°06.3	10.2	38°46.7	54.9	137°54.6	27.6	Bellatrix	278°22.6	6°22.4					
14	155°30.3	69°56.8	38.6	341°07.4	09.6	53°48.6	54.9	152°57.2	27.7	Elnath	278°01.6	28°37.7					
15	170°32.7	84°56.2	· · 38.8	356°08.5	· · 09.0	68°50.5	· · 54.8	167°59.7	· · 27.7	Alnilam	275°37.5	-1°11.0					
16	185°35.2	99°55.6	39.0	11°09.5	08.3	83°52.3	54.8	183°02.2	27.7	Betelgeuse	270°51.9	7°24.8					
17	200°37.7	114°54.9	39.2	26°10.6	07.7	98°54.2	54.7	198°04.7	27.8	Canopus	263°52.7	-52°42.3					
18	215°40.1	129°54.3	N21°39.4	41°11.7	N03°07.1	113°56.1	N22°54.7	213°07.2	S01°27.8	Sirius	258°26.1	-16°44.9					
19	230°42.6	144°53.7	39.6	56°12.7	06.4	128°58.0	54.7	228°09.7	27.8	Adhara	255°05.9	-29°00.2					
20	245°45.1	159°53.1	39.8	71°13.8	05.8	143°59.8	54.6	243°12.2	27.9	Procyon	244°50.7	5°09.6					
21	260°47.5	174°52.4	· · 40.0	86°14.9	· · 05.2	159°01.7	· · 54.6	258°14.8	· · 27.9	Pollux	243°17.1	27°57.9					
22	275°50.0	189°51.8	40.2	101°16.0	04.5	174°03.6	54.5	273°17.3	27.9	Avior	234°15.2	-59°35.4					
23	290°52.5	204°51.2	40.4	116°17.0	03.9	189°05.5	54.5	288°19.8	28.0	Suhail	222°46.4	-43°32.1					
Mer.pass. 03:40		ν -0.6' d 0.2' m -4.02			ν 1.1' d -0.6' m 1.59			ν 1.9' d -0.0' m -1.91			ν 2.5' d 0.0' m 0.83						

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	305°54.9	219°50.6	N21°40.6	131°18.1	N03°03.3	204°07.3	N22°54.5	303°22.3	S01°28.0				
1	320°57.4	234°49.9	40.8	146°19.2	02.6	219°09.2	54.4	318°24.8	28.1				
2	335°59.8	249°49.3	41.0	161°20.2	02.0	234°11.1	54.4	333°27.3	28.1				
3	351°02.3	264°48.7	· · 41.2	176°21.3	· · 01.3	249°13.0	· · 54.3	348°29.9	· · 28.1				
4	6°04.8	279°48.0	41.4	191°22.4	00.7	264°14.9	54.3	3°32.4	28.2				
5	21°07.2	294°47.4	41.6	206°23.4	03°00.1	279°16.7	54.3	18°34.9	28.2				
6	36°09.7	309°46.8	N21°41.8	221°24.5	N02°59.4	294°18.6	N22°54.2	33°37.4	S01°28.2				
7	51°12.2	324°46.2	42.0	236°25.6	58.8	309°20.5	54.2	48°39.9	28.3				
8	66°14.6	339°45.5	42.1	251°26.7	58.2	324°22.4	54.1	63°42.4	28.3				
9	81°17.1	354°44.9	· · 42.3	266°27.7	· · 57.5	339°24.2	· · 54.1	78°45.0	· · 28.3				
10	96°19.6	9°44.3	42.5	281°28.8	56.9	354°26.1	54.1	93°47.5	28.4				
11	111°22.0	24°43.6	42.7	296°29.9	56.3	9°28.0	54.0	108°50.0	28.4				
12	126°24.5	39°43.0	N21°42.9	311°30.9	N02°55.6	24°29.9	N22°54.0	123°52.5	S01°28.4				
13	141°26.9	54°42.4	43.1	326°32.0	55.0	39°31.8	53.9	138°55.0	28.5				
14	156°29.4	69°41.8	43.3	341°33.1	54.4	54°33.6	53.9	153°57.6	28.5				
15	171°31.9	84°41.1	· · 43.5	356°34.1	· · 53.7	69°35.5	· · 53.9	169°00.1	· · 28.5				
16	186°34.3	99°40.5	43.6	11°35.2	53.1	84°37.4	53.8	184°02.6	28.6				
17	201°36.8	114°39.9	43.8	26°36.3	52.5	99°39.3	53.8	199°05.1	28.6				
18	216°39.3	129°39.2	N21°44.0	41°37.4	N02°51.8	114°41.2	N22°53.7	214°07.6	S01°28.6				
19	231°41.7	144°38.6	44.2	56°38.4	51.2	129°43.0	53.7	229°10.2	28.7				
20	246°44.2	159°38.0	44.4	71°39.5	50.6	144°44.9	53.7	244°12.7	28.7				
21	261°46.7	174°37.3	· · 44.5	86°40.6	· · 49.9	159°46.8	· · 53.6	259°15.2	· · 28.8				
22	276°49.1	189°36.7	44.7	101°41.6	49.3	174°48.7	53.6	274°17.7	28.8				
23	291°51.6	204°36.1	44.9	116°42.7	48.7	189°50.5	53.5	289°20.2	28.8				
Mer.pass. 03:36		ν -0.6' d 0.2' m -4.02			ν 1.1' d -0.6' m 1.59			ν 1.9' d -0.0' m -1.91			ν 2.5' d 0.0' m 0.82		

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	306°54.1	219°35.4	N21°45.1	131°43.8	N02°48.0	204°52.4	N22°53.5	304°22.7	S01°28.9				
1	321°56.5	234°34.8	45.2	146°44.8	47.4	219°54.3	53.4	319°25.3	28.9				
2	336°59.0	249°34.2	45.4	161°45.9	46.7	234°56.2	53.4	334°27.8	28.9				
3	352°01.4	264°33.5	· · 45.6	176°47.0	· · 46.1	249°58.1	· · 53.4	349°30.3	· · 29.0				
4	7°03.9	279°32.9	45.8	191°48.0	45.5	264°59.9	53.3	4°32.8	29.0				
5	22°06.4	294°32.2	45.9	206°49.1	44.8	280°01.8	53.3	19°35.4	29.0				
6	37°08.8	309°31.6	N21°46.1	221°50.2	N02°44.2	295°03.7	N22°53.2	34°37.9	S01°29.1				
7	52°11.3	324°31.0	46.3	236°51.2	43.6	310°05.6	53.2	49°40.4	29.1				
8	67°13.8	339°30.3	46.4	251°52.3	42.9	325°07.5	53.2	64°42.9	29.2				
9	82°16.2	354°29.7	· · 46.6	266°53.4	· · 42.3	340°09.3	· · 53.1	79°45.4	· · 29.2				
10	97°18.7	9°29.1	46.8	281°54.5	41.7	355°11.2	53.1	94°48.0	29.2				
11	112°21.2	24°28.4	46.9	296°55.5	41.0	10°13.1	53.0	109°50.5	29.3				
12	127°23.6	39°27.8	N21°47.1	311°56.6	N02°40.4	25°15.0	N22°53.0	124°53.0	S01°29.3				
13	142°26.1	54°27.2	47.3	326°57.7	39.8	40°16.9	53.0	139°55.5	29.3				
14	157°28.5	69°26.5	47.4	341°58.7	39.1	55°18.7	52.9	154°58.0	29.4				
15	172°31.0	84°25.9	· · 47.6	356°59.8	· · 38.5	70°20.6	· · 52.9	170°00.6	· · 29.4				
16	187°33.5	99°25.2	47.7	12°00.9	37.8	85°22.5	52.8	185°03.1	29.4				
17	202°35.9	114°24.6	47.9	27°01.9	37.2	100°24.4	52.8	200°05.6	29.5				
18	217°38.4	129°24.0	N21°48.1	42°03.0	N02°36.6	115°26.3	N22°52.7	215°08.1	S01°29.5				
19	232°40.9	144°23.3	48.2	57°04.1	35.9	130°28.1	52.7	230°10.7	29.6				
20	247°43.3	159°22.7	48.4	72°05.1	35.3	145°30.0	52.7	245°13.2	29.6				
21	262°45.8	174°22.0	· · 48.5	87°06.2	· · 34.7	160°31.9	· · 52.6	260°15.7	· · 29.6				
22	277°48.3	189°21.4	48.7	102°07.3	34.0	175°33.8	52.6	275°18.2	29.7				
23	292°50.7	204°20.8	48.8	117°08.3	33.4	190°35.7	52.5	290°20.8	29.7				
Mer.pass. 03:32		ν -0.6' d 0.2' m -4.01			ν 1.1' d -0.6' m 1.60			ν 1.9' d -0.0' m -1.91			ν 2.5' d 0.0' m 0.82		

Jul 27 Sun	SHA	Mer.pass
Venus	275°09.7	09:20
Mars	185°56.6	15:15
Jupiter	258°26.5	10:25
Saturn	357°26.2	03:50

Jul 28 Mon	SHA	Mer.pass
Venus	273°55.6	09:21
Mars	185°23.2	15:14
Jupiter	258°12.4	10:22
Saturn	357°27.4	03:46

Jul 29 Tue	SHA	Mer.pass
Venus	272°41.4	09:22
Mars	184°49.7	15:12
Jupiter	257°58.4	10:19
Saturn	357°28.7	03:42

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

h	Sun			Moon			
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°21.6	N19°11.8	150°41.0	13.4'	N12°19.0	-14.3'	56.7'
1	193°21.6	11.3	165°13.4	13.5'	12°04.7	-14.3'	56.7'
2	208°21.6	10.7	179°45.9	13.6'	11°50.4	-14.4'	56.7'
3	223°21.6	· · 10.1	194°18.5	13.7'	11°36.0	-14.4'	56.6'
4	238°21.6	09.6	208°51.2	13.8'	11°21.6	-14.4'	56.6'
5	253°21.6	09.0	223°23.9	13.8'	11°07.1	-14.5'	56.6'
6	268°21.6	N19°08.4	237°56.8	13.9'	N10°52.7	-14.5'	56.5'
7	283°21.6	07.9	252°29.7	14.0'	09°38.1	-14.5'	56.5'
8	298°21.6	07.3	267°02.7	14.1'	10°23.6	-14.6'	56.5'
9	313°21.6	· · 06.7	281°35.8	14.1'	10°09.0	-14.6'	56.5'
10	328°21.6	06.1	296°08.9	14.2'	09°54.4	-14.6'	56.4'
11	343°21.7	05.6	310°42.1	14.3'	09°39.8	-14.7'	56.4'
12	358°21.7	N19°05.0	325°15.4	14.4'	N09°25.1	-14.7'	56.4'
13	13°21.7	04.4	339°48.8	14.4'	09°10.5	-14.7'	56.3'
14	28°21.7	03.9	354°22.3	14.5'	08°55.8	-14.7'	56.3'
15	43°21.7	· · 03.3	8°55.8	14.6'	08°41.0	-14.7'	56.3'
16	58°21.7	02.7	23°29.4	14.7'	08°26.3	-14.8'	56.3'
17	73°21.7	02.1	38°03.0	14.7'	08°11.5	-14.8'	56.2'
18	88°21.7	N19°01.6	52°36.7	14.8'	N07°56.8	-14.8'	56.2'
19	103°21.7	01.0	67°10.5	14.9'	07°42.0	-14.8'	56.2'
20	118°21.7	19°00.4	81°44.4	14.9'	07°27.2	-14.8'	56.1'
21	133°21.8	18°59.8	96°18.3	15.0'	07°12.3	-14.8'	56.1'
22	148°21.8	59.2	110°52.3	15.0'	06°57.5	-14.9'	56.1'
23	163°21.8	58.7	125°26.3	15.1'	06°42.6	-14.9'	56.1'
SD = 15.7'			SD = 15.5'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	178°21.8	N18°58.1	140°00.4	15.2'	N06°27.8	-14.9'	56.0'
1	193°21.8	57.5	154°34.5	15.2'	06°12.9	-14.9'	56.0'
2	208°21.8	56.9	169°08.8	15.3'	05°58.0	-14.9'	56.0'
3	223°21.8	· · 56.3	183°43.0	15.3'	05°43.1	-14.9'	55.9'
4	238°21.8	55.8	198°17.4	15.4'	05°28.2	-14.9'	55.9'
5	253°21.9	55.2	212°51.7	15.4'	05°13.3	-14.9'	55.9'
6	268°21.9	N18°54.6	227°26.2	15.5'	N04°58.4	-14.9'	55.9'
7	283°21.9	54.0	242°00.6	15.5'	04°43.5	-14.9'	55.8'
8	298°21.9	53.4	256°35.2	15.6'	04°28.6	-14.9'	55.8'
9	313°21.9	· · 52.8	271°09.7	15.6'	04°13.7	-14.9'	55.8'
10	328°21.9	52.3	285°44.4	15.7'	03°58.7	-14.9'	55.8'
11	343°21.9	51.7	300°19.0	15.7'	03°43.8	-14.9'	55.7'
12	358°22.0	N18°51.1	314°53.7	15.8'	N03°28.9	-14.9'	55.7'
13	13°22.0	50.5	329°28.5	15.8'	03°14.0	-14.9'	55.7'
14	28°22.0	49.9	344°03.3	15.8'	02°59.1	-14.9'	55.6'
15	43°22.0	· · 49.3	358°38.1	15.9'	02°44.1	-14.9'	55.6'
16	58°22.0	48.7	13°13.0	15.9'	02°29.2	-14.9'	55.6'
17	73°22.0	48.2	27°47.9	16.0'	02°14.3	-14.9'	55.6'
18	88°22.0	N18°47.6	42°22.9	16.0'	N01°59.4	-14.9'	55.5'
19	103°22.1	47.0	56°57.9	16.0'	01°44.5	-14.9'	55.5'
20	118°22.1	46.4	71°32.9	16.1'	01°29.7	-14.9'	55.5'
21	133°22.1	· · 45.8	86°07.9	16.1'	01°14.8	-14.9'	55.5'
22	148°22.1	45.2	100°43.0	16.1'	00°59.9	-14.9'	55.4'
23	163°22.1	44.6	115°18.2	16.1'	00°45.1	-14.8'	55.4'
SD = 15.7'			SD = 15.3'				

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	178°22.2	N18°44.0	129°53.3	16.2'	N00°30.2	-14.8'	55.4'
1	193°22.2	43.4	144°28.5	16.2'	00°15.4	-14.8'	55.4'
2	208°22.2	42.8	159°03.7	16.2'	N00°00.6	-14.8'	55.3'
3	223°22.2	· · 42.2	173°38.9	16.3'	S00°14.2	14.8'	55.3'
4	238°22.2	41.6	188°14.2	16.3'	00°29.0	14.8'	55.3'
5	253°22.2	41.1	202°49.5	16.3'	00°43.8	14.8'	55.3'
6	268°22.3	N18°40.5	217°24.8	16.3'	S00°58.6	14.7'	55.3'
7	283°22.3	39.9	232°00.1	16.3'	01°13.3	14.7'	55.2'
8	298°22.3	39.3	246°35.4	16.4'	01°28.0	14.7'	55.2'
9	313°22.3	· · 38.7	261°10.8	16.4'	01°42.7	14.7'	55.2'
10	328°22.4	38.1	275°46.2	16.4'	01°57.4	14.7'	55.2'
11	343°22.4	37.5	290°21.6	16.4'	02°12.1	14.6'	55.1'
12	358°22.4	N18°36.9	304°57.0	16.4'	S02°26.7	14.6'	55.1'
13	13°22.4	36.3	319°32.4	16.4'	02°41.3	14.6'	55.1'
14	28°22.4	35.7	334°07.8	16.4'	02°55.9	14.6'	55.1'
15	43°22.5	· · 35.1	348°43.3	16.5'	03°10.5	14.6'	55.1'
16	58°22.5	34.5	3°18.7	16.5'	03°25.1	14.5'	55.0'
17	73°22.5	33.9	17°54.2	16.5'	03°39.6	14.5'	55.0'
18	88°22.5	N18°33.3	32°29.7	16.5'	S03°54.1	14.5'	55.0'
19	103°22.6	32.7	47°05.2	16.5'	04°08.6	14.4'	55.0'
20	118°22.6	32.1	61°40.6	16.5'	04°23.0	14.4'	54.9'
21	133°22.6	· · 31.5	76°16.1	16.5'	04°37.4	14.4'	54.9'
22	148°22.6	30.9	90°51.6	16.5'	04°51.8	14.4'	54.9'
23	163°22.6	30.3	105°27.1	16.5'	05°06.2	14.3'	54.9'
SD = 15.7'			SD = 15.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	□	□	□	□	□	□
N 70°	—:—	—:—	00:41	23:17	—:—	—:—
68°	—:—	—:—	01:55	22:14	—:—	—:—
66°	—:—	—:—	02:31	21:39	—:—	—:—
64°	—:—	01:15	02:56	21:15	22:52	—:—
62°	—:—	01:59	03:16	20:55	22:10	—:—
60°	—:—	02:28	03:32	20:40	21:42	—:—
N 58°	01:08	02:50	03:45	20:26	21:22	22:59
56°	01:49	03:07	03:57	20:15	21:05	22:21
54°	02:15	03:21	04:07	20:05	20:50	21:55
52°	02:35	03:34	04:16	19:56	20:38	21:36
50°	02:52	03:45	04:24	19:48	20:27	21:20
45°	03:23	04:07	04:41	19:31	20:05	20:49
N 40°	03:46	04:24	04:55	19:18	19:48	20:26
35°	04:04	04:38	05:07	19:06	19:34	20:08
30°	04:19	04:51	05:17	18:56	19:22	19:53
20°	04:42	05:10	05:34	18:39	19:02	19:30
N 10°	05:01	05:27	05:49	18:24	18:46	19:12
0°	05:16	05:41	06:03	18:10	18:32	18:57
S 10°	05:29	05:55	06:17	17:56	18:18	18:44
20°	05:41	06:08	06:31	17:42	18:05	18:32
30°	05:54	06:23	06:48	17:25	17:51	18:20
35°	06:00	06:31	06:58	17:16	17:43	18:13
40°	06:07	06:39	07:09	17:05	17:34	18:07
45°	06:14	06:49	07:21	16:52	17:24	18:00
S 50°	06:22	07:01	07:37	16:37	17:12	17:52
52°	06:25	07:07	07:44	16:29	17:07	17:48
54°	06:29	07:12	07:52	16:21	17:01	17:45
56°	06:33	07:19	08:01	16:12	16:55	17:41
58°	06:37	07:26	08:12	16:02	16:48	17:37
S 60°	06:42	07:34	08:23	15:50	16:40	17:32

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	05:44	07:59	10:03	22:03	21:29	20:56
N 70°	06:03	08:06	10:00	21:54	21:27	21:02
68°	06:17	08:11	09:57	21:46	21:26	21:08
66°	06:29	08:15	09:55	21:39	21:25	21:12
64°	06:39	08:19	09:54	21:34	21:25	21:16
62°	06:48	08:22	09:52	21:29	21:24	21:19
60°	06:55	08:25	09:51	21:24	21:23	21:22
N 58°	07:01	08:27	09:50	21:21	21:23	21:24
56°	07:07	08:29	09:49	21:17	21:22	21:27
54°	07:12	08:31	09:48	21:14	21:22	21:29
52°	07:16	08:33	09:47	21:11	21:21	21:31
50°	07:20	08:35	09:46	21:09	21:21	21:32
45°	07:29	08:38	09:45	21:03	21:20	21:36
N 40°	07:36	08:41	09:43	20:59	21:19	21:39
35°	07:42	08:43	09:42	20:54	21:19	21:42
30°	07:48	08:46	09:41	20:51	21:18	21:45
20°	07:57	08:49	09:40	20:44	21:17	21:49
N 10°	08:05	08:53	09:38	20:39	21:16	21:53
0°	08:13	08:56	09:37	20:34	21:15	21:56
S 10°	08:20	08:59	09:35	20:28	21:15	22:00
20°	08:28	09:02	09:34	20:22	21:14	22:03
30°	08:37	09:06	09:32	20:16	21:13	22:08
35°	08:43	09:08	09:31	20:12	21:12	22:10
40°	08:49	09:10	09:30	20:08	21:11	22:13
45°	08:55	09:13	09:29	20:02	21:10	22:16
S 50°	09:04	09:16	09:28	19:56	21:09	22:20
52°	09:07	09:18	09:27	19:53	21:09	22:22
54°	09:12	09:20	09:27	19:50	21:08	22:24
56°	09:16	09:21	09:26	19:47	21:08	22:26
58°	09:21	09:23	09:25	19:43	21:07	22:29
S 60°	09:27	09:26	09:24	19:38	21:06	22:31

Day	Sun		Mer. Pass hh:mm	Moon		Age 3-5 6-19%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
27	06:34	06:33	12:07	14:23	02:01	
28	06:33	06:32	12:07	15:06	02:45	
29	06:31	06:30	12:07	15:46	03:26	

July 30, 31, 01 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	307°53.2	219°20.1	N21°49.0	132°09.4	N02°32.8	205°37.5	N22°52.5	305°23.3	S01°29.7	Alpheratz	357°34.1	29°13.9
1	322°55.7	234°19.5	49.1	147°10.5	32.1	220°39.4	52.5	320°25.8	29.8	Ankaa	353°06.4	-42°09.7
2	337°58.1	249°18.8	49.3	162°11.5	31.5	235°41.3	52.4	335°28.3	29.8	Schedar	349°30.3	56°40.5
3	353°00.6	264°18.2	· · 49.4	177°12.6	· · 30.8	250°43.2	· · 52.4	350°30.8	· · 29.9	Diphda	348°46.7	-17°50.6
4	8°03.0	279°17.5	49.6	192°13.7	30.2	265°45.1	52.3	5°33.4	29.9	Achernar	335°19.7	-57°06.1
5	23°05.5	294°16.9	49.7	207°14.7	29.6	280°47.0	52.3	20°35.9	29.9	Hamal	327°50.6	23°35.0
6	38°08.0	309°16.3	N21°49.9	222°15.8	N02°28.9	295°48.8	N22°52.2	35°38.4	S01°30.0	Polaris	313°53.1	89°22.0
7	53°10.4	324°15.6	50.0	237°16.9	28.3	310°50.7	52.2	50°40.9	30.0	Acamar	315°11.5	-40°11.8
8	68°12.9	339°15.0	50.2	252°17.9	27.7	325°52.6	52.2	65°43.5	30.0	Menkar	314°05.7	4°11.5
9	83°15.4	354°14.3	· · 50.3	267°19.0	· · 27.0	340°54.5	· · 52.1	80°46.0	· · 30.1	Mirfak	308°27.7	49°56.9
10	98°17.8	9°13.7	50.5	282°20.1	26.4	355°56.4	52.1	95°48.5	30.1	Aldebaran	290°39.3	16°33.7
11	113°20.3	24°13.0	50.6	297°21.1	25.7	10°58.2	52.0	110°51.0	30.2	Rigel	281°03.6	-8°10.2
12	128°22.8	39°12.4	N21°50.7	312°22.2	N02°25.1	26°00.1	N22°52.0	125°53.6	S01°30.2	Capella	280°21.5	46°01.3
13	143°25.2	54°11.7	50.9	327°23.3	24.5	41°02.0	52.0	140°56.1	30.2	Bellatrix	278°22.6	6°22.5
14	158°27.7	69°11.1	51.0	342°24.3	23.8	56°03.9	51.9	155°58.6	30.3	Elnath	278°01.5	28°37.7
15	173°30.2	84°10.5	· · 51.2	357°25.4	· · 23.2	71°05.8	· · 51.9	171°01.2	· · 30.3	Alnilam	275°37.5	-1°11.0
16	188°32.6	99°09.8	51.3	12°26.4	22.6	86°07.7	51.8	186°03.7	30.3	Betelgeuse	270°51.8	7°24.8
17	203°35.1	114°09.2	51.4	27°27.5	21.9	101°09.5	51.8	201°06.2	30.4	Canopus	263°52.7	-52°42.3
18	218°37.5	129°08.5	N21°51.6	42°28.6	N02°21.3	116°11.4	N22°51.7	216°08.7	S01°30.4	Sirius	258°26.1	-16°44.9
19	233°40.0	144°07.9	51.7	57°29.6	20.6	131°13.3	51.7	231°11.3	30.5	Adhara	255°05.8	-29°00.2
20	248°42.5	159°07.2	51.8	72°30.7	20.0	146°15.2	51.7	246°13.8	30.5	Procyon	244°50.7	5°09.7
21	263°44.9	174°06.6	· · 52.0	87°31.8	· · 19.4	161°17.1	· · 51.6	261°16.3	· · 30.5	Pollux	243°17.1	27°57.9
22	278°47.4	189°05.9	52.1	102°32.8	18.7	176°19.0	51.6	276°18.8	30.6	Avior	234°15.2	-59°35.4
23	293°49.9	204°05.3	52.2	117°33.9	18.1	191°20.8	51.5	291°21.4	30.6	Suhail	222°46.4	-43°32.1
Mer.pass. 03:28		ν -0.6' d 0.2' m -4.01		ν 1.1' d -0.6' m 1.60		ν 1.9' d -0.0' m -1.91		ν 2.5' d 0.0' m 0.81		Miaplacidus	221°39.2	-69°49.3
										Alphard	217°47.6	-8°46.1
										Regulus	207°34.2	11°50.6
										Dubhe	193°40.9	61°37.0
										Denebola	182°24.7	14°25.9
										Gienah	175°43.3	-17°41.1
										Acrux	173°00.0	-63°14.7
										Gacrux	171°51.4	-57°15.6
										Alioth	166°12.7	55°49.5
										Spica	158°21.9	-11°17.7
										Alkaid	152°51.7	49°11.4
										Hadar	148°35.4	-60°30.0
										Menkent	147°57.1	-36°29.9
										Arcturus	145°47.5	19°03.1
										Rigel Kent.	139°39.6	-60°56.7
										Kochab	137°19.6	74°03.3
										Zuben'ubi	136°55.5	-16°08.9
										Alphecca	126°03.2	26°37.9
										Antares	112°15.1	-26°29.4
										Atria	107°08.5	-69°04.6
										Sabik	102°02.0	-15°45.4
										Shaula	96°09.4	-37°07.4
										Rasalhague	95°57.9	12°32.6
										Eltanin	90°41.5	51°29.3
										Kaus Aust.	83°31.5	-34°22.4
										Vega	80°32.5	38°48.5
										Nunki	75°46.8	-26°15.9
										Altair	61°59.2	8°56.2
										Peacock	53°04.4	-56°39.1
										Deneb	49°25.0	45°22.3
										Enif	33°38.0	9°59.6
										Al Na'ir	27°31.9	-46°50.0
										Fomalhaut	15°13.7	-29°29.0
										Scheat	13°44.5	28°13.3
										Markab	13°29.2	15°20.6
Jul 30 Wed		SHA		Mer.pass								
Venus		271°26.9		09:23								
Mars		184°16.2		15:10								
Jupiter		257°44.4		10:16								
Saturn		357°30.1		03:38								
Jul 31 Thu		SHA		Mer.pass								
Venus		270°12.3		09:24								
Mars		183°42.6		15:09								
Jupiter		257°30.4		10:13								
Saturn		357°31.6		03:34								
Aug 01 Fri		SHA		Mer.pass								
Venus		268°57.5		09:25								
Mars		183°09.0		15:07								
Jupiter		257°16.5		10:10								
Saturn		357°33.1		03:30								
Horizontal parallax												
Venus: 0.1												
Mars: 0.1												

Sun				Moon				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	178°22.7	N18°29.7		120°02.6	16.5'	S05°20.5	14.3'	54.9'
1	193°22.7	29.0		134°38.1	16.5'	05°34.8	14.3'	54.8'
2	208°22.7	28.4		149°13.6	16.5'	05°49.1	14.2'	54.8'
3	223°22.8	· · 27.8		163°49.1	16.5'	06°03.4	14.2'	54.8'
4	238°22.8	27.2		178°24.6	16.5'	06°17.6	14.2'	54.8'
5	253°22.8	26.6		193°00.1	16.5'	06°31.7	14.1'	54.8'
6	268°22.8	N18°26.0		207°35.6	16.5'	S06°45.9	14.1'	54.8'
7	283°22.9	25.4		222°11.1	16.5'	07°00.0	14.1'	54.7'
8	298°22.9	24.8		236°46.5	16.5'	07°14.0	14.0'	54.7'
9	313°22.9	· · 24.2		251°22.0	16.4'	07°28.1	14.0'	54.7'
10	328°22.9	23.6		265°57.4	16.4'	07°42.1	14.0'	54.7'
11	343°23.0	23.0		280°32.9	16.4'	07°56.0	13.9'	54.7'
12	358°23.0	N18°22.4		295°08.3	16.4'	S08°10.0	13.9'	54.7'
13	13°23.0	21.7		309°43.7	16.4'	08°23.8	13.8'	54.6'
14	28°23.1	21.1		324°19.1	16.4'	08°37.7	13.8'	54.6'
15	43°23.1	· · 20.5		338°54.5	16.4'	08°51.5	13.8'	54.6'
16	58°23.1	19.9		353°29.8	16.3'	09°05.2	13.7'	54.6'
17	73°23.1	19.3		8°05.2	16.3'	09°19.0	13.7'	54.6'
18	88°23.2	N18°18.7		22°40.5	16.3'	S09°32.6	13.6'	54.6'
19	103°23.2	18.1		37°15.8	16.3'	09°46.3	13.6'	54.5'
20	118°23.2	17.4		51°51.1	16.3'	09°59.9	13.5'	54.5'
21	133°23.3	· · 16.8		66°26.3	16.2'	10°13.4	13.5'	54.5'
22	148°23.3	16.2		81°01.6	16.2'	10°26.9	13.5'	54.5'
23	163°23.3	15.6		95°36.8	16.2'	10°40.4	13.4'	54.5'
SD = 15.7' d = -0.6'				SD = 15.0'				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	178°23.4	N18°15.0		110°12.0	16.2'	S10°53.8	13.4'	54.5'
1	193°23.4	14.4		124°47.1	16.1'	11°07.1	13.3'	54.5'
2	208°23.4	13.7		139°22.3	16.1'	11°20.5	13.3'	54.5'
3	223°23.4	· · 13.1		153°57.4	16.1'	11°33.7	13.2'	54.4'
4	238°23.5	12.5		168°32.5	16.0'	11°46.9	13.2'	54.4'
5	253°23.5	11.9		183°07.5	16.0'	12°00.1	13.1'	54.4'
6	268°23.5	N18°11.3		197°42.5	16.0'	S12°13.2	13.1'	54.4'
7	283°23.6	10.6		212°17.5	15.9'	12°26.3	13.0'	54.4'
8	298°23.6	10.0		226°52.4	15.9'	12°39.3	13.0'	54.4'
9	313°23.6	· · 09.4		241°27.4	15.9'	12°52.3	12.9'	54.4'
10	328°23.7	08.8		256°02.2	15.8'	13°05.2	12.9'	54.4'
11	343°23.7	08.2		270°37.1	15.8'	13°18.0	12.8'	54.4'
12	358°23.8	N18°07.5		285°11.9	15.8'	S13°30.8	12.7'	54.3'
13	13°23.8	06.9		299°46.6	15.7'	13°43.5	12.7'	54.3'
14	28°23.8	06.3		314°21.4	15.7'	13°56.2	12.6'	54.3'
15	43°23.9	· · 05.7		328°56.1	15.6'	14°08.9	12.6'	54.3'
16	58°23.9	05.0		343°30.7	15.6'	14°21.4	12.5'	54.3'
17	73°23.9	04.4		358°05.3	15.6'	14°33.9	12.5'	54.3'
18	88°24.0	N18°03.8		12°39.9	15.5'	S14°46.4	12.4'	54.3'
19	103°24.0	03.2		27°14.4	15.5'	14°58.8	12.3'	54.3'
20	118°24.0	02.5		41°48.8	15.4'	15°11.1	12.3'	54.3'
21	133°24.1	· · 01.9		56°23.3	15.4'	15°23.4	12.2'	54.3'
22	148°24.1	01.3		70°57.6	15.3'	15°35.6	12.1'	54.3'
23	163°24.1	00.6		85°31.9	15.3'	15°47.7	12.1'	54.3'
SD = 15.7' d = -0.6'				SD = 14.9'				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	178°24.2	N18°00.0		100°06.2	15.2'	S15°59.8	12.0'	54.2'
1	193°24.2	17°59.4		114°40.5	15.2'	16°11.8	12.0'	54.2'
2	208°24.3	58.7		129°14.6	15.1'	16°23.8	11.9'	54.2'
3	223°24.3	· · 58.1		143°48.8	15.1'	16°35.7	11.8'	54.2'
4	238°24.3	57.5		158°22.8	15.0'	16°47.5	11.8'	54.2'
5	253°24.4	56.9		172°56.8	15.0'	16°59.3	11.7'	54.2'
6	268°24.4	N17°56.2		187°30.8	14.9'	S17°10.9	11.6'	54.2'
7	283°24.5	55.6		202°04.7	14.9'	17°22.6	11.5'	54.2'
8	298°24.5	55.0		216°38.6	14.8'	17°34.1	11.5'	54.2'
9	313°24.5	· · 54.3		231°12.4	14.7'	17°45.6	11.4'	54.2'
10	328°24.6	53.7		245°46.1	14.7'	17°57.0	11.3'	54.2'
11	343°24.6	53.0		260°19.8	14.6'	18°08.3	11.3'	54.2'
12	358°24.7	N17°52.4		274°53.4	14.6'	S18°19.6	11.2'	54.2'
13	13°24.7	51.8		289°27.0	14.5'	18°30.8	11.1'	54.2'
14	28°24.7	51.1		304°00.5	14.4'	18°41.9	11.0'	54.2'
15	43°24.8	· · 50.5		318°33.9	14.4'	18°52.9	11.0'	54.2'
16	58°24.8	49.9		333°07.3	14.3'	19°03.9	10.9'	54.2'
17	73°24.9	49.2		347°40.6	14.3'	19°14.8	10.8'	54.2'
18	88°24.9	N17°48.6		2°13.9	14.2'	S19°25.6	10.7'	54.2'
19	103°25.0	47.9		16°47.1	14.1'	19°36.3	10.7'	54.2'
20	118°25.0	47.3		31°20.2	14.1'	19°47.0	10.6'	54.2'
21	133°25.0	· · 46.7		45°53.3	14.0'	19°57.5	10.5'	54.2'
22	148°25.1	46.0		60°26.2	13.9'	20°08.0	10.4'	54.2'
23	163°25.1	45.4		74°59.2	13.9'	20°18.4	10.3'	54.2'
SD = 15.8' d = -0.6'				SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°						
N 70°	--	--	01:19	22:45	--	--
68°	--	--	02:11	21:57	--	--
66°	--	--	02:43	21:27	--	--
64°	--	01:35	03:06	21:05	22:33	--
62°	--	02:12	03:24	20:47	21:57	--
60°	--	02:38	03:39	20:32	21:33	--
N 58°	01:26	02:57	03:52	20:20	21:13	22:42
56°	02:00	03:14	04:02	20:09	20:57	22:10
54°	02:24	03:27	04:12	20:00	20:44	21:47
52°	02:43	03:39	04:21	19:51	20:33	21:28
50°	02:58	03:49	04:28	19:44	20:22	21:13
45°	03:28	04:11	04:44	19:28	20:01	20:44
N 40°	03:50	04:27	04:58	19:15	19:45	20:22
35°	04:07	04:41	05:09	19:04	19:31	20:05
30°	04:21	04:53	05:19	18:54	19:20	19:51
20°	04:44	05:12	05:35	18:37	19:01	19:29
N 10°	05:01	05:27	05:50	18:23	18:45	19:11
0°	05:16	05:41	06:03	18:10	18:32	18:57
S 10°	05:29	05:54	06:16	17:57	18:19	18:44
20°	05:40	06:07	06:30	17:43	18:06	18:33
30°	05:52	06:21	06:46	17:27	17:52	18:21
35°	05:58	06:28	06:55	17:18	17:45	18:15
40°	06:04	06:37	07:06	17:08	17:36	18:09
45°	06:11	06:46	07:18	16:55	17:27	18:03
S 50°	06:18	06:57	07:33	16:41	17:16	17:55
52°	06:21	07:02	07:40	16:34	17:11	17:52
54°	06:25	07:08	07:47	16:26	17:06	17:49
56°	06:28	07:14	07:56	16:18	17:00	17:45
58°	06:32	07:21	08:05	16:08	16:53	17:41
S 60°	06:37	07:28	08:17	15:57	16:46	17:37

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	12:06	14:26		20:19	19:26	
N 70°	11:53	13:56	16:45	20:35	19:58	18:38
68°	11:43	13:33	15:43	20:47	20:22	19:42
66°	11:34	13:16	15:09	20:58	20:41	20:17
64°	11:27	13:02	14:44	21:06	20:56	20:43
62°	11:21	12:51	14:25	21:14	21:09	21:03
60°	11:15	12:41	14:09	21:21	21:20	21:20
N 58°	11:11	12:32	13:56	21:26	21:29	21:34
56°	11:07	12:25	13:44	21:32	21:38	21:46
54°	11:03	12:18	13:34	21:36	21:45	21:56
52°	11:00	12:12	13:26	21:41	21:52	22:06
50°	10:56	12:07	13:18	21:44	21:58	22:14
45°	10:50	11:55	13:01	21:53	22:11	22:32
N 40°	10:45	11:46	12:47	22:00	22:22	22:47
35°	10:40	11:38	12:36	22:06	22:31	22:59
30°	10:36					

August 02, 03, 04 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	310°50.6	218°33.2	N21°57.4	133°26.0	N01°46.7	207°53.2	N22°49.4	308°25.4	S01°32.6	Alpheratz	357°34.1 29°13.9
1	325°53.1	233°32.5	57.4	148°27.0	46.1	222°55.1	49.4	323°27.9	32.6	Ankaa	353°06.4 -42°09.7
2	340°55.5	248°31.9	57.5	163°28.1	45.5	237°57.0	49.3	338°30.5	32.7	Schedar	349°30.2 56°40.5
3	355°58.0	263°31.2	•• 57.6	178°29.1	•• 44.8	252°58.9	•• 49.3	353°33.0	•• 32.7	Diphda	348°46.6 -17°50.6
4	11°00.5	278°30.6	57.7	193°30.2	44.2	268°00.8	49.2	8°35.5	32.8	Achernar	335°19.7 -57°06.1
5	26°02.9	293°29.9	57.8	208°31.3	43.5	283°02.7	49.2	23°38.1	32.8	Hamal	327°50.6 23°35.0
6	41°05.4	308°29.2	N21°57.8	223°32.3	N01°42.9	298°04.6	N22°49.2	38°40.6	S01°32.8	Polaris	313°51.4 89°22.0
7	56°07.9	323°28.6	57.9	238°33.4	42.3	313°06.4	49.1	53°43.1	32.9	Acamar	315°11.4 -40°11.8
8	71°10.3	338°27.9	58.0	253°34.4	41.6	328°08.3	49.1	68°45.7	32.9	Menkar	314°05.7 4°11.5
9	86°12.8	353°27.2	•• 58.1	268°35.5	•• 41.0	343°10.2	•• 49.0	83°48.2	•• 33.0	Mirfak	308°27.7 49°56.9
10	101°15.2	8°26.6	58.1	283°36.6	40.3	358°12.1	49.0	98°50.7	33.0	Aldebaran	290°39.2 16°33.7
11	116°17.7	23°25.9	58.2	298°37.6	39.7	13°14.0	48.9	113°53.3	33.1	Rigel	281°03.6 -8°10.2
12	131°20.2	38°25.2	N21°58.3	313°38.7	N01°39.0	28°15.9	N22°48.9	128°55.8	S01°33.1	Capella	280°21.5 46°01.3
13	146°22.6	53°24.6	58.3	328°39.7	38.4	43°17.8	48.8	143°58.3	33.1	Bellatrix	278°22.6 6°22.5
14	161°25.1	68°23.9	58.4	343°40.8	37.8	58°19.7	48.8	159°00.9	33.2	Elnath	278°01.5 28°37.7
15	176°27.6	83°23.3	•• 58.5	358°41.8	•• 37.1	73°21.6	•• 48.8	174°03.4	•• 33.2	Alnilam	275°37.5 -1°11.0
16	191°30.0	98°22.6	58.5	13°42.9	36.5	88°23.4	48.7	189°06.0	33.3	Betelgeuse	270°51.8 7°24.8
17	206°32.5	113°21.9	58.6	28°44.0	35.8	103°25.3	48.7	204°08.5	33.3	Canopus	263°52.6 -52°42.3
18	221°35.0	128°21.3	N21°58.7	43°45.0	N01°35.2	118°27.2	N22°48.6	219°11.0	S01°33.4	Sirius	258°26.1 -16°44.9
19	236°37.4	143°20.6	58.7	58°46.1	34.6	133°29.1	48.6	234°13.6	33.4	Adhara	255°05.8 -29°00.2
20	251°39.9	158°19.9	58.8	73°47.1	33.9	148°31.0	48.5	249°16.1	33.4	Procyon	244°50.7 5°09.7
21	266°42.4	173°19.3	•• 58.8	88°48.2	•• 33.3	163°32.9	•• 48.5	264°18.6	•• 33.5	Pollux	243°17.1 27°57.9
22	281°44.8	188°18.6	58.9	103°49.3	32.6	178°34.8	48.5	279°21.2	33.5	Avior	234°15.2 -59°35.4
23	296°47.3	203°17.9	59.0	118°50.3	32.0	193°36.7	48.4	294°23.7	33.6	Suhail	222°46.4 -43°32.1
Mer.pass. 03:16		ν -0.7' d 0.1' m -4.00		ν 1.1' d -0.6' m 1.60		ν 1.9' d -0.0' m -1.92		ν 2.5' d 0.0' m 0.79		Miaplacidus	221°39.2 -69°49.3
										Alphard	217°47.6 -8°46.1
										Regulus	207°34.2 11°50.6
										Dubhe	193°40.9 61°37.0
										Denebola	182°24.7 14°25.9
										Gienah	175°43.3 -17°41.1
										Acrux	173°00.0 -63°14.7
										Gacrux	171°51.5 -57°15.6
										Alioth	166°12.8 55°49.5
										Spica	158°21.9 -11°17.7
										Alkaid	152°51.7 49°11.4
										Hadar	148°35.4 -60°30.0
										Menkent	147°57.1 -36°29.9
										Arcturus	145°47.5 19°03.1
										Rigel Kent.	139°39.7 -60°56.7
										Kochab	137°19.7 74°03.3
										Zuben'ubi	136°55.5 -16°08.9
										Alphecca	126°03.2 26°37.9
										Antares	112°15.1 -26°29.4
										Atria	107°08.5 -69°04.6
										Sabik	102°02.0 -15°45.4
										Shaula	96°09.4 -37°07.4
										Rasalhague	95°57.9 12°32.6
										Eltanin	90°41.6 51°29.3
										Kaus Aust.	83°31.5 -34°22.4
										Vega	80°32.5 38°48.6
										Nunki	75°46.8 -26°15.9
										Altair	61°59.2 8°56.2
										Peacock	53°04.4 -56°39.1
										Deneb	49°25.0 45°22.3
										Enif	33°38.0 9°59.6
										Al Na'ir	27°31.9 -46°50.0
										Fomalhaut	15°13.7 -29°29.0
										Scheat	13°44.5 28°13.3
										Markab	13°29.2 15°20.6
										Aug 02 Sat	SHA Mer.pass
										Venus	267°42.6 09:26
										Mars	182°35.4 15:05
										Jupiter	257°02.6 10:07
										Saturn	357°34.8 03:26
										Aug 03 Sun	SHA Mer.pass
										Venus	266°27.5 09:27
										Mars	182°01.6 15:04
										Jupiter	256°48.8 10:04
										Saturn	357°36.5 03:22
										Aug 04 Mon	SHA Mer.pass
										Venus	265°12.3 09:28
										Mars	181°27.9 15:02
										Jupiter	256°35.1 10:01
										Saturn	357°38.3 03:18
										Horizontal parallax	
										Venus:	0.1
										Mars:	0.1
Mer.pass. 03:08		ν -0.7' d 0.0' m -3.99		ν 1.1' d -0.6' m 1.61		ν 1.9' d -0.0' m -1.92		ν 2.5' d 0.0' m 0.78			

h	Sun			Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°25.2	N17°44.7	89°32.0	13.8'	S20°28.8	10.2'	54.2'
1	193°25.2	44.1	104°04.8	13.7'	20°39.0	10.2'	54.2'
2	208°25.3	43.5	118°37.6	13.7'	20°49.2	10.1'	54.2'
3	223°25.3	· · 42.8	133°10.2	13.6'	20°59.2	10.0'	54.2'
4	238°25.4	42.2	147°42.8	13.5'	21°09.2	9.9'	54.2'
5	253°25.4	41.5	162°15.3	13.4'	21°19.1	9.8'	54.2'
6	268°25.4	N17°40.9	176°47.8	13.4'	S21°29.0	9.7'	54.2'
7	283°25.5	40.2	191°20.1	13.3'	21°38.7	9.6'	54.2'
8	298°25.5	39.6	205°52.4	13.2'	21°48.3	9.5'	54.2'
9	313°25.6	· · 38.9	220°24.7	13.2'	21°57.9	9.5'	54.2'
10	328°25.6	38.3	234°56.8	13.1'	22°07.3	9.4'	54.2'
11	343°25.7	37.7	249°28.9	13.0'	22°16.7	9.3'	54.2'
12	358°25.7	N17°37.0	264°00.9	12.9'	S22°26.0	9.2'	54.2'
13	13°25.8	36.4	278°32.9	12.9'	22°35.1	9.1'	54.2'
14	28°25.8	35.7	293°04.7	12.8'	22°44.2	9.0'	54.2'
15	43°25.9	· · 35.1	307°36.5	12.7'	22°53.2	8.9'	54.2'
16	58°25.9	34.4	322°08.2	12.6'	23°02.1	8.8'	54.2'
17	73°26.0	33.8	336°39.9	12.6'	23°10.9	8.7'	54.2'
18	88°26.0	N17°33.1	351°11.4	12.5'	S23°19.6	8.6'	54.2'
19	103°26.1	32.5	5°42.9	12.4'	23°28.2	8.5'	54.3'
20	118°26.1	31.8	20°14.3	12.3'	23°36.7	8.4'	54.3'
21	133°26.2	· · 31.2	34°45.7	12.3'	23°45.1	8.3'	54.3'
22	148°26.2	30.5	49°16.9	12.2'	23°53.4	8.2'	54.3'
23	163°26.3	29.8	63°48.1	12.1'	24°01.6	8.1'	54.3'
SD = 15.8'			SD = 14.8'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°26.3	N17°29.2	78°19.2	12.0'	S24°09.7	8.0'	54.3'
1	193°26.4	28.5	92°50.3	12.0'	24°17.6	7.9'	54.3'
2	208°26.4	27.9	107°21.2	11.9'	24°25.5	7.8'	54.3'
3	223°26.5	· · 27.2	121°52.1	11.8'	24°33.3	7.7'	54.3'
4	238°26.5	26.6	136°22.9	11.7'	24°41.0	7.6'	54.3'
5	253°26.6	25.9	150°53.6	11.6'	24°48.5	7.4'	54.3'
6	268°26.6	N17°25.3	165°24.3	11.6'	S24°56.0	7.3'	54.3'
7	283°26.7	24.6	179°54.8	11.5'	25°03.3	7.2'	54.4'
8	298°26.7	23.9	194°25.3	11.4'	25°10.5	7.1'	54.4'
9	313°26.8	· · 23.3	208°55.7	11.3'	25°17.7	7.0'	54.4'
10	328°26.8	22.6	223°26.1	11.3'	25°24.7	6.9'	54.4'
11	343°26.9	22.0	237°56.4	11.2'	25°31.6	6.8'	54.4'
12	358°27.0	N17°21.3	252°26.5	11.1'	S25°38.3	6.7'	54.4'
13	13°27.0	20.7	266°56.7	11.0'	25°45.0	6.5'	54.4'
14	28°27.1	20.0	281°26.7	11.0'	25°51.5	6.4'	54.4'
15	43°27.1	· · 19.3	295°56.7	10.9'	25°58.0	6.3'	54.4'
16	58°27.2	18.7	310°26.5	10.8'	26°04.3	6.2'	54.5'
17	73°27.2	18.0	324°56.4	10.7'	26°10.5	6.1'	54.5'
18	88°27.3	N17°17.3	339°26.1	10.7'	S26°16.5	5.9'	54.5'
19	103°27.3	16.7	353°55.8	10.6'	26°22.5	5.8'	54.5'
20	118°27.4	16.0	8°25.3	10.5'	26°28.3	5.7'	54.5'
21	133°27.5	· · 15.4	22°54.9	10.4'	26°34.0	5.6'	54.5'
22	148°27.5	14.7	37°24.3	10.4'	26°39.6	5.5'	54.5'
23	163°27.6	14.0	51°53.7	10.3'	26°45.1	5.3'	54.6'
SD = 15.8'			SD = 14.8'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	178°27.6	N17°13.4	66°23.0	10.2'	S26°50.4	5.2'	54.6'
1	193°27.7	12.7	80°52.2	10.2'	26°55.6	5.1'	54.6'
2	208°27.7	12.0	95°21.3	10.1'	27°00.7	5.0'	54.6'
3	223°27.8	· · 11.4	109°50.4	10.0'	27°05.6	4.8'	54.6'
4	238°27.9	10.7	124°19.4	9.9'	27°10.4	4.7'	54.6'
5	253°27.9	10.0	138°48.4	9.9'	27°15.1	4.6'	54.6'
6	268°28.0	N17°09.4	153°17.3	9.8'	S27°19.7	4.4'	54.7'
7	283°28.0	08.7	167°46.1	9.7'	27°24.1	4.3'	54.7'
8	298°28.1	08.0	182°14.8	9.7'	27°28.4	4.2'	54.7'
9	313°28.2	· · 07.4	196°43.5	9.6'	27°32.6	4.0'	54.7'
10	328°28.2	06.7	211°12.1	9.5'	27°36.6	3.9'	54.7'
11	343°28.3	06.0	225°40.7	9.5'	27°40.5	3.8'	54.7'
12	358°28.3	N17°05.3	240°09.1	9.4'	S27°44.3	3.6'	54.8'
13	13°28.4	04.7	254°37.6	9.4'	27°47.9	3.5'	54.8'
14	28°28.5	04.0	269°05.9	9.3'	27°51.4	3.4'	54.8'
15	43°28.5	· · 03.3	283°34.2	9.2'	27°54.8	3.2'	54.8'
16	58°28.6	02.7	298°02.5	9.2'	27°58.0	3.1'	54.8'
17	73°28.6	02.0	312°30.6	9.1'	28°01.0	2.9'	54.8'
18	88°28.7	N17°01.3	326°58.8	9.1'	S28°04.0	2.8'	54.9'
19	103°28.8	00.6	341°26.8	9.0'	28°06.8	2.7'	54.9'
20	118°28.8	17°00.0	355°54.8	9.0'	28°09.4	2.5'	54.9'
21	133°28.9	16°59.3	10°22.8	8.9'	28°11.9	2.4'	54.9'
22	148°29.0	58.6	24°50.7	8.8'	28°14.3	2.2'	54.9'
23	163°29.0	57.9	39°18.6	8.8'	28°16.5	2.1'	55.0'
SD = 15.8'			SD = 14.9'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	☐	☐	☐	☐	☐	☐
N 70°	—:—	—:—	01:45	22:21	—:—	—:—
68°	—:—	—:—	02:26	21:42	—:—	—:—
66°	—:—	00:56	02:54	21:15	23:06	—:—
64°	—:—	01:52	03:15	20:55	22:16	—:—
62°	—:—	02:24	03:32	20:38	21:45	—:—
60°	00:51	02:47	03:46	20:25	21:23	23:12
N 58°	01:42	03:05	03:58	20:13	21:05	22:27
56°	02:11	03:20	04:08	20:03	20:50	21:59
54°	02:32	03:33	04:17	19:54	20:38	21:38
52°	02:50	03:44	04:25	19:46	20:27	21:21
50°	03:04	03:54	04:32	19:39	20:17	21:07
45°	03:32	04:14	04:48	19:24	19:57	20:39
N 40°	03:53	04:30	05:00	19:11	19:41	20:18
35°	04:10	04:43	05:11	19:01	19:28	20:02
30°	04:24	04:55	05:20	18:52	19:17	19:48
20°	04:45	05:13	05:36	18:36	18:59	19:27
N 10°	05:02	05:28	05:50	18:22	18:44	19:10
0°	05:16	05:41	06:03	18:10	18:31	18:56
S 10°	05:28	05:53	06:15	17:57	18:19	18:44
20°	05:39	06:06	06:29	17:44	18:07	18:33
30°	05:50	06:19	06:44	17:29	17:54	18:23
35°	05:56	06:26	06:53	17:20	17:47	18:17
40°	06:01	06:34	07:03	17:10	17:39	18:11
45°	06:08	06:43	07:14	16:59	17:30	18:05
S 50°	06:14	06:53	07:28	16:45	17:20	17:59
52°	06:17	06:58	07:35	16:38	17:15	17:56
54°	06:20	07:03	07:42	16:31	17:10	17:53
56°	06:24	07:09	07:50	16:23	17:04	17:50
58°	06:27	07:15	07:59	16:14	16:58	17:46
S 60°	06:31	07:22	08:09	16:04	16:51	17:42

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	17:39	■	■	19:22	■	■
64°	16:39	■	■	20:22	■	■
62°	16:06	17:57	■	20:56	20:45	■
60°	15:41	17:16	18:47	21:21	21:27	21:44
N 58°	15:22	16:48	18:08	21:41	21:55	22:22
56°	15:06	16:27	17:41	21:58	22:17	22:50
54°	14:52	16:09	17:20	22:12	22:35	23:11
52°	14:40	15:54	17:03	22:24	22:51	23:29
50°	14:30	15:41	16:48	22:35	23:04	23:44
45°	14:08	15:14	16:18	22:58	23:31	· · ·
N 40°	13:50	14:53	15:54	23:17	23:53	· · ·
35°	13:35	14:36	15:35	23:32	· · ·	00:11
30°	13:23	14:21	15:18	23:46	· · ·	00:27
20°	13:01	13:55	14:50	· · ·	00:09	00:53
N 10°	12:42	13:33	14:26	· · ·	00:29	01:16
0°	12:25	13:13	14:04	00:02	00:48	01:37
S 10°	12:08	12:53	13:42	00:17	01:07	01:59
20°	11:50	12:31	13:18	00:34	01:27	02:22
30°	11:28	12:07	12:51	00:53	01:51	02:48
35°	11:16	11:52	12:35	01:05	02:04	03:04
40°	11:02	11:35	12:16	01:18	02:20	03:23
45°	10:46	11:15	11:53	01:33	02:40	03:45
S 50°	10:25	10:50	11:24	01:52	03:04	04:13
52°	10:16	10:38	11:10	02:01	03:16	04:27
54°	10:05	10:24	10:53	02:12	03:29	04:44
56°	09:53	10:08	10:34	02:23	03:44	05:03
58°	09:39	09:49	10:10	02:37	04:03	05:27
S 60°	09:22	09:26	09:37	02:52	04:26	05:59

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	9-11
				hh:mm	hh:mm	55-73%
02	06:19	06:17	12:06	18:36	06:13	
03	06:15	06:12	12:06	19:25	07:00	
04	06:09	06:07	12:06	20:17	07:51	

August 05, 06, 07 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	313°48.0	217°45.0	N22°00.6	134°42.0	N01°00.5	210°09.4	N22°46.2	311°28.3	S01°35.8	Alpheratz	357°34.0 29°13.9
1	328°50.5	232°44.4	00.6	149°43.1	00°59.8	225°11.3	46.2	326°30.8	35.8	Ankaa	353°06.4 -42°09.7
2	343°53.0	247°43.7	00.6	164°44.1	59.2	240°13.2	46.1	341°33.4	35.8	Schedar	349°30.2 56°40.5
3	358°55.4	262°43.0	•• 00.6	179°45.2	•• 58.5	255°15.1	•• 46.1	356°35.9	•• 35.9	Diphda	348°46.6 -17°50.6
4	13°57.9	277°42.3	00.6	194°46.3	57.9	270°17.0	46.0	11°38.4	35.9	Achernar	335°19.6 -57°06.1
5	29°00.3	292°41.6	00.6	209°47.3	57.2	285°18.9	46.0	26°41.0	36.0	Hamal	327°50.6 23°35.0
6	44°02.8	307°41.0	N22°00.6	224°48.4	N00°56.6	300°20.8	N22°45.9	41°43.5	S01°36.0	Polaris	313°49.6 89°22.0
7	59°05.3	322°40.3	00.6	239°49.4	56.0	315°22.7	45.9	56°46.1	36.1	Acamar	315°11.4 -40°11.8
8	74°07.7	337°39.6	00.6	254°50.5	55.3	330°24.6	45.8	71°48.6	36.1	Menkar	314°05.7 4°11.5
9	89°10.2	352°38.9	•• 00.6	269°51.5	•• 54.7	345°26.5	•• 45.8	86°51.2	•• 36.2	Mirfak	308°27.6 49°57.0
10	104°12.7	7°38.3	00.6	284°52.6	54.0	0°28.4	45.7	101°53.7	36.2	Aldebaran	290°39.2 16°33.7
11	119°15.1	22°37.6	00.6	299°53.6	53.4	15°30.3	45.7	116°56.3	36.3	Rigel	281°03.6 -8°10.2
12	134°17.6	37°36.9	N22°00.6	314°54.7	N00°52.7	30°32.2	N22°45.7	131°58.8	S01°36.3	Capella	280°21.5 46°01.3
13	149°20.1	52°36.2	00.6	329°55.7	52.1	45°34.0	45.6	147°01.4	36.4	Bellatrix	278°22.6 6°22.5
14	164°22.5	67°35.5	00.6	344°56.8	51.4	60°35.9	45.6	162°03.9	36.4	Elnath	278°01.5 28°37.7
15	179°25.0	82°34.9	•• 00.6	359°57.8	•• 50.8	75°37.8	•• 45.5	177°06.5	•• 36.5	Alnilam	275°37.5 -1°11.0
16	194°27.5	97°34.2	00.6	14°58.9	50.1	90°39.7	45.5	192°09.0	36.5	Betelgeuse	270°51.8 7°24.8
17	209°29.9	112°33.5	00.6	29°59.9	49.5	105°41.6	45.4	207°11.5	36.5	Canopus	263°52.6 -52°42.3
18	224°32.4	127°32.8	N22°00.6	45°01.0	N00°48.9	120°43.5	N22°45.4	222°14.1	S01°36.6	Sirius	258°26.1 -16°44.9
19	239°34.8	142°32.1	00.6	60°02.0	48.2	135°45.4	45.3	237°16.6	36.6	Adhara	255°05.8 -29°00.2
20	254°37.3	157°31.5	00.5	75°03.1	47.6	150°47.3	45.3	252°19.2	36.7	Procyon	244°50.6 5°09.7
21	269°39.8	172°30.8	•• 00.5	90°04.1	•• 46.9	165°49.2	•• 45.2	267°21.7	•• 36.7	Pollux	243°17.1 27°57.9
22	284°42.2	187°30.1	00.5	105°05.2	46.3	180°51.1	45.2	282°24.3	36.8	Avior	234°15.2 -59°35.4
23	299°44.7	202°29.4	00.5	120°06.2	45.6	195°53.0	45.1	297°26.8	36.8	Suhail	222°46.4 -43°32.1
Mer.pass. 03:04		ν -0.7' d 0.0' m -3.99		ν 1.1' d -0.6' m 1.61		ν 1.9' d -0.0' m -1.92		ν 2.5' d 0.0' m 0.78		Miaplacidus	221°39.2 -69°49.3
										Alphard	217°47.6 -8°46.1
										Regulus	207°34.2 11°50.6
										Dubhe	193°40.9 61°37.0
										Denebola	182°24.7 14°25.9
										Gienah	175°43.3 -17°41.1
										Acrux	173°00.0 -63°14.7
										Gacrux	171°51.5 -57°15.6
										Alioth	166°12.8 55°49.5
										Spica	158°21.9 -11°17.7
										Alkaid	152°51.7 49°11.4
										Hadar	148°35.5 -60°30.0
										Menkent	147°57.1 -36°29.9
										Arcturus	145°47.5 19°03.1
										Rigel Kent.	139°39.7 -60°56.7
										Kochab	137°19.7 74°03.3
										Zuben'ubi	136°55.5 -16°08.9
										Alphecca	126°03.2 26°37.9
										Antares	112°15.1 -26°29.4
										Atria	107°08.5 -69°04.6
										Sabik	102°02.0 -15°45.4
										Shaula	96°09.4 -37°07.4
										Rasalhague	95°57.9 12°32.6
										Eltanin	90°41.6 51°29.3
										Kaus Aust.	83°31.5 -34°22.4
										Vega	80°32.5 38°48.6
										Nunki	75°46.8 -26°15.9
										Altair	61°59.2 8°56.2
										Peacock	53°04.3 -56°39.1
										Deneb	49°25.0 45°22.3
										Enif	33°38.0 9°59.6
										Al Na'ir	27°31.8 -46°50.1
										Fomalhaut	15°13.7 -29°29.0
										Scheat	13°44.4 28°13.3
										Markab	13°29.1 15°20.6
										Aug 05 Tue SHA Mer.pass	
										Venus	263°57.0 09:29
										Mars	180°54.0 15:00
										Jupiter	256°21.4 09:58
										Saturn	357°40.2 03:14
										Aug 06 Wed SHA Mer.pass	
										Venus	262°41.6 09:31
										Mars	180°20.1 14:58
										Jupiter	256°07.7 09:55
										Saturn	357°42.2 03:10
										Aug 07 Thu SHA Mer.pass	
										Venus	261°26.1 09:32
										Mars	179°46.2 14:57
										Jupiter	255°54.2 09:52
										Saturn	357°44.3 03:05
										Horizontal parallax	
										Venus:	0.1
										Mars:	0.1
Mer.pass. 02:56		ν -0.7' d -0.0' m -3.99		ν 1.0' d -0.6' m 1.61		ν 1.9' d -0.0' m -1.93		ν 2.6' d 0.0' m 0.77			

Sun							Moon							
Tue	GHA	Dec	GHA	ν	Dec	d	HP							
0	178°29.1	N16°57.3	53°46.4	8.7'	S28°18.6	1.9'	55.0'							
1	193°29.1	56.6	68°14.1	8.7'	28°20.5	1.8'	55.0'							
2	208°29.2	55.9	82°41.8	8.7'	28°22.3	1.6'	55.0'							
3	223°29.3	· · 55.2	97°09.5	8.6'	28°24.0	1.5'	55.0'							
4	238°29.3	54.5	111°37.1	8.6'	28°25.5	1.4'	55.1'							
5	253°29.4	53.9	126°04.6	8.5'	28°26.8	1.2'	55.1'							
6	268°29.5	N16°53.2	140°32.1	8.5'	S28°28.0	1.1'	55.1'							
7	283°29.5	52.5	154°59.6	8.4'	28°29.1	0.9'	55.1'							
8	298°29.6	51.8	169°27.1	8.4'	28°30.0	0.8'	55.1'							
9	313°29.7	· · 51.1	183°54.4	8.4'	28°30.7	0.6'	55.2'							
10	328°29.7	50.5	198°21.8	8.3'	28°31.3	0.5'	55.2'							
11	343°29.8	49.8	212°49.1	8.3'	28°31.8	0.3'	55.2'							
12	358°29.9	N16°49.1	227°16.4	8.3'	S28°32.1	0.2'	55.2'							
13	13°29.9	48.4	241°43.7	8.2'	28°32.3	0.0'	55.3'							
14	28°30.0	47.7	256°10.9	8.2'	28°32.3	-0.1'	55.3'							
15	43°30.1	· · 47.0	270°38.1	8.2'	28°32.1	-0.3'	55.3'							
16	58°30.1	46.4	285°05.2	8.1'	28°31.8	-0.5'	55.3'							
17	73°30.2	45.7	299°32.4	8.1'	28°31.4	-0.6'	55.4'							
18	88°30.3	N16°45.0	313°59.5	8.1'	S28°30.8	-0.8'	55.4'							
19	103°30.3	44.3	328°26.5	8.1'	28°30.0	-0.9'	55.4'							
20	118°30.4	43.6	342°53.6	8.0'	28°29.1	-1.1'	55.4'							
21	133°30.5	· · 42.9	357°20.6	8.0'	28°28.0	-1.2'	55.4'							
22	148°30.6	42.2	11°47.7	8.0'	28°26.8	-1.4'	55.5'							
23	163°30.6	41.6	26°14.7	8.0'	28°25.4	-1.5'	55.5'							
SD = 15.8' $d = -0.7'$							SD = 15.0'							
Wed	GHA	Dec	GHA	ν	Dec	d	HP							
0	178°30.7	N16°40.9	40°41.6	8.0'	S28°23.9	-1.7'	55.5'							
1	193°30.8	40.2	55°08.6	7.9'	28°22.2	-1.8'	55.5'							
2	208°30.8	39.5	69°35.5	7.9'	28°20.3	-2.0'	55.6'							
3	223°30.9	· · 38.8	84°02.5	7.9'	28°18.3	-2.2'	55.6'							
4	238°31.0	38.1	98°29.4	7.9'	28°16.2	-2.3'	55.6'							
5	253°31.0	37.4	112°56.3	7.9'	28°13.9	-2.5'	55.6'							
6	268°31.1	N16°36.7	127°23.2	7.9'	S28°11.4	-2.6'	55.7'							
7	283°31.2	36.0	141°50.1	7.9'	28°08.8	-2.8'	55.7'							
8	298°31.3	35.4	156°17.0	7.9'	28°06.0	-2.9'	55.7'							
9	313°31.3	· · 34.7	170°43.9	7.9'	28°03.1	-3.1'	55.7'							
10	328°31.4	34.0	185°10.8	7.9'	28°00.0	-3.2'	55.8'							
11	343°31.5	33.3	199°37.7	7.9'	27°56.7	-3.4'	55.8'							
12	358°31.6	N16°32.6	214°04.6	7.9'	S27°53.3	-3.6'	55.8'							
13	13°31.6	31.9	228°31.5	7.9'	27°49.8	-3.7'	55.8'							
14	28°31.7	31.2	242°58.4	7.9'	27°46.1	-3.9'	55.9'							
15	43°31.8	· · 30.5	257°25.3	7.9'	27°42.2	-4.0'	55.9'							
16	58°31.9	29.8	271°52.2	7.9'	27°38.2	-4.2'	55.9'							
17	73°31.9	29.1	286°19.1	7.9'	27°34.0	-4.3'	55.9'							
18	88°32.0	N16°28.4	300°46.1	7.9'	S27°29.7	-4.5'	56.0'							
19	103°32.1	27.7	315°13.0	8.0'	27°25.2	-4.6'	56.0'							
20	118°32.2	27.0	329°40.0	8.0'	27°20.5	-4.8'	56.0'							
21	133°32.2	· · 26.3	344°06.9	8.0'	27°15.7	-5.0'	56.1'							
22	148°32.3	25.6	358°33.9	8.0'	27°10.8	-5.1'	56.1'							
23	163°32.4	24.9	13°00.9	8.0'	27°05.7	-5.3'	56.1'							
SD = 15.8' $d = -0.7'$							SD = 15.1'							
Thu	GHA	Dec	GHA	ν	Dec	d	HP							
0	178°32.5	N16°24.2	27°27.9	8.0'	S27°00.4	-5.4'	56.1'							
1	193°32.5	23.5	41°55.0	8.1'	26°55.0	-5.6'	56.2'							
2	208°32.6	22.8	56°22.0	8.1'	26°49.4	-5.7'	56.2'							
3	223°32.7	· · 22.1	70°49.1	8.1'	26°43.7	-5.9'	56.2'							
4	238°32.8	21.4	85°16.2	8.1'	26°37.8	-6.0'	56.2'							
5	253°32.8	20.7	99°43.4	8.2'	26°31.8	-6.2'	56.3'							
6	268°32.9	N16°20.0	114°10.5	8.2'	S26°25.6	-6.3'	56.3'							
7	283°33.0	19.3	128°37.7	8.2'	26°19.3	-6.5'	56.3'							
8	298°33.1	18.6	143°05.0	8.3'	26°12.8	-6.6'	56.3'							
9	313°33.2	· · 17.9	157°32.2	8.3'	26°06.2	-6.8'	56.4'							
10	328°33.2	17.2	171°59.5	8.3'	25°59.4	-6.9'	56.4'							
11	343°33.3	16.5	186°26.8	8.3'	25°52.5	-7.1'	56.4'							
12	358°33.4	N16°15.8	200°54.2	8.4'	S25°45.4	-7.2'	56.5'							
13	13°33.5	15.1	215°21.5	8.4'	25°38.2	-7.4'	56.5'							
14	28°33.6	14.4	229°49.0	8.5'	25°30.9	-7.5'	56.5'							
15	43°33.6	· · 13.7	244°16.4	8.5'	25°23.3	-7.7'	56.5'							
16	58°33.7	13.0	258°43.9	8.5'	25°15.7	-7.8'	56.6'							
17	73°33.8	12.3	273°11.4	8.6'	25°07.9	-8.0'	56.6'							
18	88°33.9	N16°11.6	287°39.0	8.6'	S24°59.9	-8.1'	56.6'							
19	103°34.0	10.9	302°06.6	8.7'	24°51.8	-8.2'	56.6'							
20	118°34.0	10.1	316°34.3	8.7'	24°43.6	-8.4'	56.7'							
21	133°34.1	· · 09.4	331°02.0	8.7'	24°35.2	-8.5'	56.7'							
22	148°34.2	08.7	345°29.7	8.8'	24°26.7	-8.7'	56.7'							
23	163°34.3	08.0	359°57.5	8.8'	24°18.0	-8.8'	56.8'							
SD = 15.8' $d = -0.7'$							SD = 15.3'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	01:02	22:57	—:—	—:—
N 70°	—:—	—:—	02:06	22:01	—:—	—:—
68°	—:—	—:—	02:41	21:27	—:—	—:—
66°	—:—	01:25	03:05	21:04	22:39	—:—
64°	—:—	02:08	03:25	20:45	22:00	—:—
62°	—:—	02:35	03:40	20:30	21:33	—:—
60°	01:17	02:56	03:53	20:17	21:13	22:49
N 58°	01:55	03:13	04:04	20:06	20:57	22:13
56°	02:21	03:27	04:14	19:57	20:43	21:48
54°	02:41	03:39	04:22	19:48	20:31	21:29
52°	02:57	03:50	04:30	19:41	20:21	21:13
50°	03:10	03:59	04:37	19:34	20:12	21:00
45°	03:37	04:18	04:51	19:20	19:53	20:34
N 40°	03:57	04:33	05:03	19:08	19:38	20:14
35°	04:13	04:46	05:13	18:58	19:25	19:58
30°	04:26	04:57	05:22	18:49	19:15	19:45
20°	04:47	05:14	05:37	18:34	18:58	19:25
N 10°	05:03	05:28	05:50	18:21	18:43	19:09
0°	05:16	05:41	06:02	18:09	18:31	18:56
S 10°	05:27	05:53	06:14	17:57	18:19	18:45
20°	05:38	06:04	06:27	17:45	18:08	18:34
30°	05:48	06:17	06:42	17:31	17:56	18:24
35°	05:53	06:23	06:50	17:22	17:49	18:19
40°	05:58	06:31	06:59	17:13	17:42	18:14
45°	06:04	06:39	07:10	17:02	17:33	18:08
S 50°	06:10	06:49	07:23	16:49	17:24	18:02
52°	06:13	06:53	07:30	16:43	17:19	18:00
54°	06:15	06:58	07:36	16:36	17:15	17:57
56°	06:18	07:03	07:44	16:29	17:09	17:54
58°	06:22	07:09	07:52	16:20	17:04	17:51
S 60°	06:25	07:15	08:02	16:10	16:57	17:48

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	■	■	■	■	■	■
64°	■	■	21:37	■	■	■
62°	■	21:11	20:56	■	23:08	· ·
60°	19:51	20:18	20:27	22:32	· ·	00:00
N 58°	19:10	19:46	20:05	23:14	· ·	00:32
56°	18:41	19:22	19:47	23:42	· ·	00:55
54°	18:19	19:03	19:32	· ·	00:04	01:15
52°	18:01	18:46	19:18	· ·	00:22	01:31
50°	17:46	18:32	19:07	· ·	00:37	01:45
45°	17:15	18:03	18:42	00:14	01:08	02:13
N 40°	16:51	17:41	18:23	00:38	01:32	02:35
35°	16:31	17:22	18:06	00:58	01:52	02:53
30°	16:14	17:06	17:52	01:14	02:09	03:09
20°	15:45	16:38	17:28	01:43	02:37	03:35
N 10°	15:21	16:15	17:07	02:07	03:02	03:58
0°	14:58	15:53	16:47	02:30	03:24	04:19
S 10°	14:35	15:31	16:28	02:53	03:47	04:40
20°	14:11	15:07	16:06	03:17	04:11	05:03
30°	13:42	14:40	15:42	03:45	04:39	05:28
35°	13:25	14:23	15:27	04:02	04:56	05:44
40°	13:05	14:04	15:10	04:22	05:15	06:01
45°	12:41	13:41	14:50	04:46	05:39	06:22
S 50°	12:11	13:11	14:24	05:16	06:09	06:49
52°	11:55	12:57	14:12	05:32	06:23	07:02
54°	11:37	12:40	13:58	05:49	06:41	07:16
56°	11:16	12:19	13:41	06:11	07:01	07:33
58°	10:48	11:54	13:21	06:39	07:27	07:54
S 60°	10:08	11:18	12:55	07:18	08:03	08:20

Day	Sun		Mer. Pass hh:mm	Moon		Age 12-14 81-94%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
05	06:04	06:01	12:06	21:11	08:44	
06	05:57	05:54	12:06	22:06	09:38	
07	05:50	05:46	12:06	23:00	10:33	

August 08, 09, 10 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	316°45.4	216°55.9	N21°58.5	135°57.6	N00°14.0	212°26.1	N22°42.9	314°31.9	S01°39.2	Alpheratz	357°34.0	29°13.9					
1	331°47.9	231°55.2	58.5	150°58.7	13.3	227°28.0	42.8	329°34.4	39.3	Ankaa	353°06.4	-42°09.7					
2	346°50.4	246°54.5	58.4	165°59.7	12.7	242°29.9	42.8	344°37.0	39.3	Schedar	349°30.2	56°40.5					
3	1°52.8	261°53.8	•• 58.3	181°00.8	•• 12.0	257°31.8	•• 42.7	359°39.6	•• 39.4	Diphda	348°46.6	-17°50.6					
4	16°55.3	276°53.2	58.3	196°01.8	11.4	272°33.7	42.7	14°42.1	39.4	Achernar	335°19.6	-57°06.0					
5	31°57.8	291°52.5	58.2	211°02.9	10.7	287°35.6	42.6	29°44.7	39.5	Hamal	327°50.6	23°35.0					
6	47°00.2	306°51.8	N21°58.1	226°03.9	N00°10.1	302°37.5	N22°42.6	44°47.2	S01°39.5	Polaris	313°47.9	89°22.0					
7	62°02.7	321°51.1	58.0	241°04.9	09.4	317°39.4	42.5	59°49.8	39.6	Acamar	315°11.4	-40°11.8					
8	77°05.2	336°50.4	58.0	256°06.0	08.8	332°41.3	42.5	74°52.3	39.6	Menkar	314°05.6	4°11.5					
9	92°07.6	351°49.7	•• 57.9	271°07.0	•• 08.1	347°43.2	•• 42.4	89°54.9	•• 39.7	Mirfak	308°27.6	49°57.0					
10	107°10.1	6°49.0	57.8	286°08.1	07.5	2°45.1	42.4	104°57.4	39.7	Aldebaran	290°39.2	16°33.7					
11	122°12.6	21°48.3	57.7	301°09.1	06.8	17°47.0	42.3	120°00.0	39.8	Rigel	281°03.6	-8°10.1					
12	137°15.0	36°47.6	N21°57.7	316°10.2	N00°06.2	32°48.9	N22°42.3	135°02.6	S01°39.8	Capella	280°21.4	46°01.3					
13	152°17.5	51°47.0	57.6	331°11.2	05.5	47°50.8	42.2	150°05.1	39.9	Bellatrix	278°22.5	6°22.5					
14	167°19.9	66°46.3	57.5	346°12.3	04.9	62°52.7	42.2	165°07.7	39.9	Elnath	278°01.5	28°37.7					
15	182°22.4	81°45.6	•• 57.4	1°13.3	•• 04.2	77°54.7	•• 42.1	180°10.2	•• 40.0	Alnilam	275°37.4	-1°11.0					
16	197°24.9	96°44.9	57.3	16°14.4	03.6	92°56.6	42.1	195°12.8	40.0	Betelgeuse	270°51.8	7°24.8					
17	212°27.3	111°44.2	57.3	31°15.4	02.9	107°58.5	42.1	210°15.3	40.1	Canopus	263°52.6	-52°42.3					
18	227°29.8	126°43.5	N21°57.2	46°16.4	N00°02.3	123°00.4	N22°42.0	225°17.9	S01°40.1	Sirius	258°26.1	-16°44.9					
19	242°32.3	141°42.8	57.1	61°17.5	01.6	138°02.3	42.0	240°20.5	40.2	Adhara	255°05.8	-29°00.2					
20	257°34.7	156°42.1	57.0	76°18.5	01.0	153°04.2	41.9	255°23.0	40.2	Procyon	244°50.6	5°09.7					
21	272°37.2	171°41.4	•• 56.9	91°19.6	N00°00.3	168°06.1	•• 41.9	270°25.6	•• 40.3	Pollux	243°17.0	27°57.9					
22	287°39.7	186°40.8	56.8	106°20.6	S00°00.3	183°08.0	41.8	285°28.1	40.3	Avior	234°15.2	-59°35.4					
23	302°42.1	201°40.1	56.7	121°21.7	00.9	198°09.9	41.8	300°30.7	40.4	Suhail	222°46.4	-43°32.1					
Mer.pass. 02:52			ν -0.7' d -0.1' m -3.98			ν 1.0' d -0.6' m 1.61			ν 1.9' d -0.0' m -1.93			ν 2.6' d 0.1' m 0.76					

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec								
0	317°44.6	216°39.4	N21°56.7	136°22.7	S00°01.6	213°11.8	N22°41.7	315°33.3	S01°40.4	Denebola	182°24.7	14°25.9					
1	332°47.0	231°38.7	56.6	151°23.7	02.2	228°13.7	41.7	330°35.8	40.5	Gienah	175°43.3	-17°41.1					
2	347°49.5	246°38.0	56.5	166°24.8	02.9	243°15.6	41.6	345°38.4	40.5	Acrux	173°00.0	-63°14.7					
3	2°52.0	261°37.3	•• 56.4	181°25.8	•• 03.5	258°17.5	•• 41.6	0°40.9	•• 40.6	Gacrux	171°51.5	-57°15.6					
4	17°54.4	276°36.6	56.3	196°26.9	04.2	273°19.4	41.5	15°43.5	40.6	Alioth	166°12.8	55°49.5					
5	32°56.9	291°35.9	56.2	211°27.9	04.8	288°21.3	41.5	30°46.0	40.7	Spica	158°21.9	-11°17.7					
6	47°59.4	306°35.2	N21°56.1	226°29.0	S00°05.5	303°23.2	N22°41.4	45°48.6	S01°40.7	Alkaid	152°51.7	49°11.4					
7	63°01.8	321°34.5	56.0	241°30.0	06.1	318°25.1	41.4	60°51.2	40.8	Hadar	148°35.5	-60°30.0					
8	78°04.3	336°33.8	55.9	256°31.0	06.8	333°27.0	41.3	75°53.7	40.8	Menkent	147°57.1	-36°29.9					
9	93°06.8	351°33.2	•• 55.8	271°32.1	•• 07.4	348°28.9	•• 41.3	90°56.3	•• 40.9	Arcturus	145°47.5	19°03.1					
10	108°09.2	6°32.5	55.7	286°33.1	08.1	3°30.8	41.2	105°58.8	40.9	Rigel Kent.	139°39.7	-60°56.7					
11	123°11.7	21°31.8	55.6	301°34.2	08.7	18°32.8	41.2	121°01.4	41.0	Kochab	137°19.8	74°03.3					
12	138°14.2	36°31.1	N21°55.5	316°35.2	S00°09.4	33°34.7	N22°41.1	136°04.0	S01°41.1	Zuben'ubi	136°55.5	-16°08.9					
13	153°16.6	51°30.4	55.4	331°36.3	10.0	48°36.6	41.1	151°06.5	41.1	Alphecca	126°03.3	26°37.9					
14	168°19.1	66°29.7	55.3	346°37.3	10.7	63°38.5	41.0	166°09.1	41.2	Antares	112°15.1	-26°29.4					
15	183°21.5	81°29.0	•• 55.2	1°38.3	•• 11.3	78°40.4	•• 41.0	181°11.7	•• 41.2	Atria	107°08.6	-69°04.6					
16	198°24.0	96°28.3	55.1	16°39.4	12.0	93°42.3	40.9	196°14.2	41.3	Sabik	102°02.0	-15°45.4					
17	213°26.5	111°27.6	55.0	31°40.4	12.6	108°44.2	40.9	211°16.8	41.3	Shaula	96°09.4	-37°07.4					
18	228°28.9	126°26.9	N21°54.9	46°41.5	S00°13.3	123°46.1	N22°40.9	226°19.3	S01°41.4	Rasalhague	95°57.9	12°32.6					
19	243°31.4	141°26.2	54.7	61°42.5	13.9	138°48.0	40.8	241°21.9	41.4	Eltanin	90°41.6	51°29.3					
20	258°33.9	156°25.5	54.6	76°43.6	14.6	153°49.9	40.8	256°24.5	41.5	Kaus Aust.	83°31.5	-34°22.4					
21	273°36.3	171°24.9	•• 54.5	91°44.6	•• 15.2	168°51.8	•• 40.7	271°27.0	•• 41.5	Vega	80°32.5	38°48.6					
22	288°38.8	186°24.2	54.4	106°45.6	15.9	183°53.7	40.7	286°29.6	41.6	Nunki	75°46.8	-26°15.9					
23	303°41.3	201°23.5	54.3	121°46.7	16.5	198°55.6	40.6	301°32.1	41.6	Altair	61°59.2	8°56.3					
Mer.pass. 02:49			ν -0.7' d -0.1' m -3.98			ν 1.0' d 0.6' m 1.61			ν 1.9' d -0.0' m -1.93			ν 2.6' d 0.1' m 0.76					

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	318°43.7	216°22.8	N21°54.2	136°47.7	S00°17.2	213°57.5	N22°40.6	316°34.7	S01°41.7		
1	333°46.2	231°22.1	54.1	151°48.8	17.8	228°59.5	40.5	331°37.3	41.7		
2	348°48.7	246°21.4	54.0	166°49.8	18.5	244°01.4	40.5	346°39.8	41.8		
3	3°51.1	261°20.7	•• 53.8	181°50.8	•• 19.1	259°03.3	•• 40.4	1°42.4	•• 41.8		
4	18°53.6	276°20.0	53.7	196°51.9	19.8	274°05.2	40.4	16°45.0	41.9		
5	33°56.0	291°19.3	53.6	211°52.9	20.4	289°07.1	40.3	31°47.5	41.9		
6	48°58.5	306°18.6	N21°53.5	226°54.0	S00°21.1	304°09.0	N22°40.3	46°50.1	S01°42.0		
7	64°01.0	321°17.9	53.4	241°55.0	21.7	319°10.9	40.2	61°52.6	42.1		
8	79°03.4	336°17.2	53.2	256°56.0	22.4	334°12.8	40.2	76°55.2	42.1		
9	94°05.9	351°16.5	•• 53.1	271°57.1	•• 23.0	349°14.7	•• 40.1	91°57.8	•• 42.2		
10	109°08.4	6°15.8	53.0	286°58.1	23.7	4°16.6	40.1	107°00.3	42.2		
11	124°10.8	21°15.1	52.9	301°59.2	24.3	19°18.5	40.0	122°02.9	42.3		
12	139°13.3	36°14.5	N21°52.7	317°00.2	S00°25.0	34°20.4	N22°40.0	137°05.5	S01°42.3		
13	154°15.8	51°13.8	52.6	332°01.2	25.6	49°22.4	39.9	152°08.0	42.4		
14	169°18.2	66°13.1	52.5	347°02.3	26.3	64°24.3	39.9	167°10.6	42.4		
15	184°20.7	81°12.4	•• 52.3	2°03.3	•• 26.9	79°26.2	•• 39.8	182°13.1	•• 42.5		
16	199°23.1	96°11.7	52.2	17°04.4	27.6	94°28.1	39.8	197°15.7	42.5		
17	214°25.6	111°11.0	52.1	32°05.4	28.2	109°30.0	39.7	212°18.3	42.6		
18	229°28.1	126°10.3	N21°51.9	47°06.4	S00°28.9	124°31.9	N22°39.7	227°20.8	S01°42.6		
19	244°30.5	141°09.6	51.8	62°07.5	29.5	139°33.8	39.6	2			

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°34.4	N16°07.3	14°25.3	8.9'	S24°09.2	-8.9'	56.8'
1	193°34.5	06.6	28°53.2	8.9'	24°00.3	-9.1'	56.8'
2	208°34.5	05.9	43°21.1	9.0'	23°51.2	-9.2'	56.8'
3	223°34.6	· · 05.2	57°49.1	9.0'	23°42.0	-9.4'	56.9'
4	238°34.7	04.5	72°17.1	9.1'	23°32.6	-9.5'	56.9'
5	253°34.8	03.8	86°45.2	9.1'	23°23.2	-9.6'	56.9'
6	268°34.9	N16°03.0	101°13.3	9.2'	S23°13.5	-9.8'	56.9'
7	283°35.0	02.3	115°41.5	9.2'	23°03.8	-9.9'	57.0'
8	298°35.0	01.6	130°09.7	9.3'	22°53.9	-10.0'	57.0'
9	313°35.1	· · 00.9	144°37.9	9.3'	22°43.9	-10.2'	57.0'
10	328°35.2	16°00.2	159°06.3	9.4'	22°33.7	-10.3'	57.1'
11	343°35.3	15°59.5	173°34.6	9.4'	22°23.4	-10.4'	57.1'
12	358°35.4	N15°58.8	188°03.0	9.5'	S22°13.0	-10.5'	57.1'
13	13°35.5	58.0	202°31.5	9.5'	22°02.5	-10.7'	57.1'
14	28°35.6	57.3	217°00.0	9.6'	21°51.8	-10.8'	57.2'
15	43°35.6	· · 56.6	231°28.6	9.6'	21°41.0	-10.9'	57.2'
16	58°35.7	55.9	245°57.2	9.7'	21°30.1	-11.0'	57.2'
17	73°35.8	55.2	260°25.9	9.7'	21°19.0	-11.2'	57.2'
18	88°35.9	N15°54.5	274°54.6	9.8'	S21°07.9	-11.3'	57.3'
19	103°36.0	53.7	289°23.4	9.8'	20°56.6	-11.4'	57.3'
20	118°36.1	53.0	303°52.3	9.9'	20°45.2	-11.5'	57.3'
21	133°36.2	· · 52.3	318°21.2	9.9'	20°33.6	-11.6'	57.3'
22	148°36.2	51.6	332°50.1	10.0'	20°22.0	-11.8'	57.4'
23	163°36.3	50.9	347°19.1	10.1'	20°10.2	-11.9'	57.4'
SD = 15.8'			SD = 15.5'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°36.4	N15°50.1	1°48.2	10.1'	S19°58.3	-12.0'	57.4'
1	193°36.5	49.4	16°17.3	10.2'	19°46.3	-12.1'	57.4'
2	208°36.6	48.7	30°46.4	10.2'	19°34.2	-12.2'	57.5'
3	223°36.7	· · 48.0	45°15.7	10.3'	19°22.0	-12.3'	57.5'
4	238°36.8	47.3	59°44.9	10.3'	19°09.7	-12.4'	57.5'
5	253°36.9	46.5	74°14.3	10.4'	18°57.2	-12.6'	57.5'
6	268°37.0	N15°45.8	88°43.6	10.4'	S18°44.7	-12.7'	57.6'
7	283°37.1	45.1	103°13.1	10.5'	18°32.0	-12.8'	57.6'
8	298°37.1	44.4	117°42.6	10.5'	18°19.2	-12.9'	57.6'
9	313°37.2	· · 43.6	132°12.1	10.6'	18°06.4	-13.0'	57.6'
10	328°37.3	42.9	146°41.7	10.6'	17°53.4	-13.1'	57.7'
11	343°37.4	42.2	161°11.3	10.7'	17°40.3	-13.2'	57.7'
12	358°37.5	N15°41.5	175°41.0	10.7'	S17°27.1	-13.3'	57.7'
13	13°37.6	40.7	190°10.8	10.8'	17°13.8	-13.4'	57.7'
14	28°37.7	40.0	204°40.6	10.8'	17°00.4	-13.5'	57.8'
15	43°37.8	· · 39.3	219°10.4	10.9'	16°47.0	-13.6'	57.8'
16	58°37.9	38.6	233°40.3	11.0'	16°33.4	-13.7'	57.8'
17	73°38.0	37.8	248°10.3	11.0'	16°19.7	-13.8'	57.8'
18	88°38.1	N15°37.1	262°40.3	11.0'	S16°05.9	-13.9'	57.9'
19	103°38.2	36.4	277°10.3	11.1'	15°52.1	-14.0'	57.9'
20	118°38.3	35.6	291°40.4	11.1'	15°38.1	-14.0'	57.9'
21	133°38.3	· · 34.9	306°10.6	11.2'	15°24.1	-14.1'	57.9'
22	148°38.4	34.2	320°40.8	11.2'	15°09.9	-14.2'	58.0'
23	163°38.5	33.5	335°11.0	11.3'	14°55.7	-14.3'	58.0'
SD = 15.8'			SD = 15.7'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	178°38.6	N15°32.7	349°41.3	11.3'	S14°41.4	-14.4'	58.0'
1	193°38.7	32.0	4°11.6	11.4'	14°27.0	-14.5'	58.0'
2	208°38.8	31.3	18°42.0	11.4'	14°12.5	-14.6'	58.1'
3	223°38.9	· · 30.5	33°12.4	11.5'	13°58.0	-14.6'	58.1'
4	238°39.0	29.8	47°42.9	11.5'	13°43.3	-14.7'	58.1'
5	253°39.1	29.1	62°13.4	11.6'	13°28.6	-14.8'	58.1'
6	268°39.2	N15°28.3	76°44.0	11.6'	S13°13.8	-14.9'	58.1'
7	283°39.3	27.6	91°14.5	11.6'	12°59.0	-14.9'	58.2'
8	298°39.4	26.9	105°45.2	11.7'	12°44.0	-15.0'	58.2'
9	313°39.5	· · 26.1	120°15.9	11.7'	12°29.0	-15.1'	58.2'
10	328°39.6	25.4	134°46.6	11.8'	12°13.9	-15.2'	58.2'
11	343°39.7	24.7	149°17.3	11.8'	11°58.7	-15.2'	58.2'
12	358°39.8	N15°23.9	163°48.1	11.8'	S11°43.5	-15.3'	58.3'
13	13°39.9	23.2	178°19.0	11.9'	11°28.2	-15.4'	58.3'
14	28°40.0	22.4	192°49.8	11.9'	11°12.8	-15.4'	58.3'
15	43°40.1	· · 21.7	207°20.7	11.9'	10°57.4	-15.5'	58.3'
16	58°40.2	21.0	221°51.7	12.0'	10°41.9	-15.6'	58.4'
17	73°40.3	20.2	236°22.6	12.0'	10°26.3	-15.6'	58.4'
18	88°40.4	N15°19.5	250°53.6	12.0'	S10°10.7	-15.7'	58.4'
19	103°40.5	18.8	265°24.7	12.1'	09°55.0	-15.7'	58.4'
20	118°40.6	18.0	279°55.7	12.1'	09°39.3	-15.8'	58.4'
21	133°40.7	· · 17.3	294°26.8	12.1'	09°23.5	-15.9'	58.4'
22	148°40.8	16.5	308°58.0	12.2'	09°07.6	-15.9'	58.5'
23	163°40.9	15.8	323°29.1	12.2'	08°51.7	-16.0'	58.5'
SD = 15.8'			SD = 15.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	01:37	22:25	—:—	—:—
N 70°	—:—	—:—	02:24	21:42	—:—	—:—
68°	—:—	00:35	02:54	21:13	23:18	—:—
66°	—:—	01:47	03:17	20:52	22:18	—:—
64°	—:—	02:22	03:34	20:35	21:45	—:—
62°	00:32	02:46	03:48	20:21	21:22	23:23
60°	01:36	03:06	04:00	20:09	21:03	22:30
N 58°	02:08	03:21	04:11	19:59	20:48	22:00
56°	02:31	03:34	04:20	19:50	20:35	21:38
54°	02:49	03:45	04:28	19:42	20:24	21:20
52°	03:04	03:55	04:35	19:35	20:14	21:06
50°	03:16	04:04	04:41	19:29	20:06	20:53
45°	03:41	04:22	04:55	19:15	19:48	20:28
N 40°	04:00	04:36	05:06	19:04	19:34	20:10
35°	04:16	04:48	05:16	18:55	19:22	19:55
30°	04:28	04:58	05:24	18:47	19:12	19:42
20°	04:48	05:15	05:38	18:33	18:56	19:23
N 10°	05:03	05:29	05:51	18:20	18:42	19:08
0°	05:16	05:41	06:02	18:09	18:30	18:55
S 10°	05:26	05:52	06:13	17:58	18:19	18:45
20°	05:36	06:03	06:25	17:46	18:09	18:35
30°	05:46	06:14	06:39	17:32	17:57	18:26
35°	05:50	06:20	06:47	17:25	17:51	18:21
40°	05:55	06:27	06:56	17:16	17:44	18:16
45°	06:00	06:35	07:06	17:06	17:36	18:11
S 50°	06:06	06:44	07:18	16:53	17:28	18:06
52°	06:08	06:48	07:24	16:47	17:24	18:04
54°	06:10	06:53	07:31	16:41	17:19	18:01
56°	06:13	06:57	07:38	16:34	17:14	17:59
58°	06:16	07:03	07:46	16:26	17:09	17:56
S 60°	06:19	07:09	07:55	16:17	17:03	17:53

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	■	22:25	21:26	■	■	03:28
N 70°	■	21:53	21:12	■	■	03:58
68°	22:18	21:29	21:01	■	01:46	04:20
66°	21:38	21:10	20:51	■	02:25	04:37
64°	21:10	20:54	20:43	00:35	02:52	04:51
62°	20:48	20:42	20:36	01:16	03:13	05:03
60°	20:30	20:31	20:30	01:44	03:30	05:13
N 58°	20:15	20:21	20:25	02:05	03:44	05:21
56°	20:02	20:13	20:20	02:23	03:56	05:29
54°	19:51	20:05	20:16	02:38	04:06	05:35
52°	19:41	19:58	20:12	02:50	04:15	05:41
50°	19:32	19:52	20:08	03:02	04:23	05:46
45°	19:14	19:39	20:00	03:25	04:41	05:58
N 40°	18:58	19:28	19:54	03:44	04:55	06:07
35°	18:45	19:18	19:48	03:59	05:07	06:15
30°	18:33	19:10	19:43	04:12	05:17	06:22
20°	18:14	18:56	19:34	04:35	05:35	06:34
N 10°	17:56	18:43	19:27	04:55	05:50	06:45
0°	17:40	18:31	19:20	05:13	06:05	06:54
S 10°	17:24	18:19	19:12	05:31	06:19	07:04
20°	17:06	18:06	19:05	05:50	06:34	07:14
30°	16:46	17:51	18:56	06:12	06:51	07:26
35°	16:34	17:43	18:51	06:25	07:01	07:32
40°	16:21	17:33	18:45	06:40	07:12	07:40
45°	16:04	17:21	18:38	06:57	07:25	07:48
S 50°	15:44	17:07	18:30	07:19	07:41	07:59
52°	15:35	17:00	18:26	07:29	07:48	08:03
54°	15:24	16:53	18:21	07:40	07:57	08:09
56°	15:12	16:45	18:17	07:53	08:06	08:14
58°	14:57	16:35	18:11	08:08	08:16	08:21
S 60°	14:40	16:24	18:06	08:26	08:28	08:28

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.		
	00 ^h	mm:ss	hh:mm	Upper	Lower	15-17
	mm:ss	mm:ss	mm:ss	hh:mm	hh:mm	98-99%
08	05:43	05:38	12:06	23:53	11:27	
09	05:34	05:30	12:05	—:—	12:18	
10	05:25	05:21	12:05	00:43	13:07	

August 11, 12, 13 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars											
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec										
0	319°42.9	216°06.1	N21°51.1	137°12.7	S00°32.8	214°43.4	N22°39.4	317°36.2	S01°43.0	Alpheratz	357°34.0										
1	334°45.3	231°05.4	51.0	152°13.7	33.4	229°45.3	39.3	332°38.8	43.0	Ankaa	353°06.3										
2	349°47.8	246°04.7	50.8	167°14.8	34.1	244°47.2	39.3	347°41.4	43.1	Schedar	349°30.1										
3	4°50.3	261°04.0	•• 50.7	182°15.8	•• 34.7	259°49.1	•• 39.2	2°43.9	•• 43.1	Diphda	348°46.6										
4	19°52.7	276°03.3	50.5	197°16.8	35.4	274°51.0	39.2	17°46.5	43.2	Achernar	335°19.5										
5	34°55.2	291°02.7	50.4	212°17.9	36.0	289°52.9	39.2	32°49.1	43.2	Hamal	327°50.5										
6	49°57.6	306°02.0	N21°50.3	227°18.9	S00°36.7	304°54.8	N22°39.1	47°51.6	S01°43.3	Polaris	313°46.5										
7	65°00.1	321°01.3	50.1	242°19.9	37.3	319°56.7	39.1	62°54.2	43.3	Acamar	315°11.4										
8	80°02.6	336°00.6	50.0	257°21.0	38.0	334°58.7	39.0	77°56.8	43.4	Menkar	314°05.6										
9	95°05.0	350°59.9	•• 49.8	272°22.0	•• 38.6	350°00.6	•• 39.0	92°59.3	•• 43.5	Mirfak	308°27.6										
10	110°07.5	5°59.2	49.7	287°23.1	39.3	5°02.5	38.9	108°01.9	43.5	Aldebaran	290°39.2										
11	125°10.0	20°58.5	49.5	302°24.1	39.9	20°04.4	38.9	123°04.5	43.6	Rigel	281°03.6										
12	140°12.4	35°57.8	N21°49.4	317°25.1	S00°40.6	35°06.3	N22°38.8	138°07.0	S01°43.6	Capella	280°21.4										
13	155°14.9	50°57.1	49.2	332°26.2	41.2	50°08.2	38.8	153°09.6	43.7	Bellatrix	278°22.5										
14	170°17.4	65°56.4	49.1	347°27.2	41.9	65°10.1	38.7	168°12.2	43.7	Elnath	278°01.4										
15	185°19.8	80°55.7	•• 48.9	2°28.2	•• 42.5	80°12.0	•• 38.7	183°14.7	•• 43.8	Alnilam	275°37.4										
16	200°22.3	95°55.0	48.7	17°29.3	43.2	95°13.9	38.6	198°17.3	43.8	Betelgeuse	270°51.8										
17	215°24.8	110°54.3	48.6	32°30.3	43.8	110°15.9	38.6	213°19.9	43.9	Canopus	263°52.6										
18	230°27.2	125°53.6	N21°48.4	47°31.3	S00°44.5	125°17.8	N22°38.5	228°22.4	S01°43.9	Sirius	258°26.1										
19	245°29.7	140°52.9	48.3	62°32.4	45.1	140°19.7	38.5	243°25.0	44.0	Adhara	255°05.8										
20	260°32.1	155°52.2	48.1	77°33.4	45.8	155°21.6	38.4	258°27.6	44.1	Procyon	244°50.6										
21	275°34.6	170°51.5	•• 47.9	92°34.5	•• 46.4	170°23.5	•• 38.4	273°30.1	•• 44.1	Pollux	243°17.0										
22	290°37.1	185°50.8	47.8	107°35.5	47.1	185°25.4	38.3	288°32.7	44.2	Avior	234°15.2										
23	305°39.5	200°50.1	47.6	122°36.5	47.7	200°27.3	38.3	303°35.3	44.2	Suhail	222°46.4										
Mer.pass. 02:41		ν -0.7' d -0.1' m -3.98		ν 1.0' d 0.6' m 1.62		ν 1.9' d -0.0' m -1.94		ν 2.6' d 0.1' m 0.75		Miaplacidus	221°39.2										
										Alphard	217°47.6										
										Regulus	207°34.2										
										Dubhe	193°41.0										
										Denebola	182°24.7										
										Gienah	175°43.3										
										Acrux	173°00.1										
										Gacrux	171°51.5										
										Alioth	166°12.8										
										Spica	158°21.9										
										Alkaid	152°51.8										
										Hadar	148°35.5										
										Menkent	147°57.1										
										Arcturus	145°47.5										
										Rigil Kent.	139°39.7										
										Kochab	137°19.8										
										Zuben'ubi	136°55.5										
										Alphecca	126°03.3										
										Antares	112°15.1										
										Atria	107°08.6										
										Sabik	102°02.0										
										Shaula	96°09.4										
										Rasalhague	95°57.9										
										Eltanin	90°41.6										
										Kaus Aust.	83°31.5										
										Vega	80°32.6										
										Nunki	75°46.8										
										Altair	61°59.2										
										Peacock	53°04.3										
										Deneb	49°25.0										
										Enif	33°38.0										
										Al Na'ir	27°31.8										
										Fomalhaut	15°13.6										
										Scheat	13°44.4										
										Markab	13°29.1										
Aug 11 Mon		SHA		Mer.pass																	
		Venus		256°23.3		09:36															
		Mars		177°29.8		14:50															
		Jupiter		255°00.5		09:40															
		Saturn		357°53.4		02:49															
Aug 12 Tue		SHA		Mer.pass																	
		Venus		255°07.4		09:37															
		Mars		176°55.6		14:48															
		Jupiter		254°47.2		09:37															
		Saturn		357°55.8		02:45															
Aug 13 Wed		SHA		Mer.pass																	
		Venus		253°51.6		09:38															
		Mars		176°21.3		14:47															
		Jupiter		254°34.1		09:34															
		Saturn		357°58.4		02:41															

h	Sun			Moon				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	178°41.0	N15°15.1		338°00.3	12.2'	S08°35.7	-16.0'	58.5'
1	193°41.1	14.3		352°31.5	12.2'	08°19.7	-16.1'	58.5'
2	208°41.2	13.6		7°02.7	12.3'	08°03.6	-16.1'	58.5'
3	223°41.3	12.8		21°34.0	12.3'	07°47.5	-16.2'	58.6'
4	238°41.4	12.1		36°05.3	12.3'	07°31.4	-16.2'	58.6'
5	253°41.5	11.3		50°36.5	12.3'	07°15.1	-16.3'	58.6'
6	268°41.6	N15°10.6		65°07.9	12.3'	S06°58.9	-16.3'	58.6'
7	283°41.7	09.9		79°39.2	12.4'	06°42.6	-16.3'	58.6'
8	298°41.8	09.1		94°10.6	12.4'	06°26.3	-16.4'	58.6'
9	313°41.9	08.4		108°41.9	12.4'	06°09.9	-16.4'	58.7'
10	328°42.0	07.6		123°13.3	12.4'	05°53.5	-16.5'	58.7'
11	343°42.1	06.9		137°44.7	12.4'	05°37.0	-16.5'	58.7'
12	358°42.2	N15°06.1		152°16.2	12.4'	S05°20.5	-16.5'	58.7'
13	13°42.3	05.4		166°47.6	12.4'	05°04.0	-16.6'	58.7'
14	28°42.4	04.6		181°19.0	12.5'	04°47.5	-16.6'	58.7'
15	43°42.5	03.9		195°50.5	12.5'	04°30.9	-16.6'	58.8'
16	58°42.6	03.1		210°22.0	12.5'	04°14.3	-16.6'	58.8'
17	73°42.7	02.4		224°53.4	12.5'	03°57.6	-16.7'	58.8'
18	88°42.8	N15°01.6		239°24.9	12.5'	S03°40.9	-16.7'	58.8'
19	103°42.9	00.9		253°56.4	12.5'	03°24.2	-16.7'	58.8'
20	118°43.0	15°00.1		268°27.9	12.5'	03°07.5	-16.7'	58.8'
21	133°43.1	14°59.4		282°59.4	12.5'	02°50.8	-16.8'	58.8'
22	148°43.3	58.6		297°30.9	12.5'	02°34.0	-16.8'	58.9'
23	163°43.4	57.9		312°02.4	12.5'	02°17.2	-16.8'	58.9'
	SD = 15.8'	$d = -0.7'$		SD = 16.0'				

Tue	GHA	Dec		GHA	ν	Dec	d	HP				
0	178°43.5	N14°57.1		326°33.9	12.5'	S02°00.4	-16.8'	58.9'				
1	193°43.6	56.4		341°05.4	12.5'	01°43.6	-16.8'	58.9'				
2	208°43.7	55.6		355°36.9	12.5'	01°26.8	-16.8'	58.9'				
3	223°43.8	54.9		10°08.4	12.5'	01°09.9	-16.9'	58.9'				
4	238°43.9	54.1		24°39.9	12.5'	00°53.1	-16.9'	58.9'				
5	253°44.0	53.4		39°11.4	12.5'	00°36.2	-16.9'	58.9'				
6	268°44.1	N14°52.6		53°42.8	12.5'	S00°19.3	-16.9'	59.0'				
7	283°44.2	51.9		68°14.3	12.5'	S00°02.5	-16.9'	59.0'				
8	298°44.3	51.1		82°45.8	12.4'	N00°14.4	16.9'	59.0'				
9	313°44.4	50.4		97°17.2	12.4'	00°31.3	16.9'	59.0'				
10	328°44.5	49.6		111°48.6	12.4'	00°48.2	16.9'	59.0'				
11	343°44.7	48.8		126°20.0	12.4'	01°05.1	16.9'	59.0'				
12	358°44.8	N14°48.1		140°51.5	12.4'	N01°22.0	16.9'	59.0'				
13	13°44.9	47.3		155°22.8	12.4'	01°38.9	16.9'	59.0'				
14	28°45.0	46.6		169°54.2	12.4'	01°55.8	16.9'	59.1'				
15	43°45.1	45.8		184°25.6	12.3'	02°12.7	16.9'	59.1'				
16	58°45.2	45.1		198°56.9	12.3'	02°29.6	16.9'	59.1'				
17	73°45.3	44.3		213°28.2	12.3'	02°46.4	16.9'	59.1'				
18	88°45.4	N14°43.5		227°59.5	12.3'	N03°03.3	16.9'	59.1'				
19	103°45.5	42.8		242°30.8	12.2'	03°20.2	16.8'	59.1'				
20	118°45.6	42.0		257°02.0	12.2'	03°37.0	16.8'	59.1'				
21	133°45.8	41.3		271°33.2	12.2'	03°53.8	16.8'	59.1'				
22	148°45.9	40.5		286°04.4	12.2'	04°10.6	16.8'	59.1'				
23	163°46.0	39.7		300°35.5	12.1'	04°27.4	16.8'	59.1'				
SD = 15.8'			$d = -0.8'$	SD = 16.1'								

Wed	GHA	Dec		GHA	ν	Dec	d	HP				
0	178°46.1	N14°39.0		315°06.7	12.1'	N04°44.2	16.8'	59.1'				
1	193°46.2	38.2		329°37.8	12.1'	05°01.0	16.7'	59.2'				
2	208°46.3	37.5		344°08.8	12.0'	05°17.7	16.7'	59.2'				
3	223°46.4	36.7		358°39.8	12.0'	05°34.4	16.7'	59.2'				
4	238°46.5	35.9		13°10.8	12.0'	05°51.1	16.7'	59.2'				
5	253°46.7	35.2		27°41.8	11.9'	06°07.8	16.6'	59.2'				
6	268°46.8	N14°34.4		42°12.7	11.9'	N06°24.4	16.6'	59.2'				
7	283°46.9	33.6		56°43.6	11.8'	06°41.0	16.6'	59.2'				
8	298°47.0	32.9		71°14.4	11.8'	06°57.6	16.5'	59.2'				
9	313°47.1	32.1		85°45.2	11.8'	07°14.1	16.5'	59.2'				
10	328°47.2	31.4		100°16.0	11.7'	07°30.6	16.5'	59.2'				
11	343°47.3	30.6		114°46.7	11.7'	07°47.1	16.4'	59.2'				
12	358°47.5	N14°29.8		129°17.3	11.6'	N08°03.5	16.4'	59.2'				
13	13°47.6	29.1		143°47.9	11.6'	08°19.9	16.4'	59.2'				
14	28°47.7	28.3		158°18.5	11.5'	08°36.3	16.3'	59.2'				
15	43°47.8	27.5		172°49.0	11.5'	08°52.6	16.3'	59.2'				
16	58°47.9	26.8		187°19.5	11.4'	09°08.9	16.2'	59.2'				
17	73°48.0	26.0		201°49.9	11.4'	09°25.1	16.2'	59.3'				
18	88°48.1	N14°25.2		216°20.3	11.3'	N09°41.3	16.1'	59.3'				
19	103°48.3	24.4		230°50.6	11.3'	09°57.4	16.1'	59.3'				
20	118°48.4	23.7		245°20.8	11.2'	10°13.5	16.0'	59.3'				
21	133°48.5	22.9		259°51.0	11.1'	10°29.5	16.0'	59.3'				
22	148°48.6	22.1		274°21.2	11.1'	10°45.5	15.9'	59.3'				
23	163°48.7	21.4		288°51.2	11.0'	11°01.4	15.9'	59.3'				
SD = 15.8'			$d = -0.8'$	SD = 16.1'								

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	02:04	22:00	—:—	—:—
N 70°	—:—	—:—	02:42	21:24	—:—	—:—
68°	—:—	01:19	03:08	20:59	22:43	—:—
66°	—:—	02:06	03:27	20:40	21:59	—:—
64°	—:—	02:35	03:43	20:24	21:31	—:—
62°	01:10	02:57	03:56	20:12	21:10	22:52
60°	01:53	03:14	04:07	20:01	20:53	22:13
N 58°	02:20	03:29	04:17	19:51	20:39	21:47
56°	02:40	03:41	04:25	19:43	20:27	21:27
54°	02:57	03:52	04:33	19:36	20:17	21:11
52°	03:10	04:01	04:39	19:29	20:08	20:58
50°	03:22	04:09	04:45	19:24	20:00	20:46
45°	03:46	04:26	04:58	19:11	19:43	20:23
N 40°	04:04	04:40	05:09	19:00	19:30	20:05
35°	04:18	04:51	05:18	18:52	19:19	19:51
30°	04:30	05:00	05:26	18:44	19:09	19:39
20°	04:49	05:16	05:39	18:31	18:54	19:21
N 10°	05:04	05:29	05:51	18:19	18:41	19:06
0°	05:15	05:40	06:02	18:08	18:30	18:55
S 10°	05:26	05:51	06:12	17:58	18:20	18:45
20°	05:35	06:01	06:24	17:47	18:09	18:36
30°	05:43	06:12	06:36	17:34	17:59	18:27
35°	05:48	06:17	06:44	17:27	17:53	18:23
40°	05:52	06:24	06:52	17:19	17:47	18:19
45°	05:56	06:31	07:02	17:09	17:40	18:14
S 50°	06:01	06:39	07:13	16:58	17:32	18:10
52°	06:03	06:43	07:19	16:52	17:28	18:08
54°	06:05	06:47	07:24	16:46	17:24	18:06
56°	06:07	06:51	07:31	16:40	17:20	18:04
58°	06:09	06:56	07:38	16:32	17:15	18:02
S 60°	06:12	07:01	07:47	16:24	17:10	17:59

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	20:46	20:09	19:27	06:10	08:33	10:58
N 70°	20:42	20:13	19:42	06:21	08:33	10:46
68°	20:38	20:17	19:54	06:30	08:32	10:36
66°	20:35	20:20	20:05	06:37	08:32	10:28
64°	20:33	20:23	20:13	06:43	08:32	10:22
62°	20:31	20:26	20:21	06:48	08:32	10:16
60°	20:29	20:28	20:27	06:53	08:31	10:11
N 58°	20:27	20:30	20:33	06:57	08:31	10:07
56°	20:26	20:32	20:38	07:00	08:31	10:03
54°	20:25	20:33	20:42	07:03	08:31	10:00
52°	20:23	20:35	20:47	07:06	08:31	09:57
50°	20:22	20:36	20:50	07:09	08:31	09:54
45°	20:20	20:39	20:59	07:14	08:31	09:48
N 40°	20:18	20:41	21:06	07:19	08:30	09:43
35°	20:16	20:43	21:12	07:23	08:30	09:38
30°	20:14	20:45	21:17	07:26	08:30	09:35
20°	20:12	20:48	21:27	07:32	08:30	09:28
N 10°	20:09	20:51	21:35	07:37	08:30	09:22
0°	20:07	20:54	21:43	07:42	08:29	09:17
S 10°	20:05	20:57	21:51	07:47	08:29	09:12
20°	20:02	21:00	21:59	07:52	08:29	09:06
30°	20:00	21:04	22:09	07:58	08:28	08:59
35°	19:58	21:06	22:15	08:01	08:28	08:56
40°	19:56	21:08	22:21	08:04	08:28	08:52
45°	19:54	21:11	22:29	08:09	08:28	08:47
S 50°	19:52	21:14	22:38	08:13	08:27	08:41
52°	19:51	21:16	22:42	08:16	08:27	08:38
54°	19:49	21:17	22:47	08:18	08:27	08:36
56°	19:48	21:19	22:52	08:21	08:27	08:32
58°	19:46	21:21	22:58	08:24	08:26	08:29
S 60°	19:45	21:24	23:05	08:27	08:26	08:25

August 14, 15, 16 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	322°40.3	215°16.0	N21°38.3	138°27.1	S01°19.6	217°01.2	N22°35.8	320°41.3	S01°47.0	335°43.8	47.0	335°43.8	47.0	Alpheratz	357°34.0	29°13.9	
1	337°42.7	230°15.3	38.1	153°28.2	20.3	232°03.1	35.8	335°43.8	47.0	335°43.8	47.0	335°43.8	47.0	Ankaa	353°06.3	-42°09.7	
2	352°45.2	245°14.6	37.9	168°29.2	20.9	247°05.1	35.7	350°46.4	47.1	350°46.4	47.1	350°46.4	47.1	Schedar	349°30.1	56°40.6	
3	7°47.7	260°13.9	•• 37.7	183°30.2	•• 21.6	262°07.0	•• 35.7	5°49.0	•• 47.2	5°49.0	•• 47.2	5°49.0	•• 47.2	Diphda	348°46.6	-17°50.6	
4	22°50.1	275°13.2	37.4	198°31.3	22.2	277°08.9	35.6	20°51.6	47.2	20°51.6	47.2	20°51.6	47.2	Achernar	335°19.5	-57°06.1	
5	37°52.6	290°12.5	37.2	213°32.3	22.9	292°10.8	35.6	35°54.1	47.3	35°54.1	47.3	35°54.1	47.3	Hamal	327°50.5	23°35.0	
6	52°55.1	305°11.8	N21°37.0	228°33.3	S01°23.5	307°12.7	N22°35.5	50°56.7	S01°47.3	50°56.7	S01°47.3	50°56.7	S01°47.3	Polaris	313°44.9	89°22.0	
7	67°57.5	320°11.1	36.8	243°34.4	24.2	322°14.7	35.5	65°59.3	47.4	65°59.3	47.4	65°59.3	47.4	Acamar	315°11.3	-40°11.8	
8	83°00.0	335°10.4	36.6	258°35.4	24.8	337°16.6	35.4	81°01.9	47.5	81°01.9	47.5	81°01.9	47.5	Menkar	314°05.6	4°11.5	
9	98°02.5	350°09.7	•• 36.3	273°36.4	•• 25.5	352°18.5	•• 35.4	96°04.5	•• 47.5	96°04.5	•• 47.5	96°04.5	•• 47.5	Mirfak	308°27.5	49°57.0	
10	113°04.9	5°09.0	36.1	288°37.4	26.1	7°20.4	35.3	111°07.0	47.6	111°07.0	47.6	111°07.0	47.6	Aldebaran	290°39.1	16°33.7	
11	128°07.4	20°08.3	35.9	303°38.5	26.8	22°22.3	35.3	126°09.6	47.6	126°09.6	47.6	126°09.6	47.6	Rigel	281°03.5	-8°10.1	
12	143°09.8	35°07.6	N21°35.6	318°39.5	S01°27.4	37°24.3	N22°35.2	141°12.2	S01°47.7	141°12.2	S01°47.7	141°12.2	S01°47.7	Capella	280°21.4	46°01.3	
13	158°12.3	50°06.9	35.4	333°40.5	28.1	52°26.2	35.2	156°14.8	47.7	156°14.8	47.7	156°14.8	47.7	Bellatrix	278°22.5	6°22.5	
14	173°14.8	65°06.2	35.2	348°41.6	28.7	67°28.1	35.1	171°17.3	47.8	171°17.3	47.8	171°17.3	47.8	Elnath	278°01.4	28°37.7	
15	188°17.2	80°05.5	•• 35.0	3°42.6	•• 29.4	82°30.0	•• 35.1	186°19.9	•• 47.9	186°19.9	•• 47.9	186°19.9	•• 47.9	Alnilam	275°37.4	-1°11.0	
16	203°19.7	95°04.8	34.7	18°43.6	30.0	97°31.9	35.0	201°22.5	47.9	201°22.5	47.9	201°22.5	47.9	Betelgeuse	270°51.7	7°24.8	
17	218°22.2	110°04.1	34.5	33°44.6	30.7	112°33.9	35.0	216°25.1	48.0	216°25.1	48.0	216°25.1	48.0	Canopus	263°52.6	-52°42.2	
18	233°24.6	125°03.4	N21°34.3	48°45.7	S01°31.3	127°35.8	N22°34.9	231°27.6	S01°48.0	231°27.6	S01°48.0	231°27.6	S01°48.0	Sirius	258°26.1	-16°44.9	
19	248°27.1	140°02.7	34.0	63°46.7	32.0	142°37.7	34.9	246°30.2	48.1	246°30.2	48.1	246°30.2	48.1	Adhara	255°05.8	-29°00.2	
20	263°29.6	155°02.0	33.8	78°47.7	32.6	157°39.6	34.8	261°32.8	48.2	261°32.8	48.2	261°32.8	48.2	Procyon	244°50.6	5°09.7	
21	278°32.0	170°01.3	•• 33.6	93°48.7	•• 33.3	172°41.5	•• 34.8	276°35.4	•• 48.2	276°35.4	•• 48.2	276°35.4	•• 48.2	Pollux	243°17.0	27°57.9	
22	293°34.5	185°00.7	33.3	108°49.8	34.0	187°43.5	34.7	291°38.0	48.3	291°38.0	48.3	291°38.0	48.3	Avior	234°15.2	-59°35.3	
23	308°36.9	200°00.0	33.1	123°50.8	34.6	202°45.4	34.7	306°40.5	48.3	306°40.5	48.3	306°40.5	48.3	Suhail	222°46.4	-43°32.0	
Mer.pass. 02:29			ν -0.7' d -0.2' m -3.97			ν 1.0' d 0.7' m 1.62			ν 1.9' d -0.1' m -1.94			ν 2.6' d 0.1' m 0.73					
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	323°39.4	214°59.3	N21°32.8	138°51.8	S01°35.3	217°47.3	N22°34.6	321°43.1	S01°48.4	331°43.1	48.4	331°43.1	48.4	Denebola	182°24.7	14°25.9	
1	338°41.9	229°58.6	32.6	153°52.9	35.9	232°49.2	34.6	336°45.7	48.4	336°45.7	48.4	336°45.7	48.4	Gienah	175°43.3	-17°41.0	
2	353°44.3	244°57.9	32.4	168°53.9	36.6	247°51.2	34.5	351°48.3	48.5	351°48.3	48.5	351°48.3	48.5	Acrux	173°00.1	-63°14.7	
3	8°46.8	259°57.2	•• 32.1	183°54.9	•• 37.2	262°53.1	•• 34.5	6°50.8	•• 48.6	6°50.8	•• 48.6	6°50.8	•• 48.6	Gacrux	171°51.5	-57°15.6	
4	23°49.3	274°56.5	31.9	198°55.9	37.9	277°55.0	34.4	21°53.4	48.6	21°53.4	48.6	21°53.4	48.6	Alioth	166°12.8	55°49.5	
5	38°51.7	289°55.8	31.6	213°57.0	38.5	292°56.9	34.4	36°56.0	48.7	36°56.0	48.7	36°56.0	48.7	Spica	158°21.9	-11°17.7	
6	53°54.2	304°55.1	N21°31.4	228°58.0	S01°39.2	307°58.8	N22°34.3	51°58.6	S01°48.7	51°58.6	S01°48.7	51°58.6	S01°48.7	Alkaid	152°51.8	49°11.4	
7	68°56.7	319°54.4	31.1	243°59.0	39.8	323°00.8	34.3	67°01.2	48.8	67°01.2	48.8	67°01.2	48.8	Hadar	148°35.5	-60°30.0	
8	83°59.1	334°53.7	30.9	259°00.0	40.5	338°02.7	34.2	82°03.7	48.9	82°03.7	48.9	82°03.7	48.9	Menkent	147°57.2	-36°29.9	
9	99°01.6	349°53.0	•• 30.6	274°01.1	•• 41.1	353°04.6	•• 34.1	97°06.3	•• 48.9	97°06.3	•• 48.9	97°06.3	•• 48.9	Arcturus	145°47.5	19°03.1	
10	114°04.1	4°52.3	30.4	289°02.1	41.8	8°06.5	34.1	112°08.9	49.0	112°08.9	49.0	112°08.9	49.0	Rigel Kent.	139°39.8	-60°56.7	
11	129°06.5	19°51.6	30.1	304°03.1	42.4	23°08.5	34.0	127°11.5	49.0	127°11.5	49.0	127°11.5	49.0	Kochab	137°19.9	74°03.3	
12	144°09.0	34°50.9	N21°29.9	319°04.1	S01°43.1	38°10.4	N22°34.0	142°14.0	S01°49.1	142°14.0	S01°49.1	142°14.0	S01°49.1	Zuben'ubi	136°55.5	-16°08.9	
13	159°11.4	49°50.2	29.6	334°05.2	43.7	53°12.3	33.9	157°16.6	49.2	157°16.6	49.2	157°16.6	49.2	Alphecca	126°03.3	26°37.9	
14	174°13.9	64°49.5	29.4	349°06.2	44.4	68°14.2	33.9	172°19.2	49.2	172°19.2	49.2	172°19.2	49.2	Antares	112°15.1	-26°29.4	
15	189°16.4	79°48.8	•• 29.1	4°07.2	•• 45.0	83°16.1	•• 33.8	187°21.8	•• 49.3	187°21.8	•• 49.3	187°21.8	•• 49.3	Atria	107°08.6	-69°04.6	
16	204°18.8	94°48.1	28.9	19°08.2	45.7	98°18.1	33.8	202°24.4	49.3	202°24.4	49.3	202°24.4	49.3	Sabik	102°02.0	-15°45.4	
17	219°21.3	109°47.4	28.6	34°09.3	46.3	113°20.0	33.7	217°26.9	49.4	217°26.9	49.4	217°26.9	49.4	Shaula	96°09.5	-37°07.4	
18	234°23.8	124°46.7	N21°28.3	49°10.3	S01°47.0	128°21.9	N22°33.7	232°29.5	S01°49.5	232°29.5	S01°49.5	232°29.5	S01°49.5	Rasalhague	95°57.9	12°32.6	
19	249°26.2	139°46.0	28.1	64°11.3	47.7	143°23.8	33.6	247°32.1	49.5	247°32.1	49.5	247°32.1	49.5	Eltanin	90°41.6	51°29.3	
20	264°28.7	154°45.3	27.8	79°12.3	48.3	158°25.8	33.6	262°34.7	49.6	262°34.7	49.6	262°34.7	49.6	Kaus Aust.	83°31.6	-34°22.4	
21	279°31.2	169°44.6	•• 27.6	94°13.4	•• 49.0	173°27.7	•• 33.5	277°37.3	•• 49.6	277°37.3	•• 49.6	277°37.3	•• 49.6	Vega	80°32.6	38°48.6	
22	294°33.6	184°43.9	27.3	109°14.4	49.6	188°29.6	33.5	292°39.8	49.7	292°39.8	49.7	292°39.8	49.7	Nunki	75°46.8	-26°15.9	
23	309°36.1	199°43.2	27.0	124°15.4	50.3	203°31.5	33.4	307°42.4	49.8	307°42.4	49.8	307°42.4	49.8	Altair	61°59.2	8°56.3	
Mer.pass. 02:25			ν -0.7' d -0.2' m -3.97			ν 1.0' d 0.7' m 1.62			ν 1.9' d -0.1' m -1.94			ν 2.6' d 0.1' m 0.73					
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	324°38.6	214°42.5	N21°26.8	139°16.4	S01°50.9	218°33.5	N22°33.4	322°45.0	S01°49.8	332°45.0	49.8	332°45.0	49.8	Fomalhaut	15°13.6	-29°29.0	
1	339°41.0	229°41.8	26.5	154°17.5	51.6	233°35.4	33.3	337°47.6	49.9	337°47.6	49.9	337°47.6	49.9	Scheat	1		

h	Sun			Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	178°48.9	N14°20.6	303°21.3	11.0'	N11°17.3	15.8'	59.3'
1	193°49.0	19.8	317°51.2	10.9'	11°33.1	15.7'	59.3'
2	208°49.1	19.1	332°21.1	10.8'	11°48.8	15.7'	59.3'
3	223°49.2	· · 18.3	346°50.9	10.8'	12°04.5	15.6'	59.3'
4	238°49.3	17.5	1°20.7	10.7'	12°20.1	15.6'	59.3'
5	253°49.4	16.7	15°50.4	10.6'	12°35.7	15.5'	59.3'
6	268°49.6	N14°16.0	30°20.0	10.6'	N12°51.2	15.4'	59.3'
7	283°49.7	15.2	44°49.6	10.5'	13°06.6	15.4'	59.3'
8	298°49.8	14.4	59°19.1	10.4'	13°21.9	15.3'	59.3'
9	313°49.9	· · 13.6	73°48.5	10.4'	13°37.2	15.2'	59.3'
10	328°50.0	12.9	88°17.9	10.3'	13°52.4	15.1'	59.3'
11	343°50.2	12.1	102°47.2	10.2'	14°07.6	15.1'	59.3'
12	358°50.3	N14°11.3	117°16.4	10.1'	N14°22.6	15.0'	59.3'
13	13°50.4	10.5	131°45.5	10.1'	14°37.6	14.9'	59.3'
14	28°50.5	09.8	146°14.6	10.0'	14°52.5	14.8'	59.3'
15	43°50.6	· · 09.0	160°43.6	9.9'	15°07.3	14.7'	59.3'
16	58°50.8	08.2	175°12.5	9.8'	15°22.1	14.7'	59.3'
17	73°50.9	07.4	189°41.3	9.8'	15°36.7	14.6'	59.3'
18	88°51.0	N14°06.7	204°10.1	9.7'	N15°51.3	14.5'	59.3'
19	103°51.1	05.9	218°38.7	9.6'	16°05.8	14.4'	59.3'
20	118°51.2	05.1	233°07.3	9.5'	16°20.2	14.3'	59.3'
21	133°51.4	· · 04.3	247°35.8	9.4'	16°34.5	14.2'	59.3'
22	148°51.5	03.5	262°04.3	9.3'	16°48.7	14.1'	59.3'
23	163°51.6	02.8	276°32.6	9.3'	17°02.8	14.0'	59.3'
SD = 15.8'			SD = 16.2'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	178°51.7	N14°02.0	291°00.9	9.2'	N17°16.8	13.9'	59.3'
1	193°51.9	01.2	305°29.1	9.1'	17°30.7	13.8'	59.3'
2	208°52.0	14°00.4	319°57.2	9.0'	17°44.5	13.7'	59.3'
3	223°52.1	13°59.6	334°25.2	8.9'	17°58.3	13.6'	59.3'
4	238°52.2	58.9	348°53.1	8.8'	18°11.9	13.5'	59.3'
5	253°52.4	58.1	3°20.9	8.8'	18°25.4	13.4'	59.3'
6	268°52.5	N13°57.3	17°48.7	8.7'	N18°38.8	13.3'	59.3'
7	283°52.6	56.5	32°16.3	8.6'	18°52.1	13.2'	59.3'
8	298°52.7	55.7	46°43.9	8.5'	19°05.3	13.1'	59.3'
9	313°52.9	· · 54.9	61°11.4	8.4'	19°18.4	13.0'	59.3'
10	328°53.0	54.2	75°38.8	8.3'	19°31.3	12.9'	59.3'
11	343°53.1	53.4	90°06.1	8.2'	19°44.2	12.7'	59.3'
12	358°53.2	N13°52.6	104°33.3	8.1'	N19°56.9	12.6'	59.3'
13	13°53.4	51.8	119°00.5	8.0'	20°09.5	12.5'	59.3'
14	28°53.5	51.0	133°27.5	7.9'	20°22.0	12.4'	59.3'
15	43°53.6	· · 50.2	147°54.4	7.9'	20°34.4	12.3'	59.3'
16	58°53.7	49.4	162°21.3	7.8'	20°46.6	12.1'	59.3'
17	73°53.9	48.7	176°48.1	7.7'	20°58.8	12.0'	59.3'
18	88°54.0	N13°47.9	191°14.7	7.6'	N21°10.8	11.9'	59.3'
19	103°54.1	47.1	205°41.3	7.5'	21°22.7	11.7'	59.3'
20	118°54.2	46.3	220°07.8	7.4'	21°34.4	11.6'	59.3'
21	133°54.4	· · 45.5	234°34.2	7.3'	21°46.0	11.5'	59.2'
22	148°54.5	44.7	249°00.5	7.2'	21°57.5	11.3'	59.2'
23	163°54.6	43.9	263°26.7	7.1'	22°08.9	11.2'	59.2'
SD = 15.8'			SD = 16.2'				

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	178°54.8	N13°43.1	277°52.9	7.0'	N22°20.1	11.1'	59.2'
1	193°54.9	42.3	292°18.9	6.9'	22°31.1	10.9'	59.2'
2	208°55.0	41.6	306°44.8	6.9'	22°42.1	10.8'	59.2'
3	223°55.1	· · 40.8	321°10.7	6.8'	22°52.9	10.7'	59.2'
4	238°55.3	40.0	335°36.5	6.7'	23°03.5	10.5'	59.2'
5	253°55.4	39.2	350°02.1	6.6'	23°14.0	10.4'	59.2'
6	268°55.5	N13°38.4	4°27.7	6.5'	N23°24.4	10.2'	59.2'
7	283°55.7	37.6	18°53.2	6.4'	23°34.6	10.1'	59.2'
8	298°55.8	36.8	33°18.6	6.3'	23°44.7	9.9'	59.2'
9	313°55.9	· · 36.0	47°43.9	6.2'	23°54.6	9.8'	59.2'
10	328°56.1	35.2	62°09.2	6.1'	24°04.4	9.6'	59.2'
11	343°56.2	34.4	76°34.3	6.1'	24°14.0	9.5'	59.2'
12	358°56.3	N13°33.6	90°59.4	6.0'	N24°23.5	9.3'	59.2'
13	13°56.4	32.8	105°24.3	5.9'	24°32.8	9.2'	59.2'
14	28°56.6	32.0	119°49.2	5.8'	24°42.0	9.0'	59.2'
15	43°56.7	· · 31.2	134°14.0	5.7'	24°51.0	8.8'	59.2'
16	58°56.8	30.5	148°38.8	5.6'	24°59.8	8.7'	59.1'
17	73°57.0	29.7	163°03.4	5.6'	25°08.5	8.5'	59.1'
18	88°57.1	N13°28.9	177°28.0	5.5'	N25°17.0	8.3'	59.1'
19	103°57.2	28.1	191°52.4	5.4'	25°25.3	8.2'	59.1'
20	118°57.4	27.3	206°16.9	5.3'	25°33.5	8.0'	59.1'
21	133°57.5	· · 26.5	220°41.2	5.3'	25°41.5	7.9'	59.1'
22	148°57.6	25.7	235°05.4	5.2'	25°49.4	7.7'	59.1'
23	163°57.8	24.9	249°29.6	5.1'	25°57.1	7.5'	59.1'
SD = 15.8'			SD = 16.2'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	02:26	21:37	—:—	—:—
N 70°	—:—	—:—	02:58	21:07	23:25	—:—
68°	—:—	01:45	03:20	20:45	22:17	—:—
66°	—:—	02:22	03:38	20:28	21:42	—:—
64°	—:—	02:48	03:53	20:14	21:18	23:30
62°	01:33	03:07	04:05	20:02	20:59	22:30
60°	02:07	03:23	04:15	19:52	20:43	21:58
N 58°	02:31	03:37	04:24	19:44	20:30	21:35
56°	02:49	03:48	04:31	19:36	20:19	21:17
54°	03:04	03:58	04:38	19:30	20:10	21:02
52°	03:17	04:06	04:44	19:23	20:01	20:50
50°	03:28	04:14	04:50	19:18	19:54	20:39
45°	03:51	04:30	05:02	19:06	19:38	20:17
N 40°	04:08	04:43	05:12	18:56	19:25	20:00
35°	04:21	04:53	05:20	18:48	19:15	19:47
30°	04:32	05:02	05:27	18:41	19:06	19:36
20°	04:50	05:17	05:40	18:29	18:51	19:18
N 10°	05:04	05:29	05:51	18:18	18:39	19:05
0°	05:15	05:40	06:01	18:08	18:29	18:54
S 10°	05:25	05:49	06:11	17:58	18:20	18:44
20°	05:33	05:59	06:22	17:48	18:10	18:36
30°	05:41	06:09	06:33	17:36	18:00	18:29
35°	05:44	06:14	06:40	17:29	17:55	18:25
40°	05:48	06:20	06:48	17:21	17:49	18:21
45°	05:52	06:27	06:57	17:12	17:43	18:18
S 50°	05:56	06:34	07:08	17:02	17:36	18:14
52°	05:58	06:37	07:13	16:57	17:32	18:12
54°	05:59	06:41	07:18	16:51	17:29	18:10
56°	06:01	06:45	07:24	16:45	17:25	18:09
58°	06:03	06:49	07:31	16:39	17:21	18:07
S 60°	06:05	06:54	07:39	16:31	17:16	18:05

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	18:22	□	□	13:50	□	□
N 70°	18:59	□	□	13:15	□	□
68°	19:25	18:25	□	12:51	15:43	□
66°	19:45	19:14	□	12:32	14:55	□
64°	20:02	19:46	19:10	12:17	14:24	16:59
62°	20:15	20:09	20:02	12:05	14:02	16:08
60°	20:27	20:28	20:34	11:55	13:43	15:37
N 58°	20:37	20:44	20:58	11:46	13:28	15:13
56°	20:46	20:58	21:17	11:38	13:15	14:55
54°	20:54	21:10	21:34	11:31	13:04	14:39
52°	21:01	21:20	21:48	11:24	12:55	14:25
50°	21:08	21:30	22:00	11:19	12:46	14:13
45°	21:22	21:50	22:26	11:07	12:27	13:49
N 40°	21:33	22:06	22:46	10:57	12:12	13:29
35°	21:43	22:20	23:03	10:48	12:00	13:13
30°	21:52	22:32	23:18	10:41	11:49	12:59
20°	22:07	22:53	23:44	10:28	11:30	12:35
N 10°	22:21	23:11	· · ·	10:17	11:14	12:14
0°	22:34	23:28	· · ·	10:06	10:59	11:55
S 10°	22:46	23:45	· · ·	09:56	10:44	11:36
20°	23:00	· · ·	00:04	09:45	10:28	11:16
30°	23:16	· · ·	00:25	09:33	10:10	10:52
35°	23:25	· · ·	00:38	09:25	09:59	10:39
40°	23:36	· · ·	00:53	09:17	09:47	10:23
45°	23:49	· · ·	01:10	09:08	09:33	10:04
S 50°	· · ·	00:04	01:32	08:57	09:16	09:41
52°	· · ·	00:12	01:43	08:51	09:08	09:30
54°	· · ·	00:20	01:55	08:46	08:59	09:18
56°	· · ·	00:29	02:08	08:39	08:49	09:03
58°	· · ·	00:39	02:24	08:32	08:37	08:46
S 60°	· · ·	00:51	02:44	08:24	08:24	08:26

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time	12 ^h		Mer. Pass.	Lower	21-23
	00 ^h	mm:ss	hh:mm	Upper	hh:mm	75-53%
14	04:45	04:39	12:05	03:54	16:20	
15	04:33	04:27	12:04	04:46	17:13	
16	04:21	04:15	12:04	05:41	18:11	

August 17, 18, 19 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	325°37.7	214°25.8	N21°20.1	139°41.0	S02°06.6	219°19.7	N22°32.1	323°47.0	S01°51.3	Alpheratz	357°34.0	29°13.9
1	340°40.2	229°25.1	19.8	154°42.0	07.2	234°21.6	32.1	338°49.6	51.3	Ankaa	353°06.3	-42°09.7
2	355°42.6	244°24.4	19.5	169°43.0	07.9	249°23.6	32.0	353°52.2	51.4	Schedar	349°30.1	56°40.6
3	10°45.1	259°23.7	•• 19.2	184°44.1	•• 08.5	264°25.5	•• 32.0	8°54.7	•• 51.5	Diphda	348°46.5	-17°50.5
4	25°47.5	274°23.0	18.9	199°45.1	09.2	279°27.4	31.9	23°57.3	51.5	Achernar	335°19.5	-57°06.1
5	40°50.0	289°22.3	18.6	214°46.1	09.9	294°29.3	31.9	38°59.9	51.6	Hamal	327°50.5	23°35.0
6	55°52.5	304°21.6	N21°18.3	229°47.1	S02°10.5	309°31.3	N22°31.8	54°02.5	S01°51.6	Polaris	313°43.1	89°22.0
7	70°54.9	319°21.0	18.0	244°48.1	11.2	324°33.2	31.8	69°05.1	51.7	Acamar	315°11.3	-40°11.8
8	85°57.4	334°20.3	17.7	259°49.2	11.8	339°35.1	31.7	84°07.7	51.8	Menkar	314°05.6	4°11.5
9	100°59.9	349°19.6	•• 17.4	274°50.2	•• 12.5	354°37.1	•• 31.7	99°10.2	•• 51.8	Mirfak	308°27.5	49°57.0
10	116°02.3	4°18.9	17.1	289°51.2	13.1	9°39.0	31.6	114°12.8	51.9	Aldebaran	290°39.1	16°33.7
11	131°04.8	19°18.2	16.8	304°52.2	13.8	24°40.9	31.6	129°15.4	51.9	Rigel	281°03.5	-8°10.1
12	146°07.3	34°17.5	N21°16.5	319°53.2	S02°14.4	39°42.8	N22°31.5	144°18.0	S01°52.0	Capella	280°21.3	46°01.3
13	161°09.7	49°16.8	16.2	334°54.3	15.1	54°44.8	31.5	159°20.6	52.1	Bellatrix	278°22.5	6°22.5
14	176°12.2	64°16.1	15.9	349°55.3	15.7	69°46.7	31.4	174°23.2	52.1	Elnath	278°01.4	28°37.7
15	191°14.7	79°15.4	•• 15.6	4°56.3	•• 16.4	84°48.6	•• 31.4	189°25.8	•• 52.2	Alnilam	275°37.4	-1°11.0
16	206°17.1	94°14.7	15.3	19°57.3	17.0	99°50.6	31.3	204°28.3	52.3	Betelgeuse	270°51.7	7°24.8
17	221°19.6	109°14.0	15.0	34°58.3	17.7	114°52.5	31.3	219°30.9	52.3	Canopus	263°52.5	-52°42.2
18	236°22.0	124°13.3	N21°14.7	49°59.3	S02°18.4	129°54.4	N22°31.2	234°33.5	S01°52.4	Sirius	258°26.0	-16°44.9
19	251°24.5	139°12.6	14.4	65°00.4	19.0	144°56.4	31.2	249°36.1	52.4	Adhara	255°05.8	-29°00.2
20	266°27.0	154°11.9	14.1	80°01.4	19.7	159°58.3	31.1	264°38.7	52.5	Procyon	244°50.6	5°09.7
21	281°29.4	169°11.2	•• 13.8	95°02.4	•• 20.3	175°00.2	•• 31.1	279°41.3	•• 52.6	Pollux	243°17.0	27°57.9
22	296°31.9	184°10.5	13.4	110°03.4	21.0	190°02.2	31.0	294°43.9	52.6	Avior	234°15.2	-59°35.3
23	311°34.4	199°09.8	13.1	125°04.4	21.6	205°04.1	31.0	309°46.4	52.7	Suhail	222°46.4	-43°32.0
Mer.pass. 02:17		ν -0.7' d -0.3' m -3.97		ν 1.0' d 0.7' m 1.62		ν 1.9' d -0.1' m -1.95		ν 2.6' d 0.1' m 0.72				

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	326°36.8	214°09.1	N21°12.8	140°05.5	S02°22.3	220°06.0	N22°30.9	324°49.0	S01°52.7		
1	341°39.3	229°08.4	12.5	155°06.5	22.9	235°07.9	30.8	339°51.6	52.8		
2	356°41.8	244°07.8	12.2	170°07.5	23.6	250°09.9	30.8	354°54.2	52.9		
3	11°44.2	259°07.1	•• 11.9	185°08.5	•• 24.2	265°11.8	•• 30.7	9°56.8	•• 52.9		
4	26°46.7	274°06.4	11.5	200°09.5	24.9	280°13.7	30.7	24°59.4	53.0		
5	41°49.2	289°05.7	11.2	215°10.5	25.5	295°15.7	30.6	40°02.0	53.1		
6	56°51.6	304°05.0	N21°10.9	230°11.6	S02°26.2	310°17.6	N22°30.6	55°04.6	S01°53.1		
7	71°54.1	319°04.3	10.6	245°12.6	26.8	325°19.5	30.5	70°07.1	53.2		
8	86°56.5	334°03.6	10.3	260°13.6	27.5	340°21.5	30.5	85°09.7	53.2		
9	101°59.0	349°02.9	•• 09.9	275°14.6	•• 28.2	355°23.4	•• 30.4	100°12.3	•• 53.3		
10	117°01.5	4°02.2	09.6	290°15.6	28.8	10°25.3	30.4	115°14.9	53.4		
11	132°03.9	19°01.5	09.3	305°16.6	29.5	25°27.3	30.3	130°17.5	53.4		
12	147°06.4	34°00.8	N21°09.0	320°17.7	S02°30.1	40°29.2	N22°30.3	145°20.1	S01°53.5		
13	162°08.9	49°00.1	08.6	335°18.7	30.8	55°31.1	30.2	160°22.7	53.6		
14	177°11.3	63°59.4	08.3	350°19.7	31.4	70°33.1	30.2	175°25.3	53.6		
15	192°13.8	78°58.7	•• 08.0	5°20.7	•• 32.1	85°35.0	•• 30.1	190°27.9	•• 53.7		
16	207°16.3	93°58.0	07.6	20°21.7	32.7	100°36.9	30.1	205°30.4	53.8		
17	222°18.7	108°57.4	07.3	35°22.7	33.4	115°38.9	30.0	220°33.0	53.8		
18	237°21.2	123°56.7	N21°07.0	50°23.8	S02°34.0	130°40.8	N22°30.0	235°35.6	S01°53.9		
19	252°23.7	138°56.0	06.6	65°24.8	34.7	145°42.7	29.9	250°38.2	53.9		
20	267°26.1	153°55.3	06.3	80°25.8	35.3	160°44.7	29.9	265°40.8	54.0		
21	282°28.6	168°54.6	•• 06.0	95°26.8	•• 36.0	175°46.6	•• 29.8	280°43.4	•• 54.1		
22	297°31.0	183°53.9	05.6	110°27.8	36.7	190°48.5	29.8	295°46.0	54.1		
23	312°33.5	198°53.2	05.3	125°28.8	37.3	205°50.5	29.7	310°48.6	54.2		
Mer.pass. 02:13		ν -0.7' d -0.3' m -3.96		ν 1.0' d 0.7' m 1.62		ν 1.9' d -0.1' m -1.95		ν 2.6' d 0.1' m 0.71			

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	327°36.0	213°52.5	N21°04.9	140°29.8	S02°38.0	220°52.4	N22°29.6	325°51.2	S01°54.3		
1	342°38.4	228°51.8	04.6	155°30.9	38.6	235°54.3	29.6	340°53.7	54.3		
2	357°40.9	243°51.1	04.3	170°31.9	39.3	250°56.3	29.5	355°56.3	54.4		
3	12°43.4	258°50.4	•• 03.9	185°32.9	•• 39.9	265°58.2	•• 29.5	10°58.9	•• 54.4		
4	27°45.8	273°49.7	03.6	200°33.9	40.6	281°00.1	29.4	26°01.5	54.5		
5	42°48.3	288°49.0	03.2	215°34.9	41.2	296°02.1	29.4	41°04.1	54.6		
6	57°50.8	303°48.4	N21°02.9	230°35.9	S02°41.9	311°04.0	N22°29.3	56°06.7	S01°54.6		
7	72°53.2	318°47.7	02.5	245°36.9	42.5	326°05.9	29.3	71°09.3	54.7		
8	87°55.7	333°47.0	02.2	260°38.0	43.2	341°07.9	29.2	86°11.9	54.8		
9	102°58.2	348°46.3	•• 01.8	275°39.0	•• 43.8	356°09.8	•• 29.2	101°14.5	•• 54.8		
10	118°00.6	3°45.6	01.5	290°40.0	44.5	11°11.7	29.1	116°17.1	54.9		
11	133°03.1	18°44.9	01.1	305°41.0	45.2	26°13.7	29.1	131°19.6	55.0		
12	148°05.5	33°44.2	N21°00.8	320°42.0	S02°45.8	41°15.6	N22°29.0	146°22.2	S01°55.0		
13	163°08.0	48°43.5	00.4	335°43.0	46.5	56°17.6	29.0	161°24.8	55.1		
14	178°10.5	63°42.8	21°00.1	350°44.0	47.1	71°19.5	28.9	176°27.4	55.1		
15	193°12.9	78°42.1	20°59.7	5°45.1	•• 47.8	86°21.4	•• 28.9	191°30.0	•• 55.2		
16	208°15.4	93°41.4	59.4	20°46.1	48.4	101°23.4	28.8	206°32.6	55.3		
17	223°17.9	108°40.8	59.0	35°47.1	49.1	116°25.3	28.8	221°35.2	55.3		
18	238°20.3	123°40.1	N20°58.6	50°48.1	S02°49.7	131°27.2	N22°28.7	236°37.8	S01°55.4		
19	253°22.8	138°39.4	58.3	65°49.1	50.4	146°29.2	28.6	251°40.4	55.5		
20	268°25.3	153°38.7	57.9	80°50.1	51.0	161°31.1	28.6	266°43.0	55.5		
21	283°27.7	168°38.0	•• 57.6	95°51.1	•• 51.7	176°33.0	•• 28.5	281°45.6	•• 55.6		
22	298°30.2	183°37.3	57.2	110°52.1	52.3	191°35.0	28.5	296°48.2	55.7		
23	313°32.6	198°36.6	56.8	125°53.1	53.0	206°36.9	28.4	311°50.7	55.7		
Mer.pass. 02:09		ν -0.7' d -0.3' m -3.96		ν 1.0' d 0.7' m 1.62		ν 1.9' d -0.1' m -1.95		ν 2.6' d 0.1' m 0.71			

Aug 17 Sun	SHA	Mer.pass
Venus	248°48.1	09:43
Mars	174°03.3	14:40
Jupiter	253°42.0	09:21
Saturn	358°09.3	02:24

Aug 18 Mon	SHA	Mer.pass
Venus	247°32.3	09:44
Mars	173°28.6	14:39
Jupiter	253°29.2	09:18
Saturn	358°12.2	02:20

Aug 19 Tue	SHA	Mer.pass
Venus	246°16.5	09:45
Mars	172°53.9	14:37
Jupiter	253°16.4	09:15
Saturn	358°15.2	02:16

Horizontal parallax	
Venus:	0.1
Mars:	0.1

Sun							Moon						
h	GHA	Dec	GHA	ν	Dec	d	HP						
0	178°57.9	N13°24.1	263°53.7	5.0'	N26°04.6	7.3'	59.1'						
1	193°58.0	23.3	278°17.7	5.0'	26°11.9	7.2'	59.1'						
2	208°58.2	22.5	292°41.7	4.9'	26°19.1	7.0'	59.1'						
3	223°58.3	· · 21.7	307°05.6	4.8'	26°26.1	6.8'	59.1'						
4	238°58.4	20.9	321°29.4	4.8'	26°32.9	6.6'	59.1'						
5	253°58.6	20.1	335°53.2	4.7'	26°39.5	6.5'	59.0'						
6	268°58.7	N13°19.3	350°16.9	4.6'	N26°46.0	6.3'	59.0'						
7	283°58.8	18.5	4°40.5	4.6'	26°52.3	6.1'	59.0'						
8	298°59.0	17.7	19°04.1	4.5'	26°58.4	5.9'	59.0'						
9	313°59.1	· · 16.9	33°27.6	4.5'	27°04.3	5.8'	59.0'						
10	328°59.2	16.1	47°51.0	4.4'	27°10.1	5.6'	59.0'						
11	343°59.4	15.3	62°14.4	4.3'	27°15.7	5.4'	59.0'						
12	358°59.5	N13°14.5	76°37.8	4.3'	N27°21.1	5.2'	59.0'						
13	13°59.7	13.7	91°01.1	4.2'	27°26.3	5.0'	59.0'						
14	28°59.8	12.9	105°24.3	4.2'	27°31.3	4.8'	59.0'						
15	43°59.9	· · 12.1	119°47.5	4.2'	27°36.1	4.6'	59.0'						
16	59°00.1	11.2	134°10.7	4.1'	27°40.8	4.5'	58.9'						
17	74°00.2	10.4	148°33.8	4.1'	27°45.2	4.3'	58.9'						
18	89°00.3	N13°09.6	162°56.8	4.0'	N27°49.5	4.1'	58.9'						
19	104°00.5	08.8	177°19.9	4.0'	27°53.6	3.9'	58.9'						
20	119°00.6	08.0	191°42.9	4.0'	27°57.5	3.7'	58.9'						
21	134°00.8	· · 07.2	206°05.8	3.9'	28°01.2	3.5'	58.9'						
22	149°00.9	06.4	220°28.7	3.9'	28°04.7	3.3'	58.9'						
23	164°01.0	05.6	234°51.6	3.9'	28°08.0	3.1'	58.9'						
SD = 15.8' $d = -0.8'$							SD = 16.1'						
Mon	GHA	Dec	GHA	ν	Dec	d	HP						
0	179°01.2	N13°04.8	249°14.5	3.8'	N28°11.2	2.9'	58.9'						
1	194°01.3	04.0	263°37.3	3.8'	28°14.1	2.8'	58.8'						
2	209°01.4	03.2	278°00.2	3.8'	28°16.9	2.6'	58.8'						
3	224°01.6	· · 02.4	292°23.0	3.8'	28°19.4	2.4'	58.8'						
4	239°01.7	01.6	306°45.7	3.8'	28°21.8	2.2'	58.8'						
5	254°01.9	13°00.8	321°08.5	3.8'	28°24.0	2.0'	58.8'						
6	269°02.0	N12°59.9	335°31.3	3.7'	N28°26.0	1.8'	58.8'						
7	284°02.1	59.1	349°54.0	3.7'	28°27.8	1.6'	58.8'						
8	299°02.3	58.3	4°16.7	3.7'	28°29.4	1.4'	58.8'						
9	314°02.4	· · 57.5	18°39.5	3.7'	28°30.8	1.2'	58.8'						
10	329°02.6	56.7	33°02.2	3.7'	28°32.0	1.0'	58.7'						
11	344°02.7	55.9	47°24.9	3.7'	28°33.0	0.8'	58.7'						
12	359°02.8	N12°55.1	61°47.7	3.7'	N28°33.9	0.6'	58.7'						
13	14°03.0	54.3	76°10.4	3.7'	28°34.5	0.4'	58.7'						
14	29°03.1	53.5	90°33.1	3.8'	28°34.9	0.2'	58.7'						
15	44°03.3	· · 52.6	104°55.9	3.8'	28°35.2	0.1'	58.7'						
16	59°03.4	51.8	119°18.6	3.8'	28°35.2	-0.1'	58.7'						
17	74°03.6	51.0	133°41.4	3.8'	28°35.1	-0.3'	58.7'						
18	89°03.7	N12°50.2	148°04.2	3.8'	N28°34.8	-0.5'	58.6'						
19	104°03.8	49.4	162°27.0	3.8'	28°34.3	-0.7'	58.6'						
20	119°04.0	48.6	176°49.9	3.9'	28°33.5	-0.9'	58.6'						
21	134°04.1	· · 47.8	191°12.8	3.9'	28°32.6	-1.1'	58.6'						
22	149°04.3	46.9	205°35.6	3.9'	28°31.5	-1.3'	58.6'						
23	164°04.4	46.1	219°58.6	4.0'	28°30.3	-1.5'	58.6'						
SD = 15.8' $d = -0.8'$							SD = 16.1'						
Tue	GHA	Dec	GHA	ν	Dec	d	HP						
0	179°04.6	N12°45.3	234°21.5	4.0'	N28°28.8	-1.7'	58.6'						
1	194°04.7	44.5	248°44.5	4.0'	28°27.1	-1.9'	58.5'						
2	209°04.8	43.7	263°07.6	4.1'	28°25.3	-2.0'	58.5'						
3	224°05.0	· · 42.9	277°30.6	4.1'	28°23.2	-2.2'	58.5'						
4	239°05.1	42.0	291°53.7	4.2'	28°21.0	-2.4'	58.5'						
5	254°05.3	41.2	306°16.9	4.2'	28°18.6	-2.6'	58.5'						
6	269°05.4	N12°40.4	320°40.1	4.3'	N28°16.0	-2.8'	58.5'						
7	284°05.6	39.6	335°03.4	4.3'	28°13.2	-3.0'	58.5'						
8	299°05.7	38.8	349°26.7	4.4'	28°10.2	-3.2'	58.4'						
9	314°05.9	· · 37.9	3°50.0	4.4'	28°07.0	-3.3'	58.4'						
10	329°06.0	37.1	18°13.5	4.5'	28°03.7	-3.5'	58.4'						
11	344°06.1	36.3	32°36.9	4.5'	28°00.2	-3.7'	58.4'						
12	359°06.3	N12°35.5	47°00.5	4.6'	N27°56.5	-3.9'	58.4'						
13	14°06.4	34.7	61°24.1	4.7'	27°52.6	-4.1'	58.4'						
14	29°06.6	33.8	75°47.7	4.7'	27°48.5	-4.2'	58.4'						
15	44°06.7	· · 33.0	90°11.5	4.8'	27°44.3	-4.4'	58.3'						
16	59°06.9	32.2	104°35.3	4.9'	27°39.9	-4.6'	58.3'						
17	74°07.0	31.4	118°59.1	4.9'	27°35.3	-4.8'	58.3'						
18	89°07.2	N12°30.6	133°23.1	5.0'	N27°30.5	-4.9'	58.3'						
19	104°07.3	29.7	147°47.1	5.1'	27°25.5	-5.1'	58.3'						
20	119°07.5	28.9	162°11.2	5.2'	27°20.4	-5.3'	58.3'						
21	134°07.6	· · 28.1	176°35.4	5.3'	27°15.1	-5.5'	58.2'						
22	149°07.8	27.3	190°59.7	5.3'	27°09.7	-5.6'	58.2'						
23	164°07.9	26.4	205°24.0	5.4'	27°04.0	-5.8'	58.2'						
SD = 15.8' $d = -0.8'$							SD = 16.0'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	02:46	21:17	—:—	—:—
N 70°	—:—	01:16	03:13	20:51	22:41	—:—
68°	—:—	02:07	03:33	20:32	21:56	—:—
66°	—:—	02:37	03:49	20:16	21:26	—:—
64°	01:07	03:00	04:02	20:04	21:05	22:52
62°	01:52	03:18	04:13	19:53	20:48	22:11
60°	02:20	03:32	04:22	19:44	20:33	21:44
N 58°	02:41	03:44	04:30	19:36	20:22	21:24
56°	02:58	03:55	04:37	19:29	20:11	21:07
54°	03:12	04:04	04:44	19:23	20:02	20:54
52°	03:24	04:12	04:49	19:17	19:55	20:42
50°	03:34	04:19	04:54	19:12	19:48	20:32
45°	03:55	04:34	05:05	19:01	19:33	20:11
N 40°	04:11	04:46	05:15	18:52	19:21	19:56
35°	04:24	04:56	05:22	18:45	19:11	19:43
30°	04:35	05:04	05:29	18:38	19:03	19:32
20°	04:51	05:18	05:41	18:26	18:49	19:16
N 10°	05:04	05:29	05:51	18:16	18:38	19:03
0°	05:15	05:39	06:00	18:07	18:28	18:53
S 10°	05:23	05:48	06:10	17:58	18:19	18:44
20°	05:31	05:57	06:19	17:48	18:11	18:37
30°	05:38	06:06	06:31	17:37	18:02	18:30
35°	05:41	06:11	06:37	17:31	17:57	18:27
40°	05:44	06:16	06:44	17:24	17:52	18:24
45°	05:48	06:22	06:52	17:16	17:46	18:21
S 50°	05:51	06:29	07:02	17:06	17:40	18:18
52°	05:52	06:32	07:07	17:02	17:37	18:16
54°	05:54	06:35	07:12	16:57	17:34	18:15
56°	05:55	06:38	07:17	16:51	17:30	18:14
58°	05:56	06:42	07:24	16:45	17:26	18:12
S 60°	05:58	06:47	07:31	16:38	17:22	18:11

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°						
N 70°						
68°						
66°						
64°						
62°	19:43		22:24	18:34		20:12
60°	20:50	21:39	23:13	17:27	18:48	19:22
N 58°	21:25	22:19	23:44	16:52	18:08	18:51
56°	21:51	22:47		16:27	17:40	18:28
54°	22:11	23:09		16:07	17:19	18:08
52°	22:28	23:27		15:50	17:01	17:52
50°	22:43	23:42		15:35	16:45	17:38
45°	23:13		00:13	15:06	16:14	17:09
N 40°	23:36		00:37	14:43	15:50	16:47
35°	23:56		00:57	14:24	15:30	16:28
30°		00:12	01:13	14:08	15:13	16:12
20°		00:40	01:42	13:41	14:45	15:44
N 10°	00:06	01:05	02:07	13:17	14:20	15:21
0°	00:26	01:27	02:29	12:55	13:57	14:59
S 10°	00:47	01:50	02:52	12:33	13:34	14:36
20°	01:09	02:15	03:17	12:10	13:09	14:12
30°	01:35	02:43	03:45	11:43	12:40	13:44
35°	01:51	03:00	04:02	11:26	12:23	13:28
40°	02:09	03:20	04:22	11:08	12:03	13:08
45°	02:30	03:44	04:46	10:45	11:39	12:45
S 50°	02:58	04:15	05:17	10:17	11:08	12:15
52°	03:12	04:31	05:32	10:03	10:52	12:00
54°	03:27	04:49	05:50	09:47	10:34	11:42
56°	03:46	05:11	06:11	09:28	10:11	11:21
58°	04:09	05:40	06:38	09:05	09:43	10:54
S 60°	04:39	06:22	07:17	08:34	09:01	10:15

August 20, 21, 22 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	328°35.1	213°35.9	N20°56.5	140°54.2	S02°53.7	221°38.9	N22°28.4	326°53.3	S01°55.8	Alpheratz	357°33.9	29°14.0
1	343°37.6	228°35.2	56.1	155°55.2	54.3	236°40.8	28.3	341°55.9	55.9	Ankaa	353°06.3	-42°09.7
2	358°40.0	243°34.5	55.7	170°56.2	55.0	251°42.7	28.3	356°58.5	55.9	Schedar	349°30.1	56°40.6
3	13°42.5	258°33.9	· · 55.4	185°57.2	· · 55.6	266°44.7	· · 28.2	12°01.1	· · 56.0	Diphda	348°46.5	-17°50.5
4	28°45.0	273°33.2	55.0	200°58.2	56.3	281°46.6	28.2	27°03.7	56.0	Achernar	335°19.5	-57°06.1
5	43°47.4	288°32.5	54.6	215°59.2	56.9	296°48.6	28.1	42°06.3	56.1	Hamal	327°50.5	23°35.0
6	58°49.9	303°31.8	N20°54.3	231°00.2	S02°57.6	311°50.5	N22°28.1	57°08.9	S01°56.2	Polaris	313°41.3	89°22.0
7	73°52.4	318°31.1	53.9	246°01.2	58.2	326°52.4	28.0	72°11.5	56.2	Acamar	315°11.3	-40°11.8
8	88°54.8	333°30.4	53.5	261°02.2	58.9	341°54.4	28.0	87°14.1	56.3	Menkar	314°05.5	4°11.5
9	103°57.3	348°29.7	· · 53.2	276°03.3	02°59.5	356°56.3	· · 27.9	102°16.7	· · 56.4	Mirfak	308°27.5	49°57.0
10	118°59.8	3°29.0	52.8	291°04.3	03°00.2	11°58.2	27.9	117°19.3	56.4	Aldebaran	290°39.1	16°33.7
11	134°02.2	18°28.3	52.4	306°05.3	00.9	27°00.2	27.8	132°21.9	56.5	Rigel	281°03.5	-8°10.1
12	149°04.7	33°27.7	N20°52.0	321°06.3	S03°01.5	42°02.1	N22°27.8	147°24.5	S01°56.6	Capella	280°21.3	46°01.3
13	164°07.1	48°27.0	51.6	336°07.3	02.2	57°04.1	27.7	162°27.0	56.6	Bellatrix	278°22.5	6°22.5
14	179°09.6	63°26.3	51.3	351°08.3	02.8	72°06.0	27.6	177°29.6	56.7	Elnath	278°01.4	28°37.7
15	194°12.1	78°25.6	· · 50.9	6°09.3	· · 03.5	87°07.9	· · 27.6	192°32.2	· · 56.8	Alnilam	275°37.3	-1°11.0
16	209°14.5	93°24.9	50.5	21°10.3	04.1	102°09.9	27.5	207°34.8	56.8	Betelgeuse	270°51.7	7°24.8
17	224°17.0	108°24.2	50.1	36°11.3	04.8	117°11.8	27.5	222°37.4	56.9	Canopus	263°52.5	-52°42.2
18	239°19.5	123°23.5	N20°49.7	51°12.3	S03°05.4	132°13.8	N22°27.4	237°40.0	S01°57.0	Sirius	258°26.0	-16°44.9
19	254°21.9	138°22.8	49.4	66°13.4	06.1	147°15.7	27.4	252°42.6	57.0	Adhara	255°05.7	-29°00.2
20	269°24.4	153°22.2	49.0	81°14.4	06.7	162°17.6	27.3	267°45.2	57.1	Procyon	244°50.6	5°09.7
21	284°26.9	168°21.5	· · 48.6	96°15.4	· · 07.4	177°19.6	· · 27.3	282°47.8	· · 57.1	Pollux	243°17.0	27°57.9
22	299°29.3	183°20.8	48.2	111°16.4	08.0	192°21.5	27.2	297°50.4	57.2	Avior	234°15.1	-59°35.3
23	314°31.8	198°20.1	47.8	126°17.4	08.7	207°23.5	27.2	312°53.0	57.3	Suhail	222°46.4	-43°32.0
Mer.pass. 02:05		ν -0.7' d -0.4' m -3.96		ν 1.0' d 0.7' m 1.62		ν 1.9' d -0.1' m -1.96		ν 2.6' d 0.1' m 0.71				

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	329°34.3	213°19.4	N20°47.4	141°18.4	S03°09.4	222°25.4	N22°27.1	327°55.6	S01°57.3
1	344°36.7	228°18.7	47.0	156°19.4	10.0	237°27.3	27.1	342°58.2	57.4
2	359°39.2	243°18.0	46.6	171°20.4	10.7	252°29.3	27.0	358°00.8	57.5
3	14°41.6	258°17.3	· · 46.2	186°21.4	· · 11.3	267°31.2	· · 27.0	13°03.4	· · 57.5
4	29°44.1	273°16.7	45.9	201°22.4	12.0	282°33.2	26.9	28°06.0	57.6
5	44°46.6	288°16.0	45.5	216°23.4	12.6	297°35.1	26.9	43°08.6	57.7
6	59°49.0	303°15.3	N20°45.1	231°24.4	S03°13.3	312°37.1	N22°26.8	58°11.2	S01°57.7
7	74°51.5	318°14.6	44.7	246°25.4	13.9	327°39.0	26.7	73°13.8	57.8
8	89°54.0	333°13.9	44.3	261°26.5	14.6	342°40.9	26.7	88°16.4	57.9
9	104°56.4	348°13.2	· · 43.9	276°27.5	· · 15.2	357°42.9	· · 26.6	103°19.0	· · 57.9
10	119°58.9	3°12.5	43.5	291°28.5	15.9	12°44.8	26.6	118°21.5	58.0
11	135°01.4	18°11.9	43.1	306°29.5	16.6	27°46.8	26.5	133°24.1	58.1
12	150°03.8	33°11.2	N20°42.7	321°30.5	S03°17.2	42°48.7	N22°26.5	148°26.7	S01°58.1
13	165°06.3	48°10.5	42.3	336°31.5	17.9	57°50.7	26.4	163°29.3	58.2
14	180°08.8	63°09.8	41.9	351°32.5	18.5	72°52.6	26.4	178°31.9	58.3
15	195°11.2	78°09.1	· · 41.5	6°33.5	· · 19.2	87°54.5	· · 26.3	193°34.5	· · 58.3
16	210°13.7	93°08.4	41.1	21°34.5	19.8	102°56.5	26.3	208°37.1	58.4
17	225°16.1	108°07.7	40.7	36°35.5	20.5	117°58.4	26.2	223°39.7	58.5
18	240°18.6	123°07.1	N20°40.2	51°36.5	S03°21.1	133°00.4	N22°26.2	238°42.3	S01°58.5
19	255°21.1	138°06.4	39.8	66°37.5	21.8	148°02.3	26.1	253°44.9	58.6
20	270°23.5	153°05.7	39.4	81°38.5	22.4	163°04.3	26.1	268°47.5	58.7
21	285°26.0	168°05.0	· · 39.0	96°39.5	· · 23.1	178°06.2	· · 26.0	283°50.1	· · 58.7
22	300°28.5	183°04.3	38.6	111°40.5	23.8	193°08.1	25.9	298°52.7	58.8
23	315°30.9	198°03.6	38.2	126°41.5	24.4	208°10.1	25.9	313°55.3	58.9
Mer.pass. 02:01		ν -0.7' d -0.4' m -3.96		ν 1.0' d 0.7' m 1.62		ν 1.9' d -0.1' m -1.96		ν 2.6' d 0.1' m 0.70	

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	330°33.4	213°03.0	N20°37.8	141°42.5	S03°25.1	223°12.0	N22°25.8	328°57.9	S01°58.9
1	345°35.9	228°02.3	37.4	156°43.6	25.7	238°14.0	25.8	344°00.5	59.0
2	0°38.3	243°01.6	37.0	171°44.6	26.4	253°15.9	25.7	359°03.1	59.1
3	15°40.8	258°00.9	· · 36.5	186°45.6	· · 27.0	268°17.9	· · 25.7	14°05.7	· · 59.1
4	30°43.2	273°00.2	36.1	201°46.6	27.7	283°19.8	25.6	29°08.3	59.2
5	45°45.7	287°59.5	35.7	216°47.6	28.3	298°21.8	25.6	44°10.9	59.3
6	60°48.2	302°58.9	N20°35.3	231°48.6	S03°29.0	313°23.7	N22°25.5	59°13.5	S01°59.3
7	75°50.6	317°58.2	34.9	246°49.6	29.6	328°25.6	25.5	74°16.1	59.4
8	90°53.1	332°57.5	34.4	261°50.6	30.3	343°27.6	25.4	89°18.7	59.5
9	105°55.6	347°56.8	· · 34.0	276°51.6	· · 30.9	358°29.5	· · 25.4	104°21.3	· · 59.5
10	120°58.0	2°56.1	33.6	291°52.6	31.6	13°31.5	25.3	119°23.9	59.6
11	136°00.5	17°55.5	33.2	306°53.6	32.3	28°33.4	25.3	134°26.5	59.7
12	151°03.0	32°54.8	N20°32.7	321°54.6	S03°32.9	43°35.4	N22°25.2	149°29.1	S01°59.7
13	166°05.4	47°54.1	32.3	336°55.6	33.6	58°37.3	25.1	164°31.7	59.8
14	181°07.9	62°53.4	31.9	351°56.6	34.2	73°39.3	25.1	179°34.3	59.9
15	196°10.4	77°52.7	· · 31.5	6°57.6	· · 34.9	88°41.2	· · 25.0	194°36.9	01°59.9
16	211°12.8	92°52.0	31.0	21°58.6	35.5	103°43.2	25.0	209°39.5	02°00.0
17	226°15.3	107°51.4	30.6	36°59.6	36.2	118°45.1	24.9	224°42.1	00.1
18	241°17.7	122°50.7	N20°30.2	52°00.6	S03°36.8	133°47.1	N22°24.9	239°44.7	S02°00.1
19	256°20.2	137°50.0	29.7	67°01.6	37.5	148°49.0	24.8	254°47.3	00.2
20	271°22.7	152°49.3	29.3	82°02.6	38.1	163°51.0	24.8	269°49.9	00.3
21	286°25.1	167°48.6	· · 28.9	97°03.6	· · 38.8	178°52.9	· · 24.7	284°52.5	· · 00.3
22	301°27.6	182°48.0	28.4	112°04.6	39.5	193°54.8	24.7	299°55.1	00.4
23	316°30.1	197°47.3	28.0	127°05.6	40.1	208°56.8	24.6	314°57.7	00.5
Mer.pass. 01:57		ν -0.7' d -0.4' m -3.96		ν 1.0' d 0.7' m 1.62		ν 1.9' d -0.1' m -1.96		ν 2.6' d 0.1' m 0.70	

Aug 20 Wed	SHA	Mer.pass
Venus	245°00.8	09:46
Mars	172°19.0	14:35
Jupiter	253°03.7	09:12
Saturn	358°18.2	02:12

Aug 21 Thu	SHA	Mer.pass
Venus	243°45.2	09:47
Mars	171°44.1	14:34
Jupiter	252°51.2	09:09
Saturn	358°21.3	02:08

Aug 22 Fri	SHA	Mer.pass
Venus	242°29.6	09:48
Mars	171°09.2	14:32
Jupiter	252°38.6	09:06
Saturn	358°24.5	02:04

Horizontal parallax	
Venus:	0.1
Mars:	0.1

Sun				Moon			
h							
Wed	GHA	Dec		GHA	ν	Dec	HP
0	179°08.1	N12°25.6		219°48.4	5.5'	N26°58.2	-6.0'
1	194°08.2	24.8		234°12.9	5.6'	26°52.2	-6.1'
2	209°08.4	24.0		248°37.5	5.7'	26°46.1	-6.3'
3	224°08.5	· · 23.1		263°02.2	5.8'	26°39.8	-6.5'
4	239°08.7	22.3		277°27.0	5.9'	26°33.4	-6.6'
5	254°08.8	21.5		291°51.8	6.0'	26°26.7	-6.8'
6	269°09.0	N12°20.7		306°16.8	6.1'	N26°19.9	-6.9'
7	284°09.1	19.8		320°41.9	6.1'	26°13.0	-7.1'
8	299°09.3	19.0		335°07.0	6.2'	26°05.9	-7.3'
9	314°09.4	· · 18.2		349°32.2	6.3'	25°58.7	-7.4'
10	329°09.6	17.4		3°57.6	6.4'	25°51.2	-7.6'
11	344°09.7	16.5		18°23.0	6.5'	25°43.7	-7.7'
12	359°09.9	N12°15.7		32°48.6	6.6'	N25°36.0	-7.9'
13	14°10.0	14.9		47°14.2	6.7'	25°28.1	-8.0'
14	29°10.2	14.0		61°39.9	6.8'	25°20.1	-8.2'
15	44°10.3	· · 13.2		76°05.8	6.9'	25°11.9	-8.3'
16	59°10.5	12.4		90°31.7	7.0'	25°03.6	-8.4'
17	74°10.6	11.6		104°57.8	7.2'	24°55.2	-8.6'
18	89°10.8	N12°10.7		119°23.9	7.3'	N24°46.6	-8.7'
19	104°10.9	09.9		133°50.2	7.4'	24°37.9	-8.9'
20	119°11.1	09.1		148°16.6	7.5'	24°29.0	-9.0'
21	134°11.2	· · 08.2		162°43.0	7.6'	24°20.0	-9.1'
22	149°11.4	07.4		177°09.6	7.7'	24°10.8	-9.3'
23	164°11.5	06.6		191°36.3	7.8'	24°01.6	-9.4'
SD = 15.8'			d = -0.8'	SD = 15.9'			

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	179° 11.7	N12° 05.7	206° 03.1	7.9'	N23° 52.1	-9.5'	57.8'
1	194° 11.8	04.9	220° 30.0	8.0'	23° 42.6	-9.7'	57.7'
2	209° 12.0	04.1	234° 57.0	8.1'	23° 32.9	-9.8'	57.7'
3	224° 12.1	· · 03.2	249° 24.1	8.2'	23° 23.1	-9.9'	57.7'
4	239° 12.3	02.4	263° 51.4	8.3'	23° 13.2	-10.0'	57.7'
5	254° 12.4	01.6	278° 18.7	8.5'	23° 03.2	-10.2'	57.7'
6	269° 12.6	N12° 00.7	292° 46.2	8.6'	N22° 53.0	-10.3'	57.6'
7	284° 12.8	11° 59.9	307° 13.7	8.7'	22° 42.7	-10.4'	57.6'
8	299° 12.9	59.1	321° 41.4	8.8'	22° 32.3	-10.5'	57.6'
9	314° 13.1	· · 58.2	336° 09.2	8.9'	22° 21.8	-10.6'	57.6'
10	329° 13.2	57.4	350° 37.1	9.0'	22° 11.1	-10.8'	57.6'
11	344° 13.4	56.6	5° 05.1	9.1'	22° 00.4	-10.9'	57.5'
12	359° 13.5	N11° 55.7	19° 33.2	9.2'	N21° 49.5	-11.0'	57.5'
13	14° 13.7	54.9	34° 01.4	9.3'	21° 38.5	-11.1'	57.5'
14	29° 13.8	54.0	48° 29.8	9.4'	21° 27.4	-11.2'	57.5'
15	44° 14.0	· · 53.2	62° 58.2	9.6'	21° 16.2	-11.3'	57.5'
16	59° 14.2	52.4	77° 26.8	9.7'	21° 04.9	-11.4'	57.4'
17	74° 14.3	51.5	91° 55.4	9.8'	20° 53.5	-11.5'	57.4'
18	89° 14.5	N11° 50.7	106° 24.2	9.9'	N20° 42.0	-11.6'	57.4'
19	104° 14.6	49.9	120° 53.1	10.0'	20° 30.4	-11.7'	57.4'
20	119° 14.8	49.0	135° 22.1	10.1'	20° 18.7	-11.8'	57.4'
21	134° 14.9	· · 48.2	149° 51.2	10.2'	20° 06.8	-11.9'	57.3'
22	149° 15.1	47.3	164° 20.4	10.3'	19° 54.9	-12.0'	57.3'
23	164° 15.3	46.5	178° 49.7	10.4'	19° 42.9	-12.1'	57.3'
	SD = 15.8'	d = -0.8'			SD = 15.8'		

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	179° 15.4	N11° 45.7	193° 19.1	10.5'	N19° 30.8	-12.2'	57.3'
1	194° 15.6	44.8	207° 48.6	10.6'	19° 18.7	-12.3'	57.2'
2	209° 15.7	44.0	222° 18.3	10.7'	19° 06.4	-12.4'	57.2'
3	224° 15.9	· · 43.1	236° 48.0	10.8'	18° 54.0	-12.4'	57.2'
4	239° 16.0	42.3	251° 17.8	10.9'	18° 41.6	-12.5'	57.2'
5	254° 16.2	41.4	265° 47.8	11.1'	18° 29.0	-12.6'	57.2'
6	269° 16.4	N11° 40.6	280° 17.8	11.2'	N18° 16.4	-12.7'	57.1'
7	284° 16.5	39.8	294° 48.0	11.3'	18° 03.7	-12.8'	57.1'
8	299° 16.7	38.9	309° 18.2	11.4'	17° 51.0	-12.9'	57.1'
9	314° 16.8	· · 38.1	323° 48.6	11.5'	17° 38.1	-12.9'	57.1'
10	329° 17.0	37.2	338° 19.1	11.6'	17° 25.2	-13.0'	57.0'
11	344° 17.2	36.4	352° 49.6	11.7'	17° 12.2	-13.1'	57.0'
12	359° 17.3	N11° 35.5	7° 20.3	11.8'	N16° 59.1	-13.2'	57.0'
13	14° 17.5	34.7	21° 51.0	11.9'	16° 45.9	-13.2'	57.0'
14	29° 17.6	33.9	36° 21.9	11.9'	16° 32.7	-13.3'	57.0'
15	44° 17.8	· · 33.0	50° 52.8	12.0'	16° 19.4	-13.4'	56.9'
16	59° 18.0	32.2	65° 23.9	12.1'	16° 06.1	-13.4'	56.9'
17	74° 18.1	31.3	79° 55.0	12.2'	15° 52.6	-13.5'	56.9'
18	89° 18.3	N11° 30.5	94° 26.2	12.3'	N15° 39.1	-13.6'	56.9'
19	104° 18.5	29.6	108° 57.6	12.4'	15° 25.6	-13.6'	56.8'
20	119° 18.6	28.8	123° 29.0	12.5'	15° 12.0	-13.7'	56.8'
21	134° 18.8	· · 27.9	138° 00.5	12.6'	14° 58.3	-13.7'	56.8'
22	149° 18.9	27.1	152° 32.1	12.7'	14° 44.6	-13.8'	56.8'
23	164° 19.1	26.2	167° 03.8	12.8'	14° 30.8	-13.8'	56.7'
	SD = 15.8'	$d = -0.8'$			SD = 15.6'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	—:—	03:04	20:58	23:25	—:—
N 70°	—:—	01:47	03:27	20:36	22:12	—:—
68°	—:—	02:25	03:45	20:18	21:36	—:—
66°	—:—	02:51	03:59	20:05	21:11	23:30
64°	01:33	03:11	04:11	19:53	20:52	22:26
62°	02:08	03:27	04:21	19:44	20:36	21:54
60°	02:33	03:40	04:29	19:35	20:24	21:30
N 58°	02:51	03:52	04:37	19:28	20:13	21:12
56°	03:07	04:01	04:43	19:22	20:03	20:57
54°	03:19	04:10	04:49	19:16	19:55	20:45
52°	03:30	04:17	04:54	19:11	19:48	20:34
50°	03:40	04:24	04:59	19:06	19:41	20:25
45°	03:59	04:38	05:09	18:56	19:27	20:06
N 40°	04:15	04:49	05:17	18:48	19:16	19:51
35°	04:27	04:58	05:25	18:41	19:07	19:39
30°	04:37	05:06	05:31	18:35	19:00	19:29
20°	04:52	05:19	05:42	18:24	18:47	19:13
N 10°	05:04	05:30	05:51	18:15	18:36	19:02
0°	05:14	05:39	06:00	18:06	18:28	18:52
S 10°	05:22	05:47	06:08	17:58	18:19	18:44
20°	05:29	05:55	06:17	17:49	18:12	18:37
30°	05:35	06:03	06:27	17:39	18:04	18:32
35°	05:38	06:07	06:33	17:33	17:59	18:29
40°	05:40	06:12	06:40	17:27	17:55	18:26
45°	05:43	06:17	06:47	17:19	17:50	18:24
S 50°	05:46	06:23	06:56	17:11	17:44	18:22
52°	05:46	06:26	07:01	17:06	17:41	18:21
54°	05:47	06:29	07:05	17:02	17:38	18:20
56°	05:48	06:32	07:10	16:57	17:35	18:19
58°	05:49	06:35	07:16	16:51	17:32	18:18
S 60°	05:50	06:39	07:22	16:45	17:29	18:17

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	☐	☐	☐	☐	☐	21:25
N 70°	☐	☐	00:07	☐	22:23	20:55
68°	☐	☐	01:12	☐	21:17	20:33
66°	☐	☐	01:47	☐	20:40	20:15
64°	23:58	· ·	02:12	20:38	20:14	20:00
62°	· ·	00:36	02:32	20:00	19:53	19:48
60°	· ·	01:02	02:48	19:33	19:37	19:37
N 58°	· ·	01:22	03:01	19:12	19:22	19:28
56°	00:07	01:39	03:13	18:54	19:10	19:20
54°	00:26	01:54	03:23	18:40	18:59	19:13
52°	00:42	02:06	03:32	18:27	18:50	19:06
50°	00:56	02:17	03:39	18:15	18:41	19:00
45°	01:24	02:40	03:56	17:51	18:23	18:47
N 40°	01:46	02:58	04:10	17:32	18:08	18:37
35°	02:04	03:13	04:22	17:16	17:55	18:27
30°	02:19	03:26	04:32	17:02	17:44	18:19
20°	02:46	03:49	04:49	16:38	17:24	18:05
N 10°	03:09	04:08	05:04	16:17	17:07	17:53
0°	03:30	04:26	05:18	15:57	16:51	17:41
S 10°	03:51	04:44	05:32	15:37	16:35	17:29
20°	04:13	05:03	05:47	15:16	16:18	17:17
30°	04:39	05:25	06:03	14:51	15:58	17:02
35°	04:55	05:38	06:13	14:37	15:46	16:53
40°	05:12	05:53	06:24	14:20	15:33	16:44
45°	05:34	06:10	06:37	13:59	15:16	16:32
S 50°	06:01	06:31	06:53	13:33	14:56	16:18
52°	06:14	06:41	07:00	13:21	14:47	16:11
54°	06:29	06:53	07:09	13:06	14:36	16:04
56°	06:46	07:06	07:18	12:49	14:24	15:56
58°	07:07	07:21	07:28	12:29	14:09	15:46
S 60°	07:34	07:39	07:40	12:02	13:52	15:36

Day	Sun		Mer. Pass hh:mm	Moon		Age 27-29 12-2%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
20	03:28	03:21	12:03	09:44	22:12	
21	03:13	03:06	12:03	10:39	23:05	
22	02:58	02:51	12:03	11:30	23:53	

August 23, 24, 25 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	331°32.5	212°46.6	N20°27.6	142°06.6	S03°40.8	223°58.7	N22°24.6	330°00.3	S02°00.5	Alpheratz	357°33.9	29°14.0
1	346°35.0	227°45.9	27.1	157°07.6	41.4	239°00.7	24.5	345°02.9	00.6	Ankaa	353°06.3	-42°09.7
2	1°37.5	242°45.2	26.7	172°08.6	42.1	254°02.6	24.4	0°05.5	00.7	Schedar	349°30.0	56°40.6
3	16°39.9	257°44.6	•• 26.2	187°09.6	•• 42.7	269°04.6	•• 24.4	15°08.1	•• 00.7	Diphda	348°46.5	-17°50.5
4	31°42.4	272°43.9	25.8	202°10.6	43.4	284°06.5	24.3	30°10.7	00.8	Achernar	335°19.4	-57°06.1
5	46°44.9	287°43.2	25.4	217°11.6	44.0	299°08.5	24.3	45°13.3	00.9	Hamal	327°50.4	23°35.1
6	61°47.3	302°42.5	N20°24.9	232°12.6	S03°44.7	314°10.4	N22°24.2	60°15.9	S02°00.9	Polaris	313°39.9	89°22.0
7	76°49.8	317°41.9	24.5	247°13.6	45.3	329°12.4	24.2	75°18.5	01.0	Acamar	315°11.2	-40°11.8
8	91°52.2	332°41.2	24.0	262°14.6	46.0	344°14.3	24.1	90°21.1	01.1	Menkar	314°05.5	4°11.6
9	106°54.7	347°40.5	•• 23.6	277°15.6	•• 46.7	359°16.3	•• 24.1	105°23.7	•• 01.1	Mirfak	308°27.4	49°57.0
10	121°57.2	2°39.8	23.1	292°16.6	47.3	14°18.2	24.0	120°26.3	01.2	Aldebaran	290°39.1	16°33.7
11	136°59.6	17°39.1	22.7	307°17.6	48.0	29°20.2	24.0	135°28.9	01.3	Rigel	281°03.5	-8°10.1
12	152°02.1	32°38.5	N20°22.2	322°18.6	S03°48.6	44°22.1	N22°23.9	150°31.5	S02°01.3	Capella	280°21.3	46°01.3
13	167°04.6	47°37.8	21.8	337°19.6	49.3	59°24.1	23.9	165°34.1	01.4	Bellatrix	278°22.4	6°22.5
14	182°07.0	62°37.1	21.3	352°20.6	49.9	74°26.0	23.8	180°36.7	01.5	Elnath	278°01.3	28°37.7
15	197°09.5	77°36.4	•• 20.9	7°21.6	•• 50.6	89°28.0	•• 23.8	195°39.3	•• 01.5	Alnilam	275°37.3	-1°11.0
16	212°12.0	92°35.8	20.4	22°22.6	51.2	104°29.9	23.7	210°41.9	01.6	Betelgeuse	270°51.7	7°24.8
17	227°14.4	107°35.1	20.0	37°23.6	51.9	119°31.9	23.6	225°44.5	01.7	Canopus	263°52.5	-52°42.2
18	242°16.9	122°34.4	N20°19.5	52°24.6	S03°52.5	134°33.8	N22°23.6	240°47.1	S02°01.8	Sirius	258°26.0	-16°44.9
19	257°19.3	137°33.7	19.0	67°25.6	53.2	149°35.8	23.5	255°49.7	01.8	Adhara	255°05.7	-29°00.2
20	272°21.8	152°33.0	18.6	82°26.6	53.8	164°37.7	23.5	270°52.3	01.9	Procyon	244°50.5	5°09.7
21	287°24.3	167°32.4	•• 18.1	97°27.6	•• 54.5	179°39.7	•• 23.4	285°54.9	•• 02.0	Pollux	243°17.0	27°57.9
22	302°26.7	182°31.7	17.7	112°28.6	55.2	194°41.6	23.4	300°57.5	02.0	Avior	234°15.1	-59°35.3
23	317°29.2	197°31.0	17.2	127°29.6	55.8	209°43.6	23.3	316°00.1	02.1	Suhail	222°46.4	-43°32.0
Mer.pass. 01:54		ν -0.7' d -0.4' m -3.96		ν 1.0' d 0.7' m 1.62		ν 1.9' d -0.1' m -1.96		ν 2.6' d 0.1' m 0.69				

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	332°31.7	212°30.3	N20°16.8	142°30.6	S03°56.5	224°45.5	N22°23.3	331°02.7	S02°02.2
1	347°34.1	227°29.7	16.3	157°31.6	57.1	239°47.5	23.2	346°05.3	02.2
2	2°36.6	242°29.0	15.8	172°32.6	57.8	254°49.4	23.2	1°07.9	02.3
3	17°39.1	257°28.3	•• 15.4	187°33.6	•• 58.4	269°51.4	•• 23.1	16°10.5	•• 02.4
4	32°41.5	272°27.6	14.9	202°34.6	59.1	284°53.3	23.1	31°13.1	02.4
5	47°44.0	287°27.0	14.4	217°35.6	03°59.7	299°55.3	23.0	46°15.7	02.5
6	62°46.5	302°26.3	N20°14.0	232°36.6	S04°00.4	314°57.3	N22°22.9	61°18.4	S02°02.6
7	77°48.9	317°25.6	13.5	247°37.6	01.0	329°59.2	22.9	76°21.0	02.6
8	92°51.4	332°25.0	13.0	262°38.6	01.7	345°01.2	22.8	91°23.6	02.7
9	107°53.8	347°24.3	•• 12.6	277°39.6	•• 02.4	0°03.1	•• 22.8	106°26.2	•• 02.8
10	122°56.3	2°23.6	12.1	292°40.6	03.0	15°05.1	22.7	121°28.8	02.8
11	137°58.8	17°22.9	11.6	307°41.6	03.7	30°07.0	22.7	136°31.4	02.9
12	153°01.2	32°22.3	N20°11.1	322°42.6	S04°04.3	45°09.0	N22°22.6	151°34.0	S02°03.0
13	168°03.7	47°21.6	10.7	337°43.6	05.0	60°10.9	22.6	166°36.6	03.1
14	183°06.2	62°20.9	10.2	352°44.6	05.6	75°12.9	22.5	181°39.2	03.1
15	198°08.6	77°20.2	•• 09.7	7°45.6	•• 06.3	90°14.8	•• 22.5	196°41.8	•• 03.2
16	213°11.1	92°19.6	09.2	22°46.6	06.9	105°16.8	22.4	211°44.4	03.3
17	228°13.6	107°18.9	08.7	37°47.6	07.6	120°18.7	22.3	226°47.0	03.3
18	243°16.0	122°18.2	N20°08.3	52°48.5	S04°08.2	135°20.7	N22°22.3	241°49.6	S02°03.4
19	258°18.5	137°17.5	07.8	67°49.5	08.9	150°22.7	22.2	256°52.2	03.5
20	273°20.9	152°16.9	07.3	82°50.5	09.5	165°24.6	22.2	271°54.8	03.5
21	288°23.4	167°16.2	•• 06.8	97°51.5	•• 10.2	180°26.6	•• 22.1	286°57.4	•• 03.6
22	303°25.9	182°15.5	06.3	112°52.5	10.9	195°28.5	22.1	302°00.0	03.7
23	318°28.3	197°14.9	05.9	127°53.5	11.5	210°30.5	22.0	317°02.6	03.7
Mer.pass. 01:50		ν -0.7' d -0.5' m -3.96		ν 1.0' d 0.7' m 1.62		ν 2.0' d -0.1' m -1.97		ν 2.6' d 0.1' m 0.69	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	333°30.8	212°14.2	N20°05.4	142°54.5	S04°12.2	225°32.4	N22°22.0	332°05.2	S02°03.8
1	348°33.3	227°13.5	04.9	157°55.5	12.8	240°34.4	21.9	347°07.8	03.9
2	3°35.7	242°12.9	04.4	172°56.5	13.5	255°36.3	21.9	2°10.4	04.0
3	18°38.2	257°12.2	•• 03.9	187°57.5	•• 14.1	270°38.3	•• 21.8	17°13.1	•• 04.0
4	33°40.7	272°11.5	03.4	202°58.5	14.8	285°40.2	21.8	32°15.7	04.1
5	48°43.1	287°10.8	02.9	217°59.5	15.4	300°42.2	21.7	47°18.3	04.2
6	63°45.6	302°10.2	N20°02.4	233°00.5	S04°16.1	315°44.2	N22°21.6	62°20.9	S02°04.2
7	78°48.1	317°09.5	01.9	248°01.5	16.7	330°46.1	21.6	77°23.5	04.3
8	93°50.5	332°08.8	01.5	263°02.5	17.4	345°48.1	21.5	92°26.1	04.4
9	108°53.0	347°08.2	•• 01.0	278°03.5	•• 18.0	0°50.0	•• 21.5	107°28.7	•• 04.4
10	123°55.4	2°07.5	00.5	293°04.5	18.7	15°52.0	21.4	122°31.3	04.5
11	138°57.9	17°06.8	20°00.0	308°05.4	19.4	30°53.9	21.4	137°33.9	04.6
12	154°00.4	32°06.2	N19°59.5	323°06.4	S04°20.0	45°55.9	N22°21.3	152°36.5	S02°04.6
13	169°02.8	47°05.5	59.0	338°07.4	20.7	60°57.9	21.3	167°39.1	04.7
14	184°05.3	62°04.8	58.5	353°08.4	21.3	75°59.8	21.2	182°41.7	04.8
15	199°07.8	77°04.2	•• 58.0	8°09.4	•• 22.0	91°01.8	•• 21.2	197°44.3	•• 04.9
16	214°10.2	92°03.5	57.5	23°10.4	22.6	106°03.7	21.1	212°46.9	04.9
17	229°12.7	107°02.8	57.0	38°11.4	23.3	121°05.7	21.0	227°49.5	05.0
18	244°15.2	122°02.2	N19°56.5	53°12.4	S04°23.9	136°07.6	N22°21.0	242°52.2	S02°05.1
19	259°17.6	137°01.5	56.0	68°13.4	24.6	151°09.6	20.9	257°54.8	05.1
20	274°20.1	152°00.8	55.5	83°14.4	25.2	166°11.6	20.9	272°57.4	05.2
21	289°22.5	167°00.2	•• 54.9	98°15.4	•• 25.9	181°13.5	•• 20.8	288°00.0	•• 05.3
22	304°25.0	181°59.5	54.4	113°16.4	26.5	196°15.5	20.8	303°02.6	05.3
23	319°27.5	196°58.8	53.9	128°17.3	27.2	211°17.4	20.7	318°05.2	05.4
Mer.pass. 01:46		ν -0.7' d -0.5' m -3.95		ν 1.0' d 0.7' m 1.62		ν 2.0' d -0.1' m -1.97		ν 2.6' d 0.1' m 0.68	

Aug 23 Sat	SHA	Mer.pass
Venus	241°14.1	09:49
Mars	170°34.1	14:31
Jupiter	252°26.2	09:03
Saturn	358°27.8	02:00

Aug 24 Sun	SHA	Mer.pass
Venus	239°58.7	09:50
Mars	169°58.9	14:29
Jupiter	252°13.9	09:00
Saturn	358°31.1	01:55

Aug 25 Mon	SHA	Mer.pass
Venus	238°43.4	09:51
Mars	169°23.7	14:27
Jupiter	252°01.6	08:57
Saturn	358°34.4	01:51

Horizontal parallax	
Venus:	0.1
Mars:	0.1

DUT1 = UT1-UTC = +0.0953 sec ΔT = TT-UT1 = +69.0887 sec

2025 August 23 to Aug. 25 UT

h	Sun			Moon			
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	179°19.3	N11°25.4	181°35.6	12.9'	N14°16.9	-13.9'	56.7'
1	194°19.4	24.5	196°07.5	13.0'	14°03.0	-14.0'	56.7'
2	209°19.6	23.7	210°39.4	13.0'	13°49.1	-14.0'	56.7'
3	224°19.8	· · 22.8	225°11.5	13.1'	13°35.1	-14.1'	56.7'
4	239°19.9	22.0	239°43.6	13.2'	13°21.0	-14.1'	56.6'
5	254°20.1	21.1	254°15.8	13.3'	13°06.9	-14.1'	56.6'
6	269°20.2	N11°20.3	268°48.1	13.4'	N12°52.8	-14.2'	56.6'
7	284°20.4	19.4	283°20.5	13.5'	12°38.6	-14.2'	56.6'
8	299°20.6	18.6	297°52.9	13.5'	12°24.3	-14.3'	56.5'
9	314°20.7	· · 17.7	312°25.4	13.6'	12°10.1	-14.3'	56.5'
10	329°20.9	16.9	326°58.1	13.7'	11°55.7	-14.4'	56.5'
11	344°21.1	16.0	341°30.7	13.8'	11°41.4	-14.4'	56.5'
12	359°21.2	N11°15.2	356°03.5	13.8'	N11°27.0	-14.4'	56.4'
13	14°21.4	14.3	10°36.4	13.9'	11°12.5	-14.5'	56.4'
14	29°21.6	13.5	25°09.3	14.0'	10°58.0	-14.5'	56.4'
15	44°21.7	· · 12.6	39°42.3	14.1'	10°43.5	-14.5'	56.4'
16	59°21.9	11.8	54°15.3	14.1'	10°29.0	-14.6'	56.3'
17	74°22.1	10.9	68°48.5	14.2'	10°14.4	-14.6'	56.3'
18	89°22.2	N11°10.1	83°21.7	14.3'	N09°59.8	-14.6'	56.3'
19	104°22.4	09.2	97°54.9	14.3'	09°45.1	-14.7'	56.3'
20	119°22.6	08.4	112°28.3	14.4'	09°30.5	-14.7'	56.3'
21	134°22.7	· · 07.5	127°01.7	14.5'	09°15.8	-14.7'	56.2'
22	149°22.9	06.7	141°35.2	14.5'	09°01.1	-14.8'	56.2'
23	164°23.1	05.8	156°08.7	14.6'	08°46.3	-14.8'	56.2'
SD = 15.8'			SD = 15.5'				

Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	179°23.2	N11°04.9	170°42.3	14.7'	N08°31.5	-14.8'	56.2'
1	194°23.4	04.1	185°15.9	14.7'	08°16.7	-14.8'	56.1'
2	209°23.6	03.2	199°49.7	14.8'	08°01.9	-14.8'	56.1'
3	224°23.7	· · 02.4	214°23.5	14.8'	07°47.1	-14.9'	56.1'
4	239°23.9	01.5	228°57.3	14.9'	07°32.2	-14.9'	56.1'
5	254°24.1	11°00.7	243°31.2	15.0'	07°17.3	-14.9'	56.0'
6	269°24.2	N10°59.8	258°05.1	15.0'	N07°02.4	-14.9'	56.0'
7	284°24.4	58.9	272°39.2	15.1'	06°47.5	-14.9'	56.0'
8	299°24.6	58.1	287°13.2	15.1'	06°32.6	-14.9'	56.0'
9	314°24.7	· · 57.2	301°47.3	15.2'	06°17.7	-15.0'	56.0'
10	329°24.9	56.4	316°21.5	15.2'	06°02.7	-15.0'	55.9'
11	344°25.1	55.5	330°55.7	15.3'	05°47.8	-15.0'	55.9'
12	359°25.2	N10°54.7	345°30.0	15.3'	N05°32.8	-15.0'	55.9'
13	14°25.4	53.8	0°04.3	15.4'	05°17.8	-15.0'	55.9'
14	29°25.6	52.9	14°38.7	15.4'	05°02.8	-15.0'	55.8'
15	44°25.7	· · 52.1	29°13.1	15.5'	04°47.8	-15.0'	55.8'
16	59°25.9	51.2	43°47.5	15.5'	04°32.8	-15.0'	55.8'
17	74°26.1	50.4	58°22.0	15.5'	04°17.8	-15.0'	55.8'
18	89°26.3	N10°49.5	72°56.6	15.6'	N04°02.8	-15.0'	55.7'
19	104°26.4	48.6	87°31.1	15.6'	03°47.8	-15.0'	55.7'
20	119°26.6	47.8	102°05.8	15.7'	03°32.7	-15.0'	55.7'
21	134°26.8	· · 46.9	116°40.4	15.7'	03°17.7	-15.0'	55.7'
22	149°26.9	46.0	131°15.1	15.7'	03°02.7	-15.0'	55.7'
23	164°27.1	45.2	145°49.9	15.8'	02°47.7	-15.0'	55.6'
SD = 15.8'			SD = 15.3'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	179°27.3	N10°44.3	160°24.6	15.8'	N02°32.7	-15.0'	55.6'
1	194°27.5	43.5	174°59.4	15.8'	02°17.7	-15.0'	55.6'
2	209°27.6	42.6	189°34.3	15.9'	02°02.6	-15.0'	55.6'
3	224°27.8	· · 41.7	204°09.2	15.9'	01°47.6	-15.0'	55.5'
4	239°28.0	40.9	218°44.1	15.9'	01°32.6	-15.0'	55.5'
5	254°28.1	40.0	233°19.0	16.0'	01°17.7	-15.0'	55.5'
6	269°28.3	N10°39.1	247°54.0	16.0'	N01°02.7	-15.0'	55.5'
7	284°28.5	38.3	262°28.9	16.0'	00°47.7	-15.0'	55.5'
8	299°28.7	37.4	277°04.0	16.0'	00°32.7	-15.0'	55.4'
9	314°28.8	· · 36.5	291°39.0	16.1'	00°17.8	-14.9'	55.4'
10	329°29.0	35.7	306°14.1	16.1'	N00°02.8	-14.9'	55.4'
11	344°29.2	34.8	320°49.2	16.1'	S00°12.1	14.9'	55.4'
12	359°29.4	N10°33.9	335°24.3	16.1'	S00°27.0	14.9'	55.3'
13	14°29.5	33.1	349°59.4	16.2'	00°41.9	14.9'	55.3'
14	29°29.7	32.2	4°34.6	16.2'	00°56.8	14.9'	55.3'
15	44°29.9	· · 31.3	19°09.7	16.2'	01°11.7	14.9'	55.3'
16	59°30.0	30.5	33°44.9	16.2'	01°26.6	14.8'	55.3'
17	74°30.2	29.6	48°20.1	16.2'	01°41.4	14.8'	55.2'
18	89°30.4	N10°28.7	62°55.3	16.2'	S01°56.2	14.8'	55.2'
19	104°30.6	27.9	77°30.6	16.2'	02°11.0	14.8'	55.2'
20	119°30.7	27.0	92°05.8	16.3'	02°25.8	14.8'	55.2'
21	134°30.9	· · 26.1	106°41.1	16.3'	02°40.6	14.7'	55.2'
22	149°31.1	25.3	121°16.4	16.3'	02°55.3	14.7'	55.1'
23	164°31.3	24.4	135°51.6	16.3'	03°10.0	14.7'	55.1'
SD = 15.8'			SD = 15.2'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	01:18	03:21	20:40	22:35	—:—
N 70°	—:—	02:10	03:41	20:20	21:48	—:—
68°	—:—	02:42	03:57	20:05	21:18	—:—
66°	01:07	03:05	04:09	19:53	20:56	22:47
64°	01:54	03:22	04:20	19:43	20:39	22:05
62°	02:23	03:37	04:29	19:34	20:25	21:38
60°	02:44	03:49	04:36	19:27	20:14	21:18
N 58°	03:01	03:59	04:43	19:20	20:04	21:01
56°	03:15	04:08	04:49	19:14	19:55	20:48
54°	03:27	04:16	04:54	19:09	19:48	20:36
52°	03:37	04:23	04:59	19:05	19:41	20:26
50°	03:46	04:29	05:03	19:00	19:35	20:17
45°	04:04	04:42	05:13	18:51	19:22	20:00
N 40°	04:18	04:52	05:20	18:44	19:12	19:46
35°	04:29	05:01	05:27	18:37	19:03	19:35
30°	04:39	05:08	05:33	18:32	18:56	19:25
20°	04:53	05:20	05:43	18:22	18:44	19:11
N 10°	05:05	05:30	05:51	18:13	18:35	19:00
0°	05:13	05:38	05:59	18:06	18:27	18:51
S 10°	05:21	05:45	06:07	17:58	18:19	18:44
20°	05:27	05:53	06:15	17:50	18:12	18:38
30°	05:32	06:00	06:24	17:41	18:05	18:33
40°	05:34	06:04	06:29	17:36	18:01	18:31
45°	05:36	06:08	06:35	17:30	17:57	18:29
50°	05:38	06:12	06:42	17:23	17:53	18:27
S 50°	05:40	06:17	06:50	17:15	17:48	18:26
52°	05:41	06:20	06:54	17:11	17:46	18:25
54°	05:41	06:22	06:58	17:07	17:43	18:25
56°	05:42	06:25	07:03	17:03	17:41	18:24
58°	05:42	06:28	07:08	16:58	17:38	18:24
S 60°	05:42	06:31	07:14	16:52	17:35	18:23

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	02:48	05:16	07:23	20:32	19:54	19:21
N 70°	03:16	05:27	07:24	20:18	19:50	19:24
68°	03:37	05:36	07:25	20:07	19:46	19:27
66°	03:53	05:43	07:26	19:57	19:43	19:29
64°	04:06	05:49	07:26	19:49	19:40	19:31
62°	04:17	05:55	07:27	19:43	19:38	19:33
60°	04:27	05:59	07:27	19:37	19:35	19:34
N 58°	04:35	06:03	07:27	19:31	19:34	19:35
56°	04:42	06:07	07:28	19:27	19:32	19:36
54°	04:48	06:10	07:28	19:22	19:30	19:37
52°	04:54	06:13	07:28	19:19	19:29	19:38
50°	04:59	06:15	07:28	19:15	19:28	19:39
45°	05:10	06:21	07:29	19:07	19:25	19:41
N 40°	05:19	06:26	07:29	19:01	19:22	19:43
35°	05:27	06:30	07:30	18:55	19:20	19:44
30°	05:34	06:33	07:30	18:50	19:19	19:45
20°	05:46	06:39	07:30	18:42	19:15	19:47
N 10°	05:56	06:45	07:31	18:34	19:12	19:49
0°	06:05	06:50	07:31	18:27	19:10	19:51
S 10°	06:15	06:55	07:32	18:20	19:07	19:53
20°	06:25	07:00	07:32	18:12	19:04	19:55
30°	06:36	07:06	07:33	18:03	19:01	19:57
35°	06:43	07:09	07:33	17:58	18:59	19:58
40°	06:51	07:13	07:34	17:52	18:56	19:59
45°	06:59	07:18	07:34	17:45	18:54	20:01
S 50°	07:10	07:23	07:35	17:36	18:51	20:03
52°	07:14	07:25	07:35	17:32	18:49	20:04
54°	07:20	07:28	07:35	17:28	18:48	20:05
56°	07:25	07:31	07:36	17:23	18:46	20:06
58°	07:32	07:34	07:36	17:18	18:44	20:07
S 60°	07:39	07:38	07:36	17:12	18:42	20:08

Day	Sun		Mer. Pass hh:mm	Moon		Age 0-2 0-3%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
23	02:43	02:35	12:03	12:16	—:—	
24	02:27	02:19	12:02	13:00	00:38	
25	02:11	02:03	12:02	13:41	01:21	

Aries		Venus		Mars		Jupiter		Saturn	
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	334° 29.9	211° 58.2	N19° 53.4	143° 18.3	S04° 27.9	226° 19.4	N22° 20.7	333° 07.8	S02° 05.5
1	349° 32.4	226° 57.5	52.9	158° 19.3	28.5	241° 21.4	20.6	348° 10.4	05.6
2	4° 34.9	241° 56.8	52.4	173° 20.3	29.2	256° 23.3	20.6	3° 13.0	05.6
3	19° 37.3	256° 56.2	•• 51.9	188° 21.3	•• 29.8	271° 25.3	•• 20.5	18° 15.6	•• 05.7
4	34° 39.8	271° 55.5	51.4	203° 22.3	30.5	286° 27.2	20.5	33° 18.2	05.8
5	49° 42.3	286° 54.8	50.9	218° 23.3	31.1	301° 29.2	20.4	48° 20.8	05.8
6	64° 44.7	301° 54.2	N19° 50.3	233° 24.3	S04° 31.8	316° 31.2	N22° 20.3	63° 23.4	S02° 05.9
7	79° 47.2	316° 53.5	49.8	248° 25.3	32.4	331° 33.1	20.3	78° 26.1	06.0
8	94° 49.7	331° 52.8	49.3	263° 26.3	33.1	346° 35.1	20.2	93° 28.7	06.0
9	109° 52.1	346° 52.2	•• 48.8	278° 27.2	•• 33.7	1° 37.0	•• 20.2	108° 31.3	•• 06.1
10	124° 54.6	1° 51.5	48.3	293° 28.2	34.4	16° 39.0	20.1	123° 33.9	06.2
11	139° 57.0	16° 50.8	47.8	308° 29.2	35.0	31° 41.0	20.1	138° 36.5	06.3
12	154° 59.5	31° 50.2	N19° 47.2	323° 30.2	S04° 35.7	46° 42.9	N22° 20.0	153° 39.1	S02° 06.3
13	170° 02.0	46° 49.5	46.7	338° 31.2	36.4	61° 44.9	20.0	168° 41.7	06.4
14	185° 04.4	61° 48.9	46.2	353° 32.2	37.0	76° 46.8	19.9	183° 44.3	06.5
15	200° 06.9	76° 48.2	•• 45.7	8° 33.2	•• 37.7	91° 48.8	•• 19.9	198° 46.9	•• 06.5
16	215° 09.4	91° 47.5	45.1	23° 34.2	38.3	106° 50.8	19.8	213° 49.5	06.6
17	230° 11.8	106° 46.9	44.6	38° 35.2	39.0	121° 52.7	19.7	228° 52.1	06.7
18	245° 14.3	121° 46.2	N19° 44.1	53° 36.1	S04° 39.6	136° 54.7	N22° 19.7	243° 54.8	S02° 06.8
19	260° 16.8	136° 45.5	43.6	68° 37.1	40.3	151° 56.6	19.6	258° 57.4	06.8
20	275° 19.2	151° 44.9	43.0	83° 38.1	40.9	166° 58.6	19.6	274° 00.0	06.9
21	290° 21.7	166° 44.2	•• 42.5	98° 39.1	•• 41.6	182° 00.6	•• 19.5	289° 02.6	•• 07.0
22	305° 24.2	181° 43.6	42.0	113° 40.1	42.2	197° 02.5	19.5	304° 05.2	07.0
23	320° 26.6	196° 42.9	41.4	128° 41.1	42.9	212° 04.5	19.4	319° 07.8	07.1
Mer.pass. 01:42		ν -0.7' d -0.5' m -3.95		ν 1.0' d 0.7' m 1.62		ν 2.0' d -0.1' m -1.97		ν 2.6' d 0.1' m 0.68	

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	335°29.1	211°42.2	N19°40.9	143°42.1	S04°43.5	227°06.5	N22°19.4	334°10.4	S02°07.2
1	350°31.5	226°41.6	40.4	158°43.1	44.2	242°08.4	19.3	349°13.0	07.2
2	5°34.0	241°40.9	39.8	173°44.0	44.8	257°10.4	19.3	4°15.6	07.3
3	20°36.5	256°40.3	· · 39.3	188°45.0	· · 45.5	272°12.4	· · 19.2	19°18.2	· · 07.4
4	35°38.9	271°39.6	38.8	203°46.0	46.2	287°14.3	19.1	34°20.9	07.5
5	50°41.4	286°38.9	38.2	218°47.0	46.8	302°16.3	19.1	49°23.5	07.5
6	65°43.9	301°38.3	N19°37.7	233°48.0	S04°47.5	317°18.2	N22°19.0	64°26.1	S02°07.6
7	80°46.3	316°37.6	37.1	248°49.0	48.1	332°20.2	19.0	79°28.7	07.7
8	95°48.8	331°37.0	36.6	263°50.0	48.8	347°22.2	18.9	94°31.3	07.7
9	110°51.3	346°36.3	· · 36.1	278°51.0	· · 49.4	2°24.1	· · 18.9	109°33.9	· · 07.8
10	125°53.7	1°35.6	35.5	293°51.9	50.1	17°26.1	18.8	124°36.5	07.9
11	140°56.2	16°35.0	35.0	308°52.9	50.7	32°28.1	18.8	139°39.1	08.0
12	155°58.6	31°34.3	N19°34.4	323°53.9	S04°51.4	47°30.0	N22°18.7	154°41.7	S02°08.0
13	171°01.1	46°33.7	33.9	338°54.9	52.0	62°32.0	18.7	169°44.4	08.1
14	186°03.6	61°33.0	33.3	353°55.9	52.7	77°34.0	18.6	184°47.0	08.2
15	201°06.0	76°32.4	· · 32.8	8°56.9	· · 53.3	92°35.9	· · 18.5	199°49.6	· · 08.2
16	216°08.5	91°31.7	32.3	23°57.8	54.0	107°37.9	18.5	214°52.2	08.3
17	231°11.0	106°31.0	31.7	38°58.8	54.6	122°39.9	18.4	229°54.8	08.4
18	246°13.4	121°30.4	N19°31.2	53°59.8	S04°55.3	137°41.8	N22°18.4	244°57.4	S02°08.5
19	261°15.9	136°29.7	30.6	69°00.8	56.0	152°43.8	18.3	260°00.0	08.5
20	276°18.4	151°29.1	30.0	84°01.8	56.6	167°45.8	18.3	275°02.6	08.6
21	291°20.8	166°28.4	· · 29.5	99°02.8	· · 57.3	182°47.7	· · 18.2	290°05.2	· · 08.7
22	306°23.3	181°27.8	28.9	114°03.8	57.9	197°49.7	18.2	305°07.9	08.7
23	321°25.8	196°27.1	28.4	129°04.7	58.6	212°51.7	18.1	320°10.5	08.8
Mer.pass. 01:38		ν -0.7' d -0.5' m -3.95		ν 1.0' d 0.7' m 1.62		ν 2.0' d -0.1' m -1.98		ν 2.6' d 0.1' m 0.67	

Th _u	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	336°28.2	211°26.5	N19°27.8	144°05.7	S04°59.2	227°53.6	N22°18.1	335°13.1	S02°08.9
1	351°30.7	226°25.8	27.3	159°06.7	04°59.9	242°55.6	18.0	350°15.7	09.0
2	6°33.1	241°25.1	26.7	174°07.7	05°00.5	257°57.6	17.9	5°18.3	09.0
3	21°35.6	256°24.5	· · 26.2	189°08.7	· · 01.2	272°59.5	· · 17.9	20°20.9	· · 09.1
4	36°38.1	271°23.8	25.6	204°09.7	01.8	288°01.5	17.8	35°23.5	09.2
5	51°40.5	286°23.2	25.0	219°10.6	02.5	303°03.5	17.8	50°26.2	09.2
6	66°43.0	301°22.5	N19°24.5	234°11.6	S05°03.1	318°05.4	N22°17.7	65°28.8	S02°09.3
7	81°45.5	316°21.9	23.9	249°12.6	03.8	333°07.4	17.7	80°31.4	09.4
8	96°47.9	331°21.2	23.3	264°13.6	04.4	348°09.4	17.6	95°34.0	09.5
9	111°50.4	346°20.6	· · 22.8	279°14.6	· · 05.1	3°11.3	· · 17.6	110°36.6	· · 09.5
10	126°52.9	1°19.9	22.2	294°15.5	05.7	18°13.3	17.5	125°39.2	09.6
11	141°55.3	16°19.3	21.7	309°16.5	06.4	33°15.3	17.4	140°41.8	09.7
12	156°57.8	31°18.6	N19°21.1	324°17.5	S05°07.1	48°17.2	N22°17.4	155°44.4	S02°09.7
13	172°00.2	46°18.0	20.5	339°18.5	07.7	63°19.2	17.3	170°47.1	09.8
14	187°02.7	61°17.3	19.9	354°19.5	08.4	78°21.2	17.3	185°49.7	09.9
15	202°05.2	76°16.7	· · 19.4	9°20.5	· · 09.0	93°23.1	· · 17.2	200°52.3	· · 10.0
16	217°07.6	91°16.0	18.8	24°21.4	09.7	108°25.1	17.2	215°54.9	10.0
17	232°10.1	106°15.4	18.2	39°22.4	10.3	123°27.1	17.1	230°57.5	10.1
18	247°12.6	121°14.7	N19°17.7	54°23.4	S05°11.0	138°29.0	N22°17.1	246°00.1	S02°10.2
19	262°15.0	136°14.1	17.1	69°24.4	11.6	153°31.0	17.0	261°02.7	10.3
20	277°17.5	151°13.4	16.5	84°25.4	12.3	168°33.0	17.0	276°05.4	10.3
21	292°20.0	166°12.8	· · 15.9	99°26.3	· · 12.9	183°35.0	· · 16.9	291°08.0	· · 10.4
22	307°22.4	181°12.1	15.4	114°27.3	13.6	198°36.9	16.8	306°10.6	10.5
23	322°24.9	196°11.5	14.8	129°28.3	14.2	213°38.9	16.8	321°13.2	10.5
Mer.pass. 01:34		ν -0.7' d -0.6' m -3.95		ν 1.0' d 0.7' m 1.62		ν 2.0' d -0.1' m -1.98		ν 2.6' d 0.1' m 0.67	

	Stars	
	SHA	Dec
Alpheratz	357° 33.9	29° 14.0
Ankaa	353° 06.2	-42° 09.7
Schedar	349° 30.0	56° 40.6
Diphda	348° 46.5	-17° 50.5
Achernar	335° 19.4	-57° 06.1
Hamal	327° 50.4	23° 35.1
Polaris	313° 38.5	89° 22.0
Acamar	315° 11.2	-40° 11.8
Menkar	314° 05.5	4° 11.6
Mirfak	308° 27.4	49° 57.0
Aldebaran	299° 39.0	16° 33.7
Rigel	281° 03.4	-8° 10.1
Capella	280° 21.2	46° 01.3
Bellatrix	278° 22.4	6° 22.5
Elnath	278° 01.3	28° 37.7
Anilam	275° 37.3	-1° 11.0
Betelgeuse	270° 51.7	7° 24.8
Canopus	263° 52.5	-52° 42.2
Sirius	258° 26.0	-16° 44.9
Adhara	255° 05.7	-29° 00.1
Procyon	244° 50.5	5° 09.7
Pollux	243° 16.9	27° 57.9
Avior	234° 15.1	-55° 35.3
Suhail	222° 46.4	-43° 32.0
Miaplacidus	221° 39.1	-69° 49.2
Alphard	217° 47.6	-8° 46.1
Regulus	207° 34.2	11° 50.6
Dubhe	193° 41.0	61° 36.9
Denebola	182° 24.7	14° 25.8
Gienah	175° 43.3	-17° 41.0
Acrux	173° 00.1	-63° 14.6
Gacrux	171° 51.6	-57° 15.5
Alioth	166° 12.9	55° 49.4
Spica	158° 22.0	-11° 17.7
Alkaid	152° 51.8	49° 11.3
Hadar	148° 35.6	-60° 30.0
Menkent	147° 57.2	-36° 29.9
Arcturus	145° 47.6	19° 03.1
Rigel Kent.	139° 39.9	-60° 56.7
Kochab	137° 20.1	74° 03.2
Zuben 'ubi	136° 55.5	-16° 08.9
Alphecca	126° 03.3	26° 37.9
Antares	112° 15.2	-26° 29.4
Atria	107° 08.8	-69° 04.7
Sabik	102° 02.1	-15° 45.4
Shaula	96° 09.5	-37° 07.4
Rasalhague	95° 57.9	12° 32.6
Eltanin	90° 41.7	51° 29.3
Kaus Aust.	83° 31.6	-34° 22.4
Vega	80° 32.6	38° 48.6
Nunki	75° 46.9	-26° 15.9
Altair	61° 59.2	8° 56.3
Peacock	53° 04.3	-56° 39.2
Deneb	49° 25.0	45° 22.5
Enif	33° 38.0	9° 59.7
Al Na'ir	27° 31.8	-46° 50.1
Fomalhaut	15° 13.6	-29° 29.0
Scheat	13° 44.4	28° 13.4
Markab	13° 29.1	15° 20.7

Aug 26 Tue	SHA	Mer.pass
Venus	237°28.2	09:53
Mars	168°48.4	14:26
Jupiter	251°49.5	08:54
Saturn	358°37.9	01:47

Aug 27 Wed	SHA	Mer.pass
Venus	236° 13.2	09:54
Mars	168° 13.0	14:24
Jupiter	251° 37.4	08:50
Saturn	358° 41.3	01:43

Aug 28 Thu	SHA	Mer.pass
Venus	234° 58.2	09:55
Mars	167° 37.5	14:23
Jupiter	251° 25.4	08:47
Saturn	358° 44.9	01:39

Horizontal parallax	
Venus:	0.1
Mars:	0.1

DUT1 = UT1-UTC = +0.0957 sec ΔT = TT-UT1 = +69.0883 sec

2025 August 26 to Aug. 28 UT

h Sun Moon						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	179°31.4	N10°23.5	150°26.9	16.3'	S03°24.7	14.7' 55.1'
1	194°31.6	22.7	165°02.2	16.3'	03°39.4	14.6' 55.1'
2	209°31.8	21.8	179°37.5	16.3'	03°54.0	14.6' 55.1'
3	224°32.0	· · 20.9	194°12.8	16.3'	04°08.7	14.6' 55.0'
4	239°32.2	20.0	208°48.2	16.3'	04°23.3	14.6' 55.0'
5	254°32.3	19.2	223°23.5	16.3'	04°37.8	14.5' 55.0'
6	269°32.5	N10°18.3	237°58.8	16.3'	S04°52.4	14.5' 55.0'
7	284°32.7	17.4	252°34.1	16.3'	05°06.9	14.5' 55.0'
8	299°32.9	16.6	267°09.4	16.3'	05°21.3	14.4' 54.9'
9	314°33.0	· · 15.7	281°44.8	16.3'	05°35.8	14.4' 54.9'
10	329°33.2	14.8	296°20.1	16.3'	05°50.2	14.4' 54.9'
11	344°33.4	13.9	310°55.4	16.3'	06°04.6	14.3' 54.9'
12	359°33.6	N10°13.1	325°30.7	16.3'	S06°18.9	14.3' 54.9'
13	14°33.7	12.2	340°06.0	16.3'	06°33.2	14.3' 54.9'
14	29°33.9	11.3	354°41.3	16.3'	06°47.5	14.2' 54.8'
15	44°34.1	· · 10.4	9°16.6	16.3'	07°01.8	14.2' 54.8'
16	59°34.3	09.6	23°51.9	16.3'	07°16.0	14.2' 54.8'
17	74°34.5	08.7	38°27.2	16.3'	07°30.1	14.1' 54.8'
18	89°34.6	N10°07.8	53°02.4	16.3'	S07°44.3	14.1' 54.8'
19	104°34.8	06.9	67°37.7	16.2'	07°58.4	14.1' 54.7'
20	119°35.0	06.1	82°12.9	16.2'	08°12.4	14.0' 54.7'
21	134°35.2	· · 05.2	96°48.2	16.2'	08°26.4	14.0' 54.7'
22	149°35.4	04.3	111°23.4	16.2'	08°40.4	13.9' 54.7'
23	164°35.5	03.4	125°58.6	16.2'	08°54.4	13.9' 54.7'
SD = 15.8' d = -0.9' SD = 15.0'						

Wed	GHA	Dec	GHA	ν	Dec	d HP
0	179°35.7	N10°02.6	140°33.8	16.2'	S09°08.2	13.8' 54.7'
1	194°35.9	01.7	155°08.9	16.1'	09°22.1	13.8' 54.7'
2	209°36.1	10°00.8	169°44.1	16.1'	09°35.9	13.8' 54.6'
3	224°36.3	09°59.9	184°19.2	16.1'	09°49.7	13.7' 54.6'
4	239°36.4	59.1	198°54.3	16.1'	10°03.4	13.7' 54.6'
5	254°36.6	58.2	213°29.4	16.1'	10°17.0	13.6' 54.6'
6	269°36.8	N09°57.3	228°04.5	16.0'	S10°30.7	13.6' 54.6'
7	284°37.0	56.4	242°39.5	16.0'	10°44.2	13.5' 54.6'
8	299°37.2	55.6	257°14.5	16.0'	10°57.8	13.5' 54.5'
9	314°37.3	· · 54.7	271°49.5	16.0'	11°11.2	13.4' 54.5'
10	329°37.5	53.8	286°24.5	15.9'	11°24.7	13.4' 54.5'
11	344°37.7	52.9	300°59.5	15.9'	11°38.0	13.3' 54.5'
12	359°37.9	N09°52.0	315°34.4	15.9'	S11°51.4	13.3' 54.5'
13	14°38.1	51.2	330°09.2	15.9'	12°04.6	13.2' 54.5'
14	29°38.3	50.3	344°44.1	15.8'	12°17.9	13.2' 54.5'
15	44°38.4	· · 49.4	359°18.9	15.8'	12°31.0	13.1' 54.5'
16	59°38.6	48.5	13°53.7	15.8'	12°44.1	13.1' 54.4'
17	74°38.8	47.6	28°28.5	15.7'	12°57.2	13.0' 54.4'
18	89°39.0	N09°46.8	43°03.2	15.7'	S13°10.2	12.9' 54.4'
19	104°39.2	45.9	57°37.9	15.7'	13°23.1	12.9' 54.4'
20	119°39.3	45.0	72°12.5	15.6'	13°36.0	12.8' 54.4'
21	134°39.5	· · 44.1	86°47.2	15.6'	13°48.9	12.8' 54.4'
22	149°39.7	43.2	101°21.8	15.5'	14°01.6	12.7' 54.4'
23	164°39.9	42.3	115°56.3	15.5'	14°14.3	12.7' 54.4'
SD = 15.8' d = -0.9' SD = 14.9'						

Thu	GHA	Dec	GHA	ν	Dec	d HP
0	179°40.1	N09°41.5	130°30.8	15.5'	S14°27.0	12.6' 54.3'
1	194°40.3	40.6	145°05.3	15.4'	14°39.6	12.5' 54.3'
2	209°40.5	39.7	159°39.7	15.4'	14°52.1	12.5' 54.3'
3	224°40.6	· · 38.8	174°14.1	15.3'	15°04.6	12.4' 54.3'
4	239°40.8	37.9	188°48.4	15.3'	15°17.0	12.3' 54.3'
5	254°41.0	37.0	203°22.7	15.3'	15°29.3	12.3' 54.3'
6	269°41.2	N09°36.2	217°57.0	15.2'	S15°41.6	12.2' 54.3'
7	284°41.4	35.3	232°31.2	15.2'	15°53.8	12.1' 54.3'
8	299°41.6	34.4	247°05.3	15.1'	16°06.0	12.1' 54.3'
9	314°41.7	· · 33.5	261°39.5	15.1'	16°18.0	12.0' 54.3'
10	329°41.9	32.6	276°13.5	15.0'	16°30.0	11.9' 54.3'
11	344°42.1	31.7	290°47.6	15.0'	16°42.0	11.9' 54.2'
12	359°42.3	N09°30.8	305°21.5	14.9'	S16°53.8	11.8' 54.2'
13	14°42.5	30.0	319°55.5	14.9'	17°05.6	11.7' 54.2'
14	29°42.7	29.1	334°29.3	14.8'	17°17.4	11.7' 54.2'
15	44°42.9	· · 28.2	349°03.2	14.8'	17°29.0	11.6' 54.2'
16	59°43.0	27.3	3°36.9	14.7'	17°40.6	11.5' 54.2'
17	74°43.2	26.4	18°10.6	14.7'	17°52.1	11.4' 54.2'
18	89°43.4	N09°25.5	32°44.3	14.6'	S18°03.6	11.4' 54.2'
19	104°43.6	24.6	47°17.9	14.6'	18°14.9	11.3' 54.2'
20	119°43.8	23.7	61°51.5	14.5'	18°26.2	11.2' 54.2'
21	134°44.0	· · 22.9	76°25.0	14.4'	18°37.4	11.1' 54.2'
22	149°44.2	22.0	90°58.4	14.4'	18°48.6	11.1' 54.2'
23	164°44.4	21.1	105°31.8	14.3'	18°59.6	11.0' 54.2'
SD = 15.8' d = -0.9' SD = 14.8'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	01:52	03:37	20:22	22:03	—:—
N 70°	—:—	02:31	03:54	20:05	21:27	—:—
68°	—:—	02:57	04:08	19:52	21:02	23:23
66°	01:36	03:17	04:19	19:41	20:42	22:20
64°	02:12	03:33	04:29	19:32	20:27	21:47
62°	02:36	03:46	04:37	19:25	20:15	21:23
60°	02:55	03:57	04:44	19:18	20:04	21:05
N 58°	03:10	04:06	04:50	19:12	19:55	20:50
56°	03:23	04:14	04:55	19:07	19:47	20:38
54°	03:34	04:22	05:00	19:02	19:40	20:27
52°	03:43	04:28	05:04	18:58	19:34	20:18
50°	03:51	04:34	05:08	18:54	19:28	20:10
45°	04:08	04:45	05:16	18:46	19:17	19:54
N 40°	04:21	04:55	05:23	18:39	19:07	19:41
35°	04:32	05:03	05:29	18:33	18:59	19:30
30°	04:41	05:10	05:34	18:28	18:53	19:22
20°	04:54	05:21	05:43	18:19	18:42	19:08
N 10°	05:05	05:30	05:51	18:12	18:33	18:58
0°	05:13	05:37	05:58	18:05	18:26	18:50
S 10°	05:19	05:44	06:05	17:58	18:19	18:44
20°	05:25	05:50	06:13	17:51	18:13	18:39
30°	05:29	05:57	06:21	17:42	18:07	18:35
35°	05:31	06:00	06:26	17:38	18:03	18:33
40°	05:32	06:04	06:31	17:33	18:00	18:31
45°	05:33	06:07	06:37	17:27	17:56	18:30
S 50°	05:34	06:12	06:44	17:19	17:52	18:30
52°	05:34	06:14	06:48	17:16	17:50	18:29
54°	05:35	06:16	06:51	17:12	17:48	18:29
56°	05:35	06:18	06:56	17:08	17:46	18:29
58°	05:35	06:20	07:00	17:04	17:44	18:30
S 60°	05:34	06:23	07:05	16:59	17:42	18:30

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	09:27	11:38	14:50	18:46	18:02	16:19
N 70°	09:18	11:16	13:37	18:58	18:26	17:34
68°	09:11	11:00	13:00	19:07	18:44	18:12
66°	09:05	10:46	12:34	19:15	18:59	18:39
64°	09:00	10:35	12:15	19:22	19:12	19:00
62°	08:56	10:26	11:59	19:27	19:22	19:16
60°	08:52	10:18	11:46	19:33	19:31	19:31
N 58°	08:49	10:11	11:34	19:37	19:39	19:43
56°	08:46	10:05	11:25	19:41	19:46	19:53
54°	08:44	10:00	11:16	19:45	19:53	20:03
52°	08:42	09:55	11:08	19:48	19:58	20:11
50°	08:39	09:50	11:01	19:51	20:04	20:18
45°	08:35	09:41	10:47	19:57	20:15	20:35
N 40°	08:31	09:33	10:35	20:03	20:24	20:48
35°	08:28	09:26	10:24	20:08	20:32	20:59
30°	08:25	09:20	10:16	20:12	20:39	21:09
20°	08:20	09:10	10:00	20:19	20:52	21:26
N 10°	08:16	09:01	09:47	20:26	21:03	21:41
0°	08:12	08:53	09:35	20:32	21:13	21:56
S 10°	08:08	08:45	09:23	20:38	21:23	22:10
20°	08:04	08:36	09:10	20:44	21:34	22:25
30°	07:59	08:26	08:55	20:52	21:47	22:43
35°	07:57	08:21	08:46	20:56	21:54	22:53
40°	07:54	08:14	08:37	21:01	22:03	23:05
45°	07:50	08:07	08:25	21:07	22:12	23:19
S 50°	07:46	07:58	08:12	21:14	22:24	23:36
52°	07:44	07:54	08:06	21:17	22:30	23:44
54°	07:42	07:50	07:59	21:20	22:36	23:53
56°	07:40	07:45	07:51	21:24	22:43	· · ·
58°	07:38	07:40	07:42	21:29	22:51	· · ·
S 60°	07:35	07:33	07:33	21:33	23:00	· · ·

Day	Sun		Mer. Pass hh:mm	Moon		Age 3-5 8-21%
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Mer. Pass. Upper hh:mm	Lower hh:mm	
26	01:54	01:46	12:02	14:22	02:02	
27	01:37	01:28	12:01	15:03	02:42	
28	01:20	01:11	12:01	15:45	03:24	

August 29, 30, 31 UT (Fri., Sat., Sun.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	337°27.4	211°10.8	N19°14.2	144°29.3	S05°14.9	228°40.9	N22°16.7	336°15.8	S02°10.6	Alpheratz	357°33.9	29°14.0											
1	352°29.8	226°10.2	13.6	159°30.3	15.5	243°42.8	16.7	351°18.4	10.7	Ankaa	353°06.2	-42°09.8											
2	7°32.3	241°09.5	13.0	174°31.2	16.2	258°44.8	16.6	6°21.0	10.8	Schedar	349°30.0	56°40.6											
3	22°34.7	256°08.9	•• 12.5	189°32.2	•• 16.8	273°46.8	•• 16.6	21°23.7	•• 10.8	Diphda	348°46.5	-17°50.5											
4	37°37.2	271°08.2	11.9	204°33.2	17.5	288°48.8	16.5	36°26.3	10.9	Achernar	335°19.4	-57°06.1											
5	52°39.7	286°07.6	11.3	219°34.2	18.1	303°50.7	16.5	51°28.9	11.0	Hamal	327°50.4	23°35.1											
6	67°42.1	301°06.9	N19°10.7	234°35.2	S05°18.8	318°52.7	N22°16.4	66°31.5	S02°11.1	Polaris	313°36.8	89°22.0											
7	82°44.6	316°06.3	10.1	249°36.1	19.5	333°54.7	16.4	81°34.1	11.1	Acamar	315°11.2	-40°11.8											
8	97°47.1	331°05.6	09.5	264°37.1	20.1	348°56.6	16.3	96°36.7	11.2	Menkar	314°05.5	4°11.6											
9	112°49.5	346°05.0	•• 09.0	279°38.1	•• 20.8	3°58.6	•• 16.2	111°39.3	•• 11.3	Mirfak	308°27.4	49°57.0											
10	127°52.0	1°04.3	08.4	294°39.1	21.4	19°00.6	16.2	126°42.0	11.3	Aldebaran	290°39.0	16°33.7											
11	142°54.5	16°03.7	07.8	309°40.0	22.1	34°02.6	16.1	141°44.6	11.4	Rigel	281°03.4	-8°10.1											
12	157°56.9	31°03.0	N19°07.2	324°41.0	S05°22.7	49°04.5	N22°16.1	156°47.2	S02°11.5	Capella	280°21.2	46°01.3											
13	172°59.4	46°02.4	06.6	339°42.0	23.4	64°06.5	16.0	171°49.8	11.6	Bellatrix	278°22.4	6°22.5											
14	188°01.9	61°01.7	06.0	354°43.0	24.0	79°08.5	16.0	186°52.4	11.6	Elnath	278°01.3	28°37.7											
15	203°04.3	76°01.1	•• 05.4	9°44.0	•• 24.7	94°10.4	•• 15.9	201°55.0	•• 11.7	Alnilam	275°37.3	-1°11.0											
16	218°06.8	91°00.5	04.8	24°44.9	25.3	109°12.4	15.9	216°57.7	11.8	Betelgeuse	270°51.6	7°24.8											
17	233°09.2	105°59.8	04.2	39°45.9	26.0	124°14.4	15.8	232°00.3	11.8	Canopus	263°52.4	-52°42.2											
18	248°11.7	120°59.2	N19°03.6	54°46.9	S05°26.6	139°16.4	N22°15.7	247°02.9	S02°11.9	Sirius	258°26.0	-16°44.8											
19	263°14.2	135°58.5	03.0	69°47.9	27.3	154°18.3	15.7	262°05.5	12.0	Adhara	255°05.7	-29°00.1											
20	278°16.6	150°57.9	02.4	84°48.8	27.9	169°20.3	15.6	277°08.1	12.1	Procyon	244°50.5	5°09.7											
21	293°19.1	165°57.2	•• 01.8	99°49.8	•• 28.6	184°22.3	•• 15.6	292°10.7	•• 12.1	Pollux	243°16.9	27°57.9											
22	308°21.6	180°56.6	01.2	114°50.8	29.2	199°24.3	15.5	307°13.4	12.2	Avior	234°15.1	-59°35.2											
23	323°24.0	195°55.9	00.6	129°51.8	29.9	214°26.2	15.5	322°16.0	12.3	Suhail	222°46.4	-43°32.0											
Mer.pass. 01:30		ν -0.6' d -0.6' m -3.95		ν 1.0' d 0.7' m 1.62		ν 2.0' d -0.1' m -1.98		ν 2.6' d 0.1' m 0.66		Miaplacidus 221°39.1 -69°49.2													
										Alphard 217°47.5 -8°46.1													
										Regulus 207°34.2 11°50.6													
										Dubhe 193°41.0 61°36.8													
										Denebola 182°24.7 14°25.8													
										Gienah 175°43.3 -17°41.0													
										Acrux 173°00.2 -63°14.6													
										Gacrux 171°51.6 -57°15.5													
										Alioth 166°12.9 55°49.4													
										Spica 158°22.0 -11°17.7													
										Alkaid 152°51.9 49°11.3													
										Hadar 148°35.6 -60°30.0													
										Menkent 147°57.2 -36°29.9													
										Arcturus 145°47.6 19°03.1													
										Rigel Kent. 139°39.9 -60°56.7													
										Kochab 137°20.2 74°03.2													
										Zuben'ubi 136°55.6 -16°08.9													
										Alphecca 126°03.4 26°37.9													
										Antares 112°15.2 -26°29.4													
										Atria 107°08.8 -69°04.7													
										Sabik 102°02.1 -15°45.4													
										Shaula 96°09.5 -37°07.4													
										Rasalhague 95°57.9 12°32.6													
										Eltanin 90°41.7 51°29.4													
										Kaus Aust. 83°31.6 -34°22.4													
										Vega 80°32.6 38°48.6													
										Nunki 75°46.9 -26°15.9													
										Altair 61°59.2 8°56.3													
										Peacock 53°04.4 -56°39.2													
										Deneb 49°25.0 45°22.5													
										Enif 33°38.0 9°59.7													
										Al Na'ir 27°31.8 -46°50.1													
										Fomalhaut 15°13.6 -29°29.0													
										Scheat 13°44.3 28°13.4													
										Markab 13°29.0 15°20.7													
Aug 29 Fri				SHA				Mer.pass															
				Venus 233°43.5				09:56															
				Mars 167°01.9				14:21															
				Jupiter 251°13.5				08:44															
				Saturn 358°48.5				01:35															
Aug 30 Sat				SHA				Mer.pass															
				Venus 232°28.8				09:57															
				Mars 166°26.3				14:20															
				Jupiter 251°01.7				08:41															
				Saturn 358°52.1				01:30															
Aug 31 Sun				SHA				Mer.pass															
				Venus 231°14.3				09:58															
				Mars 165°50.5				14:18															
				Jupiter 250°50.0				08:38															
				Saturn 358°55.8				01:26															
Horizontal parallax																							
Venus: 0.1																							
Mars: 0.1																							

h	Sun				Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP	
0	179°44.5	N09°20.2	120°05.2	14.3'	S19°10.6	10.9'	54.2'	
1	194°44.7	19.3	134°38.4	14.2'	19°21.5	10.8'	54.2'	
2	209°44.9	18.4	149°11.6	14.2'	19°32.4	10.7'	54.2'	
3	224°45.1	· · 17.5	163°44.8	14.1'	19°43.1	10.7'	54.2'	
4	239°45.3	16.6	178°17.9	14.0'	19°53.8	10.6'	54.1'	
5	254°45.5	15.7	192°50.9	14.0'	20°04.3	10.5'	54.1'	
6	269°45.7	N09°14.9	207°23.9	13.9'	S20°14.8	10.4'	54.1'	
7	284°45.9	14.0	221°56.8	13.9'	20°25.2	10.3'	54.1'	
8	299°46.0	13.1	236°29.7	13.8'	20°35.6	10.2'	54.1'	
9	314°46.2	· · 12.2	251°02.5	13.7'	20°45.8	10.2'	54.1'	
10	329°46.4	11.3	265°35.2	13.7'	20°56.0	10.1'	54.1'	
11	344°46.6	10.4	280°07.9	13.6'	21°06.0	10.0'	54.1'	
12	359°46.8	N09°09.5	294°40.5	13.5'	S21°16.0	9.9'	54.1'	
13	14°47.0	08.6	309°13.0	13.5'	21°25.9	9.8'	54.1'	
14	29°47.2	07.7	323°45.5	13.4'	21°35.7	9.7'	54.1'	
15	44°47.4	· · 06.8	338°17.9	13.3'	21°45.4	9.6'	54.1'	
16	59°47.6	05.9	352°50.2	13.3'	21°55.0	9.5'	54.1'	
17	74°47.8	05.0	7°22.5	13.2'	22°04.5	9.4'	54.1'	
18	89°47.9	N09°04.1	21°54.7	13.1'	S22°14.0	9.3'	54.1'	
19	104°48.1	03.3	36°26.8	13.1'	22°23.3	9.2'	54.1'	
20	119°48.3	02.4	50°58.9	13.0'	22°32.6	9.2'	54.1'	
21	134°48.5	· · 01.5	65°30.9	12.9'	22°41.7	9.1'	54.1'	
22	149°48.7	09°00.6	80°02.8	12.9'	22°50.8	9.0'	54.1'	
23	164°48.9	08°59.7	94°34.7	12.8'	22°59.7	8.9'	54.1'	
SD = 15.8'		d = -0.9'	SD = 14.8'					
Sat	GHA	Dec	GHA	ν	Dec	d	HP	
0	179°49.1	N08°58.8	109°06.5	12.7'	S23°08.6	8.8'	54.1'	
1	194°49.3	57.9	123°38.2	12.7'	23°17.3	8.7'	54.1'	
2	209°49.5	57.0	138°09.9	12.6'	23°26.0	8.6'	54.1'	
3	224°49.7	· · 56.1	152°41.5	12.5'	23°34.6	8.5'	54.1'	
4	239°49.9	55.2	167°13.0	12.5'	23°43.0	8.4'	54.2'	
5	254°50.1	54.3	181°44.5	12.4'	23°51.4	8.3'	54.2'	
6	269°50.2	N08°53.4	196°15.8	12.3'	S23°59.6	8.2'	54.2'	
7	284°50.4	52.5	210°47.2	12.2'	24°07.8	8.0'	54.2'	
8	299°50.6	51.6	225°18.4	12.2'	24°15.8	7.9'	54.2'	
9	314°50.8	· · 50.7	239°49.6	12.1'	24°23.8	7.8'	54.2'	
10	329°51.0	49.8	254°20.7	12.0'	24°31.6	7.7'	54.2'	
11	344°51.2	48.9	268°51.7	12.0'	24°39.3	7.6'	54.2'	
12	359°51.4	N08°48.0	283°22.7	11.9'	S24°47.0	7.5'	54.2'	
13	14°51.6	47.1	297°53.6	11.8'	24°54.5	7.4'	54.2'	
14	29°51.8	46.2	312°24.4	11.8'	25°01.9	7.3'	54.2'	
15	44°52.0	· · 45.3	326°55.1	11.7'	25°09.2	7.2'	54.2'	
16	59°52.2	44.4	341°25.8	11.6'	25°16.4	7.1'	54.2'	
17	74°52.4	43.5	355°56.4	11.5'	25°23.4	7.0'	54.2'	
18	89°52.6	N08°42.6	10°27.0	11.5'	S25°30.4	6.8'	54.2'	
19	104°52.8	41.7	24°57.4	11.4'	25°37.2	6.7'	54.2'	
20	119°53.0	40.8	39°27.8	11.3'	25°44.0	6.6'	54.2'	
21	134°53.1	· · 39.9	53°58.2	11.3'	25°50.6	6.5'	54.3'	
22	149°53.3	39.0	68°28.4	11.2'	25°57.1	6.4'	54.3'	
23	164°53.5	38.1	82°58.6	11.1'	26°03.5	6.3'	54.3'	
SD = 15.8'		d = -0.9'	SD = 14.8'					
Sun	GHA	Dec	GHA	ν	Dec	d	HP	
0	179°53.7	N08°37.2	97°28.8	11.1'	S26°09.7	6.1'	54.3'	
1	194°53.9	36.3	111°58.8	11.0'	26°15.9	6.0'	54.3'	
2	209°54.1	35.4	126°28.8	10.9'	26°21.9	5.9'	54.3'	
3	224°54.3	· · 34.5	140°58.7	10.9'	26°27.8	5.8'	54.3'	
4	239°54.5	33.6	155°28.6	10.8'	26°33.6	5.7'	54.3'	
5	254°54.7	32.7	169°58.3	10.7'	26°39.2	5.5'	54.3'	
6	269°54.9	N08°31.8	184°28.1	10.7'	S26°44.8	5.4'	54.3'	
7	284°55.1	30.9	198°57.7	10.6'	26°50.2	5.3'	54.4'	
8	299°55.3	30.0	213°27.3	10.5'	26°55.5	5.2'	54.4'	
9	314°55.5	· · 29.1	227°56.8	10.5'	27°00.7	5.0'	54.4'	
10	329°55.7	28.2	242°26.3	10.4'	27°05.7	4.9'	54.4'	
11	344°55.9	27.3	256°55.7	10.3'	27°10.6	4.8'	54.4'	
12	359°56.1	N08°26.4	271°25.0	10.3'	S27°15.4	4.7'	54.4'	
13	14°56.3	25.5	285°54.3	10.2'	27°20.1	4.5'	54.4'	
14	29°56.5	24.6	300°23.5	10.1'	27°24.6	4.4'	54.4'	
15	44°56.7	· · 23.7	314°52.6	10.1'	27°29.0	4.3'	54.5'	
16	59°56.9	22.8	329°21.7	10.0'	27°33.3	4.1'	54.5'	
17	74°57.1	21.9	343°50.7	10.0'	27°37.4	4.0'	54.5'	
18	89°57.3	N08°21.0	358°19.6	9.9'	S27°41.4	3.9'	54.5'	
19	104°57.5	20.1	12°48.5	9.8'	27°45.3	3.7'	54.5'	
20	119°57.7	19.2	27°17.4	9.8'	27°49.1	3.6'	54.5'	
21	134°57.9	· · 18.3	41°46.2	9.7'	27°52.7	3.5'	54.6'	
22	149°58.1	17.3	56°14.9	9.7'	27°56.1	3.3'	54.6'	
23	164°58.3	16.4	70°43.5	9.6'	27°59.5	3.2'	54.6'	
SD = 15.8'		d = -0.9'	SD = 14.8'					

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	02:18	03:52	20:05	21:37	—:—
N 70°	—:—	02:49	04:07	19:51	21:07	—:—
68°	01:12	03:12	04:19	19:39	20:46	22:39
66°	01:58	03:29	04:29	19:30	20:29	21:57
64°	02:27	03:44	04:38	19:22	20:15	21:30
62°	02:48	03:55	04:45	19:15	20:04	21:09
60°	03:05	04:05	04:51	19:09	19:54	20:53
N 58°	03:19	04:14	04:56	19:04	19:46	20:40
56°	03:31	04:21	05:01	18:59	19:39	20:29
54°	03:41	04:27	05:05	18:55	19:32	20:19
52°	03:49	04:33	05:09	18:51	19:27	20:10
50°	03:57	04:38	05:12	18:48	19:22	20:03
45°	04:13	04:49	05:20	18:41	19:11	19:47
N 40°	04:25	04:58	05:26	18:34	19:02	19:35
35°	04:35	05:05	05:31	18:29	18:55	19:26
30°	04:43	05:12	05:36	18:25	18:49	19:18
20°	04:55	05:22	05:44	18:17	18:39	19:05
N 10°	05:05	05:30	05:51	18:10	18:31	18:56
0°	05:12	05:36	05:57	18:04	18:25	18:49
S 10°	05:18	05:42	06:04	17:58	18:19	18:43
20°	05:22	05:48	06:10	17:51	18:14	18:39
30°	05:26	05:53	06:17	17:44	18:08	18:36
35°	05:27	05:56	06:22	17:40	18:05	18:35
40°	05:28	05:59	06:26	17:35	18:03	18:34
45°	05:28	06:02	06:32	17:30	18:00	18:34
S 50°	05:28	06:06	06:38	17:24	17:56	18:34
52°	05:28	06:07	06:41	17:21	17:55	18:34
54°	05:28	06:09	06:44	17:18	17:53	18:34
56°	05:28	06:10	06:48	17:14	17:52	18:35
58°	05:27	06:12	06:52	17:10	17:50	18:35
S 60°	05:26	06:14	06:56	17:06	17:48	18:36

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	14:42	■	■	18:03	■	■
64°	14:04	16:22	■	18:42	18:01	■
62°	13:37	15:24	17:33	19:10	19:01	18:36
60°	13:17	14:51	16:25	19:31	19:34	19:44
N 58°	13:00	14:26	15:50	19:49	19:59	20:19
56°	12:46	14:07	15:25	20:03	20:19	20:45
54°	12:33	13:51	15:05	20:16	20:36	21:05
52°	12:23	13:37	14:48	20:27	20:50	21:22
50°	12:13	13:25	14:33	20:37	21:02	21:37
45°	11:53	13:00	14:04	20:58	21:28	22:06
N 40°	11:37	12:40	13:41	21:16	21:49	22:30
35°	11:24	12:23	13:23	21:30	22:06	22:49
30°	11:12	12:09	13:07	21:43	22:21	23:05
20°	10:52	11:45	12:39	22:04	22:46	23:33
N 10°	10:35	11:24	12:16	22:23	23:08	23:57
0°	10:19	11:05	11:55	22:41	23:29	· · ·
S 10°	10:03:					

September 01, 02, 03 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	340°24.8	210°24.8	N18°30.1	145°39.4	S06°01.8	231°03.2	N22°12.8	339°24.3	S02°15.9	Alpheratz	357°33.9	29°14.0
1	355°27.2	225°24.1	29.4	160°40.4	02.5	246°05.2	12.7	354°26.9	16.0	Ankaa	353°06.2	-42°09.8
2	10°29.7	240°23.5	28.8	175°41.4	03.1	261°07.2	12.7	9°29.5	16.0	Schedar	349°30.0	56°40.6
3	25°32.2	255°22.9	•• 28.1	190°42.3	•• 03.8	276°09.1	•• 12.6	24°32.2	•• 16.1	Diphda	348°46.5	-17°50.5
4	40°34.6	270°22.2	27.5	205°43.3	04.4	291°11.1	12.6	39°34.8	16.2	Achernar	335°19.3	-57°06.1
5	55°37.1	285°21.6	26.8	220°44.3	05.1	306°13.1	12.5	54°37.4	16.3	Hamal	327°50.4	23°35.1
6	70°39.6	300°21.0	N18°26.2	235°45.2	S06°05.7	321°15.1	N22°12.4	69°40.0	S02°16.3	Polaris	313°35.1	89°22.1
7	85°42.0	315°20.3	25.5	250°46.2	06.4	336°17.1	12.4	84°42.6	16.4	Acamar	315°11.2	-40°11.8
8	100°44.5	330°19.7	24.9	265°47.2	07.0	351°19.1	12.3	99°45.3	16.5	Menkar	314°05.5	4°11.6
9	115°46.9	345°19.1	•• 24.2	280°48.1	•• 07.7	6°21.0	•• 12.3	114°47.9	•• 16.6	Mirfak	308°27.3	49°57.0
10	130°49.4	0°18.5	23.6	295°49.1	08.3	21°23.0	12.2	129°50.5	16.6	Aldebaran	290°39.0	16°33.7
11	145°51.9	15°17.8	22.9	310°50.1	09.0	36°25.0	12.2	144°53.1	16.7	Rigel	281°03.4	-8°10.1
12	160°54.3	30°17.2	N18°22.2	325°51.0	S06°09.6	51°27.0	N22°12.1	159°55.7	S02°16.8	Capella	280°21.2	46°01.3
13	175°56.8	45°16.6	21.6	340°52.0	10.3	66°29.0	12.1	174°58.4	16.9	Bellatrix	278°22.4	6°22.5
14	190°59.3	60°16.0	20.9	355°53.0	10.9	81°31.0	12.0	190°01.0	16.9	Elnath	278°01.3	28°37.7
15	206°01.7	75°15.3	•• 20.3	10°53.9	•• 11.6	96°33.0	•• 12.0	205°03.6	•• 17.0	Alnilam	275°37.3	-1°11.0
16	221°04.2	90°14.7	19.6	25°54.9	12.2	111°34.9	11.9	220°06.2	17.1	Betelgeuse	270°51.6	7°24.8
17	236°06.7	105°14.1	18.9	40°55.9	12.9	126°36.9	11.8	235°08.9	17.2	Canopus	263°52.4	-52°42.2
18	251°09.1	120°13.5	N18°18.3	55°56.8	S06°13.5	141°38.9	N22°11.8	250°11.5	S02°17.2	Sirius	258°25.9	-16°44.8
19	266°11.6	135°12.8	17.6	70°57.8	14.2	156°40.9	11.7	265°14.1	17.3	Adhara	255°05.7	-29°00.1
20	281°14.1	150°12.2	17.0	85°58.8	14.8	171°42.9	11.7	280°16.7	17.4	Procyon	244°50.5	5°09.7
21	296°16.5	165°11.6	•• 16.3	100°59.7	•• 15.5	186°44.9	•• 11.6	295°19.3	•• 17.5	Pollux	243°16.9	27°57.9
22	311°19.0	180°11.0	15.6	116°00.7	16.1	201°46.9	11.6	310°22.0	17.5	Avior	234°15.1	-59°35.2
23	326°21.4	195°10.3	15.0	131°01.6	16.8	216°48.8	11.5	325°24.6	17.6	Suhail	222°46.3	-43°32.0
Mer.pass. 01:18		ν -0.6' d -0.6' m -3.95		ν 1.0' d 0.7' m 1.62		ν 2.0' d -0.1' m -1.99		ν 2.6' d 0.1' m 0.65		Miaplacidus	221°39.1	-69°49.2
										Alphard	217°47.5	-8°46.1
										Regulus	207°34.2	11°50.6
										Dubhe	193°41.0	61°36.8
										Denebola	182°24.7	14°25.8
										Gienah	175°43.3	-17°41.0
										Acrux	173°00.2	-63°14.6
										Gacrux	171°51.6	-57°15.5
										Alioth	166°12.9	55°49.4
										Spica	158°22.0	-11°17.7
										Alkaid	152°51.9	49°11.3
										Hadar	148°35.7	-60°30.0
										Menkent	147°57.2	-36°29.9
										Arcturus	145°47.6	19°03.1
										Rigil Kent.	139°39.9	-60°56.7
										Kochab	137°20.3	74°03.2
										Zuben'ubi	136°55.6	-16°08.9
										Alphecca	126°03.4	26°37.9
										Antares	112°15.2	-26°29.4
										Atria	107°08.8	-69°04.7
										Sabik	102°02.1	-15°45.4
										Shaula	96°09.5	-37°07.4
										Rasalhague	95°58.0	12°32.6
										Eltanin	90°41.7	51°29.4
										Kaus Aust.	83°31.6	-34°22.4
										Vega	80°32.7	38°48.7
										Nunki	75°46.9	-26°15.9
										Altair	61°59.2	8°56.3
										Peacock	53°04.4	-56°39.2
										Deneb	49°25.1	45°22.5
										Enif	33°38.0	9°59.7
										Al Na'ir	27°31.8	-46°50.1
										Fomalhaut	15°13.5	-29°29.0
										Scheat	13°44.3	28°13.4
										Markab	13°29.0	15°20.7
Sep 01 Mon SHA Mer.pass												
Venus 230°00.0 09:59												
Mars 165°14.6 14:16												
Jupiter 250°38.4 08:35												
Saturn 358°59.5 01:22												
Sep 02 Tue SHA Mer.pass												
Venus 228°45.8 10:00												
Mars 164°38.7 14:15												
Jupiter 250°26.9 08:31												
Saturn 359°03.3 01:18												
Sep 03 Wed SHA Mer.pass												
Venus 227°31.8 10:01												
Mars 164°02.7 14:13												
Jupiter 250°15.5 08:28												
Saturn 359°07.1 01:14												
Horizontal parallax												
Venus: 0.1												
Mars: 0.1												
Mer.pass. 01:10		ν -0.6' d -0.7' m -3.94		ν 1.0' d 0.6' m 1.62		ν 2.0' d -0.1' m -2.00		ν 2.6' d 0.1' m 0.64				

Sun				Moon							
Mon	GHA	Dec		GHA	ν	Dec	d	HP			
0	179°58.4	N08°15.5		85°12.1	9.6'	S28°02.7	3.1'	54.6'			
1	194°58.6	14.6		99°40.7	9.5'	28°05.8	2.9'	54.6'			
2	209°58.8	13.7		114°09.2	9.5'	28°08.7	2.8'	54.6'			
3	224°59.0	•• 12.8		128°37.7	9.4'	28°11.5	2.7'	54.7'			
4	239°59.2	11.9		143°06.1	9.3'	28°14.1	2.5'	54.7'			
5	254°59.4	11.0		157°34.4	9.3'	28°16.6	2.4'	54.7'			
6	269°59.6	N08°10.1		172°02.7	9.2'	S28°19.0	2.2'	54.7'			
7	284°59.8	09.2		186°31.0	9.2'	28°21.3	2.1'	54.7'			
8	300°00.0	08.3		200°59.2	9.2'	28°23.4	2.0'	54.7'			
9	315°00.2	•• 07.4		215°27.3	9.1'	28°25.3	1.8'	54.8'			
10	330°00.4	06.5		229°55.4	9.1'	28°27.1	1.7'	54.8'			
11	345°00.6	05.5		244°23.5	9.0'	28°28.8	1.5'	54.8'			
12	0°00.8	N08°04.6		258°51.5	9.0'	S28°30.3	1.4'	54.8'			
13	15°01.0	03.7		273°19.5	8.9'	28°31.7	1.2'	54.8'			
14	30°01.2	02.8		287°47.4	8.9'	28°32.9	1.1'	54.9'			
15	45°01.4	•• 01.9		302°15.3	8.9'	28°34.0	0.9'	54.9'			
16	60°01.6	01.0		316°43.2	8.8'	28°35.0	0.8'	54.9'			
17	75°01.8	08°00.1		331°11.0	8.8'	28°35.8	0.7'	54.9'			
18	90°02.0	N07°59.2		345°38.8	8.7'	S28°36.4	0.5'	54.9'			
19	105°02.2	58.3		0°06.5	8.7'	28°36.9	0.4'	55.0'			
20	120°02.4	57.4		14°34.2	8.7'	28°37.3	0.2'	55.0'			
21	135°02.6	•• 56.4		29°01.9	8.6'	28°37.5	0.1'	55.0'			
22	150°02.8	55.5		43°29.6	8.6'	28°37.6	-0.1'	55.0'			
23	165°03.0	54.6		57°57.2	8.6'	28°37.5	-0.2'	55.1'			
SD = 15.8'				$d = -0.9'$				SD = 14.9'			
Tue	GHA	Dec		GHA	ν	Dec	d	HP			
0	180°03.2	N07°53.7		72°24.8	8.6'	S28°37.2	-0.4'	55.1'			
1	195°03.4	52.8		86°52.3	8.5'	28°36.9	-0.5'	55.1'			
2	210°03.7	51.9		101°19.9	8.5'	28°36.3	-0.7'	55.1'			
3	225°03.9	•• 51.0		115°47.4	8.5'	28°35.6	-0.8'	55.2'			
4	240°04.1	50.1		130°14.9	8.5'	28°34.8	-1.0'	55.2'			
5	255°04.3	49.1		144°42.3	8.4'	28°33.8	-1.1'	55.2'			
6	270°04.5	N07°48.2		159°09.8	8.4'	S28°32.7	-1.3'	55.2'			
7	285°04.7	47.3		173°37.2	8.4'	28°31.4	-1.4'	55.3'			
8	300°04.9	46.4		188°04.6	8.4'	28°30.0	-1.6'	55.3'			
9	315°05.1	•• 45.5		202°32.0	8.4'	28°28.4	-1.7'	55.3'			
10	330°05.3	44.6		216°59.4	8.4'	28°26.7	-1.9'	55.3'			
11	345°05.5	43.7		231°26.7	8.3'	28°24.8	-2.0'	55.4'			
12	0°05.7	N07°42.8		245°54.1	8.3'	S28°22.7	-2.2'	55.4'			
13	15°05.9	41.8		260°21.4	8.3'	28°20.5	-2.3'	55.4'			
14	30°06.1	40.9		274°48.7	8.3'	28°18.2	-2.5'	55.4'			
15	45°06.3	•• 40.0		289°16.0	8.3'	28°15.7	-2.7'	55.5'			
16	60°06.5	39.1		303°43.3	8.3'	28°13.0	-2.8'	55.5'			
17	75°06.7	38.2		318°10.6	8.3'	28°10.2	-3.0'	55.5'			
18	90°06.9	N07°37.3		332°37.9	8.3'	S28°07.3	-3.1'	55.5'			
19	105°07.1	36.3		347°05.2	8.3'	28°04.2	-3.3'	55.6'			
20	120°07.3	35.4		1°32.5	8.3'	28°00.9	-3.4'	55.6'			
21	135°07.5	•• 34.5		15°59.8	8.3'	27°57.5	-3.6'	55.6'			
22	150°07.7	33.6		30°27.1	8.3'	27°53.9	-3.7'	55.7'			
23	165°07.9	32.7		44°54.3	8.3'	27°50.2	-3.9'	55.7'			
SD = 15.8'				$d = -0.9'$				SD = 15.0'			
Wed	GHA	Dec		GHA	ν	Dec	d	HP			
0	180°08.1	N07°31.8		59°21.6	8.3'	S27°46.3	-4.0'	55.7'			
1	195°08.3	30.8		73°48.9	8.3'	27°42.3	-4.2'	55.7'			
2	210°08.5	29.9		88°16.2	8.3'	27°38.1	-4.3'	55.8'			
3	225°08.7	•• 29.0		102°43.5	8.3'	27°33.8	-4.5'	55.8'			
4	240°08.9	28.1		117°10.8	8.3'	27°29.3	-4.6'	55.8'			
5	255°09.1	27.2		131°38.2	8.3'	27°24.7	-4.8'	55.9'			
6	270°09.3	N07°26.3		146°05.5	8.3'	S27°19.9	-4.9'	55.9'			
7	285°09.6	25.3		160°32.8	8.4'	27°14.9	-5.1'	55.9'			
8	300°09.8	24.4		175°00.2	8.4'	27°09.8	-5.2'	55.9'			
9	315°10.0	•• 23.5		189°27.6	8.4'	27°04.6	-5.4'	56.0'			
10	330°10.2	22.6		203°54.9	8.4'	26°59.2	-5.5'	56.0'			
11	345°10.4	21.7		218°22.3	8.4'	26°53.6	-5.7'	56.0'			
12	0°10.6	N07°20.7		232°49.8	8.4'	S26°47.9	-5.8'	56.1'			
13	15°10.8	19.8		247°17.2	8.5'	26°42.1	-6.0'	56.1'			
14	30°11.0	18.9		261°44.7	8.5'	26°36.1	-6.1'	56.1'			
15	45°11.2	•• 18.0		276°12.1	8.5'	26°29.9	-6.3'	56.2'			
16	60°11.4	17.1		290°39.6	8.5'	26°23.7	-6.4'	56.2'			
17	75°11.6	16.1		305°07.1	8.5'	26°17.2	-6.6'	56.2'			
18	90°11.8	N07°15.2		319°34.7	8.6'	S26°10.6	-6.7'	56.2'			
19	105°12.0	14.3		334°02.3	8.6'	26°03.9	-6.9'	56.3'			
20	120°12.2	13.4		348°29.9	8.6'	25°57.0	-7.0'	56.3'			
21	135°12.4	•• 12.5		2°57.5	8.6'	25°49.9	-7.2'	56.3'			
22	150°12.6	11.5		17°25.1	8.7'	25°42.7	-7.3'	56.4'			
23	165°12.9	10.6		31°52.8	8.7'	25°35.4	-7.5'	56.4'			
SD = 15.8'				$d = -0.9'$				SD = 15.2'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	02:40	04:07	19:48	21:14	—:—
N 70°	00:24	03:06	04:20	19:36	20:49	23:10
68°	01:42	03:26	04:31	19:26	20:30	22:10
66°	02:17	03:41	04:39	19:18	20:15	21:37
64°	02:41	03:54	04:46	19:11	20:03	21:14
62°	03:00	04:04	04:52	19:05	19:53	20:56
60°	03:15	04:13	04:58	19:00	19:44	20:42
N 58°	03:28	04:21	05:02	18:55	19:37	20:30
56°	03:38	04:27	05:07	18:51	19:30	20:19
54°	03:47	04:33	05:10	18:48	19:25	20:10
52°	03:55	04:38	05:14	18:44	19:20	20:02
50°	04:02	04:43	05:17	18:41	19:15	19:56
45°	04:17	04:53	05:23	18:35	19:05	19:41
N 40°	04:28	05:01	05:29	18:30	18:57	19:30
35°	04:37	05:08	05:34	18:25	18:51	19:21
30°	04:45	05:13	05:38	18:21	18:45	19:14
20°	04:56	05:22	05:45	18:14	18:37	19:03
N 10°	05:05	05:30	05:51	18:08	18:30	18:54
0°	05:11	05:35	05:56	18:03	18:24	18:48
S 10°	05:16	05:41	06:02	17:58	18:19	18:43
20°	05:20	05:45	06:08	17:52	18:14	18:40
30°	05:22	05:50	06:14	17:46	18:10	18:38
35°	05:23	05:52	06:18	17:42	18:08	18:37
40°	05:23	05:55	06:22	17:38	18:05	18:37
45°	05:23	05:57	06:26	17:34	18:03	18:37
S 50°	05:22	06:00	06:32	17:28	18:01	18:38
52°	05:22	06:01	06:35	17:26	18:00	18:39
54°	05:21	06:02	06:37	17:23	17:58	18:39
56°	05:20	06:03	06:40	17:20	17:57	18:40
58°	05:19	06:04	06:44	17:16	17:56	18:41
S 60°	05:18	06:06	06:48	17:13	17:55	18:43

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	■	■	■	■	■	■
64°	■	■	20:40	■	■	21:05
62°	■	■	19:14	■	■	22:31
60°	17:43	18:23	18:37	20:16	21:28	23:07
N 58°	17:00	17:46	18:11	20:59	22:05	23:33
56°	16:31	17:19	17:50	21:27	22:31	23:53
54°	16:09	16:58	17:33	21:50	22:52	•• ••
52°	15:51	16:41	17:18	22:08	23:10	•• ••
50°	15:35	16:26	17:05	22:24	23:24	•• ••
45°	15:04	15:56	16:38	22:55	23:54	•• ••
N 40°	14:40	15:32	16:17	23:19	•• ••	00:18
35°	14:20	15:12	15:59	23:39	•• ••	00:37
30°	14:03	14:56	15:44	23:56	•• ••	00:53
20°	13:34	14:27	15:18	•• ••	00:25	01:21
N 10°	13:10	14:03	14:56	•• ••	00:50	01:45
0°	12:47	13:41	14:35	00:19	01:12	02:07
S 10°	12:24	13:18	14:14	00:42	01:35	02:29
20°	11:59	12:54	13:51	01:06	02:00	02:52
30°	11:31	12:25	13:25	01:34	02:29	03:19
35°	11:14	12:08	13:09	01:50	02:45	03:35
40°	10:55	11:49	12:51	02:10	03:05	03:54
45°	10:31	11:25	12:29	02:33	03:29	04:16
S 50°	10:00	10:54	12:02	03:04	04:00	04:45
52°	09:45	10:39	11:48	03:19	04:15	04:59
54°	09:28	10:21	11:32	03:36	04:33	05:15
56°	09:06	10:00	11:14	03:57	04:55	05:34
58°	08:39	09:32	10:51	04:24	05:23	05:57
S 60°	08:01	08:52	10:20	05:02	06:02	06:28

Day	Sun		Mer. Pass hh:mm	Moon		Age 9-11 57-76%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
01	00:06	00:03	12:00	19:00	06:33	
02	00:13	00:23	12:00	19:54	07:26	
03	00:32	00:42	11:59	20:48	08:21	

September 04, 05, 06 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	343°22.2	209°40.1	N17°41.2	146°48.7	S06°48.6	233°26.4	N22°08.8	342°33.2	S02°21.3	Alpheratz	357°33.9	29°14.0					
1	358°24.7	224°39.5	40.5	161°49.7	49.2	248°28.4	08.8	357°35.8	21.4	Ankaa	353°06.2	-42°09.8					
2	13°27.1	239°38.9	39.8	176°50.6	49.8	263°30.4	08.7	12°38.4	21.5	Schedar	349°29.9	56°40.7					
3	28°29.6	254°38.3	· · 39.1	191°51.6	· · 50.5	278°32.4	· · 08.6	27°41.1	· · 21.5	Diphda	348°46.4	-17°50.5					
4	43°32.0	269°37.7	38.3	206°52.5	51.1	293°34.4	08.6	42°43.7	21.6	Achernar	335°19.3	-57°06.1					
5	58°34.5	284°37.1	37.6	221°53.5	51.8	308°36.4	08.5	57°46.3	21.7	Hamal	327°50.4	23°35.1					
6	73°37.0	299°36.5	N17°36.9	236°54.5	S06°52.4	323°38.4	N22°08.5	72°48.9	S02°21.8	Polaris	313°33.6	89°22.1					
7	88°39.4	314°35.9	36.2	251°55.4	53.1	338°40.4	08.4	87°51.6	21.8	Acamar	315°11.1	-40°11.8					
8	103°41.9	329°35.3	35.5	266°56.4	53.7	353°42.4	08.4	102°54.2	21.9	Menkar	314°05.4	4°11.6					
9	118°44.4	344°34.7	· · 34.8	281°57.3	· · 54.4	8°44.4	· · 08.3	117°56.8	· · 22.0	Mirfak	308°27.3	49°57.0					
10	133°46.8	359°34.1	34.0	296°58.3	55.0	23°46.4	08.3	132°59.4	22.1	Aldebaran	290°39.0	16°33.7					
11	148°49.3	14°33.5	33.3	311°59.2	55.7	38°48.4	08.2	148°02.1	22.1	Rigel	281°03.4	-8°10.1					
12	163°51.8	29°32.9	N17°32.6	327°00.2	S06°56.3	53°50.4	N22°08.1	163°04.7	S02°22.2	Capella	280°21.1	46°01.3					
13	178°54.2	44°32.2	31.9	342°01.1	57.0	68°52.4	08.1	178°07.3	22.3	Bellatrix	278°22.3	6°22.5					
14	193°56.7	59°31.6	31.2	357°02.1	57.6	83°54.4	08.0	193°09.9	22.4	Elnath	278°01.2	28°37.7					
15	208°59.2	74°31.0	· · 30.4	12°03.0	· · 58.3	98°56.4	· · 08.0	208°12.6	· · 22.5	Alnilam	275°37.2	-1°11.0					
16	224°01.6	89°30.4	29.7	27°04.0	58.9	113°58.4	07.9	223°15.2	22.5	Betelgeuse	270°51.6	7°24.8					
17	239°04.1	104°29.8	29.0	42°05.0	06°59.6	129°00.4	07.9	238°17.8	22.6	Canopus	263°52.4	-52°42.2					
18	254°06.5	119°29.2	N17°28.3	57°05.9	S07°00.2	144°02.4	N22°07.8	253°20.5	S02°22.7	Sirius	258°25.9	-16°44.8					
19	269°09.0	134°28.6	27.5	72°06.9	00.9	159°04.4	07.8	268°23.1	22.8	Adhara	255°05.6	-29°00.1					
20	284°11.5	149°28.0	26.8	87°07.8	01.5	174°06.4	07.7	283°25.7	22.8	Procyon	244°50.5	5°09.7					
21	299°13.9	164°27.4	· · 26.1	102°08.8	· · 02.1	189°08.4	· · 07.7	298°28.3	· · 22.9	Pollux	243°16.9	27°57.9					
22	314°16.4	179°26.8	25.3	117°09.7	02.8	204°10.4	07.6	313°31.0	23.0	Avior	234°15.0	-59°35.2					
23	329°18.9	194°26.2	24.6	132°10.7	03.4	219°12.4	07.5	328°33.6	23.1	Suhail	222°46.3	-43°32.0					
Mer.pass. 01:06		ν -0.6' d -0.7' m -3.94			ν 1.0' d 0.6' m 1.62			ν 2.0' d -0.1' m -2.00			ν 2.6' d 0.1' m 0.64						

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	344°21.3	209°25.6	N17°23.9	147°11.6	S07°04.1	234°14.4	N22°07.5	343°36.2	S02°23.1			
1	359°23.8	224°25.0	23.2	162°12.6	04.7	249°16.4	07.4	358°38.8	23.2			
2	14°26.3	239°24.4	22.4	177°13.5	05.4	264°18.4	07.4	13°41.5	23.3			
3	29°28.7	254°23.8	· · 21.7	192°14.5	· · 06.0	279°20.4	· · 07.3	28°44.1	· · 23.4			
4	44°31.2	269°23.2	20.9	207°15.4	06.7	294°22.4	07.3	43°46.7	23.4			
5	59°33.7	284°22.6	20.2	222°16.4	07.3	309°24.4	07.2	58°49.4	23.5			
6	74°36.1	299°22.0	N17°19.5	237°17.3	S07°08.0	324°26.4	N22°07.2	73°52.0	S02°23.6			
7	89°38.6	314°21.4	18.7	252°18.3	08.6	339°28.4	07.1	88°54.6	23.7			
8	104°41.0	329°20.8	18.0	267°19.2	09.3	354°30.4	07.0	103°57.2	23.7			
9	119°43.5	344°20.2	· · 17.3	282°20.2	· · 09.9	9°32.4	· · 07.0	118°59.9	· · 23.8			
10	134°46.0	359°19.6	16.5	297°21.1	10.6	24°34.4	06.9	134°02.5	23.9			
11	149°48.4	14°19.0	15.8	312°22.1	11.2	39°36.4	06.9	149°05.1	24.0			
12	164°50.9	29°18.4	N17°15.0	327°23.0	S07°11.8	54°38.4	N22°06.8	164°07.8	S02°24.1			
13	179°53.4	44°17.8	14.3	342°24.0	12.5	69°40.4	06.8	179°10.4	24.1			
14	194°55.8	59°17.2	13.6	357°24.9	13.1	84°42.4	06.7	194°13.0	24.2			
15	209°58.3	74°16.6	· · 12.8	12°25.9	· · 13.8	99°44.4	· · 06.7	209°15.6	· · 24.3			
16	225°00.8	89°16.0	12.1	27°26.8	14.4	114°46.4	06.6	224°18.3	24.4			
17	240°03.2	104°15.4	11.3	42°27.8	15.1	129°48.4	06.5	239°20.9	24.4			
18	255°05.7	119°14.8	N17°10.6	57°28.7	S07°15.7	144°50.4	N22°06.5	254°23.5	S02°24.5			
19	270°08.1	134°14.2	09.8	72°29.7	16.4	159°52.4	06.4	269°26.2	24.6			
20	285°10.6	149°13.6	09.1	87°30.6	17.0	174°54.4	06.4	284°28.8	24.7			
21	300°13.1	164°13.0	· · 08.3	102°31.6	· · 17.7	189°56.4	· · 06.3	299°31.4	· · 24.7			
22	315°15.5	179°12.4	07.6	117°32.5	18.3	204°58.4	06.3	314°34.0	24.8			
23	330°18.0	194°11.9	06.8	132°33.5	19.0	220°00.4	06.2	329°36.7	24.9			
Mer.pass. 01:02		ν -0.6' d -0.7' m -3.94			ν 1.0' d 0.6' m 1.62			ν 2.0' d -0.1' m -2.01			ν 2.6' d 0.1' m 0.64	

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	345°20.5	209°11.3	N17°06.1	147°34.4	S07°19.6	235°02.4	N22°06.2	344°39.3	S02°25.0			
1	0°22.9	224°10.7	05.3	162°35.4	20.2	250°04.4	06.1	359°41.9	25.1			
2	15°25.4	239°10.1	04.6	177°36.3	20.9	265°06.4	06.1	14°44.6	25.1			
3	30°27.9	254°09.5	· · 03.8	192°37.3	· · 21.5	280°08.4	· · 06.0	29°47.2	· · 25.2			
4	45°30.3	269°08.9	03.1	207°38.2	22.2	295°10.4	05.9	44°49.8	25.3			
5	60°32.8	284°08.3	02.3	222°39.2	22.8	310°12.4	05.9	59°52.4	25.4			
6	75°35.3	299°07.7	N17°01.6	237°40.1	S07°23.5	325°14.4	N22°05.8	74°55.1	S02°25.4			
7	90°37.7	314°07.1	00.8	252°41.1	24.1	340°16.5	05.8	89°57.7	25.5			
8	105°40.2	329°06.5	17°00.0	267°42.0	24.8	355°18.5	05.7	105°00.3	25.6			
9	120°42.6	344°05.9	16°59.3	282°43.0	· · 25.4	10°20.5	· · 05.7	120°03.0	· · 25.7			
10	135°45.1	359°05.3	58.5	297°43.9	26.1	25°22.5	05.6	135°05.6	25.7			
11	150°47.6	14°04.7	57.8	312°44.8	26.7	40°24.5	05.6	150°08.2	25.8			
12	165°50.0	29°04.1	N16°57.0	327°45.8	S07°27.3	55°26.5	N22°05.5	165°10.8	S02°25.9			
13	180°52.5	44°03.6	56.2	342°46.7	28.0	70°28.5	05.4	180°13.5	26.0			
14	195°55.0	59°03.0	55.5	357°47.7	28.6	85°30.5	05.4	195°16.1	26.1			
15	210°57.4	74°02.4	· · 54.7	12°48.6	· · 29.3	100°32.5	· · 05.3	210°18.7	· · 26.1			
16	225°59.9	89°01.8	53.9	27°49.6	29.9	115°34.5	05.3	225°21.4	26.2			
17	241°02.4	104°01.2	53.2	42°50.5	30.6	130°36.5	05.2	240°24.0	26.3			
18	256°04.8	119°00.6	N16°52.4	57°51.5	S07°31.2	145°38.5	N22°05.2	255°26.6	S02°26.4			
19	271°07.3	134°00.0	51.6	72°52.4	31.9	160°40.5	05.1	270°29.3	26.4			
20	286°09.8	148°59.4	50.9	87°53.4	32.5	175°42.5	05.1	285°31.9	26.5			
21	301°12.2	163°58.8	· · 50.1	102°54.3	· · 33.1	190°44.6	· · 05.0	300°34.5	· · 26.6			
22	316°14.7	178°58.3	49.3	117°55.2	33.8	205°46.6	05.0	315°37.2	26.7			
23	331°17.1	193°57.7	48.6	132°56.2	34.4	220°48.6	04.9	330°39.8	26.8			
Mer.pass. 00:58		ν -0.6' d -0.8' m -3.94			ν 0.9' d 0.6' m 1.62			ν 2.0' d -0.1' m -2.01			ν 2.6' d 0.1' m 0.63	

Sep 04 Thu	SHA	Mer.pass
Venus	226°17.9	10:02
Mars	163°26.5	14:12
Jupiter	250°04.2	08:25
Saturn	359°11.0	01:10

Sep 05 Fri	SHA	Mer.pass
Venus	225°04.3	10:03
Mars	162°50.3	14:10
Jupiter	249°53.0	08:22
Saturn	359°14.9	01:05

Sep 06 Sat	SHA	Mer.pass
Venus	223°50.8	10:04
Mars	162°14.0	14:09
Jupiter	249°41.9	08:19
Saturn	359°18.8	01:01

Horizontal parallax	
Venus:	0.1
Mars:	0.1

h							Moon						
Sun													
Thu	GHA	Dec		GHA	ν	Dec	d	HP					
0	180°13.1	N07°09.7		46°20.5	8.7'	S25°27.9	-7.6'	56.4'					
1	195°13.3	08.8		60°48.2	8.8'	25°20.3	-7.8'	56.5'					
2	210°13.5	07.9		75°16.0	8.8'	25°12.5	-7.9'	56.5'					
3	225°13.7	· · 06.9		89°43.8	8.8'	25°04.6	-8.1'	56.5'					
4	240°13.9	06.0		104°11.6	8.9'	24°56.6	-8.2'	56.6'					
5	255°14.1	05.1		118°39.5	8.9'	24°48.4	-8.3'	56.6'					
6	270°14.3	N07°04.2		133°07.4	8.9'	S24°40.0	-8.5'	56.6'					
7	285°14.5	03.2		147°35.3	9.0'	24°31.5	-8.6'	56.7'					
8	300°14.7	02.3		162°03.3	9.0'	24°22.9	-8.8'	56.7'					
9	315°14.9	· · 01.4		176°31.3	9.0'	24°14.1	-8.9'	56.7'					
10	330°15.1	07°00.5		190°59.4	9.1'	24°05.2	-9.0'	56.8'					
11	345°15.3	06°59.5		205°27.5	9.1'	23°56.2	-9.2'	56.8'					
12	0°15.6	N06°58.6		219°55.6	9.2'	S23°47.0	-9.3'	56.8'					
13	15°15.8	57.7		234°23.7	9.2'	23°37.7	-9.5'	56.9'					
14	30°16.0	56.8		248°51.9	9.2'	23°28.2	-9.6'	56.9'					
15	45°16.2	· · 55.9		263°20.2	9.3'	23°18.6	-9.7'	56.9'					
16	60°16.4	54.9		277°48.4	9.3'	23°08.9	-9.9'	57.0'					
17	75°16.6	54.0		292°16.8	9.4'	22°59.0	-10.0'	57.0'					
18	90°16.8	N06°53.1		306°45.1	9.4'	S22°49.0	-10.1'	57.0'					
19	105°17.0	52.1		321°13.5	9.4'	22°38.9	-10.3'	57.1'					
20	120°17.2	51.2		335°42.0	9.5'	22°28.6	-10.4'	57.1'					
21	135°17.4	· · 50.3		350°10.4	9.5'	22°18.2	-10.5'	57.1'					
22	150°17.6	49.4		4°39.0	9.6'	22°07.7	-10.7'	57.2'					
23	165°17.9	48.4		19°07.5	9.6'	21°57.0	-10.8'	57.2'					
SD = 15.9' d = -0.9'							SD = 15.4'						
Fri	GHA	Dec		GHA	ν	Dec	d	HP					
0	180°18.1	N06°47.5		33°36.2	9.7'	S21°46.2	-10.9'	57.2'					
1	195°18.3	46.6		48°04.8	9.7'	21°35.3	-11.0'	57.3'					
2	210°18.5	45.7		62°33.5	9.7'	21°24.2	-11.2'	57.3'					
3	225°18.7	· · 44.7		77°02.2	9.8'	21°13.1	-11.3'	57.3'					
4	240°18.9	43.8		91°31.0	9.8'	21°01.8	-11.4'	57.4'					
5	255°19.1	42.9		105°59.9	9.9'	20°50.3	-11.5'	57.4'					
6	270°19.3	N06°42.0		120°28.7	9.9'	S20°38.8	-11.7'	57.4'					
7	285°19.5	41.0		134°57.7	10.0'	20°27.1	-11.8'	57.5'					
8	300°19.7	40.1		149°26.6	10.0'	20°15.3	-11.9'	57.5'					
9	315°20.0	· · 39.2		163°55.6	10.0'	20°03.4	-12.0'	57.5'					
10	330°20.2	38.2		178°24.7	10.1'	19°51.4	-12.2'	57.6'					
11	345°20.4	37.3		192°53.8	10.1'	19°39.2	-12.3'	57.6'					
12	0°20.6	N06°36.4		207°22.9	10.2'	S19°27.0	-12.4'	57.6'					
13	15°20.8	35.5		221°52.1	10.2'	19°14.6	-12.5'	57.7'					
14	30°21.0	34.5		236°21.3	10.3'	19°02.1	-12.6'	57.7'					
15	45°21.2	· · 33.6		250°50.6	10.3'	18°49.5	-12.7'	57.7'					
16	60°21.4	32.7		265°19.9	10.4'	18°36.7	-12.8'	57.8'					
17	75°21.6	31.7		279°49.2	10.4'	18°23.9	-13.0'	57.8'					
18	90°21.9	N06°30.8		294°18.6	10.4'	S18°10.9	-13.1'	57.8'					
19	105°22.1	29.9		308°48.1	10.5'	17°57.9	-13.2'	57.9'					
20	120°22.3	29.0		323°17.5	10.5'	17°44.7	-13.3'	57.9'					
21	135°22.5	· · 28.0		337°47.1	10.6'	17°31.4	-13.4'	57.9'					
22	150°22.7	27.1		352°16.6	10.6'	17°18.1	-13.5'	58.0'					
23	165°22.9	26.2		6°46.2	10.6'	17°04.6	-13.6'	58.0'					
SD = 15.9' d = -0.9'							SD = 15.6'						
Sat	GHA	Dec		GHA	ν	Dec	d	HP					
0	180°23.1	N06°25.2		21°15.9	10.7'	S16°51.0	-13.7'	58.0'					
1	195°23.3	24.3		35°45.6	10.7'	16°37.3	-13.8'	58.1'					
2	210°23.6	23.4		50°15.3	10.8'	16°23.5	-13.9'	58.1'					
3	225°23.8	· · 22.4		64°45.1	10.8'	16°09.6	-14.0'	58.1'					
4	240°24.0	21.5		79°14.9	10.8'	15°55.6	-14.1'	58.2'					
5	255°24.2	20.6		93°44.7	10.9'	15°41.5	-14.2'	58.2'					
6	270°24.4	N06°19.6		108°14.6	10.9'	S15°27.3	-14.3'	58.2'					
7	285°24.6	18.7		122°44.5	11.0'	15°13.0	-14.4'	58.3'					
8	300°24.8	17.8		137°14.5	11.0'	14°58.6	-14.5'	58.3'					
9	315°25.0	· · 16.8		151°44.5	11.0'	14°44.2	-14.6'	58.3'					
10	330°25.3	15.9		166°14.5	11.1'	14°29.6	-14.7'	58.4'					
11	345°25.5	15.0		180°44.6	11.1'	14°15.0	-14.7'	58.4'					
12	0°25.7	N06°14.0		195°14.7	11.1'	S14°00.2	-14.8'	58.4'					
13	15°25.9	13.1		209°44.8	11.2'	13°45.4	-14.9'	58.4'					
14	30°26.1	12.2		224°15.0	11.2'	13°30.5	-15.0'	58.5'					
15	45°26.3	· · 11.2		238°45.2	11.2'	13°15.5	-15.1'	58.5'					
16	60°26.5	10.3		253°15.4	11.3'	13°00.4	-15.2'	58.5'					
17	75°26.8	09.4		267°45.7	11.3'	12°45.2	-15.2'	58.6'					
18	90°27.0	N06°08.4		282°16.0	11.3'	S12°30.0	-15.3'	58.6'					
19	105°27.2	07.5		296°46.3	11.4'	12°14.6	-15.4'	58.6'					
20	120°27.4	06.6		311°16.7	11.4'	11°59.2	-15.5'	58.7'					
21	135°27.6	· · 05.6		325°47.0	11.4'	11°43.7	-15.6'	58.7'					
22	150°27.8	04.7		340°17.5	11.4'	11°28.2	-15.6'	58.7'					
23	165°28.0	03.8		354°47.9	11.5'	11°12.6	-15.7'	58.7'					
SD = 15.9' d = -0.9'							SD = 15.8'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	—:—	03:00	04:22	19:32	20:53	—:—
N 70°	01:20	03:22	04:33	19:22	20:32	22:27
68°	02:05	03:39	04:42	19:13	20:15	21:46
66°	02:34	03:52	04:49	19:06	20:02	21:19
64°	02:55	04:03	04:55	19:00	19:51	20:59
62°	03:11	04:13	05:00	18:55	19:42	20:43
60°	03:25	04:21	05:05	18:51	19:35	20:30
N 58°	03:36	04:28	05:09	18:47	19:28	20:19
56°	03:45	04:34	05:12	18:44	19:22	20:10
54°	03:54	04:39	05:16	18:40	19:17	20:02
52°	04:01	04:44	05:19	18:38	19:12	19:55
50°	04:08	04:48	05:21	18:35	19:08	19:48
45°	04:21	04:57	05:27	18:29	18:59	19:35
N 40°	04:31	05:04	05:32	18:25	18:52	19:25
35°	04:40	05:10	05:36	18:21	18:47	19:17
30°	04:47	05:15	05:39	18:18	18:42	19:10
20°	04:57	05:23	05:45	18:12	18:34	19:00
N 10°	05:05	05:29	05:51	18:07	18:28	18:52
0°	05:10	05:35	05:55	18:02	18:23	18:47
S 10°	05:14	05:39	06:00	17:57	18:18	18:43
20°	05:17	05:43	06:05	17:53	18:15	18:40
30°	05:19	05:46	06:10	17:47	18:11	18:39
35°	05:19	05:48	06:13	17:44	18:10	18:39
40°	05:19	05:50	06:17	17:41	18:08	18:39
45°	05:18	05:52	06:21	17:37	18:06	18:40
S 50°	05:16	05:53	06:26	17:32	18:05	18:42
52°	05:15	05:54	06:28	17:30	18:04	18:43
54°	05:14	05:55	06:30	17:28	18:04	18:44
56°	05:13	05:56	06:33	17:26	18:03	18:46
58°	05:11	05:56	06:36	17:23	18:02	18:48
S 60°	05:09	05:57	06:39	17:20	18:01	18:50

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	■	21:36	20:01	■	23:53	· ·
N 70°	■	20:34	19:40	■	■	00:53
68°	■	19:59	19:24	■	■	01:27
66°	20:13	19:33	19:11	23:23	· ·	01:51
64°	19:32	19:13	19:00	· ·	00:04	02:10
62°	19:04	18:57	18:51	· ·	00:31	02:25
60°	18:42	18:43	18:43	· ·	00:53	02:38
N 58°	18:24	18:31	18:35	· ·	01:10	02:49
56°	18:09	18:21	18:29	· ·	01:24	02:58
54°	17:56	18:11	18:23	00:10	01:37	03:06
52°	17:44	18:03	18:18	00:25	01:48	03:14
50°	17:34	17:56	18:13	00:37	01:57	03:20
45°	17:12	17:40	18:03	01:03	02:18	03:35
N 40°	16:55	17:27	17:55	01:23	02:34	

September 07, 08, 09 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	346°19.6	208°57.1	N16°47.8	147°57.1	S07°35.1	235°50.6	N22°04.8	345°42.4	S02°26.8	Alpheratz	357°33.9	29°14.0
1	1°22.1	223°56.5	47.0	162°58.1	35.7	250°52.6	04.8	0°45.0	26.9	Ankaa	353°06.2	-42°09.8
2	16°24.5	238°55.9	46.2	177°59.0	36.4	265°54.6	04.7	15°47.7	27.0	Schedar	349°29.9	56°40.7
3	31°27.0	253°55.3	•• 45.5	193°00.0	•• 37.0	280°56.6	•• 04.7	30°50.3	•• 27.1	Diphda	348°46.4	-17°50.5
4	46°29.5	268°54.7	44.7	208°00.9	37.7	295°58.6	04.6	45°52.9	27.1	Achernar	335°19.3	-57°06.1
5	61°31.9	283°54.2	43.9	223°01.9	38.3	311°00.6	04.6	60°55.6	27.2	Hamal	327°50.3	23°35.1
6	76°34.4	298°53.6	N16°43.1	238°02.8	S07°38.9	326°02.6	N22°04.5	75°58.2	S02°27.3	Polaris	313°32.3	89°22.1
7	91°36.9	313°53.0	42.4	253°03.7	39.6	341°04.6	04.5	91°00.8	27.4	Acamar	315°11.1	-40°11.8
8	106°39.3	328°52.4	41.6	268°04.7	40.2	356°06.7	04.4	106°03.5	27.4	Menkar	314°05.4	4°11.6
9	121°41.8	343°51.8	•• 40.8	283°05.6	•• 40.9	11°08.7	•• 04.3	121°06.1	•• 27.5	Mirfak	308°27.3	49°57.0
10	136°44.2	358°51.2	40.0	298°06.6	41.5	26°10.7	04.3	136°08.7	27.6	Aldebaran	290°39.0	16°33.7
11	151°46.7	13°50.6	39.2	313°07.5	42.2	41°12.7	04.2	151°11.4	27.7	Rigel	281°03.4	-8°10.1
12	166°49.2	28°50.1	N16°38.5	328°08.5	S07°42.8	56°14.7	N22°04.2	166°14.0	S02°27.8	Capella	280°21.1	46°01.3
13	181°51.6	43°49.5	37.7	343°09.4	43.4	71°16.7	04.1	181°16.6	27.8	Bellatrix	278°22.3	6°22.5
14	196°54.1	58°48.9	36.9	358°10.3	44.1	86°18.7	04.1	196°19.2	27.9	Elnath	278°01.2	28°37.7
15	211°56.6	73°48.3	•• 36.1	13°11.3	•• 44.7	101°20.7	•• 04.0	211°21.9	•• 28.0	Alnilam	275°37.2	-1°11.0
16	226°59.0	88°47.7	35.3	28°12.2	45.4	116°22.7	04.0	226°24.5	28.1	Betelgeuse	270°51.6	7°24.8
17	242°01.5	103°47.1	34.5	43°13.2	46.0	131°24.8	03.9	241°27.1	28.1	Canopus	263°52.3	-52°42.2
18	257°04.0	118°46.6	N16°33.8	58°14.1	S07°46.7	146°26.8	N22°03.9	256°29.8	S02°28.2	Sirius	258°25.9	-16°44.8
19	272°06.4	133°46.0	33.0	73°15.0	47.3	161°28.8	03.8	271°32.4	28.3	Adhara	255°05.6	-29°00.1
20	287°08.9	148°45.4	32.2	88°16.0	48.0	176°30.8	03.7	286°35.0	28.4	Procyon	244°50.5	5°09.7
21	302°11.4	163°44.8	•• 31.4	103°16.9	•• 48.6	191°32.8	•• 03.7	301°37.7	•• 28.5	Pollux	243°16.9	27°57.9
22	317°13.8	178°44.2	30.6	118°17.9	49.2	206°34.8	03.6	316°40.3	28.5	Avior	234°15.0	-59°35.2
23	332°16.3	193°43.7	29.8	133°18.8	49.9	221°36.8	03.6	331°42.9	28.6	Suhail	222°46.3	-43°32.0
Mer.pass. 00:55		ν -0.6' d -0.8' m -3.94		ν 0.9' d 0.6' m 1.62		ν 2.0' d -0.1' m -2.01		ν 2.6' d 0.1' m 0.63		Miaplacidus	221°39.0	-69°49.1
										Alphard	217°47.5	-8°46.0
										Regulus	207°34.2	11°50.6
										Dubhe	193°40.9	61°36.8
										Denebola	182°24.7	14°25.8
										Gienah	175°43.3	-17°41.0
										Acrux	173°00.2	-63°14.6
										Gacrux	171°51.6	-57°15.5
										Alioth	166°12.9	55°49.4
										Spica	158°22.0	-11°17.7
										Alkaid	152°51.9	49°11.3
										Hadar	148°35.7	-60°30.0
										Menkent	147°57.2	-36°29.8
										Arcturus	145°47.6	19°03.1
										Rigil Kent.	139°40.0	-60°56.7
										Kochab	137°20.3	74°03.2
										Zuben'ubi	136°55.6	-16°08.9
										Alphecca	126°03.4	26°37.9
										Antares	112°15.2	-26°29.4
										Atria	107°08.9	-69°04.7
										Sabik	102°02.1	-15°45.4
										Shaula	96°09.6	-37°07.4
										Rasalhague	95°58.0	12°32.6
										Eltanin	90°41.8	51°29.4
										Kaus Aust.	83°31.6	-34°22.4
										Vega	80°32.7	38°48.7
										Nunki	75°46.9	-26°15.9
										Altair	61°59.2	8°56.3
										Peacock	53°04.4	-56°39.2
										Deneb	49°25.1	45°22.5
										Enif	33°38.0	9°59.7
										Al Na'ir	27°31.7	-46°50.1
										Fomalhaut	15°13.5	-29°29.0
										Scheat	13°44.3	28°13.5
										Markab	13°29.0	15°20.7
Sep 07 Sun			SHA		Mer.pass							
Venus			222°37.5		10:05							
Mars			161°37.5		14:07							
Jupiter			249°31.0		08:16							
Saturn			359°22.8		00:57							
Sep 08 Mon			SHA		Mer.pass							
Venus			221°24.3		10:06							
Mars			161°01.0		14:06							
Jupiter			249°20.1		08:12							
Saturn			359°26.8		00:53							
Sep 09 Tue			SHA		Mer.pass							
Venus			220°11.4		10:06							
Mars			160°24.4		14:04							
Jupiter			249°09.3		08:09							
Saturn			359°30.9		00:49							
Horizontal parallax												
Venus: 0.1												
Mars: 0.1												

Sun				Moon					
Sun	GHA	Dec		GHA	ν	Dec	d	HP	
0	180°28.3	N06°02.8		9°18.4	11.5'	S10°56.9	-15.8'	58.8'	
1	195°28.5	01.9		23°48.8	11.5'	10°41.1	-15.8'	58.8'	
2	210°28.7	01.0		38°19.4	11.5'	10°25.2	-15.9'	58.8'	
3	225°28.9	06°00.0		52°49.9	11.6'	10°09.3	-16.0'	58.9'	
4	240°29.1	05°59.1		67°20.4	11.6'	09°53.3	-16.0'	58.9'	
5	255°29.3	58.2		81°51.0	11.6'	09°37.3	-16.1'	58.9'	
6	270°29.5	N05°57.2		96°21.6	11.6'	S09°21.2	-16.2'	58.9'	
7	285°29.8	56.3		110°52.3	11.6'	09°05.0	-16.2'	59.0'	
8	300°30.0	55.4		125°22.9	11.7'	08°48.8	-16.3'	59.0'	
9	315°30.2	· · 54.4		139°53.5	11.7'	08°32.5	-16.3'	59.0'	
10	330°30.4	53.5		154°24.2	11.7'	08°16.1	-16.4'	59.0'	
11	345°30.6	52.5		168°54.9	11.7'	07°59.7	-16.5'	59.1'	
12	0°30.8	N05°51.6		183°25.6	11.7'	S07°43.3	-16.5'	59.1'	
13	15°31.0	50.7		197°56.3	11.7'	07°26.8	-16.6'	59.1'	
14	30°31.3	49.7		212°27.1	11.7'	07°10.2	-16.6'	59.1'	
15	45°31.5	· · 48.8		226°57.8	11.8'	06°53.6	-16.7'	59.2'	
16	60°31.7	47.9		241°28.6	11.8'	06°36.9	-16.7'	59.2'	
17	75°31.9	46.9		255°59.3	11.8'	06°20.2	-16.8'	59.2'	
18	90°32.1	N05°46.0		270°30.1	11.8'	S06°03.5	-16.8'	59.2'	
19	105°32.3	45.0		285°00.9	11.8'	05°46.7	-16.8'	59.3'	
20	120°32.6	44.1		299°31.7	11.8'	05°29.8	-16.9'	59.3'	
21	135°32.8	· · 43.2		314°02.5	11.8'	05°12.9	-16.9'	59.3'	
22	150°33.0	42.2		328°33.3	11.8'	04°56.0	-17.0'	59.3'	
23	165°33.2	41.3		343°04.1	11.8'	04°39.0	-17.0'	59.4'	
SD = 15.9' d = -0.9'				SD = 16.0'					
Mon	GHA	Dec		GHA	ν	Dec	d	HP	
0	180°33.4	N05°40.3		357°34.9	11.8'	S04°22.0	-17.0'	59.4'	
1	195°33.6	39.4		12°05.7	11.8'	04°05.0	-17.1'	59.4'	
2	210°33.9	38.5		26°36.5	11.8'	03°47.9	-17.1'	59.4'	
3	225°34.1	· · 37.5		41°07.3	11.8'	03°30.9	-17.1'	59.4'	
4	240°34.3	36.6		55°38.1	11.8'	03°13.7	-17.2'	59.5'	
5	255°34.5	35.6		70°08.9	11.8'	02°56.6	-17.2'	59.5'	
6	270°34.7	N05°34.7		84°39.7	11.8'	S02°39.4	-17.2'	59.5'	
7	285°34.9	33.8		99°10.5	11.8'	02°22.2	-17.2'	59.5'	
8	300°35.2	32.8		113°41.2	11.8'	02°05.0	-17.2'	59.5'	
9	315°35.4	· · 31.9		128°12.0	11.8'	01°47.7	-17.3'	59.6'	
10	330°35.6	30.9		142°42.8	11.8'	01°30.4	-17.3'	59.6'	
11	345°35.8	30.0		157°13.5	11.7'	01°13.2	-17.3'	59.6'	
12	0°36.0	N05°29.1		171°44.3	11.7'	S00°55.9	-17.3'	59.6'	
13	15°36.2	28.1		186°15.0	11.7'	00°38.5	-17.3'	59.6'	
14	30°36.5	27.2		200°45.7	11.7'	00°21.2	-17.3'	59.6'	
15	45°36.7	· · 26.2		215°16.4	11.7'	S00°03.9	-17.4'	59.7'	
16	60°36.9	25.3		229°47.1	11.7'	N00°13.5	17.4'	59.7'	
17	75°37.1	24.4		244°17.8	11.6'	00°30.8	17.4'	59.7'	
18	90°37.3	N05°23.4		258°48.4	11.6'	N00°48.2	17.4'	59.7'	
19	105°37.5	22.5		273°19.0	11.6'	01°05.6	17.4'	59.7'	
20	120°37.8	21.5		287°49.6	11.6'	01°22.9	17.4'	59.7'	
21	135°38.0	· · 20.6		302°20.2	11.6'	01°40.3	17.4'	59.8'	
22	150°38.2	19.6		316°50.8	11.5'	01°57.7	17.4'	59.8'	
23	165°38.4	18.7		331°21.3	11.5'	02°15.1	17.4'	59.8'	
SD = 15.9' d = -0.9'				SD = 16.2'					
Tue	GHA	Dec		GHA	ν	Dec	d	HP	
0	180°38.6	N05°17.8		345°51.8	11.5'	N02°32.4	17.4'	59.8'	
1	195°38.8	16.8		0°22.3	11.5'	02°49.8	17.4'	59.8'	
2	210°39.1	15.9		14°52.7	11.4'	03°07.1	17.3'	59.8'	
3	225°39.3	· · 14.9		29°23.2	11.4'	03°24.5	17.3'	59.8'	
4	240°39.5	14.0		43°53.6	11.4'	03°41.8	17.3'	59.8'	
5	255°39.7	13.0		58°23.9	11.3'	03°59.1	17.3'	59.9'	
6	270°39.9	N05°12.1		72°54.2	11.3'	N04°16.4	17.3'	59.9'	
7	285°40.2	11.2		87°24.5	11.3'	04°33.7	17.3'	59.9'	
8	300°40.4	10.2		101°54.8	11.2'	04°51.0	17.2'	59.9'	
9	315°40.6	· · 09.3		116°25.0	11.2'	05°08.2	17.2'	59.9'	
10	330°40.8	08.3		130°55.2	11.1'	05°25.4	17.2'	59.9'	
11	345°41.0	07.4		145°25.3	11.1'	05°42.6	17.2'	59.9'	
12	0°41.2	N05°06.4		159°55.4	11.1'	N05°59.8	17.1'	59.9'	
13	15°41.5	05.5		174°25.5	11.0'	06°17.0	17.1'	59.9'	
14	30°41.7	04.5		188°55.5	11.0'	06°34.1	17.1'	59.9'	
15	45°41.9	· · 03.6		203°25.5	10.9'	06°51.2	17.1'	60.0'	
16	60°42.1	02.7		217°55.4	10.9'	07°08.2	17.0'	60.0'	
17	75°42.3	01.7		232°25.3	10.8'	07°25.2	17.0'	60.0'	
18	90°42.6	N05°00.8		246°55.1	10.8'	N07°42.2	16.9'	60.0'	
19	105°42.8	04°59.8		261°24.9	10.7'	07°59.2	16.9'	60.0'	
20	120°43.0	58.9		275°54.6	10.7'	08°16.1	16.9'	60.0'	
21	135°43.2	· · 57.9		290°24.3	10.6'	08°32.9	16.8'	60.0'	
22	150°43.4	57.0		304°53.9	10.6'	08°49.7	16.8'	60.0'	
23	165°43.7	56.0		319°23.5	10.5'	09°06.5	16.7'	60.0'	
SD = 15.9' d = -0.9'				SD = 16.3'					

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	00:42	03:18	04:36	19:16	20:33	22:54
N 70°	01:50	03:37	04:45	19:08	20:15	21:58
68°	02:25	03:51	04:52	19:01	20:01	21:26
66°	02:49	04:03	04:58	18:55	19:49	21:03
64°	03:07	04:13	05:04	18:50	19:40	20:45
62°	03:22	04:21	05:08	18:46	19:32	20:31
60°	03:34	04:28	05:12	18:42	19:25	20:19
N 58°	03:44	04:35	05:15	18:39	19:19	20:09
56°	03:53	04:40	05:18	18:36	19:14	20:01
54°	04:00	04:45	05:21	18:33	19:09	19:53
52°	04:07	04:49	05:23	18:31	19:05	19:47
50°	04:13	04:53	05:26	18:29	19:01	19:41
45°	04:25	05:01	05:30	18:24	18:54	19:29
N 40°	04:35	05:07	05:34	18:20	18:47	19:20
35°	04:42	05:12	05:38	18:17	18:42	19:12
30°	04:48	05:17	05:41	18:14	18:38	19:06
20°	04:58	05:24	05:46	18:09	18:31	18:57
N 10°	05:05	05:29	05:50	18:05	18:26	18:50
0°	05:09	05:34	05:54	18:01	18:22	18:46
S 10°	05:13	05:37	05:58	17:57	18:18	18:43
20°	05:15	05:40	06:02	17:53	18:15	18:41
30°	05:15	05:43	06:07	17:49	18:13	18:41
35°	05:15	05:44	06:09	17:46	18:12	18:41
40°	05:14	05:45	06:12	17:44	18:11	18:42
45°	05:12	05:46	06:15	17:41	18:10	18:44
S 50°	05:10	05:47	06:19	17:37	18:09	18:47
52°	05:08	05:47	06:21	17:35	18:09	18:48
54°	05:07	05:48	06:23	17:33	18:09	18:50
56°	05:05	05:48	06:25	17:31	18:08	18:52
58°	05:03	05:48	06:27	17:29	18:08	18:54
S 60°	05:00	05:48	06:30	17:27	18:08	18:56

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	19:14	18:36	17:56	03:14	05:45	08:12
N 70°	19:06	18:37	18:07	03:32	05:49	08:05
68°	19:00	18:38	18:16	03:45	05:53	07:59
66°	18:54	18:39	18:23	03:57	05:56	07:54
64°	18:49	18:39	18:29	04:06	05:58	07:50
62°	18:45	18:40	18:35	04:14	06:00	07:47
60°	18:42	18:40	18:39	04:20	06:02	07:44
N 58°	18:38	18:41	18:44	04:26	06:03	07:41
56°	18:35	18:41	18:47	04:32	06:05	07:39
54°	18:33	18:42	18:51	04:36	06:06	07:37
52°	18:30	18:42	18:54	04:40	06:07	07:35
50°	18:28	18:42	18:57	04:44	06:08	07:33
45°	18:24	18:43	19:03	04:52	06:10	07:29
N 40°	18:20	18:44	19:08	04:59	06:12	07:26
35°	18:16	18:44	19:13	05:05	06:14	07:23
30°	18:13	18:45	19:17	05:10	06:15	07:21
20°	18:08	18:45	19:24	05:19	06:18	07:17
N 10°	18:03	18:46	19:30	05:26	06:2	

September 10, 11, 12 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	349°17.0	208°15.6	N15°50.1	149°04.6	S08°21.3	238°15.7	N22°00.9	348°52.0	S02°32.4	Alpheratz	357°33.9	29°14.0
1	4°19.5	223°15.1	49.2	164°05.6	22.0	253°17.7	00.8	3°54.6	32.5	Ankaa	353°06.2	-42°09.8
2	19°21.9	238°14.5	48.4	179°06.5	22.6	268°19.8	00.8	18°57.2	32.6	Schedar	349°29.9	56°40.7
3	34°24.4	253°13.9	•• 47.6	194°07.4	•• 23.3	283°21.8	•• 00.7	33°59.9	•• 32.7	Diphda	348°46.4	-17°50.5
4	49°26.9	268°13.4	46.7	209°08.4	23.9	298°23.8	00.7	49°02.5	32.7	Achernar	335°19.3	-57°06.1
5	64°29.3	283°12.8	45.9	224°09.3	24.5	313°25.8	00.6	64°05.1	32.8	Hamal	327°50.3	23°35.1
6	79°31.8	298°12.3	N15°45.1	239°10.2	S08°25.2	328°27.9	N22°00.6	79°07.8	S02°32.9	Polaris	313°30.9	89°22.1
7	94°34.3	313°11.7	44.2	254°11.1	25.8	343°29.9	00.5	94°10.4	33.0	Acamar	315°11.1	-40°11.8
8	109°36.7	328°11.1	43.4	269°12.1	26.5	358°31.9	00.5	109°13.0	33.1	Menkar	314°05.4	4°11.6
9	124°39.2	343°10.6	•• 42.6	284°13.0	•• 27.1	13°33.9	•• 00.4	124°15.7	•• 33.1	Mirfak	308°27.3	49°57.0
10	139°41.7	358°10.0	41.7	299°13.9	27.7	28°36.0	00.4	139°18.3	33.2	Aldebaran	290°38.9	16°33.7
11	154°44.1	13°09.4	40.9	314°14.9	28.4	43°38.0	00.3	154°20.9	33.3	Rigel	281°03.3	-8°10.1
12	169°46.6	28°08.9	N15°40.0	329°15.8	S08°29.0	58°40.0	N22°00.2	169°23.6	S02°33.4	Capella	280°21.1	46°01.3
13	184°49.0	43°08.3	39.2	344°16.7	29.6	73°42.0	00.2	184°26.2	33.4	Bellatrix	278°22.3	6°22.5
14	199°51.5	58°07.8	38.4	359°17.6	30.3	88°44.1	00.1	199°28.8	33.5	Elnath	278°01.2	28°37.7
15	214°54.0	73°07.2	•• 37.5	14°18.6	•• 30.9	103°46.1	•• 00.1	214°31.5	•• 33.6	Alnilam	275°37.2	-1°11.0
16	229°56.4	88°06.6	36.7	29°19.5	31.6	118°48.1	00.0	229°34.1	33.7	Betelgeuse	270°51.5	7°24.8
17	244°58.9	103°06.1	35.8	44°20.4	32.2	133°50.1	22°00.0	244°36.7	33.8	Canopus	263°52.3	-52°42.2
18	260°01.4	118°05.5	N15°35.0	59°21.4	S08°32.8	148°52.2	N21°59.9	259°39.4	S02°33.8	Sirius	258°25.9	-16°44.8
19	275°03.8	133°05.0	34.1	74°22.3	33.5	163°54.2	59.9	274°42.0	33.9	Adhara	255°05.6	-29°00.1
20	290°06.3	148°04.4	33.3	89°23.2	34.1	178°56.2	59.8	289°44.6	34.0	Procyon	244°50.4	5°09.7
21	305°08.8	163°03.9	•• 32.5	104°24.1	•• 34.8	193°58.2	•• 59.8	304°47.3	•• 34.1	Pollux	243°16.8	27°57.9
22	320°11.2	178°03.3	31.6	119°25.1	35.4	209°00.3	59.7	319°49.9	34.2	Avior	234°15.0	-59°35.2
23	335°13.7	193°02.7	30.8	134°26.0	36.0	224°02.3	59.6	334°52.6	34.2	Suhail	222°46.3	-43°31.9
Mer.pass. 00:43		ν -0.6' d -0.8' m -3.94		ν 0.9' d 0.6' m 1.61		ν 2.0' d -0.1' m -2.02		ν 2.6' d 0.1' m 0.62				
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	350°16.2	208°02.2	N15°29.9	149°26.9	S08°36.7	239°04.3	N21°59.6	349°55.2	S02°34.3	Denebola	182°24.7	14°25.8
1	5°18.6	223°01.6	29.1	164°27.8	37.3	254°06.4	59.5	4°57.8	34.4	Gienah	175°43.3	-17°41.0
2	20°21.1	238°01.1	28.2	179°28.8	38.0	269°08.4	59.5	20°00.5	34.5	Acrux	173°00.2	-63°14.5
3	35°23.5	253°00.5	•• 27.4	194°29.7	•• 38.6	284°10.4	•• 59.4	35°03.1	•• 34.5	Gacrux	171°51.6	-57°15.5
4	50°26.0	268°00.0	26.5	209°30.6	39.2	299°12.4	59.4	50°05.7	34.6	Alioth	166°12.9	55°49.4
5	65°28.5	282°59.4	25.7	224°31.5	39.9	314°14.5	59.3	65°08.4	34.7	Spica	158°22.0	-11°17.7
6	80°30.9	297°58.8	N15°24.8	239°32.5	S08°40.5	329°16.5	N21°59.3	80°11.0	S02°34.8	Alkaid	152°51.9	49°11.3
7	95°33.4	312°58.3	23.9	254°33.4	41.1	344°18.5	59.2	95°13.6	34.9	Hadar	148°35.7	-60°30.0
8	110°35.9	327°57.7	23.1	269°34.3	41.8	359°20.6	59.2	110°16.3	34.9	Menkent	147°57.3	-36°29.8
9	125°38.3	342°57.2	•• 22.2	284°35.2	•• 42.4	14°22.6	•• 59.1	125°18.9	•• 35.0	Arcturus	145°47.6	19°03.1
10	140°40.8	357°56.6	21.4	299°36.2	43.1	29°24.6	59.0	140°21.5	35.1	Rigil Kent.	139°40.0	-60°56.7
11	155°43.3	12°56.1	20.5	314°37.1	43.7	44°26.6	59.0	155°24.2	35.2	Kochab	137°20.4	74°03.2
12	170°45.7	27°55.5	N15°19.7	329°38.0	S08°44.3	59°28.7	N21°58.9	170°26.8	S02°35.3	Zuben'ubi	136°55.6	-16°08.9
13	185°48.2	42°55.0	18.8	344°38.9	45.0	74°30.7	58.9	185°29.5	35.3	Alphecca	126°03.4	26°37.9
14	200°50.7	57°54.4	17.9	359°39.9	45.6	89°32.7	58.8	200°32.1	35.4	Antares	112°15.2	-26°29.4
15	215°53.1	72°53.9	•• 17.1	14°40.8	•• 46.3	104°34.8	•• 58.8	215°34.7	•• 35.5	Atria	107°09.0	-69°04.7
16	230°55.6	87°53.3	16.2	29°41.7	46.9	119°36.8	58.7	230°37.4	35.6	Sabik	102°02.1	-15°45.4
17	245°58.0	102°52.8	15.4	44°42.6	47.5	134°38.8	58.7	245°40.0	35.6	Shaula	96°09.6	-37°07.4
18	261°00.5	117°52.2	N15°14.5	59°43.6	S08°48.2	149°40.9	N21°58.6	260°42.6	S02°35.7	Rasalhague	95°58.0	12°32.6
19	276°03.0	132°51.7	13.6	74°44.5	48.8	164°42.9	58.6	275°45.3	35.8	Eltanin	90°41.8	51°29.4
20	291°05.4	147°51.1	12.8	89°45.4	49.4	179°44.9	58.5	290°47.9	35.9	Kaus Aust.	83°31.7	-34°22.4
21	306°07.9	162°50.6	•• 11.9	104°46.3	•• 50.1	194°46.9	•• 58.4	305°50.5	•• 36.0	Vega	80°32.7	38°48.7
22	321°10.4	177°50.0	11.0	119°47.2	50.7	209°49.0	58.4	320°53.2	36.0	Nunki	75°46.9	-26°15.9
23	336°12.8	192°49.5	10.2	134°48.2	51.4	224°51.0	58.3	335°55.8	36.1	Altair	61°59.2	8°56.3
Mer.pass. 00:39		ν -0.6' d -0.8' m -3.94		ν 0.9' d 0.6' m 1.61		ν 2.0' d -0.1' m -2.03		ν 2.6' d 0.1' m 0.61				
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	351°15.3	207°48.9	N15°09.3	149°49.1	S08°52.0	239°53.0	N21°58.3	350°58.5	S02°36.2	Sep 10 Wed	SHA	Mer.pass
1	6°17.8	222°48.4	08.4	164°50.0	52.6	254°55.1	58.2	6°01.1	36.3	Venus	218°58.6	10:07
2	21°20.2	237°47.8	07.6	179°50.9	53.3	269°57.1	58.2	21°03.7	36.4	Mars	159°47.6	14:03
3	36°22.7	252°47.3	•• 06.7	194°51.9	•• 53.9	284°59.1	•• 58.1	36°06.4	•• 36.4	Jupiter	248°58.7	08:06
4	51°25.1	267°46.7	05.8	209°52.8	54.5	300°01.2	58.1	51°09.0	36.5	Saturn	359°34.9	00:44
5	66°27.6	282°46.2	05.0	224°53.7	55.2	315°03.2	58.0	66°11.6	36.6			
6	81°30.1	297°45.6	N15°04.1	239°54.6	S08°55.8	330°05.2	N21°58.0	81°14.3	S02°36.7			
7	96°32.5	312°45.1	03.2	254°55.5	56.4	345°07.3	57.9	96°16.9	36.7			
8	111°35.0	327°44.5	02.3	269°56.5	57.1	0°09.3	57.8	111°19.5	36.8			
9	126°37.5	342°44.0	•• 01.5	284°57.4	•• 57.7	15°11.3	•• 57.8	126°22.2	•• 36.9			
10	141°39.9	357°43.4	15°00.6	299°58.3	58.4	30°13.4	57.7	141°24.8	37.0	Sep 11 Thu	SHA	Mer.pass
11	156°42.4	12°42.9	14°59.7	314°59.2	59.0	45°15.4	57.7	156°27.5	37.1	Venus	217°46.0	10:08
12	171°44.9	27°42.3	N14°58.8	330°00.1	S08°59.6	60°17.4	N21°57.6	171°30.1	S02°37.1	Mars	159°10.8	14:01
13	186°47.3	42°41.8	58.0	345°01.1	09°00.3	75°19.5	57.6	186°32.7	37.2	Jupiter	248°48.2	08:03
14	201°49.8	57°41.3	57.1	0°02.0	00.9	90°21.5	57.5	201°35.4	37.3	Saturn	359°39.0	00:40
15	216°52.3	72°40.7	•• 56.2	15°02.9	•• 01.5	105°23.6	•• 57.5	216°38.0	•• 37.4			
16	231°54.7	87°40.2	55.3	30°03.8	02.2	120°25.6	57.4	231°40.6	37.5			
17	246°57.2	102°39.6	54.4	45°04.7	02.8	135°27.6	57.4	246°43.3	37.5			
18	261°59.6	117°39.1	N14°53.6	60°05.6	S09°03.4	150°29.7	N21°57.3	261°45.9	S02°37.6			
19	277°02.1	132°38.5	52.7	75°06.6	04.1	165°31.7	57.3	276°48.6	37.7			
20	292°04.6	147°38.0	51.8	90°07.5	04.7	180°33.7	57.2	291°51.2	37.8			
21	307°07.0	162°37.5	•• 50.9	105°08.4	•• 05.4	195°35.8	•• 57.1	306°53.8	•• 37.9			
22	322°09.5	177°36.9	50.0	120°09.3	06.0	210°37.8	57.1	321°56.5</				

Sun				Moon											
Wed	GHA	Dec		GHA	ν	Dec	d	HP	Lat.	Twilight		Sunrise	Sunset	Twilight	
										Naut.	Civil			Civil	Naut.
0	180°43.9	N04°55.1		333°53.0	10.5'	N09°23.2	16.7'	60.0'							
1	195°44.1	54.1		348°22.5	10.4'	09°39.9	16.6'	60.0'	N 72°	01:30	03:35	04:50	19:00	20:14	22:13
2	210°44.3	53.2		2°51.9	10.3'	09°56.5	16.6'	60.0'	N 70°	02:14	03:51	04:57	18:53	19:59	21:33
3	225°44.5	•• 52.2		17°21.2	10.3'	10°13.1	16.5'	60.0'	68°	02:42	04:04	05:03	18:48	19:47	21:07
4	240°44.8	51.3		31°50.5	10.2'	10°29.6	16.5'	60.0'	66°	03:02	04:14	05:08	18:43	19:37	20:47
5	255°45.0	50.3		46°19.7	10.2'	10°46.1	16.4'	60.0'	64°	03:19	04:23	05:12	18:39	19:28	20:32
6	270°45.2	N04°49.4		60°48.9	10.1'	N11°02.5	16.3'	60.0'	62°	03:32	04:30	05:16	18:36	19:21	20:19
7	285°45.4	48.5		75°18.0	10.0'	11°18.8	16.3'	60.0'	60°	03:42	04:36	05:19	18:33	19:15	20:08
8	300°45.6	47.5		89°47.0	10.0'	11°35.1	16.2'	60.0'	N 58°	03:52	04:41	05:22	18:30	19:10	19:59
9	315°45.9	•• 46.6		104°16.0	9.9'	11°51.3	16.1'	60.0'	56°	04:00	04:46	05:24	18:28	19:06	19:52
10	330°46.1	45.6		118°44.9	9.8'	12°07.5	16.1'	60.0'	54°	04:06	04:50	05:26	18:26	19:02	19:45
11	345°46.3	44.7		133°13.8	9.8'	12°23.5	16.0'	60.0'	52°	04:13	04:54	05:28	18:24	18:58	19:39
12	0°46.5	N04°43.7		147°42.6	9.7'	N12°39.5	15.9'	60.0'	50°	04:18	04:57	05:30	18:22	18:55	19:34
13	15°46.7	42.8		162°11.3	9.6'	12°55.5	15.9'	60.0'	45°	04:29	05:04	05:34	18:18	18:48	19:23
14	30°47.0	41.8		176°39.9	9.6'	13°11.3	15.8'	60.0'	N 40°	04:38	05:10	05:37	18:15	18:42	19:15
15	45°47.2	•• 40.9		191°08.5	9.5'	13°27.1	15.7'	60.0'	35°	04:45	05:15	05:40	18:12	18:38	19:08
16	60°47.4	39.9		205°37.0	9.4'	13°42.8	15.6'	60.0'	30°	04:50	05:18	05:42	18:10	18:34	19:02
17	75°47.6	39.0		220°05.4	9.4'	13°58.5	15.5'	60.0'	20°	04:59	05:24	05:47	18:06	18:28	18:54
18	90°47.8	N04°38.0		234°33.8	9.3'	N14°14.0	15.5'	60.0'	N 10°	05:05	05:29	05:50	18:03	18:24	18:48
19	105°48.1	37.1		249°02.0	9.2'	14°29.5	15.4'	60.0'	0°	05:09	05:33	05:53	18:00	18:21	18:45
20	120°48.3	36.1		263°30.3	9.1'	14°44.9	15.3'	60.0'	S 10°	05:11	05:35	05:56	17:57	18:18	18:42
21	135°48.5	•• 35.2		277°58.4	9.1'	15°00.2	15.2'	60.0'	20°	05:12	05:38	06:00	17:54	18:16	18:41
22	150°48.7	34.2		292°26.4	9.0'	15°15.4	15.1'	60.0'	30°	05:11	05:39	06:03	17:50	18:14	18:42
23	165°48.9	33.3		306°54.4	8.9'	15°30.5	15.0'	60.0'	35°	05:11	05:40	06:05	17:49	18:14	18:43
SD = 15.9'		d = -0.9'	SD = 16.4'												
Thu	GHA	Dec		GHA	ν	Dec	d	HP							
0	180°49.2	N04°32.3		321°22.3	8.8'	N15°45.5	14.9'	60.0'							
1	195°49.4	31.4		335°50.2	8.7'	16°00.4	14.8'	60.0'							
2	210°49.6	30.4		350°17.9	8.7'	16°15.3	14.7'	60.0'							
3	225°49.8	•• 29.5		4°45.6	8.6'	16°30.0	14.6'	60.0'							
4	240°50.0	28.5		19°13.2	8.5'	16°44.6	14.5'	60.0'							
5	255°50.3	27.6		33°40.7	8.4'	16°59.2	14.4'	60.0'							
6	270°50.5	N04°26.6		48°08.1	8.3'	N17°13.6	14.3'	60.0'							
7	285°50.7	25.7		62°35.4	8.3'	17°27.9	14.2'	60.0'							
8	300°50.9	24.7		77°02.7	8.2'	17°42.2	14.1'	60.0'							
9	315°51.1	•• 23.8		91°29.9	8.1'	17°56.3	14.0'	60.0'							
10	330°51.4	22.8		105°57.0	8.0'	18°10.3	13.9'	60.0'							
11	345°51.6	21.9		120°24.0	7.9'	18°24.2	13.8'	60.0'							
12	0°51.8	N04°20.9		134°50.9	7.8'	N18°37.9	13.7'	59.9'							
13	15°52.0	20.0		149°17.8	7.8'	18°51.6	13.5'	59.9'							
14	30°52.2	19.0		163°44.5	7.7'	19°05.1	13.4'	59.9'							
15	45°52.5	•• 18.1		178°11.2	7.6'	19°18.5	13.3'	59.9'							
16	60°52.7	17.1		192°37.8	7.5'	19°31.8	13.2'	59.9'							
17	75°52.9	16.1		207°04.3	7.4'	19°45.0	13.1'	59.9'							
18	90°53.1	N04°15.2		221°30.8	7.3'	N19°58.1	12.9'	59.9'							
19	105°53.4	14.2		235°57.1	7.3'	20°11.0	12.8'	59.9'							
20	120°53.6	13.3		250°23.3	7.2'	20°23.8	12.7'	59.9'							
21	135°53.8	•• 12.3		264°49.5	7.1'	20°36.5	12.5'	59.9'							
22	150°54.0	11.4		279°15.6	7.0'	20°49.0	12.4'	59.9'							
23	165°54.2	10.4		293°41.6	6.9'	21°01.4	12.3'	59.8'							
SD = 15.9'		d = -1.0'	SD = 16.4'												
Fri	GHA	Dec		GHA	ν	Dec	d	HP							
0	180°54.5	N04°09.5		308°07.5	6.8'	N21°13.7	12.1'	59.8'							
1	195°54.7	08.5		322°33.3	6.7'	21°25.8	12.0'	59.8'							
2	210°54.9	07.6		336°59.1	6.7'	21°37.8	11.9'	59.8'							
3	225°55.1	•• 06.6		351°24.7	6.6'	21°49.6	11.7'	59.8'							
4	240°55.3	05.7		5°50.3	6.5'	22°01.3	11.6'	59.8'							
5	255°55.6	04.7		20°15.8	6.4'	22°12.9	11.4'	59.8'							
6	270°55.8	N04°03.8		34°41.2	6.3'	N22°24.3	11.3'	59.8'							
7	285°56.0	02.8		49°06.5	6.2'	22°35.6	11.1'	59.7'							
8	300°56.2	01.8		63°31.8	6.2'	22°46.7	11.0'	59.7'							
9	315°56.5	04°00.9		77°56.9	6.1'	22°57.7	10.8'	59.7'							
10	330°56.7	03°59.9		92°22.0	6.0'	23°08.5	10.7'	59.7'							
11	345°56.9	59.0		106°47.0	5.9'	23°19.2	10.5'	59.7'							
12	0°57.1	N03°58.0		121°11.9	5.8'	N23°29.7	10.4'	59.7'							
13	15°57.3	57.1		135°36.8	5.8'	23°40.1	10.2'	59.7'							
14	30°57.6	56.1		150°01.5	5.7'	23°50.3	10.0'	59.7'							
15	45°57.8	•• 55.2		164°26.2	5.6'	24°00.3	9.9'	59.6'							
16	60°58.0	54.2		178°50.8	5.5'	24°10.2	9.7'	59.6'							
17	75°58.2	53.2		193°15.3	5.5'	24°19.9	9.6'	59.6'							
18	90°58.5	N03°52.3		207°39.8	5.4'	N24°29.5	9.4'	59.6'							
19	105°58.7	51.3		222°04.1	5.3'	24°38.9	9.2'	59.6'							
20	120°58.9	50.4		236°28.5	5.2'	24°48.1	9.1'	59.6'							
21	135°59.1	•• 49.4		250°52.7	5.2'	24°57.2	8.9'	59.5'							
22	150°59.3	48.5		265°16.8	5.1'	25°06.1	8.7'	59.5'							
23	165°59.6	47.5		279°40.9	5.0'	25°14.8	8.6'	59.5'							
SD = 15.9'		d = -1.0'	SD = 16.3'												

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	01:30	03:35	04:50	19:00	20:14	22:13
N 70°	02:14	03:51	04:57	18:53	19:59	21:33
68°	02:42	04:04	05:03	18:48	19:47	21:07
66°	03:02	04:14	05:08	18:43	19:37	20:47
64°	03:19	04:23	05:12	18:39	19:28	20:32
62°	03:32	04:30	05:16	18:36	19:21	20:19
60°	03:42	04:36	05:19	18:33	19:15	20:08
N 58°	03:52	04:41	05:22	18:30	19:10	19:59
56°	04:00	04:46	05:24	18:28	19:06	19:52
54°	04:06	04:50	05:26	18:26	19:02	19:45
52°	04:13	04:54	05:28	18:24	18:58	19:39
50°	04:18	04:57	05:30	18:22	18:55	19:34
45°	04:29	05:04	05:34	18:18	18:48	19:23
N 40°	04:38	05:10	05:37	18:15	td	

September 13, 14, 15 UT (Sat., Sun., Mon.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	352°14.4	207°35.8	N14°48.3	150°11.2	S09°07.3	240°41.9	N21°57.0	352°01.7	S02°38.1					Alpheratz	357°33.9	29°14.1	
1	7°16.9	222°35.3	47.4	165°12.1	07.9	255°43.9	56.9	7°04.4	38.2					Ankaa	353°06.2	-42°09.8	
2	22°19.4	237°34.7	46.5	180°13.0	08.5	270°46.0	56.9	22°07.0	38.2					Schedar	349°29.9	56°40.7	
3	37°21.8	252°34.2	•• 45.6	195°13.9	•• 09.2	285°48.0	•• 56.8	37°09.7	•• 38.3					Diphda	348°46.4	-17°50.5	
4	52°24.3	267°33.7	44.7	210°14.8	09.8	300°50.0	56.8	52°12.3	38.4					Achernar	335°19.2	-57°06.1	
5	67°26.8	282°33.1	43.8	225°15.7	10.4	315°52.1	56.7	67°14.9	38.5					Hamal	327°50.3	23°35.1	
6	82°29.2	297°32.6	N14°42.9	240°16.7	S09°11.1	330°54.1	N21°56.7	82°17.6	S02°38.6					Polaris	313°29.2	89°22.1	
7	97°31.7	312°32.0	42.1	255°17.6	11.7	345°56.1	56.6	97°20.2	38.6					Acamar	315°11.1	-40°11.8	
8	112°34.1	327°31.5	41.2	270°18.5	12.3	0°58.2	56.5	112°22.8	38.7					Menkar	314°05.4	4°11.6	
9	127°36.6	342°31.0	•• 40.3	285°19.4	•• 13.0	16°00.2	•• 56.5	127°25.5	•• 38.8					Mirfak	308°27.2	49°57.0	
10	142°39.1	357°30.4	39.4	300°20.3	13.6	31°02.3	56.4	142°28.1	38.9					Aldebaran	290°38.9	16°33.7	
11	157°41.5	12°29.9	38.5	315°21.2	14.3	46°04.3	56.4	157°30.8	39.0					Rigel	281°03.3	-8°10.1	
12	172°44.0	27°29.4	N14°37.6	330°22.1	S09°14.9	61°06.3	N21°56.3	172°33.4	S02°39.0					Capella	280°21.0	46°01.3	
13	187°46.5	42°28.8	36.7	345°23.1	15.5	76°08.4	56.3	187°36.0	39.1					Bellatrix	278°22.3	6°22.5	
14	202°48.9	57°28.3	35.8	0°24.0	16.2	91°10.4	56.2	202°38.7	39.2					Elnath	278°01.2	28°37.7	
15	217°51.4	72°27.7	•• 34.9	15°24.9	•• 16.8	106°12.5	•• 56.2	217°41.3	•• 39.3					Alnilam	275°37.2	-1°11.0	
16	232°53.9	87°27.2	34.0	30°25.8	17.4	121°14.5	56.1	232°43.9	39.4					Betelgeuse	270°51.5	7°24.8	
17	247°56.3	102°26.7	33.1	45°26.7	18.1	136°16.5	56.1	247°46.6	39.4					Canopus	263°52.3	-52°42.2	
18	262°58.8	117°26.1	N14°32.2	60°27.6	S09°18.7	151°18.6	N21°56.0	262°49.2	S02°39.5					Sirius	258°25.9	-16°44.8	
19	278°01.3	132°25.6	31.3	75°28.5	19.3	166°20.6	56.0	277°51.9	39.6					Adhara	255°05.6	-29°00.1	
20	293°03.7	147°25.1	30.4	90°29.5	20.0	181°22.7	55.9	292°54.5	39.7					Procyon	244°50.4	5°09.7	
21	308°06.2	162°24.5	•• 29.5	105°30.4	•• 20.6	196°24.7	•• 55.8	307°57.1	•• 39.7					Pollux	243°16.8	27°57.8	
22	323°08.6	177°24.0	28.6	120°31.3	21.2	211°26.8	55.8	322°59.8	39.8					Avior	234°15.0	-59°35.2	
23	338°11.1	192°23.5	27.7	135°32.2	21.9	226°28.8	55.7	338°02.4	39.9					Suhail	222°46.3	-43°31.9	
Mer.pass. 00:31		ν -0.5' d -0.9' m -3.94			ν 0.9' d 0.6' m 1.61			ν 2.0' d -0.1' m -2.04			ν 2.6' d 0.1' m 0.61						

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	353°13.6	207°22.9	N14°26.8	150°33.1	S09°22.5	241°30.8	N21°55.7	353°05.1	S02°40.0				
1	8°16.0	222°22.4	25.9	165°34.0	23.1	256°32.9	55.6	8°07.7	40.1				
2	23°18.5	237°21.9	25.0	180°34.9	23.8	271°34.9	55.6	23°10.3	40.1				
3	38°21.0	252°21.3	•• 24.1	195°35.8	•• 24.4	286°37.0	•• 55.5	38°13.0	•• 40.2				
4	53°23.4	267°20.8	23.2	210°36.7	25.0	301°39.0	55.5	53°15.6	40.3				
5	68°25.9	282°20.3	22.3	225°37.7	25.7	316°41.1	55.4	68°18.2	40.4				
6	83°28.4	297°19.7	N14°21.4	240°38.6	S09°26.3	331°43.1	N21°55.4	83°20.9	S02°40.5				
7	98°30.8	312°19.2	20.5	255°39.5	26.9	346°45.1	55.3	98°23.5	40.5				
8	113°33.3	327°18.7	19.6	270°40.4	27.6	1°47.2	55.3	113°26.2	40.6				
9	128°35.8	342°18.1	•• 18.6	285°41.3	•• 28.2	16°49.2	•• 55.2	128°28.8	•• 40.7				
10	143°38.2	357°17.6	17.7	300°42.2	28.8	31°51.3	55.1	143°31.4	40.8				
11	158°40.7	12°17.1	16.8	315°43.1	29.5	46°53.3	55.1	158°34.1	40.9				
12	173°43.1	27°16.5	N14°15.9	330°44.0	S09°30.1	61°55.4	N21°55.0	173°36.7	S02°40.9				
13	188°45.6	42°16.0	15.0	345°44.9	30.7	76°57.4	55.0	188°39.4	41.0				
14	203°48.1	57°15.5	14.1	0°45.8	31.4	91°59.5	54.9	203°42.0	41.1				
15	218°50.5	72°14.9	•• 13.2	15°46.8	•• 32.0	107°01.5	•• 54.9	218°44.6	•• 41.2				
16	233°53.0	87°14.4	12.3	30°47.7	32.6	122°03.5	54.8	233°47.3	41.2				
17	248°55.5	102°13.9	11.3	45°48.6	33.3	137°05.6	54.8	248°49.9	41.3				
18	263°57.9	117°13.4	N14°10.4	60°49.5	S09°33.9	152°07.6	N21°54.7	263°52.5	S02°41.4				
19	279°00.4	132°12.8	09.5	75°50.2	34.5	167°09.7	54.7	278°55.2	41.5				
20	294°02.9	147°12.3	08.6	90°51.3	35.2	182°11.7	54.6	293°57.8	41.6				
21	309°05.3	162°11.8	•• 07.7	105°52.2	•• 35.8	197°13.8	•• 54.6	309°00.5	•• 41.6				
22	324°07.8	177°11.3	06.8	120°53.1	36.2	212°15.8	54.5	324°03.1	41.7				
23	339°10.2	192°10.7	05.8	135°54.0	37.1	227°17.9	54.5	339°05.7	41.8				
Mer.pass. 00:27		ν -0.5' d -0.9' m -3.94			ν 0.9' d 0.6' m 1.61			ν 2.0' d -0.1' m -2.04			ν 2.6' d 0.1' m 0.60		

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec				
0	354°12.7	207°10.2	N14°04.9	150°54.9	S09°37.7	242°19.9	N21°54.4	354°08.4	S02°41.9				
1	9°15.2	222°09.7	04.0	165°55.8	38.3	257°22.0	54.3	9°11.0	42.0				
2	24°17.6	237°09.1	03.1	180°56.7	39.0	272°24.0	54.3	24°13.7	42.0				
3	39°20.1	252°08.6	•• 02.2	195°57.7	•• 39.6	287°26.1	•• 54.2	39°16.3	•• 42.1				
4	54°22.6	267°08.1	01.2	210°58.6	40.2	302°28.1	54.2	54°18.9	42.2				
5	69°25.0	282°07.6	14°00.3	225°59.5	40.8	317°30.2	54.1	69°21.6	42.3				
6	84°27.5	297°07.0	N13°59.4	241°00.4	S09°41.5	332°32.2	N21°54.1	84°24.2	S02°42.4				
7	99°30.0	312°06.5	58.5	256°01.3	42.1	347°34.3	54.0	99°26.9	42.4				
8	114°32.4	327°06.0	57.5	271°02.2	42.7	2°36.3	54.0	114°29.5	42.5				
9	129°34.9	342°05.5	•• 56.6	286°03.1	•• 43.4	17°38.4	•• 53.9	129°32.1	•• 42.6				
10	144°37.4	357°05.0	55.7	301°04.0	44.0	32°40.4	53.9	144°34.8	42.7				
11	159°39.8	12°04.4	54.8	316°04.9	44.6	47°42.4	53.8	159°37.4	42.7				
12	174°42.3	27°03.9	N13°53.8	331°05.8	S09°45.3	62°44.5	N21°53.8	174°40.1	S02°42.8				
13	189°44.7	42°03.4	52.9	346°06.7	45.9	77°46.5	53.7	189°42.7	42.9				
14	204°47.2	57°02.9	52.0	1°07.6	46.5	92°48.6	53.6	204°45.3	43.0				
15	219°49.7	72°02.3	•• 51.0	16°08.5	•• 47.2	107°50.7	•• 53.6	219°48.0	•• 43.1				
16	234°52.1	87°01.8	50.1	31°09.4	47.8	122°52.7	53.5	234°50.6	43.1				
17	249°54.6	102°01.3	49.2	46°10.3	48.4	137°54.8	53.5	249°53.2	43.2				
18	264°57.1	117°00.8	N13°48.3	61°11.2	S09°49.1	152°56.8	N21°53.4	264°55.9	S02°43.3				
19	279°59.5	132°00.3	47.3	76°12.1	49.7	167°58.9	53.4	279°58.5	43.4				
20	295°02.0	146°59.7	46.4	91°13.0	50.3	183°00.9	53.3	295°01.2	43.5				
21	310°04.5	161°59.2	•• 45.5	106°13.9	•• 50.9	198°03.0	•• 53.3	310°03.8	•• 43.5				
22	325°06.9	176°58.7	44.5	121°14.8	51.6	213°05.0	53.2	325°06.4	43.6				
23	340°09.4	191°58.2	43.6	136°15.7	52.2	228°07.1	53.2	340°09.1	43.7				
Mer.pass. 00:23		ν -0.5' d -0.9' m -3.93			ν 0.9' d 0.6' m 1.61			ν 2.0' d -0.1' m -2.05			ν 2.6' d 0.1' m 0.60		

Sep 13 Sat	SHA	Mer.pass
Venus	215°21.4	10:10
Mars	157°56.7	13:58
Jupiter	248°27.5	07:56
Saturn	359°47.3	00:32

Sep 14 Sun	SHA	Mer.pass
Venus	214°09.3	10:11
Mars	157°19.5	13:57
Jupiter	248°17.3	07:53
Saturn	359°51.5	00:28

Sep 15 Mon	SHA	Mer.pass
Venus	212°57.5	10:12
Mars	156°42.2	13:55
Jupiter	248°07.2	07:50
Saturn	359°55.7	00:23

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

h Sun Moon						
Sat	GHA	Dec	GHA	ν	Dec	d HP
0	180°59.8	N03°46.6	294°05.0	5.0'	N25°23.4	8.4' 59.5'
1	196°00.0	45.6	308°28.9	4.9'	25°31.8	8.2' 59.5'
2	211°00.2	44.6	322°52.8	4.8'	25°40.0	8.0' 59.5'
3	226°00.5	· · 43.7	337°16.6	4.8'	25°48.0	7.9' 59.4'
4	241°00.7	42.7	351°40.4	4.7'	25°55.9	7.7' 59.4'
5	256°00.9	41.8	6°04.1	4.6'	26°03.6	7.5' 59.4'
6	271°01.1	N03°40.8	20°27.7	4.6'	N26°11.1	7.3' 59.4'
7	286°01.3	39.9	34°51.3	4.5'	26°18.4	7.2' 59.4'
8	301°01.6	38.9	49°14.8	4.5'	26°25.6	7.0' 59.4'
9	316°01.8	· · 37.9	63°38.3	4.4'	26°32.5	6.8' 59.3'
10	331°02.0	37.0	78°01.7	4.4'	26°39.3	6.6' 59.3'
11	346°02.2	36.0	92°25.1	4.3'	26°45.9	6.4' 59.3'
12	1°02.5	N03°35.1	106°48.4	4.3'	N26°52.4	6.2' 59.3'
13	16°02.7	34.1	121°11.7	4.2'	26°58.6	6.1' 59.3'
14	31°02.9	33.2	135°34.9	4.2'	27°04.7	5.9' 59.3'
15	46°03.1	· · 32.2	149°58.1	4.1'	27°10.5	5.7' 59.2'
16	61°03.3	31.2	164°21.2	4.1'	27°16.2	5.5' 59.2'
17	76°03.6	30.3	178°44.3	4.1'	27°21.7	5.3' 59.2'
18	91°03.8	N03°29.3	193°07.4	4.0'	N27°27.0	5.1' 59.2'
19	106°04.0	28.4	207°30.4	4.0'	27°32.1	4.9' 59.2'
20	121°04.2	27.4	221°53.4	4.0'	27°37.1	4.7' 59.1'
21	136°04.5	· · 26.4	236°16.3	3.9'	27°41.8	4.6' 59.1'
22	151°04.7	25.5	250°39.2	3.9'	27°46.4	4.4' 59.1'
23	166°04.9	24.5	265°02.1	3.9'	27°50.7	4.2' 59.1'
SD = 15.9' d = -1.0'			SD = 16.2'			

Sun	GHA	Dec	GHA	ν	Dec	d HP
0	181°05.1	N03°23.6	279°25.0	3.9'	N27°54.9	4.0' 59.1'
1	196°05.4	22.6	293°47.9	3.8'	27°58.9	3.8' 59.0'
2	211°05.6	21.6	308°10.7	3.8'	28°02.6	3.6' 59.0'
3	226°05.8	· · 20.7	322°33.5	3.8'	28°06.2	3.4' 59.0'
4	241°06.0	19.7	336°56.3	3.8'	28°09.6	3.2' 59.0'
5	256°06.2	18.8	351°19.1	3.8'	28°12.8	3.0' 59.0'
6	271°06.5	N03°17.8	5°41.9	3.8'	N28°15.9	2.8' 58.9'
7	286°06.7	16.9	20°04.6	3.8'	28°18.7	2.6' 58.9'
8	301°06.9	15.9	34°27.4	3.8'	28°21.3	2.4' 58.9'
9	316°07.1	· · 14.9	48°50.1	3.8'	28°23.7	2.2' 58.9'
10	331°07.4	14.0	63°12.9	3.8'	28°26.0	2.0' 58.9'
11	346°07.6	13.0	77°35.6	3.8'	28°28.0	1.9' 58.8'
12	1°07.8	N03°12.0	91°58.4	3.8'	N28°29.9	1.7' 58.8'
13	16°08.0	11.1	106°21.2	3.8'	28°31.6	1.5' 58.8'
14	31°08.2	10.1	120°43.9	3.8'	28°33.0	1.3' 58.8'
15	46°08.5	· · 09.2	135°06.7	3.8'	28°34.3	1.1' 58.8'
16	61°08.7	08.2	149°29.5	3.8'	28°35.4	0.9' 58.7'
17	76°08.9	07.2	163°52.3	3.8'	28°36.3	0.7' 58.7'
18	91°09.1	N03°06.3	178°15.1	3.8'	N28°37.0	0.5' 58.7'
19	106°09.4	05.3	192°38.0	3.9'	28°37.5	0.3' 58.7'
20	121°09.6	04.4	207°00.9	3.9'	28°37.8	0.1' 58.7'
21	136°09.8	· · 03.4	221°23.7	3.9'	28°37.9	-0.1' 58.6'
22	151°10.0	02.4	235°46.7	4.0'	28°37.8	-0.3' 58.6'
23	166°10.3	01.5	250°09.6	4.0'	28°37.6	-0.5' 58.6'
SD = 15.9' d = -1.0'			SD = 16.1'			

Mon	GHA	Dec	GHA	ν	Dec	d HP
0	181°10.5	N03°00.5	264°32.6	4.0'	N28°37.1	-0.6' 58.6'
1	196°10.7	02°59.6	278°55.6	4.1'	28°36.5	-0.8' 58.5'
2	211°10.9	58.6	293°18.7	4.1'	28°35.6	-1.0' 58.5'
3	226°11.1	· · 57.6	307°41.8	4.1'	28°34.6	-1.2' 58.5'
4	241°11.4	56.7	322°04.9	4.2'	28°33.4	-1.4' 58.5'
5	256°11.6	55.7	336°28.1	4.2'	28°32.0	-1.6' 58.5'
6	271°11.8	N02°54.7	350°51.3	4.3'	N28°30.4	-1.8' 58.4'
7	286°12.0	53.8	5°14.6	4.3'	28°28.6	-2.0' 58.4'
8	301°12.3	52.8	19°38.0	4.4'	28°26.7	-2.1' 58.4'
9	316°12.5	· · 51.9	34°01.3	4.4'	28°24.5	-2.3' 58.4'
10	331°12.7	50.9	48°24.8	4.5'	28°22.2	-2.5' 58.4'
11	346°12.9	49.9	62°48.3	4.6'	28°19.7	-2.7' 58.3'
12	1°13.2	N02°49.0	77°11.8	4.6'	N28°17.0	-2.9' 58.3'
13	16°13.4	48.0	91°35.5	4.7'	28°14.1	-3.1' 58.3'
14	31°13.6	47.0	105°59.1	4.7'	28°11.1	-3.2' 58.3'
15	46°13.8	· · 46.1	120°22.9	4.8'	28°07.8	-3.4' 58.2'
16	61°14.0	45.1	134°46.7	4.9'	28°04.4	-3.6' 58.2'
17	76°14.3	44.1	149°10.6	5.0'	28°00.8	-3.8' 58.2'
18	91°14.5	N02°43.2	163°34.5	5.0'	N27°57.1	-3.9' 58.2'
19	106°14.7	42.2	177°58.6	5.1'	27°53.1	-4.1' 58.2'
20	121°14.9	41.3	192°22.7	5.2'	27°49.0	-4.3' 58.1'
21	136°15.2	· · 40.3	206°46.8	5.3'	27°44.7	-4.5' 58.1'
22	151°15.4	39.3	221°11.1	5.3'	27°40.3	-4.6' 58.1'
23	166°15.6	38.4	235°35.4	5.4'	27°35.6	-4.8' 58.1'
SD = 15.9' d = -1.0'			SD = 16.0'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	02:01	03:51	05:04	18:45	19:56	21:43
N 70°	02:34	04:05	05:09	18:39	19:43	21:12
68°	02:58	04:15	05:14	18:35	19:33	20:49
66°	03:16	04:24	05:17	18:32	19:24	20:32
64°	03:30	04:32	05:21	18:29	19:17	20:18
62°	03:41	04:38	05:23	18:26	19:11	20:07
60°	03:51	04:43	05:26	18:24	19:06	19:58
N 58°	03:59	04:48	05:28	18:22	19:01	19:50
56°	04:06	04:52	05:30	18:20	18:57	19:43
54°	04:13	04:56	05:32	18:18	18:54	19:37
52°	04:18	04:59	05:33	18:17	18:51	19:31
50°	04:23	05:02	05:35	18:15	18:48	19:27
45°	04:33	05:08	05:38	18:13	18:42	19:17
N 40°	04:41	05:13	05:40	18:10	18:37	19:09
35°	04:47	05:17	05:42	18:08	18:34	19:03
30°	04:52	05:20	05:44	18:06	18:30	18:58
20°	04:59	05:25	05:47	18:03	18:26	18:51
N 10°	05:04	05:29	05:50	18:01	18:22	18:46
0°	05:08	05:32	05:52	17:59	18:19	18:43
S 10°	05:09	05:34	05:55	17:57	18:18	18:42
20°	05:09	05:35	05:57	17:54	18:16	18:42
30°	05:08	05:35	05:59	17:52	18:16	18:44
35°	05:06	05:36	06:01	17:51	18:16	18:45
40°	05:04	05:35	06:02	17:49	18:16	18:48
45°	05:01	05:35	06:04	17:48	18:17	18:51
S 50°	04:57	05:34	06:06	17:46	18:18	18:56
52°	04:54	05:34	06:07	17:45	18:18	18:58
54°	04:52	05:33	06:08	17:44	18:19	19:00
56°	04:49	05:32	06:09	17:43	18:20	19:03
58°	04:46	05:31	06:10	17:42	18:21	19:07
S 60°	04:42	05:30	06:12	17:40	18:22	19:11

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°						
N 70°						
68°						
66°						
64°						
62°	18:03		19:47	16:00		18:36
60°	18:55	19:31	20:53	15:09	16:43	17:29
N 58°	19:27	20:11	21:27	14:37	16:02	16:54
56°	19:51	20:39	21:52	14:14	15:34	16:29
54°	20:10	21:01	22:13	13:55	15:13	16:09
52°	20:27	21:19	22:29	13:39	14:55	15:51
50°	20:41	21:35	22:44	13:25	14:39	15:37
45°	21:10	22:06	23:13	12:56	14:08	15:07
N 40°	21:32	22:30	23:36	12:34	13:44	14:44
35°	21:51	22:50	23:55	12:16	13:24	14:24
30°	22:08	23:07	· · ·	12:00	13:07	14:08
20°	22:35	23:36	· · ·	11:33	12:39	13:40
N 10°	22:59	· · ·	00:00	11:10	12:14	13:15
0°	23:21	· · ·	00:23	10:49	11:51	12:53
S 10°	23:44	· · ·	00:46	10:28	11:28	12:30
20°	· · ·	00:08	01:11	10:05	11:04	12:06
30°	· · ·	00:36	01:40	09:39	10:35	11:37
35°	· · ·	00:53	01:57	09:23	10:18	11:20
40°	· · ·	01:12	02:17	09:06	09:58	11:00
45°	00:19	01:36	02:42	08:44	09:34	10:36
S 50°	00:45	02:07	03:13	08:17	09:03	10:05
52°	00:58	02:22	03:29	08:04	08:48	09:50
54°	01:13	02:40	03:47	07:49	08:30	09:31
56°	01:30	03:01	04:09	07:31	08:08	09:09
58°	01:51	03:29	04:38	07:09	07:40	08:41
S 60°	02:18	04:08	05:20	06:42	07:00	07:59

Day	Sun		Mer. Pass hh:mm	Moon		Age 21-23 66-44%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
13	03:59	04:10	11:56	04:35	17:05	
14	04:21	04:31	11:55	05:36	18:07	
15	04:42	04:53	11:55	06:38	19:08	

September 16, 17, 18 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	355° 11.9	206° 57.7	N13° 42.6	151° 16.6	S09° 52.8	243° 09.1	N21° 53.1	355° 11.7	S02° 43.8	Alpheratz	357° 33.8	29° 14.1
1	10° 14.3	221° 57.1	41.7	166° 17.5	53.5	258° 11.2	53.1	10° 14.4	43.9	Ankaa	353° 06.1	-42° 09.8
2	25° 16.8	236° 56.6	40.8	181° 18.4	54.1	273° 13.2	53.0	25° 17.0	43.9	Schedar	349° 29.9	56° 40.7
3	40° 19.2	251° 56.1	· · 39.8	196° 19.3	· · 54.7	288° 15.3	· · 53.0	40° 19.6	· · 44.0	Diphda	348° 46.4	-17° 50.5
4	55° 21.7	266° 55.6	38.9	211° 20.2	55.4	303° 17.3	52.9	55° 22.3	44.1	Achernar	335° 19.2	-57° 06.1
5	70° 24.2	281° 55.1	38.0	226° 21.2	56.0	318° 19.4	52.9	70° 24.9	44.2	Hamal	327° 50.3	23° 35.1
6	85° 26.6	296° 54.5	N13° 37.0	241° 22.1	S09° 56.6	333° 21.4	N21° 52.8	85° 27.6	S02° 44.3	Polaris	313° 27.7	89° 22.1
7	100° 29.1	311° 54.0	36.1	256° 23.0	57.2	348° 23.5	52.7	100° 30.2	44.3	Acamar	315° 11.1	-40° 11.8
8	115° 31.6	326° 53.5	35.1	271° 23.9	57.9	3° 25.5	52.7	115° 32.8	44.4	Menkar	314° 05.4	4° 11.6
9	130° 34.0	341° 53.0	· · 34.2	286° 24.8	· · 58.5	18° 27.6	· · 52.6	130° 35.5	· · 44.5	Mirfak	308° 27.2	49° 57.1
10	145° 36.5	356° 52.5	33.2	301° 25.7	59.1	33° 29.7	52.6	145° 38.1	44.6	Aldebaran	290° 38.9	16° 33.7
11	160° 39.0	11° 52.0	32.3	316° 26.6	09° 59.8	48° 31.7	52.5	160° 40.8	44.6	Rigel	281° 03.3	-8° 10.1
12	175° 41.4	26° 51.5	N13° 31.4	331° 27.5	S10° 00.4	63° 33.8	N21° 52.5	175° 43.4	S02° 44.7	Capella	280° 21.0	46° 01.3
13	190° 43.9	41° 50.9	30.4	346° 28.4	01.0	78° 35.8	52.4	190° 46.0	44.8	Bellatrix	278° 22.2	6° 22.5
14	205° 46.4	56° 50.4	29.5	1° 29.3	01.6	93° 37.9	52.4	205° 48.7	44.9	Elnath	278° 01.1	28° 37.7
15	220° 48.8	71° 49.9	· · 28.5	16° 30.2	· · 02.3	108° 39.9	· · 52.3	220° 51.3	· · 45.0	Alnilam	275° 37.1	-1° 11.0
16	235° 51.3	86° 49.4	27.6	31° 31.0	02.9	123° 42.0	52.3	235° 54.0	45.0	Betelgeuse	270° 51.5	7° 24.8
17	250° 53.7	101° 48.9	26.6	46° 31.9	03.5	138° 44.0	52.2	250° 56.6	45.1	Canopus	263° 52.3	-52° 42.2
18	265° 56.2	116° 48.4	N13° 25.7	61° 32.8	S10° 04.2	153° 46.1	N21° 52.2	265° 59.2	S02° 45.2	Sirius	258° 25.8	-16° 44.8
19	280° 58.7	131° 47.9	24.7	76° 33.7	04.8	168° 48.2	52.1	281° 01.9	45.3	Adhara	255° 05.5	-29° 00.1
20	296° 01.1	146° 47.3	23.8	91° 34.6	05.4	183° 50.2	52.1	296° 04.5	45.4	Procyon	244° 50.4	5° 09.7
21	311° 03.6	161° 46.8	· · 22.8	106° 35.5	· · 06.0	198° 52.3	· · 52.0	311° 07.2	· · 45.4	Pollux	243° 16.8	27° 57.8
22	326° 06.1	176° 46.3	21.9	121° 36.4	06.7	213° 54.3	51.9	326° 09.8	45.5	Avior	234° 14.9	-59° 35.2
23	341° 08.5	191° 45.8	20.9	136° 37.3	07.3	228° 56.4	51.9	341° 12.4	45.6	Suhail	222° 46.3	-43° 31.9
Mer.pass. 00:19		ν -0.5' d -0.9' m -3.93		ν 0.9' d 0.6' m 1.60		ν 2.1' d -0.1' m -2.05		ν 2.6' d 0.1' m 0.60		Miaplacidus	221° 39.0	-69° 49.1
										Alphard	217° 47.5	-8° 46.0
										Regulus	207° 34.1	11° 50.6
										Dubhe	193° 40.9	61° 36.7
										Denebola	182° 24.7	14° 25.8
										Gienah	175° 43.3	-17° 41.0
										Acrux	173° 00.2	-63° 14.5
										Gacrux	171° 51.6	-57° 15.4
										Alioth	166° 13.0	55° 49.3
										Spica	158° 22.0	-11° 17.7
										Alkaid	152° 51.9	49° 11.3
										Hadar	148° 35.7	-60° 29.9
										Menkent	147° 57.3	-36° 29.8
										Arcturus	145° 47.6	19° 03.1
										Rigil Kent.	139° 40.0	-60° 56.6
										Kochab	137° 20.5	74° 03.2
										Zuben'ubi	136° 55.6	-16° 08.9
										Alphecca	126° 03.4	26° 37.9
										Antares	112° 15.2	-26° 29.4
										Atria	107° 09.0	-69° 04.7
										Sabik	102° 02.2	-15° 45.4
										Shaula	96° 09.6	-37° 07.4
										Rasalhague	95° 58.0	12° 32.6
										Eltanin	90° 41.9	51° 29.4
										Kaus Aust.	83° 31.7	-34° 22.4
										Vega	80° 32.7	38° 48.7
										Nunki	75° 46.9	-26° 15.9
										Altair	61° 59.2	8° 56.3
										Peacock	53° 04.4	-56° 39.2
										Deneb	49° 25.1	45° 22.5
										Enif	33° 38.0	9° 59.7
										Al Na'ir	27° 31.7	-46° 50.2
										Fomalhaut	15° 13.5	-29° 29.1
										Scheat	13° 44.3	28° 13.5
										Markab	13° 29.0	15° 20.8
										Sep 16 Tue	SHA	Mer.pass
										Venus	211° 45.8	10:13
										Mars	156° 04.8	13:54
										Jupiter	247° 57.3	07:46
										Saturn	359° 59.9	00:19
										Sep 17 Wed	SHA	Mer.pass
										Venus	210° 34.3	10:13
										Mars	155° 27.2	13:53
										Jupiter	247° 47.4	07:43
										Saturn	0° 04.1	00:15
										Sep 18 Thu	SHA	Mer.pass
										Venus	209° 23.0	10:14
										Mars	154° 49.6	13:51
										Jupiter	247° 37.8	07:40
										Saturn	0° 08.3	00:11
										Horizontal parallax		
										Venus:	0.1	
										Mars:	0.1	
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	356° 11.0	206° 45.3	N13° 20.0	151° 38.2	S10° 07.9	243° 58.4	N21° 51.8	356° 15.1	S02° 45.7			
1	11° 13.5	221° 44.8	19.0	166° 39.1	08.6	259° 00.5	51.8	11° 17.7	45.8			
2	26° 15.9	236° 44.3	18.1	181° 40.0	09.2	274° 02.6	51.7	26° 20.4	45.8			
3	41° 18.4	251° 43.8	· · 17.1	196° 40.9	· · 09.8	289° 04.6	· · 51.7	41° 23.0	· · 45.9			
4	56° 20.9	266° 43.3	16.2	211° 41.8	10.4	304° 06.7	51.6	56° 25.6	46.0			
5	71° 23.3	281° 42.7	15.2	226° 42.7	11.1	319° 08.7	51.6	71° 28.3	46.1			
6	86° 25.8	296° 42.2	N13° 14.2	241° 43.6	S10° 11.7	334° 10.8	N21° 51.5	86° 30.9	S02° 46.2			
7	101° 28.2	311° 41.7	13.3	256° 44.5	12.3	349° 12.8	51.5	101° 33.6	46.2			
8	116° 30.7	326° 41.2	12.3	271° 45.4	13.0	4° 14.9	51.4	116° 36.2	46.3			
9	131° 33.2	341° 40.7	· · 11.4	286° 46.3	· · 13.6	19° 17.0	· · 51.4	131° 38.9	· · 46.4			
10	146° 35.6	356° 40.2	10.4	301° 47.2	14.2	34° 19.0	51.3	146° 41.5	46.5			
11	161° 38.1	11° 39.7	09.4	316° 48.1	14.8	49° 21.1	51.3	161° 44.1	46.5			
12	176° 40.6	26° 39.2	N13° 08.5	331° 49.0	S10° 15.5	64° 23.1	N21° 51.2	176° 46.8	S02° 46.6			
13	191° 43.0	41° 38.7	07.5	346° 49.9	16.1	79° 25.2	51.2	191° 49.4	46.7			
14	206° 45.5	56° 38.2	06.6	1° 50.8	16.7	94° 27.3	51.1	206° 52.1	46.8			
15	221° 48.0	71° 37.7	· · 05.6	16° 51.7	· · 17.3	109° 29.3	· · 51.1	221° 54.7	· · 46.9			
16	236° 50.4	86° 37.2	04.6	31° 52.6	18.0	124° 31.4	51.0	236° 57.3	46.9			
17	251° 52.9	101° 36.6	03.7	46° 53.5	18.6	139° 33.5	50.9	252° 00.0	47.0			
18	266° 55.3	116° 36.1	N13° 02.7	61° 54.3	S10° 19.2	154° 35.5	N21° 50.9	267° 02.6	S02° 47.1			
19	281° 57.8	131° 35.6	01.7	76° 55.2	19.8	169° 37.6	50.8	282° 05.3	47.2			
20	297° 00.3	146° 35.1	13° 00.8	91° 56.1	20.5	184° 39.6	50.8	297° 07.9	47.3			
21	312° 02.7	161° 34.6	12° 59.8	106° 57.0	· · 21.1	199° 41.7	· · 50.7	312° 10.5	· · 47.3			
22	327° 05.2	176° 34.1	58.8	121° 57.9	21.7	214° 43.8	50.7	327° 13.2	47.4			
23	342° 07.7	191° 33.6	57.9	136° 58.8	22.4	229° 45.8	50.6	342° 15.8	47.5			
Mer.pass. 00:15		ν -0.5' d -1.0' m -3.93		ν 0.9' d 0.6' m 1.60		ν 2.1' d -0.1' m -2.05		ν 2.6' d 0.1' m 0.59				
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	357° 10.1	206° 33.1	N12° 56.9	151° 59.7	S10° 23.0	244° 47.9	N21° 50.6	357° 18.5	S02° 47.6			
1	12° 12.6	221° 32.6	55.9	167° 00.6	23.6	259° 50.0	50.5	12° 21.1	47.7			
2	27° 15.1	236° 32.1	55.0	182° 01.5	24.2	274° 52.0	50.5	27° 23.7	47.7			
3	42° 17.5	251° 31.6	· · 54.0	197° 02.4	· · 24.9	289° 54.1	· · 50.4	42° 26.4	· · 47.8			
4	57° 20.0	266° 31.1	53.0	212° 03.3	25.5	304° 56.1	50.4	57° 29.0	47.9			
5	72° 22.5	281° 30.6	52.1	227° 04.2	26.1	319° 58.2	50.3	72° 31.7	48.0			
6	87° 24.9	296° 30.1	N12° 51.1	242° 05.1	S10° 26.7	335° 00.3	N21° 50.3	87° 34.3	S02° 48.1			
7	102° 27.4	311° 29.6	50.1	257° 05.9	27.4	350° 02.3	50.2	102° 36.9	48.1			
8	117° 29.8	326° 29.1	49.1	272° 06.8	28.0	5° 04.4	50.2	117° 39.6	48.2			
9	132° 32.3	341° 28.6	· · 48.2	287° 07.7	· · 28.6	20° 06.5	· · 50.1	132° 42.2	· · 48.3			
10	147° 34.8	356° 28.1	47.2	302° 08.6	29.2	35° 08.5	50.1</					

Sun				Moon					
Tue	GHA	Dec		GHA	ν	Dec	d	HP	
0	181°15.8	N02°37.4		249°59.9	5.5'	N27°30.8	-5.0'	58.0'	
1	196°16.1	36.4		264°24.4	5.6'	27°25.9	-5.1'	58.0'	
2	211°16.3	35.5		278°49.0	5.7'	27°20.7	-5.3'	58.0'	
3	226°16.5	· · 34.5		293°13.6	5.8'	27°15.4	-5.5'	58.0'	
4	241°16.7	33.5		307°38.4	5.9'	27°09.9	-5.6'	58.0'	
5	256°16.9	32.6		322°03.2	5.9'	27°04.3	-5.8'	57.9'	
6	271°17.2	N02°31.6		336°28.2	6.0'	N26°58.5	-6.0'	57.9'	
7	286°17.4	30.7		350°53.2	6.1'	26°52.6	-6.1'	57.9'	
8	301°17.6	29.7		5°18.4	6.2'	26°46.4	-6.3'	57.9'	
9	316°17.8	· · 28.7		19°43.6	6.3'	26°40.2	-6.4'	57.8'	
10	331°18.1	27.8		34°08.9	6.4'	26°33.7	-6.6'	57.8'	
11	346°18.3	26.8		48°34.3	6.5'	26°27.2	-6.7'	57.8'	
12	1°18.5	N02°25.8		62°59.8	6.6'	N26°20.4	-6.9'	57.8'	
13	16°18.7	24.9		77°25.4	6.7'	26°13.5	-7.0'	57.8'	
14	31°19.0	23.9		91°51.1	6.8'	26°06.5	-7.2'	57.7'	
15	46°19.2	· · 22.9		106°16.9	6.9'	25°59.3	-7.3'	57.7'	
16	61°19.4	22.0		120°42.8	7.0'	25°51.9	-7.5'	57.7'	
17	76°19.6	21.0		135°08.9	7.1'	25°44.4	-7.6'	57.7'	
18	91°19.8	N02°20.0		149°35.0	7.2'	N25°36.8	-7.8'	57.7'	
19	106°20.1	19.1		164°01.2	7.3'	25°29.0	-7.9'	57.6'	
20	121°20.3	18.1		178°27.5	7.4'	25°21.1	-8.1'	57.6'	
21	136°20.5	· · 17.1		192°53.9	7.5'	25°13.0	-8.2'	57.6'	
22	151°20.7	16.2		207°20.5	7.6'	25°04.8	-8.3'	57.6'	
23	166°21.0	15.2		221°47.1	7.7'	24°56.5	-8.5'	57.5'	
SD = 15.9'				$d = -1.0'$				SD = 15.8'	
Wed	GHA	Dec		GHA	ν	Dec	d	HP	
0	181°21.2	N02°14.2		236°13.8	7.8'	N24°48.0	-8.6'	57.5'	
1	196°21.4	13.3		250°40.7	8.0'	24°39.4	-8.7'	57.5'	
2	211°21.6	12.3		265°07.6	8.1'	24°30.7	-8.9'	57.5'	
3	226°21.9	· · 11.3		279°34.7	8.2'	24°21.8	-9.0'	57.5'	
4	241°22.1	10.4		294°01.9	8.3'	24°12.8	-9.1'	57.4'	
5	256°22.3	09.4		308°29.2	8.4'	24°03.6	-9.3'	57.4'	
6	271°22.5	N02°08.4		322°56.5	8.5'	N23°54.4	-9.4'	57.4'	
7	286°22.7	07.5		337°24.0	8.6'	23°45.0	-9.5'	57.4'	
8	301°23.0	06.5		351°51.7	8.7'	23°35.5	-9.6'	57.3'	
9	316°23.2	· · 05.5		6°19.4	8.8'	23°25.8	-9.8'	57.3'	
10	331°23.4	04.6		20°47.2	8.9'	23°16.1	-9.9'	57.3'	
11	346°23.6	03.6		35°15.1	9.0'	23°06.2	-10.0'	57.3'	
12	1°23.9	N02°02.6		49°43.2	9.1'	N22°56.2	-10.1'	57.3'	
13	16°24.1	01.7		64°11.3	9.3'	22°46.1	-10.2'	57.2'	
14	31°24.3	02°00.7		78°39.6	9.4'	22°35.9	-10.3'	57.2'	
15	46°24.5	01°59.7		93°07.9	9.5'	22°25.5	-10.5'	57.2'	
16	61°24.8	58.8		107°36.4	9.6'	22°15.1	-10.6'	57.2'	
17	76°25.0	57.8		122°05.0	9.7'	22°04.5	-10.7'	57.1'	
18	91°25.2	N01°56.8		136°33.7	9.8'	N21°53.8	-10.8'	57.1'	
19	106°25.4	55.9		151°02.5	9.9'	21°43.1	-10.9'	57.1'	
20	121°25.6	54.9		165°31.4	10.0'	21°32.2	-11.0'	57.1'	
21	136°25.9	· · 53.9		180°00.4	10.1'	21°21.2	-11.1'	57.1'	
22	151°26.1	53.0		194°29.5	10.2'	21°10.1	-11.2'	57.0'	
23	166°26.3	52.0		208°58.7	10.3'	20°58.9	-11.3'	57.0'	
SD = 15.9'				$d = -1.0'$				SD = 15.7'	
Thu	GHA	Dec		GHA	ν	Dec	d	HP	
0	181°26.5	N01°51.0		223°28.0	10.4'	N20°47.6	-11.4'	57.0'	
1	196°26.8	50.1		237°57.5	10.5'	20°36.2	-11.5'	57.0'	
2	211°27.0	49.1		252°27.0	10.6'	20°24.7	-11.6'	57.0'	
3	226°27.2	· · 48.1		266°56.7	10.7'	20°13.2	-11.7'	56.9'	
4	241°27.4	47.2		281°26.4	10.9'	20°01.5	-11.8'	56.9'	
5	256°27.7	46.2		295°56.3	11.0'	19°49.7	-11.9'	56.9'	
6	271°27.9	N01°45.2		310°26.2	11.1'	N19°37.9	-11.9'	56.9'	
7	286°28.1	44.2		324°56.3	11.2'	19°25.9	-12.0'	56.8'	
8	301°28.3	43.3		339°26.4	11.3'	19°13.9	-12.1'	56.8'	
9	316°28.5	· · 42.3		353°56.7	11.4'	19°01.8	-12.2'	56.8'	
10	331°28.8	41.3		8°27.0	11.5'	18°49.6	-12.3'	56.8'	
11	346°29.0	40.4		22°57.5	11.6'	18°37.3	-12.4'	56.8'	
12	1°29.2	N01°39.4		37°28.1	11.7'	N18°24.9	-12.4'	56.7'	
13	16°29.4	38.4		51°58.7	11.8'	18°12.5	-12.5'	56.7'	
14	31°29.7	37.5		66°29.5	11.8'	18°00.0	-12.6'	56.7'	
15	46°29.9	· · 36.5		81°00.3	11.9'	17°47.4	-12.7'	56.7'	
16	61°30.1	35.5		95°31.2	12.0'	17°34.7	-12.8'	56.7'	
17	76°30.3	34.6		110°02.3	12.1'	17°21.9	-12.8'	56.6'	
18	91°30.5	N01°33.6		124°33.4	12.2'	N17°09.1	-12.9'	56.6'	
19	106°30.8	32.6		139°04.6	12.3'	16°56.2	-13.0'	56.6'	
20	121°31.0	31.7		153°36.0	12.4'	16°43.2	-13.0'	56.6'	
21	136°31.2	· · 30.7		168°07.4	12.5'	16°30.2	-13.1'	56.5'	
22	151°31.4	29.7		182°38.9	12.6'	16°17.1	-13.2'	56.5'	
23	166°31.7	28.7		197°10.5	12.7'	16°03.9	-13.2'	56.5'	
SD = 15.9'				$d = -1.0'$				SD = 15.5'	

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	02:25	04:06	05:17	18:29	19:39	21:17
N 70°	02:53	04:18	05:21	18:25	19:28	20:52
68°	03:12	04:27	05:24	18:22	19:19	20:33
66°	03:28	04:35	05:27	18:20	19:12	20:18
64°	03:40	04:41	05:29	18:18	19:06	20:06
62°	03:51	04:46	05:31	18:16	19:01	19:56
60°	03:59	04:51	05:33	18:14	18:56	19:47
N 58°	04:07	04:55	05:34	18:13	18:53	19:40
56°	04:13	04:58	05:36	18:12	18:49	19:34
54°	04:19	05:01	05:37	18:11	18:46	19:29
52°	04:24	05:04	05:38	18:10	18:44	19:24
50°	04:28	05:07	05:39	18:09	18:41	19:20
45°	04:37	05:12	05:41	18:07	18:36	19:11
N 40°	04:44	05:16	05:43	18:05	18:32	19:04
35°	04:49	05:19	05:44	18:04	18:29	18:59
30°	04:54	05:22	05:46	18:03	18:27	18:55
20°	05:00	05:26	05:48	18:01	18:23	18:48
N 10°	05:04	05:29	05:50	17:59	18:20	18:44
0°	05:06	05:30	05:51	17:58	18:18	18:42
S 10°	05:07	05:32	05:53	17:56	18:17	18:42
20°	05:07	05:32	05:54	17:55	18:17	18:43
30°	05:04	05:32	05:56	17:54	18:18	18:45
35°	05:02	05:31	05:56	17:53	18:18	18:48
40°	04:59	05:30	05:57	17:52	18:19	18:51
45°	04:55	05:29	05:58	17:51	18:20	18:55
S 50°	04:50	05:27	06:00	17:50	18:22	19:00
52°	04:47	05:27	06:00	17:50	18:23	19:03
54°	04:44	05:26	06:01	17:49	18:24	19:06
56°	04:41	05:24	06:01	17:49	18:26	19:09
58°	04:37	05:23	06:02	17:48	18:27	19:13
S 60°	04:32	05:21	06:03	17:47	18:29	19:18

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	☐	☐	23:52	☐	☐	20:11
N 70°	☐	☐	☐	☐	☐	19:27
68°	☐	22:15	· · ·	☐	☐	18:57
66°	☐	23:10	· · ·	☐	☐	19:07
64°	21:15	23:42	· · ·	19:09	18:33	18:17
62°	22:07	· · ·	00:06	18:16	18:08	18:02
60°	22:39	· · ·	00:25	17:44	17:49	17:50
N 58°	23:02	· · ·	00:40	17:20	17:32	17:39
56°	23:21	· · ·	00:53	17:00	17:18	17:30
54°	23:37	· · ·	01:05	16:44	17:06	17:21
52°	23:51	· · ·	01:15	16:30	16:56	17:13
50°	· · ·	00:03	01:24	16:18	16:46	17:07
45°	· · ·	00:27	01:43	15:52	16:26	16:52
N 40°	· · ·	00:47	01:58	15:32	16:09	16:40
35°	· · ·	01:03	02:11	15:14	15:55	16:29
30°	00:11	01:17	02:22	15:00	15:43	16:20
20°	00:39	01:41	02:41	14:34	15:22	16:04
N 10°	01:02	02:02	02:58	14:12	15:04	15:50
0°	01:24	02:21	03:13	13:52	14:46	15:37
S 10°	01:46	02:40	03:29	13:31	14	

September 19, 20, 21 UT (Fri., Sat., Sun.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	358°09.3	206°21.1	N12°33.5	152°21.1	S10°38.0	245°37.5	N21°49.3	358°21.8	S02°49.5					Alpheratz	357°33.8	29°14.1	
1	13°11.7	221°20.6	32.5	167°21.9	38.6	260°39.5	49.3	13°24.5	49.6					Ankaa	353°06.1	-42°09.8	
2	28°14.2	236°20.1	31.5	182°22.8	39.2	275°41.6	49.2	28°27.1	49.6					Schedar	349°29.9	56°40.7	
3	43°16.7	251°19.6	· · 30.5	197°23.7	· · 39.8	290°43.7	· · 49.2	43°29.8	· · 49.7					Diphda	348°46.4	-17°50.5	
4	58°19.1	266°19.1	29.5	212°24.6	40.5	305°45.7	49.1	58°32.4	49.8					Achernar	335°19.2	-57°06.1	
5	73°21.6	281°18.6	28.5	227°25.5	41.1	320°47.8	49.1	73°35.1	49.9					Hamal	327°50.3	23°35.1	
6	88°24.1	296°18.1	N12°27.6	242°26.4	S10°41.7	335°49.9	N21°49.0	88°37.7	S02°49.9					Polaris	313°26.5	89°22.1	
7	103°26.5	311°17.6	26.6	257°27.3	42.3	350°52.0	49.0	103°40.3	50.0					Acamar	315°11.0	-40°11.8	
8	118°29.0	326°17.1	25.6	272°28.1	43.0	5°54.0	48.9	118°43.0	50.1					Menkar	314°05.3	4°11.6	
9	133°31.4	341°16.6	· · 24.6	287°29.0	· · 43.6	20°56.1	· · 48.9	133°45.6	· · 50.2					Mirfak	308°27.2	49°57.1	
10	148°33.9	356°16.1	23.6	302°29.9	44.2	35°58.2	48.8	148°48.3	50.3					Aldebaran	290°38.9	16°33.7	
11	163°36.4	11°15.7	22.6	317°30.8	44.8	51°00.2	48.8	163°50.9	50.3					Rigel	281°03.3	-8°10.1	
12	178°38.8	26°15.2	N12°21.6	332°31.7	S10°45.5	66°02.3	N21°48.7	178°53.5	S02°50.4					Capella	280°21.0	46°01.3	
13	193°41.3	41°14.7	20.6	347°32.6	46.1	81°04.4	48.6	193°56.2	50.5					Bellatrix	278°22.2	6°22.5	
14	208°43.8	56°14.2	19.6	2°33.4	46.7	96°06.5	48.6	208°58.8	50.6					Elnath	278°01.1	28°37.7	
15	223°46.2	71°13.7	· · 18.6	17°34.3	· · 47.3	111°08.5	· · 48.5	224°01.5	· · 50.7					Alnilam	275°37.1	-1°10.9	
16	238°48.7	86°13.2	17.7	32°35.2	47.9	126°10.6	48.5	239°04.1	50.7					Betelgeuse	270°51.5	7°24.8	
17	253°51.2	101°12.7	16.7	47°36.1	48.6	141°12.7	48.4	254°06.7	50.8					Canopus	263°52.2	-52°42.2	
18	268°53.6	116°12.2	N12°15.7	62°37.0	S10°49.2	156°14.7	N21°48.4	269°09.4	S02°50.9					Sirius	258°25.8	-16°44.8	
19	283°56.1	131°11.7	14.7	77°37.9	49.8	171°16.8	48.3	284°12.0	51.0					Adhara	255°05.5	-29°00.1	
20	298°58.6	146°11.2	13.7	92°38.7	50.4	186°18.9	48.3	299°14.7	51.1					Procyon	244°50.4	5°09.7	
21	314°01.0	161°10.7	· · 12.7	107°39.6	· · 51.1	201°21.0	· · 48.2	314°17.3	· · 51.1					Pollux	243°16.8	27°57.8	
22	329°03.5	176°10.2	11.7	122°40.5	51.7	216°23.0	48.2	329°20.0	51.2					Avior	234°14.9	-59°35.2	
23	344°05.9	191°09.8	10.7	137°41.4	52.3	231°25.1	48.1	344°22.6	51.3					Suhail	222°46.2	-43°31.9	
Mer.pass. 00:07			ν -0.5' d -1.0' m -3.93	ν 0.9' d 0.6' m 1.60	ν 2.1' d -0.1' m -2.06	ν 2.6' d 0.1' m 0.59											

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	359°08.4	206°09.3	N12°09.7	152°42.3	S10°52.9	246°27.2	N21°48.1	359°25.2	S02°51.4
1	14°10.9	221°08.8	08.7	167°43.2	53.5	261°29.3	48.0	14°27.9	51.4
2	29°13.3	236°08.3	07.7	182°44.0	54.2	276°31.3	48.0	29°30.5	51.5
3	44°15.8	251°07.8	· · 06.7	197°44.9	· · 54.8	291°33.4	· · 47.9	44°33.2	· · 51.6
4	59°18.3	266°07.3	05.7	212°45.8	55.4	306°35.5	47.9	59°35.8	51.7
5	74°20.7	281°06.8	04.7	227°46.7	56.0	321°37.6	47.8	74°38.4	51.8
6	89°23.2	296°06.3	N12°03.7	242°47.6	S10°56.6	336°39.6	N21°47.8	89°41.1	S02°51.8
7	104°25.7	311°05.8	02.7	257°48.4	57.3	351°41.7	47.7	104°43.7	51.9
8	119°28.1	326°05.4	01.7	272°49.3	57.9	6°43.8	47.7	119°46.4	52.0
9	134°30.6	341°04.9	12°00.7	287°50.2	· · 58.5	21°45.9	· · 47.6	134°49.0	· · 52.1
10	149°33.0	356°04.4	11°59.7	302°51.1	59.1	36°47.9	47.6	149°51.7	52.2
11	164°35.5	11°03.9	58.7	317°52.0	10°59.7	51°50.0	47.5	164°54.3	52.2
12	179°38.0	26°03.4	N11°57.7	332°52.8	S11°00.4	66°52.1	N21°47.5	179°56.9	S02°52.3
13	194°40.4	41°02.9	56.6	347°53.7	01.0	81°54.2	47.4	194°59.6	52.4
14	209°42.9	56°02.4	55.6	2°54.6	01.6	96°56.2	47.4	210°02.2	52.5
15	224°45.4	71°02.0	· · 54.6	17°55.5	· · 02.2	111°58.3	· · 47.3	225°04.9	· · 52.6
16	239°47.8	86°01.5	53.6	32°56.4	02.8	127°00.4	47.3	240°07.5	52.6
17	254°50.3	101°01.0	52.6	47°57.2	03.5	142°02.5	47.2	255°10.1	52.7
18	269°52.8	116°00.5	N11°51.6	62°58.1	S11°04.1	157°04.6	N21°47.2	270°12.8	S02°52.8
19	284°55.2	131°00.0	50.6	77°59.0	04.7	172°06.6	47.1	285°15.4	52.9
20	299°57.7	145°59.5	49.6	92°59.9	05.3	187°08.7	47.1	300°18.1	52.9
21	315°00.2	160°59.1	· · 48.6	108°00.7	· · 05.9	202°10.8	· · 47.0	315°20.7	· · 53.0
22	330°02.6	175°58.6	47.6	123°01.6	06.6	217°12.9	46.9	330°23.4	53.1
23	345°05.1	190°58.1	46.6	138°02.5	07.2	232°14.9	46.9	345°26.0	53.2
Mer.pass. 23:59			ν -0.5' d -1.0' m -3.93	ν 0.9' d 0.6' m 1.60	ν 2.1' d -0.1' m -2.07	ν 2.6' d 0.1' m 0.58			

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	0°07.5	205°57.6	N11°45.5	153°03.4	S11°07.8	247°17.0	N21°46.8	0°28.6	S02°53.3
1	15°10.0	220°57.1	44.5	168°04.2	08.4	262°19.1	46.8	15°31.3	53.3
2	30°12.5	235°56.6	43.5	183°05.1	09.0	277°21.2	46.7	30°33.9	53.4
3	45°14.9	250°56.2	· · 42.5	198°06.0	· · 09.7	292°23.3	· · 46.7	45°36.6	· · 53.5
4	60°17.4	265°55.7	41.5	213°06.9	10.3	307°25.3	46.6	60°39.2	53.6
5	75°19.9	280°55.2	40.5	228°07.8	10.9	322°27.4	46.6	75°41.8	53.7
6	90°22.3	295°54.7	N11°39.5	243°08.6	S11°11.5	337°29.5	N21°46.5	90°44.5	S02°53.7
7	105°24.8	310°54.2	38.4	258°09.5	12.1	352°31.6	46.5	105°47.1	53.8
8	120°27.3	325°53.8	37.4	273°10.4	12.8	7°33.7	46.4	120°49.8	53.9
9	135°29.7	340°53.3	· · 36.4	288°11.3	· · 13.4	22°35.8	· · 46.4	135°52.4	· · 54.0
10	150°32.2	355°52.8	35.4	303°12.1	14.0	37°37.8	46.3	150°55.1	54.0
11	165°34.6	10°52.3	34.4	318°13.0	14.6	52°39.9	46.3	165°57.7	54.1
12	180°37.1	25°51.8	N11°33.3	333°13.9	S11°15.2	67°42.0	N21°46.2	181°00.3	S02°54.2
13	195°39.6	40°51.4	32.3	348°14.7	15.8	82°44.1	46.2	196°03.0	54.3
14	210°42.0	55°50.9	31.3	3°15.6	16.5	97°46.2	46.1	211°05.6	54.4
15	225°44.5	70°50.4	· · 30.3	18°16.5	· · 17.1	112°48.2	· · 46.1	226°08.3	· · 54.4
16	240°47.0	85°49.9	29.3	33°17.4	17.7	127°50.3	46.0	241°10.9	54.5
17	255°49.4	100°49.4	28.2	48°18.2	18.3	142°52.4	46.0	256°13.5	54.6
18	270°51.9	115°49.0	N11°27.2	63°19.1	S11°18.9	157°54.5	N21°45.9	271°16.2	S02°54.7
19	285°54.4	130°48.5	26.2	78°20.0	19.5	172°56.6	45.9	286°18.8	54.8
20	300°56.8	145°48.0	25.2	93°20.9	20.2	187°58.7	45.8	301°21.5	54.8
21	315°59.3	160°47.5	· · 24.1	108°21.7	· · 20.8	203°00.8	· · 45.8	316°24.1	· · 54.9
22	331°01.8	175°47.1	23.1	123°22.6	21.4	218°02.8	45.7	331°26.8	55.0
23	346°04.2	190°46.6	22.1	138°23.5	22.0	233°04.9	45.7	346°29.4	55.1
Mer.pass. 23:56			ν -0.5' d -1.0' m -3.93	ν 0.9' d 0.6' m 1.59	ν 2.1' d -0.1' m -2.07	ν 2.6' d 0.1' m 0.58			

Sep 19 Fri	SHA	Mer.pass
Venus	208°11.8	10:15
Mars	154°11.8	13:50
Jupiter	247°28.2	07:36
Saturn	0°12.6	00:07

Sep 20 Sat	SHA	Mer.pass
Venus	207°00.9	10:16
Mars	153°33.9	13:48
Jupiter	247°18.8	07:33
Saturn	0°16.8	00:02

Sep 21 Sun	SHA	Mer.pass
Venus	205°50.1	10:16
Mars	152°55.8	13:47
Jupiter	247°09.5	07:30
Saturn	0°21.1	23:54

Horizontal parallax		
Venus:		0.1
Mars:		0.1

Sun							Moon								
Fri	GHA	Dec	GHA	ν	Dec	d	HP	GHA	ν	Dec	d	HP			
0	181°31.9	N01°27.8	211°42.1	12.8'	N15°50.7	-13.3'	56.5'	0	181°31.9	N01°27.8	211°42.1	12.8'	56.5'		
1	196°32.1	26.8	226°13.9	12.9'	15°37.4	-13.4'	56.5'	1	196°32.1	26.8	226°13.9	12.9'	56.5'		
2	211°32.3	25.8	240°45.8	12.9'	15°24.0	-13.4'	56.4'	2	211°32.3	25.8	240°45.8	12.9'	56.4'		
3	226°32.5	24.9	255°17.7	13.0'	15°10.6	-13.5'	56.4'	3	226°32.5	24.9	255°17.7	13.0'	56.4'		
4	241°32.8	23.9	269°49.7	13.1'	14°57.1	-13.5'	56.4'	4	241°32.8	23.9	269°49.7	13.1'	56.4'		
5	256°33.0	22.9	284°21.8	13.2'	14°43.6	-13.6'	56.4'	5	256°33.0	22.9	284°21.8	13.2'	56.4'		
6	271°33.2	N01°22.0	298°54.0	13.3'	N14°30.0	-13.6'	56.4'	6	271°33.2	N01°22.0	298°54.0	13.3'	56.4'		
7	286°33.4	21.0	313°26.3	13.4'	14°16.4	-13.7'	56.3'	7	286°33.4	21.0	313°26.3	13.4'	56.3'		
8	301°33.7	20.0	327°58.6	13.4'	14°02.7	-13.8'	56.3'	8	301°33.7	20.0	327°58.6	13.4'	56.3'		
9	316°33.9	19.0	342°31.1	13.5'	13°48.9	-13.8'	56.3'	9	316°33.9	19.0	342°31.1	13.5'	56.3'		
10	331°34.1	18.1	357°03.6	13.6'	13°35.1	-13.9'	56.3'	10	331°34.1	18.1	357°03.6	13.6'	56.3'		
11	346°34.3	17.1	11°36.2	13.7'	13°21.2	-13.9'	56.3'	11	346°34.3	17.1	11°36.2	13.7'	56.3'		
12	1°34.5	N01°16.1	26°08.9	13.7'	N13°07.3	-14.0'	56.2'	12	1°34.5	N01°16.1	26°08.9	13.7'	56.2'		
13	16°34.8	15.2	40°41.6	13.8'	12°53.4	-14.0'	56.2'	13	16°34.8	15.2	40°41.6	13.8'	56.2'		
14	31°35.0	14.2	55°14.4	13.9'	12°39.4	-14.0'	56.2'	14	31°35.0	14.2	55°14.4	13.9'	56.2'		
15	46°35.2	13.2	69°47.3	14.0'	12°25.3	-14.1'	56.2'	15	46°35.2	13.2	69°47.3	14.0'	56.2'		
16	61°35.4	12.3	84°20.3	14.0'	12°11.2	-14.1'	56.1'	16	61°35.4	12.3	84°20.3	14.0'	56.1'		
17	76°35.7	11.3	98°53.3	14.1'	11°57.1	-14.2'	56.1'	17	76°35.7	11.3	98°53.3	14.1'	56.1'		
18	91°35.9	N01°10.3	113°26.4	14.2'	N11°42.9	-14.2'	56.1'	18	91°35.9	N01°10.3	113°26.4	14.2'	56.1'		
19	106°36.1	09.3	127°59.6	14.2'	11°28.7	-14.3'	56.1'	19	106°36.1	09.3	127°59.6	14.2'	56.1'		
20	121°36.3	08.4	142°32.9	14.3'	11°14.5	-14.3'	56.1'	20	121°36.3	08.4	142°32.9	14.3'	56.1'		
21	136°36.5	07.4	157°06.2	14.4'	11°00.2	-14.3'	56.0'	21	136°36.5	07.4	157°06.2	14.4'	56.0'		
22	151°36.8	06.4	171°39.5	14.4'	10°45.9	-14.4'	56.0'	22	151°36.8	06.4	171°39.5	14.4'	56.0'		
23	166°37.0	05.5	186°13.0	14.5'	10°31.5	-14.4'	56.0'	23	166°37.0	05.5	186°13.0	14.5'	56.0'		
SD = 15.9' d = -1.0'							SD = 15.4'								
Sat	GHA	Dec	GHA	ν	Dec	d	HP	Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	181°37.2	N01°04.5	200°46.5	14.6'	N10°17.1	-14.4'	56.0'	0	181°37.2	N01°04.5	200°46.5	14.6'	N10°17.1	-14.4'	56.0'
1	196°37.4	03.5	215°20.1	14.6'	10°02.7	-14.5'	56.0'	1	196°37.4	03.5	215°20.1	14.6'	10°02.7	-14.5'	56.0'
2	211°37.7	02.5	229°53.7	14.7'	09°48.2	-14.5'	55.9'	2	211°37.7	02.5	229°53.7	14.7'	09°48.2	-14.5'	55.9'
3	226°37.9	01.6	244°27.4	14.8'	09°33.7	-14.5'	55.9'	3	226°37.9	01.6	244°27.4	14.8'	09°33.7	-14.5'	55.9'
4	241°38.1	01°00.6	259°01.2	14.8'	09°19.2	-14.6'	55.9'	4	241°38.1	01°00.6	259°01.2	14.8'	09°19.2	-14.6'	55.9'
5	256°38.3	00°59.6	273°35.0	14.9'	09°04.6	-14.6'	55.9'	5	256°38.3	00°59.6	273°35.0	14.9'	09°04.6	-14.6'	55.9'
6	271°38.5	N00°58.7	288°08.9	14.9'	N08°50.1	-14.6'	55.9'	6	271°38.5	N00°58.7	288°08.9	14.9'	N08°50.1	-14.6'	55.9'
7	286°38.8	57.7	302°42.8	15.0'	08°35.5	-14.6'	55.8'	7	286°38.8	57.7	302°42.8	15.0'	08°35.5	-14.6'	55.8'
8	301°39.0	56.7	317°16.8	15.0'	08°20.8	-14.7'	55.8'	8	301°39.0	56.7	317°16.8	15.0'	08°20.8	-14.7'	55.8'
9	316°39.2	55.7	331°50.8	15.1'	08°06.2	-14.7'	55.8'	9	316°39.2	55.7	331°50.8	15.1'	08°06.2	-14.7'	55.8'
10	331°39.4	54.8	346°24.9	15.1'	07°51.5	-14.7'	55.8'	10	331°39.4	54.8	346°24.9	15.1'	07°51.5	-14.7'	55.8'
11	346°39.6	53.8	0°59.1	15.2'	07°36.8	-14.7'	55.8'	11	346°39.6	53.8	0°59.1	15.2'	07°36.8	-14.7'	55.8'
12	1°39.9	N00°52.8	15°33.3	15.2'	N07°22.1	-14.7'	55.7'	12	1°39.9	N00°52.8	15°33.3	15.2'	N07°22.1	-14.7'	55.7'
13	16°40.1	51.9	30°07.5	15.3'	07°07.3	-14.8'	55.7'	13	16°40.1	51.9	30°07.5	15.3'	07°07.3	-14.8'	55.7'
14	31°40.3	50.9	44°41.8	15.3'	06°52.6	-14.8'	55.7'	14	31°40.3	50.9	44°41.8	15.3'	06°52.6	-14.8'	55.7'
15	46°40.5	49.9	59°16.1	15.4'	06°37.8	-14.8'	55.7'	15	46°40.5	49.9	59°16.1	15.4'	06°37.8	-14.8'	55.7'
16	61°40.8	48.9	73°50.5	15.4'	06°23.0	-14.8'	55.7'	16	61°40.8	48.9	73°50.5	15.4'	06°23.0	-14.8'	55.7'
17	76°41.0	48.0	88°25.0	15.5'	06°08.2	-14.8'	55.6'	17	76°41.0	48.0	88°25.0	15.5'	06°08.2	-14.8'	55.6'
18	91°41.2	N00°47.0	102°59.4	15.5'	N05°53.4	-14.8'	55.6'	18	91°41.2	N00°47.0	102°59.4	15.5'	N05°53.4	-14.8'	55.6'
19	106°41.4	46.0	117°34.0	15.6'	05°38.6	-14.8'	55.6'	19	106°41.4	46.0	117°34.0	15.6'	05°38.6	-14.8'	55.6'
20	121°41.6	45.1	132°08.5	15.6'	05°23.7	-14.9'	55.6'	20	121°41.6	45.1	132°08.5	15.6'	05°23.7	-14.9'	55.6'
21	136°41.9	44.1	146°43.1	15.6'	05°08.9	-14.9'	55.6'	21	136°41.9	44.1	146°43.1	15.6'	05°08.9	-14.9'	55.6'
22	151°42.1	43.1	161°17.8	15.7'	04°54.0	-14.9'	55.6'	22	151°42.1	43.1	161°17.8	15.7'	04°54.0	-14.9'	55.6'
23	166°42.3	42.1	175°52.5	15.7'	04°39.1	-14.9'	55.5'	23	166°42.3	42.1	175°52.5	15.7'	04°39.1	-14.9'	55.5'
SD = 15.9' d = -1.0'							SD = 15.3'								
Sun	GHA	Dec	GHA	ν	Dec	d	HP	Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	181°42.5	N00°41.2	190°27.2	15.8'	N04°24.2	-14.9'	55.5'	0	181°42.5	N00°41.2	190°27.2	15.8'	N04°24.2	-14.9'	55.5'
1	196°42.7	40.2	205°02.0	15.8'	04°09.3	-14.9'	55.5'	1	196°42.7	40.2	205°02.0	15.8'	04°09.3	-14.9'	55.5'
2	211°43.0	39.2	219°36.8	15.8'	03°54.4	-14.9'	55.5'	2	211°43.0	39.2	219°36.8	15.8'	03°54.4	-14.9'	55.5'
3	226°43.2	38.3	234°11.6	15.9'	03°39.5	-14.9'	55.5'	3	226°43.2	38.3	234°11.6	15.9'	03°39.5	-14.9'	55.5'
4	241°43.4	37.3	248°46.5	15.9'	03°24.6	-14.9'	55.4'	4	241°43.4	37.3	248°46.5	15.9'	03°24.6	-14.9'	55.4'
5	256°43.6	36.3	263°21.3	15.9'	03°09.7	-14.9'	55.4'	5	256°43.6	36.3	263°21.3	15.9'	03°09.7	-14.9'	55.4'
6	271°43.8	N00°35.3	277°56.3	16.0'	N02°54.8	-14.9'	55.4'	6	271°43.8	N00°35.3	277°56.3	16.0'	N02°54.8	-14.9'	55.4'
7	286°44.1	34.4	292°31.2	16.0'	02°39.9	-14.9'	55.4'	7	286°44.1	34.4	292°31.2	16.0'	02°39.9	-14.9'	55.4'
8	301°44.3	33.4	307°06.2	16.0'	02°25.0	-14.9'	55.4'	8	301°44.3	33.4	307°06.2	16.0'	02°25.0	-14.9'	55.4'
9	316°44.5	32.4	321°41.2	16.0'	02°10.0	-14.9'	55.3'	9	316°44.5	32.4	321°41.2	16.0'	02°10.0	-14.9'	55.3'
10	331°44.7	31.4	336°16.3	16.1'	01°55.1	-14.9'	55.3'	10	331°44.7	31.4	336°16.3	16.1'	01°55.1	-14.9'	55.3'
11	346°44.9	30.5	350°51.3	16.1'	01°40.2	-14.9'	55.3'	11	346°44.9	30.5	350°51.3	16.1'	01°40.2	-14.9'	55.3'
12	1°45.2	N00°29.5	5°26.4	16.1'	N01°25.3	-14.9'	55.3'	12	1°45.2	N00°29.5	5°26.4	16.1'	N01°25.3	-14.9'	55.3'
13	16°45.4	28.5	20°01.6	16.1'	01°10.4	-14.9'	55.3'	13	16°45.4	28.5	20°016				

September 22, 23, 24 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	1°06.7	205°46.1	N11°21.1	153°24.3	S11°22.6	248°07.0	N21°45.6	1°32.0	S02°55.1	Alpheratz	357°33.8 29°14.1
1	16°09.1	220°45.6	20.0	168°25.2	23.2	263°09.1	45.6	16°34.7	55.2	Ankaa	353°06.1 -42°09.8
2	31°11.6	235°45.2	19.0	183°26.1	23.9	278°11.2	45.5	31°37.3	55.3	Schedar	349°29.9 56°40.8
3	46°14.1	250°44.7	•• 18.0	198°27.0	•• 24.5	293°13.3	•• 45.5	46°40.0	•• 55.4	Diphda	348°46.4 -17°50.6
4	61°16.5	265°44.2	16.9	213°27.8	25.1	308°15.4	45.4	61°42.6	55.5	Achernar	335°19.2 -57°06.1
5	76°19.0	280°43.7	15.9	228°28.7	25.7	323°17.4	45.4	76°45.2	55.5	Hamal	327°50.3 23°35.1
6	91°21.5	295°43.3	N11°14.9	243°29.6	S11°26.3	338°19.5	N21°45.3	91°47.9	S02°55.6	Polaris	313°25.4 89°22.1
7	106°23.9	310°42.8	13.9	258°30.4	26.9	353°21.6	45.3	106°50.5	55.7	Acamar	315°11.0 -40°11.8
8	121°26.4	325°42.3	12.8	273°31.3	27.6	8°23.7	45.2	121°53.2	55.8	Menkar	314°05.3 4°11.6
9	136°28.9	340°41.8	•• 11.8	288°32.2	•• 28.2	23°25.8	•• 45.2	136°55.8	•• 55.9	Mirfak	308°27.1 49°57.1
10	151°31.3	355°41.4	10.8	303°33.0	28.8	38°27.9	45.1	151°58.5	55.9	Aldebaran	290°38.8 16°33.7
11	166°33.8	10°40.9	09.7	318°33.9	29.4	53°30.0	45.1	167°01.1	56.0	Rigel	281°03.2 -8°10.1
12	181°36.2	25°40.4	N11°08.7	333°34.8	S11°30.0	68°32.0	N21°45.0	182°03.7	S02°56.1	Capella	280°20.9 46°01.3
13	196°38.7	40°39.9	07.7	348°35.6	30.6	83°34.1	45.0	197°06.4	56.2	Bellatrix	278°22.2 6°22.5
14	211°41.2	55°39.5	06.6	3°36.5	31.3	98°36.2	44.9	212°09.0	56.2	Elnath	278°01.1 28°37.7
15	226°43.6	70°39.0	•• 05.6	18°37.4	•• 31.9	113°38.3	•• 44.9	227°11.7	•• 56.3	Alnilam	275°37.1 -1°10.9
16	241°46.1	85°38.5	04.6	33°38.3	32.5	128°40.4	44.8	242°14.3	56.4	Betelgeuse	270°51.5 7°24.8
17	256°48.6	100°38.1	03.5	48°39.1	33.1	143°42.5	44.8	257°16.9	56.5	Canopus	263°52.2 -52°42.2
18	271°51.0	115°37.6	N11°02.5	63°40.0	S11°33.7	158°44.6	N21°44.7	272°19.6	S02°56.6	Sirius	258°25.8 -16°44.8
19	286°53.5	130°37.1	01.4	78°40.9	34.3	173°46.7	44.7	287°22.2	56.6	Adhara	255°05.5 -29°00.1
20	301°56.0	145°36.7	11°00.4	93°41.7	34.9	188°48.8	44.6	302°24.9	56.7	Procyon	244°50.4 5°09.7
21	316°58.4	160°36.2	10°59.4	108°42.6	•• 35.6	203°50.9	•• 44.6	317°27.5	•• 56.8	Pollux	243°16.8 27°57.8
22	332°00.9	175°35.7	58.3	123°43.5	36.2	218°52.9	44.5	332°30.2	56.9	Avior	234°14.9 -59°35.2
23	347°03.4	190°35.2	57.3	138°44.3	36.8	233°55.0	44.5	347°32.8	57.0	Suhail	222°46.2 -43°31.9
Mer.pass. 23:52		ν -0.5' d -1.0' m -3.93		ν 0.9' d 0.6' m 1.59		ν 2.1' d -0.1' m -2.08		ν 2.6' d 0.1' m 0.58		Miaplacidus	221°38.9 -69°49.1
										Alphard	217°47.4 -8°46.0
										Regulus	207°34.1 11°50.6
										Dubhe	193°40.9 61°36.7
										Denebola	182°24.7 14°25.8
										Gienah	175°43.3 -17°41.0
										Acrux	173°00.2 -63°14.5
										Gacrux	171°51.6 -57°15.4
										Alioth	166°13.0 55°49.3
										Spica	158°22.0 -11°17.7
										Alkaid	152°51.9 49°11.2
										Hadar	148°35.8 -60°29.9
										Menkent	147°57.3 -36°29.8
										Arcturus	145°47.7 19°03.0
										Rigil Kent.	139°40.1 -60°56.6
										Kochab	137°20.6 74°03.1
										Zuben'ubi	136°55.6 -16°08.9
										Alphecca	126°03.5 26°37.9
										Antares	112°15.3 -26°29.4
										Atria	107°09.1 -69°04.6
										Sabik	102°02.2 -15°45.4
										Shaula	96°09.6 -37°07.4
										Rasalhague	95°58.0 12°32.6
										Eltanin	90°41.9 51°29.4
										Kaus Aust.	83°31.7 -34°22.4
										Vega	80°32.8 38°48.7
										Nunki	75°47.0 -26°15.9
										Altair	61°59.3 8°56.3
										Peacock	53°04.5 -56°39.3
										Deneb	49°25.1 45°22.6
										Enif	33°38.0 9°59.7
										Al Na'ir	27°31.8 -46°50.2
										Fomalhaut	15°13.5 -29°29.1
										Scheat	13°44.3 28°13.5
										Markab	13°29.0 15°20.8
										Sep 22 Mon SHA Mer.pass	
										Venus	204°39.4 10:17
										Mars	152°17.7 13:46
										Jupiter	247°00.3 07:26
										Saturn	0°25.4 23:50
										Sep 23 Tue SHA Mer.pass	
										Venus	203°29.0 10:18
										Mars	151°39.4 13:44
										Jupiter	246°51.3 07:23
										Saturn	0°29.6 23:45
										Sep 24 Wed SHA Mer.pass	
										Venus	202°18.6 10:19
										Mars	151°00.9 13:43
										Jupiter	246°42.4 07:20
										Saturn	0°33.9 23:41
										Horizontal parallax	
										Venus:	0.1
										Mars:	0.1
Mer.pass. 23:44		ν -0.5' d -1.1' m -3.93		ν 0.9' d 0.6' m 1.59		ν 2.1' d -0.0' m -2.09		ν 2.6' d 0.1' m 0.59			

2025 September 22 to Sep. 24 UT

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:06	04:36	05:44	17:58	19:06	20:34
N 70°	03:25	04:43	05:45	17:58	18:59	20:16
68°	03:39	04:49	05:45	17:57	18:53	20:02
66°	03:51	04:54	05:46	17:57	18:48	19:51
64°	04:01	04:59	05:46	17:57	18:44	19:42
62°	04:09	05:02	05:47	17:56	18:41	19:34
60°	04:15	05:05	05:47	17:56	18:38	19:27
N 58°	04:21	05:08	05:47	17:56	18:35	19:22
56°	04:26	05:10	05:48	17:56	18:33	19:17
54°	04:31	05:12	05:48	17:56	18:31	19:13
52°	04:34	05:14	05:48	17:56	18:29	19:09
50°	04:38	05:16	05:48	17:56	18:28	19:06
45°	04:45	05:19	05:48	17:55	18:25	18:59
N 40°	04:50	05:22	05:49	17:55	18:22	18:54
35°	04:54	05:24	05:49	17:55	18:20	18:50
30°	04:57	05:25	05:49	17:55	18:19	18:47
20°	05:02	05:27	05:49	17:55	18:17	18:43
N 10°	05:04	05:28	05:49	17:55	18:16	18:41
0°	05:04	05:28	05:49	17:56	18:16	18:40
S 10°	05:04	05:28	05:49	17:56	18:17	18:41
20°	05:01	05:27	05:49	17:56	18:18	18:44
30°	04:56	05:24	05:48	17:57	18:21	18:49
35°	04:53	05:23	05:48	17:57	18:23	18:52
40°	04:49	05:20	05:47	17:58	18:25	18:57
45°	04:43	05:18	05:47	17:58	18:28	19:02
S 50°	04:36	05:14	05:46	17:59	18:32	19:10
52°	04:33	05:12	05:46	18:00	18:33	19:13
54°	04:29	05:10	05:46	18:00	18:35	19:17
56°	04:24	05:08	05:45	18:00	18:38	19:22
58°	04:19	05:06	05:45	18:01	18:40	19:27
S 60°	04:13	05:03	05:44	18:02	18:43	19:34

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	06:51	08:57	11:28	17:13	16:34	15:31
N 70°	06:46	08:41	10:49	17:21	16:52	16:11
68°	06:42	08:29	10:23	17:27	17:06	16:39
66°	06:39	08:19	10:03	17:33	17:18	17:00
64°	06:36	08:10	09:47	17:38	17:28	17:17
62°	06:34	08:03	09:34	17:42	17:37	17:31
60°	06:31	07:57	09:23	17:45	17:44	17:43
N 58°	06:30	07:51	09:14	17:49	17:51	17:54
56°	06:28	07:46	09:06	17:51	17:56	18:03
54°	06:27	07:42	08:58	17:54	18:02	18:11
52°	06:25	07:38	08:52	17:56	18:06	18:18
50°	06:24	07:35	08:46	17:59	18:11	18:25
45°	06:22	07:27	08:33	18:03	18:20	18:39
N 40°	06:19	07:21	08:23	18:07	18:28	18:51
35°	06:18	07:16	08:14	18:11	18:35	19:01
30°	06:16	07:11	08:06	18:14	18:41	19:10
20°	06:13	07:03	07:53	18:19	18:51	19:25
N 10°	06:11	06:56	07:41	18:24	19:00	19:38
0°	06:09	06:49	07:31	18:28	19:09	19:51
S 10°	06:06	06:43	07:20	18:32	19:18	20:04
20°	06:04	06:36	07:09	18:37	19:27	20:17
30°	06:01	06:28	06:56	18:42	19:38	20:33
35°	06:00	06:23	06:48	18:46	19:44	20:42
40°	05:58	06:18	06:40	18:49	19:51	20:53
45°	05:56	06:13	06:30	18:53	19:59	21:05
S 50°	05:54	06:06	06:19	18:58	20:09	21:20
52°	05:53	06:03	06:13	19:00	20:14	21:27
54°	05:52	05:59	06:07	19:03	20:19	21:35
56°	05:51	05:55	06:01	19:06	20:24	21:44
58°	05:49	05:51	05:53	19:09	20:31	21:54
S 60°	05:48	05:46	05:45	19:12	20:38	22:06

Day	Sun			Moon		Age 1-3 0-5%
	Egn.of Time	Mer. Pass hh:mm	Mer.Pass.		Lower hh:mm	
	00 ^h mm:ss		Upper hh:mm			
22	07:11	07:22	11:53	12:18	--:-	
23	07:32	07:43	11:52	12:59	00:39	
24	07:53	08:04	11:52	13:41	01:20	

September 25, 26, 27 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	4°04.1	205°12.6	N10°05.7	154°26.5	S12°06.7	250°37.8	N21°42.0	4°42.2	S03°00.8					Alpheratz	357°33.8	29°14.1	
1	19°06.6	220°12.1	04.6	169°27.3	07.3	265°39.9	42.0	19°44.9	00.9					Ankaa	353°06.1	-42°09.8	
2	34°09.0	235°11.7	03.5	184°28.2	07.9	280°42.0	41.9	34°47.5	00.9					Schedar	349°29.8	56°40.8	
3	49°11.5	250°11.2	•• 02.5	199°29.0	•• 08.6	295°44.1	•• 41.9	49°50.2	•• 01.0					Diphda	348°46.4	-17°50.6	
4	64°13.9	265°10.8	01.4	214°29.9	09.2	310°46.2	41.8	64°52.8	01.1					Achernar	335°19.2	-57°06.2	
5	79°16.4	280°10.3	10°00.3	229°30.8	09.8	325°48.3	41.8	79°55.4	01.2					Hamal	327°50.2	23°35.1	
6	94°18.9	295°09.9	N09°59.3	244°31.6	S12°10.4	340°50.4	N21°41.7	94°58.1	S03°01.2					Polaris	313°24.0	89°22.1	
7	109°21.3	310°09.4	58.2	259°32.5	11.0	355°52.5	41.7	110°00.7	01.3					Acamar	315°11.0	-40°11.8	
8	124°23.8	325°08.9	57.1	274°33.3	11.6	10°54.6	41.6	125°03.4	01.4					Menkar	314°05.3	4°11.6	
9	139°26.3	340°08.5	•• 56.1	289°34.2	•• 12.2	25°56.7	•• 41.6	140°06.0	•• 01.5					Mirfak	308°27.1	49°57.1	
10	154°28.7	355°08.0	55.0	304°35.0	12.8	40°58.8	41.5	155°08.6	01.5					Aldebaran	290°38.8	16°33.7	
11	169°31.2	10°07.6	53.9	319°35.9	13.4	56°00.9	41.5	170°11.3	01.6					Rigel	281°03.2	-8°10.1	
12	184°33.7	25°07.1	N09°52.9	334°36.7	S12°14.0	71°03.0	N21°41.4	185°13.9	S03°01.7					Capella	280°20.9	46°01.3	
13	199°36.1	40°06.7	51.8	349°37.6	14.6	86°05.1	41.4	200°16.6	01.8					Bellatrix	278°22.2	6°22.5	
14	214°38.6	55°06.2	50.7	4°38.4	15.2	101°07.2	41.3	215°19.2	01.9					Elnath	278°01.1	28°37.7	
15	229°41.1	70°05.8	•• 49.6	19°39.3	•• 15.8	116°09.3	•• 41.3	230°21.8	•• 01.9					Alnilam	275°37.1	-1°10.9	
16	244°43.5	85°05.3	48.6	34°40.1	16.5	131°11.5	41.2	245°24.5	02.0					Betelgeuse	270°57.4	7°24.8	
17	259°46.0	100°04.9	47.5	49°41.0	17.1	146°13.6	41.2	260°27.1	02.1					Canopus	263°52.2	-52°42.2	
18	274°48.4	115°04.4	N09°46.4	64°41.8	S12°17.7	161°15.7	N21°41.1	275°29.8	S03°02.2					Sirius	258°25.8	-16°44.8	
19	289°50.9	130°04.0	45.3	79°42.7	18.3	176°17.8	41.1	290°32.4	02.2					Adhara	255°05.5	-29°00.1	
20	304°53.4	145°03.5	44.3	94°43.5	18.9	191°19.9	41.0	305°35.1	02.3					Procyon	244°50.3	5°09.7	
21	319°55.8	160°03.1	•• 43.2	109°44.4	•• 19.5	206°22.0	•• 41.0	320°37.7	•• 02.4					Pollux	243°16.7	27°57.8	
22	334°58.3	175°02.6	42.1	124°45.2	20.1	221°24.1	40.9	335°40.3	02.5					Avior	234°14.9	-59°35.2	
23	350°00.8	190°02.2	41.0	139°46.1	20.7	236°26.2	40.9	350°43.0	02.6					Suhail	222°46.2	-43°31.9	
Mer.pass. 23:40			ν -0.5' d -1.1' m -3.93	ν 0.9' d 0.6' m 1.59	ν 2.1' d -0.0' m -2.09	ν 2.6' d 0.1' m 0.59											

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	5°03.2	205°01.7	N09°39.9	154°46.9	S12°21.3	251°28.3	N21°40.9	5°45.6	S03°02.6		
1	20°05.7	220°01.3	38.9	169°47.8	21.9	266°30.4	40.8	20°48.3	02.7		
2	35°08.2	235°00.8	37.8	184°48.6	22.5	281°32.5	40.8	35°50.9	02.8		
3	50°10.6	250°00.4	•• 36.7	199°49.5	•• 23.1	296°34.6	•• 40.7	50°53.5	•• 02.9		
4	65°13.1	264°59.9	35.6	214°50.3	23.7	311°36.8	40.7	65°56.2	02.9		
5	80°15.6	279°59.5	34.6	229°51.2	24.3	326°38.9	40.6	80°58.8	03.0		
6	95°18.0	294°59.0	N09°33.5	244°52.0	S12°24.9	341°41.0	N21°40.6	96°01.5	S03°03.1		
7	110°20.5	309°58.6	32.4	259°52.9	25.5	356°43.1	40.5	111°04.1	03.2		
8	125°22.9	324°58.1	31.3	274°53.7	26.1	11°45.2	40.5	126°06.7	03.2		
9	140°25.4	339°57.7	•• 30.2	289°54.6	•• 26.7	26°47.3	•• 40.4	141°09.4	•• 03.3		
10	155°27.9	354°57.2	29.1	304°55.4	27.3	41°49.4	40.4	156°12.0	03.4		
11	170°30.3	9°56.8	28.1	319°56.3	28.0	56°51.5	40.3	171°14.7	03.5		
12	185°32.8	24°56.3	N09°27.0	334°57.1	S12°28.6	71°53.6	N21°40.3	186°17.3	S03°03.6		
13	200°35.3	39°55.9	25.9	349°58.0	29.2	86°55.7	40.2	201°20.0	03.6		
14	215°37.7	54°55.5	24.8	4°58.8	29.8	101°57.9	40.2	216°22.6	03.7		
15	230°40.2	69°55.0	•• 23.7	19°59.6	•• 30.4	117°00.0	•• 40.1	231°25.2	•• 03.8		
16	245°42.7	84°54.6	22.6	35°00.5	31.0	132°02.1	40.1	246°27.9	03.9		
17	260°45.1	99°54.1	21.6	50°01.3	31.6	147°04.2	40.0	261°30.5	03.9		
18	275°47.6	114°53.7	N09°20.5	65°02.2	S12°32.2	162°06.3	N21°40.0	276°33.2	S03°04.0		
19	290°50.0	129°53.2	19.4	80°03.0	32.8	177°08.4	39.9	291°35.8	04.1		
20	305°52.5	144°52.8	18.3	95°03.9	33.4	192°10.5	39.9	306°38.4	04.2		
21	320°55.0	159°52.3	•• 17.2	110°04.7	•• 34.0	207°12.7	•• 39.8	321°41.1	•• 04.2		
22	335°57.4	174°51.9	16.1	125°05.6	34.6	222°14.8	39.8	336°43.7	04.3		
23	350°59.9	189°51.4	15.0	140°06.4	35.2	237°16.9	39.7	351°46.4	04.4		
Mer.pass. 23:36			ν -0.4' d -1.1' m -3.93	ν 0.8' d 0.6' m 1.58	ν 2.1' d -0.0' m -2.10	ν 2.6' d 0.1' m 0.60					

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	6°02.4	204°51.0	N09°13.9	155°07.3	S12°35.8	252°19.0	N21°39.7	6°49.0	S03°04.5		
1	21°04.8	219°50.6	12.8	170°08.1	36.4	267°21.1	39.6	21°51.6	04.6		
2	36°07.3	234°50.1	11.8	185°08.9	37.0	282°23.2	39.6	36°54.3	04.6		
3	51°09.8	249°49.7	•• 10.7	200°09.8	•• 37.6	297°25.3	•• 39.5	51°56.9	•• 04.7		
4	66°12.2	264°49.2	09.6	215°10.6	38.2	312°27.5	39.5	66°59.6	04.8		
5	81°14.7	279°48.8	08.5	230°11.5	38.8	327°29.6	39.5	82°02.2	04.9		
6	96°17.2	294°48.3	N09°07.4	245°12.3	S12°39.4	342°31.7	N21°39.4	97°04.8	S03°04.9		
7	111°19.6	309°47.9	06.3	260°13.2	40.0	357°33.8	39.4	112°07.5	05.0		
8	126°22.1	324°47.5	05.2	275°14.0	40.6	12°35.9	39.3	127°10.1	05.1		
9	141°24.5	339°47.0	•• 04.1	290°14.8	•• 41.2	27°38.0	•• 39.3	142°12.8	•• 05.2		
10	156°27.0	354°46.6	03.0	305°15.7	41.8	42°40.2	39.2	157°15.4	05.2		
11	171°29.5	9°46.1	01.9	320°16.5	42.4	57°42.3	39.2	172°18.0	05.3		
12	186°31.9	24°45.7	N09°00.8	335°17.4	S12°43.0	72°44.4	N21°39.1	187°20.7	S03°05.4		
13	201°34.4	39°45.3	08°59.7	350°18.2	43.6	87°46.5	39.1	202°23.3	05.5		
14	216°36.9	54°44.8	58.6	5°19.0	44.2	102°48.6	39.0	217°26.0	05.6		
15	231°39.3	69°44.4	•• 57.5	20°19.9	•• 44.8	117°50.7	•• 39.0	232°28.6	•• 05.6		
16	246°41.8	84°43.9	56.4	35°20.7	45.4	132°52.9	38.9	247°31.2	05.7		
17	261°44.3	99°43.5	55.3	50°21.6	46.0	147°55.0	38.9	262°33.9	05.8		
18	276°46.7	114°43.1	N08°54.2	65°22.4	S12°46.6	162°57.1	N21°38.8	277°36.5	S03°05.9		
19	291°49.2	129°42.6	53.1	80°23.2	47.2	177°59.2	38.8	292°39.2	05.9		
20	306°51.7	144°42.2	52.0	95°24.1	47.8	193°01.3	38.7	307°41.8	06.0		
21	321°54.1	159°41.7	•• 51.0	110°24.9	•• 48.4	208°03.5	•• 38.7	322°44.4	•• 06.1		
22	336°56.6	174°41.3	49.9	125°25.8	49.0	223°05.6	38.6	337°47.1	06.2		
23	351°59.0	189°40.9	48.8	140°26.6	49.6	238°07.7	38.6	352°49.7	06.2		
Mer.pass. 23:32			ν -0.4' d -1.1' m -3.93	ν 0.8' d 0.6' m 1.58	ν 2.1' d -0.0' m -2.10	ν 2.6' d 0.1' m 0.60					

Stars		
Sep 25 Thu	SHA	Mer.pass
Venus	201°08.5	10:19
Mars	150°22.4	07:41
Jupiter	246°33.7	13:16
Saturn	0°38.1	23:37
Sep 26 Fri	SHA	Mer.pass
Venus	199°58.5	10:20
Mars	149°43.7	13:40
Jupiter	246°25.1	07:13
Saturn	0°42.4	23:33
Sep 27 Sat	SHA	Mer.pass
Venus	198°48.6	10:21
Mars	149°04.9	13:39
Jupiter	246°16.6	07:10
Saturn	0°46.6	23:29
Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

Sun				Moon				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	182°03.5	S00°52.3		150°19.4	14.6'	S17°45.7	11.5'	54.1'
1	197°03.7	53.2		164°53.1	14.6'	17°57.2	11.4'	54.1'
2	212°03.9	54.2		179°26.7	14.5'	18°08.6	11.3'	54.1'
3	227°04.2	· · 55.2		194°00.2	14.5'	18°20.0	11.3'	54.1'
4	242°04.4	56.2		208°33.7	14.4'	18°31.2	11.2'	54.1'
5	257°04.6	57.1		223°07.1	14.4'	18°42.4	11.1'	54.1'
6	272°04.8	S00°58.1		237°40.5	14.3'	S18°53.5	11.0'	54.1'
7	287°05.0	00°59.1		252°13.8	14.3'	19°04.6	10.9'	54.1'
8	302°05.2	01°00.1		266°47.0	14.2'	19°15.5	10.9'	54.1'
9	317°05.4	· · 01.0		281°20.2	14.1'	19°26.4	10.8'	54.1'
10	332°05.7	02.0		295°53.4	14.1'	19°37.2	10.7'	54.1'
11	347°05.9	03.0		310°26.5	14.0'	19°47.9	10.6'	54.1'
12	2°06.1	S01°04.0		324°59.5	14.0'	S19°58.5	10.5'	54.1'
13	17°06.3	04.9		339°32.5	13.9'	20°09.0	10.5'	54.0'
14	32°06.5	05.9		354°05.4	13.9'	20°19.5	10.4'	54.0'
15	47°06.7	· · 06.9		8°38.3	13.8'	20°29.8	10.3'	54.0'
16	62°06.9	07.9		23°11.1	13.7'	20°40.1	10.2'	54.0'
17	77°07.2	08.8		37°43.8	13.7'	20°50.3	10.1'	54.0'
18	92°07.4	S01°09.8		52°16.5	13.6'	S21°00.4	10.0'	54.0'
19	107°07.6	10.8		66°49.1	13.6'	21°10.4	9.9'	54.0'
20	122°07.8	11.8		81°21.7	13.5'	21°20.4	9.8'	54.0'
21	137°08.0	· · 12.7		95°54.2	13.4'	21°30.2	9.7'	54.0'
22	152°08.2	13.7		110°26.6	13.4'	21°39.9	9.7'	54.0'
23	167°08.5	14.7		124°59.0	13.3'	21°49.6	9.6'	54.0'
SD = 15.9'			$d = 1.0'$	SD = 14.8'				

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	182°08.7	S01°15.6	139°31.3	13.3'	S21°59.2	9.5'	54.0'
1	197°08.9	16.6	154°03.6	13.2'	22°08.6	9.4'	54.0'
2	212°09.1	17.6	168°35.8	13.1'	22°18.0	9.3'	54.0'
3	227°09.3	· · 18.6	183°07.9	13.1'	22°27.3	9.2'	54.0'
4	242°09.5	19.5	197°39.9	13.0'	22°36.4	9.1'	54.0'
5	257°09.7	20.5	212°11.9	12.9'	22°45.5	9.0'	54.0'
6	272°09.9	S01°21.5	226°43.9	12.9'	S22°54.5	8.9'	54.0'
7	287°10.2	22.5	241°15.8	12.8'	23°03.4	8.8'	54.0'
8	302°10.4	23.4	255°47.6	12.7'	23°12.2	8.7'	54.0'
9	317°10.6	· · 24.4	270°19.3	12.7'	23°20.9	8.6'	54.0'
10	332°10.8	25.4	284°51.0	12.6'	23°29.5	8.5'	54.0'
11	347°11.0	26.4	299°22.6	12.6'	23°37.9	8.4'	54.0'
12	2°11.2	S01°27.3	313°54.2	12.5'	S23°46.3	8.3'	54.0'
13	17°11.4	28.3	328°25.7	12.4'	23°54.6	8.2'	54.0'
14	32°11.7	29.3	342°57.1	12.4'	24°02.8	8.1'	54.0'
15	47°11.9	· · 30.3	357°28.5	12.3'	24°10.8	8.0'	54.0'
16	62°12.1	31.2	11°59.8	12.2'	24°18.8	7.9'	54.0'
17	77°12.3	32.2	26°31.0	12.2'	24°26.7	7.7'	54.0'
18	92°12.5	S01°33.2	41°02.2	12.1'	S24°34.4	7.6'	54.0'
19	107°12.7	34.1	55°33.3	12.0'	24°42.1	7.5'	54.0'
20	122°12.9	35.1	70°04.3	12.0'	24°49.6	7.4'	54.0'
21	137°13.1	· · 36.1	84°35.3	11.9'	24°57.0	7.3'	54.0'
22	152°13.4	37.1	99°06.2	11.9'	25°04.3	7.2'	54.0'
23	167°13.6	38.0	113°37.1	11.8'	25°11.5	7.1'	54.0'
SD = 15.9'			$d = 1.0'$	SD = 14.7'			

Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	182°13.8	S01°39.0	128°07.9	11.7'	S25°18.6	7.0'	54.0'
1	197°14.0	40.0	142°38.6	11.7'	25°25.6	6.9'	54.0'
2	212°14.2	41.0	157°09.2	11.6'	25°32.5	6.7'	54.0'
3	227°14.4	· · 41.9	171°39.8	11.5'	25°39.2	6.6'	54.0'
4	242°14.6	42.9	186°10.4	11.5'	25°45.8	6.5'	54.0'
5	257°14.8	43.9	200°40.9	11.4'	25°52.4	6.4'	54.0'
6	272°15.1	S01°44.9	215°11.3	11.4'	S25°58.8	6.3'	54.0'
7	287°15.3	45.8	229°41.6	11.3'	26°05.0	6.2'	54.1'
8	302°15.5	46.8	244°11.9	11.2'	26°11.2	6.0'	54.1'
9	317°15.7	· · 47.8	258°42.2	11.2'	26°17.3	5.9'	54.1'
10	332°15.9	48.7	273°12.3	11.1'	26°23.2	5.8'	54.1'
11	347°16.1	49.7	287°42.4	11.1'	26°29.0	5.7'	54.1'
12	2°16.3	S01°50.7	302°12.5	11.0'	S26°34.7	5.6'	54.1'
13	17°16.5	51.7	316°42.5	10.9'	26°40.2	5.4'	54.1'
14	32°16.7	52.6	331°12.4	10.9'	26°45.7	5.3'	54.1'
15	47°17.0	· · 53.6	345°42.3	10.8'	26°51.0	5.2'	54.1'
16	62°17.2	54.6	0°12.1	10.8'	26°56.2	5.1'	54.1'
17	77°17.4	55.6	14°41.9	10.7'	27°01.3	4.9'	54.1'
18	92°17.6	S01°56.5	29°11.6	10.6'	S27°06.2	4.8'	54.1'
19	107°17.8	57.5	43°41.2	10.6'	27°11.0	4.7'	54.1'
20	122°18.0	58.5	58°10.8	10.5'	27°15.7	4.6'	54.1'
21	137°18.2	01°59.4	72°40.4	10.5'	27°20.3	4.4'	54.2'
22	152°18.4	02°00.4	87°09.9	10.4'	27°24.8	4.3'	54.2'
23	167°18.6	01.4	101°39.3	10.4'	27°29.1	4.2'	54.2'
SD = 15.9'			$d = 1.0'$	SD = 14.7'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:23	04:50	05:57	17:43	18:50	20:15
N 70°	03:39	04:56	05:57	17:44	18:44	20:00
68°	03:52	05:00	05:56	17:45	18:40	19:48
66°	04:02	05:04	05:55	17:45	18:36	19:38
64°	04:10	05:07	05:55	17:46	18:33	19:30
62°	04:17	05:10	05:54	17:47	18:31	19:23
60°	04:23	05:13	05:54	17:47	18:28	19:18
N 58°	04:28	05:15	05:54	17:48	18:27	19:13
56°	04:33	05:16	05:53	17:48	18:25	19:09
54°	04:36	05:18	05:53	17:48	18:23	19:05
52°	04:40	05:19	05:53	17:49	18:22	19:02
50°	04:43	05:20	05:53	17:49	18:21	18:59
45°	04:49	05:23	05:52	17:50	18:19	18:53
N 40°	04:53	05:25	05:52	17:50	18:17	18:49
35°	04:56	05:26	05:51	17:51	18:16	18:45
30°	04:59	05:27	05:51	17:51	18:15	18:43
20°	05:02	05:28	05:50	17:52	18:14	18:40
N 10°	05:04	05:28	05:49	17:53	18:14	18:39
0°	05:03	05:27	05:48	17:54	18:15	18:39
S 10°	05:02	05:26	05:47	17:56	18:17	18:41
20°	04:58	05:24	05:46	17:57	18:19	18:45
30°	04:53	05:20	05:44	17:59	18:23	18:50
35°	04:49	05:18	05:43	18:00	18:25	18:55
40°	04:44	05:15	05:42	18:01	18:28	19:00
45°	04:37	05:12	05:41	18:02	18:31	19:06
S 50°	04:29	05:07	05:40	18:04	18:36	19:15
52°	04:25	05:05	05:39	18:05	18:38	19:19
54°	04:21	05:03	05:38	18:05	18:41	19:23
56°	04:15	05:00	05:37	18:06	18:44	19:29
58°	04:10	04:57	05:36	18:07	18:47	19:35
S 60°	04:03	04:54	05:35	18:09	18:51	19:42

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	12:43	■	■	15:49	■	■
66°	12:01	■	■	16:33	■	■
64°	11:32	13:33	■	17:02	16:37	■
62°	11:10	12:53	14:48	17:25	17:17	17:04
60°	10:53	12:26	14:00	17:43	17:45	17:52
N 58°	10:39	12:05	13:30	17:58	18:07	18:22
56°	10:26	11:48	13:07	18:11	18:24	18:46
54°	10:15	11:33	12:48	18:23	18:40	19:04
52°	10:06	11:20	12:33	18:33	18:53	19:20
50°	09:58	11:09	12:19	18:42	19:04	19:34
45°	09:40	10:46	11:52	19:01	19:28	20:03
N 40°	09:25	10:28	11:30	19:17	19:48	20:25
35°	09:13	10:12	11:12	19:30	20:04	20:43
30°	09:02	09:59	10:56	19:42	20:18	20:59
20°	08:44	09:37	10:30	20:01	20:42	21:26
N 10°	08:28	09:17	10:08	20:19	21:02	21:50
0°	08:14	08:59	09:47	20:35	21:22	22:11
S 10°	07:59	08:41	09:26	20:52	21:41	22:33
20°	07:44	08:22	09:04	21:09	22:02	22:56
30°	07:26	08:00	08:39	21:30	22:27	23:24
35°	07:16	07:47	08:24	21:42	22:41	23:40
40°	07:04	07:32	08:06	21:55	22:58	23:59
45°	06:51	07:15	07:46	22:12	23:18	· · ·
S 50°	06:34	06:54	07:20	22:32	23:43	· · ·
52°	06:26	06:43	07:07	22:42	23:55	· · ·
54°	06:18	06:32	06:53	22:53	· · ·	00:10
56°	06:08	06:19	06:36	23:05	· · ·	00:26
58°	05:57	06:04	06:16	23:20	· · ·	00:46
S 60°	05:45	05:46	05:50	23:37	· · ·	01:10

Day	Sun		Mer. Pass	Moon		Age
	Eqn.of Time			Mer.Pass.		
	00 ^h	12 ^h	Pass	Upper	Lower	4-6
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	9-23%
25	08:14	08:24	11:52	14:24	02:02	
26	08:35	08:45	11:51	15:10	02:47	
27	08:55	09:05	11:51	15:59	03:34	

September 28, 29, 30 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	7°01.5	204°40.4	N08°47.6	155°27.4	S12°50.2	253°09.8	N21°38.5	7°52.4	S03°06.3	Alpheratz	357°33.8	29°14.1
1	22°04.0	219°40.0	46.5	170°28.3	50.8	268°11.9	38.5	22°55.0	06.4	Ankaa	353°06.1	-42°09.8
2	37°06.4	234°39.5	45.4	185°29.1	51.4	283°14.1	38.5	37°57.6	06.5	Schedar	349°29.8	56°40.8
3	52°08.9	249°39.1	· · 44.3	200°30.0	· · 52.0	298°16.2	· · 38.4	53°00.3	· · 06.5	Diphda	348°46.4	-17°50.6
4	67°11.4	264°38.7	43.2	215°30.8	52.6	313°18.3	38.4	68°02.9	06.6	Achernar	335°19.2	-57°06.2
5	82°13.8	279°38.2	42.1	230°31.6	53.2	328°20.4	38.3	83°05.6	06.7	Hamal	327°50.2	23°35.1
6	97°16.3	294°37.8	N08°41.0	245°32.5	S12°53.8	343°22.6	N21°38.3	98°08.2	S03°06.8	Polaris	313°22.6	89°22.2
7	112°18.8	309°37.4	39.9	260°33.3	54.4	358°24.7	38.2	113°10.8	06.8	Acamar	315°11.0	-40°11.8
8	127°21.2	324°36.9	38.8	275°34.1	55.0	13°26.8	38.2	128°13.5	06.9	Menkar	314°05.3	4°11.6
9	142°23.7	339°36.5	· · 37.7	290°35.0	· · 55.6	28°28.9	· · 38.1	143°16.1	· · 07.0	Mirfak	308°27.1	49°57.1
10	157°26.2	354°36.1	36.6	305°35.8	56.2	43°31.0	38.1	158°18.8	07.1	Aldebaran	290°38.8	16°33.7
11	172°28.6	9°35.6	35.5	320°36.6	56.8	58°33.2	38.0	173°21.4	07.2	Rigel	281°03.2	-8°10.1
12	187°31.1	24°35.2	N08°34.4	335°37.5	S12°57.4	73°35.3	N21°38.0	188°24.0	S03°07.2	Capella	280°20.9	46°01.3
13	202°33.5	39°34.7	33.3	350°38.3	58.0	88°37.4	37.9	203°26.7	07.3	Bellatrix	278°22.2	6°22.5
14	217°36.0	54°34.3	32.2	5°39.2	58.6	103°39.5	37.9	218°29.3	07.4	Elnath	278°01.0	28°37.7
15	232°38.5	69°33.9	· · 31.1	20°40.0	· · 59.2	118°41.7	· · 37.8	233°32.0	· · 07.5	Alnilam	275°37.1	-1°11.0
16	247°40.9	84°33.4	30.0	35°40.8	12°59.8	133°43.8	37.8	248°34.6	07.5	Betelgeuse	270°51.4	7°24.8
17	262°43.4	99°33.0	28.9	50°41.7	13°00.4	148°45.9	37.7	263°37.2	07.6	Canopus	263°52.1	-52°42.2
18	277°45.9	114°32.6	N08°27.8	65°42.5	S13°01.0	163°48.0	N21°37.7	278°39.9	S03°07.7	Sirius	258°25.7	-16°44.8
19	292°48.3	129°32.1	26.7	80°43.3	01.6	178°50.2	37.7	293°42.5	07.8	Adhara	255°05.5	-29°00.1
20	307°50.8	144°31.7	25.5	95°44.2	02.2	193°52.3	37.6	308°45.2	07.8	Procyon	244°50.3	5°09.7
21	322°53.3	159°31.3	· · 24.4	110°45.0	· · 02.8	208°54.4	· · 37.6	323°47.8	· · 07.9	Pollux	243°16.7	27°57.8
22	337°55.7	174°30.8	23.3	125°45.8	03.4	223°56.5	37.5	338°50.4	08.0	Avior	234°14.8	-59°35.2
23	352°58.2	189°30.4	22.2	140°46.7	04.0	238°58.7	37.5	353°53.1	08.1	Suhail	222°46.2	-43°31.9
Mer.pass. 23:28		ν -0.4' d -1.1' m -3.93		ν 0.8' d 0.6' m 1.58		ν 2.1' d -0.0' m -2.11		ν 2.6' d 0.1' m 0.60				

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	8°00.6	204°30.0	N08°21.1	155°47.5	S13°04.6	254°00.8	N21°37.4	8°55.7	S03°08.1
1	23°03.1	219°29.5	20.0	170°48.3	05.2	269°02.9	37.4	23°58.4	08.2
2	38°05.6	234°29.1	18.9	185°49.2	05.8	284°05.1	37.3	39°01.0	08.3
3	53°08.0	249°28.7	· · 17.8	200°50.0	· · 06.4	299°07.2	· · 37.3	54°03.6	· · 08.4
4	68°10.5	264°28.2	16.7	215°50.8	06.9	314°09.3	37.2	69°06.3	08.4
5	83°13.0	279°27.8	15.5	230°51.7	07.5	329°11.4	37.2	84°08.9	08.5
6	98°15.4	294°27.4	N08°14.4	245°52.5	S13°08.1	344°13.6	N21°37.1	99°11.6	S03°08.6
7	113°17.9	309°27.0	13.3	260°53.3	08.7	359°15.7	37.1	114°14.2	08.7
8	128°20.4	324°26.5	12.2	275°54.1	09.3	14°17.8	37.1	129°16.8	08.7
9	143°22.8	339°26.1	· · 11.1	290°55.0	· · 09.9	29°20.0	· · 37.0	144°19.5	· · 08.8
10	158°25.3	354°25.7	10.0	305°55.8	10.5	44°22.1	37.0	159°22.1	08.9
11	173°27.8	9°25.2	08.9	320°56.6	11.1	59°24.2	36.9	174°24.8	09.0
12	188°30.2	24°24.8	N08°07.7	335°57.5	S13°11.7	74°26.3	N21°36.9	189°27.4	S03°09.0
13	203°32.7	39°24.4	06.6	350°58.3	12.3	89°28.5	36.8	204°30.0	09.1
14	218°35.1	54°23.9	05.5	5°59.1	12.9	104°30.6	36.8	219°32.7	09.2
15	233°37.6	69°23.5	· · 04.4	21°00.0	· · 13.5	119°32.7	· · 36.7	234°35.3	· · 09.3
16	248°40.1	84°23.1	03.3	36°00.8	14.1	134°34.9	36.7	249°37.9	09.3
17	263°42.5	99°22.7	02.1	51°01.6	14.7	149°37.0	36.6	264°40.6	09.4
18	278°45.0	114°22.2	N08°01.0	66°02.4	S13°15.3	164°39.1	N21°36.6	279°43.2	S03°09.5
19	293°47.5	129°21.8	07°59.9	81°03.3	15.9	179°41.3	36.5	294°45.9	09.6
20	308°49.9	144°21.4	58.8	96°04.1	16.5	194°43.4	36.5	309°48.5	09.7
21	323°52.4	159°20.9	· · 57.7	111°04.9	· · 17.1	209°45.5	· · 36.5	324°51.1	· · 09.7
22	338°54.9	174°20.5	56.6	126°05.8	17.6	224°47.7	36.4	339°53.8	09.8
23	353°57.3	189°20.1	55.4	141°06.6	18.2	239°49.8	36.4	354°56.4	09.9
Mer.pass. 23:24		ν -0.4' d -1.1' m -3.93		ν 0.8' d 0.6' m 1.58		ν 2.1' d -0.0' m -2.11		ν 2.6' d 0.1' m 0.61	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	8°59.8	204°19.7	N07°54.3	156°07.4	S13°18.8	254°51.9	N21°36.3	9°59.1	S03°10.0
1	24°02.3	219°19.2	53.2	171°08.2	19.4	269°54.1	36.3	25°01.7	10.0
2	39°04.7	234°18.8	52.1	186°09.1	20.0	284°56.2	36.2	40°04.3	10.1
3	54°07.2	249°18.4	· · 50.9	201°09.9	· · 20.6	299°58.3	· · 36.2	55°07.0	· · 10.2
4	69°09.6	264°17.9	49.8	216°10.7	21.2	315°00.5	36.1	70°09.6	10.3
5	84°12.1	279°17.5	48.7	231°11.5	21.8	330°02.6	36.1	85°12.3	10.3
6	99°14.6	294°17.1	N07°47.6	246°12.4	S13°22.4	345°04.7	N21°36.0	100°14.9	S03°10.4
7	114°17.0	309°16.7	46.4	261°13.2	23.0	0°06.9	36.0	115°17.5	10.5
8	129°19.5	324°16.2	45.3	276°14.0	23.6	15°09.0	35.9	130°20.2	10.6
9	144°22.0	339°15.8	· · 44.2	291°14.8	· · 24.2	30°11.1	· · 35.9	145°22.8	· · 10.6
10	159°24.4	354°15.4	43.1	306°15.7	24.8	45°13.3	35.9	160°25.4	10.7
11	174°26.9	9°15.0	41.9	321°16.5	25.3	60°15.4	35.8	175°28.1	10.8
12	189°29.4	24°14.5	N07°40.8	336°17.3	S13°25.9	75°17.5	N21°35.8	190°30.7	S03°10.9
13	204°31.8	39°14.1	39.7	351°18.1	26.5	90°19.7	35.7	205°33.4	10.9
14	219°34.3	54°13.7	38.6	6°19.0	27.1	105°21.8	35.7	220°36.0	11.0
15	234°36.8	69°13.3	· · 37.4	21°19.8	· · 27.7	120°24.0	· · 35.6	235°38.6	· · 11.1
16	249°39.2	84°12.8	36.3	36°20.6	28.3	135°26.1	35.6	250°41.3	11.2
17	264°41.7	99°12.4	35.2	51°21.4	28.9	150°28.2	35.5	265°43.9	11.2
18	279°44.1	114°12.0	N07°34.1	66°22.3	S13°29.5	165°30.4	N21°35.5	280°46.6	S03°11.3
19	294°46.6	129°11.6	32.9	81°23.1	30.1	180°32.5	35.4	295°49.2	11.4
20	309°49.1	144°11.1	31.8	96°23.9	30.7	195°34.6	35.4	310°51.8	11.4
21	324°51.5	159°10.7	· · 30.7	111°24.7	· · 31.2	210°36.8	· · 35.4	325°54.5	· · 11.5
22	339°54.0	174°10.3	29.5	126°25.5	31.8	225°38.9	35.3	340°57.1	11.6
23	354°56.5	189°09.9	28.4	141°26.4	32.4	240°41.1	35.3	355°59.7	11.7
Mer.pass. 23:20		ν -0.4' d -1.1' m -3.93		ν 0.8' d 0.6' m 1.57		ν 2.1' d -0.0' m -2.12		ν 2.6' d 0.1' m 0.61	

Sep 28 Sun	SHA	Mer.pass
Venus	197°38.9	10:22
Mars	148°25.9	13:37
Jupiter	246°08.3	07:06
Saturn	0°50.9	23:24

Sep 29 Mon	SHA	Mer.pass
Venus	196°29.3	10:22
Mars	147°46.8	13:36
Jupiter	246°00.2	07:03
Saturn	0°55.1	23:20

Sep 30 Tue	SHA	Mer.pass
Venus	195°19.9	10:23
Mars	147°07.6	13:35
Jupiter	245°52.1	07:00
Saturn	0°59.3	23:16

Horizontal parallax	
Venus:	0.1
Mars:	0.1

Sun				Moon				
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	182°18.8	S02°02.4		116°08.7	10.3'	S27°33.3	4.1'	54.2'
1	197°19.1	03.3		130°38.0	10.3'	27°37.3	3.9'	54.2'
2	212°19.3	04.3		145°07.3	10.2'	27°41.2	3.8'	54.2'
3	227°19.5	· · 05.3		159°36.5	10.2'	27°45.0	3.7'	54.2'
4	242°19.7	06.3		174°05.7	10.1'	27°48.7	3.5'	54.2'
5	257°19.9	07.2		188°34.8	10.1'	27°52.2	3.4'	54.2'
6	272°20.1	S02°08.2		203°03.9	10.0'	S27°55.6	3.3'	54.3'
7	287°20.3	09.2		217°32.9	10.0'	27°58.9	3.1'	54.3'
8	302°20.5	10.1		232°01.9	9.9'	28°02.0	3.0'	54.3'
9	317°20.7	· · 11.1		246°30.8	9.9'	28°05.0	2.9'	54.3'
10	332°20.9	12.1		260°59.7	9.9'	28°07.9	2.7'	54.3'
11	347°21.2	13.1		275°28.6	9.8'	28°10.6	2.6'	54.3'
12	2°21.4	S02°14.0		289°57.4	9.8'	S28°13.2	2.5'	54.3'
13	17°21.6	15.0		304°26.2	9.7'	28°15.6	2.3'	54.3'
14	32°21.8	16.0		318°54.9	9.7'	28°18.0	2.2'	54.4'
15	47°22.0	· · 17.0		333°23.6	9.6'	28°20.1	2.0'	54.4'
16	62°22.2	17.9		347°52.2	9.6'	28°22.2	1.9'	54.4'
17	77°22.4	18.9		2°20.8	9.6'	28°24.1	1.8'	54.4'
18	92°22.6	S02°19.9		16°49.4	9.5'	S28°25.9	1.6'	54.4'
19	107°22.8	20.8		31°18.0	9.5'	28°27.5	1.5'	54.4'
20	122°23.0	21.8		45°46.5	9.5'	28°29.0	1.3'	54.4'
21	137°23.2	· · 22.8		60°14.9	9.4'	28°30.3	1.2'	54.5'
22	152°23.4	23.8		74°43.4	9.4'	28°31.5	1.1'	54.5'
23	167°23.7	24.7		89°11.8	9.4'	28°32.6	0.9'	54.5'
SD = 16.0'			$d = 1.0'$	SD = 14.8'				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	182°23.9	S02°25.7		103°40.1	9.3'	S28°33.5	0.8'	54.5'
1	197°24.1	26.7		118°08.5	9.3'	28°34.3	0.6'	54.5'
2	212°24.3	27.6		132°36.8	9.3'	28°34.9	0.5'	54.6'
3	227°24.5	· · 28.6		147°05.1	9.3'	28°35.4	0.4'	54.6'
4	242°24.7	29.6		161°33.3	9.2'	28°35.8	0.2'	54.6'
5	257°24.9	30.6		176°01.6	9.2'	28°36.0	0.1'	54.6'
6	272°25.1	S02°31.5		190°29.8	9.2'	S28°36.1	-0.1'	54.6'
7	287°25.3	32.5		204°58.0	9.2'	28°36.0	-0.2'	54.6'
8	302°25.5	33.5		219°26.1	9.1'	28°35.8	-0.4'	54.7'
9	317°25.7	· · 34.4		233°54.3	9.1'	28°35.4	-0.5'	54.7'
10	332°25.9	35.4		248°22.4	9.1'	28°34.9	-0.7'	54.7'
11	347°26.1	36.4		262°50.5	9.1'	28°34.3	-0.8'	54.7'
12	2°26.3	S02°37.4		277°18.6	9.1'	S28°33.5	-0.9'	54.7'
13	17°26.6	38.3		291°46.7	9.1'	28°32.5	-1.1'	54.8'
14	32°26.8	39.3		306°14.8	9.0'	28°31.4	-1.2'	54.8'
15	47°27.0	· · 40.3		320°42.8	9.0'	28°30.2	-1.4'	54.8'
16	62°27.2	41.2		335°10.8	9.0'	28°28.8	-1.5'	54.8'
17	77°27.4	42.2		349°38.9	9.0'	28°27.3	-1.7'	54.9'
18	92°27.6	S02°43.2		4°06.9	9.0'	S28°25.6	-1.8'	54.9'
19	107°27.8	44.2		18°34.9	9.0'	28°23.8	-2.0'	54.9'
20	122°28.0	45.1		33°02.9	9.0'	28°21.9	-2.1'	54.9'
21	137°28.2	· · 46.1		47°30.9	9.0'	28°19.8	-2.3'	54.9'
22	152°28.4	47.1		61°58.8	9.0'	28°17.5	-2.4'	55.0'
23	167°28.6	48.0		76°26.8	9.0'	28°15.1	-2.5'	55.0'
SD = 16.0'			$d = 1.0'$	SD = 14.9'				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	182°28.8	S02°49.0		90°54.8	9.0'	S28°12.6	-2.7'	55.0'
1	197°29.0	50.0		105°22.8	9.0'	28°09.9	-2.8'	55.0'
2	212°29.2	50.9		119°50.8	9.0'	28°07.0	-3.0'	55.1'
3	227°29.4	· · 51.9		134°18.7	9.0'	28°04.0	-3.1'	55.1'
4	242°29.6	52.9		148°46.7	9.0'	28°00.9	-3.3'	55.1'
5	257°29.8	53.9		163°14.7	9.0'	27°57.6	-3.4'	55.1'
6	272°30.0	S02°54.8		177°42.7	9.0'	S27°54.2	-3.6'	55.2'
7	287°30.3	55.8		192°10.7	9.0'	27°50.6	-3.7'	55.2'
8	302°30.5	56.8		206°38.6	9.0'	27°46.9	-3.9'	55.2'
9	317°30.7	· · 57.7		221°06.6	9.0'	27°43.1	-4.0'	55.2'
10	332°30.9	58.7		235°34.6	9.0'	27°39.0	-4.2'	55.3'
11	347°31.1	02°59.7		250°02.7	9.0'	27°34.9	-4.3'	55.3'
12	2°31.3	S03°00.7		264°30.7	9.0'	S27°30.6	-4.4'	55.3'
13	17°31.5	01.6		278°58.7	9.0'	27°26.1	-4.6'	55.3'
14	32°31.7	02.6		293°26.8	9.1'	27°21.6	-4.7'	55.4'
15	47°31.9	· · 03.6		307°54.8	9.1'	27°16.8	-4.9'	55.4'
16	62°32.1	04.5		322°22.9	9.1'	27°11.9	-5.0'	55.4'
17	77°32.3	05.5		336°51.0	9.1'	27°06.9	-5.2'	55.5'
18	92°32.5	S03°06.5		351°19.1	9.1'	S27°01.7	-5.3'	55.5'
19	107°32.7	07.4		5°47.2	9.1'	26°56.4	-5.5'	55.5'
20	122°32.9	08.4		20°15.4	9.2'	26°51.0	-5.6'	55.5'
21	137°33.1	· · 09.4		34°43.5	9.2'	26°45.3	-5.7'	55.6'
22	152°33.3	10.3		49°11.7	9.2'	26°39.6	-5.9'	55.6'
23	167°33.5	11.3		63°39.9	9.2'	26°33.7	-6.0'	55.6'
SD = 16.0'			$d = 1.0'$	SD = 15.0'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:40	05:03	06:11	17:27	18:34	19:57
N 70°	03:53	05:08	06:08	17:30	18:30	19:44
68°	04:04	05:11	06:06	17:32	18:27	19:34
66°	04:13	05:14	06:05	17:34	18:25	19:25
64°	04:20	05:16	06:03	17:35	18:23	19:19
62°	04:26	05:18	06:02	17:37	18:21	19:13
60°	04:31	05:20	06:01	17:38	18:19	19:08
N 58°	04:35	05:21	06:00	17:39	18:18	19:04
56°	04:39	05:22	05:59	17:40	18:17	19:00
54°	04:42	05:23	05:59	17:41	18:16	18:57
52°	04:45	05:24	05:58	17:42	18:15	18:54
50°	04:47	05:25	05:57	17:42	18:15	18:52
45°	04:52	05:26	05:56	17:44	18:13	18:47
N 40°	04:56	05:27	05:54	17:45	18:12	18:44
35°	04:59	05:28	05:53	17:47	18:12	18:41
30°	05:01	05:28	05:52	17:48	18:12	18:39
20°	05:03	05:28	05:50	17:50	18:12	18:37
N 10°	05:03	05:28	05:49	17:52	18:13	18:37
0°	05:02	05:26	05:47	17:53	18:14	18:38
S 10°	05:00	05:24	05:45	17:55	18:16	18:41
20°	04:55	05:21	05:43	17:58	18:20	18:45
30°	04:49	05:17	05:41	18:00	18:24	18:52
35°	04:44	05:14	05:39	18:02	18:27	18:57
40°	04:39	05:10	05:38	18:04	18:31	19:03
45°	04:31	05:06	05:36	18:06	18:35	19:10
S 50°	04:22	05:01	05:33	18:08	18:41	19:20
52°	04:18	04:58	05:32	18:10	18:44	19:24
54°	04:13	04:55	05:31	18:11	18:47	19:29
56°	04:07	04:52	05:29	18:12	18:50	19:35
58°	04:00	04:48	05:28	18:14	18:54	19:42
S 60°	03:53	04:44	05:26	18:16	18:58	19:50

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	■	■	■	■	■	■
64°	■	■	■	■	■	■
62°	■	■	17:33	■	■	19:46
60°	15:26	16:22	16:44	18:12	19:06	20:34
N 58°	14:46	15:41	16:14	18:52	19:46	21:05
56°	14:18	15:13	15:50	19:20	20:14	21:28
54°	13:57	14:51	15:31	19:42	20:36	21:47
52°	13:39	14:33	15:15	20:00	20:54	22:02
50°	13:23	14:18	15:01	20:15	21:09	22:16
45°	12:53	13:47	14:33	20:46	21:40	22:44
N 40°	12:29	13:23	14:10	21:10	22:04	23:06
35°	12:09	13:03	13:52	21:30	22:24	23:24
30°	11:53	12:46	13:36	21:47	22:41	23:39
20°	11:24	12:18	13:08	22:16	23:09	· · ·
N 10°	11:00	11:53	12:45	22:40	23:33	· · ·
0°	10:38	11:30	12:23	23:03	23:56	· · ·
S 10°	10:15	11:07	12:01	23:26	· · ·	00:18
20°	09:51	10:43	11:38	23:50	· · ·	00:42
30°	09:23	10:14	11:11	· · ·	00:19	01:10
35°	09:07	09:57	10:54			

October 01, 02, 03 UT (Wed., Thu., Fri.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	SHA	Dec	SHA	Dec		
0	9°58.9	204°09.5	N07°27.3	156°27.2	S13°33.0	255°43.2	N21°35.2	11°02.4	S03°11.7	Alpheratz	357°33.8	29°14.1											
1	25°01.4	219°09.0	26.1	171°28.0	33.6	270°45.3	35.2	26°05.0	11.8	Ankaa	353°06.1	-42°09.8											
2	40°03.9	234°08.6	25.0	186°28.8	34.2	285°47.5	35.1	41°07.7	11.9	Schedar	349°29.8	56°40.8											
3	55°06.3	249°08.2	•• 23.9	201°29.7	•• 34.8	300°49.6	•• 35.1	56°10.3	•• 12.0	Diphda	348°46.4	-17°50.6											
4	70°08.8	264°07.8	22.7	216°30.5	35.4	315°51.8	35.0	71°12.9	12.0	Achernar	335°19.1	-57°06.2											
5	85°11.2	279°07.3	21.6	231°31.3	36.0	330°53.9	35.0	86°15.6	12.1	Hamal	327°50.2	23°35.2											
6	100°13.7	294°06.9	N07°20.5	246°32.1	S13°36.5	345°56.0	N21°35.0	101°18.2	S03°12.2	Polaris	313°21.4	89°22.2											
7	115°16.2	309°06.5	19.3	261°32.9	37.1	0°58.2	34.9	116°20.8	12.3	Acamar	315°11.0	-40°11.8											
8	130°18.6	324°06.1	18.2	276°33.8	37.7	16°00.3	34.9	131°23.5	12.3	Menkar	314°05.3	4°11.6											
9	145°21.1	339°05.7	•• 17.1	291°34.6	•• 38.3	31°02.5	•• 34.8	146°26.1	•• 12.4	Mirfak	308°27.0	49°57.1											
10	160°23.6	354°05.2	15.9	306°35.4	38.9	46°04.6	34.8	161°28.8	12.5	Aldebaran	290°38.8	16°33.7											
11	175°26.0	9°04.8	14.8	321°36.2	39.5	61°06.7	34.7	176°31.4	12.6	Rigel	281°03.2	-8°10.1											
12	190°28.5	24°04.4	N07°13.7	336°37.0	S13°40.1	76°08.9	N21°34.7	191°34.0	S03°12.6	Capella	280°20.8	46°01.3											
13	205°31.0	39°04.0	12.5	351°37.8	40.7	91°11.0	34.6	206°36.7	12.7	Bellatrix	278°22.1	6°22.5											
14	220°33.4	54°03.6	11.4	6°38.7	41.3	106°13.2	34.6	221°39.3	12.8	Elnath	278°01.0	28°37.7											
15	235°35.9	69°03.1	•• 10.3	21°39.5	•• 41.8	121°15.3	•• 34.5	236°41.9	•• 12.9	Alnilam	275°37.0	-1°11.0											
16	250°38.4	84°02.7	09.1	36°40.3	42.4	136°17.5	34.5	251°44.6	12.9	Betelgeuse	270°51.4	7°24.8											
17	265°40.8	99°02.3	08.0	51°41.1	43.0	151°19.6	34.5	266°47.2	13.0	Canopus	263°52.1	-52°42.2											
18	280°43.3	114°01.9	N07°06.9	66°41.9	S13°43.6	166°21.8	N21°34.4	281°49.9	S03°13.1	Sirius	258°25.7	-16°44.8											
19	295°45.7	129°01.5	05.7	81°42.8	44.2	181°23.9	34.4	296°52.5	13.2	Adhara	255°05.4	-29°00.1											
20	310°48.2	144°01.0	04.6	96°43.6	44.8	196°26.0	34.3	311°55.1	13.2	Procyon	244°50.3	5°09.7											
21	325°50.7	159°00.6	•• 03.4	111°44.4	•• 45.4	211°28.2	•• 34.3	326°57.8	•• 13.3	Pollux	243°16.7	27°57.8											
22	340°53.1	174°00.2	02.3	126°45.2	45.9	226°30.3	34.2	342°00.4	13.4	Avior	234°14.8	-59°35.2											
23	355°55.6	188°59.8	01.2	141°46.0	46.5	241°32.5	34.2	357°03.0	13.5	Suhail	222°46.2	-43°31.9											
Mer.pass. 23:16		ν -0.4' d -1.1' m -3.93		ν 0.8' d 0.6' m 1.57		ν 2.1' d -0.0' m -2.12		ν 2.6' d 0.1' m 0.62															

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	10°58.1	203°59.4	N07°00.0	156°46.8	S13°47.1	256°34.6	N21°34.1	12°05.7	S03°13.5	Denebola	182°24.7	14°25.8			
1	26°00.5	218°59.0	06°58.9	171°47.6	47.7	271°36.8	34.1	27°08.3	13.6	Gienah	175°43.3	-17°41.0			
2	41°03.0	233°58.5	57.7	186°48.5	48.3	286°38.9	34.1	42°11.0	13.7	Acrux	173°00.2	-63°14.5			
3	56°05.5	248°58.1	•• 56.6	201°49.3	•• 48.9	301°41.1	•• 34.0	57°13.6	•• 13.8	Gacrux	171°51.6	-57°15.4			
4	71°07.9	263°57.7	55.5	216°50.1	49.5	316°43.2	34.0	72°16.2	13.8	Alioth	166°13.0	55°49.3			
5	86°10.4	278°57.3	54.3	231°50.9	50.0	331°45.4	33.9	87°18.9	13.9	Spica	158°22.0	-11°17.7			
6	101°12.9	293°56.9	N06°53.2	246°51.7	S13°50.6	346°47.5	N21°33.9	102°21.5	S03°14.0	Alkaid	152°52.0	49°11.2			
7	116°15.3	308°56.4	52.0	261°52.5	51.2	1°49.7	33.8	117°24.1	14.0	Hadar	148°35.8	-60°29.9			
8	131°17.8	323°56.0	50.9	276°53.4	51.8	16°51.8	33.8	132°26.8	14.1	Menkent	147°57.3	-36°29.8			
9	146°20.2	338°55.6	•• 49.7	291°54.2	•• 52.4	31°53.9	•• 33.7	147°29.4	•• 14.2	Arcturus	145°47.7	19°03.0			
10	161°22.7	353°55.2	48.6	306°55.0	53.0	46°56.1	33.7	162°32.0	14.3	Rigil Kent.	139°40.1	-60°56.6			
11	176°25.2	8°54.8	47.5	321°55.8	53.6	61°58.2	33.7	177°34.7	14.3	Kochab	137°20.7	74°03.1			
12	191°27.6	23°54.4	N06°46.3	336°56.6	S13°54.1	77°00.4	N21°33.6	192°37.3	S03°14.4	Zuben'ubi	136°55.6	-16°08.9			
13	206°30.1	38°54.0	45.2	351°57.4	54.7	92°02.5	33.6	207°40.0	14.5	Alphecca	126°03.5	26°37.8			
14	221°32.6	53°53.5	44.0	6°58.2	55.3	107°04.7	33.5	222°42.6	14.6	Antares	112°15.3	-26°29.4			
15	236°35.0	68°53.1	•• 42.9	21°59.0	•• 55.9	122°06.8	•• 33.5	237°45.2	•• 14.6	Atria	107°09.2	-69°04.6			
16	251°37.5	83°52.7	41.7	36°59.9	56.5	137°09.0	33.4	252°47.9	14.7	Sabik	102°02.2	-15°45.4			
17	266°40.0	98°52.3	40.6	52°00.7	57.1	152°11.1	33.4	267°50.5	14.8	Shaula	96°09.7	-37°07.4			
18	281°42.4	113°51.9	N06°39.4	67°01.5	S13°57.6	167°13.3	N21°33.4	282°53.1	S03°14.9	Rasalhague	95°58.1	12°32.6			
19	296°44.9	128°51.5	38.3	82°02.3	58.2	182°15.4	33.3	297°55.8	14.9	Eltanin	90°42.0	51°29.4			
20	311°47.3	143°51.0	37.1	97°03.1	58.8	197°17.6	33.3	312°58.4	15.0	Kaus Aust.	83°31.7	-34°22.4			
21	326°49.8	158°50.6	•• 36.0	112°03.9	13°59.4	212°19.8	•• 33.2	328°01.0	•• 15.1	Vega	80°32.8	38°48.7			
22	341°52.3	173°50.2	34.8	127°04.7	14°00.0	227°21.9	33.2	343°03.7	15.2	Nunki	75°47.0	-26°15.9			
23	356°54.7	188°49.8	33.7	142°05.5	00.6	242°24.1	33.1	358°06.3	15.2	Altair	61°59.3	8°56.3			
Mer.pass. 23:12		ν -0.4' d -1.1' m -3.93		ν 0.8' d 0.6' m 1.57		ν 2.1' d -0.0' m -2.13		ν 2.6' d 0.1' m 0.62							

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	11°57.2	203°49.4	N06°32.5	157°06.3	S14°01.1	257°26.2	N21°33.1	13°09.0	S03°15.3	Venus	194°10.5	10:24			
1	26°59.7	218°49.0	31.4	172°07.1	01.7	272°28.4	33.0	28°11.6	15.4	Mars	146°28.3	13:33			
2	42°02.1	233°48.6	30.2	187°08.0	02.3	287°30.5	33.0	43°14.2	15.4	Jupiter	245°44.3	06:56			
3	57°04.6	248°48.1	•• 29.1	202°08.8	•• 02.9	302°32.7	•• 33.0	58°16.9	•• 15.5	Saturn	1°03.5	23:12			
4	72°07.1	263°47.7	27.9	217°09.6	03.5	317°34.8	32.9	73°19.5	15.6						
5	87°09.5	278°47.3	26.8	232°10.4	04.0	332°37.0	32.9	88°22.1	15.7						
6	102°12.0	293°46.9	N06°25.6	247°11.2	S14°04.6	347°39.1	N21°32.8	103°24.8	S03°15.7						
7	117°14.5	308°46.5	24.5	262°12.0	05.2	2°41.3	32.8	118°27.4	15.8						
8	132°16.9	323°46.1	23.3	277°12.8	05.8	17°43.4	32.7	133°30.0	15.9						
9	147°19.4	338°45.7	•• 22.2	292°13.6	•• 06.4	32°45.6	•• 32.7	148°32.7	•• 16.0						
10	162°21.8	353°45.3	21.0	307°14.4	06.9	47°47.7	32.7	163°35.3	16.0						
11	177°24.3	8°44.8	19.9	322°15.2	07.5	62°49.9	32.6	178°38.0	16.1						
12	192°26.8	23°44.4	N06°18.7	337°16.0	S14°08.1	77°52.1	N21°32.6	193°40.6	S03°16.2						
13	207°29.2	38°44.0	17.6	352°16.8	08.7	92°54.2	32.5	208°43.2	16.2						
14	222°31.7	53°43.6	16.4	7°17.7	09.3	107°56.4	32.5	223°45.9	16.3						
15	237°34.2	68°43.2	•• 15.3	22°18.5	•• 09.8	122°58.5	•• 32.4	238°48.5	•• 16.4						
16	252°36.6	83°42.8	14.1	37°19.3	10.4	138°00.7	32.4	253°51.1	16.5						
17	267°39.1	98°42.4	13.0	52°20.1	11.0	153°02.8	32.4	268°53.8	16.5						
18	282°41.6	113°42.0	N06°11.8	67°20.9	S14°11.6	168°05.0	N21°32.3	283°56.4	S03°16.6						
19	297°44.0	128°41.6	10.6	82°21.7	12.2	183°07.2	32.3	298°59.0	16.7						
20	312°46.5	143°41.1	09.5	97°22.5	12.7	198°09.3	32.2	314°01.7	16.8						
21	327°49.0	158°40.7	•• 08.3	112°23.3	•• 13.3	213°11.5	•• 32.2	329°04.3	•• 16.8						
22	342°51.4	173°40.3	07.2	127°24.1	13.9	228°13.6	32.1	344°06.9	16.9						
23	357°53.9	188°39.9	06.0	142°24.9	14.5	243°15.8	32.1	359°09.6	17.0						
Mer.pass. 23:08		ν -0.4' d -1.1' m -3.92		ν 0.8' d 0.6' m 1.56		ν 2.2' d -0.0' m -2.13		ν 2.6' d 0.1' m 0.63							

Oct 01 Wed	SHA	Mer.pass
Venus	194°10.5	10:24
Mars	146°28.3	13:33
Jupiter	245°44.3	06:56
Saturn	1°03.5	23:12

Oct 02 Thu	SHA	Mer.pass
Venus	193°01.3	10:24
Mars	145°48.8	13:32
Jupiter	245°36.6	06:53
Saturn	1°07.6	23:08

Oct 03 Fri	SHA	Mer.pass
Venus	191°52.2	10:25
Mars	145°09.1	13:31
Jupiter	245°29.0	06:49
Saturn		

Sun				Moon				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	182°33.7	S03°12.3		78°08.1	9.2'	S26°27.7	-6.2'	55.7'
1	197°33.9	13.3		92°36.4	9.3'	26°21.5	-6.3'	55.7'
2	212°34.1	14.2		107°04.6	9.3'	26°15.2	-6.5'	55.7'
3	227°34.3	15.2		121°32.9	9.3'	26°08.7	-6.6'	55.8'
4	242°34.5	16.2		136°01.2	9.3'	26°02.1	-6.7'	55.8'
5	257°34.7	17.1		150°29.6	9.4'	25°55.4	-6.9'	55.8'
6	272°34.9	S03°18.1		164°58.0	9.4'	S25°48.5	-7.0'	55.9'
7	287°35.1	19.1		179°26.3	9.4'	25°41.4	-7.2'	55.9'
8	302°35.3	20.0		193°54.8	9.4'	25°34.3	-7.3'	55.9'
9	317°35.5	21.0		208°23.2	9.5'	25°27.0	-7.4'	56.0'
10	332°35.7	22.0		222°51.7	9.5'	25°19.5	-7.6'	56.0'
11	347°35.9	22.9		237°20.2	9.5'	25°11.9	-7.7'	56.0'
12	2°36.1	S03°23.9		251°48.7	9.6'	S25°04.2	-7.9'	56.0'
13	17°36.3	24.9		266°17.3	9.6'	24°56.4	-8.0'	56.1'
14	32°36.5	25.8		280°45.9	9.6'	24°48.4	-8.1'	56.1'
15	47°36.7	26.8		295°14.5	9.7'	24°40.2	-8.3'	56.1'
16	62°36.9	27.8		309°43.2	9.7'	24°31.9	-8.4'	56.2'
17	77°37.1	28.8		324°11.9	9.7'	24°23.5	-8.5'	56.2'
18	92°37.3	S03°29.7		338°40.6	9.8'	S24°15.0	-8.7'	56.3'
19	107°37.5	30.7		353°09.3	9.8'	24°06.3	-8.8'	56.3'
20	122°37.7	31.7		7°38.1	9.8'	23°57.5	-8.9'	56.3'
21	137°37.9	32.6		22°06.9	9.9'	23°48.6	-9.1'	56.4'
22	152°38.1	33.6		36°35.8	9.9'	23°39.5	-9.2'	56.4'
23	167°38.3	34.6		51°04.7	9.9'	23°30.3	-9.3'	56.4'
SD = 16.0' d = 1.0'				SD = 15.2'				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	182°38.5	S03°35.5		65°33.6	10.0'	S23°20.9	-9.5'	56.5'
1	197°38.7	36.5		80°02.6	10.0'	23°11.4	-9.6'	56.5'
2	212°38.9	37.5		94°31.6	10.0'	23°01.8	-9.7'	56.5'
3	227°39.1	38.4		109°00.6	10.1'	22°52.1	-9.9'	56.6'
4	242°39.3	39.4		123°29.7	10.1'	22°42.2	-10.0'	56.6'
5	257°39.5	40.4		137°58.8	10.1'	22°32.2	-10.1'	56.6'
6	272°39.7	S03°41.3		152°27.9	10.2'	S22°22.1	-10.2'	56.7'
7	287°39.9	42.3		166°57.1	10.2'	22°11.9	-10.4'	56.7'
8	302°40.1	43.3		181°26.3	10.3'	22°01.5	-10.5'	56.7'
9	317°40.3	44.2		195°55.6	10.3'	21°51.0	-10.6'	56.8'
10	332°40.5	45.2		210°24.9	10.3'	21°40.4	-10.8'	56.8'
11	347°40.7	46.2		224°54.2	10.4'	21°29.6	-10.9'	56.9'
12	2°40.9	S03°47.1		239°23.5	10.4'	S21°18.7	-11.0'	56.9'
13	17°41.1	48.1		253°52.9	10.4'	21°07.7	-11.1'	56.9'
14	32°41.3	49.1		268°22.4	10.5'	20°56.6	-11.2'	57.0'
15	47°41.5	50.0		282°51.8	10.5'	20°45.4	-11.4'	57.0'
16	62°41.7	51.0		297°21.4	10.5'	20°34.0	-11.5'	57.0'
17	77°41.9	52.0		311°50.9	10.6'	20°22.5	-11.6'	57.1'
18	92°42.1	S03°52.9		326°20.5	10.6'	S20°10.9	-11.7'	57.1'
19	107°42.3	53.9		340°50.1	10.6'	19°59.2	-11.8'	57.2'
20	122°42.5	54.9		355°19.7	10.7'	19°47.4	-12.0'	57.2'
21	137°42.7	55.8		9°49.4	10.7'	19°35.4	-12.1'	57.2'
22	152°42.9	56.8		24°19.1	10.8'	19°23.4	-12.2'	57.3'
23	167°43.1	57.8		38°48.9	10.8'	19°11.2	-12.3'	57.3'
SD = 16.0' d = 1.0'				SD = 15.4'				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	182°43.3	S03°58.7		53°18.7	10.8'	S18°58.9	-12.4'	57.4'
1	197°43.5	03°59.7		67°48.5	10.9'	18°46.5	-12.5'	57.4'
2	212°43.7	04°00.7		82°18.4	10.9'	18°34.0	-12.6'	57.4'
3	227°43.9	01.6		96°48.3	10.9'	18°21.3	-12.7'	57.5'
4	242°44.1	02.6		111°18.2	11.0'	18°08.6	-12.8'	57.5'
5	257°44.3	03.6		125°48.1	11.0'	17°55.8	-13.0'	57.5'
6	272°44.5	S04°04.5		140°18.1	11.0'	S17°42.8	-13.1'	57.6'
7	287°44.7	05.5		154°48.2	11.1'	17°29.7	-13.2'	57.6'
8	302°44.9	06.4		169°18.2	11.1'	17°16.6	-13.3'	57.7'
9	317°45.1	07.4		183°48.3	11.1'	17°03.3	-13.4'	57.7'
10	332°45.2	08.4		198°18.4	11.1'	16°49.9	-13.5'	57.7'
11	347°45.4	09.3		212°48.6	11.2'	16°36.4	-13.6'	57.8'
12	2°45.6	S04°10.3		227°18.8	11.2'	S16°22.8	-13.7'	57.8'
13	17°45.8	11.3		241°49.0	11.2'	16°09.2	-13.8'	57.9'
14	32°46.0	12.2		256°19.2	11.3'	15°55.4	-13.9'	57.9'
15	47°46.2	13.2		270°49.5	11.3'	15°41.5	-14.0'	57.9'
16	62°46.4	14.2		285°19.8	11.3'	15°27.5	-14.1'	58.0'
17	77°46.6	15.1		299°50.1	11.3'	15°13.4	-14.2'	58.0'
18	92°46.8	S04°16.1		314°20.4	11.4'	S14°59.2	-14.3'	58.1'
19	107°47.0	17.1		328°50.8	11.4'	14°45.0	-14.4'	58.1'
20	122°47.2	18.0		343°21.2	11.4'	14°30.6	-14.5'	58.1'
21	137°47.4	19.0		357°51.6	11.4'	14°16.1	-14.6'	58.2'
22	152°47.6	19.9		12°22.0	11.5'	14°01.6	-14.6'	58.2'
23	167°47.8	20.9		26°52.5	11.5'	13°46.9	-14.7'	58.2'
SD = 16.0' d = 1.0'				SD = 15.6'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	03:55	05:17	06:24	17:12	18:19	19:40
N 70°	04:07	05:20	06:20	17:16	18:17	19:29
68°	04:16	05:22	06:17	17:20	18:15	19:20
66°	04:23	05:23	06:14	17:22	18:13	19:13
64°	04:29	05:25	06:12	17:25	18:12	19:08
62°	04:34	05:26	06:10	17:27	18:11	19:03
60°	04:38	05:27	06:08	17:29	18:10	18:59
N 58°	04:42	05:28	06:07	17:31	18:10	18:55
56°	04:45	05:28	06:05	17:32	18:09	18:52
54°	04:48	05:29	06:04	17:33	18:09	18:50
52°	04:50	05:29	06:03	17:35	18:08	18:47
50°	04:52	05:30	06:02	17:36	18:08	18:45
45°	04:56	05:30	05:59	17:38	18:08	18:42
N 40°	04:59	05:30	05:57	17:41	18:07	18:39
35°	05:01	05:30	05:56	17:42	18:08	18:37
30°	05:02	05:30	05:54	17:44	18:08	18:36
20°	05:04	05:29	05:51	17:47	18:09	18:35
N 10°	05:03	05:28	05:49	17:50	18:11	18:35
0°	05:01	05:25	05:46	17:53	18:13	18:37
S 10°	04:58	05:22	05:43	17:55	18:16	18:41
20°	04:53	05:18	05:40	17:58	18:20	18:46
30°	04:45	05:13	05:37	18:02	18:26	18:54
35°	04:40	05:10	05:35	18:04	18:30	19:00
40°	04:33	05:05	05:33	18:07	18:34	19:06
45°	04:25	05:00	05:30	18:09	18:39	19:14
S 50°	04:15	04:54	05:27	18:13	18:46	19:25
52°	04:10	04:51	05:25	18:15	18:49	19:30
54°	04:04	04:48	05:23	18:16	18:52	19:36
56°	03:58	04:44	05:21	18:18	18:56	19:42
58°	03:51	04:40	05:19	18:20	19:01	19:50
S 60°	03:42	04:35	05:17	18:23	19:06	19:59

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	■	■	18:45	■	■	■
N 70°	■	19:58	18:14	■	21:01	
68°	■	18:38	17:51	■	22:20	
66°	■	18:00	17:33	■	22:57	
64°	17:58	17:32	17:18	21:11	23:23	
62°	17:19	17:11	17:05	21:49	23:43	
60°	16:52	16:54	16:54	22:16	23:59	
N 58°	16:30	16:40	16:45	22:36		00:13
56°	16:13	16:27	16:37	22:54		00:24
54°	15:58	16:16	16:29	23:08		00:35
52°	15:45	16:06	16:23	23:20		00:44
50°	15:33	15:58	16:17	23:31		00:52
45°	15:09	15:39	16:04	23:54		01:09
N 40°	14:50	15:24	15:53		00:13	01:23
35°	14:34	15:11	15:44		00:28	01:35
30°	14:20	15:00	15:36		00:41	01:45
20°	13:56	14:40	15:21	00:05	01:04	02:02
N 10°	13:35	14:23	15:09	00:28	01:23	02:17
0°	13:16	14:07	14:57	00:49	01:41	02:31
S 10°	12:56	13:51	14:45	01:09	01:59	02:45
20°	12:35	13:34	14:33	01:32	02:17	03:00
30°	12:11	13:14	14:18	01:57	02:39	03:17
35°	11:57	13:02	14:10	02:12	02:52	03:26
40°	11:40	12:49	14:00	02:30	03:06	03:37
45°	11:20	12:33	13:49	02:50	03:23	03:50
S 50°	10:55	12:13	13:35	03:16	03:44	04:06
52°	10:43	12:04	13:28	03:29	03:54	04:13
54°	10:29	11:53	13:21	03:44	04:06	04:21
56°	10:12	11:41	13:13	04:00	04:18	04:30
58°	09:53	11:27	13:04	04:20	04:33	04:40
S 60°	09:28	11:10	12:53	04:46	04:50	04:52

Day	Sun		Mer. Pass	Moon		Age 10-12 60-79%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
01	10:15	10:25	11:50	19:28	07:02	
02	10:34	10:44	11:49	20:19	07:54	
03	10:53	11:03	11:49	21:09	08:44	

October 04, 05, 06 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	12°56.3	203°39.5	N06°04.9	157°25.7	S14°15.1	258°17.9	N21°32.1	14°12.2	S03°17.0	Alpheratz	357°33.8	29°14.1
1	27°58.8	218°39.1	03.7	172°26.5	15.6	273°20.1	32.0	29°14.8	17.1	Ankaa	353°06.1	-42°09.9
2	43°01.3	233°38.7	02.6	187°27.3	16.2	288°22.3	32.0	44°17.5	17.2	Schedar	349°29.8	56°40.8
3	58°03.7	248°38.3	•• 01.4	202°28.1	•• 16.8	303°24.4	•• 31.9	59°20.1	•• 17.3	Diphda	348°46.4	-17°50.6
4	73°06.2	263°37.9	06°00.2	217°28.9	17.4	318°26.6	31.9	74°22.7	17.3	Achernar	335°19.1	-57°06.2
5	88°08.7	278°37.5	05°59.1	232°29.7	18.0	333°28.7	31.8	89°25.4	17.4	Hamal	327°50.2	23°35.2
6	103°11.1	293°37.1	N05°57.9	247°30.5	S14°18.5	348°30.9	N21°31.8	104°28.0	S03°17.5	Polaris	313°20.5	89°22.2
7	118°13.6	308°36.6	56.8	262°31.3	19.1	3°33.1	31.8	119°30.7	17.6	Acamar	315°10.9	-40°11.8
8	133°16.1	323°36.2	55.6	277°32.1	19.7	18°35.2	31.7	134°33.3	17.6	Menkar	314°05.3	4°11.6
9	148°18.5	338°35.8	•• 54.4	292°32.9	•• 20.3	33°37.4	•• 31.7	149°35.9	•• 17.7	Mirfak	308°27.0	49°57.1
10	163°21.0	353°35.4	53.3	307°33.7	20.8	48°39.5	31.6	164°38.6	17.8	Aldebaran	290°38.7	16°33.7
11	178°23.4	8°35.0	52.1	322°34.5	21.4	63°41.7	31.6	179°41.2	17.8	Rigel	281°03.2	-8°10.1
12	193°25.9	23°34.6	N05°51.0	337°35.3	S14°22.0	78°43.9	N21°31.5	194°43.8	S03°17.9	Capella	280°20.8	46°01.3
13	208°28.4	38°34.2	49.8	352°36.1	22.6	93°46.0	31.5	209°46.5	18.0	Bellatrix	278°22.1	6°22.5
14	223°30.8	53°33.8	48.6	7°36.9	23.1	108°48.2	31.5	224°49.1	18.1	Elnath	278°01.0	28°37.7
15	238°33.3	68°33.4	•• 47.5	22°37.7	•• 23.7	123°50.4	•• 31.4	239°51.7	•• 18.1	Alnilam	275°37.0	-1°11.0
16	253°35.8	83°33.0	46.3	37°38.5	24.3	138°52.5	31.4	254°54.4	18.2	Betelgeuse	270°51.4	7°24.8
17	268°38.2	98°32.6	45.1	52°39.3	24.9	153°54.7	31.3	269°57.0	18.3	Canopus	263°52.1	-52°42.2
18	283°40.7	113°32.2	N05°44.0	67°40.1	S14°25.4	168°56.8	N21°31.3	284°59.6	S03°18.4	Sirius	258°25.7	-16°44.8
19	298°43.2	128°31.7	42.8	82°40.9	26.0	183°59.0	31.2	300°02.3	18.4	Adhara	255°05.4	-29°00.1
20	313°45.6	143°31.3	41.7	97°41.7	26.6	199°01.2	31.2	315°04.9	18.5	Procyon	244°50.3	5°09.7
21	328°48.1	158°30.9	•• 40.5	112°42.5	•• 27.2	214°03.3	•• 31.2	330°07.5	•• 18.6	Pollux	243°16.7	27°57.8
22	343°50.6	173°30.5	39.3	127°43.3	27.7	229°05.5	31.1	345°10.2	18.6	Avior	234°14.8	-59°35.1
23	358°53.0	188°30.1	38.2	142°44.1	28.3	244°07.7	31.1	0°12.8	18.7	Suhail	222°46.1	-43°31.9
Mer.pass. 23:04		ν -0.4' d -1.2' m -3.92		ν 0.8' d 0.6' m 1.56		ν 2.2' d -0.0' m -2.14		ν 2.6' d 0.1' m 0.63				

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	13°55.5	203°29.7	N05°37.0	157°44.9	S14°28.9	259°09.8	N21°31.0	15°15.4	S03°18.8
1	28°57.9	218°29.3	35.8	172°45.7	29.5	274°12.0	31.0	30°18.1	18.9
2	44°00.4	233°28.9	34.7	187°46.5	30.0	289°14.2	31.0	45°20.7	18.9
3	59°02.9	248°28.5	•• 33.5	202°47.3	•• 30.6	304°16.3	•• 30.9	60°23.3	•• 19.0
4	74°05.3	263°28.1	32.3	217°48.1	31.2	319°18.5	30.9	75°26.0	19.1
5	89°07.8	278°27.7	31.2	232°48.9	31.8	334°20.7	30.8	90°28.6	19.1
6	104°10.3	293°27.3	N05°30.0	247°49.7	S14°32.3	349°22.8	N21°30.8	105°31.2	S03°19.2
7	119°12.7	308°26.9	28.8	262°50.5	32.9	4°25.0	30.7	120°33.9	19.3
8	134°15.2	323°26.5	27.7	277°51.3	33.5	19°27.2	30.7	135°36.5	19.4
9	149°17.7	338°26.1	•• 26.5	292°52.1	•• 34.1	34°29.3	•• 30.7	150°39.1	•• 19.4
10	164°20.1	353°25.7	25.3	307°52.9	34.6	49°31.5	30.6	165°41.8	19.5
11	179°22.6	8°25.3	24.2	322°53.7	35.2	64°33.7	30.6	180°44.4	19.6
12	194°25.0	23°24.9	N05°23.0	337°54.5	S14°35.8	79°35.8	N21°30.5	195°47.0	S03°19.6
13	209°27.5	38°24.4	21.8	352°55.3	36.4	94°38.0	30.5	210°49.7	19.7
14	224°30.0	53°24.0	20.7	7°56.1	36.9	109°40.2	30.5	225°52.3	19.8
15	239°32.4	68°23.6	•• 19.5	22°56.9	•• 37.5	124°42.4	•• 30.4	240°54.9	•• 19.9
16	254°34.9	83°23.2	18.3	37°57.7	38.1	139°44.5	30.4	255°57.6	19.9
17	269°37.4	98°22.8	17.1	52°58.5	38.6	154°46.7	30.3	271°00.2	20.0
18	284°39.8	113°22.4	N05°16.0	67°59.2	S14°39.2	169°48.9	N21°30.3	286°02.8	S03°20.1
19	299°42.3	128°22.0	14.8	83°00.0	39.8	184°51.0	30.2	301°05.5	20.1
20	314°44.8	143°21.6	13.6	98°00.8	40.4	199°53.2	30.2	316°08.1	20.2
21	329°47.2	158°21.2	•• 12.5	113°01.6	•• 40.9	214°55.4	•• 30.2	331°10.7	•• 20.3
22	344°49.7	173°20.8	11.3	128°02.4	41.5	229°57.5	30.1	346°13.4	20.4
23	359°52.2	188°20.4	10.1	143°03.2	42.1	244°59.7	30.1	1°16.0	20.4
Mer.pass. 23:01		ν -0.4' d -1.2' m -3.92		ν 0.8' d 0.6' m 1.56		ν 2.2' d -0.0' m -2.15		ν 2.6' d 0.1' m 0.63	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	14°54.6	203°20.0	N05°08.9	158°04.0	S14°42.6	260°01.9	N21°30.0	16°18.6	S03°20.5
1	29°57.1	218°19.6	07.8	173°04.8	43.2	275°04.1	30.0	31°21.3	20.6
2	44°59.5	233°19.2	06.6	188°05.6	43.8	290°06.2	30.0	46°23.9	20.6
3	60°02.0	248°18.8	•• 05.4	203°06.4	•• 44.3	305°08.4	•• 29.9	61°26.5	•• 20.7
4	75°04.5	263°18.4	04.3	218°07.2	44.9	320°10.6	29.9	76°29.2	20.8
5	90°06.9	278°18.0	03.1	233°08.0	45.5	335°12.8	29.8	91°31.8	20.9
6	105°09.4	293°17.6	N05°01.9	248°08.8	S14°46.1	350°14.9	N21°29.8	106°34.4	S03°20.9
7	120°11.9	308°17.2	05°00.7	263°09.5	46.6	5°17.1	29.8	121°37.1	21.0
8	135°14.3	323°16.8	04°59.6	278°10.3	47.2	20°19.3	29.7	136°39.7	21.1
9	150°16.8	338°16.4	•• 58.4	293°11.1	•• 47.8	35°21.5	•• 29.7	151°42.3	•• 21.1
10	165°19.3	353°16.0	57.2	308°11.9	48.3	50°23.6	29.6	166°45.0	21.2
11	180°21.7	8°15.6	56.0	323°12.7	48.9	65°25.8	29.6	181°47.6	21.3
12	195°24.2	23°15.2	N04°54.9	338°13.5	S14°49.5	80°28.0	N21°29.6	196°50.2	S03°21.3
13	210°26.7	38°14.8	53.7	353°14.3	50.0	95°30.2	29.5	211°52.8	21.4
14	225°29.1	53°14.4	52.5	8°15.1	50.6	110°32.3	29.5	226°55.5	21.5
15	240°31.6	68°14.0	•• 51.3	23°15.9	•• 51.2	125°34.5	•• 29.4	241°58.1	•• 21.6
16	255°34.0	83°13.6	50.2	38°16.6	51.7	140°36.7	29.4	257°00.7	21.6
17	270°36.5	98°13.2	49.0	53°17.4	52.3	155°38.9	29.3	272°03.4	21.7
18	285°39.0	113°12.8	N04°47.8	68°18.2	S14°52.9	170°41.0	N21°29.3	287°06.0	S03°21.8
19	300°41.4	128°12.4	46.6	83°19.0	53.4	185°43.2	29.3	302°08.6	21.8
20	315°43.9	143°12.0	45.4	98°19.8	54.0	200°45.4	29.2	317°11.3	21.9
21	330°46.4	158°11.6	•• 44.3	113°20.6	•• 54.6	215°47.6	•• 29.2	332°13.9	•• 22.0
22	345°48.8	173°11.2	43.1	128°21.4	55.1	230°49.7	29.1	347°16.5	22.1
23	0°51.3	188°10.8	41.9	143°22.2	55.7	245°51.9	29.1	2°19.2	22.1
Mer.pass. 22:57		ν -0.4' d -1.2' m -3.92		ν 0.8' d 0.6' m 1.56		ν 2.2' d -0.0' m -2.15		ν 2.6' d 0.1' m 0.64	

Oct 04 Sat	SHA	Mer.pass
Venus	190°43.2	10:26
Mars	144°29.4	13:30
Jupiter	245°21.6	06:46
Saturn	1°15.9	22:59

Oct 05 Sun	SHA	Mer.pass
Venus	189°34.2	10:26
Mars	143°49.4	13:28
Jupiter	245°14.4	06:42
Saturn	1°20.0	22:55

Oct 06 Mon	SHA	Mer.pass
Venus	188°25.4	10:27
Mars	143°09.4	13:27
Jupiter	245°07.3	06:39
Saturn	1°24.0	22:51

Horizontal parallax		
Venus:		0.1
Mars:		0.1

Sun							Moon								
Sat	GHA	Dec	GHA	ν	Dec	d	HP	Sat	GHA	ν	Dec	d	HP		
0	182°48.0	S04°21.9	41°23.0	11.5'	S13°32.2	-14.8'	58.3'	0	182°48.0	S04°21.9	41°23.0	11.5'	S13°32.2	-14.8'	58.3'
1	197°48.2	22.8	55°53.5	11.5'	13°17.4	-14.9'	58.3'	1	197°48.2	22.8	55°53.5	11.5'	13°17.4	-14.9'	58.3'
2	212°48.3	23.8	70°24.0	11.6'	13°02.5	-15.0'	58.4'	2	212°48.3	23.8	70°24.0	11.6'	13°02.5	-15.0'	58.4'
3	227°48.5	· · 24.8	84°54.6	11.6'	12°47.5	-15.1'	58.4'	3	227°48.5	· · 24.8	84°54.6	11.6'	12°47.5	-15.1'	58.4'
4	242°48.7	25.7	99°25.2	11.6'	12°32.4	-15.2'	58.4'	4	242°48.7	25.7	99°25.2	11.6'	12°32.4	-15.2'	58.4'
5	257°48.9	26.7	113°55.8	11.6'	12°17.3	-15.2'	58.5'	5	257°48.9	26.7	113°55.8	11.6'	12°17.3	-15.2'	58.5'
6	272°49.1	S04°27.7	128°26.4	11.6'	S12°02.0	-15.3'	58.5'	6	272°49.1	S04°27.7	128°26.4	11.6'	S12°02.0	-15.3'	58.5'
7	287°49.3	28.6	142°57.0	11.6'	11°46.7	-15.4'	58.6'	7	287°49.3	28.6	142°57.0	11.6'	11°46.7	-15.4'	58.6'
8	302°49.5	29.6	157°27.6	11.7'	11°31.3	-15.5'	58.6'	8	302°49.5	29.6	157°27.6	11.7'	11°31.3	-15.5'	58.6'
9	317°49.7	· · 30.5	171°58.3	11.7'	11°15.8	-15.6'	58.6'	9	317°49.7	· · 30.5	171°58.3	11.7'	11°15.8	-15.6'	58.6'
10	332°49.9	31.5	186°28.9	11.7'	11°00.3	-15.6'	58.7'	10	332°49.9	31.5	186°28.9	11.7'	11°00.3	-15.6'	58.7'
11	347°50.1	32.5	200°59.6	11.7'	10°44.6	-15.7'	58.7'	11	347°50.1	32.5	200°59.6	11.7'	10°44.6	-15.7'	58.7'
12	2°50.3	S04°33.4	215°30.3	11.7'	S10°28.9	-15.8'	58.7'	12	2°50.3	S04°33.4	215°30.3	11.7'	S10°28.9	-15.8'	58.7'
13	17°50.5	34.4	230°01.0	11.7'	10°13.1	-15.9'	58.8'	13	17°50.5	34.4	230°01.0	11.7'	10°13.1	-15.9'	58.8'
14	32°50.7	35.4	244°31.7	11.7'	09°57.3	-15.9'	58.8'	14	32°50.7	35.4	244°31.7	11.7'	09°57.3	-15.9'	58.8'
15	47°50.8	· · 36.3	259°02.4	11.7'	09°41.3	-16.0'	58.9'	15	47°50.8	· · 36.3	259°02.4	11.7'	09°41.3	-16.0'	58.9'
16	62°51.0	37.3	273°33.2	11.7'	09°25.3	-16.1'	58.9'	16	62°51.0	37.3	273°33.2	11.7'	09°25.3	-16.1'	58.9'
17	77°51.2	38.2	288°03.9	11.7'	09°09.3	-16.1'	58.9'	17	77°51.2	38.2	288°03.9	11.7'	09°09.3	-16.1'	58.9'
18	92°51.4	S04°39.2	302°34.6	11.7'	S08°53.2	-16.2'	59.0'	18	92°51.4	S04°39.2	302°34.6	11.7'	S08°53.2	-16.2'	59.0'
19	107°51.6	40.2	317°05.4	11.8'	08°37.0	-16.3'	59.0'	19	107°51.6	40.2	317°05.4	11.8'	08°37.0	-16.3'	59.0'
20	122°51.8	41.1	331°36.1	11.8'	08°20.7	-16.3'	59.0'	20	122°51.8	41.1	331°36.1	11.8'	08°20.7	-16.3'	59.0'
21	137°52.0	· · 42.1	346°06.9	11.8'	08°04.4	-16.4'	59.1'	21	137°52.0	· · 42.1	346°06.9	11.8'	08°04.4	-16.4'	59.1'
22	152°52.2	43.0	0°37.6	11.8'	07°48.0	-16.4'	59.1'	22	152°52.2	43.0	0°37.6	11.8'	07°48.0	-16.4'	59.1'
23	167°52.4	44.0	15°08.4	11.8'	07°31.5	-16.5'	59.1'	23	167°52.4	44.0	15°08.4	11.8'	07°31.5	-16.5'	59.1'
SD = 16.0'			$d = 1.0'$		SD = 15.9'										
Sun	GHA	Dec	GHA	ν	Dec	d	HP	Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	182°52.6	S04°45.0	29°39.2	11.8'	S07°15.0	-16.6'	59.2'	0	182°52.6	S04°45.0	29°39.2	11.8'	S07°15.0	-16.6'	59.2'
1	197°52.7	45.9	44°09.9	11.8'	06°58.5	-16.6'	59.2'	1	197°52.7	45.9	44°09.9	11.8'	06°58.5	-16.6'	59.2'
2	212°52.9	46.9	58°40.7	11.8'	06°41.8	-16.7'	59.2'	2	212°52.9	46.9	58°40.7	11.8'	06°41.8	-16.7'	59.2'
3	227°53.1	· · 47.9	73°11.4	11.7'	06°25.2	-16.7'	59.3'	3	227°53.1	· · 47.9	73°11.4	11.7'	06°25.2	-16.7'	59.3'
4	242°53.3	48.8	87°42.2	11.7'	06°08.5	-16.8'	59.3'	4	242°53.3	48.8	87°42.2	11.7'	06°08.5	-16.8'	59.3'
5	257°53.5	49.8	102°12.9	11.7'	05°51.7	-16.8'	59.4'	5	257°53.5	49.8	102°12.9	11.7'	05°51.7	-16.8'	59.4'
6	272°53.7	S04°50.7	116°43.6	11.7'	S05°34.8	-16.9'	59.4'	6	272°53.7	S04°50.7	116°43.6	11.7'	S05°34.8	-16.9'	59.4'
7	287°53.9	51.7	131°14.4	11.7'	05°18.0	-16.9'	59.4'	7	287°53.9	51.7	131°14.4	11.7'	05°18.0	-16.9'	59.4'
8	302°54.1	52.7	145°45.1	11.7'	05°01.1	-17.0'	59.5'	8	302°54.1	52.7	145°45.1	11.7'	05°01.1	-17.0'	59.5'
9	317°54.3	· · 53.6	160°15.8	11.7'	04°44.1	-17.0'	59.5'	9	317°54.3	· · 53.6	160°15.8	11.7'	04°44.1	-17.0'	59.5'
10	332°54.4	54.6	174°46.5	11.7'	04°27.1	-17.1'	59.5'	10	332°54.4	54.6	174°46.5	11.7'	04°27.1	-17.1'	59.5'
11	347°54.6	55.5	189°17.2	11.7'	04°10.0	-17.1'	59.6'	11	347°54.6	55.5	189°17.2	11.7'	04°10.0	-17.1'	59.6'
12	2°54.8	S04°56.5	203°47.8	11.7'	S03°52.9	-17.1'	59.6'	12	2°54.8	S04°56.5	203°47.8	11.7'	S03°52.9	-17.1'	59.6'
13	17°55.0	57.5	218°18.5	11.6'	03°35.8	-17.2'	59.6'	13	17°55.0	57.5	218°18.5	11.6'	03°35.8	-17.2'	59.6'
14	32°55.2	58.4	232°49.1	11.6'	03°18.6	-17.2'	59.6'	14	32°55.2	58.4	232°49.1	11.6'	03°18.6	-17.2'	59.6'
15	47°55.4	04°59.4	247°19.7	11.6'	03°01.4	-17.2'	59.7'	15	47°55.4	04°59.4	247°19.7	11.6'	03°01.4	-17.2'	59.7'
16	62°55.6	05°00.3	261°50.4	11.6'	02°44.2	-17.3'	59.7'	16	62°55.6	05°00.3	261°50.4	11.6'	02°44.2	-17.3'	59.7'
17	77°55.7	01.3	276°20.9	11.6'	02°26.9	-17.3'	59.7'	17	77°55.7	01.3	276°20.9	11.6'	02°26.9	-17.3'	59.7'
18	92°55.9	S05°02.3	290°51.5	11.5'	S02°09.6	-17.3'	59.8'	18	92°55.9	S05°02.3	290°51.5	11.5'	S02°09.6	-17.3'	59.8'
19	107°56.1	03.2	305°22.0	11.5'	01°52.3	-17.4'	59.8'	19	107°56.1	03.2	305°22.0	11.5'	01°52.3	-17.4'	59.8'
20	122°56.3	04.2	319°52.6	11.5'	01°34.9	-17.4'	59.8'	20	122°56.3	04.2	319°52.6	11.5'	01°34.9	-17.4'	59.8'
21	137°56.5	· · 05.1	334°23.1	11.5'	01°17.5	-17.4'	59.9'	21	137°56.5	· · 05.1	334°23.1	11.5'	01°17.5	-17.4'	59.9'
22	152°56.7	06.1	348°53.5	11.4'	01°00.1	-17.4'	59.9'	22	152°56.7	06.1	348°53.5	11.4'	01°00.1	-17.4'	59.9'
23	167°56.9	07.0	3°24.0	11.4'	00°42.7	-17.5'	59.9'	23	167°56.9	07.0	3°24.0	11.4'	00°42.7	-17.5'	59.9'
SD = 16.0'			$d = 1.0'$		SD = 16.1'										
Mon	GHA	Dec	GHA	ν	Dec	d	HP	Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	182°57.1	S05°08.0	17°54.4	11.4'	S00°25.2	-17.5'	60.0'	0	182°57.1	S05°08.0	17°54.4	11.4'	S00°25.2	-17.5'	60.0'
1	197°57.2	09.0	32°24.8	11.4'	S00°07.7	-17.5'	60.0'	1	197°57.2	09.0	32°24.8	11.4'	S00°07.7	-17.5'	60.0'
2	212°57.4	09.9	46°55.1	11.3'	N00°09.8	17.5'	60.0'	2	212°57.4	09.9	46°55.1	11.3'	N00°09.8	17.5'	60.0'
3	227°57.6	· · 10.9	61°25.4	11.3'	00°27.3	17.5'	60.0'	3	227°57.6	· · 10.9	61°25.4	11.3'	00°27.3	17.5'	60.0'
4	242°57.8	11.8	75°55.7	11.3'	00°44.8	17.5'	60.1'	4	242°57.8	11.8	75°55.7	11.3'	00°44.8	17.5'	60.1'
5	257°58.0	12.8	90°26.0	11.2'	01°02.3	17.5'	60.1'	5	257°58.0	12.8	90°26.0	11.2'	01°02.3	17.5'	60.1'
6	272°58.2	S05°13.8	104°56.2	11.2'	N01°19.9	17.6'	60.1'	6	272°58.2	S05°13.8	104°56.2	11.2'	N01°19.9	17.6'	60.1'
7	287°58.3	14.7	119°26.4	11.1'	01°37.4	17.6'	60.1'	7	287°58.3	14.7	119°26.4	11.1'	01°37.4	17.6'	60.1'
8	302°58.5	15.7	133°56.5	11.1'	01°55.0	17.6'	60.2'	8	302°58.5	15.7	133°56.5	11.1'	01°55.0	17.6'	60.2'
9	317°58.7	· · 16.6	148°26.6	11.1'	02°12.6	17.6'	60.2'	9	317°58.7	· · 16.6	148°26.6	11.1'	02°12.6	17.6'	60.2'
10	332°58.9	17.6	162°56.7	11.0'	02°3°3										

October 07, 08, 09 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	15°53.8	203°10.4	N04°40.7	158°22.9	S14°56.3	260°54.1	N21°29.1	17°21.8	S03°22.2	Alpheratz	357°33.8	29°14.1
1	30°56.2	218°10.0	39.6	173°23.7	56.8	275°56.3	29.0	32°24.4	22.3	Ankaa	353°06.1	-42°09.9
2	45°58.7	233°09.6	38.4	188°24.5	57.4	290°58.5	29.0	47°27.1	22.3	Schedar	349°29.8	56°40.8
3	61°01.1	248°09.2	•• 37.2	203°25.3	•• 58.0	306°00.6	•• 28.9	62°29.7	•• 22.4	Diphda	348°46.4	-17°50.6
4	76°03.6	263°08.8	36.0	218°26.1	58.5	321°02.8	28.9	77°32.3	22.5	Achernar	335°19.1	-57°06.2
5	91°06.1	278°08.4	34.8	233°26.9	59.1	336°05.0	28.9	92°35.0	22.5	Hamal	327°50.2	23°35.2
6	106°08.5	293°08.0	N04°33.7	248°27.7	S14°59.7	351°07.2	N21°28.8	107°37.6	S03°22.6	Polaris	313°19.5	89°22.2
7	121°11.0	308°07.6	32.5	263°28.4	15°00.2	6°09.4	28.8	122°40.2	22.7	Acamar	315°10.9	-40°11.8
8	136°13.5	323°07.2	31.3	278°29.2	00.8	21°11.5	28.7	137°42.8	22.8	Menkar	314°05.2	4°11.6
9	151°15.9	338°06.8	•• 30.1	293°30.0	•• 01.4	36°13.7	•• 28.7	152°45.5	•• 22.8	Mirfak	308°27.0	49°57.1
10	166°18.4	353°06.4	28.9	308°30.8	01.9	51°15.9	28.7	167°48.1	22.9	Aldebaran	290°38.7	16°33.7
11	181°20.9	8°06.0	27.7	323°31.6	02.5	66°18.1	28.6	182°50.7	23.0	Rigel	281°03.1	-8°10.1
12	196°23.3	23°05.6	N04°26.6	338°32.4	S15°03.1	81°20.3	N21°28.6	197°53.4	S03°23.0	Capella	280°20.8	46°01.3
13	211°25.8	38°05.2	25.4	353°33.1	03.6	96°22.5	28.5	212°56.0	23.1	Bellatrix	278°22.1	6°22.5
14	226°28.3	53°04.8	24.2	8°33.9	04.2	111°24.6	28.5	227°58.6	23.2	Elnath	278°01.0	28°37.7
15	241°30.7	68°04.4	•• 23.0	23°34.7	•• 04.8	126°26.8	•• 28.5	243°01.3	•• 23.2	Alnilam	275°37.0	-1°11.0
16	256°33.2	83°04.0	21.8	38°35.5	05.3	141°29.0	28.4	258°03.9	23.3	Betelgeuse	270°51.3	7°24.8
17	271°35.6	98°03.6	20.6	53°36.3	05.9	156°31.2	28.4	273°06.5	23.4	Canopus	263°52.0	-52°42.2
18	286°38.1	113°03.2	N04°19.5	68°37.0	S15°06.5	171°33.4	N21°28.3	288°09.1	S03°23.5	Sirius	258°25.7	-16°44.8
19	301°40.6	128°02.8	18.3	83°37.8	07.0	186°35.6	28.3	303°11.8	23.5	Adhara	255°05.4	-29°00.1
20	316°43.0	143°02.4	17.1	98°38.6	07.6	201°37.7	28.3	318°14.4	23.6	Procyon	244°50.3	5°09.7
21	331°45.5	158°02.0	•• 15.9	113°39.4	•• 08.1	216°39.9	•• 28.2	333°17.0	•• 23.7	Pollux	243°16.6	27°57.8
22	346°48.0	173°01.6	14.7	128°40.2	08.7	231°42.1	28.2	348°19.7	23.7	Avior	234°14.7	-59°35.1
23	1°50.4	188°01.2	13.5	143°41.0	09.3	246°44.3	28.2	3°22.3	23.8	Suhail	222°46.1	-43°31.9
Mer.pass. 22:53		ν -0.4' d -1.2' m -3.92		ν 0.8' d 0.6' m 1.55		ν 2.2' d -0.0' m -2.16		ν 2.6' d 0.1' m 0.64		Miaplacidus	221°38.7	-69°49.0
										Alphard	217°47.4	-8°46.0
										Regulus	207°34.0	11°50.6
										Dubhe	193°40.8	61°36.6
										Denebola	182°24.6	14°25.8
										Gienah	175°43.3	-17°41.0
										Acrux	173°00.2	-63°14.4
										Gacrux	171°51.6	-57°15.4
										Alioth	166°13.0	55°49.2
										Spica	158°22.0	-11°17.7
										Alkaid	152°52.0	49°11.2
										Hadar	148°35.8	-60°29.9
										Menkent	147°57.3	-36°29.8
										Arcturus	145°47.7	19°03.0
										Rigil Kent.	139°40.1	-60°56.6
										Kochab	137°20.8	74°03.1
										Zuben'ubi	136°55.7	-16°08.9
										Alphecca	126°03.5	26°37.8
										Antares	112°15.3	-26°29.4
										Atria	107°09.3	-69°04.6
										Sabik	102°02.3	-15°45.4
										Shaula	96°09.7	-37°07.4
										Rasalhague	95°58.1	12°32.6
										Eltanin	90°42.1	51°29.4
										Kaus Aust.	83°31.8	-34°22.4
										Vega	80°32.9	38°48.7
										Nunki	75°47.0	-26°15.9
										Altair	61°59.3	8°56.3
										Peacock	53°04.6	-56°39.3
										Deneb	49°25.2	45°22.6
										Enif	33°38.0	9°59.8
										Al Na'ir	27°31.8	-46°50.2
										Fomalhaut	15°13.5	-29°29.1
										Scheat	13°44.3	28°13.6
										Markab	13°29.0	15°20.8
										Oct 07 Tue	SHA	Mer.pass
										Venus	187°16.6	10:28
										Mars	142°29.2	13:26
										Jupiter	245°00.4	06:35
										Saturn	1°28.0	22:47
										Oct 08 Wed	SHA	Mer.pass
										Venus	186°07.9	10:28
										Mars	141°48.8	13:25
										Jupiter	244°53.6	06:32
										Saturn	1°32.0	22:42
										Oct 09 Thu	SHA	Mer.pass
										Venus	184°59.3	10:29
										Mars	141°08.3	13:23
										Jupiter	244°47.0	06:28
										Saturn	1°36.0	22:38
										Horizontal parallax		
										Venus:	0.1	
										Mars:	0.1	

Sun				Moon				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	183°01.5	S05°31.0		5°52.6	10.3'	N06°35.4	17.4'	60.5'
1	198°01.6		31.9	20°21.9	10.2'	06°52.8	17.4'	60.5'
2	213°01.8		32.9	34°51.2	10.2'	07°10.1	17.3'	60.6'
3	228°02.0	· ·	33.8	49°20.3	10.1'	07°27.4	17.3'	60.6'
4	243°02.2		34.8	63°49.5	10.1'	07°44.7	17.3'	60.6'
5	258°02.4		35.8	78°18.5	10.0'	08°02.0	17.2'	60.6'
6	273°02.5	S05°	36.7	92°47.5	9.9'	N08°19.2	17.2'	60.6'
7	288°02.7		37.7	107°16.4	9.9'	08°36.4	17.1'	60.6'
8	303°02.9		38.6	121°45.3	9.8'	08°53.6	17.1'	60.7'
9	318°03.1	· ·	39.6	136°14.1	9.7'	09°10.7	17.1'	60.7'
10	333°03.3		40.5	150°42.8	9.6'	09°27.7	17.0'	60.7'
11	348°03.4		41.5	165°11.4	9.6'	09°44.7	17.0'	60.7'
12	3°03.6	S05°	42.4	179°40.0	9.5'	N10°01.7	16.9'	60.7'
13	18°03.8		43.4	194°08.5	9.4'	10°18.6	16.9'	60.7'
14	33°04.0		44.3	208°37.0	9.4'	10°35.5	16.8'	60.7'
15	48°04.2	· ·	45.3	223°05.3	9.3'	10°52.3	16.8'	60.7'
16	63°04.3		46.3	237°33.6	9.2'	11°09.1	16.7'	60.8'
17	78°04.5		47.2	252°01.8	9.1'	11°25.7	16.6'	60.8'
18	93°04.7	S05°	48.2	266°29.9	9.0'	N11°42.4	16.6'	60.8'
19	108°04.9		49.1	280°58.0	9.0'	11°58.9	16.5'	60.8'
20	123°05.0		50.1	295°25.9	8.9'	12°15.4	16.4'	60.8'
21	138°05.2	· ·	51.0	309°53.8	8.8'	12°31.9	16.4'	60.8'
22	153°05.4		52.0	324°21.6	8.7'	12°48.2	16.3'	60.8'
23	168°05.6		52.9	338°49.3	8.6'	13°04.5	16.2'	60.8'
SD = 16.0' d = 1.0'				SD = 16.5'				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	183°05.8	S05°	53.9	353°17.0	8.6'	N13°20.7	16.1'	60.8'
1	198°05.9		54.8	7°44.5	8.5'	13°36.9	16.1'	60.8'
2	213°06.1		55.8	22°12.0	8.4'	13°52.9	16.0'	60.8'
3	228°06.3	· ·	56.7	36°39.4	8.3'	14°08.9	15.9'	60.8'
4	243°06.5		57.7	51°06.7	8.2'	14°24.8	15.8'	60.8'
5	258°06.6		58.6	65°33.9	8.1'	14°40.6	15.7'	60.8'
6	273°06.8	S05°	59.6	80°01.0	8.0'	N14°56.4	15.6'	60.9'
7	288°07.0	06°	00.5	94°28.0	7.9'	15°12.0	15.5'	60.9'
8	303°07.2		01.5	108°55.0	7.8'	15°27.5	15.4'	60.9'
9	318°07.3	· ·	02.4	123°21.8	7.8'	15°43.0	15.4'	60.9'
10	333°07.5		03.4	137°48.6	7.7'	15°58.3	15.3'	60.9'
11	348°07.7		04.4	152°15.2	7.6'	16°13.6	15.2'	60.9'
12	3°07.9	S06°	05.3	166°41.8	7.5'	N16°28.7	15.0'	60.9'
13	18°08.0		06.3	181°08.3	7.4'	16°43.8	14.9'	60.9'
14	33°08.2		07.2	195°34.7	7.3'	16°58.7	14.8'	60.9'
15	48°08.4	· ·	08.2	210°01.0	7.2'	17°13.6	14.7'	60.9'
16	63°08.6		09.1	224°27.2	7.1'	17°28.3	14.6'	60.9'
17	78°08.7		10.1	238°53.3	7.0'	17°42.9	14.5'	60.9'
18	93°08.9	S06°	11.0	253°19.3	6.9'	N17°57.4	14.4'	60.9'
19	108°09.1		12.0	267°45.2	6.8'	18°11.8	14.3'	60.9'
20	123°09.3		12.9	282°11.1	6.7'	18°26.1	14.1'	60.9'
21	138°09.4	· ·	13.9	296°36.8	6.6'	18°40.2	14.0'	60.8'
22	153°09.6		14.8	311°02.4	6.5'	18°54.2	13.9'	60.8'
23	168°09.8		15.8	325°28.0	6.4'	19°08.1	13.8'	60.8'
SD = 16.0' d = 1.0'				SD = 16.6'				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	183°10.0	S06°	16.7	339°53.4	6.4'	N19°21.9	13.6'	60.8'
1	198°10.1		17.7	354°18.8	6.3'	19°35.6	13.5'	60.8'
2	213°10.3		18.6	8°44.0	6.2'	19°49.1	13.4'	60.8'
3	228°10.5	· ·	19.6	23°09.2	6.1'	20°02.5	13.2'	60.8'
4	243°10.6		20.5	37°34.2	6.0'	20°15.7	13.1'	60.8'
5	258°10.8		21.5	51°59.2	5.9'	20°28.8	13.0'	60.8'
6	273°11.0	S06°	22.4	66°24.1	5.8'	N20°41.8	12.8'	60.8'
7	288°11.2		23.3	80°48.9	5.7'	20°54.6	12.7'	60.8'
8	303°11.3		24.3	95°13.6	5.6'	21°07.3	12.5'	60.8'
9	318°11.5	· ·	25.2	109°38.2	5.5'	21°19.9	12.4'	60.8'
10	333°11.7		26.2	124°02.7	5.4'	21°32.3	12.2'	60.8'
11	348°11.8		27.1	138°27.1	5.3'	21°44.5	12.1'	60.7'
12	3°12.0	S06°	28.1	152°51.4	5.2'	N21°56.6	11.9'	60.7'
13	18°12.2		29.0	167°15.6	5.1'	22°08.6	11.8'	60.7'
14	33°12.3		30.0	181°39.8	5.1'	22°20.3	11.6'	60.7'
15	48°12.5	· ·	30.9	196°03.8	5.0'	22°32.0	11.5'	60.7'
16	63°12.7		31.9	210°27.8	4.9'	22°43.5	11.3'	60.7'
17	78°12.9		32.8	224°51.7	4.8'	22°54.8	11.2'	60.7'
18	93°13.0	S06°	33.8	239°15.5	4.7'	N23°05.9	11.0'	60.7'
19	108°13.2		34.7	253°39.2	4.6'	23°16.9	10.8'	60.6'
20	123°13.4		35.7	268°02.8	4.5'	23°27.8	10.7'	60.6'
21	138°13.5	· ·	36.6	282°26.3	4.4'	23°38.4	10.5'	60.6'
22	153°13.7		37.6	296°49.8	4.4'	23°48.9	10.3'	60.6'
23	168°13.9		38.5	311°13.1	4.3'	23°59.2	10.2'	60.6'
SD = 16.0' d = 0.9'				SD = 16.6'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:24	05:44	06:52	16:41	17:49	19:07
N 70°	04:32	05:43	06:45	16:49	17:50	19:00
68°	04:38	05:43	06:39	16:55	17:50	18:55
66°	04:43	05:42	06:34	17:00	17:51	18:50
64°	04:47	05:42	06:30	17:04	17:51	18:46
62°	04:50	05:41	06:26	17:08	17:52	18:43
60°	04:53	05:41	06:23	17:11	17:53	18:41
N 58°	04:55	05:41	06:20	17:14	17:53	18:38
56°	04:57	05:40	06:17	17:17	17:54	18:37
54°	04:59	05:40	06:15	17:19	17:54	18:35
52°	05:00	05:39	06:13	17:21	17:55	18:34
50°	05:01	05:39	06:11	17:23	17:55	18:33
45°	05:04	05:38	06:07	17:27	17:57	18:31
N 40°	05:05	05:36	06:03	17:31	17:58	18:29
35°	05:06	05:35	06:00	17:34	17:59	18:29
30°	05:06	05:34	05:58	17:37	18:01	18:29
20°	05:05	05:31	05:53	17:42	18:04	18:30
N 10°	05:03	05:27	05:48	17:46	18:08	18:32
0°	04:59	05:23	05:44	17:51	18:12	18:36
S 10°	04:54	05:19	05:40	17:55	18:16	18:41
20°	04:47	05:13	05:35	18:00	18:22	18:48
30°	04:37	05:06	05:30	18:06	18:30	18:58
35°	04:31	05:01	05:27	18:09	18:35	19:05
40°	04:23	04:56	05:23	18:13	18:40	19:13
45°	04:13	04:49	05:19	18:17	18:47	19:23
S 50°	04:01	04:41	05:14	18:22	18:56	19:36
52°	03:55	04:37	05:11	18:25	19:00	19:42
54°	03:48	04:32	05:09	18:28	19:04	19:49
56°	03:40	04:27	05:06	18:30	19:09	19:57
58°	03:31	04:22	05:03	18:34	19:15	20:06
S 60°	03:21	04:15	04:59	18:38	19:21	20:17

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	15:44	14:22	☐	07:44	11:00	☐
N 70°	16:01	15:10	☐	07:30	10:15	☐
68°	16:15	15:41	☐	07:19	09:45	☐
66°	16:26	16:05	15:24	07:09	09:23	12:04
64°	16:36	16:23	16:05	07:02	09:06	11:25
62°	16:44	16:39	16:33	06:55	08:52	10:58
60°	16:51	16:52	16:54	06:50	08:40	10:37
N 58°	16:58	17:03	17:12	06:45	08:30	10:20
56°	17:03	17:13	17:27	06:40	08:21	10:06
54°	17:09	17:22	17:40	06:36	08:13	09:54
52°	17:13	17:30	17:52	06:33	08:06	09:43
50°	17:17	17:37	18:02	06:30	08:00	09:33
45°	17:27	17:52	18:24	06:23	07:47	09:13
N 40°	17:34	18:05	18:41	06:17	07:36	08:57
35°	17:41	18:16	18:56	06:12	07:26	08:43
30°	17:47	18:25	19:09	06:08	07:18	08:31
20°	17:57	18:42	19:31	06:00	07:04	08:11
N 10°	18:07	18:56	19:51	05:54	06:52	07:54
0°	18:15	19:10	20:09	05:48	06:41	07:37
S 10°	18:24	19:24	20:27	05:42	06:29	07:21
20°	18:33	19:39	20:47	05:35	06:17	07:04
30°	18:44	19:56	21:10	05:28	06:04	06:44
35°	18:51	20:06	21:24	05:24	05:56	06:33
40°	18:58	20:18	21:40	05:19	05:47	06:20
45°	19:07	20:32	21:59	05:14	05:37	06:05
S 50°	19:17	20:49	22:22	05:07	05:24	05:46
52°	19:22	20:57	22:34	05:04	05:18	05:37
54°	19:27	21:06	22:47	05:01	05:12	05:27
56°	19:33	21:16	23:02	04:57	05:05	05:16
58°	19:40	21:28	23:20	04:53	04:57	05:04
S 60°	19:47	21:42	23:42	04:49	04:48	04:49

Day	Sun		Mer.	Moon		Age
	Eqn.of Time		Pass	Mer.Pass.		
	00 ^h	12 ^h		Upper	Lower	16-18
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	100-95%
07	12:06	12:14	11:48	—:—	12:01	
08	12:23	12:31	11:47	00:28	12:55	
09	12:40	12:48	11:47	01:24	13:53	

October 10, 11, 12 UT (Fri., Sat., Sun.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	18°51.2	202°41.8	N03°15.2	159°18.9	S15°36.6	263°31.7	N21°26.3	20°31.1	S03°27.2	Alpheratz	357°33.8	29°14.1											
1	33°53.6	217°41.4	14.0	174°19.6	37.2	278°33.9	26.2	35°33.7	27.2	Ankaa	353°06.1	-42°09.9											
2	48°56.1	232°41.0	12.8	189°20.4	37.7	293°36.1	26.2	50°36.3	27.3	Schedar	349°29.8	56°40.9											
3	63°58.6	247°40.6	•• 11.6	204°21.2	•• 38.3	308°38.3	•• 26.2	65°39.0	•• 27.4	Diphda	348°46.4	-17°50.6											
4	79°01.0	262°40.2	10.4	219°21.9	38.8	323°40.5	26.1	80°41.6	27.4	Achernar	335°19.1	-57°06.2											
5	94°03.5	277°39.8	09.2	234°22.7	39.4	338°42.7	26.1	95°44.2	27.5	Hamal	327°50.2	23°35.2											
6	109°06.0	292°39.4	N03°08.0	249°23.5	S15°39.9	353°44.9	N21°26.0	110°46.8	S03°27.6	Polaris	313°18.2	89°22.2											
7	124°08.4	307°39.1	06.8	264°24.2	40.5	8°47.1	26.0	125°49.5	27.6	Acamar	315°10.9	-40°11.8											
8	139°10.9	322°38.7	05.6	279°25.0	41.0	23°49.3	26.0	140°52.1	27.7	Menkar	314°05.2	4°11.6											
9	154°13.3	337°38.3	•• 04.4	294°25.8	•• 41.6	38°51.5	•• 25.9	155°54.7	•• 27.8	Mirfak	308°27.0	49°57.1											
10	169°15.8	352°37.9	03.2	309°26.5	42.1	53°53.7	25.9	170°57.3	27.8	Aldebaran	290°38.7	16°33.7											
11	184°18.3	7°37.5	02.0	324°27.3	42.7	68°55.9	25.9	186°00.0	27.9	Rigel	281°03.1	-8°10.1											
12	199°20.7	22°37.1	N03°00.8	339°28.1	S15°43.2	83°58.1	N21°25.8	201°02.6	S03°28.0	Capella	280°20.8	46°01.3											
13	214°23.2	37°36.7	02°59.6	354°28.8	43.8	99°00.3	25.8	216°05.2	28.0	Bellatrix	278°22.1	6°22.5											
14	229°25.7	52°36.3	58.4	9°29.6	44.4	114°02.5	25.7	231°07.8	28.1	Elnath	278°00.9	28°37.7											
15	244°28.1	67°35.9	•• 57.2	24°30.4	•• 44.9	129°04.8	•• 25.7	246°10.5	•• 28.2	Alnilam	275°37.0	-1°11.0											
16	259°30.6	82°35.5	56.0	39°31.1	45.5	144°07.0	25.7	261°13.1	28.2	Betelgeuse	270°51.3	7°24.8											
17	274°33.1	97°35.1	54.8	54°31.9	46.0	159°09.2	25.6	276°15.7	28.3	Canopus	263°52.0	-52°42.2											
18	289°35.5	112°34.7	N02°53.6	69°32.6	S15°46.6	174°11.4	N21°25.6	291°18.4	S03°28.4	Sirius	258°25.7	-16°44.8											
19	304°38.0	127°34.3	52.4	84°33.4	47.1	189°13.6	25.6	306°21.0	28.4	Adhara	255°05.4	-29°00.1											
20	319°40.5	142°33.9	51.2	99°34.2	47.7	204°15.8	25.5	321°23.6	28.5	Procyon	244°50.2	5°09.6											
21	334°42.9	157°33.6	•• 50.0	114°34.9	•• 48.2	219°18.0	•• 25.5	336°26.2	•• 28.6	Pollux	243°16.6	27°57.8											
22	349°45.4	172°33.2	48.8	129°35.7	48.8	234°20.2	25.5	351°28.9	28.6	Avior	234°14.7	-59°35.1											
23	4°47.8	187°32.8	47.6	144°36.5	49.3	249°22.4	25.4	6°31.5	28.7	Suhail	222°46.1	-43°31.9											
Mer.pass. 22:41				ν -0.4' d -1.2' m -3.92				ν 0.8' d 0.6' m 1.54				ν 2.2' d -0.0' m -2.17				ν 2.6' d 0.1' m 0.66							

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	19°50.3	202°32.4	N02°46.4	159°37.2	S15°49.9	264°24.6	N21°25.4	21°34.1	S03°28.8										
1	34°52.8	217°32.0	45.2	174°38.0	50.4	279°26.8	25.3	36°36.7	28.8										
2	49°55.2	232°31.6	44.0	189°38.7	51.0	294°29.0	25.3	51°39.4	28.9										
3	64°57.7	247°31.2	•• 42.8	204°39.5	•• 51.5	309°31.2	•• 25.3	66°42.0	•• 29.0										
4	80°00.2	262°30.8	41.6	219°40.3	52.0	324°33.4	25.2	81°44.6	29.0										
5	95°02.6	277°30.4	40.4	234°41.0	52.6	339°35.6	25.2	96°47.2	29.1										
6	110°05.1	292°30.0	N02°39.2	249°41.8	S15°53.1	354°37.8	N21°25.2	111°49.9	S03°29.2										
7	125°07.6	307°29.6	38.0	264°42.5	53.7	9°40.1	25.1	126°52.5	29.2										
8	140°10.0	322°29.2	36.8	279°43.3	54.2	24°42.3	25.1	141°55.1	29.3										
9	155°12.5	337°28.8	•• 35.6	294°44.1	•• 54.8	39°44.5	•• 25.1	156°57.7	•• 29.4										
10	170°15.0	352°28.5	34.4	309°44.8	55.3	54°46.7	25.0	172°00.3	29.4										
11	185°17.4	7°28.1	33.2	324°45.6	55.9	69°48.9	25.0	187°03.0	29.5										
12	200°19.9	22°27.7	N02°32.0	339°46.3	S15°56.4	84°51.1	N21°24.9	202°05.6	S03°29.6										
13	215°22.3	37°27.3	30.8	354°47.1	57.0	99°53.3	24.9	217°08.2	29.6										
14	230°24.8	52°26.9	29.5	9°47.8	57.5	114°55.5	24.9	232°10.8	29.7										
15	245°27.3	67°26.5	•• 28.3	24°48.6	•• 58.1	129°57.7	•• 24.8	247°13.5	•• 29.8										
16	260°29.7	82°26.1	27.1	39°49.4	58.6	144°59.9	24.8	262°16.1	29.8										
17	275°32.2	97°25.7	25.9	54°50.1	59.2	160°02.2	24.8	277°18.7	29.9										
18	290°34.7	112°25.3	N02°24.7	69°50.9	S15°59.7	175°04.4	N21°24.7	292°21.3	S03°30.0										
19	305°37.1	127°24.9	23.5	84°51.6	16°00.3	190°06.6	24.7	307°24.0	30.0										
20	320°39.6	142°24.5	22.3	99°52.4	00.8	205°08.8	24.7	322°26.6	30.1										
21	335°42.1	157°24.1	•• 21.1	114°53.1	•• 01.3	220°11.0	•• 24.6	337°29.2	•• 30.2										
22	350°44.5	172°23.8	19.9	129°53.9	01.9	235°13.2	24.6	352°31.8	30.2										
23	5°47.0	187°23.4	18.7	144°54.7	02.4	250°15.4	24.6	7°34.5	30.3										
Mer.pass. 22:37				ν -0.4' d -1.2' m -3.92				ν 0.8' d 0.5' m 1.54				ν 2.2' d -0.0' m -2.18				ν 2.6' d 0.1' m 0.66			

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	20°49.5	202°23.0	N02°17.5	159°55.4	S16°03.0	265°17.6	N21°24.5	22°37.1	S03°30.4										
1	35°51.9	217°22.6	16.3	174°56.2	03.5	280°19.9	24.5	37°39.7	30.4										
2	50°54.4	232°22.2	15.1	189°56.9	04.1	295°22.1	24.4	52°42.3	30.5										
3	65°56.8	247°21.8	•• 13.9	204°57.7	•• 04.6	310°24.3	•• 24.4	67°45.0	•• 30.5										
4	80°59.3	262°21.4	12.7	219°58.4	05.2	325°26.5	24.4	82°47.6	30.6										
5	96°01.8	277°21.0	11.5	234°59.2	05.7	340°28.7	24.3	97°50.2	30.7										
6	111°04.2	292°20.6	N02°10.3	249°59.9	S16°06.2	355°30.9	N21°24.3	112°52.8	S03°30.7										
7	126°06.7	307°20.2	09.0	265°00.7	06.8	10°33.1	24.3	127°55.4	30.8										
8	141°09.2	322°19.8	07.8	280°01.4	07.3	25°35.4	24.2	142°58.1	30.9										
9	156°11.6	337°19.4	•• 06.6	295°02.2	•• 07.9	40°37.6	•• 24.2	158°00.7	•• 30.9										
10	171°14.1	352°19.1	05.4	310°02.9	08.4	55°39.8	24.2	173°03.3	31.0										
11	186°16.6	7°18.7	04.2	325°03.7	09.0	70°42.0	24.1	188°05.9	31.1										
12	201°19.0	22°18.3	N02°03.0	340°04.5	S16°09.5	85°44.2	N21°24.1	203°08.6	S03°31.1										
13	216°21.5	37°17.9	01.8	355°05.2	10.0	100°46.4	24.1	218°11.2	31.2										
14	231°23.9	52°17.5	02°00.6	10°06.0	10.6	115°48.7	24.0	233°13.8	31.3										
15	246°26.4	67°17.1	01°59.4	25°06.7	•• 11.1	130°50.9	•• 24.0	248°16.4	•• 31.3										
16	261°28.9	82°16.7	58.2	40°07.5	11.7	145°53.1	24.0	263°19.0	31.4										
17	276°31.3	97°16.3	57.0	55°08.2	12.2	160°55.3	23.9	278°21.7	31.5										
18	291°33.8	112°15.9	N01°55.8	70°09.0	S16°12.8	175°57.5	N21°23.9	293°24.3	S03°31.5										
19	306°36.3	127°15.5	54.6	85°09.7	13.3	190°59.8	23.9	308°26.9	31.6										
20	321°38.7	142°15.1	53.3	100°10.5	13.8	206°02.0	23.8	323°29.5	31.7										
21	336°41.2	157°14.8	•• 52.1	115°11.2	•• 14.4	221°04.2	•• 23.8	338°32.2	•• 31.7										
22	351°43.7	172°14.4	50.9	130°12.0	14.9	236°06.4	23.7	353°34.8	31.8										
23	6°46.1	187°14.0	49.7	145°12.7	15.5	251°08.6	23.7	8°37.4	31.8										
Mer.pass. 22:33				ν -0.4' d -1.2' m -3.92				ν 0.8' d 0.5' m 1.54				ν 2.2' d -0.0' m -2.19				ν 2.6' d 0.1' m 0.67			

Oct 10 Fri			SHA	Mer.pass
Venus			183°50.6	10:29
Mars			140°27.7	13:22
Jupiter			244°40.6	06:25
Saturn			1°39.9	22:34

Oct 11 Sat			SHA	Mer.pass
Venus			182°42.1	10:30
Mars			139°46.9	13:21
Jupiter			244°34.3	06:21
Saturn			1°43.8	22:30

Oct 12 Sun			SHA	Mer.pass
Venus			181°33.5	10:31
Mars			139°06.0	13:20
Jupiter			244°28.2	06:18
Saturn			1°47.6	22:26

Horizontal parallax			
Venus:			0.1
Mars:			0.1

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	183°14.0	S06°39.5	325°36.4	4.2'	N24°09.4	10.0'	60.6'
1	198°14.2	40.4	339°59.6	4.1'	24°19.4	9.8'	60.5'
2	213°14.4	41.3	354°22.7	4.1'	24°29.2	9.6'	60.5'
3	228°14.5	· · 42.3	8°45.8	4.0'	24°38.8	9.4'	60.5'
4	243°14.7	43.2	23°08.8	3.9'	24°48.2	9.3'	60.5'
5	258°14.9	44.2	37°31.7	3.8'	24°57.5	9.1'	60.5'
6	273°15.0	S06°45.1	51°54.5	3.8'	N25°06.6	8.9'	60.5'
7	288°15.2	46.1	66°17.3	3.7'	25°15.5	8.7'	60.4'
8	303°15.4	47.0	80°40.0	3.6'	25°24.2	8.5'	60.4'
9	318°15.5	· · 48.0	95°02.6	3.6'	25°32.8	8.4'	60.4'
10	333°15.7	48.9	109°25.2	3.5'	25°41.1	8.2'	60.4'
11	348°15.9	49.8	123°47.7	3.4'	25°49.3	8.0'	60.4'
12	3°16.0	S06°50.8	138°10.1	3.4'	N25°57.2	7.8'	60.3'
13	18°16.2	51.7	152°32.5	3.3'	26°05.0	7.6'	60.3'
14	33°16.4	52.7	166°54.8	3.3'	26°12.6	7.4'	60.3'
15	48°16.5	· · 53.6	181°17.1	3.2'	26°20.0	7.2'	60.3'
16	63°16.7	54.6	195°39.3	3.2'	26°27.2	7.0'	60.3'
17	78°16.9	55.5	210°01.5	3.1'	26°34.3	6.8'	60.2'
18	93°17.0	S06°56.5	224°23.6	3.1'	N26°41.1	6.6'	60.2'
19	108°17.2	57.4	238°45.7	3.0'	26°47.7	6.4'	60.2'
20	123°17.3	58.3	253°07.7	3.0'	26°54.1	6.2'	60.2'
21	138°17.5	06°59.3	267°29.7	2.9'	27°00.4	6.0'	60.1'
22	153°17.7	07°00.2	281°51.6	2.9'	27°06.4	5.8'	60.1'
23	168°17.8	01.2	296°13.5	2.9'	27°12.2	5.6'	60.1'
SD = 16.0'			$d = 0.9'$			SD = 16.5'	
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	183°18.0	S07°02.1	310°35.4	2.8'	N27°17.8	5.4'	60.1'
1	198°18.2	03.1	324°57.3	2.8'	27°23.3	5.2'	60.0'
2	213°18.3	04.0	339°19.1	2.8'	27°28.5	5.0'	60.0'
3	228°18.5	· · 04.9	353°40.9	2.8'	27°33.5	4.8'	60.0'
4	243°18.6	05.9	8°02.6	2.7'	27°38.4	4.6'	60.0'
5	258°18.8	06.8	22°24.4	2.7'	27°43.0	4.4'	59.9'
6	273°19.0	S07°07.8	36°46.1	2.7'	N27°47.4	4.2'	59.9'
7	288°19.1	08.7	51°07.8	2.7'	27°51.6	4.0'	59.9'
8	303°19.3	09.6	65°29.5	2.7'	27°55.6	3.8'	59.9'
9	318°19.4	· · 10.6	79°51.2	2.7'	27°59.5	3.6'	59.8'
10	333°19.6	11.5	94°12.9	2.7'	28°03.1	3.4'	59.8'
11	348°19.8	12.5	108°34.6	2.7'	28°06.5	3.2'	59.8'
12	3°19.9	S07°13.4	122°56.2	2.7'	N28°09.7	3.0'	59.8'
13	18°20.1	14.3	137°17.9	2.7'	28°12.7	2.8'	59.7'
14	33°20.2	15.3	151°39.6	2.7'	28°15.4	2.6'	59.7'
15	48°20.4	· · 16.2	166°01.3	2.7'	28°18.0	2.4'	59.7'
16	63°20.6	17.2	180°23.0	2.7'	28°20.4	2.2'	59.6'
17	78°20.7	18.1	194°44.7	2.7'	28°22.6	2.0'	59.6'
18	93°20.9	S07°19.0	209°06.4	2.7'	N28°24.6	1.8'	59.6'
19	108°21.0	20.0	223°28.1	2.8'	28°26.3	1.6'	59.6'
20	123°21.2	20.9	237°49.9	2.8'	28°27.9	1.4'	59.5'
21	138°21.4	· · 21.9	252°11.7	2.8'	28°29.2	1.2'	59.5'
22	153°21.5	22.8	266°33.5	2.8'	28°30.4	1.0'	59.5'
23	168°21.7	23.7	280°55.3	2.9'	28°31.4	0.8'	59.4'
SD = 16.0'			$d = 0.9'$			SD = 16.4'	
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	183°21.8	S07°24.7	295°17.2	2.9'	N28°32.1	0.5'	59.4'
1	198°22.0	25.6	309°39.1	2.9'	28°32.7	0.3'	59.4'
2	213°22.1	26.6	324°01.0	3.0'	28°33.0	0.1'	59.4'
3	228°22.3	· · 27.5	338°23.0	3.0'	28°33.2	-0.1'	59.3'
4	243°22.5	28.4	352°45.0	3.1'	28°33.1	-0.3'	59.3'
5	258°22.6	29.4	7°07.1	3.1'	28°32.9	-0.5'	59.3'
6	273°22.8	S07°30.3	21°29.2	3.2'	N28°32.4	-0.6'	59.2'
7	288°22.9	31.2	35°51.3	3.2'	28°31.8	-0.8'	59.2'
8	303°23.1	32.2	50°13.5	3.3'	28°30.9	-1.0'	59.2'
9	318°23.2	· · 33.1	64°35.8	3.3'	28°29.9	-1.2'	59.1'
10	333°23.4	34.0	78°58.1	3.4'	28°28.6	-1.4'	59.1'
11	348°23.5	35.0	93°20.5	3.4'	28°27.2	-1.6'	59.1'
12	3°23.7	S07°35.9	107°43.0	3.5'	N28°25.6	-1.8'	59.1'
13	18°23.9	36.9	122°05.5	3.6'	28°23.7	-2.0'	59.0'
14	33°24.0	37.8	136°28.1	3.6'	28°21.7	-2.2'	59.0'
15	48°24.2	· · 38.7	150°50.7	3.7'	28°19.5	-2.4'	59.0'
16	63°24.3	39.7	165°13.4	3.8'	28°17.1	-2.6'	58.9'
17	78°24.5	40.6	179°36.2	3.9'	28°14.6	-2.8'	58.9'
18	93°24.6	S07°41.5	193°59.1	4.0'	N28°11.8	-3.0'	58.9'
19	108°24.8	42.5	208°22.0	4.0'	28°08.8	-3.1'	58.8'
20	123°24.9	43.4	222°45.1	4.1'	28°05.7	-3.3'	58.8'
21	138°25.1	· · 44.3	237°08.2	4.2'	28°02.4	-3.5'	58.8'
22	153°25.2	45.3	251°31.4	4.3'	27°58.9	-3.7'	58.7'
23	168°25.4	46.2	265°54.7	4.4'	27°55.2	-3.9'	58.7'
SD = 16.0'			$d = 0.9'$			SD = 16.2'	

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:38	05:57	07:06	16:26	17:34	18:52
N 70°	04:44	05:55	06:57	16:35	17:36	18:47
68°	04:49	05:53	06:50	16:42	17:38	18:42
66°	04:52	05:52	06:44	16:48	17:40	18:39
64°	04:55	05:50	06:38	16:54	17:41	18:36
62°	04:58	05:49	06:34	16:58	17:43	18:34
60°	05:00	05:48	06:30	17:02	17:44	18:32
N 58°	05:02	05:47	06:27	17:06	17:45	18:30
56°	05:03	05:46	06:23	17:09	17:46	18:29
54°	05:04	05:45	06:21	17:12	17:47	18:28
52°	05:05	05:44	06:18	17:14	17:48	18:27
50°	05:06	05:43	06:16	17:17	17:49	18:26
45°	05:07	05:41	06:11	17:22	17:51	18:25
N 40°	05:08	05:39	06:06	17:26	17:53	18:25
35°	05:08	05:37	06:03	17:30	17:55	18:25
30°	05:08	05:35	05:59	17:33	17:57	18:25
20°	05:06	05:31	05:54	17:39	18:02	18:27
N 10°	05:03	05:27	05:48	17:45	18:06	18:30
0°	04:58	05:23	05:43	17:50	18:11	18:35
S 10°	04:52	05:17	05:38	17:55	18:16	18:41
20°	04:44	05:10	05:33	18:01	18:23	18:49
30°	04:34	05:02	05:26	18:07	18:32	19:00
35°	04:27	04:57	05:23	18:11	18:37	19:08
40°	04:18	04:51	05:18	18:16	18:43	19:16
45°	04:07	04:43	05:13	18:21	18:51	19:27
S 50°	03:54	04:34	05:07	18:27	19:01	19:41
52°	03:47	04:30	05:04	18:30	19:05	19:48
54°	03:39	04:25	05:01	18:33	19:10	19:56
56°	03:31	04:19	04:58	18:37	19:16	20:04
58°	03:21	04:13	04:54	18:41	19:22	20:15
S 60°	03:09	04:06	04:50	18:45	19:29	20:27

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°						
N 70°						
68°						
66°						
64°						
62°	16:24			13:15		
60°	17:03	17:28	18:37	12:37	14:25	15:30
N 58°	17:30	18:07	19:14	12:11	13:47	14:52
56°	17:51	18:34	19:41	11:50	13:21	14:26
54°	18:09	18:55	20:02	11:32	12:59	14:05
52°	18:24	19:13	20:19	11:17	12:42	13:47
50°	18:38	19:28	20:34	11:05	12:27	13:32
45°	19:05	19:58	21:04	10:38	11:56	13:01
N 40°	19:27	20:22	21:27	10:17	11:33	12:38
35°	19:45	20:42	21:47	10:00	11:13	12:18
30°	20:00	20:59	22:03	09:45	10:57	12:01
20°	20:27	21:28	22:31	09:20	10:28	11:33
N 10°	20:50	21:52	22:55	08:58	10:04	11:08
0°	21:11	22:15	23:18	08:38	09:42	10:45
S 10°	21:33	22:38	23:40	08:18	09:19	10:22
20°	21:56	23:03	· · ·	07:57	08:55	09:58
30°	22:23	23:32	· · ·	07:32	08:27	09:29
35°	22:39	23:49	· · ·	07:17	08:10	09:12
40°	22:58	· · ·	00:09	07:01	07:51	08:52
45°	23:21	· · ·	00:33	06:41	07:28	08:28
S 50°	23:50	· · ·	01:05	06:16	06:58	07:56

October 13, 14, 15 UT (Mon., Tue., Wed.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	SHA	Dec	SHA	Dec		
0	21°48.6	202°13.6	N01°48.5	160°13.5	S16°16.0	266°10.9	N21°23.7	23°40.0	S03°31.9	Alpheratz	357°33.8	29°14.2											
1	36°51.1	217°13.2		175°14.2	16.5	281°13.1	23.6	38°42.6	32.0	Ankaa	353°06.1	-42°09.9											
2	51°53.5	232°12.8		190°14.9	17.1	296°15.3	23.6	53°45.3	32.0	Schedar	349°29.8	56°40.9											
3	66°56.0	247°12.4	•• 44.9	205°15.7	•• 17.6	311°17.5	•• 23.6	68°47.9	•• 32.1	Diphda	348°46.3	-17°50.6											
4	81°58.4	262°12.0		220°16.4	18.2	326°19.7	23.5	83°50.5	32.2	Achernar	335°19.1	-57°06.2											
5	97°00.9	277°11.6		235°17.2	18.7	341°22.0	23.5	98°53.1	32.2	Hamal	327°50.2	23°35.2											
6	112°03.4	292°11.2	N01°41.2	250°17.9	S16°19.2	356°24.2	N21°23.5	113°55.7	S03°32.3	Polaris	313°17.1	89°22.2											
7	127°05.8	307°10.8		265°18.7	19.8	11°26.4	23.4	128°58.4	32.4	Acamar	315°10.9	-40°11.9											
8	142°08.3	322°10.5		280°19.4	20.3	26°28.6	23.4	144°01.0	32.4	Menkar	314°05.2	4°11.6											
9	157°10.8	337°10.1	•• 37.6	295°20.2	•• 20.9	41°30.9	•• 23.4	159°03.6	•• 32.5	Mirfak	308°26.9	49°57.1											
10	172°13.2	352°09.7		310°20.9	21.4	56°33.1	23.3	174°06.2	32.6	Aldebaran	290°38.7	16°33.7											
11	187°15.7	7°09.3		325°21.7	21.9	71°35.3	23.3	189°08.8	32.6	Rigel	281°03.1	-8°10.1											
12	202°18.2	22°08.9	N01°34.0	340°22.4	S16°22.5	86°37.5	N21°23.3	204°11.5	S03°32.7	Capella	280°20.7	46°01.3											
13	217°20.6	37°08.5		355°23.2	23.0	101°39.8	23.2	219°14.1	32.7	Bellatrix	278°22.0	6°22.5											
14	232°23.1	52°08.1		10°23.9	23.5	116°42.0	23.2	234°16.7	32.8	Elnath	278°00.9	28°37.7											
15	247°25.6	67°07.7	•• 30.3	25°24.6	•• 24.1	131°44.2	•• 23.2	249°19.3	•• 32.9	Alnilam	275°36.9	-1°11.0											
16	262°28.0	82°07.3		40°25.4	24.6	146°46.4	23.1	264°21.9	32.9	Betelgeuse	270°51.3	7°24.8											
17	277°30.5	97°06.9		55°26.1	25.2	161°48.7	23.1	279°24.6	33.0	Canopus	263°52.0	-52°42.2											
18	292°32.9	112°06.6	N01°26.7	70°26.9	S16°25.7	176°50.9	N21°23.1	294°27.2	S03°33.1	Sirius	258°25.6	-16°44.8											
19	307°35.4	127°06.2		85°27.6	26.2	191°53.1	23.0	309°29.8	33.1	Adhara	255°05.3	-29°00.1											
20	322°37.9	142°05.8		100°28.4	26.8	206°55.3	23.0	324°32.4	33.2	Procyon	244°50.2	5°09.6											
21	337°40.3	157°05.4	•• 23.1	115°29.1	•• 27.3	221°57.6	•• 23.0	339°35.0	•• 33.3	Pollux	243°16.6	27°57.8											
22	352°42.8	172°05.0		130°29.9	27.8	236°59.8	22.9	354°37.7	33.3	Avior	234°14.7	-59°35.1											
23	7°45.3	187°04.6		145°30.6	28.4	252°02.0	22.9	9°40.3	33.4	Suhail	222°46.1	-43°31.9											
Mer.pass. 22:29		ν -0.4' d -1.2' m -3.92		ν 0.7' d 0.5' m 1.53		ν 2.2' d -0.0' m -2.19		ν 2.6' d 0.1' m 0.67															

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	22°47.7	202°04.2	N01°19.4	160°31.3	S16°28.9	267°04.2	N21°22.9	24°42.9	S03°33.4	Denebola	182°24.6	14°25.7			
1	37°50.2	217°03.8		175°32.1	29.4	282°06.5	22.8	39°45.5	33.5	Gienah	175°43.3	-17°41.0			
2	52°52.7	232°03.4		190°32.8	30.0	297°08.7	22.8	54°48.1	33.6	Acrux	173°00.1	-63°14.4			
3	67°55.1	247°03.0	•• 15.8	205°33.6	•• 30.5	312°10.9	•• 22.8	69°50.8	•• 33.6	Gacrux	171°51.6	-57°15.3			
4	82°57.6	262°02.6		220°34.3	31.0	327°13.2	22.7	84°53.4	33.7	Alioth	166°13.0	55°49.2			
5	98°00.1	277°02.3		235°35.0	31.6	342°15.4	22.7	99°56.0	33.8	Spica	158°22.0	-11°17.7			
6	113°02.5	292°01.9	N01°12.1	250°35.8	S16°32.1	357°17.6	N21°22.7	114°58.6	S03°33.8	Alkaid	152°52.0	49°11.1			
7	128°05.0	307°01.5		265°36.5	32.7	12°19.8	22.6	130°01.2	33.9	Hadar	148°35.8	-60°29.8			
8	143°07.4	322°01.1		280°37.3	33.2	27°22.1	22.6	145°03.9	34.0	Menkent	147°57.3	-36°29.8			
9	158°09.9	337°00.7	•• 08.5	295°38.0	•• 33.7	42°24.3	•• 22.6	160°06.5	•• 34.0	Arcturus	145°47.7	19°03.0			
10	173°12.4	352°00.3		310°38.7	34.3	57°26.5	22.5	175°09.1	34.1	Rigil Kent.	139°40.1	-60°56.5			
11	188°14.8	6°59.9		325°39.5	34.8	72°28.8	22.5	190°11.7	34.1	Kochab	137°20.8	74°03.0			
12	203°17.3	21°59.5	N01°04.9	340°40.2	S16°35.3	87°31.0	N21°22.5	205°14.3	S03°34.2	Zuben'ubi	136°55.6	-16°08.9			
13	218°19.8	36°59.1		355°41.0	35.9	102°33.2	22.4	220°17.0	34.3	Alphecca	126°03.5	26°37.8			
14	233°22.2	51°58.7		10°41.7	36.4	117°35.5	22.4	235°19.6	34.3	Antares	112°15.3	-26°29.4			
15	248°24.7	66°58.3	•• 01.2	25°42.4	•• 36.9	132°37.7	•• 22.4	250°22.2	•• 34.4	Atria	107°09.3	-69°04.6			
16	263°27.2	81°58.0	01°00.0	40°43.2	37.4	147°39.9	22.3	265°24.8	34.5	Sabik	102°02.3	-15°45.4			
17	278°29.6	96°57.6	00°58.8	55°43.9	38.0	162°42.2	22.3	280°27.4	34.5	Shaula	96°09.7	-37°07.4			
18	293°32.1	111°57.2	N00°57.6	70°44.7	S16°38.5	177°44.4	N21°22.3	295°30.0	S03°34.6	Rasalhague	95°58.1	12°32.6			
19	308°34.6	126°56.8		85°45.4	39.0	192°46.6	22.2	310°32.7	34.6	Eltanin	90°42.1	51°29.4			
20	323°37.0	141°56.4		100°46.1	39.6	207°48.9	22.2	325°35.3	34.7	Kaus Aust.	83°31.8	-34°22.4			
21	338°39.5	156°56.0	•• 53.9	115°46.9	•• 40.1	222°51.1	•• 22.2	340°37.9	•• 34.8	Vega	80°32.9	38°48.7			
22	353°41.9	171°55.6		130°47.6	40.6	237°53.3	22.1	355°40.5	34.8	Nunki	75°47.0	-26°15.9			
23	8°44.4	186°55.2		145°48.3	41.2	252°55.6	22.1	10°43.1	34.9	Altair	61°59.3	8°56.3			
Mer.pass. 22:25		ν -0.4' d -1.2' m -3.92		ν 0.7' d 0.5' m 1.53		ν 2.2' d -0.0' m -2.20		ν 2.6' d 0.1' m 0.68							

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	
0	23°46.9	201°54.8	N00°50.3	160°49.1	S16°41.7	267°57.8	N21°22.1	25°45.7	S03°35.0	Fomalhaut	15°13.5	-29°29.1		
1	38°49.3	216°54.4		175°49.8	42.2	283°00.0	22.1	40°48.4	35.0	Scheat	13°44.3	28°13.6		
2	53°51.8	231°54.1		190°50.5	42.8	298°02.3	22.0	55°51.0	35.1	Markab	13°29.0	15°20.8		
3	68°54.3	246°53.7	•• 46.6	205°51.3	•• 43.3	313°04.5	•• 22.0	70°53.6	•• 35.1	Oct 13 Mon			SHA	Mer.pass
4	83°56.7	261°53.3		220°52.0	43.8	328°06.8	22.0	85°56.2	35.2	Venus	180°25.0	10:31		
5	98°59.2	276°52.9		235°52.7	44.3	343°09.0	21.9	100°58.8	35.3	Mars	138°24.9	13:18		
6	114°01.7	291°52.5	N00°43.0	250°53.5	S16°44.9	358°11.2	N21°21.9	116°01.5	S03°35.3	Jupiter	244°22.3	06:14		
7	129°04.1	306°52.1		265°54.2	45.4	13°13.5	21.9	131°04.1	35.4	Saturn	1°51.4	22:21		
8	144°06.6	321°51.7		280°54.9	45.9	28°15.7	21.8	146°06.7	35.5	Oct 14 Tue			SHA	Mer.pass
9	159°09.0	336°51.3	•• 39.3	295°55.7	•• 46.5	43°17.9	•• 21.8	161°09.3	•• 35.5	Venus	179°16.5	10:32		
10	174°11.5	351°50.9		310°56.4	47.0	58°20.2	21.8	176°11.9	35.6	Mars	137°43.6	13:17		
11	189°14.0	6°50.5		325°57.1	47.5	73°22.4	21.7	191°14.5	35.6	Jupiter	244°16.5	06:11		
12	204°16.4	21°50.1	N00°35.7	340°57.9	S16°48.0	88°24.7	N21°21.7	206°17.2	S03°35.7	Saturn	1°55.2	22:17		
13	219°18.9	36°49.8		355°58.6	48.6	103°26.9	21.7	221°19.8	35.8	Oct 15 Wed			SHA	Mer.pass
14	234°21.4	51°49.4		10°59.3	49.1	118°29.1	21.6	236°22.4	35.8	Venus	178°08.0	10:33		
15	249°23.8	66°49.0	•• 32.0	26°00.1	•• 49.6	133°31.4	•• 21.6	251°25.0	•• 35.9	Mars	137°02.2	13:16		
16	264°26.3	81°48.6		41°00.8	50.2	148°33.6	21.6	266°27.6	36.0	Jupiter	244°10.9	06:07		
17	279°28.8	96°48.2		56°01.5	50.7	163°35.9	21.5	281°30.2	36.0	Saturn	1°58.9	22:13		
18	294°31.2	111°47.8	N00°28.4	71°02.3	S16°51.2	178°38.1	N21°21.5	296°32.8	S03°36.1	Horizontal parallax				
19	309°33.7	126°47.4		86°03.0	51.7	193°40.3	21.5	311°35.5	36.1	Venus:	0.1			
20	324°36.2	141°47.0		101°03.7	52.3	208°42.6	21.4	326°38.1	36.2	Mars:	0.1			
21	339°38.6	156°46.6	•• 24.7	116°04.5	•• 52.8	223°44.8	•• 21.4	341°40.7	•• 36.3					
22	354°41.1	171°46.2		131°05.2	53.3	238°47.1	21.4	356°43.3	36.3					
23	9°43.5	186°45.8		146°05.9	53.8	253°49.3	21.4	11°45.9	36.4					
Mer.pass. 22:21		ν -0.4' d -1.2' m -3.92		ν 0.7' d 0.5' m 1.53		ν 2.2' d -0.0' m -2.21		ν 2.6' d 0.1' m 0.68						

DUT1 = UT1-UTC = +0.1023 sec ΔT = TT-UT1 = +69.0817 sec

2025 October 13 to Oct. 15 UT

Sun			Moon			
h						
Mon	GHA	Dec	GHA	ν	Dec	HP
0	183°25.5	S07°47.1	280°18.1	4.5'	N27°51.3	-4.0' 58.7'
1	198°25.7	48.1	294°41.5	4.6'	27°47.3	-4.2' 58.6'
2	213°25.8	49.0	309°05.1	4.7'	27°43.0	-4.4' 58.6'
3	228°26.0	· · 49.9	323°28.7	4.7'	27°38.6	-4.6' 58.6'
4	243°26.1	50.9	337°52.5	4.8'	27°34.1	-4.7' 58.6'
5	258°26.3	51.8	352°16.3	4.9'	27°29.3	-4.9' 58.5'
6	273°26.4	S07°52.7	6°40.3	5.0'	N27°24.4	-5.1' 58.5'
7	288°26.6	53.7	21°04.3	5.1'	27°19.3	-5.3' 58.5'
8	303°26.7	54.6	35°28.4	5.2'	27°14.0	-5.4' 58.4'
9	318°26.9	· · 55.5	49°52.7	5.4'	27°08.6	-5.6' 58.4'
10	333°27.0	56.5	64°17.0	5.5'	27°03.0	-5.8' 58.4'
11	348°27.2	57.4	78°41.5	5.6'	26°57.3	-5.9' 58.3'
12	3°27.3	S07°58.3	93°06.1	5.7'	N26°51.3	-6.1' 58.3'
13	18°27.5	07°59.3	107°30.7	5.8'	26°45.2	-6.2' 58.3'
14	33°27.6	08°00.2	121°55.5	5.9'	26°39.0	-6.4' 58.2'
15	48°27.8	· · 01.1	136°20.4	6.0'	26°32.6	-6.6' 58.2'
16	63°27.9	02.1	150°45.4	6.1'	26°26.0	-6.7' 58.2'
17	78°28.1	03.0	165°10.5	6.2'	26°19.3	-6.9' 58.1'
18	93°28.2	S08°03.9	179°35.8	6.3'	N26°12.5	-7.0' 58.1'
19	108°28.4	04.9	194°01.1	6.5'	26°05.4	-7.2' 58.1'
20	123°28.5	05.8	208°26.6	6.6'	25°58.3	-7.3' 58.1'
21	138°28.7	· · 06.7	222°52.2	6.7'	25°51.0	-7.5' 58.0'
22	153°28.8	07.6	237°17.8	6.8'	25°43.5	-7.6' 58.0'
23	168°29.0	08.6	251°43.7	6.9'	25°35.9	-7.8' 58.0'
	SD = 16.0'	d = 0.9'			SD = 16.0'	

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	183°29.1	S08°09.5	266°09.6	7.0'	N25°28.1	-7.9'	57.9'
1	198°29.3	10.4	280°35.6	7.2'	25°20.2	-8.0'	57.9'
2	213°29.4	11.4	295°01.8	7.3'	25°12.2	-8.2'	57.9'
3	228°29.5	· · 12.3	309°28.1	7.4'	25°04.0	-8.3'	57.8'
4	243°29.7	13.2	323°54.5	7.5'	24°55.7	-8.4'	57.8'
5	258°29.8	14.1	338°21.0	7.6'	24°47.3	-8.6'	57.8'
6	273°30.0	S08°15.1	352°47.6	7.8'	N24°38.7	-8.7'	57.7'
7	288°30.1	16.0	7°14.4	7.9'	24°30.0	-8.8'	57.7'
8	303°30.3	16.9	21°41.3	8.0'	24°21.2	-9.0'	57.7'
9	318°30.4	· · 17.9	36°08.3	8.1'	24°12.2	-9.1'	57.6'
10	333°30.6	18.8	50°35.4	8.2'	24°03.1	-9.2'	57.6'
11	348°30.7	19.7	65°02.6	8.4'	23°53.9	-9.3'	57.6'
12	3°30.8	S08°20.6	79°30.0	8.5'	N23°44.5	-9.5'	57.6'
13	18°31.0	21.6	93°57.5	8.6'	23°35.1	-9.6'	57.5'
14	33°31.1	22.5	108°25.1	8.7'	23°25.5	-9.7'	57.5'
15	48°31.3	· · 23.4	122°52.8	8.8'	23°15.8	-9.8'	57.5'
16	63°31.4	24.4	137°20.7	9.0'	23°05.9	-9.9'	57.4'
17	78°31.6	25.3	151°48.6	9.1'	22°56.0	-10.1'	57.4'
18	93°31.7	S08°26.2	166°16.7	9.2'	N22°45.9	-10.2'	57.4'
19	108°31.8	27.1	180°44.9	9.3'	22°35.8	-10.3'	57.3'
20	123°32.0	28.1	195°13.2	9.4'	22°25.5	-10.4'	57.3'
21	138°32.1	· · 29.0	209°41.7	9.6'	22°15.1	-10.5'	57.3'
22	153°32.3	29.9	224°10.2	9.7'	22°04.6	-10.6'	57.3'
23	168°32.4	30.8	238°38.9	9.8'	21°54.0	-10.7'	57.2'
	SD = 16.0'	d = 0.9'			SD = 15.8'		

Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	183°32.5	S08°31.8	253°07.7	9.9'	N21°43.3	-10.8'	57.2'
1	198°32.7	32.7	267°36.6	10.0'	21°32.5	-10.9'	57.2'
2	213°32.8	33.6	282°05.6	10.1'	21°21.6	-11.0'	57.1'
3	228°33.0	· · 34.5	296°34.8	10.3'	21°10.6	-11.1'	57.1'
4	243°33.1	35.5	311°04.0	10.4'	20°59.5	-11.2'	57.1'
5	258°33.2	36.4	325°33.4	10.5'	20°48.3	-11.3'	57.1'
6	273°33.4	S08°37.3	340°02.9	10.6'	N20°37.1	-11.4'	57.0'
7	288°33.5	38.2	354°32.5	10.7'	20°25.7	-11.5'	57.0'
8	303°33.6	39.1	9°02.2	10.8'	20°14.2	-11.6'	57.0'
9	318°33.8	· · 40.1	23°32.0	10.9'	20°02.6	-11.7'	56.9'
10	333°33.9	41.0	38°02.0	11.0'	19°51.0	-11.7'	56.9'
11	348°34.1	41.9	52°32.0	11.2'	19°39.2	-11.8'	56.9'
12	3°34.2	S08°42.8	67°02.2	11.3'	N19°27.4	-11.9'	56.9'
13	18°34.3	43.8	81°32.4	11.4'	19°15.5	-12.0'	56.8'
14	33°34.5	44.7	96°02.8	11.5'	19°03.5	-12.1'	56.8'
15	48°34.6	· · 45.6	110°33.3	11.6'	18°51.4	-12.2'	56.8'
16	63°34.7	46.5	125°03.9	11.7'	18°39.3	-12.2'	56.7''
17	78°34.9	47.4	139°34.5	11.8'	18°27.1	-12.3'	56.7'
18	93°35.0	S08°48.4	154°05.3	11.9'	N18°14.8	-12.4'	56.7'
19	108°35.2	49.3	168°36.2	12.0'	18°02.4	-12.5'	56.7'
20	123°35.3	50.2	183°07.2	12.1'	17°49.9	-12.5'	56.6'
21	138°35.4	· · 51.1	197°38.3	12.2'	17°37.4	-12.6'	56.6'
22	153°35.6	52.0	212°09.5	12.3'	17°24.8	-12.7'	56.6'
23	168°35.7	53.0	226°40.8	12.4'	17°12.1	-12.7'	56.6'
	SD = 16.0'	$d = 0.9'$			SD = 15.6'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	04:52	06:10	07:20	16:10	17:20	18:37
N 70°	04:56	06:07	07:10	16:21	17:23	18:34
68°	04:59	06:04	07:01	16:30	17:26	18:31
66°	05:02	06:01	06:53	16:37	17:29	18:28
64°	05:04	05:59	06:47	16:43	17:31	18:26
62°	05:06	05:57	06:42	16:49	17:34	18:25
60°	05:07	05:55	06:37	16:53	17:35	18:23
N 58°	05:08	05:54	06:33	16:58	17:37	18:22
56°	05:09	05:52	06:30	17:01	17:39	18:22
54°	05:10	05:51	06:26	17:05	17:40	18:21
52°	05:10	05:49	06:23	17:08	17:42	18:21
50°	05:11	05:48	06:21	17:11	17:43	18:20
45°	05:11	05:45	06:15	17:17	17:46	18:20
N 40°	05:11	05:42	06:10	17:22	17:49	18:20
35°	05:10	05:40	06:05	17:26	17:52	18:21
30°	05:09	05:37	06:01	17:30	17:54	18:22
20°	05:07	05:32	05:55	17:37	17:59	18:25
N 10°	05:03	05:27	05:48	17:43	18:04	18:29
0°	04:58	05:22	05:43	17:49	18:10	18:34
S 10°	04:51	05:15	05:37	17:55	18:17	18:41
20°	04:42	05:08	05:30	18:02	18:24	18:50
30°	04:30	04:59	05:23	18:09	18:34	19:03
35°	04:22	04:53	05:19	18:14	18:40	19:10
40°	04:13	04:46	05:14	18:19	18:47	19:20
45°	04:01	04:38	05:08	18:25	18:55	19:32
S 50°	03:46	04:27	05:01	18:32	19:06	19:47
52°	03:39	04:22	04:58	18:35	19:11	19:54
54°	03:31	04:17	04:54	18:39	19:16	20:03
56°	03:22	04:11	04:50	18:43	19:23	20:12
58°	03:11	04:04	04:46	18:47	19:30	20:24
S 60°	02:58	03:56	04:41	18:53	19:38	20:37

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	□	□	20:40	□	□	19:17
N 70°	□	□	21:56	□	□	18:00
68°	□	□	22:33	□	□	17:21
66°	□	20:35	22:59	□	17:34	16:54
64°	□	21:17	23:18	□	16:51	16:33
62°	19:42	21:45	23:34	16:30	16:22	16:16
60°	20:20	22:07	23:48	15:52	15:59	16:02
N 58°	20:46	22:24	23:59	15:25	15:41	15:49
56°	21:07	22:39	· ·	15:04	15:26	15:39
54°	21:24	22:52	· ·	14:47	15:13	15:29
52°	21:39	23:03	· ·	14:32	15:01	15:21
50°	21:51	23:13	· ·	14:19	14:51	15:13
45°	22:17	23:33	· ·	13:52	14:29	14:57
N 40°	22:38	23:50	· ·	13:30	14:11	14:43
35°	22:55	· ·	00:04	13:12	13:56	14:32
30°	23:10	· ·	00:16	12:57	13:43	14:22
20°	23:35	· ·	00:36	12:30	13:21	14:04
N 10°	23:57	· ·	00:54	12:08	13:01	13:49
0°	· ·	00:17	01:10	11:46	12:43	13:34
S 10°	· ·	00:37	01:27	11:25	12:25	13:20
20°	00:04	00:58	01:44	11:02	12:05	13:04
30°	00:32	01:23	02:05	10:35	11:42	12:46
35°	00:48	01:37	02:16	10:19	11:28	12:35
40°	01:07	01:54	02:30	10:00	11:12	12:23
45°	01:31	02:13	02:45	09:38	10:53	12:09
S 50°	02:00	02:38	03:05	09:09	10:30	11:51
52°	02:15	02:50	03:14	08:55	10:18	11:42
54°	02:31	03:04	03:24	08:38	10:05	11:33
56°	02:51	03:19	03:35	08:18	09:50	11:22
58°	03:16	03:38	03:48	07:54	09:32	11:10
S 60°	03:50	04:01	04:04	07:20	09:10	10:55

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
13	13:42	13:49	11:46	05:32	18:02	
14	13:56	14:03	11:46	06:30	18:57	
15	14:10	14:17	11:46	07:23	19:47	

October 16, 17, 18 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	24°46.0	201°45.5	N00°21.1	161°06.6	S16°54.4	268°51.5	N21°21.3	26°48.5	S03°36.4	Alpheratz	357°33.8	29°14.2					
1	39°48.5	216°45.1	19.8	176°07.4	54.9	283°53.8	21.3	41°51.2	36.5	Ankaa	353°06.1	-42°09.9					
2	54°50.9	231°44.7	18.6	191°08.1	55.4	298°56.0	21.3	56°53.8	36.6	Schedar	349°29.8	56°40.9					
3	69°53.4	246°44.3	· · 17.4	206°08.8	· · 55.9	313°58.3	· · 21.2	71°56.4	· · 36.6	Diphda	348°46.3	-17°50.6					
4	84°55.9	261°43.9	16.2	221°09.6	56.5	329°00.5	21.2	86°59.0	36.7	Achernar	335°19.1	-57°06.2					
5	99°58.3	276°43.5	15.0	236°10.3	57.0	344°02.8	21.2	102°01.6	36.8	Hamal	327°50.2	23°35.2					
6	115°00.8	291°43.1	N00°13.7	251°11.0	S16°57.5	359°05.0	N21°21.1	117°04.2	S03°36.8	Polaris	313°16.3	89°22.3					
7	130°03.3	306°42.7	12.5	266°11.7	58.0	14°07.3	21.1	132°06.8	36.9	Acamar	315°10.9	-40°11.9					
8	145°05.7	321°42.3	11.3	281°12.5	58.6	29°09.5	21.1	147°09.5	36.9	Menkar	314°05.2	4°11.6					
9	160°08.2	336°41.9	· · 10.1	296°13.2	· · 59.1	44°11.7	· · 21.0	162°12.1	· · 37.0	Mirfak	308°26.9	49°57.2					
10	175°10.7	351°41.5	08.9	311°13.9	16°59.6	59°14.0	21.0	177°14.7	37.1	Aldebaran	290°38.7	16°33.7					
11	190°13.1	6°41.1	07.6	326°14.6	17°00.1	74°16.2	21.0	192°17.3	37.1	Rigel	281°03.1	-8°10.1					
12	205°15.6	21°40.8	N00°06.4	341°15.4	S17°00.7	89°18.5	N21°21.0	207°19.9	S03°37.2	Capella	280°20.7	46°01.3					
13	220°18.0	36°40.4	05.2	356°16.1	01.2	104°20.7	20.9	222°22.5	37.2	Bellatrix	278°22.0	6°22.5					
14	235°20.5	51°40.0	04.0	11°16.8	01.7	119°23.0	20.9	237°25.1	37.3	Elnath	278°00.9	28°37.7					
15	250°23.0	66°39.6	· · 02.8	26°17.5	· · 02.2	134°25.2	· · 20.9	252°27.8	· · 37.4	Alnilam	275°36.9	-1°11.0					
16	265°25.4	81°39.2	01.5	41°18.3	02.7	149°27.5	20.8	267°30.4	37.4	Betelgeuse	270°51.3	7°24.8					
17	280°27.9	96°38.8	N00°00.3	56°19.0	03.3	164°29.7	20.8	282°33.0	37.5	Canopus	263°51.9	-52°42.2					
18	295°30.4	111°38.4	S00°00.9	71°19.7	S17°03.8	179°32.0	N21°20.8	297°35.6	S03°37.5	Sirius	258°25.6	-16°44.8					
19	310°32.8	126°38.0	02.1	86°20.4	04.3	194°34.2	20.7	312°38.2	37.6	Adhara	255°05.3	-29°00.1					
20	325°35.3	141°37.6	03.3	101°21.2	04.8	209°36.5	20.7	327°40.8	37.7	Procyon	244°50.2	5°09.6					
21	340°37.8	156°37.2	· · 04.6	116°21.9	· · 05.4	224°38.7	· · 20.7	342°43.4	· · 37.7	Pollux	243°16.5	27°57.8					
22	355°40.2	171°36.8	05.8	131°22.6	05.9	239°41.0	20.7	357°46.1	37.8	Avior	234°14.6	-59°35.1					
23	10°42.7	186°36.4	07.0	146°23.3	06.4	254°43.2	20.6	12°48.7	37.8	Suhail	222°46.0	-43°31.9					
Mer.pass. 22:17			ν -0.4' d -1.2' m -3.92			ν 0.7' d 0.5' m 1.52			ν 2.2' d -0.0' m -2.21			ν 2.6' d 0.1' m 0.68					

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	25°45.1	201°36.1	S00°08.2	161°24.1	S17°06.9	269°45.5	N21°20.6	27°51.3	S03°37.9					
1	40°47.6	216°35.7	09.4	176°24.8	07.4	284°47.7	20.6	42°53.9	38.0					
2	55°50.1	231°35.3	10.7	191°25.5	08.0	299°50.0	20.5	57°56.5	38.0					
3	70°52.5	246°34.9	· · 11.9	206°26.2	· · 08.5	314°52.2	· · 20.5	72°59.1	· · 38.1					
4	85°55.0	261°34.5	13.1	221°26.9	09.0	329°54.5	20.5	88°01.7	38.1					
5	100°57.5	276°34.1	14.3	236°27.7	09.5	344°56.7	20.4	103°04.3	38.2					
6	115°59.9	291°33.7	S00°15.5	251°28.4	S17°10.0	359°59.0	N21°20.4	118°07.0	S03°38.3					
7	131°02.4	306°33.3	16.8	266°29.1	10.6	15°01.2	20.4	133°09.6	38.3					
8	146°04.9	321°32.9	18.0	281°29.8	11.1	30°03.5	20.4	148°12.2	38.4					
9	161°07.3	336°32.5	· · 19.2	296°30.6	· · 11.6	45°05.7	· · 20.3	163°14.8	· · 38.4					
10	176°09.8	351°32.1	20.4	311°31.3	12.1	60°08.0	20.3	178°17.4	38.5					
11	191°12.3	6°31.7	21.6	326°32.0	12.6	75°10.2	20.3	193°20.0	38.6					
12	206°14.7	21°31.3	S00°22.9	341°32.7	S17°13.1	90°12.5	N21°20.2	208°22.6	S03°38.6					
13	221°17.2	36°31.0	24.1	356°33.4	13.7	105°14.8	20.2	223°25.2	38.7					
14	236°19.6	51°30.6	25.3	11°34.1	14.2	120°17.0	20.2	238°27.9	38.7					
15	251°22.1	66°30.2	· · 26.5	26°34.9	· · 14.7	135°19.3	· · 20.1	253°30.5	· · 38.8					
16	266°24.6	81°29.8	27.7	41°35.6	15.2	150°21.5	20.1	268°33.1	38.9					
17	281°27.0	96°29.4	29.0	56°36.3	15.7	165°23.8	20.1	283°35.7	38.9					
18	296°29.5	111°29.0	S00°30.2	71°37.0	S17°16.2	180°26.0	N21°20.1	298°38.3	S03°39.0					
19	311°32.0	126°28.6	31.4	86°37.7	16.8	195°28.3	20.0	313°40.9	39.0					
20	326°34.4	141°28.2	32.6	101°38.5	17.3	210°30.5	20.0	328°43.5	39.1					
21	341°36.9	156°27.8	· · 33.9	116°39.2	· · 17.8	225°32.8	· · 20.0	343°46.1	· · 39.2					
22	356°39.4	171°27.4	35.1	131°39.9	18.3	240°35.1	19.9	358°48.7	39.2					
23	11°41.8	186°27.0	36.3	146°40.6	18.8	255°37.3	19.9	13°51.4	39.3					
Mer.pass. 22:13			ν -0.4' d 1.2' m -3.92			ν 0.7' d 0.5' m 1.52			ν 2.3' d -0.0' m -2.22			ν 2.6' d 0.1' m 0.69		

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	26°44.3	201°26.6	S00°37.5	161°41.3	S17°19.3	270°39.6	N21°19.9	28°54.0	S03°39.3					
1	41°46.7	216°26.2	38.7	176°42.0	19.9	285°41.8	19.9	43°56.6	39.4					
2	56°49.2	231°25.8	40.0	191°42.8	20.4	300°44.1	19.8	58°59.2	39.5					
3	71°51.7	246°25.5	· · 41.2	206°43.5	· · 20.9	315°46.3	· · 19.8	74°01.8	· · 39.5					
4	86°54.1	261°25.1	42.4	221°44.2	21.4	330°48.6	19.8	89°04.4	39.6					
5	101°56.6	276°24.7	43.6	236°44.9	21.9	345°50.9	19.7	104°07.0	39.6					
6	116°59.1	291°24.3	S00°44.8	251°45.6	S17°22.4	0°53.1	N21°19.7	119°09.6	S03°39.7					
7	132°01.5	306°23.9	46.1	266°46.3	22.9	15°55.4	19.7	134°12.2	39.7					
8	147°04.0	321°23.5	47.3	281°47.0	23.5	30°57.6	19.7	149°14.9	39.8					
9	162°06.5	336°23.1	· · 48.5	296°47.8	· · 24.0	45°59.9	· · 19.6	164°17.5	· · 39.9					
10	177°08.9	351°22.7	49.7	311°48.5	24.5	61°02.2	19.6	179°20.1	39.9					
11	192°11.4	6°22.3	51.0	326°49.2	25.0	76°04.4	19.6	194°22.7	40.0					
12	207°13.9	21°21.9	S00°52.2	341°49.9	S17°25.5	91°06.7	N21°19.5	209°25.3	S03°40.0					
13	222°16.3	36°21.5	53.4	356°50.6	26.0	106°09.0	19.5	224°27.9	40.1					
14	237°18.8	51°21.1	54.6	11°51.3	26.5	121°11.2	19.5	239°30.5	40.2					
15	252°21.2	66°20.7	· · 55.8	26°52.0	· · 27.0	136°13.5	· · 19.5	254°33.1	· · 40.2					
16	267°23.7	81°20.3	57.1	41°52.7	27.6	151°15.7	19.4	269°35.7	40.3					
17	282°26.2	96°19.9	58.3	56°53.5	28.1	166°18.0	19.4	284°38.3	40.3					
18	297°28.6	111°19.5	S00°59.5	71°54.2	S17°28.6	181°20.3	N21°19.4	299°40.9	S03°40.4					
19	312°31.1	126°19.1	01°00.7	86°54.9	29.1	196°22.5	19.4	314°43.6	40.4					
20	327°33.6	141°18.8	02.0	101°55.6	29.6	211°24.8	19.3	329°46.2	40.5					
21	342°36.0	156°18.4	· · 03.2	116°56.3	· · 30.1	226°27.1	· · 19.3	344°48.8	· · 40.6					
22	357°38.5	171°18.0	04.4	131°57.0	30.6	241°29.3	19.3	359°51.4	40.6					
23	12°41.0	186°17.6	05.6	146°57.7	31.1	256°31.6	19.2	14°54.0	40.7					
Mer.pass. 22:09			ν -0.4' d 1.2' m -3.92			ν 0.7' d 0.5' m 1.52			ν 2.3' d -0.0' m -2.23			ν 2.6' d 0.1' m 0.69		

Oct 16 Thu	SHA	Mer.pass
Venus	176°59.4	10:33
Mars	136°20.6	13:15
Jupiter	244°05.5	06:04
Saturn	2°02.5	22:09

Oct 17 Fri	SHA	Mer.pass
Venus	175°50.9	10:34
Mars	135°38.9	13:14
Jupiter	244°00.3	06:00
Saturn	2°06.1	22:05

Oct 18 Sat	SHA	Mer.pass
Venus	174°42.3	10:35
Mars	134°57.0	13:13
Jupiter	243°55.3	05:56
Saturn	2°09.7	22:01

Horizontal parallax	
Venus:	0.1
Mars:	0.1

Sun				Moon				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	183°35.8	S08°53.9		241°12.2	12.5'	N16°59.4	-12.8'	56.5'
1	198°36.0	54.8		255°43.7	12.6'	16°46.6	-12.9'	56.5'
2	213°36.1	55.7		270°15.3	12.7'	16°33.7	-12.9'	56.5'
3	228°36.2	•• 56.6		284°47.0	12.8'	16°20.8	-13.0'	56.4'
4	243°36.4	57.6		299°18.8	12.9'	16°07.8	-13.1'	56.4'
5	258°36.5	58.5		313°50.6	13.0'	15°54.7	-13.1'	56.4'
6	273°36.6	S08°59.4		328°22.6	13.1'	N15°41.6	-13.2'	56.4'
7	288°36.8	09°00.3		342°54.7	13.1'	15°28.4	-13.2'	56.3'
8	303°36.9	01.2		357°26.8	13.2'	15°15.2	-13.3'	56.3'
9	318°37.0	•• 02.2		11°59.0	13.3'	15°01.9	-13.3'	56.3'
10	333°37.1	03.1		26°31.4	13.4'	14°48.5	-13.4'	56.3'
11	348°37.3	04.0		41°03.8	13.5'	14°35.1	-13.5'	56.2'
12	3°37.4	S09°04.9		55°36.3	13.6'	N14°21.7	-13.5'	56.2'
13	18°37.5	05.8		70°08.8	13.7'	14°08.2	-13.6'	56.2'
14	33°37.7	06.7		84°41.5	13.7'	13°54.6	-13.6'	56.2'
15	48°37.8	•• 07.7		99°14.2	13.8'	13°41.0	-13.7'	56.1'
16	63°37.9	08.6		113°47.1	13.9'	13°27.4	-13.7'	56.1'
17	78°38.1	09.5		128°20.0	14.0'	13°13.7	-13.7'	56.1'
18	93°38.2	S09°10.4		142°53.0	14.1'	N12°59.9	-13.8'	56.1'
19	108°38.3	11.3		157°26.0	14.1'	12°46.1	-13.8'	56.0'
20	123°38.4	12.2		171°59.2	14.2'	12°32.3	-13.9'	56.0'
21	138°38.6	•• 13.2		186°32.4	14.3'	12°18.4	-13.9'	56.0'
22	153°38.7	14.1		201°05.6	14.4'	12°04.5	-14.0'	56.0'
23	168°38.8	15.0		215°39.0	14.4'	11°50.5	-14.0'	56.0'
SD = 16.0' d = 0.9'				SD = 15.4'				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	183°39.0	S09°15.9		230°12.4	14.5'	N11°36.5	-14.0'	55.9'
1	198°39.1	16.8		244°45.9	14.6'	11°22.5	-14.1'	55.9'
2	213°39.2	17.7		259°19.5	14.6'	11°08.4	-14.1'	55.9'
3	228°39.3	•• 18.6		273°53.1	14.7'	10°54.3	-14.1'	55.9'
4	243°39.5	19.5		288°26.8	14.8'	10°40.2	-14.2'	55.8'
5	258°39.6	20.5		303°00.6	14.8'	10°26.0	-14.2'	55.8'
6	273°39.7	S09°21.4		317°34.4	14.9'	N10°11.8	-14.2'	55.8'
7	288°39.8	22.3		332°08.3	15.0'	09°57.6	-14.3'	55.8'
8	303°40.0	23.2		346°42.3	15.0'	09°43.3	-14.3'	55.7'
9	318°40.1	•• 24.1		1°16.3	15.1'	09°29.0	-14.3'	55.7'
10	333°40.2	25.0		15°50.4	15.1'	09°14.7	-14.3'	55.7'
11	348°40.3	25.9		30°24.5	15.2'	09°00.3	-14.4'	55.7'
12	3°40.5	S09°26.8		44°58.7	15.2'	N08°46.0	-14.4'	55.7'
13	18°40.6	27.8		59°33.0	15.3'	08°31.6	-14.4'	55.6'
14	33°40.7	28.7		74°07.3	15.4'	08°17.1	-14.4'	55.6'
15	48°40.8	•• 29.6		88°41.6	15.4'	08°02.7	-14.5'	55.6'
16	63°41.0	30.5		103°16.0	15.5'	07°48.2	-14.5'	55.6'
17	78°41.1	31.4		117°50.5	15.5'	07°33.7	-14.5'	55.6'
18	93°41.2	S09°32.3		132°25.0	15.6'	N07°19.2	-14.5'	55.5'
19	108°41.3	33.2		146°59.6	15.6'	07°04.7	-14.5'	55.5'
20	123°41.5	34.1		161°34.2	15.7'	06°50.2	-14.6'	55.5'
21	138°41.6	•• 35.0		176°08.8	15.7'	06°35.6	-14.6'	55.5'
22	153°41.7	36.0		190°43.5	15.7'	06°21.0	-14.6'	55.4'
23	168°41.8	36.9		205°18.3	15.8'	06°06.4	-14.6'	55.4'
SD = 16.0' d = 0.9'				SD = 15.3'				
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	183°41.9	S09°37.8		219°53.0	15.8'	N05°51.8	-14.6'	55.4'
1	198°42.1	38.7		234°27.9	15.9'	05°37.2	-14.6'	55.4'
2	213°42.2	39.6		249°02.7	15.9'	05°22.6	-14.6'	55.4'
3	228°42.3	•• 40.5		263°37.7	15.9'	05°07.9	-14.7'	55.3'
4	243°42.4	41.4		278°12.6	16.0'	04°53.3	-14.7'	55.3'
5	258°42.6	42.3		292°47.6	16.0'	04°38.6	-14.7'	55.3'
6	273°42.7	S09°43.2		307°22.6	16.1'	N04°23.9	-14.7'	55.3'
7	288°42.8	44.1		321°57.6	16.1'	04°09.3	-14.7'	55.3'
8	303°42.9	45.0		336°32.7	16.1'	03°54.6	-14.7'	55.3'
9	318°43.0	•• 45.9		351°07.9	16.2'	03°39.9	-14.7'	55.2'
10	333°43.1	46.8		5°43.0	16.2'	03°25.2	-14.7'	55.2'
11	348°43.3	47.7		20°18.2	16.2'	03°10.5	-14.7'	55.2'
12	3°43.4	S09°48.7		34°53.4	16.2'	N02°55.8	-14.7'	55.2'
13	18°43.5	49.6		49°28.6	16.3'	02°41.1	-14.7'	55.2'
14	33°43.6	50.5		64°03.9	16.3'	02°26.4	-14.7'	55.1'
15	48°43.7	•• 51.4		78°39.2	16.3'	02°11.7	-14.7'	55.1'
16	63°43.9	52.3		93°14.5	16.3'	01°57.0	-14.7'	55.1'
17	78°44.0	53.2		107°49.8	16.4'	01°42.2	-14.7'	55.1'
18	93°44.1	S09°54.1		122°25.2	16.4'	N01°27.5	-14.7'	55.1'
19	108°44.2	55.0		137°00.6	16.4'	01°12.8	-14.7'	55.1'
20	123°44.3	55.9		151°36.0	16.4'	00°58.2	-14.7'	55.0'
21	138°44.4	•• 56.8		166°11.4	16.4'	00°43.5	-14.7'	55.0'
22	153°44.5	57.7		180°46.8	16.5'	00°28.8	-14.7'	55.0'
23	168°44.7	58.6		195°22.3	16.5'	00°14.1	-14.7'	55.0'
SD = 16.0' d = 0.9'				SD = 15.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:05	06:23	07:35	15:54	17:05	18:23
N 70°	05:08	06:18	07:22	16:07	17:10	18:21
68°	05:10	06:14	07:12	16:17	17:15	18:19
66°	05:11	06:11	07:04	16:26	17:18	18:18
64°	05:12	06:07	06:56	16:33	17:22	18:17
62°	05:13	06:05	06:50	16:39	17:25	18:16
60°	05:14	06:02	06:45	16:45	17:27	18:15
N 58°	05:15	06:00	06:40	16:50	17:29	18:15
56°	05:15	05:58	06:36	16:54	17:32	18:15
54°	05:15	05:56	06:32	16:58	17:34	18:14
52°	05:15	05:54	06:29	17:01	17:35	18:14
50°	05:15	05:53	06:25	17:04	17:37	18:15
45°	05:15	05:49	06:18	17:11	17:41	18:15
N 40°	05:14	05:45	06:13	17:17	17:45	18:16
35°	05:13	05:42	06:08	17:22	17:48	18:17
30°	05:11	05:39	06:03	17:27	17:51	18:19
20°	05:08	05:33	05:56	17:35	17:57	18:23
N 10°	05:03	05:27	05:49	17:42	18:03	18:28
0°	04:57	05:21	05:42	17:49	18:10	18:34
S 10°	04:49	05:14	05:35	17:55	18:17	18:42
20°	04:39	05:06	05:28	18:03	18:25	18:52
30°	04:26	04:55	05:20	18:11	18:36	19:05
35°	04:18	04:49	05:15	18:16	18:42	19:13
40°	04:08	04:41	05:09	18:22	18:50	19:24
45°	03:55	04:32	05:03	18:29	18:59	19:36
S 50°	03:39	04:21	04:55	18:37	19:11	19:53
52°	03:31	04:15	04:51	18:41	19:16	20:01
54°	03:22	04:09	04:47	18:45	19:23	20:10
56°	03:12	04:03	04:43	18:49	19:29	20:21
58°	03:00	03:55	04:38	18:54	19:37	20:33
S 60°	02:46	03:46	04:32	19:00	19:46	20:47

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	•• ••	00:05	02:19	17:29	16:44	16:10
N 70°	•• ••	00:25	02:27	17:06	16:33	16:07
68°	•• ••	00:41	02:33	16:48	16:25	16:05
66°	•• ••	00:54	02:39	16:33	16:17	16:04
64°	•• ••	01:05	02:43	16:21	16:11	16:02
62°	•• ••	01:14	02:47	16:11	16:06	16:01
60°	•• ••	01:22	02:50	16:02	16:01	16:00
N 58°	•• ••	01:28	02:53	15:54	15:57	15:59
56°	00:09	01:34	02:55	15:47	15:53	15:58
54°	00:18	01:40	02:58	15:41	15:50	15:57
52°	00:25	01:44	03:00	15:35	15:47	15:56
50°	00:32	01:49	03:02	15:30	15:44	15:56
45°	00:47	01:58	03:06	15:19	15:38	15:54
N 40°	00:59	02:05	03:09	15:10	15:32	15:53
35°	01:09	02:12	03:12	15:02	15:28	15:52
30°	01:18	02:18	03:15	14:55	15:24	15:51
20°	01:34	02:28	03:19	14:42	15:17	15:49
N 10°	01:47	02:36	03:23	14:32	15:11	15:48
0°	01:59	02:44	03:27	14:21	15:05	15:46
S 10°	02:12	02:53	03:30	14:11	14:59	15:45
20°	02:25	03:01	03:34	14:00	14:53	15:43
30°	02:40	03:11	03:39	13:47	14:45	15:41
35°	02:49	03:16	03:41	13:40	14:41	15:40
40°	02:58	03:23	03:44	13:31	14:36	15:39
45°	03:10	03:30	03:47	13:21	14:31	15:38
S 50°	03:24	03:39	03:51	13:09	14:24	15:36
52°	03:30	03:43	03:53	13:04	14:21	15:35
54°	03:37	03:47	03:55	12:57	14:17	15:34
56°	03:45	03:52	03:57	12:50	14:14	15:33
58°	03:54	03:57	03:59	12:42	14:09	15:32
S 60°	04:04	04:03	04:02	12:33	14:04	15:31

Day	Sun			Moon		
	Eqn.of Time		Mer. Pass	Mer.Pass.		Age
	00 ^h	12 ^h	Pass	Upper	Lower	25-27
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	27-11%
16	14:23	14:30	11:46	08:11	20:33	
17	14:36	14:42	11:45	08:55	21:16	
18	14:48	14:54	11:45	09:36	21:57	

October 19, 20, 21 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	27°43.4	201°17.2	S01°06.8	161°58.4	S17°31.6	271°33.9	N21°19.2	29°56.6	S03°40.7	Alpheratz	357°33.8	29°14.2
1	42°45.9	216°16.8	08.1	176°59.1	32.1	286°36.1	19.2	44°59.2	40.8	Ankaa	353°06.1	-42°09.9
2	57°48.3	231°16.4	09.3	191°59.8	32.7	301°38.4	19.2	60°01.8	40.8	Schedar	349°29.8	56°40.9
3	72°50.8	246°16.0	•• 10.5	207°00.5	•• 33.2	316°40.7	•• 19.1	75°04.4	•• 40.9	Diphda	348°46.3	-17°50.6
4	87°53.3	261°15.6	11.7	222°01.3	33.7	331°42.9	19.1	90°07.0	41.0	Achernar	335°19.1	-57°06.3
5	102°55.7	276°15.2	13.0	237°02.0	34.2	346°45.2	19.1	105°09.6	41.0	Hamal	327°50.1	23°35.2
6	117°58.2	291°14.8	S01°14.2	252°02.7	S17°34.7	1°47.5	N21°19.0	120°12.3	S03°41.1	Polaris	313°15.6	89°22.3
7	133°00.7	306°14.4	15.4	267°03.4	35.2	16°49.7	19.0	135°14.9	41.1	Acamar	315°10.9	-40°11.9
8	148°03.1	321°14.0	16.6	282°04.1	35.7	31°52.0	19.0	150°17.5	41.2	Menkar	314°05.2	4°11.6
9	163°05.6	336°13.6	•• 17.8	297°04.8	•• 36.2	46°54.3	•• 19.0	165°20.1	•• 41.3	Mirfak	308°26.9	49°57.2
10	178°08.1	351°13.2	19.1	312°05.5	36.7	61°56.5	18.9	180°22.7	41.3	Aldebaran	290°38.6	16°33.8
11	193°10.5	6°12.8	20.3	327°06.2	37.2	76°58.8	18.9	195°25.3	41.4	Rigel	281°03.1	-8°10.1
12	208°13.0	21°12.4	S01°21.5	342°06.9	S17°37.7	92°01.1	N21°18.9	210°27.9	S03°41.4	Capella	280°20.7	46°01.3
13	223°15.5	36°12.0	22.7	357°07.6	38.2	107°03.3	18.9	225°30.5	41.5	Bellatrix	278°22.0	6°22.5
14	238°17.9	51°11.6	24.0	12°08.3	38.8	122°05.6	18.8	240°33.1	41.5	Elnath	278°00.9	28°37.7
15	253°20.4	66°11.2	•• 25.2	27°09.0	•• 39.3	137°07.9	•• 18.8	255°35.7	•• 41.6	Alnilam	275°36.9	-1°11.0
16	268°22.8	81°10.8	26.4	42°09.7	39.8	152°10.2	18.8	270°38.3	41.7	Betelgeuse	270°51.2	7°24.8
17	283°25.3	96°10.4	27.6	57°10.4	40.3	167°12.4	18.8	285°40.9	41.7	Canopus	263°51.9	-52°42.2
18	298°27.8	111°10.1	S01°28.9	72°11.1	S17°40.8	182°14.7	N21°18.7	300°43.5	S03°41.8	Sirius	258°25.6	-16°44.9
19	313°30.2	126°09.7	30.1	87°11.8	41.3	197°17.0	18.7	315°46.1	41.8	Adhara	255°05.3	-29°00.1
20	328°32.7	141°09.3	31.3	102°12.6	41.8	212°19.2	18.7	330°48.8	41.9	Procyon	244°50.2	5°09.6
21	343°35.2	156°08.9	•• 32.5	117°13.3	•• 42.3	227°21.5	•• 18.6	345°51.4	•• 41.9	Pollux	243°16.5	27°57.8
22	358°37.6	171°08.5	33.7	132°14.0	42.8	242°23.8	18.6	0°54.0	42.0	Avior	234°14.6	-59°35.1
23	13°40.1	186°08.1	35.0	147°14.7	43.3	257°26.1	18.6	15°56.6	42.0	Suhail	222°46.0	-43°31.9
Mer.pass. 22:05		ν -0.4' d 1.2' m -3.92		ν 0.7' d 0.5' m 1.51		ν 2.3' d -0.0' m -2.23		ν 2.6' d 0.1' m 0.70				
0	28°42.6	201°07.7	S01°36.2	162°15.4	S17°43.8	272°28.3	N21°18.6	30°59.2	S03°42.1	Alphard	217°47.3	-8°46.1
1	43°45.0	216°07.3	37.4	177°16.1	44.3	287°30.6	18.5	46°01.8	42.2	Regulus	207°34.0	11°50.5
2	58°47.5	231°06.9	38.6	192°16.8	44.8	302°32.9	18.5	61°04.4	42.2	Dubhe	193°40.7	61°36.6
3	73°50.0	246°06.5	•• 39.9	207°17.5	•• 45.3	317°35.2	•• 18.5	76°07.0	•• 42.3	Denebola	182°24.6	14°25.7
4	88°52.4	261°06.1	41.1	222°18.2	45.8	332°37.4	18.5	91°09.6	42.3	Gienah	175°43.3	-17°41.0
5	103°54.9	276°05.7	42.3	237°18.9	46.3	347°39.7	18.4	106°12.2	42.4	Acrux	173°00.1	-63°14.4
6	118°57.3	291°05.3	S01°43.5	252°19.6	S17°46.8	2°42.0	N21°18.4	121°14.8	S03°42.4	Gacrux	171°51.5	-57°15.3
7	133°59.8	306°04.9	44.7	267°20.3	47.3	17°44.3	18.4	136°17.4	42.5	Alioth	166°12.9	55°49.2
8	149°02.3	321°04.5	46.0	282°21.0	47.8	32°46.5	18.4	151°20.0	42.6	Spica	158°22.0	-11°17.7
9	164°04.7	336°04.1	•• 47.2	297°21.7	•• 48.3	47°48.8	•• 18.3	166°22.6	•• 42.6	Alkaid	152°52.0	49°11.1
10	179°07.2	351°03.7	48.4	312°22.4	48.8	62°51.1	18.3	181°25.2	42.7	Hadar	148°35.8	-60°29.8
11	194°09.7	6°03.3	49.6	327°23.1	49.3	77°53.4	18.3	196°27.8	42.7	Menkent	147°57.3	-36°29.7
12	209°12.1	21°02.9	S01°50.9	342°23.8	S17°49.8	92°55.6	N21°18.3	211°30.4	S03°42.8	Arcturus	145°47.7	19°03.0
13	224°14.6	36°02.5	52.1	357°24.5	50.3	107°57.9	18.2	226°33.0	42.8	Rigil Kent.	139°40.1	-60°56.5
14	239°17.1	51°02.1	53.3	12°25.2	50.8	123°00.2	18.2	241°35.6	42.9	Kochab	137°20.9	74°03.0
15	254°19.5	66°01.7	•• 54.5	27°25.9	•• 51.3	138°02.5	•• 18.2	256°38.3	•• 42.9	Zuben'ubi	136°55.7	-16°08.9
16	269°22.0	81°01.3	55.8	42°26.6	51.8	153°04.8	18.2	271°40.9	43.0	Alphecca	126°03.5	26°37.8
17	284°24.4	96°00.9	57.0	57°27.3	52.3	168°07.0	18.1	286°43.5	43.1	Antares	112°15.4	-26°29.4
18	299°26.9	111°00.5	S01°58.2	72°28.0	S17°52.8	183°09.3	N21°18.1	301°46.1	S03°43.1	Atria	107°09.4	-69°04.6
19	314°29.4	126°00.1	01°59.4	87°28.7	53.3	198°11.6	18.1	316°48.7	43.2	Sabik	102°02.3	-15°45.4
20	329°31.8	140°59.7	02°00.6	102°29.4	53.8	213°13.9	18.1	331°51.3	43.2	Shaula	96°09.8	-37°07.4
21	344°34.3	155°59.3	•• 01.9	117°30.1	•• 54.3	228°16.1	•• 18.0	346°53.9	•• 43.3	Rasalhague	95°58.2	12°32.6
22	359°36.8	170°58.9	03.1	132°30.8	54.8	243°18.4	18.0	1°56.5	43.3	Eltanin	90°42.1	51°29.3
23	14°39.2	185°58.5	04.3	147°31.5	55.3	258°20.7	18.0	16°59.1	43.4	Kaus Aust.	83°31.8	-34°22.4
Mer.pass. 22:02		ν -0.4' d 1.2' m -3.92		ν 0.7' d 0.5' m 1.51		ν 2.3' d -0.0' m -2.24		ν 2.6' d 0.1' m 0.70				
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	29°41.7	200°58.1	S02°05.5	162°32.2	S17°55.8	273°23.0	N21°18.0	32°01.7	S03°43.4	Oct 19 Sun	SHA	Mer.pass
1	44°44.2	215°57.7	06.8	177°32.8	56.3	288°25.3	17.9	47°04.3	43.5	Venus	173°33.8	10:35
2	59°46.6	230°57.3	08.0	192°33.5	56.8	303°27.6	17.9	62°06.9	43.6	Mars	134°15.0	13:11
3	74°49.1	245°56.9	•• 09.2	207°34.2	•• 57.3	318°29.8	•• 17.9	77°09.5	•• 43.6	Jupiter	243°50.4	05:53
4	89°51.6	260°56.5	10.4	222°34.9	57.8	333°32.1	17.9	92°12.1	43.7	Saturn	2°13.2	21:56
5	104°54.0	275°56.1	11.6	237°35.6	58.3	348°34.4	17.8	107°14.7	43.7			
6	119°56.5	290°55.7	S02°12.9	252°36.3	S17°58.8	3°36.7	N21°17.8	122°17.3	S03°43.8			
7	134°58.9	305°55.3	14.1	267°37.0	59.3	18°39.0	17.8	137°19.9	43.8			
8	150°01.4	320°54.9	15.3	282°37.7	17°59.8	33°41.3	17.8	152°22.5	43.9			
9	165°03.9	335°54.5	•• 16.5	297°38.4	18°00.3	48°43.5	•• 17.7	167°25.1	•• 43.9			
10	180°06.3	350°54.1	17.8	312°39.1	00.8	63°45.8	17.7	182°27.7	44.0	Oct 20 Mon	SHA	Mer.pass
11	195°08.8	5°53.7	19.0	327°39.8	01.3	78°48.1	17.7	197°30.3	44.0	Venus	172°25.1	10:36
12	210°11.3	20°53.3	S02°20.2	342°40.5	S18°01.8	93°50.4	N21°17.7	212°32.9	S03°44.1	Mars	133°32.8	13:10
13	225°13.7	35°52.9	21.4	357°41.2	02.3	108°52.7	17.6	227°35.5	44.2	Jupiter	243°45.8	05:49
14	240°16.2	50°52.5	22.6	12°41.9	02.8	123°55.0	17.6	242°38.1	44.2	Saturn	2°16.6	21:52
15	255°18.7	65°52.1	•• 23.9	27°42.6	•• 03.3	138°57.2	•• 17.6	257°40.7	•• 44.3			
16	270°21.1	80°51.7	25.1	42°43.3	03.8	153°59.5	17.6	272°43.3	44.3	Oct 21 Tue	SHA	Mer.pass
17	285°23.6	95°51.3	26.3	57°43.9	04.3	169°01.8	17.5	287°45.9	44.4	Venus	171°16.4	10:36
18	300°26.0	110°50.9	S02°27.5	72°44.6	S18°04.8	184°04.1	N21°17.5	302°48.5	S03°44.4	Mars	132°50.5	13:09
19	315°28.5	125°50.5	28.8	87°45.3	05.3	199°06.4	17.5	317°51.1	44.5	Jupiter	243°41.3	05:46
20	330°31.0	140°50.1	30.0	102°46.0	05.8	214°08.7	17.5	332°53.7	44.5	Saturn	2°20.0	21:48
21	345°33.4	155°49.7	•• 31.2	117°46.7	•• 06.3	229°11.0	•• 17.4	347°56.3	•• 44.6			
22	0°35.9	170°49.3	32.4	132°47.4	06.8	244°13.3	17.4	2°58.9	44.6			
23	15°38.4	185°48.9	33.6	147°48.1	07.2	259°15.5	17.4	18°01.5</				

Sun				Moon				
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	183°44.8	S09°59.5		209°57.8	16.5'	S00°00.6	14.7'	55.0'
1	198°44.9	10°00.4		224°33.2	16.5'	00°15.2	14.7'	54.9'
2	213°45.0	01.3		239°08.7	16.5'	00°29.9	14.6'	54.9'
3	228°45.1	· · 02.2		253°44.3	16.5'	00°44.5	14.6'	54.9'
4	243°45.2	03.1		268°19.8	16.5'	00°59.2	14.6'	54.9'
5	258°45.3	04.0		282°55.3	16.5'	01°13.8	14.6'	54.9'
6	273°45.5	S10°04.9		297°30.8	16.5'	S01°28.4	14.6'	54.9'
7	288°45.6	05.8		312°06.4	16.6'	01°43.0	14.6'	54.8'
8	303°45.7	06.7		326°42.0	16.6'	01°57.6	14.6'	54.8'
9	318°45.8	· · 07.6		341°17.5	16.6'	02°12.1	14.6'	54.8'
10	333°45.9	08.5		355°53.1	16.6'	02°26.7	14.5'	54.8'
11	348°46.0	09.4		10°28.7	16.6'	02°41.2	14.5'	54.8'
12	3°46.1	S10°10.3		25°04.2	16.6'	S02°55.7	14.5'	54.8'
13	18°46.2	11.2		39°39.8	16.6'	03°10.2	14.5'	54.8'
14	33°46.4	12.1		54°15.4	16.6'	03°24.7	14.5'	54.7'
15	48°46.5	· · 13.0		68°50.9	16.6'	03°39.2	14.4'	54.7'
16	63°46.6	13.9		83°26.5	16.6'	03°53.6	14.4'	54.7'
17	78°46.7	14.8		98°02.1	16.6'	04°08.0	14.4'	54.7'
18	93°46.8	S10°15.7		112°37.7	16.6'	S04°22.4	14.4'	54.7'
19	108°46.9	16.6		127°13.2	16.6'	04°36.8	14.3'	54.7'
20	123°47.0	17.5		141°48.8	16.6'	04°51.2	14.3'	54.7'
21	138°47.1	· · 18.4		156°24.3	16.5'	05°05.5	14.3'	54.6'
22	153°47.2	19.3		170°59.9	16.5'	05°19.8	14.3'	54.6'
23	168°47.3	20.2		185°35.4	16.5'	05°34.0	14.2'	54.6'
SD = 16.0' d = 0.9'				SD = 15.0'				

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	183°47.5	S10°21.1	200°10.9	16.5'	S05°48.3	14.2'	54.6'
1	198°47.6	22.0	214°46.5	16.5'	06°02.5	14.2'	54.6'
2	213°47.7	22.9	229°22.0	16.5'	06°16.7	14.2'	54.6'
3	228°47.8	· · 23.8	243°57.5	16.5'	06°30.9	14.1'	54.6'
4	243°47.9	24.7	258°32.9	16.5'	06°45.0	14.1'	54.5'
5	258°48.0	25.6	273°08.4	16.4'	06°59.1	14.1'	54.5'
6	273°48.1	S10°26.5	287°43.8	16.4'	S07°13.1	14.0'	54.5'
7	288°48.2	27.4	302°19.3	16.4'	07°27.2	14.0'	54.5'
8	303°48.3	28.3	316°54.7	16.4'	07°41.2	14.0'	54.5'
9	318°48.4	· · 29.2	331°30.1	16.4'	07°55.1	13.9'	54.5'
10	333°48.5	30.0	346°05.5	16.4'	08°09.1	13.9'	54.5'
11	348°48.6	30.9	0°40.8	16.3'	08°22.9	13.9'	54.4'
12	3°48.7	S10°31.8	15°16.2	16.3'	S08°36.8	13.8'	54.4'
13	18°48.8	32.7	29°51.5	16.3'	08°50.6	13.8'	54.4'
14	33°48.9	33.6	44°26.8	16.3'	09°04.4	13.7'	54.4'
15	48°49.0	· · 34.5	59°02.1	16.2'	09°18.1	13.7'	54.4'
16	63°49.1	35.4	73°37.3	16.2'	09°31.8	13.7'	54.4'
17	78°49.2	36.3	88°12.5	16.2'	09°45.5	13.6'	54.4'
18	93°49.3	S10°37.2	102°47.7	16.2'	S09°59.1	13.6'	54.4'
19	108°49.5	38.1	117°22.9	16.1'	10°12.7	13.5'	54.4'
20	123°49.6	39.0	131°58.0	16.1'	10°26.2	13.5'	54.3'
21	138°49.7	· · 39.9	146°33.1	16.1'	10°39.7	13.4'	54.3'
22	153°49.8	40.8	161°08.2	16.0'	10°53.1	13.4'	54.3'
23	168°49.9	41.6	175°43.3	16.0'	11°06.5	13.3'	54.3'
SD = 16.1' d = 0.9'				SD = 14.9'			

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	183°50.0	S10°42.5	190°18.3	16.0'	S11°19.9	13.3'	54.3'
1	198°50.1	43.4	204°53.3	16.0'	11°33.2	13.2'	54.3'
2	213°50.2	44.3	219°28.2	15.9'	11°46.4	13.2'	54.3'
3	228°50.3	· · 45.2	234°03.1	15.9'	11°59.6	13.2'	54.3'
4	243°50.4	46.1	248°38.0	15.8'	12°12.8	13.1'	54.3'
5	258°50.5	47.0	263°12.9	15.8'	12°25.9	13.0'	54.2'
6	273°50.6	S10°47.9	277°47.7	15.8'	S12°38.9	13.0'	54.2'
7	288°50.7	48.8	292°22.5	15.7'	12°51.9	12.9'	54.2'
8	303°50.8	49.6	306°57.2	15.7'	13°04.8	12.9'	54.2'
9	318°50.9	· · 50.5	321°31.9	15.7'	13°17.7	12.8'	54.2'
10	333°51.0	51.4	336°06.6	15.6'	13°30.6	12.8'	54.2'
11	348°51.1	52.3	350°41.2	15.6'	13°43.3	12.7'	54.2'
12	3°51.2	S10°53.2	5°15.7	15.5'	S13°56.1	12.7'	54.2'
13	18°51.3	54.1	19°50.3	15.5'	14°08.7	12.6'	54.2'
14	33°51.3	55.0	34°24.8	15.5'	14°21.3	12.5'	54.2'
15	48°51.4	· · 55.9	48°59.2	15.4'	14°33.9	12.5'	54.1'
16	63°51.5	56.7	63°33.6	15.4'	14°46.3	12.4'	54.1'
17	78°51.6	57.6	78°08.0	15.3'	14°58.8	12.4'	54.1'
18	93°51.7	S10°58.5	92°42.3	15.3'	S15°11.1	12.3'	54.1'
19	108°51.8	10°59.4	107°16.6	15.2'	15°23.4	12.2'	54.1'
20	123°51.9	11°00.3	121°50.8	15.2'	15°35.7	12.2'	54.1'
21	138°52.0	· · 01.2	136°25.0	15.1'	15°47.8	12.1'	54.1'
22	153°52.1	02.0	150°59.1	15.1'	15°59.9	12.0'	54.1'
23	168°52.2	02.9	165°33.2	15.0'	16°12.0	12.0'	54.1'
SD = 16.1' d = 0.9'				SD = 14.8'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:18	06:37	07:50	15:37	16:51	18:09
N 70°	05:19	06:30	07:35	15:52	16:58	18:08
68°	05:20	06:25	07:24	16:05	17:03	18:08
66°	05:20	06:20	07:14	16:14	17:08	18:07
64°	05:21	06:16	07:05	16:23	17:12	18:07
62°	05:21	06:12	06:58	16:30	17:16	18:07
60°	05:21	06:09	06:52	16:36	17:19	18:07
N 58°	05:21	06:06	06:47	16:42	17:22	18:07
56°	05:21	06:04	06:42	16:47	17:25	18:08
54°	05:20	06:02	06:38	16:51	17:27	18:08
52°	05:20	05:59	06:34	16:55	17:29	18:08
50°	05:20	05:57	06:30	16:59	17:31	18:09
45°	05:18	05:53	06:22	17:06	17:36	18:10
N 40°	05:17	05:48	06:16	17:13	17:40	18:12
35°	05:15	05:45	06:10	17:19	17:44	18:14
30°	05:13	05:41	06:05	17:24	17:48	18:16
20°	05:08	05:34	05:57	17:33	17:55	18:21
N 10°	05:03	05:27	05:49	17:41	18:02	18:27
0°	04:56	05:20	05:41	17:48	18:09	18:34
S 10°	04:48	05:13	05:34	17:56	18:17	18:42
20°	04:37	05:03	05:26	18:04	18:27	18:53
30°	04:23	04:52	05:16	18:14	18:38	19:07
35°	04:14	04:45	05:11	18:19	18:45	19:16
40°	04:03	04:37	05:05	18:25	18:54	19:28
45°	03:49	04:27	04:58	18:33	19:04	19:41
S 50°	03:32	04:14	04:49	18:42	19:16	19:59
52°	03:24	04:09	04:45	18:46	19:22	20:08
54°	03:14	04:02	04:40	18:51	19:29	20:18
56°	03:03	03:55	04:35	18:56	19:37	20:29
58°	02:50	03:46	04:30	19:01	19:45	20:43
S 60°	02:34	03:37	04:23	19:08	19:55	20:59

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	04:22	06:24	08:40	15:37	15:01	14:12
N 70°	04:20	06:13	08:13	15:42	15:15	14:41
68°	04:19	06:03	07:53	15:47	15:27	15:03
66°	04:18	05:56	07:38	15:50	15:36	15:20
64°	04:17	05:50	07:25	15:53	15:44	15:34
62°	04:16	05:44	07:14	15:56	15:51	15:46
60°	04:15	05:39	07:05	15:58	15:57	15:56
N 58°	04:15	05:35	06:57	16:00	16:02	16:05
56°	04:14	05:32	06:50	16:02	16:07	16:13
54°	04:13	05:28	06:44	16:04	16:11	16:20
52°	04:13	05:25	06:38	16:06	16:15	16:26
50°	04:13	05:23	06:33	16:07	16:19	16:32
45°	04:12	05:17	06:22	16:10	16:27	16:45
N 40°	04:11	05:12	06:13	16:13	16:33	16:55
35°	04:10	05:08	06:06	16:15	16:39	17:04
30°	04:10	05:04	05:59	16:17	16:44	17:12
20°	04:09	04:58	05:48	16:21	16:52	17:25
N 10°	04:08	04:53	05:38	16:24	17:00	17:37
0°	04:07	04:47	05:28	16:27	17:07	17:48
S 10°	04:07	04:42	05:19	16:29	17:14	18:00
20°	04:06	04:37	05:09	16:33	17:22	18:12
30°	04:05	04:31	04:58	16:36	17:31	18:26
35°	04:04	04:28	04:52	16:38	17:36	18:34
40°	04:04	04:24	04:45	16:40	17:42	18:43
45°	04:03	04:19	04:36	16:43	17:48	18:54

October 22, 23, 24 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	30°40.8	200°48.5	S02°34.9	162°48.8	S18°07.7	274°17.8	N21°17.4	33°04.1	S03°44.8	Alpheratz	357°33.8	29°14.2
1	45°43.3	215°48.1	36.1	177°49.5	08.2	289°20.1	17.3	48°06.7	44.8	Ankaa	353°06.1	-42°09.9
2	60°45.8	230°47.7	37.3	192°50.2	08.7	304°22.4	17.3	63°09.3	44.9	Schedar	349°29.8	56°40.9
3	75°48.2	245°47.3	· · 38.5	207°50.8	· · 09.2	319°24.7	· · 17.3	78°11.9	· · 44.9	Diphda	348°46.3	-17°50.6
4	90°50.7	260°46.9	39.7	222°51.5	09.7	334°27.0	17.3	93°14.5	45.0	Achernar	335°19.1	-57°06.3
5	105°53.2	275°46.5	41.0	237°52.2	10.2	349°29.3	17.2	108°17.1	45.0	Hamal	327°50.1	23°35.2
6	120°55.6	290°46.1	S02°42.2	252°52.9	S18°10.7	4°31.6	N21°17.2	123°19.7	S03°45.1	Polaris	313°14.8	89°22.3
7	135°58.1	305°45.7	43.4	267°53.6	11.2	19°33.9	17.2	138°22.3	45.1	Acamar	315°10.9	-40°11.9
8	151°00.5	320°45.3	44.6	282°54.3	11.7	34°36.2	17.2	153°24.9	45.2	Menkar	314°05.2	4°11.6
9	166°03.0	335°44.9	· · 45.9	297°55.0	· · 12.2	49°38.5	· · 17.2	168°27.5	· · 45.2	Mirfak	308°26.9	49°57.2
10	181°05.5	350°44.5	47.1	312°55.7	12.7	64°40.7	17.1	183°30.1	45.3	Aldebaran	290°38.6	16°33.7
11	196°07.9	5°44.1	48.3	327°56.3	13.1	79°43.0	17.1	198°32.7	45.3	Rigel	281°03.0	-8°10.1
12	211°10.4	20°43.7	S02°49.5	342°57.0	S18°13.6	94°45.3	N21°17.1	213°35.3	S03°45.4	Capella	280°20.6	46°01.3
13	226°12.9	35°43.3	50.7	357°57.7	14.1	109°47.6	17.1	228°37.9	45.4	Bellatrix	278°22.0	6°22.5
14	241°15.3	50°42.9	52.0	12°58.4	14.6	124°49.9	17.0	243°40.5	45.5	Elnath	278°00.8	28°37.7
15	256°17.8	65°42.5	· · 53.2	27°59.1	· · 15.1	139°52.2	· · 17.0	258°43.1	· · 45.6	Alnilam	275°36.9	-1°11.0
16	271°20.3	80°42.1	54.4	42°59.8	15.6	154°54.5	17.0	273°45.7	45.6	Betelgeuse	270°51.2	7°24.8
17	286°22.7	95°41.7	55.6	58°00.5	16.1	169°56.8	17.0	288°48.3	45.7	Canopus	263°51.9	-52°42.2
18	301°25.2	110°41.2	S02°56.8	73°01.1	S18°16.6	184°59.1	N21°16.9	303°50.9	S03°45.7	Sirius	258°25.6	-16°44.9
19	316°27.7	125°40.8	58.1	88°01.8	17.1	200°01.4	16.9	318°53.5	45.8	Adhara	255°05.3	-29°00.1
20	331°30.1	140°40.4	02°59.3	103°02.5	17.5	215°03.7	16.9	333°56.1	45.8	Procyon	244°50.1	5°09.6
21	346°32.6	155°40.0	03°00.5	118°03.2	· · 18.0	230°06.0	· · 16.9	348°58.7	· · 45.9	Pollux	243°16.5	27°57.8
22	1°35.0	170°39.6	01.7	133°03.9	18.5	245°08.3	16.9	4°01.3	45.9	Avior	234°14.5	-59°35.1
23	16°37.5	185°39.2	02.9	148°04.6	19.0	260°10.6	16.8	19°03.9	46.0	Suhail	222°46.0	-43°31.9
Mer.pass. 21:54		ν -0.4' d 1.2' m -3.92		ν 0.7' d 0.5' m 1.50		ν 2.3' d -0.0' m -2.25		ν 2.6' d 0.1' m 0.71				
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	31°40.0	200°38.8	S03°04.2	163°05.2	S18°19.5	275°12.9	N21°16.8	34°06.5	S03°46.0	Alphard	217°47.3	-8°46.1
1	46°42.4	215°38.4	05.4	178°05.9	20.0	290°15.2	16.8	49°09.1	46.1	Regulus	207°33.9	11°50.5
2	61°44.9	230°38.0	06.6	193°06.6	20.5	305°17.5	16.8	64°11.7	46.1	Dubhe	193°40.7	61°36.5
3	76°47.4	245°37.6	· · 07.8	208°07.3	· · 21.0	320°19.8	· · 16.7	79°14.3	· · 46.2	Denebola	182°24.6	14°25.7
4	91°49.8	260°37.2	09.0	223°08.0	21.4	335°22.1	16.7	94°16.9	46.2	Gienah	175°43.2	-17°41.0
5	106°52.3	275°36.8	10.3	238°08.7	21.9	350°24.4	16.7	109°19.5	46.3	Acrux	173°00.1	-63°14.4
6	121°54.8	290°36.4	S03°11.5	253°09.3	S18°22.4	5°26.7	N21°16.7	124°22.1	S03°46.3	Gacrux	171°51.5	-57°15.3
7	136°57.2	305°36.0	12.7	268°10.0	22.9	20°29.0	16.7	139°24.7	46.4	Alioth	166°12.9	55°49.1
8	151°59.7	320°35.6	13.9	283°10.7	23.4	35°31.3	16.6	154°27.3	46.4	Spica	158°22.0	-11°17.7
9	167°02.1	335°35.2	· · 15.1	298°11.4	· · 23.9	50°33.6	· · 16.6	169°29.9	· · 46.5	Alkaid	152°52.0	49°11.1
10	182°04.6	350°34.8	16.4	313°12.1	24.4	65°35.9	16.6	184°32.5	46.5	Hadar	148°35.8	-60°29.8
11	197°07.1	5°34.4	17.6	328°12.7	24.8	80°38.2	16.6	199°35.1	46.6	Menkent	147°57.3	-36°29.7
12	212°09.5	20°33.9	S03°18.8	343°13.4	S18°25.3	95°40.5	N21°16.5	214°37.7	S03°46.7	Arcturus	145°47.7	19°03.0
13	227°12.0	35°33.5	20.0	358°14.1	25.8	110°42.8	16.5	229°40.3	46.7	Rigil Kent.	139°40.1	-60°56.5
14	242°14.5	50°33.1	21.2	13°14.8	26.3	125°45.1	16.5	244°42.9	46.8	Kochab	137°20.9	74°03.0
15	257°16.9	65°32.7	· · 22.5	28°15.5	· · 26.8	140°47.4	· · 16.5	259°45.5	· · 46.8	Zuben'ubi	136°55.7	-16°08.9
16	272°19.4	80°32.3	23.7	43°16.1	27.3	155°49.7	16.5	274°48.1	46.9	Alphecca	126°03.5	26°37.8
17	287°21.9	95°31.9	24.9	58°16.8	27.7	170°52.0	16.4	289°50.7	46.9	Antares	112°15.4	-26°29.3
18	302°24.3	110°31.5	S03°26.1	73°17.5	S18°28.2	185°54.3	N21°16.4	304°53.3	S03°47.0	Atria	107°09.4	-69°04.6
19	317°26.8	125°31.1	27.3	88°18.2	28.7	200°56.6	16.4	319°55.9	47.0	Sabik	102°02.3	-15°45.4
20	332°29.3	140°30.7	28.6	103°18.8	29.2	215°58.9	16.4	334°58.5	47.1	Shaula	96°09.8	-37°07.4
21	347°31.7	155°30.3	· · 29.8	118°19.5	· · 29.7	231°01.2	· · 16.3	350°01.1	· · 47.1	Rasalhague	95°58.2	12°32.6
22	2°34.2	170°29.9	31.0	133°20.2	30.2	246°03.5	16.3	5°03.7	47.2	Eltanin	90°42.2	51°29.3
23	17°36.6	185°29.5	32.2	148°20.9	30.6	261°05.8	16.3	20°06.3	47.2	Kaus Aust.	83°31.9	-34°22.4
Mer.pass. 21:50		ν -0.4' d 1.2' m -3.92		ν 0.7' d 0.5' m 1.50		ν 2.3' d -0.0' m -2.26		ν 2.6' d 0.1' m 0.72				
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	32°39.1	200°29.1	S03°33.4	163°21.6	S18°31.1	276°08.1	N21°16.3	35°08.9	S03°47.3	Nunki	75°47.1	-26°15.9
1	47°41.6	215°28.6	34.6	178°22.2	31.6	291°10.4	16.3	50°11.5	47.3	Altair	61°59.4	8°56.3
2	62°44.0	230°28.2	35.9	193°22.9	32.1	306°12.7	16.2	65°14.0	47.4	Peacock	53°04.7	-56°39.3
3	77°46.5	245°27.8	· · 37.1	208°23.6	· · 32.6	321°15.0	· · 16.2	80°16.6	· · 47.4	Deneb	49°25.3	45°22.6
4	92°49.0	260°27.4	38.3	223°24.3	33.0	336°17.3	16.2	95°19.2	47.5	Enif	33°38.1	9°59.8
5	107°51.4	275°27.0	39.5	238°24.9	33.5	351°19.6	16.2	110°21.8	47.5	Al Na'ir	27°31.9	-46°50.3
6	122°53.9	290°26.6	S03°40.7	253°25.6	S18°34.0	6°21.9	N21°16.2	125°24.4	S03°47.6	Fomalhaut	15°13.6	-29°29.1
7	137°56.4	305°26.2	42.0	268°26.3	34.5	21°24.2	16.1	140°27.0	47.6	Scheat	13°44.4	28°13.6
8	152°58.8	320°25.8	43.2	283°27.0	35.0	36°26.6	16.1	155°29.6	47.7	Markab	13°29.0	15°20.8
9	168°01.3	335°25.4	· · 44.4	298°27.6	· · 35.4	51°28.9	· · 16.1	170°32.2	· · 47.7			
10	183°03.8	350°25.0	45.6	313°28.3	35.9	66°31.2	16.1	185°34.8	47.8	Oct 22 Wed	SHA	Mer.pass
11	198°06.2	5°24.6	46.8	328°29.0	36.4	81°33.5	16.1	200°37.4	47.8	Venus	170°07.7	10:37
12	213°08.7	20°24.1	S03°48.0	343°29.6	S18°36.9	96°35.8	N21°16.0	215°40.0	S03°47.9	Mars	132°07.9	13:08
13	228°11.1	35°23.7	49.3	358°30.3	37.3	111°38.1	16.0	230°42.6	47.9	Jupiter	243°37.0	05:42
14	243°13.6	50°23.3	50.5	13°31.0	37.8	126°40.4	16.0	245°45.2	48.0	Saturn	2°23.3	21:44
15	258°16.1	65°22.9	· · 51.7	28°31.7	· · 38.3	141°42.7	· · 16.0	260°47.8	· · 48.0			
16	273°18.5	80°22.5	52.9	43°32.3	38.8	156°45.0	16.0	275°50.4	48.1	Oct 23 Thu	SHA	Mer.pass
17	288°21.0	95°22.1	54.1	58°33.0	39.3	171°47.3	15.9	290°53.0	48.1	Venus	168°58.8	10:38
18	303°23.5	110°21.7	S03°55.3	73°33.7	S18°39.7	186°49.7	N21°15.9	305°55.6	S03°48.2	Mars	131°25.3	13:07
19	318°25.9	125°21.3	56.6	88°34.4	40.2	201°52.0	15.9	320°58.1	48.2	Jupiter	243°32.9	05:38
20	333°28.4	140°20.9	57.8	103°35.0	40.7	216°54.3	15.9	336°00.7	48.3	Saturn	2°26.6	21:40
21	348°30.9	155°20.4	03°59.0	118°35.7	· · 41.2	231°56.6	· · 15.8	351°03.3	· · 48.3			
22	3°33.3	170°20.0	04°00.2	133°3								

Sun				Moon				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	183°52.3	S11°03.8		180°07.2	15.0'	S16°24.0	11.9'	54.1'
1	198°52.4	04.7		194°41.2	14.9'	16°35.9	11.8'	54.1'
2	213°52.5	05.6		209°15.1	14.9'	16°47.7	11.8'	54.1'
3	228°52.6	· · 06.5		223°49.0	14.8'	16°59.5	11.7'	54.0'
4	243°52.7	07.3		238°22.9	14.8'	17°11.2	11.6'	54.0'
5	258°52.8	08.2		252°56.6	14.7'	17°22.8	11.6'	54.0'
6	273°52.9	S11°09.1		267°30.4	14.7'	S17°34.3	11.5'	54.0'
7	288°53.0	10.0		282°04.0	14.6'	17°45.8	11.4'	54.0'
8	303°53.1	10.9		296°37.6	14.6'	17°57.2	11.3'	54.0'
9	318°53.1	· · 11.8		311°11.2	14.5'	18°08.6	11.3'	54.0'
10	333°53.2	12.6		325°44.7	14.5'	18°19.8	11.2'	54.0'
11	348°53.3	13.5		340°18.2	14.4'	18°31.0	11.1'	54.0'
12	3°53.4	S11°14.4		354°51.6	14.3'	S18°42.1	11.0'	54.0'
13	18°53.5	15.3		9°24.9	14.3'	18°53.1	10.9'	54.0'
14	33°53.6	16.2		23°58.2	14.2'	19°04.1	10.9'	54.0'
15	48°53.7	· · 17.0		38°31.4	14.2'	19°15.0	10.8'	54.0'
16	63°53.8	17.9		53°04.6	14.1'	19°25.7	10.7'	54.0'
17	78°53.9	18.8		67°37.7	14.1'	19°36.5	10.6'	54.0'
18	93°54.0	S11°19.7		82°10.8	14.0'	S19°47.1	10.5'	54.0'
19	108°54.0	20.5		96°43.8	13.9'	19°57.6	10.5'	54.0'
20	123°54.1	21.4		111°16.7	13.9'	20°08.1	10.4'	53.9'
21	138°54.2	· · 22.3		125°49.6	13.8'	20°18.5	10.3'	53.9'
22	153°54.3	23.2		140°22.4	13.8'	20°28.7	10.2'	53.9'
23	168°54.4	24.1		154°55.1	13.7'	20°38.9	10.1'	53.9'
SD = 16.1'			$d = 0.9'$	SD = 14.7'				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	183°54.5	S11°24.9		169°27.8	13.6'	S20°49.1	10.0'	53.9'
1	198°54.6	25.8		184°00.5	13.6'	20°59.1	9.9'	53.9'
2	213°54.7	26.7		198°33.1	13.5'	21°09.0	9.8'	53.9'
3	228°54.7	· · 27.6		213°05.6	13.5'	21°18.9	9.8'	53.9'
4	243°54.8	28.4		227°38.0	13.4'	21°28.6	9.7'	53.9'
5	258°54.9	29.3		242°10.4	13.3'	21°38.3	9.6'	53.9'
6	273°55.0	S11°30.2		256°42.7	13.3'	S21°47.9	9.5'	53.9'
7	288°55.1	31.1		271°15.0	13.2'	21°57.3	9.4'	53.9'
8	303°55.2	31.9		285°47.2	13.1'	22°06.7	9.3'	53.9'
9	318°55.3	· · 32.8		300°19.4	13.1'	22°16.0	9.2'	53.9'
10	333°55.3	33.7		314°51.4	13.0'	22°25.2	9.1'	53.9'
11	348°55.4	34.6		329°23.5	13.0'	22°34.3	9.0'	53.9'
12	3°55.5	S11°35.4		343°55.4	12.9'	S22°43.3	8.9'	53.9'
13	18°55.6	36.3		358°27.3	12.8'	22°52.2	8.8'	53.9'
14	33°55.7	37.2		12°59.1	12.8'	23°01.0	8.7'	53.9'
15	48°55.8	· · 38.0		27°30.9	12.7'	23°09.7	8.6'	53.9'
16	63°55.8	38.9		42°02.6	12.6'	23°18.3	8.5'	53.9'
17	78°55.9	39.8		56°34.3	12.6'	23°26.8	8.4'	53.9'
18	93°56.0	S11°40.7		71°05.9	12.5'	S23°35.2	8.3'	53.9'
19	108°56.1	41.5		85°37.4	12.5'	23°43.5	8.2'	53.9'
20	123°56.2	42.4		100°08.8	12.4'	23°51.7	8.1'	53.9'
21	138°56.2	· · 43.3		114°40.2	12.3'	23°59.8	8.0'	53.9'
22	153°56.3	44.1		129°11.6	12.3'	24°07.8	7.9'	53.9'
23	168°56.4	45.0		143°42.8	12.2'	24°15.7	7.8'	53.9'
SD = 16.1'			$d = 0.9'$	SD = 14.7'				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	183°56.5	S11°45.9		158°14.0	12.1'	S24°23.4	7.7'	53.9'
1	198°56.6	46.7		172°45.2	12.1'	24°31.1	7.6'	53.9'
2	213°56.6	47.6		187°16.3	12.0'	24°38.7	7.4'	53.9'
3	228°56.7	· · 48.5		201°47.3	12.0'	24°46.1	7.3'	53.9'
4	243°56.8	49.3		216°18.3	11.9'	24°53.4	7.2'	53.9'
5	258°56.9	50.2		230°49.2	11.8'	25°00.7	7.1'	53.9'
6	273°57.0	S11°51.1		245°20.0	11.8'	S25°07.8	7.0'	53.9'
7	288°57.0	51.9		259°50.8	11.7'	25°14.8	6.9'	53.9'
8	303°57.1	52.8		274°21.5	11.7'	25°21.6	6.8'	53.9'
9	318°57.2	· · 53.7		288°52.2	11.6'	25°28.4	6.7'	53.9'
10	333°57.3	54.5		303°22.8	11.5'	25°35.1	6.5'	53.9'
11	348°57.3	55.4		317°53.3	11.5'	25°41.6	6.4'	53.9'
12	3°57.4	S11°56.3		332°23.8	11.4'	S25°48.0	6.3'	53.9'
13	18°57.5	57.1		346°54.3	11.4'	25°54.3	6.2'	53.9'
14	33°57.6	58.0		1°24.6	11.3'	26°00.5	6.1'	53.9'
15	48°57.7	· · 58.9		15°55.0	11.3'	26°06.6	6.0'	53.9'
16	63°57.7	11°59.7		30°25.2	11.2'	26°12.5	5.8'	53.9'
17	78°57.8	12°00.6		44°55.4	11.2'	26°18.4	5.7'	53.9'
18	93°57.9	S12°01.5		59°25.6	11.1'	S26°24.1	5.6'	53.9'
19	108°58.0	02.3		73°55.7	11.0'	26°29.7	5.5'	53.9'
20	123°58.0	03.2		88°25.7	11.0'	26°35.2	5.3'	53.9'
21	138°58.1	· · 04.1		102°55.7	10.9'	26°40.5	5.2'	53.9'
22	153°58.2	04.9		117°25.7	10.9'	26°45.7	5.1'	53.9'
23	168°58.2	05.8		131°55.5	10.8'	26°50.8	5.0'	53.9'
SD = 16.1'			$d = 0.9'$	SD = 14.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:30	06:50	08:06	15:21	16:37	17:56
N 70°	05:30	06:42	07:49	15:38	16:45	17:56
68°	05:30	06:35	07:35	15:52	16:52	17:57
66°	05:29	06:29	07:24	16:03	16:58	17:57
64°	05:29	06:24	07:15	16:13	17:03	17:58
62°	05:28	06:20	07:07	16:21	17:07	17:59
60°	05:28	06:16	07:00	16:28	17:11	18:00
N 58°	05:27	06:13	06:54	16:34	17:15	18:00
56°	05:26	06:10	06:48	16:39	17:18	18:01
54°	05:26	06:07	06:43	16:44	17:21	18:02
52°	05:25	06:04	06:39	16:49	17:23	18:03
50°	05:24	06:02	06:35	16:53	17:26	18:04
45°	05:22	05:56	06:26	17:02	17:32	18:06
N 40°	05:20	05:52	06:19	17:09	17:36	18:08
35°	05:18	05:47	06:13	17:15	17:41	18:11
30°	05:15	05:43	06:07	17:21	17:45	18:13
20°	05:09	05:35	05:58	17:31	17:53	18:19
N 10°	05:03	05:28	05:49	17:39	18:01	18:26
0°	04:55	05:20	05:41	17:48	18:09	18:33
S 10°	04:46	05:11	05:33	17:56	18:18	18:43
20°	04:35	05:01	05:24	18:05	18:28	18:54
30°	04:19	04:49	05:13	18:16	18:40	19:10
35°	04:10	04:41	05:07	18:22	18:48	19:20
40°	03:58	04:32	05:01	18:29	18:57	19:31
45°	03:44	04:21	04:53	18:37	19:08	19:46
S 50°	03:25	04:08	04:43	18:47	19:22	20:05
52°	03:16	04:02	04:38	18:51	19:28	20:15
54°	03:05	03:55	04:33	18:56	19:36	20:25
56°	02:53	03:47	04:28	19:02	19:44	20:38
58°	02:39	03:37	04:22	19:08	19:53	20:53
S 60°	02:21	03:27	04:15	19:16	20:04	21:11

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	■	■	■	■	■	■
N 70°	10:48	■	■	13:36	■	■
68°	09:59	■	■	14:26	■	■
66°	09:28	11:46	■	14:58	14:14	■
64°	09:06	10:58	■	15:22	15:03	■
62°	08:48	10:27	12:15	15:40	15:34	15:25
60°	08:33	10:04	11:38	15:56	15:58	16:03
N 58°	08:20	09:46	11:12	16:09	16:17	16:30
56°	08:10	09:31	10:51	16:21	16:33	16:51
54°	08:00	09:18	10:34	16:31	16:46	17:08
52°	07:52	09:06	10:19	16:40	16:58	17:23
50°	07:44	08:56	10:07	16:48	17:09	17:36
45°	07:28	08:35	09:41	17:06	17:31	18:03
N 40°	07:15	08:18	09:20	17:20	17:49	18:24
35°	07:04	08:04	09:03	17:32	18:04	18:42
30°	06:55	07:51	08:48	17:43	18:17	18:57
20°	06:38	07:30	08:24	18:01	18:40	19:23
N 10°	06:24	07:12	08:02	18:17	18:59	19:45
0°	06:11	06:55	07:42	18:32	19:18	20:06
S 10°	05:57	06:38	07:23	18:47	19:36	20:27
20°	05:44	06:21	07:02	19:03	19:56	20:50
30°	05:28	06:00	06:37	19:22	20:19	21:16
35°	05:18	05:48	06:23	19:33	20:32	21:31
40°	05:08	05:35	06:07	19:45	20:48	21:49
45°	04:56	05:19	05:47	20:00	21:07	22:11
S 50°	04:41	04:59	05:23	20:19	21:30	22:39
52°	04:34	04:50	05:11	20:28	21:42	22:53
54°	04:27	04:39	04:58	20:37	21:54	23:08
56°	04:18	04:28	04:42	20:48	22:09	23:27

Aries		Venus		Mars		Jupiter		Saturn	
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	33°38.2	200°19.2	S04°02.6	163°37.7	S18°42.6	277°03.5	N21°15.8	36°11.1	S03°48.5
1	48°40.7	215°18.8	03.9	178°38.4	43.1	292°05.8	15.8	51°13.7	48.5
2	63°43.2	230°18.4	05.1	193°39.0	43.5	307°08.2	15.8	66°16.3	48.6
3	78°45.6	245°18.0	· · 06.3	208°39.7	· · 44.0	322°10.5	· · 15.7	81°18.9	· · 48.6
4	93°48.1	260°17.6	07.5	223°40.4	44.5	337°12.8	15.7	96°21.5	48.7
5	108°50.6	275°17.1	08.7	238°41.0	45.0	352°15.1	15.7	111°24.1	48.7
6	123°53.0	290°16.7	S04°09.9	253°41.7	S18°45.4	7°17.4	N21°15.7	126°26.7	S03°48.8
7	138°55.5	305°16.3	11.2	268°42.4	45.9	22°19.7	15.7	141°29.3	48.8
8	153°58.0	320°15.9	12.4	283°43.1	46.4	37°22.0	15.6	156°31.8	48.9
9	169°00.4	335°15.5	· · 13.6	298°43.7	· · 46.9	52°24.4	· · 15.6	171°34.4	· · 48.9
10	184°02.9	350°15.1	14.8	313°44.4	47.3	67°26.7	15.6	186°37.0	49.0
11	199°05.4	5°14.7	16.0	328°45.1	47.8	82°29.0	15.6	201°32.6	49.0
12	214°07.8	20°14.2	S04°17.2	343°45.7	S18°48.3	97°31.3	N21°15.6	216°42.2	S03°49.1
13	229°10.3	35°13.8	18.4	358°46.4	48.7	112°33.6	15.5	231°44.8	49.1
14	244°12.7	50°13.4	19.7	373°47.1	49.2	127°35.9	15.5	246°47.4	49.2
15	259°15.2	65°13.0	· · 20.9	28°47.7	· · 49.7	142°38.3	· · 15.5	261°50.0	· · 49.2
16	274°17.7	80°12.6	22.1	43°48.4	50.2	157°40.6	15.5	276°52.6	49.3
17	289°20.1	95°12.2	23.3	58°49.0	50.6	172°42.9	15.5	291°55.2	49.3
18	304°22.6	110°11.7	S04°24.5	73°49.7	S18°51.1	187°45.2	N21°15.4	306°57.8	S03°49.4
19	319°25.1	125°11.3	25.7	88°50.4	51.6	202°47.5	15.4	322°00.4	49.4
20	334°27.5	140°10.9	26.9	103°51.0	52.0	217°49.9	15.4	337°02.9	49.5
21	349°30.0	155°10.5	· · 28.2	118°51.7	· · 52.5	232°52.2	· · 15.4	352°05.5	· · 49.5
22	4°32.5	170°10.1	29.4	133°52.4	53.0	247°54.5	15.4	7°08.1	49.6
23	19°34.9	185°09.7	30.6	148°53.0	53.4	262°56.8	15.3	22°10.7	49.6
Mer.pass. 21:42		ν -0.4' d 1.2' m -3.92		ν 0.7' d 0.5' m 1.49		ν 2.3' d -0.0' m -2.27		ν 2.6' d 0.0' m 0.73	

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	34° 37.4	200° 09.2	S04° 31.8	163° 53.7	S18° 53.9	277° 59.1	N21° 15.3	37° 13.3	S03° 49.7
1	49° 39.9	215° 08.8	33.0	178° 54.4	54.4	293° 01.5	15.3	52° 15.9	49.7
2	64° 42.3	230° 08.4	34.2	193° 55.0	54.8	308° 03.8	15.3	67° 18.5	49.7
3	79° 44.8	245° 08.0	· · 35.4	208° 55.7	· · 55.3	323° 06.1	· · 15.3	82° 21.1	· · 49.8
4	94° 47.2	260° 07.6	36.6	223° 56.3	55.8	338° 08.4	15.3	97° 23.7	49.8
5	109° 49.7	275° 07.2	37.9	238° 57.0	56.3	353° 10.8	15.2	112° 26.2	49.9
6	124° 52.2	290° 06.7	S04° 39.1	253° 57.7	S18° 56.7	8° 13.1	N21° 15.2	127° 28.8	S03° 49.9
7	139° 54.6	305° 06.3	40.3	268° 58.3	57.2	23° 15.4	15.2	142° 31.4	50.0
8	154° 57.1	320° 05.9	41.5	283° 59.0	57.7	38° 17.7	15.2	157° 34.0	50.0
9	169° 59.6	335° 05.5	· · 42.7	298° 59.6	· · 58.1	53° 20.0	· · 15.2	172° 36.6	· · 50.1
10	185° 02.0	350° 05.1	43.9	314° 00.3	58.6	68° 22.4	15.1	187° 39.2	50.1
11	200° 04.5	5° 04.7	45.1	329° 01.0	59.1	83° 24.7	15.1	202° 41.8	50.2
12	215° 07.0	20° 04.2	S04° 46.3	344° 01.6	S18° 59.5	98° 27.0	N21° 15.1	217° 44.4	S03° 50.2
13	230° 09.4	35° 03.8	47.6	359° 02.3	19° 00.0	113° 29.3	15.1	232° 47.0	50.3
14	245° 11.9	50° 03.4	48.8	14° 02.9	00.4	128° 31.7	15.1	247° 49.5	50.3
15	260° 14.4	65° 03.0	· · 50.0	29° 03.6	· · 00.9	143° 34.0	· · 15.1	262° 52.1	· · 50.4
16	275° 16.8	80° 02.6	51.2	44° 04.3	01.4	158° 36.3	15.0	277° 54.7	50.4
17	290° 19.3	95° 02.1	52.4	59° 04.9	01.8	173° 38.7	15.0	292° 57.3	50.5
18	305° 21.7	110° 01.7	S04° 53.6	74° 05.6	S19° 02.3	188° 41.0	N21° 15.0	307° 59.9	S03° 50.5
19	320° 24.2	125° 01.3	54.8	89° 06.2	02.8	203° 43.3	15.0	323° 02.5	50.6
20	335° 26.7	140° 00.9	56.0	104° 06.9	03.2	218° 45.6	15.0	338° 05.1	50.6
21	350° 29.1	155° 00.5	· · 57.2	119° 07.6	· · 03.7	233° 48.0	· · 15.0	353° 07.7	· · 50.7
22	5° 31.6	170° 00.0	58.5	134° 08.2	04.2	248° 50.3	14.9	8° 10.2	50.7
23	20° 34.1	184° 59.6	59.7	149° 08.9	04.6	263° 52.6	14.9	23° 12.8	50.7
Mer.pass. 21:38 ν -0.4' d 1.2' m -3.91 ν 0.7' d 0.5' m 1.48 ν 2.3' d -0.0' m -2.28 ν 2.6' d 0.0' m 0.74									

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	35°36.5	199°59.2	S05°00.9	164°09.5	S19°05.1	278°55.0	N21°14.9	38°15.4	S03°50.8
1	50°39.0	214°58.8	02.1	179°10.2	05.5	293°57.3	14.9	53°18.0	50.8
2	65°41.5	229°58.3	03.3	194°10.8	06.0	308°59.6	14.9	68°20.6	50.9
3	80°43.9	244°57.9	· · 04.5	209°11.5	· · 06.5	324°01.9	· · 14.8	83°23.2	· · 50.9
4	95°46.4	259°57.5	05.7	224°12.2	06.9	339°04.3	14.8	98°25.8	51.0
5	110°48.9	274°57.1	06.9	239°12.8	07.4	354°06.6	14.8	113°28.4	51.0
6	125°51.3	289°56.7	S05°08.1	254°13.5	S19°07.9	9°08.9	N21°14.8	128°30.9	S03°51.1
7	140°53.8	304°56.2	09.3	269°14.1	08.3	24°11.3	14.8	143°33.5	51.1
8	155°56.2	319°55.8	10.5	284°14.8	08.8	39°13.6	14.8	158°36.1	51.2
9	170°58.7	334°55.4	· · 11.8	299°15.4	· · 09.2	54°15.9	· · 14.7	173°38.7	· · 51.2
10	186°01.2	349°55.0	13.0	314°16.1	09.7	69°18.3	14.7	188°41.3	51.3
11	201°03.6	4°54.5	14.2	329°16.7	10.2	84°20.6	14.7	203°43.9	51.3
12	216°06.1	19°54.1	S05°15.4	344°17.4	S19°10.6	99°22.9	N21°14.7	218°46.5	S03°51.3
13	231°08.6	34°53.7	16.6	359°18.0	11.1	114°25.3	14.7	233°49.0	51.4
14	246°11.0	49°53.3	17.8	14°18.7	11.5	129°27.6	14.7	248°51.6	51.4
15	261°13.5	64°52.8	· · 19.0	29°19.3	· · 12.0	144°29.9	· · 14.7	263°54.2	· · 51.5
16	276°16.0	79°52.4	20.2	44°20.0	12.5	159°32.3	14.6	278°56.8	51.5
17	291°18.4	94°52.0	21.4	59°20.6	12.9	174°34.6	14.6	293°59.4	51.6
18	306°20.9	109°51.6	S05°22.6	74°21.3	S19°13.4	189°36.9	N21°14.6	309°02.0	S03°51.6
19	321°23.3	124°51.1	23.8	89°21.9	13.8	204°39.3	14.6	324°04.5	51.7
20	336°25.8	139°50.7	25.0	104°22.6	14.3	219°41.6	14.6	339°07.1	51.7
21	351°28.3	154°50.3	· · 26.2	119°23.3	· · 14.7	234°43.9	· · 14.6	354°09.7	· · 51.8
22	6°30.7	169°49.9	27.4	134°23.9	15.2	249°46.3	14.5	9°12.3	51.8
23	21°33.2	184°49.4	28.7	149°24.6	15.7	264°48.6	14.5	24°14.9	51.8
Mer.pass. 21:34 ν -0.4' d 1.2' m -3.91 ν 0.7' d 0.5' m 1.48 ν 2.3' d -0.0' m -2.29 ν 2.6' d 0.0' m 0.74									

Stars		
	SHA	Dec
Alpheratz	357° 33.8	29° 14.2
Ankaa	353° 06.1	-42° 09.9
Schedar	349° 29.8	56° 40.9
Diphda	348° 46.3	-17° 50.6
Achernar	335° 19.1	-57° 06.3
Hamal	327° 50.1	23° 35.2
Polaris	313° 13.9	89° 22.3
Acamar	315° 10.9	-40° 11.1
Menkar	314° 05.2	4° 11.6
Mirfak	308° 26.9	49° 57.2
Aldebaran	290° 38.6	16° 33.7
Rigel	281° 03.0	-8° 10.1
Capella	280° 20.6	46° 01.3
Bellatrix	278° 22.0	6° 22.5
Elnath	278° 00.8	28° 37.7
Anilam	275° 36.9	-1° 11.0
Betelgeuse	270° 51.2	7° 24.8
Canopus	263° 51.9	-52° 42.2
Sirius	258° 25.5	-16° 44.9
Adhara	255° 05.2	-29° 00.1
Procyon	244° 50.1	5° 09.6
Pollux	243° 16.5	27° 57.8
Avior	234° 14.5	-59° 35.1
Suhail	222° 46.0	-43° 31.9
Miaplacidus	221° 38.4	-69° 49.0
Alphard	217° 47.2	-8° 46.1
Regulus	207° 33.9	11° 50.5
Dubhe	193° 40.6	61° 36.5
Denebola	182° 24.6	14° 25.7
Gienah	175° 43.2	-17° 41.0
Acrux	173° 00.1	-63° 14.4
Gacrux	171° 51.5	-57° 15.3
Alioth	166° 12.9	55° 49.1
Spica	158° 22.0	-11° 17.7
Alkaid	152° 52.0	49° 11.1
Hadar	148° 35.8	-60° 29.8
Menkent	147° 57.3	-36° 29.7
Arcturus	145° 47.7	19° 02.9
Rigil Kent.	139° 40.1	-60° 56.5
Kochab	137° 27.9	74° 02.2
Zuben'ubi	136° 55.7	-16° 08.9
Alphecca	126° 03.5	26° 37.8
Antares	112° 15.4	-26° 29.3
Atria	107° 09.4	-69° 04.6
Sabik	102° 02.3	-15° 45.4
Shaula	96° 09.8	-37° 07.4
Rasalhague	95° 58.2	12° 32.6
Eltanin	90° 42.2	51° 29.3
Kaus Aust.	83° 31.9	-34° 22.4
Vega	80° 33.0	38° 48.7
Nunki	75° 47.1	-26° 15.9
Altair	61° 59.4	8° 56.3
Peacock	53° 04.7	-56° 39.3
Deneb	49° 25.3	45° 22.6
Enif	33° 38.1	9° 59.8
Al Na'ir	27° 31.9	-46° 50.3
Fomalhaut	15° 13.6	-29° 29.1
Scheat	13° 44.4	-28° 13.6
Markab	13° 29.1	15° 20.2

Oct 25 Sat	SHA	Mer.pass
Venus	166° 41.0	10:39
Mars	129° 59.5	13:05
Jupiter	243° 25.3	05:31
Saturn	2° 32.9	21:32

Oct 26 Sun	SHA	Mer.pass
Venus	165° 31.9	10:40
Mars	129° 16.3	13:04
Jupiter	243° 21.8	05:27
Saturn	2° 35.9	21:27

Oct 27 Mon	SHA	Mer.pass
Venus	164° 22.7	10:40
Mars	128° 33.0	13:03
Jupiter	243° 18.4	05:23
Saturn	2° 38.9	21:23

Horizontal parallax	
Venus:	0.1
Mars:	0.1

Sun				Moon				
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	183°58.3	S12°06.6		146°25.4	10.8'	S26°55.8	4.8'	53.9'
1	198°58.4	07.5		160°55.2	10.7'	27°00.6	4.7'	53.9'
2	213°58.5	08.4		175°24.9	10.7'	27°05.4	4.6'	53.9'
3	228°58.5	09.2		189°54.6	10.6'	27°10.0	4.5'	54.0'
4	243°58.6	10.1		204°24.2	10.6'	27°14.4	4.3'	54.0'
5	258°58.7	10.9		218°53.8	10.5'	27°18.8	4.2'	54.0'
6	273°58.7	S12°11.8		233°23.4	10.5'	S27°23.0	4.1'	54.0'
7	288°58.8	12.7		247°52.9	10.5'	27°27.1	4.0'	54.0'
8	303°58.9	13.5		262°22.3	10.4'	27°31.0	3.8'	54.0'
9	318°59.0	14.4		276°51.7	10.4'	27°34.9	3.7'	54.0'
10	333°59.0	15.2		291°21.1	10.3'	27°38.6	3.6'	54.0'
11	348°59.1	16.1		305°50.4	10.3'	27°42.1	3.4'	54.0'
12	3°59.2	S12°16.9		320°19.7	10.2'	S27°45.6	3.3'	54.0'
13	18°59.2	17.8		334°48.9	10.2'	27°48.9	3.2'	54.0'
14	33°59.3	18.7		349°18.1	10.2'	27°52.0	3.0'	54.0'
15	48°59.4	19.5		3°47.3	10.1'	27°55.1	2.9'	54.0'
16	63°59.4	20.4		18°16.4	10.1'	27°58.0	2.8'	54.0'
17	78°59.5	21.2		32°45.5	10.1'	28°00.8	2.6'	54.1'
18	93°59.6	S12°22.1		47°14.6	10.0'	S28°03.4	2.5'	54.1'
19	108°59.6	22.9		61°43.6	10.0'	28°05.9	2.4'	54.1'
20	123°59.7	23.8		76°12.6	10.0'	28°08.3	2.2'	54.1'
21	138°59.8	24.7		90°41.5	9.9'	28°10.5	2.1'	54.1'
22	153°59.8	25.5		105°10.5	9.9'	28°12.6	2.0'	54.1'
23	168°59.9	26.4		119°39.3	9.9'	28°14.5	1.8'	54.1'
SD = 16.1'			$d = 0.9'$	SD = 14.7'				
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	184°00.0	S12°27.2		134°08.2	9.8'	S28°16.4	1.7'	54.1'
1	199°00.0	28.1		148°37.0	9.8'	28°18.1	1.6'	54.1'
2	214°00.1	28.9		163°05.8	9.8'	28°19.6	1.4'	54.1'
3	229°00.2	29.8		177°34.6	9.8'	28°21.0	1.3'	54.1'
4	244°00.2	30.6		192°03.4	9.7'	28°22.3	1.1'	54.2'
5	259°00.3	31.5		206°32.1	9.7'	28°23.4	1.0'	54.2'
6	274°00.4	S12°32.3		221°00.8	9.7'	S28°24.4	0.9'	54.2'
7	289°00.4	33.2		235°29.5	9.7'	28°25.3	0.7'	54.2'
8	304°00.5	34.0		249°58.2	9.6'	28°26.0	0.6'	54.2'
9	319°00.5	34.9		264°26.8	9.6'	28°26.6	0.4'	54.2'
10	334°00.6	35.7		278°55.5	9.6'	28°27.0	0.3'	54.2'
11	349°00.7	36.6		293°24.1	9.6'	28°27.3	0.2'	54.2'
12	4°00.7	S12°37.4		307°52.7	9.6'	S28°27.5	0.0'	54.3'
13	19°00.8	38.3		322°21.2	9.6'	28°27.5	-0.1'	54.3'
14	34°00.9	39.1		336°49.8	9.6'	28°27.4	-0.3'	54.3'
15	49°00.9	40.0		351°18.4	9.5'	28°27.2	-0.4'	54.3'
16	64°01.0	40.8		5°46.9	9.5'	28°26.8	-0.5'	54.3'
17	79°01.0	41.7		20°15.5	9.5'	28°26.2	-0.7'	54.3'
18	94°01.1	S12°42.5		34°44.0	9.5'	S28°25.5	-0.8'	54.3'
19	109°01.2	43.4		49°12.5	9.5'	28°24.7	-1.0'	54.4'
20	124°01.2	44.2		63°41.0	9.5'	28°23.8	-1.1'	54.4'
21	139°01.3	45.1		78°09.5	9.5'	28°22.7	-1.2'	54.4'
22	154°01.3	45.9		92°38.0	9.5'	28°21.4	-1.4'	54.4'
23	169°01.4	46.7		107°06.5	9.5'	28°20.0	-1.5'	54.4'
SD = 16.1'			$d = 0.9'$	SD = 14.8'				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	184°01.4	S12°47.6		121°35.0	9.5'	S28°18.5	-1.7'	54.4'
1	199°01.5	48.4		136°03.5	9.5'	28°16.8	-1.8'	54.5'
2	214°01.6	49.3		150°32.0	9.5'	28°15.0	-1.9'	54.5'
3	229°01.6	50.1		165°00.5	9.5'	28°13.1	-2.1'	54.5'
4	244°01.7	51.0		179°29.0	9.5'	28°11.0	-2.2'	54.5'
5	259°01.7	51.8		193°57.5	9.5'	28°08.8	-2.4'	54.5'
6	274°01.8	S12°52.7		208°26.0	9.5'	S28°06.4	-2.5'	54.5'
7	289°01.8	53.5		222°54.6	9.5'	28°03.9	-2.7'	54.6'
8	304°01.9	54.3		237°23.1	9.5'	28°01.2	-2.8'	54.6'
9	319°01.9	55.2		251°51.6	9.5'	27°58.4	-2.9'	54.6'
10	334°02.0	56.0		266°20.2	9.6'	27°55.5	-3.1'	54.6'
11	349°02.1	56.9		280°48.7	9.6'	27°52.4	-3.2'	54.6'
12	4°02.1	S12°57.7		295°17.3	9.6'	S27°49.2	-3.4'	54.6'
13	19°02.2	58.6		309°45.8	9.6'	27°45.8	-3.5'	54.7'
14	34°02.2	12°59.4		324°14.4	9.6'	27°42.3	-3.6'	54.7'
15	49°02.3	13°00.2		338°43.0	9.6'	27°38.7	-3.8'	54.7'
16	64°02.3	01.1		353°11.7	9.6'	27°34.9	-3.9'	54.7'
17	79°02.4	01.9		7°40.3	9.7'	27°31.0	-4.1'	54.7'
18	94°02.4	S13°02.8		22°08.9	9.7'	S27°27.0	-4.2'	54.8'
19	109°02.5	03.6		36°37.6	9.7'	27°22.8	-4.3'	54.8'
20	124°02.5	04.4		51°06.3	9.7'	27°18.4	-4.5'	54.8'
21	139°02.6	05.3		65°35.0	9.7'	27°14.0	-4.6'	54.8'
22	154°02.6	06.1		80°03.7	9.8'	27°09.4	-4.7'	54.9'
23	169°02.7	06.9		94°32.5	9.8'	27°04.6	-4.9'	54.9'
SD = 16.1'			$d = 0.8'$	SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:43	07:03	08:23	15:04	16:23	17:43
N 70°	05:41	06:53	08:03	15:24	16:33	17:45
68°	05:40	06:45	07:47	15:39	16:41	17:46
66°	05:38	06:39	07:35	15:52	16:48	17:48
64°	05:37	06:33	07:24	16:03	16:54	17:49
62°	05:36	06:28	07:15	16:12	16:59	17:51
60°	05:35	06:23	07:07	16:19	17:03	17:52
N 58°	05:33	06:19	07:01	16:26	17:07	17:53
56°	05:32	06:16	06:55	16:32	17:11	17:55
54°	05:31	06:12	06:49	16:38	17:15	17:56
52°	05:30	06:09	06:44	16:43	17:18	17:57
50°	05:29	06:07	06:40	16:47	17:21	17:58
45°	05:26	06:00	06:30	16:57	17:27	18:01
N 40°	05:23	05:55	06:22	17:05	17:33	18:04
35°	05:20	05:50	06:16	17:12	17:38	18:07
30°	05:17	05:45	06:10	17:18	17:42	18:11
20°	05:10	05:36	05:59	17:29	17:51	18:17
N 10°	05:03	05:28	05:50	17:38	18:00	18:25
0°	04:55	05:19	05:41	17:47	18:09	18:33
S 10°	04:45	05:10	05:32	17:57	18:18	18:43
20°	04:32	04:59	05:22	18:06	18:29	18:56
30°	04:16	04:46	05:11	18:18	18:43	19:12
35°	04:06	04:37	05:04	18:25	18:51	19:23
40°	03:53	04:28	04:56	18:32	19:01	19:35
45°	03:38	04:16	04:48	18:41	19:13	19:51
S 50°	03:18	04:02	04:37	18:52	19:27	20:12
52°	03:08	03:55	04:32	18:57	19:34	20:22
54°	02:57	03:47	04:27	19:02	19:42	20:33
56°	02:43	03:39	04:21	19:09	19:51	20:47
58°	02:27	03:29	04:14	19:16	20:01	21:03
S 60°	02:08	03:17	04:06	19:24	20:13	21:24

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	■		■	■		■
N 70°	■		■	■		■
68°	■		■	■		■
66°	■		■	■		■
64°	■		■	■		■
62°	■		15:50	■		17:12
60°	13:07	14:15	14:48	■		18:14
N 58°	12:31	13:34	14:14	16:54	17:39	18:47
56°	12:05	13:06	13:49	17:20	18:07	19:12
54°	11:45	12:44	13:28	17:41	18:28	19:32
52°	11:28	12:26	13:12	17:58	18:46	19:49
50°	11:13	12:11	12:57	18:13	19:02	20:03
45°	10:43	11:40	12:28	18:43	19:33	20:32
N 40°	10:20	11:16	12:05	19:06	19:57	20:54
35°	10:01	10:56	11:46	19:26	20:16	21:13
30°	09:45	10:39	11:29	19:42	20:33	21:29
20°	09:17	10:11	11:02	20:10	21:02	21:56
N 10°	08:54	09:46	10:38	20:34	21:26	22:19
0°	08:32	09:23	10:16	20:57	21:49	22:41
S 10°	08:10	09:01	09:53	21:19	22:11	23:02
20°	07:47	08:36	09:30	21:43	22:36	23:25
30°	07:20	08:08	09:02	22:11	23:04	23:52
35°	07:04	07:51	08:45	22:28	23:20	
40°	06:45	07:32	08:26</			

October 28, 29, 30 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	36°35.7	199°49.0	S05°29.9	164°25.2	S19°16.1	279°51.0	N21°14.5	39°17.5	S03°51.9	Alpheratz	357°33.8	29°14.2
1	51°38.1	214°48.6	31.1	179°25.9	16.6	294°53.3	14.5	54°20.0	51.9	Ankaa	353°06.1	-42°09.9
2	66°40.6	229°48.2	32.3	194°26.5	17.0	309°55.6	14.5	69°22.6	52.0	Schedar	349°29.8	56°40.9
3	81°43.1	244°47.7	· · 33.5	209°27.2	· · 17.5	324°58.0	· · 14.5	84°25.2	· · 52.0	Diphda	348°46.3	-17°50.6
4	96°45.5	259°47.3	34.7	224°27.8	17.9	340°00.3	14.4	99°27.8	52.1	Achernar	335°19.1	-57°06.3
5	111°48.0	274°46.9	35.9	239°28.4	18.4	355°02.7	14.4	114°30.4	52.1	Hamal	327°50.1	23°35.2
6	126°50.5	289°46.5	S05°37.1	254°29.1	S19°18.8	10°05.0	N21°14.4	129°33.0	S03°52.2	Polaris	313°13.2	89°22.3
7	141°52.9	304°46.0	38.3	269°29.7	19.3	25°07.3	14.4	144°35.5	52.2	Acamar	315°10.8	-40°11.9
8	156°55.4	319°45.6	39.5	284°30.4	19.7	40°09.7	14.4	159°38.1	52.3	Menkar	314°05.1	4°11.6
9	171°57.8	334°45.2	· · 40.7	299°31.0	· · 20.2	55°12.0	· · 14.4	174°40.7	· · 52.3	Mirfak	308°26.8	49°57.2
10	187°00.3	349°44.7	41.9	314°31.7	20.7	70°14.4	14.4	189°43.3	52.3	Aldebaran	290°38.6	16°33.7
11	202°02.8	4°44.3	43.1	329°32.3	21.1	85°16.7	14.3	204°45.9	52.4	Rigel	281°03.0	-8°10.1
12	217°05.2	19°43.9	S05°44.3	344°33.0	S19°21.6	100°19.0	N21°14.3	219°48.5	S03°52.4	Capella	280°20.6	46°01.4
13	232°07.7	34°43.4	45.5	359°33.6	22.0	115°21.4	14.3	234°51.0	52.5	Bellatrix	278°21.9	6°22.5
14	247°10.2	49°43.0	46.7	14°34.3	22.5	130°23.7	14.3	249°53.6	52.5	Elnath	278°00.8	28°37.7
15	262°12.6	64°42.6	· · 47.9	29°34.9	· · 22.9	145°26.1	· · 14.3	264°56.2	· · 52.6	Alnilam	275°36.8	-1°11.0
16	277°15.1	79°42.2	49.1	44°35.6	23.4	160°28.4	14.3	279°58.8	52.6	Betelgeuse	270°51.2	7°24.8
17	292°17.6	94°41.7	50.3	59°36.2	23.8	175°30.7	14.2	295°01.4	52.7	Canopus	263°51.8	-52°42.2
18	307°20.0	109°41.3	S05°51.5	74°36.9	S19°24.3	190°33.1	N21°14.2	310°04.0	S03°52.7	Sirius	258°25.5	-16°44.9
19	322°22.5	124°40.9	52.7	89°37.5	24.7	205°35.4	14.2	325°06.5	52.7	Adhara	255°05.2	-29°00.1
20	337°25.0	139°40.4	53.9	104°38.1	25.2	220°37.8	14.2	340°09.1	52.8	Procyon	244°50.1	5°09.6
21	352°27.4	154°40.0	· · 55.1	119°38.8	· · 25.6	235°40.1	· · 14.2	355°11.7	· · 52.8	Pollux	243°16.4	27°57.8
22	7°29.9	169°39.6	56.3	134°39.4	26.1	250°42.5	14.2	10°14.3	52.9	Avior	234°14.5	-59°35.1
23	22°32.3	184°39.1	57.5	149°40.1	26.5	265°44.8	14.2	25°16.9	52.9	Suhail	222°45.9	-43°31.9
Mer.pass. 21:30		ν -0.4' d 1.2' m -3.91		ν 0.6' d 0.5' m 1.48		ν 2.3' d -0.0' m -2.29		ν 2.6' d 0.0' m 0.75		Miaplacidus	221°38.4	-69°49.0
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	37°34.8	199°38.7	S05°58.7	164°40.7	S19°27.0	280°47.2	N21°14.1	40°19.4	S03°53.0	Alphard	217°47.2	-8°46.1
1	52°37.3	214°38.3	05°59.9	179°41.4	27.4	295°49.5	14.1	55°22.0	53.0	Regulus	207°33.9	11°50.5
2	67°39.7	229°37.8	06°01.1	194°42.0	27.9	310°51.9	14.1	70°24.6	53.1	Dubhe	193°40.6	61°36.5
3	82°42.2	244°37.4	· · 02.3	209°42.6	· · 28.3	325°54.2	· · 14.1	85°27.2	· · 53.1	Denebola	182°24.5	14°25.7
4	97°44.7	259°37.0	03.5	224°43.3	28.8	340°56.5	14.1	100°29.8	53.1	Gienah	175°43.2	-17°41.0
5	112°47.1	274°36.5	04.7	239°43.9	29.2	355°58.9	14.1	115°32.3	53.2	Acrux	173°00.0	-63°14.3
6	127°49.6	289°36.1	S06°06.0	254°44.6	S19°29.7	11°01.2	N21°14.1	130°34.9	S03°53.2	Gacrux	171°51.5	-57°15.3
7	142°52.1	304°35.7	07.2	269°45.2	30.1	26°03.6	14.0	145°37.5	53.3	Alioth	166°12.9	55°49.1
8	157°54.5	319°35.2	08.3	284°45.9	30.6	41°05.9	14.0	160°40.1	53.3	Spica	158°21.9	-11°17.7
9	172°57.0	334°34.8	· · 09.5	299°46.5	· · 31.0	56°08.3	· · 14.0	175°42.7	· · 53.4	Alkaid	152°52.0	49°11.0
10	187°59.5	349°34.4	10.7	314°47.1	31.5	71°10.6	14.0	190°45.2	53.4	Hadar	148°35.8	-60°29.8
11	203°01.9	4°33.9	11.9	329°47.8	31.9	86°13.0	14.0	205°47.8	53.4	Menkent	147°57.3	-36°29.7
12	218°04.4	19°33.5	S06°13.1	344°48.4	S19°32.3	101°15.3	N21°14.0	220°50.4	S03°53.5	Arcturus	145°47.7	19°02.9
13	233°06.8	34°33.1	14.3	359°49.1	32.8	116°17.7	14.0	235°53.0	53.5	Rigil Kent.	139°40.1	-60°56.5
14	248°09.3	49°32.6	15.5	14°49.7	33.2	131°20.0	14.0	250°55.6	53.6	Kochab	137°20.9	74°02.9
15	263°11.8	64°32.2	· · 16.7	29°50.3	· · 33.7	146°22.4	· · 13.9	265°58.1	· · 53.6	Zuben'ubi	136°55.6	-16°08.9
16	278°14.2	79°31.8	17.9	44°51.0	34.1	161°24.7	13.9	281°00.7	53.7	Alphecca	126°03.5	26°37.8
17	293°16.7	94°31.3	19.1	59°51.6	34.6	176°27.1	13.9	296°03.3	53.7	Antares	112°15.4	-26°29.3
18	308°19.2	109°30.9	S06°20.3	74°52.3	S19°35.0	191°29.4	N21°13.9	311°05.9	S03°53.7	Atria	107°09.4	-69°04.5
19	323°21.6	124°30.5	21.5	89°52.9	35.5	206°31.8	13.9	326°08.4	53.8	Sabik	102°02.3	-15°45.4
20	338°24.1	139°30.0	22.7	104°53.5	35.9	221°34.2	13.9	341°11.0	53.8	Shaula	96°09.8	-37°07.4
21	353°26.6	154°29.6	· · 23.9	119°54.2	· · 36.4	236°36.5	· · 13.9	356°13.6	· · 53.9	Rasalhague	95°58.2	12°32.6
22	8°29.0	169°29.1	25.1	134°54.8	36.8	251°38.9	13.8	11°16.2	53.9	Eltanin	90°42.2	51°29.3
23	23°31.5	184°28.7	26.3	149°55.4	37.2	266°41.2	13.8	26°18.8	54.0	Kaus Aust.	83°31.9	-34°22.4
Mer.pass. 21:26		ν -0.4' d 1.2' m -3.91		ν 0.6' d 0.4' m 1.47		ν 2.3' d -0.0' m -2.30		ν 2.6' d 0.0' m 0.75		Vega	80°33.0	38°48.7
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	38°33.9	199°28.3	S06°27.5	164°56.1	S19°37.7	281°43.6	N21°13.8	41°21.3	S03°54.0	Nunki	75°47.1	-26°15.9
1	53°36.4	214°27.8	28.7	179°56.7	38.1	296°45.9	13.8	56°23.9	54.0	Altair	61°59.4	8°56.3
2	68°38.9	229°27.4	29.9	194°57.4	38.6	311°48.3	13.8	71°26.5	54.1	Peacock	53°04.7	-56°39.3
3	83°41.3	244°27.0	· · 31.1	209°58.0	· · 39.0	326°50.6	· · 13.8	86°29.1	· · 54.1	Deneb	49°25.4	45°22.6
4	98°43.8	259°26.5	32.3	224°58.6	39.4	341°53.0	13.8	101°31.6	54.2	Enif	33°38.1	9°59.8
5	113°46.3	274°26.1	33.5	239°59.3	39.9	356°55.3	13.8	116°34.2	54.2	Al Na'ir	27°31.9	-46°50.3
6	128°48.7	289°25.6	S06°34.7	254°59.9	S19°40.3	11°57.7	N21°13.7	131°36.8	S03°54.2	Fomalhaut	15°13.6	-29°29.1
7	143°51.2	304°25.2	35.9	270°00.5	40.8	27°00.1	13.7	146°39.4	54.3	Scheat	13°44.4	28°13.6
8	158°53.7	319°24.8	37.1	285°01.2	41.2	42°02.4	13.7	161°42.0	54.3	Markab	13°29.1	15°20.8
9	173°56.1	334°24.3	· · 38.3	300°01.8	· · 41.7	57°04.8	· · 13.7	176°44.5	· · 54.4			
10	188°58.6	349°23.9	39.5	315°02.4	42.1	72°07.1	13.7	191°47.1	54.4			
11	204°01.1	4°23.4	40.7	330°03.1	42.5	87°09.5	13.7	206°49.7	54.5			
12	219°03.5	19°23.0	S06°41.9	345°03.7	S19°43.0	102°11.8	N21°13.7	221°52.3	S03°54.5			
13	234°06.0	34°22.6	43.1	0°04.3	43.4	117°14.2	13.7	236°54.8	54.5			
14	249°08.4	49°22.1	44.2	15°05.0	43.9	132°16.6	13.6	251°57.4	54.6			
15	264°10.9	64°21.7	· · 45.4	30°05.6	· · 44.3	147°18.9	· · 13.6	267°00.0	· · 54.6			
16	279°13.4	79°21.2	46.6	45°06.2	44.7	162°21.3	13.6	282°02.6	54.7			
17	294°15.8	94°20.8	47.8	60°06.9	45.2	177°23.6	13.6	297°05.1	54.7			
18	309°18.3	109°20.4	S06°49.0	75°07.5	S19°45.6	192°26.0	N21°13.6	312°07.7	S03°54.7			
19	324°20.8	124°19.9	50.2	90°08.1	46.0	207°28.4	13.6	327°10.3	54.8			
20	339°23.2	139°19.5	51.4	105°08.8	46.5	222°30.7	13.6	342°12.9	54.8			
21	354°25.7	154°19.0	· · 52.6	120°09.4	· · 46.9	237°33.1	· · 13.6	357°15.4	· · 54.9			
22	9°28.2	169°18.6	53.8	135°10.0	47.4	252°35.4	13.5	12°18.0	54.9			
23	24°30.6	184°18										

DUT1 = UT1-UTC = +0.1043 sec $\Delta T = TT-UT1 = +69.0797$ sec

2025 October 28 to Oct. 30 UT

h Sun Moon						
Tue	GHA	Dec	GHA	ν	Dec	d HP
0	184°02.7	S13°07.8	109°01.3	9.8'	S26°59.7	-5.0' 54.9'
1	199°02.8	08.6	123°30.1	9.8'	26°54.7	-5.2' 54.9'
2	214°02.8	09.4	137°58.9	9.8'	26°49.5	-5.3' 54.9'
3	229°02.9	· · 10.3	152°27.7	9.9'	26°44.2	-5.4' 55.0'
4	244°02.9	11.1	166°56.6	9.9'	26°38.8	-5.6' 55.0'
5	259°03.0	12.0	181°25.5	9.9'	26°33.2	-5.7' 55.0'
6	274°03.0	S13°12.8	195°54.5	10.0'	S26°27.5	-5.8' 55.0'
7	289°03.1	13.6	210°23.4	10.0'	26°21.7	-6.0' 55.1'
8	304°03.1	14.5	224°52.4	10.0'	26°15.7	-6.1' 55.1'
9	319°03.2	· · 15.3	239°21.4	10.0'	26°09.6	-6.2' 55.1'
10	334°03.2	16.1	253°50.5	10.1'	26°03.4	-6.4' 55.1'
11	349°03.3	17.0	268°19.5	10.1'	25°57.0	-6.5' 55.2'
12	4°03.3	S13°17.8	282°48.6	10.1'	S25°50.5	-6.6' 55.2'
13	19°03.4	18.6	297°17.8	10.2'	25°43.9	-6.8' 55.2'
14	34°03.4	19.5	311°47.0	10.2'	25°37.1	-6.9' 55.2'
15	49°03.4	· · 20.3	326°16.2	10.2'	25°30.2	-7.0' 55.3'
16	64°03.5	21.1	340°45.4	10.3'	25°23.1	-7.2' 55.3'
17	79°03.5	21.9	355°14.7	10.3'	25°16.0	-7.3' 55.3'
18	94°03.6	S13°22.8	9°44.0	10.3'	S25°08.7	-7.4' 55.4'
19	109°03.6	23.6	24°13.3	10.4'	25°01.2	-7.6' 55.4'
20	124°03.7	24.4	38°42.7	10.4'	24°53.7	-7.7' 55.4'
21	139°03.7	· · 25.3	53°12.1	10.4'	24°46.0	-7.8' 55.4'
22	154°03.8	26.1	67°41.6	10.5'	24°38.2	-7.9' 55.5'
23	169°03.8	26.9	82°11.1	10.5'	24°30.2	-8.1' 55.5'
SD = 16.1' $d = 0.8'$ SD = 15.0'						
Wed	GHA	Dec	GHA	ν	Dec	d HP
0	184°03.8	S13°27.8	96°40.6	10.6'	S24°22.2	-8.2' 55.5'
1	199°03.9	28.6	111°10.1	10.6'	24°14.0	-8.3' 55.5'
2	214°03.9	29.4	125°39.7	10.6'	24°05.6	-8.5' 55.6'
3	229°04.0	· · 30.2	140°09.4	10.7'	23°57.2	-8.6' 55.6'
4	244°04.0	31.1	154°39.0	10.7'	23°48.6	-8.7' 55.6'
5	259°04.0	31.9	169°08.8	10.7'	23°39.9	-8.8' 55.7'
6	274°04.1	S13°32.7	183°38.5	10.8'	S23°31.1	-8.9' 55.7'
7	289°04.1	33.5	198°08.3	10.8'	23°22.1	-9.1' 55.7'
8	304°04.2	34.4	212°38.1	10.9'	23°13.1	-9.2' 55.8'
9	319°04.2	· · 35.2	227°08.0	10.9'	23°03.9	-9.3' 55.8'
10	334°04.2	36.0	241°37.9	10.9'	22°54.6	-9.4' 55.8'
11	349°04.3	36.8	256°07.8	11.0'	22°45.1	-9.6' 55.9'
12	4°04.3	S13°37.7	270°37.8	11.0'	S22°35.6	-9.7' 55.9'
13	19°04.4	38.5	285°07.8	11.1'	22°25.9	-9.8' 55.9'
14	34°04.4	39.3	299°37.9	11.1'	22°16.1	-9.9' 55.9'
15	49°04.4	· · 40.1	314°08.0	11.1'	22°06.2	-10.0' 56.0'
16	64°04.5	41.0	328°38.1	11.2'	21°56.2	-10.1' 56.0'
17	79°04.5	41.8	343°08.3	11.2'	21°46.0	-10.3' 56.0'
18	94°04.5	S13°42.6	357°38.5	11.3'	S21°35.8	-10.4' 56.1'
19	109°04.6	43.4	12°08.8	11.3'	21°25.4	-10.5' 56.1'
20	124°04.6	44.2	26°39.0	11.3'	21°14.9	-10.6' 56.1'
21	139°04.7	· · 45.1	41°09.4	11.4'	21°04.3	-10.7' 56.2'
22	154°04.7	45.9	55°39.7	11.4'	20°53.6	-10.8' 56.2'
23	169°04.7	46.7	70°10.1	11.4'	20°42.7	-10.9' 56.2'
SD = 16.1' $d = 0.8'$ SD = 15.1'						
Thu	GHA	Dec	GHA	ν	Dec	d HP
0	184°04.8	S13°47.5	84°40.6	11.5'	S20°31.8	-11.1' 56.3'
1	199°04.8	48.3	99°11.1	11.5'	20°20.7	-11.2' 56.3'
2	214°04.8	49.2	113°41.6	11.6'	20°09.6	-11.3' 56.4'
3	229°04.9	· · 50.0	128°12.1	11.6'	19°58.3	-11.4' 56.4'
4	244°04.9	50.8	142°42.7	11.6'	19°46.9	-11.5' 56.4'
5	259°04.9	51.6	157°13.3	11.7'	19°35.4	-11.6' 56.5'
6	274°05.0	S13°52.4	171°44.0	11.7'	S19°23.8	-11.7' 56.5'
7	289°05.0	53.2	186°14.7	11.7'	19°12.1	-11.8' 56.5'
8	304°05.0	54.1	200°45.4	11.8'	19°00.3	-11.9' 56.6'
9	319°05.1	· · 54.9	215°16.2	11.8'	18°48.3	-12.0' 56.6'
10	334°05.1	55.7	229°47.0	11.8'	18°36.3	-12.1' 56.6'
11	349°05.1	56.5	244°17.8	11.9'	18°24.2	-12.2' 56.7'
12	4°05.1	S13°57.3	258°48.7	11.9'	S18°11.9	-12.3' 56.7'
13	19°05.2	58.1	273°19.5	11.9'	17°59.6	-12.4' 56.7'
14	34°05.2	58.9	287°50.5	12.0'	17°47.2	-12.5' 56.8'
15	49°05.2	13°59.8	302°21.4	12.0'	17°34.6	-12.6' 56.8'
16	64°05.3	14°00.6	316°52.4	12.0'	17°22.0	-12.7' 56.9'
17	79°05.3	01.4	331°23.4	12.0'	17°09.2	-12.8' 56.9'
18	94°05.3	S14°02.2	345°54.5	12.1'	S16°56.4	-12.9' 56.9'
19	109°05.4	03.0	0°25.5	12.1'	16°43.5	-13.0' 57.0'
20	124°05.4	03.8	14°56.7	12.1'	16°30.4	-13.1' 57.0'
21	139°05.4	· · 04.6	29°27.8	12.2'	16°17.3	-13.2' 57.1'
22	154°05.4	05.4	43°58.9	12.2'	16°04.1	-13.3' 57.1'
23	169°05.5	06.2	58°30.1	12.2'	15°50.8	-13.4' 57.1'
SD = 16.1' $d = 0.8'$ SD = 15.3'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	05:55	07:17	08:40	14:46	16:09	17:30
N 70°	05:52	07:05	08:17	15:09	16:21	17:34
68°	05:50	06:56	07:59	15:27	16:30	17:36
66°	05:47	06:48	07:45	15:41	16:38	17:39
64°	05:45	06:41	07:33	15:53	16:45	17:41
62°	05:43	06:36	07:24	16:03	16:51	17:43
60°	05:41	06:30	07:15	16:11	16:56	17:45
N 58°	05:40	06:26	07:08	16:19	17:01	17:47
56°	05:38	06:22	07:01	16:26	17:05	17:49
54°	05:36	06:18	06:55	16:31	17:09	17:50
52°	05:35	06:14	06:50	16:37	17:12	17:52
50°	05:33	06:11	06:45	16:42	17:15	17:53
45°	05:30	06:04	06:35	16:52	17:23	17:57
N 40°	05:26	05:58	06:26	17:01	17:29	18:01
35°	05:23	05:52	06:18	17:09	17:35	18:04
30°	05:19	05:47	06:12	17:15	17:40	18:08
20°	05:12	05:38	06:00	17:27	17:50	18:16
N 10°	05:03	05:28	05:50	17:37	17:59	18:24
0°	04:54	05:19	05:40	17:47	18:08	18:33
S 10°	04:44	05:09	05:31	17:57	18:19	18:44
20°	04:30	04:57	05:20	18:08	18:31	18:58
30°	04:13	04:43	05:08	18:20	18:45	19:15
35°	04:02	04:34	05:01	18:27	18:54	19:26
40°	03:49	04:24	04:53	18:36	19:05	19:40
45°	03:32	04:11	04:43	18:45	19:17	19:56
S 50°	03:11	03:56	04:32	18:57	19:33	20:18
52°	03:00	03:48	04:26	19:02	19:40	20:29
54°	02:48	03:40	04:20	19:08	19:49	20:42
56°	02:33	03:31	04:14	19:15	19:58	20:56
58°	02:16	03:20	04:06	19:23	20:10	21:15
S 60°	01:54	03:07	03:58	19:31	20:22	21:37

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	■	■	17:42	■	■	20:39
N 70°	■	■	16:50	■	■	21:29
68°	■	17:35	16:17	■	19:01	22:00
66°	■	16:26	15:53	■	20:08	22:23
64°	16:27	15:50	15:34	18:23	20:44	22:41
62°	15:32	15:24	15:18	19:17	21:09	22:55
60°	14:59	15:04	15:05	19:49	21:28	23:07
N 58°	14:35	14:47	14:54	20:13	21:45	23:18
56°	14:15	14:32	14:44	20:32	21:58	23:27
54°	13:59	14:20	14:35	20:48	22:10	23:35
52°	13:45	14:09	14:27	21:02	22:21	23:42
50°	13:33	13:59	14:20	21:14	22:30	23:49
45°	13:07	13:38	14:04	21:38	22:49	· · ·
N 40°	12:47	13:22	13:52	21:58	23:05	· · ·
35°	12:29	13:07	13:41	22:14	23:18	· · ·
30°	12:15	12:55	13:31	22:28	23:30	· · ·
20°	11:50	12:34	13:15	22:52	23:49	· · ·
N 10°	11:28	12:15	13:00	23:13	· · ·	00:06
0°	11:07	11:58	12:47	23:32	· · ·	00:22
S 10°	10:47	11:40	12:33	23:51	· · ·	00:37
20°	10:25	11:22	12:19	· · ·	00:11	00:54
30°	10:00	11:00	12:02	· · ·	00:34	01:12
35°	09:45	10:47	11:52	00:07	00:48	01:23
40°	09:27	10:33	11:41	00:25	01:04	01:36
45°	09:06	10:15	11:27	00:47	01:22	01:50
S 50°	08:39	09:53	11:11	01:14	01:45	02:08
52°	08:26	09:42	11:03	01:28	01:56	02:17
54°	08:11	09:31	10:54	01:43	02:08	02:26
56°	07:53	09:17	10:45	02:01	02:23	02:36
58°	07:32	09:01	10:34	02:23	02:39	02:48
S 60°	07:04	08:42	10:21	02:51	02:59	03:02

Day	Sun		Mer. Pass hh:mm	Moon		Age 7-9 34-53%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
28	16:11	16:13	11:44	17:20	04:54	
29	16:15	16:17	11:44	18:10	05:45	
30	16:19	16:21	11:44	18:58	06:34	

October 31, 01, 02 UT (Fri., Sat., Sun.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	39°33.1	199°17.7	S06°56.2	165°11.3	S19°48.2	282°40.2	N21°13.5	42°23.2	S03°55.0	Alpheratz	357°33.8 29°14.2
1	54°35.6	214°17.2	57.4	180°11.9	48.7	297°42.5	13.5	57°25.7	55.0	Ankaa	353°06.1 -42°10.0
2	69°38.0	229°16.8	58.5	195°12.5	49.1	312°44.9	13.5	72°28.3	55.1	Schedar	349°29.8 56°41.0
3	84°40.5	244°16.4	06°59.7	210°13.2	· · 49.5	327°47.3	· · 13.5	87°30.9	· · 55.1	Diphda	348°46.3 -17°50.6
4	99°42.9	259°15.9	07°00.9	225°13.8	50.0	342°49.6	13.5	102°33.5	55.2	Achernar	335°19.1 -57°06.3
5	114°45.4	274°15.5	02.1	240°14.4	50.4	357°52.0	13.5	117°36.0	55.2	Hamal	327°50.1 23°35.2
6	129°47.9	289°15.0	S07°03.3	255°15.1	S19°50.8	12°54.4	N21°13.5	132°38.6	S03°55.2	Polaris	313°12.8 89°22.3
7	144°50.3	304°14.6	04.5	270°15.7	51.3	27°56.7	13.4	147°41.2	55.3	Acamar	315°10.8 -40°11.9
8	159°52.8	319°14.1	05.7	285°16.3	51.7	42°59.1	13.4	162°43.7	55.3	Menkar	314°05.1 4°11.6
9	174°55.3	334°13.7	· · 06.9	300°16.9	· · 52.1	58°01.4	· · 13.4	177°46.3	· · 55.4	Mirfak	308°26.8 49°57.2
10	189°57.7	349°13.2	08.1	315°17.6	52.6	73°03.8	13.4	192°48.9	55.4	Aldebaran	290°38.6 16°33.8
11	205°00.2	4°12.8	09.2	330°18.2	53.0	88°06.2	13.4	207°51.5	55.4	Rigel	281°03.0 -8°10.1
12	220°02.7	19°12.3	S07°10.4	345°18.8	S19°53.4	103°08.5	N21°13.4	222°54.0	S03°55.5	Capella	280°20.6 46°01.4
13	235°05.1	34°11.9	11.6	0°19.5	53.9	118°10.9	13.4	237°56.6	55.5	Bellatrix	278°21.9 6°22.5
14	250°07.6	49°11.4	12.8	15°20.1	54.3	133°13.3	13.4	252°59.2	55.6	Elnath	278°00.8 28°37.7
15	265°10.0	64°11.0	· · 14.0	30°20.7	· · 54.7	148°15.6	· · 13.4	268°01.8	· · 55.6	Alnilam	275°36.8 -1°11.0
16	280°12.5	79°10.6	15.2	45°21.3	55.2	163°18.0	13.4	283°04.3	55.6	Betelgeuse	270°51.2 7°24.8
17	295°15.0	94°10.1	16.4	60°22.0	55.6	178°20.4	13.3	298°06.9	55.7	Canopus	263°51.8 -52°42.2
18	310°17.4	109°09.7	S07°17.6	75°22.6	S19°56.0	193°22.8	N21°13.3	313°09.5	S03°55.7	Sirius	258°25.5 -16°44.9
19	325°19.9	124°09.2	18.7	90°23.2	56.5	208°25.1	13.3	328°12.0	55.8	Adhara	255°05.2 -29°00.2
20	340°22.4	139°08.8	19.9	105°23.8	56.9	223°27.5	13.3	343°14.6	55.8	Procyon	244°50.1 5°09.6
21	355°24.8	154°08.3	· · 21.1	120°24.5	· · 57.3	238°29.9	· · 13.3	358°17.2	· · 55.8	Pollux	243°16.4 27°57.8
22	10°27.3	169°07.9	22.3	135°25.1	57.7	253°32.2	13.3	13°19.8	55.9	Avior	234°14.4 -59°35.1
23	25°29.8	184°07.4	23.5	150°25.7	58.2	268°34.6	13.3	28°22.3	55.9	Suhail	222°45.9 -43°31.9
Mer.pass. 21:18		ν -0.4' d 1.2' m -3.91		ν 0.6' d 0.4' m 1.46		ν 2.4' d -0.0' m -2.31		ν 2.6' d 0.0' m 0.76		Miaplacidus	221°38.3 -69°49.0
										Alphard	217°47.2 -8°46.1
										Regulus	207°33.9 11°50.5
										Dubhe	193°40.6 61°36.5
										Denebola	182°24.5 14°25.7
										Gienah	175°43.2 -17°41.0
										Acrux	173°00.0 -63°14.3
										Gacrux	171°51.5 -57°15.3
										Alioth	166°12.9 55°49.1
										Spica	158°21.9 -11°17.7
										Alkaid	152°52.0 49°11.0
										Hadar	148°35.8 -60°29.8
										Menkent	147°57.3 -36°29.7
										Arcturus	145°47.7 19°02.9
										Rigel Kent.	139°40.1 -60°56.5
										Kochab	137°20.9 74°02.9
										Zuben'ubi	136°55.6 -16°08.9
										Alphecca	126°03.5 26°37.7
										Antares	112°15.4 -26°29.3
										Atria	107°09.5 -69°04.5
										Sabik	102°02.3 -15°45.4
										Shaula	96°09.8 -37°07.4
										Rasalhague	95°58.2 12°32.6
										Eltanin	90°42.2 51°29.3
										Kaus Aust.	83°31.9 -34°22.4
										Vega	80°33.0 38°48.7
										Nunki	75°47.1 -26°15.9
										Altair	61°59.4 8°56.3
										Peacock	53°04.7 -56°39.3
										Deneb	49°25.4 45°22.6
										Enif	33°38.1 9°59.8
										Al Na'ir	27°31.9 -46°50.3
										Fomalhaut	15°13.6 -29°29.2
										Scheat	13°44.4 28°13.6
										Markab	13°29.1 15°20.8
										Oct 31 Fri	
										Venus	SHA 159°44.6 Mer.pass 10:43
										Mars	125°38.2 12:59
										Jupiter	243°07.1 05:09
										Saturn	2°50.1 21:07
										Nov 01 Sat	
										Venus	SHA 158°34.7 Mer.pass 10:44
										Mars	124°54.1 12:58
										Jupiter	243°04.7 05:05
										Saturn	2°52.7 21:03
										Nov 02 Sun	
										Venus	SHA 157°24.7 Mer.pass 10:45
										Mars	124°09.9 12:57
										Jupiter	243°02.6 05:01
										Saturn	2°55.2 20:59
										Horizontal parallax	
										Venus:	0.1
										Mars:	0.1
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	41°31.4	198°56.1	S07°53.0	165°41.2	S20°08.8	284°34.0	N21°13.0	44°26.6	S03°56.9		
1	56°33.8	213°55.6	54.2	180°41.8	09.2	299°36.4	13.0	59°29.1	56.9		
2	71°36.3	228°55.1	55.4	195°42.5	09.7	314°38.7	13.0	74°31.7	56.9		
3	86°38.8	243°54.7	· · 56.6	210°43.1	· · 10.1	329°41.1	· · 13.0	89°34.3	· · 57.0		
4	101°41.2	258°54.2	57.7	225°43.7	10.5	344°43.5	13.0	104°36.8	57.0		
5	116°43.7	273°53.8	07°58.9	240°44.3	10.9	359°45.9	13.0	119°39.4	57.1		
6	131°46.1	288°53.3	S08°00.1	255°44.9	S20°11.3	14°48.3	N21°13.0	134°42.0	S03°57.1		
7	146°48.6	303°52.8	01.3	270°45.5	11.8	29°50.6	13.0	149°44.5	57.1		
8	161°51.1	318°52.4	02.4	285°46.2	12.2	44°53.0	13.0	164°47.1	57.2		
9	176°53.5	333°51.9	· · 03.6	300°46.8	· · 12.6	59°55.4	· · 13.0	179°49.7	· · 57.2		
10	191°56.0	348°51.5	04.8	315°47.4	13.0	74°57.8	13.0	194°52.2	57.2		
11	206°58.5	3°51.0	06.0	330°48.0	13.4	90°00.2	13.0	209°54.8	57.3		
12	222°00.9	18°50.5	S08°07.1	345°48.6	S20°13.9	105°02.6	N21°12.9	224°57.4	S03°57.3		
13	237°03.4	33°50.1	08.3	0°49.2	14.3	120°04.9	12.9	239°59.9	57.4		
14	252°05.9	48°49.6	09.5	15°49.8	14.7	135°07.3	12.9	255°02.5	57.4		
15	267°08.3	63°49.2	· · 10.7	30°50.4	· · 15.1	150°09.7	· · 12.9	270°05.1	· · 57.4		
16	282°10.8	78°48.7	11.8	45°51.1	15.5	165°12.1	12.9	285°07.6	57.5		
17	297°13.3	93°48.2	13.0	60°51.7	15.9	180°14.5	12.9	300°10.2	57.5		
18	312°15.7	108°47.8	S08°14.2	75°52.3	S20°16.4	195°16.9	N21°12.9	315°12.8	S03°57.5		
19	327°18.2	123°47.3	15.3	90°52.9	16.8	210°19.3	12.9	330°15.3	57.6		
20	342°20.6	138°46.8	16.5	105°53.5	17.2	225°21.6	12.9	345°17.9	57.6		
21	357°23.1	153°46.4	· · 17.7	120°54.1	· · 17.6	240°24.0	· · 12.9	0°20.5	· · 57.6		
22	12°25.6	168°45.9	18.9	135°54.7	18.0	255°26.4	12.9	15°23.0	57.7		
23	27°28.0	183°45.4	20.0	150°55.3	18.4	270°28.8	12.9	30°25.6	57.7		
Mer.pass. 21:10		ν -0.5' d 1.2' m -3.91		ν 0.6' d 0.4' m 1.45		ν 2.4' d -0.0' m -2.33		ν 2.6' d 0.0' m 0.77			

h	Sun				Moon			
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	184°05.5	S14°07.1		73°01.3	12.2'	S15°37.4	-13.5'	57.2'
1	199°05.5	07.9		87°32.6	12.3'	15°23.9	-13.6'	57.2'
2	214°05.5	08.7		102°03.8	12.3'	15°10.3	-13.7'	57.2'
3	229°05.6	· · 09.5		116°35.1	12.3'	14°56.6	-13.8'	57.3'
4	244°05.6	10.3		131°06.4	12.3'	14°42.8	-13.9'	57.3'
5	259°05.6	11.1		145°37.7	12.3'	14°28.9	-14.0'	57.4'
6	274°05.6	S14°11.9		160°09.1	12.4'	S14°15.0	-14.0'	57.4'
7	289°05.7	12.7		174°40.4	12.4'	14°00.9	-14.1'	57.4'
8	304°05.7	13.5		189°11.8	12.4'	13°46.8	-14.2'	57.5'
9	319°05.7	· · 14.3		203°43.2	12.4'	13°32.6	-14.3'	57.5'
10	334°05.7	15.1		218°14.6	12.4'	13°18.3	-14.4'	57.6'
11	349°05.8	15.9		232°46.1	12.4'	13°03.9	-14.5'	57.6'
12	4°05.8	S14°16.7		247°17.5	12.5'	S12°49.5	-14.5'	57.6'
13	19°05.8	17.5		261°49.0	12.5'	12°34.9	-14.6'	57.7'
14	34°05.8	18.4		276°20.4	12.5'	12°20.3	-14.7'	57.7'
15	49°05.8	· · 19.2		290°51.9	12.5'	12°05.6	-14.8'	57.8'
16	64°05.9	20.0		305°23.4	12.5'	11°50.8	-14.9'	57.8'
17	79°05.9	20.8		319°54.9	12.5'	11°36.0	-14.9'	57.8'
18	94°05.9	S14°21.6		334°26.5	12.5'	S11°21.0	-15.0'	57.9'
19	109°05.9	22.4		348°58.0	12.5'	11°06.0	-15.1'	57.9'
20	124°05.9	23.2		3°29.5	12.5'	10°51.0	-15.2'	58.0'
21	139°06.0	· · 24.0		18°01.1	12.5'	10°35.8	-15.2'	58.0'
22	154°06.0	24.8		32°32.6	12.6'	10°20.6	-15.3'	58.0'
23	169°06.0	25.6		47°04.2	12.6'	10°05.3	-15.4'	58.1'
SD = 16.1'		$d = 0.8'$	SD = 15.6'					
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	184°06.0	S14°26.4		61°35.7	12.6'	S09°49.9	-15.4'	58.1'
1	199°06.0	27.2		76°07.3	12.6'	09°34.4	-15.5'	58.2'
2	214°06.1	28.0		90°38.9	12.6'	09°18.9	-15.6'	58.2'
3	229°06.1	· · 28.8		105°10.4	12.6'	09°03.4	-15.6'	58.3'
4	244°06.1	29.6		119°42.0	12.6'	08°47.7	-15.7'	58.3'
5	259°06.1	30.4		134°13.5	12.6'	08°32.0	-15.8'	58.3'
6	274°06.1	S14°31.2		148°45.1	12.6'	S08°16.2	-15.8'	58.4'
7	289°06.1	32.0		163°16.6	12.5'	08°00.4	-15.9'	58.4'
8	304°06.2	32.8		177°48.2	12.5'	07°44.5	-16.0'	58.5'
9	319°06.2	· · 33.6		192°19.7	12.5'	07°28.6	-16.0'	58.5'
10	334°06.2	34.3		206°51.3	12.5'	07°12.5	-16.1'	58.5'
11	349°06.2	35.1		221°22.8	12.5'	06°56.5	-16.1'	58.6'
12	4°06.2	S14°35.9		235°54.3	12.5'	S06°40.3	-16.2'	58.6'
13	19°06.2	36.7		250°25.8	12.5'	06°24.2	-16.2'	58.7'
14	34°06.2	37.5		264°57.3	12.5'	06°07.9	-16.3'	58.7'
15	49°06.3	· · 38.3		279°28.7	12.5'	05°51.6	-16.3'	58.7'
16	64°06.3	39.1		294°00.2	12.4'	05°35.3	-16.4'	58.8'
17	79°06.3	39.9		308°31.6	12.4'	05°18.9	-16.4'	58.8'
18	94°06.3	S14°40.7		323°03.1	12.4'	S05°02.4	-16.5'	58.9'
19	109°06.3	41.5		337°34.5	12.4'	04°45.9	-16.5'	58.9'
20	124°06.3	42.3		352°05.9	12.4'	04°29.4	-16.6'	58.9'
21	139°06.3	· · 43.1		6°37.2	12.3'	04°12.8	-16.6'	59.0'
22	154°06.3	43.9		21°08.6	12.3'	03°56.2	-16.7'	59.0'
23	169°06.4	44.7		35°39.9	12.3'	03°39.5	-16.7'	59.1'
SD = 16.1'		$d = 0.8'$	SD = 15.9'					
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	184°06.4	S14°45.4		50°11.2	12.3'	S03°22.8	-16.8'	59.1'
1	199°06.4	46.2		64°42.4	12.2'	03°06.0	-16.8'	59.1'
2	214°06.4	47.0		79°13.7	12.2'	02°49.2	-16.8'	59.2'
3	229°06.4	· · 47.8		93°44.9	12.2'	02°32.4	-16.9'	59.2'
4	244°06.4	48.6		108°16.1	12.1'	02°15.5	-16.9'	59.3'
5	259°06.4	49.4		122°47.2	12.1'	01°58.6	-16.9'	59.3'
6	274°06.4	S14°50.2		137°18.3	12.1'	S01°41.7	-17.0'	59.3'
7	289°06.4	51.0		151°49.4	12.0'	01°24.7	-17.0'	59.4'
8	304°06.4	51.8		166°20.4	12.0'	01°07.7	-17.0'	59.4'
9	319°06.4	· · 52.5		180°51.5	12.0'	00°50.7	-17.1'	59.5'
10	334°06.4	53.3		195°22.4	11.9'	00°33.7	-17.1'	59.5'
11	349°06.5	54.1		209°53.3	11.9'	S00°16.6	-17.1'	59.5'
12	4°06.5	S14°54.9		224°24.2	11.8'	N00°00.5	17.1'	59.6'
13	19°06.5	55.7		238°55.1	11.8'	00°17.7	17.1'	59.6'
14	34°06.5	56.5		253°25.9	11.8'	00°34.8	17.2'	59.6'
15	49°06.5	· · 57.3		267°56.6	11.7'	00°52.0	17.2'	59.7'
16	64°06.5	58.0		282°27.3	11.7'	01°09.2	17.2'	59.7'
17	79°06.5	58.8		296°58.0	11.6'	01°26.4	17.2'	59.8'
18	94°06.5	S14°59.6		311°28.6	11.6'	N01°43.6	17.2'	59.8'
19	109°06.5	15°00.4		325°59.2	11.5'	02°00.8	17.2'	59.8'
20	124°06.5	01.2		340°29.7	11.5'	02°18.0	17.3'	59.9'
21	139°06.5	· · 01.9		355°00.1	11.4'	02°35.3	17.3'	59.9'
22	154°06.5	02.7		9°30.5	11.3'	02°52.6	17.3'	59.9'
23	169°06.5	03.5		24°00.9	11.3'	03°09.8	17.3'	60.0'
SD = 16.1'		$d = 0.8'$	SD = 16.1'					

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:07	07:30	08:58	14:27	15:55	17:18
N 70°	06:03	07:17	08:32	14:54	16:09	17:23
68°	05:59	07:06	08:12	15:14	16:19	17:27
66°	05:56	06:57	07:56	15:30	16:28	17:30
64°	05:53	06:50	07:43	15:43	16:36	17:33
62°	05:50	06:43	07:32	15:54	16:43	17:36
60°	05:48	06:37	07:23	16:04	16:49	17:38
N 58°	05:46	06:32	07:15	16:12	16:54	17:40
56°	05:44	06:28	07:07	16:19	16:59	17:43
54°	05:42	06:23	07:01	16:25	17:03	17:45
52°	05:40	06:20	06:55	16:31	17:07	17:47
50°	05:38	06:16	06:50	16:37	17:10	17:49
45°	05:33	06:08	06:39	16:48	17:19	17:53
N 40°	05:29	06:01	06:29	16:57	17:26	17:58
35°	05:25	05:55	06:21	17:06	17:32	18:02
30°	05:21	05:49	06:14	17:13	17:38	18:06
20°	05:13	05:39	06:02	17:25	17:48	18:14
N 10°	05:04	05:29	05:51	17:36	17:58	18:23
0°	04:54	05:19	05:40	17:47	18:08	18:33
S 10°	04:42	05:08	05:30	17:58	18:19	18:45
20°	04:28	04:55	05:18	18:09	18:32	18:59
30°	04:10	04:40	05:05	18:22	18:48	19:18
35°	03:58	04:31	04:58	18:30	18:57	19:30
40°	03:44	04:20	04:49	18:39	19:08	19:44
45°	03:27	04:06	04:39	18:49	19:22	20:02
S 50°	03:04	03:50	04:26	19:02	19:39	20:25
52°	02:52	03:42	04:20	19:08	19:47	20:37
54°	02:39	03:33	04:14	19:15	19:56	20:50
56°	02:23	03:23	04:07	19:22	20:06	21:06
58°	02:04	03:11	03:59	19:30	20:18	21:26
S 60°	01:39	02:57	03:49	19:39	20:32	21:53

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	16:21	15:37	15:00	23:41	· · ·	02:07
N 70°	16:01	15:29	15:01	23:58	· · ·	02:11
68°	15:46	15:22	15:01	· · ·	00:11	02:15
66°	15:33	15:17	15:02	· · ·	00:22	02:18
64°	15:22	15:12	15:03	· · ·	00:31	02:20
62°	15:13	15:08	15:03	· · ·	00:39	02:22
60°	15:05	15:04	15:03	· · ·	00:45	02:24
N 58°	14:58	15:01	15:04	· · ·	00:51	02:25
56°	14:52	14:58	15:04	· · ·	00:56	02:27
54°	14:46	14:56	15:05	· · ·	01:01	02:28
52°	14:41	14:53	15:05	· · ·	01:05	02:29
50°	14:36	14:51	15:05	· · ·	01:09	02:30
45°	14:26	14:46	15:06	00:02	01:17	02:33
N 40°	14:18	14:42	15:06	00:14	01:23	02:34
35°	14:11	14:39	15:07	00:23	01:29	02:36
30°	14:04	14:36	15:07	00:31	01:34	02:37
20°	13:53	14:31	15:08	00:46	01:42	02:40
N 10°	13:44	14:26	15:09	00:58	01:50	02:42

November 03, 04, 05 UT (Mon., Tue., Wed.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	42°30.5	198°45.0	S08°21.2	165°56.0	S20°18.9	285°31.2	N21°12.9	45°28.2	S03°57.7					Alpheratz	357°33.8 29°14.2		
1	57°33.0	213°44.5	22.4	180°56.6	19.3	300°33.6	12.8	60°30.7	57.8					Ankaa	353°06.1 -42°10.0		
2	72°35.4	228°44.1	23.5	195°57.2	19.7	315°36.0	12.8	75°33.3	57.8					Schedar	349°29.8 56°41.0		
3	87°37.9	243°43.6	· · 24.7	210°57.8	· · 20.1	330°38.4	· · 12.8	90°35.8	· · 57.9					Diphda	348°46.3 -17°50.6		
4	102°40.4	258°43.1	25.9	225°58.4	20.5	345°40.7	12.8	105°38.4	57.9					Achernar	335°19.1 -57°06.3		
5	117°42.8	273°42.7	27.1	240°59.0	20.9	0°43.1	12.8	120°41.0	57.9					Hamal	327°50.1 23°35.2		
6	132°45.3	288°42.2	S08°28.2	255°59.6	S20°21.3	15°45.5	N21°12.8	135°43.5	S03°58.0					Polaris	313°12.5 89°22.4		
7	147°47.7	303°41.7	29.4	271°00.2	21.8	30°47.9	12.8	150°46.1	58.0					Acamar	315°10.8 -40°11.9		
8	162°50.2	318°41.3	30.6	286°00.8	22.2	45°50.3	12.8	165°48.7	58.0					Menkar	314°05.1 4°11.6		
9	177°52.7	333°40.8	· · 31.7	301°01.4	· · 22.6	60°52.7	· · 12.8	180°51.2	· · 58.1					Mirfak	308°26.8 49°57.2		
10	192°55.1	348°40.3	32.9	316°02.1	23.0	75°55.1	12.8	195°53.8	58.1					Aldebaran	290°38.6 16°33.8		
11	207°57.6	3°39.8	34.1	331°02.7	23.4	90°57.5	12.8	210°56.4	58.1					Rigel	281°03.0 -8°10.1		
12	223°00.1	18°39.4	S08°35.2	346°03.3	S20°23.8	105°59.9	N21°12.8	225°58.9	S03°58.2					Capella	280°20.5 46°01.4		
13	238°02.5	33°38.9	36.4	1°03.9	24.2	121°02.3	12.8	241°01.5	58.2					Bellatrix	278°21.9 6°22.5		
14	253°05.0	48°38.4	37.6	16°04.5	24.6	136°04.7	12.8	256°04.0	58.2					Elnath	278°00.7 28°37.8		
15	268°07.5	63°38.0	· · 38.7	31°05.1	· · 25.0	151°07.0	· · 12.8	271°06.6	· · 58.3					Alnilam	275°36.8 -1°11.0		
16	283°09.9	78°37.5	39.9	46°05.7	25.5	166°09.4	12.8	286°09.2	58.3					Betelgeuse	270°51.1 7°24.8		
17	298°12.4	93°37.0	41.1	61°06.3	25.9	181°11.8	12.7	301°11.7	58.4					Canopus	263°51.8 -52°42.2		
18	313°14.9	108°36.6	S08°42.2	76°06.9	S20°26.3	196°14.2	N21°12.7	316°14.3	S03°58.4					Sirius	258°25.5 -16°44.9		
19	328°17.3	123°36.1	43.4	91°07.5	26.7	211°16.6	12.7	331°16.8	58.4					Adhara	255°05.2 -29°00.2		
20	343°19.8	138°35.6	44.6	106°08.1	27.1	226°19.0	12.7	346°19.4	58.5					Procyon	244°50.0 5°09.6		
21	358°22.2	153°35.1	· · 45.7	121°08.7	· · 27.5	241°21.4	· · 12.7	1°22.0	· · 58.5					Pollux	243°16.4 27°57.8		
22	13°24.7	168°34.7	46.9	136°09.3	27.9	256°23.8	12.7	16°24.5	58.5					Avior	234°14.4 -59°35.2		
23	28°27.2	183°34.2	48.1	151°09.9	28.3	271°26.2	12.7	31°27.1	58.6					Suhail	222°45.9 -43°31.9		
Mer.pass. 21:06			ν -0.5' d 1.2' m -3.91			ν 0.6' d 0.4' m 1.45			ν 2.4' d -0.0' m -2.34			ν 2.6' d 0.0' m 0.78					

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	43°29.6	198°33.7	S08°49.2	166°10.5	S20°28.7	286°28.6	N21°12.7	46°29.7	S03°58.6					
1	58°32.1	213°33.3	50.4	181°11.1	29.1	301°31.0	12.7	61°32.2	58.6					
2	73°34.6	228°32.8	51.5	196°11.7	29.5	316°33.4	12.7	76°34.8	58.7					
3	88°37.0	243°32.3	· · 52.7	211°12.3	· · 29.9	331°35.8	· · 12.7	91°37.3	· · 58.7					
4	103°39.5	258°31.8	53.9	226°13.0	30.4	346°38.2	12.7	106°39.9	58.7					
5	118°42.0	273°31.4	55.0	241°13.6	30.8	1°40.6	12.7	121°42.5	58.8					
6	133°44.4	288°30.9	S08°56.2	256°14.2	S20°31.2	16°43.0	N21°12.7	136°45.0	S03°58.8					
7	148°46.9	303°30.4	57.3	271°14.8	31.6	31°45.4	12.7	151°47.6	58.8					
8	163°49.3	318°29.9	58.5	286°15.4	32.0	46°47.8	12.7	166°50.1	58.9					
9	178°51.8	333°29.5	08°59.7	301°16.0	· · 32.4	61°50.2	· · 12.7	181°52.7	· · 58.9					
10	193°54.3	348°29.0	09°00.8	316°16.6	32.8	76°52.6	12.6	196°55.3	58.9					
11	208°56.7	3°28.5	02.0	331°17.2	33.2	91°55.0	12.6	211°57.8	59.0					
12	223°59.2	18°28.0	S09°03.1	346°17.8	S20°33.6	106°57.4	N21°12.6	227°00.4	S03°59.0					
13	239°01.7	33°27.6	04.3	1°18.4	34.0	121°59.8	12.6	242°02.9	59.0					
14	254°04.1	48°27.1	05.5	16°19.0	34.4	137°02.2	12.6	257°05.5	59.1					
15	269°06.6	63°26.6	· · 06.6	31°19.6	· · 34.8	152°04.6	· · 12.6	272°08.1	· · 59.1					
16	284°09.1	78°26.1	07.8	46°20.2	35.2	167°07.0	12.6	287°10.6	59.1					
17	299°11.5	93°25.6	08.9	61°20.8	35.6	182°09.4	12.6	302°13.2	59.2					
18	314°14.0	108°25.2	S09°10.1	76°21.4	S20°36.0	197°11.8	N21°12.6	317°15.7	S03°59.2					
19	329°16.5	123°24.7	11.3	91°22.0	36.4	212°14.2	12.6	332°18.3	59.2					
20	344°18.9	138°24.2	12.4	106°22.6	36.8	227°16.6	12.6	347°20.8	59.3					
21	359°21.4	153°23.7	· · 13.6	121°23.2	· · 37.2	242°19.0	· · 12.6	2°23.4	· · 59.3					
22	14°23.8	168°23.2	14.7	136°23.8	37.6	257°21.4	12.6	17°26.0	59.3					
23	29°26.3	183°22.8	15.9	151°24.4	38.0	272°23.8	12.6	32°28.5	59.4					
Mer.pass. 21:03			ν -0.5' d 1.2' m -3.91			ν 0.6' d 0.4' m 1.44			ν 2.4' d -0.0' m -2.34			ν 2.6' d 0.0' m 0.78		

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	44°28.8	198°22.3	S09°17.0	166°25.0	S20°38.4	287°26.2	N21°12.6	47°31.1	S03°59.4					
1	59°31.2	213°21.8	18.2	181°25.6	38.8	302°28.6	12.6	62°33.6	59.4					
2	74°33.7	228°21.3	19.3	196°26.2	39.2	317°31.0	12.6	77°36.2	59.5					
3	89°36.2	243°20.8	· · 20.5	211°26.8	· · 39.6	332°33.4	· · 12.6	92°38.7	· · 59.5					
4	104°38.6	258°20.4	21.7	226°27.3	40.0	347°35.8	12.6	107°41.3	59.5					
5	119°41.1	273°19.9	22.8	241°27.9	40.4	2°38.2	12.6	122°43.9	59.6					
6	134°43.6	288°19.4	S09°24.0	256°28.5	S20°40.8	17°40.7	N21°12.6	137°46.4	S03°59.6					
7	149°46.0	303°18.9	25.1	271°29.1	41.2	32°43.1	12.6	152°49.0	59.6					
8	164°48.5	318°18.4	26.3	286°29.7	41.6	47°45.5	12.6	167°51.5	59.7					
9	179°51.0	333°17.9	· · 27.4	301°30.3	· · 42.0	62°47.9	· · 12.5	182°54.1	· · 59.7					
10	194°53.4	348°17.5	28.6	316°30.9	42.4	77°50.3	12.5	197°56.6	59.7					
11	209°55.9	3°17.0	29.7	331°31.5	42.8	92°52.7	12.5	212°59.2	59.8					
12	224°58.3	18°16.5	S09°30.9	346°32.1	S20°43.2	107°55.1	N21°12.5	228°01.8	S03°59.8					
13	240°00.8	33°16.0	32.0	1°32.7	43.6	122°57.5	12.5	243°04.3	59.8					
14	255°03.3	48°15.5	33.2	16°33.3	44.0	137°59.9	12.5	258°06.9	59.8					
15	270°05.7	63°15.0	· · 34.3	31°33.9	· · 44.4	153°02.3	· · 12.5	273°09.4	· · 59.9					
16	285°08.2	78°14.5	35.5	46°34.5	44.8	168°04.7	12.5	288°12.0	59.9					
17	300°10.7	93°14.1	36.6	61°35.1	45.2	183°07.2	12.5	303°14.5	03°59.9					
18	315°13.1	108°13.6	S09°37.8	76°35.7	S20°45.6	198°09.6	N21°12.5	318°17.1	S04°00.0					
19	330°15.6	123°13.1	38.9	91°36.3	46.0	213°12.0	12.5	333°19.6	00.0					
20	345°18.1	138°12.6	40.1	106°36.9	46.4	228°14.4	12.5	348°22.2	00.0					
21	0°20.5	153°12.1	· · 41.2	121°37.5	· · 46.7	243°16.8	· · 12.5	3°24.8	· · 00.1					
22	15°23.0	168°11.6	42.4	136°38.0	47.1	258°19.2	12.5	18°27.3	00.1					
23	30°25.5	183°11.1	43.5	151°38.6	47.5	273°21.6	12.5	33°29.9	00.1					
Mer.pass. 20:59			ν -0.5' d 1.2' m -3.91			ν 0.6' d 0.4' m 1.44			ν 2.4' d -0.0' m -2.35			ν 2.6' d 0.0' m 0.79		

Stars		
	SHA	Dec
Alpheratz	357°33.8	29°14.2
Ankaa	353°06.1	-42°10

Sun				Moon				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	184°06.5	S15°04.3		38°31.1	11.2'	N03°27.1	17.3'	60.0'
1	199°06.5	05.1		53°01.4	11.2'	03°44.4	17.3'	60.0'
2	214°06.5	05.8		67°31.5	11.1'	04°01.6	17.3'	60.1'
3	229°06.5	· · 06.6		82°01.6	11.0'	04°18.9	17.3'	60.1'
4	244°06.5	07.4		96°31.7	11.0'	04°36.2	17.3'	60.1'
5	259°06.5	08.2		111°01.6	10.9'	04°53.4	17.3'	60.2'
6	274°06.5	S15°09.0		125°31.5	10.8'	N05°10.7	17.3'	60.2'
7	289°06.5	09.7		140°01.4	10.8'	05°27.9	17.2'	60.2'
8	304°06.5	10.5		154°31.1	10.7'	05°45.2	17.2'	60.3'
9	319°06.5	· · 11.3		169°00.8	10.6'	06°02.4	17.2'	60.3'
10	334°06.5	12.1		183°30.5	10.6'	06°19.6	17.2'	60.3'
11	349°06.5	12.8		198°00.0	10.5'	06°36.8	17.2'	60.4'
12	4°06.5	S15°13.6		212°29.5	10.4'	N06°54.0	17.2'	60.4'
13	19°06.5	14.4		226°58.9	10.3'	07°11.2	17.1'	60.4'
14	34°06.5	15.2		241°28.2	10.2'	07°28.3	17.1'	60.5'
15	49°06.5	· · 15.9		255°57.5	10.2'	07°45.4	17.1'	60.5'
16	64°06.5	16.7		270°26.7	10.1'	08°02.5	17.1'	60.5'
17	79°06.5	17.5		284°55.8	10.0'	08°19.6	17.0'	60.6'
18	94°06.5	S15°18.3		299°24.8	9.9'	N08°36.6	17.0'	60.6'
19	109°06.5	19.0		313°53.7	9.8'	08°53.6	17.0'	60.6'
20	124°06.5	19.8		328°22.5	9.8'	09°10.6	16.9'	60.6'
21	139°06.5	· · 20.6		342°51.3	9.7'	09°27.5	16.9'	60.7'
22	154°06.5	21.3		357°19.9	9.6'	09°44.4	16.9'	60.7'
23	169°06.5	22.1		11°48.5	9.5'	10°01.3	16.8'	60.7'
SD = 16.1' d = 0.8'				SD = 16.4'				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	184°06.5	S15°22.9		26°17.0	9.4'	N10°18.1	16.8'	60.7'
1	199°06.4	23.7		40°45.4	9.3'	10°34.9	16.7'	60.8'
2	214°06.4	24.4		55°13.7	9.2'	10°51.6	16.7'	60.8'
3	229°06.4	· · 25.2		69°41.9	9.1'	11°08.3	16.6'	60.8'
4	244°06.4	26.0		84°10.0	9.0'	11°24.9	16.6'	60.8'
5	259°06.4	26.7		98°38.0	8.9'	11°41.4	16.5'	60.9'
6	274°06.4	S15°27.5		113°06.0	8.8'	N11°58.0	16.5'	60.9'
7	289°06.4	28.3		127°33.8	8.7'	12°14.4	16.4'	60.9'
8	304°06.4	29.0		142°01.5	8.6'	12°30.8	16.3'	60.9'
9	319°06.4	· · 29.8		156°29.1	8.5'	12°47.1	16.3'	61.0'
10	334°06.4	30.6		170°56.7	8.4'	13°03.4	16.2'	61.0'
11	349°06.4	31.3		185°24.1	8.3'	13°19.6	16.1'	61.0'
12	4°06.3	S15°32.1		199°51.4	8.2'	N13°35.7	16.1'	61.0'
13	19°06.3	32.8		214°18.6	8.1'	13°51.8	16.0'	61.0'
14	34°06.3	33.6		228°45.7	8.0'	14°07.8	15.9'	61.1'
15	49°06.3	· · 34.4		243°12.7	7.9'	14°23.7	15.8'	61.1'
16	64°06.3	35.1		257°39.6	7.8'	14°39.5	15.7'	61.1'
17	79°06.3	35.9		272°06.4	7.7'	14°55.3	15.7'	61.1'
18	94°06.3	S15°36.7		286°33.1	7.6'	N15°10.9	15.6'	61.1'
19	109°06.3	37.4		300°59.7	7.5'	15°26.5	15.5'	61.1'
20	124°06.2	38.2		315°26.1	7.4'	15°42.0	15.4'	61.2'
21	139°06.2	· · 38.9		329°52.5	7.2'	15°57.4	15.3'	61.2'
22	154°06.2	39.7		344°18.7	7.1'	16°12.7	15.2'	61.2'
23	169°06.2	40.5		358°44.9	7.0'	16°27.9	15.1'	61.2'
SD = 16.1' d = 0.8'				SD = 16.6'				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	184°06.2	S15°41.2		13°10.9	6.9'	N16°43.0	15.0'	61.2'
1	199°06.2	42.0		27°36.8	6.8'	16°58.1	14.9'	61.2'
2	214°06.2	42.7		42°02.6	6.7'	17°13.0	14.8'	61.2'
3	229°06.1	· · 43.5		56°28.2	6.6'	17°27.8	14.7'	61.3'
4	244°06.1	44.3		70°53.8	6.4'	17°42.5	14.6'	61.3'
5	259°06.1	45.0		85°19.3	6.3'	17°57.1	14.5'	61.3'
6	274°06.1	S15°45.8		99°44.6	6.2'	N18°11.5	14.4'	61.3'
7	289°06.1	46.5		114°09.8	6.1'	18°25.9	14.2'	61.3'
8	304°06.1	47.3		128°34.9	6.0'	18°40.1	14.1'	61.3'
9	319°06.0	· · 48.0		142°59.9	5.9'	18°54.3	14.0'	61.3'
10	334°06.0	48.8		157°24.8	5.8'	19°08.3	13.9'	61.3'
11	349°06.0	49.5		171°49.5	5.6'	19°22.1	13.8'	61.3'
12	4°06.0	S15°50.3		186°14.2	5.5'	N19°35.9	13.6'	61.3'
13	19°06.0	51.0		200°38.7	5.4'	19°49.5	13.5'	61.3'
14	34°05.9	51.8		215°03.1	5.3'	20°03.0	13.4'	61.4'
15	49°05.9	· · 52.6		229°27.4	5.2'	20°16.4	13.2'	61.4'
16	64°05.9	53.3		243°51.6	5.1'	20°29.6	13.1'	61.4'
17	79°05.9	54.1		258°15.7	5.0'	20°42.7	12.9'	61.4'
18	94°05.9	S15°54.8		272°39.6	4.8'	N20°55.6	12.8'	61.4'
19	109°05.8	55.6		287°03.5	4.7'	21°08.4	12.6'	61.4'
20	124°05.8	56.3		301°27.2	4.6'	21°21.0	12.5'	61.4'
21	139°05.8	· · 57.1		315°50.8	4.5'	21°33.5	12.3'	61.4'
22	154°05.8	57.8		330°14.3	4.4'	21°45.9	12.2'	61.4'
23	169°05.7	58.6		344°37.7	4.3'	21°58.1	12.0'	61.4'
SD = 16.1' d = 0.8'				SD = 16.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:19	07:44	09:18	14:08	15:42	17:07
N 70°	06:13	07:29	08:47	14:39	15:57	17:12
68°	06:09	07:17	08:24	15:02	16:09	17:17
66°	06:04	07:07	08:07	15:19	16:19	17:21
64°	06:01	06:58	07:53	15:34	16:28	17:25
62°	05:57	06:51	07:41	15:46	16:35	17:29
60°	05:54	06:44	07:30	15:56	16:42	17:32
N 58°	05:52	06:39	07:22	16:05	16:48	17:34
56°	05:49	06:34	07:14	16:13	16:53	17:37
54°	05:47	06:29	07:07	16:20	16:58	17:40
52°	05:44	06:25	07:01	16:26	17:02	17:42
50°	05:42	06:21	06:55	16:32	17:06	17:44
45°	05:37	06:12	06:43	16:44	17:15	17:50
N 40°	05:32	06:04	06:33	16:54	17:22	17:54
35°	05:28	05:58	06:24	17:03	17:29	17:59
30°	05:23	05:51	06:16	17:11	17:35	18:04
20°	05:14	05:40	06:03	17:24	17:47	18:13
N 10°	05:04	05:30	05:51	17:36	17:58	18:23
0°	04:54	05:19	05:40	17:47	18:08	18:33
S 10°	04:41	05:07	05:29	17:58	18:20	18:46
20°	04:26	04:54	05:17	18:11	18:34	19:01
30°	04:07	04:37	05:03	18:25	18:50	19:21
35°	03:55	04:28	04:55	18:33	19:00	19:33
40°	03:40	04:16	04:45	18:42	19:12	19:48
45°	03:22	04:02	04:34	18:54	19:26	20:07
S 50°	02:57	03:44	04:21	19:07	19:44	20:32
52°	02:45	03:36	04:15	19:14	19:53	20:44
54°	02:30	03:26	04:08	19:21	20:03	20:59
56°	02:13	03:15	04:00	19:28	20:14	21:17
58°	01:52	03:03	03:51	19:37	20:27	21:39
S 60°	01:22	02:48	03:41	19:48	20:42	22:10

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	14:21	13:27	□	04:32	07:16	□
N 70°	14:31	13:53	12:28	04:25	06:53	10:16
68°	14:40	14:13	13:28	04:20	06:35	09:18
66°	14:47	14:29	14:03	04:15	06:21	08:45
64°	14:53	14:42	14:28	04:11	06:10	08:21
62°	14:58	14:53	14:48	04:08	06:00	08:02
60°	15:03	15:03	15:05	04:05	05:52	07:46
N 58°	15:07	15:12	15:19	04:02	05:44	07:33
56°	15:11	15:19	15:31	04:00	05:38	07:22
54°	15:14	15:26	15:42	03:58	05:32	07:12
52°	15:17	15:32	15:51	03:56	05:27	07:03
50°	15:20	15:38	16:00	03:55	05:23	06:55
45°	15:26	15:50	16:18	03:51	05:13	06:39
N 40°	15:31	16:00	16:33	03:48	05:05	06:25
35°	15:36	16:08	16:46	03:45	04:58	06:14
30°	15:40	16:16	16:57	03:43	04:52	06:04
20°	15:47	16:29	17:17	03:39	04:41	05:47
N 10°	15:53	16:41	17			

November 06, 07, 08 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec			
0	45°27.9	198°10.6	S09°44.7	166°39.2	S20°47.9	288°24.0	N21°12.5	48°32.4	S04°00.2	Alpheratz	357°33.8	29°14.2					
1	60°30.4	213°10.2	45.8	181°39.8	48.3	303°26.5	12.5	63°35.0	00.2	Ankaa	353°06.1	-42°10.0					
2	75°32.8	228°09.7	46.9	196°40.4	48.7	318°28.9	12.5	78°37.5	00.2	Schedar	349°29.8	56°41.0					
3	90°35.3	243°09.2	· · 48.1	211°41.0	· · 49.1	333°31.3	· · 12.5	93°40.1	· · 00.3	Diphda	348°46.4	-17°50.6					
4	105°37.8	258°08.7	49.2	226°41.6	49.5	348°33.7	12.5	108°42.6	00.3	Achernar	335°19.1	-57°06.4					
5	120°40.2	273°08.2	50.4	241°42.2	49.9	3°36.1	12.5	123°45.2	00.3	Hamal	327°50.1	23°35.2					
6	135°42.7	288°07.7	S09°51.5	256°42.8	S20°50.3	18°38.5	N21°12.5	138°47.7	S04°00.3	Polaris	313°11.8	89°22.4					
7	150°45.2	303°07.2	52.7	271°43.4	50.7	33°40.9	12.5	153°50.3	00.4	Acamar	315°10.8	-40°12.0					
8	165°47.6	318°06.7	53.8	286°43.9	51.1	48°43.4	12.5	168°52.8	00.4	Menkar	314°05.1	4°11.6					
9	180°50.1	333°06.2	· · 55.0	301°44.5	· · 51.4	63°45.8	· · 12.5	183°55.4	· · 00.4	Mirfak	308°26.8	49°57.2					
10	195°52.6	348°05.7	56.1	316°45.1	51.8	78°48.2	12.5	198°57.9	00.5	Aldebaran	290°38.5	16°33.7					
11	210°55.0	3°05.2	57.2	331°45.7	52.2	93°50.6	12.5	214°00.5	00.5	Rigel	281°02.9	-8°10.2					
12	225°57.5	18°04.7	S09°58.4	346°46.3	S20°52.6	108°53.0	N21°12.5	229°03.1	S04°00.5	Capella	280°20.5	46°01.4					
13	240°59.9	33°04.2	09°59.5	1°46.9	53.0	123°55.4	12.5	244°05.6	00.6	Bellatrix	278°21.9	6°22.5					
14	256°02.4	48°03.8	10°00.7	16°47.5	53.4	138°57.9	12.5	259°08.2	00.6	Elnath	278°00.7	28°37.7					
15	271°04.9	63°03.3	· · 01.8	31°48.1	· · 53.8	154°00.3	· · 12.5	274°10.7	· · 00.6	Alnilam	275°36.8	-1°11.0					
16	286°07.3	78°02.8	02.9	46°48.7	54.2	169°02.7	12.5	289°13.3	00.6	Betelgeuse	270°51.1	7°24.8					
17	301°09.8	93°02.3	04.1	61°49.2	54.6	184°05.1	12.5	304°15.8	00.7	Canopus	263°51.7	-52°42.2					
18	316°12.3	108°01.8	S10°05.2	76°49.8	S20°54.9	199°07.5	N21°12.5	319°18.4	S04°00.7	Sirius	258°25.5	-16°44.9					
19	331°14.7	123°01.3	06.4	91°50.4	55.3	214°10.0	12.5	334°20.9	00.7	Adhara	255°05.1	-29°00.2					
20	346°17.2	138°00.8	07.5	106°51.0	55.7	229°12.4	12.5	349°23.5	00.8	Procyon	244°50.0	5°09.6					
21	1°19.7	153°00.3	· · 08.6	121°51.6	· · 56.1	244°14.8	· · 12.5	4°26.0	· · 00.8	Pollux	243°16.4	27°57.8					
22	16°22.1	167°59.8	09.8	136°52.2	56.5	259°17.2	12.5	19°28.6	00.8	Avior	234°14.4	-59°35.2					
23	31°24.6	182°59.3	10.9	151°52.8	56.9	274°19.6	12.5	34°31.1	00.9	Suhail	222°45.8	-43°31.9					
Mer.pass. 20:55			ν -0.5' d 1.1' m -3.91			ν 0.6' d 0.4' m 1.43			ν 2.4' d -0.0' m -2.36			ν 2.6' d 0.0' m 0.79					

Fri	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	46°27.1	197°58.8	S10°12.1	166°53.3	S20°57.3	289°22.1	N21°12.5	49°33.7	S04°00.9	289°22.1	N21°12.5	49°33.7	S04°00.9	357°33.8	29°14.2	Alpheratz	357°33.8	29°14.2	357°33.8	29°14.2	357°33.8	29°14.2		
1	61°29.5	212°58.3	13.2	181°53.9	57.6	304°24.5	12.5	64°36.2	00.9	181°53.9	57.6	64°36.2	00.9	Ankaa	353°06.1	-42°10.0	Ankaa	353°06.1	-42°10.0	353°06.1	-42°10.0			
2	76°32.0	227°57.8	14.3	196°54.5	58.0	319°26.9	12.4	79°38.8	00.9	196°54.5	58.0	79°38.8	00.9	Schedar	349°29.8	56°41.0	Schedar	349°29.8	56°41.0	349°29.8	56°41.0			
3	91°34.4	242°57.3	· · 15.5	211°55.1	· · 58.4	334°29.3	· · 12.4	94°41.3	· · 01.0	211°55.1	· · 58.4	94°41.3	· · 01.0	Diphda	348°46.4	-17°50.6	Diphda	348°46.4	-17°50.6	348°46.4	-17°50.6			
4	106°36.9	257°56.8	16.6	226°55.7	58.8	349°31.8	12.4	109°43.9	01.0	226°55.7	58.8	109°43.9	01.0	Achernar	335°19.1	-57°06.4	Achernar	335°19.1	-57°06.4	335°19.1	-57°06.4			
5	121°39.4	272°56.3	17.7	241°56.3	59.2	4°34.2	12.4	124°46.4	01.0	241°56.3	59.2	124°46.4	01.0	Hamal	327°50.1	23°35.2	Hamal	327°50.1	23°35.2	327°50.1	23°35.2			
6	136°41.8	287°55.8	S10°18.9	256°56.8	S20°59.6	19°36.6	N21°12.4	139°49.0	S04°01.1	256°56.8	S20°59.6	19°36.6	N21°12.4	Polaris	313°11.8	89°22.4	Polaris	313°11.8	89°22.4	313°11.8	89°22.4			
7	151°44.3	302°55.3	20.0	271°57.4	20°59.9	34°39.0	12.4	154°51.5	01.1	271°57.4	20°59.9	34°39.0	12.4	Acamar	315°10.8	-40°12.0	Acamar	315°10.8	-40°12.0	315°10.8	-40°12.0			
8	166°46.8	317°54.8	21.1	286°58.0	21°00.3	49°41.5	12.4	169°54.1	01.1	286°58.0	21°00.3	49°41.5	12.4	Menkar	314°05.1	4°11.6	Menkar	314°05.1	4°11.6	314°05.1	4°11.6			
9	181°49.2	332°54.3	· · 22.3	301°58.6	· · 00.7	64°43.9	· · 12.4	184°56.6	· · 01.1	301°58.6	· · 00.7	64°43.9	· · 12.4	Mirfak	308°26.8	49°57.2	Mirfak	308°26.8	49°57.2	308°26.8	49°57.2			
10	196°51.7	347°53.8	23.4	316°59.2	01.1	79°46.3	12.4	199°59.2	01.2	316°59.2	01.1	79°46.3	12.4	Aldebaran	290°38.5	16°33.7	Aldebaran	290°38.5	16°33.7	290°38.5	16°33.7			
11	211°54.2	2°53.3	24.5	331°59.8	01.5	94°48.7	12.4	215°01.7	01.2	331°59.8	01.5	94°48.7	12.4	Rigel	281°02.9	-8°10.2	Rigel	281°02.9	-8°10.2	281°02.9	-8°10.2			
12	226°56.6	17°52.8	S10°25.7	347°00.3	S21°01.9	109°51.2	N21°12.4	230°04.3	S04°01.2	347°00.3	S21°01.9	109°51.2	N21°12.4	Capella	280°20.5	46°01.4	Capella	280°20.5	46°01.4	280°20.5	46°01.4			
13	241°59.1	32°52.3	26.8	2°00.9	02.2	124°53.6	12.4	245°06.8	01.3	2°00.9	02.2	124°53.6	12.4	Bellatrix	278°21.9	6°22.5	Bellatrix	278°21.9	6°22.5	278°21.9	6°22.5			
14	257°01.6	47°51.8	27.9	17°01.5	02.6	139°56.0	12.4	260°09.4	01.3	17°01.5	02.6	139°56.0	12.4	Elnath	278°00.7	28°37.7	Elnath	278°00.7	28°37.7	278°00.7	28°37.7			
15	272°04.0	62°51.3	· · 29.1	32°02.1	· · 03.0	154°58.4	· · 12.4	275°11.9	· · 01.3	32°02.1	· · 03.0	154°58.4	· · 12.4	Alnilam	275°36.8	-1°11.0	Alnilam	275°36.8	-1°11.0	275°36.8	-1°11.0			
16	287°06.5	77°50.8	30.2	47°02.7	03.4	170°00.9	12.4	290°14.5	01.3	47°02.7	03.4	170°00.9	12.4	Betelgeuse	270°51.1	7°24.8	Betelgeuse	270°51.1	7°24.8	270°51.1	7°24.8			
17	302°08.9	92°50.3	31.3	62°03.2	03.8	185°03.3	12.4	305°17.0	01.4	62°03.2	03.8	185°03.3	12.4	Canopus	263°51.7	-52°42.2	Canopus	263°51.7	-52°42.2	263°51.7	-52°42.2			
18	317°11.4	107°49.8	S10°32.5	77°03.8	S21°04.1	200°05.7	N21°12.4	320°19.6	S04°01.4	77°03.8	S21°04.1	200°05.7	N21°12.4	Sirius	258°25.5	-16°44.9	Sirius	258°25.5	-16°44.9	258°25.5	-16°44.9			
19	332°13.9	122°49.2	33.6	92°04.4	04.5	215°08.2	12.4	335°22.1	01.4	92°04.4	04.5	215°08.2	12.4	Adhara	255°05.1	-29°00.2	Adhara	255°05.1	-29°00.2	255°05.1	-29°00.2			
20	347°16.3	137°48.7	34.7	107°05.0	04.9	230°10.6	12.4	350°24.7	01.5	107°05.0	04.9	230°10.6	12.4	Procyon	244°50.0	5°09.6	Procyon	244°50.0	5°09.6	244°50.0	5°09.6			
21	2°18.8	152°48.2	· · 35.8	122°05.6	· · 05.3	245°13.0	· · 12.4	5°27.2	· · 01.5	122°05.6	· · 05.3	245°13.0	· · 12.4	Pollux	243°16.4	27°57.8	Pollux	243°16.4	27°57.8	243°				

Only use these tags ['table', 'tr', 'td', 'th>, 'td',td

h	Sun				Moon			
Thu	GHA	Dec	GHA	ν	Dec	d	HP	
0	184°05.7	S15°59.3	359°01.0	4.2'	N22°10.1	11.9'	61.4'	
1	199°05.7	16°00.1	13°24.2	4.1'	22°22.0	11.7'	61.4'	
2	214°05.7	00.8	27°47.2	4.0'	22°33.7	11.6'	61.4'	
3	229°05.7	· · 01.5	42°10.2	3.8'	22°45.3	11.4'	61.4'	
4	244°05.6	02.3	56°33.0	3.7'	22°56.7	11.2'	61.4'	
5	259°05.6	03.0	70°55.8	3.6'	23°07.9	11.1'	61.4'	
6	274°05.6	S16°03.8	85°18.4	3.5'	N23°19.0	10.9'	61.4'	
7	289°05.5	04.5	99°40.9	3.4'	23°29.9	10.7'	61.4'	
8	304°05.5	05.3	114°03.4	3.3'	23°40.6	10.5'	61.3'	
9	319°05.5	· · 06.0	128°25.7	3.2'	23°51.1	10.4'	61.3'	
10	334°05.5	06.8	142°47.9	3.1'	24°01.5	10.2'	61.3'	
11	349°05.4	07.5	157°10.1	3.0'	24°11.7	10.0'	61.3'	
12	4°05.4	S16°08.2	171°32.1	2.9'	N24°21.7	9.8'	61.3'	
13	19°05.4	09.0	185°54.1	2.9'	24°31.5	9.6'	61.3'	
14	34°05.4	09.7	200°15.9	2.8'	24°41.2	9.5'	61.3'	
15	49°05.3	· · 10.5	214°37.7	2.7'	24°50.6	9.3'	61.3'	
16	64°05.3	11.2	228°59.4	2.6'	24°59.9	9.1'	61.3'	
17	79°05.3	12.0	243°21.0	2.5'	25°09.0	8.9'	61.3'	
18	94°05.2	S16°12.7	257°42.5	2.4'	N25°17.9	8.7'	61.3'	
19	109°05.2	13.4	272°03.9	2.3'	25°26.6	8.5'	61.2'	
20	124°05.2	14.2	286°25.2	2.3'	25°35.1	8.3'	61.2'	
21	139°05.1	· · 14.9	300°46.5	2.2'	25°43.4	8.1'	61.2'	
22	154°05.1	15.6	315°07.7	2.1'	25°51.5	7.9'	61.2'	
23	169°05.1	16.4	329°28.8	2.0'	25°59.4	7.7'	61.2'	
SD = 16.1'		$d = 0.7'$	SD = 16.7'					
Fri	GHA	Dec	GHA	ν	Dec	d	HP	
0	184°05.0	S16°17.1	343°49.9	2.0'	N26°07.1	7.5'	61.2'	
1	199°05.0	17.9	358°10.8	1.9'	26°14.6	7.3'	61.2'	
2	214°05.0	18.6	12°31.8	1.9'	26°21.9	7.1'	61.2'	
3	229°04.9	· · 19.3	26°52.6	1.8'	26°29.0	6.9'	61.1'	
4	244°04.9	20.1	41°13.4	1.7'	26°35.9	6.7'	61.1'	
5	259°04.9	20.8	55°34.1	1.7'	26°42.6	6.5'	61.1'	
6	274°04.8	S16°21.5	69°54.8	1.6'	N26°49.1	6.3'	61.1'	
7	289°04.8	22.3	84°15.4	1.6'	26°55.4	6.1'	61.1'	
8	304°04.8	23.0	98°36.0	1.5'	27°01.4	5.8'	61.0'	
9	319°04.7	· · 23.7	112°56.6	1.5'	27°07.3	5.6'	61.0'	
10	334°04.7	24.5	127°17.0	1.5'	27°12.9	5.4'	61.0'	
11	349°04.7	25.2	141°37.5	1.4'	27°18.3	5.2'	61.0'	
12	4°04.6	S16°25.9	155°57.9	1.4'	N27°23.5	5.0'	61.0'	
13	19°04.6	26.7	170°18.3	1.4'	27°28.5	4.8'	60.9'	
14	34°04.6	27.4	184°38.6	1.3'	27°33.3	4.6'	60.9'	
15	49°04.5	· · 28.1	198°59.0	1.3'	27°37.9	4.3'	60.9'	
16	64°04.5	28.9	213°19.3	1.3'	27°42.2	4.1'	60.9'	
17	79°04.4	29.6	227°39.5	1.3'	27°46.3	3.9'	60.9'	
18	94°04.4	S16°30.3	241°59.8	1.2'	N27°50.3	3.7'	60.8'	
19	109°04.4	31.0	256°20.0	1.2'	27°53.9	3.5'	60.8'	
20	124°04.3	31.8	270°40.3	1.2'	27°57.4	3.3'	60.8'	
21	139°04.3	· · 32.5	285°00.5	1.2'	28°00.7	3.0'	60.8'	
22	154°04.2	33.2	299°20.7	1.2'	28°03.7	2.8'	60.7'	
23	169°04.2	33.9	313°41.0	1.2'	28°06.5	2.6'	60.7'	
SD = 16.1'		$d = 0.7'$	SD = 16.7'					
Sat	GHA	Dec	GHA	ν	Dec	d	HP	
0	184°04.2	S16°34.7	328°01.2	1.2'	N28°09.2	2.4'	60.7'	
1	199°04.1	35.4	342°21.4	1.2'	28°11.5	2.2'	60.7'	
2	214°04.1	36.1	356°41.7	1.3'	28°13.7	1.9'	60.6'	
3	229°04.0	· · 36.8	11°01.9	1.3'	28°15.7	1.7'	60.6'	
4	244°04.0	37.6	25°22.2	1.3'	28°17.4	1.5'	60.6'	
5	259°04.0	38.3	39°42.5	1.3'	28°18.9	1.3'	60.5'	
6	274°03.9	S16°39.0	54°02.8	1.3'	N28°20.2	1.1'	60.5'	
7	289°03.9	39.7	68°23.2	1.4'	28°21.3	0.9'	60.5'	
8	304°03.8	40.5	82°43.6	1.4'	28°22.1	0.6'	60.5'	
9	319°03.8	· · 41.2	97°04.0	1.4'	28°22.8	0.4'	60.4'	
10	334°03.7	41.9	111°24.4	1.5'	28°23.2	0.2'	60.4'	
11	349°03.7	42.6	125°44.9	1.5'	28°23.4	-0.0'	60.4'	
12	4°03.6	S16°43.3	140°05.4	1.6'	N28°23.4	-0.2'	60.3'	
13	19°03.6	44.1	154°26.0	1.6'	28°23.2	-0.4'	60.3'	
14	34°03.5	44.8	168°46.7	1.7'	28°22.8	-0.6'	60.3'	
15	49°03.5	· · 45.5	183°07.3	1.7'	28°22.1	-0.9'	60.3'	
16	64°03.5	46.2	197°28.1	1.8'	28°21.3	-1.1'	60.2'	
17	79°03.4	46.9	211°48.9	1.9'	28°20.2	-1.3'	60.2'	
18	94°03.4	S16°47.6	226°09.7	1.9'	N28°19.0	-1.5'	60.2'	
19	109°03.3	48.4	240°30.7	2.0'	28°17.5	-1.7'	60.1'	
20	124°03.3	49.1	254°51.7	2.1'	28°15.8	-1.9'	60.1'	
21	139°03.2	· · 49.8	269°12.7	2.1'	28°13.9	-2.1'	60.1'	
22	154°03.2	50.5	283°33.9	2.2'	28°11.8	-2.3'	60.0'	
23	169°03.1	51.2	297°55.1	2.3'	28°09.5	-2.5'	60.0'	
SD = 16.1'		$d = 0.7'$	SD = 16.6'					

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:30	07:58	09:40	13:46	15:28	16:55
N 70°	06:24	07:41	09:03	14:23	15:45	17:02
68°	06:18	07:27	08:37	14:49	15:59	17:08
66°	06:13	07:16	08:18	15:08	16:10	17:13
64°	06:08	07:07	08:02	15:24	16:20	17:18
62°	06:04	06:58	07:49	15:37	16:28	17:22
60°	06:01	06:51	07:38	15:48	16:35	17:25
N 58°	05:58	06:45	07:29	15:58	16:42	17:29
56°	05:55	06:39	07:20	16:06	16:47	17:32
54°	05:52	06:34	07:13	16:14	16:52	17:35
52°	05:49	06:30	07:06	16:21	16:57	17:38
50°	05:47	06:25	07:00	16:27	17:01	17:40
45°	05:41	06:16	06:47	16:40	17:11	17:46
N 40°	05:35	06:08	06:36	16:51	17:19	17:52
35°	05:30	06:00	06:27	17:00	17:27	17:57
30°	05:25	05:54	06:19	17:08	17:33	18:02
20°	05:15	05:42	06:05	17:23	17:46	18:12
N 10°	05:05	05:30	05:52	17:35	17:57	18:22
0°	04:54	05:19	05:40	17:47	18:09	18:34
S 10°	04:41	05:06	05:28	17:59	18:21	18:47
20°	04:25	04:52	05:15	18:12	18:35	19:03
30°	04:04	04:35	05:01	18:27	18:53	19:23
35°	03:52	04:25	04:52	18:36	19:03	19:36
40°	03:36	04:12	04:42	18:46	19:16	19:52
45°	03:16	03:57	04:30	18:58	19:31	20:12
S 50°	02:50	03:39	04:16	19:12	19:50	20:39
52°	02:37	03:30	04:09	19:19	19:59	20:52
54°	02:22	03:19	04:02	19:27	20:09	21:08
56°	02:03	03:08	03:54	19:35	20:21	21:27
58°	01:39	02:54	03:44	19:45	20:35	21:53
S 60°	01:04	02:38	03:34	19:56	20:52	22:30

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	☐	☐	☐	☐	☐	☐
N 70°	☐	☐	☐	☐	☐	☐
68°	☐	☐	☐	☐	☐	☐
66°	☐	☐	☐	☐	☐	☐
64°	14:02	☐	☐	10:55	☐	☐
62°	14:43	14:33	☐	10:15	12:42	☐
60°	15:11	15:28	16:18	09:48	11:47	13:16
N 58°	15:33	16:01	16:57	09:26	11:15	12:38
56°	15:51	16:25	17:24	09:09	10:51	12:11
54°	16:06	16:45	17:45	08:54	10:31	11:50
52°	16:19	17:02	18:03	08:41	10:15	11:32
50°	16:31	17:16	18:18	08:30	10:01	11:17
45°	16:56	17:45	18:48	08:07	09:32	10:46
N 40°	17:15	18:08	19:12	07:48	09:09	10:22
35°	17:32	18:27	19:32	07:33	08:51	10:02
30°	17:46	18:44	19:49	07:19	08:35	09:46
20°	18:11	19:12	20:17	06:57	08:08	09:17
N 10°	18:32	19:36	20:41	06:37	07:45	08:52
0°	18:52	19:58	21:04	06:19	07:23	08:30
S 10°	19:12	20:21	21:27	06:01	07:02	08:07
20°	19:34	20:45	21:51	05:41	06:39	07:42
30°	19:59	21:13	22:20	05:19	06:12	07:13
35°	20:14	21:30	22:36	05:06	05:56	06:56
40°	20:32	21:50	22:56	04:51	05:38	

November 09, 10, 11 UT (Sun., Mon., Tue.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	48°25.3	197°34.4	S11°06.2	167°21.1	S21°15.4	291°18.7	N21°12.5	51°35.9	S04°02.2	Alpheratz	357°33.8	29°14.2
1	63°27.8	212°33.9	07.3	182°21.7	15.7	306°21.2	12.5	66°38.5	02.2	Ankaa	353°06.1	-42°10.0
2	78°30.3	227°33.4	08.4	197°22.2	16.1	321°23.6	12.5	81°41.0	02.3	Schedar	349°29.8	56°41.0
3	93°32.7	242°32.9	· · 09.5	212°22.8	· · 16.5	336°26.1	· · 12.5	96°43.6	· · 02.3	Diphda	348°46.3	-17°50.6
4	108°35.2	257°32.3	10.6	227°23.4	16.8	351°28.5	12.5	111°46.1	02.3	Achernar	335°19.1	-57°06.4
5	123°37.7	272°31.8	11.7	242°24.0	17.2	6°30.9	12.5	126°48.6	02.3	Hamal	327°50.1	23°35.2
6	138°40.1	287°31.3	S11°12.8	257°24.5	S21°17.6	21°33.4	N21°12.5	141°51.2	S04°02.4	Polaris	313°11.2	89°22.4
7	153°42.6	302°30.8	14.0	272°25.1	18.0	36°35.8	12.5	156°53.7	02.4	Acamar	315°10.8	-40°12.0
8	168°45.1	317°30.3	15.1	287°25.7	18.3	51°38.3	12.5	171°56.3	02.4	Menkar	314°05.1	4°11.6
9	183°47.5	332°29.7	· · 16.2	302°26.2	· · 18.7	66°40.7	· · 12.5	186°58.8	· · 02.4	Mirfak	308°26.8	49°57.2
10	198°50.0	347°29.2	17.3	317°26.8	19.1	81°43.2	12.5	202°01.3	02.5	Aldebaran	290°38.5	16°33.7
11	213°52.4	2°28.7	18.4	332°27.4	19.4	96°45.6	12.5	217°03.9	02.5	Rigel	281°02.9	-8°10.2
12	228°54.9	17°28.2	S11°19.5	347°27.9	S21°19.8	111°48.0	N21°12.5	232°06.4	S04°02.5	Capella	280°20.5	46°01.4
13	243°57.4	32°27.7	20.6	2°28.5	20.2	126°50.5	12.5	247°09.0	02.5	Bellatrix	278°21.9	6°22.5
14	258°59.8	47°27.1	21.7	17°29.1	20.5	141°52.9	12.5	262°11.5	02.6	Elnath	278°00.7	28°37.8
15	274°02.3	62°26.6	· · 22.9	32°29.7	· · 20.9	156°55.4	· · 12.5	277°14.1	· · 02.6	Alnilam	275°36.8	-1°11.0
16	289°04.8	77°26.1	24.0	47°30.2	21.2	171°57.8	12.5	292°16.6	02.6	Betelgeuse	270°51.1	7°24.8
17	304°07.2	92°25.6	25.1	62°30.8	21.6	187°00.3	12.5	307°19.1	02.6	Canopus	263°51.7	-52°42.3
18	319°09.7	107°25.0	S11°26.2	77°31.4	S21°22.0	202°02.7	N21°12.5	322°21.7	S04°02.7	Sirius	258°25.4	-16°44.9
19	334°12.2	122°24.5	27.3	92°31.9	22.3	217°05.2	12.5	337°24.2	02.7	Adhara	255°05.1	-29°00.2
20	349°14.6	137°24.0	28.4	107°32.5	22.7	232°07.6	12.5	352°26.8	02.7	Procyon	244°50.0	5°09.6
21	4°17.1	152°23.5	· · 29.5	122°33.1	· · 23.1	247°10.0	· · 12.5	7°29.3	· · 02.7	Pollux	243°16.3	27°57.8
22	19°19.6	167°22.9	30.6	137°33.6	23.4	262°12.5	12.5	22°31.8	02.8	Avior	234°14.3	-59°35.2
23	34°22.0	182°22.4	31.7	152°34.2	23.8	277°14.9	12.5	37°34.4	02.8	Suhail	222°45.8	-43°31.9
Mer.pass. 20:43		ν -0.5' d 1.1' m -3.91		ν 0.6' d 0.4' m 1.42		ν 2.4' d 0.0' m -2.38		ν 2.5' d 0.0' m 0.81				

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	49°24.5	197°21.9	S11°32.8	167°34.8	S21°24.1	292°17.4	N21°12.5	52°36.9	S04°02.8
1	64°26.9	212°21.4	33.9	182°35.3	24.5	307°19.8	12.5	67°39.5	02.8
2	79°29.4	227°20.8	35.0	197°35.9	24.9	322°22.3	12.5	82°42.0	02.9
3	94°31.9	242°20.3	· · 36.1	212°36.4	· · 25.2	337°24.7	· · 12.5	97°44.5	· · 02.9
4	109°34.3	257°19.8	37.2	227°37.0	25.6	352°27.2	12.5	112°47.1	02.9
5	124°36.8	272°19.2	38.4	242°37.6	26.0	7°29.6	12.5	127°49.6	02.9
6	139°39.3	287°18.7	S11°39.5	257°38.1	S21°26.3	22°32.1	N21°12.5	142°52.2	S04°03.0
7	154°41.7	302°18.2	40.6	272°38.7	26.7	37°34.5	12.6	157°54.7	03.0
8	169°44.2	317°17.7	41.7	287°39.3	27.0	52°37.0	12.6	172°57.2	03.0
9	184°46.7	332°17.1	· · 42.8	302°39.8	· · 27.4	67°39.4	· · 12.6	187°59.8	· · 03.0
10	199°49.1	347°16.6	43.9	317°40.4	27.7	82°41.9	12.6	203°02.3	03.1
11	214°51.6	2°16.1	45.0	332°41.0	28.1	97°44.3	12.6	218°04.8	03.1
12	229°54.1	17°15.5	S11°46.1	347°41.5	S21°28.5	112°46.8	N21°12.6	233°07.4	S04°03.1
13	244°56.5	32°15.0	47.2	2°42.1	28.8	127°49.2	12.6	248°09.9	03.1
14	259°59.0	47°14.5	48.3	17°42.6	29.2	142°51.7	12.6	263°12.5	03.2
15	275°01.4	62°13.9	· · 49.4	32°43.2	· · 29.5	157°54.2	· · 12.6	278°15.0	· · 03.2
16	290°03.9	77°13.4	50.5	47°43.8	29.9	172°56.6	12.6	293°17.5	03.2
17	305°06.4	92°12.9	51.6	62°44.3	30.2	187°59.1	12.6	308°20.1	03.2
18	320°08.8	107°12.3	S11°52.7	77°44.9	S21°30.6	203°01.5	N21°12.6	323°22.6	S04°03.2
19	335°11.3	122°11.8	53.8	92°45.4	31.0	218°04.0	12.6	338°25.1	03.3
20	350°13.8	137°11.3	54.9	107°46.0	31.3	233°06.4	12.6	353°27.7	03.3
21	5°16.2	152°10.7	· · 55.9	122°46.6	· · 31.7	248°08.9	· · 12.6	8°30.2	· · 03.3
22	20°18.7	167°10.2	57.0	137°47.1	32.0	263°11.3	12.6	23°32.8	03.3
23	35°21.2	182°09.7	58.1	152°47.7	32.4	278°13.8	12.6	38°35.3	03.4
Mer.pass. 20:39		ν -0.5' d 1.1' m -3.91		ν 0.6' d 0.4' m 1.42		ν 2.4' d 0.0' m -2.39		ν 2.5' d 0.0' m 0.81	

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	50°23.6	197°09.1	S11°59.2	167°48.2	S21°32.7	293°16.2	N21°12.6	53°37.8	S04°03.4
1	65°26.1	212°08.6	12°00.3	182°48.8	33.1	308°18.7	12.6	68°40.4	03.4
2	80°28.5	227°08.0	01.4	197°49.4	33.4	323°21.2	12.6	83°42.9	03.4
3	95°31.0	242°07.5	· · 02.5	212°49.9	· · 33.8	338°23.6	· · 12.6	98°45.4	· · 03.4
4	110°33.5	257°07.0	03.6	227°50.5	34.1	353°26.1	12.6	113°48.0	03.5
5	125°35.9	272°06.4	04.7	242°51.0	34.5	8°28.5	12.6	128°50.5	03.5
6	140°38.4	287°05.9	S12°05.8	257°51.6	S21°34.8	23°31.0	N21°12.7	143°53.0	S04°03.5
7	155°40.9	302°05.3	06.9	272°52.2	35.2	38°33.5	12.7	158°55.6	03.5
8	170°43.3	317°04.8	08.0	287°52.7	35.6	53°35.9	12.7	173°58.1	03.6
9	185°45.8	332°04.3	· · 09.1	302°53.3	· · 35.9	68°38.4	· · 12.7	189°00.6	· · 03.6
10	200°48.3	347°03.7	10.2	317°53.8	36.3	83°40.8	12.7	204°03.2	03.6
11	215°50.7	2°03.2	11.2	332°54.4	36.6	98°43.3	12.7	219°05.7	03.6
12	230°53.2	17°02.6	S12°12.3	347°54.9	S21°37.0	113°45.8	N21°12.7	234°08.2	S04°03.6
13	245°55.7	32°02.1	13.4	2°55.5	37.3	128°48.2	12.7	249°10.8	03.7
14	260°58.1	47°01.6	14.5	17°56.0	37.7	143°50.7	12.7	264°13.3	03.7
15	276°00.6	62°01.0	· · 15.6	32°56.6	· · 38.0	158°53.1	· · 12.7	279°15.8	· · 03.7
16	291°03.0	77°00.5	16.7	47°57.2	38.3	173°55.6	12.7	294°18.4	03.7
17	306°05.5	91°59.9	17.8	62°57.7	38.7	188°58.1	12.7	309°20.9	03.8
18	321°08.0	106°59.4	S12°18.9	77°58.3	S21°39.0	204°00.5	N21°12.7	324°23.4	S04°03.8
19	336°10.4	121°58.8	19.9	92°58.8	39.4	219°03.0	12.7	339°26.0	03.8
20	351°12.9	136°58.3	21.0	107°59.4	39.7	234°05.5	12.7	354°28.5	03.8
21	6°15.4	151°57.7	· · 22.1	122°59.9	· · 40.1	249°07.9	· · 12.7	9°31.0	· · 03.8
22	21°17.8	166°57.2	23.2	138°00.5	40.4	264°10.4	12.7	24°33.6	03.9
23	36°20.3	181°56.6	24.3	153°01.0	40.8	279°12.9	12.8	39°36.1	03.9
Mer.pass. 20:35		ν -0.5' d 1.1' m -3.91		ν 0.6' d 0.4' m 1.41		ν 2.5' d 0.0' m -2.39		ν 2.5' d 0.0' m 0.82	

Nov 09 Sun	SHA	Mer.pass
Venus	149°09.1	10:50
Mars	118°55.8	12:50
Jupiter	242°53.4	04:34
Saturn	3°10.6	20:30

Nov 10 Mon	SHA	Mer.pass
Venus	147°57.4	10:51
Mars	118°10.3	12:49
Jupiter	242°52.9	04:30
Saturn	3°12.4	20:26

Nov 11 Tue	SHA	Mer.pass
Venus	146°45.5	10:52
Mars	117°24.6	12:48
Jupiter	242°52.6	04:26
Saturn	3°14.2	20:22

Horizontal parallax	
Venus:	0.1
Mars:	0.1

Sun				Moon											
Sun	GHA	Dec		GHA	ν	Dec	d	HP	Lat.	Twilight		Sunrise	Sunset	Twilight	
0	184°03.1	S16°51.9		312°16.4	2.4'	N28°07.0	-2.7'	60.0'	N 72°	Naut.	Civil			Civil	Naut.
1	199°03.0	52.7		326°37.8	2.5'	28°04.3	-2.9'	59.9'		06:42	08:12	10:05	13:22	15:15	16:45
2	214°03.0	53.4		340°59.3	2.6'	28°01.4	-3.1'	59.9'	N 70°	06:34	07:53	09:20	14:07	15:34	16:53
3	229°02.9	· · 54.1		355°20.8	2.7'	27°58.3	-3.3'	59.9'	68°	06:27	07:38	08:51	14:36	15:49	17:00
4	244°02.9	54.8		9°42.5	2.7'	27°55.0	-3.5'	59.8'	66°	06:21	07:25	08:29	14:58	16:02	17:06
5	259°02.8	55.5		24°04.2	2.8'	27°51.5	-3.7'	59.8'	64°	06:16	07:15	08:12	15:15	16:12	17:11
6	274°02.8	S16°56.2		38°26.1	2.9'	N27°47.8	-3.9'	59.7'	62°	06:11	07:06	07:58	15:29	16:21	17:16
7	289°02.7	56.9		52°48.0	3.0'	27°43.9	-4.1'	59.7'	60°	06:07	06:58	07:46	15:41	16:29	17:20
8	304°02.6	57.6		67°10.1	3.1'	27°39.8	-4.3'	59.7'	N 58°	06:03	06:51	07:35	15:52	16:36	17:24
9	319°02.6	· · 58.3		81°32.2	3.2'	27°35.5	-4.5'	59.6'	56°	06:00	06:45	07:26	16:01	16:42	17:27
10	334°02.5	59.1		95°54.4	3.4'	27°31.1	-4.6'	59.6'	54°	05:57	06:40	07:19	16:09	16:48	17:30
11	349°02.5	16°59.8		110°16.8	3.5'	27°26.4	-4.8'	59.6'	52°	05:54	06:35	07:11	16:16	16:53	17:33
12	4°02.4	S17°00.5		124°39.3	3.6'	N27°21.6	-5.0'	59.5'	50°	05:51	06:30	07:05	16:22	16:57	17:36
13	19°02.4	01.2		139°01.8	3.7'	27°16.6	-5.2'	59.5'	45°	05:44	06:20	06:51	16:36	17:08	17:43
14	34°02.3	01.9		153°24.5	3.8'	27°11.4	-5.4'	59.5'	N 40°	05:38	06:11	06:40	16:48	17:17	17:49
15	49°02.3	· · 02.6		167°47.3	3.9'	27°06.0	-5.6'	59.4'	35°	05:33	06:03	06:30	16:58	17:24	17:55
16	64°02.2	03.3		182°10.3	4.0'	27°00.5	-5.7'	59.4'	30°	05:27	05:56	06:21	17:06	17:32	18:00
17	79°02.2	04.0		196°33.3	4.2'	26°54.7	-5.9'	59.3'	20°	05:17	05:43	06:06	17:22	17:45	18:11
18	94°02.1	S17°04.7		210°56.5	4.3'	N26°48.8	-6.1'	59.3'	N 10°	05:06	05:31	05:53	17:35	17:57	18:22
19	109°02.0	05.4		225°19.7	4.4'	26°42.8	-6.2'	59.3'	0°	04:54	05:19	05:40	17:47	18:09	18:34
20	124°02.0	06.1		239°43.1	4.5'	26°36.5	-6.4'	59.2'	S 10°	04:40	05:06	05:28	18:00	18:22	18:48
21	139°01.9	· · 06.8		254°06.7	4.7'	26°30.1	-6.6'	59.2'	20°	04:23	04:51	05:14	18:14	18:37	19:05
22	154°01.9	07.5		268°30.3	4.8'	26°23.5	-6.7'	59.2'	30°	04:02	04:33	04:59	18:30	18:55	19:26
23	169°01.8	08.2		282°54.1	4.9'	26°16.8	-6.9'	59.1'	35°	03:49	04:22	04:49	18:39	19:07	19:40
SD = 16.1' d = 0.7'				SD = 16.4'											
Mon	GHA	Dec		GHA	ν	Dec	d	HP	Lat.	Moonrise		Moonset			
0	184°01.7	S17°08.9		297°18.0	5.0'	N26°09.9	-7.1'	59.1'	N 72°						
1	199°01.7	09.6		311°42.0	5.2'	26°02.8	-7.2'	59.0'	N 70°			19:03			16:44
2	214°01.6	10.3		326°06.2	5.3'	25°55.6	-7.4'	59.0'	68°			19:59			15:47
3	229°01.6	· · 11.0		340°30.5	5.4'	25°48.2	-7.5'	59.0'	66°			20:32			15:13
4	244°01.5	11.7		354°54.9	5.6'	25°40.6	-7.7'	58.9'	64°		17:37	20:55		15:09	14:48
5	259°01.4	12.4		9°19.5	5.7'	25°32.9	-7.8'	58.9'	62°	17:08	19:19	21:14	14:41	14:33	14:28
6	274°01.4	S17°13.1		23°44.2	5.8'	N25°25.1	-8.0'	58.9'	60°	17:54	19:44	21:30	13:55	14:07	14:11
7	289°01.3	13.8		38°09.0	6.0'	25°17.1	-8.1'	58.8'	N 58°	18:24	20:04	21:43	13:25	13:47	13:58
8	304°01.3	14.5		52°34.0	6.1'	25°08.9	-8.3'	58.8'	56°	18:47	20:21	21:54	13:01	13:30	13:46
9	319°01.2	· · 15.2		66°59.1	6.2'	25°00.7	-8.4'	58.7'	54°	19:05	20:35	22:03	12:43	13:15	13:35
10	334°01.1	15.9		81°24.3	6.4'	24°52.2	-8.6'	58.7'	52°	19:21	20:47	22:12	12:27	13:02	13:26
11	349°01.1	16.6		95°49.7	6.5'	24°43.6	-8.7'	58.7'	50°	19:34	20:58	22:20	12:13	12:51	13:17
12	4°01.0	S17°17.3		110°15.2	6.6'	N24°34.9	-8.9'	58.6'	45°	20:02	21:20	22:36	11:44	12:28	12:59
13	19°00.9	18.0		124°40.9	6.8'	24°26.1	-9.0'	58.6'	N 40°	20:24	21:38	22:50	11:22	12:09	12:45
14	34°00.9	18.7		139°06.6	6.9'	24°17.1	-9.1'	58.5'	35°	20:42	21:53	23:01	11:03	11:53	12:32
15	49°00.8	· · 19.4		153°32.6	7.1'	24°08.0	-9.3'	58.5'	30°	20:57	22:06	23:11	10:47	11:39	12:21
16	64°00.7	20.1		167°58.6	7.2'	23°58.7	-9.4'	58.5'	20°	21:23	22:28	23:28	10:20	11:15	12:02
17	79°00.7	20.8		182°24.8	7.3'	23°49.3	-9.5'	58.4'	N 10°	21:46	22:47	23:42	09:57	10:54	11:45
18	94°00.6	S17°21.5		196°51.2	7.5'	N23°39.8	-9.6'	58.4'	0°	22:07	23:04	23:56	09:35	10:35	11:30
19	109°00.5	22.2		211°17.6	7.6'	23°30.2	-9.8'	58.3'	S 10°	22:28	23:22	· · ·	09:13	10:16	11:14
20	124°00.5	22.9		225°44.2	7.7'	23°20.4	-9.9'	58.3'	20°	22:50	23:41	· · ·	08:49	09:55	10:57
21	139°00.4	· · 23.5		240°11.0	7.9'	23°10.5	-10.0'	58.3'	30°	23:16	· · ·	00:02	08:21	09:30	10:37
22	154°00.3	24.2		254°37.9	8.0'	23°00.5	-10.1'	58.2'	35°	23:31	· · ·	00:15	08:04	09:16	10:26
23	169°00.3	24.9		269°04.9	8.2'	22°50.4	-10.2'	58.2'	40°	23:49	· · ·	00:29	07:45	08:59	10:12
SD = 16.1' d = 0.7'				SD = 16.1'											
Tue	GHA	Dec		GHA	ν	Dec	d	HP	Day	Sun		Mer.	Moon		Age
0	184°00.2	S17°25.6		283°32.0	8.3'	N22°40.1	-10.4'	58.2'		Eqn.of Time	Mer.	Pass	Upper	Lower	19-21
1	199°00.1	26.3		297°59.3	8.4'	22°29.8	-10.5'	58.1'		00 ^h	Pass		hh:mm	hh:mm	83-63%
2	214°00.1	27.0		312°26.8	8.6'	22°19.3	-10.6'	58.1'		mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	
3	229°00.0	· · 27.7		326°54.3	8.7'	22°08.7	-10.7'	58.0'	09	16:12	16:10	11:44	03:19	15:51	
4	243°59.9	28.4		341°22.0	8.8'	21°58.0	-10.8'	58.0'	10	16:07	16:04	11:44	04:21	16:50	
5	258°59.9	29.1		355°49.9	9.0'	21°47.3	-10.9'	58.0'	11	16:01	15:57	11:44	05:17	17:43	
6	273°59.8	S17°29.7		10°17.8	9.1'	N21°36.4	-11.0'	57.9'							
7	288°59.7	30.4		24°45.9	9.2'	21°25.3	-11.1'	57.9'							
8	303°59.7	31.1		39°14.2	9.4'	21°14.2	-11.2'	57.8'							
9	318°59.6	· · 31.8		53°42.5	9.5'	21°03.0	-11.3'	57.8'							
10	333°59.5	32.5		68°11.0	9.6'	20°51.7	-11.4'	57.8'							
11	348°59.4	33.2		82°39.7	9.8'	20°40.3	-11.5'	57.7'							
12	3°59.4	S17°33.9		97°08.4	9.9'	N20°28.8	-11.6'	57.7'							
13	18°59.3	34.5		111°37.3	10.0'	20°17.3	-11.7'	57.7'							
14	33°59.2	35.2		126°06.3	10.1'	20°05.6	-11.8'	57.6'							
15	48°59.1	· · 35.9		140°35.5	10.3'	19°53.8	-11.9'	57.6'							
16	63°59.1	36.6		155°04.8	10.4'	19°42.0	-11.9'	57.5'							
17	78°59.0	37.3		169°34.2	10.5'	19°30.0	-12.0'	57.5'							
18	93°58.9	S17°37.9		184°03.7	10.6'	N19°18.0	-12.1'	57.5'							
19	108°58.8	38.6		198°33.3	10.8'	19°05.9	-12.2'	57.4'							
20	123°58.8	39.3		213°03.1	10.9'	18°1									

November 12, 13, 14 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	51°22.8	196°56.1	S12°25.4	168°01.6	S21°41.1	294°15.3	N21°12.8	54°38.6	S04°03.9	Alpheratz	357°33.8	29°14.2
1	66°25.2	211°55.5	26.4	183°02.1	41.5	309°17.8	12.8	69°41.2	03.9	Ankaa	353°06.2	-42°10.0
2	81°27.7	226°55.0	27.5	198°02.7	41.8	324°20.3	12.8	84°43.7	03.9	Schedar	349°29.8	56°41.0
3	96°30.2	241°54.4	•• 28.6	213°03.2	•• 42.2	339°22.7	•• 12.8	99°46.2	•• 04.0	Diphda	348°46.3	-17°50.6
4	111°32.6	256°53.9	29.7	228°03.8	42.5	354°25.2	12.8	114°48.8	04.0	Achernar	335°19.1	-57°06.4
5	126°35.1	271°53.4	30.8	243°04.4	42.8	9°27.7	12.8	129°51.3	04.0	Hamal	327°50.1	23°35.2
6	141°37.5	286°52.8	S12°31.8	258°04.9	S21°43.2	24°30.1	N21°12.8	144°53.8	S04°04.0	Polaris	313°11.1	89°22.4
7	156°40.0	301°52.3	32.9	273°05.5	43.5	39°32.6	12.8	159°56.4	04.0	Acamar	315°10.8	-40°12.0
8	171°42.5	316°51.7	34.0	288°06.0	43.9	54°35.1	12.8	174°58.9	04.1	Menkar	314°05.1	4°11.6
9	186°44.9	331°51.1	•• 35.1	303°06.6	•• 44.2	69°37.5	•• 12.8	190°01.4	•• 04.1	Mirfak	308°26.8	49°57.3
10	201°47.4	346°50.6	36.2	318°07.1	44.6	84°40.0	12.8	205°04.0	04.1	Aldebaran	290°38.5	16°33.7
11	216°49.9	1°50.0	37.2	333°07.7	44.9	99°42.5	12.8	220°06.5	04.1	Rigel	281°02.9	-8°10.2
12	231°52.3	16°49.5	S12°38.3	348°08.2	S21°45.2	114°44.9	N21°12.8	235°09.0	S04°04.1	Capella	280°20.5	46°01.4
13	246°54.8	31°48.9	39.4	3°08.8	45.6	129°47.4	12.8	250°11.5	04.2	Bellatrix	278°21.9	6°22.5
14	261°57.3	46°48.4	40.5	18°09.3	45.9	144°49.9	12.9	265°14.1	04.2	Elnath	278°00.7	28°37.8
15	276°59.7	61°47.8	•• 41.5	33°09.9	•• 46.3	159°52.3	•• 12.9	280°16.6	•• 04.2	Alnilam	275°36.8	-1°11.0
16	292°02.2	76°47.3	42.6	48°10.4	46.6	174°54.8	12.9	295°19.1	04.2	Betelgeuse	270°51.1	7°24.8
17	307°04.6	91°46.7	43.7	63°10.9	46.9	189°57.3	12.9	310°21.7	04.2	Canopus	263°51.7	-52°42.3
18	322°07.1	106°46.2	S12°44.8	78°11.5	S21°47.3	204°59.8	N21°12.9	325°24.2	S04°04.3	Sirius	258°25.4	-16°44.9
19	337°09.6	121°45.6	45.8	93°12.0	47.6	220°02.2	12.9	340°26.7	04.3	Adhara	255°05.1	-29°00.2
20	352°12.0	136°45.1	46.9	108°12.6	48.0	235°04.7	12.9	355°29.3	04.3	Procyon	244°50.0	5°09.6
21	7°14.5	151°44.5	•• 48.0	123°13.1	•• 48.3	250°07.2	•• 12.9	10°31.8	•• 04.3	Pollux	243°16.3	27°57.7
22	22°17.0	166°43.9	49.0	138°13.7	48.6	265°09.7	12.9	25°34.3	04.3	Avior	234°14.3	-59°35.2
23	37°19.4	181°43.4	50.1	153°14.2	49.0	280°12.1	12.9	40°36.8	04.4	Suhail	222°45.8	-43°31.9
Mer.pass. 20:31		ν -0.5' d 1.1' m -3.91		ν 0.6' d 0.3' m 1.41		ν 2.5' d 0.0' m -2.40		ν 2.5' d 0.0' m 0.82				

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	52°21.9	196°42.8	S12°51.2	168°14.8	S21°49.3	295°14.6	N21°12.9	55°39.4	S04°04.4	Denebola	182°24.4	14°25.6
1	67°24.4	211°42.3	52.3	183°15.3	49.7	310°17.1	12.9	70°41.9	04.4	Gienah	175°43.1	-17°41.0
2	82°26.8	226°41.7	53.3	198°15.9	50.0	325°19.6	12.9	85°44.4	04.4	Acrux	172°59.9	-63°14.3
3	97°29.3	241°41.1	•• 54.4	213°16.4	•• 50.3	340°22.0	•• 13.0	100°47.0	•• 04.4	Gacrux	171°51.3	-57°15.3
4	112°31.8	256°40.6	55.5	228°17.0	50.7	355°24.5	13.0	115°49.5	04.5	Alioth	166°12.8	55°49.0
5	127°34.2	271°40.0	56.5	243°17.5	51.0	10°27.0	13.0	130°52.0	04.5	Spica	158°21.9	-11°17.7
6	142°36.7	286°39.5	S12°57.6	258°18.1	S21°51.3	25°29.5	N21°13.0	145°54.5	S04°04.5	Alkaid	152°51.9	49°10.9
7	157°39.1	301°38.9	58.7	273°18.6	51.7	40°31.9	13.0	160°57.1	04.5	Hadar	148°35.7	-60°29.7
8	172°41.6	316°38.3	12°59.8	288°19.1	52.0	55°34.4	13.0	175°59.6	04.5	Menkent	147°57.2	-36°29.7
9	187°44.1	331°37.8	13°00.8	303°19.7	•• 52.3	70°36.9	•• 13.0	191°02.1	•• 04.6	Arcturus	145°47.6	19°02.9
10	202°46.5	346°37.2	01.9	318°20.2	52.7	85°39.4	13.0	206°04.6	04.6	Rigel Kent.	139°40.0	-60°56.4
11	217°49.0	1°36.6	02.9	333°20.8	53.0	100°41.8	13.0	221°07.2	04.6	Kochab	137°21.0	74°02.8
12	232°51.5	16°36.1	S13°04.0	348°21.3	S21°53.3	115°44.3	N21°13.0	236°09.7	S04°04.6	Zuben'ubi	136°55.6	-16°08.9
13	247°53.9	31°35.5	05.0	3°21.9	53.7	130°46.8	13.0	251°12.2	04.6	Alphecca	126°03.5	26°37.7
14	262°56.4	46°35.0	06.1	18°22.4	54.0	145°49.3	13.1	266°14.8	04.6	Antares	112°15.4	-26°29.3
15	277°58.9	61°34.4	•• 07.2	33°22.9	•• 54.3	160°51.8	•• 13.1	281°17.3	•• 04.7	Atria	107°09.5	-69°04.5
16	293°01.3	76°33.8	08.2	48°23.5	54.7	175°54.2	13.1	296°19.8	04.7	Sabik	102°02.3	-15°45.4
17	308°03.8	91°33.3	09.3	63°24.0	55.0	190°56.7	13.1	311°22.3	04.7	Shaula	96°09.8	-37°07.4
18	323°06.3	106°32.7	S13°10.4	78°24.6	S21°55.3	205°59.2	N21°13.1	326°24.9	S04°04.7	Rasalhague	95°58.2	12°32.6
19	338°08.7	121°32.1	11.4	93°25.1	55.7	221°01.7	13.1	341°27.4	04.7	Eltanin	90°42.3	51°29.3
20	353°11.2	136°31.6	12.5	108°25.7	56.0	236°04.2	13.1	356°29.9	04.8	Kaus Aust.	83°31.9	-34°22.4
21	8°13.6	151°31.0	•• 13.5	123°26.2	•• 56.3	251°06.6	•• 13.1	11°32.4	•• 04.8	Vega	80°33.1	38°48.6
22	23°16.1	166°30.4	14.6	138°26.7	56.6	266°09.1	13.1	26°35.0	04.8	Nunki	75°47.1	-26°15.9
23	38°18.6	181°29.9	15.6	153°27.3	57.0	281°11.6	13.1	41°37.5	04.8	Altair	61°59.4	8°56.3
Mer.pass. 20:27		ν -0.6' d 1.1' m -3.91		ν 0.5' d 0.3' m 1.40		ν 2.5' d 0.0' m -2.41		ν 2.5' d 0.0' m 0.83				

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec			
0	53°21.0	196°29.3	S13°16.7	168°27.8	S21°57.3	296°14.1	N21°13.1	56°40.0	S04°04.8	0		
1	68°23.5	211°28.7	17.8	183°28.4	57.6	311°16.6	13.2	71°42.5	04.8	1		
2	83°26.0	226°28.1	18.8	198°28.9	58.0	326°19.1	13.2	86°45.1	04.9	2		
3	98°28.4	241°27.6	•• 19.9	213°29.4	•• 58.3	341°21.5	•• 13.2	101°47.6	•• 04.9	3		
4	113°30.9	256°27.0	20.9	228°30.0	58.6	356°24.0	13.2	116°50.1	04.9	4		
5	128°33.4	271°26.4	22.0	243°30.5	58.9	11°26.5	13.2	131°52.6	04.9	5		
6	143°35.8	286°25.9	S13°23.0	258°31.1	S21°59.3	26°29.0	N21°13.2	146°55.2	S04°04.9	6		
7	158°38.3	301°25.3	24.1	273°31.6	59.6	41°31.5	13.2	161°57.7	04.9	7		
8	173°40.7	316°24.7	25.1	288°32.1	21°59.9	56°34.0	13.2	177°00.2	05.0	8		
9	188°43.2	331°24.1	•• 26.2	303°32.7	22°00.3	71°36.5	•• 13.2	192°02.7	•• 05.0	9		
10	203°45.7	346°23.6	27.2	318°33.2	00.6	86°38.9	13.2	207°05.2	05.0	10		
11	218°48.1	1°23.0	28.3	333°33.7	00.9	101°41.4	13.3	222°07.8	05.0	11		
12	233°50.6	16°22.4	S13°29.3	348°34.3	S22°01.2	116°43.9	N21°13.3	237°10.3	S04°05.0	12		
13	248°53.1	31°21.8	30.4	3°34.8	01.6	131°46.4	13.3	252°12.8	05.0	13		
14	263°55.5	46°21.3	31.4	18°35.4	01.9	146°48.9	13.3	267°15.3	05.1	14		
15	278°58.0	61°20.7	•• 32.5	33°35.9	•• 02.2	161°51.4	•• 13.3	282°17.9	•• 05.1	15		
16	294°00.5	76°20.1	33.5	48°36.4	02.5	176°53.9	13.3	297°20.4	05.1	16		
17	309°02.9	91°19.5	34.6	63°37.0	02.8	191°56.4	13.3	312°22.9	05.1	17		
18	324°05.4	106°19.0	S13°35.6	78°37.5	S22°03.2	206°58.9	N21°13.3	327°25.4	S04°05.1	18		
19	339°07.9	121°18.4	36.7	93°38.0	03.5	222°01.3	13.3	342°27.9	05.1	19		
20	354°10.3	136°17.8	37.7	108°38.6	03.8	237°03.8	13.3	357°30.5	05.2	20		
21	9°12.8	151°17.2	•• 38.8	123°39.1	•• 04.1	252°06.3	•• 13.4	12°33.0	•• 05.2	21		
22	24°15.2	166°16.6	39.8	138°39.6	04.5	267°08.8	13.4	27°35.5	05.2	22		
23	39°17.7	181°16.1	40.9	153°40.2	04.8	282°11.3	13.4	42°38.0	05.2	23		
Mer.pass. 20:23		ν -0.6' d 1.1' m -3.91		ν 0.5' d 0.3' m 1.40		ν 2.5' d 0.0' m -2.42		ν 2.5' d 0.0' m 0.83				

Nov 12 Wed		SHA	Mer.pass
Venus		145°33.3	10:53
Mars		116°38.8	12:47
Jupiter		242°52.6	04:22
Saturn		3°15.9	20:18

Nov 13 Thu		SHA	Mer.pass
Venus		144°20.9	10:54
Mars		115°52.9	12:47
Jupiter		242°52.7	04:18
Saturn		3°17.5	20:14

Nov 14 Fri		SHA	Mer.pass
Venus		143°08.2	10:54
Mars		115°06.8	12:46
Jupiter		242°53.1	04:14
Saturn		3°19.0	20:10

Horizontal parallax		
Venus:		0.1
Mars:		0.1

Sun							Moon						
Wed	GHA	Dec	GHA	ν	Dec	d	HP						
0	183°58.5	S17°42.0	271°03.4	11.4'	N18°04.2	-12.6'	57.3'						
1	198°58.4	42.7	285°33.7	11.5'	17°51.6	-12.6'	57.2'						
2	213°58.3	43.4	300°04.2	11.6'	17°39.0	-12.7'	57.2'						
3	228°58.2	44.0	314°34.8	11.7'	17°26.3	-12.8'	57.1'						
4	243°58.2	44.7	329°05.5	11.8'	17°13.5	-12.8'	57.1'						
5	258°58.1	45.4	343°36.4	11.9'	17°00.7	-12.9'	57.1'						
6	273°58.0	S17°46.1	358°07.3	12.0'	N16°47.8	-13.0'	57.0'						
7	288°57.9	46.7	12°38.3	12.2'	16°34.8	-13.0'	57.0'						
8	303°57.8	47.4	27°09.5	12.3'	16°21.8	-13.1'	57.0'						
9	318°57.8	48.1	41°40.8	12.4'	16°08.7	-13.2'	56.9'						
10	333°57.7	48.8	56°12.1	12.5'	15°55.5	-13.2'	56.9'						
11	348°57.6	49.4	70°43.6	12.6'	15°42.3	-13.3'	56.9'						
12	3°57.5	S17°50.1	85°15.2	12.7'	N15°29.0	-13.3'	56.8'						
13	18°57.4	50.8	99°46.9	12.8'	15°15.7	-13.4'	56.8'						
14	33°57.3	51.4	114°18.7	12.9'	15°02.3	-13.4'	56.8'						
15	48°57.3	52.1	128°50.6	13.0'	14°48.9	-13.5'	56.7'						
16	63°57.2	52.8	143°22.5	13.1'	14°35.4	-13.5'	56.7'						
17	78°57.1	53.4	157°54.6	13.2'	14°21.9	-13.6'	56.7'						
18	93°57.0	S17°54.1	172°26.8	13.3'	N14°08.3	-13.6'	56.6'						
19	108°56.9	54.8	186°59.1	13.4'	13°54.7	-13.7'	56.6'						
20	123°56.8	55.4	201°31.5	13.5'	13°41.0	-13.7'	56.6'						
21	138°56.8	56.1	216°03.9	13.6'	13°27.3	-13.8'	56.5'						
22	153°56.7	56.8	230°36.5	13.6'	13°13.6	-13.8'	56.5'						
23	168°56.6	57.4	245°09.1	13.7'	12°59.8	-13.8'	56.5'						
SD = 16.1'			d = 0.7'				SD = 15.6'						

Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	183°56.5	S17°58.1	259°41.9	13.8'	N12°45.9	-13.9'	56.4'
1	198°56.4	58.8	274°14.7	13.9'	12°32.1	-13.9'	56.4'
2	213°56.3	17°59.4	288°47.6	14.0'	12°18.1	-14.0'	56.4'
3	228°56.2	18°00.1	303°20.6	14.1'	12°04.2	-14.0'	56.3'
4	243°56.1	00.8	317°53.7	14.2'	11°50.2	-14.0'	56.3'
5	258°56.1	01.4	332°26.8	14.2'	11°36.2	-14.1'	56.3'
6	273°56.0	S18°02.1	347°00.1	14.3'	N11°22.1	-14.1'	56.2'
7	288°55.9	02.7	1°33.4	14.4'	11°08.0	-14.1'	56.2'
8	303°55.8	03.4	16°06.8	14.5'	10°53.9	-14.2'	56.2'
9	318°55.7	04.1	30°40.2	14.5'	10°39.7	-14.2'	56.1'
10	333°55.6	04.7	45°13.8	14.6'	10°25.5	-14.2'	56.1'
11	348°55.5	05.4	59°47.4	14.7'	10°11.3	-14.2'	56.1'
12	3°55.4	S18°06.0	74°21.1	14.8'	N09°57.1	-14.3'	56.1'
13	18°55.3	06.7	88°54.9	14.8'	09°42.8	-14.3'	56.0'
14	33°55.2	07.4	103°28.7	14.9'	09°28.5	-14.3'	56.0'
15	48°55.2	08.0	118°02.6	15.0'	09°14.2	-14.3'	56.0'
16	63°55.1	08.7	132°36.6	15.0'	08°59.9	-14.4'	55.9'
17	78°55.0	09.3	147°10.6	15.1'	08°45.5	-14.4'	55.9'
18	93°54.9	S18°10.0	161°44.7	15.2'	N08°31.1	-14.4'	55.9'
19	108°54.8	10.6	176°18.9	15.2'	08°16.7	-14.4'	55.8'
20	123°54.7	11.3	190°53.1	15.3'	08°02.3	-14.4'	55.8'
21	138°54.6	11.9	205°27.4	15.3'	07°47.9	-14.5'	55.8'
22	153°54.5	12.6	220°01.8	15.4'	07°33.4	-14.5'	55.8'
23	168°54.4	13.2	234°36.2	15.5'	07°18.9	-14.5'	55.7'
SD = 16.2'			d = 0.7'				SD = 15.4'

Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	183°54.3	S18°13.9	249°10.6	15.5'	N07°04.5	-14.5'	55.7'
1	198°54.2	14.5	263°45.1	15.6'	06°49.9	-14.5'	55.7'
2	213°54.1	15.2	278°19.7	15.6'	06°35.4	-14.5'	55.7'
3	228°54.0	15.8	292°54.3	15.7'	06°20.9	-14.5'	55.6'
4	243°53.9	16.5	307°29.0	15.7'	06°06.4	-14.5'	55.6'
5	258°53.8	17.1	322°03.7	15.8'	05°51.8	-14.6'	55.6'
6	273°53.7	S18°17.8	336°38.5	15.8'	N05°37.3	-14.6'	55.5'
7	288°53.6	18.4	351°13.3	15.9'	05°22.7	-14.6'	55.5'
8	303°53.5	19.1	5°48.2	15.9'	05°08.1	-14.6'	55.5'
9	318°53.4	19.7	20°23.1	16.0'	04°53.5	-14.6'	55.5'
10	333°53.3	20.4	34°58.0	16.0'	04°38.9	-14.6'	55.4'
11	348°53.2	21.0	49°33.0	16.0'	04°24.3	-14.6'	55.4'
12	3°53.1	S18°21.7	64°08.1	16.1'	N04°09.7	-14.6'	55.4'
13	18°53.0	22.3	78°43.1	16.1'	03°55.1	-14.6'	55.4'
14	33°52.9	22.9	93°18.2	16.1'	03°40.5	-14.6'	55.3'
15	48°52.8	23.6	107°53.4	16.2'	03°25.9	-14.6'	55.3'
16	63°52.7	24.2	122°28.6	16.2'	03°11.3	-14.6'	55.3'
17	78°52.6	24.9	137°03.8	16.3'	02°56.7	-14.6'	55.3'
18	93°52.5	S18°25.5	151°39.0	16.3'	N02°42.1	-14.6'	55.3'
19	108°52.4	26.2	166°14.3	16.3'	02°27.5	-14.6'	55.2'
20	123°52.3	26.8	180°49.6	16.3'	02°12.9	-14.6'	55.2'
21	138°52.2	27.4	195°25.0	16.4'	01°58.3	-14.6'	55.2'
22	153°52.1	28.1	210°00.4	16.4'	01°43.7	-14.6'	55.2'
23	168°52.0	28.7	224°35.8	16.4'	01°29.1	-14.6'	55.1'
SD = 16.2'			d = 0.7'				SD = 15.2'

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	06:53	08:26	10:36	12:52	15:02	16:34
N 70°	06:44	08:05	09:38	13:49	15:23	16:44
68°	06:36	07:48	09:05	14:23	15:40	16:52
66°	06:29	07:34	08:40	14:47	15:53	16:58
64°	06:23	07:23	08:22	15:06	16:05	17:04
62°	06:18	07:13	08:06	15:22	16:14	17:10
60°	06:13	07:05	07:53	15:34	16:23	17:14
N 58°	06:09	06:57	07:42	15:45	16:30	17:19
56°	06:05	06:51	07:33	15:55	16:37	17:23
54°	06:02	06:45	07:24	16:04	16:43	17:26
52°	05:58	06:39	07:17	16:11	16:49	17:30
50°	05:55	06:34	07:10	16:18	16:54	17:33
45°	05:48	06:23	06:55	16:33	17:05	17:40
N 40°	05:41	06:14	06:43	16:45	17:14	17:47
35°	05:35	06:06	06:33	16:56	17:22	17:53
30°	05:29	05:58	06:24	17:05	17:30	17:59
20°	05:18	05:45	06:08	17:21	17:44	18:10
N 10°	05:06	05:32	05:54	17:35	17:57	18:22
0°	04:54	05:19	05:41	17:48	18:10	18:35
S 10°	04:39	05:05	05:28	18:01	18:23	18:49
20°	04:22	04:50	05:13	18:16	18:39	19:07
30°	04:00	04:31	04:57	18:32	18:58	19:29
35°	03:46	04:19	04:47	18:42	19:10	19:44
40°	03:29	04:06	04:36	18:53	19:23	20:01
45°	03:07	03:49	04:23	19:06	19:40	20:23
S 50°	02:38	03:28	04:07	19:22	20:01	20:53
52°	02:23	03:18	04:00	19:30	20:12	21:08
54°	02:04	03:07	03:51	19:39	20:23	21:27
56°	01:41	02:53	03:42	19:48	20:37	21:50
58°	01:09	02:38	03:31	19:59	20:53	22:24
S 60°	--	02:18	03:19	20:12	21:13	--

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	21:32	23:55		15:57	15:06	14:30
N 70°	21:58		00:05	15:28	14:53	14:26
68°	22:18		00:14	15:06	14:42	14:22
66°	22:34		00:21	14:49	14:33	14:19
64°	22:47		00:27	14:35	14:25	14:16
62°	22:57		00:32	14:23	14:18	14:14
60°	23:07		00:37	14:13	14:12	14:11
N 58°	23:14		00:41	14:04	14:07	14:10
56°	23:21		00:44	13:56	14:03	14:08
54°	23:28		00:47	13:49	13:58	14:06
52°	23:33		00:50	13:42	13:55	14:05
50°	23:38		00:52	13:36	13:51	14:04
45°	23:49		00:58	13:24	13:44	14:01
N 40°	23:58		01:03	13:13	13:37	13:59
35°		00:06	01:07	13:04	13:32	13:57
30°		00:12	01:10	12:56	13:27	13:55
20°		00:24	01:16	12:43	13:19	13:52
N 10°		00:34	01:21	12:31	13:11	13:49
0°		00:43	01:26	12:19	13:04	13:46
S 10°	00:10	00:52	01:31	12:08	12:57	13:43
20°	00:24	01:02	01:36	11:55	12:49	13:40
30°	00:41	01:13	01:42	11:41	12:40	13:37
35°	00:50	01:20	01:46	11:32	12:35	13:35
40°	01:01	01:27	01:49	11:23	12:29	13:33
45°	01:14	01:36	01:54	11:12	12:23	13:30
S 50°	01:29	01:46	01:59	10:58	12:14	13:27
52°	01:37	01:50	02:01	10:51	12:10	13:26
54°	01:44	01:56	02:04	10:44	12:06	13:24
56°	01:53	02:01	02:07	10:36	12:02	13:22
58°	02:03	02:08	02:10	10:27	11:56	13:20
S 60°	02:15	02:15	02:14	10:17	11:50	13:18

Day	Sun		Mer. Pass	Moon		Age
	Eqn.of Time	Mer. Pass		Upper	Lower	
	00 ^h	12 ^h	hh:mm	hh:mm	hh:mm	22-24
	mm:ss	mm:ss				53-32%
12	15:54	15:50	11:44	06:08	18:31	
13	15:46	15:42	11:44	06:54	19:15	
14	15:37	15:33	11:44	07:36	19:57	

November 15, 16, 17 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	54°20.2	196°15.5	S13°41.9	168°40.7	S22°05.1	297°13.8	N21°13.4	57°40.6	S04°05.2	Alpheratz	357°33.8 29°14.2
1	69°22.6	211°14.9	42.9	183°41.2	05.4	312°16.3	13.4	72°43.1	05.2	Ankaa	353°06.2 -42°10.0
2	84°25.1	226°14.3	44.0	198°41.8	05.7	327°18.8	13.4	87°45.6	05.3	Schedar	349°29.9 56°41.0
3	99°27.6	241°13.7	· · 45.0	213°42.3	· · 06.1	342°21.3	· · 13.4	102°48.1	· · 05.3	Diphda	348°46.4 -17°50.7
4	114°30.0	256°13.2	46.1	228°42.8	06.4	357°23.8	13.4	117°50.6	05.3	Achernar	335°19.1 -57°06.4
5	129°32.5	271°12.6	47.1	243°43.4	06.7	12°26.3	13.4	132°53.2	05.3	Hamal	327°50.1 23°35.2
6	144°35.0	286°12.0	S13°48.1	258°43.9	S22°07.0	27°28.8	N21°13.5	147°55.7	S04°05.3	Polaris	313°11.1 89°22.4
7	159°37.4	301°11.4	49.2	273°44.4	07.3	42°31.3	13.5	162°58.2	05.3	Acamar	315°10.8 -40°12.0
8	174°39.9	316°10.8	50.2	288°45.0	07.6	57°33.7	13.5	178°00.7	05.3	Menkar	314°05.1 4°11.6
9	189°42.4	331°10.2	· · 51.3	303°45.5	· · 08.0	72°36.2	· · 13.5	193°03.2	· · 05.4	Mirfak	308°26.8 49°57.3
10	204°44.8	346°09.7	52.3	318°46.0	08.3	87°38.7	13.5	208°05.8	05.4	Aldebaran	290°38.5 16°33.7
11	219°47.3	1°09.1	53.3	333°46.6	08.6	102°41.2	13.5	223°08.3	05.4	Rigel	281°02.9 -8°10.2
12	234°49.7	16°08.5	S13°54.4	348°47.1	S22°08.9	117°43.7	N21°13.5	238°10.8	S04°05.4	Capella	280°20.4 46°01.4
13	249°52.2	31°07.9	55.4	3°47.6	09.2	132°46.2	13.5	253°13.3	05.4	Bellatrix	278°21.8 6°22.5
14	264°54.7	46°07.3	56.4	18°48.2	09.5	147°48.7	13.6	268°15.8	05.4	Elnath	278°00.7 28°37.8
15	279°57.1	61°06.7	· · 57.5	33°48.7	· · 09.9	162°51.2	· · 13.6	283°18.3	· · 05.4	Alnilam	275°36.7 -1°11.0
16	294°59.6	76°06.1	58.5	48°49.2	10.2	177°53.7	13.6	298°20.9	05.5	Betelgeuse	270°51.1 7°24.8
17	310°02.1	91°05.5	13°59.5	63°49.8	10.5	192°56.2	13.6	313°23.4	05.5	Canopus	263°51.7 -52°42.3
18	325°04.5	106°05.0	S14°00.6	78°50.3	S22°10.8	207°58.7	N21°13.6	328°25.9	S04°05.5	Sirius	258°25.4 -16°44.9
19	340°07.0	121°04.4	01.6	93°50.8	11.1	223°01.2	13.6	343°28.4	05.5	Adhara	255°05.1 -29°00.2
20	355°09.5	136°03.8	02.6	108°51.3	11.4	238°03.7	13.6	358°30.9	05.5	Procyon	244°49.9 5°09.6
21	10°11.9	151°03.2	· · 03.7	123°51.9	· · 11.7	253°06.2	· · 13.6	13°33.5	· · 05.5	Pollux	243°16.3 27°57.7
22	25°14.4	166°02.6	04.7	138°52.4	12.1	268°08.7	13.6	28°36.0	05.5	Avior	234°14.2 -59°35.2
23	40°16.8	181°02.0	05.7	153°52.9	12.4	283°11.2	13.7	43°38.5	05.6	Suhail	222°45.8 -43°31.9
Mer.pass. 20:19		ν -0.6' d 1.0' m -3.91		ν 0.5' d 0.3' m 1.39		ν 2.5' d 0.0' m -2.42		ν 2.5' d 0.0' m 0.84		Miaplacidus	221°38.0 -69°49.0
										Alphard	217°47.1 -8°46.1
										Regulus	207°33.7 11°50.4
										Dubhe	193°40.4 61°36.4
										Denebola	182°24.4 14°25.6
										Gienah	175°43.1 -17°41.0
										Acrux	172°59.9 -63°14.3
										Gacrux	171°51.3 -57°15.3
										Alioth	166°12.8 55°49.0
										Spica	158°21.9 -11°17.7
										Alkaid	152°51.9 -49°10.9
										Hadar	148°35.7 -60°29.7
										Menkent	147°57.2 -36°29.7
										Arcturus	145°47.6 19°02.9
										Rigel Kent.	139°40.0 -60°56.4
										Kochab	137°21.0 74°02.8
										Zuben'ubi	136°55.6 -16°08.9
										Alphecca	126°03.5 26°37.7
										Antares	112°15.4 -26°29.3
										Atria	107°09.5 -69°04.5
										Sabik	102°02.3 -15°45.4
										Shaula	96°09.8 -37°07.4
										Rasalhague	95°58.2 12°32.5
										Eltanin	90°42.3 51°29.3
										Kaus Aust.	83°31.9 -34°22.4
										Vega	80°33.1 38°48.6
										Nunki	75°47.2 -26°15.9
										Altair	61°59.5 8°56.3
										Peacock	53°04.8 -56°39.3
										Deneb	49°25.5 45°22.6
										Enif	33°38.1 9°59.8
										Al Na'ir	27°32.0 -46°50.3
										Fomalhaut	15°13.6 -29°29.2
										Scheat	13°44.4 28°13.6
										Markab	13°29.1 15°20.8
										Nov 15 Sat	SHA Mer.pass
										Venus	141°55.3 10:55
										Mars	114°20.5 12:45
										Jupiter	242°53.6 04:10
										Saturn	3°20.4 20:06
										Nov 16 Sun	SHA Mer.pass
										Venus	140°42.1 10:56
										Mars	113°34.1 12:44
										Jupiter	242°54.4 04:06
										Saturn	3°21.7 20:02
										Nov 17 Mon	SHA Mer.pass
										Venus	139°28.6 10:57
										Mars	112°47.6 12:43
										Jupiter	242°55.4 04:02
										Saturn	3°22.9 19:58
										Horizontal parallax	
										Venus:	0.1
										Mars:	0.1

h Sun Moon						
Sat	GHA	Dec	GHA	ν	Dec	d HP
0	183°51.9	S18°29.4	239°11.2	16.4'	N01°14.5	-14.6' 55.1'
1	198°51.8	30.0	253°46.6	16.5'	00°59.9	-14.6' 55.1'
2	213°51.7	30.6	268°22.1	16.5'	00°45.3	-14.6' 55.1'
3	228°51.6	· · 31.3	282°57.6	16.5'	00°30.8	-14.6' 55.0'
4	243°51.5	31.9	297°33.1	16.5'	00°16.2	-14.5' 55.0'
5	258°51.4	32.5	312°08.6	16.6'	N00°01.7	-14.5' 55.0'
6	273°51.3	S18°33.2	326°44.2	16.6'	S00°12.9	14.5' 55.0'
7	288°51.2	33.8	341°19.8	16.6'	00°27.4	14.5' 55.0'
8	303°51.1	34.4	355°55.3	16.6'	00°41.9	14.5' 54.9'
9	318°51.0	· · 35.1	10°30.9	16.6'	00°56.4	14.5' 54.9'
10	333°50.8	35.7	25°06.6	16.6'	01°10.9	14.5' 54.9'
11	348°50.7	36.3	39°42.2	16.6'	01°25.4	14.5' 54.9'
12	3°50.6	S18°37.0	54°17.8	16.6'	S01°39.9	14.4' 54.9'
13	18°50.5	37.6	68°53.5	16.7'	01°54.3	14.4' 54.8'
14	33°50.4	38.2	83°29.1	16.7'	02°08.7	14.4' 54.8'
15	48°50.3	· · 38.9	98°04.8	16.7'	02°23.1	14.4' 54.8'
16	63°50.2	39.5	112°40.5	16.7'	02°37.5	14.4' 54.8'
17	78°50.1	40.1	127°16.1	16.7'	02°51.9	14.4' 54.8'
18	93°50.0	S18°40.7	141°51.8	16.7'	S03°06.3	14.3' 54.8'
19	108°49.9	41.4	156°27.5	16.7'	03°20.6	14.3' 54.7'
20	123°49.7	42.0	171°03.2	16.7'	03°34.9	14.3' 54.7'
21	138°49.6	· · 42.6	185°38.9	16.7'	03°49.2	14.3' 54.7'
22	153°49.5	43.2	200°14.6	16.7'	04°03.5	14.3' 54.7'
23	168°49.4	43.9	214°50.3	16.7'	04°17.8	14.2' 54.7'
SD = 16.2' d = 0.6'			SD = 15.0'			
Sun	GHA	Dec	GHA	ν	Dec	d HP
0	183°49.3	S18°44.5	229°25.9	16.7'	S04°32.0	14.2' 54.6'
1	198°49.2	45.1	244°01.6	16.7'	04°46.2	14.2' 54.6'
2	213°49.1	45.7	258°37.3	16.7'	05°00.4	14.2' 54.6'
3	228°49.0	· · 46.4	273°13.0	16.7'	05°14.5	14.1' 54.6'
4	243°48.8	47.0	287°48.6	16.7'	05°28.7	14.1' 54.6'
5	258°48.7	47.6	302°24.3	16.6'	05°42.8	14.1' 54.6'
6	273°48.6	S18°48.2	316°59.9	16.6'	S05°56.8	14.0' 54.5'
7	288°48.5	48.8	331°35.6	16.6'	06°10.9	14.0' 54.5'
8	303°48.4	49.5	346°11.2	16.6'	06°24.9	14.0' 54.5'
9	318°48.3	· · 50.1	0°46.8	16.6'	06°38.9	14.0' 54.5'
10	333°48.1	50.7	15°22.4	16.6'	06°52.8	13.9' 54.5'
11	348°48.0	51.3	29°58.0	16.6'	07°06.8	13.9' 54.5'
12	3°47.9	S18°51.9	44°33.6	16.6'	S07°20.7	13.9' 54.5'
13	18°47.8	52.5	59°09.1	16.5'	07°34.5	13.8' 54.4'
14	33°47.7	53.2	73°44.7	16.5'	07°48.3	13.8' 54.4'
15	48°47.6	· · 53.8	88°20.2	16.5'	08°02.1	13.8' 54.4'
16	63°47.4	54.4	102°55.7	16.5'	08°15.9	13.7' 54.4'
17	78°47.3	55.0	117°31.2	16.5'	08°29.6	13.7' 54.4'
18	93°47.2	S18°55.6	132°06.6	16.4'	S08°43.3	13.6' 54.4'
19	108°47.1	56.2	146°42.1	16.4'	08°56.9	13.6' 54.4'
20	123°47.0	56.8	161°17.5	16.4'	09°10.5	13.6' 54.3'
21	138°46.8	· · 57.5	175°52.9	16.4'	09°24.1	13.5' 54.3'
22	153°46.7	58.1	190°28.2	16.3'	09°37.6	13.5' 54.3'
23	168°46.6	58.7	205°03.6	16.3'	09°51.1	13.4' 54.3'
SD = 16.2' d = 0.6'			SD = 14.9'			
Mon	GHA	Dec	GHA	ν	Dec	d HP
0	183°46.5	S18°59.3	219°38.9	16.3'	S10°04.5	13.4' 54.3'
1	198°46.3	18°59.9	234°14.2	16.3'	10°17.9	13.4' 54.3'
2	213°46.2	19°00.5	248°49.4	16.2'	10°31.3	13.3' 54.3'
3	228°46.1	· · 01.1	263°24.7	16.2'	10°44.6	13.3' 54.3'
4	243°46.0	01.7	277°59.8	16.2'	10°57.9	13.2' 54.2'
5	258°45.9	02.3	292°35.0	16.1'	11°11.1	13.2' 54.2'
6	273°45.7	S19°02.9	307°10.1	16.1'	S11°24.3	13.1' 54.2'
7	288°45.6	03.5	321°45.2	16.1'	11°37.4	13.1' 54.2'
8	303°45.5	04.2	336°20.3	16.0'	11°50.5	13.0' 54.2'
9	318°45.4	· · 04.8	350°55.3	16.0'	12°03.5	13.0' 54.2'
10	333°45.2	05.4	5°30.3	16.0'	12°16.5	12.9' 54.2'
11	348°45.1	06.0	20°05.3	15.9'	12°29.4	12.9' 54.2'
12	3°45.0	S19°06.6	34°40.2	15.9'	S12°42.3	12.8' 54.2'
13	18°44.8	07.2	49°15.1	15.8'	12°55.1	12.8' 54.1'
14	33°44.7	07.8	63°49.9	15.8'	13°07.9	12.7' 54.1'
15	48°44.6	· · 08.4	78°24.7	15.8'	13°20.7	12.7' 54.1'
16	63°44.5	09.0	92°59.5	15.7'	13°33.3	12.6' 54.1'
17	78°44.3	09.6	107°34.2	15.7'	13°45.9	12.6' 54.1'
18	93°44.2	S19°10.2	122°08.9	15.6'	S13°58.5	12.5' 54.1'
19	108°44.1	10.8	136°43.5	15.6'	14°11.0	12.4' 54.1'
20	123°43.9	11.4	151°18.1	15.5'	14°23.5	12.4' 54.1'
21	138°43.8	· · 12.0	165°52.6	15.5'	14°35.9	12.3' 54.1'
22	153°43.7	12.6	180°27.1	15.4'	14°48.2	12.3' 54.1'
23	168°43.6	13.2	195°01.6	15.4'	15°00.5	12.2' 54.1'
SD = 16.2' d = 0.6'			SD = 14.8'			

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:04	08:40	■	■	14:48	16:25
N 70°	06:53	08:16	09:58	13:31	15:12	16:35
68°	06:44	07:58	09:19	14:10	15:31	16:44
66°	06:37	07:43	08:52	14:37	15:45	16:52
64°	06:30	07:31	08:31	14:58	15:58	16:58
62°	06:25	07:21	08:15	15:14	16:08	17:04
60°	06:19	07:11	08:01	15:28	16:17	17:09
N 58°	06:15	07:04	07:49	15:40	16:25	17:14
56°	06:11	06:56	07:39	15:50	16:32	17:18
54°	06:07	06:50	07:30	15:59	16:39	17:22
52°	06:03	06:44	07:22	16:07	16:45	17:26
50°	05:59	06:39	07:15	16:14	16:50	17:30
45°	05:52	06:27	06:59	16:30	17:02	17:37
N 40°	05:45	06:17	06:46	16:43	17:12	17:45
35°	05:38	06:09	06:36	16:54	17:21	17:51
30°	05:32	06:01	06:26	17:03	17:29	17:58
20°	05:20	05:46	06:09	17:20	17:43	18:10
N 10°	05:07	05:33	05:55	17:35	17:57	18:22
0°	04:54	05:19	05:41	17:48	18:10	18:36
S 10°	04:39	05:05	05:27	18:02	18:25	18:51
20°	04:21	04:49	05:13	18:17	18:41	19:09
30°	03:58	04:29	04:55	18:35	19:01	19:32
35°	03:43	04:17	04:45	18:45	19:13	19:47
40°	03:25	04:03	04:34	18:57	19:27	20:05
45°	03:02	03:46	04:20	19:10	19:45	20:28
S 50°	02:32	03:24	04:03	19:27	20:07	21:00
52°	02:15	03:13	03:55	19:35	20:18	21:16
54°	01:56	03:01	03:46	19:44	20:30	21:36
56°	01:30	02:47	03:36	19:55	20:45	22:03
58°	00:52	02:30	03:25	20:06	21:02	22:44
S 60°	—:—	02:09	03:12	20:20	21:24	—:—

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	01:58	03:59	06:07	13:58	13:25	12:42
N 70°	01:59	03:50	05:46	14:01	13:36	13:05
68°	02:00	03:44	05:30	14:04	13:45	13:23
66°	02:01	03:38	05:17	14:06	13:52	13:37
64°	02:01	03:33	05:07	14:08	13:59	13:49
62°	02:02	03:29	04:58	14:09	14:04	14:00
60°	02:02	03:26	04:50	14:10	14:09	14:09
N 58°	02:03	03:23	04:43	14:12	14:14	14:16
56°	02:03	03:20	04:37	14:13	14:18	14:23
54°	02:03	03:18	04:32	14:14	14:21	14:29
52°	02:03	03:15	04:27	14:15	14:24	14:35
50°	02:04	03:13	04:23	14:15	14:27	14:40
45°	02:04	03:09	04:14	14:17	14:33	14:51
N 40°	02:05	03:06	04:06	14:19	14:39	15:00
35°	02:05	03:02	04:00	14:20	14:43	15:08
30°	02:05	03:00	03:54	14:21	14:47	15:15
20°	02:06	02:55	03:44	14:23	14:54	15:27
N 10°	02:07	02:51	03:35	14:25	15:01	15:37
0°	02:07	02:47	03:27	14:26	15:06	15:47
S 10°	02:08	02:43	03:20	14:28	15:12	15:57
20°	02:08	02:39	03:11	14:30	15:19	16:08
30°	02:09	02:35	03:02	14:32	15:26	16:20
35°	02:09	02:32	02:56	14:33	15:30	16:28
40°	02:10	02:30	02:50	14:34	15:35	16:36
45°	02:10	02:26	02:43	14:36	15:40	16:45
S 50°	02:11	02:22	02:34	14:37	15:47	16:57
52°	02:11	02:21	02:31	14:38	15:50	17:02
54°	02:11	02:19	02:26	14:39	15:53	17:08
56°	02:12	02:16	02:22	14:40	15:57	17:15
58°	02:12	02:14	02:16	14:41	16:01	17:23
S 60°	02:13	02:11	02:10	14:42	16:06	17:31

Day	Sun		Mer. Pass	Moon		Age
	Eqn. of Time 00 ^h mm:ss	12 ^h mm:ss		Upper hh:mm	Lower hh:mm	
15	15:28	15:23	11:45	08:17	20:37	
16	15:17	15:12	11:45	08:57	21:17	
17	15:06	15:00	11:45	09:37	21:58	

November 18, 19, 20 UT (Tue., Wed., Thu.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	57°17.6	195°32.4	S14°55.4	169°18.5	S22°27.2	300°14.2	N21°14.3	60°41.6	S04°06.2	Alpheratz	357°33.9	29°14.2											
1	72°20.1	210°31.8	56.4	184°19.0	27.5	315°16.7	14.4	75°44.1	06.2	Ankaa	353°06.2	-42°10.0											
2	87°22.5	225°31.2	57.4	199°19.5	27.8	330°19.2	14.4	90°46.6	06.2	Schedar	349°29.9	56°41.0											
3	102°25.0	240°30.6	•• 58.4	214°20.1	•• 28.1	345°21.7	•• 14.4	105°49.2	•• 06.2	Diphda	348°46.4	-17°50.7											
4	117°27.4	255°30.0	14°59.4	229°20.6	28.4	0°24.3	14.4	120°51.7	06.2	Achernar	335°19.1	-57°06.4											
5	132°29.9	270°29.3	15°00.4	244°21.1	28.7	15°26.8	14.4	135°54.2	06.2	Hamal	327°50.1	23°35.2											
6	147°32.4	285°28.7	S15°01.4	259°21.6	S22°29.0	30°29.3	N21°14.4	150°56.7	S04°06.2	Polaris	313°10.9	89°22.5											
7	162°34.8	300°28.1	02.4	274°22.1	29.3	45°31.8	14.5	165°59.2	06.2	Acamar	315°10.8	-40°12.0											
8	177°37.3	315°27.5	03.4	289°22.6	29.6	60°34.3	14.5	181°01.7	06.2	Menkar	314°05.1	4°11.6											
9	192°39.8	330°26.9	•• 04.4	304°23.1	•• 29.9	75°36.9	•• 14.5	196°04.2	•• 06.2	Mirfak	308°26.8	49°57.3											
10	207°42.2	345°26.2	05.4	319°23.7	30.1	90°39.4	14.5	211°06.7	06.3	Aldebaran	290°38.5	16°33.7											
11	222°44.7	0°25.6	06.4	334°24.2	30.4	105°41.9	14.5	226°09.2	06.3	Rigel	281°02.9	-8°10.2											
12	237°47.2	15°25.0	S15°07.3	349°24.7	S22°30.7	120°44.4	N21°14.5	241°11.7	S04°06.3	Capella	280°20.4	46°01.4											
13	252°49.6	30°24.4	08.3	4°25.2	31.0	135°47.0	14.6	256°14.2	06.3	Bellatrix	278°21.8	6°22.5											
14	267°52.1	45°23.8	09.3	19°25.7	31.3	150°49.5	14.6	271°16.7	06.3	Elnath	278°00.6	28°37.8											
15	282°54.5	60°23.1	•• 10.3	34°26.2	•• 31.6	165°52.0	•• 14.6	286°19.2	•• 06.3	Alnilam	275°36.7	-1°11.0											
16	297°57.0	75°22.5	11.3	49°26.7	31.9	180°54.5	14.6	301°21.7	06.3	Betelgeuse	270°51.0	7°24.8											
17	312°59.5	90°21.9	12.3	64°27.2	32.2	195°57.1	14.6	316°24.3	06.3	Canopus	263°51.6	-52°42.3											
18	328°01.9	105°21.3	S15°13.3	79°27.8	S22°32.5	210°59.6	N21°14.6	331°26.8	S04°06.3	Sirius	258°25.4	-16°44.9											
19	343°04.4	120°20.6	14.3	94°28.3	32.7	226°02.1	14.7	346°29.3	06.3	Adhara	255°05.1	-29°00.2											
20	358°06.9	135°20.0	15.2	109°28.8	33.0	241°04.6	14.7	1°31.8	06.3	Procyon	244°49.9	5°09.6											
21	13°09.3	150°19.4	•• 16.2	124°29.3	•• 33.3	256°07.2	•• 14.7	16°34.3	•• 06.4	Pollux	243°16.3	27°57.7											
22	28°11.8	165°18.8	17.2	139°29.8	33.6	271°09.7	14.7	31°36.8	06.4	Avior	234°14.2	-59°35.2											
23	43°14.3	180°18.1	18.2	154°30.3	33.9	286°12.2	14.7	46°39.3	06.4	Suhail	222°45.7	-43°31.9											
Mer.pass. 20:08		ν -0.6' d 1.0' m -3.91		ν 0.5' d 0.3' m 1.38		ν 2.5' d 0.0' m -2.45		ν 2.5' d 0.0' m 0.85															

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	58°16.7	195°17.5	S15°19.2	169°30.8	S22°34.2	301°14.7	N21°14.7	61°41.8	S04°06.4	Dubhe	193°40.4	61°36.4	
1	73°19.2	210°16.9	20.2	184°31.3	34.5	316°17.3	14.8	76°44.3	06.4	Denebola	182°24.4	14°25.6	
2	88°21.7	225°16.3	21.1	199°31.8	34.7	331°19.8	14.8	91°46.8	06.4	Gienah	175°43.1	-17°41.0	
3	103°24.1	240°15.6	•• 22.1	214°32.4	•• 35.0	346°22.3	•• 14.8	106°49.3	•• 06.4	Acrux	172°59.8	-63°14.3	
4	118°26.6	255°15.0	23.1	229°32.9	35.3	1°24.8	14.8	121°51.8	06.4	Gacrux	171°51.3	-57°15.2	
5	133°29.0	270°14.4	24.1	244°33.4	35.6	16°27.4	14.8	136°54.3	06.4	Alioth	166°12.8	55°49.0	
6	148°31.5	285°13.7	S15°25.1	259°33.9	S22°35.9	31°29.9	N21°14.8	151°56.8	S04°06.4	Spica	158°21.8	-11°17.7	
7	163°34.0	300°13.1	26.0	274°34.4	36.2	46°32.4	14.9	166°59.3	06.4	Alkaid	152°51.9	49°10.9	
8	178°36.4	315°12.5	27.0	289°34.9	36.4	61°35.0	14.9	182°01.8	06.4	Hadar	148°35.6	-60°29.7	
9	193°38.9	330°11.8	•• 28.0	304°35.4	•• 36.7	76°37.5	•• 14.9	197°04.3	•• 06.5	Menkent	147°57.2	-36°29.7	
10	208°41.4	345°11.2	29.0	319°35.9	37.0	91°40.0	14.9	212°06.8	06.5	Arcturus	145°47.6	19°02.8	
11	223°43.8	0°10.6	29.9	334°36.4	37.3	106°42.6	14.9	227°09.3	06.5	Rigel Kent.	139°40.0	-60°56.4	
12	238°46.3	15°09.9	S15°30.9	349°36.9	S22°37.6	121°45.1	N21°15.0	242°11.8	S04°06.5	Kochab	137°21.0	74°02.8	
13	253°48.8	30°09.3	31.9	4°37.4	37.8	136°47.6	15.0	257°14.3	06.5	Zuben'ubi	136°55.6	-16°08.9	
14	268°51.2	45°08.7	32.9	19°37.9	38.1	151°50.2	15.0	272°16.8	06.5	Alphecca	126°03.5	26°37.7	
15	283°53.7	60°08.0	•• 33.8	34°38.5	•• 38.4	166°52.7	•• 15.0	287°19.3	•• 06.5	Antares	112°15.4	-26°29.3	
16	298°56.2	75°07.4	34.8	49°39.0	38.7	181°55.2	15.0	302°21.9	06.5	Atria	107°09.5	-69°04.5	
17	313°58.6	90°06.8	35.8	64°39.5	39.0	196°57.8	15.0	317°24.4	06.5	Sabik	102°02.3	-15°45.4	
18	329°01.1	105°06.1	S15°36.8	79°40.0	S22°39.2	212°00.3	N21°15.1	332°26.9	S04°06.5	Shaula	96°09.8	-37°07.4	
19	344°03.5	120°05.5	37.7	94°40.5	39.5	227°02.8	15.1	347°29.4	06.5	Rasalhague	95°58.2	12°32.5	
20	359°06.0	135°04.9	38.7	109°41.0	39.8	242°05.4	15.1	2°31.9	06.5	Eltanin	90°42.3	51°29.2	
21	14°08.5	150°04.2	•• 39.7	124°41.5	•• 40.1	257°07.9	•• 15.1	17°34.4	•• 06.5	Kaus Aust.	83°31.9	-34°22.4	
22	29°10.9	165°03.6	40.6	139°42.0	40.4	272°10.4	15.1	32°36.9	06.6	Vega	80°33.1	38°48.6	
23	44°13.4	180°02.9	41.6	154°42.5	40.6	287°13.0	15.2	47°39.4	06.6	Nunki	75°47.2	-26°15.9	
Mer.pass. 20:04		ν -0.6' d 1.0' m -3.91		ν 0.5' d 0.3' m 1.37		ν 2.5' d 0.0' m -2.45		ν 2.5' d 0.0' m 0.86					

Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	59°15.9	195°02.3	S15°42.6	169°43.0	S22°40.9	302°15.5	N21°15.2	62°41.9	S04°06.6	Altair	61°59.5	8°56.3	
1	74°18.3	210°01.7	43.5	184°43.5	41.2	317°18.0	15.2	77°44.4	06.6	Peacock	53°04.9	-56°39.3	
2	89°20.8	225°01.0	44.5	199°44.0	41.5	332°20.6	15.2	92°46.9	06.6	Deneb	49°25.5	45°22.6	
3	104°23.3	240°00.4	•• 45.5	214°44.5	•• 41.7	347°23.1	•• 15.2	107°49.4	•• 06.6	Enif	33°38.1	9°59.8	
4	119°25.7	254°59.7	46.4	229°45.0	42.0	2°25.6	15.2	122°51.9	06.6	Al Na'ir	27°32.0	-46°50.3	
5	134°28.2	269°59.1	47.4	244°45.5	42.3	17°28.2	15.3	137°54.4	06.6	Fomalhaut	15°13.7	-29°29.2	
6	149°30.7	284°58.5	S15°48.3	259°46.0	S22°42.6	32°30.7	N21°15.3	152°56.9	S04°06.6	Scheat	13°44.4	28°13.6	
7	164°33.1	299°57.8	49.3	274°46.5	42.8	47°33.3	15.3	167°59.4	06.6	Markab	13°29.1	15°20.8	
8	179°35.6	314°57.2	50.3	289°47.0	43.1	62°35.8	15.3	183°01.9	06.6				
9	194°38.0	329°56.5	•• 51.2	304°47.5	•• 43.4	77°38.3	•• 15.3	198°04.4	•• 06.6				
10	209°40.5	344°55.9	52.2	319°48.0	43.7	92°40.9	15.4	213°06.9	06.6				
11	224°43.0	359°55.2	53.1	334°48.5	43.9	107°43.4	15.4	228°09.4	06.6				
12	239°45.4	14°54.6	S15°54.1	349°49.0	S22°44.2	122°46.0	N21°15.4	243°11.9	S04°06.6				
13	254°47.9	29°53.9	55.1	4°49.5	44.5	137°48.5	15.4	258°14.4	06.6				
14	269°50.4	44°53.3	56.0	19°50.0	44.7	152°51.0	15.4	273°16.9	06.7				
15	284°52.8	59°52.6	•• 57.0	34°50.5	•• 45.0	167°53.6	•• 15.5	288°19.4	•• 06.7				
16	299°55.3	74°52.0	57.9	49°51.0	45.3	182°56.1	15.5	303°21.9	06.7				
17	314°57.8	89°51.4	58.9	64°51.5	45.5	197°58.7	15.5	318°24.4	06.7				
18	330°00.2	104°50.7	S15°59.8	79°52.0	S22°45.8	213°01.2	N21°15.5	333°26.9	S04°06.7				
19	345°02.7	119°50.1	16°00.8	94°52.5	46.1	228°03.8	15.5	348°29.4	06.7				
20	0°05.1	134°49.4	01.7	109°53.0	46.4	243°06.3	15.6	3°31.9	06.7				
21	15°07.6	149°48.8	•• 02.7	124°53.5	•• 46.6	258°08.8	•• 15.6	18°34.3					

DUT1 = UT1-UTC = +0.1071 sec $\Delta T = TT-UT1 = +69.0769$ sec

2025 November 18 to Nov. 20 UT

h Sun Moon							
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	183°43.4	S19°13.8	209°36.0	15.4'	S15°12.7	12.2'	54.0'
1	198°43.3	14.4	224°10.3	15.3'	15°24.8	12.1'	54.0'
2	213°43.2	15.0	238°44.6	15.3'	15°36.9	12.0'	54.0'
3	228°43.0	· · 15.5	253°18.9	15.2'	15°48.9	12.0'	54.0'
4	243°42.9	16.1	267°53.1	15.2'	16°00.9	11.9'	54.0'
5	258°42.8	16.7	282°27.2	15.1'	16°12.8	11.8'	54.0'
6	273°42.6	S19°17.3	297°01.3	15.1'	S16°24.6	11.8'	54.0'
7	288°42.5	17.9	311°35.4	15.0'	16°36.4	11.7'	54.0'
8	303°42.4	18.5	326°09.4	14.9'	16°48.1	11.6'	54.0'
9	318°42.2	· · 19.1	340°43.3	14.9'	16°59.7	11.6'	54.0'
10	333°42.1	19.7	355°17.2	14.8'	17°11.3	11.5'	54.0'
11	348°42.0	20.3	9°51.1	14.8'	17°22.8	11.4'	54.0'
12	3°41.8	S19°20.9	24°24.9	14.7'	S17°34.2	11.4'	54.0'
13	18°41.7	21.5	38°58.6	14.7'	17°45.6	11.3'	54.0'
14	33°41.6	22.0	53°32.2	14.6'	17°56.9	11.2'	54.0'
15	48°41.4	· · 22.6	68°05.9	14.6'	18°08.1	11.1'	53.9'
16	63°41.3	23.2	82°39.4	14.5'	18°19.2	11.1'	53.9'
17	78°41.1	23.8	97°12.9	14.4'	18°30.2	11.0'	53.9'
18	93°41.0	S19°24.4	111°46.4	14.4'	S18°41.2	10.9'	53.9'
19	108°40.9	25.0	126°19.7	14.3'	18°52.1	10.8'	53.9'
20	123°40.7	25.6	140°53.1	14.3'	19°03.0	10.8'	53.9'
21	138°40.6	· · 26.1	155°26.3	14.2'	19°13.7	10.7'	53.9'
22	153°40.5	26.7	169°59.5	14.1'	19°24.4	10.6'	53.9'
23	168°40.3	27.3	184°32.7	14.1'	19°35.0	10.5'	53.9'
SD = 16.2' $d = 0.6'$ SD = 14.7'							
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	183°40.2	S19°27.9	199°05.7	14.0'	S19°45.5	10.4'	53.9'
1	198°40.0	28.5	213°38.8	14.0'	19°55.9	10.3'	53.9'
2	213°39.9	29.0	228°11.7	13.9'	20°06.3	10.3'	53.9'
3	228°39.8	· · 29.6	242°44.6	13.8'	20°16.5	10.2'	53.9'
4	243°39.6	30.2	257°17.4	13.8'	20°26.7	10.1'	53.9'
5	258°39.5	30.8	271°50.2	13.7'	20°36.8	10.0'	53.9'
6	273°39.3	S19°31.4	286°22.9	13.6'	S20°46.8	9.9'	53.9'
7	288°39.2	31.9	300°55.6	13.6'	20°56.7	9.8'	53.9'
8	303°39.0	32.5	315°28.1	13.5'	21°06.6	9.7'	53.9'
9	318°38.9	· · 33.1	330°00.7	13.5'	21°16.3	9.7'	53.9'
10	333°38.8	33.7	344°33.1	13.4'	21°26.0	9.6'	53.9'
11	348°38.6	34.2	359°05.5	13.3'	21°35.5	9.5'	53.9'
12	3°38.5	S19°34.8	13°37.8	13.3'	S21°45.0	9.4'	53.9'
13	18°38.3	35.4	28°10.1	13.2'	21°54.4	9.3'	53.9'
14	33°38.2	36.0	42°42.3	13.1'	22°03.7	9.2'	53.9'
15	48°38.0	· · 36.5	57°14.4	13.1'	22°12.9	9.1'	53.9'
16	63°37.9	37.1	71°46.5	13.0'	22°22.0	9.0'	53.9'
17	78°37.8	37.7	86°18.5	12.9'	22°31.0	8.9'	53.9'
18	93°37.6	S19°38.2	100°50.4	12.9'	S22°39.9	8.8'	53.9'
19	108°37.5	38.8	115°22.3	12.8'	22°48.7	8.7'	53.9'
20	123°37.3	39.4	129°54.1	12.7'	22°57.4	8.6'	53.9'
21	138°37.2	· · 40.0	144°25.8	12.7'	23°06.0	8.5'	53.9'
22	153°37.0	40.5	158°57.5	12.6'	23°14.6	8.4'	53.9'
23	168°36.9	41.1	173°29.1	12.5'	23°23.0	8.3'	53.9'
SD = 16.2' $d = 0.6'$ SD = 14.7'							
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	183°36.7	S19°41.7	188°00.7	12.5'	S23°31.3	8.2'	53.9'
1	198°36.6	42.2	202°32.1	12.4'	23°39.5	8.1'	53.8'
2	213°36.4	42.8	217°03.5	12.4'	23°47.6	8.0'	53.8'
3	228°36.3	· · 43.4	231°34.9	12.3'	23°55.6	7.9'	53.8'
4	243°36.1	43.9	246°06.2	12.2'	24°03.5	7.8'	53.8'
5	258°36.0	44.5	260°37.4	12.2'	24°11.3	7.7'	53.8'
6	273°35.8	S19°45.0	275°08.6	12.1'	S24°19.0	7.6'	53.9'
7	288°35.7	45.6	289°39.7	12.0'	24°26.5	7.5'	53.9'
8	303°35.5	46.2	304°10.7	12.0'	24°34.0	7.4'	53.9'
9	318°35.4	· · 46.7	318°41.7	11.9'	24°41.4	7.2'	53.9'
10	333°35.2	47.3	333°12.6	11.8'	24°48.6	7.1'	53.9'
11	348°35.1	47.9	347°43.4	11.8'	24°55.8	7.0'	53.9'
12	3°34.9	S19°48.4	2°14.2	11.7'	S25°02.8	6.9'	53.9'
13	18°34.8	49.0	16°44.9	11.7'	25°09.7	6.8'	53.9'
14	33°34.6	49.5	31°15.6	11.6'	25°16.5	6.7'	53.9'
15	48°34.5	· · 50.1	45°46.2	11.5'	25°23.2	6.6'	53.9'
16	63°34.3	50.6	60°16.7	11.5'	25°29.8	6.5'	53.9'
17	78°34.1	51.2	74°47.2	11.4'	25°36.2	6.3'	53.9'
18	93°34.0	S19°51.8	89°17.6	11.4'	S25°42.6	6.2'	53.9'
19	108°33.8	52.3	103°48.0	11.3'	25°48.8	6.1'	53.9'
20	123°33.7	52.9	118°18.3	11.2'	25°54.9	6.0'	53.9'
21	138°33.5	· · 53.4	132°48.5	11.2'	26°00.9	5.9'	53.9'
22	153°33.4	54.0	147°18.7	11.1'	26°06.8	5.7'	53.9'
23	168°33.2	54.5	161°48.9	11.1'	26°12.5	5.6'	53.9'
SD = 16.2' $d = 0.6'$ SD = 14.7'							

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:14	08:54	■	■	14:35	16:15
N 70°	07:02	08:28	10:20	13:10	15:02	16:27
68°	06:53	08:08	09:33	13:57	15:22	16:37
66°	06:44	07:52	09:03	14:27	15:38	16:46
64°	06:37	07:39	08:41	14:49	15:51	16:53
62°	06:31	07:28	08:23	15:07	16:03	16:59
60°	06:25	07:18	08:08	15:22	16:12	17:05
N 58°	06:20	07:09	07:56	15:34	16:21	17:10
56°	06:16	07:02	07:45	15:45	16:28	17:15
54°	06:11	06:55	07:36	15:55	16:35	17:19
52°	06:07	06:49	07:27	16:03	16:41	17:23
50°	06:04	06:43	07:19	16:11	16:47	17:27
45°	05:55	06:31	07:03	16:27	16:59	17:35
N 40°	05:48	06:21	06:50	16:41	17:10	17:43
35°	05:41	06:11	06:38	16:52	17:19	17:50
30°	05:34	06:03	06:28	17:02	17:28	17:57
20°	05:21	05:48	06:11	17:19	17:43	18:10
N 10°	05:08	05:34	05:56	17:35	17:57	18:23
0°	04:54	05:20	05:42	17:49	18:11	18:37
S 10°	04:39	05:05	05:27	18:04	18:26	18:52
20°	04:20	04:48	05:12	18:19	18:43	19:11
30°	03:56	04:28	04:54	18:37	19:04	19:35
35°	03:41	04:15	04:43	18:48	19:16	19:51
40°	03:22	04:00	04:31	19:00	19:31	20:10
45°	02:58	03:42	04:17	19:15	19:49	20:34
S 50°	02:26	03:19	03:59	19:32	20:13	21:07
52°	02:09	03:08	03:51	19:41	20:24	21:24
54°	01:47	02:55	03:42	19:50	20:37	21:46
56°	01:18	02:40	03:31	20:01	20:52	22:16
58°	00:29	02:22	03:19	20:13	21:11	23:12
S 60°	—:—	01:59	03:05	20:28	21:35	—:—

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	09:02	■	■	11:16	■	■
N 70°	08:02	■	■	12:17	■	■
68°	07:28	10:14	■	12:53	11:39	■
66°	07:03	09:06	■	13:19	12:47	■
64°	06:44	08:31	10:39	13:38	13:23	12:52
62°	06:29	08:06	09:50	13:55	13:49	13:43
60°	06:16	07:46	09:19	14:08	14:10	14:14
N 58°	06:05	07:30	08:55	14:20	14:27	14:38
56°	05:56	07:16	08:37	14:30	14:41	14:57
54°	05:48	07:04	08:21	14:40	14:53	15:13
52°	05:40	06:54	08:08	14:48	15:04	15:27
50°	05:33	06:45	07:56	14:55	15:14	15:39
45°	05:19	06:25	07:31	15:11	15:35	16:04
N 40°	05:07	06:10	07:12	15:24	15:51	16:25
35°	04:58	05:56	06:56	15:35	16:06	16:41
30°	04:49	05:45	06:42	15:45	16:18	16:56
20°	04:34	05:25	06:18	16:01	16:39	17:21
N 10°	04:21	05:09	05:58	16:16	16:58	17:42
0°	04:09	04:53	05:39	16:30	17:15	18:03
S 10°	03:57	04:37	05:20	16:44	17:32	18:23
20°	03:45	04:21	05:00	16:59	17:51	18:44
30°	03:30	04:02	04:37	17:16	18:12	19:10
35°	03:22	03:51	04:24	17:26	18:25	19:24
40°	03:12	03:38	04:08	17:37	18:40	19:42
45°	03:01	03:23	03:50	17:51	18:57	20:03
S 50°	02:48	03:05	03:27	18:08	19:19	20:29
52°	02:42	02:57	03:16	18:16	19:30	20:42
54°	02:35	02:47	03:04	18:24	19:41	20:57
56°	02:28	02:37	02:50	18:34	19:55	21:14
58°	02:20	02:25	02:33	18:46	20:11	21:35
S 60°	02:10	02:11	02:14	18:59	20:30	22:02

Day	Sun		Mer. Pass	Moon		Age 28-0 5-0%
	Eqn.of Time 00 ^h mm:ss	12 ^h mm:ss		Mer.Pass. Upper hh:mm	Lower hh:mm	
18	14:54	14:47	11:45	10:19	22:41	
19	14:41	14:34	11:45	11:04	23:27	
20	14:27	14:20	11:46	11:51	—:—	

November 21, 22, 23 UT (Fri., Sat., Sun.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	60°15.0	194°46.8	S16°05.5	169°55.0	S22°47.4	303°16.5	N21°15.6	63°41.8	S04°06.7	Alpheratz	357°33.9 29°14.2
1	75°17.5	209°46.2	06.5	184°55.5	47.7	318°19.0	15.7	78°44.3	06.7	Ankaa	353°06.2 -42°10.0
2	90°19.9	224°45.5	07.4	199°56.0	48.0	333°21.6	15.7	93°46.8	06.7	Schedar	349°29.9 56°41.0
3	105°22.4	239°44.8	•• 08.4	214°56.5	•• 48.2	348°24.1	•• 15.7	108°49.3	•• 06.7	Diphda	348°46.4 -17°50.7
4	120°24.9	254°44.2	09.3	229°57.0	48.5	3°26.7	15.7	123°51.8	06.7	Achernar	335°19.2 -57°06.4
5	135°27.3	269°43.5	10.3	244°57.5	48.8	18°29.2	15.7	138°54.3	06.7	Hamal	327°50.1 23°35.2
6	150°29.8	284°42.9	S16°11.2	259°58.0	S22°49.0	33°31.7	N21°15.8	153°56.8	S04°06.7	Polaris	313°10.7 89°22.5
7	165°32.3	299°42.2	12.2	274°58.5	49.3	48°34.3	15.8	168°59.3	06.7	Acamar	315°10.8 -40°12.0
8	180°34.7	314°41.6	13.1	289°59.0	49.6	63°36.8	15.8	184°01.8	06.7	Menkar	314°05.1 4°11.6
9	195°37.2	329°40.9	•• 14.0	304°59.5	•• 49.8	78°39.4	•• 15.8	199°04.3	•• 06.7	Mirfak	308°26.7 49°57.3
10	210°39.6	344°40.3	15.0	320°00.0	50.1	93°41.9	15.8	214°06.8	06.7	Aldebaran	290°38.5 16°33.7
11	225°42.1	359°39.6	15.9	335°00.5	50.3	108°44.5	15.9	229°09.3	06.8	Rigel	281°02.9 -8°10.2
12	240°44.6	14°38.9	S16°16.9	350°01.0	S22°50.6	123°47.0	N21°15.9	244°11.8	S04°06.8	Capella	280°20.4 46°01.4
13	255°47.0	29°38.3	17.8	5°01.5	50.9	138°49.6	15.9	259°14.3	06.8	Bellatrix	278°21.8 6°22.5
14	270°49.5	44°37.6	18.7	20°02.0	51.1	153°52.1	15.9	274°16.8	06.8	Elnath	278°00.6 28°37.8
15	285°52.0	59°37.0	•• 19.7	35°02.5	•• 51.4	168°54.7	•• 15.9	289°19.3	•• 06.8	Alnilam	275°36.7 -1°11.0
16	300°54.4	74°36.3	20.6	50°03.0	51.6	183°57.2	16.0	304°21.8	06.8	Betelgeuse	270°51.0 7°24.8
17	315°56.9	89°35.6	21.6	65°03.5	51.9	198°59.8	16.0	319°24.3	06.8	Canopus	263°51.6 -52°42.3
18	330°59.4	104°35.0	S16°22.5	80°04.0	S22°52.2	214°02.3	N21°16.0	334°26.8	S04°06.8	Sirius	258°25.3 -16°45.0
19	346°01.8	119°34.3	23.4	95°04.5	52.4	229°04.9	16.0	349°29.3	06.8	Adhara	255°05.0 -29°00.2
20	1°04.3	134°33.7	24.4	110°05.0	52.7	244°07.4	16.1	4°31.7	06.8	Procyon	244°49.9 5°09.6
21	16°06.8	149°33.0	•• 25.3	125°05.5	•• 52.9	259°10.0	•• 16.1	19°34.2	•• 06.8	Pollux	243°16.2 27°57.7
22	31°09.2	164°32.3	26.2	140°06.0	53.2	274°12.5	16.1	34°36.7	06.8	Avior	234°14.2 -59°35.2
23	46°11.7	179°31.7	27.2	155°06.4	53.5	289°15.1	16.1	49°39.2	06.8	Suhail	222°45.7 -43°31.9
Mer.pass. 19:56		ν -0.7' d 0.9' m -3.90		ν 0.5' d 0.3' m 1.36		ν 2.5' d 0.0' m -2.47		ν 2.5' d 0.0' m 0.87		Miaplacidus	221°37.9 -69°49.0
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	Alphard	217°47.0 -8°46.1
0	61°14.1	194°31.0	S16°28.1	170°06.9	S22°53.7	304°17.7	N21°16.1	64°41.7	S04°06.8	Regulus	207°33.7 11°50.4
1	76°16.6	209°30.3	29.0	185°07.4	54.0	319°20.2	16.2	79°44.2	06.8	Dubhe	193°40.3 61°36.4
2	91°19.1	224°29.7	30.0	200°07.9	54.2	334°22.8	16.2	94°46.7	06.8	Denebola	182°24.4 14°25.6
3	106°21.5	239°29.0	•• 30.9	215°08.4	•• 54.5	349°25.3	•• 16.2	109°49.2	•• 06.8	Gienah	175°43.1 -17°41.0
4	121°24.0	254°28.4	31.8	230°08.9	54.8	4°27.9	16.2	124°51.7	06.8	Acrux	172°59.8 -63°14.3
5	136°26.5	269°27.7	32.7	245°09.4	55.0	19°30.4	16.3	139°54.2	06.8	Gacrux	171°51.2 -57°15.2
6	151°28.9	284°27.0	S16°33.7	260°09.9	S22°55.3	34°33.0	N21°16.3	154°56.7	S04°06.8	Alioth	166°12.7 55°49.0
7	166°31.4	299°26.3	34.6	275°10.4	55.5	49°35.5	16.3	169°59.2	06.8	Spica	158°21.8 -11°17.7
8	181°33.9	314°25.7	35.5	290°10.9	55.8	64°38.1	16.3	185°01.7	06.8	Alkaid	152°51.9 49°10.9
9	196°36.3	329°25.0	•• 36.4	305°11.4	•• 56.0	79°40.7	•• 16.3	200°04.1	•• 06.8	Hadar	148°35.6 -60°29.7
10	211°38.8	344°24.3	37.4	320°11.9	56.3	94°43.2	16.4	215°06.6	06.8	Menkent	147°57.2 -36°29.7
11	226°41.3	359°23.7	38.3	335°12.4	56.5	109°45.8	16.4	230°09.1	06.8	Arcturus	145°47.6 19°02.8
12	241°43.7	14°23.0	S16°39.2	350°12.8	S22°56.8	124°48.3	N21°16.4	245°11.6	S04°06.8	Rigel Kent.	139°40.0 -60°56.4
13	256°46.2	29°22.3	40.1	5°13.3	57.0	139°50.9	16.4	260°14.1	06.8	Kochab	137°20.9 74°02.8
14	271°48.6	44°21.7	41.1	20°13.8	57.3	154°53.4	16.5	275°16.6	06.8	Zuben'ubi	136°55.6 -16°08.9
15	286°51.1	59°21.0	•• 42.0	35°14.3	•• 57.5	169°56.0	•• 16.5	290°19.1	•• 06.8	Alphecca	126°03.5 26°37.6
16	301°53.6	74°20.3	42.9	50°14.8	57.8	184°58.6	16.5	305°21.6	06.8	Antares	112°15.4 -26°29.3
17	316°56.0	89°19.6	43.8	65°15.3	58.0	200°01.1	16.5	320°24.1	06.8	Atria	107°09.5 -69°04.4
18	331°58.5	104°19.0	S16°44.7	80°15.8	S22°58.3	215°03.7	N21°16.5	335°26.6	S04°06.8	Sabik	102°02.3 -15°45.4
19	347°01.0	119°18.3	45.6	95°16.3	58.6	230°06.2	16.6	350°29.1	06.8	Shaula	96°09.8 -37°07.4
20	2°03.4	134°17.6	46.6	110°16.8	58.8	245°08.8	16.6	5°31.5	06.8	Rasalhague	95°58.2 12°32.5
21	17°05.9	149°17.0	•• 47.5	125°17.2	•• 59.1	260°11.4	•• 16.6	20°34.0	•• 06.8	Eltanin	90°42.3 51°29.2
22	32°08.4	164°16.3	48.4	140°17.7	59.3	275°13.9	16.6	35°36.5	06.8	Kaus Aust.	83°31.9 -34°22.4
23	47°10.8	179°15.6	49.3	155°18.2	59.5	290°16.5	16.7	50°39.0	06.8	Vega	80°33.1 38°48.6
Mer.pass. 19:52		ν -0.7' d 0.9' m -3.90		ν 0.5' d 0.3' m 1.36		ν 2.6' d 0.0' m -2.47		ν 2.5' d 0.0' m 0.87		Nunki	75°47.2 -26°15.9
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	Altair	61°59.5 8°56.3
0	62°13.3	194°14.9	S16°50.2	170°18.7	S22°59.8	305°19.0	N21°16.7	65°41.5	S04°06.8	Peacock	53°04.9 -56°39.3
1	77°15.8	209°14.2	51.1	185°19.2	23°00.0	320°21.6	16.7	80°44.0	06.8	Deneb	49°25.5 45°22.6
2	92°18.2	224°13.6	52.0	200°19.7	00.3	335°24.2	16.7	95°46.5	06.8	Enif	33°38.1 9°59.8
3	107°20.7	239°12.9	•• 53.0	215°20.2	•• 00.5	350°26.7	•• 16.7	110°49.0	•• 06.9	Al Na'ir	27°32.0 -46°50.3
4	122°23.1	254°12.2	53.9	230°20.7	00.8	5°29.3	16.8	125°51.5	06.9	Fomalhaut	15°13.7 -29°29.2
5	137°25.6	269°11.5	54.8	245°21.1	01.0	20°31.9	16.8	140°53.9	06.9	Scheat	13°44.4 28°13.6
6	152°28.1	284°10.9	S16°55.7	260°21.6	S23°01.3	35°34.4	N21°16.8	155°56.4	S04°06.9	Markab	13°29.1 15°20.8
7	167°30.5	299°10.2	56.6	275°22.1	01.5	50°37.0	16.8	170°58.9	06.9		
8	182°33.0	314°09.5	57.5	290°22.6	01.8	65°39.6	16.9	186°01.4	06.9	Nov 21 Fri	SHA Mer.pass
9	197°35.5	329°08.8	•• 58.4	305°23.1	•• 02.0	80°42.1	•• 16.9	201°03.9	•• 06.9	Venus	134°31.8 11:01
10	212°37.9	344°08.1	16°59.3	320°23.6	02.3	95°44.7	16.9	216°06.4	06.9	Mars	109°40.0 12:40
11	227°40.4	359°07.4	17°00.2	335°24.1	02.5	110°47.3	16.9	231°08.9	06.9	Jupiter	243°01.5 03:46
12	242°42.9	14°06.8	S17°01.1	350°24.5	S23°02.8	125°49.8	N21°17.0	246°11.4	S04°06.9	Saturn	3°26.8 19:42
13	257°45.3	29°06.1	02.0	5°25.0	03.0	140°52.4	17.0	261°13.8	06.9		
14	272°47.8	44°05.4	02.9	20°25.5	03.2	155°55.0	17.0	276°16.3	06.9	Nov 22 Sat	SHA Mer.pass
15	287°50.2	59°04.7	•• 03.8	35°26.0	•• 03.5	170°57.5	•• 17.0	291°18.8	•• 06.9	Venus	133°16.9 11:02
16	302°52.7	74°04.0	04.7	50°26.5	03.7	186°00.1	17.1	306°21.3	06.9	Mars	108°52.8 12:39
17	317°55.2	89°03.3	05.6	65°27.0	04.0	201°02.7	17.1	321°23.8	06.9	Jupiter	243°03.5 03:42
18	332°57.6	104°02.7	S17°06.5	80°27.5	S23°04.2	216°05.2	N21°17.1	336°26.3	S04°06.9	Saturn	3°27.6 19:38
19	348°00.1	119°02.0	07.4	95°27.9	04.4	231°07.8	17.1	351°28.8	06.9		
20	3°02.6	134°01.3	08.3	110°28.4	04.7	246°10.4	17.2	6°31.2	06.9	Nov 23 Sun	SHA Mer.pass
21	18°05.0	149°00.6	•• 09.2	125°28.9	•• 04.9	261°12.9	•• 17.2	21°33.7	•• 06.9	Venus	132°01.6 11:04
22	33°07.5	163°59.9	10.1	140°29.4	05.2	276°15.5	17.2	36°36.2	06.9	Mars	108°05.4 12:38
23	48°10.0	178°59.2	11.0	155°29.9	05.4	291°18.1	17.2	51°38.7	06.9	Jupiter	243°05.8 03:38
Mer.pass. 19:48		ν -0.7' d 0.9' m -3.90		ν 0.5' d 0.2' m 1.35		ν 2.6' d 0.0' m -2.48		ν 2.5' d 0.0' m 0.88		Saturn	3°28.2 19:34
										Horizontal parallax	
										Venus:	0.1
										Mars:	0.1

h	Sun			Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP
0	183°33.1	S19°55.1	176°18.9	11.0'	S26°18.1	5.5'	53.9'
1	198°32.9	55.6	190°49.0	11.0'	26°23.6	5.4'	53.9'
2	213°32.7	56.2	205°18.9	10.9'	26°29.0	5.3'	53.9'
3	228°32.6	· · 56.7	219°48.8	10.9'	26°34.3	5.1'	53.9'
4	243°32.4	57.3	234°18.7	10.8'	26°39.4	5.0'	53.9'
5	258°32.3	57.8	248°48.5	10.8'	26°44.4	4.9'	53.9'
6	273°32.1	S19°58.4	263°18.3	10.7'	S26°49.3	4.8'	53.9'
7	288°32.0	58.9	277°48.0	10.7'	26°54.1	4.6'	53.9'
8	303°31.8	19°59.5	292°17.6	10.6'	26°58.7	4.5'	53.9'
9	318°31.6	20°00.0	306°47.2	10.6'	27°03.2	4.4'	53.9'
10	333°31.5	00.6	321°16.8	10.5'	27°07.6	4.3'	53.9'
11	348°31.3	01.1	335°46.3	10.5'	27°11.9	4.1'	53.9'
12	3°31.2	S20°01.6	350°15.8	10.4'	S27°16.0	4.0'	53.9'
13	18°31.0	02.2	4°45.2	10.4'	27°20.0	3.9'	53.9'
14	33°30.8	02.7	19°14.6	10.3'	27°23.9	3.7'	53.9'
15	48°30.7	· · 03.3	33°43.9	10.3'	27°27.6	3.6'	53.9'
16	63°30.5	03.8	48°13.2	10.3'	27°31.2	3.5'	54.0'
17	78°30.3	04.4	62°42.5	10.2'	27°34.7	3.3'	54.0'
18	93°30.2	S20°04.9	77°11.7	10.2'	S27°38.1	3.2'	54.0'
19	108°30.0	05.4	91°40.9	10.1'	27°41.3	3.1'	54.0'
20	123°29.9	06.0	106°10.0	10.1'	27°44.4	2.9'	54.0'
21	138°29.7	· · 06.5	120°39.1	10.1'	27°47.3	2.8'	54.0'
22	153°29.5	07.0	135°08.2	10.0'	27°50.1	2.7'	54.0'
23	168°29.4	07.6	149°37.2	10.0'	27°52.8	2.5'	54.0'
SD = 16.2' d = 0.6'			SD = 14.7'				
Sat	GHA	Dec	GHA	ν	Dec	d	HP
0	183°29.2	S20°08.1	164°06.2	10.0'	S27°55.4	2.4'	54.0'
1	198°29.0	08.7	178°35.1	9.9'	27°57.8	2.3'	54.0'
2	213°28.9	09.2	193°04.1	9.9'	28°00.0	2.1'	54.0'
3	228°28.7	· · 09.7	207°33.0	9.9'	28°02.2	2.0'	54.0'
4	243°28.5	10.3	222°01.9	9.8'	28°04.2	1.9'	54.0'
5	258°28.4	10.8	236°30.7	9.8'	28°06.1	1.7'	54.0'
6	273°28.2	S20°11.3	250°59.5	9.8'	S28°07.8	1.6'	54.0'
7	288°28.0	11.9	265°28.3	9.8'	28°09.4	1.5'	54.1'
8	303°27.9	12.4	279°57.1	9.7'	28°10.8	1.3'	54.1'
9	318°27.7	· · 12.9	294°25.8	9.7'	28°12.2	1.2'	54.1'
10	333°27.5	13.5	308°54.6	9.7'	28°13.3	1.0'	54.1'
11	348°27.4	14.0	323°23.3	9.7'	28°14.4	0.9'	54.1'
12	3°27.2	S20°14.5	337°52.0	9.7'	S28°15.3	0.8'	54.1'
13	18°27.0	15.0	352°20.6	9.7'	28°16.1	0.6'	54.1'
14	33°26.9	15.6	6°49.3	9.6'	28°16.7	0.5'	54.1'
15	48°26.7	· · 16.1	21°17.9	9.6'	28°17.2	0.3'	54.1'
16	63°26.5	16.6	35°46.6	9.6'	28°17.5	0.2'	54.1'
17	78°26.3	17.1	50°15.2	9.6'	28°17.7	0.1'	54.1'
18	93°26.2	S20°17.7	64°43.8	9.6'	S28°17.8	-0.1'	54.2'
19	108°26.0	18.2	79°12.4	9.6'	28°17.7	-0.2'	54.2'
20	123°25.8	18.7	93°40.9	9.6'	28°17.5	-0.3'	54.2'
21	138°25.7	· · 19.2	108°09.5	9.6'	28°17.2	-0.5'	54.2'
22	153°25.5	19.8	122°38.1	9.6'	28°16.7	-0.6'	54.2'
23	168°25.3	20.3	137°06.7	9.6'	28°16.0	-0.8'	54.2'
SD = 16.2' d = 0.5'			SD = 14.7'				
Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	183°25.1	S20°20.8	151°35.2	9.6'	S28°15.3	-0.9'	54.2'
1	198°25.0	21.3	166°03.8	9.6'	28°14.4	-1.0'	54.2'
2	213°24.8	21.8	180°32.3	9.6'	28°13.3	-1.2'	54.2'
3	228°24.6	· · 22.4	195°00.9	9.6'	28°12.1	-1.3'	54.2'
4	243°24.4	22.9	209°29.5	9.6'	28°10.8	-1.5'	54.3'
5	258°24.3	23.4	223°58.0	9.6'	28°09.3	-1.6'	54.3'
6	273°24.1	S20°23.9	238°26.6	9.6'	S28°07.7	-1.7'	54.3'
7	288°23.9	24.4	252°55.2	9.6'	28°06.0	-1.9'	54.3'
8	303°23.7	24.9	267°23.8	9.6'	28°04.1	-2.0'	54.3'
9	318°23.6	· · 25.5	281°52.4	9.6'	28°02.1	-2.2'	54.3'
10	333°23.4	26.0	296°21.0	9.6'	27°59.9	-2.3'	54.3'
11	348°23.2	26.5	310°49.6	9.6'	27°57.6	-2.4'	54.3'
12	3°23.0	S20°27.0	325°18.2	9.6'	S27°55.1	-2.6'	54.4'
13	18°22.9	27.5	339°46.8	9.6'	27°52.5	-2.7'	54.4'
14	33°22.7	28.0	354°15.5	9.7'	27°49.8	-2.9'	54.4'
15	48°22.5	· · 28.5	8°44.1	9.7'	27°47.0	-3.0'	54.4'
16	63°22.3	29.1	23°12.8	9.7'	27°43.9	-3.1'	54.4'
17	78°22.1	29.6	37°41.5	9.7'	27°40.8	-3.3'	54.4'
18	93°22.0	S20°30.1	52°10.2	9.7'	S27°37.5	-3.4'	54.4'
19	108°21.8	30.6	66°39.0	9.8'	27°34.1	-3.6'	54.5'
20	123°21.6	31.1	81°07.7	9.8'	27°30.6	-3.7'	54.5'
21	138°21.4	· · 31.6	95°36.5	9.8'	27°26.9	-3.8'	54.5'
22	153°21.3	32.1	110°05.3	9.8'	27°23.0	-4.0'	54.5'
23	168°21.1	32.6	124°34.1	9.9'	27°19.1	-4.1'	54.5'
SD = 16.2' d = 0.5'			SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:24	09:09	■	■	14:23	16:07
N 70°	07:11	08:39	10:47	12:44	14:52	16:20
68°	07:01	08:18	09:48	13:43	15:14	16:31
66°	06:52	08:00	09:15	14:17	15:31	16:40
64°	06:44	07:46	08:50	14:41	15:45	16:48
62°	06:37	07:34	08:31	15:01	15:57	16:55
60°	06:31	07:24	08:16	15:16	16:08	17:01
N 58°	06:25	07:15	08:02	15:29	16:17	17:06
56°	06:20	07:07	07:51	15:41	16:25	17:11
54°	06:16	07:00	07:41	15:51	16:32	17:16
52°	06:12	06:54	07:32	16:00	16:38	17:20
50°	06:08	06:48	07:24	16:08	16:44	17:24
45°	05:59	06:35	07:07	16:25	16:57	17:33
N 40°	05:51	06:24	06:53	16:39	17:08	17:41
35°	05:43	06:14	06:41	16:51	17:18	17:49
30°	05:36	06:05	06:31	17:01	17:27	17:56
20°	05:23	05:50	06:13	17:19	17:43	18:09
N 10°	05:09	05:35	05:57	17:35	17:57	18:23
0°	04:55	05:21	05:43	17:50	18:12	18:37
S 10°	04:39	05:05	05:28	18:05	18:27	18:54
20°	04:19	04:48	05:12	18:21	18:45	19:13
30°	03:55	04:26	04:53	18:40	19:06	19:38
35°	03:39	04:13	04:42	18:51	19:19	19:54
40°	03:19	03:58	04:29	19:03	19:35	20:14
45°	02:55	03:39	04:14	19:19	19:54	20:39
S 50°	02:20	03:15	03:56	19:37	20:18	21:14
52°	02:02	03:04	03:47	19:46	20:30	21:32
54°	01:39	02:50	03:38	19:56	20:44	21:56
56°	01:06	02:34	03:27	20:07	21:00	22:31
58°	—:—	02:15	03:14	20:20	21:20	—:—
S 60°	—:—	01:50	02:59	20:35	21:46	—:—

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	■	■	■	■	■	■
64°	■	■	■	■	■	■
62°	11:44	■	14:04	13:31	■	14:47
60°	10:50	12:05	12:48	14:26	14:57	16:03
N 58°	10:17	11:26	12:13	14:59	15:37	16:38
56°	09:53	10:59	11:47	15:23	16:04	17:04
54°	09:34	10:37	11:26	15:42	16:26	17:24
52°	09:18	10:20	11:09	15:59	16:43	17:41
50°	09:04	10:04	10:54	16:13	16:58	17:56
45°	08:35	09:34	10:25	16:42	17:29	18:25
N 40°	08:13	09:10	10:01	17:05	17:53	18:48
35°	07:55	08:51	09:42	17:24	18:12	19:07
30°	07:39	08:34	09:25	17:40	18:29	19:24
20°	07:12	08:06	08:58	18:07	18:57	19:51
N 10°	06:49	07:42	08:34	18:31	19:22	20:14
0°	06:28	07:19	08:11	18:53	19:44	20:36
S 10°	06:07	06:57	07:49	19:15	20:07	20:58
20°	05:44	06:33	07:25	19:38	20:31	21:21
30°	05:18	06:05	06:57	20:06	20:59	21:48
35°	05:03	05:48	06:40	20:22	21:16	22:04
40°	04:45	05:29	06:21	20:41	21:35	22:23
45°	04:23	05:06	05:58	21:04	21:59	22:45
S 50°	03:56	04:36	05:27	21:34	22:29	23:13
52°	03:43	04:21	05:13	21:48	22:44	23:27
54°	03:28	04:04	04:55	22:05	23:02	23:43
56°	03:11	03:44	04:34	22:26	23:23	· · ·
58°	02:49	03:18	04:08	22:51	23:49	· · ·
S 60°	02:22	02:42	03:30	23:26	· · ·	00:27

Day	Sun			Moon		
	Eqn.of Time		Mer. Pass	Mer.Pass.		Age
	00 ^h	12 ^h		Upper	Lower	
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	1-3 1-7%
21	14:12	14:05	11:46	12:40	00:15	
22	13:57	13:49	11:46	13:32	01:06	
23	13:41	13:32	11:46	14:24	01:58	

November 24, 25, 26 UT (Mon., Tue., Wed.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	63°12.4	193°58.5	S17°11.9	170°30.4	S23°05.6	306°20.6	N21°17.3	66°41.2	S04°06.9	Alpheratz	357°33.9	29°14.2
1	78°14.9	208°57.8	12.8	185°30.8	05.9	321°23.2	17.3	81°43.7	06.9	Ankaa	353°06.2	-42°10.0
2	93°17.4	223°57.2	13.7	200°31.3	06.1	336°25.8	17.3	96°46.2	06.9	Schedar	349°29.9	56°41.1
3	108°19.8	238°56.5	•• 14.6	215°31.8	•• 06.4	351°28.4	•• 17.3	111°48.6	•• 06.9	Diphda	348°46.4	-17°50.7
4	123°22.3	253°55.8	15.5	230°32.3	06.6	6°30.9	17.4	126°51.1	06.9	Achernar	335°19.2	-57°06.4
5	138°24.7	268°55.1	16.4	245°32.8	06.8	21°33.5	17.4	141°53.6	06.9	Hamal	327°50.1	23°35.2
6	153°27.2	283°54.4	S17°17.2	260°33.2	S23°07.1	36°36.1	N21°17.4	156°56.1	S04°06.8	Polaris	313°10.7	89°22.5
7	168°29.7	298°53.7	18.1	275°33.7	07.3	51°38.7	17.4	171°58.6	06.8	Acamar	315°10.8	-40°12.0
8	183°32.1	313°53.0	19.0	290°34.2	07.5	66°41.2	17.5	187°01.1	06.8	Menkar	314°05.1	4°11.6
9	198°34.6	328°52.3	•• 19.9	305°34.7	•• 07.8	81°43.8	•• 17.5	202°03.5	•• 06.8	Mirfak	308°26.7	49°57.3
10	213°37.1	343°51.6	20.8	320°35.2	08.0	96°46.4	17.5	217°06.0	06.8	Aldebaran	290°38.5	16°33.7
11	228°39.5	358°50.9	21.7	335°35.6	08.3	111°48.9	17.5	232°08.5	06.8	Rigel	281°02.9	-8°10.2
12	243°42.0	13°50.2	S17°22.6	350°36.1	S23°08.5	126°51.5	N21°17.6	247°11.0	S04°06.8	Capella	280°20.4	46°01.4
13	258°44.5	28°49.5	23.5	5°36.6	08.7	141°54.1	17.6	262°13.5	06.8	Bellatrix	278°21.8	6°22.5
14	273°46.9	43°48.8	24.3	20°37.1	09.0	156°56.7	17.6	277°16.0	06.8	Elnath	278°00.6	28°37.8
15	288°49.4	58°48.1	•• 25.2	35°37.6	•• 09.2	171°59.3	•• 17.6	292°18.4	•• 06.8	Alnilam	275°36.7	-1°11.0
16	303°51.9	73°47.4	26.1	50°38.0	09.4	187°01.8	17.7	307°20.9	06.8	Betelgeuse	270°51.0	7°24.8
17	318°54.3	88°46.7	27.0	65°38.5	09.7	202°04.4	17.7	322°23.4	06.8	Canopus	263°51.6	-52°42.3
18	333°56.8	103°46.1	S17°27.9	80°39.0	S23°09.9	217°07.0	N21°17.7	337°25.9	S04°06.8	Sirius	258°25.3	-16°45.0
19	348°59.2	118°45.4	28.7	95°39.5	10.1	232°09.6	17.7	352°28.4	06.8	Adhara	255°05.0	-29°00.2
20	4°01.7	133°44.7	29.6	110°40.0	10.3	247°12.1	17.8	7°30.8	06.8	Procyon	244°49.9	5°09.6
21	19°04.2	148°44.0	•• 30.5	125°40.4	•• 10.6	262°14.7	•• 17.8	22°33.3	•• 06.8	Pollux	243°16.2	27°57.7
22	34°06.6	163°43.3	31.4	140°40.9	10.8	277°17.3	17.8	37°35.8	06.8	Avior	234°14.1	-59°35.2
23	49°09.1	178°42.6	32.2	155°41.4	11.0	292°19.9	17.8	52°38.3	06.8	Suhail	222°45.7	-43°31.9
Mer.pass. 19:44		ν -0.7' d 0.9' m -3.90		ν 0.5' d 0.2' m 1.34		ν 2.6' d 0.0' m -2.49		ν 2.5' d -0.0' m 0.88				

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	64°11.6	193°41.9	S17°33.1	170°41.9	S23°11.3	307°22.5	N21°17.9	67°40.8	S04°06.8
1	79°14.0	208°41.1	34.0	185°42.3	11.5	322°25.0	17.9	82°43.2	06.8
2	94°16.5	223°40.4	34.9	200°42.8	11.7	337°27.6	17.9	97°45.7	06.8
3	109°19.0	238°39.7	•• 35.7	215°43.3	•• 12.0	352°30.2	•• 17.9	112°48.2	•• 06.8
4	124°21.4	253°39.0	36.6	230°43.8	12.2	7°32.8	18.0	127°50.7	06.8
5	139°23.9	268°38.3	37.5	245°44.3	12.4	22°35.4	18.0	142°53.2	06.8
6	154°26.4	283°37.6	S17°38.4	260°44.7	S23°12.6	37°37.9	N21°18.0	157°55.6	S04°06.8
7	169°28.8	298°36.9	39.2	275°45.2	12.9	52°40.5	18.0	172°58.1	06.8
8	184°31.3	313°36.2	40.1	290°45.7	13.1	67°43.1	18.1	188°00.6	06.8
9	199°33.7	328°35.5	•• 41.0	305°46.2	•• 13.3	82°45.7	•• 18.1	203°03.1	•• 06.8
10	214°36.2	343°34.8	41.8	320°46.6	13.5	97°48.3	18.1	218°05.6	06.8
11	229°38.7	358°34.1	42.7	335°47.1	13.8	112°50.9	18.2	233°08.0	06.8
12	244°41.1	13°33.4	S17°43.6	350°47.6	S23°14.0	127°53.4	N21°18.2	248°10.5	S04°06.8
13	259°43.6	28°32.7	44.4	5°48.1	14.2	142°56.0	18.2	263°13.0	06.8
14	274°46.1	43°32.0	45.3	20°48.5	14.4	157°58.6	18.2	278°15.5	06.8
15	289°48.5	58°31.3	•• 46.2	35°49.0	•• 14.7	173°01.2	•• 18.3	293°18.0	•• 06.8
16	304°51.0	73°30.6	47.0	50°49.5	14.9	188°03.8	18.3	308°20.4	06.8
17	319°53.5	88°29.9	47.9	65°49.9	15.1	203°06.4	18.3	323°22.9	06.8
18	334°55.9	103°29.1	S17°48.7	80°50.4	S23°15.3	218°08.9	N21°18.3	338°25.4	S04°06.8
19	349°58.4	118°28.4	49.6	95°50.9	15.6	233°11.5	18.4	353°27.9	06.8
20	5°00.9	133°27.7	50.5	110°51.4	15.8	248°14.1	18.4	8°30.3	06.8
21	20°03.3	148°27.0	•• 51.3	125°51.8	•• 16.0	263°16.7	•• 18.4	23°32.8	•• 06.7
22	35°05.8	163°26.3	52.2	140°52.3	16.2	278°19.3	18.5	38°35.3	06.7
23	50°08.2	178°25.6	53.0	155°52.8	16.4	293°21.9	18.5	53°37.8	06.7
Mer.pass. 19:40		ν -0.7' d 0.9' m -3.90		ν 0.5' d 0.2' m 1.34		ν 2.6' d 0.0' m -2.49		ν 2.5' d -0.0' m 0.89	

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	65°10.7	193°24.9	S17°53.9	170°53.3	S23°16.7	308°24.5	N21°18.5	68°40.3	S04°06.7
1	80°13.2	208°24.2	54.7	185°53.7	16.9	323°27.1	18.5	83°42.7	06.7
2	95°15.6	223°23.4	55.6	200°54.2	17.1	338°29.6	18.6	98°45.2	06.7
3	110°18.1	238°22.7	•• 56.4	215°54.7	•• 17.3	353°32.2	•• 18.6	113°47.7	•• 06.7
4	125°20.6	253°22.0	57.3	230°55.1	17.5	8°34.8	18.6	128°50.2	06.7
5	140°23.0	268°21.3	58.1	245°55.6	17.8	23°37.4	18.6	143°52.6	06.7
6	155°25.5	283°20.6	S17°59.0	260°56.1	S23°18.0	38°40.0	N21°18.7	158°55.1	S04°06.7
7	170°28.0	298°19.9	17°59.8	275°56.6	18.2	53°42.6	18.7	173°57.6	06.7
8	185°30.4	313°19.1	18°00.7	290°57.0	18.4	68°45.2	18.7	189°00.1	06.7
9	200°32.9	328°18.4	•• 01.5	305°57.5	•• 18.6	83°47.8	•• 18.8	204°02.5	•• 06.7
10	215°35.3	343°17.7	02.4	320°58.0	18.8	98°50.4	18.8	219°05.0	06.7
11	230°37.8	358°17.0	03.2	335°58.4	19.1	113°53.0	18.8	234°07.5	06.7
12	245°40.3	13°16.3	S18°04.1	350°58.9	S23°19.3	128°55.5	N21°18.8	249°10.0	S04°06.7
13	260°42.7	28°15.5	04.9	5°59.4	19.5	143°58.1	18.9	264°12.4	06.7
14	275°45.2	43°14.8	05.8	20°59.8	19.7	159°00.7	18.9	279°14.9	06.7
15	290°47.7	58°14.1	•• 06.6	36°00.3	•• 19.9	174°03.3	•• 18.9	294°17.4	•• 06.7
16	305°50.1	73°13.4	07.4	51°00.8	20.1	189°05.9	19.0	309°19.9	06.7
17	320°52.6	88°12.7	08.3	66°01.3	20.3	204°08.5	19.0	324°22.3	06.7
18	335°55.1	103°11.9	S18°09.1	81°01.7	S23°20.5	219°11.1	N21°19.0	339°24.8	S04°06.6
19	350°57.5	118°11.2	10.0	96°02.2	20.8	234°13.7	19.0	354°27.3	06.6
20	6°00.0	133°10.5	10.8	111°02.7	21.0	249°16.3	19.1	9°29.7	06.6
21	21°02.5	148°09.8	•• 11.6	126°03.1	•• 21.2	264°18.9	•• 19.1	24°32.2	•• 06.6
22	36°04.9	163°09.0	12.5	141°03.6	21.4	279°21.5	19.1	39°34.7	06.6
23	51°07.4	178°08.3	13.3	156°04.1	21.6	294°24.1	19.2	54°37.2	06.6
Mer.pass. 19:36		ν -0.7' d 0.9' m -3.90		ν 0.5' d 0.2' m 1.33		ν 2.6' d 0.0' m -2.50		ν 2.5' d -0.0' m 0.89	

Nov 24 Mon	SHA	Mer.pass
Venus	130°46.1	11:05
Mars	107°17.9	12:38
Jupiter	243°08.2	03:34
Saturn	3°28.8	19:30

Nov 25 Tue	SHA	Mer.pass
Venus	129°30.3	11:06
Mars	106°30.3	12:37
Jupiter	243°10.9	03:30
Saturn	3°29.2	19:26

Nov 26 Wed	SHA	Mer.pass
Venus	128°14.2	11:07
Mars	105°42.6	12:36
Jupiter	243°13.8	03:26
Saturn	3°29.5	19:22

Horizontal parallax	
Venus:	0.1
Mars:	0.1

Sun							Moon						
Mon	GHA	Dec	GHA	ν	Dec	d	HP						
0	183°20.9	S20°33.1	139°03.0	9.9'	S27°15.0	-4.2'	54.5'						
1	198°20.7	33.6	153°31.9	9.9'	27°10.7	-4.4'	54.5'						
2	213°20.5	34.1	168°00.8	9.9'	27°06.4	-4.5'	54.6'						
3	228°20.3	· · 34.6	182°29.7	10.0'	27°01.8	-4.6'	54.6'						
4	243°20.2	35.1	196°58.7	10.0'	26°57.2	-4.8'	54.6'						
5	258°20.0	35.6	211°27.7	10.0'	26°52.4	-4.9'	54.6'						
6	273°19.8	S20°36.1	225°56.7	10.1'	S26°47.5	-5.0'	54.6'						
7	288°19.6	36.6	240°25.8	10.1'	26°42.5	-5.2'	54.6'						
8	303°19.4	37.1	254°54.8	10.1'	26°37.3	-5.3'	54.7'						
9	318°19.2	· · 37.6	269°24.0	10.2'	26°32.0	-5.4'	54.7'						
10	333°19.1	38.1	283°53.1	10.2'	26°26.5	-5.6'	54.7'						
11	348°18.9	38.6	298°22.3	10.2'	26°21.0	-5.7'	54.7'						
12	3°18.7	S20°39.1	312°51.6	10.3'	S26°15.2	-5.8'	54.7'						
13	18°18.5	39.6	327°20.8	10.3'	26°09.4	-6.0'	54.7'						
14	33°18.3	40.1	341°50.1	10.3'	26°03.4	-6.1'	54.8'						
15	48°18.1	· · 40.6	356°19.5	10.4'	25°57.3	-6.2'	54.8'						
16	63°18.0	41.1	10°48.8	10.4'	25°51.1	-6.4'	54.8'						
17	78°17.8	41.6	25°18.3	10.5'	25°44.8	-6.5'	54.8'						
18	93°17.6	S20°42.1	39°47.7	10.5'	S25°38.3	-6.6'	54.8'						
19	108°17.4	42.6	54°17.2	10.5'	25°31.7	-6.7'	54.8'						
20	123°17.2	43.1	68°46.8	10.6'	25°24.9	-6.9'	54.9'						
21	138°17.0	· · 43.6	83°16.3	10.6'	25°18.1	-7.0'	54.9'						
22	153°16.8	44.1	97°46.0	10.7'	25°11.1	-7.1'	54.9'						
23	168°16.6	44.5	112°15.6	10.7'	25°03.9	-7.2'	54.9'						
SD = 16.2' d = 0.5'							SD = 14.9'						
Tue	GHA	Dec	GHA	ν	Dec	d	HP						
0	183°16.4	S20°45.0	126°45.3	10.8'	S24°56.7	-7.4'	54.9'						
1	198°16.3	45.5	141°15.1	10.8'	24°49.3	-7.5'	55.0'						
2	213°16.1	46.0	155°44.9	10.8'	24°41.8	-7.6'	55.0'						
3	228°15.9	· · 46.5	170°14.7	10.9'	24°34.2	-7.7'	55.0'						
4	243°15.7	47.0	184°44.6	10.9'	24°26.5	-7.9'	55.0'						
5	258°15.5	47.5	199°14.6	11.0'	24°18.6	-8.0'	55.1'						
6	273°15.3	S20°48.0	213°44.5	11.0'	S24°10.7	-8.1'	55.1'						
7	288°15.1	48.4	228°14.6	11.1'	24°02.6	-8.2'	55.1'						
8	303°14.9	48.9	242°44.6	11.1'	23°54.4	-8.3'	55.1'						
9	318°14.7	· · 49.4	257°14.8	11.2'	23°46.0	-8.5'	55.1'						
10	333°14.5	49.9	271°44.9	11.2'	23°37.6	-8.6'	55.2'						
11	348°14.4	50.4	286°15.1	11.3'	23°29.0	-8.7'	55.2'						
12	3°14.2	S20°50.9	300°45.4	11.3'	S23°20.3	-8.8'	55.2'						
13	18°14.0	51.3	315°15.7	11.4'	23°11.5	-8.9'	55.2'						
14	33°13.8	51.8	329°46.0	11.4'	23°02.6	-9.0'	55.2'						
15	48°13.6	· · 52.3	344°16.4	11.4'	22°53.5	-9.2'	55.3'						
16	63°13.4	52.8	358°46.9	11.5'	22°44.4	-9.3'	55.3'						
17	78°13.2	53.2	13°17.4	11.5'	22°35.1	-9.4'	55.3'						
18	93°13.0	S20°53.7	27°47.9	11.6'	S22°25.7	-9.5'	55.3'						
19	108°12.8	54.2	42°18.5	11.6'	22°16.3	-9.6'	55.4'						
20	123°12.6	54.7	56°49.1	11.7'	22°06.7	-9.7'	55.4'						
21	138°12.4	· · 55.2	71°19.8	11.7'	21°56.9	-9.8'	55.4'						
22	153°12.2	55.6	85°50.5	11.8'	21°47.1	-9.9'	55.4'						
23	168°12.0	56.1	100°21.3	11.8'	21°37.2	-10.0'	55.5'						
SD = 16.2' d = 0.5'							SD = 15.0'						
Wed	GHA	Dec	GHA	ν	Dec	d	HP						
0	183°11.8	S20°56.6	114°52.1	11.9'	S21°27.2	-10.1'	55.5'						
1	198°11.6	57.0	129°23.0	11.9'	21°17.0	-10.3'	55.5'						
2	213°11.4	57.5	143°53.9	12.0'	21°06.8	-10.4'	55.5'						
3	228°11.2	· · 58.0	158°24.9	12.0'	20°56.4	-10.5'	55.6'						
4	243°11.0	58.5	172°55.9	12.1'	20°45.9	-10.6'	55.6'						
5	258°10.8	58.9	187°27.0	12.1'	20°35.4	-10.7'	55.6'						
6	273°10.6	S20°59.4	201°58.1	12.1'	S20°24.7	-10.8'	55.6'						
7	288°10.4	20°59.9	216°29.2	12.2'	20°13.9	-10.9'	55.7'						
8	303°10.2	21°00.3	231°00.4	12.2'	20°03.0	-11.0'	55.7'						
9	318°10.0	· · 00.8	245°31.6	12.3'	19°52.1	-11.1'	55.7'						
10	333°09.8	01.3	260°02.9	12.3'	19°41.0	-11.2'	55.8'						
11	348°09.6	01.7	274°34.2	12.4'	19°29.8	-11.3'	55.8'						
12	3°09.4	S21°02.2	289°05.6	12.4'	S19°18.5	-11.4'	55.8'						
13	18°09.2	02.7	303°37.0	12.4'	19°07.1	-11.5'	55.8'						
14	33°09.0	03.1	318°08.4	12.5'	18°55.7	-11.6'	55.9'						
15	48°08.8	· · 03.6	332°39.9	12.5'	18°44.1	-11.7'	55.9'						
16	63°08.6	04.0	347°11.5	12.6'	18°32.4	-11.8'	55.9'						
17	78°08.4	04.5	1°43.0	12.6'	18°20.7	-11.9'	55.9'						
18	93°08.2	S21°05.0	16°14.7	12.7'	S18°08.8	-12.0'	56.0'						
19	108°08.0	05.4	30°46.3	12.7'	17°56.9	-12.0'	56.0'						
20	123°07.8	05.9	45°18.0	12.7'	17°44.8	-12.1'	56.0'						
21	138°07.6	· · 06.3	59°49.7	12.8'	17°32.7	-12.2'	56.1'						
22	153°07.4	06.8	74°21.5	12.8'	17°20.4	-12.3'	56.1'						
23	168°07.2	07.3	88°53.3	12.8'	17°08.1	-12.4'	56.1'						
SD = 16.2' d = 0.5'							SD = 15.1'						

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:34	09:23	■	■	14:10	15:59
N 70°	07:20	08:51	■	■	14:43	16:13
68°	07:08	08:27	10:04	13:30	15:06	16:25
66°	06:58	08:09	09:26	14:08	15:25	16:35
64°	06:50	07:54	08:59	14:34	15:40	16:43
62°	06:43	07:41	08:39	14:54	15:53	16:51
60°	06:36	07:30	08:22	15:11	16:03	16:57
N 58°	06:30	07:21	08:09	15:25	16:13	17:03
56°	06:25	07:12	07:57	15:37	16:21	17:08
54°	06:20	07:05	07:46	15:47	16:29	17:13
52°	06:16	06:58	07:37	15:57	16:36	17:18
50°	06:12	06:52	07:29	16:05	16:42	17:22
45°	06:02	06:38	07:11	16:23	16:55	17:32
N 40°	05:53	06:27	06:56	16:37	17:07	17:40
35°	05:46	06:17	06:44	16:50	17:17	17:48
30°	05:38	06:08	06:33	17:00	17:26	17:56
20°	05:24	05:51	06:15	17:19	17:43	18:10
N 10°	05:10	05:36	05:59	17:35	17:58	18:24
0°	04:56	05:21	05:43	17:51	18:13	18:39
S 10°	04:39	05:05	05:28	18:06	18:29	18:55
20°	04:19	04:47	05:11	18:23	18:47	19:15
30°	03:53	04:25	04:52	18:42	19:09	19:41
35°	03:37	04:12	04:41	18:54	19:22	19:58
40°	03:17	03:56	04:28	19:07	19:38	20:18
45°	02:51	03:37	04:12	19:22	19:58	20:44
S 50°	02:15	03:12	03:53	19:42	20:23	21:20
52°	01:56	03:00	03:44	19:51	20:36	21:40
54°	01:30	02:45	03:34	20:01	20:50	22:06
56°	00:52	02:28	03:22	20:13	21:07	22:46
58°	—:—	02:07	03:09	20:27	21:29	—:—
S 60°	—:—	01:40	02:53	20:43	21:57	—:—

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	■	■	■	■	■	■
N 70°	■	■	15:26	■	■	18:40
68°	■	■	14:41	■	■	19:24
66°	■	14:53	14:11	■	17:31	19:53
64°	■	14:06	13:48	■	18:17	20:14
62°	13:43	13:35	13:29	16:56	18:47	20:32
60°	13:05	13:12	13:14	17:33	19:09	20:46
N 58°	12:39	12:53	13:01	17:59	19:27	20:58
56°	12:18	12:37	12:50	18:19	19:43	21:08
54°	12:01	12:24	12:40	18:36	19:56	21:18
52°	11:46	12:12	12:31	18:51	20:07	21:26
50°	11:33	12:02	12:23	19:03	20:17	21:33
45°	11:06	11:39	12:06	19:29	20:38	21:49
N 40°	10:45	11:22	11:53	19:50	20:55	22:01
35°	10:27	11:07	11:41	20:07	21:09	22:12
30°	10:12	10:53	11:30	20:22	21:21	22:21
20°	09:46	10:31	11:12	20:46	21:42	22:37
N 10°	09:24	10:12	10:56	21:08	22:00	22:51
0°	09:03	09:53	10:42	21:27	22:17	23:04
S 10°	08:42	09:35	10:27	21:47	22:33	23:17
20°	08:19	09:15	10:11	22:08	22:51	23

November 27, 28, 29 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec			
0	66°09.8	193°07.6	S18°14.1	171°04.5	S23°21.8	309°26.7	N21°19.2	69°39.6	S04°06.6	Alpheratz	357°33.9	29°14.2					
1	81°12.3	208°06.9	15.0	186°05.0	22.0	324°29.3	19.2	84°42.1	06.6	Ankaa	353°06.2	-42°10.0					
2	96°14.8	223°06.1	15.8	201°05.5	22.2	339°31.9	19.2	99°44.6	06.6	Schedar	349°29.9	56°41.1					
3	111°17.2	238°05.4	•• 16.6	216°05.9	•• 22.4	354°34.5	•• 19.3	114°47.1	•• 06.6	Diphda	348°46.4	-17°50.7					
4	126°19.7	253°04.7	17.5	231°06.4	22.7	9°37.1	19.3	129°49.5	06.6	Achernar	335°19.2	-57°06.4					
5	141°22.2	268°03.9	18.3	246°06.9	22.9	24°39.7	19.3	144°52.0	06.6	Hamal	327°50.1	23°35.2					
6	156°24.6	283°03.2	S18°19.1	261°07.3	S23°23.1	39°42.3	N21°19.4	159°54.5	S04°06.6	Polaris	313°10.9	89°22.5					
7	171°27.1	298°02.5	20.0	276°07.8	23.3	54°44.9	19.4	174°56.9	06.6	Acamar	315°10.8	-40°12.0					
8	186°29.6	313°01.8	20.8	291°08.3	23.5	69°47.5	19.4	189°59.4	06.6	Menkar	314°05.1	4°11.6					
9	201°32.0	328°01.0	•• 21.6	306°08.7	•• 23.7	84°50.1	•• 19.4	205°01.9	•• 06.6	Mirfak	308°26.7	49°57.3					
10	216°34.5	343°00.3	22.4	321°09.2	23.9	99°52.7	19.5	220°04.4	06.5	Aldebaran	290°38.4	16°33.7					
11	231°37.0	357°59.6	23.3	336°09.7	24.1	114°55.3	19.5	235°06.8	06.5	Rigel	281°02.8	-8°10.2					
12	246°39.4	12°58.8	S18°24.1	351°10.1	S23°24.3	129°57.9	N21°19.5	250°09.3	S04°06.5	Capella	280°20.3	46°01.4					
13	261°41.9	27°58.1	24.9	6°10.6	24.5	145°00.5	19.6	265°11.8	06.5	Bellatrix	278°21.8	6°22.5					
14	276°44.3	42°57.4	25.7	21°11.0	24.7	160°03.1	19.6	280°14.2	06.5	Elnath	278°00.6	28°37.8					
15	291°46.8	57°56.6	•• 26.6	36°11.5	•• 24.9	175°05.7	•• 19.6	295°16.7	•• 06.5	Alnilam	275°36.7	-1°11.0					
16	306°49.3	72°55.9	27.4	51°12.0	25.1	190°08.3	19.7	310°19.2	06.5	Betelgeuse	270°51.0	7°24.8					
17	321°51.7	87°55.2	28.2	66°12.4	25.3	205°10.9	19.7	325°21.6	06.5	Canopus	263°51.6	-52°42.3					
18	336°54.2	102°54.4	S18°29.0	81°12.9	S23°25.5	220°13.5	N21°19.7	340°24.1	S04°06.5	Sirius	258°25.3	-16°45.0					
19	351°56.7	117°53.7	29.8	96°13.4	25.7	235°16.1	19.7	355°26.6	06.5	Adhara	255°05.0	-29°00.3					
20	6°59.1	132°53.0	30.7	111°13.8	25.9	250°18.7	19.8	10°29.1	06.5	Procyon	244°49.9	5°09.5					
21	22°01.6	147°52.2	•• 31.5	126°14.3	•• 26.1	265°21.3	•• 19.8	25°31.5	•• 06.5	Pollux	243°16.2	27°57.7					
22	37°04.1	162°51.5	32.3	141°14.8	26.3	280°23.9	19.8	40°34.0	06.5	Avior	234°14.1	-59°35.2					
23	52°06.5	177°50.7	33.1	156°15.2	26.5	295°26.5	19.9	55°36.5	06.5	Suhail	222°45.6	-43°32.0					
Mer.pass. 19:32			ν -0.7' d 0.8' m -3.90			ν 0.5' d 0.2' m 1.33			ν 2.6' d 0.0' m -2.51			ν 2.5' d -0.0' m 0.90					

Fri	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	67°09.0	192°50.0	S18°33.9	171°15.7	S23°26.7	310°29.1	N21°19.9	70°38.9	S04°06.4					
1	82°11.5	207°49.3	34.7	186°16.1	26.9	325°31.7	19.9	85°41.4	06.4					
2	97°13.9	222°48.5	35.5	201°16.6	27.1	340°34.3	20.0	100°43.9	06.4					
3	112°16.4	237°47.8	•• 36.4	216°17.1	•• 27.3	355°36.9	•• 20.0	115°46.3	•• 06.4					
4	127°18.8	252°47.0	37.2	231°17.5	27.5	10°39.5	20.0	130°48.8	06.4					
5	142°21.3	267°46.3	38.0	246°18.0	27.7	25°42.1	20.1	145°51.3	06.4					
6	157°23.8	282°45.6	S18°38.8	261°18.5	S23°27.9	40°44.7	N21°20.1	160°53.7	S04°06.4					
7	172°26.2	297°44.8	39.6	276°18.9	28.1	55°47.3	20.1	175°56.2	06.4					
8	187°28.7	312°44.1	40.4	291°19.4	28.3	70°50.0	20.1	190°58.7	06.4					
9	202°31.2	327°43.3	•• 41.2	306°19.8	•• 28.5	85°52.6	•• 20.2	206°01.1	•• 06.4					
10	217°33.6	342°42.6	42.0	321°20.3	28.7	100°55.2	20.2	221°03.6	06.4					
11	232°36.1	357°41.8	42.8	336°20.8	28.9	115°57.8	20.2	236°06.1	06.4					
12	247°38.6	12°41.1	S18°43.6	351°21.2	S23°29.1	131°00.4	N21°20.3	251°08.5	S04°06.3					
13	262°41.0	27°40.4	44.4	6°21.7	29.3	146°03.0	20.3	266°11.0	06.3					
14	277°43.5	42°39.6	45.2	21°22.1	29.5	161°05.6	20.3	281°13.5	06.3					
15	292°45.9	57°38.9	•• 46.0	36°22.6	•• 29.7	176°08.2	•• 20.4	296°15.9	•• 06.3					
16	307°48.4	72°38.1	46.8	51°23.1	29.9	191°10.8	20.4	311°18.4	06.3					
17	322°50.9	87°37.4	47.6	66°23.5	30.1	206°13.4	20.4	326°20.9	06.3					
18	337°53.3	102°36.6	S18°48.4	81°24.0	S23°30.3	221°16.0	N21°20.5	341°23.3	S04°06.3					
19	352°55.8	117°35.9	49.2	96°24.4	30.5	236°18.7	20.5	356°25.8	06.3					
20	7°58.3	132°35.1	50.0	111°24.9	30.7	251°21.3	20.5	11°28.3	06.3					
21	23°00.7	147°34.4	•• 50.8	126°25.3	•• 30.9	266°23.9	•• 20.5	26°30.7	•• 06.3					
22	38°03.2	162°33.6	51.6	141°25.8	31.0	281°26.5	20.6	41°33.2	06.3					
23	53°05.7	177°32.9	52.4	156°26.3	31.2	296°29.1	20.6	56°35.7	06.2					
Mer.pass. 19:28			ν -0.7' d 0.8' m -3.90			ν 0.5' d 0.2' m 1.32			ν 2.6' d 0.0' m -2.52			ν 2.5' d -0.0' m 0.90		

Sat	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec						
0	68°08.1	192°32.1	S18°53.2	171°26.7	S23°31.4	311°31.7	N21°20.6	71°38.1	S04°06.2					
1	83°10.6	207°31.4	54.0	186°27.2	31.6	326°34.3	20.7	86°40.6	06.2					
2	98°13.1	222°30.6	54.8	201°27.6	31.8	341°36.9	20.7	101°43.0	06.2					
3	113°15.5	237°29.9	•• 55.6	216°28.1	•• 32.0	356°39.6	•• 20.7	116°45.5	•• 06.2					
4	128°18.0	252°29.1	56.4	231°28.5	32.2	11°42.2	20.8	131°48.0	06.2					
5	143°20.4	267°28.4	57.1	246°29.0	32.4	26°44.8	20.8	146°50.4	06.2					
6	158°22.9	282°27.6	S18°57.9	261°29.5	S23°32.6	41°47.4	N21°20.8	161°52.9	S04°06.2					
7	173°25.4	297°26.8	58.7	276°29.9	32.7	56°50.0	20.9	176°55.4	06.2					
8	188°27.8	312°26.1	18°59.5	291°30.4	32.9	71°52.6	20.9	191°57.8	06.2					
9	203°30.3	327°25.3	19°00.3	306°30.8	•• 33.1	86°55.3	•• 20.9	207°00.3	•• 06.1					
10	218°32.8	342°24.6	01.1	321°31.3	33.3	101°57.9	21.0	222°02.8	06.1					
11	233°35.2	357°23.8	01.8	336°31.7	33.5	117°00.5	21.0	237°05.2	06.1					
12	248°37.7	12°23.1	S19°02.6	351°32.2	S23°33.7	132°03.1	N21°21.0	252°07.7	S04°06.1					
13	263°40.2	27°22.3	03.4	6°32.7	33.9	147°05.7	21.1	267°10.1	06.1					
14	278°42.6	42°21.6	04.2	21°33.1	34.0	162°08.3	21.1	282°12.6	06.1					
15	293°45.1	57°20.8	•• 05.0	36°33.6	•• 34.2	177°11.0	•• 21.1	297°15.1	•• 06.1					
16	308°47.5	72°20.0	05.7	51°34.0	34.4	192°13.6	21.2	312°17.5	06.1					
17	323°50.0	87°19.3	06.5	66°34.5	34.6	207°16.2	21.2	327°20.0	06.1					
18	338°52.5	102°18.5	S19°07.3	81°34.9	S23°34.8	222°18.8	N21°21.2	342°22.4	S04°06.0					
19	353°54.9	117°17.8	08.1	96°35.4	35.0	237°21.4	21.3	357°24.9	06.0					
20	8°57.4	132°17.0	08.8	111°35.8	35.1	252°24.1	21.3	12°27.4	06.0					
21	23°59.9	147°16.2	•• 09.6	126°36.3	•• 35.3	267°26.7	•• 21.3	27°29.8	•• 06.0					
22	39°02.3	162°15.5	10.4	141°36.7	35.5	282°29.3	21.4	42°32.3	06.0					
23	54°04.8	177°14.7	11.2	156°37.2	35.7	297°31.9	21.4	57°34.7	06.0					
Mer.pass. 19:24			ν -0.8' d 0.8' m -3.90			ν 0.5' d 0.2' m 1.32			ν 2.6' d 0.0' m -2.52			ν 2.5' d -0.0' m 0.91		

Nov 27 Thu	SHA	Mer.pass
Venus	126°57.7	11:08
Mars	104°54.7	12:35
Jupiter	243°16.8	03:22
Saturn	3°29.8	19:18

Nov 28 Fri	SHA	Mer.pass
Venus	125°41.0	11:09
Mars	104°06.7	12:35
Jupiter	243°20.1	03:17
Saturn	3°29.9	19:14

Nov 29 Sat	SHA	Mer.pass
Venus	124°24.0	11:10
Mars	103°18.6	12:34
Jupiter	243°23.6	03:13
Saturn	3°30.0	19:10

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

Sun				Moon				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	183°07.0	S21°07.7		103°25.2	12.9'	S16°55.7	-12.5'	56.2'
1	198°06.8	08.2		117°57.0	12.9'	16°43.2	-12.6'	56.2'
2	213°06.6	08.6		132°29.0	13.0'	16°30.6	-12.7'	56.2'
3	228°06.4	· · 09.1		147°00.9	13.0'	16°18.0	-12.8'	56.2'
4	243°06.2	09.5		161°32.9	13.0'	16°05.2	-12.8'	56.3'
5	258°06.0	10.0		176°04.9	13.0'	15°52.4	-12.9'	56.3'
6	273°05.8	S21°10.4		190°37.0	13.1'	S15°39.4	-13.0'	56.3'
7	288°05.6	10.9		205°09.0	13.1'	15°26.4	-13.1'	56.4'
8	303°05.4	11.3		219°41.1	13.1'	15°13.3	-13.2'	56.4'
9	318°05.2	· · 11.8		234°13.3	13.2'	15°00.2	-13.3'	56.4'
10	333°05.0	12.2		248°45.5	13.2'	14°46.9	-13.3'	56.5'
11	348°04.8	12.7		263°17.7	13.2'	14°33.6	-13.4'	56.5'
12	3°04.5	S21°13.1		277°49.9	13.3'	S14°20.1	-13.5'	56.5'
13	18°04.3	13.6		292°22.1	13.3'	14°06.6	-13.6'	56.6'
14	33°04.1	14.0		306°54.4	13.3'	13°53.1	-13.7'	56.6'
15	48°03.9	· · 14.5		321°26.7	13.3'	13°39.4	-13.7'	56.6'
16	63°03.7	14.9		335°59.1	13.4'	13°25.7	-13.8'	56.7'
17	78°03.5	15.4		350°31.4	13.4'	13°11.9	-13.9'	56.7'
18	93°03.3	S21°15.8		5°03.8	13.4'	S12°58.0	-14.0'	56.7'
19	108°03.1	16.3		19°36.2	13.4'	12°44.0	-14.0'	56.8'
20	123°02.9	16.7		34°08.6	13.4'	12°30.0	-14.1'	56.8'
21	138°02.7	· · 17.1		48°41.0	13.5'	12°15.9	-14.2'	56.8'
22	153°02.5	17.6		63°13.5	13.5'	12°01.8	-14.2'	56.9'
23	168°02.2	18.0		77°46.0	13.5'	11°47.5	-14.3'	56.9'
SD = 16.2' $d = 0.5'$				SD = 15.3'				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	183°02.0	S21°18.5		92°18.4	13.5'	S11°33.2	-14.4'	56.9'
1	198°01.8	18.9		106°50.9	13.5'	11°18.8	-14.4'	57.0'
2	213°01.6	19.3		121°23.5	13.5'	11°04.4	-14.5'	57.0'
3	228°01.4	· · 19.8		135°56.0	13.5'	10°49.9	-14.6'	57.0'
4	243°01.2	20.2		150°28.5	13.6'	10°35.3	-14.6'	57.1'
5	258°01.0	20.7		165°01.1	13.6'	10°20.7	-14.7'	57.1'
6	273°00.8	S21°21.1		179°33.6	13.6'	S10°05.9	-14.8'	57.1'
7	288°00.6	21.5		194°06.2	13.6'	09°51.2	-14.8'	57.2'
8	303°00.3	22.0		208°38.8	13.6'	09°36.3	-14.9'	57.2'
9	318°00.1	· · 22.4		223°11.4	13.6'	09°21.4	-15.0'	57.2'
10	332°59.9	22.8		237°44.0	13.6'	09°06.5	-15.0'	57.3'
11	347°59.7	23.3		252°16.6	13.6'	08°51.5	-15.1'	57.3'
12	2°59.5	S21°23.7		266°49.2	13.6'	S08°36.4	-15.1'	57.4'
13	17°59.3	24.1		281°21.8	13.6'	08°21.3	-15.2'	57.4'
14	32°59.1	24.5		295°54.4	13.6'	08°06.1	-15.2'	57.4'
15	47°58.8	· · 25.0		310°27.0	13.6'	07°50.9	-15.3'	57.5'
16	62°58.6	25.4		324°59.6	13.6'	07°35.6	-15.3'	57.5'
17	77°58.4	25.8		339°32.1	13.6'	07°20.2	-15.4'	57.5'
18	92°58.2	S21°26.3		354°04.7	13.6'	S07°04.8	-15.5'	57.6'
19	107°58.0	26.7		8°37.3	13.6'	06°49.4	-15.5'	57.6'
20	122°57.8	27.1		23°09.9	13.6'	06°33.9	-15.6'	57.6'
21	137°57.5	· · 27.5		37°42.5	13.6'	06°18.3	-15.6'	57.7'
22	152°57.3	28.0		52°15.0	13.6'	06°02.7	-15.7'	57.7'
23	167°57.1	28.4		66°47.6	13.5'	05°47.0	-15.7'	57.8'
SD = 16.2' $d = 0.4'$				SD = 15.5'				
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	182°56.9	S21°28.8		81°20.1	13.5'	S05°31.3	-15.7'	57.8'
1	197°56.7	29.2		95°52.6	13.5'	05°15.6	-15.8'	57.8'
2	212°56.4	29.6		110°25.2	13.5'	04°59.8	-15.8'	57.9'
3	227°56.2	· · 30.1		124°57.7	13.5'	04°44.0	-15.9'	57.9'
4	242°56.0	30.5		139°30.1	13.5'	04°28.1	-15.9'	57.9'
5	257°55.8	30.9		154°02.6	13.4'	04°12.2	-16.0'	58.0'
6	272°55.6	S21°31.3		168°35.0	13.4'	S03°56.2	-16.0'	58.0'
7	287°55.3	31.7		183°07.5	13.4'	03°40.2	-16.0'	58.1'
8	302°55.1	32.2		197°39.8	13.4'	03°24.2	-16.1'	58.1'
9	317°54.9	· · 32.6		212°12.2	13.3'	03°08.1	-16.1'	58.1'
10	332°54.7	33.0		226°44.6	13.3'	02°52.0	-16.1'	58.2'
11	347°54.5	33.4		241°16.9	13.3'	02°35.9	-16.2'	58.2'
12	2°54.2	S21°33.8		255°49.2	13.3'	S02°19.7	-16.2'	58.2'
13	17°54.0	34.2		270°21.5	13.2'	02°03.5	-16.2'	58.3'
14	32°53.8	34.7		284°53.7	13.2'	01°47.3	-16.3'	58.3'
15	47°53.6	· · 35.1		299°25.9	13.2'	01°31.0	-16.3'	58.4'
16	62°53.4	35.5		313°58.1	13.1'	01°14.7	-16.3'	58.4'
17	77°53.1	35.9		328°30.2	13.1'	00°58.4	-16.4'	58.4'
18	92°52.9	S21°36.3		343°02.3	13.1'	S00°42.0	-16.4'	58.5'
19	107°52.7	36.7		357°34.3	13.0'	00°25.7	-16.4'	58.5'
20	122°52.5	37.1		12°06.4	13.0'	S00°09.3	-16.4'	58.6'
21	137°52.2	· · 37.5		26°38.3	12.9'	N00°07.2	16.4'	58.6'
22	152°52.0	37.9		41°10.3	12.9'	00°23.6	16.5'	58.6'
23	167°51.8	38.3		55°42.2	12.8'	00°40.1	16.5'	58.7'
SD = 16.2' $d = 0.4'$				SD = 15.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:43	09:37	■	■	13:58	15:52
N 70°	07:28	09:01	■	■	14:34	16:07
68°	07:15	08:36	10:20	13:15	14:59	16:20
66°	07:05	08:16	09:37	13:59	15:19	16:30
64°	06:56	08:00	09:08	14:27	15:35	16:39
62°	06:48	07:47	08:47	14:49	15:48	16:47
60°	06:41	07:36	08:29	15:06	16:00	16:54
N 58°	06:35	07:26	08:15	15:21	16:10	17:00
56°	06:30	07:17	08:02	15:34	16:18	17:06
54°	06:24	07:09	07:51	15:44	16:26	17:11
52°	06:20	07:02	07:42	15:54	16:33	17:16
50°	06:15	06:56	07:33	16:03	16:40	17:20
45°	06:05	06:42	07:15	16:21	16:54	17:31
N 40°	05:56	06:30	07:00	16:36	17:06	17:39
35°	05:48	06:19	06:47	16:49	17:16	17:48
30°	05:41	06:10	06:36	17:00	17:26	17:55
20°	05:26	05:53	06:17	17:19	17:43	18:10
N 10°	05:12	05:38	06:00	17:36	17:58	18:24
0°	04:56	05:22	05:44	17:52	18:14	18:40
S 10°	04:39	05:06	05:28	18:08	18:30	18:57
20°	04:19	04:47	05:11	18:25	18:49	19:18
30°	03:52	04:25	04:52	18:45	19:12	19:44
35°	03:36	04:11	04:40	18:57	19:25	20:01
40°	03:15	03:55	04:26	19:10	19:42	20:22
45°	02:48	03:35	04:10	19:26	20:02	20:49
S 50°	02:11	03:09	03:51	19:46	20:28	21:27
52°	01:50	02:56	03:41	19:56	20:41	21:48
54°	01:22	02:41	03:31	20:06	20:56	22:17
56°	00:37	02:23	03:18	20:19	21:15	23:05
58°	—:—	02:01	03:04	20:33	21:37	—:—
S 60°	—:—	01:31	02:48	20:50	22:08	—:—

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	14:51	14:02	13:25	20:55	23:20	· · ·
N 70°	14:25	13:50	13:22	21:18	23:29	· · ·
68°	14:05	13:41	13:20	21:36	23:36	· · ·
66°	13:49	13:33	13:18	21:51	23:42	· · ·
64°	13:36	13:26	13:17	22:02	23:47	· · ·
62°	13:24	13:20	13:15	22:12	23:51	· · ·
60°	13:15	13:15	13:14	22:21	23:55	· · ·
N 58°	13:06	13:10	13:13	22:28	23:58	· · ·
56°	12:59	13:06	13:12	22:34	· · ·	00:01
54°	12:52	13:02	13:11	22:40	· · ·	00:03
52°	12:46	12:59	13:10	22:45	· · ·	00:06
50°	12:41	12:56	13:09	22:50	· · ·	00:08
45°	12:29	12:49	13:08	23:00	· · ·	00:12
N 40°	12:19	12:43	13:06	23:08	· · ·	00:16
35°	12:11	12:38	13:05	23:15	· · ·	00:19
30°	12:03	12:34	13:04	23:22	· · ·	00:22
20°	11:50	12:27	13:02	23:32	· · ·	00:27
N 10°	11:39	12:20	13:01	23:41	· · ·	00:31
0°	11:28	12:14	12:59	23:50		

November 30, 01, 02 UT (Sun., Mon., Tue.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Sun	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec		GHA	Dec		SHA	Dec	
0	69°07.3	192°13.9	S19°11.9	171°37.6	S23°35.9	312°34.5	N21°21.4	72°37.2	S04°06.0						Alpheratz	357°33.9	29°14.2
1	84°09.7	207°13.2	12.7	186°38.1	36.1	327°37.2	21.5	87°39.7	06.0						Ankaa	353°06.2	-42°10.0
2	99°12.2	222°12.4	13.5	201°38.6	36.2	342°39.8	21.5	102°42.1	06.0						Schedar	349°29.9	56°41.1
3	114°14.7	237°11.6	•• 14.2	216°39.0	•• 36.4	357°42.4	•• 21.5	117°44.6	•• 05.9						Diphda	348°46.4	-17°50.7
4	129°17.1	252°10.9	15.0	231°39.5	36.6	12°45.0	21.6	132°47.0	05.9						Achernar	335°19.2	-57°06.5
5	144°19.6	267°10.1	15.8	246°39.9	36.8	27°47.7	21.6	147°49.5	05.9						Hamal	327°50.1	23°35.2
6	159°22.0	282°09.4	S19°16.5	261°40.4	S23°36.9	42°50.3	N21°21.6	162°52.0	S04°05.9						Polaris	313°11.3	89°22.5
7	174°24.5	297°08.6	17.3	276°40.8	37.1	57°52.9	21.7	177°54.4	05.9						Acamar	315°10.8	-40°12.1
8	189°27.0	312°07.8	18.1	291°41.3	37.3	72°55.5	21.7	192°56.9	05.9						Menkar	314°05.1	4°11.6
9	204°29.4	327°07.0	•• 18.8	306°41.7	•• 37.5	87°58.1	•• 21.7	207°59.3	•• 05.9						Mirfak	308°26.7	49°57.3
10	219°31.9	342°06.3	19.6	321°42.2	37.7	103°00.8	21.8	223°01.8	05.9						Aldebaran	290°38.4	16°33.7
11	234°34.4	357°05.5	20.4	336°42.6	37.8	118°03.4	21.8	238°04.3	05.9						Rigel	281°02.8	-8°10.2
12	249°36.8	12°04.7	S19°21.1	351°43.1	S23°38.0	133°06.0	N21°21.8	253°06.7	S04°05.8						Capella	280°20.3	46°01.4
13	264°39.3	27°04.0	21.9	6°43.5	38.2	148°08.6	21.9	268°09.2	05.8						Bellatrix	278°21.8	6°22.4
14	279°41.8	42°03.2	22.6	21°44.0	38.4	163°11.3	21.9	283°11.6	05.8						Elnath	278°00.6	28°37.8
15	294°44.2	57°02.4	•• 23.4	36°44.4	•• 38.5	178°13.9	•• 21.9	298°14.1	•• 05.8						Alnilam	275°36.7	-1°11.1
16	309°46.7	72°01.7	24.1	51°44.9	38.7	193°16.5	22.0	313°16.5	05.8						Betelgeuse	270°51.0	7°24.8
17	324°49.2	87°00.9	24.9	66°45.3	38.9	208°19.2	22.0	328°19.0	05.8						Canopus	263°51.6	-52°42.4
18	339°51.6	102°00.1	S19°25.6	81°45.8	S23°39.0	223°21.8	N21°22.0	343°21.5	S04°05.8						Sirius	258°25.3	-16°45.0
19	354°54.1	116°59.3	26.4	96°46.2	39.2	238°24.4	22.1	358°23.9	05.8						Adhara	255°05.0	-29°00.3
20	9°56.5	131°58.6	27.2	111°46.7	39.4	253°27.0	22.1	13°26.4	05.7						Procyon	244°49.8	5°09.5
21	24°59.0	146°57.8	•• 27.9	126°47.1	•• 39.6	268°29.7	•• 22.1	28°28.8	•• 05.7						Pollux	243°16.2	27°57.7
22	40°01.5	161°57.0	28.7	141°47.6	39.7	283°32.3	22.2	43°31.3	05.7						Avior	234°14.1	-59°35.2
23	55°03.9	176°56.2	29.4	156°48.0	39.9	298°34.9	22.2	58°33.7	05.7						Suhail	222°45.6	-43°32.0
Mer.pass. 19:20			ν -0.8' d 0.8' m -3.90	ν 0.5' d 0.2' m 1.31	ν 2.6' d 0.0' m -2.53	ν 2.5' d -0.0' m 0.91											

Mon	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec	
0	70°06.4	191°55.5	S19°30.1	171°48.5	S23°40.1	313°37.6	N21°22.2	73°36.2	S04°05.7		
1	85°08.9	206°54.7	30.9	186°48.9	40.2	328°40.2	22.3	88°38.7	05.7		
2	100°11.3	221°53.9	31.6	201°49.4	40.4	343°42.8	22.3	103°41.1	05.7		
3	115°13.8	236°53.1	•• 32.4	216°49.8	•• 40.6	358°45.4	•• 22.3	118°43.6	•• 05.6		
4	130°16.3	251°52.4	33.1	231°50.3	40.8	13°48.1	22.4	133°46.0	05.6		
5	145°18.7	266°51.6	33.9	246°50.7	40.9	28°50.7	22.4	148°48.5	05.6		
6	160°21.2	281°50.8	S19°34.6	261°51.1	S23°41.1	43°53.3	N21°22.4	163°50.9	S04°05.6		
7	175°23.6	296°50.0	35.4	276°51.6	41.3	58°56.0	22.5	178°53.4	05.6		
8	190°26.1	311°49.2	36.1	291°52.0	41.4	73°58.6	22.5	193°55.8	05.6		
9	205°28.6	326°48.5	•• 36.8	306°52.5	•• 41.6	89°01.2	•• 22.5	208°58.3	•• 05.6		
10	220°31.0	341°47.7	37.6	321°52.9	41.8	104°03.9	22.6	224°00.8	05.5		
11	235°33.5	356°46.9	38.3	336°53.4	41.9	119°06.5	22.6	239°03.2	05.5		
12	250°36.0	11°46.1	S19°39.1	351°53.8	S23°42.1	134°09.1	N21°22.6	254°05.7	S04°05.5		
13	265°38.4	26°45.3	39.8	6°54.3	42.3	149°11.8	22.7	269°08.1	05.5		
14	280°40.9	41°44.6	40.5	21°54.7	42.4	164°14.4	22.7	284°10.6	05.5		
15	295°43.4	56°43.8	•• 41.3	36°55.2	•• 42.6	179°17.0	•• 22.8	299°13.0	•• 05.5		
16	310°45.8	71°43.0	42.0	51°55.6	42.7	194°19.7	22.8	314°15.5	05.5		
17	325°48.3	86°42.2	42.7	66°56.1	42.9	209°22.3	22.8	329°17.9	05.4		
18	340°50.8	101°41.4	S19°43.5	81°56.5	S23°43.1	224°24.9	N21°22.9	344°20.4	S04°05.4		
19	355°53.2	116°40.6	44.2	96°56.9	43.2	239°27.6	22.9	359°22.8	05.4		
20	10°55.7	131°39.9	44.9	111°57.4	43.4	254°30.2	22.9	14°25.3	05.4		
21	25°58.1	146°39.1	•• 45.6	126°57.8	•• 43.6	269°32.9	•• 23.0	29°27.7	•• 05.4		
22	41°00.6	161°38.3	46.4	141°58.3	43.7	284°35.5	23.0	44°30.2	05.4		
23	56°03.1	176°37.5	47.1	156°58.7	43.9	299°38.1	23.0	59°32.6	05.4		
Mer.pass. 19:16			ν -0.8' d 0.7' m -3.90	ν 0.4' d 0.2' m 1.31	ν 2.6' d 0.0' m -2.54	ν 2.5' d -0.0' m 0.91					

Tue	GHA	Dec	GHA	Dec		GHA	Dec		GHA	Dec	
0	71°05.5	191°36.7	S19°47.8	171°59.2	S23°44.0	314°40.8	N21°23.1	74°35.1	S04°05.3		
1	86°08.0	206°35.9	48.5	186°59.6	44.2	329°43.4	23.1	89°37.5	05.3		
2	101°10.5	221°35.1	49.3	202°00.1	44.4	344°46.0	23.1	104°40.2	05.3		
3	116°12.9	236°34.3	•• 50.0	217°00.5	•• 44.5	359°48.7	•• 23.2	119°42.4	•• 05.3		
4	131°15.4	251°33.5	50.7	232°00.9	44.7	14°51.3	23.2	134°44.9	05.3		
5	146°17.9	266°32.8	51.4	247°01.4	44.8	29°54.0	23.2	149°47.4	05.3		
6	161°20.3	281°32.0	S19°52.1	262°01.8	S23°45.0	44°56.6	N21°23.3	164°49.8	S04°05.3		
7	176°22.8	296°31.2	52.9	277°02.3	45.2	59°59.2	23.3	179°52.3	05.2		
8	191°25.3	311°30.4	53.6	292°02.7	45.3	75°01.9	23.4	194°54.7	05.2		
9	206°27.7	326°29.6	•• 54.3	307°03.2	•• 45.5	90°04.5	•• 23.4	209°57.2	•• 05.2		
10	221°30.2	341°28.8	55.0	322°03.6	45.6	105°07.2	23.4	224°59.6	05.2		
11	236°32.6	356°28.0	55.7	337°04.0	45.8	120°09.8	23.5	240°02.1	05.2		
12	251°35.1	11°27.2	S19°56.4	352°04.5	S23°45.9	135°12.4	N21°23.5	255°04.5	S04°05.2		
13	266°37.6	26°26.4	57.2	7°04.9	46.1	150°15.1	23.5	270°07.0	05.1		
14	281°40.0	41°25.6	57.9	22°05.4	46.2	165°17.7	23.6	285°09.4	05.1		
15	296°42.5	56°24.8	•• 58.6	37°05.8	•• 46.4	180°20.4	•• 23.6	300°11.9	•• 05.1		
16	311°45.0	71°24.0	19°59.3	52°06.3	46.5	195°23.0	23.6	315°14.3	05.1		
17	326°47.4	86°23.2	20°00.0	67°06.7	46.7	210°25.7	23.7	330°16.8	05.1		
18	341°49.9	101°22.4	S20°00.7	82°07.1	S23°46.9	225°28.3	N21°23.7	345°19.2	S04°05.1		
19	356°52.4	116°21.6	01.4	97°07.6	47.0	240°30.9	23.8	0°21.6	05.0		
20	11°54.8	131°20.8	02.1	112°08.0	47.2	255°33.6	23.8	15°24.1	05.0		
21	26°57.3	146°20.0	•• 02.8	127°08.5	•• 47.3	270°36.2	•• 23.8	30°26.5	•• 05.0		
22	41°59.7	161°19.3	03.5	142°08.9	47.5	285°38.9	23.9	45°29.0	05.0		
23	57°02.2	176°18.5	04.2	157°09.3	47.6	300°41.5	23.9	60°31.4	05.0		
Mer.pass. 19:12			ν -0.8' d 0.7' m -3.90	ν 0.4' d 0.2' m 1.30	ν 2.6' d 0.0' m -2.54	ν 2.5' d -0.0' m 0.92					

Nov 30 Sun	SHA	Mer.pass
Venus	123°06.7	11:12
Mars	102°30.4	12:33
Jupiter	243°27.3	03:09
Saturn	3°29.9	19:06

Dec 01 Mon	SHA	Mer.pass
Venus	121°49.1	11:13
Mars	101°42.1	12:32
Jupiter	243°31.2	03:05
Saturn	3°29.8	19:02

Dec 02 Tue	SHA	Mer.pass
Venus	120°31.2	11:14
Mars	100°53.6	12:32
Jupiter	243°35.2	03:01
Saturn	3°29.6	18:59

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

Sun							Moon								
Sun	GHA	Dec	GHA	ν	Dec	d	HP	Sun	GHA	Dec	GHA	ν	Dec	d	HP
0	182°51.6	S21°38.7	70°14.0	12.8'	N00°56.6	16.5'	58.7'	0	182°51.6	S21°38.7	70°14.0	12.8'	N00°56.6	16.5'	58.7'
1	197°51.3	39.1	84°45.8	12.8'	01°13.1	16.5'	58.7'	1	197°51.3	39.1	84°45.8	12.8'	01°13.1	16.5'	58.7'
2	212°51.1	39.5	99°17.6	12.7'	01°29.6	16.5'	58.8'	2	212°51.1	39.5	99°17.6	12.7'	01°29.6	16.5'	58.8'
3	227°50.9	· · 40.0	113°49.3	12.7'	01°46.1	16.5'	58.8'	3	227°50.9	· · 40.0	113°49.3	12.7'	01°46.1	16.5'	58.8'
4	242°50.7	40.4	128°20.9	12.6'	02°02.7	16.6'	58.9'	4	242°50.7	40.4	128°20.9	12.6'	02°02.7	16.6'	58.9'
5	257°50.4	40.8	142°52.5	12.5'	02°19.2	16.6'	58.9'	5	257°50.4	40.8	142°52.5	12.5'	02°19.2	16.6'	58.9'
6	272°50.2	S21°41.2	157°24.1	12.5'	N02°35.8	16.6'	58.9'	6	272°50.2	S21°41.2	157°24.1	12.5'	N02°35.8	16.6'	58.9'
7	287°50.0	41.6	171°55.6	12.4'	02°52.4	16.6'	59.0'	7	287°50.0	41.6	171°55.6	12.4'	02°52.4	16.6'	59.0'
8	302°49.8	42.0	186°27.0	12.4'	03°09.0	16.6'	59.0'	8	302°49.8	42.0	186°27.0	12.4'	03°09.0	16.6'	59.0'
9	317°49.5	· · 42.4	200°58.4	12.3'	03°25.5	16.6'	59.0'	9	317°49.5	· · 42.4	200°58.4	12.3'	03°25.5	16.6'	59.0'
10	332°49.3	42.8	215°29.7	12.3'	03°42.1	16.6'	59.1'	10	332°49.3	42.8	215°29.7	12.3'	03°42.1	16.6'	59.1'
11	347°49.1	43.2	230°01.0	12.2'	03°58.8	16.6'	59.1'	11	347°49.1	43.2	230°01.0	12.2'	03°58.8	16.6'	59.1'
12	2°48.8	S21°43.6	244°32.2	12.1'	N04°15.4	16.6'	59.2'	12	2°48.8	S21°43.6	244°32.2	12.1'	N04°15.4	16.6'	59.2'
13	17°48.6	43.9	259°03.3	12.1'	04°32.0	16.6'	59.2'	13	17°48.6	43.9	259°03.3	12.1'	04°32.0	16.6'	59.2'
14	32°48.4	44.3	273°34.4	12.0'	04°48.6	16.6'	59.2'	14	32°48.4	44.3	273°34.4	12.0'	04°48.6	16.6'	59.2'
15	47°48.2	· · 44.7	288°05.4	11.9'	05°05.2	16.6'	59.3'	15	47°48.2	· · 44.7	288°05.4	11.9'	05°05.2	16.6'	59.3'
16	62°47.9	45.1	302°36.3	11.9'	05°21.8	16.6'	59.3'	16	62°47.9	45.1	302°36.3	11.9'	05°21.8	16.6'	59.3'
17	77°47.7	45.5	317°07.2	11.8'	05°38.4	16.6'	59.3'	17	77°47.7	45.5	317°07.2	11.8'	05°38.4	16.6'	59.3'
18	92°47.5	S21°45.9	331°37.9	11.7'	N05°55.0	16.6'	59.4'	18	92°47.5	S21°45.9	331°37.9	11.7'	N05°55.0	16.6'	59.4'
19	107°47.2	46.3	346°08.7	11.6'	06°11.5	16.6'	59.4'	19	107°47.2	46.3	346°08.7	11.6'	06°11.5	16.6'	59.4'
20	122°47.0	46.7	0°39.3	11.6'	06°28.1	16.6'	59.5'	20	122°47.0	46.7	0°39.3	11.6'	06°28.1	16.6'	59.5'
21	137°46.8	· · 47.1	15°09.9	11.5'	06°44.7	16.5'	59.5'	21	137°46.8	· · 47.1	15°09.9	11.5'	06°44.7	16.5'	59.5'
22	152°46.5	47.5	29°40.4	11.4'	07°01.2	16.5'	59.5'	22	152°46.5	47.5	29°40.4	11.4'	07°01.2	16.5'	59.5'
23	167°46.3	47.9	44°10.8	11.3'	07°17.7	16.5'	59.6'	23	167°46.3	47.9	44°10.8	11.3'	07°17.7	16.5'	59.6'
SD = 16.2'			d = 0.4'		SD = 16.0'										

Mon	GHA	Dec	GHA	ν	Dec	d	HP
0	182°46.1	S21°48.3	58°41.1	11.2'	N07°34.2	16.5'	59.6'
1	197°45.9	48.6	73°11.3	11.2'	07°50.7	16.5'	59.6'
2	212°45.6	49.0	87°41.5	11.1'	08°07.2	16.5'	59.7'
3	227°45.4	· · 49.4	102°11.6	11.0'	08°23.7	16.4'	59.7'
4	242°45.2	49.8	116°41.6	10.9'	08°40.1	16.4'	59.7'
5	257°44.9	50.2	131°11.5	10.8'	08°56.5	16.4'	59.8'
6	272°44.7	S21°50.6	145°41.3	10.7'	N09°12.9	16.3'	59.8'
7	287°44.5	51.0	160°11.0	10.6'	09°29.2	16.3'	59.8'
8	302°44.2	51.3	174°40.6	10.5'	09°45.5	16.3'	59.9'
9	317°44.0	· · 51.7	189°10.1	10.4'	10°01.8	16.2'	59.9'
10	332°43.8	52.1	203°39.6	10.3'	10°18.0	16.2'	59.9'
11	347°43.5	52.5	218°08.9	10.2'	10°34.3	16.2'	60.0'
12	2°43.3	S21°52.9	232°38.2	10.1'	N10°50.4	16.1'	60.0'
13	17°43.1	53.2	247°07.3	10.0'	11°06.5	16.1'	60.0'
14	32°42.8	53.6	261°36.4	9.9'	11°22.6	16.0'	60.1'
15	47°42.6	· · 54.0	276°05.3	9.8'	11°38.7	16.0'	60.1'
16	62°42.3	54.4	290°34.1	9.7'	11°54.7	15.9'	60.1'
17	77°42.1	54.8	305°02.9	9.6'	12°10.6	15.9'	60.2'
18	92°41.9	S21°55.1	319°31.5	9.5'	N12°26.5	15.8'	60.2'
19	107°41.6	55.5	334°00.0	9.4'	12°42.4	15.8'	60.2'
20	122°41.4	55.9	348°28.5	9.3'	12°58.1	15.7'	60.3'
21	137°41.2	· · 56.2	2°56.8	9.2'	13°13.9	15.7'	60.3'
22	152°40.9	56.6	17°24.9	9.1'	13°29.5	15.6'	60.3'
23	167°40.7	57.0	31°53.0	9.0'	13°45.2	15.5'	60.4'
SD = 16.2'			d = 0.4'		SD = 16.3'		

Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	182°40.4	S21°57.4	46°21.0	8.9'	N14°00.7	15.5'	60.4'
1	197°40.2	57.7	60°48.9	8.7'	14°16.2	15.4'	60.4'
2	212°40.0	58.1	75°16.6	8.6'	14°31.6	15.3'	60.5'
3	227°39.7	· · 58.5	89°44.2	8.5'	14°46.9	15.3'	60.5'
4	242°39.5	58.8	104°11.7	8.4'	15°02.2	15.2'	60.5'
5	257°39.3	59.2	118°39.1	8.3'	15°17.4	15.1'	60.5'
6	272°39.0	S21°59.6	133°06.4	8.1'	N15°32.5	15.0'	60.6'
7	287°38.8	21°59.9	147°33.5	8.0'	15°47.6	15.0'	60.6'
8	302°38.5	22°00.3	162°00.5	7.9'	16°02.5	14.9'	60.6'
9	317°38.3	· · 00.7	176°27.4	7.8'	16°17.4	14.8'	60.6'
10	332°38.1	01.0	190°54.2	7.7'	16°32.2	14.7'	60.7'
11	347°37.8	01.4	205°20.9	7.5'	16°46.9	14.6'	60.7'
12	2°37.6	S22°01.8	219°47.4	7.4'	N17°01.5	14.5'	60.7'
13	17°37.3	02.1	234°13.8	7.3'	17°16.0	14.4'	60.7'
14	32°37.1	02.5	248°40.1	7.1'	17°30.5	14.3'	60.8'
15	47°36.8	· · 02.8	263°06.2	7.0'	17°44.8	14.2'	60.8'
16	62°36.6	03.2	277°32.2	6.9'	17°59.0	14.1'	60.8'
17	77°36.4	03.6	291°58.1	6.8'	18°13.1	14.0'	60.8'
18	92°36.1	S22°03.9	306°23.9	6.6'	N18°27.1	13.9'	60.9'
19	107°35.9	04.3	320°49.5	6.5'	18°41.1	13.8'	60.9'
20	122°35.6	04.6	335°15.0	6.4'	18°54.9	13.7'	60.9'
21	137°35.4	· · 05.0	349°40.4	6.2'	19°08.5	13.6'	60.9'
22	152°35.1	05.3	4°05.7	6.1'	19°22.1	13.5'	61.0'
23	167°34.9	05.7	18°30.8	6.0'	19°35.6	13.3'	61.0'
SD = 16.2'			d = 0.4'		SD = 16.5'		

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:52	09:51	■	■	13:46	15:46
N 70°	07:35	09:12	■	■	14:26	16:02
68°	07:22	08:44	10:38	13:00	14:53	16:16
66°	07:11	08:24	09:48	13:50	15:14	16:27
64°	07:02	08:07	09:17	14:21	15:31	16:36
62°	06:53	07:53	08:54	14:44	15:45	16:44
60°	06:46	07:41	08:35	15:02	15:57	16:52
N 58°	06:40	07:31	08:20	15:18	16:07	16:58
56°	06:34	07:22	08:07	15:31	16:16	17:04
54°	06:28	07:14	07:56	15:42	16:24	17:09
52°	06:23	07:06	07:46	15:52	16:32	17:14
50°	06:19	07:00	07:37	16:01	16:38	17:19
45°	06:08	06:45	07:18	16:20	16:53	17:30
N 40°	05:59	06:33	07:03	16:35	17:05	17:39
35°	05:51	06:22	06:50	16:48	17:16	17:47
30°	05:43	06:12	06:38	17:00	17:26	17:55
20°	05:28	05:55	06:19	17:19	17:43	18:10
N 10°	05:13	05:39	06:02	17:37	17:59	18:25
0°	04:57	05:23	05:45	17:53	18:15	18:41
S 10°	04:40	05:06	05:29	18:09	18:32	18:59
20°	04:19	04:47	05:12	18:27	18:51	19:20
30°	03:52	04:24	04:51	18:47	19:14	19:

December 03, 04, 05 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars	
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec
0	72°04.7	191°17.7	S20°04.9	172°09.8	S23°47.8	315°44.2	N21°23.9	75°33.9	S04°05.0	Alpheratz	357°33.9 29°14.2
1	87°07.1	206°16.9	05.6	187°10.2	47.9	330°46.8	24.0	90°36.3	04.9	Ankaa	353°06.2 -42°10.1
2	102°09.6	221°16.1	06.3	202°10.7	48.1	345°49.5	24.0	105°38.8	04.9	Schedar	349°29.9 56°41.1
3	117°12.1	236°15.3	· · 07.0	217°11.1	· · 48.2	0°52.1	· · 24.1	120°41.2	· · 04.9	Diphda	348°46.4 -17°50.7
4	132°14.5	251°14.4	07.7	232°11.5	48.4	15°54.8	24.1	135°43.7	04.9	Achernar	335°19.2 -57°06.5
5	147°17.0	266°13.6	08.4	247°12.0	48.5	30°57.4	24.1	150°46.1	04.9	Hamal	327°50.1 23°35.2
6	162°19.5	281°12.8	S20°09.1	262°12.4	S23°48.6	46°00.0	N21°24.2	165°48.6	S04°04.9	Polaris	313°11.4 89°22.5
7	177°21.9	296°12.0	09.8	277°12.9	48.8	61°02.7	24.2	180°51.0	04.8	Acamar	315°10.8 -40°12.1
8	192°24.4	311°11.2	10.5	292°13.3	48.9	76°05.3	24.2	195°53.5	04.8	Menkar	314°05.1 4°11.6
9	207°26.9	326°10.4	· · 11.2	307°13.7	· · 49.1	91°08.0	· · 24.3	210°55.9	· · 04.8	Mirfak	308°26.7 49°57.3
10	222°29.3	341°09.6	11.9	322°14.2	49.2	106°10.6	24.3	225°58.4	04.8	Aldebaran	290°38.4 16°33.7
11	237°31.8	356°08.8	12.6	337°14.6	49.4	121°13.3	24.3	241°00.8	04.8	Rigel	281°02.8 -8°10.2
12	252°34.2	11°08.0	S20°13.3	352°15.0	S23°49.5	136°15.9	N21°24.4	256°03.2	S04°04.8	Capella	280°20.3 46°01.4
13	267°36.7	26°07.2	14.0	7°15.5	49.7	151°18.6	24.4	271°05.7	04.7	Bellatrix	278°21.8 6°22.4
14	282°39.2	41°06.4	14.6	22°15.9	49.8	166°21.2	24.5	286°08.1	04.7	Elnath	278°00.6 28°37.8
15	297°41.6	56°05.6	· · 15.3	37°16.4	· · 50.0	181°23.9	· · 24.5	301°10.6	· · 04.7	Alnilam	275°36.7 -1°11.1
16	312°44.1	71°04.8	16.0	52°16.8	50.1	196°26.5	24.5	316°13.0	04.7	Betelgeuse	270°51.0 7°24.8
17	327°46.6	86°04.0	16.7	67°17.2	50.2	211°29.2	24.6	331°15.5	04.7	Canopus	263°51.6 -52°42.4
18	342°49.0	101°03.2	S20°17.4	82°17.7	S23°50.4	226°31.8	N21°24.6	346°17.9	S04°04.6	Sirius	258°25.3 -16°45.0
19	357°51.5	116°02.4	18.1	97°18.1	50.5	241°34.5	24.7	1°20.4	04.6	Adhara	255°05.0 -29°00.3
20	12°54.0	131°01.6	18.8	112°18.5	50.7	256°37.1	24.7	16°22.8	04.6	Procyon	244°49.8 5°09.5
21	27°56.4	146°00.7	· · 19.4	127°19.0	· · 50.8	271°39.8	· · 24.7	31°25.3	· · 04.6	Pollux	243°16.1 27°57.7
22	42°58.9	160°59.9	20.1	142°19.4	51.0	286°42.4	24.8	46°27.7	04.6	Avior	234°14.0 -59°35.3
23	58°01.4	175°59.1	20.8	157°19.8	51.1	301°45.1	24.8	61°30.1	04.6	Suhail	222°45.6 -43°32.0
Mer.pass. 19:09		ν -0.8' d 0.7' m -3.90		ν 0.4' d 0.1' m 1.29		ν 2.6' d 0.0' m -2.55		ν 2.4' d -0.0' m 0.92		Miaplacidus	221°37.7 -69°49.1
										Alphard	217°46.9 -8°46.2
										Regulus	207°33.6 11°50.4
										Dubhe	193°40.1 61°36.4
										Denebola	182°24.3 12°25.6
										Gienah	175°43.0 -17°41.0
										Acrux	172°59.6 -63°14.3
										Gacrux	171°51.1 -57°15.2
										Alioth	166°12.6 55°48.9
										Spica	158°21.7 -11°17.7
										Alkaid	152°51.8 49°10.8
										Hadar	148°35.5 -60°29.7
										Menkent	147°57.1 -36°29.7
										Arcturus	145°47.5 19°02.8
										Rigel Kent.	139°39.9 -60°56.4
										Kochab	137°20.9 74°02.7
										Zuben'ubi	136°55.5 -16°08.9
										Alphecca	126°03.5 26°37.6
										Antares	112°15.3 -26°29.3
										Atria	107°09.5 -69°04.4
										Sabik	102°02.3 -15°45.4
										Shaula	96°09.8 -37°07.3
										Rasalhague	95°58.2 12°32.5
										Eltanin	90°42.4 51°29.2
										Kaus Aust.	83°31.9 -34°22.4
										Vega	80°33.1 38°48.5
										Nunki	75°47.2 -26°15.9
										Altair	61°59.5 8°56.3
										Peacock	53°04.9 -56°39.3
										Deneb	49°25.6 45°22.6
										Enif	33°38.2 9°59.7
										Al Na'ir	27°32.1 -46°50.3
										Fomalhaut	15°13.7 -29°29.2
										Scheat	13°44.5 28°13.6
										Markab	13°29.2 15°20.8
										Dec 03 Wed	SHA
										Venus	119°13.0 11:15
										Mars	100°05.1 12:31
										Jupiter	243°39.5 02:57
										Saturn	3°29.2 18:55
										Dec 04 Thu	SHA
										Venus	117°54.5 11:17
										Mars	99°16.5 12:30
										Jupiter	243°43.9 02:52
										Saturn	3°28.8 18:51
										Dec 05 Fri	SHA
										Venus	116°35.7 11:18
										Mars	98°27.7 12:30
										Jupiter	243°48.6 02:48
										Saturn	3°28.2 18:47
										Horizontal parallax	
										Venus:	0.1
										Mars:	0.1
Mer.pass. 19:01		ν -0.8' d 0.7' m -3.90		ν 0.4' d 0.1' m 1.28		ν 2.7' d 0.0' m -2.56		ν 2.4' d -0.0' m 0.93			

Sun				Moon				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	182°34.7	S22°06.0		32°55.8	5.9'	N19°48.9	13.2'	61.0'
1	197°34.4	06.4		47°20.6	5.7'	20°02.1	13.1'	61.0'
2	212°34.2	06.8		61°45.3	5.6'	20°15.2	13.0'	61.0'
3	227°33.9	· · 07.1		76°09.9	5.5'	20°28.2	12.8'	61.0'
4	242°33.7	07.5		90°34.4	5.3'	20°41.0	12.7'	61.1'
5	257°33.4	07.8		104°58.7	5.2'	20°53.7	12.6'	61.1'
6	272°33.2	S22°08.2		119°22.9	5.1'	N21°06.3	12.4'	61.1'
7	287°32.9	08.5		133°47.0	4.9'	21°18.7	12.3'	61.1'
8	302°32.7	08.8		148°10.9	4.8'	21°31.0	12.2'	61.1'
9	317°32.4	· · 09.2		162°34.7	4.7'	21°43.2	12.0'	61.1'
10	332°32.2	09.5		176°58.4	4.5'	21°55.2	11.9'	61.2'
11	347°32.0	09.9		191°22.0	4.4'	22°07.1	11.7'	61.2'
12	2°31.7	S22°10.2		205°45.4	4.3'	N22°18.8	11.6'	61.2'
13	17°31.5	10.6		220°08.7	4.2'	22°30.3	11.4'	61.2'
14	32°31.2	10.9		234°31.8	4.0'	22°41.8	11.3'	61.2'
15	47°31.0	· · 11.3		248°54.9	3.9'	22°53.0	11.1'	61.2'
16	62°30.7	11.6		263°17.8	3.8'	23°04.1	10.9'	61.2'
17	77°30.5	11.9		277°40.6	3.7'	23°15.0	10.8'	61.3'
18	92°30.2	S22°12.3		292°03.2	3.5'	N23°25.8	10.6'	61.3'
19	107°30.0	12.6		306°25.8	3.4'	23°36.4	10.4'	61.3'
20	122°29.7	13.0		320°48.2	3.3'	23°46.9	10.3'	61.3'
21	137°29.5	· · 13.3		335°10.5	3.2'	23°57.1	10.1'	61.3'
22	152°29.2	13.6		349°32.7	3.1'	24°07.2	9.9'	61.3'
23	167°29.0	14.0		3°54.8	3.0'	24°17.2	9.7'	61.3'
SD = 16.2' d = 0.4'				SD = 16.6'				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	182°28.7	S22°14.3		18°16.7	2.8'	N24°26.9	9.6'	61.3'
1	197°28.5	14.6		32°38.6	2.7'	24°36.5	9.4'	61.3'
2	212°28.2	15.0		47°00.3	2.6'	24°45.9	9.2'	61.3'
3	227°28.0	· · 15.3		61°21.9	2.5'	24°55.1	9.0'	61.3'
4	242°27.7	15.6		75°43.4	2.4'	25°04.1	8.8'	61.3'
5	257°27.5	16.0		90°04.8	2.3'	25°12.9	8.6'	61.3'
6	272°27.2	S22°16.3		104°26.1	2.2'	N25°21.6	8.5'	61.3'
7	287°27.0	16.6		118°47.3	2.1'	25°30.0	8.3'	61.3'
8	302°26.7	17.0		133°08.4	2.0'	25°38.3	8.1'	61.3'
9	317°26.5	· · 17.3		147°29.4	1.9'	25°46.4	7.9'	61.3'
10	332°26.2	17.6		161°50.3	1.8'	25°54.3	7.7'	61.3'
11	347°26.0	17.9		176°11.1	1.7'	26°01.9	7.5'	61.3'
12	2°25.7	S22°18.3		190°31.8	1.6'	N26°09.4	7.3'	61.3'
13	17°25.4	18.6		204°52.4	1.5'	26°16.7	7.1'	61.3'
14	32°25.2	18.9		219°13.0	1.5'	26°23.7	6.9'	61.3'
15	47°24.9	· · 19.2		233°33.4	1.4'	26°30.6	6.7'	61.3'
16	62°24.7	19.6		247°53.8	1.3'	26°37.3	6.5'	61.3'
17	77°24.4	19.9		262°14.1	1.2'	26°43.7	6.2'	61.3'
18	92°24.2	S22°20.2		276°34.3	1.2'	N26°50.0	6.0'	61.3'
19	107°23.9	20.5		290°54.5	1.1'	26°56.0	5.8'	61.3'
20	122°23.7	20.9		305°14.6	1.0'	27°01.8	5.6'	61.3'
21	137°23.4	· · 21.2		319°34.6	1.0'	27°07.4	5.4'	61.3'
22	152°23.2	21.5		333°54.6	0.9'	27°12.8	5.2'	61.3'
23	167°22.9	21.8		348°14.5	0.8'	27°18.0	5.0'	61.3'
SD = 16.2' d = 0.3'				SD = 16.7'				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	182°22.6	S22°22.1		2°34.3	0.8'	N27°23.0	4.7'	61.3'
1	197°22.4	22.4		16°54.1	0.7'	27°27.7	4.5'	61.3'
2	212°22.1	22.8		31°13.9	0.7'	27°32.2	4.3'	61.3'
3	227°21.9	· · 23.1		45°33.6	0.7'	27°36.5	4.1'	61.3'
4	242°21.6	23.4		59°53.2	0.6'	27°40.6	3.9'	61.3'
5	257°21.4	23.7		74°12.9	0.6'	27°44.5	3.6'	61.3'
6	272°21.1	S22°24.0		88°32.5	0.6'	N27°48.1	3.4'	61.2'
7	287°20.8	24.3		102°52.0	0.5'	27°51.5	3.2'	61.2'
8	302°20.6	24.6		117°11.6	0.5'	27°54.7	3.0'	61.2'
9	317°20.3	· · 25.0		131°31.1	0.5'	27°57.7	2.7'	61.2'
10	332°20.1	25.3		145°50.6	0.5'	28°00.4	2.5'	61.2'
11	347°19.8	25.6		160°10.0	0.5'	28°03.0	2.3'	61.2'
12	2°19.5	S22°25.9		174°29.5	0.5'	N28°05.3	2.1'	61.2'
13	17°19.3	26.2		188°49.0	0.5'	28°07.3	1.8'	61.2'
14	32°19.0	26.5		203°08.4	0.5'	28°09.2	1.6'	61.1'
15	47°18.8	· · 26.8		217°27.9	0.5'	28°10.8	1.4'	61.1'
16	62°18.5	27.1		231°47.3	0.5'	28°12.2	1.2'	61.1'
17	77°18.3	27.4		246°06.8	0.5'	28°13.4	0.9'	61.1'
18	92°18.0	S22°27.7		260°26.3	0.5'	N28°14.3	0.7'	61.1'
19	107°17.7	28.0		274°45.7	0.5'	28°15.1	0.5'	61.1'
20	122°17.5	28.3		289°05.3	0.5'	28°15.6	0.3'	61.0'
21	137°17.2	· · 28.6		303°24.8	0.6'	28°15.8	0.0'	61.0'
22	152°16.9	28.9		317°44.3	0.6'	28°15.9	-0.2'	61.0'
23	167°16.7	29.2		332°03.9	0.6'	28°15.7	-0.4'	61.0'
SD = 16.2' d = 0.3'				SD = 16.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	07:59	10:05	■	■	13:35	15:41
N 70°	07:42	09:21	■	■	14:19	15:58
68°	07:28	08:52	10:57	12:43	14:48	16:12
66°	07:17	08:30	09:58	13:42	15:10	16:24
64°	07:07	08:13	09:25	14:16	15:27	16:33
62°	06:58	07:58	09:00	14:40	15:42	16:42
60°	06:51	07:46	08:41	14:59	15:54	16:50
N 58°	06:44	07:35	08:25	15:15	16:05	16:56
56°	06:38	07:26	08:12	15:28	16:14	17:03
54°	06:32	07:18	08:00	15:40	16:23	17:08
52°	06:27	07:10	07:50	15:50	16:30	17:13
50°	06:22	07:03	07:41	16:00	16:37	17:18
45°	06:11	06:48	07:21	16:19	16:52	17:29
N 40°	06:02	06:35	07:06	16:35	17:05	17:39
35°	05:53	06:24	06:52	16:48	17:16	17:47
30°	05:45	06:15	06:41	17:00	17:26	17:55
20°	05:30	05:57	06:21	17:20	17:44	18:11
N 10°	05:14	05:40	06:03	17:37	18:00	18:26
0°	04:58	05:24	05:47	17:54	18:16	18:42
S 10°	04:40	05:07	05:30	18:11	18:34	19:00
20°	04:19	04:48	05:12	18:29	18:53	19:22
30°	03:51	04:24	04:51	18:50	19:17	19:50
35°	03:34	04:10	04:39	19:02	19:31	20:07
40°	03:12	03:53	04:25	19:16	19:48	20:29
45°	02:44	03:31	04:08	19:33	20:10	20:58
S 50°	02:03	03:04	03:47	19:54	20:37	21:39
52°	01:39	02:50	03:37	20:04	20:51	22:03
54°	01:06	02:34	03:26	20:16	21:08	22:37
56°	—:—	02:14	03:13	20:29	21:28	—:—
58°	—:—	01:49	02:57	20:44	21:53	—:—
S 60°	—:—	01:14	02:39	21:03	22:30	—:—

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	□	□	□	□	□	□
N 70°	□	□	□	□	□	□
68°	10:39	□	□	09:27	□	□
66°	11:53	□	□	08:14	□	□
64°	12:30	11:35	□	07:38	10:46	□
62°	12:57	12:52	□	07:12	09:30	□
60°	13:18	13:29	14:00	06:52	08:54	10:44
N 58°	13:36	13:55	14:36	06:36	08:28	10:08
56°	13:50	14:16	15:02	06:22	08:07	09:42
54°	14:03	14:33	15:23	06:10	07:50	09:21
52°	14:14	14:48	15:40	05:59	07:36	09:04
50°	14:24	15:01	15:55	05:50	07:23	08:49
45°	14:45	15:28	16:26	05:30	06:57	08:19
N 40°	15:03	15:50	16:49	05:15	06:37	07:55
35°	15:17	16:08	17:09	05:01	06:20	07:36
30°	15:30	16:23	17:25	04:50	06:05	07:20
20°	15:52	16:49	17:54	04:30	05:40	06:52
N 10°	16:11	17:12	18:18	04:13	05:19	06:28
0°	16:29	17:33	18:41	03:57	04:59	06:05
S 10°	16:47	17:55	19:04	03:41	04:39	05:43
20°	17:06	18:18	19:28	03:24	04:18	05:19
30°	17:29	18:45	19:57	03:05	03:53	04:51
35°	17:43	19:01	20:14	02:54	03:39	04:35
40°	17:58	19:19	20:34	02:41	03:22	04:15
45°	18:17	19:42	20:58	02:26	03:03	03:52
S 50°	18:40	20:11	21:29	02:07	02:38	03:23
52°	18:51	20:25	21:44	01:59	02:26	03:08
54°	19:04	20:42	22:02	01:49	02:13	02:51
56°	19:19	21:02	22:24	01:39	01:57	02:31
58°	19:36	21:26	22:52	01:26	01:39	02:06
S 60°	19:58	22:00	23:32	01:12	01:17	01:33

Day	Sun			Moon		
	Eqn.of Time		Mer. Pass	Mer.Pass.		Age
	00 ^h	12 ^h		Upper	Lower	
	mm:ss	mm:ss	hh:mm	hh:mm	hh:mm	94-100%
03	10:19	10:07	11:50	22:44	10:13	
04	09:55	09:43	11:50	23:49	11:16	
05	09:31	09:18	11:51	—:—	12:23	

December 06, 07, 08 UT (Sat., Sun., Mon.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	75°02.1	190°18.8	S20°52.8	172°41.0	S23°57.4	318°55.5	N21°26.7	78°29.7	S04°03.5					Alpheratz	357°33.9	29°14.2	
1	90°04.6	205°18.0	53.4	187°41.4	57.5	333°58.2	26.8	93°32.1	03.5					Ankaa	353°06.2	-42°10.1	
2	105°07.0	220°17.1	54.1	202°41.9	57.7	349°00.8	26.8	108°34.6	03.5					Schedar	349°29.9	56°41.1	
3	120°09.5	235°16.3	•• 54.7	217°42.3	•• 57.8	4°03.5	•• 26.8	123°37.0	•• 03.5					Diphda	348°46.4	-17°50.7	
4	135°12.0	250°15.5	55.3	232°42.7	57.9	19°06.2	26.9	138°39.4	03.5					Achernar	335°19.2	-57°06.5	
5	150°14.4	265°14.6	55.9	247°43.1	58.0	34°08.8	26.9	153°41.9	03.4					Hamal	327°50.1	23°35.3	
6	165°16.9	280°13.8	S20°56.6	262°43.6	S23°58.1	49°11.5	N21°27.0	168°44.3	S04°03.4					Polaris	313°11.5	89°22.6	
7	180°19.4	295°12.9	57.2	277°44.0	58.3	64°14.2	27.0	183°46.7	03.4					Acamar	315°10.8	-40°12.1	
8	195°21.8	310°12.1	57.8	292°44.4	58.4	79°16.9	27.1	198°49.2	03.4					Menkar	314°05.1	4°11.6	
9	210°24.3	325°11.3	•• 58.4	307°44.8	•• 58.5	94°19.5	•• 27.1	213°51.6	•• 03.3					Mirfak	308°26.7	49°57.3	
10	225°26.7	340°10.4	59.0	322°45.3	58.6	109°22.2	27.1	228°54.0	03.3					Aldebaran	290°38.4	16°33.7	
11	240°29.2	355°09.6	20°59.7	337°45.7	58.7	124°24.9	27.2	243°56.5	03.3					Rigel	281°02.8	-8°10.2	
12	255°31.7	10°08.7	S21°00.3	352°46.1	S23°58.8	139°27.6	N21°27.2	258°58.9	S04°03.3					Capella	280°20.3	46°01.4	
13	270°34.1	25°07.9	00.9	7°46.5	58.9	154°30.2	27.3	274°01.3	03.2					Bellatrix	278°21.7	6°22.4	
14	285°36.6	40°07.1	01.5	22°47.0	59.1	169°32.9	27.3	289°03.8	03.2					Elnath	278°00.5	28°37.8	
15	300°39.1	55°06.2	•• 02.1	37°47.4	•• 59.2	184°35.6	•• 27.3	304°06.2	•• 03.2					Alnilam	275°36.6	-1°11.1	
16	315°41.5	70°05.4	02.7	52°47.8	59.3	199°38.2	27.4	319°08.6	03.2					Betelgeuse	270°50.9	7°24.7	
17	330°44.0	85°04.5	03.3	67°48.2	59.4	214°40.9	27.4	334°11.1	03.2					Canopus	263°51.5	-52°42.4	
18	345°46.5	100°03.7	S21°04.0	82°48.7	S23°59.5	229°43.6	N21°27.5	349°13.5	S04°03.1					Sirius	258°25.3	-16°45.0	
19	0°48.9	115°02.8	04.6	97°49.1	59.6	244°46.3	27.5	4°15.9	03.1					Adhara	255°04.9	-29°00.3	
20	15°51.4	130°02.0	05.2	112°49.5	59.7	259°48.9	27.5	19°18.4	03.1					Procyon	244°49.8	5°09.5	
21	30°53.9	145°01.2	•• 05.8	127°49.9	•• 59.8	274°51.6	•• 27.6	34°20.8	•• 03.1					Pollux	243°16.1	27°57.7	
22	45°56.3	160°00.3	06.4	142°50.4	23°59.9	289°54.3	27.6	49°23.2	03.0					Avior	234°14.0	-59°35.3	
23	60°58.8	174°59.5	07.0	157°50.8	24°00.0	304°57.0	27.7	64°25.6	03.0					Suhail	222°45.5	-43°32.0	
Mer.pass. 18:57			ν -0.8' d 0.6' m -3.90	ν 0.4' d 0.1' m 1.28	ν 2.7' d 0.0' m -2.57	ν 2.4' d -0.0' m 0.93											

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	76°01.2	189°58.6	S21°07.6	172°51.2	S24°00.2	319°59.6	N21°27.7	79°28.1	S04°03.0		
1	91°03.7	204°57.8	08.2	187°51.6	00.3	335°02.3	27.8	94°30.5	03.0		
2	106°06.2	219°56.9	08.8	202°52.1	00.4	350°05.0	27.8	109°32.9	02.9		
3	121°08.6	234°56.1	•• 09.4	217°52.5	•• 00.5	5°07.7	•• 27.8	124°35.4	•• 02.9		
4	136°11.1	249°55.2	10.0	232°52.9	00.6	20°10.4	27.9	139°37.8	02.9		
5	151°13.6	264°54.4	10.6	247°53.3	00.7	35°13.0	27.9	154°40.2	02.9		
6	166°16.0	279°53.5	S21°11.2	262°53.8	S24°00.8	50°15.7	N21°28.0	169°42.7	S04°02.8		
7	181°18.5	294°52.7	11.8	277°54.2	00.9	65°18.4	28.0	184°45.1	02.8		
8	196°21.0	309°51.8	12.4	292°54.6	01.0	80°21.1	28.0	199°47.5	02.8		
9	211°23.4	324°51.0	•• 13.0	307°55.0	•• 01.1	95°23.7	•• 28.1	214°50.0	•• 02.8		
10	226°25.9	339°50.1	13.6	322°55.4	01.2	110°26.4	28.1	229°52.4	02.7		
11	241°28.4	354°49.3	14.2	337°55.9	01.3	125°29.1	28.2	244°54.8	02.7		
12	256°30.8	9°48.4	S21°14.7	352°56.3	S24°01.4	140°31.8	N21°28.2	259°57.2	S04°02.7		
13	271°33.3	24°47.6	15.3	7°56.7	01.5	155°34.5	28.3	274°59.7	02.7		
14	286°35.7	39°46.7	15.9	22°57.1	01.6	170°37.2	28.3	290°02.1	02.6		
15	301°38.2	54°45.9	•• 16.5	37°57.6	•• 01.7	185°39.8	•• 28.3	305°04.5	•• 02.6		
16	316°40.7	69°45.0	17.1	52°58.0	01.8	200°42.5	28.4	320°07.0	02.6		
17	331°43.1	84°44.2	17.7	67°58.4	01.9	215°45.2	28.4	335°09.4	02.6		
18	346°45.6	99°43.3	S21°18.3	82°58.8	S24°02.0	230°47.9	N21°28.5	350°11.8	S04°02.5		
19	1°48.1	114°42.5	18.9	97°59.2	02.1	245°50.6	28.5	5°14.2	02.5		
20	16°50.5	129°41.6	19.4	112°59.7	02.2	260°53.2	28.6	20°16.7	02.5		
21	31°53.0	144°40.8	•• 20.0	128°00.1	•• 02.3	275°55.9	•• 28.6	35°19.1	•• 02.5		
22	46°55.5	159°39.9	20.6	143°00.5	02.4	290°58.6	28.6	50°21.5	02.4		
23	61°57.9	174°39.0	21.2	158°00.9	02.5	306°01.3	28.7	65°24.0	02.4		
Mer.pass. 18:53			ν -0.8' d 0.6' m -3.90	ν 0.4' d 0.1' m 1.27	ν 2.7' d 0.0' m -2.57	ν 2.4' d -0.0' m 0.94					

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec		
0	77°00.4	189°38.2	S21°21.8	173°01.3	S24°02.6	321°04.0	N21°28.7	80°26.4	S04°02.4		
1	92°02.9	204°37.3	22.3	188°01.8	02.7	336°06.7	28.8	95°28.8	02.4		
2	107°05.3	219°36.5	22.9	203°02.2	02.8	351°09.3	28.8	110°31.2	02.3		
3	122°07.8	234°35.6	•• 23.5	218°02.6	•• 02.9	6°12.0	•• 28.9	125°33.7	•• 02.3		
4	137°10.2	249°34.8	24.1	233°03.0	03.0	21°14.7	28.9	140°36.1	02.3		
5	152°12.7	264°33.9	24.6	248°03.4	03.1	36°17.4	28.9	155°38.5	02.3		
6	167°15.2	279°33.0	S21°25.2	263°03.9	S24°03.2	51°20.1	N21°29.0	170°40.9	S04°02.2		
7	182°17.6	294°32.2	25.8	278°04.3	03.3	66°22.8	29.0	185°43.4	02.2		
8	197°20.1	309°31.3	26.3	293°04.7	03.4	81°25.5	29.1	200°45.8	02.2		
9	212°22.6	324°30.5	•• 26.9	308°05.1	•• 03.5	96°28.1	•• 29.1	215°48.2	•• 02.2		
10	227°25.0	339°29.6	27.5	323°05.5	03.6	111°30.8	29.2	230°50.6	02.1		
11	242°27.5	354°28.7	28.0	338°06.0	03.7	126°33.5	29.2	245°53.1	02.1		
12	257°30.0	9°27.9	S21°28.6	353°06.4	S24°03.8	141°36.2	N21°29.2	260°55.5	S04°02.1		
13	272°32.4	24°27.0	29.2	8°06.8	03.9	156°38.9	29.3	275°57.9	02.0		
14	287°34.9	39°26.2	29.7	23°07.2	04.0	171°41.6	29.3	291°00.3	02.0		
15	302°37.3	54°25.3	•• 30.3	38°07.6	•• 04.0	186°44.3	•• 29.4	306°02.8	•• 02.0		
16	317°39.8	69°24.4	30.9	53°08.0	04.1	201°47.0	29.4	321°05.2	02.0		
17	332°42.3	84°23.6	31.4	68°08.5	04.2	216°49.7	29.5	336°07.6	01.9		
18	347°44.7	99°22.7	S21°32.0	83°08.9	S24°04.3	231°52.3	N21°29.5	351°10.0	S04°01.9		
19	2°47.2	114°21.8	32.5	98°09.3	04.4	246°55.0	29.5	6°12.5	01.9		
20	17°49.7	129°21.0	33.1	113°09.7	04.5	261°57.7	29.6	21°14.9	01.9		
21	32°52.1	144°20.1	•• 33.6	128°10.1	•• 04.6	277°00.4	•• 29.6	36°17.3	•• 01.8		
22	47°54.6	159°19.2	34.2	143°10.6	04.7	292°03.1	29.7	51°19.7	01.8		
23	62°57.1	174°18.4	34.7	158°11.0	04.8	307°05.8	29.7	66°22.2	01.8		
Mer.pass. 18:49			ν -0.9' d 0.6' m -3.90	ν 0.4' d 0.1' m 1.27	ν 2.7' d 0.0' m -2.58	ν 2.4' d -0.0' m 0.94					

Dec 06 Sat	SHA	Mer.pass
Venus	115°16.7	11:19
Mars	97°38.9	12:29
Jupiter	243°53.4	02:44
Saturn	3°27.6	18:43

Dec 07 Sun	SHA	Mer.pass
Venus	113°57.4	11:21
Mars	96°50.0	12:28
Jupiter	243°58.4	02:40
Saturn	3°26.8	18:39

Dec 08 Mon	SHA	Mer.pass
Venus	112°37.8	11:22
Mars	96°01.0	12:28
Jupiter	244°03.6	02:35
Saturn	3°26.0	18:35

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

Sun							Moon							
Sat	GHA	Dec	GHA	ν	Dec	d	HP	Lat.	Twilight		Sunrise	Sunset	Twilight	
									Naut.	Civil			Civil	Naut.
0	182°16.4	S22°29.5	346°23.6	0.7'	N28°15.3	-0.6'	61.0'	N 72°	08:06	10:18	■	■	13:24	15:36
1	197°16.2	29.8	0°43.2	0.7'	28°14.7	-0.8'	60.9'	N 70°	07:48	09:30	■	■	14:13	15:54
2	212°15.9	30.1	15°02.9	0.7'	28°13.8	-1.1'	60.9'	68°	07:34	08:59	11:21	12:21	14:43	16:09
3	227°15.6	· · 30.4	29°22.7	0.8'	28°12.8	-1.3'	60.9'	66°	07:22	08:36	10:08	13:35	15:06	16:21
4	242°15.4	30.7	43°42.5	0.9'	28°11.5	-1.5'	60.9'	64°	07:11	08:18	09:32	14:11	15:25	16:31
5	257°15.1	31.0	58°02.3	0.9'	28°10.0	-1.7'	60.8'	62°	07:02	08:03	09:06	14:36	15:40	16:40
6	272°14.8	S22°31.3	72°22.2	1.0'	N28°08.3	-1.9'	60.8'	60°	06:55	07:50	08:46	14:56	15:52	16:48
7	287°14.6	31.6	86°42.2	1.0'	28°06.3	-2.2'	60.8'	N 58°	06:48	07:39	08:30	15:13	16:03	16:55
8	302°14.3	31.9	101°02.2	1.1'	28°04.2	-2.4'	60.8'	56°	06:41	07:30	08:16	15:27	16:13	17:02
9	317°14.1	· · 32.2	115°22.3	1.2'	28°01.8	-2.6'	60.7'	54°	06:35	07:21	08:04	15:39	16:22	17:07
10	332°13.8	32.5	129°42.5	1.2'	27°59.2	-2.8'	60.7'	52°	06:30	07:13	07:54	15:49	16:29	17:13
11	347°13.5	32.8	144°02.7	1.3'	27°56.4	-3.0'	60.7'	50°	06:25	07:06	07:44	15:59	16:37	17:18
12	2°13.3	S22°33.1	158°23.0	1.4'	N27°53.4	-3.2'	60.7'	45°	06:14	06:51	07:25	16:18	16:52	17:29
13	17°13.0	33.3	172°43.4	1.5'	27°50.2	-3.4'	60.6'	N 40°	06:04	06:38	07:08	16:34	17:05	17:39
14	32°12.7	33.6	187°03.9	1.6'	27°46.7	-3.6'	60.6'	35°	05:55	06:27	06:55	16:48	17:16	17:48
15	47°12.5	· · 33.9	201°24.5	1.7'	27°43.1	-3.8'	60.6'	30°	05:47	06:17	06:43	17:00	17:26	17:56
16	62°12.2	34.2	215°45.1	1.7'	27°39.3	-4.1'	60.6'	20°	05:31	05:59	06:22	17:21	17:44	18:12
17	77°11.9	34.5	230°05.9	1.8'	27°35.2	-4.3'	60.5'	N 10°	05:16	05:42	06:05	17:38	18:01	18:27
18	92°11.7	S22°34.8	244°26.7	1.9'	N27°30.9	-4.5'	60.5'	0°	04:59	05:25	05:48	17:55	18:18	18:44
19	107°11.4	35.1	258°47.7	2.0'	27°26.5	-4.7'	60.5'	S 10°	04:41	05:08	05:31	18:12	18:35	19:02
20	122°11.1	35.4	273°08.7	2.1'	27°21.8	-4.9'	60.4'	20°	04:19	04:48	05:13	18:31	18:55	19:24
21	137°10.9	· · 35.6	287°29.8	2.2'	27°17.0	-5.1'	60.4'	30°	03:51	04:24	04:51	18:52	19:19	19:52
22	152°10.6	35.9	301°51.1	2.4'	27°11.9	-5.2'	60.4'	35°	03:33	04:10	04:39	19:04	19:34	20:10
23	167°10.3	36.2	316°12.4	2.5'	27°06.7	-5.4'	60.3'	40°	03:11	03:52	04:24	19:19	19:51	20:32
	SD = 16.2'	d = 0.3'			SD = 16.6'			45°	02:42	03:30	04:07	19:36	20:13	21:02
Sun	GHA	Dec	GHA	ν	Dec	d	HP	S 50°	02:00	03:02	03:46	19:58	20:41	21:44
0	182°10.1	S22°36.5	330°33.9	2.6'	N27°01.2	-5.6'	60.3'	52°	01:35	02:48	03:35	20:08	20:56	22:09
1	197°09.8	36.8	344°55.5	2.7'	26°55.6	-5.8'	60.3'	54°	00:59	02:31	03:24	20:20	21:13	22:46
2	212°09.5	37.0	359°17.2	2.8'	26°49.8	-6.0'	60.3'	56°	—:—	02:11	03:10	20:33	21:33	—:—
3	227°09.3	· · 37.3	13°39.0	2.9'	26°43.8	-6.2'	60.2'	58°	—:—	01:45	02:55	20:49	22:00	—:—
4	242°09.0	37.6	28°00.9	3.1'	26°37.6	-6.4'	60.2'	S 60°	—:—	01:06	02:36	21:08	22:40	—:—
5	257°08.7	37.9	42°23.0	3.2'	26°31.2	-6.6'	60.2'							
6	272°08.5	S22°38.2	56°45.1	3.3'	N26°24.6	-6.7'	60.1'							
7	287°08.2	38.4	71°07.4	3.4'	26°17.9	-6.9'	60.1'							
8	302°07.9	38.7	85°29.9	3.6'	26°11.0	-7.1'	60.1'							
9	317°07.7	· · 39.0	99°52.4	3.7'	26°03.9	-7.3'	60.0'							
10	332°07.4	39.2	114°15.1	3.8'	25°56.6	-7.4'	60.0'							
11	347°07.1	39.5	128°37.9	4.0'	25°49.1	-7.6'	59.9'							
12	2°06.8	S22°39.8	143°00.9	4.1'	N25°41.5	-7.8'	59.9'							
13	17°06.6	40.1	157°24.0	4.2'	25°33.8	-7.9'	59.9'							
14	32°06.3	40.3	171°47.2	4.4'	25°25.8	-8.1'	59.8'							
15	47°06.0	· · 40.6	186°10.6	4.5'	25°17.7	-8.3'	59.8'							
16	62°05.8	40.9	200°34.0	4.6'	25°09.4	-8.4'	59.8'							
17	77°05.5	41.1	214°57.7	4.8'	25°01.0	-8.6'	59.7'							
18	92°05.2	S22°41.4	229°21.5	4.9'	N24°52.4	-8.7'	59.7'							
19	107°05.0	41.7	243°45.4	5.1'	24°43.7	-8.9'	59.7'							
20	122°04.7	41.9	258°09.4	5.2'	24°34.8	-9.0'	59.6'							
21	137°04.4	· · 42.2	272°33.6	5.3'	24°25.7	-9.2'	59.6'							
22	152°04.1	42.5	286°58.0	5.5'	24°16.6	-9.3'	59.5'							
23	167°03.9	42.7	301°22.5	5.6'	24°07.2	-9.5'	59.5'							
	SD = 16.2'	d = 0.3'			SD = 16.4'									
Mon	GHA	Dec	GHA	ν	Dec	d	HP	S 10°	20:10	21:09	22:02	06:50	07:57	09:00
0	182°03.6	S22°43.0	315°47.1	5.8'	N23°57.7	-9.6'	59.5'	20°	20:33	21:30	22:18	06:26	07:35	08:42
1	197°03.3	43.3	330°11.9	5.9'	23°48.1	-9.8'	59.4'	30°	21:00	21:53	22:36	05:58	07:09	08:20
2	212°03.0	43.5	344°36.8	6.1'	23°38.4	-9.9'	59.4'	35°	21:16	22:07	22:47	05:41	06:53	08:07
3	227°02.8	· · 43.8	359°01.9	6.2'	23°28.5	-10.0'	59.4'	40°	21:35	22:23	22:59	05:21	06:35	07:52
4	242°02.5	44.0	13°27.1	6.4'	23°18.4	-10.2'	59.3'	45°	21:57	22:41	23:14	04:57	06:13	07:35
5	257°02.2	44.3	27°52.5	6.5'	23°08.3	-10.3'	59.3'	S 50°	22:26	23:05	23:31	04:26	05:46	07:12
6	272°02.0	S22°44.6	42°18.0	6.7'	N22°58.0	-10.4'	59.2'	52°	22:40	23:16	23:39	04:11	05:32	07:02
7	287°01.7	44.8	56°43.6	6.8'	22°47.6	-10.5'	59.2'	54°	22:56	23:29	23:49	03:53	05:16	06:50
8	302°01.4	45.1	71°09.5	6.9'	22°37.0	-10.7'	59.2'	56°	23:15	23:43	23:59	03:31	04:58	06:36
9	317°01.1	· · 45.3	85°35.4	7.1'	22°26.4	-10.8'	59.1'	58°	23:38	· · ·	00:00	03:03	04:35	06:20
10	332°00.9	45.6	100°01.5	7.2'	22°15.6	-10.9'	59.1'	S 60°	· · ·	00:09	00:20	02:23	04:04	06:00
11	347°00.6	45.8	114°27.7	7.4'	22°04.7	-11.0'	59.0'							
12	2°00.3	S22°46.1	128°54.1	7.5'	N21°53.6	-11.1'	59.0'							
13	17°00.0	46.3	143°20.7	7.7'	21°42.5	-11.3'	59.0'							
14	31°59.8	46.6	157°47.3	7.8'	21°31.2	-11.4'	58.9'							
15	46°59.5	· · 46.8	172°14.2	8.0'	21°19.9	-11.5'	58.9'							
16	61°59.2	47.1	186°41.1	8.1'	21°08.4	-11.6'	58.8'							
17	76°58.9	47.3	201°08.2	8.3'	20°56.8	-11.7'	58.8'							
18	91°58.7	S22°47.6	215°35.5	8.4'	N20°45.1	-11.8'	58.8'							
19	106°58.4	47.8	230°02.9	8.5'	20°33.3	-11.9'	58.7'							
20	121°58.1	48.1	244°30.4	8.7'	20°21.5	-12.0'	58.7'							
21	136°57.8	· · 48.3	258°58.1	8.8'	20°09.5	-12.1'	58.6'							
22	151°57.5	48.6	273°25.9	9.0'	19°57.4	-12.2'	58.6'							
23	166°57.3	48.8	287°53.9	9.1'	19°45.2	-12.3'	58.5'							
	SD = 16.2'	d = 0.3'			SD = 16.2'									
		Sun			Moon									
Day		Eqn.of Time		Mer.	Mer.Pass.		Age							
		00 ^h	12 ^h	Pass	Upper		Lower							
		mm:ss	mm:ss	hh:mm	hh:mm		hh:mm							
06		09:06	08:53	11:51	00:57		13:30							
07		08:40	08:27	11:52	02:03		14:34							
08		08:14	08:01	11:52	03:04		15:32							

December 09, 10, 11 UT (Tue., Wed., Thu.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Tue	GHA	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	SHA Dec	SHA Dec		
0	77°59.5	189°17.5	S21°35.3	173°11.4	S24°04.8	322°08.5	N21°29.8	81°24.6	S04°01.8	Alpheratz	357°33.9	29°14.2					
1	93°02.0	204°16.6	35.8	188°11.8	04.9	337°11.2	29.8	96°27.0	01.7	Ankaa	353°06.3	-42°10.1					
2	108°04.5	219°15.8	36.4	203°12.2	05.0	352°13.9	29.9	111°29.4	01.7	Schedar	349°30.0	56°41.1					
3	123°06.9	234°14.9	· · 36.9	218°12.6	· · 05.1	7°16.6	· · 29.9	126°31.9	· · 01.7	Diphda	348°46.4	-17°50.7					
4	138°09.4	249°14.0	37.5	233°13.1	05.2	22°19.3	29.9	141°34.3	01.6	Achernar	335°19.2	-57°06.5					
5	153°11.8	264°13.2	38.0	248°13.5	05.3	37°21.9	30.0	156°36.7	01.6	Hamal	327°50.1	23°35.3					
6	168°14.3	279°12.3	S21°38.6	263°13.9	S24°05.4	52°24.6	N21°30.0	171°39.1	S04°01.6	Polaris	313°12.0	89°22.6					
7	183°16.8	294°11.4	39.1	278°14.3	05.4	67°27.3	30.1	186°41.5	01.6	Acamar	315°10.8	-40°12.1					
8	198°19.2	309°10.5	39.7	293°14.7	05.5	82°30.0	30.1	201°44.0	01.5	Menkar	314°05.1	4°11.6					
9	213°21.7	324°09.7	· · 40.2	308°15.1	· · 05.6	97°32.7	· · 30.2	216°46.4	· · 01.5	Mirfak	308°26.7	49°57.3					
10	228°24.2	339°08.8	40.8	323°15.5	05.7	112°35.4	30.2	231°48.8	01.5	Aldebaran	290°38.4	16°33.7					
11	243°26.6	354°07.9	41.3	338°16.0	05.8	127°38.1	30.3	246°51.2	01.4	Rigel	281°02.8	-8°10.2					
12	258°29.1	9°07.1	S21°41.8	353°16.4	S24°05.9	142°40.8	N21°30.3	261°53.7	S04°01.4	Capella	280°20.3	46°01.4					
13	273°31.6	24°06.2	42.4	8°16.8	05.9	157°43.5	30.3	276°56.1	01.4	Bellatrix	278°21.7	6°22.4					
14	288°34.0	39°05.3	42.9	23°17.2	06.0	172°46.2	30.4	291°58.5	01.4	Elnath	278°00.5	28°37.8					
15	303°36.5	54°04.4	· · 43.4	38°17.6	· · 06.1	187°48.9	· · 30.4	307°00.9	· · 01.3	Alnilam	275°36.6	-1°11.1					
16	318°39.0	69°03.6	44.0	53°18.0	06.2	202°51.6	30.5	322°03.3	01.3	Betelgeuse	270°50.9	7°24.7					
17	333°41.4	84°02.7	44.5	68°18.4	06.3	217°54.3	30.5	337°05.8	01.3	Canopus	263°51.5	-52°42.4					
18	348°43.9	99°01.8	S21°45.0	83°18.9	S24°06.3	232°57.0	N21°30.6	352°08.2	S04°01.2	Sirius	258°25.2	-16°45.0					
19	3°46.3	114°00.9	45.6	98°19.3	06.4	247°59.7	30.6	7°10.6	01.2	Adhara	255°04.9	-29°00.3					
20	18°48.8	129°00.1	46.1	113°19.7	06.5	263°02.4	30.7	22°13.0	01.2	Procyon	244°49.8	5°09.5					
21	33°51.3	143°59.2	· · 46.6	128°20.1	· · 06.6	278°05.1	· · 30.7	37°15.4	· · 01.2	Pollux	243°16.1	27°57.7					
22	48°53.7	158°58.3	47.2	143°20.5	06.7	293°07.8	30.7	52°17.9	01.1	Avior	234°14.0	-59°35.3					
23	63°56.2	173°57.4	47.7	158°20.9	06.7	308°10.5	30.8	67°20.3	01.1	Suhail	222°45.5	-43°32.0					
Mer.pass. 18:45			ν -0.9' d 0.6' m -3.90			ν 0.4' d 0.1' m 1.26			ν 2.7' d 0.0' m -2.58			ν 2.4' d -0.0' m 0.94					
Wed	GHA	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec					
0	78°58.7	188°56.6	S21°48.2	173°21.3	S24°06.8	323°13.2	N21°30.8	82°22.7	S04°01.1	Denebola	182°24.2	14°25.5					
1	94°01.1	203°55.7	48.7	188°21.8	06.9	338°15.9	30.9	97°25.1	01.0	Gienah	175°42.9	-17°41.1					
2	109°03.6	218°54.8	49.3	203°22.2	07.0	353°18.6	30.9	112°27.5	01.0	Acrux	172°59.5	-63°14.3					
3	124°06.1	233°53.9	· · 49.8	218°22.6	· · 07.0	8°21.3	· · 31.0	127°29.9	· · 01.0	Gacrux	171°51.0	-57°15.2					
4	139°08.5	248°53.0	50.3	233°23.0	07.1	23°24.0	31.0	142°32.4	01.0	Alioth	166°12.5	55°48.9					
5	154°11.0	263°52.2	50.8	248°23.4	07.2	38°26.7	31.1	157°34.8	00.9	Spica	158°21.7	-11°17.8					
6	169°13.5	278°51.3	S21°51.3	263°23.8	S24°07.3	53°29.4	N21°31.1	172°37.2	S04°00.9	Alkaid	152°51.7	49°10.8					
7	184°15.9	293°50.4	51.9	278°24.2	07.3	68°32.1	31.1	187°39.6	00.9	Hadar	148°35.4	-60°29.7					
8	199°18.4	308°49.5	52.4	293°24.6	07.4	83°34.8	31.2	202°42.0	00.8	Menkent	147°57.0	-36°29.7					
9	214°20.8	323°48.6	· · 52.9	308°25.1	· · 07.5	98°37.5	· · 31.2	217°44.5	· · 00.8	Arcturus	145°47.5	19°02.7					
10	229°23.3	338°47.8	53.4	323°25.5	07.5	113°40.2	31.3	232°46.9	00.8	Rigel Kent.	139°39.8	-60°56.4					
11	244°25.8	353°46.9	53.9	338°25.9	07.6	128°42.9	31.3	247°49.3	00.7	Kochab	137°20.8	74°02.7					
12	259°28.2	8°46.0	S21°54.4	353°26.3	S24°07.7	143°45.6	N21°31.4	262°51.7	S04°00.7	Zuben'ubi	136°55.5	-16°08.9					
13	274°30.7	23°45.1	54.9	8°26.7	07.8	158°48.3	31.4	277°54.1	00.7	Alphecca	126°03.4	26°37.6					
14	289°33.2	38°44.2	55.4	23°27.1	07.8	173°51.0	31.5	292°56.5	00.7	Antares	112°15.3	-26°29.3					
15	304°35.6	53°43.3	· · 56.0	38°27.5	· · 07.9	188°53.7	· · 31.5	307°59.0	· · 00.6	Atria	107°09.4	-69°04.4					
16	319°38.1	68°42.5	56.5	53°27.9	08.0	203°56.4	31.6	323°01.4	00.6	Sabik	102°02.3	-15°45.4					
17	334°40.6	83°41.6	57.0	68°28.3	08.0	218°59.1	31.6	338°03.8	00.6	Shaula	96°09.8	-37°07.3					
18	349°43.0	98°40.7	S21°57.5	83°28.8	S24°08.1	234°01.8	N21°31.6	353°06.2	S04°00.5	Rasalhague	95°58.2	12°32.5					
19	4°45.5	113°39.8	58.0	98°29.2	08.2	249°04.5	31.7	8°08.6	00.5	Eltanin	90°42.4	51°29.1					
20	19°47.9	128°38.9	58.5	113°29.6	08.2	264°07.2	31.7	23°11.0	00.5	Kaus Aust.	83°31.9	-34°22.4					
21	34°50.4	143°38.0	· · 59.0	128°30.0	· · 08.3	279°09.9	· · 31.8	38°13.5	· · 00.4	Vega	80°33.1	38°48.5					
22	49°52.9	158°37.1	21°59.5	143°30.4	08.4	294°12.6	31.8	53°15.9	00.4	Nunki	75°47.2	-26°15.9					
23	64°55.3	173°36.3	22°00.0	158°30.8	08.4	309°15.3	31.9	68°18.3	00.4	Altair	61°59.5	8°56.3					
Mer.pass. 18:41			ν -0.9' d 0.5' m -3.90			ν 0.4' d 0.1' m 1.25			ν 2.7' d 0.0' m -2.59			ν 2.4' d -0.0' m 0.95					
Thu	GHA	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec	GHA Dec					
0	79°57.8	188°35.4	S22°00.5	173°31.2	S24°08.5	324°18.0	N21°31.9	83°20.7	S04°00.3	Dec 09 Tue	SHA	Mer.pass					
1	95°00.3	203°34.5	01.0	188°31.6	08.6	339°20.7	32.0	98°23.1	00.3	Venus	111°18.0	11:23					
2	110°02.7	218°33.6	01.5	203°32.0	08.6	354°23.4	32.0	113°25.5	00.3	Mars	95°11.9	12:27					
3	125°05.2	233°32.7	· · 02.0	218°32.4	· · 08.7	9°26.1	· · 32.1	128°27.9	· · 00.3	Jupiter	244°09.0	02:31					
4	140°07.7	248°31.8	02.5	233°32.9	08.8	24°28.8	32.1	143°30.4	00.2	Saturn	3°25.1	18:31					
5	155°10.1	263°30.9	02.9	248°33.3	08.8	39°31.6	32.2	158°32.8	00.2								
6	170°12.6	278°30.0	S22°03.4	263°33.7	S24°08.9	54°34.3	N21°32.2	173°35.2	S04°00.2								
7	185°15.1	293°29.1	03.9	278°34.1	09.0	69°37.0	32.2	188°37.6	00.1								
8	200°17.5	308°28.3	04.4	293°34.5	09.0	84°39.7	32.3	203°40.0	00.1								
9	215°20.0	323°27.4	· · 04.9	308°34.9	· · 09.1	99°42.4	· · 32.3	218°42.4	· · 00.1								
10	230°22.4	338°26.5	05.4	323°35.3	09.1	114°45.1	32.4	233°44.8	00.0								
11	245°24.9	353°25.6	05.9	338°35.7	09.2	129°47.8	32.4	248°47.3	00.0								
12	260°27.4	8°24.7	S22°06.4	353°36.1	S24°09.3	144°50.5	N21°32.5	263°49.7	S04°00.0								
13	275°29.8	23°23.8	06.8	8°36.5	09.3	159°53.2	32.5	278°52.1	03°59.9								
14	290°32.3	38°22.9	07.3	23°36.9	09.4	174°55.9	32.6	293°54.5	59.9								
15	305°34.8	53°22.0	· · 07.8	38°37.4	· · 09.4	189°58.6	· · 32.6	308°56.9	· · 59.9								
16	320°37.2	68°21.1	08.3	53°37.8	09.5	205°01.4	32.7	323°59.3	59.8								
17	335°39.7	83°20.2	08.8	68°38.2	09.6	220°04.1	32.7	339°01.7	59.8								
18	350°42.2	98°19.3	S22°09.3	83°38.6	S24°09.6	235°06.8	N21°32.8	354°04.1	S03°59.8								
19	5°44.6	113°18.4	09.7	98°39.0	09.7	250°09.5	32.8	9°06.6	59.7								
20	20°47.1	128°17.5	10.2	113°39.4	09.7	265°12.2	32.9	24°09.0	59.7								
21	35°49.6	143°16.6	· · 10.7	128°39.8	· · 09.8	280°											

Sun			Moon				
Tue	GHA	Dec	GHA	ν	Dec	d	HP
0	181°57.0	S22°49.1	302°22.0	9.2'	N19°33.0	-12.4'	58.5'
1	196°56.7	49.3	316°50.2	9.4'	19°20.6	-12.4'	58.5'
2	211°56.4	49.6	331°18.6	9.5'	19°08.2	-12.5'	58.4'
3	226°56.2	· · 49.8	345°47.1	9.6'	18°55.6	-12.6'	58.4'
4	241°55.9	50.0	0°15.7	9.8'	18°43.0	-12.7'	58.3'
5	256°55.6	50.3	14°44.5	9.9'	18°30.3	-12.8'	58.3'
6	271°55.3	S22°50.5	29°13.4	10.0'	N18°17.5	-12.9'	58.3'
7	286°55.0	50.8	43°42.5	10.2'	18°04.7	-12.9'	58.2'
8	301°54.8	51.0	58°11.6	10.3'	17°51.8	-13.0'	58.2'
9	316°54.5	· · 51.2	72°40.9	10.4'	17°38.7	-13.1'	58.1'
10	331°54.2	51.5	87°10.4	10.6'	17°25.7	-13.1'	58.1'
11	346°53.9	51.7	101°39.9	10.7'	17°12.5	-13.2'	58.1'
12	1°53.6	S22°51.9	116°09.6	10.8'	N16°59.3	-13.3'	58.0'
13	16°53.4	52.2	130°39.4	10.9'	16°46.0	-13.4'	58.0'
14	31°53.1	52.4	145°09.4	11.1'	16°32.7	-13.4'	57.9'
15	46°52.8	· · 52.6	159°39.4	11.2'	16°19.2	-13.5'	57.9'
16	61°52.5	52.9	174°09.6	11.3'	16°05.8	-13.5'	57.8'
17	76°52.2	53.1	188°39.9	11.4'	15°52.2	-13.6'	57.8'
18	91°52.0	S22°53.3	203°10.3	11.5'	N15°38.6	-13.7'	57.8'
19	106°51.7	53.6	217°40.9	11.7'	15°25.0	-13.7'	57.7'
20	121°51.4	53.8	232°11.5	11.8'	15°11.3	-13.8'	57.7'
21	136°51.1	· · 54.0	246°42.3	11.9'	14°57.5	-13.8'	57.6'
22	151°50.8	54.2	261°13.2	12.0'	14°43.7	-13.9'	57.6'
23	166°50.6	54.5	275°44.2	12.1'	14°29.8	-13.9'	57.6'
	SD = 16.2'	$d = 0.2'$	SD = 16.0'				
Wed	GHA	Dec	GHA	ν	Dec	d	HP
0	181°50.3	S22°54.7	290°15.3	12.2'	N14°15.9	-14.0'	57.5'
1	196°50.0	54.9	304°46.6	12.3'	14°01.9	-14.0'	57.5'
2	211°49.7	55.1	319°17.9	12.4'	13°47.9	-14.1'	57.4'
3	226°49.4	· · 55.4	333°49.3	12.5'	13°33.9	-14.1'	57.4'
4	241°49.1	55.6	348°20.9	12.7'	13°19.8	-14.1'	57.4'
5	256°48.9	55.8	2°52.5	12.8'	13°05.6	-14.2'	57.3'
6	271°48.6	S22°56.0	17°24.3	12.9'	N12°51.4	-14.2'	57.3'
7	286°48.3	56.2	31°56.1	13.0'	12°37.2	-14.3'	57.2'
8	301°48.0	56.5	46°28.1	13.1'	12°22.9	-14.3'	57.2'
9	316°47.7	· · 56.7	61°00.1	13.2'	12°08.6	-14.3'	57.2'
10	331°47.4	56.9	75°32.3	13.2'	11°54.3	-14.4'	57.1'
11	346°47.2	57.1	90°04.5	13.3'	11°39.9	-14.4'	57.1'
12	1°46.9	S22°57.3	104°36.9	13.4'	N11°25.5	-14.4'	57.1'
13	16°46.6	57.6	119°09.3	13.5'	11°11.1	-14.5'	57.0'
14	31°46.3	57.8	133°41.8	13.6'	10°56.7	-14.5'	57.0'
15	46°46.0	· · 58.0	148°14.5	13.7'	10°42.2	-14.5'	56.9'
16	61°45.7	58.2	162°47.2	13.8'	10°27.7	-14.5'	56.9'
17	76°45.5	58.4	177°20.0	13.9'	10°13.1	-14.6'	56.9'
18	91°45.2	S22°58.6	191°52.8	14.0'	N09°58.5	-14.6'	56.8'
19	106°44.9	58.8	206°25.8	14.0'	09°44.0	-14.6'	56.8'
20	121°44.6	59.0	220°58.9	14.1'	09°29.3	-14.6'	56.7'
21	136°44.3	· · 59.2	235°32.0	14.2'	09°14.7	-14.7'	56.7'
22	151°44.0	59.5	250°05.2	14.3'	09°00.1	-14.7'	56.7'
23	166°43.7	59.7	264°38.5	14.4'	08°45.4	-14.7'	56.6'
	SD = 16.2'	$d = 0.2'$	SD = 15.7'				
Thu	GHA	Dec	GHA	ν	Dec	d	HP
0	181°43.5	S22°59.9	279°11.9	14.4'	N08°30.7	-14.7'	56.6'
1	196°43.2	23°00.1	293°45.3	14.5'	08°16.0	-14.7'	56.6'
2	211°42.9	00.3	308°18.8	14.6'	08°01.3	-14.7'	56.5'
3	226°42.6	· · 00.5	322°52.4	14.7'	07°46.5	-14.7'	56.5'
4	241°42.3	00.7	337°26.1	14.7'	07°31.8	-14.8'	56.5'
5	256°42.0	00.9	351°59.8	14.8'	07°17.0	-14.8'	56.4'
6	271°41.7	S23°01.1	6°33.6	14.9'	N07°02.3	-14.8'	56.4'
7	286°41.4	01.3	21°07.4	14.9'	06°47.5	-14.8'	56.3'
8	301°41.2	01.5	35°41.4	15.0'	06°32.7	-14.8'	56.3'
9	316°40.9	· · 01.7	50°15.4	15.1'	06°17.9	-14.8'	56.3'
10	331°40.6	01.9	64°49.4	15.1'	06°03.1	-14.8'	56.2'
11	346°40.3	02.1	79°23.5	15.2'	05°48.3	-14.8'	56.2'
12	1°40.0	S23°02.3	93°57.7	15.2'	N05°33.4	-14.8'	56.2'
13	16°39.7	02.5	108°32.0	15.3'	05°18.6	-14.8'	56.1'
14	31°39.4	02.7	123°06.2	15.3'	05°03.8	-14.8'	56.1'
15	46°39.1	· · 02.9	137°40.6	15.4'	04°49.0	-14.8'	56.1'
16	61°38.8	03.1	152°15.0	15.5'	04°34.1	-14.8'	56.0'
17	76°38.6	03.3	166°49.5	15.5'	04°19.3	-14.8'	56.0'
18	91°38.3	S23°03.4	181°24.0	15.6'	N04°04.5	-14.8'	56.0'
19	106°38.0	03.6	195°58.5	15.6'	03°49.7	-14.8'	55.9'
20	121°37.7	03.8	210°33.1	15.7'	03°34.8	-14.8'	55.9'
21	136°37.4	· · 04.0	225°07.8	15.7'	03°20.0	-14.8'	55.9'
22	151°37.1	04.2	239°42.5	15.7'	03°05.2	-14.8'	55.8'
23	166°36.8	04.4	254°17.2	15.8'	02°50.4	-14.8'	55.8'
	SD = 16.2'	$d = 0.2'$	SD = 15.4'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:12	10:30	■	■	13:15	15:33
N 70°	07:54	09:38	■	■	14:08	15:52
68°	07:38	09:05	■	■	14:40	16:07
66°	07:26	08:42	10:16	13:29	15:04	16:19
64°	07:15	08:23	09:38	14:07	15:23	16:30
62°	07:06	08:07	09:12	14:34	15:38	16:39
60°	06:58	07:54	08:51	14:54	15:51	16:47
N 58°	06:51	07:43	08:34	15:11	16:02	16:54
56°	06:44	07:33	08:20	15:25	16:12	17:01
54°	06:39	07:24	08:08	15:38	16:21	17:07
52°	06:33	07:17	07:57	15:48	16:29	17:12
50°	06:28	07:09	07:48	15:58	16:36	17:18
45°	06:17	06:54	07:27	16:18	16:52	17:29
N 40°	06:07	06:41	07:11	16:35	17:05	17:39</

December 12, 13, 14 UT (Fri., Sat., Sun.)

Aries				Venus				Mars				Jupiter				Saturn				Stars			
Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec						
0	80°56.9	188°13.9	S22°12.1	173°41.0	S24°10.0	325°23.0	N21°33.0	84°18.6	S03°59.6	Alpheratz	357°33.9	29°14.2											
1	95°59.4	203°13.1	12.6	188°41.4	10.0	340°25.8	33.1	99°21.0	59.5	Ankaa	353°06.3	-42°10.1											
2	111°01.9	218°12.2	13.0	203°41.8	10.1	355°28.5	33.1	114°23.4	59.5	Schedar	349°30.0	56°41.1											
3	126°04.3	233°11.3	•• 13.5	218°42.2	•• 10.1	10°31.2	•• 33.2	129°25.8	•• 59.5	Diphda	348°46.4	-17°50.7											
4	141°06.8	248°10.4	14.0	233°42.6	10.2	25°33.9	33.2	144°28.3	59.4	Achernar	335°19.3	-57°06.5											
5	156°09.3	263°09.5	14.4	248°43.0	10.2	40°36.6	33.3	159°30.7	59.4	Hamal	327°50.1	23°35.3											
6	171°11.7	278°08.6	S22°14.9	263°43.5	S24°10.3	55°39.3	N21°33.3	174°33.1	S03°59.4	Polaris	313°12.7	89°22.6											
7	186°14.2	293°07.7	15.4	278°43.9	10.3	70°42.0	33.4	189°35.5	59.3	Acamar	315°10.8	-40°12.1											
8	201°16.7	308°06.8	15.8	293°44.3	10.4	85°44.8	33.4	204°37.9	59.3	Menkar	314°05.1	4°11.6											
9	216°19.1	323°05.9	•• 16.3	308°44.7	•• 10.4	100°47.5	•• 33.5	219°40.3	•• 59.3	Mirfak	308°26.7	49°57.4											
10	231°21.6	338°05.0	16.7	323°45.1	10.5	115°50.2	33.5	234°42.7	59.2	Aldebaran	290°38.4	16°33.7											
11	246°24.0	353°04.1	17.2	338°45.5	10.5	130°52.9	33.6	249°45.1	59.2	Rigel	281°02.8	-8°10.3											
12	261°26.5	8°03.2	S22°17.7	353°45.9	S24°10.6	145°55.6	N21°33.6	264°47.5	S03°59.2	Capella	280°20.3	46°01.5											
13	276°29.0	23°02.3	18.1	8°46.3	10.6	160°58.3	33.7	279°49.9	59.1	Bellatrix	278°21.7	6°22.4											
14	291°31.4	38°01.4	18.6	23°46.7	10.7	176°01.1	33.7	294°52.3	59.1	Elnath	278°00.5	28°37.8											
15	306°33.9	53°00.4	•• 19.0	38°47.1	•• 10.7	191°03.8	•• 33.7	309°54.8	•• 59.1	Alnilam	275°36.6	-1°11.1											
16	321°36.4	67°59.5	19.5	53°47.5	10.8	206°06.5	33.8	324°57.2	59.0	Betelgeuse	270°50.9	7°24.7											
17	336°38.8	82°58.6	19.9	68°47.9	10.8	221°09.2	33.8	339°59.6	59.0	Canopus	263°51.5	-52°42.4											
18	351°41.3	97°57.7	S22°20.4	83°48.3	S24°10.9	236°11.9	N21°33.9	355°02.0	S03°59.0	Sirius	258°25.2	-16°45.0											
19	6°43.8	112°56.8	20.8	98°48.7	10.9	251°14.6	33.9	10°04.4	58.9	Adhara	255°04.9	-29°00.3											
20	21°46.2	127°55.9	21.3	113°49.1	11.0	266°17.4	34.0	25°06.8	58.9	Procyon	244°49.8	5°09.5											
21	36°48.7	142°55.0	•• 21.7	128°49.5	•• 11.0	281°20.1	•• 34.0	40°09.2	•• 58.9	Pollux	243°16.1	27°57.7											
22	51°51.2	157°54.1	22.2	143°49.9	11.1	296°22.8	34.1	55°11.6	58.8	Avior	234°14.0	-59°35.3											
23	66°53.6	172°53.2	22.6	158°50.3	11.1	311°25.5	34.1	70°14.0	58.8	Suhail	222°45.5	-43°32.0											
Mer.pass. 18:33				ν -0.9' d 0.5' m -3.90				ν 0.4' d 0.1' m 1.24				ν 2.7' d 0.0' m -2.60				ν 2.4' d -0.0' m 0.95							

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	81°56.1	187°52.3	S22°23.0	173°50.7	S24°11.1	326°28.2	N21°34.2	85°16.4	S03°58.8										
1	96°58.5	202°51.4	23.5	188°51.1	11.2	341°31.0	34.2	100°18.8	58.7										
2	112°01.0	217°50.5	23.9	203°51.5	11.2	356°33.7	34.3	115°21.2	58.7										
3	127°03.5	232°49.6	•• 24.4	218°52.0	•• 11.3	11°36.4	•• 34.3	130°23.6	•• 58.7										
4	142°05.9	247°48.7	24.8	233°52.4	11.3	26°39.1	34.4	145°26.1	58.6										
5	157°08.4	262°47.8	25.2	248°52.8	11.4	41°41.8	34.4	160°28.5	58.6										
6	172°10.9	277°46.9	S22°25.7	263°53.2	S24°11.4	56°44.6	N21°34.5	175°30.9	S03°58.6										
7	187°13.3	292°46.0	26.1	278°53.6	11.4	71°47.3	34.5	190°33.3	58.5										
8	202°15.8	307°45.0	26.6	293°54.0	11.5	86°50.0	34.6	205°35.7	58.5										
9	217°18.3	322°44.1	•• 27.0	308°54.4	•• 11.5	101°52.7	•• 34.6	220°38.1	•• 58.5										
10	232°20.7	337°43.2	27.4	323°54.8	11.6	116°55.4	34.7	235°40.5	58.4										
11	247°23.2	352°42.3	27.8	338°55.2	11.6	131°58.2	34.7	250°42.9	58.4										
12	262°25.6	7°41.4	S22°28.3	353°55.6	S24°11.6	147°00.9	N21°34.8	265°45.3	S03°58.4										
13	277°28.1	22°40.5	28.7	8°56.0	11.7	162°03.6	34.8	280°47.7	58.3										
14	292°30.6	37°39.6	29.1	23°56.4	11.7	177°06.3	34.8	295°50.1	58.3										
15	307°33.0	52°38.7	•• 29.6	38°56.8	•• 11.8	192°09.1	•• 34.9	310°52.5	•• 58.2										
16	322°35.5	67°37.8	30.0	53°57.2	11.8	207°11.8	34.9	325°54.9	58.2										
17	337°38.0	82°36.8	30.4	68°57.6	11.8	222°14.5	35.0	340°57.3	58.2										
18	352°40.4	97°35.9	S22°30.8	83°58.0	S24°11.9	237°17.2	N21°35.0	355°59.7	S03°58.1										
19	7°42.9	112°35.0	31.3	98°58.4	11.9	252°20.0	35.1	11°02.1	58.1										
20	22°45.4	127°34.1	31.7	113°58.8	11.9	267°22.7	35.1	26°04.5	58.1										
21	37°47.8	142°33.2	•• 32.1	128°59.2	•• 12.0	282°25.4	•• 35.2	41°06.9	•• 58.0										
22	52°50.3	157°32.3	32.5	143°59.6	12.0	297°28.1	35.2	56°09.3	58.0										
23	67°52.8	172°31.4	32.9	159°00.0	12.0	312°30.9	35.3	71°11.7	58.0										
Mer.pass. 18:29				ν -0.9' d 0.4' m -3.90				ν 0.4' d 0.0' m 1.24				ν 2.7' d 0.0' m -2.61				ν 2.4' d -0.0' m 0.96			

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec										
0	82°55.2	187°30.5	S22°33.3	174°00.4	S24°12.1	327°33.6	N21°35.3	86°14.1	S03°57.9										
1	97°57.7	202°29.5	33.8	189°00.8	12.1	342°36.3	35.4	101°16.5	57.9										
2	113°00.1	217°28.6	34.2	204°01.2	12.1	357°39.0	35.4	116°18.9	57.8										
3	128°02.6	232°27.7	•• 34.6	219°01.6	•• 12.2	12°41.8	•• 35.5	131°21.3	•• 57.8										
4	143°05.1	247°26.8	35.0	234°02.0	12.2	27°44.5	35.5	146°23.7	57.8										
5	158°07.5	262°25.9	35.4	249°02.4	12.2	42°47.2	35.6	161°26.2	57.7										
6	173°10.0	277°25.0	S22°35.8	264°02.8	S24°12.3	57°49.9	N21°35.6	176°28.6	S03°57.7										
7	188°12.5	292°24.0	36.2	279°03.2	12.3	72°52.7	35.7	191°31.0	57.7										
8	203°14.9	307°23.1	36.6	294°03.6	12.3	87°55.4	35.7	206°33.4	57.6										
9	218°17.4	322°22.2	•• 37.0	309°04.0	•• 12.3	102°58.1	•• 35.8	221°35.8	•• 57.6										
10	233°19.9	337°21.3	37.4	324°04.4	12.4	118°00.9	35.8	236°38.2	57.6										
11	248°22.3	352°20.4	37.8	339°04.8	12.4	133°03.6	35.9	251°40.6	57.5										
12	263°24.8	7°19.5	S22°38.2	354°05.2	S24°12.4	148°06.3	N21°35.9	266°43.0	S03°57.5										
13	278°27.3	22°18.5	38.6	9°05.6	12.5	163°09.1	36.0	281°45.4	57.4										
14	293°29.7	37°17.6	39.0	24°06.0	12.5	178°11.8	36.0	296°47.8	57.4										
15	308°32.2	52°16.7	•• 39.4	39°06.4	•• 12.5	193°14.5	•• 36.1	311°50.2	•• 57.4										
16	323°34.6	67°15.8	39.8	54°06.8	12.5	208°17.2	36.1	326°52.6	57.3										
17	338°37.1	82°14.9	40.2	69°07.2	12.6	223°20.0	36.2	341°55.0	57.3										
18	353°39.6	97°13.9	S22°40.6	84°07.6	S24°12.6	238°22.7	N21°36.2	356°57.4	S03°57.3										
19	8°42.0	112°13.0	41.0	99°08.0	12.6	253°25.4	36.3	11°59.8	57.2										
20	23°44.5	127°12.1	41.4	114°08.4	12.6	268°28.2	36.3	27°02.2	57.2										
21	38°47.0	142°11.2	•• 41.8	129°08.8	•• 12.7	283°30.9	•• 36.4	42°04.6	•• 57.1										
22	53°49.4	157°10.2	42.2	144°09.2	12.7	298°33.6	36.4	57°07.0	57.1										
23	68°51.9	172°09.3	42.6	159°09.6	12.7	313°36.4	36.5	72°09.4	57.1										
Mer.pass. 18:25				ν -0.9' d 0.4' m -3.90				ν 0.4' d 0.0' m 1.23				ν 2.7' d 0.0' m -2.61				ν 2.4' d -0.0' m 0.96			

Dec 12 Fri	SHA	Mer.pass
Venus	107°17.0	11:28
Mars	92°44.1	12:25
Jupiter	244°26.1	02:18
Saturn	3°21.7	18:20

Dec 13 Sat	SHA	Mer.pass
Venus	105°56.2	11:29
Mars	91°54.7	12:24
Jupiter	244°32.2	02:14
Saturn	3°20.3	18:16

Dec 14 Sun	SHA	Mer.pass
Venus	104°35.2	11:31
Mars	91°05.2	12:24
Jupiter	244°38.4	02:09
Saturn	3°18.9	18:12

Horizontal parallax		
Venus:		0.1
Mars:		0.1

h	Sun				Moon			
Fri	GHA	Dec	GHA	ν	Dec	d	HP	
0	181°36.5	S23°04.6	268°52.0	15.8'	N02°35.6	-14.8'	55.8'	
1	196°36.2	04.8	283°26.8	15.9'	02°20.8	-14.8'	55.8'	
2	211°35.9	05.0	298°01.7	15.9'	02°06.0	-14.8'	55.7'	
3	226°35.7	· · 05.1	312°36.6	15.9'	01°51.2	-14.8'	55.7'	
4	241°35.4	05.3	327°11.5	16.0'	01°36.4	-14.8'	55.7'	
5	256°35.1	05.5	341°46.5	16.0'	01°21.6	-14.8'	55.6'	
6	271°34.8	S23°05.7	356°21.5	16.0'	N01°06.9	-14.7'	55.6'	
7	286°34.5	05.9	10°56.6	16.1'	00°52.1	-14.7'	55.6'	
8	301°34.2	06.1	25°31.7	16.1'	00°37.4	-14.7'	55.5'	
9	316°33.9	· · 06.2	40°06.8	16.1'	00°22.7	-14.7'	55.5'	
10	331°33.6	06.4	54°41.9	16.2'	N00°08.0	-14.7'	55.5'	
11	346°33.3	06.6	69°17.1	16.2'	S00°06.7	14.7'	55.5'	
12	1°33.0	S23°06.8	83°52.3	16.2'	S00°21.4	14.7'	55.4'	
13	16°32.7	06.9	98°27.5	16.2'	00°36.1	14.6'	55.4'	
14	31°32.4	07.1	113°02.8	16.3'	00°50.7	14.6'	55.4'	
15	46°32.1	· · 07.3	127°38.0	16.3'	01°05.3	14.6'	55.3'	
16	61°31.9	07.5	142°13.3	16.3'	01°19.9	14.6'	55.3'	
17	76°31.6	07.6	156°48.6	16.3'	01°34.5	14.6'	55.3'	
18	91°31.3	S23°07.8	171°24.0	16.3'	S01°49.1	14.5'	55.3'	
19	106°31.0	08.0	185°59.3	16.4'	02°03.7	14.5'	55.2'	
20	121°30.7	08.2	200°34.7	16.4'	02°18.2	14.5'	55.2'	
21	136°30.4	· · 08.3	215°10.1	16.4'	02°32.7	14.5'	55.2'	
22	151°30.1	08.5	229°45.5	16.4'	02°47.2	14.5'	55.2'	
23	166°29.8	08.7	244°20.9	16.4'	03°01.6	14.4'	55.1'	
SD = 16.2'		d = 0.2'	SD = 15.2'					
Sat	GHA	Dec	GHA	ν	Dec	d	HP	
0	181°29.5	S23°08.8	258°56.3	16.4'	S03°16.1	14.4'	55.1'	
1	196°29.2	09.0	273°31.7	16.4'	03°30.5	14.4'	55.1'	
2	211°28.9	09.2	288°07.2	16.4'	03°44.9	14.4'	55.1'	
3	226°28.6	· · 09.3	302°42.6	16.5'	03°59.2	14.3'	55.0'	
4	241°28.3	09.5	317°18.1	16.5'	04°13.6	14.3'	55.0'	
5	256°28.0	09.7	331°53.5	16.5'	04°27.9	14.3'	55.0'	
6	271°27.7	S23°09.8	346°29.0	16.5'	S04°42.1	14.2'	55.0'	
7	286°27.4	10.0	1°04.5	16.5'	04°56.4	14.2'	54.9'	
8	301°27.1	10.2	15°39.9	16.5'	05°10.6	14.2'	54.9'	
9	316°26.8	· · 10.3	30°15.4	16.5'	05°24.8	14.2'	54.9'	
10	331°26.6	10.5	44°50.9	16.5'	05°39.0	14.1'	54.9'	
11	346°26.3	10.6	59°26.3	16.5'	05°53.1	14.1'	54.8'	
12	1°26.0	S23°10.8	74°01.8	16.5'	S06°07.2	14.1'	54.8'	
13	16°25.7	11.0	88°37.3	16.5'	06°21.2	14.0'	54.8'	
14	31°25.4	11.1	103°12.7	16.5'	06°35.3	14.0'	54.8'	
15	46°25.1	· · 11.3	117°48.2	16.4'	06°49.3	14.0'	54.8'	
16	61°24.8	11.4	132°23.6	16.4'	07°03.2	13.9'	54.7'	
17	76°24.5	11.6	146°59.1	16.4'	07°17.1	13.9'	54.7'	
18	91°24.2	S23°11.7	161°34.5	16.4'	S07°31.0	13.8'	54.7'	
19	106°23.9	11.9	176°09.9	16.4'	07°44.9	13.8'	54.7'	
20	121°23.6	12.0	190°45.3	16.4'	07°58.7	13.8'	54.7'	
21	136°23.3	· · 12.2	205°20.7	16.4'	08°12.5	13.7'	54.6'	
22	151°23.0	12.3	219°56.1	16.4'	08°26.2	13.7'	54.6'	
23	166°22.7	12.5	234°31.4	16.3'	08°39.9	13.7'	54.6'	
SD = 16.2'		d = 0.2'	SD = 15.0'					
Sun	GHA	Dec	GHA	ν	Dec	d	HP	
0	181°22.4	S23°12.6	249°06.8	16.3'	S08°53.5	13.6'	54.6'	
1	196°22.1	12.8	263°42.1	16.3'	09°07.2	13.6'	54.6'	
2	211°21.8	12.9	278°17.4	16.3'	09°20.7	13.5'	54.5'	
3	226°21.5	· · 13.1	292°52.7	16.3'	09°34.3	13.5'	54.5'	
4	241°21.2	13.2	307°28.0	16.2'	09°47.7	13.4'	54.5'	
5	256°20.9	13.4	322°03.2	16.2'	10°01.2	13.4'	54.5'	
6	271°20.6	S23°13.5	336°38.5	16.2'	S10°14.6	13.4'	54.5'	
7	286°20.3	13.7	351°13.7	16.2'	10°27.9	13.3'	54.5'	
8	301°20.0	13.8	5°48.8	16.2'	10°41.2	13.3'	54.4'	
9	316°19.7	· · 13.9	20°24.0	16.1'	10°54.5	13.2'	54.4'	
10	331°19.4	14.1	34°59.1	16.1'	11°07.7	13.2'	54.4'	
11	346°19.1	14.2	49°34.2	16.1'	11°20.9	13.1'	54.4'	
12	1°18.8	S23°14.4	64°09.3	16.0'	S11°34.0	13.1'	54.4'	
13	16°18.5	14.5	78°44.3	16.0'	11°47.0	13.0'	54.4'	
14	31°18.2	14.6	93°19.3	16.0'	12°00.1	13.0'	54.4'	
15	46°17.9	· · 14.8	107°54.3	15.9'	12°13.0	12.9'	54.3'	
16	61°17.6	14.9	122°29.3	15.9'	12°25.9	12.9'	54.3'	
17	76°17.3	15.1	137°04.2	15.9'	12°38.8	12.8'	54.3'	
18	91°17.0	S23°15.2	151°39.1	15.8'	S12°51.6	12.8'	54.3'	
19	106°16.7	15.3	166°13.9	15.8'	13°04.3	12.7'	54.3'	
20	121°16.4	15.5	180°48.7	15.8'	13°17.0	12.6'	54.3'	
21	136°16.1	· · 15.6	195°23.5	15.7'	13°29.7	12.6'	54.3'	
22	151°15.8	15.7	209°58.2	15.7'	13°42.3	12.5'	54.2'	
23	166°15.5	15.9	224°32.9	15.7'	13°54.8	12.5'	54.2'	
SD = 16.2'		d = 0.1'	SD = 14.9'					

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:18	10:41	■	■	13:07	15:31
N 70°	07:58	09:44	■	■	14:04	15:50
68°	07:43	09:11	■	■	14:38	16:06
66°	07:30	08:46	10:24	13:25	15:02	16:18
64°	07:19	08:27	09:43	14:05	15:21	16:29
62°	07:10	08:11	09:16	14:32	15:37	16:39
60°	07:01	07:58	08:55	14:53	15:50	16:47
N 58°	06:54	07:46	08:38	15:10	16:02	16:54
56°	06:47	07:36	08:23	15:25	16:12	17:01
54°	06:41	07:27	08:11	15:37	16:21	17:07
52°	06:36	07:19	08:00	15:48	16:29	17:13
50°	06:31	07:12	07:50	15:58	16:36	17:18
45°	06:19	06:56	07:30	16:19	16:52	17:29
N 40°	06:09	06:43	07:13	16:35	17:06	17:40
35°	06:00	06:31	06:59	16:49	17:17	17:49
30°	05:51	06:21	06:47	17:01	17:28	17:58
20°	05:35	06:02	06:26	17:22	17:46	18:14
N 10°	05:19	05:45	06:08	17:41	18:04	18:30
0°	05:02	05:28	05:51	17:58	18:21	18:47
S 10°	04:43	05:10	05:33	18:15	18:38	19:05
20°	04:21	04:50	05:14	18:34	18:59	19:28
30°	03:52	04:25	04:53	18:56	19:23	19:57
35°	03:34	04:10	04:40	19:09	19:39	20:15
40°	03:11	03:52	04:25	19:24	19:57	20:38
45°	02:41	03:30	04:07	19:42	20:19	21:08
S 50°	01:56	03:01	03:45	20:04	20:48	21:53
52°	01:30	02:46	03:34	20:15	21:03	22:20
54°	00:47	02:28	03:22	20:27	21:21	23:03
56°	—:—	02:07	03:08	20:41	21:43	—:—
58°	—:—	01:38	02:52	20:57	22:11	—:—
S 60°	—:—	00:52	02:32	21:17	22:58	—:—

Lat.	Moonrise			Moonset		
	Fri	Sat	Sun	Fri	Sat	Sun
N 72°	· · ·	01:33	03:38	12:18	11:46	11:07
N 70°	· · ·	01:28	03:22	12:19	11:54	11:26
68°	· · ·	01:23	03:09	12:20	12:01	11:41
66°	· · ·	01:20	02:58	12:20	12:07	11:53
64°	· · ·	01:17	02:50	12:21	12:12	12:03
62°	· · ·	01:14	02:42	12:21	12:16	12:12
60°	· · ·	01:12	02:36	12:21	12:20	12:20
N 58°	· · ·	01:10	02:30	12:21	12:24	12:26
56°	· · ·	01:08	02:25	12:22	12:27	12:32
54°	· · ·	01:06	02:21	12:22	12:29	12:37
52°	· · ·	01:05	02:17	12:22	12:32	12:42
50°	· · ·	01:03	02:13	12:22	12:34	12:47
45°	· · ·	01:01	02:06	12:23	12:39	12:56
N 40°	· · ·	00:58	01:59	12:23	12:43	13:04
35°	· · ·	00:56	01:54	12:23	12:47	13:11
30°	· · ·	00:54	01:49	12:24	12:50	13:17
20°	00:01	00:51	01:40	12:24	12:56	13:28
N 10°	00:03	00:48	01:33	12:24	13:01	13:37
0°	00:05	00:46	01:26	12:25	13:05	13:46
S 10°	00:07	00:43	01:20	12:25	13:10	13:55
20°	00:09	00:41	01:12	12:25	13:15	14:04
30°	00:11	00:38	01:04	12:26	13:21	14:15
35°	00:13	00:36	01:00	12:26	13:24	14:21
40°	00:14	00:34	00:55	12:26	13:28	14:28
45°	00:16	00:32	00:49	12:27	13:32	14:37
S 50°	00:18	00:30	00:42	12:27	13:37	14:47
52°	00:19	00:28	00:38	12:27	13:40	14:52
54°	00:20	00:27	00:35	12:27	13:42	14:57
56°	00:21	00:26	00:31	12:27	13:45	15:03

December 15, 16, 17 UT (Mon., Tue., Wed.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	83°54.4	187°08.4	S22°42.9	174°10.0	S24°12.7	328°39.1	N21°36.5	87°11.8	S03°57.0					Alpheratz	357°33.9	29°14.2	
1	98°56.8	202°07.5	43.3	189°10.4	12.8	343°41.8	36.6	102°14.2	57.0					Ankaa	353°06.3	-42°10.1	
2	113°59.3	217°06.6	43.7	204°10.8	12.8	358°44.6	36.6	117°16.6	57.0					Schedar	349°30.0	56°41.1	
3	129°01.7	232°05.6	· · 44.1	219°11.2	· · 12.8	13°47.3	· · 36.7	132°19.0	· · 56.9					Diphda	348°46.4	-17°50.7	
4	144°04.2	247°04.7	44.5	234°11.6	12.8	28°50.0	36.7	147°21.4	56.9					Achernar	335°19.3	-57°06.5	
5	159°06.7	262°03.8	44.9	249°12.0	12.8	43°52.8	36.8	162°23.7	56.8					Hamal	327°50.1	23°35.3	
6	174°09.1	277°02.9	S22°45.2	264°12.4	S24°12.9	58°55.5	N21°36.8	177°26.1	S03°56.8					Polaris	313°13.3	89°22.6	
7	189°11.6	292°01.9	45.6	279°12.8	12.9	73°58.2	36.9	192°28.5	56.8					Acamar	315°10.8	-40°12.1	
8	204°14.1	307°01.0	46.0	294°13.2	12.9	89°01.0	36.9	207°30.9	56.7					Menkar	314°05.1	4°11.5	
9	219°16.5	322°00.1	· · 46.4	309°13.6	· · 12.9	104°03.7	· · 37.0	222°33.3	· · 56.7					Mirfak	308°26.7	49°57.4	
10	234°19.0	336°59.2	46.8	324°14.0	12.9	119°06.4	37.0	237°35.7	56.6					Aldebaran	290°38.4	16°33.7	
11	249°21.5	351°58.2	47.1	339°14.4	12.9	134°09.2	37.1	252°38.1	56.6					Rigel	281°02.8	-8°10.3	
12	264°23.9	6°57.3	S22°47.5	354°14.8	S24°13.0	149°11.9	N21°37.1	267°40.5	S03°56.6					Capella	280°20.3	46°01.5	
13	279°26.4	21°56.4	47.9	9°15.2	13.0	164°14.6	37.2	282°42.9	56.5					Bellatrix	278°21.7	6°22.4	
14	294°28.9	36°55.4	48.2	24°15.6	13.0	179°17.4	37.2	297°45.3	56.5					Elnath	278°00.5	28°37.8	
15	309°31.3	51°54.5	· · 48.6	39°15.9	· · 13.0	194°20.1	· · 37.3	312°47.7	· · 56.5					Alnilam	275°36.6	-1°11.1	
16	324°33.8	66°53.6	49.0	54°16.3	13.0	209°22.9	37.3	327°50.1	56.4					Betelgeuse	270°50.9	7°24.7	
17	339°36.2	81°52.7	49.3	69°16.7	13.0	224°25.6	37.4	342°52.5	56.4					Canopus	263°51.5	-52°42.5	
18	354°38.7	96°51.7	S22°49.7	84°17.1	S24°13.1	239°28.3	N21°37.4	357°54.9	S03°56.3					Sirius	258°25.2	-16°45.0	
19	9°41.2	111°50.8	50.1	99°17.5	13.1	254°31.1	37.5	12°57.3	56.3					Adhara	255°04.9	-29°00.3	
20	24°43.6	126°49.9	50.4	114°17.9	13.1	269°33.8	37.5	27°59.7	56.3					Procyon	244°49.7	5°09.5	
21	39°46.1	141°48.9	· · 50.8	129°18.3	· · 13.1	284°36.5	· · 37.6	43°02.1	· · 56.2					Pollux	243°16.0	27°57.7	
22	54°48.6	156°48.0	51.2	144°18.7	13.1	299°39.3	37.6	58°04.5	56.2					Avior	234°13.9	-59°35.3	
23	69°51.0	171°47.1	51.5	159°19.1	13.1	314°42.0	37.7	73°06.9	56.1					Suhail	222°45.5	-43°32.0	
Mer.pass. 18:21			ν -0.9' d 0.4' m -3.90			ν 0.4' d 0.0' m 1.23			ν 2.7' d 0.0' m -2.62			ν 2.4' d -0.0' m 0.96					

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec								
0	84°53.5	186°46.2	S22°51.9	174°19.5	S24°13.1	329°44.8	N21°37.7	88°09.3	S03°56.1								
1	99°56.0	201°45.2	52.2	189°19.9	13.1	344°47.5	37.8	103°11.7	56.1								
2	114°58.4	216°44.3	52.6	204°20.3	13.1	359°50.2	37.8	118°14.1	56.0								
3	130°00.9	231°43.4	· · 52.9	219°20.7	· · 13.2	14°53.0	· · 37.9	133°16.5	· · 56.0								
4	145°03.4	246°42.4	53.3	234°21.1	13.2	29°55.7	37.9	148°18.9	55.9								
5	160°05.8	261°41.5	53.6	249°21.5	13.2	44°58.5	38.0	163°21.3	55.9								
6	175°08.3	276°40.6	S22°54.0	264°21.9	S24°13.2	60°01.2	N21°38.0	178°23.6	S03°55.9								
7	190°10.7	291°39.6	54.3	279°22.3	13.2	75°03.9	38.1	193°26.0	55.8								
8	205°13.2	306°38.7	54.7	294°22.7	13.2	90°06.7	38.1	208°28.4	55.8								
9	220°15.7	321°37.8	· · 55.0	309°23.1	· · 13.2	105°09.4	· · 38.2	223°30.8	· · 55.7								
10	235°18.1	336°36.8	55.4	324°23.5	13.2	120°12.2	38.2	238°33.2	55.7								
11	250°20.6	351°35.9	55.7	339°23.9	13.2	135°14.9	38.3	253°35.6	55.7								
12	265°23.1	6°35.0	S22°56.1	354°24.3	S24°13.2	150°17.6	N21°38.3	268°38.0	S03°55.6								
13	280°25.5	21°34.0	56.4	9°24.6	13.2	165°20.4	38.4	283°40.4	55.6								
14	295°28.0	36°33.1	56.8	24°25.0	13.2	180°23.1	38.4	298°42.8	55.5								
15	310°30.5	51°32.2	· · 57.1	39°25.4	· · 13.2	195°25.9	· · 38.5	313°45.2	· · 55.5								
16	325°32.9	66°31.2	57.4	54°25.8	13.2	210°28.6	38.5	328°47.6	55.5								
17	340°35.4	81°30.3	57.8	69°26.2	13.3	225°31.4	38.6	343°50.0	55.4								
18	355°37.9	96°29.4	S22°58.1	84°26.6	S24°13.3	240°34.1	N21°38.6	358°52.4	S03°55.4								
19	10°40.3	111°28.4	58.5	99°27.0	13.3	255°36.9	38.7	13°54.7	55.3								
20	25°42.8	126°27.5	58.8	114°27.4	13.3	270°39.6	38.7	28°57.1	55.3								
21	40°45.2	141°26.6	· · 59.1	129°27.8	· · 13.3	285°42.3	· · 38.8	43°59.5	· · 55.2								
22	55°47.7	156°25.6	59.5	144°28.2	13.3	300°45.1	38.8	59°01.9	55.2								
23	70°50.2	171°24.7	59.8	159°28.6	13.3	315°47.8	38.9	74°04.3	55.2								
Mer.pass. 18:17			ν -0.9' d 0.4' m -3.90			ν 0.4' d 0.0' m 1.22			ν 2.7' d 0.1' m -2.62			ν 2.4' d -0.0' m 0.97					

Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec								
0	85°52.6	186°23.7	S23°00.1	174°29.0	S24°13.3	330°50.6	N21°38.9	89°06.7	S03°55.1								
1	100°55.1	201°22.8	00.4	189°29.4	13.3	345°53.3	39.0	104°09.1	55.1								
2	115°57.6	216°21.9	00.8	204°29.8	13.3	0°56.1	39.0	119°11.5	55.0								
3	131°00.0	231°20.9	· · 01.1	219°30.2	· · 13.3	15°58.8	· · 39.1	134°13.9	· · 55.0								
4	146°02.5	246°20.0	01.4	234°30.6	13.3	31°01.6	39.1	149°16.3	55.0								
5	161°05.0	261°19.0	01.7	249°30.9	13.3	46°04.3	39.2	164°18.7	54.9								
6	176°07.4	276°18.1	S23°02.1	264°31.3	S24°13.3	61°07.1	N21°39.2	179°21.0	S03°54.9								
7	191°09.9	291°17.2	02.4	279°31.7	13.3	76°09.8	39.3	194°23.4	54.8								
8	206°12.3	306°16.2	02.7	294°32.1	13.2	91°12.5	39.3	209°25.8	54.8								
9	221°14.8	321°15.3	· · 03.0	309°32.5	· · 13.2	106°15.3	· · 39.4	224°28.2	· · 54.7								
10	236°17.3	336°14.3	03.3	324°32.9	13.2	121°18.0	39.4	239°30.6	54.7								
11	251°19.7	351°13.4	03.7	339°33.3	13.2	136°20.8	39.5	254°33.0	54.7								
12	266°22.2	6°12.5	S23°04.0	354°33.7	S24°13.2	151°23.5	N21°39.5	269°35.4	S03°54.6								
13	281°24.7	21°11.5	04.3	9°34.1	13.2	166°26.3	39.6	284°37.8	54.6								
14	296°27.1	36°10.6	04.6	24°34.5	13.2	181°29.0	39.7	299°40.2	54.5								
15	311°29.6	51°09.6	· · 04.9	39°34.9	· · 13.2	196°31.8	· · 39.7	314°42.5	· · 54.5								
16	326°32.1	66°08.7	05.2	54°35.3	13.2	211°34.5	39.8	329°44.9	54.5								
17	341°34.5	81°07.8	05.5	69°35.7	13.2	226°37.3	39.8	344°47.3	54.4								
18	356°37.0	96°06.8	S23°05.8	84°36.0	S24°13.2	241°40.0	N21°39.9	359°49.7	S03°54.4								
19	11°39.5	111°05.9	06.1	99°36.4	13.2	256°42.8	39.9	14°52.1	54.3								
20	26°41.9	126°04.9	06.4	114°36.8	13.2	271°45.5	40.0	29°54.5	54.3								
21	41°44.4	141°04.0	· · 06.7	129°37.2	· · 13.2	286°48.3	· · 40.0	44°56.9	· · 54.2								
22	56°46.8	156°03.0	07.0	144°37.6	13.2	301°51.0	40.1	59°59.3	54.2								
23	71°49.3	171°02.1	07.3	159°38.0	13.1	316°53.8	40.1	75°01.6	54.2								
Mer.pass. 18:13			ν -0.9' d 0.3' m -3.90			ν 0.4' d -0.0' m 1.21			ν 2.7' d 0.1' m -2.63			ν 2.4' d -0.0' m 0.97					

Dec 15 Mon	SHA	Mer.pass
Venus	103°14.0	11:32
Mars	90°15.6	12:23
Jupiter	244°44.7	02:05
Saturn	3°17.4	18:08

Dec 16 Tue	SHA	Mer.pass
Venus	101°52.7	11:34
Mars	89°26.0	12:22
Jupiter	244°51.3	02:01
Saturn	3°15.8	18:04

Dec 17 Wed	SHA	Mer.pass
Venus	100°31.1	11:35
Mars	88°36.3	12:22
Jupiter	244°57.9	01:56
Saturn	3°14.1	18:01

Horizontal parallax		
Venus:	0.1	
Mars:	0.1	

Sun				Moon				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	181°15.2	S23°16.0		239°07.5	15.6'	S14°07.3	12.4'	54.2'
1	196°14.9	16.1		253°42.2	15.6'	14°19.7	12.4'	54.2'
2	211°14.6	16.2		268°16.7	15.5'	14°32.1	12.3'	54.2'
3	226°14.3	· · 16.4		282°51.3	15.5'	14°44.4	12.2'	54.2'
4	241°14.0	16.5		297°25.7	15.4'	14°56.6	12.2'	54.2'
5	256°13.7	16.6		312°00.2	15.4'	15°08.8	12.1'	54.2'
6	271°13.4	S23°16.7		326°34.6	15.3'	S15°20.9	12.1'	54.1'
7	286°13.1	16.9		341°08.9	15.3'	15°32.9	12.0'	54.1'
8	301°12.8	17.0		355°43.2	15.3'	15°44.9	11.9'	54.1'
9	316°12.5	· · 17.1		10°17.5	15.2'	15°56.8	11.9'	54.1'
10	331°12.2	17.2		24°51.7	15.2'	16°08.7	11.8'	54.1'
11	346°11.9	17.4		39°25.9	15.1'	16°20.5	11.7'	54.1'
12	1°11.6	S23°17.5		54°00.0	15.1'	S16°32.2	11.7'	54.1'
13	16°11.3	17.6		68°34.0	15.0'	16°43.9	11.6'	54.1'
14	31°11.0	17.7		83°08.0	15.0'	16°55.5	11.5'	54.1'
15	46°10.7	· · 17.8		97°42.0	14.9'	17°07.0	11.5'	54.1'
16	61°10.4	17.9		112°15.9	14.8'	17°18.5	11.4'	54.1'
17	76°10.1	18.1		126°49.7	14.8'	17°29.8	11.3'	54.0'
18	91°09.8	S23°18.2		141°23.5	14.7'	S17°41.1	11.2'	54.0'
19	106°09.5	18.3		155°57.3	14.7'	17°52.4	11.2'	54.0'
20	121°09.2	18.4		170°31.0	14.6'	18°03.6	11.1'	54.0'
21	136°08.9	· · 18.5		185°04.6	14.6'	18°14.7	11.0'	54.0'
22	151°08.6	18.6		199°38.2	14.5'	18°25.7	10.9'	54.0'
23	166°08.3	18.7		214°11.7	14.5'	18°36.6	10.9'	54.0'
SD = 16.2' d = 0.1'				SD = 14.8'				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	181°08.0	S23°18.9		228°45.1	14.4'	S18°47.5	10.8'	54.0'
1	196°07.7	19.0		243°18.5	14.3'	18°58.3	10.7'	54.0'
2	211°07.4	19.1		257°51.9	14.3'	19°09.0	10.6'	54.0'
3	226°07.0	· · 19.2		272°25.2	14.2'	19°19.6	10.6'	54.0'
4	241°06.7	19.3		286°58.4	14.2'	19°30.2	10.5'	54.0'
5	256°06.4	19.4		301°31.5	14.1'	19°40.7	10.4'	54.0'
6	271°06.1	S23°19.5		316°04.6	14.0'	S19°51.1	10.3'	54.0'
7	286°05.8	19.6		330°37.7	14.0'	20°01.4	10.2'	54.0'
8	301°05.5	19.7		345°10.7	13.9'	20°11.6	10.2'	54.0'
9	316°05.2	· · 19.8		359°43.6	13.9'	20°21.8	10.1'	53.9'
10	331°04.9	19.9		14°16.4	13.8'	20°31.8	10.0'	53.9'
11	346°04.6	20.0		28°49.2	13.7'	20°41.8	9.9'	53.9'
12	1°04.3	S23°20.1		43°21.9	13.7'	S20°51.7	9.8'	53.9'
13	16°04.0	20.2		57°54.6	13.6'	21°01.5	9.7'	53.9'
14	31°03.7	20.3		72°27.2	13.5'	21°11.3	9.6'	53.9'
15	46°03.4	· · 20.4		86°59.7	13.5'	21°20.9	9.5'	53.9'
16	61°03.1	20.5		101°32.2	13.4'	21°30.4	9.5'	53.9'
17	76°02.8	20.6		116°04.6	13.3'	21°39.9	9.4'	53.9'
18	91°02.5	S23°20.7		130°36.9	13.3'	S21°49.2	9.3'	53.9'
19	106°02.2	20.8		145°09.2	13.2'	21°58.5	9.2'	53.9'
20	121°01.9	20.9		159°41.4	13.1'	22°07.7	9.1'	53.9'
21	136°01.6	· · 21.0		174°13.6	13.1'	22°16.8	9.0'	53.9'
22	151°01.3	21.1		188°45.6	13.0'	22°25.8	8.9'	53.9'
23	166°01.0	21.2		203°17.7	12.9'	22°34.7	8.8'	53.9'
SD = 16.2' d = 0.1'				SD = 14.7'				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	181°00.6	S23°21.3		217°49.6	12.9'	S22°43.5	8.7'	53.9'
1	196°00.3	21.4		232°21.5	12.8'	22°52.2	8.6'	53.9'
2	211°00.0	21.4		246°53.3	12.7'	23°00.8	8.5'	53.9'
3	225°59.7	· · 21.5		261°25.0	12.7'	23°09.3	8.4'	53.9'
4	240°59.4	21.6		275°56.7	12.6'	23°17.7	8.3'	53.9'
5	255°59.1	21.7		290°28.3	12.5'	23°26.0	8.2'	53.9'
6	270°58.8	S23°21.8		304°59.8	12.5'	S23°34.2	8.1'	53.9'
7	285°58.5	21.9		319°31.3	12.4'	23°42.3	8.0'	53.9'
8	300°58.2	22.0		334°02.7	12.3'	23°50.3	7.9'	53.9'
9	315°57.9	· · 22.0		348°34.1	12.3'	23°58.2	7.8'	53.9'
10	330°57.6	22.1		3°05.3	12.2'	24°06.0	7.7'	53.9'
11	345°57.3	22.2		17°36.6	12.1'	24°13.7	7.6'	53.9'
12	0°57.0	S23°22.3		32°07.7	12.1'	S24°21.3	7.5'	53.9'
13	15°56.7	22.4		46°38.8	12.0'	24°28.8	7.4'	53.9'
14	30°56.4	22.4		61°09.8	11.9'	24°36.1	7.3'	53.9'
15	45°56.0	· · 22.5		75°40.7	11.9'	24°43.4	7.2'	53.9'
16	60°55.7	22.6		90°11.6	11.8'	24°50.6	7.0'	53.9'
17	75°55.4	22.7		104°42.4	11.8'	24°57.6	6.9'	53.9'
18	90°55.1	S23°22.8		119°13.2	11.7'	S25°04.5	6.8'	53.9'
19	105°54.8	22.8		133°43.9	11.6'	25°11.3	6.7'	53.9'
20	120°54.5	22.9		148°14.5	11.6'	25°18.1	6.6'	53.9'
21	135°54.2	· · 23.0		162°45.0	11.5'	25°24.6	6.5'	53.9'
22	150°53.9	23.1		177°15.5	11.4'	25°31.1	6.4'	53.9'
23	165°53.6	23.1		191°46.0	11.4'	25°37.5	6.2'	53.9'
SD = 16.2' d = 0.1'				SD = 14.7'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:22	10:50	■	■	13:01	15:30
N 70°	08:02	09:50	■	■	14:02	15:49
68°	07:46	09:15	■	■	14:36	16:05
66°	07:33	08:50	10:29	13:22	15:01	16:18
64°	07:22	08:30	09:48	14:04	15:21	16:29
62°	07:12	08:14	09:20	14:31	15:37	16:39
60°	07:04	08:01	08:58	14:53	15:51	16:47
N 58°	06:57	07:49	08:41	15:10	16:02	16:55
56°	06:50	07:39	08:26	15:25	16:12	17:01
54°	06:44	07:30	08:14	15:38	16:21	17:08
52°	06:38	07:22	08:03	15:49	16:30	17:13
50°	06:33	07:14	07:53	15:59	16:37	17:19
45°	06:21	06:58	07:32	16:19	16:53	17:30
N 40°	06:11	06:45	07:15	16:36	17:06	17:41
35°	06:01	06:33	07:01	16:50	17:18	17:50
30°	05:53	06:23	06:49	17:02	17:29	17:59
20°	05:36	06:04	06:28	17:24	17:48	18:15
N 10°	05:20	05:47	06:09	17:42	18:05	18:31
0°	05:03	05:29	05:52	17:59	18:22	18:48
S 10°	04:44	05:11	05:34	18:17	18:40	19:07
20°	04:22	04:51	05:16	18:36	19:00	19:30
30°	03:53	04:26	04:54	18:58	19:25	19:59
35°	03:34	04:11	04:41	19:11	19:41	20:17
40°	03:11	03:53	04:26	19:26	19:59	20:40
45°	02:41	03:30	04:08	19:44	20:21	21:11
S 50°	01:56	03:01	03:45	20:06	20:51	21:56
52°	01:28	02:46	03:34	20:17	21:06	22:24
54°	00:43	02:28	03:22	20:29	21:24	23:09
56°	—:—	02:06	03:08	20:44	21:46	—:—
58°	—:—	01:36	02:51	21:00	22:16	—:—
S 60°	—:—	00:48	02:31	21:21	23:05	—:—

Lat.	Moonrise			Moonset		
	Mon	Tue	Wed	Mon	Tue	Wed
N 72°	06:07	■	■	10:06	■	■
N 70°	05:28	■	■	10:46	■	■
68°	05:02	07:20	■	11:14	10:27	■
66°	04:42	06:37	■	11:36	11:11	■
64°	04:26	06:09	08:06	11:53	11:40	11:18
62°	04:12	05:47	07:28	12:07	12:02	11:57
60°	04:01	05:30	07:01	12:19	12:20	12:24
N 58°	03:52	05:15	06:40	12:30	12:36	12:45
56°	03:43	05:03	06:23	12:39	12:49	13:03
54°	03:36	04:52	06:09	12:47	13:00	13:18
52°	03:29	04:43	05:56	12:55	13:10	13:31
50°	03:23	04:34	05:45	13:01	13:19	13:42
45°	03:11	04:16	05:22	13:15	13:38	14:06
N 40°	03:00	04:02	05:04	13:27	13:54	14:25
35°	02:51	03:50	04:49	13:37	14:07	14:41
30°	02:43	03:39	04:36	13:46	14:18	14:55
20°	02:30	03:21	04:13	14:02	14:38	15:19
N 10°	02:18	03:05	03:54	14:15	14:56	15:39
0°	02:07	02:50	03:36	14:28	15:12	15:59
S 10°	01:57	02:36	03:18	14:41	15:28	16:18
20°	01:45	02:20	02:59	14:54	15:46	16:39
30°	01:32	02:03	02:37	15:10	16:06	17:03
35°	01:25	01:53	02:24	15:19	16:18	17:17
40°	01:16	01:41	02:10	15:30	16:32	17:34
45°	01:07	01:27	01:52	15:42	16:48	17:54
S 50°	00:55	01:11	01:31	15:57	17:08	18:19
52°	00:49	01:03	01:21	16:05	17:18	18:31
54°	00:43	00:54	01:10	16:12	17:29	18:45
56°	00:37	00:45	00:57	16:21	17:41	19:01
58°	00:29	00:34	00:420			

December 18, 19, 20 UT (Thu., Fri., Sat.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Thu	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	86°51.8	186°01.2	S23°07.6	174°38.4	S24°13.1	331°56.5	N21°40.2	90°04.0	S03°54.1	Alpheratz 357°33.9 29°14.2 Ankaa 353°06.3 -42°10.1 Schedar 349°30.0 56°41.1 Diphda 348°46.4 -17°50.7 Achernar 335°19.3 -57°06.5 Hamal 327°50.1 23°35.3 Polaris 313°13.9 89°22.6 Acamar 315°10.8 -40°12.1 Menkar 314°05.1 4°11.5 Mirfak 308°26.7 49°57.4 Aldebaran 290°38.4 16°33.7 Rigel 281°02.8 -8°10.3 Capella 280°20.2 46°01.5 Bellatrix 278°21.7 6°22.4 Elnath 278°00.5 28°37.8 Alnilam 275°36.6 -1°11.1 Betelgeuse 270°50.9 7°24.7 Canopus 263°51.5 -52°42.5 Sirius 258°25.2 -16°45.1 Adhara 255°04.9 -29°00.4 Procyon 244°49.7 5°09.5 Pollux 243°16.0 27°57.7 Avior 234°13.9 -59°35.3 Suhail 222°45.4 -43°32.0 Miaplacidus 221°37.5 -69°49.1 Alphard 217°46.8 -8°46.2 Regulus 207°33.5 11°50.3 Dubhe 193°39.9 61°36.3 Denebola 182°24.2 14°25.5 Gienah 175°42.8 -17°41.1 Acrux 172°59.4 -63°14.3 Gacrux 171°50.9 -57°15.3 Alioth 166°12.5 55°48.8 Spica 158°21.6 -11°17.8 Alkaid 152°51.6 49°10.7 Hadar 148°35.3 -60°29.7 Menkent 147°56.9 -36°29.7 Arcturus 145°47.4 19°02.7 Rigel Kent. 139°39.7 -60°56.3 Kochab 137°20.7 74°02.6 Zuben'ubi 136°55.4 -16°08.9 Alphecca 126°03.4 26°37.5 Antares 112°15.2 -26°29.3 Atria 107°09.3 -69°04.3 Sabik 102°02.2 -15°45.4 Shaula 96°09.7 -37°07.3 Rasalhague 95°58.2 12°32.4 Eltanin 90°42.4 51°29.1 Kaus Aust. 83°31.9 -34°22.3 Vega 80°33.1 38°48.5 Nunki 75°47.2 -26°15.9 Altair 61°59.5 8°56.2 Peacock 53°05.0 -56°39.2 Deneb 49°25.6 45°22.5 Enif 33°38.2 9°59.7 Al Na'ir 27°32.2 -46°50.3 Fomalhaut 15°13.8 -29°29.2 Scheat 13°44.5 28°13.6 Markab 13°29.2 15°20.8							
1	101°54.2	201°00.2	07.9	189°38.8	13.1	346°59.3	40.2	105°06.4	54.1								
2	116°56.7	215°59.3	08.2	204°39.2	13.1	2°02.0	40.3	120°08.8	54.0								
3	131°59.2	230°58.3	· · 08.5	219°39.6	· · 13.1	17°04.8	· · 40.3	135°11.2	· · 54.0								
4	147°01.6	245°57.4	08.8	234°40.0	13.1	32°07.5	40.4	150°13.6	53.9								
5	162°04.1	260°56.4	09.1	249°40.4	13.1	47°10.3	40.4	165°16.0	53.9								
6	177°06.6	275°55.5	S23°09.4	264°40.7	S24°13.1	62°13.0	N21°40.5	180°18.4	S03°53.8								
7	192°09.0	290°54.5	09.7	279°41.1	13.0	77°15.8	40.5	195°20.7	53.8								
8	207°11.5	305°53.6	10.0	294°41.5	13.0	92°18.5	40.6	210°23.1	53.8								
9	222°14.0	320°52.7	· · 10.3	309°41.9	· · 13.0	107°21.3	· · 40.6	225°25.5	· · 53.7								
10	237°16.4	335°51.7	10.6	324°42.3	13.0	122°24.1	40.7	240°27.9	53.7								
11	252°18.9	350°50.8	10.9	339°42.7	13.0	137°26.8	40.7	255°30.3	53.6								
12	267°21.3	5°49.8	S23°11.2	354°43.1	S24°13.0	152°29.6	N21°40.8	270°32.7	S03°53.6								
13	282°23.8	20°48.9	11.4	9°43.5	12.9	167°32.3	40.8	285°35.0	53.5								
14	297°26.3	35°47.9	11.7	24°43.9	12.9	182°35.1	40.9	300°37.4	53.5								
15	312°28.7	50°47.0	· · 12.0	39°44.3	· · 12.9	197°37.8	· · 41.0	315°39.8	· · 53.5								
16	327°31.2	65°46.0	12.3	54°44.6	12.9	212°40.6	41.0	330°42.2	53.4								
17	342°33.7	80°45.1	12.6	69°45.0	12.9	227°43.3	41.1	345°44.6	53.4								
18	357°36.1	95°44.1	S23°12.8	84°45.4	S24°12.9	242°46.1	N21°41.1	0°47.0	S03°53.3								
19	12°38.6	110°43.2	13.1	99°45.8	12.8	257°48.8	41.2	15°49.4	53.3								
20	27°41.1	125°42.2	13.4	114°46.2	12.8	272°51.6	41.2	30°51.7	53.2								
21	42°43.5	140°41.3	· · 13.7	129°46.6	· · 12.8	287°54.4	· · 41.3	45°54.1	· · 53.2								
22	57°46.0	155°40.3	13.9	144°47.0	12.8	302°57.1	41.3	60°56.5	53.1								
23	72°48.5	170°39.4	14.2	159°47.4	12.8	317°59.9	41.4	75°58.9	53.1								
Mer.pass. 18:10			ν -0.9' d 0.3' m -3.90			ν 0.4' d -0.0' m 1.21			ν 2.8' d 0.1' m -2.63			ν 2.4' d -0.0' m 0.97					

Fri	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	87°50.9	185°38.4	S23°14.5	174°47.8	S24°12.7	333°02.6	N21°41.4	91°01.3	S03°53.1					
1	102°53.4	200°37.5	14.7	189°48.1	12.7	348°05.4	41.5	106°03.7	53.0					
2	117°55.8	215°36.5	15.0	204°48.5	12.7	3°08.1	41.5	121°06.0	53.0					
3	132°58.3	230°35.6	· · 15.3	219°48.9	· · 12.7	18°10.9	· · 41.6	136°08.4	· · 52.9					
4	148°00.8	245°34.6	15.5	234°49.3	12.6	33°13.7	41.6	151°10.8	52.9					
5	163°03.2	260°33.7	15.8	249°49.7	12.6	48°16.4	41.7	166°13.2	52.8					
6	178°05.7	275°32.7	S23°16.1	264°50.1	S24°12.6	63°19.2	N21°41.7	181°15.6	S03°52.8					
7	193°08.2	290°31.8	16.3	279°50.5	12.6	78°21.9	41.8	196°17.9	52.7					
8	208°10.6	305°30.8	16.6	294°50.9	12.5	93°24.7	41.8	211°20.3	52.7					
9	223°13.1	320°29.9	· · 16.9	309°51.3	· · 12.5	108°27.5	· · 41.9	226°22.7	· · 52.6					
10	238°15.6	335°28.9	17.1	324°51.6	12.5	123°30.2	42.0	241°25.1	52.6					
11	253°18.0	350°28.0	17.4	339°52.0	12.5	138°33.0	42.0	256°27.5	52.6					
12	268°20.5	5°27.0	S23°17.6	354°52.4	S24°12.4	153°35.7	N21°42.1	271°29.8	S03°52.5					
13	283°23.0	20°26.1	17.9	9°52.8	12.4	168°38.5	42.1	286°32.2	52.5					
14	298°25.4	35°25.1	18.1	24°53.2	12.4	183°41.3	42.2	301°34.6	52.4					
15	313°27.9	50°24.2	· · 18.4	39°53.6	· · 12.3	198°44.0	· · 42.2	316°37.0	· · 52.4					
16	328°30.3	65°23.2	18.6	54°54.0	12.3	213°46.8	42.3	331°39.4	52.3					
17	343°32.8	80°22.2	18.9	69°54.4	12.3	228°49.5	42.3	346°41.8	52.3					
18	358°35.3	95°21.3	S23°19.1	84°54.8	S24°12.3	243°52.3	N21°42.4	1°44.1	S03°52.2					
19	13°37.7	110°20.3	19.4	99°55.1	12.2	258°55.1	42.4	16°46.5	52.2					
20	28°40.2	125°19.4	19.6	114°55.5	12.2	273°57.8	42.5	31°48.9	52.1					
21	43°42.7	140°18.4	· · 19.9	129°55.9	· · 12.2	289°00.6	· · 42.5	46°51.3	· · 52.1					
22	58°45.1	155°17.5	20.1	144°56.3	12.1	304°03.3	42.6	61°53.6	52.0					
23	73°47.6	170°16.5	20.4	159°56.7	12.1	319°06.1	42.6	76°56.0	52.0					
Mer.pass. 18:06			ν -1.0' d 0.3' m -3.90			ν 0.4' d -0.0' m 1.20			ν 2.8' d 0.1' m -2.63			ν 2.4' d -0.0' m 0.98		

Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	88°50.1	185°15.6	S23°20.6	174°57.1	S24°12.1	334°08.9	N21°42.7	91°58.4	S03°52.0
1	103°52.5	200°14.6	20.8	189°57.5	12.0	349°11.6	42.7	107°00.8	51.9
2	118°55.0	215°13.7	21.1	204°57.9	12.0	4°14.4	42.8	122°03.2	51.9
3	133°57.5	230°12.7	· · 21.3	219°58.2	· · 12.0	19°17.2	· · 42.9	137°05.5	· · 51.8
4	148°59.9	245°11.7	21.6	234°58.6	11.9	34°19.9	42.9	152°07.9	51.8
5	164°02.4	260°10.8	21.8	249°59.0	11.9	49°22.7	43.0	167°10.3	51.7
6	179°04.8	275°09.8	S23°22.0	264°59.4	S24°11.9	64°25.4	N21°43.0	182°12.7	S03°51.7
7	194°07.3	290°08.9	22.3	279°59.8	11.8	79°28.2	43.1	197°15.1	51.6
8	209°09.8	305°07.9	22.5	295°00.2	11.8	94°31.0	43.1	212°17.4	51.6
9	224°12.2	320°07.0	· · 22.7	310°00.6	· · 11.8	109°33.7	· · 43.2	227°19.8	· · 51.5
10	239°14.7	335°06.0	22.9	325°01.0	11.7	124°36.5	43.2	242°22.2	51.5
11	254°17.2	350°05.0	23.2	340°01.3	11.7	139°39.3	43.3	257°24.6	51.4
12	269°19.6	5°04.1	S23°23.4	355°01.7	S24°11.6	154°42.0	N21°43.3	272°26.9	S03°51.4
13	284°22.1	20°03.1	23.6	10°02.1	11.6	169°44.8	43.4	287°29.3	51.3
14	299°24.6	35°02.2	23.8	25°02.5	11.6	184°47.6	43.4	302°31.7	51.3
15	314°27.0	50°01.2	· · 24.1	40°02.9	· · 11.5	199°50.3	· · 43.5	317°34.1	· · 51.2
16	329°29.5	65°00.3	24.3	55°03.3	11.5	214°53.1	43.5	332°36.4	51.2
17	344°31.9	79°59.3	24.5	70°03.7	11.4	229°55.9	43.6	347°38.8	51.2
18	359°34.4	94°58.3	S23°24.7	85°04.0	S24°11.4	244°58.6	N21°43.7	2°41.2	S03°51.1
19	14°36.9	109°57.4	24.9	100°04.4	11.4	260°01.4	43.7	17°43.6	51.1
20	29°39.3	124°56.4	25.2	115°04.8	11.3	275°04.2	43.8	32°46.0	51.0
21	44°41.8	139°55.5	· · 25.4	130°05.2	· · 11.3	290°06.9	· · 43.8	47°48.3	· · 51.0
22	59°44.3	154°54.5	25.6	145°05.6	11.2	305°09.7	43.9	62°50.7	50.9

Sun				Moon				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	180°53.3	S23°23.2		206°16.3	11.3'	S25°43.7	6.1'	53.9'
1	195°53.0	23.3		220°46.7	11.2'	25°49.9	6.0'	53.9'
2	210°52.7	23.3		235°16.9	11.2'	25°55.9	5.9'	53.9'
3	225°52.4	•• 23.4		249°47.1	11.1'	26°01.8	5.8'	53.9'
4	240°52.1	23.5		264°17.2	11.1'	26°07.5	5.7'	53.9'
5	255°51.7	23.5		278°47.3	11.0'	26°13.2	5.5'	53.9'
6	270°51.4	S23°23.6		293°17.3	10.9'	S26°18.7	5.4'	54.0'
7	285°51.1	23.7		307°47.2	10.9'	26°24.1	5.3'	54.0'
8	300°50.8	23.7		322°17.1	10.8'	26°29.4	5.2'	54.0'
9	315°50.5	•• 23.8		336°47.0	10.8'	26°34.6	5.0'	54.0'
10	330°50.2	23.9		351°16.7	10.7'	26°39.7	4.9'	54.0'
11	345°49.9	23.9		5°46.5	10.7'	26°44.6	4.8'	54.0'
12	0°49.6	S23°24.0		20°16.1	10.6'	S26°49.4	4.7'	54.0'
13	15°49.3	24.1		34°45.7	10.6'	26°54.0	4.5'	54.0'
14	30°49.0	24.1		49°15.3	10.5'	26°58.6	4.4'	54.0'
15	45°48.7	•• 24.2		63°44.8	10.5'	27°03.0	4.3'	54.0'
16	60°48.3	24.2		78°14.3	10.4'	27°07.3	4.2'	54.0'
17	75°48.0	24.3		92°43.7	10.4'	27°11.5	4.0'	54.0'
18	90°47.7	S23°24.3		107°13.0	10.3'	S27°15.5	3.9'	54.0'
19	105°47.4	24.4		121°42.3	10.3'	27°19.4	3.8'	54.0'
20	120°47.1	24.5		136°11.6	10.2'	27°23.2	3.6'	54.0'
21	135°46.8	•• 24.5		150°40.8	10.2'	27°26.8	3.5'	54.0'
22	150°46.5	24.6		165°10.0	10.1'	27°30.3	3.4'	54.0'
23	165°46.2	24.6		179°39.1	10.1'	27°33.7	3.2'	54.0'
SD = 16.2'			$d = 0.1'$	SD = 14.7'				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	180°45.9	S23°24.7		194°08.2	10.0'	S27°37.0	3.1'	54.1'
1	195°45.6	24.7		208°37.2	10.0'	27°40.1	3.0'	54.1'
2	210°45.3	24.8		223°06.2	10.0'	27°43.1	2.8'	54.1'
3	225°44.9	•• 24.8		237°35.2	9.9'	27°45.9	2.7'	54.1'
4	240°44.6	24.9		252°04.1	9.9'	27°48.6	2.6'	54.1'
5	255°44.3	24.9		266°33.0	9.8'	27°51.2	2.4'	54.1'
6	270°44.0	S23°25.0		281°01.9	9.8'	S27°53.6	2.3'	54.1'
7	285°43.7	25.0		295°30.7	9.8'	27°55.9	2.2'	54.1'
8	300°43.4	25.1		309°59.4	9.7'	27°58.1	2.0'	54.1'
9	315°43.1	•• 25.1		324°28.2	9.7'	28°00.2	1.9'	54.1'
10	330°42.8	25.1		338°56.9	9.7'	28°02.1	1.8'	54.1'
11	345°42.5	25.2		353°25.6	9.7'	28°03.8	1.6'	54.1'
12	0°42.2	S23°25.2		7°54.2	9.6'	S28°05.4	1.5'	54.1'
13	15°41.9	25.3		22°22.9	9.6'	28°06.9	1.3'	54.2'
14	30°41.5	25.3		36°51.4	9.6'	28°08.3	1.2'	54.2'
15	45°41.2	•• 25.4		51°20.0	9.5'	28°09.5	1.1'	54.2'
16	60°40.9	25.4		65°48.6	9.5'	28°10.5	0.9'	54.2'
17	75°40.6	25.4		80°17.1	9.5'	28°11.5	0.8'	54.2'
18	90°40.3	S23°25.5		94°45.6	9.5'	S28°12.3	0.6'	54.2'
19	105°40.0	25.5		109°14.1	9.5'	28°12.9	0.5'	54.2'
20	120°39.7	25.5		123°42.6	9.4'	28°13.4	0.4'	54.2'
21	135°39.4	•• 25.6		138°11.0	9.4'	28°13.8	0.2'	54.2'
22	150°39.1	25.6		152°39.4	9.4'	28°14.0	0.1'	54.2'
23	165°38.7	25.6		167°07.9	9.4'	28°14.1	-0.1'	54.2'
SD = 16.2'			$d = 0.1'$	SD = 14.7'				
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	180°38.4	S23°25.7		181°36.3	9.4'	S28°14.1	-0.2'	54.3'
1	195°38.1	25.7		196°04.7	9.4'	28°13.9	-0.3'	54.3'
2	210°37.8	25.7		210°33.0	9.4'	28°13.5	-0.5'	54.3'
3	225°37.5	•• 25.8		225°01.4	9.4'	28°13.0	-0.6'	54.3'
4	240°37.2	25.8		239°29.8	9.4'	28°12.4	-0.8'	54.3'
5	255°36.9	25.8		253°58.1	9.4'	28°11.7	-0.9'	54.3'
6	270°36.6	S23°25.8		268°26.5	9.4'	S28°10.8	-1.0'	54.3'
7	285°36.3	25.9		282°54.9	9.4'	28°09.7	-1.2'	54.3'
8	300°36.0	25.9		297°23.2	9.4'	28°08.5	-1.3'	54.3'
9	315°35.6	•• 25.9		311°51.6	9.4'	28°07.2	-1.5'	54.4'
10	330°35.3	26.0		326°19.9	9.4'	28°05.7	-1.6'	54.4'
11	345°35.0	26.0		340°48.3	9.4'	28°04.1	-1.8'	54.4'
12	0°34.7	S23°26.0		355°16.6	9.4'	S28°02.4	-1.9'	54.4'
13	15°34.4	26.0		9°45.0	9.4'	28°00.5	-2.0'	54.4'
14	30°34.1	26.0		24°13.4	9.4'	27°58.4	-2.2'	54.4'
15	45°33.8	•• 26.1		38°41.7	9.4'	27°56.3	-2.3'	54.4'
16	60°33.5	26.1		53°10.1	9.4'	27°53.9	-2.5'	54.4'
17	75°33.2	26.1		67°38.5	9.4'	27°51.5	-2.6'	54.4'
18	90°32.8	S23°26.1		82°06.9	9.4'	S27°48.9	-2.7'	54.5'
19	105°32.5	26.1		96°35.4	9.4'	27°46.1	-2.9'	54.5'
20	120°32.2	26.1		111°03.8	9.5'	27°43.3	-3.0'	54.5'
21	135°31.9	•• 26.2		125°32.3	9.5'	27°40.2	-3.2'	54.5'
22	150°31.6	26.2		140°00.7	9.5'	27°37.1	-3.3'	54.5'
23	165°31.3	26.2		154°29.2	9.5'	27°33.8	-3.4'	54.5'
SD = 16.2'			$d = 0.0'$	SD = 14.8'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:25	10:56	■	■	12:58	15:30
N 70°	08:04	09:53	■	■	14:01	15:50
68°	07:49	09:18	■	■	14:36	16:06
66°	07:35	08:53	10:33	13:21	15:02	16:19
64°	07:24	08:33	09:51	14:03	15:22	16:30
62°	07:15	08:17	09:23	14:32	15:38	16:40
60°	07:06	08:03	09:01	14:53	15:51	16:48
N 58°	06:59	07:51	08:43	15:11	16:03	16:56
56°	06:52	07:41	08:29	15:26	16:13	17:02
54°	06:46	07:32	08:16	15:38	16:22	17:09
52°	06:40	07:24	08:05	15:50	16:31	17:14
50°	06:35	07:16	07:55	15:59	16:38	17:20
45°	06:23	07:00	07:34	16:20	16:54	17:31
N 40°	06:13	06:47	07:17	16:37	17:08	17:42
35°	06:03	06:35	07:03	16:51	17:19	17:51
30°	05:54	06:24	06:51	17:04	17:30	18:00
20°	05:38	06:05	06:29	17:25	17:49	18:16
N 10°	05:22	05:48	06:11	17:43	18:06	18:33
0°	05:05	05:31	05:53	18:01	18:23	18:50
S 10°	04:46	05:13	05:36	18:19	18:42	19:09
20°	04:23	04:52	05:17	18:38	19:02	19:31
30°	03:54	04:27	04:55	19:00	19:27	20:00
35°	03:35	04:12	04:42	19:13	19:42	20:19
40°	03:12	03:54	04:27	19:28	20:01	20:42
45°	02:41	03:31	04:09	19:46	20:23	21:13
S 50°	01:56	03:01	03:46	20:08	20:53	21:59
52°	01:28	02:46	03:35	20:19	21:08	22:27
54°	00:41	02:28	03:23	20:32	21:26	23:14
56°	—:—	02:06	03:09	20:46	21:49	—:—
58°	—:—	01:36	02:52	21:03	22:18	—:—
S 60°	—:—	00:45	02:31	21:23	23:10	—:—

Lat.	Moonrise			Moonset		
	Thu	Fri	Sat	Thu	Fri	Sat
N 72°	■	■	■	■	■	■
N 70°	■	■	■	■	■	■
68°	■	■	■	■	■	■
66°	■	■	■	■	■	■
64°	■	■	■	■	■	■
62°	09:17	■	■	11:48	■	■
60°	08:33	09:55	10:49	12:33	12:57	13:53
N 58°	08:04	09:18	10:11	13:03	13:35	14:30
56°	07:42	08:51	09:45	13:25	14:02	14:57
54°	07:23	08:30	09:24	13:44	14:23	15:17
52°	07:08	08:13	09:06	14:00	14:40	15:35
50°	06:54	07:58	08:51	14:13	14:55	15:50
45°	06:27	07:28	08:21	14:41	15:25	16:20
N 40°	06:06	07:04	07:58	15:03	15:49	16:43
35°	05:48	06:45	07:38	15:21	16:08	17:02
30°	05:33	06:29	07:22	15:37	16:25	17:19
20°	05:07	06:01	06:54	16:04	16:53	17:46
N 10°	04:45	05:37	06:29	16:27	17:17	18:10
0°	04:24	05:15	06:07	16:48	17:40	18:32
S 10°	04:04	04:53	05:44	17:10	18:02	18:54
20°	03:42	04:29	05:20	17:33	18:26	19:18
30°	03:16	04:01	04:52	18:		

December 21, 22, 23 UT (Sun., Mon., Tue.)

Aries			Venus			Mars			Jupiter			Saturn			Stars		
Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec		
0	89°49.2	184°52.6	S23°26.0	175°06.4	S24°11.1	335°15.2	N21°44.0	92°55.5	S03°50.8	Alpheratz	357°33.9	29°14.2					
1	104°51.7	199°51.6	26.2	190°06.7	11.1	350°18.0	44.0	107°57.8	50.8	Ankaa	353°06.3	-42°10.1					
2	119°54.1	214°50.7	26.4	205°07.1	11.0	5°20.8	44.1	123°00.2	50.7	Schedar	349°30.0	56°41.1					
3	134°56.6	229°49.7	•• 26.6	220°07.5	•• 11.0	20°23.5	•• 44.1	138°02.6	•• 50.7	Diphda	348°46.4	-17°50.7					
4	149°59.1	244°48.7	26.8	235°07.9	11.0	35°26.3	44.2	153°04.9	50.6	Achernar	335°19.3	-57°06.5					
5	165°01.5	259°47.8	27.0	250°08.3	10.9	50°29.1	44.2	168°07.3	50.6	Hamal	327°50.1	23°35.3					
6	180°04.0	274°46.8	S23°27.2	265°08.7	S24°10.9	65°31.8	N21°44.3	183°09.7	S03°50.5	Polaris	313°14.5	89°22.6					
7	195°06.4	289°45.9	27.4	280°09.1	10.8	80°34.6	44.4	198°12.1	50.5	Acamar	315°10.9	-40°12.1					
8	210°08.9	304°44.9	27.6	295°09.4	10.8	95°37.4	44.4	213°14.4	50.4	Menkar	314°05.1	4°11.5					
9	225°11.4	319°43.9	•• 27.8	310°09.8	•• 10.7	110°40.1	•• 44.5	228°16.8	•• 50.4	Mirfak	308°26.7	49°57.4					
10	240°13.8	334°43.0	28.0	325°10.2	10.7	125°42.9	44.5	243°19.2	50.3	Aldebaran	290°38.4	16°33.7					
11	255°16.3	349°42.0	28.2	340°10.6	10.6	140°45.7	44.6	258°21.6	50.3	Rigel	281°02.8	-8°10.3					
12	270°18.8	4°41.0	S23°28.4	355°11.0	S24°10.6	155°48.5	N21°44.6	273°23.9	S03°50.2	Capella	280°20.2	46°01.5					
13	285°21.2	19°40.1	28.6	10°11.4	10.5	170°51.2	44.7	288°26.3	50.2	Bellatrix	278°21.7	6°22.4					
14	300°23.7	34°39.1	28.8	25°11.8	10.5	185°54.0	44.7	303°28.7	50.1	Elnath	278°00.5	28°37.8					
15	315°26.2	49°38.2	•• 29.0	40°12.1	•• 10.4	200°56.8	•• 44.8	318°31.1	•• 50.1	Alnilam	275°36.6	-1°11.1					
16	330°28.6	64°37.2	29.2	55°12.5	10.4	215°59.5	44.8	333°33.4	50.0	Betelgeuse	270°50.9	7°24.7					
17	345°31.1	79°36.2	29.4	70°12.9	10.3	231°02.3	44.9	348°35.8	50.0	Canopus	263°51.5	-52°42.5					
18	0°33.6	94°35.3	S23°29.6	85°13.3	S24°10.3	246°05.1	N21°45.0	3°38.2	S03°49.9	Sirius	258°25.2	-16°45.1					
19	15°36.0	109°34.3	29.8	100°13.7	10.2	261°07.9	45.0	18°40.5	49.9	Adhara	255°04.9	-29°00.4					
20	30°38.5	124°33.3	30.0	115°14.1	10.2	276°10.6	45.1	33°42.9	49.8	Procyon	244°49.7	5°09.5					
21	45°40.9	139°32.4	•• 30.1	130°14.4	•• 10.1	291°13.4	•• 45.1	48°45.3	•• 49.8	Pollux	243°16.0	27°57.7					
22	60°43.4	154°31.4	30.3	145°14.8	10.1	306°16.2	45.2	63°47.7	49.7	Avior	234°13.9	-59°35.4					
23	75°45.9	169°30.5	30.5	160°15.2	10.0	321°18.9	45.2	78°50.0	49.7	Suhail	222°45.4	-43°32.1					
Mer.pass. 17:58			ν -1.0' d 0.2' m -3.90			ν 0.4' d -0.0' m 1.19			ν 2.8' d 0.1' m -2.64			ν 2.4' d -0.0' m 0.98					

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	90°48.3	184°29.5	S23°30.7	175°15.6	S24°09.9	336°21.7	N21°45.3	93°52.4	S03°49.6					
1	105°50.8	199°28.5	30.9	190°16.0	09.9	351°24.5	45.3	108°54.8	49.6					
2	120°53.3	214°27.6	31.0	205°16.4	09.8	6°27.3	45.4	123°57.1	49.5					
3	135°55.7	229°26.6	•• 31.2	220°16.8	•• 09.8	21°30.0	•• 45.4	138°59.5	•• 49.5					
4	150°58.2	244°25.6	31.4	235°17.1	09.7	36°32.8	45.5	154°01.9	49.4					
5	166°00.7	259°24.7	31.6	250°17.5	09.7	51°35.6	45.5	169°04.3	49.4					
6	181°03.1	274°23.7	S23°31.8	265°17.9	S24°09.6	66°38.4	N21°45.6	184°06.6	S03°49.3					
7	196°05.6	289°22.7	31.9	280°18.3	09.5	81°41.1	45.7	199°09.0	49.3					
8	211°08.1	304°21.8	32.1	295°18.7	09.5	96°43.9	45.7	214°11.4	49.2					
9	226°10.5	319°20.8	•• 32.3	310°19.1	•• 09.4	111°46.7	•• 45.8	229°13.7	•• 49.2					
10	241°13.0	334°19.8	32.4	325°19.4	09.4	126°49.5	45.8	244°16.1	49.1					
11	256°15.4	349°18.9	32.6	340°19.8	09.3	141°52.2	45.9	259°18.5	49.1					
12	271°17.9	4°17.9	S23°32.8	355°20.2	S24°09.2	156°55.0	N21°45.9	274°20.8	S03°49.0					
13	286°20.4	19°16.9	32.9	10°20.6	09.2	171°57.8	46.0	289°23.2	49.0					
14	301°22.8	34°16.0	33.1	25°21.0	09.1	187°00.6	46.0	304°25.6	48.9					
15	316°25.3	49°15.0	•• 33.3	40°21.4	•• 09.1	202°03.3	•• 46.1	319°27.9	•• 48.9					
16	331°27.8	64°14.1	33.4	55°21.7	09.0	217°06.1	46.2	334°30.3	48.8					
17	346°30.2	79°13.1	33.6	70°22.1	08.9	232°08.9	46.2	349°32.7	48.8					
18	1°32.7	94°12.1	S23°33.7	85°22.5	S24°08.9	247°11.7	N21°46.3	4°35.1	S03°48.7					
19	16°35.2	109°11.2	33.9	100°22.9	08.8	262°14.4	46.3	19°37.4	48.7					
20	31°37.6	124°10.2	34.0	115°23.3	08.7	277°17.2	46.4	34°39.8	48.6					
21	46°40.1	139°09.2	•• 34.2	130°23.7	•• 08.7	292°20.0	•• 46.4	49°42.2	•• 48.6					
22	61°42.6	154°08.3	34.4	145°24.0	08.6	307°22.8	46.5	64°44.5	48.5					
23	76°45.0	169°07.3	34.5	160°24.4	08.5	322°25.6	46.5	79°46.9	48.5					
Mer.pass. 17:54			ν -1.0' d 0.2' m -3.90			ν 0.4' d -0.1' m 1.18			ν 2.8' d 0.1' m -2.65			ν 2.4' d -0.0' m 0.98		

Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec					
0	91°47.5	184°06.3	S23°34.7	175°24.8	S24°08.5	337°28.3	N21°46.6	94°49.3	S03°48.4					
1	106°49.9	199°05.3	34.8	190°25.2	08.4	352°31.1	46.6	109°51.6	48.4					
2	121°52.4	214°04.4	35.0	205°25.6	08.3	7°33.9	46.7	124°54.0	48.3					
3	136°54.9	229°03.4	•• 35.1	220°26.0	•• 08.3	22°36.7	•• 46.8	139°56.4	•• 48.3					
4	151°57.3	244°02.4	35.2	235°26.3	08.2	37°39.4	46.8	154°58.7	48.2					
5	166°59.8	259°01.5	35.4	250°26.7	08.1	52°42.2	46.9	170°01.1	48.2					
6	182°02.3	274°00.5	S23°35.5	265°27.1	S24°08.1	67°45.0	N21°46.9	185°03.5	S03°48.1					
7	197°04.7	288°59.5	35.7	280°27.5	08.0	82°47.8	47.0	200°05.8	48.1					
8	212°07.2	303°58.6	35.8	295°27.9	07.9	97°50.6	47.0	215°08.2	48.0					
9	227°09.7	318°57.6	•• 36.0	310°28.3	•• 07.9	112°53.3	•• 47.1	230°10.6	•• 48.0					
10	242°12.1	333°56.6	36.1	325°28.6	07.8	127°56.1	47.1	245°12.9	47.9					
11	257°14.6	348°55.7	36.2	340°29.0	07.7	142°58.9	47.2	260°15.3	47.9					
12	272°17.1	3°54.7	S23°36.4	355°29.4	S24°07.6	158°01.7	N21°47.2	275°17.7	S03°47.8					
13	287°19.5	18°53.7	36.5	10°29.8	07.6	173°04.5	47.3	290°20.0	47.8					
14	302°22.0	33°52.8	36.6	25°30.2	07.5	188°07.2	47.4	305°22.4	47.7					
15	317°24.4	48°51.8	•• 36.8	40°30.6	•• 07.4	203°10.0	•• 47.4	320°24.7	•• 47.6					
16	332°26.9	63°50.8	36.9	55°30.9	07.3	218°12.8	47.5	335°27.1	47.6					
17	347°29.4	78°49.9	37.0	70°31.3	07.3	233°15.6	47.5	350°29.5	47.5					
18	2°31.8	93°48.9	S23°37.1	85°31.7	S24°07.2	248°18.4	N21°47.6	5°31.8	S03°47.5					
19	17°34.3	108°47.9	37.3	100°32.1	07.1	263°21.1	47.6	20°34.2	47.4					
20	32°36.8	123°46.9	37.4	115°32.5	07.0	278°23.9	47.7	35°36.6	47.4					
21	47°39.2	138°46.0	•• 37.5	130°32.8	•• 07.0	293°26.7	•• 47.7	50°38.9	•• 47.3					
22	62°41.7	153°45.0	37.6	145°33.2	06.9	308°29.5	47.8	65°41.3	47.3					
23	77°44.2	168°44.0	37.8	160°33.6	06.8	323°32.3	47.9	80°43.7	47.2					
Mer.pass. 17:50			ν -1.0' d 0.1' m -3.90			ν 0.4' d -0.1' m 1.18			ν 2.8' d 0.1' m -2.65			ν 2.4' d -0.1' m 0.99		

Dec 21 Sun	SHA	Mer.pass
Venus	95°03.4	11:41
Mars	85°17.2	12:19
Jupiter	245°26.0	01:39
Saturn	3°06.3	17:45

Dec 22 Mon	SHA	Mer.pass
Venus	93°41.2	11:43
Mars	84°27.3	12:19
Jupiter	245°33.4	01:34
Saturn	3°04.1	17:42

Dec 23 Tue	SHA	Mer.pass
Venus	92°18.8	11:44
Mars	83°37.3	12:18
Jupiter	245°40.8	01:30
Saturn	3°01.8	17:38

Sun				Moon				
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	180°31.0	S23°26.2		168°57.7	9.5'	S27°30.3	-3.6'	54.5'
1	195°30.7	26.2		183°26.3	9.6'	27°26.7	-3.7'	54.6'
2	210°30.4	26.2		197°54.8	9.6'	27°23.0	-3.9'	54.6'
3	225°30.0	· · 26.2		212°23.4	9.6'	27°19.2	-4.0'	54.6'
4	240°29.7	26.2		226°52.0	9.6'	27°15.2	-4.1'	54.6'
5	255°29.4	26.3		241°20.6	9.7'	27°11.0	-4.3'	54.6'
6	270°29.1	S23°26.3		255°49.3	9.7'	S27°06.8	-4.4'	54.6'
7	285°28.8	26.3		270°18.0	9.7'	27°02.3	-4.5'	54.6'
8	300°28.5	26.3		284°46.7	9.7'	26°57.8	-4.7'	54.6'
9	315°28.2	· · 26.3		299°15.4	9.8'	26°53.1	-4.8'	54.7'
10	330°27.9	26.3		313°44.2	9.8'	26°48.3	-5.0'	54.7'
11	345°27.6	26.3		328°13.0	9.8'	26°43.3	-5.1'	54.7'
12	0°27.2	S23°26.3		342°41.8	9.9'	S26°38.2	-5.2'	54.7'
13	15°26.9	26.3		357°10.7	9.9'	26°33.0	-5.4'	54.7'
14	30°26.6	26.3		11°39.6	9.9'	26°27.7	-5.5'	54.7'
15	45°26.3	· · 26.3		26°08.6	10.0'	26°22.2	-5.6'	54.7'
16	60°26.0	26.3		40°37.5	10.0'	26°16.5	-5.8'	54.8'
17	75°25.7	26.3		55°06.6	10.1'	26°10.8	-5.9'	54.8'
18	90°25.4	S23°26.3		69°35.6	10.1'	S26°04.9	-6.0'	54.8'
19	105°25.1	26.3		84°04.7	10.1'	25°58.9	-6.2'	54.8'
20	120°24.7	26.3		98°33.9	10.2'	25°52.7	-6.3'	54.8'
21	135°24.4	· · 26.3		113°03.0	10.2'	25°46.4	-6.4'	54.8'
22	150°24.1	26.3		127°32.3	10.3'	25°40.0	-6.5'	54.9'
23	165°23.8	26.3		142°01.5	10.3'	25°33.5	-6.7'	54.9'
SD = 16.2'			d = 0.0'	SD = 14.9'				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	180°23.5	S23°26.3		156°30.9	10.4'	S25°26.8	-6.8'	54.9'
1	195°23.2	26.3		171°00.2	10.4'	25°20.0	-6.9'	54.9'
2	210°22.9	26.2		185°29.6	10.4'	25°13.1	-7.1'	54.9'
3	225°22.6	· · 26.2		199°59.1	10.5'	25°06.0	-7.2'	54.9'
4	240°22.3	26.2		214°28.6	10.5'	24°58.9	-7.3'	54.9'
5	255°21.9	26.2		228°58.1	10.6'	24°51.6	-7.4'	55.0'
6	270°21.6	S23°26.2		243°27.7	10.6'	S24°44.1	-7.5'	55.0'
7	285°21.3	26.2		257°57.3	10.7'	24°36.6	-7.7'	55.0'
8	300°21.0	26.2		272°27.0	10.7'	24°28.9	-7.8'	55.0'
9	315°20.7	· · 26.2		286°56.8	10.8'	24°21.1	-7.9'	55.0'
10	330°20.4	26.1		301°26.5	10.8'	24°13.2	-8.0'	55.0'
11	345°20.1	26.1		315°56.4	10.9'	24°05.2	-8.2'	55.1'
12	0°19.8	S23°26.1		330°26.3	10.9'	S23°57.0	-8.3'	55.1'
13	15°19.5	26.1		344°56.2	11.0'	23°48.7	-8.4'	55.1'
14	30°19.1	26.1		359°26.2	11.0'	23°40.3	-8.5'	55.1'
15	45°18.8	· · 26.1		13°56.3	11.1'	23°31.8	-8.6'	55.1'
16	60°18.5	26.0		28°26.4	11.2'	23°23.2	-8.7'	55.2'
17	75°18.2	26.0		42°56.5	11.2'	23°14.4	-8.9'	55.2'
18	90°17.9	S23°26.0		57°26.7	11.3'	S23°05.6	-9.0'	55.2'
19	105°17.6	26.0		71°57.0	11.3'	22°56.6	-9.1'	55.2'
20	120°17.3	26.0		86°27.3	11.4'	22°47.5	-9.2'	55.2'
21	135°17.0	· · 25.9		100°57.6	11.4'	22°38.3	-9.3'	55.2'
22	150°16.6	25.9		115°28.1	11.5'	22°29.0	-9.4'	55.3'
23	165°16.3	25.9		129°58.5	11.5'	22°19.6	-9.5'	55.3'
SD = 16.3'			d = -0.0'	SD = 15.0'				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	180°16.0	S23°25.8		144°29.0	11.6'	S22°10.0	-9.6'	55.3'
1	195°15.7	25.8		158°59.6	11.6'	22°00.4	-9.8'	55.3'
2	210°15.4	25.8		173°30.3	11.7'	21°50.6	-9.9'	55.3'
3	225°15.1	· · 25.8		188°00.9	11.7'	21°40.8	-10.0'	55.4'
4	240°14.8	25.7		202°31.7	11.8'	21°30.8	-10.1'	55.4'
5	255°14.5	25.7		217°02.5	11.8'	21°20.7	-10.2'	55.4'
6	270°14.2	S23°25.7		231°33.3	11.9'	S21°10.6	-10.3'	55.4'
7	285°13.8	25.6		246°04.2	12.0'	21°00.3	-10.4'	55.4'
8	300°13.5	25.6		260°35.2	12.0'	20°49.9	-10.5'	55.4'
9	315°13.2	· · 25.6		275°06.2	12.1'	20°39.4	-10.6'	55.5'
10	330°12.9	25.5		289°37.2	12.1'	20°28.8	-10.7'	55.5'
11	345°12.6	25.5		304°08.4	12.2'	20°18.1	-10.8'	55.5'
12	0°12.3	S23°25.5		318°39.5	12.2'	S20°07.3	-10.9'	55.5'
13	15°12.0	25.4		333°10.7	12.3'	19°56.4	-11.0'	55.5'
14	30°11.7	25.4		347°42.0	12.3'	19°45.4	-11.1'	55.6'
15	45°11.4	· · 25.4		2°13.3	12.4'	19°34.3	-11.2'	55.6'
16	60°11.0	25.3		16°44.7	12.4'	19°23.2	-11.3'	55.6'
17	75°10.7	25.3		31°16.1	12.5'	19°11.9	-11.4'	55.6'
18	90°10.4	S23°25.2		45°47.6	12.5'	S19°00.5	-11.5'	55.6'
19	105°10.1	25.2		60°19.1	12.6'	18°49.0	-11.6'	55.7'
20	120°09.8	25.1		74°50.7	12.6'	18°37.5	-11.7'	55.7'
21	135°09.5	· · 25.1		89°22.3	12.7'	18°25.8	-11.7'	55.7'
22	150°09.2	25.1		103°54.0	12.7'	18°14.1	-11.8'	55.7'
23	165°08.9	25.0		118°25.8	12.8'	18°02.2	-11.9'	55.8'
SD = 16.3'			d = -0.0'	SD = 15.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:26	10:58	■	■	12:59	15:31
N 70°	08:06	09:55	■	■	14:02	15:51
68°	07:50	09:20	■	■	14:38	16:07
66°	07:37	08:54	10:35	13:22	15:03	16:20
64°	07:26	08:35	09:53	14:05	15:23	16:31
62°	07:16	08:18	09:24	14:33	15:39	16:41
60°	07:08	08:05	09:03	14:55	15:53	16:50
N 58°	07:00	07:53	08:45	15:12	16:04	16:57
56°	06:54	07:43	08:30	15:27	16:15	17:04
54°	06:47	07:34	08:18	15:40	16:24	17:10
52°	06:42	07:25	08:06	15:51	16:32	17:16
50°	06:36	07:18	07:56	16:01	16:39	17:21
45°	06:25	07:02	07:36	16:22	16:56	17:33
N 40°	06:14	06:48	07:19	16:39	17:09	17:43
35°	06:05	06:36	07:05	16:53	17:21	17:53
30°	05:56	06:26	06:52	17:05	17:32	18:01
20°	05:39	06:07	06:31	17:26	17:50	18:18
N 10°	05:23	05:50	06:12	17:45	18:08	18:34
0°	05:06	05:32	05:55	18:02	18:25	18:51
S 10°	04:47	05:14	05:37	18:20	18:43	19:10
20°	04:24	04:54	05:18	18:39	19:04	19:33
30°	03:55	04:29	04:56	19:01	19:29	20:02
35°	03:37	04:13	04:43	19:14	19:44	20:21
40°	03:13	03:55	04:28	19:29	20:02	20:44
45°	02:43	03:33	04:10	19:47	20:25	21:15
S 50°	01:57	03:03	03:47	20:10	20:55	22:00
52°	01:29	02:48	03:37	20:21	21:10	22:28
54°	00:42	02:30	03:24	20:33	21:28	23:15
56°	—:—	02:07	03:10	20:47	21:50	—:—
58°	—:—	01:37	02:53	21:04	22:20	—:—
S 60°	—:—	00:46	02:33	21:25	23:11	—:—

Lat.	Moonrise			Moonset		
	Sun	Mon	Tue	Sun	Mon	Tue
N 72°	■	■	■	■	■	■
N 70°	■	■	14:17	■	■	15:44
68°	■	■	13:05	■	■	16:54
66°	■	13:33	12:28	■	14:44	17:30
64°	■	12:23	12:02	■	15:54	17:55
62°	11:55	11:47	11:41	14:36	16:29	18:15
60°	11:12	11:21	11:24	15:18	16:54	18:31
N 58°	10:43	11:00	11:10	15:47	17:14	18:45
56°	10:21	10:43	10:57	16:09	17:31	18:56
54°	10:02	10:29	10:47	16:27	17:45	19:06
52°	09:47	10:16	10:37	16:42	17:57	19:15
50°	09:33	10:05	10:28	16:55	18:08	19:23
45°	09:06	09:41	10:10	17:22	18:30	19:40
N 40°	08:44	09:23	09:55	17:43	18:48	19:54
35°	08:26	09:07	09:42	18:01	19:03	20:06
30°	08:10	08:53	09:31	18:16	19:16	20:16
20°	07:44	08:30	09:12	18:42	19:38	20:33
N 10°	07:21	08:09	08:55	19:04	19:57	20:48
0°	06:59	07:50	08:39	19:24	20:14	21:02
S 10°	06:38	07:31	08:24	19:44	20:32	21:16
20°	06:15	07:11	08:07	20:06	20:50	21:31
30°	05:48	06:47	07:47	20:31	21:12	21:47
35°	05:32	06:33	07:35	20:46	21:24	21:57
40°	05:14	06:17	07:22	21:03	21:38	22:08
45°	04:51	05:57	07:06	21:23	21:55	22:21
S 50°	04:23	05:32	06:47	21:49	22:16	22:37
52°	04:09	05:21	06:38	22:01	22:26	22:44
54°	03:53	0				

December 24, 25, 26 UT (Wed., Thu., Fri.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Wed	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	92°46.6	183°43.1	S23°37.9	175°34.0	S24°06.7	338°35.1	N21°47.9	95°46.0	S03°47.2	Alpheratz	357°34.0	29°14.2
1	107°49.1	198°42.1	38.0	190°34.4	06.7	353°37.8	48.0	110°48.4	47.1	Ankaa	353°06.3	-42°10.1
2	122°51.5	213°41.1	38.1	205°34.8	06.6	8°40.6	48.0	125°50.7	47.1	Schedar	349°30.1	56°41.1
3	137°54.0	228°40.2	· · 38.2	220°35.1	· · 06.5	23°43.4	· · 48.1	140°53.1	· · 47.0	Diphda	348°46.4	-17°50.7
4	152°56.5	243°39.2	38.4	235°35.5	06.4	38°46.2	48.1	155°55.5	47.0	Achernar	335°19.4	-57°06.5
5	167°58.9	258°38.2	38.5	250°35.9	06.3	53°49.0	48.2	170°57.8	46.9	Hamal	327°50.1	23°35.3
6	183°01.4	273°37.2	S23°38.6	265°36.3	S24°06.3	68°51.8	N21°48.2	186°00.2	S03°46.9	Polaris	313°15.5	89°22.7
7	198°03.9	288°36.3	38.7	280°36.7	06.2	83°54.5	48.3	201°02.6	46.8	Acamar	315°10.9	-40°12.2
8	213°06.3	303°35.3	38.8	295°37.0	06.1	98°57.3	48.4	216°04.9	46.7	Menkar	314°05.1	4°11.5
9	228°08.8	318°34.3	· · 38.9	310°37.4	· · 06.0	114°00.1	· · 48.4	231°07.3	· · 46.7	Mirfak	308°26.7	49°57.4
10	243°11.3	333°33.4	39.0	325°37.8	05.9	129°02.9	48.5	246°09.6	46.6	Aldebaran	290°38.4	16°33.7
11	258°13.7	348°32.4	39.1	340°38.2	05.9	144°05.7	48.5	261°12.0	46.6	Rigel	281°02.8	-8°10.3
12	273°16.2	3°31.4	S23°39.2	355°38.6	S24°05.8	159°08.5	N21°48.6	276°14.4	S03°46.5	Capella	280°20.2	46°01.5
13	288°18.7	18°30.4	39.3	10°39.0	05.7	174°11.2	48.6	291°16.7	46.5	Bellatrix	278°21.7	6°22.4
14	303°21.1	33°29.5	39.4	25°39.3	05.6	189°14.0	48.7	306°19.1	46.4	Elnath	278°00.5	28°37.8
15	318°23.6	48°28.5	· · 39.5	40°39.7	· · 05.5	204°16.8	· · 48.7	321°21.5	· · 46.4	Alnilam	275°36.6	-1°11.1
16	333°26.0	63°27.5	39.6	55°40.1	05.4	219°19.6	48.8	336°23.8	46.3	Betelgeuse	270°50.9	7°24.7
17	348°28.5	78°26.6	39.7	70°40.5	05.4	234°22.4	48.9	351°26.2	46.3	Canopus	263°51.5	-52°42.5
18	3°31.0	93°25.6	S23°39.8	85°40.9	S24°05.3	249°25.2	N21°48.9	6°28.5	S03°46.2	Sirius	258°25.2	-16°45.1
19	18°33.4	108°24.6	39.9	100°41.2	05.2	264°28.0	49.0	21°30.9	46.2	Adhara	255°04.9	-29°00.4
20	33°35.9	123°23.6	40.0	115°41.6	05.1	279°30.7	49.0	36°33.3	46.1	Procyon	244°49.7	5°09.5
21	48°38.4	138°22.7	· · 40.1	130°42.0	· · 05.0	294°33.5	· · 49.1	51°35.6	· · 46.0	Pollux	243°16.0	27°57.7
22	63°40.8	153°21.7	40.2	145°42.4	04.9	309°36.3	49.1	66°38.0	46.0	Avior	234°13.9	-59°35.4
23	78°43.3	168°20.7	40.3	160°42.8	04.8	324°39.1	49.2	81°40.3	45.9	Suhail	222°45.4	-43°32.1
Mer.pass. 17:46		ν -1.0' d 0.1' m -3.90		ν 0.4' d -0.1' m 1.17		ν 2.8' d 0.1' m -2.65		ν 2.4' d -0.1' m 0.99		Miaplacidus	221°37.4	-69°49.2
										Alphard	217°46.7	-8°46.3
										Regulus	207°33.4	11°50.3
										Dubhe	193°39.8	61°36.3
										Denebola	182°24.1	14°25.5
										Gienah	175°42.8	-17°41.1
										Acrux	172°59.3	-63°14.3
										Gacrux	171°50.8	-57°15.3
										Alioth	166°12.4	55°48.8
										Spica	158°21.6	-11°17.8
										Alkaid	152°51.6	49°10.7
										Hadar	148°35.2	-60°29.7
										Menkent	147°56.9	-36°29.7
										Arcturus	145°47.4	19°02.7
										Rigel Kent.	139°39.6	-60°56.3
										Kochab	137°20.6	74°02.6
										Zuben'ubi	136°55.4	-16°08.9
										Alphecca	126°03.4	26°37.5
										Antares	112°15.2	-26°29.3
										Atria	107°09.3	-69°04.3
										Sabik	102°02.2	-15°45.4
										Shaula	96°09.7	-37°07.3
										Rasalhague	95°58.2	12°32.4
										Eltanin	90°42.4	51°29.0
										Kaus Aust.	83°31.9	-34°22.3
										Vega	80°33.1	38°48.4
										Nunki	75°47.1	-26°15.9
										Altair	61°59.5	8°56.2
										Peacock	53°05.0	-56°39.2
										Deneb	49°25.7	45°22.5
										Enif	33°38.2	9°59.7
										Al Na'ir	27°32.2	-46°50.3
										Fomalhaut	15°13.8	-29°29.2
										Scheat	13°44.5	28°13.6
										Markab	13°29.2	15°20.8
										Dec 24 Wed	SHA	Mer.pass
										Venus	90°56.4	11:46
										Mars	82°47.4	12:17
										Jupiter	245°48.4	01:25
										Saturn	2°59.4	17:34
										Dec 25 Thu	SHA	Mer.pass
										Venus	89°34.0	11:47
										Mars	81°57.4	12:17
										Jupiter	245°56.1	01:21
										Saturn	2°56.9	17:30
										Dec 26 Fri	SHA	Mer.pass
										Venus	88°11.5	11:49
										Mars	81°07.4	12:16
										Jupiter	246°03.9	01:17
										Saturn	2°54.4	17:27
										Horizontal parallax		
										Venus:	0.1	
										Mars:	0.1	
Mer.pass. 17:38		ν -1.0' d 0.1' m -3.90		ν 0.4' d -0.1' m 1.16		ν 2.8' d 0.1' m -2.66		ν 2.4' d -0.1' m 0.99				

Sun				Moon				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	180°08.5	S23°25.0		132°57.5	12.8'	S17°50.3	-12.0'	55.8'
1	195°08.2	24.9		147°29.4	12.9'	17°38.3	-12.1'	55.8'
2	210°07.9	24.9		162°01.2	12.9'	17°26.2	-12.2'	55.8'
3	225°07.6	· · 24.8		176°33.1	13.0'	17°14.0	-12.3'	55.8'
4	240°07.3	24.8		191°05.1	13.0'	17°01.7	-12.4'	55.9'
5	255°07.0	24.7		205°37.1	13.1'	16°49.4	-12.4'	55.9'
6	270°06.7	S23°24.7		220°09.2	13.1'	S16°36.9	-12.5'	55.9'
7	285°06.4	24.6		234°41.3	13.1'	16°24.4	-12.6'	55.9'
8	300°06.1	24.6		249°13.4	13.2'	16°11.8	-12.7'	55.9'
9	315°05.7	· · 24.5		263°45.6	13.2'	15°59.1	-12.8'	56.0'
10	330°05.4	24.5		278°17.8	13.3'	15°46.4	-12.8'	56.0'
11	345°05.1	24.4		292°50.1	13.3'	15°33.5	-12.9'	56.0'
12	0°04.8	S23°24.3		307°22.4	13.4'	S15°20.6	-13.0'	56.0'
13	15°04.5	24.3		321°54.8	13.4'	15°07.6	-13.1'	56.1'
14	30°04.2	24.2		336°27.1	13.4'	14°54.5	-13.1'	56.1'
15	45°03.9	· · 24.2		350°59.6	13.5'	14°41.4	-13.2'	56.1'
16	60°03.6	24.1		5°32.0	13.5'	14°28.2	-13.3'	56.1'
17	75°03.3	24.1		20°04.6	13.5'	14°14.9	-13.4'	56.2'
18	90°03.0	S23°24.0		34°37.1	13.6'	S14°01.5	-13.4'	56.2'
19	105°02.6	23.9		49°09.7	13.6'	13°48.1	-13.5'	56.2'
20	120°02.3	23.9		63°42.3	13.6'	13°34.5	-13.6'	56.2'
21	135°02.0	· · 23.8		78°14.9	13.7'	13°21.0	-13.7'	56.2'
22	150°01.7	23.7		92°47.6	13.7'	13°07.3	-13.7'	56.3'
23	165°01.4	23.7		107°20.3	13.7'	12°53.6	-13.8'	56.3'
SD = 16.3' d = -0.0'				SD = 15.2'				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	180°01.1	S23°23.6		121°53.1	13.8'	S12°39.8	-13.9'	56.3'
1	195°00.8	23.5		136°25.8	13.8'	12°26.0	-13.9'	56.3'
2	210°00.5	23.5		150°58.7	13.8'	12°12.0	-14.0'	56.4'
3	225°00.2	· · 23.4		165°31.5	13.9'	11°58.1	-14.0'	56.4'
4	239°59.8	23.3		180°04.3	13.9'	11°44.0	-14.1'	56.4'
5	254°59.5	23.3		194°37.2	13.9'	11°29.9	-14.2'	56.4'
6	269°59.2	S23°23.2		209°10.1	13.9'	S11°15.7	-14.2'	56.5'
7	284°58.9	23.1		223°43.1	14.0'	11°01.5	-14.3'	56.5'
8	299°58.6	23.1		238°16.0	14.0'	10°47.2	-14.3'	56.5'
9	314°58.3	· · 23.0		252°49.0	14.0'	10°32.9	-14.4'	56.5'
10	329°58.0	22.9		267°22.0	14.0'	10°18.5	-14.5'	56.6'
11	344°57.7	22.8		281°55.1	14.0'	10°04.0	-14.5'	56.6'
12	359°57.4	S23°22.8		296°28.1	14.1'	S09°49.5	-14.6'	56.6'
13	14°57.1	22.7		311°01.2	14.1'	09°34.9	-14.6'	56.6'
14	29°56.8	22.6		325°34.2	14.1'	09°20.3	-14.7'	56.7'
15	44°56.4	· · 22.5		340°07.3	14.1'	09°05.6	-14.7'	56.7'
16	59°56.1	22.4		354°40.5	14.1'	08°50.9	-14.8'	56.7'
17	74°55.8	22.4		9°13.6	14.1'	08°36.1	-14.8'	56.8'
18	89°55.5	S23°22.3		23°46.7	14.2'	S08°21.3	-14.9'	56.8'
19	104°55.2	22.2		38°19.9	14.2'	08°06.4	-14.9'	56.8'
20	119°54.9	22.1		52°53.0	14.2'	07°51.5	-15.0'	56.8'
21	134°54.6	· · 22.0		67°26.2	14.2'	07°36.5	-15.0'	56.9'
22	149°54.3	22.0		81°59.4	14.2'	07°21.5	-15.1'	56.9'
23	164°54.0	21.9		96°32.6	14.2'	07°06.4	-15.1'	56.9'
SD = 16.3' d = -0.1'				SD = 15.4'				
Fri	GHA	Dec		GHA	ν	Dec	d	HP
0	179°53.7	S23°21.8		111°05.8	14.2'	S06°51.3	-15.2'	56.9'
1	194°53.3	21.7		125°39.0	14.2'	06°36.2	-15.2'	57.0'
2	209°53.0	21.6		140°12.2	14.2'	06°21.0	-15.2'	57.0'
3	224°52.7	· · 21.5		154°45.4	14.2'	06°05.7	-15.3'	57.0'
4	239°52.4	21.4		169°18.6	14.2'	05°50.5	-15.3'	57.0'
5	254°52.1	21.3		183°51.8	14.2'	05°35.1	-15.4'	57.1'
6	269°51.8	S23°21.3		198°25.0	14.2'	S05°19.8	-15.4'	57.1'
7	284°51.5	21.2		212°58.2	14.2'	05°04.4	-15.4'	57.1'
8	299°51.2	21.1		227°31.4	14.2'	04°49.0	-15.5'	57.2'
9	314°50.9	· · 21.0		242°04.6	14.2'	04°33.5	-15.5'	57.2'
10	329°50.6	20.9		256°37.8	14.2'	04°18.0	-15.5'	57.2'
11	344°50.3	20.8		271°11.0	14.2'	04°02.5	-15.6'	57.2'
12	359°50.0	S23°20.7		285°44.1	14.2'	S03°46.9	-15.6'	57.3'
13	14°49.6	20.6		300°17.3	14.1'	03°31.3	-15.6'	57.3'
14	29°49.3	20.5		314°50.4	14.1'	03°15.7	-15.7'	57.3'
15	44°49.0	· · 20.4		329°23.6	14.1'	03°00.0	-15.7'	57.4'
16	59°48.7	20.3		343°56.7	14.1'	02°44.4	-15.7'	57.4'
17	74°48.4	20.2		358°29.8	14.1'	02°28.7	-15.7'	57.4'
18	89°48.1	S23°20.1		13°02.9	14.1'	S02°12.9	-15.8'	57.4'
19	104°47.8	20.0		27°36.0	14.1'	01°57.2	-15.8'	57.5'
20	119°47.5	19.9		42°09.0	14.0'	01°41.4	-15.8'	57.5'
21	134°47.2	· · 19.8		56°42.1	14.0'	01°25.6	-15.8'	57.5'
22	149°46.9	19.7		71°15.1	14.0'	01°09.7	-15.8'	57.6'
23	164°46.6	19.6		85°48.1	14.0'	00°53.9	-15.9'	57.6'
SD = 16.3' d = -0.1'				SD = 15.5'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:27	10:57	■	■	13:04	15:34
N 70°	08:07	09:55	■	■	14:05	15:53
68°	07:51	09:20	■	■	14:40	16:09
66°	07:38	08:55	10:35	13:25	15:05	16:22
64°	07:27	08:35	09:53	14:07	15:25	16:34
62°	07:17	08:19	09:25	14:35	15:41	16:43
60°	07:09	08:06	09:03	14:57	15:55	16:51
N 58°	07:01	07:54	08:46	15:14	16:06	16:59
56°	06:55	07:44	08:31	15:29	16:17	17:06
54°	06:49	07:35	08:19	15:42	16:26	17:12
52°	06:43	07:27	08:08	15:53	16:34	17:18
50°	06:38	07:19	07:58	16:03	16:41	17:23
45°	06:26	07:03	07:37	16:23	16:57	17:35
N 40°	06:15	06:50	07:20	16:40	17:11	17:45
35°	06:06	06:38	07:06	16:54	17:23	17:54
30°	05:57	06:27	06:54	17:07	17:33	18:03
20°	05:41	06:08	06:32	17:28	17:52	18:20
N 10°	05:25	05:51	06:14	17:46	18:09	18:36
0°	05:08	05:34	05:56	18:04	18:26	18:53
S 10°	04:49	05:16	05:39	18:21	18:45	19:12
20°	04:26	04:55	05:20	18:40	19:05	19:34
30°	03:57	04:30	04:58	19:02	19:30	20:03
35°	03:38	04:15	04:45	19:15	19:45	20:22
40°	03:15	03:57	04:30	19:30	20:03	20:45
45°	02:45	03:34	04:12	19:48	20:26	21:16
S 50°	01:59	03:05	03:49	20:11	20:56	22:01
52°	01:31	02:50	03:38	20:22	21:11	22:29
54°	00:45	02:32	03:26	20:34	21:28	23:14
56°	—:—	02:09	03:12	20:48	21:51	—:—
58°	—:—	01:40	02:55	21:05	22:20	—:—
S 60°	—:—	00:49	02:35	21:25	23:10	—:—

Lat.	Moonrise			Moonset		
	Wed	Thu	Fri	Wed	Thu	Fri
N 72°	13:19	12:24	11:46	18:20	20:51	23:04
N 70°	12:47	12:09	11:41	18:51	21:03	23:06
68°	12:23	11:57	11:36	19:13	21:13	23:07
66°	12:04	11:47	11:33	19:30	21:21	23:08
64°	11:49	11:39	11:30	19:44	21:27	23:09
62°	11:36	11:31	11:27	19:56	21:33	23:10
60°	11:25	11:25	11:24	20:06	21:38	23:10
N 58°	11:16	11:19	11:22	20:14	21:42	23:11
56°	11:07	11:14	11:20	20:22	21:46	23:11
54°	11:00	11:10	11:19	20:28	21:50	23:12
52°	10:53	11:06	11:17	20:34	21:53	23:12
50°	10:47	11:02	11:16	20:39	21:56	23:12
45°	10:34	10:54	11:12	20:51	22:02	23:13
N 40°	10:23	10:47	11:10	21:00	22:07	23:14
35°	10:13	10:41	11:07	21:09	22:11	23:14
30°	10:05	10:36	11:05	21:16	22:15	23:15
20°	09:51	10:27	11:02	21:28	22:21	23:15
N 10°	09:38	10:19	10:59	21:38	22:27	23:16
0°	09:26	10:11	10:56	21:4		

December 27, 28, 29 UT (Sat., Sun., Mon.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Sat	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	95°44.0	182°33.0	S23°43.2	176°01.4	S23°59.9	341°55.9	N21°51.9	98°35.8	S03°43.2	Alpheratz	357°34.0	29°14.2
1	110°46.5	197°32.1	43.2	191°01.8	59.8	356°58.7	52.0	113°38.1	43.1	Ankaa	353°06.3	-42°10.1
2	125°49.0	212°31.1	43.3	206°02.2	59.7	12°01.4	52.0	128°40.5	43.1	Schedar	349°30.1	56°41.1
3	140°51.4	227°30.1	· · 43.3	221°02.6	· · 59.6	27°04.2	· · 52.1	143°42.8	· · 43.0	Diphda	348°46.5	-17°50.7
4	155°53.9	242°29.1	43.3	236°02.9	59.5	42°07.0	52.2	158°45.2	43.0	Achernar	335°19.4	-57°06.5
5	170°56.4	257°28.2	43.3	251°03.3	59.4	57°09.8	52.2	173°47.5	42.9	Hamal	327°50.1	23°35.3
6	185°58.8	272°27.2	S23°43.3	266°03.7	S23°59.3	72°12.6	N21°52.3	188°49.9	S03°42.8	Polaris	313°16.6	89°22.7
7	201°01.3	287°26.2	43.4	281°04.1	59.1	87°15.4	52.3	203°52.2	42.8	Acamar	315°10.9	-40°12.2
8	216°03.7	302°25.2	43.4	296°04.5	59.0	102°18.2	52.4	218°54.6	42.7	Menkar	314°05.1	4°11.5
9	231°06.2	317°24.3	· · 43.4	311°04.8	· · 58.9	117°21.0	· · 52.4	233°56.9	· · 42.7	Mirfak	308°26.7	49°57.4
10	246°08.7	332°23.3	43.4	326°05.2	58.8	132°23.8	52.5	248°59.3	42.6	Aldebaran	290°38.4	16°33.7
11	261°11.1	347°22.3	43.4	341°05.6	58.7	147°26.6	52.6	264°01.6	42.5	Rigel	281°02.8	-8°10.3
12	276°13.6	2°21.3	S23°43.4	356°06.0	S23°58.6	162°29.4	N21°52.6	279°04.0	S03°42.5	Capella	280°20.2	46°01.5
13	291°16.1	17°20.4	43.5	11°06.4	58.4	177°32.2	52.7	294°06.3	42.4	Bellatrix	278°21.7	6°22.4
14	306°18.5	32°19.4	43.5	26°06.7	58.3	192°35.0	52.7	309°08.7	42.4	Elnath	278°00.5	28°37.8
15	321°21.0	47°18.4	· · 43.5	41°07.1	· · 58.2	207°37.8	· · 52.8	324°11.0	· · 42.3	Alnilam	275°36.6	-1°11.1
16	336°23.5	62°17.4	43.5	56°07.5	58.1	222°40.6	52.8	339°13.4	42.2	Betelgeuse	270°50.9	7°24.7
17	351°25.9	77°16.5	43.5	71°07.9	58.0	237°43.4	52.9	354°15.7	42.2	Canopus	263°51.5	-52°42.5
18	6°28.4	92°15.5	S23°43.5	86°08.3	S23°57.8	252°46.2	N21°53.0	9°18.1	S03°42.1	Sirius	258°25.2	-16°45.1
19	21°30.9	107°14.5	43.5	101°08.6	57.7	267°49.0	53.0	24°20.4	42.1	Adhara	255°04.8	-29°00.4
20	36°33.3	122°13.5	43.5	116°09.0	57.6	282°51.8	53.1	39°22.8	42.0	Procyon	244°49.7	5°09.5
21	51°35.8	137°12.6	· · 43.5	131°09.4	· · 57.5	297°54.6	· · 53.1	54°25.1	· · 42.0	Pollux	243°16.0	27°57.7
22	66°38.2	152°11.6	43.5	146°09.8	57.4	312°57.4	53.2	69°27.5	41.9	Avior	234°13.8	-59°35.4
23	81°40.7	167°10.6	43.5	161°10.1	57.2	328°00.2	53.2	84°29.8	41.8	Suhail	222°45.4	-43°32.1
Mer.pass. 17:34		ν -1.0' d 0.0' m -3.90		ν 0.4' d -0.1' m 1.15		ν 2.8' d 0.1' m -2.66		ν 2.4' d -0.1' m 1.00				

Sun	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	96°43.2	182°09.7	S23°43.5	176°10.5	S23°57.1	343°03.0	N21°53.3	99°32.2	S03°41.8
1	111°45.6	197°08.7	43.5	191°10.9	57.0	358°05.8	53.3	114°34.5	41.7
2	126°48.1	212°07.7	43.5	206°11.3	56.9	13°08.6	53.4	129°36.9	41.7
3	141°50.6	227°06.7	· · 43.5	221°11.7	· · 56.7	28°11.4	· · 53.5	144°39.2	· · 41.6
4	156°53.0	242°05.8	43.5	236°12.0	56.6	43°14.2	53.5	159°41.6	41.5
5	171°55.5	257°04.8	43.5	251°12.4	56.5	58°17.0	53.6	174°43.9	41.5
6	186°58.0	272°03.8	S23°43.5	266°12.8	S23°56.4	73°19.8	N21°53.6	189°46.2	S03°41.4
7	202°00.4	287°02.8	43.5	281°13.2	56.2	88°22.6	53.7	204°48.6	41.4
8	217°02.9	302°01.9	43.4	296°13.6	56.1	103°25.4	53.7	219°50.9	41.3
9	232°05.4	317°00.9	· · 43.4	311°13.9	· · 56.0	118°28.2	· · 53.8	234°53.3	· · 41.2
10	247°07.8	331°59.9	43.4	326°14.3	55.9	133°31.0	53.9	249°55.6	41.2
11	262°10.3	346°58.9	43.4	341°14.7	55.7	148°33.8	53.9	264°58.0	41.1
12	277°12.7	1°58.0	S23°43.4	356°15.1	S23°55.6	163°36.6	N21°54.0	280°00.3	S03°41.1
13	292°15.2	16°57.0	43.4	11°15.5	55.5	178°39.4	54.0	295°02.7	41.0
14	307°17.7	31°56.0	43.3	26°15.8	55.4	193°42.2	54.1	310°05.0	40.9
15	322°20.1	46°55.0	· · 43.3	41°16.2	· · 55.2	208°45.0	· · 54.1	325°07.4	· · 40.9
16	337°22.6	61°54.1	43.3	56°16.6	55.1	223°47.8	54.2	340°09.7	40.8
17	352°25.1	76°53.1	43.3	71°17.0	55.0	238°50.6	54.3	355°12.1	40.8
18	7°27.5	91°52.1	S23°43.2	86°17.4	S23°54.8	253°53.4	N21°54.3	10°14.4	S03°40.7
19	22°30.0	106°51.1	43.2	101°17.7	54.7	268°56.2	54.4	25°16.7	40.6
20	37°32.5	121°50.2	43.2	116°18.1	54.6	283°59.0	54.4	40°19.1	40.6
21	52°34.9	136°49.2	· · 43.2	131°18.5	· · 54.4	299°01.8	· · 54.5	55°21.4	· · 40.5
22	67°37.4	151°48.2	43.1	146°18.9	54.3	314°04.6	54.5	70°23.8	40.5
23	82°39.8	166°47.2	43.1	161°19.3	54.2	329°07.4	54.6	85°26.1	40.4
Mer.pass. 17:30		ν -1.0' d -0.0' m -3.90		ν 0.4' d -0.1' m 1.15		ν 2.8' d 0.1' m -2.66		ν 2.3' d -0.1' m 1.00	

Mon	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec
0	97°42.3	181°46.3	S23°43.1	176°19.6	S23°54.0	344°10.2	N21°54.7	100°28.5	S03°40.3
1	112°44.8	196°45.3	43.0	191°20.0	53.9	359°13.0	54.7	115°30.8	40.3
2	127°47.2	211°44.3	43.0	206°20.4	53.8	14°15.8	54.8	130°33.2	40.2
3	142°49.7	226°43.4	· · 43.0	221°20.8	· · 53.6	29°18.6	· · 54.8	145°35.5	· · 40.2
4	157°52.2	241°42.4	42.9	236°21.2	53.5	44°21.4	54.9	160°37.8	40.1
5	172°54.6	256°41.4	42.9	251°21.5	53.4	59°24.2	54.9	175°40.2	40.0
6	187°57.1	271°40.4	S23°42.8	266°21.9	S23°53.2	74°27.0	N21°55.0	190°42.5	S03°40.0
7	202°59.6	286°39.5	42.8	281°22.3	53.1	89°29.8	55.0	205°44.9	39.9
8	218°02.0	301°38.5	42.8	296°22.7	53.0	104°32.6	55.1	220°47.2	39.8
9	233°04.5	316°37.5	· · 42.7	311°23.1	· · 52.8	119°35.4	· · 55.2	235°49.6	· · 39.8
10	248°07.0	331°36.5	42.7	326°23.4	52.7	134°38.2	55.2	250°51.9	39.7
11	263°09.4	346°35.6	42.6	341°23.8	52.5	149°41.0	55.3	265°54.2	39.7
12	278°11.9	1°34.6	S23°42.6	356°24.2	S23°52.4	164°43.8	N21°55.3	280°56.6	S03°39.6
13	293°14.3	16°33.6	42.5	11°24.6	52.3	179°46.6	55.4	295°58.9	39.5
14	308°16.8	31°32.6	42.5	26°24.9	52.1	194°49.4	55.4	311°01.3	39.5
15	323°19.3	46°31.7	· · 42.4	41°25.3	· · 52.0	209°52.2	· · 55.5	326°03.6	· · 39.4
16	338°21.7	61°30.7	42.4	56°25.7	51.8	224°55.0	55.6	341°06.0	39.4
17	353°24.2	76°29.7	42.3	71°26.1	51.7	239°57.8	55.6	356°08.3	39.3
18	8°26.7	91°28.8	S23°42.3	86°26.5	S23°51.6	255°00.7	N21°55.7	11°10.6	S03°39.2
19	23°29.1	106°27.8	42.2	101°26.8	51.4	270°03.5	55.7	26°13.0	39.2
20	38°31.6	121°26.8	42.1	116°27.2	51.3	285°06.3	55.8	41°15.3	39.1
21	53°34.1	136°25.8	· · 42.1	131°27.6	· · 51.1	300°09.1	· · 55.8	56°17.7	· · 39.0
22	68°36.5	151°24.9	42.0	146°28.0	51.0	315°11.9	55.9	71°20.0	39.0
23	83°39.0	166°23.9	42.0	161°28.4	50.8	330°14.7	56.0	86°22.4	38.9
Mer.pass. 17:26		ν -1.0' d -0.0' m -3.90		ν 0.4' d -0.1' m 1.14		ν 2.8' d 0.1' m -2.67		ν 2.3' d -0.1' m 1.00	

Dec 27 Sat	SHA	Mer.pass
Venus	86°49.0	11:51
Mars	80°17.4	12:16
Jupiter	246°11.8	01:12
Saturn	2°51.7	17:23

Dec 28 Sun	SHA	Mer.pass
Venus	85°26.5	11:52
Mars	79°27.4	12:15
Jupiter	246°19.8	01:08
Saturn	2°49.0	17:19

Dec 29 Mon	SHA	Mer.pass
Venus	84°04.0	11:54
Mars	78°37.3	12:14
Jupiter	246°27.9	01:03
Saturn	2°46.2	17:15

Horizontal parallax	
Venus:	0.1
Mars:	0.1

Sun				Moon				
Sat	GHA	Dec		GHA	ν	Dec	d	HP
0	179°46.3	S23°19.5		100°21.0	13.9'	S00°38.0	-15.9'	57.6'
1	194°45.9	19.4		114°54.0	13.9'	00°22.1	-15.9'	57.6'
2	209°45.6	19.3		129°26.9	13.9'	S00°06.2	-15.9'	57.7'
3	224°45.3	· · 19.2		143°59.7	13.8'	N00°09.7	15.9'	57.7'
4	239°45.0	19.1		158°32.6	13.8'	00°25.6	16.0'	57.7'
5	254°44.7	19.0		173°05.4	13.8'	00°41.6	16.0'	57.8'
6	269°44.4	S23°18.8		187°38.2	13.7'	N00°57.5	16.0'	57.8'
7	284°44.1	18.7		202°10.9	13.7'	01°13.5	16.0'	57.8'
8	299°43.8	18.6		216°43.7	13.7'	01°29.5	16.0'	57.9'
9	314°43.5	· · 18.5		231°16.3	13.6'	01°45.5	16.0'	57.9'
10	329°43.2	18.4		245°49.0	13.6'	02°01.5	16.0'	57.9'
11	344°42.9	18.3		260°21.6	13.6'	02°17.5	16.0'	57.9'
12	359°42.6	S23°18.2		274°54.1	13.5'	N02°33.5	16.0'	58.0'
13	14°42.3	18.1		289°26.6	13.5'	02°49.6	16.0'	58.0'
14	29°42.0	17.9		303°59.1	13.4'	03°05.6	16.0'	58.0'
15	44°41.7	· · 17.8		318°31.5	13.4'	03°21.6	16.0'	58.1'
16	59°41.3	17.7		333°03.9	13.3'	03°37.6	16.0'	58.1'
17	74°41.0	17.6		347°36.2	13.3'	03°53.7	16.0'	58.1'
18	89°40.7	S23°17.5		2°08.5	13.2'	N04°09.7	16.0'	58.2'
19	104°40.4	17.3		16°40.7	13.2'	04°25.7	16.0'	58.2'
20	119°40.1	17.2		31°12.9	13.1'	04°41.8	16.0'	58.2'
21	134°39.8	· · 17.1		45°45.0	13.1'	04°57.8	16.0'	58.2'
22	149°39.5	17.0		60°17.1	13.0'	05°13.8	16.0'	58.3'
23	164°39.2	16.9		74°49.1	12.9'	05°29.8	16.0'	58.3'
SD = 16.3' d = -0.1'				SD = 15.7'				
Sun	GHA	Dec		GHA	ν	Dec	d	HP
0	179°38.9	S23°16.7		89°21.0	12.9'	N05°45.8	16.0'	58.3'
1	194°38.6	16.6		103°52.9	12.8'	06°01.8	16.0'	58.4'
2	209°38.3	16.5		118°24.7	12.8'	06°17.7	16.0'	58.4'
3	224°38.0	· · 16.4		132°56.5	12.7'	06°33.7	15.9'	58.4'
4	239°37.7	16.2		147°28.1	12.6'	06°49.6	15.9'	58.5'
5	254°37.4	16.1		161°59.8	12.6'	07°05.6	15.9'	58.5'
6	269°37.1	S23°16.0		176°31.3	12.5'	N07°21.5	15.9'	58.5'
7	284°36.8	15.8		191°02.8	12.4'	07°37.4	15.9'	58.6'
8	299°36.5	15.7		205°34.2	12.3'	07°53.2	15.9'	58.6'
9	314°36.2	· · 15.6		220°05.6	12.3'	08°09.1	15.8'	58.6'
10	329°35.8	15.4		234°36.8	12.2'	08°24.9	15.8'	58.6'
11	344°35.5	15.3		249°08.0	12.1'	08°40.7	15.8'	58.7'
12	359°35.2	S23°15.2		263°39.1	12.0'	N08°56.5	15.8'	58.7'
13	14°34.9	15.0		278°10.2	12.0'	09°12.3	15.7'	58.7'
14	29°34.6	14.9		292°41.1	11.9'	09°28.0	15.7'	58.8'
15	44°34.3	· · 14.8		307°12.0	11.8'	09°43.7	15.7'	58.8'
16	59°34.0	14.6		321°42.8	11.7'	09°59.3	15.6'	58.8'
17	74°33.7	14.5		336°13.5	11.6'	10°15.0	15.6'	58.9'
18	89°33.4	S23°14.4		350°44.1	11.5'	N10°30.5	15.6'	58.9'
19	104°33.1	14.2		5°14.7	11.4'	10°46.1	15.5'	58.9'
20	119°32.8	14.1		19°45.1	11.4'	11°01.6	15.5'	58.9'
21	134°32.5	· · 13.9		34°15.5	11.3'	11°17.1	15.4'	59.0'
22	149°32.2	13.8		48°45.7	11.2'	11°32.5	15.4'	59.0'
23	164°31.9	13.6		63°15.9	11.1'	11°47.9	15.3'	59.0'
SD = 16.3' d = -0.1'				SD = 15.9'				
Mon	GHA	Dec		GHA	ν	Dec	d	HP
0	179°31.6	S23°13.5		77°46.0	11.0'	N12°03.3	15.3'	59.1'
1	194°31.3	13.4		92°16.0	10.9'	12°18.6	15.2'	59.1'
2	209°31.0	13.2		106°45.8	10.8'	12°33.8	15.2'	59.1'
3	224°30.7	· · 13.1		121°15.6	10.7'	12°49.0	15.1'	59.2'
4	239°30.4	12.9		135°45.3	10.6'	13°04.2	15.1'	59.2'
5	254°30.1	12.8		150°14.9	10.5'	13°19.2	15.0'	59.2'
6	269°29.8	S23°12.6		164°44.4	10.4'	N13°34.3	15.0'	59.2'
7	284°29.5	12.5		179°13.8	10.3'	13°49.3	14.9'	59.3'
8	299°29.2	12.3		193°43.0	10.2'	14°04.2	14.9'	59.3'
9	314°28.9	· · 12.2		208°12.2	10.1'	14°19.1	14.8'	59.3'
10	329°28.6	12.0		222°41.3	10.0'	14°33.9	14.7'	59.4'
11	344°28.3	11.9		237°10.2	9.8'	14°48.6	14.7'	59.4'
12	359°28.0	S23°11.7		251°39.1	9.7'	N15°03.3	14.6'	59.4'
13	14°27.7	11.6		266°07.8	9.6'	15°17.9	14.5'	59.4'
14	29°27.4	11.4		280°36.4	9.5'	15°32.4	14.5'	59.5'
15	44°27.1	· · 11.2		295°04.9	9.4'	15°46.8	14.4'	59.5'
16	59°26.8	11.1		309°33.3	9.3'	16°01.2	14.3'	59.5'
17	74°26.4	10.9		324°01.6	9.2'	16°15.5	14.2'	59.5'
18	89°26.1	S23°10.8		338°29.8	9.0'	N16°29.8	14.2'	59.6'
19	104°25.8	10.6		352°57.8	8.9'	16°43.9	14.1'	59.6'
20	119°25.5	10.5		7°25.8	8.8'	16°58.0	14.0'	59.6'
21	134°25.2	· · 10.3		21°53.6	8.7'	17°12.0	13.9'	59.7'
22	149°24.9	10.1		36°21.3	8.6'	17°25.9	13.8'	59.7'
23	164°24.6	10.0		50°48.8	8.4'	17°39.7	13.7'	59.7'
SD = 16.3' d = -0.1'				SD = 16.1'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:26	10:52	■	■	13:11	15:37
N 70°	08:07	09:54	■	■	14:10	15:57
68°	07:51	09:20	■	■	14:44	16:12
66°	07:38	08:55	10:33	13:30	15:09	16:25
64°	07:27	08:35	09:52	14:11	15:28	16:36
62°	07:18	08:20	09:25	14:39	15:44	16:46
60°	07:09	08:06	09:03	15:00	15:57	16:54
N 58°	07:02	07:55	08:46	15:17	16:09	17:01
56°	06:55	07:44	08:32	15:32	16:19	17:08
54°	06:49	07:35	08:19	15:44	16:28	17:14
52°	06:44	07:27	08:08	15:55	16:36	17:20
50°	06:38	07:20	07:58	16:05	16:43	17:25
45°	06:27	07:04	07:38	16:26	16:59	17:37
N 40°	06:17	06:51	07:21	16:42	17:13	17:47
35°	06:07	06:39	07:07	16:56	17:24	17:56
30°	05:59	06:28	06:55	17:09	17:35	18:05
20°	05:42	06:10	06:34	17:30	17:54	18:21
N 10°	05:26	05:52	06:15	17:48	18:11	18:37
0°	05:09	05:35	05:58	18:05	18:28	18:54
S 10°	04:50	05:17	05:40	18:23	18:46	19:13
20°	04:28	04:57	05:22	18:42	19:06	19:35
30°	03:59	04:32	05:00	19:04	19:31	20:04
35°	03:40	04:17	04:47	19:16	19:46	20:23
40°	03:17	03:59	04:32	19:31	20:04	20:46
45°	02:47	03:37	04:14	19:49	20:26	21:16
S 50°	02:02	03:07	03:52	20:12	20:56	22:01
52°	01:35	02:52	03:41	20:22	21:11	22:28
54°	00:51	02:34	03:29	20:34	21:28	23:11
56°	—:—	02:12	03:14	20:48	21:50	—:—
58°	—:—	01:43	02:58	21:05	22:19	—:—
S 60°	—:—	00:55	02:38	21:25	23:06	—:—

Lat.	Moonrise			Moonset		
	Sat	Sun	Mon	Sat	Sun	Mon
N 72°	11:12	10:34	09:41	· · ·	01:20	03:56
N 70°	11:15	10:46	10:09	· · ·	01:11	03:30
68°	11:17	10:56	10:31	· · ·	01:04	03:11
66°	11:19	11:05	10:48	· · ·	00:58	02:56
64°	11:21	11:12	11:02	· · ·	00:53	02:43
62°	11:22	11:18	11:14	· · ·	00:48	02:33
60°	11:24	11:23	11:24	· · ·	00:45	02:24
N 58°	11:25	11:28	11:33	· · ·	00:41	02:16
56°	11:26	11:32	11:40	· · ·	00:38	02:10
54°	11:27	11:36	11:48	· · ·	00:36	02:04
52°	11:28	11:40	11:54	· · ·	00:33	01:58
50°	11:29	11:43	12:00	· · ·	00:31	01:53
45°	11:31	11:50	12:12	· · ·	00:26	01:43
N 40°	11:32	11:56	12:23	· · ·	00:22	01:34
35°	11:33	12:01	12:31	· · ·	00:19	01:27
30°	11:35	12:05	12:39	· · ·	00:16	01:20
20°	11:37	12:13	12:53	· · ·	00:11	01:09
N 10°	11:39	12:20	13:05	· · ·	00:06	00:59
0°	11:40	12:27	13:17	· · ·	00:02	00:50
S 10°	11:42					

December 30, 31, 01 UT (Tue., Wed., Thu.)

Aries		Venus		Mars		Jupiter		Saturn		Stars		
Tue	GHA	GHA	Dec	GHA	Dec	GHA	Dec	GHA	Dec	SHA	Dec	
0	98°41.5	181°22.9	S23°41.9	176°28.7	S23°50.7	345°17.5	N21°56.0	101°24.7	S03°38.9	Alpheratz	357°34.0	29°14.2
1	113°43.9	196°22.0	41.8	191°29.1	50.6	0°20.3	56.1	116°27.0	38.8	Ankaa	353°06.4	-42°10.1
2	128°46.4	211°21.0	41.8	206°29.5	50.4	15°23.1	56.1	131°29.4	38.7	Schedar	349°30.1	56°41.1
3	143°48.8	226°20.0	· · 41.7	221°29.9	· · 50.3	30°25.9	· · 56.2	146°31.7	· · 38.7	Diphda	348°46.5	-17°50.7
4	158°51.3	241°19.0	41.6	236°30.3	50.1	45°28.7	56.2	161°34.1	38.6	Achernar	335°19.4	-57°06.5
5	173°53.8	256°18.1	41.6	251°30.6	50.0	60°31.5	56.3	176°36.4	38.5	Hamal	327°50.1	23°35.3
6	188°56.2	271°17.1	S23°41.5	266°31.0	S23°49.8	75°34.3	N21°56.4	191°38.7	S03°38.5	Polaris	313°17.4	89°22.7
7	203°58.7	286°16.1	41.4	281°31.4	49.7	90°37.1	56.4	206°41.1	38.4	Acamar	315°10.9	-40°12.2
8	219°01.2	301°15.1	41.3	296°31.8	49.5	105°39.9	56.5	221°43.4	38.4	Menkar	314°05.1	4°11.5
9	234°03.6	316°14.2	· · 41.3	311°32.2	· · 49.4	120°42.7	· · 56.5	236°45.8	· · 38.3	Mirfak	308°26.7	49°57.4
10	249°06.1	331°13.2	41.2	326°32.5	49.2	135°45.5	56.6	251°48.1	38.2	Aldebaran	290°38.4	16°33.7
11	264°08.6	346°12.2	41.1	341°32.9	49.1	150°48.3	56.6	266°50.4	38.2	Rigel	281°02.8	-8°10.3
12	279°11.0	1°11.3	S23°41.0	356°33.3	S23°48.9	165°51.2	N21°56.7	281°52.8	S03°38.1	Capella	280°20.2	46°01.5
13	294°13.5	16°10.3	41.0	11°33.7	48.8	180°54.0	56.8	296°55.1	38.0	Bellatrix	278°21.7	6°22.4
14	309°16.0	31°09.3	40.9	26°34.1	48.6	195°56.8	56.8	311°57.4	38.0	Elnath	278°00.5	28°37.8
15	324°18.4	46°08.3	· · 40.8	41°34.4	· · 48.5	210°59.6	· · 56.9	326°59.8	· · 37.9	Alnilam	275°36.6	-1°11.1
16	339°20.9	61°07.4	40.7	56°34.8	48.3	226°02.4	56.9	342°02.1	37.8	Betelgeuse	270°50.9	7°24.7
17	354°23.3	76°06.4	40.6	71°35.2	48.2	241°05.2	57.0	357°04.5	37.8	Canopus	263°51.5	-52°42.5
18	9°25.8	91°05.4	S23°40.5	86°35.6	S23°48.0	256°08.0	N21°57.0	12°06.8	S03°37.7	Sirius	258°25.2	-16°45.1
19	24°28.3	106°04.5	40.4	101°35.9	47.9	271°10.8	57.1	27°09.1	37.7	Adhara	255°04.8	-29°00.4
20	39°30.7	121°03.5	40.4	116°36.3	47.7	286°13.6	57.2	42°11.5	37.6	Procyon	244°49.7	5°09.5
21	54°33.2	136°02.5	· · 40.3	131°36.7	· · 47.5	301°16.4	· · 57.2	57°13.8	· · 37.5	Pollux	243°16.0	27°57.7
22	69°35.7	151°01.5	40.2	146°37.1	47.4	316°19.2	57.3	72°16.2	37.5	Avior	234°13.8	-59°35.4
23	84°38.1	166°00.6	40.1	161°37.5	47.2	331°22.0	57.3	87°18.5	37.4	Suhail	222°45.4	-43°32.1
Mer.pass. 17:22		ν -1.0' d -0.1' m -3.90		ν 0.4' d -0.1' m 1.14		ν 2.8' d 0.1' m -2.67		ν 2.3' d -0.1' m 1.00		Miaplacidus	221°37.4	-69°49.2
										Alphard	217°46.7	-8°46.3
										Regulus	207°33.4	11°50.3
										Dubhe	193°39.7	61°36.3
										Denebola	182°24.1	14°25.5
										Gienah	175°42.7	-17°41.1
										Acrux	172°59.2	-63°14.3
										Gacrux	171°50.7	-57°15.3
										Alioth	166°12.3	55°48.8
										Spica	158°21.5	-11°17.8
										Alkaid	152°51.5	49°10.7
										Hadar	148°35.1	-60°29.7
										Menkent	147°56.8	-36°29.7
										Arcturus	145°47.3	19°02.7
										Rigil Kent.	139°39.5	-60°56.3
										Kochab	137°20.5	74°02.6
										Zuben'ubi	136°55.3	-16°09.0
										Alphecca	126°03.3	26°37.5
										Antares	112°15.2	-26°29.3
										Atria	107°09.2	-69°04.3
										Sabik	102°02.2	-15°45.4
										Shaula	96°09.7	-37°07.3
										Rasalhague	95°58.1	12°32.4
										Eltanin	90°42.3	51°29.0
										Kaus Aust.	83°31.9	-34°22.3
										Vega	80°33.1	38°48.4
										Nunki	75°47.1	-26°15.9
										Altair	61°59.5	8°56.2
										Peacock	53°05.0	-56°39.2
										Deneb	49°25.7	45°22.5
										Enif	33°38.2	9°59.7
										Al Na'ir	27°32.2	-46°50.3
										Fomalhaut	15°13.8	-29°29.2
										Scheat	13°44.6	28°13.6
										Markab	13°29.2	15°20.8
										Dec 30 Tue		
										Venus	82°41.5	11:55
										Mars	77°47.3	12:14
										Jupiter	246°36.0	00:59
										Saturn	2°43.2	17:12
										Dec 31 Wed		
										Venus	81°19.0	11:57
										Mars	76°57.3	12:13
										Jupiter	246°44.3	00:54
										Saturn	2°40.2	17:08
										Jan 01 Thu		
										Venus	79°56.6	11:58
										Mars	76°07.2	12:13
										Jupiter	246°52.5	00:50
										Saturn	2°37.1	17:04
										Horizontal parallax		
										Venus:	0.1	
										Mars:	0.1	
Mer.pass. 17:15		ν -1.0' d -0.1' m -3.90		ν 0.4' d -0.2' m 1.12		ν 2.8' d 0.1' m -2.67		ν 2.3' d -0.1' m 1.01				

Sun				Moon				
Tue	GHA	Dec		GHA	ν	Dec	d	HP
0	179°24.3	S23°09.8		65°16.3	8.3'	N17°53.4	13.6'	59.7'
1	194°24.0	09.6		79°43.6	8.2'	18°07.0	13.5'	59.8'
2	209°23.7	09.5		94°10.8	8.1'	18°20.6	13.4'	59.8'
3	224°23.4	· · 09.3		108°37.8	7.9'	18°34.0	13.3'	59.8'
4	239°23.1	09.1		123°04.8	7.8'	18°47.3	13.2'	59.8'
5	254°22.8	09.0		137°31.6	7.7'	19°00.6	13.1'	59.9'
6	269°22.5	S23°08.8		151°58.3	7.6'	N19°13.7	13.0'	59.9'
7	284°22.2	08.6		166°24.9	7.4'	19°26.8	12.9'	59.9'
8	299°21.9	08.5		180°51.3	7.3'	19°39.7	12.8'	59.9'
9	314°21.6	· · 08.3		195°17.6	7.2'	19°52.5	12.7'	60.0'
10	329°21.3	08.1		209°43.8	7.1'	20°05.2	12.6'	60.0'
11	344°21.0	08.0		224°09.9	6.9'	20°17.8	12.5'	60.0'
12	359°20.7	S23°07.8		238°35.8	6.8'	N20°30.3	12.4'	60.0'
13	14°20.4	07.6		253°01.6	6.7'	20°42.6	12.2'	60.1'
14	29°20.1	07.4		267°27.2	6.5'	20°54.9	12.1'	60.1'
15	44°19.8	· · 07.3		281°52.8	6.4'	21°07.0	12.0'	60.1'
16	59°19.5	07.1		296°18.2	6.3'	21°19.0	11.9'	60.1'
17	74°19.2	06.9		310°43.4	6.1'	21°30.9	11.7'	60.1'
18	89°18.9	S23°06.7		325°08.6	6.0'	N21°42.6	11.6'	60.2'
19	104°18.6	06.6		339°33.6	5.9'	21°54.2	11.5'	60.2'
20	119°18.4	06.4		353°58.5	5.7'	22°05.7	11.3'	60.2'
21	134°18.1	· · 06.2		8°23.2	5.6'	22°17.0	11.2'	60.2'
22	149°17.8	06.0		22°47.8	5.5'	22°28.2	11.1'	60.3'
23	164°17.5	05.8		37°12.3	5.4'	22°39.3	10.9'	60.3'
SD = 16.3'			$d = -0.2'$	SD = 16.3'				
Wed	GHA	Dec		GHA	ν	Dec	d	HP
0	179°17.2	S23°05.7		51°36.7	5.2'	N22°50.2	10.8'	60.3'
1	194°16.9	05.5		66°00.9	5.1'	23°01.0	10.6'	60.3'
2	209°16.6	05.3		80°25.0	5.0'	23°11.6	10.5'	60.3'
3	224°16.3	· · 05.1		94°49.0	4.8'	23°22.1	10.3'	60.3'
4	239°16.0	04.9		109°12.8	4.7'	23°32.4	10.2'	60.4'
5	254°15.7	04.7		123°36.5	4.6'	23°42.6	10.0'	60.4'
6	269°15.4	S23°04.5		138°00.1	4.5'	N23°52.6	9.9'	60.4'
7	284°15.1	04.4		152°23.6	4.3'	24°02.5	9.7'	60.4'
8	299°14.8	04.2		166°46.9	4.2'	24°12.2	9.5'	60.4'
9	314°14.5	· · 04.0		181°10.2	4.1'	24°21.8	9.4'	60.5'
10	329°14.2	03.8		195°33.2	4.0'	24°31.1	9.2'	60.5'
11	344°13.9	03.6		209°56.2	3.9'	24°40.4	9.1'	60.5'
12	359°13.6	S23°03.4		224°19.1	3.7'	N24°49.4	8.9'	60.5'
13	14°13.3	03.2		238°41.8	3.6'	24°58.3	8.7'	60.5'
14	29°13.0	03.0		253°04.4	3.5'	25°07.0	8.5'	60.5'
15	44°12.7	· · 02.8		267°26.9	3.4'	25°15.5	8.4'	60.5'
16	59°12.4	02.6		281°49.3	3.3'	25°23.9	8.2'	60.6'
17	74°12.1	02.4		296°11.6	3.2'	25°32.1	8.0'	60.6'
18	89°11.8	S23°02.2		310°33.8	3.1'	N25°40.1	7.8'	60.6'
19	104°11.5	02.0		324°55.8	2.9'	25°47.9	7.6'	60.6'
20	119°11.2	01.8		339°17.8	2.8'	25°55.5	7.5'	60.6'
21	134°10.9	· · 01.6		353°39.6	2.7'	26°03.0	7.3'	60.6'
22	149°10.6	01.4		8°01.3	2.6'	26°10.3	7.1'	60.6'
23	164°10.3	01.2		22°23.0	2.5'	26°17.3	6.9'	60.6'
SD = 16.3'			$d = -0.2'$	SD = 16.4'				
Thu	GHA	Dec		GHA	ν	Dec	d	HP
0	179°10.1	S23°01.0		36°44.5	2.4'	N26°24.2	6.7'	60.7'
1	194°09.8	00.8		51°06.0	2.3'	26°30.9	6.5'	60.7'
2	209°09.5	00.6		65°27.3	2.3'	26°37.4	6.3'	60.7'
3	224°09.2	· · 00.4		79°48.6	2.2'	26°43.7	6.1'	60.7'
4	239°08.9	00.2		94°09.7	2.1'	26°49.8	5.9'	60.7'
5	254°08.6	23°00.0		108°30.8	2.0'	26°55.7	5.7'	60.7'
6	269°08.3	S22°59.8		122°51.8	1.9'	N27°01.5	5.5'	60.7'
7	284°08.0	59.6		137°12.7	1.8'	27°07.0	5.3'	60.7'
8	299°07.7	59.4		151°33.5	1.8'	27°12.3	5.1'	60.7'
9	314°07.4	· · 59.2		165°54.3	1.7'	27°17.4	4.9'	60.7'
10	329°07.1	59.0		180°14.9	1.6'	27°22.3	4.7'	60.7'
11	344°06.8	58.8		194°35.6	1.5'	27°27.0	4.5'	60.7'
12	359°06.5	S22°58.6		208°56.1	1.5'	N27°31.4	4.3'	60.7'
13	14°06.2	58.3		223°16.6	1.4'	27°35.7	4.1'	60.8'
14	29°05.9	58.1		237°37.0	1.4'	27°39.8	3.9'	60.8'
15	44°05.6	· · 57.9		251°57.3	1.3'	27°43.6	3.6'	60.8'
16	59°05.4	57.7		266°17.6	1.2'	27°47.3	3.4'	60.8'
17	74°05.1	57.5		280°37.9	1.2'	27°50.7	3.2'	60.8'
18	89°04.8	S22°57.3		294°58.1	1.2'	N27°53.9	3.0'	60.8'
19	104°04.5	57.1		309°18.2	1.1'	27°56.9	2.8'	60.8'
20	119°04.2	56.8		323°38.3	1.1'	27°59.7	2.6'	60.8'
21	134°03.9	· · 56.6		337°58.4	1.0'	28°02.2	2.3'	60.8'
22	149°03.6	56.4		352°18.5	1.0'	28°04.6	2.1'	60.8'
23	164°03.3	56.2		6°38.5	1.0'	28°06.7	1.9'	60.8'
SD = 16.3'			$d = -0.2'$	SD = 16.5'				

Lat.	Twilight		Sunrise	Sunset	Twilight	
	Naut.	Civil			Civil	Naut.
N 72°	08:25	10:45	■	■	13:21	15:42
N 70°	08:06	09:51	■	■	14:16	16:01
68°	07:50	09:18	■	■	14:49	16:16
66°	07:38	08:54	10:29	13:37	15:13	16:29
64°	07:27	08:35	09:50	14:16	15:32	16:40
62°	07:18	08:19	09:23	14:43	15:47	16:49
60°	07:09	08:06	09:03	15:04	16:01	16:57
N 58°	07:02	07:54	08:46	15:21	16:12	17:04
56°	06:56	07:44	08:31	15:35	16:22	17:11
54°	06:50	07:36	08:19	15:47	16:31	17:17
52°	06:44	07:28	08:08	15:58	16:39	17:22
50°	06:39	07:20	07:59	16:08	16:46	17:27
45°	06:27	07:05	07:38	16:28	17:02	17:39
N 40°	06:17	06:51	07:22	16:45	17:15	17:49
35°	06:08	06:40	07:08	16:58	17:27	17:58
30°	06:00	06:29	06:56	17:11	17:37	18:07
20°	05:43	06:11	06:35	17:31	17:55	18:23
N 10°	05:27	05:54	06:17	17:50	18:12	18:39
0°	05:11	05:37	05:59	18:07	18:29	18:55
S 10°	04:52	05:19	05:42	18:24	18:47	19:14
20°	04:30	04:59	05:23	18:43	19:07	19:36
30°	04:01	04:34	05:02	19:04	19:32	20:05
35°	03:43	04:19	04:49	19:17	19:47	20:23
40°	03:20	04:01	04:34	19:32	20:05	20:46
45°	02:50	03:39	04:16	19:50	20:27	21:16
S 50°	02:06	03:10	03:54	20:12	20:56	22:00
52°	01:39	02:55	03:44	20:22	21:10	22:26
54°	00:58	02:38	03:32	20:34	21:28	23:06
56°	—:—	02:16	03:18	20:48	21:49	—:—
58°	—:—	01:48	03:01	21:04	22:17	—:—
S 60°	—:—	01:03	02:42	21:24	23:01	—:—

Lat.	Moonrise			Moonset		
	Tue	Wed	Thu	Tue	Wed	Thu
N 72°	□	□	□	□	□	□
N 70°	08:43	□	□	06:48	□	□
68°	09:47	□	□	05:46	□	□
66°	10:23	□	□	05:11	□	□
64°	10:49	10:25	□	04:46	07:12	□
62°	11:09	11:04	10:58	04:27	06:34	08:53
60°	11:26	11:32	11:50	04:11	06:07	08:02
N 58°	11:40	11:54	12:21	03:58	05:45	07:31
56°	11:52	12:12	12:45	03:47	05:28	07:07
54°	12:03	12:27	13:05	03:37	05:14	06:48
52°	12:13	12:40	13:21	03:28	05:01	06:32
50°	12:21	12:52	13:36	03:20	04:50	06:18
45°	12:40	13:16	14:05	03:03	04:27	05:49
N 40°	12:55	13:35	14:27	02:50	04:08	05:27
35°	13:08	13:52	14:46	02:38	03:53	05:09
30°	13:19	14:06	15:03	02:28	03:40	04:53
20°	13:38	14:30	15:30	02:11	03:17	04:26
N 10°	13:55	14:52	15:54	01:56	02:57	04:03
0°	14:11	15:11	16:16	01:42	02:39	03:42
S 1						